Algebraic Geometry

— ***EYNTKA*** **Series** —

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Contents

Contents **3**

Contents **11**

Preliminary Chapter **i**

0.1 Recall Classical Algebraic Geometry i

0.2 Recall Projective Spaces x

0.3 Noetherian Induction or Well-founded Induction xvii

0.4 Categorical Reminder xviii

0.4.1 Summary: Categorical Tools for Schemes xxvii

0.5 *Key Results from Differential Geometry xxvii

0.5.1 Summary: Differential vs. Algebraic Geometry xxxviii

0.6 *Key Results from Complex Geometry xxxviii

0.6.1 Summary: Complex Geometry and Algebraic Geometry l

1 Sheaves **1**

1.1 Definitions and Examples 4

1.1.1 Stalks and Sheaves 14

1.1.2 Sheaf from a Topological Base 16

1.1.3 Gluing Sheaves on an Open Cover 18

1.2 Morphisms and Exactness 20

1.3 Grothendieck's Functors: Pushforward, Pullback, and more 43

1.3.1 Pushforward and Pullback 43

1.3.2 Proper Direct and Inverse Image . . . 49
1.3.3 Hom and Tensor . . . 51
1.4 Ringed Space and Locally Ringed Space . . . 52
1.5⋆ Diffeology . . . 55

2 Schemes I: Basics **57**
2.1 Defining Spec . . . 59
2.1.1 Zariski Topology . . . 68
2.1.2 (Topological) Basis: Distinguished Open Sets . . . 73
2.1.3 Topological Properties . . . 76
2.1.4 Ring to Top Functor: Induced Spectra Maps . . . 80
2.1.5 *Subsets of Spectra via quotient and Localization . . . 84
2.2 Schemes . . . 87
2.2.1 The Structure Sheaf for Affine Schemes . . . 88
2.2.2 Ideal Correspondence in Affine Schemes . . . 95
2.2.3 Schemes . . . 102
2.2.4 Comparing Schemes and Real/Complex Manifolds . . . 109
2.2.5 Evaluation and Stalks on Schemes . . . 113
2.2.6 Independence of Affine Covering . . . 118
2.2.7 *Affine Atlases . . . 123
2.2.8 Subschemes: Open . . . 127
2.2.9 Gluing (Coproduct) of Schemes . . . 130
2.2.10 *Gluing Affine Schemes and Fiber Products of Rings . . . 135
2.3 Projective Schemes . . . 137
2.4 Properties of Schemes . . . 146
2.4.1 Reducedness and Integrality . . . 153
2.4.2 Normality and Factoriality . . . 159
2.5 Associated Points . . . 162
2.5.1 Associated Primes of Modules . . . 162
2.5.2 Associated Points of Schemes . . . 164
2.6 Dimension . . . 168
2.6.1 Codimension . . . 170
2.6.2 Transcendence Dimension . . . 171
2.6.3 Krull's Height Theorem and Algebraic Hartogs' Lemma . . . 172
2.7 Regular and Smooth Schemes . . . 175
2.7.1 Unramified and Étale Morphisms . . . 182

2.7.2 Properties of Regular Schemes . . . 186
2.7.3 Dimension and Regularity for Schemes Over $\mathbb{Z}$ . . . 187
2.8⋆ *How Much Does the Topology Know? . . . 188

3 Schemes II: Morphisms and Constructions 193
3.1 Elementary Concepts . . . 195
3.1.1 Scheme Morphisms . . . 201
3.1.2 S-Schemes . . . 207
3.1.3 Subschemes: Closed, Locally Closed, and More . . . 211
3.1.4 More on Projective Schemes . . . 221
3.1.5 The Category of Schemes . . . 224
3.2 (Fiber) Product and Fibers . . . 228
3.2.1 Fibers and family of Schemes . . . 234
3.2.2 *Monomorphisms in Sch . . . 239
3.3 The Functor of Points . . . 240
3.3.1 From Evaluation to Morphisms . . . 241
3.3.2 Z-valued Points . . . 242
3.3.3 The Yoneda Embedding for Schemes . . . 245
3.3.4 Key Examples . . . 246
3.3.5 Representable Functors in Algebraic Geometry . . . 249
3.3.6 Hilbert Schemes . . . 251
3.3.7 Schemes as Functors . . . 253
3.3.8 Moduli Problems and the Limits of Representability . . . 255
3.3.9 *The Fourier–Mukai Transform . . . 256
3.4 Finiteness and Affine Conditions . . . 259
3.5 Separated and Proper Schemes . . . 282
3.5.1 Separated Morphisms . . . 282
3.5.2 Valuation Criterion for Separatedness . . . 290
3.5.3 Universally Closed Morphisms . . . 298
3.5.4 Proper Morphisms . . . 300
3.6 Rational Maps . . . 315
3.7 Abstract Varieties . . . 319
3.8⋆ Bertini Theorems . . . 330
3.9⋆ The Category of Affine Schemes as Ringed Spaces . . . 358
3.10⋆Limits of Schemes . . . 360
3.11⋆Finite Presentation and Limits . . . 368

4⋆ Schemes II.5: Group Schemes **381**
4.1 Definition and First Examples . . . 382
4.1.1 The Hopf Algebra Perspective . . . 387
4.1.2 Finite and Infinitesimal Group Schemes . . . 389
4.2 Homomorphisms, Kernels, and Exact Sequences . . . 391
4.3 Actions of Group Schemes on Schemes . . . 392
4.4 Quotients by Group Actions . . . 393
4.5 The Lie Algebra of a Group Scheme . . . 395
4.6 Cartier Duality . . . 397
4.7 Rational Maps into Group Schemes . . . 398

5 Schemes III: Sheaves of Modules, Divisors, and Differentials **411**
5.1 Definition and Basic Properties . . . 412
5.1.1 Quasi-Coherent and Coherent Sheaves . . . 414
5.1.2 Properties of Quasi-Coherent Sheaves . . . 419
5.1.3 Locally Free Sheaves and Vector Bundles . . . 423
5.1.4 Ideal Sheaves and Closed Subschemes . . . 428
5.1.5 Relative Spectrum . . . 431
5.1.6 Regular Embeddings and Local Complete Intersections . . . 433
5.2 Images of Scheme Morphisms . . . 437
5.2.1 Chevalley's Theorem . . . 438
5.2.2 Application: Fiber Dimension . . . 447
5.2.3 The Scheme-Theoretic Image . . . 451
5.3 Q.C. Sheaves on Projective Schemes . . . 456
5.3.1 The Tilde Construction on Proj . . . 456
5.3.2 Characterizing Projective Subschemes . . . 466
5.3.3 Maps to Projective Space . . . 469
5.3.4 Ample Sheaves . . . 474
5.3.5 Characterizing Morphisms to Projective Space . . . 482
5.3.6 Relative Proj and Projective Bundles . . . 486
5.3.7 The Grassmannian and Its Tautological Sequence . . . 492
5.4 Divisors: Geometric Representation of Line Bundles . . . 496
5.4.1 Weil Divisors . . . 497
5.4.2 Cartier Divisors . . . 512
5.4.3 The Picard Group . . . 522
5.4.4⋆ Families of Weil Divisors and Birational Transforms . . . 528

5.4.5⋆ Birational Transforms of Linear Systems . . . 540
5.5 Differentials . . . 551
5.5.1 Kähler Differentials . . . 552
5.5.2 Differentials and Local Rings . . . 557
5.5.3 The Sheaf of Differentials . . . 559
5.5.4 Smoothness and the Cotangent Sheaf . . . 563
5.5.5 Tangent Sheaf, Canonical Sheaf, and the Adjunction Formula . . . 566
5.5.6 *Invariant Differentials on Group Schemes . . . 569
5.6⋆ *Formal Schemes and Local Geometry . . . 570

6 Cohomology of Schemes 575
6.1 Definition and Basic Properties . . . 576
6.1.1 Computation: Čech Cohomology . . . 583
6.1.2 Refinement and the Degree-One Comparison . . . 590
6.1.3 Soft Sheaves and the Paracompact Comparison . . . 592
6.2 Vanishing Theorems . . . 595
6.2.1 Dimension Theorem for Schemes . . . 595
6.2.2 Vanishing Theorem for Affine Schemes . . . 599
6.3 Application: Sheaf Cohomology of Projective Spaces . . . 603
6.3.1 Cohomology of Twisted Sheaves . . . 603
6.3.2 Cohomology of Twisted Sheaves of Differentials . . . 607
6.3.3⋆ Separating Points and Jets . . . 609
6.4 Ext Groups and Sheaf Ext . . . 613
6.4.1 Global and Local Ext . . . 613
6.4.2 The Local-to-Global Spectral Sequence . . . 617
6.4.3 Computations on Projective Space . . . 619
6.5 Serre Duality Theorem . . . 622
6.5.1 Duality on Projective Space . . . 622
6.5.2 Duality for Projective Schemes . . . 626
6.6 Higher Direct Images . . . 631
6.6.1 Definition and First Properties . . . 631
6.6.2 The Finiteness Theorem . . . 634
6.6.3 The Proper Case . . . 636
6.6.4 Computation of Higher Direct Images . . . 639
6.7 Euler Characteristics and Hilbert Polynomials . . . 643
6.7.1 The Euler Characteristic . . . 644

6.7.2 Hilbert Polynomials . . . 646
6.7.3 Riemann–Roch for Curves . . . 649
6.8 Flat Morphisms and Semicontinuity . . . 652
6.8.1 Flat Families . . . 653
6.8.2 The Flat Base Change Theorem . . . 659
6.8.3 Semicontinuity . . . 661
6.8.4 Grauert's Theorem and Cohomology and Base Change . . . 664
6.8.5 Recognizing Flatness Cohomologically . . . 667
6.9 *GAGA . . . 669
6.9.1 Analytification . . . 669
6.9.2 Statement of the GAGA Theorems . . . 671
6.9.3 Consequences . . . 673
6.9.4 Summary . . . 678

7 Application: Algebraic Curves 681
7.1 Curves and Their Function Fields . . . 682
7.2 The Riemann-Roch Theorem . . . 694
7.2.1 Divisors on Curves: Degree, Linear Systems, and $\ell(D)$ . . . 694
7.2.2 The Canonical Divisor and Genus . . . 696
7.2.3 Statement and Proof of Riemann-Roch . . . 700
7.3 Applications of Riemann-Roch . . . 704
7.4 Ramification and Hurwitz's Formula . . . 712
7.4.1 Ramification . . . 712
7.4.2 Hurwitz's Formula . . . 715
7.4.3 Applications of Hurwitz's Formula . . . 717
7.5 Hyperelliptic Curves . . . 721
7.6 Elliptic Curves . . . 724
7.7 Further Topics on Curves . . . 732
7.7.1 Riemann-Roch in Families and Semicontinuity . . . 733
7.7.2 Curves over Non-Algebraically-Closed Fields . . . 735

8 Application: Algebraic Surfaces 737
8.1 Surfaces and the Néron-Severi Group . . . 738
8.2 Intersection Theory on Surfaces . . . 742
8.3 Riemann-Roch and the Hodge Index Theorem . . . 749
8.3.1 The Riemann-Roch Theorem for Surfaces . . . 750

8.3.2 Noether's Formula . . . 751
8.3.3 The Hodge Index Theorem . . . 753
8.3.4 The Picard Scheme and Algebraic Equivalence . . . 757
8.4 Blow-ups . . . 758
8.4.1 Hironaka's Example . . . 770
8.5 Ruled Surfaces and Rational Surfaces . . . 772
8.5.1 Rational Surfaces . . . 774
8.6 The Enriques-Kodaira Classification . . . 775
8.6.1 Kodaira Dimension . . . 776
8.6.2 Minimal Models . . . 776
8.6.3 Overview of the Classification . . . 778
8.6.4 The Classification Table . . . 780
8.6.5 Vanishing Theorems . . . 782
8.7 Application: Counting Points on Curves over Finite Fields . . . 783
8.7.1 Arithmetic Surfaces and Integral Points . . . 795

9 What Next? 799

Index 802

Contents

Algebraic Geometry originated as the study of the zero-sets of polynomials, namely it was (and in many ways behind all the abstraction predominantly still is) the solution to a system of polynomials. By explicitly working with the zero sets of polynomials, we can discover a great multitude of geometric results within algebra, however as the field progressed it was discovered zero sets are in many ways insufficient to capture all the geometric data. For some quick examples:

1. it was discovered in the mid 20th century that results in number theory seem eerily similar to the study of 1 dimensional curves.
2. The galois group of a field share many of the same properties of as the fundamental group of a topological space.
3. The fact that a polynomial may have a root of high multiplicity is lost in the classical geometric picture $V(f)$, but this is often key information.
4. Number theorists do not work over algebraically closed fields as they are looking for integer (or integral) solutions to polynomial equations. However, these do not have the same nice geometric correspondence within classical algebraic geometry as the Nullstellensatz works well when the field is algebraically closed.
5. All rings can be represented as the quotient of $\mathbb{Z}[x_i, \mid i \in \mathcal{I}]$ or $k[x_i, \mid i \in \mathcal{I}]$ for some indexing set I, hence all ring (not just reduced finitely generated rings over algebraically closed fields) have some notion of being a function with evaluation
6. When studying a family of varieties (ex. $x^2 + ty^2 = 1$ as t varies) the resulting object will often have "bad" singularities which varieties are poorly equipped to work with.

Such realizations lead to the need to ground algebraic geometry with a new object that encapsulates all of these ideas. this new object is the *scheme*. As compared to varieties that requires an ambient space and working over polynomial rings over fields, schemes are *intrinsic* in that they do not require any ambient space to be defined in, and they generalize classical algebraic geometry to be able to capture the geometry of both Diophantine equations (for number theory) as well as the geometry of (quasi-projective) varieties and other more novel. Perhaps key in their success over other variations is how well they also remember "multiplicity" information of roots of polynomials. This data shall be paramount in many results we shall introduce[1].

This book shall be written with graduate or well-read undergraduates in mind. The reason is to show how inter-connected schemes are to various other fields of mathematics. For example:

1. The geometry of schemes draws many parallels to complex geometry and some results from complex analysis (ex. Hartog's lemma from Complex analysis over several variables shall have a strong appearance when discussing normal schemes).
2. A sheaf draws many similarities to the bundles that are put on manifolds; in fact a manifold can be defined equivalently by using a sheaf of differentiable functions instead of an atlas. Some results such as Poincaré duality shall be proven for schemes (see 6.5)
3. Chapter 6 shall strongly rely on the readers intuitions from algebraic topology and use modern homological algebra.

[1]We shall see for Elliptic Curves that the finite-generation of certain torsion groups shall rely on Schemes in a key way to remember multiplicity

4. Many of the properties in commutative algebra shall become interesting geometric properties in algebraic geometry.

Essentially, anything non-optional material from EYNTKA Classical Algebraic Geometry ([22]), EYNTKA Differential Topology ([59]), and EYNTKA Algebraic Topology ([57]) will be considered as assumed prerequisites. Some results from EYNTKA complex analysis ([58]) will be drawn on as well given the strong connection between complex geometry and the geometry of schemes.

All essential category theory will be assumed; it can be reviewed in EYNTKA Undergraduate Algebra.

Throughout this book, we shall use the following terminology:

- Let k be a field, $\overline{k}$ the algebraic closure of k
- Let R an arbitrary ring, $M_n(R)$ the matrix ring over R, $R_{\mathfrak{p}}$ the localization of R by $R \setminus \mathfrak{p}$, R_f the localization of R by $\{f, f^2, ...\}$, $\mathrm{Frac}(R)$ the field of fractions of R is R is an integral domain
- A is an arbitrary R-algebra.

see [22].

adding new citation: [52]

The material covered in the preliminary chapter is at a break-neck pace and is only there for expository purposes for the reader to see if they have the right intuitions to cover algebraic geometry in its greater generality. The first main Part, Scheme Theory, focuses heavily on building up all the necessary theoretical framework to have a comfortable grasp of the material and is heavily inspired by the many famous texts covering the material (ex. Hartshorne, Shafarevich, Vakil, Serre, Eisenbud, and so forth).

Preliminary Chapter

In this chapter, we shall quickly review elementary concepts linking algebra and geometry without doing any proofs, that is we are going to explore some classical algebraic geometry. For all the important details, see [22].

0.1 Recall Classical Algebraic Geometry

If A is an R-algebra, then by the universal property of free R-algebras we have if S is a set of generators for A,

$$\begin{array}{ccc} R[S] & \xrightarrow{\varphi} & A \\ {\scriptstyle\iota}\uparrow & \nearrow_{\iota} & \\ S & & \end{array}$$

If S is finite, then $R[S] \cong R[x_1, x_2, \ldots, x_n]$ by relabelling S. Since every ring is a $\mathbb{Z}$-algebra, we see that polynomial rings are central to the study of general rings. It is thus fruitful to realize that polynomial rings can naturally be interpreted as functions, that is:

$$f \in R[S],\ R^S \to R[S]^\vee \qquad f \mapsto f(s_i)_{i \in S}$$

In the finite case:

$$f \in R[x_1, x_2, \ldots, x_n], \qquad R^n \to R[x_1, x_2, \ldots, x_n]^\vee \qquad (a_1, a_2, \ldots, a_n) \mapsto (f \mapsto f(a_1, a_2, \ldots, a_n))$$

We do not want to consider R^S as having any structure besides being an affine space, hence we shall label it as $\mathbb{A}^S_R$ or in the finite case $\mathbb{A}^n_R$. If the ring is evident in context, we shall write $\mathbb{A}^n$.

0.1.1 Conventions in Number Theory In a number-theoretic context, $\mathbb{A}^n$ often implicitly means $\mathbb{A}^n_{\mathbb{C}}$ (or $\mathbb{A}^n_{\overline{\mathbb{Q}}}$), with $\mathbb{A}^n(R)$ denoting the R-points. This convention arises because polynomials over $\mathbb{C}$ always have solutions, which is not guaranteed over $\mathbb{R}$ or $\mathbb{Q}$.

For now, we shall stick to the finitely generated case. Note that in $\mathbb{A}^n_R$, the point $(0,\dots,0)$ has no special property, as we are treating this space as simply a collection of points. To emphasize this geometric interpretation, any $p \in \mathbb{A}^n_R$ will be called a *point*, and for

$$p = (r_1, r_2, \dots, r_n)$$

each r_i is called a *coordinate* of p. Hence, we may view any $f \in R[x_1, x_2, \dots, x_n]$ as a function:

$$f : \mathbb{A}^n_R \to R \qquad (r_1, r_2, \dots, r_n) \mapsto f(r_1, r_2, \dots, r_n)$$

Algebraic Sets and the Zariski Topology

The zeros of polynomials form many interesting algebraic shapes. A prototypical example is $V(x^2 + y^2 - 1) \subseteq \mathbb{A}^2_{\mathbb{R}}$, the unit circle.

Definition 0.1.2: Zero Set

For $S \subseteq R[x_1, \dots, x_n]$, the *zero set* (or *vanishing locus*) of S is

$$\mathcal{Z}(S) = \{p \in \mathbb{A}^n_R \mid f(p) = 0,\ \forall f \in S\}$$

Definition 0.1.3: Algebraic Set

A subset $X \subseteq \mathbb{A}^n_R$ is called an *algebraic set* (or *closed set*) if there exists $S \subseteq R[x_1, \dots, x_n]$ such that $\mathcal{Z}(S) = X$.

The collection of algebraic sets satisfies the axioms for closed sets of a topology:

1. $\mathbb{A}^n_R = \mathcal{Z}(0)$ and $\emptyset = \mathcal{Z}(1)$ are algebraic.
2. Finite unions: $\mathcal{Z}(S_1) \cup \mathcal{Z}(S_2) = \mathcal{Z}(S_1 \cdot S_2)$ where $S_1 \cdot S_2 = \{fg : f \in S_1, g \in S_2\}$.
3. Arbitrary intersections: $\bigcap_\alpha \mathcal{Z}(S_\alpha) = \mathcal{Z}\big(\bigcup_\alpha S_\alpha\big)$.

Definition 0.1.4: Zariski Topology

The *Zariski topology* on $\mathbb{A}^n_R$ is the topology whose closed sets are precisely the algebraic sets.

0.1.5 Properties of the Zariski Topology The Zariski topology is quite coarse compared to the Euclidean topology on $\mathbb{R}^n$ or $\mathbb{C}^n$:

1. It is generally *not* Hausdorff: in $\mathbb{A}^1_k$, the only proper closed subsets are finite sets of points, so any two nonempty open sets intersect.
2. Every nonempty open set is dense.
3. Despite this, the Zariski topology is well-suited to algebraic geometry because its closed sets are precisely the sets defined by polynomial equations.

The Ideal–Variety Correspondence

Conversely, given a subset $S \subseteq \mathbb{A}^n_R$, we can ask which polynomials vanish on S:

Definition 0.1.6: Ideal of a Set

For $S \subseteq \mathbb{A}^n_R$, the *ideal of* S is

$$\mathcal{I}(S) = \{f \in R[x_1, \ldots, x_n] \mid f(p) = 0,\ \forall p \in S\}$$

One readily checks that $\mathcal{I}(S)$ is always an ideal in $R[x_1, \ldots, x_n]$. Moreover, $\mathcal{I}(S)$ is always a *radical* ideal, i.e., $f^n \in \mathcal{I}(S)$ implies $f \in \mathcal{I}(S)$. This is immediate: if f^n vanishes on S, then f vanishes on S.

The operators $\mathcal{Z}$ and $\mathcal{I}$ form an order-reversing correspondence:

$$S_1 \subseteq S_2 \implies \mathcal{I}(S_1) \supseteq \mathcal{I}(S_2), \qquad I_1 \subseteq I_2 \implies \mathcal{Z}(I_1) \supseteq \mathcal{Z}(I_2)$$

However, this correspondence is not bijective in general. The precise relationship requires the notion of radical ideals and irreducibility.

Definition 0.1.7: Radical Ideal

An ideal $I \subseteq R$ is *radical* if $f^n \in I$ implies $f \in I$ for all $f \in R$ and $n \geq 1$. Equivalently, $I = \sqrt{I}$ where $\sqrt{I} = \{f \in R \mid \exists n \geq 1,\ f^n \in I\}$.

Definition 0.1.8: Irreducible Set

An algebraic set V is *irreducible* if whenever $V = V_1 \cup V_2$ with V_1, V_2 closed, then $V = V_1$ or $V = V_2$. Equivalently, V is irreducible if and only if any two nonempty open subsets of V intersect.

Definition 0.1.9: Variety (Classical)

An algebraic set $V \subseteq \mathbb{A}^n_k$ is an (affine) *variety* if V is irreducible.

The key algebraic characterization of irreducibility is:

Proposition 0.1.10: Irreducibility and Prime Ideals

An algebraic set V is irreducible if and only if $\mathcal{I}(V)$ is a prime ideal.

***Proof*:**
See [22, Coordinate Ideal of Algebraic Variety].

Example 0.1.11: Reducible vs. Irreducible
Consider $V(xy) \subseteq \mathbb{A}^2_k$. This is the union of the two coordinate axes:

$$V(xy) = V(x) \cup V(y)$$

The ideal (xy) is radical but not prime (since $xy \in (xy)$ but $x \notin (xy)$ and $y \notin (xy)$). Hence $V(xy)$

is an algebraic set but *not* a variety.

In contrast, $V(y - x^2) \subseteq \mathbb{A}^2_k$ (a parabola) is irreducible: the ideal $(y - x^2)$ is prime.

We can now state the Nullstellensatz precisely. From here on, we assume k is an algebraically closed field.

Theorem 0.1.12: Hilbert's Nullstellensatz

Let k be algebraically closed. Then:

1. **(Weak form)** If $I \subsetneq k[x_1, \ldots, x_n]$ is a proper ideal, then $\mathcal{Z}(I) \neq \emptyset$.
2. **(Strong form)** For any ideal $I \subseteq k[x_1, \ldots, x_n]$, we have $\mathcal{I}(\mathcal{Z}(I)) = \sqrt{I}$.
3. **(Maximal ideal form)** The maximal ideals of $k[x_1, \ldots, x_n]$ are precisely those of the form $\mathfrak{m}_p = (x_1 - a_1, \ldots, x_n - a_n)$ for points $p = (a_1, \ldots, a_n) \in \mathbb{A}^n_k$.

***Proof*:**

See [22, Hilbert's Nullstellensatz, Hilbert's Nullstellensatz - Weak Version].

The strong form immediately gives:

Corollary 0.1.13: The Ideal–Variety Correspondence

Let k be algebraically closed. The maps $\mathcal{Z}$ and $\mathcal{I}$ induce inclusion-reversing bijections:

$$\{\text{radical ideals in } k[x_1, \ldots, x_n]\} \overset{\mathcal{Z}}{\longleftrightarrow} \{\text{algebraic sets in } \mathbb{A}^n_k\}$$

Under this correspondence:

Ideals	$\longleftrightarrow$	**Algebraic Sets**
radical ideals	$\longleftrightarrow$	algebraic sets
prime ideals	$\longleftrightarrow$	varieties (irreducible)
maximal ideals	$\longleftrightarrow$	points

The Coordinate Ring

The passage from geometry to algebra is achieved through the coordinate ring:

Definition 0.1.14: Coordinate Ring

Let $V \subseteq \mathbb{A}^n_k$ be an algebraic set. The *coordinate ring* (or *ring of regular functions*) of V is

$$k[V] := k[x_1, \ldots, x_n]/\mathcal{I}(V)$$

The elements of $k[V]$ are equivalence classes of polynomials, where two polynomials are equivalent if they agree on V. Thus $k[V]$ can be interpreted as the ring of *polynomial functions* $V \to k$.

Proposition 0.1.15: Properties of the Coordinate Ring

Let $V \subseteq \mathbb{A}_k^n$ be an algebraic set with coordinate ring $k[V]$. Then:

1. $k[V]$ is a finitely generated k-algebra.
2. $k[V]$ is reduced (i.e., has no nilpotents) if and only if $\mathcal{I}(V)$ is radical.
3. $k[V]$ is an integral domain if and only if V is irreducible.
4. Points $p \in V$ correspond to maximal ideals $\mathfrak{m}_p \subseteq k[V]$.

***Proof*:**
See [22, Hilbert's Nullstellensatz, Equivalent characterization].

Example 0.1.16: Coordinate Rings

1. $k[\mathbb{A}^n] = k[x_1, \ldots, x_n]$, since $\mathcal{I}(\mathbb{A}^n) = (0)$.
2. For the parabola $V = V(y - x^2) \subseteq \mathbb{A}^2$, we have $k[V] = k[x, y]/(y - x^2) \cong k[x]$, reflecting that $V \cong \mathbb{A}^1$.
3. For the circle $V = V(x^2 + y^2 - 1) \subseteq \mathbb{A}^2_{\mathbb{C}}$, we have $k[V] = \mathbb{C}[x, y]/(x^2 + y^2 - 1)$. This is an integral domain since $(x^2 + y^2 - 1)$ is prime over $\mathbb{C}$.
4. For the nodal cubic $V = V(y^2 - x^3 - x^2) \subseteq \mathbb{A}^2$, the coordinate ring is $k[V] = k[x, y]/(y^2 - x^3 - x^2)$. This is an integral domain (the curve is irreducible), but V is singular at the origin.

Morphisms of Affine Varieties

To form a category of varieties, we need morphisms.

Definition 0.1.17: Regular Map / Morphism

Let $V \subseteq \mathbb{A}^n$ and $W \subseteq \mathbb{A}^m$ be algebraic sets. A *regular map* (or *morphism*) $\varphi : V \to W$ is a map of the form

$$\varphi(p) = (f_1(p), f_2(p), \ldots, f_m(p))$$

where each $f_i \in k[V]$. That is, φ is given by polynomial functions in coordinates.

Proposition 0.1.18: Morphisms and Algebra Homomorphisms

A regular map $\varphi : V \to W$ induces a k-algebra homomorphism

$$\varphi^* : k[W] \to k[V], \qquad g \mapsto g \circ \varphi$$

called the *pullback*. Conversely, every k-algebra homomorphism $k[W] \to k[V]$ arises as φ^* for a unique regular map $\varphi : V \to W$.

***Proof*:**
See [22, Regular Map Algebraic Correspondence].

Note the contravariance: a map $V \to W$ gives a map $k[W] \to k[V]$ in the opposite direction. This leads to the fundamental equivalence:

Theorem 0.1.19: Equivalence of Categories

Let k be algebraically closed. There is a contravariant equivalence of categories:

$$\text{Aff-Var}_k \underset{\text{mSpec}}{\overset{V \mapsto k[V]}{\rightleftarrows}} \left(\text{Alg}_k^{\text{f.g., red}}\right)^{\text{op}}$$

where Aff-Var_k is the category of affine varieties over k with regular maps, and $\text{Alg}_k^{\text{f.g., red}}$ is the category of finitely generated reduced k-algebras with k-algebra homomorphisms.

***Proof*:**
The functor $V \mapsto k[V]$ sends varieties to their coordinate rings and sends morphisms $\varphi : V \to W$ to pullbacks $\varphi^* : k[W] \to k[V]$. The quasi-inverse sends a finitely generated reduced k-algebra A to its maximal spectrum $\text{mSpec}(A)$, the set of maximal ideals with the Zariski topology. The Nullstellensatz guarantees that for $A = k[V]$, the points of V are in bijection with maximal ideals of A. See [22, Regular Map Algebraic Correspondence] for details.

0.1.20 Why Reduced Algebras? The restriction to reduced algebras (no nilpotents) corresponds to the fact that classical varieties are determined by their point-sets. Modern algebraic geometry (via schemes) removes this restriction, allowing nilpotent elements to encode infinitesimal data – this is one of the key generalizations.

Local Rings and Regularity

To study the local behavior of a variety at a point, we first introduce the algebraic machinery of localization. This process allows us to systematically make elements of a ring invertible, essentially allowing us to study rings "locally" by focusing on fractions whose denominators do not vanish at the point of interest.

Definition 0.1.21: Localization of a Ring

Let R be a commutative ring. A subset $S \subseteq R$ is *multiplicatively closed* if $1 \in S$ and for all $a, b \in S$, we have $ab \in S$. The *localization* of R at S, denoted $S^{-1}R$, is the set of equivalence classes of fractions $\frac{r}{s}$ with $r \in R$ and $s \in S$, modulo the equivalence relation:

$$\frac{r}{s} \sim \frac{r'}{s'} \iff \text{there exists } t \in S \text{ such that } t(rs' - r's) = 0$$

This becomes a ring under the standard addition and multiplication of fractions.

Localization is completely characterized by the following universal property, which states that $S^{-1}R$ is the "most efficient" way to force the elements of S to become units.

Proposition 0.1.22: Universal Property of Localization

Let S be a multiplicative set of a ring R. The natural map $\iota : R \to S^{-1}R$ given by $\iota(r) = \frac{r}{1}$ is a ring homomorphism that maps elements of S to units in $S^{-1}R$. Furthermore, if A is any ring and $g : R \to A$ is a ring homomorphism such that $g(S) \subseteq A^\times$ (the group of units of A), then there exists a unique ring homomorphism $h : S^{-1}R \to A$ such that $h \circ \iota = g$.

***Proof*:**

We define $h\left(\frac{r}{s}\right) = g(r)g(s)^{-1}$. To see that h is well-defined, suppose $\frac{r}{s} = \frac{r'}{s'}$. Then there is some $t \in S$ such that $t(rs' - r's) = 0$. Applying g gives $g(t)(g(r)g(s') - g(r')g(s)) = 0$. Since $t \in S$, $g(t)$ is a unit in A, so we can multiply by $g(t)^{-1}$ to obtain $g(r)g(s') = g(r')g(s)$. Multiplying by $g(s)^{-1}g(s')^{-1}$ yields $g(r)g(s)^{-1} = g(r')g(s')^{-1}$, confirming h is well-defined.

It is straightforward to verify that h is a ring homomorphism. For uniqueness, if $h' : S^{-1}R \to A$ satisfies $h' \circ \iota = g$, then $h'\left(\frac{r}{1}\right) = g(r)$ and $h'\left(\frac{1}{s}\right) = h'\left(\frac{s}{1}\right)^{-1} = g(s)^{-1}$. Thus, $h'\left(\frac{r}{s}\right) = g(r)g(s)^{-1} = h\left(\frac{r}{s}\right)$.

An important consequence of this universal property is that localization is a functor.

Proposition 0.1.23: Functoriality of Localization

Let R, R' be rings with multiplicative sets S, S' respectively. If $\varphi : R \to R'$ is a ring homomorphism such that $\varphi(S) \subseteq S'$, then there is a unique induced ring homomorphism $\widetilde{\varphi} : S^{-1}R \to (S')^{-1}R'$ making the following diagram commute:

$$\widetilde{\varphi} \circ \iota = \iota' \circ \varphi$$

where ι and ι' are the natural localization maps.

***Proof*:**

Consider the composition $\iota' \circ \varphi : R \to R' \to (S')^{-1}R'$. For any $s \in S$, we have $\varphi(s) \in S'$. Therefore, $\iota'(\varphi(s))$ is a unit in $(S')^{-1}R'$. By the universal property of $S^{-1}R$ applied to the homomorphism $\iota' \circ \varphi$, there exists a unique ring homomorphism $\widetilde{\varphi} : S^{-1}R \to (S')^{-1}R'$ satisfying

$\widetilde{\varphi} \circ \iota = \iota' \circ \varphi$. Explicitly, this map is given by $\widetilde{\varphi}\left(\frac{r}{s}\right) = \frac{\varphi(r)}{\varphi(s)}$.

We now apply this general algebraic construction to the coordinate ring of a variety to study its local geometry.

Definition 0.1.24: Local Ring at a Point

Let V be an affine variety and $p \in V$ a point with corresponding maximal ideal $\mathfrak{m}_p \subseteq k[V]$. The set $S = k[V] \setminus \mathfrak{m}_p$ is multiplicatively closed. The *local ring of* V *at* p is the localization of $k[V]$ at S:

$$\mathcal{O}_{V,p} := k[V]_{\mathfrak{m}_p} = \left\{ \frac{f}{g} \;\middle|\; f, g \in k[V],\ g(p) \neq 0 \right\}$$

This is a local ring with unique maximal ideal $\mathfrak{m}_{V,p} = \mathfrak{m}_p \cdot \mathcal{O}_{V,p}$.

The local ring $\mathcal{O}_{V,p}$ captures the "germs" of regular functions near p: two functions are identified if they agree on some neighborhood of p.

Definition 0.1.25: Zariski Tangent Space

The *Zariski tangent space* to V at p is the k-vector space

$$T_pV := \left(\mathfrak{m}_{V,p}/\mathfrak{m}_{V,p}^2\right)^\vee = \mathrm{Hom}_k\left(\mathfrak{m}_{V,p}/\mathfrak{m}_{V,p}^2, k\right)$$

The *Zariski cotangent space* is $T_p^*V := \mathfrak{m}_{V,p}/\mathfrak{m}_{V,p}^2$.

0.1.26 Tangent Space via Derivations Equivalently, T_pV can be defined as the space of k-derivations $\mathcal{O}_{V,p} \to k$ at p, or as the space of "ε-points": morphisms $\operatorname{Spec} k[\varepsilon]/(\varepsilon^2) \to V$ centered at p. This latter perspective generalizes naturally to schemes.

Definition 0.1.27: Smooth Point, Singular Point

Let V be a variety of dimension d (see below). A point $p \in V$ is:

- *smooth* (or *regular*) if $\dim_k T_pV = d$;
- *singular* if $\dim_k T_pV > d$.

A variety is *smooth* (or *nonsingular*) if every point is smooth.

Example 0.1.28: Singularities

1. The parabola $V(y - x^2)$ is smooth everywhere. At any point, $\dim T_pV = 1 = \dim V$.
2. The cuspidal cubic $V(y^2 - x^3)$ has a singularity (cusp) at the origin. One computes $\dim T_0V = 2 > 1 = \dim V$.
3. The nodal cubic $V(y^2 - x^2(x+1))$ has a node at the origin where two branches cross.

Proposition 0.1.29: Regularity and Local Rings

A point $p \in V$ is smooth if and only if $\mathcal{O}_{V,p}$ is a *regular local ring*, i.e., $\dim_k(\mathfrak{m}_{V,p}/\mathfrak{m}_{V,p}^2) = \dim \mathcal{O}_{V,p}$ (Krull dimension).

***Proof*:**
See [22, Jacobian Criterion and Regular Local Rings].

Rational Functions and Dimension

For an irreducible variety, we can form the field of fractions of its coordinate ring:

Definition 0.1.30: Function Field

Let V be an affine variety (irreducible). The *function field* of V is

$$k(V) := \mathrm{Frac}(k[V])$$

Elements of $k(V)$ are called *rational functions* on V. A rational function f/g is defined on the open set $\{p \in V : g(p) \neq 0\}$.

Definition 0.1.31: Dimension

The *dimension* of an affine variety V is

$$\dim V := \mathrm{tr.deg}_k\ k(V)$$

the transcendence degree of the function field over k. Equivalently, $\dim V = \dim k[V]$, the Krull dimension of the coordinate ring.

Example 0.1.32: Dimensions

1. $\dim \mathbb{A}^n = n$, since $k(\mathbb{A}^n) = k(x_1, \ldots, x_n)$ has transcendence degree n.
2. Any plane curve $V(f) \subseteq \mathbb{A}^2$ with f irreducible has dimension 1.
3. A hypersurface $V(f) \subseteq \mathbb{A}^n$ with f irreducible has dimension $n - 1$.

Proposition 0.1.33: Dimension and Chains of Primes

For an affine variety V, we have $\dim V = \dim k[V]$, where the Krull dimension of $k[V]$ is the supremum of lengths of chains of prime ideals

$$\mathfrak{p}_0 \subsetneq \mathfrak{p}_1 \subsetneq \cdots \subsetneq \mathfrak{p}_d$$

in $k[V]$.

***Proof*:**
See [22, Transcendental and Krull Dimension].

0.2 Recall Projective Spaces

Projective space is the natural setting for algebraic geometry. Many phenomena that are awkward or incomplete in affine space become clean and uniform in projective space: intersection theory works properly (Bézout's theorem), curves have well-defined degrees, and there are no "missing points at infinity." This section reviews the essential constructions.

Definition and Homogeneous Coordinates

Definition 0.2.1: Projective Space

Let k be a field. *Projective n-space* over k is

$$\mathbb{P}^n_k := \left(k^{n+1} \setminus \{0\}\right) / \sim$$

where $(a_0, a_1, \ldots, a_n) \sim (\lambda a_0, \lambda a_1, \ldots, \lambda a_n)$ for all $\lambda \in k^\times$. The equivalence class of $(a_0, \ldots, a_n)$ is denoted $[a_0 : a_1 : \cdots : a_n]$ and called a *point* in $\mathbb{P}^n$. The a_i are *homogeneous coordinates*.

0.2.2 Projective Space as Lines Equivalently, $\mathbb{P}^n_k$ is the set of 1-dimensional linear subspaces (lines through the origin) in k^{n+1}. This is sometimes denoted $\mathbb{P}(k^{n+1})$ or $\mathrm{Proj}(k^{n+1})$.

Example 0.2.3: Low-Dimensional Projective Spaces

1. $\mathbb{P}^0$ is a single point.
2. $\mathbb{P}^1$ is the *projective line*. Points are $[a_0 : a_1]$ with $(a_0, a_1) \neq (0, 0)$. If $a_1 \neq 0$, we can normalize to $[t : 1]$ for $t = a_0/a_1 \in k$. The remaining point $[1 : 0]$ is the "point at infinity." Thus $\mathbb{P}^1 \cong \mathbb{A}^1 \sqcup \{\infty\}$.
3. $\mathbb{P}^2$ is the *projective plane*, the natural home for plane curves.

Affine Charts

Projective space is covered by affine open sets:

Definition 0.2.4: Standard Affine Charts

For each $i \in \{0, 1, \ldots, n\}$, define the *standard affine open*

$$U_i := \{[a_0 : \cdots : a_n] \in \mathbb{P}^n \mid a_i \neq 0\}$$

There is an isomorphism $\varphi_i : U_i \xrightarrow{\sim} \mathbb{A}^n$ given by

$$\varphi_i([a_0 : \cdots : a_n]) = \left(\frac{a_0}{a_i}, \ldots, \widehat{\frac{a_i}{a_i}}, \ldots, \frac{a_n}{a_i}\right)$$

where the hat denotes omission.

Proposition 0.2.5: Affine Cover of Projective Space

The standard affine opens cover $\mathbb{P}^n$:

$$\mathbb{P}^n = U_0 \cup U_1 \cup \cdots \cup U_n$$

The complement of U_i is the hyperplane $H_i = \{[a_0 : \cdots : a_n] : a_i = 0\} \cong \mathbb{P}^{n-1}$.

Example 0.2.6: The Projective Line Revisited

For $\mathbb{P}^1$, we have two charts:

- $U_0 = \{[a_0 : a_1] : a_0 \neq 0\} \cong \mathbb{A}^1$ via $[a_0 : a_1] \mapsto a_1/a_0$
- $U_1 = \{[a_0 : a_1] : a_1 \neq 0\} \cong \mathbb{A}^1$ via $[a_0 : a_1] \mapsto a_0/a_1$

On the overlap $U_0 \cap U_1$, if $t = a_1/a_0$ is the coordinate on U_0 and $s = a_0/a_1$ on U_1, then $s = 1/t$. This is the standard gluing of two copies of $\mathbb{A}^1$ to form $\mathbb{P}^1$.

Homogeneous Polynomials and Graded Rings

Polynomials do not define functions on $\mathbb{P}^n$ directly, since $f(\lambda a_0, \ldots, \lambda a_n) \neq f(a_0, \ldots, a_n)$ in general. However, the *vanishing* of homogeneous polynomials is well-defined.

Definition 0.2.7: Homogeneous Polynomial

A polynomial $f \in k[x_0, x_1, \ldots, x_n]$ is *homogeneous of degree d* if every monomial in f has total degree d. Equivalently, $f(\lambda x_0, \ldots, \lambda x_n) = \lambda^d f(x_0, \ldots, x_n)$ for all $\lambda \in k$.

Definition 0.2.8: Graded Ring

The polynomial ring $S = k[x_0, \ldots, x_n]$ has a *grading* $S = \bigoplus_{d \geq 0} S_d$, where S_d is the k-vector space of homogeneous polynomials of degree d (including 0). We have $S_d \cdot S_e \subseteq S_{d+e}$.

Definition 0.2.9: Homogeneous Ideal

An ideal $I \subseteq k[x_0, \ldots, x_n]$ is *homogeneous* if it is generated by homogeneous polynomials. Equivalently, $I = \bigoplus_{d \geq 0} (I \cap S_d)$.

Proposition 0.2.10: Characterization of Homogeneous Ideals

An ideal I is homogeneous if and only if for every $f \in I$, all homogeneous components of f are also in I.

***Proof*:**
See [12, Characterizing Graded Ideals].

Projective Varieties

Definition 0.2.11: Projective Algebraic Set

For a set T of homogeneous polynomials in $k[x_0, \ldots, x_n]$, the *projective zero set* is

$$\mathcal{Z}_+(T) := \{[a_0 : \cdots : a_n] \in \mathbb{P}^n \mid f(a_0, \ldots, a_n) = 0, \ \forall f \in T\}$$

This is well-defined since f homogeneous implies $f(\lambda \mathbf{a}) = \lambda^d f(\mathbf{a})$, so $f(\mathbf{a}) = 0 \iff f(\lambda \mathbf{a}) = 0$. A subset $V \subseteq \mathbb{P}^n$ is a *projective algebraic set* if $V = \mathcal{Z}_+(T)$ for some set T of homogeneous polynomials.

Definition 0.2.12: Homogeneous Ideal of a Set

For $V \subseteq \mathbb{P}^n$, define

$$\mathcal{I}_+(V) := \text{ideal generated by } \{f \in k[x_0, \ldots, x_n] \text{ homogeneous} \mid f(p) = 0, \ \forall p \in V\}$$

This is always a homogeneous radical ideal.

The projective algebraic sets form the closed sets of the *Zariski topology on* $\mathbb{P}^n$.

Definition 0.2.13: Projective Variety

A projective algebraic set $V \subseteq \mathbb{P}^n$ is a *projective variety* if it is irreducible in the Zariski topology.

0.2.14 The Irrelevant Ideal The ideal $\mathfrak{m}_+ = (x_0, x_1, \ldots, x_n) \subseteq k[x_0, \ldots, x_n]$ is called the *irrelevant ideal*. It satisfies $\mathcal{Z}_+(\mathfrak{m}_+) = \emptyset$, since the only common zero of all x_i in k^{n+1} is the origin, which does not correspond to a point in $\mathbb{P}^n$. This is an important subtlety in the projective Nullstellensatz.

Theorem 0.2.15: Projective Nullstellensatz

Let k be algebraically closed and let $I \subseteq k[x_0, \dots, x_n]$ be a homogeneous ideal. Then:

1. $\mathcal{Z}_+(I) = \emptyset$ if and only if $\sqrt{I} \supseteq \mathfrak{m}_+^N$ for some $N \geq 1$ (equivalently, $\sqrt{I} = \mathfrak{m}_+$ or $\sqrt{I} = (1)$).
2. If $\mathcal{Z}_+(I) \neq \emptyset$, then $\mathcal{I}_+(\mathcal{Z}_+(I)) = \sqrt{I}$.

Hence $\mathcal{Z}_+$ and $\mathcal{I}_+$ give a bijection between:

$$\left\{\text{homogeneous radical ideals } I \text{ with } \sqrt{I} \neq \mathfrak{m}_+\right\} \longleftrightarrow \{\text{projective algebraic sets in } \mathbb{P}^n\}$$

***Proof*:**
See [22, Characterizing Irrelevant Ideals, Cone Identities, Hilbert's Nullstellensatz].

Homogeneous Coordinate Ring and the Proj Construction

Definition 0.2.16: Homogeneous Coordinate Ring

For a projective variety $V = \mathcal{Z}_+(I) \subseteq \mathbb{P}^n$, the *homogeneous coordinate ring* is the graded ring

$$S(V) := k[x_0, \dots, x_n]/I$$

with the induced grading $S(V) = \bigoplus_{d \geq 0} S(V)_d$.

0.2.17 Homogeneous Coordinate Ring is Not Intrinsic Unlike the affine case, the homogeneous coordinate ring $S(V)$ depends on the embedding $V \hookrightarrow \mathbb{P}^n$, not just on V as an abstract variety. Different embeddings of the same variety can give non-isomorphic graded rings.

Definition 0.2.18: The Proj Construction

For a graded ring $S = \bigoplus_{d \geq 0} S_d$, we define

$$\operatorname{Proj}(S) := \{\mathfrak{p} \in \operatorname{Spec}(S) \mid \mathfrak{p} \text{ is homogeneous and } \mathfrak{p} \not\supseteq S_+\}$$

where $S_+ = \bigoplus_{d>0} S_d$ is the irrelevant ideal. This set is equipped with a natural topology and structure sheaf making it a scheme. For $S = k[x_0, \dots, x_n]$, we recover $\operatorname{Proj}(S) = \mathbb{P}^n_k$.

Relation Between Affine and Projective

The affine charts give a way to study projective varieties locally:

Definition 0.2.19: Affine Cone

For a projective variety $V \subseteq \mathbb{P}^n$, the *affine cone* over V is

$$C(V) := \mathcal{Z}(\mathcal{I}_+(V)) \subseteq \mathbb{A}^{n+1}$$

This is an affine variety in $\mathbb{A}^{n+1}$, consisting of all lines through the origin that correspond to points of V, together with the origin itself.

Proposition 0.2.20: Affine Pieces of Projective Varieties

Let $V \subseteq \mathbb{P}^n$ be a projective variety and $U_i \cong \mathbb{A}^n$ a standard affine chart. Then $V \cap U_i$ is an affine variety in $\mathbb{A}^n$.
Explicitly, if $V = \mathcal{Z}_+(f_1, \dots, f_r)$ with each f_j homogeneous of degree d_j, then $V \cap U_i = \mathcal{Z}(f_1^{(i)}, \dots, f_r^{(i)})$ where

$$f_j^{(i)}(y_0, \dots, \widehat{y_i}, \dots, y_n) := f_j(y_0, \dots, y_{i-1}, 1, y_{i+1}, \dots, y_n)$$

is the *dehomogenization* of f_j with respect to x_i.

Definition 0.2.21: Homogenization

Conversely, given $g \in k[y_1, \dots, y_n]$ of degree d, its *homogenization* with respect to x_0 is

$$g^h(x_0, x_1, \dots, x_n) := x_0^d \cdot g\left(\frac{x_1}{x_0}, \dots, \frac{x_n}{x_0}\right)$$

This is homogeneous of degree d.

Example 0.2.22: Projective Closure

The parabola $V(y - x^2) \subseteq \mathbb{A}^2$ has homogenization $yz - x^2 \in k[x, y, z]$. Its projective closure is $\overline{V} = \mathcal{Z}_+(yz - x^2) \subseteq \mathbb{P}^2$. The "new" point $[0 : 1 : 0]$ (where $z = 0$ and $x = 0$) is the point at infinity where the parabola "closes up."

Quasi-Projective Varieties

Definition 0.2.23: Quasi-Projective Variety

A *quasi-projective variety* is a locally closed subset of $\mathbb{P}^n$, i.e., an open subset of a projective variety. Equivalently, it is the intersection of an open set and a closed set in $\mathbb{P}^n$.

Proposition 0.2.24: Affine and Projective are Quasi-Projective

Both affine varieties and projective varieties are quasi-projective:

1. A projective variety $V \subseteq \mathbb{P}^n$ is quasi-projective (it is closed, hence locally closed).
2. An affine variety $V \subseteq \mathbb{A}^n \cong U_0 \subseteq \mathbb{P}^n$ is quasi-projective (it is closed in the open set U_0).

The category of quasi-projective varieties with regular maps is the classical category of "algebraic varieties" in much of the literature.

Morphisms of Projective Varieties

Defining morphisms of projective varieties requires care, since homogeneous polynomials do not define functions.

Definition 0.2.25: Morphism of Projective Varieties

Let $V \subseteq \mathbb{P}^n$ and $W \subseteq \mathbb{P}^m$ be projective varieties. A *morphism* $\varphi : V \to W$ is a map that is locally given by homogeneous polynomials of the same degree. That is, for each $p \in V$, there exist homogeneous polynomials $f_0, \dots, f_m \in k[x_0, \dots, x_n]$ of the same degree d such that:

1. $f_0, \dots, f_m$ do not simultaneously vanish on a neighborhood of p in V.
2. $\varphi(q) = [f_0(q) : \cdots : f_m(q)]$ for all q in this neighborhood.

0.2.26 Local Nature of Morphisms Different homogeneous polynomials may be needed on different open subsets of V. On overlaps, they must agree as maps to $\mathbb{P}^m$. This local definition is necessary because no single tuple of homogeneous polynomials may work globally.

Example 0.2.27: The Veronese Embedding

The *Veronese embedding* of degree d is

$$\nu_d : \mathbb{P}^n \to \mathbb{P}^N, \qquad [x_0 : \cdots : x_n] \mapsto [\cdots : x_0^{i_0} \cdots x_n^{i_n} : \cdots]$$

where the coordinates on $\mathbb{P}^N$ are indexed by monomials of degree d, so $N = \binom{n+d}{d} - 1$.

For example, $\nu_2 : \mathbb{P}^1 \to \mathbb{P}^2$ is given by $[s : t] \mapsto [s^2 : st : t^2]$. The image is the conic $\mathcal{Z}_+(xz - y^2)$, showing that $\mathbb{P}^1$ is isomorphic to this conic.

Example 0.2.28: The Segre Embedding

The *Segre embedding* is

$$\sigma : \mathbb{P}^n \times \mathbb{P}^m \to \mathbb{P}^{(n+1)(m+1)-1}, \qquad ([x_i], [y_j]) \mapsto [x_i y_j]_{0 \le i \le n,\, 0 \le j \le m}$$

For example, $\sigma : \mathbb{P}^1 \times \mathbb{P}^1 \to \mathbb{P}^3$ is given by

$$([s : t], [u : v]) \mapsto [su : sv : tu : tv]$$

The image is the quadric surface $\mathcal{Z}_+(xw - yz) \subseteq \mathbb{P}^3$.

The Segre embedding shows that $\mathbb{P}^n \times \mathbb{P}^m$ can be given the structure of a projective variety.

Products of Projective Varieties

Proposition 0.2.29: Products of Projective Varieties

Let $V \subseteq \mathbb{P}^n$ and $W \subseteq \mathbb{P}^m$ be projective varieties. Then $V \times W$, embedded via the Segre embedding $\mathbb{P}^n \times \mathbb{P}^m \hookrightarrow \mathbb{P}^{(n+1)(m+1)-1}$, is a projective variety.

***Proof*:**
See [22, Segre Embedding is Closed, A Product of Irreducibles Is Irreducible].

0.2.30 Contrast with Affine Products For affine varieties, products are straightforward: $V \times W \subseteq \mathbb{A}^n \times \mathbb{A}^m = \mathbb{A}^{n+m}$, and $k[V \times W] \cong k[V] \otimes_k k[W]$. The projective case requires the Segre embedding because $\mathbb{P}^n \times \mathbb{P}^m \ncong \mathbb{P}^{n+m}$.

Completeness and Properness

Projective varieties enjoy a crucial property not shared by affine varieties:

Definition 0.2.31: Complete Variety

A variety V is *complete* if for every variety W, the projection $\pi : V \times W \to W$ is a closed map (sends closed sets to closed sets).

Theorem 0.2.32: Projective Varieties are Complete

Every projective variety is complete.

***Proof*:**
See [22, Projection of Projective Variety].

0.2.33 Completeness as Compactness Completeness is the algebraic analogue of compactness. Over $\mathbb{C}$, a projective variety is compact in the classical (Euclidean) topology. Affine varieties of positive dimension are never complete: for example, the projection $\mathbb{A}^1 \times \mathbb{A}^1 \to \mathbb{A}^1$ does not send the closed hyperbola $\mathcal{Z}(xy - 1)$ to a closed set.

Proposition 0.2.34: Affine and Complete Implies Finite

If V is both affine and complete, then V is a finite set of points.

***Proof*:**
See [22].

0.3 Noetherian Induction or Well-founded Induction

This form of induction is used in later proofs, for example in theorem 5.2.7 and theorem 6.6.13. It generalizes the usual induction, where the well-ordered set is $\mathbb{N}$. Recall that a binary relation on a set A is a subset of $A \times A$. When (a, b) belongs to the relation we write $a \prec b$ or $b \succ a$. A binary relation is a strict partial order if it is:

1. Irreflexive: $\forall x \in A, \neg(x \prec x)$
2. Transitive: $\forall x, y, z \in A, (x \prec y \wedge y \prec z) \implies x \prec z$
3. Asymmetric: $\forall x, y \in A, x \prec y \implies \neg(y \prec x)$

Given a set A with a strict partial order $\prec$, a descending chain is a sequence $a_1, a_2, a_3, \ldots$ of elements of A such that $a_1 \succ a_2 \succ a_3 \succ \cdots$.

Definition 0.3.1: Well-founded Relation

A strict partial order $\prec$ on a set A is well-founded if there exist no infinite descending chains in A with respect to $\prec$.

Definition 0.3.2: Minimal Element

Given a subset S of a well-founded set A, an element $m \in S$ is minimal if there is no element $x \in S$ such that $x \prec m$.

Theorem 0.3.3: Noetherian Induction

Let $(A, \prec)$ be a set with a well-founded relation $\prec$, and let P be a property defined on A. If for all $x \in A : [\forall y \in A, (y \prec x \implies P(y))] \implies P(x)$ then $P(x)$ holds for all $x \in A$.

***Proof*:**
Step 1: every nonempty subset $B \subseteq A$ has a minimal element. Suppose B is nonempty and has no minimal element in the sense of definition 0.3.2. Then every $b \in B$ admits some $b' \in B$ with $b' \prec b$, so the axiom of dependent choice gives a sequence $b_1, b_2, b_3, \ldots$ in B with $b_{k+1} \prec b_k$ for every k. This is an infinite descending chain $b_1 \succ b_2 \succ b_3 \succ \cdots$ in A, which definition 0.3.1 excludes.

Step 2: induction. Let $B = \{x \in A \mid P(x) \text{ fails}\}$ and suppose $B \neq \emptyset$. By Step 1, B has a minimal element m. Every $y \in A$ with $y \prec m$ lies outside B by minimality, so $P(y)$ holds for every such y, and the hypothesis gives $P(m)$. This contradicts $m \in B$. Hence $B = \emptyset$, that is, $P(x)$ holds for every $x \in A$, as claimed.

The following may be an insightful alternative:

Theorem 0.3.4: Noetherian Induction - Alternative Form

Let $(A, \prec)$ be a set with a well-founded relation $\prec$. If $S \subseteq A$ is a subset such that for all minimal elements m of $A\backslash S$, we have $m \in S$, then $S = A$.

***Proof*:**
Let $P(x)$ be the property $x \in S$, and let $x \in A$ be such that $P(y)$ holds for every $y \prec x$. If $x \notin S$, then $x \in A\backslash S$, and no $z \in A\backslash S$ satisfies $z \prec x$, since every such z lies in S. So x is a minimal element of $A\backslash S$, and the hypothesis on S gives $x \in S$, a contradiction. Hence $P(x)$ holds whenever $P(y)$ holds for all $y \prec x$, and theorem 0.3.3 gives $P(x)$ for every $x \in A$, that is, $S = A$, as we sought to show.

The connection between these forms lies in taking S to be the set of elements where P holds, and showing that if S is not all of A, then any minimal element of $A\backslash S$ must be in S, giving a contradiction.

Corollary 0.3.5: Noetherian Induction on Closed Subsets

Let X be a Noetherian topological space, that is, every descending chain $Z_1 \supseteq Z_2 \supseteq \cdots$ of closed subsets of X is eventually constant. Let P be a property of closed subsets of X such that $P(Z)$ holds for every closed $Z \subseteq X$ for which $P(Z')$ holds for all closed $Z' \subsetneq Z$. Then $P(Z)$ holds for every closed subset $Z \subseteq X$.

***Proof*:**
Let A be the set of closed subsets of X, with $Z' \prec Z$ meaning $Z' \subsetneq Z$. Strict inclusion is irreflexive, transitive and asymmetric, so $\prec$ is a strict partial order. An infinite descending chain $Z_1 \succ Z_2 \succ \cdots$ would be a descending chain of closed subsets that is never constant, which the hypothesis on X excludes, so $\prec$ is well-founded (definition 0.3.1). The hypothesis on P is the hypothesis of theorem 0.3.3 for $(A, \prec)$, which therefore gives $P(Z)$ for every closed Z, as claimed.

0.4 Categorical Reminder

Category theory provides the language and organizational framework for modern algebraic geometry. Schemes are defined as certain locally ringed spaces, morphisms are defined via universal properties, and many constructions (fiber products, sheafification, etc.) are instances of general categorical machinery. This section collects the essential concepts.

Categories, Functors, and Natural Transformations

Definition 0.4.1: Category

A *category* $\mathcal{C}$ consists of:

1. A class $\mathrm{Ob}(\mathcal{C})$ of *objects*.
2. For each pair of objects X, Y, a set of *morphisms* from X to Y denoted $\mathrm{Hom}_{\mathcal{C}}(X, Y)$ or $\mathrm{Mor}_{\mathbb{C}}(X, Y)$.
3. For each triple X, Y, Z, a *composition* map $\mathrm{Hom}(Y, Z) \times \mathrm{Hom}(X, Y) \to \mathrm{Hom}(X, Z)$.
4. For each object X, an *identity* morphism $\mathrm{id}_X \in \mathrm{Hom}(X, X)$.

These satisfy associativity of composition and the identity laws.

Example 0.4.2: Categories in Algebra and Geometry

1. **Set**: sets and functions.
2. **Grp**, **Ab**, **Ring**, **CRing**: groups, abelian groups, rings, commutative rings.
3. $\mathbf{Mod}_R$: R-modules and R-linear maps.
4. **Top**: topological spaces and continuous maps.
5. $\mathbf{Aff}_k$: affine varieties over k and regular maps.

Definition 0.4.3: Functor

A *(covariant) functor* $F : \mathcal{C} \to \mathcal{D}$ consists of:

1. A map $F : \mathrm{Ob}(\mathcal{C}) \to \mathrm{Ob}(\mathcal{D})$.
2. For each $X, Y \in \mathcal{C}$, a map $F : \mathrm{Hom}_{\mathcal{C}}(X, Y) \to \mathrm{Hom}_{\mathcal{D}}(F(X), F(Y))$.

These must satisfy $F(\mathrm{id}_X) = \mathrm{id}_{F(X)}$ and $F(g \circ f) = F(g) \circ F(f)$.
A *contravariant functor* $F : \mathcal{C} \to \mathcal{D}$ reverses arrows: $F : \mathrm{Hom}_{\mathcal{C}}(X, Y) \to \mathrm{Hom}_{\mathcal{D}}(F(Y), F(X))$ with $F(g \circ f) = F(f) \circ F(g)$. Equivalently, it is a covariant functor $\mathcal{C}^{\mathrm{op}} \to \mathcal{D}$.

Example 0.4.4: Functors in Algebraic Geometry

1. The coordinate ring functor $V \mapsto k[V]$ is a contravariant functor $\mathbf{Aff}_k \to \mathbf{CRing}$.
2. Forgetful functors: $\mathbf{Grp} \to \mathbf{Set}$, etc.
3. We shall define the spectrum functor $A \mapsto \mathrm{Spec}\,(A)$ is a contravariant functor $\mathbf{CRing} \to \mathbf{Sch}$.
4. We shall define the global sections functor $X \mapsto \Gamma(X, \mathcal{O}_X)$ is a contravariant functor $\mathbf{Sch} \to \mathbf{CRing}$.

Definition 0.4.5: Natural Transformation

Let $F, G : \mathcal{C} \to \mathcal{D}$ be functors. A *natural transformation* $\eta : F \Rightarrow G$ is a family of morphisms $\eta_X : F(X) \to G(X)$ for each $X \in \mathcal{C}$, such that for every morphism $f : X \to Y$ in $\mathcal{C}$, the following diagram commutes:

$$\begin{array}{ccc} F(X) & \xrightarrow{\eta_X} & G(X) \\ {\scriptstyle F(f)}\downarrow & & \downarrow{\scriptstyle G(f)} \\ F(Y) & \xrightarrow[\eta_Y]{} & G(Y) \end{array}$$

A *natural isomorphism* is a natural transformation where each η_X is an isomorphism.

Definition 0.4.6: Equivalence of Categories

An *equivalence of categories* between $\mathcal{C}$ and $\mathcal{D}$ consists of functors $F : \mathcal{C} \to \mathcal{D}$ and $G : \mathcal{D} \to \mathcal{C}$ together with natural isomorphisms $G \circ F \cong \mathrm{id}_{\mathcal{C}}$ and $F \circ G \cong \mathrm{id}_{\mathcal{D}}$.
A functor F is an equivalence if and only if it is:

- *Fully faithful*: $F : \mathrm{Hom}_{\mathcal{C}}(X, Y) \to \mathrm{Hom}_{\mathcal{D}}(F(X), F(Y))$ is a bijection for all X, Y.
- *Essentially surjective*: every object of $\mathcal{D}$ is isomorphic to $F(X)$ for some $X \in \mathcal{C}$.

Example 0.4.7: The Classical Equivalence

The equivalence $\mathbf{Aff}_k \simeq (\mathbf{Alg}_k^{\text{f.g., red}})^{\text{op}}$ from section 0.1 is an equivalence of categories: the coordinate ring functor is fully faithful and essentially surjective.

Universal Properties: Limits and Colimits

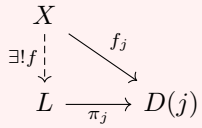
Many constructions in mathematics are characterized by universal properties. Limits and colimits provide a unified framework.

Definition 0.4.8: Diagram and Cone

Let J be a small category (the "index category"). A *diagram* of shape J in $\mathcal{C}$ is a functor $D : J \to \mathcal{C}$.
A *cone* over D with vertex $X \in \mathcal{C}$ is a collection of morphisms $\pi_j : X \to D(j)$ for each $j \in J$, such that for every morphism $\alpha : j \to j'$ in J, we have $D(\alpha) \circ \pi_j = \pi_{j'}$.

Definition 0.4.9: Limit

The *limit* of a diagram $D : J \to \mathcal{C}$, denoted $\varprojlim D$ or $\lim_J D$, is a cone $(\pi_j : L \to D(j))_{j \in J}$ that is universal: for any other cone $(f_j : X \to D(j))$, there exists a unique morphism $f : X \to L$ such that $\pi_j \circ f = f_j$ for all j.



Definition 0.4.10: Colimit

Dually, a *cocone* under D with vertex X is a collection $\iota_j : D(j) \to X$ compatible with the diagram. The *colimit* $\varinjlim D$ is the universal cocone.

Example 0.4.11: Specific Limits and Colimits

Shape J	**Limit**	**Colimit**
Discrete (no arrows)	Product $\prod_i X_i$	Coproduct $\coprod_i X_i$
$\bullet \rightrightarrows \bullet$	Equalizer	Coequalizer
$\bullet \to \bullet \leftarrow \bullet$	Pullback (fiber product)	—
$\bullet \leftarrow \bullet \to \bullet$	—	Pushout
Empty	Terminal object $*$	Initial object $\emptyset$
Directed poset	Directed (inverse) limit	Directed (direct) limit

Products and Coproducts

Definition 0.4.12: Product

The *product* of objects $\{X_i\}_{i \in I}$ is an object $\prod_i X_i$ together with projection morphisms $\pi_j : \prod_i X_i \to X_j$ such that for any object Y with morphisms $f_j : Y \to X_j$, there exists a unique $f : Y \to \prod_i X_i$ with $\pi_j \circ f = f_j$.

Definition 0.4.13: Coproduct

Dually, the *coproduct* $\coprod_i X_i$ has inclusion morphisms $\iota_j : X_j \to \coprod_i X_i$ universal for morphisms out of the X_j.

Example 0.4.14: Products and Coproducts in Various Categories

Category	Product	Coproduct
Set	Cartesian product	Disjoint union
Grp	Direct product	Free product
Ab, $\mathbf{Mod}_R$	Direct product	Direct sum
CRing	Direct product	Tensor product $A \otimes_{\mathbb{Z}} B$
Top	Product topology	Disjoint union topology
Sch	Fiber product over $\operatorname{Spec}\mathbb{Z}$	Disjoint union

Fiber products are ubiquitous in algebraic geometry: base change, intersections, and fiber constructions all use them.

Definition 0.4.15: Fiber Product / Pullback

Given morphisms $f : X \to S$ and $g : Y \to S$, the *fiber product* (or *pullback*) is an object $X \times_S Y$ with morphisms $p : X \times_S Y \to X$ and $q : X \times_S Y \to Y$ such that $f \circ p = g \circ q$, universal with this property:

$$\begin{array}{ccc} X \times_S Y & \xrightarrow{q} & Y \\ {\scriptstyle p}\downarrow & \square & \downarrow{\scriptstyle g} \\ X & \xrightarrow[f]{} & S \end{array}$$

For any Z with morphisms to X and Y making the square commute, there is a unique morphism $Z \to X \times_S Y$.

Example 0.4.16: Fiber Products in Sets

In **Set**, the fiber product is

$$X \times_S Y = \{(x, y) \in X \times Y : f(x) = g(y)\}$$

If $S = *$ is a point, this is the ordinary Cartesian product.

Adjoint Functors

Adjunctions capture many important relationships between categories.

Definition 0.4.17: Adjoint Pair

Let $F : \mathcal{C} \to \mathcal{D}$ and $G : \mathcal{D} \to \mathcal{C}$ be functors. We say F is *left adjoint* to G (and G is *right adjoint* to F), written $F \dashv G$, if there is a natural bijection

$$\operatorname{Hom}_{\mathcal{D}}(F(X), Y) \cong \operatorname{Hom}_{\mathcal{C}}(X, G(Y))$$

for all $X \in \mathcal{C}$ and $Y \in \mathcal{D}$.

Proposition 0.4.18: Characterizations of Adjunctions

The following are equivalent to $F \dashv G$:

1. Natural bijection $\mathrm{Hom}_{\mathcal{D}}(F(X), Y) \cong \mathrm{Hom}_{\mathcal{C}}(X, G(Y))$.
2. Existence of a *unit* $\eta : \mathrm{id}_{\mathcal{C}} \Rightarrow G \circ F$ and *counit* $\varepsilon : F \circ G \Rightarrow \mathrm{id}_{\mathcal{D}}$ satisfying the triangle identities.
3. F preserves colimits and satisfies a solution set condition (Freyd's adjoint functor theorem).

***Proof*:**
See any standard reference on category theory, e.g., Mac Lane [9].

Example 0.4.19: Adjunctions in Algebra and Geometry

1. **Free-Forgetful**: The free group functor $F : \mathbf{Set} \to \mathbf{Grp}$ is left adjoint to the forgetful functor $U : \mathbf{Grp} \to \mathbf{Set}$:
$$\mathrm{Hom}_{\mathbf{Grp}}(F(S), G) \cong \mathrm{Hom}_{\mathbf{Set}}(S, U(G))$$
2. **Tensor-Hom**: For R-modules, $(-) \otimes_R M \dashv \mathrm{Hom}_R(M, -)$:
$$\mathrm{Hom}_R(N \otimes_R M, P) \cong \mathrm{Hom}_R(N, \mathrm{Hom}_R(M, P))$$
3. **Inverse and Direct Image**: For a continuous map $f : X \to Y$, the inverse image functor f^{-1} on sheaves is left adjoint to the direct image f_*.

Two adjunctions we shall encounter are:

1. **Spec-Global Sections**: The functor $\mathrm{Spec} : \mathbf{CRing}^{\mathrm{op}} \to \mathbf{Sch}$ is left adjoint to $\Gamma : \mathbf{Sch} \to \mathbf{CRing}^{\mathrm{op}}$:
$$\mathrm{Hom}_{\mathbf{Sch}}(\mathrm{Spec}\,(A), X) \cong \mathrm{Hom}_{\mathbf{CRing}}(\Gamma(X, \mathcal{O}_X), A)$$
2. **Sheafification**: The sheafification functor is left adjoint to the inclusion of sheaves into presheaves.

Proposition 0.4.20: Adjoints and Limits

1. Left adjoints preserve colimits.
2. Right adjoints preserve limits.

In particular, if $F \dashv G$, then F preserves coproducts, coequalizers, and pushouts, while G preserves products, equalizers, and pullbacks.

***Proof*:**
See [9, Right Adjoints Preserve Limits].

Representable Functors and the Yoneda Lemma

The Yoneda perspective is fundamental to modern algebraic geometry: schemes can be understood through their "functor of points."

Definition 0.4.21: Representable Functor

A functor $F : \mathcal{C}^{\mathrm{op}} \to \mathbf{Set}$ is *representable* if there exists an object $X \in \mathcal{C}$ and a natural isomorphism

$$F \cong \mathrm{Hom}_{\mathcal{C}}(-, X) =: h_X$$

The object X is called the *representing object*. We say "X represents F" or "F is represented by X."

Theorem 0.4.22: Yoneda Lemma

Let $\mathcal{C}$ be a category, $X \in \mathcal{C}$, and $F : \mathcal{C}^{\mathrm{op}} \to \mathbf{Set}$ a functor. There is a natural bijection

$$\mathrm{Nat}(h_X, F) \cong F(X)$$

where $\mathrm{Nat}(h_X, F)$ is the set of natural transformations from h_X to F. The bijection sends $\eta : h_X \Rightarrow F$ to $\eta_X(\mathrm{id}_X) \in F(X)$.

***Proof*:**
See [9, The Yoneda Lemma, Presheaf Form].

Corollary 0.4.23: Yoneda Embedding

The functor $h : \mathcal{C} \to \mathbf{Fun}(\mathcal{C}^{\mathrm{op}}, \mathbf{Set})$ given by $X \mapsto h_X = \mathrm{Hom}(-, X)$ is fully faithful. In particular:

1. If $h_X \cong h_Y$, then $X \cong Y$ (representing objects are unique up to unique isomorphism).
2. $\mathrm{Hom}_{\mathcal{C}}(X, Y) \cong \mathrm{Nat}(h_X, h_Y)$.

Example 0.4.24: Moduli Problems as Functors
Many moduli problems are phrased as functors $F : \mathbf{Sch}^{\mathrm{op}} \to \mathbf{Set}$:

- $F(S) = \{\text{families of elliptic curves over } S\}$
- $F(S) = \{\text{closed subschemes of } \mathbb{P}^n_S \text{ flat over } S\}$ (Hilbert functor)

If F is representable by a scheme M, then M is the *moduli space*. Often F is not representable by a scheme but by an algebraic stack.

Filtered Colimits

Filtered colimits arise naturally when taking stalks of sheaves and germs of functions.

Definition 0.4.25: Filtered Category

A category J is *filtered* if:

1. J is nonempty.
2. For any $i, j \in J$, there exists $k \in J$ with morphisms $i \to k$ and $j \to k$.
3. For any parallel morphisms $f, g : i \to j$, there exists $h : j \to k$ with $h \circ f = h \circ g$.

A *directed set* (poset where every pair has an upper bound) is filtered when viewed as a category.

Definition 0.4.26: Filtered Colimit

A *filtered colimit* (or *direct limit*) is a colimit over a filtered index category.

Example 0.4.27: Localization as Filtered Colimit

For a ring A and multiplicative set S, the localization is

$$S^{-1}A = \varinjlim_{s \in S} A_s$$

where $A_s = A[1/s]$ and the colimit is over S ordered by divisibility.

Proposition 0.4.28: Filtered Colimits Commute with Finite Limits

In **Set** (and more generally in any *finitely presentable* category), filtered colimits commute with finite limits. In particular, filtered colimits are exact in module categories.

***Proof*:**

See [9].

Abelian Categories

For homological algebra and cohomology, we need abelian categories.

Definition 0.4.29: Abelian Category

A category $\mathcal{A}$ is *abelian* if:

1. Every set $\mathrm{Hom}_{\mathcal{A}}(A, B)$ is an abelian group and composition is bilinear.
2. $\mathcal{A}$ has a zero object (both initial and terminal).
3. $\mathcal{A}$ has all binary products and coproducts (which coincide: biproducts).
4. Every morphism has a kernel and cokernel.
5. Every monomorphism is a kernel; every epimorphism is a cokernel.

Example 0.4.30: Abelian Categories

1. $\mathbf{Ab}$, $\mathbf{Mod}_R$ for any ring R.
2. $\mathbf{Sh}(X, \mathbf{Ab})$: sheaves of abelian groups on X.
3. $\mathbf{QCoh}(X)$: quasi-coherent sheaves on a scheme X.
4. $\mathbf{Coh}(X)$: coherent sheaves on a Noetherian scheme X.

0.4.31 Homological Algebra in Abelian Categories In an abelian category, one can define:

- Exact sequences: $A \to B \to C$ is exact at B if $\ker(B \to C) = \mathrm{im}(A \to B)$.
- Chain complexes and homology.
- Derived functors (if there are enough injectives or projectives).

Sheaf cohomology $H^i(X, \mathcal{F})$ is defined as the right derived functors of the global sections functor $\Gamma(X, -)$.

Definition 0.4.32: Exact Functor

Let $\mathcal{A}$ be a category with a zero object, whose hom-sets are abelian groups with bilinear composition, and in which every morphism has a kernel and a cokernel, for example an abelian category. A sequence $A' \xrightarrow{f} A \xrightarrow{g} A''$ in $\mathcal{A}$ with $g \circ f = 0$ is *exact at* A if $\ker g$ and $\mathrm{im} f = \ker(\mathrm{coker} f)$ are the same subobject of A, and $0 \to A' \to A \to A'' \to 0$ is *short exact* if it is exact at A', A and A''. Let $\mathcal{B}$ be a second such category. A functor $F \colon \mathcal{A} \to \mathcal{B}$ is *additive* if every map $\mathrm{Hom}_{\mathcal{A}}(A, B) \to \mathrm{Hom}_{\mathcal{B}}(FA, FB)$ is a group homomorphism. An additive functor F is *left exact* if for every short exact sequence $0 \to A' \to A \to A'' \to 0$ the sequence $0 \to FA' \to FA \to FA''$ is exact at FA' and at FA, *right exact* if $FA' \to FA \to FA'' \to 0$ is exact at FA and at FA'', and *exact* if it is both.

0.4.1 Summary: Categorical Tools for Schemes

Concept	Use in Algebraic Geometry
Equivalence of categories	Classical: $\mathbf{Aff}_k \simeq (\mathbf{Alg}_k^{\text{f.g.,red}})^{\text{op}}$
Fiber products	Base change, fibers, intersections
Adjunctions	$\Gamma \dashv \operatorname{Spec}$, $f^{-1} \dashv f_*$, sheafification
Representable functors	Schemes via functor of points
Yoneda lemma	Universal properties; moduli problems
Filtered colimits	Stalks, localizations
Abelian categories	Sheaf cohomology, derived categories

0.5 *Key Results from Differential Geometry

Many constructions in algebraic geometry—tangent bundles, differential forms, cohomology, metrics—have their origins in differential geometry. This section reviews the essential concepts from the smooth category that will inform and motivate their algebraic counterparts.

Throughout this section, "smooth" means C^∞ (infinitely differentiable).

Smooth Manifolds

Definition 0.5.1: Smooth Manifold

A *topological manifold* of dimension n is a Hausdorff, second-countable topological space M that is locally homeomorphic to $\mathbb{R}^n$. A *chart* is a pair (U, φ) where $U \subseteq M$ is open and $\varphi : U \xrightarrow{\sim} \varphi(U) \subseteq \mathbb{R}^n$ is a homeomorphism.
A *smooth structure* on M is a maximal atlas $\{(U_\alpha, \varphi_\alpha)\}$ such that all transition maps

$$\varphi_\beta \circ \varphi_\alpha^{-1} : \varphi_\alpha(U_\alpha \cap U_\beta) \to \varphi_\beta(U_\alpha \cap U_\beta)$$

are smooth (as maps between open subsets of $\mathbb{R}^n$).
A *smooth manifold* is a topological manifold equipped with a smooth structure.

Example 0.5.2: Examples of Smooth Manifolds

1. $\mathbb{R}^n$ with the standard atlas (single chart $\mathrm{id} : \mathbb{R}^n \to \mathbb{R}^n$).
2. Open subsets of $\mathbb{R}^n$ (or of any smooth manifold).
3. The n-sphere $S^n = \{x \in \mathbb{R}^{n+1} : |x| = 1\}$.
4. Real and complex projective spaces $\mathbb{RP}^n$, $\mathbb{CP}^n$.
5. Lie groups: $\mathrm{GL}_n(\mathbb{R})$, $\mathrm{O}(n)$, $\mathrm{U}(n)$, $\mathrm{SL}_n(\mathbb{R})$, etc.
6. Any smooth algebraic variety over $\mathbb{R}$ or $\mathbb{C}$ (with its classical topology).

Definition 0.5.3: Smooth Map

A map $f : M \to N$ between smooth manifolds is *smooth* if for every $p \in M$ and charts (U, φ) around p and (V, ψ) around $f(p)$, the composition

$$\psi \circ f \circ \varphi^{-1} : \varphi(U \cap f^{-1}(V)) \to \psi(V)$$

is smooth as a map between open subsets of Euclidean spaces.
A *diffeomorphism* is a smooth bijection with smooth inverse.

Definition 0.5.4: Tangent Space (via Derivations)

Let M be a smooth manifold and $p \in M$. A *derivation at* p is an $\mathbb{R}$-linear map $v : C^\infty(M) \to \mathbb{R}$ satisfying the Leibniz rule:

$$v(fg) = f(p) \cdot v(g) + g(p) \cdot v(f)$$

The *tangent space* T_pM is the vector space of all derivations at p.

Proposition 0.5.5: Dimension of Tangent Space

If $\dim M = n$, then $\dim T_pM = n$ for all $p \in M$. In local coordinates $(x^1, \ldots, x^n)$ centered at p, a basis for T_pM is given by the partial derivative operators:

$$\left\{ \frac{\partial}{\partial x^1}\bigg|_p, \ldots, \frac{\partial}{\partial x^n}\bigg|_p \right\}$$

Definition 0.5.6: Differential / Pushforward

Let $f : M \to N$ be smooth and $p \in M$. The *differential* (or *pushforward*) of f at p is the linear map

$$df_p = f_{*,p} : T_pM \to T_{f(p)}N, \qquad (df_p(v))(g) := v(g \circ f)$$

for $v \in T_pM$ and $g \in C^\infty(N)$.

Definition 0.5.7: Tangent Bundle

The *tangent bundle* of M is the disjoint union

$$TM := \bigsqcup_{p \in M} T_pM = \{(p, v) : p \in M,\ v \in T_pM\}$$

equipped with the natural projection $\pi : TM \to M$, $(p, v) \mapsto p$. This is a smooth manifold of dimension $2n$ and a vector bundle of rank n over M.

Definition 0.5.8: Vector Field

A *vector field* on M is a smooth section of the tangent bundle, i.e., a smooth map $X : M \to TM$ such that $\pi \circ X = \mathrm{id}_M$. Equivalently, X assigns to each $p \in M$ a tangent vector $X_p \in T_pM$, varying smoothly with p.
The space of vector fields is denoted $\mathfrak{X}(M)$ or $\Gamma(TM)$.

Proposition 0.5.9: Vector Fields as Derivations

A vector field $X \in \mathfrak{X}(M)$ defines a derivation $X : C^\infty(M) \to C^\infty(M)$ by $(Xf)(p) := X_p(f)$. This gives an isomorphism between $\mathfrak{X}(M)$ and the $C^\infty(M)$-module of derivations of $C^\infty(M)$.

Definition 0.5.10: Lie Bracket

The *Lie bracket* of vector fields $X, Y \in \mathfrak{X}(M)$ is the vector field $[X, Y]$ defined by

$$[X, Y](f) := X(Y(f)) - Y(X(f))$$

This makes $\mathfrak{X}(M)$ into a Lie algebra over $\mathbb{R}$.

Cotangent Spaces and Differential Forms

Definition 0.5.11: Cotangent Space

The *cotangent space* at $p \in M$ is the dual vector space

$$T_p^*M := (T_pM)^* = \mathrm{Hom}_{\mathbb{R}}(T_pM, \mathbb{R})$$

Elements of T_p^*M are called *cotangent vectors* or *covectors*.

Definition 0.5.12: Differential of a Function

For $f \in C^\infty(M)$ and $p \in M$, the *differential* $df_p \in T_p^*M$ is defined by

$$df_p(v) := v(f) \quad \text{for } v \in T_pM$$

In local coordinates, $df = \sum_{i=1}^{n} \frac{\partial f}{\partial x^i} dx^i$, where $\{dx^1, \ldots, dx^n\}$ is the dual basis to $\{\partial/\partial x^1, \ldots, \partial/\partial x^n\}$.

Definition 0.5.13: Cotangent Bundle

The *cotangent bundle* is $T^*M := \bigsqcup_{p \in M} T_p^*M$, a vector bundle of rank n over M.

Definition 0.5.14: Differential k-Forms

A *differential k-form* (or simply *k-form*) on M is a smooth section of $\bigwedge^k T^*M$. That is, a k-form ω assigns to each $p \in M$ an alternating k-linear map

$$\omega_p : \underbrace{T_pM \times \cdots \times T_pM}_{k \text{ times}} \to \mathbb{R}$$

varying smoothly with p.
The space of k-forms is denoted $\Omega^k(M)$. We set $\Omega^0(M) = C^\infty(M)$.

Example 0.5.15: Differential Forms in Coordinates

In local coordinates $(x^1, \ldots, x^n)$, every k-form can be written as

$$\omega = \sum_{i_1 < \cdots < i_k} f_{i_1 \ldots i_k} \, dx^{i_1} \wedge \cdots \wedge dx^{i_k}$$

where $f_{i_1 \ldots i_k} \in C^\infty(M)$ and $\wedge$ denotes the wedge product.

Definition 0.5.16: Wedge Product

The *wedge product* $\wedge : \Omega^k(M) \times \Omega^\ell(M) \to \Omega^{k+\ell}(M)$ is the unique bilinear, associative operation satisfying:

1. $df \wedge dg = -dg \wedge df$ for $f, g \in C^\infty(M)$.
2. $(f\omega) \wedge \eta = f(\omega \wedge \eta) = \omega \wedge (f\eta)$ for $f \in C^\infty(M)$.
3. $\omega \wedge \eta = (-1)^{k\ell} \eta \wedge \omega$ for $\omega \in \Omega^k$, $\eta \in \Omega^\ell$.

This makes $\Omega^*(M) := \bigoplus_{k \geq 0} \Omega^k(M)$ into a graded-commutative algebra.

Definition 0.5.17: Pullback of Forms

Let $f : M \to N$ be smooth. The *pullback* $f^* : \Omega^k(N) \to \Omega^k(M)$ is defined by

$$(f^*\omega)_p(v_1, \ldots, v_k) := \omega_{f(p)}(df_p(v_1), \ldots, df_p(v_k))$$

The pullback satisfies $f^*(\omega \wedge \eta) = f^*\omega \wedge f^*\eta$ and $(g \circ f)^* = f^* \circ g^*$.

Exterior Derivative and de Rham Cohomology

Definition 0.5.18: Exterior Derivative

The *exterior derivative* is the unique family of $\mathbb{R}$-linear maps $d : \Omega^k(M) \to \Omega^{k+1}(M)$ satisfying:

1. On $\Omega^0(M) = C^\infty(M)$, d is the differential: $(df)(X) = X(f)$.
2. $d \circ d = 0$.
3. $d(\omega \wedge \eta) = d\omega \wedge \eta + (-1)^k \omega \wedge d\eta$ for $\omega \in \Omega^k$ (graded Leibniz rule).

Example 0.5.19: Exterior Derivative in Coordinates

For $\omega = \sum_I f_I \, dx^I$ (where I is a multi-index), we have

$$d\omega = \sum_I \sum_{j=1}^{n} \frac{\partial f_I}{\partial x^j} dx^j \wedge dx^I$$

For instance, if $\omega = f\,dx + g\,dy$ on $\mathbb{R}^2$, then

$$d\omega = \left(\frac{\partial g}{\partial x} - \frac{\partial f}{\partial y} \right) dx \wedge dy$$

Proposition 0.5.20: Naturality of d

The exterior derivative commutes with pullback: for $f : M \to N$ smooth,

$$f^* \circ d = d \circ f^*$$

That is, $f^*(d\omega) = d(f^*\omega)$ for all $\omega \in \Omega^*(N)$.

Definition 0.5.21: de Rham Complex

The *de Rham complex* of M is the cochain complex

$$0 \to \Omega^0(M) \xrightarrow{d} \Omega^1(M) \xrightarrow{d} \Omega^2(M) \xrightarrow{d} \cdots \xrightarrow{d} \Omega^n(M) \to 0$$

A form ω is *closed* if $d\omega = 0$, and *exact* if $\omega = d\eta$ for some η. Since $d^2 = 0$, every exact form is closed.

Definition 0.5.22: de Rham Cohomology

The *k-th de Rham cohomology* of M is

$$H^k_{\mathrm{dR}}(M) := \frac{\ker(d : \Omega^k \to \Omega^{k+1})}{\mathrm{im}(d : \Omega^{k-1} \to \Omega^k)} = \frac{\{\text{closed } k\text{-forms}\}}{\{\text{exact } k\text{-forms}\}}$$

This is a real vector space measuring the "failure of closed forms to be exact."

Theorem 0.5.23: de Rham's Theorem

For a smooth manifold M, there is a natural isomorphism

$$H^k_{\mathrm{dR}}(M) \cong H^k(M; \mathbb{R})$$

where $H^k(M; \mathbb{R})$ is the singular cohomology of M with real coefficients. The isomorphism is given by integration of forms over cycles.

***Proof*:**
See [59]

Example 0.5.24: de Rham Cohomology Examples

1. $H^k_{\mathrm{dR}}(\mathbb{R}^n) = 0$ for $k > 0$, and $H^0_{\mathrm{dR}}(\mathbb{R}^n) = \mathbb{R}$ (Poincaré lemma).
2. $H^0_{\mathrm{dR}}(M) = \mathbb{R}^{\pi_0(M)}$, where $\pi_0(M)$ is the number of connected components.
3. $H^1_{\mathrm{dR}}(S^1) = \mathbb{R}$, generated by $[d\theta]$.
4. $H^k_{\mathrm{dR}}(S^n) = \mathbb{R}$ for $k = 0, n$, and 0 otherwise.

0.5.25 Algebraic de Rham Cohomology For a smooth algebraic variety X over a field k of characteristic zero, one can define *algebraic de Rham cohomology* using the complex of algebraic differential forms $\Omega^*_{X/k}$. Over $\mathbb{C}$, this agrees with singular cohomology by Grothendieck's algebraic de Rham theorem.

Vector Bundles

Definition 0.5.26: Vector Bundle

A *(real) vector bundle of rank r* over a smooth manifold M is a smooth manifold E together with:

1. A smooth surjection $\pi : E \to M$.
2. For each $p \in M$, the fiber $E_p := \pi^{-1}(p)$ has the structure of an r-dimensional real vector space.
3. Local triviality: each $p \in M$ has a neighborhood U with a diffeomorphism $\varphi : \pi^{-1}(U) \xrightarrow{\sim} U \times \mathbb{R}^r$ such that φ is linear on each fiber.

Example 0.5.27: Vector Bundle Examples

1. The *trivial bundle* $M \times \mathbb{R}^r \to M$.
2. The tangent bundle TM and cotangent bundle T^*M.
3. The *tautological line bundle* $\mathcal{O}(-1)$ over $\mathbb{RP}^n$ or $\mathbb{CP}^n$: the fiber over a line ℓ is ℓ itself.
4. Exterior powers $\bigwedge^k T^*M$ (whose sections are k-forms).
5. The *canonical bundle* $K_M := \bigwedge^n T^*M$ (top exterior power).

Definition 0.5.28: Section of a Vector Bundle

A *section* of $\pi : E \to M$ is a smooth map $s : M \to E$ with $\pi \circ s = \mathrm{id}_M$. The space of sections is denoted $\Gamma(E)$ or $\Gamma(M, E)$. This is a $C^\infty(M)$-module.

Definition 0.5.29: Operations on Vector Bundles

Given vector bundles E, F over M:

- *Direct sum*: $(E \oplus F)_p = E_p \oplus F_p$.
- *Tensor product*: $(E \otimes F)_p = E_p \otimes_{\mathbb{R}} F_p$.
- *Dual*: $E_p^* = (E_p)^*$.
- *Exterior powers*: $(\bigwedge^k E)_p = \bigwedge^k E_p$.
- *Pullback*: for $f : N \to M$, $(f^*E)_q = E_{f(q)}$.

Definition 0.5.30: Transition Functions

If $\{U_\alpha\}$ is an open cover of M with local trivializations $\varphi_\alpha : E|_{U_\alpha} \xrightarrow{\sim} U_\alpha \times \mathbb{R}^r$, the *transition functions* are

$$g_{\alpha\beta} : U_\alpha \cap U_\beta \to \mathrm{GL}_r(\mathbb{R}), \qquad \varphi_\alpha \circ \varphi_\beta^{-1}(p, v) = (p, g_{\alpha\beta}(p) \cdot v)$$

These satisfy the cocycle condition: $g_{\alpha\beta} g_{\beta\gamma} = g_{\alpha\gamma}$ on triple overlaps.

0.5.31 Line Bundles and Divisors A *line bundle* is a vector bundle of rank 1. Line bundles are classified by $H^1(M; C^\infty(M, \mathbb{R}^\times))$ via transition functions. In algebraic geometry, line bundles on a variety X correspond to Cartier divisors and form the Picard group $\mathrm{Pic}(X)$.

Connections and Curvature

Definition 0.5.32: Connection on a Vector Bundle

A *connection* (or *covariant derivative*) on a vector bundle $E \to M$ is an $\mathbb{R}$-linear map

$$\nabla : \Gamma(E) \to \Gamma(T^*M \otimes E) = \Omega^1(M, E)$$

satisfying the Leibniz rule: for $f \in C^\infty(M)$ and $s \in \Gamma(E)$,

$$\nabla(fs) = df \otimes s + f \nabla s$$

For a vector field X, we write $\nabla_X s := \iota_X(\nabla s) \in \Gamma(E)$ for the covariant derivative of s in direction X.

0.5.33 Connections Exist Every vector bundle over a paracompact manifold admits a connection. Connections form an affine space over $\Omega^1(M, \mathrm{End}(E))$: if ∇ is a connection and $A \in \Omega^1(M, \mathrm{End}(E))$, then $\nabla + A$ is also a connection.

Definition 0.5.34: Curvature

The *curvature* of a connection ∇ is the $\mathrm{End}(E)$-valued 2-form $F_\nabla \in \Omega^2(M, \mathrm{End}(E))$ defined by

$$F_\nabla(X, Y) := \nabla_X \nabla_Y - \nabla_Y \nabla_X - \nabla_{[X,Y]}$$

for vector fields X, Y. A connection is *flat* if $F_\nabla = 0$.

Proposition 0.5.35: Curvature as Obstruction

A connection ∇ is flat if and only if parallel transport is path-independent (depends only on the homotopy class of the path). Flat connections on a trivial bundle correspond to representations $\pi_1(M) \to \mathrm{GL}_r(\mathbb{R})$.

Riemannian Metrics

Definition 0.5.36: Riemannian Metric

A *Riemannian metric* on M is a smooth assignment of an inner product $g_p : T_pM \times T_pM \to \mathbb{R}$ to each $p \in M$. In local coordinates, $g = \sum_{i,j} g_{ij}\, dx^i \otimes dx^j$ where $(g_{ij}(p))$ is a positive definite symmetric matrix for each p.
A *Riemannian manifold* is a pair (M, g).

Example 0.5.37: Riemannian Metrics

1. The standard metric on $\mathbb{R}^n$: $g = dx^1 \otimes dx^1 + \cdots + dx^n \otimes dx^n$.
2. The round metric on S^n (induced from $\mathbb{R}^{n+1}$).
3. The hyperbolic metric on the upper half-plane: $g = \frac{dx^2+dy^2}{y^2}$.
4. The Fubini–Study metric on $\mathbb{CP}^n$.

Proposition 0.5.38: Existence of Riemannian Metrics

Every smooth manifold admits a Riemannian metric. (This uses partitions of unity.)

Definition 0.5.39: Levi-Civita Connection

On a Riemannian manifold (M, g), there is a unique connection ∇ on TM that is:

1. *Metric-compatible*: $X(g(Y, Z)) = g(\nabla_X Y, Z) + g(Y, \nabla_X Z)$.
2. *Torsion-free*: $\nabla_X Y - \nabla_Y X = [X, Y]$.

This is called the *Levi-Civita connection*.

Definition 0.5.40: Geodesics

A *geodesic* is a curve $\gamma : I \to M$ satisfying $\nabla_{\dot\gamma}\dot\gamma = 0$. Geodesics are locally length-minimizing curves.

Integration and Orientation

Definition 0.5.41: Orientation

An *orientation* on an n-manifold M is a choice of smoothly varying orientation on each tangent space T_pM. Equivalently, it is a nowhere-vanishing n-form $\omega \in \Omega^n(M)$, or a trivialization of the canonical bundle $K_M = \bigwedge^n T^*M$.
A manifold is *orientable* if it admits an orientation, and *oriented* if one has been chosen.

Example 0.5.42: Orientability

1. $\mathbb{R}^n$, S^n, and $\mathbb{CP}^n$ are orientable.
2. $\mathbb{RP}^n$ is orientable if and only if n is odd.
3. The Möbius strip is not orientable.

Definition 0.5.43: Integration of Top Forms

Let M be an oriented n-manifold and $\omega \in \Omega_c^n(M)$ a compactly supported n-form. The *integral* $\int_M \omega$ is defined by:

1. In a single oriented chart (U, φ) with $\operatorname{supp}(\omega) \subseteq U$: write $\omega = f\, dx^1 \wedge \cdots \wedge dx^n$ and set
$$\int_M \omega := \int_{\varphi(U)} f(\varphi^{-1}(x))\, dx^1 \cdots dx^n$$
2. For general ω, use a partition of unity to reduce to the local case.

Theorem 0.5.44: Stokes' Theorem

Let M be an oriented n-manifold with boundary ∂M (given the induced orientation), and let $\omega \in \Omega_c^{n-1}(M)$. Then
$$\int_M d\omega = \int_{\partial M} \omega$$
In particular, if M is closed (compact without boundary), then $\int_M d\omega = 0$.

***Proof*:**
See [59, Stokes' Theorem]

0.5.45 Classical Theorems as Special Cases Stokes' theorem generalizes:

- The fundamental theorem of calculus ($n = 1$).
- Green's theorem ($n = 2$, planar region).
- The classical Stokes theorem ($n = 3$, surface in $\mathbb{R}^3$).
- The divergence theorem ($n = 3$, solid region in $\mathbb{R}^3$).

Partitions of Unity

Definition 0.5.46: Partition of Unity

Let $\{U_\alpha\}_{\alpha\in A}$ be an open cover of M. A *partition of unity subordinate to* $\{U_\alpha\}$ is a collection of smooth functions $\{\rho_\alpha : M \to [0,1]\}_{\alpha\in A}$ such that:

1. $\text{supp}(\rho_\alpha) \subseteq U_\alpha$ for each α.
2. The collection $\{\text{supp}(\rho_\alpha)\}$ is locally finite.
3. $\sum_\alpha \rho_\alpha(p) = 1$ for all $p \in M$.

Theorem 0.5.47: Existence of Partitions of Unity

Every open cover of a smooth manifold admits a subordinate partition of unity.

***Proof*:**
See [59, Existence of Smooth Partitions of Unity].

0.5.48 Partitions of Unity in Algebraic Geometry Partitions of unity are a crucial tool in differential geometry, enabling:

- Existence of Riemannian metrics.
- Existence of connections.
- Extension of local constructions to global ones.
- Proof that smooth manifolds are paracompact.

Importantly, partitions of unity do not exist in algebraic geometry. Polynomial or algebraic functions cannot be "bump functions." This is one reason why sheaf cohomology becomes essential in algebraic geometry: it replaces the role of partitions of unity in extending local to global.

0.5.1 Summary: Differential vs. Algebraic Geometry

Concept	Differential Geometry	Algebraic Geometry
Local model	$\mathbb{R}^n$ or $\mathbb{C}^n$	$\mathrm{Spec}\,(A)$
Functions	$C^\infty(M)$	$\mathcal{O}_X(X)$ (regular functions)
Tangent space	T_pM (derivations)	$(\mathfrak{m}_p/\mathfrak{m}_p^2)^\vee$
Cotangent space	T_p^*M	$\mathfrak{m}_p/\mathfrak{m}_p^2$
Vector bundle	Smooth fiber bundle	Locally free sheaf
Differential forms	$\Omega^k(M)$	$\Omega^k_{X/k}$ (Kähler differentials)
Cohomology	de Rham $H^*_{\mathrm{dR}}(M)$	Sheaf cohomology $H^*(X, \mathcal{F})$
Partitions of unity	Exist	Do not exist
Gluing	Smooth	Via sheaf axioms

The absence of partitions of unity in the algebraic world makes sheaf-theoretic methods indispensable. Many constructions that are "automatic" in differential geometry (e.g., existence of metrics) require careful sheaf-cohomological arguments in algebraic geometry.

0.6 *Key Results from Complex Geometry

Complex geometry sits at the intersection of differential geometry and algebraic geometry. Over $\mathbb{C}$, algebraic varieties carry both a Zariski topology (algebraic) and a classical topology (analytic), and Serre's GAGA theorems establish connections between these perspectives. This section reviews the essential concepts from complex analytic geometry that inform and enrich algebraic geometry over $\mathbb{C}$.

Complex Manifolds

Definition 0.6.1: Complex Manifold

A *complex manifold* of dimension n is a smooth manifold M of real dimension $2n$ equipped with an atlas $\{(U_\alpha, \varphi_\alpha)\}$ where each $\varphi_\alpha : U_\alpha \xrightarrow{\sim} \varphi_\alpha(U_\alpha) \subseteq \mathbb{C}^n$ is a homeomorphism onto an open subset of $\mathbb{C}^n$, and all transition maps

$$\varphi_\beta \circ \varphi_\alpha^{-1} : \varphi_\alpha(U_\alpha \cap U_\beta) \to \varphi_\beta(U_\alpha \cap U_\beta)$$

are *holomorphic* (i.e., complex analytic).

Definition 0.6.2: Holomorphic Function

A function $f : U \to \mathbb{C}$ on an open set $U \subseteq \mathbb{C}^n$ is *holomorphic* if it satisfies the Cauchy–Riemann equations in each variable, or equivalently, if it is complex differentiable at every point. In coordinates $z_j = x_j + iy_j$, this means

$$\frac{\partial f}{\partial \bar{z}_j} = 0 \quad \text{for all } j$$

where $\frac{\partial}{\partial \bar{z}_j} = \frac{1}{2}\left(\frac{\partial}{\partial x_j} + i\frac{\partial}{\partial y_j}\right)$.

Definition 0.6.3: Holomorphic Map

A map $f : M \to N$ between complex manifolds is *holomorphic* if it is holomorphic in local coordinates. A *biholomorphism* is a holomorphic bijection with holomorphic inverse.

Example 0.6.4: Complex Manifolds

1. $\mathbb{C}^n$ with the standard complex structure.
2. Open subsets of $\mathbb{C}^n$ (or of any complex manifold).
3. Complex projective space $\mathbb{CP}^n$: covered by $n+1$ charts $U_i = \{[z_0 : \cdots : z_n] : z_i \neq 0\} \cong \mathbb{C}^n$.
4. Smooth projective varieties over $\mathbb{C}$ (with the classical topology).
5. Complex tori $\mathbb{C}^n/\Lambda$ where $\Lambda \cong \mathbb{Z}^{2n}$ is a lattice.
6. Riemann surfaces: complex manifolds of dimension 1.

0.6.5 Smooth vs. Complex vs. Algebraic We have strict inclusions of categories:

$$\{\text{smooth projective varieties}/\mathbb{C}\} \subsetneq \{\text{compact Kähler manifolds}\} \subsetneq \{\text{compact complex manifolds}\}$$

Not every complex manifold is algebraic (e.g., generic complex tori of dimension ≥ 2), and not every compact complex manifold is Kähler (e.g., most Hopf surfaces).

Almost Complex and Complex Structures

Definition 0.6.6: Almost Complex Structure

An *almost complex structure* on a smooth manifold M is a smooth bundle endomorphism $J : TM \to TM$ satisfying $J^2 = -\mathrm{id}$. This makes each tangent space T_pM into a complex vector space via $(a+bi) \cdot v := av + bJ(v)$.

Proposition 0.6.7: Complex Manifolds Have Almost Complex Structures

Every complex manifold has a natural almost complex structure: in holomorphic coordinates $(z_1, \dots, z_n)$ with $z_j = x_j + iy_j$,

$$J\left(\frac{\partial}{\partial x_j}\right) = \frac{\partial}{\partial y_j}, \qquad J\left(\frac{\partial}{\partial y_j}\right) = -\frac{\partial}{\partial x_j}$$

The converse is not automatic:

Definition 0.6.8: Integrable Almost Complex Structure

An almost complex structure J is *integrable* if it arises from a complex manifold structure. The *Nijenhuis tensor* N_J measures the obstruction to integrability.

Theorem 0.6.9: Newlander–Nirenberg Theorem

An almost complex structure J is integrable if and only if the Nijenhuis tensor vanishes: $N_J = 0$.

***Proof*:**
The argument is analytic rather than algebro-geometric and is not proved here; nothing below depends on it. See [41] or [73].

Complexified Tangent Bundle and (p, q)-Forms

Definition 0.6.10: Complexified Tangent Bundle

For a complex manifold M, the *complexified tangent bundle* is $T_{\mathbb{C}}M := TM \otimes_{\mathbb{R}} \mathbb{C}$. The almost complex structure J extends $\mathbb{C}$-linearly and has eigenvalues $\pm i$. This gives a decomposition

$$T_{\mathbb{C}}M = T^{1,0}M \oplus T^{0,1}M$$

where $T^{1,0}M$ is the $+i$ eigenspace and $T^{0,1}M$ is the $-i$ eigenspace.
In local coordinates, $T^{1,0}M$ is spanned by $\frac{\partial}{\partial z_j} := \frac{1}{2}\left(\frac{\partial}{\partial x_j} - i\frac{\partial}{\partial y_j}\right)$ and $T^{0,1}M$ by $\frac{\partial}{\partial \bar{z}_j} := \frac{1}{2}\left(\frac{\partial}{\partial x_j} + i\frac{\partial}{\partial y_j}\right)$.

Definition 0.6.11: (p,q)-Forms

The complexified cotangent bundle similarly decomposes: $T^*_{\mathbb{C}}M = (T^{1,0}M)^* \oplus (T^{0,1}M)^*$. A *differential form of type* (p,q) is a section of

$$\bigwedge^{p,q} T^*_{\mathbb{C}}M := \bigwedge^{p}(T^{1,0}M)^* \otimes \bigwedge^{q}(T^{0,1}M)^*$$

The space of (p,q)-forms is denoted $\Omega^{p,q}(M)$ or $\mathcal{A}^{p,q}(M)$.

Proposition 0.6.12: Decomposition of Forms

The space of complex k-forms decomposes as

$$\Omega^k(M,\mathbb{C}) = \bigoplus_{p+q=k} \Omega^{p,q}(M)$$

In local coordinates, a (p,q)-form can be written as

$$\omega = \sum_{|I|=p,|J|=q} f_{I,J}\, dz^I \wedge d\bar{z}^J$$

where $dz^I = dz_{i_1} \wedge \cdots \wedge dz_{i_p}$ and $d\bar{z}^J = d\bar{z}_{j_1} \wedge \cdots \wedge d\bar{z}_{j_q}$.

The Dolbeault Operators

Definition 0.6.13: Dolbeault Operators

The exterior derivative $d : \Omega^k(M,\mathbb{C}) \to \Omega^{k+1}(M,\mathbb{C})$ decomposes as $d = \partial + \bar{\partial}$ where:

$$\partial : \Omega^{p,q}(M) \to \Omega^{p+1,q}(M)$$
$$\bar{\partial} : \Omega^{p,q}(M) \to \Omega^{p,q+1}(M)$$

These satisfy $\partial^2 = 0$, $\bar{\partial}^2 = 0$, and $\partial\bar{\partial} + \bar{\partial}\partial = 0$.

Example 0.6.14: Dolbeault Operators in Coordinates

For $f \in C^\infty(M,\mathbb{C}) = \Omega^{0,0}(M)$:

$$\partial f = \sum_{j=1}^{n} \frac{\partial f}{\partial z_j} dz_j, \qquad \bar{\partial} f = \sum_{j=1}^{n} \frac{\partial f}{\partial \bar{z}_j} d\bar{z}_j$$

A function is holomorphic if and only if $\bar{\partial} f = 0$.

Definition 0.6.15: Dolbeault Complex

For each p, the *Dolbeault complex* is

$$0 \to \Omega^{p,0}(M) \xrightarrow{\bar\partial} \Omega^{p,1}(M) \xrightarrow{\bar\partial} \Omega^{p,2}(M) \xrightarrow{\bar\partial} \cdots \xrightarrow{\bar\partial} \Omega^{p,n}(M) \to 0$$

Definition 0.6.16: Dolbeault Cohomology

The *Dolbeault cohomology groups* are

$$H^{p,q}_{\bar\partial}(M) := \frac{\ker(\bar\partial : \Omega^{p,q} \to \Omega^{p,q+1})}{\mathrm{im}(\bar\partial : \Omega^{p,q-1} \to \Omega^{p,q})}$$

These are complex vector spaces measuring the failure of the $\bar\partial$-Poincaré lemma.

Theorem 0.6.17: Dolbeault's Theorem

For a complex manifold M, there is a natural isomorphism

$$H^{p,q}_{\bar\partial}(M) \cong H^q(M, \Omega^p_M)$$

where Ω^p_M is the sheaf of holomorphic p-forms.

***Proof Cited*:**
The $\bar\partial$-Poincaré lemma makes the Dolbeault complex $\mathcal{A}^{p,0} \to \mathcal{A}^{p,1} \to \cdots$ a resolution of Ω^p_M by sheaves of smooth forms, which are acyclic because they admit partitions of unity; so the resolution computes $H^\bullet(M, \Omega^p_M)$, and its global sections are by definition the Dolbeault groups. The details are in [41].

Hermitian and Kähler Metrics

Definition 0.6.18: Hermitian Metric

A *Hermitian metric* on a complex manifold M is a smoothly varying Hermitian inner product h_p on each $T^{1,0}_p M$. Equivalently, it is a Riemannian metric g such that $g(JX, JY) = g(X, Y)$ for all tangent vectors X, Y.
In local coordinates, $h = \sum_{j,k} h_{j\bar k}\, dz_j \otimes d\bar z_k$ with $(h_{j\bar k})$ a positive definite Hermitian matrix.

Definition 0.6.19: Fundamental Form

The *fundamental form* (or *associated* $(1,1)$*-form*) of a Hermitian metric h is

$$\omega := -\operatorname{Im}(h) = \frac{i}{2}\sum_{j,k} h_{j\bar{k}}\, dz_j \wedge d\bar{z}_k$$

This is a real $(1,1)$-form satisfying $\omega(X, JX) > 0$ for nonzero X.

Definition 0.6.20: Kähler Manifold

A Hermitian metric h is *Kähler* if its fundamental form ω is closed: $d\omega = 0$. A *Kähler manifold* is a complex manifold equipped with a Kähler metric. The form ω is then called the *Kähler form*.

Proposition 0.6.21: Equivalent Characterizations of Kähler

The following are equivalent for a Hermitian manifold (M, h):

1. $d\omega = 0$ (the Kähler condition).
2. The Levi-Civita connection ∇ of $g = \operatorname{Re}(h)$ satisfies $\nabla J = 0$.
3. Locally, $\omega = i\partial\bar{\partial}\varphi$ for some real function φ (a *Kähler potential*).
4. The holonomy of g is contained in $\mathrm{U}(n) \subset \mathrm{SO}(2n)$.

***Proof Cited*:**
Each condition says the metric agrees with the flat one to second order at every point. The equivalence of the first two is a computation with the Levi-Civita connection, and the local potential in the third is produced from $d\omega = 0$ by the $\partial\bar{\partial}$-lemma on a polydisc. See [41] or [31].

Example 0.6.22: Kähler Manifolds

1. $\mathbb{C}^n$ with the standard metric $\omega = \frac{i}{2}\sum_j dz_j \wedge d\bar{z}_j$.
2. $\mathbb{CP}^n$ with the *Fubini–Study metric*:
$$\omega_{\mathrm{FS}} = \frac{i}{2\pi}\partial\bar{\partial}\log\left(|z_0|^2 + \cdots + |z_n|^2\right)$$
3. Any smooth projective variety over $\mathbb{C}$ (inherits Kähler structure from $\mathbb{CP}^n$).
4. Complex submanifolds of Kähler manifolds are Kähler.
5. Complex tori $\mathbb{C}^n/\Lambda$ with the flat metric.

Example 0.6.23: Non-Kähler Manifolds

1. The *Hopf surface* $(\mathbb{C}^2 \setminus \{0\})/\langle z \mapsto 2z \rangle \cong S^1 \times S^3$ is a compact complex surface with $b_2 = 0$, hence not Kähler (Kähler manifolds have $b_2 \geq 1$ by the existence of $[\omega]$).
2. Most compact complex manifolds are not Kähler.

Hodge Theory

Hodge theory provides a bridge between the topology, complex structure, and Riemannian geometry of a Kähler manifold.

Definition 0.6.24: Hodge Star and Laplacian

On a Riemannian manifold (M, g) of dimension n, the *Hodge star* $* : \Omega^k(M) \to \Omega^{n-k}(M)$ is defined by

$$\alpha \wedge *\beta = g(\alpha, \beta)\,\mathrm{vol}_g$$

The *Hodge Laplacian* is $\Delta := dd^* + d^*d$ where $d^* := (-1)^{n(k+1)+1} * d*$ is the formal adjoint of d.
A form is *harmonic* if $\Delta\omega = 0$.

Theorem 0.6.25: Hodge Theorem (Riemannian)

On a compact oriented Riemannian manifold M:

1. Every de Rham cohomology class contains a unique harmonic representative.
2. $H^k_{\mathrm{dR}}(M) \cong \mathcal{H}^k(M) := \{\omega \in \Omega^k(M) : \Delta\omega = 0\}$.

***Proof*:**
The proof rests on the elliptic theory of the Laplacian and is not proved here; nothing below depends on it. See [73].

On a Kähler manifold, there are additional Laplacians:

Definition 0.6.26: Dolbeault Laplacians

On a Hermitian manifold, define

$$\Delta_\partial := \partial\partial^* + \partial^*\partial, \qquad \Delta_{\bar\partial} := \bar\partial\bar\partial^* + \bar\partial^*\bar\partial$$

Theorem 0.6.27: Kähler Identities

On a Kähler manifold:

$$\Delta = 2\Delta_\partial = 2\Delta_{\bar\partial}$$

In particular, a form is harmonic if and only if it is ∂-harmonic if and only if it is $\bar\partial$-harmonic.

***Proof*:**
These identities are established by a local computation in normal coordinates and are not proved here; nothing below depends on them. See [41] or [73].

Theorem 0.6.28: Hodge Decomposition

On a compact Kähler manifold M:

1. There is a decomposition

$$H^k(M,\mathbb{C}) = \bigoplus_{p+q=k} H^{p,q}(M)$$

   where $H^{p,q}(M) \cong H^{p,q}_{\bar\partial}(M) \cong H^q(M,\Omega^p_M)$.

2. Complex conjugation gives $\overline{H^{p,q}(M)} = H^{q,p}(M)$.

3. The decomposition is independent of the choice of Kähler metric.

***Proof Cited*:**
Hodge theory on the compact manifold M represents each class by a unique harmonic form, and on a Kähler manifold the Kähler identities give $\Delta_d = 2\Delta_{\bar\partial}$. That single identity is what forces the Laplacian to preserve bidegree, so the harmonic forms split by type and the decomposition descends to cohomology. See [31] or [73].

Definition 0.6.29: Hodge Numbers

The *Hodge numbers* of a compact Kähler manifold are

$$h^{p,q}(M) := \dim_{\mathbb{C}} H^{p,q}(M)$$

These satisfy:

1. $h^{p,q} = h^{q,p}$ (complex conjugation).
2. $h^{p,q} = h^{n-p,n-q}$ (Serre duality).
3. $b_k = \sum_{p+q=k} h^{p,q}$ (Betti numbers).

The Hodge numbers are often displayed in the *Hodge diamond*.

Example 0.6.30: Hodge Diamond of $\mathbb{CP}^n$
For $\mathbb{CP}^n$, the only nonzero Hodge numbers are $h^{p,p} = 1$ for $0 \leq p \leq n$:

$$\begin{array}{ccccccc} & & & 1 & & & \\ & & 0 & & 0 & & \\ & 0 & & 1 & & 0 & \\ & & \cdots & & \cdots & & \end{array}$$

This reflects $H^{2k}(\mathbb{CP}^n, \mathbb{C}) = \mathbb{C}$ and $H^{2k+1}(\mathbb{CP}^n, \mathbb{C}) = 0$.

Example 0.6.31: Hodge Diamond of a Curve
For a smooth projective curve C of genus g:

$$\begin{array}{ccc} & 1 & \\ g & & g \\ & 1 & \end{array}$$

Here $h^{1,0} = h^{0,1} = g = \dim H^0(C, \Omega^1_C) = \dim H^1(C, \mathcal{O}_C)$.

Holomorphic Vector Bundles and Sheaves

Definition 0.6.32: Holomorphic Vector Bundle

A *holomorphic vector bundle* of rank r over a complex manifold M is a complex manifold E with a holomorphic projection $\pi : E \to M$ such that:

1. Each fiber $E_p = \pi^{-1}(p)$ is a $\mathbb{C}$-vector space of dimension r.
2. There exist local holomorphic trivializations: $\pi^{-1}(U) \cong U \times \mathbb{C}^r$.

Equivalently, the transition functions $g_{\alpha\beta} : U_\alpha \cap U_\beta \to \mathrm{GL}_r(\mathbb{C})$ are holomorphic.

Example 0.6.33: Holomorphic Vector Bundles

1. The *holomorphic tangent bundle* $T^{1,0}M$ (sections are holomorphic vector fields).
2. The *canonical bundle* $K_M := \bigwedge^n (T^{1,0}M)^* = \Omega^n_M$.
3. The *tautological line bundle* $\mathcal{O}(-1)$ on $\mathbb{CP}^n$: the fiber over $[\ell]$ is the line $\ell \subset \mathbb{C}^{n+1}$.
4. $\mathcal{O}(k) := \mathcal{O}(-1)^{\otimes(-k)}$ for $k \in \mathbb{Z}$ (the twisting sheaves on $\mathbb{CP}^n$).

Definition 0.6.34: Sheaf of Holomorphic Sections

For a holomorphic vector bundle $E \to M$, the *sheaf of holomorphic sections* $\mathcal{E}$ is defined by

$$\mathcal{E}(U) := \{s : U \to E \text{ holomorphic} : \pi \circ s = \mathrm{id}_U\}$$

This is a sheaf of $\mathcal{O}_M$-modules, where $\mathcal{O}_M$ is the sheaf of holomorphic functions.

Definition 0.6.35: Locally Free Sheaf

A sheaf $\mathcal{F}$ of $\mathcal{O}_M$-modules is *locally free of rank* r if every point has a neighborhood U with $\mathcal{F}|_U \cong \mathcal{O}_U^{\oplus r}$.

Proposition 0.6.36: Equivalence: Bundles and Locally Free Sheaves

The functor $E \mapsto \mathcal{E}$ gives an equivalence of categories:

$$\{\text{holomorphic vector bundles on } M\} \simeq \{\text{locally free } \mathcal{O}_M\text{-modules}\}$$

***Proof Cited*:**
See [41].

0.6.37 Coherent Sheaves More generally, *coherent sheaves* on M are $\mathcal{O}_M$-modules that are locally finitely presented. These include locally free sheaves but also sheaves with torsion (e.g., skyscraper sheaves supported on subvarieties). Coherent sheaves form an abelian category $\mathbf{Coh}(M)$.

Line Bundles and Divisors

Definition 0.6.38: Divisor (Complex Manifold)

A *divisor* on a complex manifold M is a formal $\mathbb{Z}$-linear combination $D = \sum_i n_i V_i$ where each V_i is an irreducible analytic hypersurface (codimension 1) and $n_i \in \mathbb{Z}$. The group of divisors is denoted $\mathrm{Div}(M)$.

Definition 0.6.39: Line Bundle Associated to a Divisor

To a divisor D, one associates a holomorphic line bundle $\mathcal{O}(D)$ (or $\mathcal{L}(D)$) whose sections over U are meromorphic functions f with $(f) + D|_U \geq 0$, i.e., f may have poles along negative components of D and must vanish along positive components.

Proposition 0.6.40: Divisors and Line Bundles

On a compact complex manifold M:

1. Every holomorphic line bundle L is isomorphic to $\mathcal{O}(D)$ for some divisor D.
2. $\mathcal{O}(D_1) \cong \mathcal{O}(D_2)$ if and only if $D_1 - D_2 = (f)$ for some meromorphic function f (linear equivalence).
3. The *Picard group* $\mathrm{Pic}(M) = \{\text{line bundles}\}/\cong$ satisfies $\mathrm{Pic}(M) \cong \mathrm{Div}(M)/\sim$.

Definition 0.6.41: First Chern Class

The *first Chern class* of a line bundle L is a cohomology class $c_1(L) \in H^2(M, \mathbb{Z})$ (or $H^{1,1}(M) \cap H^2(M, \mathbb{Z})$ for Kähler manifolds). It can be computed as:

$$c_1(L) = \frac{i}{2\pi}[F_\nabla]$$

where F_∇ is the curvature of any connection on L.

Proposition 0.6.42: Chern Class Properties

1. $c_1(L_1 \otimes L_2) = c_1(L_1) + c_1(L_2)$.
2. $c_1(L^*) = -c_1(L)$.
3. $c_1(\mathcal{O}(D)) = [D]$ (the cohomology class of D).
4. On $\mathbb{CP}^n$: $c_1(\mathcal{O}(1)) = [\omega_{\mathrm{FS}}]$, the class of a hyperplane.

Sheaf Cohomology

Definition 0.6.43: Sheaf Cohomology

For a sheaf $\mathcal{F}$ on a topological space X, the *sheaf cohomology groups* $H^i(X, \mathcal{F})$ are the right derived functors of the global sections functor $\Gamma(X, -)$.
Concretely, $H^i(X, \mathcal{F})$ can be computed via:

- Čech cohomology (for suitable covers).
- Injective resolutions.
- The Dolbeault resolution (for $\mathcal{O}_M$-modules on complex manifolds).

Theorem 0.6.44: Dolbeault Resolution

On a complex manifold M, the sheaf Ω^p_M of holomorphic p-forms has a resolution by fine sheaves:

$$0 \to \Omega^p_M \to \mathcal{A}^{p,0} \xrightarrow{\bar\partial} \mathcal{A}^{p,1} \xrightarrow{\bar\partial} \cdots \xrightarrow{\bar\partial} \mathcal{A}^{p,n} \to 0$$

where $\mathcal{A}^{p,q}$ is the sheaf of smooth (p,q)-forms. This gives $H^q(M, \Omega^p_M) \cong H^{p,q}_{\bar\partial}(M)$.

***Proof Cited*:**
The $\bar\partial$-Poincaré lemma gives exactness, and the sheaves $\mathcal{A}^{p,q}$ are acyclic, admitting partitions of unity. See [41].

Example 0.6.45: Important Cohomology Groups

1. $H^0(M, \mathcal{O}_M) =$ holomorphic functions on M. If M is compact and connected, $H^0(M, \mathcal{O}_M) = \mathbb{C}$.
2. $H^0(M, \Omega^n_M) =$ holomorphic n-forms. For a Calabi–Yau manifold, $h^{n,0} = 1$.
3. $H^1(M, \mathcal{O}^*_M) \cong \mathrm{Pic}(M)$ (line bundles via transition functions).
4. $H^0(M, \mathcal{O}(D)) =$ meromorphic functions with poles bounded by D.

GAGA: Algebraic and Analytic Geometry

The GAGA theorems (Géométrie Algébrique et Géométrie Analytique) of Serre establish a connection between algebraic geometry and complex analytic geometry.

Definition 0.6.46: Analytification

Let X be a scheme of finite type over $\mathbb{C}$. The *analytification* X^{an} is a complex analytic space with:

- The same underlying set as the $\mathbb{C}$-points $X(\mathbb{C})$.
- The classical (Euclidean) topology.
- The structure sheaf $\mathcal{O}_{X^{\mathrm{an}}}$ which, where $X = \operatorname{Spec} \mathbb{C}[x_1, \ldots, x_n]/(f_1, \ldots, f_r)$, is $\mathcal{O}_{\mathbb{C}^n}/(f_1, \ldots, f_r)\mathcal{O}_{\mathbb{C}^n}$ restricted to the common zero set of the f_j, with $\mathcal{O}_{\mathbb{C}^n}$ the sheaf of holomorphic functions on $\mathbb{C}^n$.

When X is smooth of dimension n, the analytification X^{an} is a complex manifold of dimension n, and conversely [36, App. B, §1]. The forward direction is the implicit function theorem: near each point, X is cut out of some $\mathbb{A}^N$ by $N - n$ polynomials whose Jacobian matrix has rank $N - n$, and the holomorphic implicit function theorem turns that rank condition into a holomorphic chart.

Definition 0.6.47: Algebraic vs. Analytic Sheaves

For a coherent algebraic sheaf $\mathcal{F}$ on X, there is an associated analytic sheaf $\mathcal{F}^{\mathrm{an}}$ on X^{an}:

$$\mathcal{F}^{\mathrm{an}} := \mathcal{F} \otimes_{\mathcal{O}_X} \mathcal{O}_{X^{\mathrm{an}}}$$

Theorem 0.6.48: Serre's GAGA Theorems (Preview)

Let X be a projective scheme over $\mathbb{C}$. Then:

1. **(Equivalence of categories)** The functor $\mathcal{F} \mapsto \mathcal{F}^{\mathrm{an}}$ gives an equivalence
$$\mathbf{Coh}(X) \xrightarrow{\sim} \mathbf{Coh}(X^{\mathrm{an}})$$
between algebraic and analytic coherent sheaves.
2. **(Cohomology comparison)** For any coherent sheaf $\mathcal{F}$ on X:
$$H^i(X, \mathcal{F}) \xrightarrow{\sim} H^i(X^{\mathrm{an}}, \mathcal{F}^{\mathrm{an}})$$
3. **(Chow's theorem)** Every closed analytic subspace of X^{an} is algebraic.
4. **(Holomorphic = algebraic)** Every holomorphic map between projective varieties is algebraic.

***Proof Cited*:**
This is Serre's original theorem, [66]; a modern account is [37, Appendix B].

0.6.49 Importance of Projectivity GAGA requires projectivity (or at least properness). For affine varieties, the theorems fail dramatically:

- $\mathbb{C}$ has many holomorphic functions (e.g., e^z) but only polynomial algebraic functions.
- The analytic structure of $\mathbb{C}$ is much richer than its algebraic structure.

0.6.50 Applications of GAGA GAGA allows one to:

- Compute algebraic invariants using analytic methods (Hodge theory, $\bar{\partial}$-techniques).
- Prove algebraic results by analytic means (e.g., Kodaira embedding theorem).
- Transfer between the Zariski and classical topologies on projective varieties.

0.6.1 Summary: Complex Geometry and Algebraic Geometry

This table has more than what was covered for the purpose of reference:

Concept	**Complex Analytic**	**Algebraic**
Space	Complex manifold	Smooth variety / scheme
Functions	Holomorphic	Regular (polynomial)
Topology	Classical (Euclidean)	Zariski
Local model	$\mathbb{C}^n$	$\operatorname{Spec}(A)$
Differential forms	(p,q)-forms, $\bar{\partial}$	Kähler differentials $\Omega^p_{X/k}$
Cohomology	Dolbeault $H^{p,q}_{\bar{\partial}}$	Sheaf cohomology $H^q(X, \Omega^p)$
Vector bundles	Holomorphic bundles	Locally free sheaves
Divisors	Analytic hypersurfaces	Weil/Cartier divisors
Line bundles	$\operatorname{Pic}(M)$	$\operatorname{Pic}(X)$
Metrics	Hermitian, Kähler	(No direct analogue)
Key theorem	Hodge decomposition	Serre duality, GAGA

0.6.51 Foreshadowing GAGA For projective varieties over $\mathbb{C}$, one can freely pass between algebraic and analytic viewpoints via GAGA:

- **Algebraic methods**: Work over any field, use scheme theory, étale cohomology.
- **Analytic methods**: Use Hodge theory, L^2-methods, Kähler geometry, transcendental techniques.

Many results (e.g., Hodge conjecture, Torelli theorems) involve both perspectives essentially.

1

Sheaves

Roughly speaking, Sheaf Theory is the theory of augmenting a topological space[1] that adds information to each open set (adds "local" information) which can be "glued" to get information in larger open sets, and can be "restricted" to get information on smaller sets. The prototypical example of a sheaf is the set of smooth functions on a manifold. Smooth functions on a manifold can be restricted to open subsets of a manifold which are homeomorphic to $U \subseteq \mathbb{R}^n$ and if a smooth function on each part of the manifold is defined in a compatible way (i.e. they agree on intersections), they can be glued back together to form a function on the entire manifold. As an open covering of M can always be chosen so that each U is simply connected connected, the non-trivial information is in the ability or choice of gluing of functions.

A key motivator behind sheaves is that they can hold the "geometric" information of a space. The reader may recall that a vector-space and subspaces can be characterized via their dual spaces, that is all functions from the vector space to $\mathbb{F}$. A similar result holds for Varieties with the coordinate ring uniquely characterizing a Variety. This duality was found to exist everywhere in mathematics, to name three more examples:

- A classical example that started the duality between algebra and geometry is that of $C(X)$ and X where X is a compact space and $C(X)$ all continuous function on X. By taking the stone-Cech compactification of a space X, the space can be achieved for any space (see [44, p.144] for details)
- Similarly, a radon measure's can be characterized using the duals of $C_c(X)$ where X is a locally compact space (classical result can be found in EYTNKA at [61])
- Spin manifolds can be characterized by spectral triples (see [23])

[1] more generally, a topoi, which concretely translates in our case to a category whose objects are open sets and morphisms are inclusions

- Sets with set-function can be thought of as the most "abstract" geometric object (they only have information about containment). There is an appropriate dual called the *complete atomic Boolean algebras*. The reader may be interested in knowing that Boolean algebra's characterize logical operations, which gives an interesting geometric interpretation to logic[2].

There is a long list of such examples (the reader may be curious to checkout this link). This gave rise to the idea that a geometric space can be understood by studying some subset of function on this space with certain. In algebraic cases, there are relatively few functions that are defined globally on a space, but many that are defined on local components (the prototypical example of this would be $\mathbb{C}$[3]).[4]

Why do functions capture this geometric information? One can think of functions as tools for representing relations. If our function is of the form $f : X \to \mathbb{R}$, then this is relating X to numbers, i.e. quantification, usually X being a topological space and f being continuous for some notion of closeness. If we have a continuous function of the form $f : X \to \mathbb{R}^n$ (or $\mathbb{C}^n$), with X a topological space, then we may pullback a lot of geometric results from $\mathbb{R}^n$ (or $\mathbb{C}^n$) onto X, depending on how f is defined (ex. it can be shown that f is a diffeomorphism if and only if it pulls back the differential structure of $\mathbb{R}^n$). Very often, it is not possible to define a diffeomorphism $f : X \to \mathbb{R}^n$, however we may define it on components. Functions have the property that we may restrict them or construct them by defining what happening at each point (or in the continuous case, on each open set where they agree on intersections[5]). But you may say that morphisms in a category already capture this intuition. This is indeed the case, what makes functions special is that you can construct larger morphisms from smaller morphisms with an appropriate notion of *gluing* and *restricting*. In particular, the closeness given by the notion of topological spaces and the locality properties inspired by functions shall be the two main properties of sheaves. As is the course of mathematics, it may be asked if these properties of functions can be generalized, and we may ask if we can take special categories which have these features; this is called a *site*. For an introduction to algebraic geometry, limiting to the sites defined on topological spaces is plenty sufficient and shall be the focus of this book.

A particularly important realization will be that commutative rings will in many ways behave like functions on a space, that is if R is a ring, there is a sense in which the elements of R will be the "functions" on a special space (namely, the space $\mathrm{Spec}\,(R)$). This shall be elaborated on in chapter 2 and was commented on in the front-matter. At this stage it is motivating to know that there is a reason why we may care about sheaves other than the sheaf of functions, namely the interpolation between commutative rings and functions.

Motivating Example from Vakil

Before exploring sheaves in generality, it is useful to have the prototypical example of a sheaf in mind in more detail:

[2] This connection is further refined using Topoi

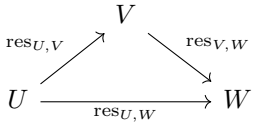
[3] entire holomorphic are necessarily power series, while holomorphic on (not necessarily simply connected) open sets are analytic and may even have non-trivial Laurent expansion around holes

[4] see this link for more details

[5] by the gluing lemma

Example 1.0.1: Sheaf Of Differentiable Functions On Manifold

Let M be a smooth manifold. Then the we shall constuct the sheaf of differentiable function on M. For each open subset $U \subseteq M$, we shall assocaite to it the set of smooth function $C^\infty(U)$. Denote this collection by $\mathcal{O}(U)$ (this will be common notation for sheaves). Then if we have an open subset of U, $V \subseteq U$, we may restrict the function from U to function on V, namely we have a map $\text{res}_{U,V} : \mathcal{O}(U) \to \mathcal{O}(V)$ mapping $f \mapsto f|_V$. Naturally, if we have $W \hookrightarrow V \hookrightarrow U$, then the following commutative diagram.



This collection of maps satisfy two important properties. In the following let and $U = \bigcup_i U_i$.

1. *Defining Locally*: Let's say $f_1, f_2 \in \mathcal{O}(U)$, Then if f_1 and f_2 agree on all restriction, that is $\text{res}_{U,U_i} f_1 = \text{res}_{U,U_j} f_2$, then $f_1 = f_2$
2. *Gluing*: If there exists a collection $f_i \in \mathcal{O}(U_i)$ such that f_i and f_j agree on intersections, that is $\text{res}_{U_i,U_i\cap U_j} f_i = \text{res}_{U_j,U_i\cap U_j} f_j$, then there is a $f \in \mathcal{O}(U)$ such that $\text{res}_{U,U_i} f = f_i$ for all i.

Then a sheaf will respect these properties. Note that in fact, we may define a manifold M to be $(M, \mathcal{O}_M)$ where M Is a topological space and $\mathcal{O}_M$ is a sheaf of smooth functions on M. If instead $\mathcal{O}(U)$ was the collection of continuous functions, we would get a topological manifold. Even more generally, we may let the functions be set functions, for set functions work well under restrictions and we may define set functions locally and construct set function via gluing.

Note that Diffeology can be thought of as a generalization of this result or of a special case of sheaf theory, and we shall explore it briefly at the end of this chapter.

Example 1.0.2: Sheaf On Set Functions

Take $\text{Hom}_{\mathbf{Set}}(X, Y)$ to be the collection of arbitrary set functions (or equivalently, let X, Y have the discrete topology so that all set functions are continuous). Then given an arbitrary collection of sets $\{U_i\}_{i\in I}$ of X with functions $f_i : U_I \to Y$, if the collection of functions agree on all pair-wise intersections $U_i \cap U_j$, $i, j \in I$, then there exists a unique function f on $\bigcup_{i\in I} U_i$ that restricts to all these functions. Notice the lack of essentially any structure. This shows that the heavy-lifter insuring the gluability of local data is the *associativity* axiom functions must satisfy.

So if even set functions can form a sheaf, then why did I write "nice" function earlier? As it turns out, what we shall do is abstract the notion of function! In particular, we shall look at systems where we may have the behavior of localizing and gluing "functions" on a topological space. Hence, in the following abstraction, we will consider $\mathcal{O}_M$ to be the data that stores a collection of objects that *behave like functions*.

A key property that distinguishes functions from general morphisms is the notion of evaluation (taking $f(x)$ for some element x and function f). Since we are abstracting the notion of function, we shall need a notion of *evaluation*. To motivate this generalization, let's come back to $(M, \mathcal{O}_M)$ and consider some $\mathcal{O}(U) = C^\infty(U)$ for some open $U \subseteq M$ and take $p \in U$. Recall that $\mathcal{O}(U)$ is a ring

and that the *germ* at p is collection of equivalence relations:

$$[f] = \{(f, U) \mid p \in U,\ f \in \mathcal{O}(U)\}$$
$$(f, U) \sim (g, V) \ \Leftrightarrow W \subseteq U, W \subseteq V, p \in W,\ f|_W = g|_W$$

that is, $[f] = [g]$ if and only if there is an open neighborhood $W \subseteq U, V$ containing p ($p \in W$) such that $f|_W = g|_W$ (in the language of sheaf theory, $\text{res}_{U,W} f = \text{res}_{V,W} g$). Denote this collection $\mathcal{O}(U)_p$. The key is that the germ of $\mathcal{O}(U)$ at p is a *local ring* with maximal ideal $\mathfrak{m}_p$ of germs such that $[f](p) = 0$ (or simply $f(p) = 0$ if unambiguous). Then:

$$0 \to \mathfrak{m}_p \to \mathcal{O}(U)_p \xrightarrow{f \mapsto f(p)} \mathbb{R} \to 0$$

and all elements $\mathcal{O}(U) \setminus \mathfrak{m}_p$ are invertible. Hence, we shall see that we may thing of "evalutation" of f at some point $p \in U$ as taking first the germ $\mathcal{O}(U)_p$ and quotient it by the maximal ideal $\mathfrak{m}_p$, $\mathcal{O}(U)_p/\mathfrak{m}_p$, and then taking $\overline{f} \in \mathcal{O}(U)_p/\mathfrak{m}_p$ to be the value of f. Note that the above construction required that $\mathcal{O}(U)$ is a *local ring*. This is the case when $\mathcal{O}(U) = C^\infty(U)$, however in general sheaf theory this ring will not in general be local. In fact, $\mathcal{O}(U)$ will not even need to be a ring! When each $\mathcal{O}(U)$ is a ring, then we shall say that X is a *ringed space*, and when all the germs are local rings, then we shall say that X is a *locally ringed space*.

A final word on generalization: we have been working over a topological space, however we may abstract further and work over a general category; in this way, we would define the notion of a "topology" on a category, known as a *Grothendeick topology* (a category with this structure is called a *site*), and develop sheaf theory accordingly. Though this is fruitful, this generlization shall be ommited for now to concentrate on the geoemtric properties at hand when using a topological space.

1.1 Definitions and Examples

We start by formalizing the above intuitions.

Definition 1.1.1: Presheaf Axiomatic Definition

Let X be a topological space. Then a *presheaf* over a category $\mathbf{C}$ on X, denoted $\mathscr{F}_X$ contains the following data:

1. for each open set $U \subseteq X$, there is an object c of $\mathbf{C}$ and denote it by $\mathscr{F}(U)$. If C is a category whose objects are sets with additioanl structure, then the elements of $\mathscr{F}(U)$ are called the *sections* of $\mathscr{F}$ over U. If we say that s is a section of $\mathscr{F}$, then we mean that s is a section of $\mathscr{F}$ over X.
2. For each inclusion map $U \hookrightarrow V$, there is a morphism $\mathrm{res}_{V,U} : \mathscr{F}(V) \to \mathscr{F}(U)$ called the *restriction* morphism (or *restriction map*) such that
   (a) $\mathrm{res}_{U,U} = \mathrm{id}_U$
   (b) If $W \hookrightarrow V \hookrightarrow U$, then the following diagram commutes

$$\begin{array}{ccc} & \mathscr{F}(V) & \\ {\scriptstyle \mathrm{res}_{U,V}}\nearrow & & \searrow{\scriptstyle \mathrm{res}_{V,W}} \\ \mathscr{F}(U) & \xrightarrow[\mathrm{res}_{U,W}]{} & \mathscr{F}(W) \end{array}$$

The collection of all presheaves on a topological space X is denoted $\mathbf{PSh}_{\mathbf{C}}(X)$ (or $\mathbf{PSh}(X)$ if unambiguous). If U is clear from context, we shall often write $\mathrm{res}_{U,V} f$ as $f|_V$.

The nomenclature "section" will be made clear in proposition 1.1.13. When it is no ambiguous the image of the restriction morphism for $s \in \mathscr{F}(U)$ shall be denoted:

$$\mathrm{res}_{U,V}(s) = s|_V$$

This notation can be ambiguous as it is not explicit which $U \subseteq X$ is taken from which $s \in \mathscr{F}(U)$, however it will be unambiguous in-context.

When constructing sections of non-affine scheme's in chapter 2, it shall be important to have internalized the following consequence of the restriction maps. Let $U = U_1 \cup U_2$ and $W = U_1 \cap U_2$. Then $W \subseteq U_1 \subseteq U$ and $W \subseteq U_2 \subseteq U$. Then by definition of the restriction map, the following diagram commutes:

$$\begin{array}{ccccc} & & \mathscr{F}(U_1) & & \\ & {\scriptstyle \mathrm{res}_{U,U_1}}\nearrow & & \searrow{\scriptstyle \mathrm{res}_{U_1,W}} & \\ F(U) & & \xrightarrow[\mathrm{res}_{U,W}]{} & & F(W) \\ & {\scriptstyle \mathrm{res}_{U,U_2}}\searrow & & \nearrow{\scriptstyle \mathrm{res}_{U_2,W}} & \\ & & \mathscr{F}(U_2) & & \end{array}$$

Importantly:

$$f|_W = (f|_{U_1})|_W = (f|_{U_2})|_W$$

The key take-away is that a larger section on U can be restricted to different sections on an open cover of U, but these sections map to the same restriction on intersections. In this way, presheaves preserve elements "going down". We shall see example of how "going up" fails after introducing some more basic terminology.

For practical computational purposes, it shall be important to re-interpret a presheaf in more categorical language, Notice that it is equivalent to define a presheaf as a *contravariant functor*. Namely

Definition 1.1.2: Presheaf Functorial Definition

Let X be a topological space and let $O(X)$ be the category whose objects are the open sets of X and whose morphism are $\subseteq$ (i.e. the category induced by the partial order given by $\subseteq$). Then a *presheaf* over $\mathbf{C}$ is a contravariant functor from $O(X)$ to $\mathbf{C}$, that is a functor $\mathscr{F} : O(X)^{op} \to \mathbf{C}$.

Usually, our presheaves shall be over **Set**, **Ab**, Mod_k or **CRing**[6].

An important notion for geometric objects is their behavior locally. The extreme (or limit) of the notation of locality is the behavior around a point p. This idea originated when studying germs of differential functions in Differential Geometry. These are well-defined on presheaves;

Definition 1.1.3: Stalk as Equivalence Relation

let $(X, \mathcal{O}_X)$ be a presheaf. Then the *stalks* at $p \in X$ is the collection of equivalence class:

$$[f] = \{(f, U) \mid p \in U,\ f \in \mathcal{O}(U)\}$$
$$(f, U) \sim (g, V) \qquad \Leftrightarrow \qquad W \subseteq U, W \subseteq V, p \in W,\ \mathrm{res}_{U,W} f = \mathrm{res}_{V,W} g$$

The stalks at p is usually denoted $\mathcal{O}_{X,p}$ or $\mathcal{O}_p$ if unambiguous in context.

If $\mathbf{C}$ is a category whose objects are augmented sets[a], then if $p \in U$ and $f \in \mathscr{F}(U)$, $[f] \in \mathscr{F}_p$ is called the *germ* of f at p.

[a] ex. **Set**, **Ring**, **Grp**, etc.

Note that in general, the notion of "evaluation" of a stalk/germ as it is usually defined on differential geometry will not make sense; though the symbol f is written the open sets need not contain functions or objects that act like function. We shall later define nice presheaves in which we may recover the notion of evaluation and treat the elements like functions. There is again a categorical way of interpreting stalks that is useful to us:

Definition 1.1.4: Stalks As Colimits

Let $\mathscr{F} \in \mathbf{PSh}_{\mathbf{C}}(X)$ be a presheaf where $\mathbf{C}$ is a category with small filtered colimits. Then a *stalk* at $p \in X$ is:

$$\mathscr{F}_p := \varinjlim \mathscr{F}(U)$$

that is $\mathscr{F}_p$ is the colimit over all $\mathscr{F}(U)$ where $p \in U$.

The adjectives "small filtered colimit" are only there to guarantee that the colimit exists. Most base category that will be worked over shall have the stronger property of bieng *cocompelte* (having all colimits) and often also being *complete* (having all limits). Categories that are complete and cocomplete include:

[6] Technically, **Ab** is $\mathbf{Mod}_{\mathbb{Z}}$, however distinguishing these two is useful as we usually mean k to be a field

1. **Set** (Category of Sets)
2. **Grp** (Category of Groups)
3. **Ab** (Category of Abelian Groups)
4. **Ring** (Category of Rings)
5. R-**Mod** (Category of modules over a ring R)
6. **Top** (Category of Topological Spaces)

Hence, this shows us that as long as the colimit exist in our category, we may define the notion of stalks for presheaves over that category without needing to reference the notion of elements. Note that it is the colimit and not the limit since we need $\mathscr{F}_p$ to "represent" the behaviour around p rather than "embed" germs.

One of the most important properties of stalks is their ability to be defined locally. First, if $U \subseteq X$ is an open set, we can define the sub-presheaf on U by letting $\mathcal{F}|_U(V) = \mathcal{F}(V \cap U)$. The following proposition will be really useful later when we need to work with stalks between larger and smaller (pre)sheaves, and mirrors the property in differential geometry (recall [59, chapter 2.1])

Proposition 1.1.5: Stalk Isomorphism

Let $\mathbf{C}$ be a category with small filtered colimits, $\mathscr{F} \in \mathbf{PSh}_{\mathbf{C}}(X)$ a presheaf, and $U \subseteq X$ an open set. Write $\mathscr{F}_{U,x}$ for the stalk at x of the restriction $\mathscr{F}|_U$, that is, the colimit of $\mathscr{F}(V)$ over the open sets V with $x \in V \subseteq U$. Then for any $x \in U$:

$$\mathscr{F}_{X,x} \cong_{\mathbf{C}} \mathscr{F}_{U,x}$$

***Proof*:**
Let N_X be the set of open neighbourhoods of x in X and $N_U \subseteq N_X$ the subset of those contained in U, both directed by reverse inclusion. Write $\lambda_V \colon \mathscr{F}(V) \to \mathscr{F}_{X,x}$ for $V \in N_X$ and $\mu_V \colon \mathscr{F}(V) \to \mathscr{F}_{U,x}$ for $V \in N_U$ for the colimit maps, so that $\lambda_{V'} \circ \mathrm{res}_{V,V'} = \lambda_V$ and $\mu_{V'} \circ \mathrm{res}_{V,V'} = \mu_V$ whenever $V' \subseteq V$.

Step 1: the map $\varphi \colon \mathscr{F}_{X,x} \to \mathscr{F}_{U,x}$. For $V \in N_X$ the set $V \cap U$ lies in N_U. Put $\varphi_V = \mu_{V \cap U} \circ \mathrm{res}_{V,V\cap U}$. If $V' \subseteq V$ in N_X, then

$$\varphi_{V'} \circ \mathrm{res}_{V,V'} = \mu_{V'\cap U} \circ \mathrm{res}_{V,V'\cap U} = \mu_{V'\cap U} \circ \mathrm{res}_{V\cap U,V'\cap U} \circ \mathrm{res}_{V,V\cap U} = \varphi_V$$

by the composition axiom of definition 1.1.1 and the compatibility of the maps μ. The universal property of the colimit $\mathscr{F}_{X,x}$ gives a unique morphism φ with $\varphi \circ \lambda_V = \varphi_V$ for every $V \in N_X$.

Step 2: the map $\psi \colon \mathscr{F}_{U,x} \to \mathscr{F}_{X,x}$. The maps λ_V for $V \in N_U$ are compatible with restriction, so the universal property of $\mathscr{F}_{U,x}$ gives a unique morphism ψ with $\psi \circ \mu_V = \lambda_V$ for every $V \in N_U$.

Step 3: φ and ψ are mutually inverse. For $V \in N_X$,

$$\psi \circ \varphi \circ \lambda_V = \psi \circ \mu_{V\cap U} \circ \mathrm{res}_{V,V\cap U} = \lambda_{V\cap U} \circ \mathrm{res}_{V,V\cap U} = \lambda_V$$

so $\psi \circ \varphi$ and id agree after composition with every λ_V, and the uniqueness in the universal property of $\mathscr{F}_{X,x}$ gives $\psi \circ \varphi = \mathrm{id}$. For $V \in N_U$ we have $V \cap U = V$ and $\mathrm{res}_{V,V} = \mathrm{id}$, so

$$\varphi \circ \psi \circ \mu_V = \varphi \circ \lambda_V = \mu_V$$

and the uniqueness in the universal property of $\mathscr{F}_{U,x}$ gives $\varphi \circ \psi = \mathrm{id}$. Hence φ is an isomorphism in $\mathbf{C}$. For $\mathbf{C} = \mathbf{Set}$, with stalks described by germs as in definition 1.1.3, it is $[(f, V)] \mapsto [(f|_{U\cap V}, U \cap V)]$, with inverse $[(f, V)] \mapsto [(f, V)]$, completing the proof.

Now, presheaves allow for the notion of restricting and localizing, that is there is structure preserved "going down". But there are no condition on any structure being preserved "going up". In particular, not all presheaves are amenable to gluing local information to create more global information (think of bounded holomorphic functions on open subsets of $\mathbb{C}$). We must add this condition axiomatically:

Definition 1.1.6: Sheaf using Axioms

Let $\mathscr{F} \in \mathbf{PSh}_C(X)$ be a presheaf over a category $\mathbf{C}$ whose objects are sets possibly with additional structure. Then $\mathscr{F}$ is also a sheaf if it satisfies the following two axioms;

1. *Gluing Axiom (Existence)*: Let $U = \bigcup_i U_i$. If there exists a collection $f_i \in \mathscr{F}(U_i)$ such that $\mathrm{res}_{U_i, U_i\cap U_j} f_i = \mathrm{res}_{U_j, U_i\cap U_j} f_j$, then there is a $f \in \mathscr{F}(U)$ such that $\mathrm{res}_{U,U_i} f = f_i$ for all i.
2. *Locality Axiom (Uniqueness)*. Let $U = \bigcup_i U_i$. If $f_1, f_2 \in \mathscr{F}(U)$, then if $\mathrm{res}_{U,U_i} f_1 = \mathrm{res}_{U,U_i} f_2$ for all i, then $f_1 = f_2$
3. $\mathscr{F}(\emptyset)$ is associated with the terminal object (if we are working over set, $\mathscr{F}(\emptyset)$ is a singleton).

If only the locality axiom is satisfied, then our presheaf is called a *separated presheaf*. The collection of all sheaves on a topological space X is denoted $\mathbf{Sh}_C(X)$ (or $\mathbf{Sh}(X)$ if unambiguous)

We may think of the locality axiom telling us there is *at most* one way to glue, while the gluing axiom is telling us there is *at least* one way to glue. A separated presheaf is so named as it allows sections to "separate" points (there a cannot be two sections identical on restrictions of an open covering).

1.1.7 Separation and Evaluation Later, the notion of "evaluation" shall be defined for sections where the objects of $\mathbf{C}$ behave like sets of functions. Being separated *does not* imply a function is defined by the evaluation at every point as the intuition of separated algebra for the Stone-Weierstrass Theorem would imply. This is because evaluation on locally ringed spaces will carry more data that needs to be taken into account (ex. see lemma 2.2.6).

The following is sometimes (redundantly) included in the definition of a presheaf:

Lemma 1.1.8: Sheaf and Zero Section

Let $\mathbf{C}$ be a category with the zero object 0 and $\mathscr{F}$ a sheaf (or particularly, a separated presheaf) on X. Let $\bigcup_{i\in I} U_i = X$ be an open cover and $s \in \mathscr{F}(X)$. Then if $s|_{U_i} = 0$ for all i, then $s = 0$.

***Proof*:**
Write $0_U \in \mathscr{F}(U)$ for the zero section over an open set U, the distinguished element given by the unique morphism $0 \to \mathscr{F}(U)$ from the zero object. Restriction preserves it: res_{X,U_i} is a morphism in $\mathbf{C}$, so the composite $0 \to \mathscr{F}(X) \to \mathscr{F}(U_i)$ is a morphism out of the zero object, hence the unique one, and $0_X|_{U_i} = 0_{U_i}$.

The two sections s and 0_X of $\mathscr{F}(X)$ therefore satisfy

$$s|_{U_i} = 0_{U_i} = 0_X|_{U_i}$$

for every i. The locality axiom of definition 1.1.6, applied to the cover $\bigcup_{i\in I} U_i = X$, gives $s = 0_X$.

Notice that we are taking elements of each $\mathscr{F}(U)$ because $\mathscr{F}(U)$ is an object in a category whose objects are sets with possibly additional structure. This is a deviation from the generalization presented in the definition of presheaves as $\mathbf{C}$ may be more general. There is in fact a way to define a sheaf for general categories:

Proposition 1.1.9: Sheaf Categorical Interpretation

Let $\mathbf{C}$ be **Set**, **Ab**, Mod_R or **CRing**, so that products in $\mathbf{C}$ are cartesian products with componentwise operations. Let $\mathscr{F} \in \mathbf{PSh}_{\mathbf{C}}(X)$, let $U = \bigcup_{i\in I} U_i$ be an open cover of an open set U, and consider

$$\mathscr{F}(U) \xrightarrow{\alpha} \prod_{i\in I} \mathscr{F}(U_i) \underset{\gamma}{\overset{\beta}{\rightrightarrows}} \prod_{(i,j)\in I\times I} \mathscr{F}(U_i \cap U_j)$$

where $\alpha(s) = (s|_{U_i})_i$, $\beta((s_i)_i) = (s_i|_{U_i\cap U_j})_{(i,j)}$ and $\gamma((s_i)_i) = (s_j|_{U_i\cap U_j})_{(i,j)}$. Let $\mathrm{Eq}(\beta,\gamma)$ be the set of elements on which β and γ agree. Then:

1. $\mathscr{F}$ satisfies the locality axiom for this cover if and only if α is injective.
2. $\mathscr{F}$ satisfies the gluing axiom for this cover if and only if the image of α is $\mathrm{Eq}(\beta,\gamma)$.
3. α is an equalizer of β and γ in $\mathbf{C}$ if and only if α is injective with image $\mathrm{Eq}(\beta,\gamma)$.

Hence $\mathscr{F}$ is a sheaf if and only if $\mathscr{F}(\emptyset)$ is terminal and α is an equalizer of β and γ for every open cover of every open set. For $\mathbf{C} = \mathbf{Ab}$ or Mod_R the last condition says that $0 \to \mathscr{F}(U) \xrightarrow{\alpha} \prod_i \mathscr{F}(U_i) \xrightarrow{\beta-\gamma} \prod_{(i,j)} \mathscr{F}(U_i \cap U_j)$ is exact.

***Proof*:**
The composition axiom of definition 1.1.1 gives $(s|_{U_i})|_{U_i\cap U_j} = s|_{U_i\cap U_j} = (s|_{U_j})|_{U_i\cap U_j}$, so the image of α always lies in $\mathrm{Eq}(\beta,\gamma)$.

1. The locality axiom of definition 1.1.6 for this cover says that $s,t \in \mathscr{F}(U)$ with $s|_{U_i} = t|_{U_i}$ for all i are equal, that is, $\alpha(s) = \alpha(t)$ implies $s = t$.
2. A family $(s_i)_i$ satisfies $s_i|_{U_i\cap U_j} = s_j|_{U_i\cap U_j}$ for all i,j exactly when it lies in $\mathrm{Eq}(\beta,\gamma)$, and a section s with $s|_{U_i} = s_i$ for all i is a preimage of $(s_i)_i$ under α. So the gluing axiom for

this cover says that $\mathrm{Eq}(\beta, \gamma)$ lies in the image of α, which with the opening remark means equality.

3. The maps β and γ are morphisms of $\mathbf{C}$, so $\mathrm{Eq}(\beta, \gamma)$ is a subobject (a subset, subgroup, submodule or subring), and a morphism $h\colon T \to \prod_i \mathscr{F}(U_i)$ with $\beta h = \gamma h$ takes values in it and factors uniquely through its inclusion. Hence the inclusion of $\mathrm{Eq}(\beta, \gamma)$ is an equalizer of β and γ. The morphism α satisfies $\beta\alpha = \gamma\alpha$ by the opening remark, so it factors as the inclusion composed with a morphism $\alpha'\colon \mathscr{F}(U) \to \mathrm{Eq}(\beta, \gamma)$, and α is an equalizer exactly when α' is an isomorphism. In each of the four categories a morphism is an isomorphism if and only if it is bijective, so α is an equalizer if and only if α is injective with image $\mathrm{Eq}(\beta, \gamma)$.

By definition 1.1.6, $\mathscr{F}$ is a sheaf when $\mathscr{F}(\emptyset)$ is terminal and the locality and gluing axioms hold for every open cover of every open set, which by (1) to (3) is the stated equalizer condition. In **Ab** and Mod_R the map $\beta - \gamma$ is a homomorphism with kernel $\mathrm{Eq}(\beta, \gamma)$, so injectivity of α with image $\mathrm{Eq}(\beta, \gamma)$ is exactness at $\mathscr{F}(U)$ and at $\prod_i \mathscr{F}(U_i)$, as claimed.

For a presheaf with values in a general category with products, the equalizer condition of proposition 1.1.9 is taken as the definition of a sheaf.

1.1.10 Foreshadowing Čech Cohomology

What if we extend this to have further equalizers? Sheaves that satisfy higher-order intersection-exactness conditions are a proper subset of sheaves (over non-trivial categories and topological spaces). Such sheaves can be thought of as never having any gluing obstruction, and hence have trivial cohomology. We shall return to these ideas when exploring Čech cohomology in section 6.1.1.

Example 1.1.11: Presheaves failing exactness (i.e. gluability and locality)

Let X be any topological space and define:

$$\mathcal{F}(X) = \mathbb{Z} \qquad \mathcal{F}(U) = 0 \qquad (\text{for } U \subsetneq X)$$

Then this sheaf is gluable, but not separated. If $U \subsetneq X$ is covered, then gluing must hold. If $U = X = \cup_i U_i$, then that restrictions holds as any $z \in \mathbb{Z}$ has $z|_{U_i} = 0$ and so they certainly agree on all intersections. Hence, we see that each $z \in \mathbb{Z}$ is glued by a bunch of 0-elements!

Example 1.1.12: Presheaves And Sheaves

Most of these examples will be on sheaves of sets.

1. Verify that the sheaf of differentiable functions over M, namely $\mathcal{O}_M$, is indeed a sheaf. More generally, if X is any topological space, then the data defined by $\mathscr{F}(U) = \mathrm{Hom}_{\mathbf{C}}(U, \mathbb{R})$ where $\mathbf{C}$ is either **Top** or **Set** is a sheaf.

   Note how in a sheaf, we are "forced" to have interesting global behaviour if there are local section on non-intersecting sets. Consider $U_1 = (0, 1)$ and $U_2 = (3, 4)$. Then $U_1 \cap U_2 = 0$, so $\mathscr{F}(U_1 \cap U_2) = \{*\}$ is the terminal object (a singleton). If there is a function $f \in \mathscr{F}(U_1)$ and

$g \in \mathscr{F}(U_2)$, then the restriction map

$$\mathrm{res}_{U_1,\emptyset} f = * = \mathrm{res}_{U_2,\emptyset}$$

Meaning they *have* to agree on intersection. Thus, by the gluability axiom, there must exist a function F on $U_1 \cup U_2$ that restricts to f, g. For functions, this is obvious as it is the function defined by:

$$F(x) = \begin{cases} f(x) & x \in (0,1) \\ g(x) & x \in (3,4) \end{cases}$$

This becomes more interesting when the category in question is not obviously interpret able as functions on a space: a sheaf of truth values assigns to each open set the propositions that hold on it, and restriction is implication.

2. (*Restriction of Sheaves*) Let $\mathscr{F} \in \mathbf{Sh}(X)$ and $U \subseteq X$ an open subset of X. Then the *restriction* of $\mathscr{F}$ to U, denoted $\mathscr{F}|_U$ has as data for each open $V \subseteq U$ $\mathscr{F}|_U(V) = \mathscr{F}(V)$ (note hos it is important that U is open for well-definedness). We shall later see how we may define restrictions of sheaves to arbitrary subsets, as the inverse image along the inclusion (section 1.3)

3. (*Skyscraper Sheaf*) Let X be a topological space, $p \in X$, and S be any set. Let $\iota_p : \{p\} \to X$ be the inclusion map. Then define the sheaf of sets $(\iota_p)_* S$ on X (not on S!) to be:

$$(\iota_p)_* S(U) = \begin{cases} S & p \in U \\ \{e\} & p \notin U \end{cases}$$

This sheaf is called the *skyscraper sheaf*. The name comes from the fact that the informal picture for this sheaf looks like a skyscraper at p. For sheaves of Abelian groups, $\{e\}$ is the trivial group. More generally, $(\iota_p)_*(U)$ would be the final object if $p \notin U$.

4. (*Constant Presheaf*) Let X be a topological space and S be any set. Define $\underline{S}_{\mathrm{pre}}(U) = S$ for all open $U \subseteq X$ Then it should be verified that $\underline{S}_{\mathrm{pre}}$ forms a presheaf on X, called the *constant presheaf associated to* S. This set is not in general a sheaf if $|S| > 1$: take $X = U_1 \coprod U_2$ to be the disjoint union of two (open) sets, take the constant presheaf so that all restriction maps are constant maps. Chose $s_1 \in S$ and $s_2 \in S$ which are distinct: $s_1 \neq s_2$. Then by the gluing axiom, there would have to exist a $s \in S$ such that $s|_{U_1} = s_1$ and $s|_{U_2} = s_2$ and their intersection is empty and hence tautologically follow the gluing axiom, however as we have the identity map we have $s_1 = s = s_2$ however $s_1 \neq s_2$.

   This is fixed by a process called *sheafification* which will add the appropriate section equivalent to (s_1, s_2) that would restrict to both of these elements.

   What makes the sky-scrapper sheaf special is the "distinguishing" of a point making it act like a generalization of an indicator function. In the above example, one of the two sets will not have the point p.

5. (*Constant Sheaf*) Again let X be a topological space and S be any set. For every open $U \subseteq X$, let $\mathscr{F}(U)$ be the collection of all *locally constant* continuous functions from U to S, that is if $f \in \mathscr{F}(U)$, for each $p \in U$ there is an open neighborhood $V \subseteq U$ containing p, $p \in V \subseteq U$, such that $f|_V : V \to S$ is constant (on euclidean space, this is equivalent to

having a constant function on each connected component). Then this sheaf is called the *constant sheaf associated to* S, and is denoted $\underline{S}$.

Another way of describing this sheaf is if we endow S with the discrete topology and let $\mathscr{F}(U)$ be the continuous maps $U \to S$.

6. A great example of a pre-sheaf on $\mathbb{C}$ that is not a sheaf is the sheaf of bounded holomorphic functions on $\mathbb{C}$. Recall that bounded entire holomorphic functions (defined on all of $\mathbb{C}$) must be constant by Louisville's Theorem, however there do exist bounded functions on open subsets of $\mathbb{C}$. For example , on each disk of radius r, $\mathbb{D}_r^2$, the holomorphic function z^2 is bounded but $\mathbb{C} = \cup_r \mathbb{D}_r^2$ has no bounded holomorphic function for which the restriction to any disk $\mathbb{D}_r^2$ is z^2; hence the gluing axiom fails.

   Another example would be the sheaf composed of holomorphic functions admitting holomorphic square roots, as there is no global square-root function on $\mathbb{C}$ due to non-trivial monodromy.

7. (*Sheaf of continuous maps*) Let X and Y be topological spaces. Define $\mathscr{F}(U) = \mathrm{Hom}_{\mathbf{Top}}(U, Y)$. This sheaf is a generalization sheaf of differentiable functions and the constant sheaf, namely

   - The sheaf of differentiable functions has $Y = \mathbb{R}$
   - the constant sheaf has $Y = S$ with the discrete topology.

   If Y is also a topological group, then the continuous maps to Y form a sheaf of groups.

8. (*Sheaf of section of a maps*) This is an important special case of the above sheaf. Let X, Y be topological spaces and let $\mu : Y \to X$ be a continuous map. Then the sections of μ (in the sense of the existence of a map $s : X \to Y$ such that $\mu \circ s = \mathrm{id}$) form a sheaf. In particular, for each open $U \subseteq X$, $\mathscr{F}(U)$ is the set of section maps $s : U \to Y$ (continuous maps $s : U \to Y$ such that $\mu \circ s = \mathrm{id}|_U$). Vector bundles and vector fields are a good example of such a sheaf.

9. The following examples shows how different sheaves can be put on the same object to characterize different type of information. Throughout this example, let $X = \mathbb{A}_k^1$ be the (classical) affine line (see definition 0.1.9). The open subsets in the Zariski topology are all cofinite subsets. If $U \subseteq \mathbb{A}_k^1$ is open let $a_1, ..., a_n$ be the points that are omitted from U.

   (a) (Sheaf of regular function) Then letting $f_i = (x - a_i)$ and $F = \{f_1, ..., f_n\}$, define $\mathscr{F}(U) = k[x]_F$ i.e. $\mathscr{F}(U)$ is the ring generated by $k[x]$ and $1/f_i$ for each i. The reader should verify that this is indeed a pre-sheaf, and even a sheaf.

   (b) (Sheaf of invertible functions) Let X be as above, and let $\mathscr{F}(U) = (k[x]_F)^*$, that is consider only the invertible functions. Then X is again a sheaf.

   (c) (sheaf of roots of unit) Again consider X as above. Then let $\mathscr{F}(U)$ be the subgroup of $k[x]_F^*$ consisting of all roots of unity present in $\mathscr{F}(U)$. This forms a sheaf

   (d) (Sheaf of differential forms) If the reader is familiar with differential geometry, let $k = \mathbb{R}$ and let $\mathscr{F}(U)$ be the collection of all closed forms on U (or more generally, all smooth forms on U). Note that only $\mathscr{F}(\mathbb{A}_{\mathbb{R}}^1)$ are all the closed forms exact.

If $\mathscr{F} \in \mathbf{PSh}(U)$ and $f \in \mathscr{F}(U)$ for some open $U \subseteq X$, then f is called a section. The following proposition motivates this idea

Proposition 1.1.13: Space of Sections of (Pre)Sheaf: étale spaces

Let $\mathscr{F}$ be a presheaf of sets. Then there exists a topological space F and a local homeomorphism $\pi : F \to X$ so that $\mathscr{F}$ can be represented as the sheaf of sections from F to X. The space F is called the *étalé space*. In particular, F is the *étalé space* of $\mathscr{F}$.

Furthermore, there is an equivalence between the category $\mathbf{Sh}(X)$ and the category of étalé spaces over X (with bundle morphisms as morphisms).

The idea is that (pre)sheaves correspond to a space F that is it is locally homeomorphic to X. Thus, geometrically sheaves could be thought of as local homeomorphisms, though the total space F may be rather ugly as shall be shown in example 1.1.14. Note that local homeomorphisms need not be surjective; this is a common misconceptions as many examples of surjective local homeomorphisms are covering spaces, but examples such as the sky-scrapper sheaf should show why this need not be the case.

***Proof*:**
Let F be the disjoint union of the stalks on F.

$$F = \coprod_{x \in X} \mathscr{F}_x$$

Then $\pi : F \to X$ is naturally a set map. Define a basis for F as follows: for each $s \in \mathscr{F}(U)$, take the collection

$$\{[s]_x \in \mathscr{F}_x \mid x \in U\}$$

These form a basis for F (note that each stalk has the discrete topology). With the setup, it is now easy to see the equivalence of categories.

The étale space may seem like it ought to be pretty similar to X being locally homeomorphic, however this can be far from the case:

Example 1.1.14: Non-Hausdorff Étale Space
Take $X = \mathbb{R}$ be the real numbers and take $\mathcal{F}$ to be the smooth functions on $\mathbb{R}$. Take the constant germ around 0 as well as the germ which can be represented by the function:

$$g(x) = \begin{cases} x\sin(1/x) & x\sin(1/x) > 0 \\ 0 & x\sin(1/x) \leq 0 \\ 0 & x < 0 \end{cases}$$

In other words, g is zero when x is negative and when the function $x\sin(1/x)$ would be negative; we want to keep when $x\sin(1/x)$ is positive. Then these two are different points in the étale space, but *cannot* be separated by open sets.

Note that if we were working with $X = \mathbb{C}$ and took analytic or holomorphic functions, then this cannot happen as we have a non-isolated 0 without all of g being zero. This turns out to be the main impingement for the étale space being Hausdorff. In general sheaves of analytic functions shall form Hausdorff étale spaces.

1.1.1 Stalks and Sheaves

As mentioned earlier, presheaves induce stalks. It was mentioned that sheaves but *not* presheaves can be characterized on the level of stalks, meaning the stalks of a sheaf shall give us all the information about the morphism a sheaf. This in fact almost characterizes sheaves from presheaves, as it in fact characterizes *separated presheaves* (recall they satisfy the locality but not gluing axiom). The following shows the value of working with separated presheaves:

Lemma 1.1.15: Sections Determined By Germs

Let $\mathscr{F}$ be a sheaf (more specifically, a separated presheaf). Then the map:

$$\mathscr{F}(U) \to \prod_{p \in U} (\mathscr{F}_p)$$

is injective

This map being injective justifies the term "separated", namely the points are separated by the sections.

***Proof*:**
Let f be the map and consider $f(s) = (s_x)_{x \in U}$ Say:

$$(s_x)_{x \in U} = (t_x)_{x \in U} \qquad \text{equiv.} \qquad [(s, V_x)]_{x \in U} = [(t, W_x)]_{x \in U}$$

Then by definition there exists a $C_x \subseteq V_x \cap W_x$ such that

$$s|_{C_x} = t|_{C_x}$$

Now as $x \in C_x$ for each $x \in U$ we have $U = \bigcup_{x \in U} C_x$ and furthermore for for any pair C_x, C_y in the open cover we have:

$$\text{res}_{C_x, C_x \cap C_y} s = \text{res}_{C_y, C_x \cap C_y} t$$

hence, by the gluing axiom $s = t$, as we sought to show.

Note that we can show the locality axiom form this argument:

Corollary 1.1.16: Global Section Over Local Sections

Let $\mathscr{F}$ be a sheaf on X and $U_i \subseteq X$ is an open cover. Then:

$$\mathscr{F}(X) \to \prod_i \mathscr{F}(U_i)$$

is injective

***Proof*:**
Map $s \mapsto \left(s|_{U_i}\right)_i$. Consider

$$\left(s|_{U_i}\right)_i = \left(t|_{U_i}\right)_i$$

then s and t agree on intersections of U_i, hence there exists a unique element in $\mathcal{F}(X)$ for both these elements, namely $s = t$

In example 1.1.11, the sheaf $\mathcal{F}(X) = \mathbb{Z}$ and $\mathcal{F}(U) = 0$ for $U \subsetneq X$ was not a separated presheaf, and certainly $\mathcal{F}(X) \to \prod_{x\in X} \mathcal{F}_x$ is *not* injective, as each $\mathcal{F}_x$ consist of a single equivalence class representing the zero element (and hence $\prod_{x\in X} \mathcal{F}_x \equiv \{[0]\}$, and $\mathcal{F}(X) = \mathbb{Z}$. This example shows that we can somehow construct a nonzero object from a bunch of zero-objects. This can be thought of as being the key-blocker, especially in the case of abelian categories where most of the time the pre-image of 0 is only 0. This motivate the following:

Definition 1.1.17: Support of Sections

Let $\mathscr{F}$ be a sheaf of an abelian group on X, and s a global section of $\mathscr{F}$. Then the *support* of s, denoted $\operatorname{Supp}(s)$, is the points $p \in X$ where s has a nonzero germ:

$$\operatorname{Supp}(s) := \{p \in X \mid s_p \neq 0 \text{ in } \mathscr{F}_p\}$$

This notion differs a bit from the usual notion of support in analysis, in particular notice that it is the germ itself that is nonzero and not that the evaluation of the germ s_p at the point p that is nonzero.

Lemma 1.1.18: Support is Closed

Let $\mathscr{F}$ be a sheaf of abelian groups on X and $s \in \mathscr{F}(X)$. Then $\operatorname{Supp}(s)$ is closed

***Proof*:**
We will show $\operatorname{Supp}(s)^c$ is open, that is the set of all points $p \in X$ such that $s_p = 0$ is open. If $s_p = 0$ in the stalk, there exists an open subset $W_p \subseteq X$ such that $s|_{W_p} = 0$. Taking the union of arbitrary open sets is open, and hence taking the union of all W_p cover $\operatorname{Supp}(s)^c$ making $\operatorname{Supp}(s)$ closed, completing the proof.

The equivalent in analysis is the definition of the *closed support* of a function. Due to this, the compliment of the support of an element s is the 0-section. The section can also be defined on presheaves, however there are examples of presheaves where there are nonzero sections that "live nowhere" because it is 0 in every stalk[7]

Let us now take a closer look at $\prod_{p\in U}(\mathscr{F}_p)$. This set is much bigger than sections of a sheaf. We thus take the special sub-collection that associates with $\mathscr{F}(U)$ (essentially characterizing the image of $\mathscr{F}(U)$:

[7]Take $\mathscr{F}$ to be the sheaf of holomorphic functions on $\mathbb{C}$, $\mathscr{G}$ the sheaf of bounded holomorphic functions. Then take the sheaf $\mathcal{H} = \mathscr{F}/\mathscr{G}$. Then show that $\mathcal{H}_x = 0$, however $\mathcal{H}(\mathbb{C}) = \{\text{entire functions}\}/\mathbb{C}$

Definition 1.1.19: Compatible Germs

Let $\mathscr{F}$ be a sheaf on X and $U \subseteq X$ an open set. Then we say an elements $(s_p)_{p\in U} \in \prod_{p\in U} \mathscr{F}_p$ consists of *compatible germs* if there exists some representative open sets U_p for s_p:

$$U_p \subseteq U \text{ open}, \ \tilde{s}_p \in \mathscr{F}(U_p)$$

such that the germ of $\tilde{s}_p$ for all $q \in U_p$ is s_q. In other words, if $\{U_i\}$ is an open cover of U where $p \in U_i$ for all i, then s_p is the germ of f_i at p.

Hence, a compatible germ is the collection of germs that are "compatible" when taking subsets, that is *restrictions*. Using the gluing axiom, it is a good exercise to show that any choice of compatible germs for a sheaf $\mathscr{F}$ over U is the image of a section of $\mathscr{F}$ over U, that is we have successfully identified the right hand side of the equation in lemma 1.1.15. Note how important it is that we use sheaves (or at least separated presheaves) for the proof. The proof also motivate the agricultural terminology of "sheaf", for in agriculture a sheaf is a collection of "Stalks" (which is the main stem of a plant).

1.1.20 Remark When working with sheaves of sets or some structures on sets, the elements of the section can literally be thought of as functions using the above definition, namely function of the form $s : U \to \coprod_{p\in U} \mathscr{F}_p$. The generalization presented above allows to generalize the notion of "open set U" so that the topological space X is replaced by a category called a *site*, which is developed in *Étale Cohomology* [16].

1.1.2 Sheaf from a Topological Base

When defining a differential or topological structure, one of the strategies that is taken is to define pieces of the manifold in a compatible way, or said differently define it on some basis of the manifold. This section endeavours to do this for sheaves.

Let X be a topological space with a sheaf $\mathscr{F}$. Take a basis $\{B_i\}$ for X and take the subset of $\mathscr{F}$ consisting of $\mathscr{F}(B_i)$ along with all the appropriate restriction maps. Then there is enough Stalk information to recover all of $\mathscr{F}$ (show this), and hence we my want to start with such a collection to generate a sheaf. This collection alone will generate a presheaf, we will require the basis to satisfy the locality and gluability axioms:

Definition 1.1.21: Sheaf On A Base

Let X be a topological space and $\mathcal{B}$ a base of its topology. A *presheaf* F *on* $\mathcal{B}$ assigns to each $B \in \mathcal{B}$ a set, abelian group, ring or module $F(B)$, and to each inclusion $B' \subseteq B$ of members of $\mathcal{B}$ a morphism $\mathrm{res}_{B,B'} \colon F(B) \to F(B')$, with $\mathrm{res}_{B,B} = \mathrm{id}$ and $\mathrm{res}_{B',B''} \circ \mathrm{res}_{B,B'} = \mathrm{res}_{B,B''}$. It is a *sheaf on* $\mathcal{B}$ if for every $B \in \mathcal{B}$ and every cover $B = \bigcup_i B_i$ by members of $\mathcal{B}$:

1. **Base locality axiom**: if $f, g \in F(B)$ satisfy $\mathrm{res}_{B,B_i} f = \mathrm{res}_{B,B_i} g$ for all i, then $f = g$.
2. **Base gluability**: if $f_i \in F(B_i)$ are such that f_i and f_j have the same restriction to every member of $\mathcal{B}$ contained in $B_i \cap B_j$, then there exists $f \in F(B)$ such that $\mathrm{res}_{B,B_i} f = f_i$ for all i.

Theorem 1.1.22: Define Sheaf From Base

Let $\mathcal{B}$ be a base of the topology of X and F a sheaf on $\mathcal{B}$ (definition 1.1.21). Then there are a sheaf $\mathscr{F}$ on X, with values of the same kind as F, and isomorphisms $\theta_B \colon F(B) \to \mathscr{F}(B)$ for $B \in \mathcal{B}$ such that $\mathrm{res}_{B,B'} \circ \theta_B = \theta_{B'} \circ \mathrm{res}_{B,B'}$ whenever $B' \subseteq B$ in $\mathcal{B}$. If $\mathscr{G}$ is a sheaf with isomorphisms $\theta'_B \colon F(B) \to \mathscr{G}(B)$ satisfying the same condition, there are unique isomorphisms $\Theta_U \colon \mathscr{F}(U) \to \mathscr{G}(U)$, for $U \subseteq X$ open, that commute with restriction and satisfy $\Theta_B \circ \theta_B = \theta'_B$ for every $B \in \mathcal{B}$.

***Proof*:**
For $p \in X$, the members of $\mathcal{B}$ containing p, ordered by reverse inclusion, form a directed set, since for two of them $\mathcal{B}$ contains a third containing p inside their intersection. Let F_p be the set of classes f_p of pairs (f, B) with $p \in B \in \mathcal{B}$ and $f \in F(B)$, where $(f, B) \sim (f', B')$ when f and f' have the same restriction to some member of $\mathcal{B}$ that contains p and lies in $B \cap B'$. In the algebraic cases the operations of F_p are computed on representatives restricted to a common smaller member of $\mathcal{B}$, as for stalks (definition 1.1.3), and each map $F(B) \to F_p$ is a homomorphism.

Step 1: the sheaf $\mathscr{F}$. For $U \subseteq X$ open let $\mathscr{F}(U)$ be the set of tuples $(h_p)_{p \in U}$ with $h_p \in F_p$ such that every $p \in U$ has a *witness*, a pair (f, B) with $p \in B \subseteq U$, $B \in \mathcal{B}$ and $f \in F(B)$, such that $f_q = h_q$ for all $q \in B$. For $U' \subseteq U$, restriction sends $(h_p)_{p \in U}$ to $(h_p)_{p \in U'}$: a witness (f, B) at $p \in U'$ gives the witness $(\mathrm{res}_{B,B'} f, B')$ for any $B' \in \mathcal{B}$ with $p \in B' \subseteq B \cap U'$. Shortening tuples satisfies the identity and composition axioms, and $\mathscr{F}(\emptyset)$ consists of the empty tuple. Two tuples on U that agree on every member of an open cover agree at every point, which is locality. If tuples on the members U_i of an open cover of U agree on overlaps, they have equal components at each point of $U_i \cap U_j$, so they define a tuple on U, and a witness at $p \in U_i$ for the tuple on U_i is a witness for it. This is gluing, so $\mathscr{F}$ is a sheaf. In the algebraic cases $\mathscr{F}(U)$ is closed under componentwise operations, because witnesses (f, B) and (f', B') at p restrict to a common $B'' \in \mathcal{B}$ with $p \in B'' \subseteq B \cap B'$ and the restricted sums and products are witnesses, and the restriction maps are homomorphisms.

Step 2: the maps θ_B. For $B \in \mathcal{B}$ put $\theta_B(f) = (f_q)_{q \in B}$, which has the witness (f, B) at every point. For $B' \subseteq B$ in $\mathcal{B}$ and $q \in B'$ the pairs $(\mathrm{res}_{B,B'} f, B')$ and (f, B) have the same class, so $\mathrm{res}_{B,B'}(\theta_B(f)) = \theta_{B'}(\mathrm{res}_{B,B'} f)$. In the algebraic cases θ_B is a homomorphism, since the maps

$F(B) \to F_q$ are.

Step 3: θ_B is injective. If $f_q = g_q$ for every $q \in B$, each $q \in B$ has a member $B_q \in \mathcal{B}$ with $q \in B_q \subseteq B$ on which f and g have the same restriction. The sets B_q cover B, so $f = g$ by base locality.

Step 4: θ_B is surjective. Let $(h_q)_{q\in B} \in \mathscr{F}(B)$ and choose witnesses (f^q, B_q) at the points $q \in B$. Let $B'' \in \mathcal{B}$ lie in $B_q \cap B_{q'}$. The restrictions of f^q and $f^{q'}$ to B'' have the class h_r at every $r \in B''$, so each $r \in B''$ lies in a member of $\mathcal{B}$ inside B'' on which they agree, and base locality on B'' shows that they agree on B''. Base gluability gives $f \in F(B)$ with $\mathrm{res}_{B,B_q} f = f^q$ for all q, and then $f_r = (f^q)_r = h_r$ for $r \in B_q$, so $\theta_B(f) = (h_r)_{r\in B}$. A bijective homomorphism is an isomorphism, so each θ_B is an isomorphism.

Step 5: uniqueness. The maps $\varphi_B = \theta'_B \circ \theta_B^{-1} \colon \mathscr{F}(B) \to \mathscr{G}(B)$ are isomorphisms commuting with restriction between members of $\mathcal{B}$. Let U be open and $s \in \mathscr{F}(U)$. For members B, B' of $\mathcal{B}$ inside U, the sections $\varphi_B(s|_B)$ and $\varphi_{B'}(s|_{B'})$ both restrict to $\varphi_{B''}(s|_{B''})$ on every member B'' of $\mathcal{B}$ inside $B \cap B'$, and these cover $B \cap B'$, so they agree on $B \cap B'$ by locality of $\mathscr{G}$. The members of $\mathcal{B}$ inside U cover U, so gluing in $\mathscr{G}$ gives a unique $\Theta_U(s) \in \mathscr{G}(U)$ with $\Theta_U(s)|_B = \varphi_B(s|_B)$ for all of them. For $U' \subseteq U$, the section $\Theta_U(s)|_{U'}$ has the defining property of $\Theta_{U'}(s|_{U'})$, so the Θ_U commute with restriction, and taking $U = B$ gives $\Theta_B = \varphi_B$, that is, $\Theta_B \circ \theta_B = \theta'_B$. The same construction applied to the φ_B^{-1} gives maps $\mathscr{G}(U) \to \mathscr{F}(U)$, and both composites are identities, because a section is determined by its restrictions to the members of $\mathcal{B}$ inside U and there the composites are identities. In the algebraic cases each Θ_U is a homomorphism, since $\Theta_U(s + s')$ and $\Theta_U(s) + \Theta_U(s')$ agree on each member of $\mathcal{B}$ inside U, and likewise for products and scalars. Finally, any family Θ'_U commuting with restriction with $\Theta'_B \circ \theta_B = \theta'_B$ satisfies $\Theta'_U(s)|_B = \varphi_B(s|_B)$ for every member B of $\mathcal{B}$ inside U, so $\Theta'_U = \Theta_U$ by locality of $\mathscr{G}$, completing the proof.

Lots more to cover here that is worth going over

1.1.3 Gluing Sheaves on an Open Cover

Just as we can construct a sheaf from data on a topological base, we can construct a sheaf from sheaves defined on the pieces of an open cover, provided the local sheaves are compatible on overlaps. This result is foundational: it will be used in chapter 2 to construct the structure sheaf when gluing schemes.

Theorem 1.1.23: Gluing Sheaves On An Open Cover

Let X be a topological space and $\{U_i\}_{i\in I}$ an open cover. Suppose we are given:

1. a sheaf $\mathscr{F}_i$ on each U_i,
2. isomorphisms of sheaves $\varphi_{ij} : \mathscr{F}_i|_{U_i\cap U_j} \xrightarrow{\sim} \mathscr{F}_j|_{U_i\cap U_j}$ for all i, j

satisfying the cocycle condition on triple overlaps:

$$\varphi_{jk} \circ \varphi_{ij} = \varphi_{ik} \quad \text{on } U_i \cap U_j \cap U_k$$

Then there exists a sheaf $\mathscr{F}$ on X, unique up to unique isomorphism, together with isomorphisms $\psi_i : \mathscr{F}|_{U_i} \xrightarrow{\sim} \mathscr{F}_i$ such that $\varphi_{ij} \circ \psi_i = \psi_j$ on $U_i \cap U_j$.

***Proof*:**
For each open set $V \subseteq X$, define:

$$\mathscr{F}(V) = \left\{(s_i)_{i\in I} \;\middle|\; s_i \in \mathscr{F}_i(V \cap U_i), \quad \varphi_{ij}\big(s_i|_{V\cap U_i\cap U_j}\big) = s_j|_{V\cap U_i\cap U_j} \;\forall\, i, j\right\}$$

with restriction maps defined componentwise: if $W \subseteq V$ then $\mathrm{res}_{V,W}((s_i)_i) = (s_i|_{W\cap U_i})_i$.

$\mathscr{F}$ is a presheaf. The restriction of a compatible tuple is compatible (as φ_{ij} commutes with restriction, being a sheaf morphism), and the identity and composition axioms are inherited componentwise.

Locality axiom. Let $V = \bigcup_\alpha V_\alpha$ and suppose $(s_i), (t_i) \in \mathscr{F}(V)$ satisfy $(s_i)|_{V_\alpha} = (t_i)|_{V_\alpha}$ for all α. Then for each i, we have $s_i|_{V_\alpha\cap U_i} = t_i|_{V_\alpha\cap U_i}$ for all α. Since $\{V_\alpha \cap U_i\}_\alpha$ is an open cover of $V \cap U_i$ and $\mathscr{F}_i$ is a sheaf (satisfying locality), $s_i = t_i$ for all i.

Gluing axiom. Let $V = \bigcup_\alpha V_\alpha$ and suppose $(s_i^{(\alpha)})_i \in \mathscr{F}(V_\alpha)$ agree on overlaps, i.e. $s_i^{(\alpha)}|_{V_\alpha\cap V_\beta\cap U_i} = s_i^{(\beta)}|_{V_\alpha\cap V_\beta\cap U_i}$ for all α, β, i. For each fixed i, the sections $s_i^{(\alpha)} \in \mathscr{F}_i(V_\alpha \cap U_i)$ agree on overlaps $V_\alpha \cap V_\beta \cap U_i$. As $\mathscr{F}_i$ is a sheaf, there exists a unique $s_i \in \mathscr{F}_i(V \cap U_i)$ with $s_i|_{V_\alpha\cap U_i} = s_i^{(\alpha)}$ for all α.

It remains to check the compatibility condition for the glued tuple $(s_i)_i$: we need $\varphi_{ij}(s_i|_{V\cap U_i\cap U_j}) = s_j|_{V\cap U_i\cap U_j}$. Both sides are sections of $\mathscr{F}_j(V \cap U_i \cap U_j)$ that agree on each $V_\alpha \cap U_i \cap U_j$ (since $(s_i^{(\alpha)})_i \in \mathscr{F}(V_\alpha)$ satisfies the compatibility condition). By the locality axiom for $\mathscr{F}_j$, they are equal.

Hence $\mathscr{F}$ is a sheaf. For $V \subseteq U_i$ let $\psi_i(V)\colon \mathscr{F}(V) \to \mathscr{F}_i(V)$ be the projection $(s_j)_j \mapsto s_i$, which commutes with restriction. The cocycle condition with $i = j = k$ gives $\varphi_{ii} \circ \varphi_{ii} = \varphi_{ii}$, so $\varphi_{ii} = \mathrm{id}$, as φ_{ii} is an isomorphism. For $t \in \mathscr{F}_i(V)$ the tuple with components $\varphi_{ij}(t|_{V\cap U_j})$ is compatible, since $\varphi_{jk}(\varphi_{ij}(t|)) = \varphi_{ik}(t|)$ on $V \cap U_j \cap U_k$ by the cocycle condition, and its i-th component is $\varphi_{ii}(t) = t$. Conversely, the compatibility condition with $V \cap U_i = V$ says that a tuple $(s_j)_j \in \mathscr{F}(V)$ has $s_j = \varphi_{ij}(s_i|_{V\cap U_j})$ for every j. So $\psi_i(V)$ is bijective, and $\psi_i\colon \mathscr{F}|_{U_i} \to \mathscr{F}_i$ is an isomorphism. For $V \subseteq U_i \cap U_j$ and $(s_k)_k \in \mathscr{F}(V)$, compatibility gives $\varphi_{ij}(s_i) = s_j$, that is, $\varphi_{ij} \circ \psi_i = \psi_j$. When the $\mathscr{F}_i$ take values in abelian groups, rings or modules, so do the $\mathscr{F}(V)$ under componentwise operations, because the φ_{ij} and the restriction maps are homomorphisms, and the ψ_i are then isomorphisms of such sheaves.

Uniqueness. Let $\mathscr{G}$ be a sheaf with isomorphisms $\psi_i' \colon \mathscr{G}|_{U_i} \to \mathscr{F}_i$ satisfying $\varphi_{ij} \circ \psi_i' = \psi_j'$. On $U_i \cap U_j$ we have $\psi_j = \varphi_{ij} \circ \psi_i$ and $\psi_j' = \varphi_{ij} \circ \psi_i'$, hence $\psi_j'^{-1} \circ \psi_j = \psi_i'^{-1} \circ \psi_i$. For V open and $s \in \mathscr{F}(V)$, the sections $\psi_i'^{-1}(\psi_i(s|_{V \cap U_i})) \in \mathscr{G}(V \cap U_i)$ therefore agree on the overlaps $V \cap U_i \cap U_j$, and gluing in $\mathscr{G}$ gives a unique $\Theta_V(s) \in \mathscr{G}(V)$ restricting to them. By the uniqueness of gluing, the maps Θ_V commute with restriction, and $\psi_i' \circ \Theta|_{U_i} = \psi_i$. In the algebraic cases each Θ_V is a homomorphism, since $\Theta_V(s+s')$ and $\Theta_V(s) + \Theta_V(s')$ agree on each $V \cap U_i$, and likewise for products and scalars. The same construction with the roles of $(\mathscr{F}, \psi_i)$ and $(\mathscr{G}, \psi_i')$ exchanged gives an inverse, both composites restricting to identities on each $V \cap U_i$. Any Θ' with $\psi_i' \circ \Theta'|_{U_i} = \psi_i$ for all i satisfies $\Theta'_V(s)|_{V \cap U_i} = \psi_i'^{-1}(\psi_i(s|_{V \cap U_i}))$, so $\Theta' = \Theta$ by locality of $\mathscr{G}$. Hence $\mathscr{F}$ is unique up to unique isomorphism, completing the proof.

1.2 Morphisms and Exactness

Now that we have defined (pre)sheaves, it is natural to define morphism between sheaves to show that $\mathbf{PSh}(X)$ and $\mathbf{Sh}(X)$ form a category. Sheaf morphisms shall be over the same object, and hence can be thought of as the equivalent of looking at local homeomorphisms between two étale spaces:

Definition 1.2.1: Morphism Of Sheaves

Let $\mathscr{F}, \mathscr{G} \in \mathbf{PSh}(X)$ (note that $\mathbf{Sh}(X) \subseteq \mathbf{PSh}(X)$). Then a *morphism of (pre)sheaves* (over the chosen category) $\varphi : \mathscr{F} \to \mathscr{G}$ is a natural transformation between the two functors. Concretely, it contains the data $\varphi(U) : \mathscr{F}(U) \to \mathscr{G}(U)$ such that if $V \subseteq U$ then the following diagram commutes:

$$\begin{array}{ccc} \mathscr{F}(U) & \xrightarrow{\varphi(U)} & \mathscr{G}(U) \\ {\scriptstyle \mathrm{res}_{U,V}}\downarrow & & \downarrow{\scriptstyle \mathrm{res}_{U,V}} \\ \mathscr{F}(V) & \xrightarrow[\varphi(V)]{} & \mathscr{G}(V) \end{array}$$

that is

$$\mathrm{res}_{U,V}(\varphi(U)(-)) = \varphi(V)(\mathrm{res}_{U,V}(-))$$

The collection of morphisms between two sheaves is denoted $\mathrm{Hom}_{\mathbf{Sh}(X)}(\mathscr{F}, \mathscr{G})$ or $\mathrm{Mor}_{\mathbf{Sh}(X)}(\mathscr{F}, \mathscr{G})$

To emphasize when the domain and codomain of a presheaf morphism is a sheaf, the morphism shall be called a *sheaf morphism.* A useful geometric intuition of sheaf morphisms is that they apply a transformation to each map (ex. $\exp(f)(z) = e^{f(z)}$). With the morphisms and objects defined, it is easy to show that they form a category. We shall denote the category of sheaves over a category $\mathbf{C}$ on a space X as $\mathbf{C}_X$. Thus:

$$\mathbf{Set}_X \qquad \mathbf{Ab}_X \qquad \mathbf{Ring}_X$$

are the categories of sheaves over sets, abelian groups, and rings respectively. Since the morphisms are the same, the category $\mathbf{Sh}(X)$ is a full subcategory of $\mathbf{PSh}(X)$. For presheaves, we shall denote the category with an extra superscript:

$$\mathbf{Set}_X^{\mathrm{pre}} \qquad \mathbf{Ab}_X^{\mathrm{pre}} \qquad \mathbf{Ring}_X^{\mathrm{pre}}$$

Example 1.2.2: Morphism of Presheaves and Sheaves

(If the reader is familiar with Riemann surface) Let X be a Riemann surface and $\mathcal{O}_X(-P)$ be the collection of meromorphic function $f \in K^*(X)$ such that $\operatorname{div}(f) + (-p) \geq 0$ ($P \in X$ a point). Then if $\mathcal{O}_X$ is the sheaf of holomorphic functions on X there are sheaf morphisms:

$$0 \to \mathcal{O}_K(-P) \to \mathcal{O}_X \to \kappa_P \to 0$$

where κ_P is the sky-scrapper sheaf at P, namely:

$$\kappa_P = \begin{cases} (0) & x \neq P \\ \mathbb{C} & x = P \end{cases}$$

This is even an exact sequence of sheaves, but we need to build-up to show how to properly define exactness in $\mathbf{C}_X$.

From now on, we shall be working with *abelian categories* unless stated otherwise: they all have 0 objects, they are closed under finite products and coproducts, and their kernel/cokernels behave well with epimorphisms and monomorphisms; see section 0.4 for a refresher or [9] for more details. We shall build-up to showing that $\mathbf{C}_X$ and $\mathbf{C}_X^{\text{pre}}$ are abelian categories.

As a presheaf morphism is a natural transformation, it important to emphasize that it does not make sense to say that a sheaf morphism is injective or surjective. Instead, it can be a *monomorphism* or an *epimorphism* and have corresponding kernel, cokernels, and images. Let us find objects in $\mathbf{PSh}(X)$ and $\mathbf{Sh}(X)$ that can represent these kernel, cokernel, and image. Starting with the kernel of a presheaf morphism $\varphi : \mathscr{F} \to \mathscr{G}$ which is defined to be be object $\mathcal{K}$ with the morphism $K : \mathcal{K} \to \mathcal{F}$ such that $\varphi \circ K = 0$ universally.

Proposition 1.2.3: Kernels in Presheaf Category

Let $\varphi : \mathscr{F} \to \mathscr{G}$ be a morphism of presheaves of R-modules ($R = \mathbb{Z}$ gives abelian groups). Then $\ker \varphi$ can be defined to be the kernel of each $\varphi(U)$:

$$\mathcal{K}(U) = \ker(\varphi(U))$$

The resulting presheaf is denoted $\ker_{\text{pre}}(\varphi)$. Furthermore, if $\mathscr{F}$ is a sheaf and $\mathscr{G}$ satisfies at least the locality axiom (in particular if it is also a sheaf), then $\mathcal{K}$ is a sheaf.

***Proof*:**

Each $\mathcal{K}(U) = \ker(\varphi(U))$ is a submodule of $\mathscr{F}(U)$, and all maps below are R-linear.

Step 1: $\mathcal{K}$ is a presheaf. Let $V \subseteq U$ and $s \in \mathcal{K}(U)$. Naturality of φ (definition 1.2.1) gives $\varphi(V)(s|_V) = (\varphi(U)(s))|_V = 0|_V = 0$, so $\operatorname{res}_{U,V}$ carries $\mathcal{K}(U)$ into $\mathcal{K}(V)$. The restricted maps $\mathcal{K}(U) \to \mathcal{K}(V)$ satisfy the identity and composition axioms of definition 1.1.1 because those of $\mathscr{F}$ do. The inclusions $\iota(U)\colon \mathcal{K}(U) \to \mathscr{F}(U)$ commute with restriction, so they form a presheaf morphism $\iota\colon \mathcal{K} \to \mathscr{F}$ with $\varphi \circ \iota = 0$.

Step 2: $(\mathcal{K}, \iota)$ is a kernel of φ among presheaves. Let $\chi\colon \mathscr{H} \to \mathscr{F}$ be a presheaf morphism with $\varphi \circ \chi = 0$. For each U we have $\varphi(U) \circ \chi(U) = 0$, so $\chi(U)$ takes values in $\mathcal{K}(U)$ and defines

$\chi'(U)\colon \mathscr{H}(U) \to \mathcal{K}(U)$ with $\iota(U) \circ \chi'(U) = \chi(U)$. It is the only map with this property, as $\iota(U)$ is injective. For $V \subseteq U$ and $h \in \mathscr{H}(U)$,

$$\iota(V)\big(\chi'(V)(h|_V)\big) = \chi(V)(h|_V) = \chi(U)(h)|_V = \iota(V)\big(\chi'(U)(h)|_V\big)$$

since restriction in $\mathcal{K}$ is restriction in $\mathscr{F}$, and injectivity of $\iota(V)$ gives $\chi'(V)(h|_V) = \chi'(U)(h)|_V$. So χ' is the unique presheaf morphism with $\iota \circ \chi' = \chi$. Steps 1 and 2 use only the presheaf axioms.

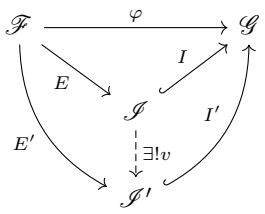
Step 3: the sheaf case. Let $\mathscr{F}$ be a sheaf and $\mathscr{G}$ satisfy the locality axiom of definition 1.1.6. The group $\mathscr{F}(\emptyset)$ is zero, hence so is $\mathcal{K}(\emptyset)$. Locality: two sections of $\mathcal{K}(U)$ agreeing on an open cover of U agree as sections of $\mathscr{F}$, hence are equal by the locality axiom for $\mathscr{F}$. Gluing: let $U = \bigcup_i U_i$ and $s_i \in \mathcal{K}(U_i)$ with $s_i|_{U_i \cap U_j} = s_j|_{U_i \cap U_j}$. The gluing axiom for $\mathscr{F}$ gives $s \in \mathscr{F}(U)$ with $s|_{U_i} = s_i$ for all i. Then $\varphi(U)(s)|_{U_i} = \varphi(U_i)(s_i) = 0 = 0|_{U_i}$ for every i, so $\varphi(U)(s) = 0$ by the locality axiom for $\mathscr{G}$, that is, $s \in \mathcal{K}(U)$. Hence $\mathcal{K}$ is a sheaf. Since $\mathbf{Sh}(X)$ is a full subcategory of $\mathbf{PSh}(X)$ (definition 1.2.1), Step 2 with $\mathscr{H}$ a sheaf shows that $(\mathcal{K}, \iota)$ is also a kernel of φ in $\mathbf{Sh}(X)$, as claimed.

Note that as the representation of the kernel for sheaves is the same as the characterization of presheaves, we can just write $\ker(\varphi)$ without any ambiguity. As the monomorphism condition can be checked on open sets, the following usual definition of injectivity is recovered on the level of open sets:

Definition 1.2.4: Injective Presheaf and Sheaf Morphism

Let $\varphi : \mathscr{F} \to \mathscr{G}$ be a presheaf morphism between shaves or presheaves. Then φ is *injective* if $\ker \varphi(U) = 0$ for all open $U \subseteq X$.

The image of a presheaf morphism $\varphi : \mathscr{F} \to \mathscr{G}$ is defined to be be object $\mathscr{I}$ with the monomorphism $I\colon \mathscr{I} \to \mathscr{G}$ and a morphism $E\colon \mathscr{F} \to \mathscr{I}$ such that $\varphi = I \circ E$ universally, namely:



The reader familiar with category may be more familiar with images in abelian categories defined as:

$$\mathrm{im}(\varphi) = \ker(\mathrm{coker}(\varphi))$$

As we are working with abelian categories, $\ker(\mathrm{coker}(\varphi)) \cong \mathrm{coker}(\ker(\varphi))$, and hence there is no need of defining the coimage.

Proposition 1.2.5: Images in Presheaf and Sheaf Category

Let $\varphi : \mathscr{F} \to \mathscr{G}$ be a morphism of presheaves of R-modules ($R = \mathbb{Z}$ gives abelian groups). Then $\mathrm{im}_{\mathrm{pre}}\varphi$ can be defined to be the image of each $\varphi(U)$:

$$\mathrm{im}_{\mathrm{pre}}\varphi(U) = \mathrm{im}(\varphi(U))$$

and similarly for cokernels:

$$\mathrm{coker}_{\mathrm{pre}}(\varphi(U)) = \frac{\mathscr{G}(U)}{\mathrm{im}(\varphi(U))}$$

The resulting $\mathrm{im}_{\mathrm{pre}}(\varphi)$ and $\mathrm{coker}_{\mathrm{pre}}(\varphi)$ is called the *image presheaf* and *cokernel presheaf* respectively.

***Proof*:**
All maps below are R-linear.

Step 1: the image presheaf. Let $V \subseteq U$ and $t = \varphi(U)(s) \in \mathrm{im}(\varphi(U))$. Naturality of φ gives $t|_V = \varphi(V)(s|_V) \in \mathrm{im}(\varphi(V))$, so the restriction maps of $\mathscr{G}$ carry $\mathrm{im}_{\mathrm{pre}}\varphi(U)$ into $\mathrm{im}_{\mathrm{pre}}\varphi(V)$, and $\mathrm{im}_{\mathrm{pre}}\varphi$ is a presheaf whose identity and composition axioms are inherited from $\mathscr{G}$. The inclusions form a presheaf morphism $I\colon \mathrm{im}_{\mathrm{pre}}\varphi \to \mathscr{G}$, injective on every open set. The maps $E(U)\colon \mathscr{F}(U) \to \mathrm{im}_{\mathrm{pre}}\varphi(U)$, $s \mapsto \varphi(U)(s)$, commute with restriction because φ does, and $\varphi = I \circ E$.

Step 2: the cokernel presheaf. By Step 1, $\mathrm{res}_{U,V}\colon \mathscr{G}(U) \to \mathscr{G}(V)$ carries $\mathrm{im}(\varphi(U))$ into $\mathrm{im}(\varphi(V))$, so it induces a homomorphism $\mathscr{G}(U)/\mathrm{im}(\varphi(U)) \to \mathscr{G}(V)/\mathrm{im}(\varphi(V))$, $[t] \mapsto [t|_V]$. The identity and composition axioms descend from $\mathscr{G}$, so $\mathrm{coker}_{\mathrm{pre}}\varphi$ is a presheaf, and the quotient maps $Q(U)\colon \mathscr{G}(U) \to \mathrm{coker}_{\mathrm{pre}}\varphi(U)$ form a presheaf morphism Q with $Q \circ \varphi = 0$.

Step 3: Q is a cokernel of φ among presheaves. Let $\chi\colon \mathscr{G} \to \mathscr{H}$ satisfy $\chi \circ \varphi = 0$. Each $\chi(U)$ vanishes on $\mathrm{im}(\varphi(U))$, so it factors uniquely as $\chi(U) = \chi'(U) \circ Q(U)$ with $\chi'(U)\colon \mathrm{coker}_{\mathrm{pre}}\varphi(U) \to \mathscr{H}(U)$ a homomorphism. For $t \in \mathscr{G}(U)$,

$$\chi'(V)\big([t]|_V\big) = \chi'(V)([t|_V]) = \chi(V)(t|_V) = \chi(U)(t)|_V = \chi'(U)([t])|_V$$

so χ' is a presheaf morphism with $\chi' \circ Q = \chi$, and it is unique because every $Q(U)$ is surjective.

Step 4: $(\mathrm{im}_{\mathrm{pre}}\varphi, I, E)$ is an image of φ among presheaves. A monomorphism of presheaves $I'\colon \mathscr{I}' \to \mathscr{G}$ is injective on every open set: for the kernel inclusion $\iota\colon \ker_{\mathrm{pre}}(I') \to \mathscr{I}'$ of proposition 1.2.3 we have $I' \circ \iota = 0 = I' \circ 0$, so $\iota = 0$, which says $\ker(I'(U)) = 0$ for every U. Now let $\varphi = I' \circ E'$ with I' a monomorphism. For each U, $\mathrm{im}(\varphi(U)) = I'(U)\big(E'(U)(\mathscr{F}(U))\big)$ lies in the image of the injective map $I'(U)$, so there is a unique map $v(U)\colon \mathrm{im}(\varphi(U)) \to \mathscr{I}'(U)$ with $I'(U) \circ v(U)$ the inclusion, namely $v(U)(t) = I'(U)^{-1}(t)$, and it is a homomorphism. For $V \subseteq U$ and $t \in \mathrm{im}(\varphi(U))$,

$$I'(V)\big(v(V)(t|_V)\big) = t|_V = \big(I'(U)(v(U)(t))\big)|_V = I'(V)\big(v(U)(t)|_V\big)$$

so injectivity of $I'(V)$ shows that v commutes with restriction. Thus $v\colon \mathrm{im}_{\mathrm{pre}}\varphi \to \mathscr{I}'$ is the unique presheaf morphism with $I' \circ v = I$, and $I' \circ v \circ E = I \circ E = \varphi = I' \circ E'$ gives $v \circ E = E'$

because I' is a monomorphism. This is the universal property of the image displayed above. On each open set $\mathrm{im}_{\mathrm{pre}}\varphi(U) = \ker(Q(U))$, so the image presheaf is also the kernel of the cokernel. No step used a sheaf axiom, completing the proof.

When $\mathscr{F}$ and $\mathscr{G}$ are sheaves, $\mathrm{im}_{\mathrm{pre}}\varphi$ and $\mathrm{coker}_{\mathrm{pre}}\varphi$ need not be sheaves, so images and cokernels in $\mathbf{C}_X$ require more work (example 1.2.7).

Definition 1.2.6: Surjective Presheaf Morphism

Let $\varphi : \mathscr{F} \to \mathscr{G}$ be a presheaf morphism between presheaves. Then φ is *surjective* if $\mathrm{im}(\varphi(U)) = \mathscr{G}(U)$ for all open $U \subseteq X$.

As noted earlier, images of sheaves need not be sheaves:

Example 1.2.7: Sheaves: difficulty With Cokernels and Images

1. Let $X = \mathbb{C}$ with the usual topology, $\underline{\mathbb{Z}}$ be the constant sheaf on X, $\mathcal{O}_X$ the sheaf of holomorphic functions, and $\mathscr{F}$ the *presheaf* of functions admitting a holomorphic logarithm. Then there is an exact sequence of presheaves on X:

$$0 \to \underline{\mathbb{Z}} \to \mathcal{O}_X \to \mathscr{F} \to 0$$

where $\underline{\mathbb{Z}} \to \mathcal{O}_X$ is the natural inclusion on each open set, and $\mathcal{O}_X \to \mathscr{F}$ is given by $f \mapsto \exp(2\pi i f)$ (again, over each open set). This is indeed a short exact sequence, for if $U \subseteq V \subseteq \mathbb{C}$, then:

$$\begin{array}{ccccccccc} 0 & \longrightarrow & \underline{\mathbb{Z}}(V) & \longrightarrow & \mathcal{O}_X(V) & \longrightarrow & \mathscr{F}(V) & \longrightarrow & 0 \\ & & \downarrow{\scriptstyle \mathrm{res}} & & \downarrow{\scriptstyle \mathrm{res}} & & \downarrow{\scriptstyle \mathrm{res}} & & \\ 0 & \longrightarrow & \underline{\mathbb{Z}}(U) & \longrightarrow & \mathcal{O}_X(U) & \longrightarrow & \mathscr{F}(U) & \longrightarrow & 0 \end{array}$$

commutes. Certainly each $\underline{\mathbb{Z}}(V) \to \mathcal{O}_X(V)$ is an inclusion, and $\mathcal{O}_X(V) \to \mathscr{F}(V)$ is surjective (by definition) with kernel being the constant functions $\mathbb{Z}$ since if f admits a logarithmic inverse, then $f = \log(g)$, so $g \mapsto \exp(2\pi i \log g) = f$.

Then $\mathscr{F}$ is *not* a sheaf: take $U = \mathbb{C}^*$, cover it with $U_1 = \mathbb{C} \setminus \mathbb{R}_{\geq 0}$ and $U_2 = \mathbb{C} \setminus \mathbb{R}_{\leq 0}$. Then $\mathscr{F}(U)$, $\mathscr{F}(U_1)$, and $\mathscr{F}(U_2)$ each have multiple non-zero functions admitting logarithms (see [58]). If $\mathscr{F}$ were a sheaf, then as z admits logarithmic inverse on U_1 and U_2 and tautologically $z|_{U_1 \cap U_2} = z|_{U_1 \cap U_2}$ there should be a function defined on U restricting to these two, that restricts to both of these, but no such function exists (consider that such a function would have to be continuous, but going around the origin introduces non-trivial monodromy).

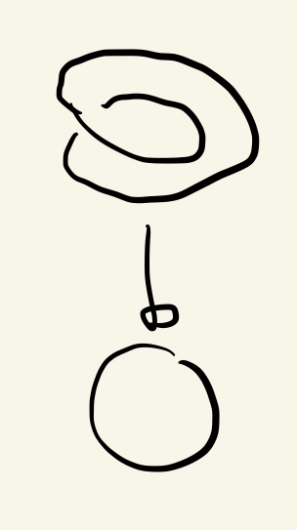
2. Consider the following similar but subtly different example. Consider $\mathcal{O}_{\mathbb{C}}^*$ the sheaf of all non-vanishing holomorphic functions (this sheaf is strictly "bigger" than the above $\mathscr{F}$, ex. $\mathscr{F}(\mathbb{C}^*) \subsetneq \mathcal{O}_{\mathbb{C}}^*(\mathbb{C}^*)$). Consider the following short exact sequence of *sheaves*:

$$0 \to \underline{\mathbb{Z}} \xrightarrow{\iota} \mathcal{O}_{\mathbb{C}} \xrightarrow{\exp} \mathcal{O}_{\mathbb{C}}^* \xrightarrow{!} 0$$

Notice that on $U = \mathbb{C}^*$, z does not vanish and so $z \in \mathcal{O}_{\mathbb{C}}^*(U)$. However, z does not admit a logarithmic inverse (again see [58]). This would imply that $\exp(\mathbb{C}^*)$ is *not* surjective and

so exp is not surjective on all open sets. And yet it is indeed the case that there exists a morphism $\xrightarrow{!}$, namely exp is an *epimorphism* on sheaves. This will be an example of an epimorphism that is *not* characterize by surjectivity on open sets, as shall be explained by defining sheafification after these examples.

3. For the reader less comfortable with complex analysis, here is a more rudimentary example: take the sheaf of continuous sections of the circle S^1. For our first sheaf of sections, let $\pi : S^1 \to S^1$ be the identity (the trivial bundle), and let $\mathcal{F}_1$ be the sheaf of sections of this bundle. The second sheaf of section will with the projection from the twisted circle, which can be visualized as:



Label this shape S_2^1. The second sheaf $\mathcal{F}_2$ is all the continuous sections of all the open subsets of the codomain for the map $\pi : S_2^1 \to S^1$ with a reasonable definition of projection map mirroring the above image (over the complex numbers, we can take $z \mapsto z^2$). Note that $\mathcal{F}_2(S^2) = 0$, i.e. it has no global sections. Then there is certainly a surjective continuous map φ satisfying the commutative diagram:

$$\begin{array}{ccc} S_2^1 & \xrightarrow{\varphi} & S^1 \\ & \searrow \quad \swarrow & \\ & S^1 & \end{array}$$

However, there is no surjective maps:

$$\mathcal{F}_2(X) \to \mathcal{F}_1(X)$$

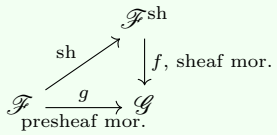
namely since there no continuous section from S^1 to S_2^1 that would satisfy the projection condition.

In the above, the first example shows that exactness has no issue with pre-sheaves, and in the second example we claimed the short exact sequence of sheaves was still exact in the category of sheaves.

Notice that the image of exp in $\mathcal{O}_{\mathbb{C}}^*$ was the presheaf $\mathscr{F}$ in the first example. What we shall show is that this image uniquely determines $\mathcal{O}_{\mathbb{C}}^*$. In the same vein in which we can from an abelian group via a abelian semi-group using groupification, or a normal subgroup by normalizing a subgroup, we can from a sheaf from a presheaf via sheafification. As is usual, this is done in a universal way that is adjoint to the forgetful functor. This sheafification shall be the solution to defining right notion of surjection for sheaves:

Theorem 1.2.8: Sheafification

Let $\mathscr{F}$ be a presheaf on X. Then there exists a sheaf $\mathscr{F}^{\text{sh}}$ and a map $\text{sh} : \mathscr{F} \to \mathscr{F}^{\text{sh}}$ such that for any sheaf $\mathscr{G}$ and any presheaf morphism $g : \mathscr{F} \to \mathscr{G}$, there exists a unique morphism of sheaves $f : \mathscr{F}^{\text{sh}} \to \mathscr{G}$ such that the following diagram commutes:



The sheafification of the presheaf $\mathscr{F}$ is often denoted $\mathscr{F}^+$.

The idea will be to remove all section that violate locality, and add sections to respect gluability. Another good take is the following post here.

***Proof*:**
The presheaf $\mathscr{F}$ takes values in sets, abelian groups, rings or modules. For $p \in X$, the stalk $\mathscr{F}_p$ is the set of germs of definition 1.1.3, and in the algebraic cases its operations are computed on representatives over a common smaller neighbourhood, $[(a, W)] + [(b, W')] = [(a|_{W\cap W'} + b|_{W\cap W'}, W \cap W')]$ and likewise for products and scalars. This does not depend on the representatives, because changing them changes a and b only by restriction to smaller neighbourhoods and the restriction maps are homomorphisms, and it makes each germ map $\mathscr{F}(W) \to \mathscr{F}_p$ a homomorphism.

Step 1: the presheaf $\mathscr{F}^{\text{sh}}$, with componentwise operations. For $U \subseteq X$ open and a tuple $(f_p)_{p\in U} \in \prod_{p\in U} \mathscr{F}_p$, a *witness* at $p \in U$ is a pair (V, s) with V open, $p \in V \subseteq U$, and $s \in \mathscr{F}(V)$ such that $s_q = f_q$ for all $q \in V$. Put

$$\mathscr{F}^{\text{sh}}(U) = \left\{ (f_p)_{p\in U} \in \prod_{p\in U} \mathscr{F}_p \;\middle|\; \text{there is a witness at every } p \in U \right\}$$

For $U' \subseteq U$, restriction sends $(f_p)_{p\in U}$ to $(f_p)_{p\in U'}$, which lies in $\mathscr{F}^{\text{sh}}(U')$, a witness (V, s) at $p \in U'$ giving the witness $(V \cap U', s|_{V\cap U'})$. These maps satisfy the identity and composition axioms because they shorten tuples. In the algebraic cases $\mathscr{F}^{\text{sh}}(U)$ is closed under the componentwise operations of $\prod_{p\in U} \mathscr{F}_p$: if (V, s) and (V', s') are witnesses for (f_p) and (g_p) at p, then $(V \cap V', s|_{V\cap V'} + s'|_{V\cap V'})$ and $(V \cap V', s|_{V\cap V'} \cdot s'|_{V\cap V'})$ are witnesses for $(f_p + g_p)$ and $(f_p g_p)$ at p, since germ maps are homomorphisms, (V, rs) is a witness for (rf_p) when r is a scalar, and $(U, 0)$ and $(U, 1)$ are witnesses for the constant tuples 0 and 1. So each $\mathscr{F}^{\text{sh}}(U)$ is a group, ring or module under componentwise operations, and the restriction maps are homomorphisms.

Step 2: $\mathscr{F}^{\mathrm{sh}}$ is a sheaf. The set $\mathscr{F}^{\mathrm{sh}}(\emptyset)$ has exactly one element, the empty tuple, so it is terminal. Locality: two tuples in $\mathscr{F}^{\mathrm{sh}}(U)$ that agree on every member of an open cover of U agree at every point of U. Gluing: let $U = \bigcup_i U_i$ and $f^{(i)} \in \mathscr{F}^{\mathrm{sh}}(U_i)$ with $f^{(i)}|_{U_i \cap U_j} = f^{(j)}|_{U_i \cap U_j}$. For $p \in U_i \cap U_j$ this gives $f^{(i)}_p = f^{(j)}_p$, so $f_p = f^{(i)}_p$ for $p \in U_i$ defines a tuple on U. A witness for $f^{(i)}$ at $p \in U_i$ is a witness for f at p, so $f \in \mathscr{F}^{\mathrm{sh}}(U)$, and $f|_{U_i} = f^{(i)}$.

Step 3: the morphism sh. For $\sigma \in \mathscr{F}(U)$ put $\mathrm{sh}(\sigma) = (\sigma_p)_{p \in U}$, which lies in $\mathscr{F}^{\mathrm{sh}}(U)$ with witness (U, σ) at every point. Since $(\sigma|_{U'})_p = \sigma_p$ for $p \in U' \subseteq U$, the maps $\mathrm{sh}(U)$ commute with restriction, and they are homomorphisms because germ maps are.

Step 4: construction of f. Let $\mathscr{G}$ be a sheaf and $g \colon \mathscr{F} \to \mathscr{G}$ a presheaf morphism. Let $s \in \mathscr{F}^{\mathrm{sh}}(U)$, and choose an open cover $U = \bigcup_{i \in I} V_i$ with $\sigma_i \in \mathscr{F}(V_i)$ such that $(\sigma_i)_q = s_q$ for all $q \in V_i$, as the definition allows. For $q \in V_i \cap V_j$ we have $(\sigma_i)_q = s_q = (\sigma_j)_q$, so there is an open W_q with $q \in W_q \subseteq V_i \cap V_j$ and $\sigma_i|_{W_q} = \sigma_j|_{W_q}$, whence $g(\sigma_i)|_{W_q} = g(\sigma_i|_{W_q}) = g(\sigma_j|_{W_q}) = g(\sigma_j)|_{W_q}$. The sets W_q cover $V_i \cap V_j$, so $g(\sigma_i)$ and $g(\sigma_j)$ agree on $V_i \cap V_j$ by the locality axiom for $\mathscr{G}$, and the gluing axiom gives a unique $T \in \mathscr{G}(U)$ with $T|_{V_i} = g(\sigma_i)$ for all i. If (V'_k, σ'_k) is another such choice with glued section T', the same argument shows that $g(\sigma_i)$ and $g(\sigma'_k)$ agree on $V_i \cap V'_k$, so T and T' agree on every member of the cover $\{V_i \cap V'_k\}$ of U, and $T = T'$ by locality. Put $f(U)(s) = T$.

Step 5: f is a sheaf morphism with $f \circ \mathrm{sh} = g$. For $U' \subseteq U$, the sets $V_i \cap U'$ with the sections $\sigma_i|_{V_i \cap U'}$ form a valid choice for $s|_{U'}$, and $T|_{U'}$ restricts to $g(\sigma_i)|_{V_i \cap U'} = g(\sigma_i|_{V_i \cap U'})$ on each of them, so $f(U')(s|_{U'}) = T|_{U'}$. In the algebraic cases, given $s, s' \in \mathscr{F}^{\mathrm{sh}}(U)$ with choices (V_i, σ_i) and (V'_k, σ'_k), the sections $\sigma_i|_{V_i \cap V'_k} + \sigma'_k|_{V_i \cap V'_k}$ form a choice for $s + s'$ by Step 1, so $f(U)(s + s')$ and $f(U)(s) + f(U)(s')$ agree on every $V_i \cap V'_k$ and are equal by locality. Replacing the sum by the product $\sigma_i|_{V_i \cap V'_k} \cdot \sigma'_k|_{V_i \cap V'_k}$, by a scalar multiple $r\sigma_i$, or by the section $1 \in \mathscr{F}(U)$ witnessing the tuple 1, the same comparison on each $V_i \cap V'_k$ shows that $f(U)$ preserves products, scalars and 1, since g does. For $\sigma \in \mathscr{F}(U)$ the cover $\{U\}$ with σ itself is a choice for $\mathrm{sh}(\sigma)$, so $f(U)(\mathrm{sh}(\sigma)) = g(\sigma)$.

Step 6: uniqueness. Let $f' \colon \mathscr{F}^{\mathrm{sh}} \to \mathscr{G}$ be a sheaf morphism with $f' \circ \mathrm{sh} = g$, and let s and (V_i, σ_i) be as in Step 4. Then $s|_{V_i} = \mathrm{sh}(\sigma_i)$, both being the tuple $(s_q)_{q \in V_i}$, so $f'(U)(s)|_{V_i} = f'(V_i)(\mathrm{sh}(\sigma_i)) = g(\sigma_i)$ for every i, and $f'(U)(s) = T = f(U)(s)$ by the locality axiom for $\mathscr{G}$, completing the proof.

The "failure" of a presheaf being a sheaf can be measured as the lack of exactness of at the $\prod_i F(U_i)$ in proposition 1.1.9.

Example 1.2.9: Sheafification
To gain intuition, show the following sheafifications:

1. Show that $(\underline{S}_{\mathrm{pre}})^{\mathrm{sh}} = \underline{S}$, that is the presheaf of constant function sheafifies to the sheaf of constant functions.
2. Let $\mathscr{F}$ on $\mathbb{R}$ be all bounded continuous functions on each open set. Show that $\mathscr{F}^{\mathrm{sh}}$ is *all* continuous functions on each open subset.

3. Show that $\mathcal{H}$ on $\mathbb{C}$ of non-vanishing holomorphic functions that admit a logarithm sheafify to all non-vanishing holomorphic functions.

4. Let $\mathscr{F}$ be a presheaf that is not the zero presheaf but all of whose stalks vanish. Show that $\mathscr{F}^{\mathrm{sh}}$ is trivial, and give an example of such an $\mathscr{F}$.

Corollary 1.2.10: Sheafification and Stalks

Let $\mathcal{F} \in \mathbf{PSh}(X)$ be a presheaf. Then for all $p \in X$ the map sending the germ of σ at p to the germ of $\mathrm{sh}(\sigma)$ at p is an isomorphism

$$\mathcal{F}_p \cong \mathcal{F}_p^{\mathrm{sh}}$$

***Proof*:**
Here $\mathcal{F}_p^{\mathrm{sh}}$ denotes the stalk $(\mathcal{F}^{\mathrm{sh}})_p$, and $\mathcal{F}^{\mathrm{sh}}$ is the sheaf of compatible germs constructed in the proof of theorem 1.2.8. A section $s \in \mathcal{F}^{\mathrm{sh}}(U)$ is a tuple $(s_q)_{q\in U}$ of germs $s_q \in \mathcal{F}_q$, restriction shortens the tuple, and $\mathrm{sh}(U)$ sends $\sigma \in \mathcal{F}(U)$ to $(\sigma_q)_{q\in U}$.

Step 1: the map $\varepsilon\colon \mathcal{F}_p^{\mathrm{sh}} \to \mathcal{F}_p$. Send the class of (s, U) to the component s_p. If $(s, U) \sim (t, V)$, there is an open W with $p \in W \subseteq U \cap V$ and $s|_W = t|_W$, and restriction keeps components, so $s_p = (s|_W)_p = (t|_W)_p = t_p$. Hence ε is well defined.

Step 2: the map $\eta\colon \mathcal{F}_p \to \mathcal{F}_p^{\mathrm{sh}}$. Send the class of (σ, U) to the class of $(\mathrm{sh}(\sigma), U)$. If $(\sigma, U) \sim (\tau, V)$, there is an open $W \ni p$ in $U \cap V$ with $\sigma|_W = \tau|_W$, and since sh commutes with restriction, $\mathrm{sh}(\sigma)|_W = \mathrm{sh}(\sigma|_W) = \mathrm{sh}(\tau|_W) = \mathrm{sh}(\tau)|_W$. Hence η is well defined.

Step 3: ε *and* η *are mutually inverse.* For a germ $[(\sigma, U)] \in \mathcal{F}_p$, $\varepsilon(\eta[(\sigma, U)])$ is the p-component of $\mathrm{sh}(\sigma)$, which is $\sigma_p = [(\sigma, U)]$. Conversely let $s \in \mathcal{F}^{\mathrm{sh}}(U)$ with $p \in U$. By the definition of $\mathcal{F}^{\mathrm{sh}}(U)$ there are an open V with $p \in V \subseteq U$ and $\sigma \in \mathcal{F}(V)$ with $\sigma_q = s_q$ for every $q \in V$. Then $\varepsilon[(s, U)] = s_p = [(\sigma, V)]$, and $\eta[(\sigma, V)] = [(\mathrm{sh}(\sigma), V)]$ with $\mathrm{sh}(\sigma) = (\sigma_q)_{q\in V} = (s_q)_{q\in V} = s|_V$, so $\eta(\varepsilon[(s, U)]) = [(s|_V, V)] = [(s, U)]$.

When $\mathcal{F}$ is a presheaf of abelian groups, rings or modules, sections of $\mathcal{F}^{\mathrm{sh}}(U)$ are added and multiplied componentwise, each $\mathrm{sh}(U)$ is a homomorphism, and the operations on stalks are computed on representatives over a common smaller open set (Steps 1 and 3 and the opening paragraph of the proof of theorem 1.2.8). The maps ε and η are induced on representatives by the homomorphisms $s \mapsto s_p$ and $\mathrm{sh}(U)$, so they are mutually inverse homomorphisms and $\mathcal{F}_p \cong \mathcal{F}_p^{\mathrm{sh}}$, as claimed.

Let us now put together the universal properties we have shown so far to get that the category of sheaves also have cokernels:

Proposition 1.2.11: Sheaf Morphisms Have Cokernels

Let $\varphi\colon \mathcal{F} \to \mathcal{G}$ be a morphism of sheaves of R-modules, let $Q\colon \mathcal{G} \to \operatorname{coker}_{\mathrm{pre}}\varphi$ be the cokernel presheaf of proposition 1.2.5, and put $q = \mathrm{sh} \circ Q\colon \mathcal{G} \to (\operatorname{coker}_{\mathrm{pre}}\varphi)^{\mathrm{sh}}$ (theorem 1.2.8). Then q is a cokernel of φ in the category of sheaves of R-modules.

***Proof*:**
First, $q \circ \varphi = \mathrm{sh} \circ Q \circ \varphi = 0$. Let $\chi\colon \mathcal{G} \to \mathcal{A}$ be a morphism to a sheaf of R-modules with $\chi \circ \varphi = 0$. By proposition 1.2.5 there is a unique presheaf morphism $\chi'\colon \operatorname{coker}_{\mathrm{pre}}\varphi \to \mathcal{A}$ with $\chi' \circ Q = \chi$, and by theorem 1.2.8 a unique sheaf morphism $\chi''\colon (\operatorname{coker}_{\mathrm{pre}}\varphi)^{\mathrm{sh}} \to \mathcal{A}$ with $\chi'' \circ \mathrm{sh} = \chi'$. Then $\chi'' \circ q = \chi$. If β also satisfies $\beta \circ q = \chi$, then $(\beta \circ \mathrm{sh}) \circ Q = \chi' \circ Q$, and every $Q(U)$ is surjective, so $\beta \circ \mathrm{sh} = \chi'$ and $\beta = \chi''$ by the uniqueness in theorem 1.2.8, as claimed.

Corollary 1.2.12: Image and Sheafification

Let $\varphi\colon \mathcal{F} \to \mathcal{G}$ be a morphism of sheaves of R-modules and $q\colon \mathcal{G} \to \operatorname{coker}\varphi$ its cokernel (proposition 1.2.11). Then the kernel $\operatorname{im}\varphi = \ker q$, computed as in proposition 1.2.3, is the subsheaf of $\mathcal{G}$ with

$$(\operatorname{im}\varphi)(U) = \{t \in \mathcal{G}(U) \mid \text{each } p \in U \text{ has an open } W \ni p \text{ in } U \text{ with } t|_W \in \operatorname{im}(\varphi(W))\}$$

and the morphism $j^{\mathrm{sh}}\colon (\operatorname{im}_{\mathrm{pre}}\varphi)^{\mathrm{sh}} \to \mathcal{G}$ with $j^{\mathrm{sh}} \circ \mathrm{sh} = j$, for the inclusion $j\colon \operatorname{im}_{\mathrm{pre}}\varphi \to \mathcal{G}$ (theorem 1.2.8), is an isomorphism onto $\operatorname{im}\varphi$. If every $\varphi(V)$ is injective, then $(\operatorname{im}\varphi)(U) = \operatorname{im}(\varphi(U))$ for every open U.

***Proof*:**
A section of a sheaf of R-modules is zero if all of its germs are zero, by the locality axiom. So $t \in \mathcal{G}(U)$ lies in $\ker q(U)$ if and only if the germ of $\mathrm{sh}(Q(t))$ at every $p \in U$ is zero. By corollary 1.2.10 this germ is the image of the germ of $Q(t)$ under an isomorphism, so the condition is that every $p \in U$ has an open $W \ni p$ in U with $Q(t)|_W = Q(t|_W) = 0$, that is, with $t|_W \in \operatorname{im}(\varphi(W))$.

By the same corollary, for every section s of $(\operatorname{im}_{\mathrm{pre}}\varphi)^{\mathrm{sh}}$ over U each point of U has an open neighbourhood $W \subseteq U$ and a section σ of $\operatorname{im}_{\mathrm{pre}}\varphi$ over W with $s|_W = \mathrm{sh}(\sigma)$, and then $j^{\mathrm{sh}}(s)|_W = j^{\mathrm{sh}}(\mathrm{sh}(\sigma)) = j(\sigma) = \sigma$. So $j^{\mathrm{sh}}(s)$ lies in $\operatorname{im}\varphi(U)$, and if $j^{\mathrm{sh}}(s) = 0$, then each such σ is zero, $s|_W = \mathrm{sh}(0) = 0$, and $s = 0$ by locality. Conversely let $t \in \operatorname{im}\varphi(U)$, with opens $W_p \ni p$ such that $\sigma_p = t|_{W_p}$ lies in $\operatorname{im}_{\mathrm{pre}}\varphi(W_p)$. The sections $\mathrm{sh}(\sigma_p)$ agree on overlaps, since σ_p and $\sigma_{p'}$ have the same restriction to $W_p \cap W_{p'}$, so they glue to some s with $j^{\mathrm{sh}}(s)|_{W_p} = \sigma_p = t|_{W_p}$, and $j^{\mathrm{sh}}(s) = t$. Hence j^{sh} is injective with image $\operatorname{im}\varphi$.

Finally let every $\varphi(V)$ be injective and $t \in \operatorname{im}\varphi(U)$, with $t|_{W_p} = \varphi(a_p)$. On $W_p \cap W_{p'}$ the restrictions of a_p and $a_{p'}$ both map to the restriction of t, so they agree, and gluing in $\mathcal{F}$ gives $a \in \mathcal{F}(U)$ with $a|_{W_p} = a_p$. Then $\varphi(a)$ and t agree on every W_p, so $\varphi(a) = t$ by locality, completing the proof.

Definition 1.2.13: Surjective Sheaf Morphism

Let $\varphi : \mathscr{F} \to \mathscr{G}$ be a morphism of sheaves of R-modules. Its cokernel and image are those of proposition 1.2.11 and corollary 1.2.12. The morphism φ is *surjective* if $\mathrm{im}\varphi = \mathscr{G}$, that is, if every section of $\mathscr{G}$ is locally in the image of φ, equivalently if $j^{\mathrm{sh}} : (\mathrm{im}_{\mathrm{pre}}\varphi)^{\mathrm{sh}} \to \mathscr{G}$ is an isomorphism.

Using this, we see that the map $\mathcal{O}_{\mathbb{C}} \xrightarrow{\exp} \mathcal{O}_{\mathbb{C}}^*$ is *surjective* in the category of sheaves, even though it is not surjective in the category of presheaves. Hence, exactness in presheaves and sheaves is *different*. The following example using simpler categories should demonstrate why this is not so strange:

Example 1.2.14: Ab Vs TF-Ab

Let **Ab** be the category of abelian groups and TF-Ab be the category of torsion-free abelian groups. Consider the morphism $\varphi : \mathbb{Z} \to \mathbb{Z}$, $\varphi(n) = 2n$. This morphism exists in both categories. The cokernel of φ in **Ab** is $\mathbb{Z}/2\mathbb{Z}$, which is torsion. To find the cokernel in TF-AB, the "best" torsion-free group must be found so that its composition with φ is universal. Any homomorphism $\psi : \mathbb{Z} \to T$ into a torsion-free group with $\psi \circ \varphi = 0$ has $2\psi(1) = 0$ and hence $\psi = 0$, so the cokernel is 0:

$$\mathrm{coker}_{\mathbf{Ab}}(\varphi) = \mathbb{Z}/2\mathbb{Z} \neq 0 = \mathrm{coker}_{\text{TF-Ab}}(\varphi)$$

Thus, though the morphism in each category are the same, the epimorphism differ.

Thus, a theoretical understanding of epimorphisms in $\mathbf{C}_X$ is done. However, it is computationally intractable to always sheafify the image sheaf to check for surjectivity. As sheaves are created to be "local", we shall be able to shall be able to check their condition on *stalks*. Proving this shall fit in better into the broader framework on when the functor taking $\mathscr{F} \mapsto \mathscr{F}(U)$ and $\mathscr{F} \mapsto \mathscr{F}_p$ is exact.

Functors from (pre)sheaves: simplifying checking exactness

As presheaves and sheaves store all local information, we may want that there are some natural functors that preserve some exactness to extract that information. The two natural candidates for functors to would be $-(U)$ taking a (pre)sheaf to its sections over an open set U, and $(-)_p$ the functor taking a (pre)sheaf to it's stalk at p.

The assignment $-(U) : \mathbf{C}_X \to \mathbf{C}$ given by

$$\mathscr{F} \mapsto \mathscr{F}(U) \qquad (\varphi : \mathscr{F} \to \mathscr{G}) \mapsto (\varphi(U) : \mathscr{F}(U) \to \mathscr{G}(U))$$

preserves composition and identities, so it is a functor. Let us show $(-)_p$ is also a functor:

Proposition 1.2.15: Morphism Of (Pre)Sheaves Induces Morphism On Stalks

Let $\varphi : \mathscr{F} \to \mathscr{G}$ be a morphism of (pre)sheaves over any category $\mathbf{C}$ having colimits (so that stalks are well-defined). Then there is an induced morphism on stalks $\varphi_p : \mathscr{F}_p \to \mathscr{G}_p$, that is, it induced a functor $\mathbf{C}_X \to \mathbf{C}$.

***Proof*:**
This is definition pushing: For each open $U \subseteq X$ we have the morphism $\varphi(U) : \mathscr{F}(U) \to \mathscr{G}(U)$. Define the map:

$$[(f, U)]_p \mapsto [(\varphi(U)(f), U]_p$$

For this map to be well-defined if $(f, U) \sim (f', V)$ then we need:

$$[(f', V)]_p \mapsto [(\varphi(V)(f'), V]_p = [(\varphi(U)(f), U]_p$$

Hence we need that $((\varphi(V)(f'), V) \sim ((\varphi(U)(f), U)$. As $(f, U) \sim (f', V)$ we have there exists $W \subseteq U, V$ such that

$$\mathrm{res}_{U,W} f = \mathrm{res}_{V,W} f' = g$$

Then as φ is a pre-sheaf morphism we have:

$$\begin{array}{ccc} \varphi(U) : f & \longmapsto & \varphi(U)(f) \\ \downarrow{\scriptstyle \mathrm{res}_{U,W}} & & \downarrow{\scriptstyle \mathrm{res}_{U,W}} \\ \varphi(W) : \mathrm{res}_{U,W}(f) = g & \longmapsto & \varphi(W)(g) \end{array} \qquad \begin{array}{ccc} \varphi(V) : f' & \longmapsto & \varphi(U)(f') \\ \downarrow{\scriptstyle \mathrm{res}_{V,W}} & & \downarrow{\scriptstyle \mathrm{res}_{V,W}} \\ \varphi(W) : \mathrm{res}_{V,W}(f) = g & \longmapsto & \varphi(W)(g) \end{array}$$

which implies

$$\mathrm{res}_{U,W} \varphi(U)(f) = \mathrm{res}_{V,W} \varphi(V)(f')$$

But that means they are similar, $((\varphi(V)(f'), V) \sim ((\varphi(U)(f), U)$, as we sought to show

From here on the (pre)sheaves are presheaves and sheaves of R-modules for a ring R, with $R = \mathbb{Z}$ giving abelian groups.

Proposition 1.2.16: On Presheaf: Open Set Functor is Exact

Let R be a ring.

1. A sequence $\mathscr{F}' \xrightarrow{\varphi} \mathscr{F} \xrightarrow{\psi} \mathscr{F}''$ of presheaves of R-modules is exact at $\mathscr{F}$ if and only if $\ker(\psi(V)) = \mathrm{im}(\varphi(V))$ for every open $V \subseteq X$.
2. For $U \subseteq X$ open, the functor $\mathscr{F} \mapsto \mathscr{F}(U)$ from presheaves of R-modules to R-modules is exact (definition 0.4.32).
3. On sheaves of R-modules the functor $\mathscr{F} \mapsto \mathscr{F}(U)$ is left exact, and it carries monomorphisms of sheaves to injective maps.

***Proof*:**
Morphisms of presheaves of R-modules are added on each open set, $(\varphi + \psi)(V) = \varphi(V) + \psi(V)$, which commutes with restriction because the restriction maps are homomorphisms, and composition is bilinear. Together with the zero presheaf, the kernels and cokernels of proposition 1.2.3 and proposition 1.2.5, and for sheaves the cokernels of proposition 1.2.11, this is the structure that definition 0.4.32 uses.

1. By proposition 1.2.3 and proposition 1.2.5, $\ker\psi$ and $\mathrm{im}\varphi$ are the subpresheaves $V \mapsto \ker(\psi(V))$ and $V \mapsto \mathrm{im}(\varphi(V))$ of $\mathscr{F}$, with the inclusions. Two subpresheaves of $\mathscr{F}$ are

the same subobject exactly when they are equal, since an isomorphism between them commuting with the inclusions is the identity on every open set.

2. The functor sends $\varphi + \psi$ to $\varphi(U) + \psi(U)$, so it is additive. A short exact sequence $0 \to \mathscr{F}' \to \mathscr{F} \to \mathscr{F}'' \to 0$ is exact at its three terms, and by (1) with $V = U$ so is $0 \to \mathscr{F}'(U) \to \mathscr{F}(U) \to \mathscr{F}''(U) \to 0$.

3. Let φ be a monomorphism of sheaves and $\iota\colon \ker\varphi \to \mathscr{F}'$ the kernel inclusion of proposition 1.2.3. Then $\varphi \circ \iota = 0 = \varphi \circ 0$, so $\iota = 0$, and as ι is injective on every open set, $\ker(\varphi(U)) = 0$. Now let $0 \to \mathscr{F}' \xrightarrow{\varphi} \mathscr{F} \xrightarrow{\psi} \mathscr{F}''$ be exact in the category of sheaves. Exactness at $\mathscr{F}'$ says that φ has zero kernel, so every $\varphi(V)$ is injective, as kernels of sheaf morphisms are the presheaf kernels. Then corollary 1.2.12 gives $(\mathrm{im}\varphi)(U) = \mathrm{im}(\varphi(U))$, and exactness at $\mathscr{F}$ gives $\ker(\psi(U)) = \mathrm{im}(\varphi(U))$. So $0 \to \mathscr{F}'(U) \to \mathscr{F}(U) \to \mathscr{F}''(U)$ is exact, completing the proof.

The functor $\mathscr{F} \mapsto \mathscr{F}(U)$ is not right exact on sheaves: the exponential sequence of example 1.2.7(2) is exact as a sequence of sheaves, while exp is not surjective on $\mathbb{C}^*$.

1.2.17 Remark For sheaves of sets, which have no kernels, Vakil [71] uses "indicator sheaves" to see that monomorphisms are injective on sections. In categorical terms, to test the value of a section $s \in \mathcal{F}(U)$, one uses the Yoneda Lemma. The indicator sheaf mentioned corresponds to the representable sheaf h_U, with one section over each open subset of U and none over other open sets. Mapping out of h_U allows one to probe the algebraic properties of $\mathcal{F}(U)$ element by element.

Proposition 1.2.18: Exactness of Presheaves On The Level of Open Sets

A sequence of presheaves of R-modules

$$0 \to \mathscr{F}_1 \to \mathscr{F}_2 \to \cdots \to \mathscr{F}_n \to 0$$

is exact if and only if the sequence

$$0 \to \mathscr{F}_1(U) \to \mathscr{F}_2(U) \to \cdots \to \mathscr{F}_n(U) \to 0$$

is exact for all U.

***Proof*:**
Exactness at each term, including the conditions at $\mathscr{F}_1$ and $\mathscr{F}_n$ coming from the zero presheaves at the ends, is by proposition 1.2.16(1) the equality of the kernel of the outgoing map and the image of the incoming map on every open set, which is exactness of every sequence of sections, as claimed.

Corollary 1.2.19: Exactness of Stalk For Presheaves

Let $\varphi\colon \mathscr{F} \to \mathscr{G}$ be a morphism of presheaves of R-modules and $p \in X$. The stalk maps of the inclusions $\ker_{\text{pre}}\varphi \to \mathscr{F}$ and $\text{im}_{\text{pre}}\varphi \to \mathscr{G}$ are injective with images $\ker(\varphi_p)$ and $\text{im}(\varphi_p)$, and the stalk map of the quotient $Q\colon \mathscr{G} \to \text{coker}_{\text{pre}}\varphi$ induces an isomorphism $\text{coker}(\varphi_p) \cong (\text{coker}_{\text{pre}}\varphi)_p$. Consequently, if $\mathscr{F}' \xrightarrow{\varphi} \mathscr{F} \xrightarrow{\psi} \mathscr{F}''$ is exact at $\mathscr{F}$ in the category of presheaves of R-modules, then $\mathscr{F}'_p \to \mathscr{F}_p \to \mathscr{F}''_p$ is exact.

***Proof*:**
Write s_p for the germ of a section s at p, so that $\varphi_p(s_p) = (\varphi(s))_p$ (proposition 1.2.15).

Kernel. A germ of $\ker_{\text{pre}}\varphi$ at p is represented by $s \in \mathscr{F}(U)$ with $\varphi(s) = 0$, and it maps to $s_p \in \ker(\varphi_p)$. If $s_p = 0$ in $\mathscr{F}_p$, then $s|_W = 0$ for an open $W \ni p$, so the germ is zero in $(\ker_{\text{pre}}\varphi)_p$. If $\varphi_p(s_p) = 0$ for $s \in \mathscr{F}(U)$, then $\varphi(s)|_W = 0$ for an open $W \ni p$ in U, so $s|_W \in \ker(\varphi(W))$ and s_p is the image of its germ.

Image. A germ of $\text{im}_{\text{pre}}\varphi$ at p is represented by $\varphi(s)$ for some $s \in \mathscr{F}(U)$, and it maps to $(\varphi(s))_p = \varphi_p(s_p)$. Injectivity is proved as for the kernel, and every element $\varphi_p(s_p)$ of $\text{im}(\varphi_p)$ is the image of the germ of $\varphi(s)$.

Cokernel. The stalk map $Q_p\colon \mathscr{G}_p \to (\text{coker}_{\text{pre}}\varphi)_p$ is surjective, since every germ of $\text{coker}_{\text{pre}}\varphi$ is $(Q(t))_p = Q_p(t_p)$, and it vanishes on $\text{im}(\varphi_p)$, as $Q \circ \varphi = 0$. If $Q_p(t_p) = 0$, then $Q(t|_W) = 0$ for an open $W \ni p$, so $t|_W \in \text{im}(\varphi(W))$ and $t_p \in \text{im}(\varphi_p)$. So Q_p induces an isomorphism $\text{coker}(\varphi_p) \cong (\text{coker}_{\text{pre}}\varphi)_p$.

Exactness. By proposition 1.2.16(1), exactness at $\mathscr{F}$ says that $\ker_{\text{pre}}\psi$ and $\text{im}_{\text{pre}}\varphi$ are the same subpresheaf of $\mathscr{F}$, so their stalks, viewed inside $\mathscr{F}_p$, agree, and these are $\ker(\psi_p)$ and $\text{im}(\varphi_p)$, completing the proof.

Note that for presheaves, exactness on all stalks does not imply exactness on presheaves as example 1.2.7(2) shows.

Theorem 1.2.20: Presheaves Over Ab Category Form Ab Category

The category $\mathbf{PSh}_{\mathbf{C}}(X)$ over an abelian category C is an abelian category

***Proof*:**
By definition 0.4.29, the hom-sets of $\mathbf{C}$ are abelian groups, composition is bilinear, and finite biproducts exist. Morphisms in $\mathbf{PSh}_{\mathbf{C}}(X)$ are natural transformations (definition 1.2.1).

Step 1: additive structure. For $\varphi, \psi\colon \mathcal{F} \to \mathcal{G}$ put $(\varphi + \psi)(U) = \varphi(U) + \psi(U)$. For $V \subseteq U$, bilinearity gives $\text{res}_{U,V} \circ (\varphi(U) + \psi(U)) = \varphi(V) \circ \text{res}_{U,V} + \psi(V) \circ \text{res}_{U,V} = (\varphi(V) + \psi(V)) \circ \text{res}_{U,V}$, so $\varphi + \psi$ is a morphism. With the zero morphism and $-\varphi$ this makes each hom-set an abelian group, and composition is bilinear because it is computed componentwise. The presheaf $U \mapsto 0$

is a zero object. For presheaves $\mathcal{F}, \mathcal{G}$ let $(\mathcal{F} \oplus \mathcal{G})(U) = \mathcal{F}(U) \oplus \mathcal{G}(U)$, with restriction the map $\mathrm{res}_{U,V} \oplus \mathrm{res}_{U,V}$ induced by the biproduct structure. The inclusions i_1, i_2 and projections p_1, p_2 of these biproducts commute with restriction, and $p_1 i_1 = \mathrm{id}$, $p_2 i_2 = \mathrm{id}$ and $i_1 p_1 + i_2 p_2 = \mathrm{id}$ hold componentwise. Given $\alpha\colon \mathcal{H} \to \mathcal{F}$ and $\beta\colon \mathcal{H} \to \mathcal{G}$, the morphism $i_1\alpha + i_2\beta$ has components α and β, and any γ with $p_1\gamma = \alpha$ and $p_2\gamma = \beta$ equals $(i_1p_1 + i_2p_2)\gamma = i_1\alpha + i_2\beta$. So $\mathcal{F} \oplus \mathcal{G}$ is a product. Given $\alpha\colon \mathcal{F} \to \mathcal{H}$ and $\beta\colon \mathcal{G} \to \mathcal{H}$, the morphism $\alpha p_1 + \beta p_2$ restricts to α and β along i_1 and i_2, and any γ with $\gamma i_1 = \alpha$ and $\gamma i_2 = \beta$ equals $\gamma(i_1p_1 + i_2p_2) = \alpha p_1 + \beta p_2$, so $\mathcal{F} \oplus \mathcal{G}$ is also a coproduct.

Step 2: kernels. Let $\varphi\colon \mathcal{F} \to \mathcal{G}$. For each U choose a kernel $k_U\colon \mathcal{K}(U) \to \mathcal{F}(U)$ of $\varphi(U)$ in $\mathbf{C}$. For $V \subseteq U$, $\varphi(V) \circ \mathrm{res}_{U,V} \circ k_U = \mathrm{res}_{U,V} \circ \varphi(U) \circ k_U = 0$, so $\mathrm{res}_{U,V} \circ k_U = k_V \circ \rho_{U,V}$ for a unique $\rho_{U,V}\colon \mathcal{K}(U) \to \mathcal{K}(V)$. Since each k_W is a monomorphism, $\rho_{U,U} = \mathrm{id}$ and $\rho_{V,W} \circ \rho_{U,V} = \rho_{U,W}$, both sides composed with k_W being $\mathrm{res}_{U,W} \circ k_U$. So $\mathcal{K}$ is a presheaf and $k\colon \mathcal{K} \to \mathcal{F}$ a morphism with $\varphi \circ k = 0$. If $\chi\colon \mathcal{H} \to \mathcal{F}$ satisfies $\varphi \circ \chi = 0$, each $\chi(U)$ factors uniquely as $k_U \circ \chi'(U)$, and $k_V \circ \chi'(V) \circ \mathrm{res}_{U,V} = \mathrm{res}_{U,V} \circ \chi(U) = k_V \circ \rho_{U,V} \circ \chi'(U)$ shows that χ' commutes with restriction, k_V being a monomorphism. Hence k is a kernel of φ. For presheaves of R-modules it is the kernel presheaf of proposition 1.2.3.

Step 3: cokernels. Choose cokernels $q_U\colon \mathcal{G}(U) \to \mathcal{Q}(U)$ of the $\varphi(U)$. For $V \subseteq U$, $q_V \circ \mathrm{res}_{U,V} \circ \varphi(U) = q_V \circ \varphi(V) \circ \mathrm{res}_{U,V} = 0$, so $q_V \circ \mathrm{res}_{U,V} = \sigma_{U,V} \circ q_U$ for a unique $\sigma_{U,V}\colon \mathcal{Q}(U) \to \mathcal{Q}(V)$. Since each q_U is an epimorphism, $\sigma_{U,U} = \mathrm{id}$ and $\sigma_{V,W} \circ \sigma_{U,V} = \sigma_{U,W}$, both sides composed with q_U being $q_W \circ \mathrm{res}_{U,W}$. If $\chi\colon \mathcal{G} \to \mathcal{H}$ satisfies $\chi \circ \varphi = 0$, each $\chi(U)$ factors uniquely as $\chi'(U) \circ q_U$, and $\chi'(V) \circ \sigma_{U,V} \circ q_U = \chi'(V) \circ q_V \circ \mathrm{res}_{U,V} = \mathrm{res}_{U,V} \circ \chi(U) = \mathrm{res}_{U,V} \circ \chi'(U) \circ q_U$ gives $\chi'(V) \circ \sigma_{U,V} = \mathrm{res}_{U,V} \circ \chi'(U)$, q_U being an epimorphism. Hence $q\colon \mathcal{G} \to \mathcal{Q}$ is a cokernel of φ. For presheaves of R-modules it is the cokernel presheaf of proposition 1.2.5.

Step 4: every monomorphism is a kernel. Let $\varphi\colon \mathcal{F} \to \mathcal{G}$ be a monomorphism and k its kernel from Step 2. Then $\varphi \circ k = 0 = \varphi \circ 0$ gives $k = 0$, so $k_U \circ \mathrm{id}_{\mathcal{K}(U)} = 0 = k_U \circ 0$ and, k_U being a monomorphism, $\mathcal{K}(U) = 0$. So $\varphi(U)$ has zero kernel and is a monomorphism of $\mathbf{C}$: if $\varphi(U) \circ a = \varphi(U) \circ b$, then $a - b$ factors through $\mathcal{K}(U) = 0$. In the abelian category $\mathbf{C}$, the monomorphism $\varphi(U)$ is the kernel of some morphism [18, Abelian Category], and every kernel is the kernel of its cokernel [18, Abelian Category: Kernels And Cokernels], so $\varphi(U)$ is a kernel of q_U, where $q\colon \mathcal{G} \to \mathcal{Q}$ is the cokernel of φ from Step 3. Hence $q \circ \varphi = 0$, and a morphism $\chi\colon \mathcal{H} \to \mathcal{G}$ with $q \circ \chi = 0$ gives unique maps $\chi'(U)$ with $\chi(U) = \varphi(U) \circ \chi'(U)$. Then $\varphi(V) \circ \chi'(V) \circ \mathrm{res}_{U,V} = \chi(V) \circ \mathrm{res}_{U,V} = \mathrm{res}_{U,V} \circ \chi(U) = \mathrm{res}_{U,V} \circ \varphi(U) \circ \chi'(U) = \varphi(V) \circ \mathrm{res}_{U,V} \circ \chi'(U)$, so χ' commutes with restriction, $\varphi(V)$ being a monomorphism. So φ is a kernel of q.

Step 5: every epimorphism is a cokernel. Let $\psi\colon \mathcal{G} \to \mathcal{H}$ be an epimorphism and q its cokernel from Step 3. Then $q \circ \psi = 0 = 0 \circ \psi$ gives $q = 0$, and q_U being an epimorphism forces its target to be 0, so $\psi(U)$ has zero cokernel and is an epimorphism of $\mathbf{C}$. So $\psi(U)$ is the cokernel of some morphism [18, Abelian Category], and every cokernel is the cokernel of its kernel [18, Abelian Category: Kernels And Cokernels], so $\psi(U)$ is a cokernel of k_U, where k is the kernel of ψ from Step 2. A morphism $\chi\colon \mathcal{G} \to \mathcal{E}$ with $\chi \circ k = 0$ gives unique maps $\chi'(U)$ with $\chi(U) = \chi'(U) \circ \psi(U)$, and $\chi'(V) \circ \mathrm{res}_{U,V} \circ \psi(U) = \chi'(V) \circ \psi(V) \circ \mathrm{res}_{U,V} = \chi(V) \circ \mathrm{res}_{U,V} = \mathrm{res}_{U,V} \circ \chi(U) = \mathrm{res}_{U,V} \circ \chi'(U) \circ \psi(U)$ shows that χ' commutes with restriction, $\psi(U)$ being an epimorphism. So ψ is a cokernel of k, and all five axioms of definition 0.4.29 hold, completing the proof.

To keep track of all morphisms between open sets, we box the following definition:

Definition 1.2.21: Sheaf Hom

Let $\mathscr{F},\mathscr{G} \in \mathbf{Sh}_{\mathbf{C}}(X)$. Then the *sheaf hom* is defined to be:

$$Hom(\mathscr{F},\mathscr{G})(U) := \mathrm{Mor}(\mathscr{F}_U, \mathscr{G}|_U)$$

where $\mathrm{Mor}(\mathscr{F}|_U, \mathscr{G}|_U)$ are all morphism of (pre)sheaves. The dual of $\mathscr{F}$ is:

$$\mathscr{F}^\vee = Hom_{\mathcal{O}_X}(\mathscr{F}, \mathcal{O}_X)$$

Just like Hom, *Hom* is a (co/contra)variant functor, depending on which position you choose.

The situation is not so clean for $(-)_p$, namely the functor on $\mathbf{Sh}_{\mathbf{C}}(X)$ is not always exact for *any* abelian category; it need not be the case that filtered colimits preserve exactness, and hence that $(-)_p$ preserve exactness (think of the category of finite abelian groups which does not have $\varinjlim_p \mathbb{Z}/p^n\mathbb{Z}$). Grothendieck axiomatically added this condition to abelian categories (such categories are called *Grothendieck categories*). Fortunately for us, this will never be an issue as all the categories of interest to us shall have this property, namely:

1. abelian groups, more generally R-Mod
2. Sheaves of abelian groups (more generally $\mathcal{O}_X$-modules which are defined later)
3. the product of Grothendieck categories
4. In section 1.3.3, it shall be important to have internal Hom's: given a small category $\mathbf{C}$ and a Grothendieck category $\mathbf{G}$, all functors from $\mathbf{C}$ to $\mathbf{G}$ (with morphism being natural transformation) form Grothendieck categories.

Proposition 1.2.22: Sheaves and exactness For Stalk

Let R be a ring and $p \in X$.

1. For a morphism $\varphi\colon \mathscr{F}' \to \mathscr{F}$ of sheaves of R-modules, the stalk map of the inclusion $\mathrm{im}\varphi \to \mathscr{F}$ (corollary 1.2.12) is injective with image $\mathrm{im}(\varphi_p)$.
2. Let $\mathscr{F}' \xrightarrow{\varphi} \mathscr{F} \xrightarrow{\psi} \mathscr{F}''$ be morphisms of sheaves of R-modules with $\psi\circ\varphi = 0$. The sequence is exact at $\mathscr{F}$ in the category of sheaves of R-modules if and only if $\mathscr{F}'_q \to \mathscr{F}_q \to \mathscr{F}''_q$ is exact for every $q \in X$.
3. The stalk functor $(-)_p$ from sheaves of R-modules to R-modules is exact.

***Proof*:**
A germ at p of a section of a subsheaf A of $\mathscr{F}$ that vanishes in $\mathscr{F}_p$ vanishes on an open neighbourhood of p, so the stalk map $A_p \to \mathscr{F}_p$ is injective, and we view A_p inside $\mathscr{F}_p$.

1. A section t of $\mathrm{im}\varphi$ agrees near p with $\varphi(s)$ for some section s of $\mathscr{F}'$, so $t_p = (\varphi(s))_p = \varphi_p(s_p)$. Conversely $\varphi_p(s_p)$ is the germ of $\varphi(s)$, a section of $\mathrm{im}\varphi$.

2. Two subsheaves A and B of $\mathscr{F}$ with $A_q = B_q$ for every q are equal. For $s \in A(U)$, each $q \in U$ has an open $W_q \ni q$ in U and a section of B over W_q with the same germ at q as s, hence equal to s on a smaller open $W'_q \ni q$. So $s|_{W'_q} \in B(W'_q)$ for all q, these sections agree on overlaps, and gluing in B gives a section of $B(U)$ equal to s by locality of $\mathscr{F}$. By symmetry $A = B$. Exactness at $\mathscr{F}$ says that the subsheaves $\ker\psi$ (proposition 1.2.3) and $\mathrm{im}\varphi$ of $\mathscr{F}$ are equal, and their stalks are $\ker(\psi_q)$ (corollary 1.2.19) and $\mathrm{im}(\varphi_q)$ by (1). So exactness at $\mathscr{F}$ is equivalent to $\ker(\psi_q) = \mathrm{im}(\varphi_q)$ for every q.

3. The stalk functor is additive, as germs of sums are sums of germs. A short exact sequence of sheaves is exact at each of its three terms, so by (2) its stalks at p form a short exact sequence, completing the proof.

With the right notion of compatible stalks defined, we may show how to determine morphisms of sheaves at the level of stalks, namely a scheme morphisms is determined by stalks, and a scheme morphism that is an isomorphism on stalks induces an inverse scheme morphism:

Proposition 1.2.23: Morphisms Determined By Stalks

Let φ_1, φ_2 be morphisms from a presheaf of sets $\mathscr{F}$ to a *sheaf* of sets $\mathscr{G}$ that induce the same maps on each stalk. Then $\varphi_1 = \varphi_2$

***Proof*:**
Let U be open and $s \in \mathscr{F}(U)$. Since $\mathscr{G}$ is a sheaf, the section $t = \varphi_1(U)(s)$ is uniquely determined by its germs $(t_p)_{p\in U}$.

For any $p \in U$, the germ of the image is the image of the germ:

$$(\varphi_1(s))_p = (\varphi_1)_p(s_p)$$

By assumption, $(\varphi_1)_p = (\varphi_2)_p$, so:

$$(\varphi_1)_p(s_p) = (\varphi_2)_p(s_p) = (\varphi_2(s))_p$$

Since $\varphi_1(s)$ and $\varphi_2(s)$ have identical germs at every point in U, and $\mathscr{G}$ is a sheaf (satisfying the identity axiom), the sections must be identical. Thus $\varphi_1(U)(s) = \varphi_2(U)(s)$.

The above proof can be summarized by the following diagram:

$$\begin{array}{ccc} \mathscr{F}(U) & \longrightarrow & \mathscr{G}(U) \\ \downarrow & & \downarrow \\ \prod_{p\in U} \mathscr{F}_p & \longrightarrow & \prod_{p\in U} \mathscr{G}_p \end{array}$$

Proposition 1.2.24: Stalks-Local Isomorphism induces Isomorphism

Let $\varphi\colon \mathscr{F} \to \mathscr{G}$ be a morphism of sheaves of sets, abelian groups, rings or modules. Then φ is an isomorphism if and only if φ_p is an isomorphism for every $p \in X$.

***Proof*:**
If φ has an inverse ψ, then ψ_p is inverse to φ_p. Conversely, let every φ_p be bijective and fix an open U. For injectivity of $\varphi(U)$, let $s, s' \in \mathscr{F}(U)$ with $\varphi(s) = \varphi(s')$. Then $\varphi_p(s_p) = \varphi_p(s'_p)$, so $s_p = s'_p$ for every $p \in U$, and $s = s'$ by lemma 1.1.15.

For surjectivity, let $t \in \mathscr{G}(U)$. For $p \in U$ choose a germ with image t_p under φ_p, represented by $s(p) \in \mathscr{F}(V_p)$ with $p \in V_p \subseteq U$. Then $\varphi(s(p))$ and $t|_{V_p}$ have the same germ at p, so after shrinking V_p we have $\varphi(s(p)) = t|_{V_p}$. On $V_p \cap V_q$ the sections $s(p)$ and $s(q)$ both map to the restriction of t, so they are equal by injectivity of $\varphi(V_p \cap V_q)$. The gluing axiom gives $s \in \mathscr{F}(U)$ with $s|_{V_p} = s(p)$ for all p, and $\varphi(s)$ and t agree on every V_p, so $\varphi(s) = t$ by the locality axiom.

So every $\varphi(U)$ is bijective, and a bijective homomorphism is an isomorphism. The inverses commute with restriction because the $\varphi(U)$ do, so they form an inverse morphism and φ is an isomorphism, completing the proof.

This shows that a *morphism* $\varphi : \mathscr{F} \to \mathscr{G}$ can be determined on the level of stalks. This does not imply that if two *sheaves* have isomorphic stalks, then they are isomorphic! You need to already have a sheaf homomorphism which in turn induces the isomorphic stalks. Conceptually, this is saying that being locally isomorphic does not imply it is globally isomorphic; there are plenty concrete example form differential geometry to choose from.

Example 1.2.25: Isomorphic Stalks, Not Isomorphic Sheaves

1. Let X be a Hausdorff space. Consider the constant sheaf $\underline{\mathbb{Z}}$ and the sky-scrapper sheaf $\iota_*(\mathbb{Z})$. Show that the stalks at each point is $\mathbb{Z}$, however these two sheaves are not isomorphic.
2. The following will rely on the reader's intuitions of varieties (see [22]). Take the variety associated to the coordinate ring:

   $$A = \mathbb{R}[x, y, z]/(x^2 + y^2 + z^2 - 1)$$

   which is the 2-sphere in $\mathbb{R}^3$. Take the tangent bundle, i.e. A-module, of all derivations $E = \mathrm{Der}_{\mathbb{R}}(A)$. Then E is a rank 2 locally trivial module since for any open subset or distinguished open subset, $E_x \cong \mathbb{R}^2$, showing each stalk is isomorphic. However, $E \ncong A^2$.
3. Another famous example would be the trivial line bundle on S^1 vs. the möbius bundle.

As stalks define maps on some open neighbourhood, we can use this condition to find the closest condition that allows to check if a map between topological spaces is sheaf map on the level of open sets:

Proposition 1.2.26: Characterizing Stalks Inducing Sheaf Morphism

Let X be a topological space with two sheaves $\mathscr{F},\mathscr{G}$. Let $\varphi_p : \mathscr{F}_p \to \mathscr{G}_p$ be a collection of stalk morphisms for all $p \in X$. Then this gives a map of sheaves $\Phi : \mathscr{F} \to \mathscr{G}$ if and only if for all open subsets $U \subseteq X$, for all sections $s \in \mathscr{F}(U)$, there is an open cover $U = \cup_i U_i$ and sections $t_i \in \mathscr{G}(U_i)$ such that

$$\varphi_p(s_p) = (t_i)_p$$

for all $p \in U_i$ and all i.

***Proof*:**
($\Rightarrow$) Assume $\Phi : \mathscr{F} \to \mathscr{G}$ is a morphism of sheaves. Then for any open U and section $s \in \mathscr{F}(U)$, we can simply take the trivial cover $U_1 = U$. Define $t_1 = \Phi(U)(s) \in \mathscr{G}(U)$. By the definition of the induced map on stalks (proposition 1.2.15), for any $p \in U$, the map φ_p sends the germ of s to the germ of $\Phi(U)(s)$. Thus, $\varphi_p(s_p) = (t_1)_p$, satisfying the condition.

($\Leftarrow$) Conversely, assume the local condition holds. We must construct a morphism $\Phi(U) : \mathscr{F}(U) \to \mathscr{G}(U)$ for every open set U. Let $s \in \mathscr{F}(U)$. By hypothesis, there exists a cover $\{U_i\}$ and sections $t_i \in \mathscr{G}(U_i)$ such that for every $p \in U_i$, the germ of t_i at p is $\varphi_p(s_p)$.

We first show that these local sections t_i glue together. Consider the intersection $U_{ij} = U_i \cap U_j$. For any point $p \in U_{ij}$, we have:

$$(t_i)_p = \varphi_p(s_p) = (t_j)_p$$

Since $\mathscr{G}$ is a sheaf (specifically, a separated presheaf), sections are determined by their germs (lemma 1.1.15). Because t_i and t_j have identical germs at every point in U_{ij}, they must be equal as sections:

$$\mathrm{res}_{U_i,U_{ij}}(t_i) = \mathrm{res}_{U_j,U_{ij}}(t_j)$$

Since the sections agree on overlaps, the Gluing Axiom for $\mathscr{G}$ implies there exists a unique section $t \in \mathscr{G}(U)$ such that $t|_{U_i} = t_i$. We define $\Phi(U)(s) = t$.

To ensure Φ is well-defined, suppose there was another cover $\{V_j\}$ and sections $\{r_j\}$ satisfying the condition, yielding a glued section t'. For any $p \in U$, $t_p = \varphi_p(s_p)$ (by the construction on the cover U_i) and $t'_p = \varphi_p(s_p)$ (by the construction on V_j). Thus t and t' have the same germs everywhere, and since $\mathscr{G}$ is a sheaf, $t = t'$.

Finally, Φ commutes with restrictions naturally because the construction is defined pointwise via germs: for $V \subseteq U$, the germs of $\Phi(V)(s|_V)$ are $\varphi_p((s|_V)_p) = \varphi_p(s_p)$, which matches the germs of $(\Phi(U)(s))|_V$.

Summarizing the results for morphisms of sheaves:

Theorem 1.2.27: Equivalence of Mono- and Epimorphisms

Let $\varphi : \mathcal{F} \to \mathcal{G}$ be a morphism of sheaves of R-modules on X.

The following conditions are equivalent:

1. φ is a monomorphism in the category of sheaves of R-modules.
2. φ is injective on the level of stalks: $\varphi_p : \mathcal{F}_p \to \mathcal{G}_p$ is injective for all $p \in X$.
3. φ is injective on the level of open sets: $\varphi(U) : \mathcal{F}(U) \to \mathcal{G}(U)$ is injective for all open $U \subseteq X$.

The following conditions are equivalent:

1. φ is an epimorphism in the category of sheaves of R-modules.
2. φ is surjective on the level of stalks: $\varphi_p : \mathcal{F}_p \to \mathcal{G}_p$ is surjective for all $p \in X$.
3. φ is surjective in the sense of definition 1.2.13.

If $\varphi(U)$ is surjective for all open sets U, then φ is an epimorphism (surjectivity on sections implies surjectivity on stalks). The converse is *false*: an epimorphism of sheaves need not be surjective on global sections (as seen in the exponential sequence).

***Proof*:**
Monomorphisms.

(1) ⇒ (3) This is proposition 1.2.16(3).

(3) ⇒ (1) If $\varphi \circ a = \varphi \circ b$, then $\varphi(U) \circ a(U) = \varphi(U) \circ b(U)$, so $a(U) = b(U)$ for every U.

(3) ⇒ (2) If $\varphi_p(s_p) = 0$ for $s \in \mathcal{F}(U)$, then $\varphi(s)|_W = 0$ for an open $W \ni p$ in U, so $s|_W = 0$ by injectivity of $\varphi(W)$, and $s_p = 0$.

(2) ⇒ (3) If $\varphi(s) = 0$ for $s \in \mathcal{F}(U)$, then $\varphi_p(s_p) = 0$, so $s_p = 0$ for every $p \in U$, and $s = 0$ by lemma 1.1.15.

Epimorphisms. Let $q \colon \mathcal{G} \to \operatorname{coker}\varphi$ be the cokernel of proposition 1.2.11.

(1) ⇒ (3) The equality $q \circ \varphi = 0 = 0 \circ \varphi$ gives $q = 0$, so $\ker q = \mathcal{G}$, and corollary 1.2.12 identifies $\ker q$ with $\operatorname{im}\varphi$.

(3) ⇒ (2) A germ $t_p \in \mathcal{G}_p$ of $t \in \mathcal{G}(U)$ has an open $W \ni p$ and $s \in \mathcal{F}(W)$ with $t|_W = \varphi(s)$, so $t_p = \varphi_p(s_p)$.

(2) ⇒ (1) Let $a, b \colon \mathcal{G} \to \mathcal{H}$ satisfy $a \circ \varphi = b \circ \varphi$, and let $t \in \mathcal{G}(U)$. Each $p \in U$ has $t_p = \varphi_p(s_p)$ for a section s of $\mathcal{F}$, so t and $\varphi(s)$ agree on an open $W \ni p$, and $a(t)|_W = a(\varphi(s|_W)) = b(\varphi(s|_W)) = b(t)|_W$. By locality $a(t) = b(t)$, so $a = b$.

If every $\varphi(U)$ is surjective, then (3) holds. The exponential sequence of example 1.2.7(2) shows that the converse fails, completing the proof.

Theorem 1.2.28: Sheaves Over Ab Category Form Ab Category

For a ring R, the category $\mathbf{Sh}_{\mathrm{Mod}_R}(X)$ of sheaves of R-modules on X is an abelian category. Kernels are the presheaf kernels of proposition 1.2.3, cokernels are those of proposition 1.2.11, and for a morphism φ and $p \in X$

$$(\ker\varphi)_p = \ker(\varphi_p) \qquad (\mathrm{im}\varphi)_p = \mathrm{im}(\varphi_p) \qquad (\mathrm{coker}\varphi)_p \cong \mathrm{coker}(\varphi_p)$$

***Proof*:**
Additive structure. $\mathbf{Sh}_{\mathrm{Mod}_R}(X)$ is a full subcategory of the presheaf category (definition 1.2.1), so its hom-sets inherit the abelian group structure and bilinear composition of theorem 1.2.20. The zero presheaf is a sheaf, giving a zero object. The open-set-wise biproduct $(\mathscr{F}\oplus\mathscr{G})(U) = \mathscr{F}(U)\oplus\mathscr{G}(U)$ of two sheaves is a sheaf, both sheaf axioms being checked componentwise because restrictions act componentwise. It is a biproduct in the presheaf category, and it and its structure maps lie in $\mathbf{Sh}_{\mathrm{Mod}_R}(X)$, so it is a biproduct there as well.

Kernels, cokernels and stalks. The presheaf kernel of a morphism φ of sheaves is a sheaf and is a kernel in the category of sheaves (proposition 1.2.3), with stalk $\ker(\varphi_p)$ by corollary 1.2.19, and cokernels exist by proposition 1.2.11. The stalk of $\mathrm{im}\varphi$ is $\mathrm{im}(\varphi_p)$ by proposition 1.2.22(1). The cokernel is $(\mathrm{coker}_{\mathrm{pre}}\varphi)^{\mathrm{sh}}$ with the map $\mathrm{sh}\circ Q$, and by corollary 1.2.10 and corollary 1.2.19 its stalk map is surjective with kernel $\mathrm{im}(\varphi_p)$, which gives $(\mathrm{coker}\varphi)_p \cong \mathrm{coker}(\varphi_p)$.

Every monomorphism is a kernel. Let $\varphi\colon \mathscr{F}\to\mathscr{G}$ be a monomorphism and $q\colon \mathscr{G}\to\mathrm{coker}\varphi$ its cokernel. Every $\varphi(U)$ is injective (theorem 1.2.27), so $(\ker q)(U) = (\mathrm{im}\varphi)(U) = \mathrm{im}(\varphi(U))$ by corollary 1.2.12. Hence the morphism $\mathscr{F}\to\ker q$ through which φ factors is bijective on every open set, an isomorphism, and φ is a kernel of q.

Every epimorphism is a cokernel. Let $\varphi\colon \mathscr{F}\to\mathscr{G}$ be an epimorphism, $\iota\colon \ker\varphi\to\mathscr{F}$ the kernel inclusion and $c\colon \mathscr{F}\to\mathrm{coker}\iota$ its cokernel. Since $\varphi\circ\iota = 0$, there is $\chi\colon \mathrm{coker}\iota\to\mathscr{G}$ with $\chi\circ c = \varphi$. On stalks, c_p is surjective with kernel $\mathrm{im}(\iota_p) = \ker(\varphi_p)$ by the stalk formulas, and $\varphi_p = \chi_p\circ c_p$ is surjective (theorem 1.2.27), so χ_p is bijective. By proposition 1.2.24, χ is an isomorphism, so φ is a cokernel of ι.

All five axioms of definition 0.4.29 hold, so $\mathbf{Sh}_{\mathrm{Mod}_R}(X)$ is abelian, as claimed.

Morphisms and Stalks

Corollary 1.2.29: Commuting Kernel and Stalk

Let $\varphi : \mathcal{F} \to \mathcal{G}$ be a morphism of sheaves of R-modules. Then the stalk of the kernel is the kernel of the stalk:

$$(\ker(\varphi))_p \cong \ker(\varphi_p)$$

***Proof*:**
This is one of the stalk formulas of theorem 1.2.28.

Corollary 1.2.30: Commuting Cokernel and Stalk

Let $\varphi : \mathcal{F} \to \mathcal{G}$ be a morphism of sheaves of R-modules. Then the stalk of the cokernel is the cokernel of the stalk:

$$(\operatorname{coker}(\varphi))_p \cong \operatorname{coker}(\varphi_p)$$

***Proof*:**
This is one of the stalk formulas of theorem 1.2.28.

Due to sheafification, the surjectivity of a sheaf morphism is no longer conditional on the surjectivity of sections on all open sets. This leads to the following counter-intuitive examples.

Example 1.2.31: Surjective on Stalks $\neq$ Surjective on Global Sections

1. (if you know Complex Analysis): Let $X = \mathbb{C} \setminus \{0\}$. Let $\mathcal{O}_X$ be the sheaf of holomorphic functions and Ω^1_X be the sheaf of holomorphic differential 1-forms. Define $\varphi : \mathcal{O}_X \to \Omega^1_X$ by $f \mapsto df$.
   - *Locally:* On any simply connected disk U, every closed form is exact. Since holomorphic forms are closed, $\omega = df$ has a solution. Thus φ_p is surjective.
   - *Globally:* The form $\omega = \frac{dz}{z}$ is a global section of $\Omega^1_X(X)$. However, its primitive is $\log(z)$, which is not a function on $\mathbb{C} \setminus \{0\}$. Thus $\frac{dz}{z} \notin \operatorname{im}(\varphi(X))$.
2. Let $X = \mathbb{R}$, $\mathscr{F} = \underline{\mathbb{Z}}$ (constant sheaf), and $\mathscr{G} = i_{p*}(\mathbb{Z}) \oplus i_{q*}(\mathbb{Z})$ (skyscrapers at distinct points p, q). Define $\varphi : \mathscr{F} \to \mathscr{G}$ by restriction.
   - *Stalks:* At p, $\mathscr{F}_p = \mathbb{Z}$ and $\mathscr{G}_p = \mathbb{Z} \oplus 0 \cong \mathbb{Z}$. The map is identity (surjective). Similarly at q. Elsewhere, both are 0.
   - *Globally:* $\mathscr{F}(X) = \mathbb{Z}$ (constant functions). $\mathscr{G}(X) = \mathbb{Z} \oplus \mathbb{Z}$. The map sends $n \mapsto (n, n)$. The image is the diagonal, which is not the whole plane $\mathbb{Z} \oplus \mathbb{Z}$.

Corollary 1.2.32: Commuting Image and Stalk

Let $\varphi : \mathcal{F} \to \mathcal{G}$ be a morphism of sheaves of R-modules. Then the stalk of the image is the image of the stalk:

$$(\mathrm{im}(\varphi))_p \cong \mathrm{im}(\varphi_p)$$

***Proof*:**
This is one of the stalk formulas of theorem 1.2.28.

1.2.33 Summary

- **Presheaves** form an abelian category because computation is done on the level of open sets.
- **Sheaves** form an abelian category because computation is done on the level of stalks.

Exercise 1.2.1

1. Show that $\varphi : \mathscr{F} \to \mathscr{G}$ is surjective if and only if for every $s \in \mathscr{G}(U)$, there exists an open covering $\bigcup_i U_i = U$ and elements $t_i \in \mathscr{F}(U_i)$ such that $\varphi(t_i) = s|_{U_i}$ (hint: recall φ is surjective if and only if it is surjective on stalks).

   This exercise will come back in proposition 5.1.16

2. Let $(X, \mathcal{O}_X)$ be a ringed space and $U \subseteq X$ an open subset. Show that we can define non-vanishing distinguished open subset by:

$$\begin{array}{ccc} U_f & \longrightarrow & \mathcal{O}_X^* \\ \downarrow & & \downarrow \\ U & \xrightarrow{f} & \mathcal{O}_X \end{array}$$

   that is, $U_f = f^{-1}(\mathcal{O}_X^*) = \left\{x \in U \;\middle|\; f(x) \in \mathcal{O}_{X,x}^*\right\}$. Show these form a sheaf.

3. Define a *local ringed space* to be a ringed space $(X, \mathcal{O}_X)$ such that for all open $U \subseteq X$ and all $f \in \mathcal{O}_X(U)$, $U = U_f \cup U_{1-f}$, namely for all $x \in U$, f or $1 - f$ is invertible.

   a) Show that if X, $\mathrm{mSpec}\, \mathcal{O}_X(X)$ is a singleton.

   b) Let $C(X)$ be all continuous real function on X. Show that $(X, \mathbb{C}(X))$ is local.

4. Let $(X, \mathcal{O}_X)$ is a local ringed space, $U \subseteq X$ open, and $f \in \mathcal{O}_X(U)$. Then if $f_1 + \cdots + f_r = 1$, then $\bigcup U_i = U$.

5. Let $\mathscr{F}$ be a sheaf of sets, show that $Hom(\underline{\{p\}}, \mathscr{F}) \cong \mathscr{F}$

6. Let $\mathscr{F}$ be a sheaf of groups, show that $Hom(\underline{\mathbb{Z}}, \mathscr{F}) \cong \mathscr{F}$

7. It is tempting to then think that $Hom(\mathscr{F}, \mathscr{G})_p \cong \mathrm{Hom}(\mathscr{F}_p, \mathscr{G}_p)$. However this is *not* the case: think of a counter-example. There is a map from one of these sets to another, what is it?

1.3 Grothendieck's Functors: Pushforward, Pullback, and more

We next endeavor to move sheaves around from one topological space to another given a continuous map $f : X \to Y$. Grothendieck identified 4 key ways of doing so that come in adjoint pairs:

1. The pushforward and pullback: f_* and f^*
2. the proper direct image and proper inverse image: $f_!$ and $f^!$

These are some of the key functors in Grothendieck *6 functor formalism* where the adjoint pair Hom and $\otimes$ is missing. These functor's are more naturally introduced in a cohomological setting and will be present in chapter 6.

1.3.1 Pushforward and Pullback

The easiest of these is the pushforward. If $U \subseteq V \subseteq Y$ and $f : X \to Y$, then $f^{-1}(U) \subseteq f^{-1}(V)$. Hence, f^{-1} can be thought of as a contravariant functor preserving $\subseteq$. In fact, more is true: f^{-1} also preserves the properties in proposition 1.1.9, and hence works on sheaves as well:

Definition 1.3.1: Pushforward Of (Pre)sheaf

Let $f : X \to Y$ be a continuous map and $\mathscr{F} \in \mathbf{PSh}(X)$. Then the *pushforward* or *direct image* of $\mathscr{F}$ through f is defined on every open set $U \subseteq Y$ to be

$$f_*\mathscr{F}(U) = \mathscr{F}(f^{-1}(U))$$

and induces a (pre)sheaf on Y. The pushforward defines a [covariant] functor $f_* : \mathbf{Sh}(X) \to \mathbf{Sh}(Y)$ sending:

$$f_* : \mathscr{F} \to f_*(\mathscr{F})$$

If working with a particular category, for example $\mathbf{Set}$, we can write $f_* : \mathbf{Set}_X \to \mathbf{Set}_Y$.

The term "pushforward" comes from the fact that we are taking a sheaf on X and defining a sheaf on Y through the map $f : X \to Y$. The pullback shall do the same thing in reverse.

It is perhaps unsurprising that f^{-1} preserves pre-sheaves, however it is a little less obvious that it preserves sheaves as well; it is worth thinking this through. To me, this shows that sheaves are a very natural concept, as the usually topological function f^{-1} naturally preserves them. This also motivates the Grothendieck topology that will be introduced much later. We have already seen an example of a pushforward: the skyscraper sheaf is induced that pushing forward along the inclusion map $\iota_p : \{p\} \to X$ the constant sheaf $\underline{S}$.

Proposition 1.3.2: Pushforward Induces Map On Stalks

Let $f : X \to Y$ be a continuous map and $\mathscr{F} \in \mathbf{Sh}(X)$. Then for each $p \in X$ and $q = f(p)$, there exists a map

$$(f_*\mathscr{F})_q \to \mathscr{F}_p$$

between the stalks of $f_*\mathscr{F}$ and $\mathscr{F}$.

***Proof*:**
Write $\lambda_W \colon \mathscr{F}(W) \to \mathscr{F}_p$ for the colimit maps of the stalk at p (definition 1.1.4), indexed by the open neighbourhoods W of p. Let V be an open neighbourhood of q in Y. Continuity of f makes $f^{-1}(V)$ an open neighbourhood of p, so there is a morphism

$$\alpha_V = \lambda_{f^{-1}(V)} \colon (f_*\mathscr{F})(V) = \mathscr{F}(f^{-1}(V)) \longrightarrow \mathscr{F}_p$$

For $V' \subseteq V$, the restriction map of $f_*\mathscr{F}$ is $\mathrm{res}_{f^{-1}(V),f^{-1}(V')}$ by definition 1.3.1, so $\alpha_{V'} \circ \mathrm{res}_{f^{-1}(V),f^{-1}(V')} = \alpha_V$ by the compatibility of the maps λ. The universal property of the colimit $(f_*\mathscr{F})_q$ gives a unique morphism $(f_*\mathscr{F})_q \to \mathscr{F}_p$ whose composite with the colimit map at each V is α_V. On germs it sends the class of (s, V), with $s \in \mathscr{F}(f^{-1}(V))$, to the class of $(s, f^{-1}(V))$. Only the presheaf structure of $\mathscr{F}$ was used, as we sought to show.

Proposition 1.3.3: Sheaf Morphisms Determined by a Basis

Let X be a topological space and $\mathcal{B}$ a basis for the topology. Let $\mathcal{F}$ and $\mathcal{G}$ be sheaves on X. Suppose that for every basis set $B \in \mathcal{B}$, we have a morphism $\varphi_B : \mathcal{F}(B) \to \mathcal{G}(B)$ such that for any inclusion of basis sets $V \subseteq U$ (where $U, V \in \mathcal{B}$), the maps commute with restriction:

$$\varphi_V(s|_V) = (\varphi_U(s))|_V$$

Then there exists a unique sheaf morphism $\Phi : \mathcal{F} \to \mathcal{G}$ extending this collection.

***Proof*:**
The sheaves take values in sets, abelian groups, rings or modules, and the φ_B are morphisms in that category.

Step 1: the section $\Phi_U(s)$. Let $U \subseteq X$ be open and $s \in \mathcal{F}(U)$. Let $\mathcal{B}_U$ be the set of basis sets contained in U, which cover U because $\mathcal{B}$ is a basis. For $B, B' \in \mathcal{B}_U$, the intersection $B \cap B'$ is covered by basis sets $B'' \subseteq B \cap B'$, and on each of them

$$\varphi_B(s|_B)|_{B''} = \varphi_{B''}(s|_{B''}) = \varphi_{B'}(s|_{B'})|_{B''}$$

by the compatibility hypothesis, so $\varphi_B(s|_B)$ and $\varphi_{B'}(s|_{B'})$ agree on $B \cap B'$ by the locality axiom for $\mathcal{G}$. The gluing axiom gives $t \in \mathcal{G}(U)$ with $t|_B = \varphi_B(s|_B)$ for every $B \in \mathcal{B}_U$, unique by locality. Put $\Phi_U(s) = t$. If U is itself a basis set, then $\varphi_U(s)$ satisfies $\varphi_U(s)|_B = \varphi_B(s|_B)$ for $B \in \mathcal{B}_U$, so $\Phi_U = \varphi_U$.

Step 2: Φ *is a morphism of sheaves.* For $V \subseteq U$ and a basis set $B \subseteq V$, $\Phi_U(s)|_V$ restricts to

$\Phi_U(s)|_B = \varphi_B(s|_B) = \varphi_B((s|_V)|_B)$, so $\Phi_U(s)|_V = \Phi_V(s|_V)$ by the uniqueness in Step 1. In the algebraic cases, $\Phi_U(s+s')$ and $\Phi_U(s) + \Phi_U(s')$ both restrict to $\varphi_B(s|_B) + \varphi_B(s'|_B)$ on every $B \in \mathcal{B}_U$, the maps φ_B and the restriction maps being homomorphisms, so they are equal by locality. Likewise $\Phi_U(ss')$ and $\Phi_U(s)\Phi_U(s')$ both restrict to $\varphi_B(s|_B)\varphi_B(s'|_B)$, $\Phi_U(rs)$ and $r\Phi_U(s)$ to $r\varphi_B(s|_B)$, and $\Phi_U(1)$ and 1 to 1, so these pairs are equal as well, and each Φ_U is a morphism in the category of values.

Step 3: uniqueness. If Φ' is a sheaf morphism with $\Phi'_B = \varphi_B$ for every $B \in \mathcal{B}$, then for $s \in \mathcal{F}(U)$ and $B \in \mathcal{B}_U$ we have $\Phi'_U(s)|_B = \Phi'_B(s|_B) = \varphi_B(s|_B)$, so $\Phi'_U(s) = \Phi_U(s)$ by locality, completing the proof.

Corollary 1.3.4: Sheaf Isomorphism Determined By A Basis

Let X be a topological space and $\mathcal{B}$ a basis for the topology. Let $\mathcal{F}$ and $\mathcal{G}$ be sheaves on X. Suppose that for every basis set $B \in \mathcal{B}$, we have an isomorphism $\varphi_B : \mathcal{F}(B) \to \mathcal{G}(B)$ such that for any inclusion of basis sets $V \subseteq U$ (where $U, V \in \mathcal{B}$), the maps commute with restriction:

$$\varphi_V(s|_V) = (\varphi_U(s))|_V$$

Then there exists a unique sheaf isomorphism $\Phi : \mathcal{F} \to \mathcal{G}$ extending this collection.

***Proof*:**
By proposition 1.3.3 there is a unique sheaf morphism $\Phi\colon \mathcal{F} \to \mathcal{G}$ with $\Phi_B = \varphi_B$ for every basis set B. The inverses $\varphi_B^{-1}\colon \mathcal{G}(B) \to \mathcal{F}(B)$ commute with restriction: for basis sets $V \subseteq U$ and $t \in \mathcal{G}(U)$,

$$\varphi_V\left(\varphi_U^{-1}(t)|_V\right) = \varphi_U\left(\varphi_U^{-1}(t)\right)|_V = t|_V$$

so $\varphi_U^{-1}(t)|_V = \varphi_V^{-1}(t|_V)$. By proposition 1.3.3 they extend to a sheaf morphism $\Psi\colon \mathcal{G} \to \mathcal{F}$. The morphisms $\Psi \circ \Phi$ and $\mathrm{id}_{\mathcal{F}}$ both extend the family $\{\mathrm{id}_{\mathcal{F}(B)}\}_{B\in\mathcal{B}}$, so they are equal by the uniqueness in proposition 1.3.3, and likewise $\Phi \circ \Psi = \mathrm{id}_{\mathcal{G}}$. Hence Φ is an isomorphism, and it is the unique morphism extending the φ_B, as claimed.

Pullback

So far, most of the discussion of sheaves where with the assumption of a topological X. In example 1.1.12, there were examples of sheaves being pushed forward (Definition 1.3.1) via a continuous map $f : X \to Y$. Namely the sheaf $\mathscr{F}$ maps to the sheaf $f_*\mathscr{F}$ on Y defined via:

$$\pi_*\mathscr{F}(U) = \mathscr{F}\left(f^{-1}(U)\right)$$

There is an adjoint, the *pullback*, called the inverse image sheaf. It is called a pullback as it takes a sheaf $\mathscr{F}$ and Y and defines a sheaf $f^*(\mathscr{F})$ on X. It is not as clear as the case for the push-forward as the image of an open set need not be open, hence the existence is more subtle (similar to the existence of cokernels for groups). The adjoint will be important not only for having more than one perspective on sheaves, but for definition morphisms of schemes.

Definition 1.3.5: Inverse Image Sheaf

Let $f\colon X \to Y$ be a continuous map and $\mathscr{G}$ a sheaf on Y. For $U \subseteq X$ open, the open sets $V \subseteq Y$ containing $f(U)$, ordered by reverse inclusion, form a directed set, since the intersection of two of them is a third. Put

$$f^*_{\mathrm{pre}}\mathscr{G}(U) = \varinjlim_{V \supseteq f(U)} \mathscr{G}(V)$$

with colimit maps $\kappa_{U,V}\colon \mathscr{G}(V) \to f^*_{\mathrm{pre}}\mathscr{G}(U)$. For $U' \subseteq U$, every open $V \supseteq f(U)$ also contains $f(U')$, and the maps $\kappa_{U',V}$ for $V \supseteq f(U)$ are compatible with the restrictions of $\mathscr{G}$, so they induce the restriction map $\mathrm{res}_{U,U'}\colon f^*_{\mathrm{pre}}\mathscr{G}(U) \to f^*_{\mathrm{pre}}\mathscr{G}(U')$, the unique morphism with $\mathrm{res}_{U,U'} \circ \kappa_{U,V} = \kappa_{U',V}$ for every $V \supseteq f(U)$. By this uniqueness $\mathrm{res}_{U,U} = \mathrm{id}$ and $\mathrm{res}_{U',U''} \circ \mathrm{res}_{U,U'} = \mathrm{res}_{U,U''}$, so $f^*_{\mathrm{pre}}\mathscr{G}$ is a presheaf on X. The *inverse image sheaf* or *pullback* of $\mathscr{G}$ is its sheafification

$$f^*\mathscr{G} = (f^*_{\mathrm{pre}}\mathscr{G})^{\mathrm{sh}}$$

Giving us a map $f^* : \mathbf{Sh}(Y) \to \mathbf{Sh}(X)$.

1.3.6 On Vakil's Notation Vakil uses the f^{-1} notation intend of f^* as he wants to reserve the f^* notation for pulling back $\mathcal{O}_X$-modules:

$$f^*\mathcal{G} := f^{-1}\mathcal{G} \otimes_{f^{-1}\mathcal{O}_Y} \mathcal{O}_X$$

This is because we still want pullback as defined in this section to apply to $\mathcal{O}_X$-modules. We shall instead keep track of which category we are performing the operations.

For presheaves of sets or of algebraic structures, the colimit is computed as for stalks (definition 1.1.3): an element of $f^*_{\mathrm{pre}}\mathscr{G}(U)$ is the class of a pair (t, V) with $V \supseteq f(U)$ open and $t \in \mathscr{G}(V)$, where $(t, V) \sim (t', V')$ when $t|_W = t'|_W$ for some open W with $f(U) \subseteq W \subseteq V \cap V'$, and $\mathrm{res}_{U,U'}$ sends the class of (t, V) to the class of the same pair.

The sheafification is crucial:

Example 1.3.7: Presheaf Inverse Image

Let $Y = X \coprod X$, let $f\colon Y \to X$ be the map that is the identity on each copy of X, and let $\mathscr{G}$ be a sheaf on X. For $V \subseteq X$ open, its preimage $U = V \coprod V$ has $f(U) = V$, which is open, so $f^*_{\mathrm{pre}}\mathscr{G}(U) = \mathscr{G}(V)$. On each copy of X the map f is a homeomorphism onto X, so there $f^*_{\mathrm{pre}}\mathscr{G}$ restricts to $\mathscr{G}$. A section of $f^*\mathscr{G}$ over an open subset of one copy is a tuple of compatible germs of $\mathscr{G}$ (proof of theorem 1.2.8), and since $\mathscr{G}$ is a sheaf these are the germs of a unique section of $\mathscr{G}$, so $f^*\mathscr{G}$ restricts to $\mathscr{G}$ on each copy. As U is the disjoint union of two open copies of V, the sheaf axioms give

$$f^*\mathscr{G}(U) = \mathscr{G}(V) \times \mathscr{G}(V)$$

and the map $\mathrm{sh}\colon f^*_{\mathrm{pre}}\mathscr{G}(U) \to f^*\mathscr{G}(U)$ is the diagonal $s \mapsto (s, s)$, which is not surjective once $\mathscr{G}(V)$ has two elements.

Proposition 1.3.8: push-forward Sheaf Adjoint

Let $f : X \to Y$ be a continuous functions and $\mathscr{F}$,$\mathscr{G}$ be sheaves on X and Y respectfully. Then there is a bijection:

$$\mathrm{Mor}_X(f^*\mathscr{G}, \mathscr{F}) \leftrightarrow \mathrm{Mor}_Y(\mathscr{G}, f_*\mathscr{F})$$

That is, they are adjoints

***Proof*:**
At the level of presheaves there is a bijection

$$\mathrm{Mor}_{\mathbf{PSh}(X)}(f^*_{\mathrm{pre}}\mathscr{G}, \mathscr{F}) \leftrightarrow \mathrm{Mor}_{\mathbf{PSh}(Y)}(\mathscr{G}, f_*\mathscr{F})$$

Given a presheaf morphism $\alpha : f^*_{\mathrm{pre}}\mathscr{G} \to \mathscr{F}$, define $\eta : \mathscr{G} \to f_*\mathscr{F}$ as follows. For an open $V \subseteq Y$ we have $f(f^{-1}(V)) \subseteq V$, so V is one of the indices of the colimit defining $f^*_{\mathrm{pre}}\mathscr{G}(f^{-1}(V))$ (definition 1.3.5), giving a canonical map $c_V : \mathscr{G}(V) \to f^*_{\mathrm{pre}}\mathscr{G}(f^{-1}(V))$. Set

$$\eta(V) = \alpha(f^{-1}(V)) \circ c_V : \mathscr{G}(V) \to \mathscr{F}(f^{-1}(V)) = f_*\mathscr{F}(V)$$

This commutes with restriction, since α does and the c_V are compatible with the transition maps of the colimits.

Conversely, given a presheaf morphism $\eta : \mathscr{G} \to f_*\mathscr{F}$, fix an open $U \subseteq X$. For every open $V \supseteq f(U)$ we have $U \subseteq f^{-1}(V)$, so there is a composite

$$\mathscr{G}(V) \xrightarrow{\eta(V)} \mathscr{F}(f^{-1}(V)) \xrightarrow{\mathrm{res}} \mathscr{F}(U)$$

For $V' \supseteq V \supseteq f(U)$ these composites are compatible with the restriction $\mathscr{G}(V') \to \mathscr{G}(V)$, because η commutes with restriction and restrictions compose. By the universal property of the colimit they induce a map $\alpha(U) : f^*_{\mathrm{pre}}\mathscr{G}(U) \to \mathscr{F}(U)$, and these maps commute with restriction in U, again by uniqueness of maps out of a colimit.

The two constructions invert each other. Starting from η and forming α, the composite $\alpha(f^{-1}(V)) \circ c_V$ is the branch of the colimit at the index V, which is the composite of $\eta(V)$ with the identity restriction, that is $\eta(V)$. Starting from α and forming η, the map induced on $f^*_{\mathrm{pre}}\mathscr{G}(U)$ agrees with $\alpha(U)$ on every branch of the colimit, so it equals $\alpha(U)$.

Now pass to sheaves. Morphisms of sheaves are morphisms of the underlying presheaves (definition 1.2.1), and $f_*\mathscr{F}$ is a sheaf (definition 1.3.1), so $\mathrm{Mor}_{\mathbf{PSh}(Y)}(\mathscr{G}, f_*\mathscr{F}) = \mathrm{Mor}_Y(\mathscr{G}, f_*\mathscr{F})$. On the other side, since $\mathscr{F}$ is a sheaf, composition with sh is a bijection

$$\mathrm{Mor}_X(f^*\mathscr{G}, \mathscr{F}) = \mathrm{Mor}_X((f^*_{\mathrm{pre}}\mathscr{G})^{\mathrm{sh}}, \mathscr{F}) \leftrightarrow \mathrm{Mor}_{\mathbf{PSh}(X)}(f^*_{\mathrm{pre}}\mathscr{G}, \mathscr{F})$$

by the universal property of sheafification (theorem 1.2.8). Combining the two bijections gives the claim. The bijection is natural in $\mathscr{F}$ and $\mathscr{G}$, because each of its constituents is composition with a fixed canonical map (c_V, a restriction, or sh), and such compositions commute with post-composition by morphisms of $\mathscr{F}$ and pre-composition by morphisms of $\mathscr{G}$.

One advantage of the inverse image sheaf is the computation of stalk

Proposition 1.3.9: Stalks Via Inverse Image Sheaf

Let $f : X \to Y$ be a continuous function with $\mathscr{G}$ a sheaf on Y. Then for every $p \in X$:

$$(f^*\mathscr{G})_p \cong \mathscr{G}_{f(p)}$$

***Proof*:**
For $U \subseteq X$ open and $V \supseteq f(U)$ open, write $\kappa_{U,V} : \mathscr{G}(V) \to f^*_{\text{pre}}\mathscr{G}(U)$ for the colimit maps of definition 1.3.5, so that $\text{res}_{U,U'} \circ \kappa_{U,V} = \kappa_{U',V}$ for $U' \subseteq U$ by the definition of the restriction maps there. Write $\lambda_U : f^*_{\text{pre}}\mathscr{G}(U) \to (f^*_{\text{pre}}\mathscr{G})_p$ and $\gamma_V : \mathscr{G}(V) \to \mathscr{G}_{f(p)}$ for the colimit maps of the two stalks.

Step 1: sheafification. By definition 1.3.5, $f^*\mathscr{G} = (f^*_{\text{pre}}\mathscr{G})^{\text{sh}}$, so $(f^*\mathscr{G})_p \cong (f^*_{\text{pre}}\mathscr{G})_p$ by corollary 1.2.10.

Step 2: a map $\alpha : \mathscr{G}_{f(p)} \to (f^*_{\text{pre}}\mathscr{G})_p$. For an open $V \ni f(p)$, the set $f^{-1}(V)$ is an open neighbourhood of p with $f(f^{-1}(V)) \subseteq V$, so $\alpha_V = \lambda_{f^{-1}(V)} \circ \kappa_{f^{-1}(V),V}$ is defined. For $V' \subseteq V$,

$$\alpha_{V'} \circ \text{res}_{V,V'} = \lambda_{f^{-1}(V')} \circ \kappa_{f^{-1}(V'),V} = \lambda_{f^{-1}(V')} \circ \text{res}_{f^{-1}(V),f^{-1}(V')} \circ \kappa_{f^{-1}(V),V} = \alpha_V$$

using the compatibility of the colimit maps in $f^*_{\text{pre}}\mathscr{G}(f^{-1}(V'))$, then the formula for the restriction, then the compatibility of the λ. The universal property of $\mathscr{G}_{f(p)}$ gives α with $\alpha \circ \gamma_V = \alpha_V$.

Step 3: a map $\beta : (f^*_{\text{pre}}\mathscr{G})_p \to \mathscr{G}_{f(p)}$. Fix an open $U \ni p$. Every open $V \supseteq f(U)$ contains $f(p)$, and the maps γ_V are compatible with restriction, so they induce $\beta_U : f^*_{\text{pre}}\mathscr{G}(U) \to \mathscr{G}_{f(p)}$ with $\beta_U \circ \kappa_{U,V} = \gamma_V$. For $U' \subseteq U$ we have $\beta_{U'} \circ \text{res}_{U,U'} \circ \kappa_{U,V} = \beta_{U'} \circ \kappa_{U',V} = \gamma_V = \beta_U \circ \kappa_{U,V}$ for all $V \supseteq f(U)$, so $\beta_{U'} \circ \text{res}_{U,U'} = \beta_U$ by uniqueness. The universal property of $(f^*_{\text{pre}}\mathscr{G})_p$ gives β with $\beta \circ \lambda_U = \beta_U$.

Step 4: α and β are mutually inverse. For an open $V \ni f(p)$, $\beta \circ \alpha \circ \gamma_V = \beta_{f^{-1}(V)} \circ \kappa_{f^{-1}(V),V} = \gamma_V$, so $\beta \circ \alpha = \text{id}$ by uniqueness. For an open $U \ni p$ and an open $V \supseteq f(U)$ we have $U \subseteq f^{-1}(V)$, hence

$$\alpha \circ \beta \circ \lambda_U \circ \kappa_{U,V} = \alpha \circ \gamma_V = \lambda_{f^{-1}(V)} \circ \kappa_{f^{-1}(V),V} = \lambda_U \circ \text{res}_{f^{-1}(V),U} \circ \kappa_{f^{-1}(V),V} = \lambda_U \circ \kappa_{U,V}$$

The uniqueness in the universal property of $f^*_{\text{pre}}\mathscr{G}(U)$ gives $\alpha \circ \beta \circ \lambda_U = \lambda_U$ for every U, and then the uniqueness for $(f^*_{\text{pre}}\mathscr{G})_p$ gives $\alpha \circ \beta = \text{id}$. With Step 1, $(f^*\mathscr{G})_p \cong (f^*_{\text{pre}}\mathscr{G})_p \cong \mathscr{G}_{f(p)}$, as claimed.

In fact, the adjoint of this map is the *skyscraper sheaf*, namely given $f : \{p\} \to X$ where $p \in X$, then $f^*\mathscr{G} = \mathscr{G}_p$. Conversely, if we define the sheaf $\mathscr{F}(\{p\}) = S$, then $f_*\mathscr{F}(U)$ is the skyscraper sheaf.

will get back to this when it is more needed

1.3.2 Proper Direct and Inverse Image

While the standard pushforward f_* and pullback f^* are sufficient for many geometric operations, they lack certain "clean" properties regarding the supports of sections:

Example 1.3.10: Smeared Section

Consider $X = \mathbb{R} - \{0\}$ with the euclidean topology and take the inclusion $j : X \hookrightarrow \mathbb{R}$. Then let $\mathscr{F}$ on X be the $\mathbb{Z}$-constant sheaf. Then what is $j_*\mathscr{F}$ at 0? Consider that on any open neighbourhood $(-\epsilon, \epsilon)$, the pushforward is $(-\epsilon, 0) \cup (0, \epsilon)$. We can choose two arbitrary choices of constant functions. Hence:

$$j_*(\mathscr{F})_0 \cong \mathbb{Z} \oplus \mathbb{Z}$$

it is *not* 0!

For those who already encountered the Zariski topology on $\mathbb{A}^1_k$ (k infinite), consider $X = \mathbb{A}^1_k \setminus \{0\}$ and the inclusion $j : X \hookrightarrow \mathbb{A}^1_k$. Then as the open sets of $\mathbb{A}^1_k$ are cofinite, hence any open subset U containing 0 is cofinite; it is easy to show that $X \cap U$ is always *connected*. Hence, if we put the $\mathbb{Z}$-constant sheaf on X, $j_*(\mathscr{F})_0 = \mathbb{Z}$. The functor $j_!$ that will be defined soon will be characterized by $j_!(\mathscr{F})_x = 0$ if $x \notin X$.

To address this, Grothendieck introduced the "shriek" functors (denoted with an exclamation mark in the subscript, ex. $j_!$, or in Hartshorne with a tick in the subscript, ex. $j_\prime$), which handle supports more strictly. These functors behave differently depending on whether the map is an open immersion or a closed immersion.

Extension by Zero (Open Immersion)

Let $j : U \hookrightarrow X$ be an inclusion of an open set. The standard pushforward j_* "smears" the information from U up to the boundary in X. Often, we wish to extend a sheaf from U to X such that it is exactly zero outside of U.

Definition 1.3.11: Extension by Zero

Let $j : U \to X$ be an open immersion and $\mathscr{F}$ a sheaf on U. The *extension by zero*, denoted $j_!\mathscr{F}$, is the sheaf on X associated to the presheaf:

$$j_!(V) = \begin{cases} \mathscr{F}(V) & \text{if } V \subseteq U \\ 0 & \text{otherwise} \end{cases}$$

This functor $j_! : \mathbf{Sh}(U) \to \mathbf{Sh}(X)$ is also called the *proper direct image* or *lower shriek*.

The behavior of this functor is best understood at the level of stalks. Unlike j_*, which may have non-zero stalks on the boundary ∂U, $j_!$ is "clean":

$$(j_!\mathscr{F})_p = \begin{cases} \mathscr{F}_p & \text{if } p \in U \\ 0 & \text{if } p \notin U \end{cases}$$

Just as f^* is the left adjoint to f_*, the extension by zero provides a *left* adjoint to the restriction.

Proposition 1.3.12: Adjunction of Extension by Zero

Let $j : U \to X$ be an open immersion. Then $j_!$ is the left adjoint to the pullback (restriction) j^*. That is, for $\mathscr{F} \in \mathbf{Sh}(U)$ and $\mathscr{G} \in \mathbf{Sh}(X)$:

$$\mathrm{Mor}_X(j_!\mathscr{F}, \mathscr{G}) \cong \mathrm{Mor}_U(\mathscr{F}, j^*\mathscr{G})$$

***Proof*:**
The sheaves take values in abelian groups, the setting in which the zero group of definition 1.3.11 makes sense. Let $\mathscr{P}$ be the presheaf of definition 1.3.11, so that $\mathscr{P}(V) = \mathscr{F}(V)$ when $V \subseteq U$, $\mathscr{P}(V) = 0$ otherwise, the restriction maps between open subsets of U are those of $\mathscr{F}$, the remaining ones are zero, and $j_!\mathscr{F} = \mathscr{P}^{\mathrm{sh}}$.

Step 1: $j^*\mathscr{G} \cong \mathscr{G}|_U$. Let $V \subseteq U$ be open. Then $j(V) = V$ is open in X, so V itself is the smallest of the open sets $W \supseteq j(V)$ indexing the colimit $j^*_{\mathrm{pre}}\mathscr{G}(V) = \varinjlim_{W \supseteq V} \mathscr{G}(W)$. Let $\kappa_W : \mathscr{G}(W) \to j^*_{\mathrm{pre}}\mathscr{G}(V)$ be the colimit maps and $\rho : j^*_{\mathrm{pre}}\mathscr{G}(V) \to \mathscr{G}(V)$ the map induced by the compatible restrictions $\mathrm{res}_{W,V}$. Then $\rho \circ \kappa_V = \mathrm{res}_{V,V} = \mathrm{id}$, and $\kappa_V \circ \rho \circ \kappa_W = \kappa_V \circ \mathrm{res}_{W,V} = \kappa_W$ for all W, so $\kappa_V \circ \rho = \mathrm{id}$ by uniqueness. These isomorphisms ρ commute with restriction in V, as both composites agree after composition with each κ_W. Hence $j^*_{\mathrm{pre}}\mathscr{G} \cong \mathscr{G}|_U$ as presheaves on U, and $j^*_{\mathrm{pre}}\mathscr{G}$ is a sheaf because $\mathscr{G}|_U$ is one (an open cover of an open $V \subseteq U$ is an open cover in X). For a sheaf $\mathscr{E}$ the map $\mathrm{sh} : \mathscr{E} \to \mathscr{E}^{\mathrm{sh}}$ is an isomorphism: theorem 1.2.8 applied to $\mathrm{id}_{\mathscr{E}}$ gives a sheaf morphism r with $r \circ \mathrm{sh} = \mathrm{id}_{\mathscr{E}}$, and $\mathrm{sh} \circ r$ and id are two sheaf morphisms g with $g \circ \mathrm{sh} = \mathrm{sh}$, so they coincide by the uniqueness in theorem 1.2.8. With $\mathscr{E} = j^*_{\mathrm{pre}}\mathscr{G}$ this gives $j^*\mathscr{G} \cong j^*_{\mathrm{pre}}\mathscr{G} \cong \mathscr{G}|_U$.

Step 2: presheaf morphisms out of $\mathscr{P}$. Let $\theta : \mathscr{P} \to \mathscr{G}$ be a morphism of presheaves on X. For $V \not\subseteq U$ the map $\theta(V)$ is the zero map out of the zero group. The components $\theta(V)$ for $V \subseteq U$ commute with the restrictions between open subsets of U, so they form a morphism $\theta|_U : \mathscr{F} \to \mathscr{G}|_U$. Conversely, a morphism $\eta : \mathscr{F} \to \mathscr{G}|_U$ extends to $\mathscr{P}$ by zero on the open sets not contained in U. For $V' \subseteq V$ the naturality square is a square of η when $V \subseteq U$, and when $V \not\subseteq U$ both composites out of $\mathscr{P}(V) = 0$ are zero. The two constructions are inverse to each other, so $\theta \mapsto \theta|_U$ is a bijection from $\mathrm{Mor}_{\mathbf{PSh}(X)}(\mathscr{P}, \mathscr{G})$ onto $\mathrm{Mor}_U(\mathscr{F}, \mathscr{G}|_U)$.

Step 3: conclusion. By theorem 1.2.8, composition with $\mathrm{sh} : \mathscr{P} \to j_!\mathscr{F}$ is a bijection from $\mathrm{Mor}_X(j_!\mathscr{F}, \mathscr{G})$ onto $\mathrm{Mor}_{\mathbf{PSh}(X)}(\mathscr{P}, \mathscr{G})$. With Steps 2 and 1 we obtain bijections

$$\mathrm{Mor}_X(j_!\mathscr{F}, \mathscr{G}) \leftrightarrow \mathrm{Mor}_{\mathbf{PSh}(X)}(\mathscr{P}, \mathscr{G}) \leftrightarrow \mathrm{Mor}_U(\mathscr{F}, \mathscr{G}|_U) \leftrightarrow \mathrm{Mor}_U(\mathscr{F}, j^*\mathscr{G})$$

Each is composition with a fixed morphism (sh, or the isomorphism of Step 1) or restriction of components to the open subsets of U, and these operations commute with pre-composition by a morphism $\mathscr{F}' \to \mathscr{F}$ and post-composition by a morphism $\mathscr{G} \to \mathscr{G}'$. So the composite bijection is natural in $\mathscr{F}$ and $\mathscr{G}$, and $j_!$ is left adjoint to j^*, as claimed.

Sections with Support (Closed Immersion)

Conversely, let $i : Z \hookrightarrow X$ be a closed immersion. The standard pullback i^* restricts sections to Z, effectively "forgetting" everything outside. The right adjoint to the pushforward i_* is the functor that finds sections supported *only* on Z.

Definition 1.3.13: Exceptional Inverse Image

Let $i : Z \to X$ be a closed immersion. The *exceptional inverse image*, denoted $i^!\mathscr{F}$ (or sometimes $\mathcal{H}^0_Z(\mathscr{F})$), is the functor $i^! : \mathbf{Sh}(X) \to \mathbf{Sh}(Z)$ defined by the subgroup of sections:

$$\Gamma(V, i^!\mathscr{F}) = \{s \in \Gamma(V, i^*\mathscr{F}) \big| \operatorname{supp}(s) \subseteq Z\}$$

It serves as the right adjoint to the pushforward i_*.

1.3.3 Hom and Tensor

here (I think this is a good place for it, to round off the 6-functor formalism.)

Summary of the Six Operations

It is common to group these functors into a "Six Operation" formalism. While defined generally for derived categories, for sheaves on topological spaces, they exhibit a specific symmetry between open and closed maps.

1. **For an Open Immersion j:** The exceptional inverse image collapses to the standard pullback ($j^! \cong j^*$), leaving us with the triple $(j_!, j^*, j_*)$.
2. **For a Closed Immersion i:** The proper direct image collapses to the standard pushforward ($i_! \cong i_*$), leaving us with the triple $(i^*, i_*, i^!)$.

These functors allow for the decomposition of a sheaf $\mathscr{F}$ on X into its parts on an open set U and the closed complement Z via the fundamental exact sequence:

$$0 \to j_!j^*\mathscr{F} \to \mathscr{F} \to i_*i^*\mathscr{F} \to 0$$

Exercise 1.3.1

1. Let $U \subseteq Y$ open and $\iota : U \hookrightarrow Y$ be the inclusion. Show that $\iota^*\mathscr{G} \cong \mathscr{G}|_U$.
2. Show that f^* is an exact functor (naturally on abelian categories, however is just as good to show it for any ${}_R\mathrm{Mod}$)
3. Let $U \subseteq X$ be an open subset of a topological space so that $V = X \setminus U$ is closed. Let $\mathscr{F}$ be a sheaf on V, and $\iota : V \hookrightarrow X$ the inclusion.
   a) Show that $(\iota_*\mathscr{F})_p \cong \mathscr{F}_p$ if $p \in V$ and 0 otherwise.

b) Let $j_!(\mathscr{F})$ be a sheaf on X given by

$$W \mapsto \mathscr{F}(W) \qquad W \mapsto 0$$

where the first happens when $W \subseteq V$, and the second in all other cases. Show that $(j_!\mathscr{F})_p \cong \mathscr{F}_p$ if $p \in U$ and 0 otherwise. .

c) conclude there is a short exact sequence $0 \to j_!(\mathscr{F}|_U) \to \mathscr{F} \to \iota_*(\mathscr{F}|_V) \to 0$.

1.4 Ringed Space and Locally Ringed Space

Having defined the morphisms in $\mathbf{Sh}(X)$, an important class of sheaves to distinguish are sheaves over rings, as one of the ultimate goals in this Part of the textbook is the generalize the Nullstellensatz to arbitrary (commutative) rings:

Theorem 1.4.1: (Commutative) Ringed Objects in Sheaf

let $\mathbf{Sh}_{\mathbf{C}}(X)$ be the category of sheaves over a category $\mathbf{C}$ that has finite products and a terminal object. Then commutative ringed object exists, and the subcategory $\mathcal{R} \subseteq \mathbf{Sh}_{\mathbf{C}}(X)$ of commutative ringed object is equivalent to the category of sheaves $\mathbf{Sh}_{\mathscr{R}(C)}(X)$ on the *commutative ringed objects* of $\mathbf{C}$.

***Proof*:**
This is a lot of diagram chasing. With some of the proofs later in this section, this result can be proved more easily on the level of stalks.

We shall almost always work in the case of $\mathbf{C} = \mathbf{Set}$ so that $\mathscr{R}(\mathbf{C}) = \mathbf{Ring}$:

Definition 1.4.2: Ringed Space And Structure Sheaf

Let $\mathbf{Sh}_{\mathbf{Ring}}(X)$ be the category of sheaves over **Ring** on X. Then given a sheaf of [commutative] rings over X, denote it $\mathcal{O}_X$, the tuple $(X, \mathcal{O}_X)$ is called a *ringed space*. The sheaf $\mathcal{O}_X$ is called the *structure sheaf*, and the sections of the structure sheaf $\mathcal{O}_X$ over an open subset U are called *[regular] functions on* U.

If it is unambiguous, the $\mathcal{O}_X$ is dropped and X is simply called the ringed space.

1.4.3 Warning Though these are called functions, they are elements of an arbitrary [commutative] ring. The reader should have regular function from [22, Regular Function [on Affine Space]] in mind.

Before defining morphisms formally, it is useful to understand why the sheaf morphism appears to go in the "opposite" direction. Geometrically, if we have a continuous map $f : X \to Y$ and a "function" φ on an open set $V \subseteq Y$ (i.e., $\varphi \in \mathcal{O}_Y(V)$), we naturally obtain a function on X by *pulling back* φ via composition: $\varphi \circ f$. The domain of this new function is $f^{-1}(V)$.

In the language of sheaves, this operation assigns to a section in $\mathcal{O}_Y(V)$ a section in $\mathcal{O}_X(f^{-1}(V))$. Recall that by definition of the pushforward sheaf, $(f_*\mathcal{O}_X)(V) := \mathcal{O}_X(f^{-1}(V))$. Therefore, the

geometric notion of pulling back functions induces an algebraic map $\mathcal{O}_Y(V) \to (f_*\mathcal{O}_X)(V)$ for every open set V, or globally, a sheaf morphism $\mathcal{O}_Y \to f_*\mathcal{O}_X$.

Definition 1.4.4: Ringed Space Morphism

A map $(f, f^\#) : (X, \mathcal{O}_X) \to (Y, \mathcal{O}_X)$ is a *ringed space morphism* if $f : X \to Y$ is a continuous map and $f^\# : \mathcal{O}_Y \to f_*(\mathcal{O}_X)$ is a sheaf morphism.

These form a category and shall often be denoted **RingedSp**.

Example 1.4.5: Ringed Space

1. The simplest and prototypical ringed space is again the sheaf of [real/complex] continuous/smooth/holomorphic functions on a manifold/variety with component-wise addition and multiplication.
2. Schemes in the next part shall form a large class of ringed spaces
3. vector fields on a manifold do not form a ringed space as they are not closed under multiplication (they are closed under the lie bracket operation). However, they shall form a $\mathcal{O}_X$-module over manifold, as shall be explored in chapter 5.

An important sheaf that shall often be used is the sheaf of units:

Definition 1.4.6: Unit Sheaf

Let $(X, \mathcal{O}_X)$ be a ringed space. Then $\mathcal{O}_X^*$ is the sheaf given by the fibered product:

$$\begin{array}{ccc} \mathcal{O}_X^* & \longrightarrow & \{1\} \\ \downarrow & & \downarrow \\ \mathcal{O}_X \times \mathcal{O}_X & \xrightarrow{\cdot} & \mathcal{O}_X \end{array}$$

We shall almost never refer to the categories interpretations of the sheaf of rings in this Part of the book (it shall be important in section 3.3), however ringed spaces are the central object of study in algebraic geometry.

As elements the sections in $\mathcal{O}_X(U)$ are called functions, there should be a notion of evaluation. When working over coordinate rings, evaluation of f at a point a is equivalent to modding f by the (maximal) ideal $(x - a)$. This works over coordinate rings since by the Nullstellensatz the maximal ideals correspond to points. For a general (commutative) ring, there can be many more pathological behaviours, namely it is not clear how to associate to each point $x \in X$ a maximal ideal $\mathfrak{m} \subseteq R$.

The first natural place to consider is the behaviour at the stalk, namely each point x is associated to the ring $\mathcal{O}_{X,x}$. Is there a natural choice of ideal in this ring? In this case, a natural choice would mean there is only *one* choice of ideal, in which case any $f \in \mathcal{O}_X(U)$ where $x \in U$ maps into $\mathcal{O}_{X,x}$ and can be modded by this maximal ideal. In general the answer is no:

Example 1.4.7: Stalk that is not a Local Ring
Take $X = \{p, q\}$ with the topology given by $\mathcal{T}_X = \{\emptyset, \{p\}, X\}$. Let:

$$\mathcal{O}_X(\emptyset) = \mathbf{0} \qquad \mathcal{O}_X(\{p\}) = k \qquad \mathcal{O}_X(X) = k \times k$$

where k is a field with the only non-trivial restriction map being

$$\mathcal{O}_X(X) \to \mathcal{O}_X(\{p\}) \qquad (a, b) \mapsto a$$

Then the stalk at p and q are:

$$\mathcal{O}_{X,p} = k \qquad \text{and} \qquad \mathcal{O}_{X,q} = k \times k$$

showing that the stalk at q has more than one maximal ideal (namely $k \times \{0\}$ and $\{0\} \times k$).

When working over coordinate rings of varieties, all stalks will be local rings (exercise 1.4.1(1.4.1 (1))). In fact, the same holds for (smooth, complex, topological) manifolds [59, section 2.1]. This motivate focusing on ringed spaces whose stalks are local rings:

Definition 1.4.8: Locally Ringed Space

Let X be a ringed space. Then X is a *locally ringed space* if its stalks are local rings.

Definition 1.4.9: Evaluation on Locally Ringed Space

Let $(X, \mathcal{O}_X)$ be a locally ringed space. Then as every stalk is a local ring, it has a unique maximal ideal, hence a unique field

$$\mathcal{O}_{X,x}/\mathfrak{m}_x = \kappa(x)$$

called the *residue field*. Define *evaluation* of an element $f \in \mathcal{O}_X(U)$ where $x \in U$ by defining:

$$f(x) := \overline{f} \bmod \mathfrak{m}_x$$

where $\overline{f}$ is the image of f in $\mathcal{O}_X(U) \to \mathcal{O}_{X,x}$.

In general, ringed space morphisms *do not* preserve the local ring structure of the stalks (an example is worked through at the beginning of chapter 3, see example 3.1.1). This condition must be imposed:

Definition 1.4.10: Locally Ringed Space Morphism (Preliminary)

Let X, Y be locally ringed spaces. Then a morphism of locally ringed spaces $\left(\pi, \pi^{\#}\right) : X \to Y$ is a morphism of ringed spaces such that the induced map of stalks $\pi^{\sharp} : \mathcal{O}_{Y,q} \to \pi_* \mathcal{O}_{X,p}$ (where $\pi(p) = q$) are local ring homomorphisms (the pre-image of the maximal ideal is contained in the maximal ideal).

It can be checked that locally ringed spaces with locally ringed morphisms form a category which shall often be denoted **LocRingedSp** or **LRS**.

Example 1.4.11: Manifolds and Locally ringed sheaves

The two basic examples are varieties and manifolds. Abstracting, topological spaces with continuous function form locally ringed spaces. Complex analytic space (which generalize varieties by taking the vanishing set of holomorphic functions) also form locally ringed spaces. p-adic spaces (central to number theory and modern representation theory) form locally ringed spaces. More generally, Algebraic Spaces and Stacks, developed in *Moduli, Deformation Theory, and Stacks* [20], form locally ringed spaces.

Overall, locally ringed spaces are the "right" over-arching category when considering the duality between algebra and geometry.

Definition 1.4.12: Isomorphism Of Ringed Spaces

Let $(X, \mathcal{O}_X)$ and $(Y, \mathcal{O}_Y)$ be two ringed spaces. Then an isomorphism or ringed space is a map $\pi : X \to Y$ where:

1. π is a homeomorphism
2. There is an isomorphism of sheaves Between $\mathcal{O}_X$ and $\mathcal{O}_Y$ where two two sheaves are given via the pushforward π_* (i.e. $\mathcal{O}_Y \cong \pi_* \mathcal{O}_X$ on Y)[a]

[a] By proposition 1.2.24, it is enough to check this isomorphism on stalks

Exercise 1.4.1

1. Show that the stalks of coordinate rings at maximal ideals are all local rings

1.5⋆ Diffeology

⋆ *Optional. This section is an aside; nothing later in the book depends on it.*

(would love to add a word here)

Pontificating about Points

(I think it is worth writing here my thoughts on what really constitutes a point. Points are fundamental in geometry, you can say there are the necessary indivisible object, and Treating points as secial collections of functions *I think* makes sense because of the result on locally compact spaces withe the upsisde down pi isomorphism, and the more elementary result on compact groups)

(Also, This may veer into "functor of points" as well as "Yoneda's lemma" as well as Distributions as well as Mitchell's theorem were that any abelian category can be seen as an R-module, and as R-modules have elements, we get something reminiscent to recovering points)

2

Schemes I: Basics

Recall the following correspondence:

$$\left\{\text{Affine Varieties}\right\} \underset{\mathcal{Z}(S)}{\overset{\mathcal{I}(V)}{\rightleftarrows}} \left\{\begin{array}{c}\text{Finitely Generated,}\\ \text{Reduced rings}\\ \text{over algebraically closed fields}\end{array}\right\} \tag{2.1}$$

This setups an algebraic-geometric correspondence, where the algebraic object can be thought of as functions on the geometric object, or conversely that the algebraic objects can induce a new geometric space with a notion of "points" (or more generally local information).

finish this above!!

The question naturally arises then if this duality can be extended to general commutative rings. In particular, does there exist some topological space such that:

$$\left\{???\right\} \underset{\mathcal{Z}(S)}{\overset{\mathcal{I}(V)}{\rightleftarrows}} \left\{\text{Commutative Ring}\right\}$$

The necessity of commutativity stems from the need of prime ideals being well-defined[1]. For a more geometric motivation, smooth/holomorphic functions over a real/complex manifold $\mathbb{C}^\infty(M)/\mathcal{H}(M)$ form a commutative ring since they map into $\mathbb{R}$ or $\mathbb{C}$, and hence we copy this behaviour.

Why do this? This abstraction is not simply for abstractions-sake. It is because the dictionary between algebra and geometry that has been built up to is incredibly useful, and will fix some of the problems of algebraic varieties, for example:

[1]The theory of "non-commutative geometry" using non-commutative rings can be explored in *Spectral Theory* in a class on functional analysis (see [60])

1. There are many algebras over more general rings that have geometric interpretations. For a rather trivial but illustrative example, the coordinate ring of a line whose points are in some arbitrary (commutative) ring R (the coordinate ring being $R[x]$) currently does not have a good representation since the Nullstellensatz requires k to be an algebraically closed fields. Another example that may at first seem "un-geometric" but turns out to have many strong geometric properties are rings such as $\mathbb{Z}$, $\mathbb{Z}[i]$ (the Gaussian integers), and more generally *algebraic number rings* (see [7]). Mathematicians have found that these corresponds naturally to looking for integer solutions (or integral solutions when working over an algebraic ring) of a curves (see [7]). In particular, the element of $\mathbb{Z}[i] = \mathbb{Z}[x]/(x^2+1)$ should be functions that "evaluate" at points that would find information mod $\mathfrak{p}$ to solutions of x^2+1.

2. As we saw in front-matter, localization gives us powerful tools to explore the behavior of a variety at a point. In the theoretical framework given by the Nullstellensatz, we cannot consider such rings, for example, if we take the coordinate ring of an affine line, $k[x]$, takes the closure $k(x)$, or study $k[x]_{(x)}$, we loose finite generation. However, as we saw these certainly still have interpretations as functions on a geometric space (the former being the rational functions on a variety, the latter being the "germs"[2] around $0 \in \mathbb{A}^1_k$).

3. Varieties loose multiplicity information. Take for example two varieties over $\mathbb{C}$ defined by the equations
$$y = x^2 \qquad \text{and} \qquad y = 0$$
then if we take their intersection, we get a point $(0,0)$. The corresponding affine variety is given by taking the union of quotient of the two coordinate rings, namely:
$$\frac{k[x,y]}{(y, x^2-y)} = \frac{k[x]}{(x^2)}$$
Usually, we would omit the x^2 term and just use $k[x]/(x)$ for the coordinate ring, however this eliminate the multiplicity information, namely that the variety $y = x^2$ "doubly intersects" the variety $y = 0$. The ring $k[x]/(x^2)$ keeps track that we have a "double-point", while $k[x]/(x)$ does not. This is a achieved by the fact that the ring has nilpotent elements.

4. (If you know differential geometry) A geometrical object that has many algebraic parallels is the vector bundle. Most vector bundles are very algebraic, however many are not projective (cannot be embedded into $\mathbb{P}^n_k$, or even $\mathbb{P}^n_R$ for some ring R). We thus need need to generalize better our notion of projective space. This shall lead to different sheaves on spectra and will be studied in chapter 5.

This build-up will introduce to us the *affine scheme*. Many familiar geometric objects shall look like affine schemes, but not all familiar geometric objects shall look like them (notably, projective space). We will need to generalize a bit more and define a *scheme* which is locally an affine scheme. This shall complete the liaison that algebraic geometry promises between algebra and geometry.

We shall then focus on how what properties of commutative rings translate to geometric properties on a scheme, and conversely when imposing desirable geometric properties how does this translate as algebraic properties. Properties such as being an integral domain, PID, UFD, Noetherian, reduced, and so forth will have geometric interpretations, and conversely notions like dimension, smoothness, open/closed immersions, and so forth shall have algebraic counter-parts.

[2] Somewhat surprisingly, germs are also a "purely algebraic" notion given by nilpotent elements and localization, and hence appear as schemes

2.1 Defining Spec

Recall that the maximal ideals of $k[x_1, x_2, \cdots, x_n]/V$ correspond to points in a variety V (theorem 0.1.12). It may then be natural to say that the collection of maximal ideals should be considered the "points" of an arbitrary ring R. These collection were called the spectrum (plural spectra). For such an association to work, there would need to be a functor translating maps between rings to maps between the collection of maximal ideals. In particular, we would want that the functor translating a ring homomorphism $\varphi : R \to S$ to the associated continuous function mapping the maximal ideal $\mathfrak{m} \subseteq S$ to a maximal ideal $\varphi^{-1}(\mathfrak{m})$ (recall that regular function corresponds contra-variantly to k-algebra homomorphisms), However, the pre-image of a maximal ideal need not be maximal breaking functoriality (ex. take $\mathbb{Z} \hookrightarrow \mathbb{Q}$. Then (0) is maximal in $\mathbb{Q}$, but certainly not maximal in $\mathbb{Z}$).

When the theory was originally being developed within spectral theory from functional analysis (see [60]), the spaces were compact Hausdorff spaces, in which case it would have sufficed to work with the maximal ideals as within this setting the pre-image of maximal ideals (of the ring of continuous functions) is usually maximal. However, as demonstrated in the example above this does not hold for general rings.

Fortunately, if we loosen our requirement a bit and consider prime ideals, then it is indeed the case that $\varphi^{-1}(P)$ is prime, making it the more natural choice of "points" for functorial purposes, even though it shall produce some more "complicated" points (see example 2.1.3). Working with primes also brings the added benefit of incurring a lot more structure which shall turn out to be very insightful and make the addition of primes not only necessary for the definition to work, but to explain a lot of phenomena[3].

Definition 2.1.1: [Prime] Spectrum

Let R be an [arbitrary] commutative ring. Then $\operatorname{Spec} R$ is the collection of prime ideals of R. Let $\operatorname{mSpec}(R)$ represent the collection of maximal ideals of R.

We call an element of $\operatorname{Spec} R$ a *point*, and we call an element of R a *regular functions*

In the special case where R is a coordinate ring $k[x_1, ..., x_n]/I(V)$, $\operatorname{Spec} R$ corresponds to points of an affine variety as well as "extra" points associated to primes that are not maximal (ex. $(y - x^2) \subseteq k[x, y]$) along with the zero ideal which is prime in integral coordinate rings; the intuition behind having these extra points shall be elucidated throughout this section.

To avoid confusion between points of $\operatorname{Spec} R$ and ideals of R, we shall write points in square brackets, for example $[\mathfrak{p}] \in \operatorname{Spec} R$ for $\mathfrak{p} \subseteq R$. There is some merit in writing it in the same notation as the equivalence relation notation, as points on in the topology are defined up to the radical of an ideal, for example for $(x^2) \subseteq \mathbb{C}[x]$, $V(\{(x^2)\}) = [(x)]$ (the $V(-)$ function will be defined in section 2.1.1, or see [22]). As we shall see, once a structure sheaf is introduced, the sets $\operatorname{Spec} \mathbb{C}[x]/(x^2)$ and $\operatorname{Spec} \mathbb{C}[x]/(x)$ which each have one element shall not be have the same structure sheaf added to them. We shall see that in most cases, we can interpret the maximal ideals as points and can write an ideal of the form $[(x_1 - a_1, ..., x_n - a_n)]$ as $(a_1, ..., a_n)$ without ambiguity, however this is not always the case (we shall define non-trivial schemes that shall have no such point[4]). In example 2.1.3, we shall

[3] For example, recall that if k was not algebraically closed, there exists regular functions that are not polynomials. Another is the distinct characterization of divisors in number theory which unifies describing the same cohomological properties of schemes

[4] In particular, there shall be no closed points

see that we can write any point associated to a maximal ideal from a coordinate ring in this way.

Another advantage of working with spectra is that there is no need to reference an ambient space for the algebraic subsets, the information is "within" the space! In particular, every quasi-projective variety in [22] was a subset of $\mathbb{P}^n_k$, while a spectra need not be a subset of $\mathbb{P}^n_k$! In fact, there shall be schemes that are not embed-able in projective space, though they are few and far between[5]. We shall later see under which condition we can embed a scheme into $\mathbb{P}^n_R$ for a ring R, and naturally this shall always be be the case for schemes associated to quasi-projective varieties (covered in section 5.3.3).

When working with coordinate rings R of a variety V, we may consider R to be the set of functions on V, since we may have a notion of "evaluation" for each element of R. We can do the same for a general commutative ring R, motivating the notation f for a general element of a commutative ring R ($f \in R$) and for calling such elements *regular functions*. Notice that if $p(x) \in k[x]$, then we may find the value of the evaluation $p(a)$ for some $a \in k$ by doing

$$\frac{p(x)}{(x-a)} = p(a) \bmod (x-a)$$

To represent the above more formally, every element $f \in k[x]$ (for some algebraically closed field k) is associated to an element of $\mathrm{Hom}_{\mathbf{Set}}\left(\mathrm{Spec}\,(k[x]), \coprod_{(x-a)\in \mathrm{Spec}\,(k[x])} k[x]/(x-a)\right)$, in particular:

$$k[x] \to \mathrm{Hom}_{\mathbf{Set}}\left(\mathrm{Spec}\,(k[x]), \coprod_{(x-a)\in \mathrm{Spec}\,(k[x])} \frac{k[x]}{(x-a)}\right)$$

$$f \mapsto ((x-a) \mapsto f \bmod (x-a))$$

Stare at the above until it makes sense; this is essentially a result of the Nullstellensatz ([22, Hilbert's Nullstellensatz - Weak Version]). Notably, we can think of a function as holding local information (indeed, the "most local" as it gives point-wise information rather than open-set local, component local, germ local, and so forth) at each point of the spectrum[6]. We extract this point-wise information by evaluating, or in the case of spectra by modding.

Recall that if we take subsets of the classical affine space, we can often define more regular functions[7], for example by taking the (classical) affine line minus the origin, $\mathbb{A}^{\setminus}_k\{0\}$, we have $f(x) = 1/x$ is a well-defined regular function in the classical sense. We shall thus want to focus on all possible functions from $\mathrm{Frac}(R)$[8] that are regular at a point $\mathfrak{p} \in \mathrm{Spec}\, R$; we saw in [10] that this collection is given by localizing at the prime: $R_{\mathfrak{p}}$ (for $R = k[x]$ with $\mathfrak{p} = (x-a)$ we had $k[x]_{(x-a)}$). This collection is called the *stalk* at the point $\mathfrak{p}$. Notice that the codomain of the hom-set can be thought of as the disjoint union of field of fractions of the germs $k[x]_{(x-a)}$ as

$$\frac{k[x]_{(x-a)}}{\mathfrak{m}_{(x-a)}} \cong \mathrm{Frac}\left(\frac{k[x]}{(x-a)}\right) \cong \mathrm{Frac}(k) = k$$

which show that we recover the same notion of evaluation when working over the localization. This greater generality shall be important when we define the sheaf on $\mathrm{Spec}\, R$ namely since the ring $R_{\mathfrak{p}}$

[5] Chow's lemma (theorem 3.5.34) shows that all proper schemes are covered by a projective scheme, hence we shall require some "awkward blow-ups" to forcefully construct a scheme that cannot be embedded in projective space

[6] or more generally, at each point of a geometric space, like a manifold

[7] though not always, for example the punctured plane has the same algebraic functions as the plane, see example 2.2.53

[8] Technically, it is required that R be an integral domain as this corresponds to irreducible subsets. Each irreducible subsets will have its own $\mathrm{Frac}(R_i)$ of possible regular functions

shall be a *local ring* that has only a single maximal ideal $\mathfrak{m}_\mathfrak{p}$ making it unambiguous which ideal to mod by (as explored in section 1.4), and so we shall use it for our definition:

Definition 2.1.2: Value Of Regular Function

Let $f \in R$. Then the *value* of f at the point $\mathfrak{p} \in \operatorname{Spec}(R)$ is:

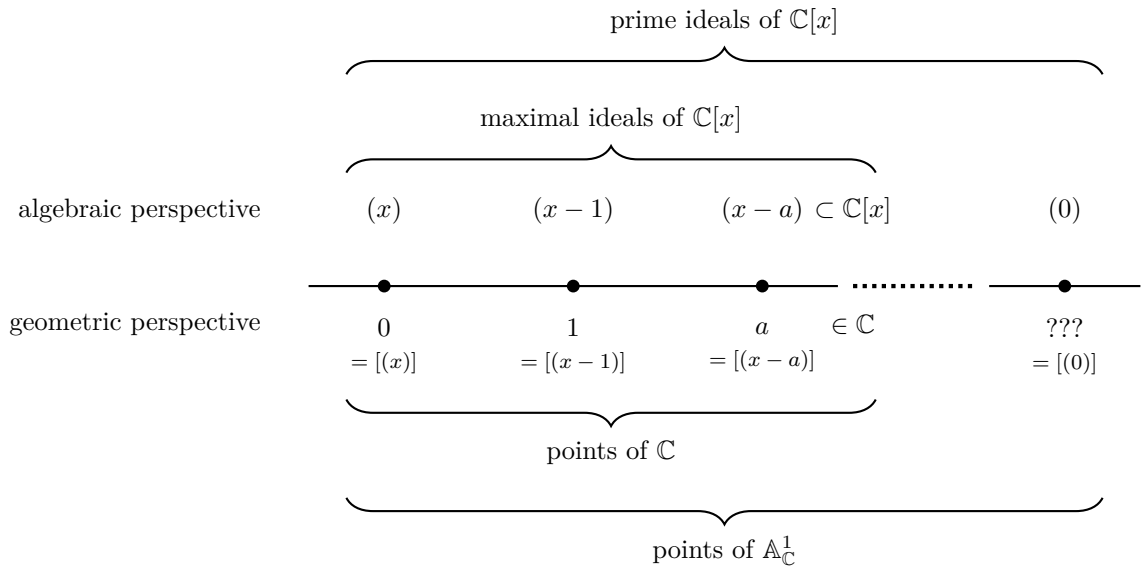
$$f(\mathfrak{p}) := \overline{f} \bmod \mathfrak{m}_\mathfrak{p} \in R_\mathfrak{p}/\mathfrak{m}_\mathfrak{p}$$

Let us now give multiple examples to get some grasp on spec and the functions on Spec

Example 2.1.3: Spectra and [regular] functions

This is a long set of examples, the reader should cover as many as they need to feel comfortable with the notion of spectrum; they may return here later for further elucidation on spectra.

1. Take $\operatorname{Spec}(\mathbb{C}[x])$ which consists of all prime (in fact maximal) ideals $(x-a)$ for $a \in \mathbb{C}$ and the prime ideal (0). This spectrum is denoted $\mathbb{A}^1_\mathbb{C}$; more generally the spectrum of $R[x_1, \ldots, x_n]$ is denoted $\mathbb{A}^n_R$. As $\mathbb{C}[x]$ is a finitely generated reduced ring over an algebraically closed field, the maximal ideals correspond exactly to the classical points of $\mathbb{A}^1_\mathbb{C}$ by the Nullstellensatz, with the exception of the special point (0) called the *generic point* which we shall return to soon. To visualize this set, imagine a real line (to remember that it is a 1-dimensional object over $\mathbb{C}$):



The generic point was located "away" from the line, but that is not in fact accurate. The problem is that it is a single point that is near every other point, while still being a single point. What it means for points to be near shall be defined shortly when the Zariski topology on spectra is introduced, but for now you can imagine that this causes issues when trying to draw such a point on a Euclidean piece of paper. There are many ways to visualize it: some write it as a cloud of points in the corner, others at some edge of the visual. Personally, I

find it fruitful to think of it as a "higher-dimensional" point that lives over all other points of $\mathbb{A}^1_{\mathbb{C}}$. This perspective shall be corroborated in section 2.5.

Let us evaluate some regular functions $f \in \mathbb{C}[x]$. Let's say $f = x^2+2x+1$ and input $[(x-3)]$. Then the answer is

$$16 \bmod (x-3) = f(3) \bmod (x-3)$$

which can be thought of as our usual answer $f(3)$! Notice that $\mathbb{C}[x]/(x-a) \cong \mathbb{C}$, and hence the codomain when evaluating at each point is the same. We shall see this is not the case soon. We can think of the fact that all the codomains of evaluation coincide as being a special symmetry for varieties over algebraically closed fields (that the evaluation information locally can all be embedded into a single global function-space).

Let us now plug in the generic point (0) into our function f (i.e. the point that is not in our usual visualization of the complex line). We get:

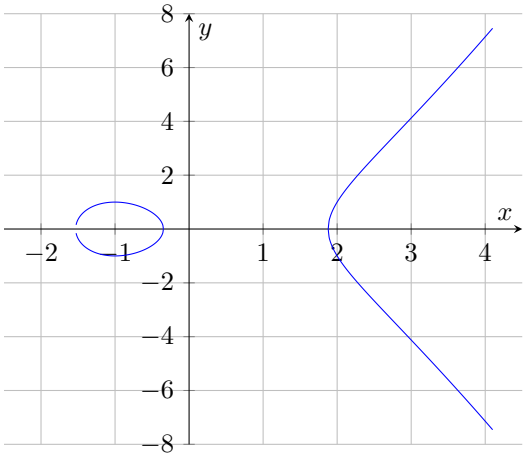
$$f \bmod 0 \in \mathbb{C}[x]$$

that is, it gives back the function itself, and is the only point for which the codomain of f is not isomorphic to $\mathbb{C}$. You can think of (0) as a "1"-[complex]-dimensional point. When we have spectra of higher dimensional (irreducible) schemes, say dimension n, then the generic point will be "n"-dimensional. We cannot yet define dimension as we are still working generically with sets; this shall be rectified in section 2.6.

More generally, this entire discussion works just as well over any algebraically closed field $k = \overline{k}$, as the completeness of $\mathbb{C}$ did not factor into the above results.

Often when working with $\overline{k}$-polynomials over several variables, we shall write the maximal ideals as tuples of points, e.g. $[(x-2, y-3)] \in \mathbb{A}^2_k$ will be represented as $(2,3) \in \mathbb{A}^2_k$. We shall often then denote the generic point $[(0)]$ as η to distinguish it from the point 0 which is associated to the ideal (x).

2. Take $y^2 - x^3 + 3x - 1 \in \mathbb{C}[x,y]$ and consider Spec $\frac{\mathbb{C}[x,y]}{(y^2-x^3+3x-1)}$. This spectrum corresponds to the elliptic curve $y^2 = x^3 - 3x + 1$. The spectrum will contain all the maximal ideals of the ring $\frac{\mathbb{C}[x,y]}{(y^2-x^3+3x-1)}$ as well as a generic point η corresponding to $[(0)]$ representing the entire space, and hence can be visualized as:



More generally, all the points of an affine algebraic set correspond to the maximal ideals of a spectrum, and so a subset of the geometry of spectra includes the classical geometry of curves, surfaces, and so forth!

3. Here is a Spectra that has no corresponding algebraic set in the classical sense: $\text{Spec}(\mathbb{Z}) = \{(2), (5), (7), \cdots, (p), ..., (0)\}$. In other words, $\text{Spec}(\mathbb{Z}) = \{(p) \big| p \text{ is prime}\} \cup \{0\}$ and $\text{mSpec}(\mathbb{Z}) = \text{Spec}(\mathbb{Z}) \backslash \{(0)\}$.

   Let us evaluate some of the regular function $n \in \mathbb{Z}$ on the points $\text{Spec}\,(\mathbb{Z})$. The regular function 15 at the point $[(7)] \in \text{Spec}\,(\mathbb{Z})$ is $15 \bmod (7) = 1 \bmod (7)$. The regular function 9 evaluated at $[(3)]$ is $0 \bmod (3)$. We may even say it has a double-zero at 3.

   Hence, functions on $\text{Spec}\,(\mathbb{Z})$ return some information about divisibility by primes.

4. For any field k, $\text{Spec}\,k = \text{mSpec}\,k = \{(0)\}$ (or $\text{Spec}\,k = \{\eta\}$). For example, $\text{Spec}\,\mathbb{Z}/3\mathbb{Z} = \{(0)\}$. Thus, fields are like "points" within algebraic geometry. An advantage of adding a structure sheaf will be to remember the original field k: we shall see that as schemes $\text{Spec}\,K \ncong \text{Spec}\,L$ if $K \ncong L$, as their sheaves will differ. Though the geometric picture as spectra is trivial, they shall still hold valuable non-trivial information.

   The evaluation of each function at the single point returns the function modulo (0), namely the function itself as an element of k:

$$3 \bmod 0 = 3 \in k$$

5. Take $\text{Spec}\,\mathbb{R}[x]$. This consists of $[(0)]$, all prime (even maximal) ideals of the form $[(x-a)]$, and all prime (even maximal) ideals that can be represented by an irreducible 2nd degree polynomial $\left[\left(x^2+bx+c\right)\right]$, for example $\left[\left(x^2+1\right)\right]$. Evaluating at any of these points (besides $[(0)]$) will give a field: if evaluating points of the form $(x-a)$ the field will be isomorphic to $\mathbb{R}$, and it evaluating points of the form $\left(x^2+ax+b\right)$, the field will be isomorphic to $\mathbb{C}$. As every irreducible quadratic splits into conjugate pairs of linear terms over $\mathbb{C}$, we can think of $\text{Spec}(\mathbb{R}[x])$ as being a copy of $\mathbb{C}$ folded over itself attaching the conjugate pairs (do not continue reading until this is clear)
   Let us take the function $x^3 - 1$ and evaluate it as some point. At the point $(x-2)$ we get $7 \bmod (x-2)$, or in our classical interpretation, $f(2)$. At the point $\left(x^2+1\right)$, we get:

$$x^3 - 1 \equiv -x - 1 \bmod \left(x^2+1\right)$$

   which we can think of as being $-i-1$. Hence, the Spectra of polynomials over fields that are not algebraically closed will behave more like algebraically closed fields as the non-linear maximal primes shall play the role of the required "additional points"! Hence, one piece of the puzzle for the "additional points" in a spectra is that they allow us to think of the spectrum as being "algebraically closed"!

6. Generalizing the above result, show that the points of $\text{Spec}\,\mathbb{Q}[x]$ (minus the generic point 0) are in bijective correspondence with the points of $\overline{\mathbb{Q}}$, the algebraic closure of $\mathbb{Q}$, where the points of $\overline{\mathbb{Q}}$ are glue to their galois-conjugates: $\text{Gal}_{\mathbb{Q}}(\overline{\mathbb{Q}})$.

7. Consider $\text{Spec}\,\mathbb{Z}[x]$. Unlike $\mathbb{Q}[x]$, the ring $\mathbb{Z}[x]$ is not a PID and has Krull dimension 2, As it is not a PID, it has prime ideals of the form $(p, f(x))$ where $f(x)$ and $f(x) \bmod p$ is irreducible. This is the first example where the irreducible functions of a polynomial ring

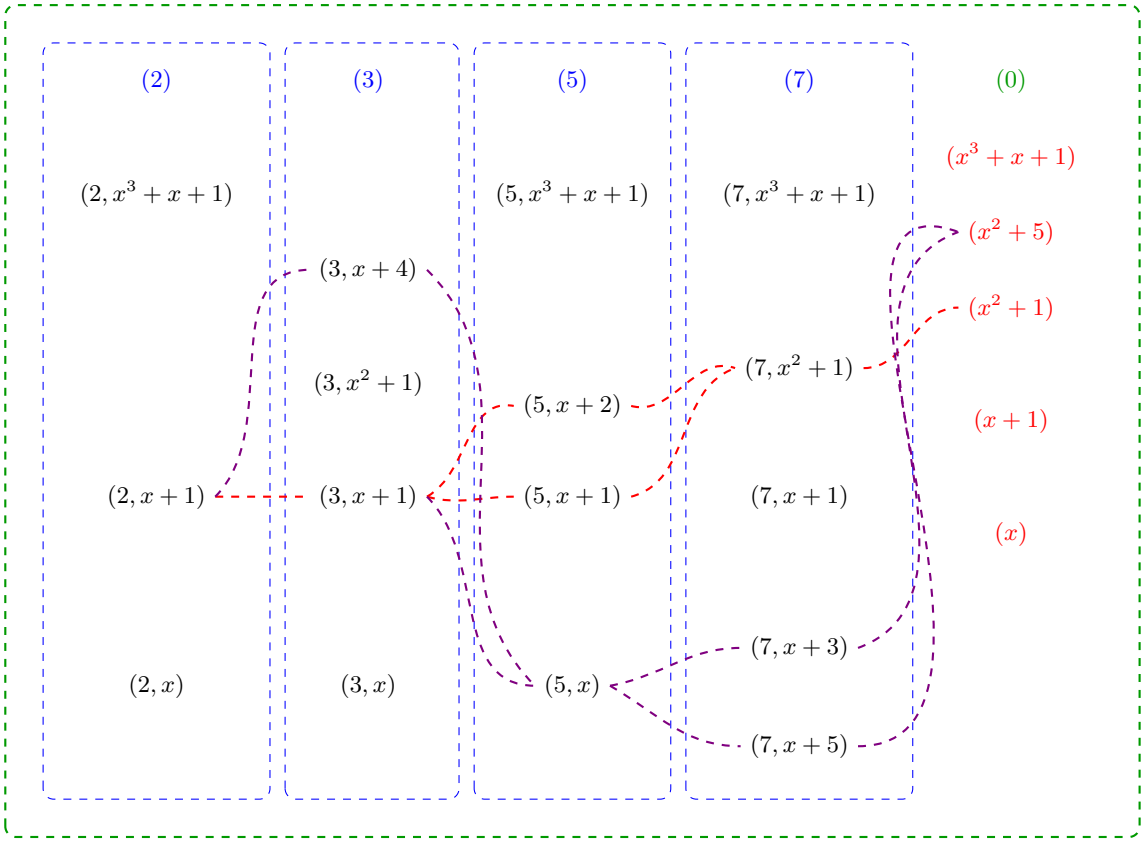
are not in one-to-one correspondence with the points, there are more points than there are irreducible functions[a]! The maximal ideals are of the form

$$(p, f(x))$$

while the prime ideals are of the form:

$$(p) \quad (f(x)) \quad (p, f(x))$$

There are now non-maximal prime ideals, namely the principal ideals (p) for prime p and principal ideals $(f(x))$ where $f(x)$ is irreducible over $\mathbb{Z}$ (or equivalently by Gauss's lemma over $\mathbb{Q}$). To understand this spectrum better, the following is a part of a visual of Spec $\mathbb{Z}[x]$:



Let us parse this image: the maximal points are all of the form $(p, f(x))$ and they are the black points. The prime (p) are contained in each maximal ideal $(p, f(x))$. They are like a more specialized generic point, where they are not near all the points, but near every point for which the ideal representing the point contains (p). Due to this, each column is surround by dotted blue lines that indicating the generic location of each prime (p). The entire image is surround by a green dotted line which represents the generic point $[(0)]$ and how it is close to every point. Notice that there are gaps in the columns exactly where the irreducible polynomial $f(x)$ is reducible mod$p$. For example $x^2 + 1$ mod 2 is reducible with solution $x = 1$, and so $\left(2, x^2 + 1\right)$ is not prime, namely given $\mathbb{Z}[x]/\left(2, x^2 + 1\right) = (\mathbb{Z}/2\mathbb{Z})[\omega]$ we have

$$(\omega+1)(\omega+1)=\omega^2+2\omega+1=1+1=0$$

showing that $\left(2,x^2+1\right)$ is not prime, and hence not a point.
Next, each point $(f(x))$ is also a generic point, however their "shape" is not a simple as the points (p). The simplest case is when a point $(p,g(x))$ contains $f(x)$, namely $f(x)=g(x)$. In the image, we see that $\left[\left(3,x^2+1\right)\right]$ is a point and hence $\left[\left(x^2+1\right)\right]$ is near this point. On the other hand, both $(5,x+2),(5,x+1)$ are contained in $\left(x^2+1\right)$, precisely because x^2+1 factors into a product of those two linear terms mod 5 . Hence $\left[\left(x^2+1\right)\right]$ is near $[(5,x+2)]$ and $[(5,x+1)]$. This closeness is shown via a red-line passing near these points and represents the fact that $\left[\left(x^2+1\right)\right]$ is near each of these points. Essentially, this represents the quadratic residues of x^2+1 !
The same thing has been done for the point $\left[\left(x^2+5\right)\right]$. Some extra points have been added to the image to demonstrate more clearly the branching behavior. Notice that $\left[\left(x^2+5\right)\right]$ and $\left[\left(x^2+1\right)\right]$ "intersect", for example at $[(2,x+1)]$ and at $[(13,x+5)]$. These give us some number-theoretic information about these equations!
The way I like to interpret these "fuzzy" points corresponding to the irreducible polynomials from some intuition given by complex analysis. This is not unreasonable as the spectrum in many ways mirror the behavior of algebraically closed fields and complex analysis is the analysis of functions that are locally infinite polynomials, and indeed these two fields are very closed linked, something section 6.9 takes up.
For now, consider the following: if we have a "germ" around a single point of a holomorphic function (i.e. the information given by its power-series representation around an arbitrarily small neighborhood around the point), we in fact have information about the entire holomorphic function. From this perspective, it does not matter at which point we choose to take the power-series representation, it will return equivalent information. Thus, the information does not live at any point point in particular but "generically" at all these points. If we choose to "specialize" and specify which data we want, we shall evaluate further to a single maximal ideal.
Notice that $\mathbb{Z}[x]$ is seen on a two-dimensional grid. We shall later see that $\operatorname{Spec}\mathbb{Z}[x]$ is naturally seen as a 2 -dimensional scheme, while $\operatorname{Spec}\mathbb{Q}[x]$ shall be 1-dimensional. Furthermore, the red-lines and blue dotted lines will be seen as 1-dimensional points, the generic point will be seen as a 2 -dimensional point, while all the maximal ideals will be 0 -dimensional points.

Finally, let us consider evaluation of functions $f\in\mathbb{Z}[x]$ at points. At maximal points, we shall get values in the field $\mathbb{F}_p(\omega)$[b] where ω is a root of the irreducible polynomial. For example, take $[(7,x^3+x+1)]\in\operatorname{Spec}\mathbb{Z}[x]$ and $x^2-4\in\mathbb{Z}[x]$. Then the value will be in $(\mathbb{Z}/7\mathbb{Z})(\omega)$ where ω is an element such that $\omega^3+\omega+1=0$. Then we would get that the value of x^2-4 is $\omega^2+3\in\mathbb{F}_7(\omega)$.

Another example would be evaluating at the point $[(3,x^2+1)]$ the function x^4-1. Working in $\mathbb{Z}[x]/(3,x^2+1)\cong\mathbb{F}_3[x]/(x^2+1)\cong\mathbb{F}_9$, we use $x^2\equiv-1$ to compute:

$$x^4-1\equiv(x^2)^2-1\equiv(-1)^2-1=1-1=0\pmod{3,\,x^2+1}$$

In other words, $f([(3,x^2+1)])=0\in\mathbb{F}_3(i)$. As we are evaluating at maximal ideals, we shall always get a "number" as the output; in particular, since we can always embed finite fields into an algebraic closure, the output of a function at a maximal point always lies within an algebraically closed field.

Let us now evaluate at point that corresponds to a non-maximal prime ideal, say at 3. The output of each function $f\in\mathbb{Z}[x]$ will then be:

$$\bar{f} \in (\mathbb{Z}/3\mathbb{Z})[x] = \mathbb{F}_3[x]$$

The way I like to think about these output is that we are asking for some of the information stored in f, namely it shall return enough information about f to determine the generic behavior of that function around $[(3)]$. If we wanted the generic behavior around a different generic point, for example $\left[\left(x^2+1\right)\right]$, then we would get:

$$f(i) \in \mathbb{Z}[i]$$

Namely, if we choose any irreducible polynomial, we are choosing which new element of an integral extension we are adding that we then evaluate the function f at. We may then extract further information by modding by different primes. It shall often be the case that evaluating at a generic point corresponds to reducing the information of a function down to a smaller geometric object or to modding information.

8. If you know some number theory, $\operatorname{Spec}\mathbb{Z}[\sqrt{-5}]$ is the quintessential example of a ring that has primes that are *not* principal, for example $(2, 1+\sqrt{-5})$ is prime but is provably not principal (see [7, Class Groups]). Evaluating at (2) and $(1+\sqrt{-5})$ will return 0, however the zero set of both of these function's shall be larger than $(2, 1+\sqrt{-5})$; we shall require the vanishing set to contain both of these.

9. The following is another example of a spectrum with a single element. Take $\operatorname{Spec}(k[x]/(x^2))$, the spectrum of a ring with nilpotent elements. This ring can be thought of as an extension of k by an element $\epsilon = x \bmod x^2$ that is thought to be "infinitely small". It has the interesting property that $\epsilon^2 = 0$, even though ϵ itself is nonzero. As the spectrum only contains one point[c], namely $[(\epsilon)]$, this can be thought of as a "thicker" point (as shall be explored in greater detail in section 2.2.2). The ring $k[x]/(x^2)$ is called the *dual numbers* and shall be a recurring ring to study the behaviour of nilpotent elements. This ring has the interesting property of having the element $\epsilon \in k[\epsilon]$ which always evaluates to 0 even though it is not the zero element! This gives us our first example of a function[d] that is not defined by what it does at every point, namely because the functions 0 and ϵ have the same outputs.

   More generally, $k[x]/(x^n)$ has a single element in its spectrum. As k and $k[x]/(x^n)$ have the same underlying topological space, what shall distinguish them will be the sheaf structure associated to these two spectra (they shall not be isomorphic as schemes).

10. The above ring only had one point in its spectrum. There are many spectrum's with only two points: a generic point and a non-generic point. A large class of such rings shall be *local rings*. For example $k[x]_{(x-a)}$ will have (0) and $(x-a)$ as the two points of the spectrum. Another example example of a spectrum with only two points is $k[[x]]$ which has (x) and (0) as its two points. As spectra, $k[x]_{(x)}$ and $k[[x]]$ are indistinguishable. What shall distinguish these two spectra shall be the sheaf-structure associated to these two spectra.

11. Let us take a look at the number theoretic equivalent of localization: take $\mathbb{Z}_{(p)}$, the localization of $\mathbb{Z}$ at $\mathbb{Z}\backslash(p)$. This spectrum contains two points, $[(0)]$ and $[(p)]$. In particular, $\operatorname{Spec}\mathbb{Z}_{(p)}$ is not a single point like $\operatorname{Spec}\mathbb{F}_p$, it somehow has some more "fuzz" around it's single point $[(p)]$. This can be thought of as $\operatorname{Spec}\mathbb{Z}_{(p)}$ "remembering" that there is some arbitrarily small neighborhood around the point $[(p)]$ that it must keep track of. Once we introduce a topology on spectra's, we will see that the closure of $[(p)]$ is itself (it is a closed point), while

the closure of $[(0)]$ is $\{[(0)], [(p)]\}$, motivating the idea that $[(0)]$ "remembers" that there is some more information outside of just the single point! This shall be even more clear once we introduce the notion of dimension: the point $[(p)]$ will be zero-dimensional, while $[(0)]$ will be one-dimensional!

12. Let $R = k[x] \times k[x]$. What is $\operatorname{Spec} R$? Notice that $\operatorname{Spec} R$ does not contain just one generic point, but two! Importantly, $[(0)]$ is not prime as $(1,0) \cdot (0,1) \in (0)$ but $(1,0), (0,1) \notin (0)$.

    In general, the primes of a product ring $A \times B$ are all of the form $\mathfrak{p} \times B$ or $A \times \mathfrak{q}$ where $\mathfrak{p} \in \operatorname{Spec}(A)$ and $\mathfrak{q} \in \operatorname{Spec}(B)$ (exercise). Hence $\operatorname{Spec}(A \times B) \cong \operatorname{Spec}(A) \sqcup \operatorname{Spec}(B)$: the spectrum of a product is the disjoint union of the spectra. In our case, $\operatorname{Spec}(k[x] \times k[x]) \cong \mathbb{A}^1_k \sqcup \mathbb{A}^1_k$, two disjoint copies of the affine line, each with its own generic point.

13. Take $\operatorname{Spec} \mathbb{F}_p[x] = \mathbb{A}^1_{\mathbb{F}_p}$. As it is a Euclidean domain, the points are (0) and $(f(x))$ for all irreducible polynomials over $\mathbb{F}_p$. One of the interesting facets of $\mathbb{A}^1_{\mathbb{F}_p}$ is that its closed points are far richer than the p elements of $\mathbb{F}_p$: for each $n \geq 1$, each irreducible polynomial of degree n over $\mathbb{F}_p$ gives a closed point whose residue field is $\mathbb{F}_{p^n}$. Since there are irreducible polynomials of every degree, $\mathbb{A}^1_{\mathbb{F}_p}$ has infinitely many closed points despite $\mathbb{F}_p$ being finite. This richness in points will allow us to distinguish functions on $\mathbb{F}_p$ by their evaluation: recall that a polynomial is not distinguished by evaluation at all points in $\mathbb{F}_p$ (e.g. $x^p - x$ vanishes everywhere on $\mathbb{F}_p$ but is nonzero), however a polynomial *will* be distinguished by its evaluation at all points in $\operatorname{Spec}(\mathbb{F}_p[x])$, as we shall show using proposition 2.1.4.

14. Take $\operatorname{Spec}(\overline{k}[x,y]) = \mathbb{A}^2_{\overline{k}}$ (e.g. $\overline{k} = \mathbb{C}$). Then the points are:

$$(0) \qquad (x-a, y-b) \qquad (f(x,y))$$

    where $f(x,y)$ is an irreducible polynomial in $\overline{k}[x,y]$. Visually, all the primes of the form $(x-a, y-b)$ correspond to the usual points in $\mathbb{C}^2$. The points $(f(x,y))$ can be thought of as tracing out their curve in $\mathbb{C}^2$; for example $(y - x^2)$ is the point that is the curve $y = x^2$. Notice how the maximal ideals correspond to "0"-dimensional points, while the ideal $(y - x^2)$ corresponds to a "1"-dimensional point. Similarly, (0) is a "2"-dimensional point. We shall come back to this through Krull dimensions in section 2.6.

    More generally, by the Nullstellensatz the maximal points of $\overline{k}[x_1, \ldots, x_n]$ are the ideals of the form:

$$(x_1 - a_1, x_2 - a_2, \ldots, x_n - a_n)$$

    The prime ideals of this ring do not lend themselves to a nice classification. Each prime ideal corresponds to an irreducible variety of a certain dimension, and hence the classification of the prime ideals is equivalent to the classification of irreducible varieties, which is a rather difficult task that is carried out in special cases, and more generally leads to the theory of moduli spaces. Some context on this classification in a classical setting was covered in [22].

15. Find the points of $\operatorname{Spec}(k[x_1, \ldots, x_n])$ where k is not algebraically closed, ex. $\mathbb{Q}$; generalize example 5[e].

[a] This sort of exploration on the correspondence between objects that "principal" (or locally principal) and how much they are not will be the focus of section 5.4

[b] Recall that $k[\omega] = k(\omega)$, or re-prove this

[c] Prove that $[(0)]$ is not a prime ideal: $\epsilon \cdot \epsilon = 0 \in (0)$ but $\epsilon \notin (0)$

[d] granted, a generalization of the notion of function

[e] If you solved it, note that $(\sqrt{2}, \sqrt{2})$ is glued to $(-\sqrt{2}, -\sqrt{2})$ but *not* to $(\sqrt{2}, -\sqrt{2})$

2.1.1 Zariski Topology

Having defined the set that represents the space, the next thing to do is give a topology to this set. This topology would have to be a generalization of the Zariski topology defined on $k[x_1, ..., x_n]$ using the definition of regular function that was just introduced:

Definition 2.1.4: Zariski Topology

Let R be a commutative ring and $\text{Spec}\,(R)$ the collection of prime ideals of R. Then define the *closed sets* for the Zariski topology on $\text{Spec}\,(R)$ to be:

$$V(S) = \{x \in \text{Spec}\,(R) \mid f(x) = 0,\ \forall f \in S\} = \{[\mathfrak{p}] \in \text{Spec}\,(R) \mid S \subseteq \mathfrak{p}\}$$

Take a moment to see how this parallels the condition that all the functions in the coordinate ring must equal zero when evaluated at $\mathfrak{p}$. the notation $f(x) = 0$ will eventually be correctly re-interpreted when we show the sheaf naturally induced by R is a *locally ringed space*, and hence evaluation is well-defined for any ring element. These sets are the closed sets of a topology:

Lemma 2.1.5: Zariski-closed Sets Form Topology

Let $\text{Spec}\,(R)$ be the collection of primes and let:

$$\mathcal{T} = \{V(S) \mid S \subseteq R\}$$

Then:

1. $\mathcal{T}$ contain the empty set and $\text{Spec}\,(R)$ (these are both the vanishing set of single elements)
2. $\mathcal{T}$ is closed under arbitrary intersections and finite unions, namely, for ideals I_i and an arbitrary indexing set:
$$\cap_i V(I_i) = V\left(\sum_i I_i\right)$$
and
$$V(I_1) \cup V(I_2) = V(I_1 I_2)$$

Furthermore, for any ideal $I \subseteq R$ with quotient map $\pi\colon R \to R/I$, the map $\text{Spec}\,(R/I) \to \text{Spec}\,(R)$, $\mathfrak{q} \mapsto \pi^{-1}(\mathfrak{q})$, is a homeomorphism onto the closed subset $V(I)$, so that $V(I) \cong \text{Spec}\,R/I$ as topological spaces[a].

[a] and as sheaves, which we will see later

***Proof*:**
A prime contains a subset S exactly when it contains the ideal (S) generated by S, so $V(S) = V((S))$ and every member of $\mathcal{T}$ has the form $V(I)$ for an ideal I.

1. No prime contains 1 and every prime contains 0, so $V(\{1\}) = \emptyset$ and $V(\{0\}) = \text{Spec}\,(R)$.
2. A prime contains every I_i if and only if it contains the ideal $\sum_i I_i$ they generate, which gives

the first equality. For the second, $I_1 I_2 \subseteq I_1$ and $I_1 I_2 \subseteq I_2$, so $V(I_1) \cup V(I_2) \subseteq V(I_1 I_2)$. Conversely, if a prime $\mathfrak{p}$ contains neither I_1 nor I_2, choose $a \in I_1 \setminus \mathfrak{p}$ and $b \in I_2 \setminus \mathfrak{p}$. Then $ab \in I_1 I_2 \setminus \mathfrak{p}$, so $\mathfrak{p} \notin V(I_1 I_2)$.

For the last assertion, write $\pi^a(\mathfrak{q}) = \pi^{-1}(\mathfrak{q})$. This is a prime ideal of R: it is proper because $\pi(1) = 1 \notin \mathfrak{q}$, and $ab \in \pi^{-1}(\mathfrak{q})$ gives $\pi(a)\pi(b) \in \mathfrak{q}$, so a or b lies in $\pi^{-1}(\mathfrak{q})$. It contains $I = \pi^{-1}(0)$, so π^a maps into $V(I)$.

Step 1: π^a is a bijection onto $V(I)$. Since π is surjective, $\pi(\pi^{-1}(\mathfrak{q})) = \mathfrak{q}$, so π^a is injective. Let $\mathfrak{p} \in V(I)$. Then $\pi(\mathfrak{p})$ is an ideal of R/I, π being surjective, and $\pi^{-1}(\pi(\mathfrak{p})) = \mathfrak{p} + I = \mathfrak{p}$. It is proper, since $1 \in \pi(\mathfrak{p})$ would give $1 \in \pi^{-1}(\pi(\mathfrak{p})) = \mathfrak{p}$, and if $\pi(a)\pi(b) \in \pi(\mathfrak{p})$ then $ab \in \pi^{-1}(\pi(\mathfrak{p})) = \mathfrak{p}$, so $a \in \mathfrak{p}$ or $b \in \mathfrak{p}$. Hence $\pi(\mathfrak{p})$ is a prime of R/I with $\pi^a(\pi(\mathfrak{p})) = \mathfrak{p}$.

Step 2: π^a is continuous. For $S \subseteq R$, $\pi^{-1}(\mathfrak{q}) \supseteq S$ if and only if $\mathfrak{q} \supseteq \pi(S)$, so $(\pi^a)^{-1}(V(S)) = V(\pi(S))$ is closed.

Step 3: π^a maps closed sets onto closed sets. For $T \subseteq R/I$, $\mathfrak{q} \supseteq T$ if and only if $\pi^{-1}(\mathfrak{q}) \supseteq \pi^{-1}(T)$, the reverse implication because $T = \pi(\pi^{-1}(T)) \subseteq \pi(\pi^{-1}(\mathfrak{q})) = \mathfrak{q}$. With Step 1 this gives $\pi^a(V(T)) = V(\pi^{-1}(T)) \cap V(I) = V(\pi^{-1}(T) \cup I)$, a closed subset of $\operatorname{Spec}(R)$.

By Steps 1 to 3, π^a is a continuous bijection onto $V(I)$ that carries closed sets to closed sets, so it is a homeomorphism onto $V(I)$, completing the proof.

Note that the vanishing set $V(-)$ defined for the spectra of rings usually have more points than their counterpart for coordinate rings in classical algebraic geometry as they will contains primes that are not maximal ideals (particularly for rings of "dimension" ≥ 2 as we have briefly discussed), and these primes will be within $V(S)$ for appropriate S.

The reader should check how this topology compares to the topology for classical algebraic sets. For example, as there exist generic points in most spectra we have that this space is no longer T_1 (it shall soon be shown that generic points are not closed). If R is Noetherian, then $\operatorname{Spec} R$ is a Noetherian topological space (lemma 2.4.2). See section 2.1.3 for more comparisons.

Just like for varieties, we may define the inverse function $I(-)$, that is for $S \subseteq \operatorname{Spec}(R)$ (with $I(\emptyset)$ the empty intersection, which is all of R),

$$I(S) \subseteq R \qquad I(S) = \bigcap_{[\mathfrak{p}] \in S} \mathfrak{p}$$

Hence $I(S)$ is a radical ideal (recall the intersection of prime ideals is a radical ideal, think $2\mathbb{Z} \cap 3\mathbb{Z} = 6\mathbb{Z}$). Just like in classical algebraic geometry:

Proposition 2.1.6: Properties of $I(-)$ and $V(-)$

1. $I(S)$ is an ideal of R, and furthermore it is always *radical*.
2. $I(-)$ is inclusion-reversing
3. $I(A \cup B) = I(A) \cap I(B)$, and, for closed A and B, $I(A \cap B) = \sqrt{I(A) + I(B)}$.
4. $I(\overline{S}) = I(S)$
5. $V(I(S)) = \overline{S}$ and $I(V(J)) = \sqrt{J}$.
6. If S is closed, then $V(I(S)) = S$. If J is radical, then $I(V(J)) = J$.
7. $I(-)$ and $V(-)$ are an inclusion-reversing bijection between closed subsets of $\operatorname{Spec}(R)$ and radical ideals of R. This is called a *galois correspondence*.

Hence:

$$\text{closed subset} \subseteq \operatorname{Spec}(R) \quad \Leftrightarrow \quad \text{radical ideal of } R$$

***Proof*:**
For any ideal J:

$$I(V(J)) = \bigcap_{\mathfrak{p} \supseteq J} \mathfrak{p} = \sqrt{J}$$

For proper J this is [10, The Radical as an Intersection of Primes]; for $J = R$ both sides equal R, as $V(R) = \emptyset$.

1. $I(S)$ is an intersection of ideals, hence an ideal. It is radical: if $f^n \in I(S)$, then $f^n \in \mathfrak{p}$ for every $[\mathfrak{p}] \in S$, and primes are radical, so $f \in \mathfrak{p}$ for every such $\mathfrak{p}$, so $f \in I(S)$.
2. If $S \subseteq T$, the intersection defining $I(T)$ runs over more primes than that defining $I(S)$, so $I(T) \subseteq I(S)$.
3. $I(A \cup B) = \bigcap_{[\mathfrak{p}] \in A \cup B} \mathfrak{p} = I(A) \cap I(B)$. Now let A and B be closed, say $A = V(\mathfrak{a})$ and $B = V(\mathfrak{b})$ for ideals $\mathfrak{a}, \mathfrak{b}$ (definition 2.1.4). Then $A \cap B = V(\mathfrak{a} + \mathfrak{b})$ (lemma 2.1.5), so the opening computation gives
   $$I(A \cap B) = \sqrt{\mathfrak{a} + \mathfrak{b}}$$
   On the other side, the opening computation gives $I(A) = \sqrt{\mathfrak{a}}$ and $I(B) = \sqrt{\mathfrak{b}}$, and
   $$\sqrt{\sqrt{\mathfrak{a}} + \sqrt{\mathfrak{b}}} = \sqrt{\mathfrak{a} + \mathfrak{b}}$$
   since $\mathfrak{a} + \mathfrak{b} \subseteq \sqrt{\mathfrak{a}} + \sqrt{\mathfrak{b}} \subseteq \sqrt{\mathfrak{a} + \mathfrak{b}}$, taking radicals is monotone, and $\sqrt{\sqrt{\mathfrak{c}}} = \sqrt{\mathfrak{c}}$ for any ideal $\mathfrak{c}$.
4. Since $S \subseteq \overline{S}$, item 2 gives $I(\overline{S}) \subseteq I(S)$. Conversely let $f \in I(S)$. Then $S \subseteq V((f))$, a closed set, so $\overline{S} \subseteq V((f))$, meaning $f \in \mathfrak{p}$ for every $[\mathfrak{p}] \in \overline{S}$. So $f \in I(\overline{S})$.
5. The set $V(I(S))$ is closed and contains S, since each $[\mathfrak{p}] \in S$ contains the intersection $I(S)$. If $V(J)$ is any closed set containing S, then $J \subseteq \mathfrak{p}$ for every $[\mathfrak{p}] \in S$, so $J \subseteq I(S)$, so

$V(I(S)) \subseteq V(J)$. Hence $V(I(S))$ is the smallest closed set containing S, which is $\overline{S}$. The identity $I(V(J)) = \sqrt{J}$ is the opening computation.

6. If S is closed, then $\overline{S} = S$, so item 5 gives $V(I(S)) = S$. If J is radical, then item 5 gives $I(V(J)) = \sqrt{J} = J$.

7. By item 1, $I(-)$ maps closed subsets to radical ideals, where the empty set goes to R, and $V(-)$ maps radical ideals to closed subsets, where R goes to the empty set. Item 6 states that the two composites are the identities, so the maps are mutually inverse bijections. Both reverse inclusion, by item 2 and by the definition of $V(-)$.

Thus, we get the (surjective) correspondence:

$$\{\text{Radical ideals of } R\} \underset{V(-)}{\overset{I(-)}{\rightleftarrows}} \{\text{closed sets of Spec}\,(R)\} \tag{2.2}$$

Notice that prime ideals are radical and hence are part of this association. This correspondence does not extend to all ideals until a sheaf, the structure sheaf, is introduced (see section 2.2.2).

As we see, the Zariski topology on Spec (R) is a generalization of the Zariski topology on $k[x_1, ..., x_n]$. The closed sets of Spec (R) will contain the "traditional" points as well as the new points, for example $V(xy, yz)$ will contain all the "traditional" points where $y = 0$ or $x = z = 0$, as well as the generic point that vanish on $V(xy)$ and $V(yz)$ (i.e. $[(y)]$ vanishes on this set). Just like for affine varieties, Spec (R) is almost never Hausdorff. This is even clearer in Spec (R), for if there are primes that are not maximal, these are *not* closed points (in a Hausdorff or T_1 space, every point is closed)

Example 2.1.7: Non-closed Points
Take $\mathbb{A}^2_{\mathbb{C}}$ with $[(y^2 - x)]$ and $[(x - 2, y - 4)]$. The point $[(y^2 - x)]$ is not closed. Notice that:

$$[(y - 4, x - 2)] \in V(I([(y^2 - x)])) = V(y^2 - x) \overset{\text{pr. } 2.1.6}{=} \overline{\{[(y^2 - x)]\}}$$

Showing that the point is *not* closed, namely points associated to non-maximal prime ideals are not closed. However, notice the set $V(I([(y^2 - x)]))$ is closed by definition. In a moment, the point $[(y^2 - x)]$ will be seen as the generic point of the above closed set.

In fact, due to Hilbert's Nullstellensatz, we know that in $\mathbb{A}^n_k$ the maximal ideals correspond to the "traditional points", that is the closed points. From here on, we shall call the "traditional points" (more generally points associated to maximal ideals) the *closed points*, and the "extra" or "bonus" points that were being referred to shall be called the *non-closed points* or *generic points*.

With a topology, we can now link points like $[(x - 2, y - 4)]$ to points like $[(y - x^2)]$. As these are point, it does not make sense to say one point contains another. However we we be able to link these points using closure:

Definition 2.1.8: Specialization And Generization

Let $x, y \in X$ be two points in a topological space. Then we say that x is a *specialization* of y and y a *generization* of x if

$$x \in \overline{\{y\}}$$

Algebraically, $[\mathfrak{q}]$ is a specialization of $[\mathfrak{p}]$ if and only if $\mathfrak{p} \subseteq \mathfrak{q}$, hence $V(\mathfrak{p}) = \overline{\{[\mathfrak{p}]\}}$.

We can now define generic point more precisely:

Definition 2.1.9: Generic Point

Let X be a topological space. Then $p \in X$ is a *generic point* for a closed subset K iff

$$\overline{\{p\}} = K$$

By this definition, closed points are technically generic points associated the closet set of a maximal idea; we shall only use the term generic point for closed sets that are not given by a maximal ideal. With generic points and the Zariski topology defined, the idea that generic points of a closed set (including all of $\operatorname{Spec}(R)$) are close to all the points in the closed subset (it is contained in all the open subsets (or all the open subsets of $V(S) \cap U$).

Nilpotent elements do not change the topology:

Proposition 2.1.10: Nilpotent Spectrum Homeomorphism

Let R be a ring and I an ideal of nilpotent elements. Then

$$\operatorname{Spec} R/I \to \operatorname{Spec} R$$

is a homeomorphism.

***Proof*:**
Let $\mathfrak{p}$ be a prime of R and $x \in I$. Then $x^n = 0 \in \mathfrak{p}$ for some n, and primality of $\mathfrak{p}$ gives $x \in \mathfrak{p}$. So every prime contains I, that is, $V(I) = \operatorname{Spec}(R)$. By lemma 2.1.5, the map $\operatorname{Spec}(R/I) \to \operatorname{Spec}(R)$, $\mathfrak{q} \mapsto \pi^{-1}(\mathfrak{q})$, where $\pi\colon R \to R/I$ is the quotient map, is a homeomorphism onto $V(I) = \operatorname{Spec}(R)$, as claimed.

Non-closed points shall have their use in maps such as $k[x, y] \to k[x](y)$ as well as in intersection theory where the intersection or "creases" of schemes shall often be generic points. We shall see that there is a "local-ring" associated to both closed and non-closed points alike and their reduction behaviour is the same (though naturally being in different dimensions means that the behaviour may differ in exactitude's).

2.1.2 (Topological) Basis: Distinguished Open Sets

As the topology of $|\operatorname{Spec}(R)|$ is generated by the closed sets $V(S)$, and these sets are the intersection of:

$$V(S) = \bigcap_{f \in S} V(f)$$

the sets $V(f)$ generate the topology of $\operatorname{Spec}(R)$. We may want to call these sets "distinguished closed sets", however we shall favor their open counterpart due to their simpler behaviour when introducing sheaf theory to spectra[9]:

Definition 2.1.11: Distinguished Open Sets

The sets $D(f)$ are called the *distinguished open sets*, *basic open sets*, or *principal open sets*, and are defined to be

$$D(f) = X_f = |\operatorname{Spec}(R)| \setminus V(f) = \{\mathfrak{p} \in \operatorname{Spec}(R) \mid f \notin \mathfrak{p}\}$$

that is, it is the collection of prime ideals of R that do not contain f. Any open set U can be written as:

$$U = \operatorname{Spec}(R) \setminus V(S) = \operatorname{Spec}(R) \setminus \bigcap_{f \in S} V(f) = \bigcup_{f \in S} D(f)$$

Writing it as $D(f)$ may be helpful; Prof. Ravi Vakil calls these "does not-vanish sets" which may serve as a helpful mnemonic. These sets are usually not closed (especially if $\operatorname{Spec}(R)$ is connected, in which case unless $D(f) = \emptyset$ it cannot be), and hence they are not the vanishing set for any functions in R. However, they are still isomorphic to a spectrum:

Proposition 2.1.12: Distinguished Open Sets and Localization

Let $D(f) \subseteq \operatorname{Spec}(R)$ be a distinguished open set. Then:

$$D(f) \cong_{\mathbf{Top}} \operatorname{Spec}(R_f)$$

justifying the X_f notation. Furthermore, if $g = f_1 f_2 \cdots f_n$, then

$$\bigcap_{i=1,\cdots,n} (\operatorname{Spec} R)_{f_i} = (\operatorname{Spec} R)_g$$

or as distinguished open sets if $D(f_1), \ldots, D(f_n) \subseteq \operatorname{Spec}(R)$, then their intersection is the distinguished open

$$D(f_1 \cdots f_n)$$

***Proof*:**
Let $\iota \colon R \to R_f$ be the localization map and $\varphi \colon \operatorname{Spec}(R_f) \to \operatorname{Spec}(R)$ the map $\mathfrak{q} \mapsto \iota^{-1}(\mathfrak{q})$, which lands in $\operatorname{Spec}(R)$ because the preimage of a prime ideal under a ring map is prime.

[9] There shall be a natural unique open subscheme structure up to isomorphism, but infinitely many natural closed subscheme structures up to isomorphism due to the possibility of nilpotents.

Step 1: φ is a bijection onto $D(f)$. By [10, Ideals of Localization], part (3), applied to the multiplicative set $\{1, f, f^2, \dots\}$, contraction $\mathfrak{q} \mapsto \iota^{-1}(\mathfrak{q})$ is a bijection from the primes of R_f onto the primes of R containing no power of f. A prime contains a power of f if and only if it contains f, so these are exactly the points of $D(f)$.

Step 2: φ is continuous. For $S \subseteq R$, a prime $\mathfrak{q}$ of R_f satisfies $S \subseteq \iota^{-1}(\mathfrak{q})$ if and only if $\iota(S) \subseteq \mathfrak{q}$, so $\varphi^{-1}(V(S)) = V(\iota(S))$ is closed.

Step 3: φ maps closed sets to closed subsets of $D(f)$. Every element of R_f has the form a/f^n with $a \in R$ and $n \geq 0$. For $T \subseteq R_f$ put $T' = \{a \in R \mid a/f^n \in T \text{ for some } n \geq 0\}$. We claim $\varphi(V(T)) = V(T') \cap D(f)$. If $T \subseteq \mathfrak{q}$ and $a \in T'$, say $a/f^n \in T$, then $a/1 = (a/f^n)(f^n/1) \in \mathfrak{q}$, so $a \in \iota^{-1}(\mathfrak{q})$, and $\varphi(\mathfrak{q}) \in D(f)$ by Step 1. Conversely let $\mathfrak{p} \in V(T') \cap D(f)$. By Step 1, $\mathfrak{p} = \iota^{-1}(\mathfrak{q})$ for a prime $\mathfrak{q}$ of R_f. For $a/f^n \in T$ we have $a \in T' \subseteq \mathfrak{p}$, so $a/1 \in \mathfrak{q}$ and $a/f^n = (a/1)(1/f^n) \in \mathfrak{q}$. Hence $T \subseteq \mathfrak{q}$ and $\mathfrak{p} \in \varphi(V(T))$.

Step 4: conclusion of the first part. By Steps 1 to 3, $\varphi\colon \operatorname{Spec}(R_f) \to D(f)$ is a continuous bijection carrying closed sets to closed subsets of $D(f)$, so its inverse is continuous and φ is a homeomorphism.

Step 5: intersections. Let $\mathfrak{p}$ be a prime of R. Since $\mathfrak{p}$ is prime, induction on n shows $f_1 \cdots f_n \in \mathfrak{p}$ if and only if $f_i \in \mathfrak{p}$ for some i. Passing to complements, $\mathfrak{p} \in D(f_1 \cdots f_n)$ if and only if $\mathfrak{p} \in D(f_i)$ for every i, that is,

$$D(f_1) \cap \cdots \cap D(f_n) = D(f_1 \cdots f_n)$$

which in the notation $X_f = D(f)$ reads $\bigcap_i (\operatorname{Spec} R)_{f_i} = (\operatorname{Spec} R)_g$ for $g = f_1 \cdots f_n$, completing the proof.

Corollary 2.1.13: Distinguished Open Sets Form Topological Basis

The collection of distinguished open sets $\{D(f)\}_{f \in R}$ from a topological basis for $\operatorname{Spec} R$ with the Zariski topology

***Proof*:**
Distinguished opens cover $\operatorname{Spec} R$ by definition, and proposition 2.1.12 show they are closed under intersections.

A mistake the reader may make is to believe that each $D(f)$ can be thought of as the spectrum minus some points, but this need not be the case:

Example 2.1.14: Spectrum and Distinguished Open Sets
Take $X = \operatorname{Spec}(k[x, y])$ and take $U = X \setminus \{[(x, y)]\}$, i.e. X minus the origin (which is a closed set as it is a maximal ideal). Show that $U \ncong D(f)$ for any f (hint: notice that (x, y) is not principal). This shows an important distinction between points and functions.

Show further that U can be represented as the union of two distinguished open sets.

Another result we may want to conclude is that each open set $U \subseteq \operatorname{Spec}(R)$ corresponds to $\operatorname{Spec}(R_S)$

for an appropriate S, i.e. $U \cong_{\mathbf{Top}} \operatorname{Spec}(R_S)$. This shall be shown to be *false*; the curious reader may jump to example 2.2.53 for a counter-example.

The following result, while elementary, is used repeatedly in gluing arguments and in the proof of the Affine Communication Lemma. It says that a distinguished open inside a distinguished open is itself distinguished in the ambient spectrum.

Proposition 2.1.15: Distinguished Open of a Distinguished Open

Let $D(f) \subseteq \operatorname{Spec}(R)$ be a distinguished open, so that $D(f) \cong \operatorname{Spec}(R_f)$. Let $g' \in R_f$ and consider the distinguished open $D(g') \subseteq \operatorname{Spec}(R_f)$. Then $D(g')$, viewed as a subset of $\operatorname{Spec}(R)$, is itself a distinguished open of $\operatorname{Spec}(R)$.
More precisely, if $g' = h/f^n$ with $h \in R$ and $n \geq 0$, then:

$$\operatorname{Spec}((R_f)_{g'}) \;=\; \operatorname{Spec}(R_{fh}).$$

In particular, $D(g') = D(fh)$ as subsets of $\operatorname{Spec}(R)$.

***Proof*:**
Since f is already a unit in R_f, inverting $g' = h/f^n$ in R_f is the same as inverting h in R_f:

$$(R_f)_{h/f^n} = (R_f)_h.$$

Now $(R_f)_h$ is the localization of R at the multiplicative set generated by f and h, which is the same as localizing at fh:

$$(R_f)_h = R_{fh}.$$

(Formally: $(R_f)_h = R[f^{-1}][h^{-1}] = R[(fh)^{-1}] = R_{fh}$, since inverting fh inverts both f and h, and conversely.) The corresponding open set is $D(fh) = D(f) \cap D(h) \subseteq \operatorname{Spec}(R)$.

2.1.16 Remark The proposition says that the collection of distinguished open sets is closed under "refinement": passing to a smaller distinguished open inside a distinguished open stays within the class. This is what makes distinguished opens a good basis for defining sheaves—one never needs to leave the class when restricting.

Let us record two further useful consequences.

Corollary 2.1.17: Unit Ideal and Distinguished Cover

Elements $f_1, \ldots, f_n \in R$ generate the unit ideal $(f_1, \ldots, f_n) = R$ if and only if $\operatorname{Spec}(R) = \bigcup_{i=1}^{n} D(f_i)$.

***Proof*:**
A prime $\mathfrak{p}$ is not in $\bigcup D(f_i)$ if and only if $f_i \in \mathfrak{p}$ for all i, if and only if $(f_1, \ldots, f_n) \subseteq \mathfrak{p}$. Thus $\bigcup D(f_i) = \operatorname{Spec}(R)$ if and only if no prime contains all the f_i, if and only if $(f_1, \ldots, f_n) = R$.

Corollary 2.1.18: Powers Generate the Unit Ideal

If $f_1, \ldots, f_n$ generate the unit ideal in R, then for any positive integers $k_1, \ldots, k_n$, the powers $f_1^{k_1}, \ldots, f_n^{k_n}$ also generate the unit ideal.

***Proof*:**
We have $D(f_i) = D(f_i^{k_i})$ (a prime contains f_i if and only if it contains $f_i^{k_i}$), so $\bigcup D(f_i^{k_i}) = \bigcup D(f_i) = \operatorname{Spec}(R)$, which by the previous corollary means $(f_1^{k_1}, \ldots, f_n^{k_n}) = R$.

2.1.3 Topological Properties

Let us see what classical topological properties applies to $\operatorname{Spec} R$. An important consequence is that for any open cover of $\operatorname{Spec}(R)$ given by distinguished open sets, there exists a finite open sub-cover. To see this, notice that $\cup_{i\in I} D(f_i) = \operatorname{Spec}(R)$ For some indexing set I if and only if $(\{f_i\}_{i\in I}) = R$ that is the ideal generated by the set $\{f_i\}_{i\in I}$. Then 1 must be inside that ideal, and must be a finite linear combination, and the result follows by the algebraic/geometric correspondence. This shows that $\operatorname{Spec}(R)$ is always compact in the topological sense. We may wish that this implies that $\operatorname{Spec}(R)$ has the usual nice properties of compactness, but this is very much not the case (notice this implies $\mathbb{A}^n_{\mathbb{C}}$ is compact). This turns out to be a consequence of lacking Hausdroffness. For this reason, a different name is given to compactness:

Definition 2.1.19: Quasicompact

Let X be a topological space. Then X is *quasicompact* if every open cover of X has a finite subcover, with no Hausdorff condition imposed.

Lemma 2.1.20: Spectra are Quasicompact

For every ring R, the space $\operatorname{Spec}(R)$ is quasicompact. A finite union of quasicompact subsets of a topological space is quasicompact.

***Proof*:**
Let $\{W_\lambda\}$ be an open cover of $\operatorname{Spec}(R)$ and let Λ' be the set of $t \in R$ such that $D(t)$ lies in some W_λ. The distinguished opens form a basis (corollary 2.1.13), so the sets $D(t)$ with $t \in \Lambda'$ cover $\operatorname{Spec}(R)$. Then no prime contains Λ', so the ideal generated by Λ' has radical $I(\emptyset) = R$ by proposition 2.1.6(5) and contains 1. Hence 1 is a combination of finitely many $t_1, \ldots, t_m \in \Lambda'$, the sets $D(t_1), \ldots, D(t_m)$ cover $\operatorname{Spec}(R)$ by corollary 2.1.17, and members $W_{\lambda_k} \supseteq D(t_k)$ form a finite subcover. For the second assertion, an open cover of a finite union covers each piece, and the union of finite subcovers of the pieces is a finite subcover, completing the proof.

As an exercise, show that if R is Noetherian, every subset of $\operatorname{Spec}(R)$ is quasicompact. In section 3.5.4, we shall show how the notion of proper maps from analysis is the right way of defining the notion of compactness which recovers many of the nice results expected from compactness.

The companion property of [quasi]compactness is connectedness. Recall a set is connected if it cannot be written as a disjoint of two nonempty open sets. Disconnectedness of Spec is equivalent to R

being a product $R_1 \times R_2$ of two nonzero rings, or, if the reader remembers the brief discussion in Representation theory (see [12]), to R having an idempotent other than 0 and 1:

Proposition 2.1.21: Spectrum Connectedness And Idempotent Elements

1. If $R = R_1 \times \cdots \times R_n = \prod_{i=1}^n R_i$ is a finite product of rings, then
$$\operatorname{Spec}\left(\prod_{i=1}^n R_i\right) \cong_{\mathbf{Top}} \coprod_{i=1}^n \operatorname{Spec}(R_i)$$
2. For a ring R the following are equivalent:
   (a) $\operatorname{Spec}(R)$ is not connected.
   (b) R is isomorphic to a product $R_1 \times R_2$ of two nonzero rings.
   (c) There exist nonzero $a_1, a_2 \in R$ with $a_1^2 = a_1$, $a_2^2 = a_2$ and $a_1 + a_2 = 1$.

***Proof*:**

1. Let $e_i \in R$ be the element with i-th coordinate 1 and all others 0, so that $e_i^2 = e_i$, $e_i e_j = 0$ for $i \neq j$, and $e_1 + \cdots + e_n = 1$. Let $\mathfrak{p}$ be a prime of R. Since $1 \notin \mathfrak{p}$, some $e_i \notin \mathfrak{p}$, and then $e_i e_j = 0 \in \mathfrak{p}$ forces $e_j \in \mathfrak{p}$ for every $j \neq i$. So each point of $\operatorname{Spec}(R)$ lies in exactly one of the open sets $D(e_1), \ldots, D(e_n)$. A subset $W \subseteq \operatorname{Spec}(R)$ is therefore open if and only if each $W \cap D(e_i)$ is open in $D(e_i)$, which says that $\operatorname{Spec}(R)$ is the topological disjoint union of the $D(e_i)$. By proposition 2.1.12, $D(e_i) \cong_{\mathbf{Top}} \operatorname{Spec}(R_{e_i})$.

   Next, $R_{e_i} \cong R_i$. The projection $\pi_i \colon R \to R_i$ sends e_i to 1, so by [10, Universal Property of Localization] it induces a ring map $\Pi_i \colon R_{e_i} \to R_i$ with $\Pi_i(a/e_i^k) = \pi_i(a)$. It is surjective because π_i is. If $\pi_i(a) = 0$, then $e_i a = 0$, as $e_i a$ has i-th coordinate $\pi_i(a)$ and all other coordinates 0, so $a/e_i^k = 0$ in R_{e_i}. Hence Π_i is an isomorphism.

   Finally, a ring isomorphism $\theta \colon A \to B$ induces a homeomorphism $\operatorname{Spec}(B) \to \operatorname{Spec}(A)$, $\mathfrak{q} \mapsto \theta^{-1}(\mathfrak{q})$: it is a bijection with inverse $\mathfrak{p} \mapsto \theta(\mathfrak{p})$, the preimage of $V(T)$ is $V(\theta(T))$, and the preimage of $V(T')$ under the inverse is $V(\theta^{-1}(T'))$. Applied to Π_i this gives $\operatorname{Spec}(R_i) \cong_{\mathbf{Top}} \operatorname{Spec}(R_{e_i}) \cong_{\mathbf{Top}} D(e_i)$, and so $\operatorname{Spec}(R) \cong_{\mathbf{Top}} \coprod_{i=1}^n \operatorname{Spec}(R_i)$.

2. *(a) implies (c).* Write $\operatorname{Spec}(R) = U_1 \cup U_2$ with U_1, U_2 open, nonempty and disjoint. Each U_k is the complement of the other, hence closed, so $U_k = V(J_k)$ for an ideal J_k (a prime contains a set if and only if it contains the ideal it generates). By lemma 2.1.5, $V(J_1 + J_2) = U_1 \cap U_2 = \emptyset$ and $V(J_1 J_2) = U_1 \cup U_2 = \operatorname{Spec}(R)$. By proposition 2.1.6(5), $\sqrt{J_1 + J_2} = I(\emptyset) = R$, so $1 = b_1 + b_2$ with $b_k \in J_k$. Also $b_1 b_2 \in J_1 J_2$ lies in every prime, so it lies in $I(V((0))) = \sqrt{(0)}$, again by proposition 2.1.6(5), and $(b_1 b_2)^N = 0$ for some $N \geq 1$. Since $b_1 + b_2 = 1$, the elements b_1, b_2 generate the unit ideal, hence so do b_1^N, b_2^N by corollary 2.1.18. Choose $u, v \in R$ with $u b_1^N + v b_2^N = 1$, and put $a_1 = u b_1^N$ and $a_2 = v b_2^N$. Then $a_1 + a_2 = 1$ and $a_1 a_2 = uv(b_1 b_2)^N = 0$, so $a_1^2 = a_1(1 - a_2) = a_1$ and likewise $a_2^2 = a_2$. Since $a_1 \in J_1$, it lies in every point of $U_1 \neq \emptyset$, so $a_1 \neq 1$ and $a_2 \neq 0$. Symmetrically $a_2 \in J_2$ gives $a_1 \neq 0$.

(c) implies (b). We have $a_1a_2 = a_1(1-a_1) = 0$. Consider $\Theta\colon R \to R/a_2R \times R/a_1R$, $r \mapsto (r + a_2R, r + a_1R)$. If $r = a_1s = a_2t$, then $r = ra_1 + ra_2 = a_1a_2t + a_2a_1s = 0$, so Θ is injective. Given $x, y \in R$, the element $r = a_1x + a_2y$ satisfies $r - x = a_2(y - x) \in a_2R$ and $r - y = a_1(x - y) \in a_1R$, so Θ is surjective. If $R/a_2R = 0$, then a_2 is a unit, and an idempotent unit equals 1, forcing $a_1 = 0$. So $R/a_2R \neq 0$, and likewise $R/a_1R \neq 0$.

(b) implies (c). If $\theta\colon R \to R_1 \times R_2$ is an isomorphism with R_1, R_2 nonzero, then $a_1 = \theta^{-1}(1, 0)$ and $a_2 = \theta^{-1}(0, 1)$ are idempotents with $a_1 + a_2 = 1$, and they are nonzero because $1 \neq 0$ in R_1 and in R_2.

(c) implies (a). A prime contains $a_1a_2 = 0$, hence contains a_1 or a_2, and it does not contain both, since $a_1 + a_2 = 1$. So $\operatorname{Spec}(R)$ is the disjoint union of the open sets $D(a_1)$ and $D(a_2)$. An idempotent $a_k \neq 0$ is not nilpotent, as $a_k^m = a_k$ for all $m \geq 1$, so by [10, The Radical as an Intersection of Primes] some prime does not contain it, and $D(a_k) \neq \emptyset$. Hence $\operatorname{Spec}(R)$ is not connected, completing the proof.

Note that for the homeomorphism, the product must be finite, if the product is infinite then $\operatorname{Spec}(\prod_i R_i)$ is larger than the right hand side. In fact, an infinite disjoint union of $\operatorname{Spec}(R_i)$ cannot be represented as $\operatorname{Spec}(R)$ for an appropriate R; it can be seen as the "simplest" example of a scheme that is not an affine scheme (see example 2.2.12(6)).

A notion similar to connectedness is *irreducibility*.

Definition 2.1.22: Irreducible Topological Space

A topological space X is *irreducible* if it is nonempty and it is not the union of two proper closed subsets: whenever $X = Y \cup Z$ with Y, Z closed in X, then $Y = X$ or $Z = X$. Equivalently, X is nonempty and every nonempty open subset of X is dense in X (lemma 2.1.24(3) proves the equivalence). The sets Y and Z may have nonempty intersection.

In the euclidean topology, this notion is much less important (think $\mathbb{C}$), however in the Zariski topology we shall see that many interesting sets are irreducible. The intuition stems from traditional varieties where irreducibility corresponds to irreducible polynomials, and indeed this intuition when properly generalized translates over to irreducible [closed] sets in the Zariski topology of Spectra.

Proposition 2.1.23: Prime Ideals and Irreducible Closed Sets

Let R be a ring. For a prime $\mathfrak{p}$ of R the closed set $V(\mathfrak{p})$ is irreducible and equals the closure of the point $[\mathfrak{p}]$. For an irreducible closed set $Z \subseteq \operatorname{Spec}(R)$ the ideal $I(Z)$ is prime. The maps $\mathfrak{p} \mapsto V(\mathfrak{p})$ and $Z \mapsto I(Z)$ are mutually inverse bijections between the prime ideals of R and the irreducible closed subsets of $\operatorname{Spec}(R)$. In particular every irreducible closed subset has exactly one generic point (definition 2.1.9).

***Proof*:**
Step 1: $V(\mathfrak{p})$ is irreducible. The closure of $\{[\mathfrak{p}]\}$ is $V(I(\{[\mathfrak{p}]\})) = V(\mathfrak{p})$ by proposition 2.1.6(5). The set $V(\mathfrak{p})$ is nonempty, as it contains $[\mathfrak{p}]$. If $V(\mathfrak{p}) = Y \cup Z$ with Y, Z closed in $V(\mathfrak{p})$, hence closed in $\operatorname{Spec}(R)$, then $[\mathfrak{p}]$ lies in Y or in Z, and that closed set then contains the closure $V(\mathfrak{p})$

of $[\mathfrak{p}]$. So $V(\mathfrak{p})$ is irreducible (definition 2.1.22).

Step 2: $I(Z)$ is prime. Let Z be closed and irreducible. The ideal $I(Z)$ is contained in every point of $Z \neq \emptyset$, so it is proper. If $ab \in I(Z)$, every point of Z contains a or b, so Z is the union of the closed sets $Z \cap V(a)$ and $Z \cap V(b)$ (lemma 2.1.5), and irreducibility puts Z inside one of them, that is, $a \in I(Z)$ or $b \in I(Z)$. So $I(Z)$ is prime.

Step 3: the bijection and generic points. Primes are radical, so $I(V(\mathfrak{p})) = \mathfrak{p}$ by proposition 2.1.6(6), and $V(I(Z)) = Z$ for closed Z by the same item. Hence the two maps are mutually inverse. If $[\mathfrak{q}]$ is a generic point of an irreducible closed set Z, then Z is the closure $V(\mathfrak{q})$ of $[\mathfrak{q}]$, so $\mathfrak{q} = I(V(\mathfrak{q})) = I(Z)$ is determined by Z, and $[I(Z)]$ is a generic point of Z by Step 1. This proves the last assertion, completing the proof.

Hence, irreducibility is in correspondence with prime ideals:

$$\{\text{prime ideals of } R\} \underset{V(-)}{\overset{I(-)}{\rightleftarrows}} \{\text{closed irreducible sets of Spec}\,(R)\} \tag{2.3}$$

If we look at irreducible closed sets, then $V(-)$ and $I(-)$ gives a bijection between the irreducible closed subsets of Spec (R) and the prime ideals of R, in other words there is a bijection between the *points* of Spec (R) and the irreducible closed subsets of Spec (R). Since it is a bijection, every irreducible closed subset of Spec (R) has *exactly* one generic point associated to it.

Irreducibility is preserved by the two operations that arise most often, closure and continuous image.

Lemma 2.1.24: Images and Closures of Irreducible Sets

Let X and Y be topological spaces, subsets carrying the subspace topology.

1. If $Z \subseteq X$ is irreducible, then $\overline{Z}$ is irreducible.
2. If $f : X \to Y$ is continuous and $Z \subseteq X$ is irreducible, then $f(Z)$ is irreducible. In particular, the continuous image of an irreducible space is irreducible.
3. A subset $Z \subseteq X$ is irreducible if and only if Z is nonempty and every nonempty relatively open subset of Z is dense in Z. Consequently, in an irreducible Z any two nonempty relatively open subsets intersect.

***Proof*:**
(3) Suppose Z is irreducible and $V \subseteq Z$ is nonempty and open in Z. If V were not dense in Z, then the closure of V in Z and the complement $Z \setminus V$ would be two proper closed subsets of Z covering Z, contradicting irreducibility. Conversely, suppose Z is nonempty with every nonempty relatively open subset dense, and let $Z = C_1 \cup C_2$ with C_1, C_2 closed in Z. If either C_i equals Z there is nothing to prove, so suppose C_1 is proper. Then $Z \setminus C_1$ is nonempty, open in Z, and contained in C_2, so its closure in Z lies in the closed set C_2. Density forces $C_2 = Z$, so Z is irreducible. For the consequence, a dense subset meets every nonempty open subset, and conversely a subset meeting every nonempty open subset is dense, no point having a neighborhood that avoids it.

(1) The set $\overline{Z}$ is nonempty. Let $\overline{Z} = C_1 \cup C_2$ with each C_i closed in $\overline{Z}$, hence closed in X. Then $Z = (Z \cap C_1) \cup (Z \cap C_2)$ expresses Z as a union of two closed subsets of Z, so irreducibility of Z gives $Z \subseteq C_i$ for some i. As C_i is closed in X, it contains $\overline{Z}$, so $C_i = \overline{Z}$.

(2) The image $f(Z)$ is nonempty. Let V_1, V_2 be nonempty relatively open subsets of $f(Z)$, say $V_i = W_i \cap f(Z)$ with W_i open in Y. Each $f^{-1}(W_i) \cap Z$ is open in Z and nonempty, since some point of Z maps into V_i. By (3) these two sets intersect, and the image of a common point lies in $V_1 \cap V_2$. So any two nonempty relatively open subsets of $f(Z)$ meet, which makes every nonempty relatively open subset dense, and (3) gives irreducibility of $f(Z)$.

In an integral scheme, every irreducible closed subset is the closure of a unique point (lemma 2.4.21).

As usual, we define the Noetherian property on Spec (R) to mean that the descending chain condition for closed sets is satisfied (to mirror the ascending chain of ideals in R). In the Noetherian case, all closed sets can be decomposed into irreducibles:

Proposition 2.1.25: Noetherian Decomposition

Let X be a Noetherian topological space. Then every nonempty closed subset Z can be expressed uniquely as a finite union

$$Z = Z_1 \cup \cdots \cup Z_n$$

of irreducible closed subsets, none contained in another.That is , any closed subset Z has a finite number of pieces.

***Proof*:**
See [22, Irreducible Decomposition], whose proof uses only the minimal-element property of closed sets in a Noetherian space together with irreducibility, and so applies verbatim here.

The reader should check the equivalent result for prime ideals

2.1.4 Ring to Top Functor: Induced Spectra Maps

As spectra are given by rings, we shall want to translate over ring homomorphisms to maps between spectra:

Definition 2.1.26: Induced Spectra Map

let $Y = \text{Spec}\,(S)$ and $X = \text{Spec}\,(R)$ be two spectrum. Then given the ring homomorphism: $\varphi : R \to S$, the induced *spectrum map* $\varphi^a : \text{Spec}\,(S) \to \text{Spec}\,(R)$ is given by:

$$\varphi^a(\mathfrak{p}) = \varphi^{-1}(\mathfrak{p})$$

The preimage of a prime ideal under a ring homomorphism is prime, so φ^a is well defined, and it is continuous and functorial (lemma 2.1.28). We now have two functions happening: φ^a given by ring homomorphisms, and regular functions given by ring elements. Often, φ^a is called a map, while the elements of R are called functions (following a similar convention from differential geometry).

Example 2.1.27: Maps between Spectra: φ^a

1. Show that the surjective map $R \to R/I$ induce an injective map $\operatorname{Spec}(R/I) \to \operatorname{Spec}(R)$. Algebraically, the surjective map adds relations. Geometrically, we are taking the points with the extra algebraic relations given by I and associating them to the points in R.

2. Show that the injective map $R \to S^{-1}R$ induce a injective map $\operatorname{Spec}(S^{-1}R) \to \operatorname{Spec}(R)$. Algebraically, the map $R \to S^{-1}R$ takes (integral) elements[a] and maps it to units.

3. Here is a concrete example generalizing the usual notion of a regular morphism as introduced in [22, Regular Map on Affine Space]: Take $\mathbb{A}^1_{\mathbb{C}} = \operatorname{Spec}\mathbb{C}[x]$ which by Nullstellensatz we can think of as the traditional points $\mathbb{C}$ along with the generic point (0). Consider the ring homomorphism between these rings defined by:

$$\varphi : \mathbb{C}[y] \to \mathbb{C}[x] \qquad y \mapsto x^2$$

Note this is not the squaring map, as that is not a ring morphism, but rather the map that maps the variable y to x^2 and then extends linearly. Then the map on the sets Spec 's is given by:

$$\varphi^a : \operatorname{Spec}(\mathbb{C}[x]) \to \operatorname{Spec}(\mathbb{C}[y])$$

$$\begin{aligned}
\varphi^a(\mathfrak{p}) &= \varphi^{-1}((x-a)) \\
&= \left\{ f(y) \in \mathbb{C}[y] \;\middle|\; f(x^2) \in (x-a) \right\} \\
&= \left\{ f(y) \in \mathbb{C}[y] \;\middle|\; x-a \middle| f(x^2) \right\} \\
&= \left\{ f(y) \in \mathbb{C}[y] \;\middle|\; f(x^2) = (x-a)q(x) \right\} \\
&= (y-a^2)
\end{aligned}$$

In one line:

$$\varphi^a([x-a]) = [(y-a^2)]$$

verify that

$$(\varphi^a)^{-1}((y-a)) = \{(x-\sqrt{a}), (x+\sqrt{a})\}$$

Perhaps more intuitively, this is the Spec-equivalent of the more usual map $\mathbb{C} \to \mathbb{C}$ given by $z \to z^2$ which can be thought of as the double-covering of $\mathbb{C}$ with branching point 0. Note the importance of visualizing schemes using complex numbers, or more generally an algebraically closed field, even if we replaced $\mathbb{C}$ with $\mathbb{R}$; for example the image of (x^2+1) is $(y+1)$ (i.e. $-i^2 = -1$) and $(\varphi^a)^{-1}\left((y^2+1)\right) = \{(x^2+1)\}$ where the information about the 2 points is captured in the degree of the (irreducible) polynomial in the pre-image (recall they are glued together in $\operatorname{Spec}\mathbb{R}[x]$).

4. More generally, any collection of regular functions can produce a spectra map, a "regular map" (in particular, this mirror's regular maps between affine varieties). Let k be any field and take polynomial $f_1, ..., f_n \in k[x_1, ..., x_m]$. Then define:

$$\varphi : k[y_1, ..., y_n] \to k[x_1, ..., x_m] \qquad y_i \to f_i$$

Let A represent the domain and B the codomain. Then show that for any ideal $I \subseteq A$, $J \subseteq B$ such that $\varphi(I) \subseteq J$, φ induces a map of sets:

$$\operatorname{Spec}(k[x_1, ..., x_m]/J) \to \operatorname{Spec}(k[y_1, ..., y_n]/I)$$

More particularly, show that this maps sends the point $[(x_1 - a_1, ..., x_m - a_m)]$ (traditionally, $(a_1, ..., a_m) \in k^m$ to the point traditionally represented as:

$$(f_1(a_1, ..., a_m), ..., f_n(a_1, ..., a_m)) \in k^n$$

From now on, we shall sometimes take liberties in spectra maps between spectra resembling varieties by defining polynomial maps on the maximal ideals (this shall pose no problems with extending the definition to the prime ideals or to the later sheaf structure that shall be added by theorem 3.7.1).

5. The map $\mathbb{A}^1_k \to \mathbb{A}^1_k$ given by $a \mapsto p(a)$ can always be represented as a map to the graph:

$$\mathbb{A}^1_k \to \mathbb{A}^2_k \qquad a \mapsto (a, p(a))$$

Furthermore, by the projection map $\pi : \mathbb{A}^2_k \to \mathbb{A}^1_k$ given by $(a, b) \mapsto a$, we shall see that this maps $\mathbb{A}^1_k$ isomorphic to all of its graphs (this works even for higher dimensions). Note that this does not extend once no coordinate is the identity, for example:

$$\mathbb{A}^1_k \to \mathbb{A}^2_k \qquad t \mapsto (t^2 - 1, t(t^2 - 1))$$

maps $\mathbb{A}^1_k$ onto the nodal cubic $\mathcal{Z}(y^2 - x^2 - x^3)$, which in [22, Rational Mapping and Birational Maps] is shown to be birational to $\mathbb{A}^1_k$ but not isomorphic to it. Another example that is not even rational is a map from $\mathbb{P}^1_k$ to an elliptic curve E.

6. This example gives a way of thinking of $\mathbb{A}^n_{\mathbb{Z}} = \operatorname{Spec}(\mathbb{Z}[x_1, ..., x_n])$ as an "$\mathbb{A}^n$-bundle" on $\operatorname{Spec}(\mathbb{Z})$ and is an important example for number theory. Define the map $\pi : \mathbb{A}^n_{\mathbb{Z}} \to \operatorname{Spec}(\mathbb{Z})$ given by the inclusion $\mathbb{Z} \hookrightarrow \mathbb{Z}[x_1, ..., x_n]$. Then show there is a bijection between $\pi^{-1}([(p)])$ where $(p) \neq (0)$and $\mathbb{A}^n_{\mathbb{F}_p}$! If $(p) = (0)$ then the pre-image is $\mathbb{A}^n_k$. The picture of this bundle can be seen as follows:

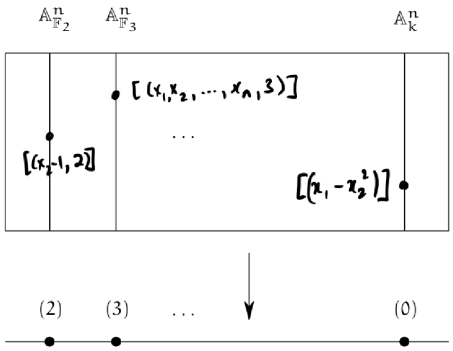


Figure 2.1: Bundle on $\operatorname{Spec}\mathbb{Z}$

7. Consider $\varphi : \mathbb{Z} \hookrightarrow k[x]$. Then $\varphi^a : \operatorname{Spec}\mathbb{C}[x] \to \operatorname{Spec}\mathbb{Z}$ is the zero map, namely $\varphi^a(\operatorname{Spec} k[x]) = \{[(0)]\}$. From this perspective, we can see that classical field theory has trivial number theory as it has no interesting fibers over $\operatorname{Spec}\mathbb{Z}$.

8. Let $\varphi : R \to S$ be an integral extension. Show that $\text{Spec}(S) \to \text{Spec}(R)$ is surjective (hint: Lying Over). This results also motivates the terminology "lying over" geometrically. Recall that is R or S is a field, the other must be too. Show that S being a field and the map being surjective can be used to conclude R is a field. What is the geometric argument if R was a field?

9. Let us see why it is important to work with primes instead of maximal ideals. Consider Spec $k(x)[y] \to$ Spec $k[x]$ induced by $k[x] \to k(x)[y]$ where $k[x]$ maps to itself (notably, all elements become units). Then as $k(x)[y]$ is a PID, all of its primes are maximal and are of the form $(y - f(x))$ for some $f(x) \in k(x)$. Then the pre-image of this ideal under the inclusion map must be the ideal (0), which is certainly not maximal (in fact it is the generic point). Hence the map Spec $k(x)[y] \to$ Spec $k[x]$ will map $[(y - f(x))] \mapsto \eta$, notably a 0 -dimensional point mapped to a 1-dimensional point!

10. Let R be a ring with nilpotence and $I = \sqrt{(0)}$ the nilradical. Take $\varphi : R \to R/I$ to be the usual projection. Then φ^a is a bijection; the key is to recall that the nilradical intersection of all prime ideals, and hence any prime $\overline{\mathfrak{p}} \subseteq R/I$ which is of the for $\mathfrak{p} + I$ must in fact equal itself: $\mathfrak{p} + I = \mathfrak{p}$ (as $I \subseteq \mathfrak{p}$). The example to keep in mind should be $R = \mathbb{C}[x]/(x^2)$ with $I = (x)$.

[a]Note that if an element is nilpotent, localizing it will map it to zero

Lemma 2.1.28: Induced Spectra Map Is Continuous

Let $\varphi : R \to S$ be a ring homomorphism. Then $\varphi^a \colon \text{Spec}\,(S) \to \text{Spec}\,(R)$ is a well-defined continuous map, with

$$(\varphi^a)^{-1}(V(T)) = V(\varphi(T)) \qquad (\varphi^a)^{-1}(D(f)) = D(\varphi(f))$$

for $T \subseteq R$ and $f \in R$. Moreover $(\text{id}_R)^a = \text{id}$, and $(\psi \circ \varphi)^a = \varphi^a \circ \psi^a$ for a ring homomorphism $\psi \colon S \to S'$.

***Proof*:**
Let $\mathfrak{q}$ be a prime of S. Then $\varphi^{-1}(\mathfrak{q})$ is an ideal of R, it is proper because $\varphi(1) = 1 \notin \mathfrak{q}$, and $ab \in \varphi^{-1}(\mathfrak{q})$ gives $\varphi(a)\varphi(b) \in \mathfrak{q}$, so $\varphi(a) \in \mathfrak{q}$ or $\varphi(b) \in \mathfrak{q}$. So $\varphi^{-1}(\mathfrak{q})$ is prime and φ^a is well defined. For $T \subseteq R$, $\mathfrak{q} \in (\varphi^a)^{-1}(V(T))$ means $T \subseteq \varphi^{-1}(\mathfrak{q})$, that is, $\varphi(T) \subseteq \mathfrak{q}$, so $(\varphi^a)^{-1}(V(T)) = V(\varphi(T))$. Preimages of closed sets are therefore closed, and φ^a is continuous. Taking $T = \{f\}$ and complements gives $(\varphi^a)^{-1}(D(f)) = D(\varphi(f))$. Finally $\text{id}_R^{-1}(\mathfrak{p}) = \mathfrak{p}$ and $(\psi \circ \varphi)^{-1}(\mathfrak{r}) = \varphi^{-1}(\psi^{-1}(\mathfrak{r}))$ for a prime $\mathfrak{r}$ of S', as claimed.

Hence Spec is a [contravariant] functor from **CRing** to **Top**. On the basis of distinguished open sets the preimages are:

$$(\varphi^a)^{-1}\,(D(f)) = D(\varphi(f))$$

Note that there will be continuous maps between the spectrum of rings that are not φ^a,

Example 2.1.29: Continuous Not Induced By Ring Homomorphism

The map from $\mathbb{A}^1_{\mathbb{C}}$ to itself that is the identity on all points except $[(x)]$ and $[(x-1)]$ (which will be represented with 0 and 1) where it swaps these two values is *not* a polynomial and hence cannot have been induced by a ring homomorphism, however it *is* continuous. If $0, 1 \notin V(S)$, then $\varphi^{-1}(V(S)) = V(S)$. Similarly if both values are inside. If 0 or 1 is in $V(S)$, say $0 \in V(S)$, then $\varphi^{-1}(V(S)) = V(S) \setminus \{0\} \cup \{1\}$ which is closed as all finite subsets of $\mathbb{A}^1_{\mathbb{C}}$ are closed.

Can you now find two ring homomorphisms that induce the same Spectra map? See exercise 2.1.1(2.1.1 (13))

Exercise 2.1.1(2.1.1 (14)) goes over some of the properties preserved by continuous maps. Most of the properties that we shall be want for schemes to have (ex. separatedness, properness, smoothness, multiplicity information, finiteness conditions) shall require more than continuity.

The correspondence is terse as Spec does not carry enough information to fully and faithfully correspond to ring homomorphism. In chapter 3 it shall be shown that scheme morphisms are the right object to carry this information (namely, they shall insure that around every point, that is each stalk, the spectra map behaves like a polynomial).

2.1.5 *Subsets of Spectra via quotient and Localization

Two common constructions that came up in a few example is the process of taking the quotient and the localization of a ring R. These form important subsets of $\operatorname{Spec}(R)$, namely by the 4th isomorphism theorem and the prime-correspondence theorem for prime ideals in localization (see [12] and [10]). By these theorems, $\operatorname{Spec}(R/I), \operatorname{Spec}(S^{-1}R) \subseteq \operatorname{Spec}(R)$. If $I = \mathfrak{p}$ and $S = R \setminus \mathfrak{p}$,the first corresponds to all prime ideals containing $\mathfrak{p}$, while the second corresponds to all prime ideals that are contained in the prime ideal $\mathfrak{p}$[10] These two sets need not union to the entire $\operatorname{Spec}(R)$ (take $\mathbb{Z}$ and 5), however they do correspond to two "opposite" ways of carving out a part of $\operatorname{Spec}(R)$. In the following visual is a very common illustration:

[10]More generally it is all prime ideals not having trivial intersection with the multiplicative set.

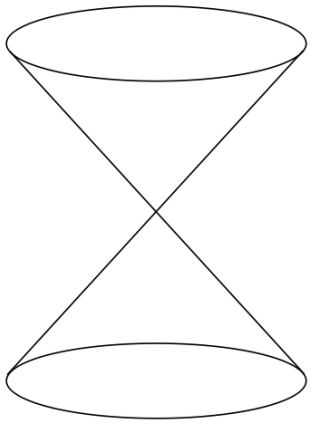


Figure 2.2: partial decomposition of $\mathrm{Spec}\,(R)$

Geometrically, $\mathrm{Spec}\,(R/I)$ can be thought of as "carving out" a part of $\mathrm{Spec}\,(R)$. For example, take $R = \mathbb{C}[x, y, z]$ and $\mathfrak{p} = (x^2 + y^2 - z)$. If we were working with affine complex varieties, then we would say this is the 3D-parabola in $\mathbb{C}^3$, however as we are on $\mathrm{Spec}\,(R/\mathfrak{p})$ this would come with some extra points (which ones?). For $\mathrm{Spec}\,(S^{-1}R)$, we may think it as the opposite (we shall cover the special case of $R_\mathfrak{p}$ afterwards). For example let's say $S = \{1, f, f^2, ...\}$ and take $S^{-1}R$. Then the primes of $\mathrm{Spec}\,(S^{-1}R)$ would be all primes that do not containing f, that is all points where f does not vanish! As an illustration, the set $\mathrm{Spec}\,(\mathbb{C}[x, y]_{y-x^2})$ are the points in the affine plane, minus all the points on the parabola $y - x^2$, namely all the singletons (a, a^2) for $a \in \mathbb{C}$ (so $[(x - a, y - a^2)]$), as well as the point $[(y - x^2)]$.

Proposition 2.1.30: Closed And Open Subsets

Let R be a commutative ring, $I \subseteq R$ an ideal, $f \in R$, and $S \subseteq R$ a multiplicative set. Each spectrum below is mapped to $\mathrm{Spec}\,(R)$ by the map induced by the canonical ring map out of R (definition 2.1.26). Then:

1. $\mathrm{Spec}\,(R/I) \to \mathrm{Spec}\,(R)$ is a homeomorphism onto the closed subset $V(I)$.
2. $\mathrm{Spec}\,(R_f) \to \mathrm{Spec}\,(R)$ is a homeomorphism onto the open subset $D(f)$.
3. $\mathrm{Spec}\,(S^{-1}R) \to \mathrm{Spec}\,(R)$ is injective, continuous, and a homeomorphism onto its image $\{\mathfrak{p} \mid \mathfrak{p} \cap S = \emptyset\}$, which need not be open or closed. In particular $\mathrm{Spec}\,(R_\mathfrak{p}) \to \mathrm{Spec}\,(R)$ need not have open or closed image.

***Proof*:**

1. This is the last assertion of lemma 2.1.5.
2. This is proposition 2.1.12.

3. Let $\iota\colon R \to S^{-1}R$ be the localization map. By [10, Ideals of Localization], part (3), $\mathfrak{q} \mapsto \iota^{-1}(\mathfrak{q})$ is a bijection from the primes of $S^{-1}R$ onto the primes $\mathfrak{p}$ of R with $\mathfrak{p} \cap S = \emptyset$, and it is continuous by lemma 2.1.28. For $T \subseteq S^{-1}R$ put $T' = \{a \in R \mid a/s \in T \text{ for some } s \in S\}$. If $T \subseteq \mathfrak{q}$ and $a/s \in T$, then $a/1 = (a/s)(s/1) \in \mathfrak{q}$, so $T' \subseteq \iota^{-1}(\mathfrak{q})$. Conversely, if $\mathfrak{p} = \iota^{-1}(\mathfrak{q})$ contains T' and $a/s \in T$, then $a \in \mathfrak{p}$ gives $a/s = (a/1)(1/s) \in \mathfrak{q}$, so $T \subseteq \mathfrak{q}$. Hence the image of $V(T)$ is $V(T')$ intersected with the image, and the map is a homeomorphism onto its image.

   For the examples, recall that the closure of a subset W is $V(I(W))$ (proposition 2.1.6(5)), and that a nonzero integer lies in only finitely many prime ideals of $\mathbb{Z}$, so the intersection of infinitely many nonzero primes of $\mathbb{Z}$ is (0). The map $\operatorname{Spec}(\mathbb{Q}) \to \operatorname{Spec}(\mathbb{Z})$ has image $\{(0)\}$. Its closure is $V((0)) = \operatorname{Spec}(\mathbb{Z})$, so it is not closed. Its complement, the set of nonzero primes, has closure $V((0)) = \operatorname{Spec}(\mathbb{Z})$ by the same computation, so the complement is not closed and the image is not open. For a prime number p, the map $\operatorname{Spec}(\mathbb{Z}_{(p)}) \to \operatorname{Spec}(\mathbb{Z})$ has image $\{(0), (p)\}$, whose closure is $V((0)) = \operatorname{Spec}(\mathbb{Z})$ and whose complement $\{(q) \mid q \neq p\}$ also has closure $\operatorname{Spec}(\mathbb{Z})$. So this image is neither closed nor open, completing the proof.

For the case of $S = R - \mathfrak{p}$, we can think of $\operatorname{Spec}(R_{\mathfrak{p}})$ as keeping near the $\mathfrak{p}$, namely since the prime ideals away from $\mathfrak{p}$ will be turned into units, the primes intersecting $R \setminus \mathfrak{p}$ will be eliminated. For example, if $R = \mathbb{C}[x, y]$ and $\mathfrak{p} = (x, y)$, then we keep all the points corresponding to being "near" the origin, that is the point $[(x, y)]$, i.e. $(0, 0)$, and the 1-dimensional points $[(f(x, y))]$ where $f(0, 0) = 0$ (i.e. where $f \equiv 0 \bmod (x, y)$) that is irreducible curves that pass through the origin. In the special case where $R = k[x]$, then the localization will always only have two points, usually called the classical point and the generic point. These shall eventually be seen as geometric interpretations of DVR.

Using these geometric results, they can lend intuition towards algebraic manipulation:

Example 2.1.31: Illustration Of Quotient And Localization
Using the above intuitions, we can quickly come up with some intuition isomorphisms. Take $R = \mathbb{C}[x, y]/(xy)$. Visually, this is the union of the y and x-axis. If we localize this at x, then by our above intuition this should eliminated all points where the "function" x vanishes. The function x vanishes at the y-axis, and so we must be left with the x-axis minus the origin. This happens to be $\operatorname{Spec} \mathbb{C}[x]_x$. In fact, these two rings are isomorphic!

$$\left(\frac{\mathbb{C}[x, y]}{(xy)}\right)_x \cong \mathbb{C}[x]_x$$

which the reader should check.

Exercise 2.1.1

1. Show that $D(f) \cap D(g) = D(fg)$.
2. $D(f) \subseteq D(g)$ iff $f^n \in (g)$ iff g is invertible in R_f. How would you prove this via "geometric explanation"? For an algebraic explanation, change to $V(f) \supseteq V(g)$ and take $I(-)$ to inverse and conclude.
3. $D(f) = \emptyset$ iff f is in the Nilradical.

4. Show that any connected component is the union of irreducible components (think of the + shape, i.e. $\operatorname{Spec}(\mathbb{C}[x,y]/(xy))$).
5. Show that $\mathbb{A}^1_{\mathbb{C}}$ is irreducible.
6. Show that in an irreducible topological space, any nonempty open set is dense.
7. If X is a topological and Z an irreducible subspace, then $\overline{Z}$ is irreducible as well
8. Show that an irreducible space is connected.
9. If A is a integral domain, $\operatorname{Spec}(A)$ is irreducible.
10. Show that $\operatorname{Spec}(\mathbb{C}[x,y]/(xy))$ is connected but reducible.
11. Show that there is a natural bijection irreducible components (maximal irreducible subsets) of $\operatorname{Spec}(R)$ and the *minimal* prime ideals of R.
12. these next two exercises show us how the theory of classical varieties persists with Schemes
    a) Let A be a finitely generated k-algebra. Then show that the closed points of $\operatorname{Spec}(A)$ are dense; show that if $f \in A$ and $D(f)$ is nonempty, then $D(f)$ contains a closed point of $\operatorname{Spec}(A)$[11]
    b) Let $A = \overline{k}[x_1, ..., x_n]/I$ where the Nilradical is trivial, and let $X = \operatorname{Spec}(A) \subseteq \mathbb{A}^n_{\overline{k}}$. Show that [regular] functions on X are determined by their valued on the closed points. Hint: if f, g are different [regular] functions on X, then $f - g$ is nowhere zero on an open subset of X. Use the above result.
13. Find two ring homomorphism that induce the same spectra map.
14. Let $f : X \to Y$ be a continuous map.
    a) Show that irreducible sets map to irreducible sets under continuous maps.
    b) Show that if X is Noetherian, then the image of f is Noetherian
    c) Show that the dimension of X is greater than or equal to the dimension of the image of Y (also see 2.6).
15. Show there does not exist a $\operatorname{Spec}(\mathbb{Z}) \to \operatorname{Spec}(\mathbb{Q})$ induced by a ring homomorphism, however there is a continuous map with respect to the Zariski topology (in chapter 3 we will show there is no scheme morphism $\operatorname{Spec}(\mathbb{Z}) \to \operatorname{Spec}(\mathbb{Q})$).

2.2 Schemes

The spectrum of a ring with the Zariski Topology does not hold all data provided by a commutative ring, for example nilpotence is forgotten. The *structure sheaf* will be added to $\operatorname{Spec} R$ in order to add this extra information (namely, we shall have the collection of regular function defined on each open set as part of the data we are tracking). The resulting object is called the *affine scheme*.

After defining the affine scheme, we shall require one more crucial abstraction that links it to differential geometry: we shall want our object to the glued-together open sets. In differential

[11] Hint: A_f is also a finitely generated k-algebra use generalized Nullstellensatz. Show the closed points of the Spec of a finitely generated k-algebra are those for which the residue field is a finite extension of k

geometry, these would all be open subsets of euclidean space, usually simply connected and all isomorphic. A key distinguishing feature in algebraic geometry is that the sets that can be glued can be very different, namely $\operatorname{Spec} R$ for all commutative rings R will be the sets from which we can select, and as the reader has already seen these sets can have very different topologies.

2.2.1 The Structure Sheaf for Affine Schemes

When working with geometric objects (varieties, real and complex manifolds, etc), we study the local behaviour of the space, namely by studying some given information on open sets or germs/stalks. In the theory of differential manifold, the information at each local was the "linear" equation (recall that one way of defining the tangent bundle is by taking all the derivations of smooth functions, which can be shown to "preserve" the linear part of each smooth function). For an affine scheme, we are interested in the functions functions are *regular* on open sets. By section 2.1.2, it is enough to define the regular functions on distinguished open sets. For those motivated by analytic parallels, Complex manifold have a sheaf of meromorphic functions which are defined on the open subsets of the complex manifold, and we usually have holomorphic functions on the entire manifold.

To motivate the following sheaf, recall that on $\mathbb{C}^n$, if we take open subsets there are many more holomorphic functions that could be defined on that open subset, namely that rational functions are well-defined when the roots of the denominator (which are not cancel by the numerator) are outside the open set. For example, on $\mathbb{C}^2 \setminus \{x\text{-axis} \cup y\text{-axis}\}$, the function

$$\frac{x^2 + 4x + y + 1}{x^5 y^4}$$

is well-defined. Moving this to Spec, in $\mathbb{A}^2_{\mathbb{C}}$, we ought to have the above function to be well-defined on $D(xy)$. Furthermore, we want the collection of global sections, $\mathcal{O}_X$, to be R, mirroring the "global section" on $\mathbb{C}^2$ being the (non-rational) holomorphic functions in two variables.

As we saw in theorem 1.1.22, it suffices to define a sheaf on a base of $\operatorname{Spec}(R)$, namely on the distinguished open sets. For a distinguished open set W let $S_W = \{g \in R \mid W \subseteq D(g)\}$, the multiplicative set of elements that vanish at no point of W, and put

$$\mathcal{O}_X(W) = S_W^{-1} R$$

which depends only on the open set W, not on an element describing it (note how this is subtly different than inverting a chosen f with $W = D(f)$: what if f is nilpotent, or $D(f) = D(g)$?). For $W = D(f)$, the map $R_f \to S_W^{-1} R$ induced by $f \in S_W$ is an isomorphism of R-algebras. If $g \in S_W$, then $V(g) \subseteq V(f)$, so $f \in I(V(f)) \subseteq I(V(g)) = \sqrt{(g)}$ (proposition 2.1.6(5)), say $f^n = rg$, and g is invertible in R_f. So every element of S_W becomes a unit in R_f, and [10, Universal Property of Localization] gives R-algebra maps in both directions, whose composites are the identities by the same uniqueness. Hence

$$\mathcal{O}_X(D(f)) = \mathcal{O}_X(X_f) \cong R_f$$

and if $D(f) = D(g)$, these isomorphisms give a canonical isomorphism $R_f \cong R_g$ of R-algebras.

For distinguished opens $W' \subseteq W$ we have $S_W \subseteq S_{W'}$, so the elements of S_W are units in $S_{W'}^{-1} R$, and [10, Universal Property of Localization] gives a unique R-algebra map $\operatorname{res}_{W,W'} \colon \mathcal{O}_X(W) \to \mathcal{O}_X(W')$. By this uniqueness, $\operatorname{res}_{W,W} = \mathrm{id}$ and $\operatorname{res}_{W',W''} \circ \operatorname{res}_{W,W'} = \operatorname{res}_{W,W''}$, both sides being R-algebra

maps out of $S_W^{-1}R$. In terms of $R_f \to R_g$ for $D(g) \subseteq D(f)$: as above, $g^n = rf$ for some $n \geq 1$ and $r \in R$, and $f \cdot r/g^n = 1$ in R_g, so the restriction map is

$$\operatorname{res}_{X_f, X_g} : R_f \to R_g \qquad \frac{s}{f^m} \mapsto \frac{r^m s}{g^{nm}}$$

which by uniqueness does not depend on the choice of n and r. Hence $\mathcal{O}_X$ is a presheaf on the base of distinguished open sets (definition 1.1.21).

Proposition 2.2.1: Distinguished Opens form Sheaf-Basis for $\operatorname{Spec} R$

Let $X = \operatorname{Spec} R$, and suppose that $D(f)$ is covered by distinguished open sets $D(f_a) \subseteq D(f)$. Then:

1. If $g, h \in R_f$ are equal in R_{f_a} when passing through the restriction map for every a, then $g = h$.
2. If for each a, there exists a $g_a \in R_{f_a}$ such that for each pair a, b, the images of g_a and g_b in $R_{f_a f_b}$ are equal, then there is an element $g \in R_f$ whose image in R_{f_a} is g_a.

In other words, $\mathcal{O}_X$ defines a $\mathcal{B}$-sheaf on the distinguished base, and by theorem 1.1.22, $\mathcal{O}_X$ extends to a sheaf on X.

***Proof*:**
Let $\operatorname{Spec}(R) = \bigcup_{i \in I} D(f_i)$, which is equivalent to saying the f_i generate the unit ideal: $(f_i)_{i \in I} = R$. We show the locality and gluability axioms, or in categorical terms the exactness of:

$$0 \longrightarrow R \longrightarrow \prod_{i \in I} R_{f_i} \rightrightarrows \prod_{i \neq j \in I} R_{f_i f_j}$$

Starting with locality. As $s_1 = s_2$ is equivalent to $s_1 - s_2 = 0$, it suffices to show the result in the case where $s = 0$. By quasicompactness we may pass to a finite subcover:

$$\operatorname{Spec}(R) = \bigcup_{i=1}^{n} D(f_i) \qquad (f_1, \ldots, f_n) = R$$

Suppose that for some $s \in R$, $\operatorname{res}_{\operatorname{Spec}(R), D(f_i)} s = 0$ for all i. We must show $s = 0$. By definition of R_{f_i}:

$$s|_{D(f_i)} = 0 \qquad \text{if and only if} \qquad f_i^{m_i} s = 0 \quad \text{for some } m_i \geq 0$$

Taking $m = \max_i(m_i)$, we have $f_i^m s = 0$ for all i. Now, as $D(f_i) = D(f_i^m)$, the cover $\bigcup D(f_i^m) = \operatorname{Spec}(R)$ still holds, so $(f_1^m, \ldots, f_n^m) = R$. Thus, for appropriate choices $r_i \in R$ with $\sum_{i=1}^n r_i f_i^m = 1$, we get:

$$s = 1 \cdot s = \left(\sum_i r_i f_i^m \right) s = \sum_i r_i (f_i^m s) = 0$$

For general $D(f)$, repeat the above argument replacing R with R_f (quasi-compactness of $D(f)$ ensures the cover can still be taken to be finite, and the same algebraic manipulations apply in R_f).

We next show gluability[a]. Starting with the case $\operatorname{Spec}(R) = \bigcup_i D(f_i)$, suppose there are elements in each R_{f_i} that agree on overlaps $R_{f_i f_j}$ (note that intersections of distinguished open sets are distinguished open sets). As there may be infinitely many sets on which there are overlaps, it cannot be assumed that I is finite. However, the finite case is a good start, say $I = \{1, \dots, n\}$. Then we have elements $\frac{a_i}{f_i^{l_i}} \in R_{f_i}$ agreeing on the overlaps $R_{f_i f_j}$, meaning:

$$(f_i^{l_i} f_j^{l_j})^{m_{ij}} (f_j^{l_j} a_i - f_i^{l_i} a_j) = 0$$

Simplifying notation by writing $g_i = f_i^{l_i}$:

$$(g_i g_j)^{m_{ij}} (g_j a_i - g_i a_j) = 0$$

As there are finitely many m_{ij}, take $m = \max_{i,j}(m_{ij})$:

$$(g_i g_j)^m (g_j a_i - g_i a_j) = 0$$

Next, convert to a different set of distinguished opens: take $b_i = a_i g_i^m$ and $h_i = g_i^{m+1}$ so that $D(h_i) = D(g_i)$. Then on each $D(h_i)$, we have the section b_i/h_i, and the overlap condition simplifies to:

$$h_j b_i = h_i b_j$$

Noting that $\bigcup_i D(h_i) = \operatorname{Spec}(R)$ gives $1 = \sum_{i=1}^n r_i h_i$ for appropriate $r_i \in R$, define:

$$r = \sum_i r_i b_i$$

Then r is the element of R that restricts to b_j/h_j on each $D(h_j)$:

$$r h_j = \sum_i r_i b_i h_j = \sum_i r_i h_i b_j = \left(\sum_i r_i h_i \right) b_j = b_j$$

showing gluability in the finite case. For the infinite case, we use the previously shown base-locality axiom. By quasi-compactness, we can choose a finite subcover $D(f_1), \dots, D(f_n)$ and construct r as above. For any $z \in I \setminus \{1, \dots, n\}$, repeat the argument on $\{1, \dots, n, z\}$ to get $r' \in R$ which restricts to a_i for $i \in \{1, \dots, n, z\}$. By base locality, $r' = r$. Hence r restricts to $a_z/f_z^{l_z}$ for all z.

Finally let $D(f)$ be covered by sets $D(f_a) \subseteq D(f)$. Under the homeomorphism $D(f) \cong \operatorname{Spec}(R_f)$ of proposition 2.1.12, $D(f_a)$ corresponds to $D(f_a/1)$. The elements of $S_{D(f_a)}$ become units in $(R_f)_{f_a/1}$, and f and f_a lie in $S_{D(f_a)}$, so $\mathcal{O}_X(D(f_a)) \cong (R_f)_{f_a/1}$ as R-algebras, and likewise for $D(f)$ and $D(f_a f_b)$. These isomorphisms commute with the restriction maps, all of which are the unique R-algebra maps between localizations of R. So the case of $\operatorname{Spec}(R_f)$ covered by the sets $D(f_a/1)$, proved above, gives both axioms for $D(f)$, as claimed.

[a] Vakil pointed out that Serre called this a "partition of unity argument," which is the local-to-global principle physically manifested in differential geometry – a rather good perspective on what is about to happen.

It may be tempting to say that $\mathcal{O}_{\operatorname{Spec}(R)}(U)$ for any open subset U is a localization of R at some multiplicatively closed subset S, however this is not true:

Example 2.2.2: Subtle Local Section Of Affine Scheme

Let $R = k[x, y, z, w]/(xw - yz)$; $\operatorname{Spec} R$ is called the *quadric cone*. Note that R is an integral domain as $(xw - yz)$ is prime[a]. Let $U = D(y) \cup D(w)$. Then a section $s \in \mathcal{O}(U)$ is determined by a choice of $f \in R_y$ and $g \in R_w$ that agree on R_{yw}, namely $\mathcal{O}(U)$ is determined by the left exact sequence:

$$0 \to \mathcal{O}(U) \hookrightarrow R_y \oplus R_w \xrightarrow{(f,g) \mapsto f-g} R_{yw}$$

The kernel of the difference map is $\mathcal{O}(U)$. Take $(x/y,\ z/w)$ and consider:

$$\frac{x}{y} = \frac{z}{w} \qquad \text{if and only if} \qquad xw - zy = 0$$

where there is no need for a factor of $(yw)^n$ in front, as there are no zero-divisors. In R, this is exactly the defining relation of the quotient, hence this equality holds. Therefore there is an element $t \in \mathcal{O}(U)$ such that:

$$t \mapsto \left(\frac{x}{y},\ \frac{z}{w}\right)$$

Next, for any $r \in R$, certainly $r \in R_y, R_w$ and $r - r = 0$, so $r \in \mathcal{O}(U)$. Is it possible that $t = r$ for some $r \in R$? As the map $\mathcal{O}(U) \hookrightarrow R_y \oplus R_w$ is injective, if $t = r$ then since $r \mapsto (r, r)$ we would need $r = x/y = z/w$, but y and w are not invertible in R, and hence $t \notin R$. Thus the ring $R[t] \neq R$. Certainly $R[t] \subseteq \mathcal{O}(U)$ and any R-polynomial $f(t) \in R[t]$ maps to:

$$f(t) \mapsto \left(f\left(\frac{x}{y}\right),\ f\left(\frac{z}{w}\right)\right)$$

Let us show that $yt - x = 0$ (equivalently $wt - z = 0$). By definition of a sheaf, this can be checked on the open cover $\{D(y), D(w)\}$. By construction (recall that restriction maps are ring homomorphisms and that $r \mapsto r$ under these maps):

$$(yt - x)|_{D(y)} = y \cdot \frac{x}{y} - x = x - x = 0$$

and

$$(yt - x)|_{D(w)} = y \cdot \frac{z}{w} - x = \frac{yz - xw}{w} = \frac{0}{w} = 0$$

giving $yt - x = 0$. Thus:

$$R[t] \cong \frac{R[T]}{(yT - x)} \cong R\left[\frac{x}{y}\right]$$

Note that y is *not* a unit in $R[x/y]$. Let us show $\mathcal{O}(U) \cong R[t]$. If $s \in \mathcal{O}(U)$,

$$s|_{D(y)} = \frac{a}{y^n} \qquad \text{and} \qquad s|_{D(w)} = \frac{b}{w^m}$$

for some $a, b \in R$ and $n, m \in \mathbb{N}$. They must agree on $D(yw)$:

$$\left.\frac{a}{y^n}\right|_{D(yw)} = \frac{aw^n}{(yw)^n} = \frac{by^m}{(yw)^m} = \left.\frac{b}{w^m}\right|_{D(yw)}$$

Since R is an integral domain, this simplifies to $aw^m = by^n$ in R.

To show that $s \in R[t]$, we use the key relations $x = ty$ and $z = tw$ (which hold in $R[t]$ since $yt = x$ and $wt = z$). In R_y, the element a/y^n can be rewritten by substituting $x = ty$ and $z = tw$: every monomial in a involving x or z can be expressed using t, y, and w. More precisely, any element of R_y can be written as a polynomial in t with coefficients in $k[y, y^{-1}, w]$ (after eliminating x and z using the relations). Similarly, any element of R_w is a polynomial in t with coefficients in $k[y, w, w^{-1}]$. The overlap condition $aw^m = by^n$ ensures that the two expressions, when restricted to R_{yw}, give the same polynomial in t. By the locality axiom (already proved), this polynomial is unique, and it lies in $R[t]$ (no negative powers of y or w are needed, as one can verify by comparing degrees). Hence $s \in R[t]$, and so $\mathcal{O}_X(U) \cong R[x/y]$.

Now, $R[x/y]$ is *not* a localization of R. Indeed, $R[x/y] \ncong R$ (since $x/y \notin R$), yet $R[x/y]$ has the same units as R (namely k^*): no element becomes invertible. Since any nontrivial localization R_S contains units not present in R, and isomorphisms must preserve units, no localization of R is isomorphic to $R[x/y]$.

More generally, given an arbitrary open subset U, we can cover it by affine opens $\{U_i\}_{i\in I}$ and cover the intersections $U_i \cap U_j$ by affine opens $\{U_{ijk}\}_{k\in K_{ij}}$. Then the sheaf condition gives that $\mathcal{O}_X(U)$ is the equalizer of:

$$\prod_{i\in I} \mathcal{O}_X(U_i) \rightrightarrows \prod_{i,j\in I,\, k\in K_{ij}} \mathcal{O}_X(U_{ijk})$$

The general idea is that $\mathcal{O}_X(U)$ may contain rational functions f/g where g is nonzero on all of U. In practice, we shall often only require the local sections on distinguished open sets or on affine open sets, and many computations reduce to these two cases.

Note that even the following weaker hope fails: given $U \subseteq \operatorname{Spec} R$, is $\mathcal{O}_X(U)$ isomorphic to the localization of R at the multiplicative set S of all elements that do not vanish at any point of U? Consider $U = D(x) \cup D(y) \subseteq \mathbb{A}^2_k = \operatorname{Spec} k[x, y]$. We will show in example 2.2.53 that U is not affine. However, $\mathcal{O}_{\mathbb{A}^2_k}(D(x) \cup D(y)) \cong k[x, y]$ (a Hartogs-type phenomenon: regular functions extend across a codimension-2 locus). This is just $k[x, y]$ itself, yet $D(x) \cup D(y) \ncong \mathbb{A}^2_k = \operatorname{Spec} k[x, y]$.

This shows that the relationship between local geometry and algebra is more nuanced than the affine case suggests.

[a] As $k[x, y, z, w]$ is a UFD, a principal ideal (p) where p is irreducible is prime. It is tedious but rudimentary to show that $xw - yz$ is irreducible.

Keeping track of the structure sheaf $\mathcal{O}_{\operatorname{Spec} R}$ along with $\operatorname{Spec}(R)$ gives us our first definition we need:

Definition 2.2.3: Affine Scheme

Let $X = \operatorname{Spec}(R)$ be the spectrum of the ring R with the Zariski topology, and let $\mathcal{O}_{\operatorname{Spec}(R)}$ be the structure sheaf. Then $(\operatorname{Spec}(R), \mathcal{O}_{\operatorname{Spec}(R)})$ is called an *affine scheme*.

In some textbooks, an affine scheme is defined as a locally ringed space (recall definition 1.4.2) $(X, \mathcal{O}_X)$ that is an *affine scheme* if it is locally-ringed-space isomorphic to $(\operatorname{Spec}(R), \mathcal{O}_{\operatorname{Spec}(R)})$ for appropriate ring R.

With the sheaf defined, we have now "justified" thinking of elements on an arbitrary ring R as functions, as

$$\mathcal{O}_{\operatorname{Spec}(R)}(\operatorname{Spec} R) = \mathcal{O}_{\operatorname{Spec}(R)}(D(1)) = R_1 = R$$

Even more is true, due to the section's being constructed through localization of R, they may be

ratio's of functions, and we give elements of section a particular name:

Definition 2.2.4: Regular Functions

Let $X = \operatorname{Spec} R$ be an and $U \subseteq X$ an open subset. Then the set $\mathcal{O}_X(U)$ is called the set of *regular functions* on U.

Note that like complex manifolds (and unlike general smooth manifolds), there are in general relatively few functions in the global section of an as compared to the global section (in the global section of an affine scheme, all rational functions are regular, that is they are R). On the other hand, $\mathcal{O}_X(U)$ may have many interesting rational functions. In fact, in many cases all open subsets of X are dense meaning the local behavior can give a lot of fruitful information on the entire scheme (as shall be the case for abstract varieties, see section 3.7)

Example 2.2.5: Affine Schemes

All spectra can be equipped with a structure sheaf by definition and so they are all affine schemes. For less obvious examples:

1. If $k \ncong k'$ are distinct fields fields, then as schemes $\operatorname{Spec}(k) \ncong \operatorname{Spec}(k')$ as their sheaves are different. Since there is only one point $[(0)]$ an all such Spectra, the functions (elements of the sheaf) will all return themselves, which the reader can think of as there only being constant functions, where each sheaf have different constant functions.

2. Let $R = k[x]_{(x)}$. Then $\operatorname{Spec} R$ contains two points, $[(x)]$ and $[(0)]$ and has 3 open sets, namely

$$\emptyset \quad \{[(0)]\} \quad \{[(0)], [(x)]\} = \operatorname{Spec} R$$

   Note that U and $\emptyset$ are distinguished open sets (since $\{(0)\} = X_x$). The corresponding structure sheaf is:

$$\mathcal{O}_X(X) = R = k[x]_{(x)}$$
$$\mathcal{O}_X(U) = k(x)$$

   The restriction map is the natural inclusion. This schemes is an example of a *local scheme*.

3. Let $R = k[[x]]$, the completion of $k[x]$ at x. The only points of Spec R are again $[(0)]$ and $[(x)]$, and so it has the same open sets. The corresponding structure sheaf is:

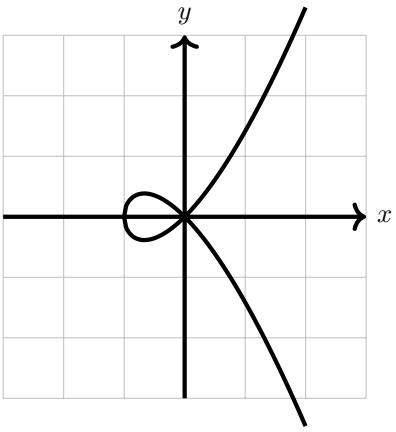
$$\mathcal{O}_X(X) = R = k[[x]]$$
$$\mathcal{O}_X(U) = \operatorname{Frac}(k[[x]])$$

   Thus, we see that there can be schemes with very similar sheaves and identical spectra, but are distinct.

   This scheme is considered to be *even more local* than localization. Consider $\mathbb{A}^2_k$. Then the sequence of maps:

$$k[x, y] \hookrightarrow k[x, y]_{(x,y)} \hookrightarrow k[[x, y]]$$

induces a sequence of inclusion spectra maps $Y \to X \to \mathbb{A}^2_k$, showing how the completion at the point (x, y) is "smaller" than the localization at that point. To demonstrate this, assume k does not have characteristic 2 or 3 and consider the affine scheme given by $y^2 = x^2 + x^3$:



Note that $y^2 - x^2 - x^3$ is irreducible, and localizing at (x, y) keeps the corresponding $(\frac{k[x,y]}{(y^2-x^2-x^3)})_{(x,y)}$ with only two points. However, in

$$\frac{k[[x, y]]}{(y^2 - x^2 - x^3)}$$

we have the access to power series expansion around $(0, 0)$, and we have that the power series:

$$x + \frac{1}{2}x^2 - \frac{1}{8}x^3 + \cdots$$

belongs to this ring, and notably this power series is the square-root of $x^2 + x^3$. Thus, we may write:

$$y^2 - x^3 - x^2 = (y - \sqrt{x^3 + x^2})(y + \sqrt{x^3 + x^2})$$

and so we get that $\frac{k[[x,y]]}{(y^2-x^3-x^2)}$ is *not* an integral domain, and in particular we shall see that Spec $\frac{k[[x,y]]}{(y^2-x^3-x^2)}$ has *two* irreducible components corresponding to the two branches at $(0, 0)$.

4. The reader may recall that most power series do not satisfy an algebraic equation. For this reason, the above ring is often considered too big, and instead we take an intermediary ring H which contains all power series that satisfy algebraic equations over $\mathrm{Frac}(R)$. This is called the *Henselization* of $\operatorname{Spec} R$ at a point $\mathfrak{p}$.

5. If X, Y are then the product of X and Y are:

$$\operatorname{Spec}(R) \times \operatorname{Spec}(S) = \operatorname{Spec}(R \otimes_{\mathbb{Z}} S) = \operatorname{Spec}(R \otimes S)$$

where the tensor product with no base ring by default assume it is over the initial ring. The generalization of this concept shall bee shown in section 3.2.

2.2.2 Ideal Correspondence in Affine Schemes

So far, the dictionary between certain types of ideals of R and the geometric equivalent in $\operatorname{Spec}(R)$ has been establish, namely through $V(-)$ and $I(-)$:

$$\begin{aligned} \text{maximal ideal of } R &\Leftrightarrow \text{closed points of } \operatorname{Spec}(R) \\ \text{minimal prime ideals of } R &\Leftrightarrow \text{irreducible components of } \operatorname{Spec}(R) \\ \text{prime ideal of } R &\Leftrightarrow \text{irreducible closed subset of } \operatorname{Spec}(R) \\ \text{radical ideals of } R &\Leftrightarrow \text{closed subsets of } \operatorname{Spec}(R) \end{aligned}$$

What of a general ideal $J \subseteq R$? In section 2.1.1, this information was "lost" as $I(V(J)) = \sqrt{J}$. However, a sheaf would capture this information!

Let's start with the simplest case of $R = \mathbb{C}[x]/(x^2)$ with $X = \operatorname{Spec}(R)$ that was mentioned in example 2.1.3(6). If we consider the set $\operatorname{Spec}(R)$, then this is just the origin, i.e. $[(x)]$. Namely, by proposition 2.1.10[12],

$$\operatorname{Spec}(\mathbb{C}[x]/(x^2)) \cong_{\mathbf{Top}} \operatorname{Spec}(\mathbb{C}[x]/(x)) \cong_{\mathbf{Top}} \{\overline{(x)}\} \equiv \{0\}$$

Note that (0) is no longer prime since $x^2 \in (0)$ but $x \notin (0)$, and thus $[(0)] \notin \operatorname{Spec}(\mathbb{C}[x]/(x^2))$. To show that the $\left(\operatorname{Spec}\left(\mathbb{C}[x]/\left(x^2\right)\right), \mathcal{O}_{\operatorname{Spec}(\mathbb{C}[x]/(x^2))}\right)$ is different from $\left(\operatorname{Spec}(\mathbb{C}[x]/(x)), \mathcal{O}_{\operatorname{Spec}\,(\mathbb{C}[x]/(x)))}\right)$, let us first label:

$$(X, \mathcal{O}_X) = \left(\operatorname{Spec} \frac{\mathbb{C}[x]}{(x^2)}, \mathcal{O}_{\operatorname{Spec}\mathbb{C}[x]/(x^2)}\right)$$

Consider the image of a polynomial $f(x)$ (i.e. an element in the global section) in $\mathbb{C}[x]/\left(x^2\right)$. For example if $\overline{f(x)} = \overline{(x-a)^2} \in \mathbb{C}[x]/\left(x^2\right)$, then:

$$\bar{f} = \overline{(x-a)^2} = \overline{x^2 - 2ax + a^2} = \overline{-2ax + a^2} \in \mathbb{C}[x]/\left(x^2\right) = \mathcal{O}_X(X)$$

Evaluating at the single point $[(x)] = \epsilon$ we get

$$\bar{f}(\epsilon) = a^2$$

showing that evaluating at ϵ acts like plugging in zero. as x is a function, we get:

$$x(\epsilon) = 0$$

even though x is *not* the zero-function. In fact, one very important algebraic fact that is "obvious" for traditional functions but is not necessarily the case for ring with nilpotents is that a function that is zero everywhere will have to be zero:

Lemma 2.2.6: Nilpotence and Nonzero "Zero" Functions

Let R be a ring with nilpotence (i.e. a nonreduced ring). Then $\mathcal{O}_{\operatorname{Spec}(R)}(\operatorname{Spec}(R))$ contains a function that evaluate to 0 everywhere but is not zero.

[12] Note that this is a topological isomorphism, these are not scheme isomorphic as we shall see in section 3.1.1!

***Proof*:**
Let $f \in R$ be a nilpotence element so that $f \neq 0$ but $f^n = 0$. Then f is contained in each prime ideal, for $ff^{n-1} = f^n = 0 \in \mathfrak{p}$, and so either $f, f^{n-1} \in \mathfrak{p}$, which we can then reduce to having $f \in \mathfrak{p}$. Hence it is in the Nilradical, and so as a function

$$f(\mathfrak{p}) = 0$$

However $f \neq 0$, as we sought to show.

The homeomorphism $\operatorname{Spec} R/I \to \operatorname{Spec} R$ for an ideal I of nilpotent elements is proposition 2.1.10, whose proof was outlined in example 2.1.27(10). The failure of a regular function to be determined by its values at points is caused exactly by nilpotents:

Proposition 2.2.7: Spectrum And Nilpotent Elements

Let R be a ring and $f, g \in R$. Then f and g have the same value at every point of $\operatorname{Spec}(R)$ if and only if $f - g$ is nilpotent.

***Proof*:**
The value of f at $[\mathfrak{p}]$ is its image in $\operatorname{Frac}(R/\mathfrak{p})$, and $R/\mathfrak{p} \to \operatorname{Frac}(R/\mathfrak{p})$ is injective, so f and g have the same value at $[\mathfrak{p}]$ if and only if $f - g \in \mathfrak{p}$. If $R = 0$, then $\operatorname{Spec}(R) = \emptyset$ and $f - g = 0$, so both conditions hold. Otherwise (0) is a proper ideal, and by [10, The Radical as an Intersection of Primes] the intersection of all primes of R is the nilradical. So $f - g$ lies in every prime if and only if it is nilpotent, as claimed.

Nilpotents have no effect on the topological space (proposition 2.1.10). Their presence is felt within the sheaf we put on spectra (section 2.2.1).

Hence if the nilradical is zero, $\sqrt{(0)} = (0)$, functions (in global sections of affine schemes) are determined by their points[13]! In the new language we developed, this say that a function vanishes at every point of an affine scheme if and only if it is nilpotent! Note how this contrasts to the case of (classical) varieties where we required k to be algebraically closed for the same result, showing the greater generality achieved by having the extra prime points!

How to then visualize the $\operatorname{Spec}(R)$ along with its structure sheaf? It is usually done by picturing a "cloud" or "fuzz" around the point:

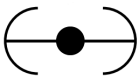


Figure 2.3: Picture of $(X, \mathcal{O}_X)$, not just $X = \operatorname{Spec}(R)$

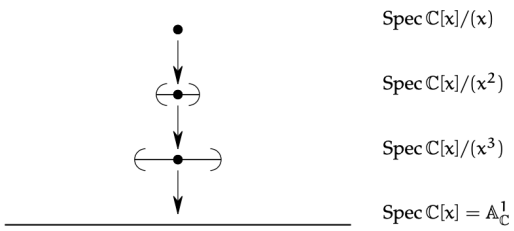
This can be thought of as the "first degree germ" information around the point (recall that in $\mathbb{C}[\epsilon]$, $f(a + \epsilon) = f(a) + f'(a)\epsilon$, showing that the output of f at a plus the "fudge" of ϵ gives $f(a)$ plus an infinitesimal amount in the direction $f'(a)$. The sequence of restrictions $\mathbb{C}[x] \to \mathbb{C}[x]/(x^2) \to \mathbb{C}[x]/(x)$ should be thought of as inching closer to $f(0)$:

[13] Proposition 2.4.6 shall push this result to its maximal generalization on quasicompact schemes

$$\mathbb{C}[x] \twoheadrightarrow \mathbb{C}[x]/(x^2) \twoheadrightarrow \mathbb{C}[x]/(x)$$

$$f(x) \longmapsto f(0),$$

The reason the point is visualized to have the fuzz is that it is the point that creates this "effect" when a function evaluates at this point, in other words remembering the functions behaviour changes the geometric interpretation of the point. If we take, $\mathbb{C}[x]/(x^3)$, then then the spectrum will again be a single point, and when finding an element from the global section, the value, 1st derivative, and 2nd derivative will be remembered. This can either be thought of a larger fuzz or a fuzz that is "bending" with more degree's of precision. We can have the following chain of inclusions:



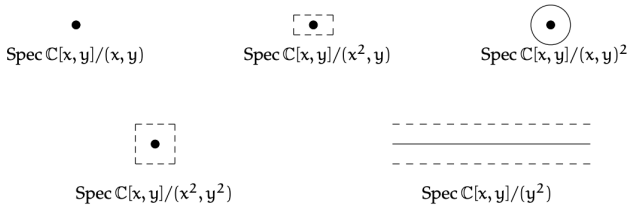
Going to higher dimensions and considering:

$$\operatorname{Spec}\left(\frac{\mathbb{C}[x,y]}{(x,y)^2}\right)$$

this can be thought of as remembering the derivative in all directions, and can be represented by a circle.

Example 2.2.8: Geometric Interpretation of Nilpotence

1. The reader should stare at this image to make sure it makes sense:



2. consider,
$$\operatorname{Spec}\left(\frac{\mathbb{C}[x,y]}{(y^2,xy)}\right)$$
Then this can be thought of as the intersection of a thickened x-axis and the "+" created by xy, that is this can be thought of as the x-axis as well as the first-order differential information around the origin.

Once we introduce stalks in section 2.2.5, these points will be more detectable. Notice that $[(x)]$ is *no longer* a generic point since $y^2 = 0 \in \overline{(x)}$ but $y \notin \overline{(x)}$, and hence the ideal is not prime, thus no longer a point. On the other hand, the function y when evaluating at the points $\overline{[(x-a,y-b)]} = \overline{[(x-a,y)]}$ is 0 at each point

3. Consider
$$\operatorname{Spec}\left(\frac{\mathbb{C}[x,y]}{(y^2)}\right)$$
By proposition 2.1.10, this is homeomorphic to $\mathbb{A}^1_{\mathbb{C}}$. As this is a rather simple example, this can be seen concretely: if $\mathfrak{p} \subseteq \mathbb{C}[x,y]/(y^2)$ is prime and contains y^2, then by primality $y \in \mathfrak{p}$, and hence the prime ideals correspond to those of $\mathbb{C}[x,y]/(y^2,y) \cong \mathbb{C}[x]$, i.e. those of $\mathbb{A}^1_{\mathbb{C}}$.

Now, in $\mathbb{C}[x,y]/(y^2)$, there cannot be y-terms of degree higher than 1: if $F(x,y) \in \mathbb{C}[x,y]/(y^2)$, then:
$$F(x,y) = f(x) + g(x)\,y$$
This can be thought of as $F(x,y)$ "remembering" some first-order information in the y-direction, as though it were the restriction of a function on $\mathbb{A}^2_{\mathbb{C}}$ to an infinitesimal neighborhood of the x-axis. In this decomposition, $f(x)$ is the value on the x-axis and $g(x)$ plays the role of $\partial F/\partial y|_{y=0}$ – a tangent direction at each point. Algebraic operations respect this: for instance,
$$(f_1 + g_1 y)(f_2 + g_2 y) = f_1 f_2 + (f_1 g_2 + f_2 g_1)\,y$$
which mirrors the product rule for derivatives.

Another behaviour that may seem strange is the addition of new units! Consider $1 + xy$. Since y is nilpotent ($y^2 = 0$), the element xy is also nilpotent: $(xy)^2 = x^2y^2 = 0$. We can verify directly that $1 + xy$ is a unit:
$$(1+xy)(1-xy) = 1 - (xy)^2 = 1 - 0 = 1$$
More generally, an element $f(x) + g(x)\,y$ is a unit in $\mathbb{C}[x,y]/(y^2)$ if and only if $f(x)$ is a unit in $\mathbb{C}[x]$, i.e. $f(x) \in \mathbb{C}^*$ is a nonzero constant. This is because the nilpotent part $g(x)\,y$ can always be "absorbed" by the geometric series trick $(1+n)^{-1} = 1 - n$ when $n^2 = 0$, but the non-nilpotent part $f(x)$ must already be invertible.

Geometrically, this means the unit group of $\mathbb{C}[x,y]/(y^2)$ is *much larger* than that of $\mathbb{C}[x]$. For $\mathbb{C}[x]$, the only units are the nonzero constants $\mathbb{C}^*$. For the "fuzzy line" $\mathbb{C}[x,y]/(y^2)$, every element $a + g(x)\,y$ with $a \in \mathbb{C}^*$ is a unit – regardless of the choice of $g(x)$. The unit group is

thus $\mathbb{C}^* \times \mathbb{C}[x]$, an infinite-dimensional group. All of these units evaluate to the constant a at every point (since y maps to 0 in every residue field), yet they are genuinely distinct elements of the ring, distinguished by their "tangent data" $g(x)$. In this sense, $1+g(x)\,y$ is an invertible function that is infinitesimally close to 1: it equals 1 on the underlying topological space but carries a nontrivial first-order perturbation. This is the multiplicative counterpart of the earlier observation that y is a nonzero function that evaluates to 0 everywhere – here, $1 + g(x)\,y$ is a non-constant unit that evaluates to the *same* constant everywhere.

Observe that $D(x)$ is the "punctured fuzzy line": it removes the point $[(x)] \in \operatorname{Spec}(\mathbb{C}[x,y]/(y^2))$ but retains the nilpotent thickening at every remaining point. On the other hand, $D(y) = \emptyset$, since every prime ideal contains y (as $y^2 = 0 \in \mathfrak{p}$ implies $y \in \mathfrak{p}$ by primality). Hence the underlying topological space $|\operatorname{Spec} R|$ does not detect the nilpotent "germ" information carried by y – but the structure sheaf does, as y is a nonzero section of $\mathcal{O}_X$ that vanishes at every point.

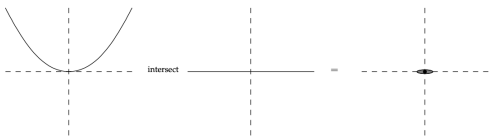
Intersection Theory and Family of Schemes

Intersection theory is covered extensively in *Intersection Theory* [19]. The intersection theory of curves as covered in [22, Intersection Number] gives insightful ways in which the discussion above has practical applications. Recall that the multiplicity information of a curve is defined as an equivalence class of elements of $k[x,y]$ up to scaling. This was to distinguish f and f^2 in $k[x,y]$, something varieties cannot do in classical algebraic geometry. Schemes naturally capture multiplicity information.

For example, consider the following:

$$\begin{aligned}\operatorname{Spec}(\mathbb{C}[x,y]/(y-x^2)) \cap \operatorname{Spec}(\mathbb{C}[x,y]/(y)) &= \operatorname{Spec}(\mathbb{C}[x,y]/(y-x^2,y)) \\ &= \operatorname{Spec}(\mathbb{C}[x,y]/(y,x^2))\end{aligned}$$

That is, intersecting the parabola $y = x^2$ with the x-axis produces a fuzzy point that remembers the multiplicity:



Let us push this idea further and identify the local subscheme associated to a curve passing through the origin with a given tangent direction. Let $R = k[\epsilon]/(\epsilon^2)$ be the dual numbers, and consider a surjection:

$$\varphi : k[x,y] \to R$$

that sends the origin to the unique point of $\operatorname{Spec} R$, i.e. $\varphi(x), \varphi(y) \in (\epsilon)$. The most general such map sends $x \mapsto b\epsilon$ and $y \mapsto -a\epsilon$ for some $a, b \in k$ not both zero (different choices of (a,b) up to scaling give different tangent directions). As $\epsilon^2 = 0$, the map φ vanishes on $(x,y)^2 = (x^2, xy, y^2)$, and hence φ factors through:

$$\overline{\varphi} : \frac{k[x,y]}{(x^2, xy, y^2)} \to R$$

As R is 2-dimensional while $k[x,y]/(x^2,xy,y^2)$ is 3-dimensional (with basis $\{1,x,y\}$), the map $\overline{\varphi}$ has a 1-dimensional kernel. One can check that this kernel is generated by $ax+by$ (the linear form vanishing in the chosen tangent direction), and hence we get the isomorphism:

$$X_{a,b} = \operatorname{Spec} \frac{k[x,y]}{(x^2,xy,y^2,ax+by)} \cong \operatorname{Spec} R$$

How can we interpret this scheme?

- Algebraically, we can interpret $X_{a,b} \subseteq \mathbb{A}^2_k$ as the subscheme associated to the ideal containing all functions $f \in k[x,y]$ that vanish at the origin *and* have partial derivatives satisfying:

  $$b\left.\frac{\partial f}{\partial x}\right|_0 - a\left.\frac{\partial f}{\partial y}\right|_0 = 0$$

  This says that the gradient of f at the origin is proportional to (a,b), or equivalently, $f = c(ax+by)+g(x,y)$ where $g(x,y) \in (x,y)^2$ consists of higher-order terms.

- Geometrically, $X_{a,b}$ is the first-order neighborhood of the origin along the line $ax+by=0$. It arises as the composite

  $$\operatorname{Spec} \frac{k[t]}{(t^2)} \hookrightarrow \mathbb{A}^1_k \xrightarrow{t \mapsto (bt,-at)} \mathbb{A}^2_k$$

  which embeds a "fuzzy point" into $\mathbb{A}^2$ along the tangent direction $(b,-a)$.

We interpret $X_{a,b}$ as containing the point $(0,0)$ together with the infinitesimally near information in the direction specified by the line $ax+by=0$ (the "tangent-line" information!). Such a scheme arises naturally when intersecting two curves non-transversally (the example with the parabola and the line is the simplest such case).

Another place where such schemes naturally arise is when looking at the *limit of families of schemes*. To understand this, take the case of two points $(0,0),(a,b) \in \mathbb{A}^2_k$. These two points together form a scheme:

$$X_1 = \operatorname{Spec} R = \operatorname{Spec} \frac{k[x,y]}{(x,y)\cap(x-a,\,y-b)} = \operatorname{Spec} \frac{k[x,y]}{(x^2-ax,\,xy-bx,\,xy-ay,\,y^2-by)}$$

It should not be hard to see (e.g. via the Chinese Remainder Theorem) that $R \cong k \times k$.

Now, suppose (a,b) moves towards $(0,0)$ along a polynomial curve. In particular, say there are two polynomial functions $a(t), b(t)$ such that $(a(0),b(0)) = (0,0)$. Let $X_t = \{(0,0),\,(a(t),b(t))\}$. Then what scheme would we expect to get when $t \to 0$? The advantage of using schemes for intersection theory, as outlined above, suggests that this limit should remember what direction the points came from – and indeed it does. Even more than this, the limit shall still be a length-2 scheme: two points that have "collided" but retained a memory of their approach direction.

Let us elucidate the details. Take

$$I_t = (x,y)\cap(x-a(t),\,y-b(t)) = (x^2-a(t)x,\;xy-b(t)x,\;xy-a(t)y,\;y^2-b(t)y)$$

As $t \to 0$, the ideal I_0 must contain x^2, xy, and y^2. Furthermore, I_t always contains a linear form:

$$a(t)y - b(t)x = (xy-b(t)x) - (xy-a(t)y)$$

Writing $a(t) = a_1 t + a_2 t^2 + \cdots$ and $b(t) = b_1 t + b_2 t^2 + \cdots$ (since $a(0) = b(0) = 0$), we can divide by t to obtain an element of I_t for $t \neq 0$:

$$\frac{a(t)y - b(t)x}{t} = a_1 y - b_1 x + t \cdot (a_2 y - b_2 x + \cdots)$$

Taking $t \to 0$, the limit ideal contains:

$$(x^2,\, xy,\, y^2,\, a_1 y - b_1 x) \subseteq I_0$$

These are in fact equal. To see this, note that the quotient $k[x,y]/I_t$ is a k-vector space of dimension 2 for each $t \neq 0$ (since I_t cuts out two distinct points). By a continuity argument (in the sense of flat limits), the limiting quotient $k[x,y]/I_0$ also has dimension 2[14]. Since the quotient by the left-hand side $(x^2, xy, y^2, a_1 y - b_1 x)$ already has dimension 2 (with basis $\{1, b_1 x\}$ if $b_1 \neq 0$, or $\{1, a_1 y\}$ if $a_1 \neq 0$), the inclusion must be an equality. Labelling $\alpha = a_1$, $\beta = b_1$ and $X_0 = X_{\alpha,\beta}$, we may write:

$$\lim_{t \to 0} X_t = X_{\alpha,\beta}$$

In higher dimensions, things get more complicated, and when our schemes are no longer local even more so. We shall see in section 6.8 that what we have been examining is a *flat family* of schemes. Flatness provides the correct conditions for a parameterized family of schemes to have well-behaved limits, generalizing the "continuity" argument used above.

Exercise 2.2.1

1. Show that for any polynomial $f \in k[x]$, if $a + b\epsilon \in \mathbb{C}(\epsilon)$, then

$$f(a + b\epsilon) = f(a) + bf'(a)\epsilon$$

2. let k be algebraically closed, R be a local k-algebra of dimension 2 with maximal ideal $\mathfrak{m}$. Show that $R \cong k[x]/(x^2)$ (hint: use some dimension argument and Nakayama's lemma to conclude $\mathfrak{m}^2 = 0$).

3. Let k be algebraically closed. Find two non-isomorphic local k-algebras of dimension 3. Geometrically, we can think of this as representing the extra flexibility in directions or local schemes can capture.

4. Let k be algebraically closed and $\operatorname{Spec} R = \operatorname{Spec} k[x_1, ..., x_n]/I \subseteq \mathbb{A}^n_k$ be a local subscheme of dimension 0 and degree 3 (R has dimension 3 over k) supported at the origin. Then $\operatorname{Spec} R$ is either isomorphic to $\operatorname{Spec} k[x]/(x^3)$ or $\operatorname{Spec} k[x,y]/(x^2, xy, y^2)$, and these two schemes are not isomorphic to each other.

5. Let R be a ring that is not a field. Show that the restriction map on the structure sheaf is almost never surjective. Hence, there are more local functions than global function.

6. Let R be a reduced ring (contains no nilpotent elements). Show that the restriction maps are injective, that is each function on a larger open set can only restrict to one function. Give a counter-example for a ring with nilpotent elements.

[14] This argument is more subtle without the finite-dimensionality hypothesis, which shall be addressed when we cover this in fuller generality in section 6.8.

2.2.3 Schemes

The last step in the definition of schemes is to observe that many geometric objects are not the spectrum of a ring, but are *locally* the spectrum of a ring. The prototypical examples are projective spaces, their closed subsets, and sequences of blow ups—none of which are in general affine.

Definition 2.2.9: Scheme

A *scheme* is a locally ringed space $(X, \mathcal{O}_X)$ that is locally affine: there exists an open cover $X = \bigcup_i U_i$ such that for each i, there is a ring R_i and an isomorphism of locally ringed spaces

$$(U_i,\ \mathcal{O}_X|_{U_i}) \cong_{\mathbf{LRS}} (\operatorname{Spec}(R_i),\ \mathcal{O}_{\operatorname{Spec}(R_i)}).$$

Such a cover is called an *affine open cover* of X.

The requirement that $(X, \mathcal{O}_X)$ be a locally ringed space—that is, that each stalk $\mathcal{O}_{X,x}$ be a local ring—is essential. Without it, the "evaluation" of a section at a point (via the residue field $\kappa(x) = \mathcal{O}_{X,x}/\mathfrak{m}_x$) would not be well-defined, and the connection to geometry would be lost. We saw in section 1.4 that locally ringed spaces are the correct generalization of the spaces that arise in both differential and algebraic geometry. Since the stalk of $\mathcal{O}_X$ at a point $x \in X$ is a local ring (by the locally ringed space condition), we denote it $\mathcal{O}_{X,x}$, its maximal ideal $\mathfrak{m}_x$, and its residue field $\kappa(x) = \mathcal{O}_{X,x}/\mathfrak{m}_x$.

The topology of affine schemes is generated by distinguished open sets. Lemma 2.2.30 shall show that given $\operatorname{Spec} R, \operatorname{Spec} S \subseteq X$ where their intersection is non-empty, the intersection can be covered by distinguished open sets that are distinguished on both $\operatorname{Spec} R$ and $\operatorname{Spec} S$. Hence the topology of schemes is also generated by distinguished open sets. See section 2.8⋆ for a deeper discussion about the purely topological properties of schemes.

The way to interpret the agreement on intersections is that locally a scheme respects the algebraic relations imposed by a ring R_i, and the compatibility on $U_i \cap U_j$ (when nonempty) corresponds to the algebraic relations between R_i and R_j being consistent. A detailed example is presented in example 2.2.57.

2.2.10 Lack of $V(-)$ and $I(-)$? The topology of an affine scheme is characterized by the ideals of its global ring. General schemes are no longer tethered to the global sections (as shall be momentarily demonstrated). However, there is a generalization of $V(-)$ and $I(-)$ that works with the (quasi-coherent) *ideal sheaves* of a scheme X; the details are covered in section 5.1.4. From this perspective, the structure sheaf $\mathcal{O}_X$ can be thought of as generalizing the notion of a ring to a "local" object, remembering information that no single ring can capture.

Definition 2.2.11: Morphism of Schemes

Let $(X, \mathcal{O}_X)$ and $(Y, \mathcal{O}_Y)$ be schemes. Then a *morphism* of schemes is a morphism of locally ringed spaces $(\varphi, \varphi^\#) : (X, \mathcal{O}_X) \to (Y, \mathcal{O}_Y)$.

A *isomorphism of schemes* is an isomorphism of locally ringed spaces: a homeomorphism $f : X \to Y$ together with a sheaf isomorphism $f^\# : \mathcal{O}_Y \xrightarrow{\sim} f_*\mathcal{O}_X$.

Not every morphism of ringed spaces preserves local rings (see example 3.1.1). The full exploration

of the nuances shall be covered in section 3.1.1.

Example 2.2.12: Schemes

1. All affine schemes are schemes, covered by a single open set.
2. The key non-affine scheme is projective space $\mathbb{P}^n_k$. We shall construct the general projective scheme $\mathbb{P}^n_R$ in section 2.3; for now, the reader can verify that the usual projective space as defined in [22, Projective Space] is a scheme by exhibiting the standard affine charts $U_i = \{x_i \neq 0\} \cong \mathbb{A}^n_k$. Any closed subvariety of $\mathbb{P}^n_k$ (with the appropriate structure sheaf) is also a scheme, typically not affine.
3. The following is the smallest non-affine scheme. Take $X = \{p, q_1, q_2\}$ with the topology generated by $X_1 = \{p, q_1\}$ and $X_2 = \{p, q_2\}$ as a sub-base. Define the sheaf of rings by
$$\mathcal{O}(X) = \mathcal{O}(X_1) = \mathcal{O}(X_2) = K[x]_{(x)}, \qquad \mathcal{O}(\{p\}) = K(x),$$
with restriction maps $\mathcal{O}(X) \to \mathcal{O}(X_i)$ the identity and $\mathcal{O}(X_i) \to \mathcal{O}(\{p\})$ the natural localization. This presheaf is a sheaf, and $(X, \mathcal{O})$ is a scheme but *not* an affine scheme: the ring $K[x]_{(x)}$ has two points in its spectrum, but X has three. Geometrically, this is the "germ of a doubled point"—the generic point of $\mathbb{A}^1$ has been split into two.
4. By proposition 2.1.21, a finite disjoint union of schemes is a scheme. An arbitrary (infinite) disjoint union of affine schemes is a scheme but not an affine scheme, as it fails to be quasi-compact.
5. All affine schemes are "separated" (a property to be defined in section 3.5.1). There exist non-separated schemes; the simplest is the "line with doubled origin," constructed by gluing two copies of $\mathbb{A}^1_k$ along $\mathbb{A}^1_k \setminus \{0\}$.
6. If the reader has already encountered blow-ups, the schemes associated to a blow-up are typically not affine.
7. The open subset $D(x) \cup D(y) \subseteq \mathbb{A}^2_k$ is a scheme that is quasi-compact, separated, and a subset of an affine scheme, yet is not itself affine. The details are in Example 2.2.53.
8. Let $X = \{*\}$ be a singleton and $\mathcal{O}_X(X) = R$ for a commutative ring R. Is $(X, \mathcal{O}_X)$ a scheme? First, it must be affine so $X = \operatorname{Spec}(R)$. Next, $\operatorname{Spec} R$ must have a single prime ideal. This must be a 0-dimensional is a local ring. For example:
$$R \in \{k, \mathbb{Z}/p^n\mathbb{Z}, k[x]/(x^m)\}$$
Note there are local rings with more than one point, for example $k[[x]]$.

It is not always immediate whether a scheme is affine. For example, $\mathbb{A}^1_{\mathbb{C}} \setminus \{[(x)]\}$ may seem non-affine since it is not an affine variety in the classical sense, but it is isomorphic to $\operatorname{Spec}(\mathbb{C}[x, x^{-1}])$. On the other hand, $\mathbb{A}^2_{\mathbb{C}} \setminus \{[(x, y)]\}$ is genuinely not affine (Example 2.2.53). We shall establish tools for detecting affineness shortly.

Given a scheme X, we write $|X|$ for the underlying topological space. The elements of $\mathcal{O}_X(U)$ are

called *sections* over U, and sections of $\mathcal{O}_X(X)$ are called *global sections*. These are variously denoted

$$\mathcal{O}_X(U), \qquad \Gamma(U, \mathcal{O}_X), \qquad H^0(U, \mathcal{O}_X),$$

depending on context: the first is convenient when only the structure sheaf is present; the second will be useful when multiple sheaves on X are in play; and the third hints at the fact that the global section functor $\Gamma(X, -)$ is left-exact but generally not exact, with its right-derived functors $H^i(X, -)$ forming the cohomology theory developed in chapter 6.

Recovering the ring from an affine scheme

A natural question arises: if a scheme X happens to be affine, can we recover the ring R from the locally ringed space $(X, \mathcal{O}_X)$? The answer is yes, and the ring is simply the global sections.

Proposition 2.2.13: Distinguished Open Sets and Schemes

Let R be a ring and $f \in R$. Then:

$$(D(f),\ \mathcal{O}_{\mathrm{Spec}\,(R)}|_{D(f)}) \ \cong\ (\mathrm{Spec}\,(R_f),\ \mathcal{O}_{\mathrm{Spec}\,(R_f)}).$$

In particular, taking $f = 1$: if $(X, \mathcal{O}_X)$ is an affine scheme, meaning $(X, \mathcal{O}_X) \cong (\mathrm{Spec}\,(R), \mathcal{O}_{\mathrm{Spec}\,(R)})$ for some ring R, then R is recovered as the global sections:

$$\mathcal{O}_X(X) \ \cong\ R.$$

***Proof*:**
Let $\iota\colon R \to R_f$ be the localization map and $\varphi\colon \mathrm{Spec}\,(R_f) \to D(f)$, $\mathfrak{q} \mapsto \iota^{-1}(\mathfrak{q})$, the homeomorphism of proposition 2.1.12. For $g \in R$ we have $\varphi^{-1}(D(fg)) = D(\iota(fg)) = D(\iota(g))$, as $\iota(f)$ is a unit, and every distinguished open $D(a/f^n)$ of $\mathrm{Spec}\,(R_f)$ equals $D(\iota(a))$. Since $D(h) \cap D(f) = D(fh)$, the sets $D(fg)$ with $g \in R$ form a basis of $D(f)$ (corollary 2.1.13).

For $g \in R$ the rings $\mathcal{O}_{\mathrm{Spec}\,(R)}(D(fg)) = R_{fg}$ and $(\varphi_*\mathcal{O}_{\mathrm{Spec}\,(R_f)})(D(fg)) = (R_f)_{\iota(g)}$ are R-algebras in which fg becomes a unit, and each of them receives a unique R-algebra map from the other: [10, Universal Property of Localization] applied to $R \to (R_f)_{\iota(g)}$ gives $\lambda_g\colon R_{fg} \to (R_f)_{\iota(g)}$, and applied first to $R \to R_{fg}$, which inverts f, and then to $R_f \to R_{fg}$, which inverts $\iota(g)$, it gives a map back. Both composites are R-algebra endomorphisms of localizations of R, hence identities by the same uniqueness, so λ_g is an isomorphism. The restriction maps of both structure sheaves between these rings are R-algebra maps out of localizations of R, so they commute with the λ_g, again by uniqueness, and λ_g depends only on the open set $D(fg)$. By corollary 1.3.4 the λ_g extend to a sheaf isomorphism $\varphi^{\#}\colon \mathcal{O}_{\mathrm{Spec}\,(R)}|_{D(f)} \to \varphi_*\mathcal{O}_{\mathrm{Spec}\,(R_f)}$, and with the homeomorphism φ it is an isomorphism of locally ringed spaces (definition 2.2.11).

Let $(\Phi, \Phi^{\#})\colon (X, \mathcal{O}_X) \to (\mathrm{Spec}\,(R), \mathcal{O}_{\mathrm{Spec}\,(R)})$ be an isomorphism and write $Y = \mathrm{Spec}\,(R)$. Then $\Phi^{\#}(Y)\colon \mathcal{O}_Y(Y) \to \mathcal{O}_X(\Phi^{-1}(Y)) = \mathcal{O}_X(X)$ is a ring isomorphism, and $\mathcal{O}_Y(Y) = \mathcal{O}_Y(D(1)) = R_1 \cong R$ by the definition of the structure sheaf, completing the proof.

The motivation behind this proposition is that for affine schemes, the global sections completely determine the scheme. This is emphatically *not* the case for general schemes: the global sections of

$\mathbb{P}^n_k$ are just k (for $n \geq 1$), which carries no information about the geometry of projective space. The structure sheaf, not the global sections, is the fundamental invariant.

Lemma 2.2.14: Germs and Stalk Maps of Isomorphisms

1. Let T be a ring, $t \in T$ and $\mathfrak{q} \in \operatorname{Spec}(T)$. The germ $t_\mathfrak{q}$ is a unit of $\mathcal{O}_{\operatorname{Spec}(T),\mathfrak{q}}$ if and only if $t \notin \mathfrak{q}$.
2. Let $(\Phi, \Phi^\#) \colon (X, \mathcal{O}_X) \to (Y, \mathcal{O}_Y)$ be an isomorphism of ringed spaces (definition 1.4.12) and $x \in X$. The stalk map $\Phi^\#_x \colon \mathcal{O}_{Y,\Phi(x)} \to \mathcal{O}_{X,x}$, sending the class of (s, V) to the class of $(\Phi^\#(V)(s), \Phi^{-1}(V))$, is a ring isomorphism.

***Proof*:**

1. If $t \notin \mathfrak{q}$, then $\mathfrak{q} \in D(t)$ and t is invertible in $\mathcal{O}_{\operatorname{Spec}(T)}(D(t)) = T_t$, so $t_\mathfrak{q}$ is a unit. Conversely, if $t_\mathfrak{q}$ is a unit, then by corollary 2.1.13 there are $h \in T$ with $\mathfrak{q} \in D(h)$ and $a/h^n \in T_h$ with $t \cdot a/h^n = 1$ in T_h, that is, $h^m(ta - h^n) = 0$ for some m. If $t \in \mathfrak{q}$, then $h^{m+n} = h^m ta \in \mathfrak{q}$ and $h \in \mathfrak{q}$, which contradicts $\mathfrak{q} \in D(h)$. So $t \notin \mathfrak{q}$.
2. As Φ is a homeomorphism, $\Phi(W)$ is open for $W \subseteq X$ open, and $\Phi^\#(\Phi(W)) \colon \mathcal{O}_Y(\Phi(W)) \to \mathcal{O}_X(W)$ is a ring isomorphism. The inverses of the $\Phi^\#(V)$ commute with restriction because the $\Phi^\#(V)$ do, so sending the class of (t, W) to the class of $((\Phi^\#(\Phi(W)))^{-1}(t), \Phi(W))$ is well defined on germs at x. It is inverse to $\Phi^\#_x$ on representatives, and $\Phi^\#_x$ is a ring homomorphism, as the $\Phi^\#(V)$ are, completing the proof.

Proposition 2.2.15: Characterization of Affine Schemes

Let $(X, \mathcal{O}_X)$ be a locally ringed space with $R = \mathcal{O}_X(X)$. For $x \in X$ and $f \in R$ write f_x for the germ of f in $\mathcal{O}_{X,x}$, and put $U_f = \{x \in X : f_x \notin \mathfrak{m}_x\}$. The set U_f is open and the restriction of f to it is a unit, so restriction induces a ring map $c_f \colon R_f \to \mathcal{O}_X(U_f)$. Then $(X, \mathcal{O}_X) \cong (\operatorname{Spec}(R), \mathcal{O}_{\operatorname{Spec}(R)})$ if and only if:

1. For every $f \in R$, the map $c_f \colon R_f \to \mathcal{O}_X(U_f)$ is an isomorphism.
2. The natural map $X \to |\operatorname{Spec}(R)|$, sending x to the prime $\mathfrak{p}_x = \{f \in R \mid f_x \in \mathfrak{m}_x\}$ (the preimage of $\mathfrak{m}_x$ under $R \to \mathcal{O}_{X,x}$), is a homeomorphism.

***Proof*:**

Step 1: U_f is open, $f|_{U_f}$ is a unit, and c_f exists. In the local ring $\mathcal{O}_{X,x}$ the complement of $\mathfrak{m}_x$ is the group of units. Let $x \in U_f$. The inverse of f_x is represented by some $g \in \mathcal{O}_X(W)$ on an open $W \ni x$, and after shrinking W we have $f|_W \cdot g = 1$ in $\mathcal{O}_X(W)$. Every $y \in W$ then has $f_y g_y = 1$, so $W \subseteq U_f$, and U_f is open. Inverses in a ring are unique, so on overlaps of two such sets W the local inverses agree, and the gluing axiom gives $g \in \mathcal{O}_X(U_f)$ restricting to each of them. Then $f|_{U_f} \cdot g$ and 1 agree on every W, so $f|_{U_f} \cdot g = 1$ by the locality axiom. By [10, Universal Property of Localization], restriction $R \to \mathcal{O}_X(U_f)$ factors uniquely through a ring map $c_f \colon R_f \to \mathcal{O}_X(U_f)$, and $c_f(s/1) = s|_{U_f}$.

Step 2: germs of global sections of an affine scheme. For a ring T, $t \in T$ and $\mathfrak{q} \in \operatorname{Spec}(T)$, the germ $t_{\mathfrak{q}}$ is a unit of $\mathcal{O}_{\operatorname{Spec}(T),\mathfrak{q}}$ if and only if $t \notin \mathfrak{q}$ (lemma 2.2.14(1)).

Step 3: the forward direction. Let $(\Phi, \Phi^{\#}) \colon (X, \mathcal{O}_X) \to (\operatorname{Spec}(R), \mathcal{O}_{\operatorname{Spec}(R)})$ be an isomorphism of locally ringed spaces (definition 2.2.11). On global sections $\Phi^{\#}$ gives a ring isomorphism $\alpha \colon R = \mathcal{O}_{\operatorname{Spec}(R)}(\operatorname{Spec}(R)) \to \mathcal{O}_X(X) = R$. For $x \in X$, the stalk map $\Phi^{\#}_x \colon \mathcal{O}_{\operatorname{Spec}(R),\Phi(x)} \to \mathcal{O}_{X,x}$ sends the class of (s, V) to the class of $(\Phi^{\#}(V)(s), \Phi^{-1}(V))$, and it is a ring isomorphism (lemma 2.2.14(2)). It sends $r_{\Phi(x)}$ to $\alpha(r)_x$ for $r \in R$. A ring isomorphism preserves units, so by Step 2, $\alpha(r)_x \in \mathfrak{m}_x$ if and only if $r \in \Phi(x)$. Hence $\mathfrak{p}_x = \alpha(\Phi(x))$. The map $\operatorname{Spec}(R) \to \operatorname{Spec}(R)$, $\mathfrak{q} \mapsto \alpha(\mathfrak{q})$, is a bijection carrying each $V(T)$ onto $V(\alpha(T))$, and its inverse $\mathfrak{q} \mapsto \alpha^{-1}(\mathfrak{q})$ carries $V(T)$ onto $V(\alpha^{-1}(T))$, so it is a homeomorphism. The map $x \mapsto \mathfrak{p}_x$ is its composite with Φ, a homeomorphism, which is (2).

For (1), fix $f \in R$ and put $r = \alpha^{-1}(f)$. Then $U_f = \{x \mid f \notin \mathfrak{p}_x\} = \{x \mid r \notin \Phi(x)\} = \Phi^{-1}(D(r))$, and $\Phi^{\#}(D(r)) \colon R_r = \mathcal{O}_{\operatorname{Spec}(R)}(D(r)) \to \mathcal{O}_X(U_f)$ is a ring isomorphism with $\Phi^{\#}(D(r))(s/1) = \alpha(s)|_{U_f}$, by compatibility of $\Phi^{\#}$ with restriction. Let $\alpha_r \colon R_r \to R_f$ be the isomorphism $s/r^n \mapsto \alpha(s)/f^n$ induced by α, whose inverse is induced by α^{-1}. The ring maps $c_f \circ \alpha_r$ and $\Phi^{\#}(D(r))$ both send $s/1$ to $\alpha(s)|_{U_f}$, so they coincide by the uniqueness in [10, Universal Property of Localization]. Hence $c_f = \Phi^{\#}(D(r)) \circ \alpha_r^{-1}$ is an isomorphism.

Step 4: the reverse direction. Assume (1) and (2), and let $\psi \colon X \to \operatorname{Spec}(R)$, $x \mapsto \mathfrak{p}_x$, a homeomorphism by (2). For $f \in R$ we have $\psi^{-1}(D(f)) = \{x \mid f \notin \mathfrak{p}_x\} = U_f$. On the distinguished opens put

$$\psi^{\#}(D(f)) = c_f \colon \mathcal{O}_{\operatorname{Spec}(R)}(D(f)) = R_f \longrightarrow \mathcal{O}_X(U_f) = (\psi_* \mathcal{O}_X)(D(f))$$

If $D(g) \subseteq D(f)$, then $U_g \subseteq U_f$, and the ring maps $\operatorname{res}_{U_f,U_g} \circ c_f$ and $c_g \circ \operatorname{res}_{D(f),D(g)}$ from R_f to $\mathcal{O}_X(U_g)$ both send $s/1$ to $s|_{U_g}$, because the restriction $R_f \to R_g$ of the structure sheaf sends $s/1$ to $s/1$. They coincide by the uniqueness in [10, Universal Property of Localization]. This applies in particular when $D(g) = D(f)$, so $\psi^{\#}(D(f))$ does not depend on the element chosen to describe the open set, and these maps commute with restriction. Each c_f is an isomorphism by (1), so corollary 1.3.4 extends them to a sheaf isomorphism $\psi^{\#} \colon \mathcal{O}_{\operatorname{Spec}(R)} \to \psi_* \mathcal{O}_X$. Thus $(\psi, \psi^{\#})$ is a homeomorphism together with a sheaf isomorphism, an isomorphism of locally ringed spaces by definition 2.2.11. Its stalk maps are ring isomorphisms (lemma 2.2.14(2)), so they carry the maximal ideal onto the maximal ideal and are local homomorphisms, as we sought to show.

Theorem 2.2.16: Ring Homomorphisms and Affine Scheme Isomorphisms

Let R and S be rings. An isomorphism of locally ringed spaces $\pi : \operatorname{Spec}(R) \to \operatorname{Spec}(S)$ induces an isomorphism $\pi^{\#} : S \xrightarrow{\sim} R$ on global sections. Conversely, an isomorphism $\varphi : S \xrightarrow{\sim} R$ induces an isomorphism $\operatorname{Spec}(\varphi) : \operatorname{Spec}(R) \xrightarrow{\sim} \operatorname{Spec}(S)$, and these two constructions are inverse to each other.

***Proof*:**
For a ring T we identify $\mathcal{O}_{\mathrm{Spec}\,(T)}(\mathrm{Spec}\,(T)) = T$ (proposition 2.2.13 with $f = 1$).

Step 1: global sections. If $(\pi, \pi^\#)$ is an isomorphism of locally ringed spaces, then $\pi^\#$ is an isomorphism of sheaves (definition 2.2.11), so $\pi^\#(\mathrm{Spec}\,(S))\colon S \to \mathcal{O}_{\mathrm{Spec}\,(R)}(\pi^{-1}(\mathrm{Spec}\,(S))) = R$ is a ring isomorphism.

Step 2: the isomorphism $\mathrm{Spec}\,(\varphi)$. Let $\varphi\colon S \to R$ be a ring isomorphism and $\varphi^a\colon \mathrm{Spec}\,(R) \to \mathrm{Spec}\,(S)$, $\mathfrak{p} \mapsto \varphi^{-1}(\mathfrak{p})$. It is a bijection with inverse $\mathfrak{q} \mapsto \varphi(\mathfrak{q})$, the preimage of $V(T)$ under φ^a is $V(\varphi(T))$, and the preimage of $V(T')$ under the inverse is $V(\varphi^{-1}(T'))$, so φ^a is a homeomorphism with $(\varphi^a)^{-1}(D(g)) = D(\varphi(g))$ for $g \in S$. On the distinguished opens put

$$\varphi^\#(D(g)) = \varphi_g\colon S_g \longrightarrow R_{\varphi(g)} = (\varphi^a_*\mathcal{O}_{\mathrm{Spec}\,(R)})(D(g))$$

where $\varphi_g(s/g^n) = \varphi(s)/\varphi(g)^n$ is the ring map given by [10, Universal Property of Localization]. It is an isomorphism, its inverse being induced in the same way by φ^{-1}. If $D(h) \subseteq D(g)$, the ring maps $\mathrm{res}_{D(\varphi(g)),D(\varphi(h))} \circ \varphi_g$ and $\varphi_h \circ \mathrm{res}_{D(g),D(h)}$ from S_g to $R_{\varphi(h)}$ both send $s/1$ to $\varphi(s)/1$, so they coincide by the uniqueness in the universal property, and in particular φ_g does not depend on the element describing $D(g)$. By corollary 1.3.4 these isomorphisms extend to a sheaf isomorphism $\varphi^\#\colon \mathcal{O}_{\mathrm{Spec}\,(S)} \to \varphi^a_*\mathcal{O}_{\mathrm{Spec}\,(R)}$, so $\mathrm{Spec}\,(\varphi) = (\varphi^a, \varphi^\#)$ is an isomorphism of locally ringed spaces (definition 2.2.11).

Step 3: the two constructions are mutually inverse. The map induced by $\mathrm{Spec}\,(\varphi)$ on global sections is $\varphi^\#(D(1)) = \varphi_1 = \varphi$. Conversely, let $(\pi, \pi^\#)$ be an isomorphism of locally ringed spaces and $\varphi = \pi^\#(\mathrm{Spec}\,(S))$. For $\mathfrak{p} \in \mathrm{Spec}\,(R)$ the stalk map $\pi^\#_{\mathfrak{p}}\colon \mathcal{O}_{\mathrm{Spec}\,(S),\pi(\mathfrak{p})} \to \mathcal{O}_{\mathrm{Spec}\,(R),\mathfrak{p}}$, sending the class of (s, V) to the class of $(\pi^\#(V)(s), \pi^{-1}(V))$, is a ring isomorphism (lemma 2.2.14(2)). It sends the germ of $s \in S$ at $\pi(\mathfrak{p})$ to the germ of $\varphi(s)$ at $\mathfrak{p}$. Ring isomorphisms preserve units, so by lemma 2.2.14(1), $s \notin \pi(\mathfrak{p})$ if and only if $\varphi(s) \notin \mathfrak{p}$, that is, $\pi(\mathfrak{p}) = \varphi^{-1}(\mathfrak{p}) = \varphi^a(\mathfrak{p})$. Hence $\pi = \varphi^a$ and $\pi^{-1}(D(g)) = D(\varphi(g))$. By compatibility of $\pi^\#$ with restriction from $\mathrm{Spec}\,(S)$, the ring map $\pi^\#(D(g))\colon S_g \to R_{\varphi(g)}$ sends $s/1$ to $\varphi(s)/1$, so it equals φ_g by the uniqueness in the universal property. Two sheaf morphisms agreeing on a basis are equal (proposition 1.3.3), so $\pi^\# = \varphi^\#$ and $(\pi, \pi^\#) = \mathrm{Spec}\,(\varphi)$, completing the proof.

Given a scheme X, the collection of distinguished open sets from any affine cover forms a basis for the topology on X. This is important because the intersection of two affine opens need not itself be affine (see section 3.5.1), but distinguished opens within each affine chart suffice.

Proposition 2.2.17: Distinguished Open Sets Form a Basis

Let X be a scheme and $\{U_i = \mathrm{Spec}\,(R_i)\}$ an affine open cover. Then the collection of all distinguished open sets $D(f) \subseteq U_i$, as i ranges over the index set and f ranges over R_i, forms a basis for the topology on X.

***Proof*:**
Every open subset of X restricts to an open subset of each U_i, which is a union of distinguished opens in U_i (since distinguished opens form a basis for the Zariski topology on $\operatorname{Spec}(R_i)$). The union over all i covers the original open set.

Finiteness conditions and base ring

In algebra, every ring is a $\mathbb{Z}$-algebra, and additional structure arises by specifying a base ring. The scheme-theoretic counterpart is the following.

Definition 2.2.18: R-scheme and Finite Type Schemes

An *R-scheme* (or a *scheme over R*) is a scheme X together with a morphism $X \to \operatorname{Spec}(R)$, called the *structure morphism*. Equivalently, the sections of $\mathcal{O}_X$ are R-algebras and all restriction maps are R-algebra homomorphisms.
An R-scheme X is *locally of finite type over R* if it can be covered by affine opens $\operatorname{Spec}(S_i)$ where each S_i is a finitely generated R-algebra. If X is also quasi-compact, then X is *of finite type over R*.

In number theory, integral extensions S/R (where S is a finitely generated R-module) play a central role. The scheme-theoretic analogue is:

Definition 2.2.19: Finite R-Schemes

An R-scheme X is *locally finite over R* if it can be covered by affine opens $\operatorname{Spec}(S_i)$ where each S_i is a finitely generated R-module. If X is quasi-compact, then X is *finite over R*.

If $R = k$ is a field (or more generally an Artinian ring), then a finite R-scheme is a finite collection of "fat points"—spectra of Artinian local k-algebras.

2.2.20 Terminology Warning Do not confuse *finite type* with *finite*. A scheme is locally of finite type if its coordinate rings are finitely generated *R-algebras* (finitely many generators as a ring); it is locally finite if its coordinate rings are finitely generated *R-modules* (finitely many generators as a module). The latter is a much stronger condition: $k[x]$ is a finitely generated k-algebra but not a finitely generated k-module.

Example 2.2.21: Finite Type R-schemes

1. Affine n-space $\mathbb{A}^n_k = \operatorname{Spec}(k[x_1, \ldots, x_n])$ is of finite type over k.
2. For polynomials $f_1, \ldots, f_m \in R[x_1, \ldots, x_n]$, the scheme $\operatorname{Spec}(R[x_1, \ldots, x_n]/(f_1, \ldots, f_m))$ is of finite type over R. Most classical intuitions from algebraic sets concern $\overline{k}$-schemes of finite type over $\overline{k}$ (abstract varieties shall require further properties; see section 3.7).
3. If R is the localization of a polynomial ring over k at a prime, then $\operatorname{Spec}(R)$ is generally not of finite type over k. Intuitively, local rings live in a "different dimension" from the global picture.

4. If the fibers of $X \to \operatorname{Spec}(R)$ are infinite-dimensional, then X is not of finite type over R.

2.2.4 Comparing Schemes and Real/Complex Manifolds

Before proceeding to further explore properties of schemes, it is worth taking a moment to catalogue how schemes and real smooth manifolds have some fundamentally different geometric properties:

1. The connected scheme given by $V(x) \cup V(y,z) \subseteq \mathbb{A}^3_k$ (the plane $x = 0$ together with the x-axis, meeting at the origin) has two irreducible components of different dimensions (as we shall more rigorously show in section 2.6). A connected manifold (smooth or even topological) is of a single, well-defined dimension at every point.

2. An abstract manifold cannot self-intersect[15], while schemes (even affine schemes, even varieties) can: consider $V(y^2 - x^3 - x^2) \subseteq \mathbb{A}^2_k$, which has a node at the origin.

3. The patches of a manifold are very regular, being diffeomorphic to $\mathbb{R}^n$; in this sense manifold patches all carry "linear smooth" information, and there is essentially no local diversity from one patch to another. The affine patches of a scheme, by contrast, may be "pathological". For example, the local rings can be non-reduced, and the coordinate ring of a patch may be non-Noetherian, in which case Spec of it may have uncountably many connected components. (The *local* rings never do: a local ring has only the idempotents 0 and 1, so its spectrum is connected.) This reflects how affine schemes represent "local algebraic information," which is far richer than the "local smooth linear information" that real manifolds encode. Some of the conditions and properties we shall introduce later (e.g. Noetherianity, regularity) will restore some order.

4. In Manifold Theory, the notion of irreducible component is trivial, since in any Hausdorff space the only irreducible subsets are singletons. From an analytic perspective, this existence of non-trivial irreducible components in schemes can be thought of as a consequence of the Zariski topology failing to be Hausdorff, together with the fact that a function vanishing at a non-closed point vanishes on the whole of its closure; part of this phenomenon is the existence of "large" (i.e. non-closed) points for schemes. More generally, the correspondence between irreducible closed sets and their generic points (which holds in every scheme, all of them being sober) shows that schemes encode much more "global" information about codimension subsets than real smooth manifolds do.

5. The topology of $\mathbb{R}^n$ (and hence of $\mathbb{C}^n$ as a real manifold) is the product topology: the sets

$$\prod_{i=1}^{n}(a_i, b_i), \qquad a_i, b_i \in \mathbb{R},\ a_i < b_i,$$

form a topological basis for $\mathbb{R}^n$. This is far from the case for $\mathbb{A}^n_k$: the Zariski topology on $\mathbb{A}^n_k$ is *not* the product of the Zariski topologies on $\mathbb{A}^1_k$. The product is strictly coarser: its closed sets are the finite unions of horizontal and vertical lines and points, so a curve such as $V(y - x^2)$, Zariski-closed in $\mathbb{A}^2_k$, is not closed in the product. However, there will still be a meaningful way to construct the scheme $\mathbb{A}^n_k$ from $\mathbb{A}^1_k$ via the *fibered product* (see section 3.2).

[15] An *immersion* of a manifold may of course have self-intersections in the target, but the source manifold itself is not singular at such points.

6. If a scheme X is irreducible, all its nonempty open sets are dense. Many of the schemes we shall work with will be unions of irreducible subschemes, where for each irreducible subscheme every nonempty affine open subset is dense in that subscheme. Dense open subsets are far from typical for manifolds (e.g. a disjoint union of two manifolds has open subsets contained in a single component, which are not dense).

7. More generally, the behaviour of schemes matches more closely with the behaviour of complex analytic geometry than with smooth real geometry. We shall draw such analogies often: for example, *Hartogs's Extension Lemma* ([58, chapter 8]) will have parallels with normal schemes, the existence of certain holomorphic/meromorphic functions will tie into divisor theory, compact complex Riemann surfaces will correspond to projective curves, and so forth.

8. (embedding properties) Any smooth real manifold M can be embedded into $\mathbb{R}^N$ for appropriate N (Whitney Embedding Theorem). For classical affine (resp. projective) varieties of dimension n, they essentially by definition embed into $\mathbb{A}^N_k$ (resp. $\mathbb{P}^N_k$) for some N. For complex manifolds, it is *not* the case that they all embed in $\mathbb{C}^N$[16], and in particular no compact complex manifold of positive dimension embeds into $\mathbb{C}^N$. This is because complex manifolds possess much more rigidity than real manifolds [17][17]. Schemes exhibit this same type of rigidity. The only types of schemes that have a universal embedding theorem into one of the classical spaces are the complete nonsingular curves, which all embed into a single $\mathbb{P}^3$ however large their genus (see proposition 7.3.9). Smooth complete surfaces are still projective, by a theorem of Zariski, and so embed in $\mathbb{P}^5$; it is in dimension three that completeness stops implying projectivity at all: proposition 8.4.12 is a smooth complete threefold admitting no embedding into any $\mathbb{P}^N_k$.

   By the Kodaira Embedding Theorem, proved in *Complex Analytic Geometry* [11], a compact Kähler manifold with a positive line bundle can be embedded into projective space. Proper schemes (the algebraic analogue of compact complex manifolds) with an *ample line bundle* (the algebraic analogue of a positive line bundle) similarly embed into projective space. In fact, this means that compact Kähler manifolds with positive line bundle are (as complex analytic spaces) isomorphic to a smooth proper scheme with an ample bundle. This is the closest to a general embedding theorem that is widely used, and will be proven in section 5.3.3.

   For a closed immersion $X \hookrightarrow \mathbb{A}^n_R$ so that $X \cong V \subseteq \mathbb{A}^n_R$ where V is a closed subscheme, X is topologically a sub-variety, though it may have a more complicated sheaf structure (for more details see section 3.7).

   For an open embedding $X \hookrightarrow \mathbb{A}^n_R$ so that $X \cong U \subseteq \mathbb{A}^n_R$ where U is open, X is a *quasi-affine subscheme*, which by definition is an open subscheme of affine space. Their characterization is more subtle: the map $X \to \operatorname{Spec}(\mathcal{O}_X(X))$ must be an open embedding or, equivalently for quasi-compact separated X and as will be discussed in section 5.1.1, every quasi-coherent sheaf of $\mathcal{O}_X$-modules must be generated by global sections. Verifying that every $\mathcal{O}_X$-module is generated by global sections is difficult; there is a simpler characterization given by the Goodman-Hartshorne Theorem [38], but it is still a nontrivial condition, illustrating the greater complexities that arise when working with non-proper schemes.

[16] If a compact connected complex manifold does embed holomorphically, it must reduce to a point, since any global holomorphic function on a compact complex manifold is constant. Non-compact complex manifolds that embed *properly* in $\mathbb{C}^N$, equivalently as closed submanifolds, are called *Stein manifolds*, and share many properties of quasi-projective varieties. Properness is essential: $\mathbb{C}^2 \setminus \{0\}$ sits in $\mathbb{C}^2$ as an open submanifold yet is not Stein, by Hartogs.

[17] For instance, the complex torus $\mathbb{C}/\Lambda$ depends on the lattice Λ up to biholomorphism, giving a one-complex-parameter family of non-isomorphic structures.

More generally, as *Toric Geometry* [21] develops, there are other interesting ambient spaces into which certain types of schemes may be embedded (e.g. toric varieties, Grassmannians, flag varieties, etc.), each serving different purposes depending on the moduli problem under study.

9. Schemes can have "doubled points": recall that the line with two origins is a standard counterexample for manifolds, as it fails to be Hausdorff at the origin. In scheme theory the analogous phenomenon is permitted; indeed, many moduli spaces that are schemes are naturally non-separated schemes (see section 3.3.6).

10. A scheme remembers multiplicity. For example, on a classical point there is little room for interesting functions, but scheme-theoretic functions (i.e. regular functions on Spec $k[x]/(x^2)$) can encode multiplicity and infinitesimal information. A smooth manifold $(M, \mathcal{A})$ with its maximal atlas does not retain such data.

11. There can be multiple structure sheaves on the same underlying topological space, defining genuinely different schemes (think of the different fields that can serve as the structure sheaf over a single point). In contrast, manifolds of dimension ≤ 3 have unique differential structures up to diffeomorphism (by work of Radó in dimension 2 and Moise in dimension 3); in fact, the categories of piecewise-linear, topological, and smooth manifolds coincide in these dimensions, showing how "linear" manifold theory is in low dimensions. In section 3.2 we shall introduce the process of *base change*, by which one can change the structure sheaf without changing the underlying topological space.

12. Though we have not defined an atlas for schemes, theorem 2.2.56 will show that schemes can be constructed by gluing affine patches, just as a manifold is built from its atlas, see section 2.2.7. (want to say why not as useful here).

13. The natural additional structure on a smooth manifold is a *tangent bundle*, whose transition functions take the form $D(\mathbf{y} \circ \mathbf{x}^{-1})$; that is, they are not just smooth maps between open subsets but must respect a $\mathrm{GL}_n(\mathbb{R})$-action given by the Jacobian. We shall define the algebraic analogue for schemes (the sheaf of Kähler differentials and the tangent sheaf) in chapter 5, section 5.5, though more care is required since schemes are not "locally linear" everywhere in the way smooth manifolds are. Much of the intuition from Poincaré duality (see [57, section 4.3]) will inform the generalization to Serre duality.

14. Schemes can better capture different levels of *local* behaviour. For example, a germ of a smooth manifold at a point is not itself a smooth manifold, but the analogous object for a scheme (the spectrum of a local ring) is again a scheme, called a *local scheme*. This is because commutative rings faithfully encode this local information, and (affine) schemes are the geometric realization of commutative rings.

15. Scheme functions are more "rigid": on a smooth manifold, partitions of unity exist thanks to the flexibility of the Euclidean topology, but no exact analogue holds for schemes. This rigidity, however, also adds structure and sometimes simplifies definitions (e.g. closed subschemes, introduced in section 3.1.3, are a rather "natural" concept compared to the work that goes into verifying that closed submanifolds are locally diffeomorphic to $\mathbb{R}^k \subseteq \mathbb{R}^n$).

    As smooth functions on each $U \subseteq M$ form a sheaf, sheaves that admit a "partition of unity" are called *fine sheaves*. A weaker notion of *soft sheaf* will be introduced in section 6.1; this is the closest scheme-theoretic analogue of the partition-of-unity property.

16. The two theories compute cohomology by different mechanisms, and the previous item is why. On a smooth manifold M the de Rham complex $\Omega^\bullet_M$ resolves the constant sheaf $\underline{\mathbb{R}}$, and partitions of unity make each Ω^p_M fine, hence acyclic; sheaf cohomology of $\underline{\mathbb{R}}$ therefore computes de Rham and singular cohomology alike, and the topology of M is recovered from a resolution of a constant sheaf. Neither half of that survives for schemes.

    First, there are no fine resolutions available, for want of partitions of unity. Second, the object being resolved is already trivial: on an *irreducible* scheme any two nonempty opens meet, so a locally constant function is constant, every constant sheaf is flasque, and

    $$H^i(X, \underline{A}) = 0 \qquad \text{for all } i > 0$$

    The Zariski topology has too few open sets for a constant sheaf to carry any cohomology at all.

    Cohomology is therefore rebuilt in two directions, answering two different questions (chapter 6):

    - *Coherent cohomology* keeps the topology and changes the sheaf, taking $H^i(X, \mathscr{F})$ for $\mathscr{F}$ quasi-coherent. This measures the scheme structure rather than the underlying space, and is the analogue of Dolbeault cohomology rather than of singular cohomology. Affine schemes are acyclic for quasi-coherent sheaves, so on a separated scheme the Čech complex of any affine cover computes it (section 6.1.1). The acyclicity is a property of the open sets; on a manifold it runs the other way, coming from softness of the sheaf rather than from the shape of the cover.
    - *Étale cohomology* keeps the sheaf and changes the *site*, replacing Zariski opens by étale maps. For n invertible on X this returns the singular answer: over $\mathbb{C}$, $H^i_{\text{ét}}(X, \mathbb{Z}/n) \cong H^i_{\text{sing}}(X(\mathbb{C}), \mathbb{Z}/n)$. See [16].

    The finiteness statements match: for X proper over a field and $\mathscr{F}$ coherent, every $H^i(X, \mathscr{F})$ is finite-dimensional, as the Betti numbers of a compact manifold are finite, and both vanish above the dimension.

    The two duality theorems are indexed differently. Serre duality (theorem 6.5.9) pairs H^i with H^{n-i} for $n = \dim X$, where Poincaré duality ([57, section 4.3]) pairs H^i with H^{2n-i}, the real dimension; Hodge theory [17] accounts for the difference by splitting each singular degree into the $H^q(X, \Omega^p)$ with $p + q$ fixed.

    Despite the vanishing above, Grothendieck proved that for smooth X over $\mathbb{C}$ the *hyper*cohomology of the algebraic de Rham complex $\Omega^\bullet_{X/\mathbb{C}}$, taken in the Zariski topology, computes $H^*_{\text{sing}}(X(\mathbb{C}); \mathbb{C})$. The Poincaré lemma fails Zariski-locally, so $\Omega^\bullet_{X/\mathbb{C}}$ is not a resolution of $\underline{\mathbb{C}}$ and the statement is about the complex rather than about any one sheaf. section 6.9.4 collects the comparisons.

17. Schemes unify "arithmetic" and "geometry" in a way that manifold theory is not equipped for. As examples, there are no manifold analogues of $\operatorname{Spec}\mathbb{Z}$, $\operatorname{Spec}\mathcal{O}_K$ (where $\mathcal{O}_K$ is a ring of integers of a number field), or $\operatorname{Spec}\mathbb{Z}[t]$. In this way, scheme theory gives geometric meaning to objects that are not tractable in differential geometry. A consequence is that different points of a single scheme can have residue fields of different characteristics (e.g. on Spec $\mathbb{Z}$, the residue field at the prime (p) is $\mathbb{F}_p$ while the residue field at the generic point is $\mathbb{Q}$), a phenomenon with no manifold analogue. To work in a more "purely geometric" setting, one often restricts to $\overline{k}$-schemes where $\overline{k}$ denotes the algebraic closure of a field of characteristic 0. In fact, when defining Z-valued points in section 3.3.2 for a scheme Z, if $Z = \operatorname{Spec}\overline{k}$ then such points are called *geometric points*. Working over a non-algebraically closed field often signals that the scheme carries arithmetic information.

18. Schemes have non-closed ("generic") points. Such points are essential: for instance, the ring map $k[x,y] \to k(x)[y]$ (inverting every nonzero polynomial in x) induces a morphism Spec $k(x)[y] \to$ Spec $k[x,y]$ whose image consists of points lying over the generic point of Spec $k[x]$. We shall later see that the image of such a morphism is *dense*.

19. Though the above list highlights many differences, many of the usual geometric tools remain applicable in the study of schemes. For example, $\mathbb{C}[x,y]/(x^2-y^3)$ is not a normal ring. This can be hard to verify algebraically, but viewing Spec $\mathbb{C}[x,y]/(x^2-y^3)$ as a curve in $\mathbb{C}^2$ (in the Euclidean topology), the fact that it has a cusp at the origin, and hence is singular there at a codimension-1 point of the curve, is precisely why it fails to be normal. (For a Noetherian ring, normality is Serre's condition R_1, regularity in codimension 1, together with S_2.) This type of geometric reasoning can be powerful: one can study the smoothness (or more specifically the normality) of a variety to determine the integral closure of its coordinate ring.

Though this is a long list consisting mostly of differences, very often in practice (especially in the setting of k-schemes, or when studying curves, surfaces, moduli spaces, etc.) we work with schemes that are proper (the correct generalization of compact manifolds) and have many methods to "resolve" singularities (e.g. eliminate nodes, cusps, etc.) and make the resulting scheme regular[18]. In fact, as we shall see in section 6.9, a result known as *GAGA* (Géométrie Algébrique et Géométrie Analytique) shows that proper schemes over $\mathbb{C}$ with appropriate coherent sheaves are equivalent, as categories, to compact complex analytic spaces with their coherent analytic sheaves. Naturally, arithmetic schemes will not be part of this manifold-scheme correspondence.

2.2.5 Evaluation and Stalks on Schemes

In differential geometry, the stalks of a manifold can be used define the evaluation of a function. This same trick is endeavored for scheme:

Proposition 2.2.22: Stalks of Schemes

1. Let R be a ring and $\mathfrak{p}$ a prime of R. The stalk of $\mathcal{O}_{\text{Spec}\,(R)}$ at $[\mathfrak{p}]$ is isomorphic to the local ring $R_{\mathfrak{p}}$, by an isomorphism sending the germ of $a \in R = \mathcal{O}_{\text{Spec}\,(R)}(\text{Spec}\,(R))$ to $a/1$:
$$\mathcal{O}_{\text{Spec}\,(R),\mathfrak{p}} \cong R_{\mathfrak{p}}$$

2. Let X be a scheme and $x \in X$, lying in an open set U with an isomorphism $U \cong \text{Spec}\,(R_i)$ carrying x to $[\mathfrak{p}]$. Then
$$\mathcal{O}_{X,x} \cong \mathcal{O}_{\text{Spec}\,(R_i),[\mathfrak{p}]} \cong (R_i)_{\mathfrak{p}}$$

3. Let $g\colon \text{Spec}\,(A) \to \text{Spec}\,(B)$ be a morphism of ringed spaces with map on global sections $\gamma\colon B \to A$, let $\mathfrak{q}$ be a prime of A, and let $\mathfrak{p}$ be the prime of B with $g([\mathfrak{q}]) = [\mathfrak{p}]$. Under the isomorphisms of (1), the stalk map of g at $[\mathfrak{q}]$ is the map $B_{\mathfrak{p}} \to A_{\mathfrak{q}}$, $b/c \mapsto \gamma(b)/\gamma(c)$.

[18]Though as we shall see, there is no universal resolution of singularities in all characteristics and dimensions. Resolution has been completely proved in characteristic 0 (Hironaka) and in low dimensions, but in positive characteristic the general case remains open. Moreover, the singularity information itself is sometimes important and should not always be discarded.

***Proof*:**

1. *Step 1: the maps λ_f.* For $f \notin \mathfrak{p}$ the image of f in $R_\mathfrak{p}$ is a unit, so [10, Universal Property of Localization] gives a unique ring map $\lambda_f \colon R_f \to R_\mathfrak{p}$ with $\lambda_f(a/1) = a/1$. If $D(g) \subseteq D(f)$ with $g \notin \mathfrak{p}$, the restriction $R_f \to R_g$ sends $a/1$ to $a/1$, so $\lambda_g \circ \mathrm{res}_{D(f),D(g)} = \lambda_f$ by uniqueness.

   Step 2: the map $\lambda \colon \mathcal{O}_{\mathrm{Spec}\,(R),\mathfrak{p}} \to R_\mathfrak{p}$. By corollary 2.1.13, every open neighbourhood of $\mathfrak{p}$ contains some $D(f)$ with $f \notin \mathfrak{p}$, so every germ at $\mathfrak{p}$ is the class of a pair $(s, D(f))$ with $f \notin \mathfrak{p}$ and $s \in R_f$. Put $\lambda[(s, D(f))] = \lambda_f(s)$. If $(s, D(f)) \sim (s', D(f'))$, the two sections agree on an open neighbourhood of $\mathfrak{p}$, hence on some $D(g) \ni \mathfrak{p}$ inside $D(f) \cap D(f')$, and Step 1 gives $\lambda_f(s) = \lambda_g(s|_{D(g)}) = \lambda_g(s'|_{D(g)}) = \lambda_{f'}(s')$. So λ is well defined, and it is a ring homomorphism because the operations on germs are computed on a common $D(g)$ and each λ_g is a ring map. The germ of $a \in R = R_1$ is sent to $\lambda_1(a/1) = a/1$.

   Step 3: λ is bijective. Every element of $R_\mathfrak{p}$ has the form a/s with $s \notin \mathfrak{p}$, and $a/s = \lambda_s(a/s) = \lambda[(a/s, D(s))]$, so λ is surjective. Suppose $\lambda[(a/f^n, D(f))] = 0$, that is, $a/f^n = 0$ in $R_\mathfrak{p}$. Then $ta = 0$ for some $t \notin \mathfrak{p}$. The element ft lies outside $\mathfrak{p}$ and $D(ft) \subseteq D(f)$. In R_{ft} we have $a/1 = 0$, since $(ft)a = f(ta) = 0$, so the restriction $R_f \to R_{ft}$, a ring map sending $a/1$ to $a/1$, sends $a/f^n = (a/1)(1/f^n)$ to 0. The germ is therefore zero, and λ is injective.

2. By proposition 1.1.5, $\mathcal{O}_{X,x} \cong \mathcal{O}_{U,x}$, and the isomorphism $U \cong \mathrm{Spec}\,(R_i)$ of locally ringed spaces (definition 2.2.9) has a ring isomorphism as its stalk map at x (lemma 2.2.14(2)). With (1) this gives $\mathcal{O}_{X,x} \cong \mathcal{O}_{\mathrm{Spec}\,(R_i),[\mathfrak{p}]} \cong (R_i)_\mathfrak{p}$.

3. Stalk maps are compatible with the maps on sections, so the stalk map of g at $[\mathfrak{q}]$ sends the germ of $b \in B$ at $[\mathfrak{p}]$ to the germ of $\gamma(b)$ at $[\mathfrak{q}]$. By (1), as a map $B_\mathfrak{p} \to A_\mathfrak{q}$ it is a ring homomorphism sending $b/1$ to $\gamma(b)/1$. For $c \notin \mathfrak{p}$ the element $c/1$ is a unit of $B_\mathfrak{p}$, so its image $\gamma(c)/1$ is a unit of $A_\mathfrak{q}$, and $b/c = (b/1)(c/1)^{-1}$ is sent to $(\gamma(b)/1)(\gamma(c)/1)^{-1} = \gamma(b)/\gamma(c)$, completing the proof.

Definition 2.2.23: Local Scheme

Let X be a scheme. Then X is a *local scheme* if it has a unique closed points.

The scheme $(\mathrm{Spec}\,\mathcal{O}_{\mathrm{Spec}\,R,\mathfrak{p}}, \mathcal{O}_{\mathrm{Spec}\,R,\mathfrak{p}})$ is called a *local scheme* at the point $\mathfrak{p}$.

The reader should show that the local scheme at the point $\mathfrak{p}$ is the intersection of all open subsets containing $\mathfrak{p}$. Hence, local schemes are the germs of schemes. Note that as there are "points within points", namely as there are non-closed points in schemes, then there is also stalks that are subsets of stalks, in particular if $\mathfrak{p}$ and $\mathfrak{m}$ is a prime and maximal ideal respectively where $\mathfrak{p} \subsetneq \mathfrak{m} \in \mathrm{Spec}\,R$ so that $[\mathfrak{m}] \in \overline{[\mathfrak{p}]}$, then $\mathcal{O}_{X,\mathfrak{p}} \cong R_\mathfrak{p}$ is a localization of $\mathcal{O}_{X,\mathfrak{m}} \cong R_\mathfrak{m}$. Due to this, notice that the local schemes in higher dimensions (section 2.6) can have local schemes contained within them. For example, the scheme $\mathrm{Spec}\,k[x,y]_{(x,y)}$ is the local scheme at the origin, it contain a single closed and a non-closed point for every irreducible curve in $\mathbb{A}^2_k$. Notice too how the local schemes are still schemes, as compared to the collection of germs at a point along with the point is not considered to be a

manifold.[19]

The main idea is that the single maximal ideal represents that there is one point at which we can evaluate for each germ. As all local rings have a maximal ideal, this motivates remembering the following notion form commutative algebra:

Definition 2.2.24: Residue Field

Let R be a commutative ring and $\mathfrak{p} \subseteq R$ a prime ideal. Then the ring of fractions of the integral domain domain $R/\mathfrak{p}$ is called the *residue field*, i.e. $\text{Frac}(R/\mathfrak{p})$ is the residue field of $\mathfrak{p}$ (with respect to R). This field is often denoted $\kappa(\mathfrak{p})$.

The use of kappa in $\kappa(\mathfrak{p})$ is due to the fact that there can be many different residue fields at different points. For example in $\text{Spec}\,\mathbb{R}[x]$, all residue fields are isomorphic to either $\mathbb{R}$ or $\mathbb{C}$ (depending if the localization is being taken at a non-maximal or maximal prime ideal).

Using residue fields, we can generalize the evaluation of a regular function (i.e. the evaluation of any element in the sections of $\mathcal{O}_X$). Recall that:

$$R_{\mathfrak{p}}/\mathfrak{m}_{\mathfrak{p}} \cong \text{Frac}(R/\mathfrak{p})$$

as localization and taking the quotient commutes (see [10]). This makes it much easier to evaluate a function if we can determine R. Then as the valuation of a function is usually in $R/\mathfrak{p}$, as rational functions are now permitted on (proper) open subsets of X, the evaluation will be in $\text{Frac}(R/\mathfrak{p})$. For a general point $x \in X$ in a scheme, we may diminish to the to define evaluation.

Definition 2.2.25: Evaluation of Function [on a Scheme]

Let X be a scheme, $U \subseteq X$ an open subset, and $f \in \mathcal{O}_X(U)$ a section. Then the value of f at the point $[\mathfrak{p}] \in U$ is the element:

$$f(\mathfrak{p}) := \bar{f} \in \mathcal{O}_{X,\mathfrak{p}}/\mathfrak{m}_{\mathfrak{p}} \cong \text{Frac}(R/\mathfrak{p}) = \kappa(\mathfrak{p})$$

where $\text{Spec}(R)$ is some affine local neighborhood of $\mathfrak{p}$.

As the quotient of any ring by a prime ideal is an integral domain, by taking its field of fraction we are forcing the codomain of each function to be a disjoint union of fields:

$$f : U \to \coprod_{\mathfrak{p}} \mathcal{O}_{U,\mathfrak{p}} \quad (\text{ and in affine case }) \quad f : U \to \coprod_{\mathfrak{p} \in U} \text{Frac}(R/\mathfrak{p})$$

If $\mathfrak{p}$ is maximal, then $\text{Frac}(R/\mathfrak{p}) = R/\mathfrak{m}$. For certain rings, it suffices to evaluate sections only over maximal points (namely Jacobson Schemes, see corollary 3.7.6). In the case of affine Jacobson schemes, we can consider:

$$f : U \to \coprod_{\mathfrak{p} \in U} R/\mathfrak{m}$$

[19] In fact, germs not being manifolds can be seen as a lacking property in the category of manifolds as this shows manifolds are not closed under filtered colimits. Synthetic differential geometry is an attempt to rectify this.

Varieties will be an example of Jacobson schemes. If $R/\mathfrak{m}_i \cong R/\mathfrak{m}_j$ for all maximal ideals (which shall be shown to be the case for varieties), then we can simply write:

$$f : U \to R/\mathfrak{m} \cong k$$

which recovers the notion of evaluation from classical algebraic geometry!

2.2.26 Foreshadowing: What Type of Function? As f is defined above, it is a set function between a relatively nice space U and a bad codomain. Can this function be characterized, or represented, by a known nicer function? In fact, it shall be representable with relatively simple scheme morphisms, recovering how in classical algebraic geometry sections are regular functions; details in proposition 3.3.9.

Example 2.2.27: Evaluation On Affine Schemes

The key idea is that the stalk $R_{\mathfrak{p}}$ gives all function's that will not vanish at $\mathfrak{p}$, and hence can divide by the function at that point, and then taking the quotient by the maximal ideal evaluate the function at that point, equivalently considering the value of the rational function in $\text{Frac}(R/\mathfrak{p})$:

1. Take $X = \mathbb{C}[x]$. Then each $f/g \in \mathcal{O}_X(U)$ is a rational polynomial where $g(\mathfrak{p}) \neq 0$ for all $\mathfrak{p} \in U$. For example if $U = D(x)$, then $h = \frac{x+1}{x} \in \mathcal{O}_X(D(x))$ and:

$$h(3) = \frac{3+1}{3} = \frac{4}{3} \in \mathbb{C} = \frac{\mathbb{C}[x]}{(x-3)} \cong \mathcal{O}_{X,3}/(x-3)$$
$$h(-1) = \frac{-1+1}{-1} = 0 \in \mathbb{C} = \frac{\mathbb{C}[x]}{(x+1)} \cong \mathcal{O}_{X,-1}(x+1)$$

   As the points of $\mathbb{C}[x]$ correspond to the variety, we see that the evaluation will always be in $\mathbb{C}$, and there is no need to take the field of fractions as $\text{Frac}(\mathbb{C}) = \mathbb{C}$.
   Another important example, take (want the vanishing at 0 , but something feels off)

2. Take $\mathbb{A}^2_k$ for a field k where char $\neq 2$. Then in the open set U away from the y-axis and the curve $y^2 - x^5$ has in it's local section $\mathcal{O}_{\mathbb{A}^2_k}(U)$ the rational function $\left(x^2+y^2\right)/x\left(y^2-x^5\right) \in \mathcal{O}_{\mathbb{A}^2_k}(U)$. It's value at the point $(2,4)$ (i.e. $[(x-2, y-4)]$) give a values in $\text{Frac}(R/(x-2, y-4)) = \text{Frac}(k[x,y]/(x-2,y-4)) \cong R_{(x-2,y-4)}/\overline{(x-2,y-4)}$. Then:

$$\frac{x^2+y^2}{x\left(y^2-x^5\right)} \bmod (x-2, y-4) = \frac{2^2+4^2}{2\left(4^2-2^5\right)} \in \mathbb{C} = \frac{\mathbb{C}[x,y]}{(x-2,y-4)} \cong \mathcal{O}_{\mathbb{A}^2_k,(2,4)}$$

   The value at the point $[(y)]$, which is the generic point of the x axis, is

$$\frac{x^2+y^2}{x\left(y^2-x^5\right)} \bmod (y) = \frac{x^2}{-x^6} = -1\frac{1}{x^4} \in \mathbb{C}(x) \cong \text{Frac}(\mathbb{C}[x,y]/(y)) \cong \mathcal{O}_{\mathbb{A}^2_k,[(y)]}$$

3. Take $X = \text{Spec}(\mathbb{Z})$ and take the open set $U = \text{Spec}(\mathbb{Z})\backslash\{(2),(7)\}$. This open set has many rational function, but let's take $27/4$. Then evaluating this function at different points of U we get:

$$(5) : 2/(-1) \equiv -2 \equiv 3 \bmod 5$$
$$(0) : 27/4$$
$$(3) : 0/1 = 0 \bmod 3$$

Notice that at 3, the function became 0. A (rational) function whose output is 0 is said to *vanish* at that point. If $\mathfrak{m}_\mathfrak{p}$ represents the maximal ideal of $\mathcal{O}_{X,\mathfrak{p}}$, then functions on an open subset U (i.e. section in $\mathscr{F}(U)$) have values at each point of U whose value lies in $\kappa(\mathfrak{p})$. A function *vanishes* at $\mathfrak{p}$ if its value at $\mathfrak{p}$ is 0. This idea is not well-defined on ringed spaces in general! There is furthermore the following properties mirroring manifold theory:

Proposition 2.2.28: Properties of Functions on Locally Ringed Spaces

Let f be function on a locally ringed space X. Then:

1. The subset of X where f vanishes is closed
2. if f vanishes nowhere, then f is invertible

Note that the property "if f vanishes everywhere, then $f = 0$" is not one of the properties. This will be addressed in proposition 2.4 .10 .

***Proof*:**

1. the subset of X where the germ of f is invertible is open
2. If $f \not\equiv 0 \bmod \mathfrak{p}$, then in particular $f \not\equiv 0 \bmod \mathfrak{m}$ for all maximal ideal, meaning f is contained in no maximal ideal. But then f must be a unit. If f where not a unit, then there would exist an $\mathfrak{m}$ that f belongs to, namely the maximal ideal containing f. But only units are not contained in any maximal ideal (as the ring is non-trivial)

Note how this is is *not* true on a variety defined in the classical sense:

Example 2.2.29: On Variety: Vanishes Nowhere, Not Invertible
Take

$$\overline{x - 100} \in \frac{\mathbb{R}[x, y]}{(x^2 + y^2 - 1)}$$

The reader should check that this function vanishes nowhere on $V(x^2 + y^2 - 1)$ as a variety, but it is *not* in invertible (i.e it is not a unit). However, $\overline{x - 100}$ *does* vanish on Spec $\mathbb{R}[x, y]/(x^2 + y^2 - 1)$; can the reader find what prime ideal it vanishes on?

This ties in with the intuition that the spectrum of $\mathbb{R}[x, y]/(x^2 + y^2 - 1)$ would be some "gluing" of $\mathbb{C}[x, y]/(x^2 + y^2 - 1)$ via galois conjugate.

Finally, a moment should be taken to interpret evaluating at generic points. Let us consider $X = \mathbb{A}^1_k$ and let η be the generic point. Then looking at the fiber at η, we see that:

$$\mathcal{O}_{\mathbb{A}^1_k,\eta} = (k[x])_{(0)} = k(x)$$

that is, everything is inverted. The earlier intuition about $R_\mathfrak{p}$ is that we only evaluate functions *near* the point $\mathfrak{p}$ and every other $\mathfrak{q} \in \operatorname{Spec} R$ can be inverted, so this means that η is near *no* points as we

can invert every point! This contrasts with the intuition that $\overline{\eta} = \mathbb{A}^1_k$ which seems to suggest that it is *close* to every point. This is where the nomenclature *generic* can be of use: it does not "see" what global sections capture, and can be thought of as "ignoring" any constraints given by points. This also explains the evaluation at the generic point, it returns the function itself motivating how it is not "particularly" extracting any information.

2.2.6 Independence of Affine Covering

A scheme X can be covered by affine open subsets in many different ways. Unlike the situation for smooth manifolds in low dimensions—where Moise's theorem [53] guarantees a unique smooth structure on topological manifolds of dimension ≤ 3—schemes admit no such rigidity. Even in dimension zero, the topological space underlying a field is always a single closed point, yet different fields give non-isomorphic affine schemes. It is therefore essential to know when a property of a scheme can be checked on *any* affine covering, rather than depending on a particular choice.

The tool for this is the *Affine Communication Lemma*. It identifies a class of properties—those that are stable under localization and can be recovered from a generating cover—for which checking on a single affine cover suffices. Such properties are called *affine-local*, and they include most of the "geometric" properties of schemes (reducedness, Noetherianness, integrality, normality, etc.). Properties that depend on "arithmetic" or global information—such as the number of connected components—are typically not affine-local.

The key preliminary observation is that the intersection of two affine opens, while generally not affine, can always be covered by open sets that are simultaneously distinguished in both.

Lemma 2.2.30: Build-up Lemma

Let $\operatorname{Spec}(R)$ and $\operatorname{Spec}(S)$ be affine open subschemes of a scheme X. Then $\operatorname{Spec}(R) \cap \operatorname{Spec}(S)$ can be covered by open sets that are simultaneously distinguished open in $\operatorname{Spec}(R)$ and distinguished open in $\operatorname{Spec}(S)$.

***Proof*:**
Let $[\mathfrak{p}] \in \operatorname{Spec}(R) \cap \operatorname{Spec}(S)$. Since $\operatorname{Spec}(R) \cap \operatorname{Spec}(S)$ is open in $\operatorname{Spec}(R)$, and the distinguished open sets form a basis for the Zariski topology (section 2.1.2), there exists $f \in R$ with $[\mathfrak{p}] \in \operatorname{Spec}(R_f) \subseteq \operatorname{Spec}(R) \cap \operatorname{Spec}(S)$. Now, as

$$\operatorname{Spec}(R_f) \subseteq \operatorname{Spec}(R) \cap \operatorname{Spec}(S) \subseteq \operatorname{Spec}(S)$$

$\operatorname{Spec}(R_f)$ is an open subset of $\operatorname{Spec}(S)$. Thus, by construction of the topology on $\operatorname{Spec}(S)$ using distinguished opens (given by S), there exists $g \in S$ with

$$[\mathfrak{p}] \in \operatorname{Spec}(S_g) \subseteq \operatorname{Spec}(R_f) \cap \operatorname{Spec}(S) \subseteq \operatorname{Spec}(R) \cap \operatorname{Spec}(S).$$

In particular, $\operatorname{Spec}(S_g) \subseteq \operatorname{Spec}(R_f)$.

Now, $g \in \mathcal{O}_X(\operatorname{Spec}(S))$ restricts to an element $g' \in \mathcal{O}_X(\operatorname{Spec}(R_f)) = R_f$, and the locus in $\operatorname{Spec}(R_f)$ where g is nonvanishing is exactly $\operatorname{Spec}(S_g)$, so:

$$\operatorname{Spec}(S_g) = \operatorname{Spec}((R_f)_{g'}).$$

By definition of R_f, write $g' = h/f^n$ for some $n \geq 0$ and $h \in R$. By Proposition 2.1.15, a distinguished open of a distinguished open is distinguished:

$$(R_f)_{g'} = (R_f)_{h/f^n} = R_{fh}$$

where the last equality holds since f is already invertible in R_f. Hence:

$$\operatorname{Spec}(S_g) = \operatorname{Spec}(R_{fh}) = D(fh) \subseteq \operatorname{Spec}(R).$$

This is a distinguished open in $\operatorname{Spec}(R)$, and it is simultaneously equal to $\operatorname{Spec}(S_g) = D(g) \subseteq \operatorname{Spec}(S)$, a distinguished open in $\operatorname{Spec}(S)$. Since $[\mathfrak{p}]$ was arbitrary, these sets cover $\operatorname{Spec}(R) \cap \operatorname{Spec}(S)$.

With this in hand, we can state and prove the main tool.

Theorem 2.2.31: Affine Communication Lemma

Let X be a scheme and let Q be a property enjoyed by some affine open subschemes of X. Suppose Q satisfies:

1. *(Localization)* If $\operatorname{Spec}(A) \subseteq X$ satisfies Q and $f \in A$, then $\operatorname{Spec}(A_f)$ satisfies Q.
2. *(Gluing)* If $f_1, \ldots, f_n \in A$ generate the unit ideal $(f_1, \ldots, f_n) = A$ and each $\operatorname{Spec}(A_{f_i})$ satisfies Q, then $\operatorname{Spec}(A)$ satisfies Q.

If there exists an affine open cover $X = \bigcup_i \operatorname{Spec}(R^{(i)})$ such that each $\operatorname{Spec}(R^{(i)})$ satisfies Q, then *every* affine open subscheme of X satisfies Q.

***Proof*:**
Let $\operatorname{Spec}(A) \subseteq X$ be an arbitrary affine open subscheme. We must show $\operatorname{Spec}(A)$ satisfies Q. For each i, the intersection $\operatorname{Spec}(A) \cap \operatorname{Spec}(R^{(i)})$ is an open subset of both $\operatorname{Spec}(A)$ and $\operatorname{Spec}(R^{(i)})$. By the Build-up Lemma (lemma 2.2.30), this intersection can be covered by open sets that are simultaneously distinguished in $\operatorname{Spec}(A)$ and distinguished in $\operatorname{Spec}(R^{(i)})$.

Since $\operatorname{Spec}(R^{(i)})$ satisfies Q, condition (1) implies that each of these distinguished opens (viewed as localizations of $R^{(i)}$) also satisfies Q. But these same opens are also distinguished opens of $\operatorname{Spec}(A)$, say of the form $\operatorname{Spec}(A_{g_j})$ for various $g_j \in A$.

Now, $\operatorname{Spec}(A)$ is quasi-compact, and the union $\bigcup_i (\operatorname{Spec}(A) \cap \operatorname{Spec}(R^{(i)})) = \operatorname{Spec}(A)$ covers it. Extracting a finite subcover from among the $\operatorname{Spec}(A_{g_j})$, we obtain elements $g_1, \ldots, g_m \in A$ such that:

(a) each $\operatorname{Spec}(A_{g_j})$ satisfies Q (inherited from the $R^{(i)}$ via localization), and

(b) $\operatorname{Spec}(A) = \bigcup_{j=1}^{m} \operatorname{Spec}(A_{g_j})$, which means $g_1, \ldots, g_m$ generate the unit ideal in A.

By condition (2), $\operatorname{Spec}(A)$ satisfies Q.

The power of the lemma is that once the two conditions (localization and gluing) are verified for a property Q—which is typically a purely algebraic statement about rings—the conclusion is that Q holds for *every* affine open, not just those in the chosen cover. This means the property can be

attributed to the scheme X itself, independent of the presentation.

The Affine Communication Lemma gives a uniform method for proving that properties of schemes are well-defined. We collect some important instances;

Example 2.2.32: Affine-local Properties

1. *Reducedness*: Let Q be the property "A is a reduced ring" (i.e., A has no nonzero nilpotents). We verify the two conditions:
   (a) *Localization:* If A is reduced, then A_f is reduced. Indeed, if $(a/f^n)^m = 0$ in A_f, then $f^k a^m = 0$ in A for some k, hence $(f^j a)^m = 0$ for j large enough, so $f^j a = 0$, hence $a/f^n = 0$ in A_f.
   (b) *Gluing:* If $(f_1, \ldots, f_n) = A$ and each A_{f_i} is reduced, then A is reduced. Indeed, if $a^m = 0$ in A, then $a^m = 0$ in each A_{f_i}, so $a = 0$ in each A_{f_i} (as A_{f_i} is reduced), meaning $f_i^{k_i} a = 0$ in A for some k_i. Since the f_i generate the unit ideal, so do the $f_i^{k_i}$ (see section 2.1.2), hence $a = 0$.

   By the Affine Communication Lemma, a scheme X is reduced if and only if some (equivalently, every) affine open cover consists of reduced rings. This shall be the foundation for definition given in section 2.4.1.

2. *Noetherian*: Let Q be the property "A is a Noetherian ring."
   (a) *Localization:* If A is Noetherian, then A_f is Noetherian (a standard result in commutative algebra: every ideal of A_f is an extension of an ideal of A, and extensions of finitely generated ideals are finitely generated).
   (b) *Gluing:* Suppose $(f_1, \ldots, f_n) = A$ and each A_{f_i} is Noetherian. Let $I \subseteq A$ be an ideal. Each extension $I \cdot A_{f_i}$ is finitely generated, say by elements $a_{i,1}/f_i^{k_1}, \ldots, a_{i,r_i}/f_i^{k_{r_i}}$. Clearing denominators, we obtain finitely many elements of I whose images generate $I \cdot A_{f_i}$ for each i. Since the f_i generate the unit ideal, these elements generate I itself.

   A scheme is *locally Noetherian* if it admits an affine cover by spectra of Noetherian rings. By the Affine Communication Lemma, this is equivalent to every affine open being the spectrum of a Noetherian ring.

3. *Normality*: Let Q be the property "A is a normal ring" (integrally closed in its total ring of fractions). This can be checked via the Affine Communication Lemma: if A is normal then A_f is normal, and normality can be recovered from a generating cover. Thus, a scheme is *normal* if and only if every affine open is the spectrum of a normal ring.

Local Properties and the Affine Communication Lemma

As more properties of schemes are defined, we must be precise about what it means for a scheme to "have" a property and how to check for it. There at least two natural levels at which a property can be detected, and they form a hierarchy.

Definition 2.2.33: Property of Schemes

A *property* P of schemes is a class of schemes closed under isomorphism: if X satisfies P and $Y \cong X$, then Y satisfies P.

Definition 2.2.34: Affine-local and Stalk-local Properties

Let P be a property of schemes.

1. P is *affine-local* if for any scheme X and any affine open cover $X = \bigcup U_i$, the scheme X satisfies P if and only if every U_i satisfies P.
2. P is *Zariski-local* if for any scheme X and any (Zariski) open cover $X = \bigcup U_i$, the scheme X satisfies P if and only if every U_i satisfies P.
3. P is *stalk-local* if for any scheme X, X satisfies P if and only if for every point $x \in X$, the affine scheme $\operatorname{Spec} \mathcal{O}_{X,x}$ satisfies P.

The Affine Communication Lemma provides a practical test for affine-locality: it suffices to verify the localization and gluing conditions for the corresponding ring property. As sheaf maps can always be verified stalk-locally, stalk-locality is another natural condition to consider. We shall see that morphism properties are not always verified affine-locally; those that are give rise to *affine morphisms* (see definition 3.4.5).

Note that $\operatorname{Spec} \mathcal{O}_{X,x}$ *need not* be open in X, hence $\{\operatorname{Spec} \mathcal{O}_{X,x}\}_{x \in X}$ need not be an affine open cover. For example $X = \mathbb{A}^1_{\mathbb{C}}$ and $\mathcal{O}_{X,2} \cong \operatorname{Spec} k[x]_{(x-2)}$ so that $\operatorname{Spec} \mathcal{O}_{X,2}$ has only 2 points but any open subscheme of X has cofinitely many points plus the generic point. Thus, a stalk-local is not the condition that it suffices to check the specific "stalk-cover".

Zariski-local properties are another natural choice, but as any scheme has a topological basis of distinguished open sets, these are identical to affine-locals:

Proposition 2.2.35: Affine-local Iff Zariski-local

Let P be a property of schemes. Then P is affine-local if and only if it is Zariski-local.

***Proof*:**
Suppose P is Zariski-local. An affine open cover is in particular an open cover, so the defining condition of affine-locality (definition 2.2.34) is a special case of the condition of Zariski-locality, and P is affine-local.

Suppose P is affine-local, and let $X = \bigcup_i U_i$ be an arbitrary open cover of a scheme X. Each U_i is itself a scheme, covered by affine opens of X: for $x \in U_i$, choose an affine open $V \cong \operatorname{Spec}(R)$ of X containing x (definition 2.2.9). The distinguished opens form a basis of V (corollary 2.1.13), so there is a distinguished open $D(f)$ with $x \in D(f) \subseteq U_i \cap V$, and $D(f)$ with the restricted structure sheaf is the affine scheme $\operatorname{Spec}(R_f)$ (proposition 2.1.12, proposition 2.2.1). Collect such charts into affine open covers $U_i = \bigcup_j V_{ij}$, where each V_{ij} is simultaneously an affine open subscheme of U_i and of X.

If X satisfies P: the family $\{V_{ij}\}_{i,j}$ is an affine open cover of X, so each V_{ij} satisfies P by

affine-locality applied to X. Fixing i, the family $\{V_{ij}\}_j$ is an affine open cover of the scheme U_i whose members all satisfy P, so U_i satisfies P by affine-locality applied to U_i.

If every U_i satisfies P: each V_{ij} satisfies P by affine-locality applied to U_i, and then X satisfies P by affine-locality applied to X with the cover $\{V_{ij}\}_{i,j}$.

Hence X satisfies P if and only if every U_i does, which is Zariski-locality (definition 2.2.34).

It is natural to ask whether stalk-locality is a stronger or weaker condition than affine-locality. The answer is that it is strictly stronger.

Proposition 2.2.36: Stalk-local implies Affine-local

If a property of a scheme is stalk-local, then it is affine-local.

***Proof*:**
Let P be stalk-local and let $X = \bigcup U_i$ be an affine open cover.

($\Rightarrow$) Suppose X satisfies P. For any $x \in U_i$, the stalk $\mathcal{O}_{U_i,x} \cong \mathcal{O}_{X,x}$ by proposition 1.1.5, and $\operatorname{Spec} \mathcal{O}_{X,x}$ satisfies property P since X satisfies P. As this holds for all $x \in U_i$, the scheme U_i satisfies P by stalk-locality.

($\Leftarrow$) Suppose every U_i satisfies P. For any $x \in X$, choose some U_i containing x. Then $\mathcal{O}_{X,x} \cong \mathcal{O}_{U_i,x}$ satisfies the local-ring property since U_i satisfies P. As x was arbitrary, X satisfies P.

Common examples are:

1. "X is reduced" is stalk-local with respect to "$\mathcal{O}_{X,x}$ has no nonzero nilpotents"
2. "X is regular" is stalk-local with respect to "$\mathcal{O}_{X,x}$ is a regular local ring"
3. "X is normal" is stalk-local with respect to "$\mathcal{O}_{X,x}$ is an integrally closed domain."

The converse is false: there exist affine-local properties that cannot be detected at the level of stalks. The issue is that stalks capture "infinitely close" behavior but lose information about finiteness conditions that involve the global structure of a neighborhood.

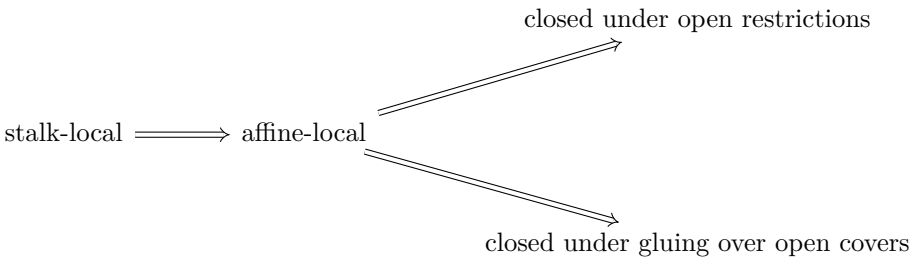
Example 2.2.37: Affine-local Does Not Imply Stalk-local
Consider the property "locally Noetherian." As shown in example 2.2.32, this is affine-local. We show it is not stalk-local.

Let $R = \prod_{i=1}^{\infty} \mathbb{F}_2$, the countably infinite product of copies of $\mathbb{F}_2$. The ring R is not Noetherian: the ideals $I_n = \{(a_1, a_2, \dots) : a_1 = \cdots = a_n = 0\}$ form a strictly ascending chain. Thus $\operatorname{Spec}(R)$ is not locally Noetherian.

However, every stalk of $\mathcal{O}_{\operatorname{Spec}(R)}$ is Noetherian. For any prime ideal $\mathfrak{p} \subseteq R$, the localization $R_{\mathfrak{p}}$ is isomorphic to $\mathbb{F}_2$. To see this, recall that R is a Boolean ring ($a^2 = a$ for all a), so every prime ideal is maximal with residue field $\mathbb{F}_2$. Since R is zero-dimensional and Boolean, the localization at any prime is a field.

If "locally Noetherian" were stalk-local, the fact that all stalks are Noetherian fields would force R to be Noetherian—a contradiction. Thus stalk-locality is strictly stronger than affine-locality.

The upshot is a hierarchy of "locality" conditions for scheme properties:



The first implication is strict, as the example above shows. In practice, most "algebraic" properties (reduced, normal, regular, Cohen-Macaulay) are stalk-local, while "finiteness" properties (Noetherian, finite type) are affine-local but not stalk-local.

2.2.38 Advanced Foreshadowing: Other Covers Notice how the collection of maps $\operatorname{Spec} \mathcal{O}_{X,x} \to X$ for all $x \in X$ can still be seen as a cover which is neither open nor closed. Such a cover still has important properties; in this case it fpqc (fidèlement plate et quasi-compacte). We shall not address such covers in the textbook, however the idea of replacing open sets with covering maps with "desirable" properties will be a key intuition in defining the *étale topology* using *étale coverings*, which is *Étale Cohomology* [16].
Readers interested in Descent Theory [72] may be interested to know that the above hierarchy is often extended to the following:

$$\text{fpqc-local} \implies \text{fppf-local} \implies \text{étale-local} \implies \text{Zariski-local}$$

where stalk-local becomes a part of fpqc-local.

2.2.7 *Affine Atlases

By definition, a scheme is a locally ringed space that admits some affine open cover. The structure sheaf $\mathcal{O}_X$ is part of the data: it is given, and the affine cover witnesses that it is "locally affine." But there is an equivalent formulation—closer in spirit to the definition of a smooth manifold—in which the structure sheaf is not given a priori, but is *recovered* from an atlas. This perspective clarifies the sense in which a scheme is "independent of coordinates."

Recall that a smooth manifold of dimension n can be defined in two equivalent ways:

1. A topological space M equipped with a sheaf C^∞_M of smooth functions such that (M, C^∞_M) is locally isomorphic to $(\mathbb{R}^n, C^\infty_{\mathbb{R}^n})$.
2. A topological space M equipped with a *maximal smooth atlas*: a maximal collection of charts (U_i, φ_i) where $\varphi_i : U_i \xrightarrow{\sim} V_i \subseteq \mathbb{R}^n$ is a homeomorphism, and the transition maps $\varphi_j \circ \varphi_i^{-1}$ are smooth (see section 0.5)

The two definitions are equivalent because the sheaf C^∞_M can be recovered from the atlas: a function $f : U \to \mathbb{R}$ is smooth if and only if $f \circ \varphi_i^{-1}$ is smooth for every chart (U_i, φ_i) with $U_i \subseteq U$. Conversely, the atlas is recoverable from the sheaf: a chart is precisely a local isomorphism of the sheaf with the standard model.

In the manifold setting, most textbooks adopt definition (2) because the transition maps are concrete and checkable. In scheme theory, definition (1) has historically been preferred because the structure sheaf interacts well with the algebraic machinery of localization, sheaf cohomology, and base change. Nevertheless, the atlas formulation works equally well and is illuminating.

Affine atlases for schemes

Definition 2.2.39: Affine Chart

Let X be a topological space. An *affine chart* on X is a pair (U, φ) where $U \subseteq X$ is open and $\varphi : U \xrightarrow{\sim} |\mathrm{Spec}\,(R)|$ is a homeomorphism for some commutative ring R. The ring R is called the *coordinate ring* of the chart.

Two charts (U_i, φ_i) with coordinate ring R_i and (U_j, φ_j) with coordinate ring R_j must be compatible on their overlap. In the manifold case, compatibility means the transition map is smooth. For schemes, it means the transition map is "algebraic" in the following sense.

Definition 2.2.40: Compatibility of Charts

Two affine charts (U_i, φ_i) and (U_j, φ_j) on a topological space X are *compatible* if for every point $x \in U_i \cap U_j$, there exist elements $f \in R_i$ and $g \in R_j$ such that:

1. $x \in \varphi_i^{-1}(D(f)) \cap \varphi_j^{-1}(D(g))$,
2. $\varphi_i^{-1}(D(f)) = \varphi_j^{-1}(D(g))$ as subsets of X, and
3. The composite homeomorphism

$$D(f) \xrightarrow{\varphi_i^{-1}} \varphi_i^{-1}(D(f)) \;=\; \varphi_j^{-1}(D(g)) \xrightarrow{\varphi_j} D(g)$$

is induced by a ring isomorphism $(R_j)_g \xrightarrow{\sim} (R_i)_f$.

Condition (3) is the key algebraic content: the transition "map" between two charts must not be a homeomorphism of topological spaces, but must arise from an isomorphism of the corresponding localized rings. This is the condition that fails for arbitrary homeomorphisms between spectra—and it is exactly what the structure sheaf encodes in the locally ringed space definition.

Definition 2.2.41: Affine Atlas

An *affine atlas* on a topological space X is a collection $\mathcal{A} = \{(U_i, \varphi_i)\}_{i \in I}$ of pairwise compatible affine charts such that $X = \bigcup_{i \in I} U_i$.

Definition 2.2.42: Maximal Affine Atlas

An affine atlas $\mathcal{A}$ is *maximal* if it contains every affine chart on X that is compatible with all charts already in $\mathcal{A}$. That is, if (V, ψ) is an affine chart compatible with every $(U_i, \varphi_i) \in \mathcal{A}$, then $(V, \psi) \in \mathcal{A}$.

Definition 2.2.43: Scheme via Atlas

A *scheme* is a pair $(X, \mathcal{A})$ where X is a topological space and $\mathcal{A}$ is a maximal affine atlas on X.

This definition makes no reference to sheaves, locally ringed spaces, or stalks. The algebraic data is encoded entirely in the transition isomorphisms between charts: the atlas definition is equivalent to definition 2.2.9. The equivalence rests on two constructions: recovering the structure sheaf from an atlas, and recovering the maximal atlas from a locally ringed space.

Proposition 2.2.44: From Atlas to Structure Sheaf

Let $(X, \mathcal{A})$ be a scheme in the sense of definition 2.2.43. Then there exists a unique sheaf of rings $\mathcal{O}_X$ on X such that $(X, \mathcal{O}_X)$ is a scheme in the sense of definition 2.2.9, and for every chart $(U, \varphi) \in \mathcal{A}$ with coordinate ring R:

$$(U,\ \mathcal{O}_X|_U) \ \cong\ (\operatorname{Spec}(R),\ \mathcal{O}_{\operatorname{Spec}(R)}).$$

***Proof*:**
Define $\mathcal{O}_X$ on the basis of opens $\{U : (U, \varphi) \in \mathcal{A} \text{ for some } \varphi\}$ by setting $\mathcal{O}_X(U) = R$ whenever $\varphi : U \xrightarrow{\sim} \operatorname{Spec}(R)$. For a distinguished open $D(f) \subseteq \operatorname{Spec}(R)$, the chart $(\varphi^{-1}(D(f)),\ \varphi|_{D(f)})$ is in $\mathcal{A}$ (by maximality) with coordinate ring R_f, so we set $\mathcal{O}_X(\varphi^{-1}(D(f))) = R_f$. The restriction maps are the natural localization maps $R \to R_f$.

The compatibility condition on charts ensures that this is well-defined on overlaps: if two charts (U_i, φ_i) and (U_j, φ_j) both contain a point x, the transition isomorphism $(R_j)_g \cong (R_i)_f$ identifies the sections. By the sheaf axiom for the structure sheaf of $\operatorname{Spec}(R)$, this basis assignment extends uniquely to a sheaf on all of X.

The stalks of $\mathcal{O}_X$ are local rings: at a point $x \in U_i$, the stalk $\mathcal{O}_{X,x}$ is the localization $(R_i)_{\mathfrak{p}}$ where $\mathfrak{p} = \varphi_i(x)$, which is local. Hence $(X, \mathcal{O}_X)$ is a locally ringed space, and each chart gives an affine open, so it is a scheme.

Proposition 2.2.45: From Locally Ringed Space to Maximal Atlas

Let $(X, \mathcal{O}_X)$ be a scheme in the sense of definition 2.2.9. Then the collection

$$\mathcal{A}_{\max} = \left\{ (U, \varphi) : \begin{array}{l} U \subseteq X \text{ is open, } \varphi : (U, \mathcal{O}_X|_U) \xrightarrow{\sim} (\operatorname{Spec}(R), \mathcal{O}_{\operatorname{Spec}(R)}) \\ \text{is an isomorphism of locally ringed spaces, for some } R \end{array} \right\}$$

is a maximal affine atlas on X. The atlas $\mathcal{A}_{\max}$ depends only on $(X, \mathcal{O}_X)$, not on any particular affine open cover.

***Proof*:**
The charts in $\mathcal{A}_{\max}$ are pairwise compatible: if (U_i, φ_i) and (U_j, φ_j) are in $\mathcal{A}_{\max}$, then on $U_i \cap U_j$ the isomorphisms φ_i and φ_j are both isomorphisms of locally ringed spaces, and the transition map is an isomorphism of the appropriate localizations (this is the content of the Build-up Lemma, lemma 2.2.30). The atlas covers X because the definition of a scheme guarantees the

existence of at least one affine cover. It is maximal by construction: every affine open is included.

These two constructions are inverse to each other: starting from an atlas, building the sheaf, and taking the maximal atlas of the resulting scheme recovers the original atlas; starting from a scheme, taking $\mathcal{A}_{\max}$, and building the sheaf from it recovers the original structure sheaf.

Just as for manifolds, the practical consequence is that one need not specify a maximal atlas. Any atlas suffices.

Proposition 2.2.46: Uniqueness of Maximal Atlas

Let $\mathcal{A}$ be any affine atlas on a topological space X. Then there exists a unique maximal affine atlas $\mathcal{A}_{\max}$ containing $\mathcal{A}$. Explicitly,

$$\mathcal{A}_{\max} = \{(V, \psi) : (V, \psi) \text{ is compatible with every } (U, \varphi) \in \mathcal{A}\}.$$

***Proof*:**
One must verify that any two charts in $\mathcal{A}_{\max}$ are compatible with each other (not just with the charts in $\mathcal{A}$). Let (V_1, ψ_1) and (V_2, ψ_2) be in $\mathcal{A}_{\max}$, and let $x \in V_1 \cap V_2$. Choose a chart $(U, \varphi) \in \mathcal{A}$ with $x \in U$. By compatibility of (V_1, ψ_1) with (U, φ), there exist distinguished opens around x where the transition is an isomorphism of localizations. By compatibility of (V_2, ψ_2) with (U, φ), the same holds. Composing (and using that distinguished opens of distinguished opens are distinguished, by proposition 2.1.15), we obtain a transition isomorphism between localizations of V_1 and V_2.

Uniqueness is immediate: any maximal atlas containing $\mathcal{A}$ must contain every compatible chart, hence must equal $\mathcal{A}_{\max}$.

The upshot is that defining a scheme requires only specifying a topological space together with *any* affine atlas—the maximal atlas is then uniquely determined. The Affine Communication Lemma (theorem 2.2.31) is the tool that ensures properties of the scheme can be checked on the given atlas rather than on the full maximal atlas: its localization and gluing conditions propagate a property from any covering atlas to all affine opens.

The atlas viewpoint also gives an explicit construction of schemes from purely algebraic data, analogous to constructing a manifold by specifying charts and transition maps. Suppose we are given:

1. Rings $R_1, \ldots, R_n$.
2. For each pair (i, j), elements $f_{ij} \in R_i$ specifying open subsets $U_{ij} = D(f_{ij}) \subseteq \operatorname{Spec}(R_i)$.
3. Ring isomorphisms $\varphi_{ij} : (R_i)_{f_{ij}} \xrightarrow{\sim} (R_j)_{f_{ji}}$.

Subject to the *cocycle condition*: for each triple (i, j, k), the composition $\varphi_{jk} \circ \varphi_{ij}$ agrees with φ_{ik} on the appropriate localizations.

These data determine a unique scheme X (up to unique isomorphism) by the gluing construction of section 2.2.9: the underlying set is the disjoint union $\coprod_i |\operatorname{Spec}(R_i)|$ modulo the identifications given by the φ_{ij}, the topology is the quotient topology, and the affine atlas consists of the images of the $\operatorname{Spec}(R_i)$. Conversely, every scheme with a given finite affine cover arises in this way: the rings are the coordinate rings of the charts, and the transition isomorphisms are read off from the structure sheaf on overlaps.

2.2.8 Subschemes: Open

We seek the natural notions of subschemes induced by the parent scheme. As a scheme has a topology, a natural starting point is to look for open or closed subsets with a sheaf naturally induced by the parent scheme[20]. To define the topology of open and closed subschemes is easy. The difficult part is to insure the subscheme has a sheaf structure that is in a natural way compatible with the parent scheme. This shall require a deeper discussion on monomorphic scheme morphisms as the sheaf structure of subschemes is "non-trivial", and hence this discussion is mostly delayed till section 3.1.3. There is one type of subscheme that shall be used in this chapter and requires no use of scheme morphism since it is directly compatible with sheaves: *open subschemes*.

Proposition 2.2.47: Open Subscheme

Let X be a scheme and $U \subseteq X$ an open subset. Then $(U, \mathcal{O}_X|_U)$ is a scheme.

***Proof*:**
U is certainly a subspace and a subsheaf with local rings as stalks. To show it is locally affine, for any $x \in U$ choose affine open $\operatorname{Spec} R \cong V \subseteq X$ where $x \in U \cap V$. As $D(f)$ forms a (topological) basis for $\operatorname{Spec} R$, there exists an f_0 such that $x \in D(f_0) \subseteq U \cap V \subseteq U$, showing that U is covered in affine, completing the proof.

Definition 2.2.48: Open Subscheme

Let X be a scheme. Then $(U, \mathcal{O}_X|_U)$ is called an *open subscheme*. If U is also an affine scheme, then U is often said to be an *affine open subscheme* or *affine open subset*.

If a scheme is isomorphic to an open subscheme, we shall say that the isomorphism map (onto the image) is an open embedding:

Definition 2.2.49: Open Embedding

A map of schemes $\pi : X \to Y$ is an open embedding if:

$$(X, \mathcal{O}_X) \xrightarrow{\sim} (U, \mathcal{O}_Y|_U) \hookrightarrow (Y, \mathcal{O}_Y)$$

2.2.50 Remark about Hartshorne's Definition of Open Subscheme Note that in Hartshorne, an open subscheme was technically not defined as $(U, \mathcal{O}_X|_U)$ but as $(U, [\mathcal{O}_X|_U])$ where $[\mathcal{O}_X|_U]$ is the equivalence class of all isomorphic sheaves. This is in fact not correct: isomorphic sheaves shall give technically different open subschemes (which are isomorphic). Hence, Hartshorne claims that an open subscheme has a unique subscheme structure, not unique up to isomorphism.

[20] Categorically, a natural starting point is to look for subobjects, but surprisingly these are not so useful in **Sch**; see section 3.1.3

Corollary 2.2.51: Subspace and Zariski Topology On Open Subscheme

Let $U = \operatorname{Spec}(R) \subseteq X$ be an affine open subset and take $(U, \mathcal{O}_X|_U)$. Then the Zariski topology on U is the subspace topology on U with respect to X

***Proof*:**
Saying that $U \cong \operatorname{Spec}(R)$ is an affine open subset means that there is an isomorphism of locally ringed spaces $\varphi : (U, \mathcal{O}_X|_U) \to (\operatorname{Spec}(R), \mathcal{O}_{\operatorname{Spec}(R)})$ (definition 2.2.9, definition 2.2.3), and the topology carried by the ringed space $(U, \mathcal{O}_X|_U)$ is the subspace topology inherited from X. The underlying map of an isomorphism of locally ringed spaces is a homeomorphism, so φ identifies the subspace topology on U with the Zariski topology on $\operatorname{Spec}(R)$.

Concretely: since U is open in X, the subspace-open subsets of U are exactly the open subsets of X contained in U. Under φ these correspond to the Zariski-open subsets of $\operatorname{Spec}(R)$, so a subset $V \subseteq U$ is open in X if and only if it corresponds to a Zariski-open subset of $\operatorname{Spec}(R)$, and by corollary 2.1.13 each such V is a union of the sets $\varphi^{-1}(D(g))$ with $g \in R$.

Lemma 2.2.52: Morphisms into an Open Subscheme

Let B be a scheme, $V \subseteq B$ an open subscheme (definition 2.2.48) with inclusion $j\colon V \to B$, and $g\colon T \to B$ a morphism of schemes with $g(T) \subseteq V$. Then there is a unique morphism $g_V : T \to V$ with $j \circ g_V = g$. Open embeddings (definition 2.2.49) are monomorphisms of schemes.

***Proof*:**
For morphisms $a\colon T \to A$ and $i\colon A \to B$ of ringed spaces and an open $W' \subseteq B$, the sheaf map of the composite is $(i \circ a)^\#(W') = a^\#(i^{-1}W') \circ i^\#(W')$. For $W' \subseteq B$ open, $j^{-1}(W') = W' \cap V$ and $j^\#(W')$ is the restriction $\mathcal{O}_B(W') \to \mathcal{O}_B(W' \cap V) = \mathcal{O}_V(W' \cap V)$, which is the identity when $W' \subseteq V$.

Monomorphisms. Let $a, b\colon T \to V$ satisfy $j \circ a = j \circ b$. As j is injective, $a = b$ as maps, and for $W \subseteq V$ open, $a^\#(W) = (j \circ a)^\#(W) = (j \circ b)^\#(W) = b^\#(W)$. So j is a monomorphism. An open embedding is an isomorphism onto an open subscheme followed by its inclusion, and isomorphisms and composites of monomorphisms are monomorphisms, so open embeddings are monomorphisms.

Factoring. Let g_V be g regarded as a map $T \to V$, continuous because V carries the subspace topology, and put $g_V^\#(W) = g^\#(W)\colon \mathcal{O}_V(W) = \mathcal{O}_B(W) \to \mathcal{O}_T(g^{-1}(W))$ for $W \subseteq V$ open. These maps commute with restriction because those of $g^\#$ do, so $g_V^\#\colon \mathcal{O}_V \to (g_V)_*\mathcal{O}_T$ is a morphism of sheaves. For $t \in T$ the open neighbourhoods of $g(t)$ inside V are cofinal among all of its open neighbourhoods, so $\mathcal{O}_{V,g(t)} = \mathcal{O}_{B,g(t)}$ and the stalk map of g_V at t is that of g, a local homomorphism. Hence g_V is a morphism of schemes (definition 2.2.11). For $W' \subseteq B$ open, $g^{-1}(W' \cap V) = g^{-1}(W')$ because $g(T) \subseteq V$, so the restriction of $g_*\mathcal{O}_T$ from W' to $W' \cap V$ is the identity, and since $g^\#$ commutes with restriction,

$$(j \circ g_V)^\#(W') = g^\#(W' \cap V) \circ \operatorname{res}_{W', W' \cap V} = g^\#(W')$$

So $j \circ g_V = g$, and g_V is unique because j is a monomorphism, as claimed.

The prototypical example of an open subscheme would be the open sets $D_+(x_i) \subseteq \mathbb{P}^n_k$ of projective space or $D(f) \subseteq \operatorname{Spec} R$ of affine space. Note that an open sub-scheme of an affine sub-scheme need not be affine:

Example 2.2.53: Open Sub-Scheme need not be Affine

Let $R = k[x, y]$, $X = \mathbb{A}^2_k = \operatorname{Spec}(k[x, y])$, and consider $U = \mathbb{A}^2_k \setminus \{(x, y)\}$ which may be considered the plane without the origin. Before seeing why this is a scheme, try to show that U is indeed an open subset (it is the compliment of intersection of certain closed subsets, equivalently the union of certain open sets, there is more than one way of creating it).

Now, there does not exist an R such that $\operatorname{Spec}(R) \cong U$. It may be tempting to say that there ought to be a distinguished open set of $\mathbb{A}^2_k$ that produces U, but the reader should think through why this is not the case (hint: (x, y) is not principle). However, U can certainly be described as the union of two distinguished open sets, namely:

$$U = D(x) \cup D(y)$$

What is the global section on the open sub-scheme U? This would be all functions in $D(x) \cup D(y)$ that agree on the intersection, $D(x) \cap D(y) = D(xy) = R_{xy} = k[x, y]_{xy}$. Computing the local sections, as $D(x)$ is all points in $\mathbb{A}^2_k$ that do not vanish in the function x, and $D(y)$ is all the points that do not vanish in the function y, we get:

$$\mathcal{O}_U(D(x)) = R_x = k\left[x, y, \frac{1}{x}\right] \qquad \mathcal{O}_U(D(y)) = R_y = k\left[x, y, \frac{1}{y}\right]$$

Now consider the functions that agree on the intersection: $R_x \cap R_y \subseteq R_{xy}$. These are all rationals functions with only powers of x in the denominator, and only powers of y in the denominator (as it must be in the intersection). But this leaves only the elements of R: $R_x \cap R_y = R$ (where R is identified with it's canonical embedding in R_{xy}). Thus:

$$\mathcal{O}_{\mathbb{A}^2_k}(U) = \mathcal{O}_U(U) = k[x, y]$$

in other words, the global section of U is identical to the global section of $\mathbb{A}^2_k$. Now, if U was affine, then by proposition 2.2.13, we would have that the is determined by the global ring, that is:

$$U \cong \mathbb{A}^2_k$$

that is, this is a scheme isomorphism, and hence also a homeomorphic map between prime ideals that must respect $V(-)$ and $I(-)$. But in U we have:

$$V(x) \cap V(y) = \emptyset$$

while this is certainly not the case in $\mathbb{A}^2_k$ (as it gives $[(x - 0, y - 0)]$). Hence U is *not* an affine scheme!

The main idea is that $D(f)$ misses a codimension 1 subset if R is Noetherian integral domain (by Krull's height theorem, [10, Krull's Height Theorem]), where Noetherian is to avoid infinite pathologies and integral domain is to insure there is only one irreducible component (see section 2.4). More generally, given R is Noetherian and an integral domain, $\operatorname{Spec} R_S$ for some $S \subseteq R$ not containing units misses a (pure) codimension 1 subset (as it is open). The reason that higher codimensional sets "do not matter" when considering global sections very reminiscent of complex geometry, namely this is an algebraic version of *Hartog's Lemma* (see [58, chapter 8]).

It is possible to have an affine open subscheme of an affine scheme, and hence by dualizing we may look for algebraic conditions on the ring homomorphisms. We state the result for those familiar with flatness (see [10]) but leave the proof till flatness is introduced in section 6.8:

Theorem 2.2.54: Characterizing Affine Open Immersions

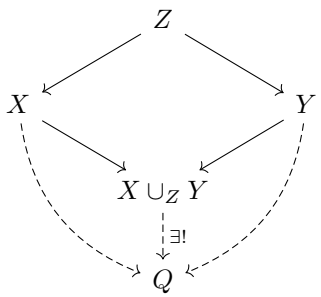
Spec $(S) \to$ Spec (R) is an open immersion if and only $R \to S$ is flat, of finite presentation, and an epimorphism.

***Proof*:**
see https://mathoverflow.net/questions/20782/ring-theoretic-characterization-of-open-affines.

2.2.55 For the expert: Cover Condition The reader well-versed in descent theory may have noticed this is almost the condition of an fppf covering, with faithful (i.e. surjective) the epimorphism condition. This motivates how fppf covers are the local open covers and appropriately generalize Zariski covers, which is developed in *Moduli, Deformation Theory, and Stacks* [20].

2.2.9 Gluing (Coproduct) of Schemes

In Differential Geometry, one of the most common way of constructing a manifold, and indeed what is at the heart of geometry, is the process of gluing pieces together with some amount of consistency (see [59, chapter 1]). More generally, gluing manifolds, and gluing schemes, is a special case of a pushout (or fibered coproduct) $X \cup_Z Y$:



The fibered product $X \times_Z Y$ can be thought of as taking a product of spaces $X \times Y$ and restricting to where the maps agree (i.e., carving out a subspace), while the fibered coproduct can be thought of as taking the disjoint union of spaces and doing some gluing. The fibered product shall be more nuanced as defining $X \times Y$ does not sense and hence is left for section 3.2.

Just like for manifolds, we cannot glue schemes any way we want and get another scheme, we must be a bit more careful. Example 3.2.7 shall give an example of "bad" gluing.

The key idea for coproducts of schemes is to ensure that the gluing preserves the local affine structure (note that for a manifold it also has to preserve Hausdorffness). Hence, it is marginally easier to construct schemes via gluing:

Theorem 2.2.56: Coproduct (Pushout) of Schemes

Let $\{X_i\}_{i\in I}$ be a collection of schemes. Suppose there are:

1. open subsets $X_{ij} \subseteq X_i$, viewed as open subschemes, not necessarily covering X_i, with $X_{ii} = X_i$, and
2. isomorphisms of schemes $f_{ij} : X_{ij} \to X_{ji}$ with $f_{ii} = \mathrm{id}$ and

$$f_{ij}(X_{ij} \cap X_{ik}) \subseteq X_{ji} \cap X_{jk}$$

such that the isomorphisms agree on triple intersection[a]:

$$f_{ik}|_{X_{ij}\cap X_{ik}} = f_{jk}|_{X_{ji}\cap X_{jk}} \circ f_{ij}|_{X_{ij}\cap X_{ik}}$$

Then there are a scheme X and open embeddings $\iota_i \colon X_i \to X$ (definition 2.2.49) whose images U_i cover X, with $U_i \cap U_j = \iota_i(X_{ij})$ and $\iota_j \circ f_{ij} = \iota_i$ on X_{ij}. If X' and ι_i' have the same properties, there is a unique isomorphism $h \colon X \to X'$ with $h \circ \iota_i = \iota_i'$ for all i.

[a] this is called the *cocycle condition*

This is a translation of the techniques from differential geometry (this condition may look more familiar if written as $\mathbf{y} \circ \mathbf{x}^{-1} : \mathbf{x}(U \cap V) \to \mathbf{y}(U \cap V)$). The cocycle condition can be visualized like so:

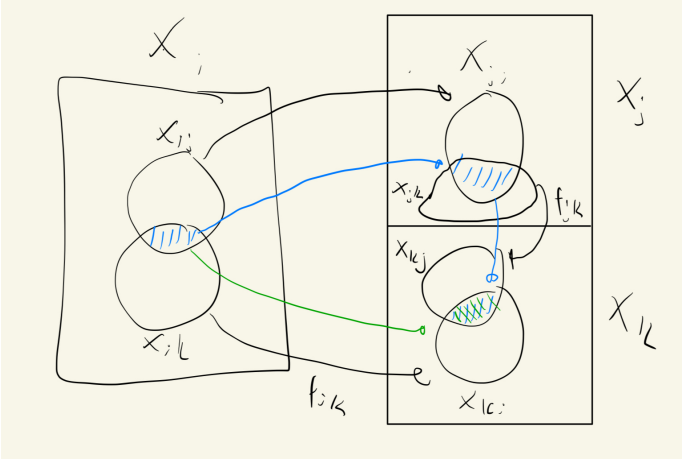


Figure 2.4: Green map must match blue map

Note the cocycle condition forces:

1. when $i = j = k$ that $f_{ii} = f_{ii} \circ f_{ii}$, so $f_{ii} = \mathrm{id}$
2. when $i = k$ that $f_{ji} = f_{ij}^{-1}$

***Proof*:**
Step 1: the space. The cocycle condition with $k = i$ gives $\mathrm{id} = f_{ji} \circ f_{ij}$ on X_{ij}, so $f_{ji} = f_{ij}^{-1}$. Applying the hypothesis on images to f_{ij} and to f_{ji} gives $f_{ij}(X_{ij} \cap X_{ik}) = X_{ji} \cap X_{jk}$. On the disjoint union $\coprod_{i\in I} X_i$ let $x \sim y$ when $x \in X_{ij}$ and $y = f_{ij}(x) \in X_{ji}$ for some i, j. This relation is reflexive as $f_{ii} = \mathrm{id}$ and symmetric as $f_{ji} = f_{ij}^{-1}$. If $y = f_{ij}(x)$ and $z = f_{jk}(y)$, then $y \in X_{ji} \cap X_{jk} = f_{ij}(X_{ij} \cap X_{ik})$, so $x \in X_{ij} \cap X_{ik}$ and $z = f_{jk}(f_{ij}(x)) = f_{ik}(x)$ by the cocycle condition. So $\sim$ is an equivalence relation. Let X be the quotient, with the quotient topology for $\pi\colon \coprod_i X_i \to X$, and let ι_i be the restriction of π to X_i, which is continuous.

Two points of X_i are equivalent only if they are equal, as $f_{ii} = \mathrm{id}$, so ι_i is injective. For $W \subseteq X_i$ open, the points of X_j equivalent to points of W form $f_{ij}(W \cap X_{ij})$, which is open in X_{ji} and hence in X_j. So $\pi^{-1}(\iota_i(W))$ is open and $\iota_i(W)$ is open. Hence ι_i is a homeomorphism onto the open set $U_i = \iota_i(X_i)$, the sets U_i cover X, and $\iota_i(x) = \iota_j(y)$ holds exactly when $x \in X_{ij}$ and $y = f_{ij}(x)$. So $U_i \cap U_j = \iota_i(X_{ij})$ and $\iota_j \circ f_{ij} = \iota_i$ on X_{ij} as maps.

Step 2: the structure sheaf. On U_i let $\mathscr{F}_i = (\iota_i)_*\mathcal{O}_{X_i}$, transported along the homeomorphism $\iota_i\colon X_i \to U_i$. For $V \subseteq U_i \cap U_j$ open, $\iota_j^{-1}(V) = f_{ij}(\iota_i^{-1}(V))$, and the sheaf isomorphism $f_{ij}^{\#}$ gives a ring isomorphism $f_{ij}^{\#}(\iota_j^{-1}(V))\colon \mathscr{F}_j(V) \to \mathscr{F}_i(V)$. Let $\varphi_{ij}(V)$ be its inverse. These commute with restriction, so they form isomorphisms $\varphi_{ij}\colon \mathscr{F}_i|_{U_i\cap U_j} \to \mathscr{F}_j|_{U_i\cap U_j}$. The sheaf map of a composite is $(f_{jk} \circ f_{ij})^{\#} = (f_{jk})_*(f_{ij}^{\#}) \circ f_{jk}^{\#}$, so the cocycle condition for the f_{ij} gives $\varphi_{jk} \circ \varphi_{ij} = \varphi_{ik}$ on $U_i \cap U_j \cap U_k$. By theorem 1.1.23 there is a sheaf of rings $\mathcal{O}_X$ on X with isomorphisms $\psi_i\colon \mathcal{O}_X|_{U_i} \to \mathscr{F}_i$ satisfying $\varphi_{ij} \circ \psi_i = \psi_j$.

Step 3: X is a scheme and the ι_i are open embeddings. The homeomorphism $\iota_i\colon X_i \to U_i$ and the sheaf isomorphism $\psi_i\colon \mathcal{O}_X|_{U_i} \to (\iota_i)_*\mathcal{O}_{X_i}$ form an isomorphism of ringed spaces $(X_i, \mathcal{O}_{X_i}) \cong (U_i, \mathcal{O}_X|_{U_i})$, and ι_i denotes this isomorphism followed by the inclusion of U_i. So the stalks of $\mathcal{O}_X$ are local rings (lemma 2.2.14(2)) and every point of X has an affine open neighbourhood, and X is a scheme (definition 2.2.9), each ι_i being an isomorphism onto the open subscheme U_i, that is, an open embedding. For $V \subseteq U_i \cap U_j$ open, the sheaf map of $\iota_j \circ f_{ij}$ at V is $f_{ij}^{\#}(\iota_j^{-1}(V)) \circ \psi_j(V) = \varphi_{ij}(V)^{-1} \circ \varphi_{ij}(V) \circ \psi_i(V) = \psi_i(V)$, the sheaf map of ι_i at V. So $\iota_j \circ f_{ij} = \iota_i$ as morphisms of schemes.

Step 4: uniqueness. Let X' and ι_i' be as in the statement, with $U_i' = \iota_i'(X_i)$. The isomorphisms $h_i = \iota_i' \circ \iota_i^{-1}\colon U_i \to U_i'$ agree on $U_i \cap U_j$, since there $\iota_j^{-1} = f_{ij} \circ \iota_i^{-1}$ and $\iota_j' \circ f_{ij} = \iota_i'$. So they define a map $h\colon X \to X'$, which is a bijection, since $\iota_i'(x) = \iota_j'(y)$ forces $x \in X_{ij}$ and $y = f_{ij}(x)$ as for X, and a homeomorphism, being one on each U_i. The opens of X' contained in some U_i' form a basis, and on such an open W' put $h^{\#}(W') = h_i^{\#}(W')$, which does not depend on i because the h_i agree on overlaps. These ring isomorphisms commute with restriction, so by proposition 1.3.3 and corollary 1.3.4 they extend uniquely to a sheaf isomorphism $h^{\#}\colon \mathcal{O}_{X'} \to h_*\mathcal{O}_X$. Then $(h, h^{\#})$ is an isomorphism with $h \circ \iota_i = \iota_i'$. Any h' with $h' \circ \iota_i = \iota_i'$ for all i restricts to h_i on each U_i, so it has the same underlying map and the same sheaf map on the basis, and $h' = h$ by the uniqueness in proposition 1.3.3, completing the proof.

What would the resulting global sections of a glued scheme look like? Consider the case of two schemes X_1, X_2 glued along $X_{12} \subseteq X_1$ and $X_{21} \subseteq X_2$ via an isomorphism $f_{12} : X_{12} \xrightarrow{\sim} X_{21}$, with

inclusion maps $\iota_i : X_i \to X$. Then:

$$\mathcal{O}_X(U) = \left\{ (s_1, s_2) \;\middle|\; s_1 \in \mathcal{O}_{X_1}(\iota_1^{-1}(U)),\ s_2 \in \mathcal{O}_{X_2}(\iota_2^{-1}(U)) \text{ such that } s_1|_{\iota_1^{-1}(U)\cap X_{12}} = f_{12}^{\#}\left(s_2|_{\iota_2^{-1}(U)\cap X_{21}}\right) \right\}$$

This formula is common enough to be given a name: we call it the *gluing formula*.

For three schemes X_1, X_2, X_3, the formula extends naturally: a global section is a triple $(s_1, s_2, s_3) \in \mathcal{O}_{X_1}(X_1) \times \mathcal{O}_{X_2}(X_2) \times \mathcal{O}_{X_3}(X_3)$ satisfying the pairwise compatibility conditions on all overlaps simultaneously:

$$s_i\big|_{X_{ij}} = f_{ij}^{\#}\left(s_j\big|_{X_{ji}}\right) \qquad \text{for all } i, j \in \{1, 2, 3\}$$

Note that one does not glue two schemes first and then attach the third; the cocycle condition ensures that all pairwise constraints are mutually consistent, so all three (or finitely many) pieces are handled at once. The pattern for n schemes is identical: global sections are n-tuples satisfying $\binom{n}{2}$ compatibility conditions.

When thinking about global sections of glued schemes, it is helpful to view the gluing maps as imposing additional *relations*: the global sections of the glued scheme are the elements of the product $\prod_i \mathcal{O}_{X_i}(X_i)$ that are compatible under all the transition isomorphisms. The more gluing constraints there are, the smaller the ring of global sections becomes.

Example 2.2.57: Gluing [affine] Schemes

Let $X = \operatorname{Spec}(k[x])$ and $Y = \operatorname{Spec}(k[y])$. Take $D(x) \subseteq X$ and $D(y) \subseteq Y$, which is the collection of all prime ideals that do not contain x and y respectively. As shown in proposition 2.1.12, these subsets are isomorphic to:

$$D(x) \cong \operatorname{Spec}\left(k\left[x, \frac{1}{x}\right]\right) \qquad D(y) \cong \operatorname{Spec}\left(k\left[y, \frac{1}{y}\right]\right)$$

As a diagram, we have:

$$\begin{array}{ccc} \operatorname{Spec}(k[x]) & & \operatorname{Spec}(k[y]) \\ & \nwarrow \quad \nearrow & \\ & \operatorname{Spec}(k[x, 1/x]) & \end{array} \qquad \text{and} \qquad \begin{array}{ccc} k[x] & & k[y] \\ & \searrow \quad \swarrow & \\ & k[x, 1/x] & \end{array}$$

or it may be more helpful to see this as:

$$\begin{array}{ccc} \operatorname{Spec} k[x] & & \operatorname{Spec} k[y] \\ & \nwarrow \quad \nearrow & \\ & \operatorname{Spec} k[x, 1/x] \cong \operatorname{Spec} k[y, 1/y] & \end{array}$$

and

$$\begin{array}{ccc} k[x] & & k[y] \\ & \searrow \quad \swarrow & \\ & k[x, 1/x] \cong k[y, 1/y] & \end{array}$$

where the cocycle condition must be satisfied by the isomorphisms. Notice how these are similar to restriction conditions: we are "manually" figuring out how the restrictions should work so that by uniqueness of gluing local sections on a sheaf we can get the global sections. These two open subsets can be glued in at least two different ways to produce two non-affine schemes that will be familiar to the reader:

1. (Line with double origin). Define the isomorphism $D(x) \cong D(y)$ via the ring isomorphism $\varphi^{\#} : k[y, 1/y] \to k[x, 1/x]$ given by $y \mapsto x$. This is a scheme by theorem 2.2.56. Label this scheme S_1. We claim S_1 is *not* affine, even though its global sections coincide with those of $\mathbb{A}^1_k$.

   By the gluing formula, a global section of $\mathcal{O}_{S_1}$ is a pair $(f(x), g(y)) \in k[x] \times k[y]$ satisfying the compatibility condition:

   $$f(x) = \varphi^{\#}(g(y)) = g(x) \qquad \text{in } k[x, 1/x]$$

   Since $k[x] \hookrightarrow k[x, 1/x]$ is injective, this simply says $f(x) = g(x)$ as polynomials (after identifying the variable names). In particular, any polynomial satisfies this condition, and so:

   $$\mathcal{O}_{S_1}(S_1) \cong k[x]$$

   By proposition 2.2.13, an affine scheme is determined by its global sections, so if S_1 were affine it would be isomorphic to $\operatorname{Spec}(k[x]) = \mathbb{A}^1_k$. But this is impossible: S_1 has two distinct closed points (the two origins) that are not separated by any open set, while $\mathbb{A}^1_k$ does not. Hence S_1 is not affine. The same construction works in higher dimensions (e.g. $\mathbb{A}^n_k$ with double origin).

   In differential geometry, such spaces are examples of non-Hausdorff manifolds, however all non-trivial schemes already fail to be Hausdorff. The relevant condition for schemes is *separatedness*: spaces with double-origins fail to be separated (as a hint: show that the intersection of the two copies of $\mathbb{A}^1_k$ inside S_1 is $D(x)$, which is not affine), while $\mathbb{A}^n_k$ is separated.

2. (Projective space). Define the isomorphism $D(x) \cong D(y)$ via the ring isomorphism $\varphi^{\#} : k[y, 1/y] \to k[x, 1/x]$ defined by $y \mapsto 1/x$ (equivalently, $1/y \mapsto x$). The reader will recognize the resulting scheme as $\mathbb{P}^1_k$. When $k \in \{\mathbb{R}, \mathbb{C}\}$, this is isomorphic (as a manifold) to a circle and a sphere respectively, while in dimension n, $\mathbb{P}^n_k$ cannot be embedded in $\mathbb{A}^n_k$. For schemes, one must account for the generic point (0) and the non-closed points given by irreducible polynomials (recall that linear factors that are Galois conjugates are identified).

   Let us show that $\mathbb{P}^1_k$ is not affine by computing $\mathcal{O}_{\mathbb{P}^1}(\mathbb{P}^1)$. A global section is a pair $(f(x), g(y)) \in k[x] \times k[y]$ satisfying:

   $$f(x) = \varphi^{\#}(g(y)) = g(1/x) \qquad \text{in } k[x, 1/x]$$

   Write $f(x) = \sum_{n=0}^{N} a_n x^n$ and $g(y) = \sum_{m=0}^{M} b_m y^m$, so that:

   $$g(1/x) = \sum_{m=0}^{M} b_m \, x^{-m}$$

   In $k[x, 1/x]$, the Laurent monomials $\{x^n\}_{n \in \mathbb{Z}}$ are linearly independent over k. The left-hand side $f(x)$ involves only non-negative powers of x, while the right-hand side $g(1/x)$ involves

only non-positive powers. For these to be equal, both must be constant:

$$a_n = 0 \text{ for } n \geq 1, \qquad b_m = 0 \text{ for } m \geq 1, \qquad a_0 = b_0$$

Hence f and g are both the same constant, giving:

$$\mathcal{O}_{\mathbb{P}^1}(\mathbb{P}^1) = k$$

By proposition 2.2.13, if $\mathbb{P}^1$ were affine then $\mathbb{P}^1 \cong \operatorname{Spec}(\mathcal{O}_{\mathbb{P}^1}(\mathbb{P}^1)) = \operatorname{Spec}(k)$, which is a single point. But $\mathbb{P}^1_k$ has more than one point, so $\mathbb{P}^1_k$ is not affine[a].

[a] Those familiar with complex analysis may recall that the only holomorphic functions on $\mathbb{P}^1_{\mathbb{C}}$ (the Riemann sphere) are constants, which is a manifestation of the same phenomenon.

2.2.10 *Gluing Affine Schemes and Fiber Products of Rings

Let us now specialize to the case where the pieces being glued are affine. Recall that the functors $\operatorname{Spec}(-)$ and $\Gamma(-)$ are contravariant and form an adjunction that exchanges limits and colimits. Hence, the cofibered product (pushout) of schemes dualizes to the fiber product (pullback) of rings. If $X_1 = \operatorname{Spec}(A)$, $X_2 = \operatorname{Spec}(B)$, and the overlap maps factor through a ring C, then the gluing formula for global sections becomes:

$$\mathcal{O}_X(X) = A \times_C B := \{(a, b) \in A \times B \mid \text{images of } a \text{ and } b \text{ agree in } C\}$$

This is dual to the familiar fact that the fibered product of affine schemes corresponds to the tensor product of rings:

$$\operatorname{Spec} A \times_{\operatorname{Spec} C} \operatorname{Spec} B \cong \operatorname{Spec}(A \otimes_C B)$$

In both cases, the contravariance of Spec exchanges limits and colimits, and hence exchanges tensor products (colimits in **CRing**) with fiber products (limits in **CRing**). More generally, for an arbitrary number of affine pieces, the global sections are the limit of the diagram of rings formed by all the overlap maps.

It is natural to ask: since **CRing** is both complete and cocomplete, does $\operatorname{Spec}(A \times_C B)$ give the pushout of $\operatorname{Spec}(A)$ and $\operatorname{Spec}(B)$ over $\operatorname{Spec}(C)$ in the category of schemes? The answer is *no* in general. The subtlety is that the universal property must be tested against different classes of objects depending on the ambient category:

- *In **Aff***: the object $\operatorname{Spec}(A \times_C B)$ satisfies the universal property of the pushout with respect to maps into affine schemes, and this pushout always exists since **CRing** is complete (so fiber products of rings always exist).
- *In **Sch***: the pushout must satisfy the universal property for maps into *all* schemes. A morphism from an affine scheme to a non-affine target Q is not determined by a ring homomorphism out of $\Gamma(Q, \mathcal{O}_Q)$ (the adjunction $\Gamma \dashv \operatorname{Spec}$ describes morphisms into affine schemes, and Q is not affine), so the affine pushout may fail the universal property.

In categorical language, the inclusion $\mathbf{Aff} \hookrightarrow \mathbf{Sch}$ does not preserve pushouts. The pushout computed in **Sch** can be genuinely non-affine: the line with doubled origin and the projective line from example 2.2.57 are both pushouts of affine schemes in **Sch** that are not affine. In each case,

$\mathrm{Spec}\,(A \times_C B)$ exists as an affine scheme, but it is *not* isomorphic to the glued scheme X – it is only the "best affine approximation" to the pushout.

The preceding discussion raises a natural question: if $\mathrm{Spec}\,(A \times_C B)$ is not the pushout in **Sch**, what *is* it? The answer is given by the adjunction between global sections and Spec: the global sections functor $\Gamma : \mathbf{Sch} \to \mathbf{CRing}^{op}$ is left adjoint to $\mathrm{Spec} : \mathbf{CRing}^{op} \to \mathbf{Sch}$,

$$\mathrm{Hom}_{\mathbf{Sch}}(X, \mathrm{Spec}\,(R)) \cong \mathrm{Hom}_{\mathbf{CRing}}(R, \Gamma(X, \mathcal{O}_X))$$

for any scheme X and any ring R, as proposition 3.1.12 proves.

Since left adjoints preserve colimits and right adjoints preserve limits, this single adjunction encodes both dualities between schemes and rings:

- Γ (left adjoint) sends pushouts in **Sch** to colimits in $\mathbf{CRing}^{op}$, i.e. limits (fiber products) in **CRing**:
$$\Gamma(X_1 \cup_Z X_2) \cong \Gamma(X_1) \times_{\Gamma(Z)} \Gamma(X_2)$$
This is why the gluing formula computes global sections as a fiber product of rings – it is a formal consequence of the adjunction.
- Spec (right adjoint) sends limits in $\mathbf{CRing}^{op}$ (i.e. colimits in **CRing**, such as tensor products) to limits (fiber products) in **Sch**:
$$\mathrm{Spec}\,(A \otimes_C B) \cong \mathrm{Spec}\,(A) \times_{\mathrm{Spec}\,(C)} \mathrm{Spec}\,(B)$$
- Spec, being a right adjoint, does *not* preserve colimits in general. The fiber product $A \times_C B$ is a limit in **CRing**, hence a colimit in $\mathbf{CRing}^{op}$, and there is no reason for Spec to preserve it.

Definition 2.2.58: Affinization

Let X be a scheme. The *affinization* of X is the affine scheme $X^{\mathbf{Aff}} := \mathrm{Spec}\,(\Gamma(X, \mathcal{O}_X))$, together with the canonical morphism

$$\eta_X : X \to \mathrm{Spec}\,(\Gamma(X, \mathcal{O}_X))$$

given by the unit of the $\Gamma \dashv \mathrm{Spec}$ adjunction.

By the universal property of adjunctions, η_X is the universal map from X to an affine scheme: any morphism $X \to \mathrm{Spec}\,(R)$ factors uniquely through η_X. For the pushout $X = X_1 \cup_Z X_2$ of affine schemes, since $\Gamma(X) \cong A \times_C B$, this gives:

$$X_1 \cup_Z X_2 \xrightarrow{\eta} \mathrm{Spec}\,(A \times_C B)$$

Hence $\mathrm{Spec}\,(A \times_C B)$ is not the pushout in **Sch**, but rather the *best affine approximation* to the pushout – the closest affine scheme that the glued scheme maps to.

Example 2.2.59: Affinization of Glued Schemes

Returning to the two gluings from example 2.2.57:

1. *Line with doubled origin.* Here $A \times_C B \cong k[x]$, so the affinization is $\mathbb{A}^1_k$. The map $\eta : S_1 \to \mathbb{A}^1_k$ identifies the two origins to a single point, collapsing exactly the non-affine pathology.

2. *Projective line.* Here $A \times_C B \cong k$, so the affinization is $\operatorname{Spec}(k)$, a single point. The map $\eta : \mathbb{P}^1_k \to \operatorname{Spec}(k)$ collapses the entire projective line to a point, reflecting the fact that $\mathbb{P}^1_k$ has only constant global functions.

In both cases, the affinization map η "kills" precisely the geometric information that cannot be captured by global sections.

Exercise 2.2.2

1. Let X be the line with two origins, $Y = \mathbb{A}^1_k$ the affine line, and $\pi : X \to Y$ the natural projection. Compute $f_*(\mathcal{O}_X)_0$, the talk of the pushforward at 0.
2. Let X be a scheme. Show that the affine open subsets form a base.
3. A subset Z of a scheme X is closed if and only if there exists an open cover of X by affine subschemes, $\{U_i = \operatorname{Spec}(A_i)\}_{i \in I}$, such that for every $i \in I$, the intersection $Z \cap U_i$ is a closed subset of the affine scheme U_i. In other words, a closed subset of X is characterized as being locally closed on an affine open covering.

2.3 Projective Schemes

Recall projective space from [59]. In [22] a few key properties of projective space $\mathbb{P}^n_{\mathbb{C}}$ were proved, namely:

1. Projective space was constructed by gluing together affine spaces. The cocycle condition is straightforward to verify over any ring (the transition maps are rational functions that do not require algebraic closure), though understanding the closed points is simpler when $k = \overline{k}$. We shall construct Proj in a more "natural" way without any choice of coordinates.
2. There shall be a homogeneous coordinate ring S from which we can construct the scheme $\operatorname{Proj} S$. Just like in the classical setting, S shall not be the ring of regular functions on $\operatorname{Proj} S$: the global sections of $\mathbb{P}^n_R$ shall be constant, and $\operatorname{Proj} S$ shall behave like the "compact" sets in algebraic geometry (namely the proper sets, as mentioned in [22]).

   We shall show that the algebraic object S will be associated to a different sheaf over $\operatorname{Proj} S$ that is not the structure sheaf, see section 5.3.
3. Bézout's theorem for plane curves is a special case of the intersection pairing on a surface (section 8.2).

Classically, n-dimensional projective space $\mathbb{R}\mathrm{P}^n$ or $\mathbb{C}\mathrm{P}^n$ is defined on $\mathbb{R}^{n+1} \setminus \{0\}$ or $\mathbb{C}^{n+1} \setminus \{0\}$ by taking an equivalence class:

$$(x_0, \ldots, x_n) \sim (y_0, \ldots, y_n) \qquad \Leftrightarrow \qquad (x_0, \ldots, x_n) = (\lambda y_0, \ldots, \lambda y_n) \qquad \text{for some } \lambda \neq 0$$

This produces an n-dimensional smooth manifold with the usual patches defined as:

$$U_i = D(x_i) = \{[x_0 : \cdots : x_n] \in \mathbb{P}^n \mid x_i \neq 0\}$$

where, using the notation $x_{k,j} = x_k/x_j$ for the affine coordinates on U_j, the transition maps were defined as (see [59, chapter 1]):

$$\varphi_i \circ \varphi_j^{-1}(x_{0,j}, \ldots, \widehat{x_{j,j}}, \ldots, x_{n,j}) = \left(\frac{x_{0,j}}{x_{i,j}}, \ldots, \frac{1}{x_{i,j}}, \ldots, \frac{\widehat{x_{i,j}}}{x_{i,j}}, \ldots, \frac{x_{n,j}}{x_{i,j}} \right)$$

In [22], we saw we can do the same for any field k, namely since inverses exist. Let us do the same for an $\mathbb{N}$-graded ring $S = \bigoplus_{d \geq 0} S_d$, the main example being the graded polynomial ring $S = R[x_0, x_1, \ldots, x_n]$ over a ring R with the standard grading ($\deg x_i = 1$, R in degree 0), and let $S_+ = \bigoplus_{d>0} S_d$ be the irrelevant ideal. Though S is $\mathbb{N}$-graded, we shall work in the setting of $\mathbb{Z}$-graded rings since localizations introduce negative degrees[21]: for f homogeneous of degree e, the localization S_f is graded by $\deg(a/f^k) = \deg a - ke$ for homogeneous a.

As a set, define:

Definition 2.3.1: Projective Space – as a Set

$$\operatorname{Proj} S = \{[\mathfrak{p}] \mid \mathfrak{p} \subseteq S \text{ is a homogeneous prime with } S_+ \not\subseteq \mathfrak{p}\}$$

The condition $S_+ \not\subseteq \mathfrak{p}$ excludes the "irrelevant" prime consisting of all positive-degree elements, which would correspond to the non-existent point $[0 : \cdots : 0]$ in projective space. For a set T of homogeneous elements of positive degree and for f homogeneous of positive degree put

$$V_+(T) = \{[\mathfrak{p}] \in \operatorname{Proj} S \mid T \subseteq \mathfrak{p}\} \qquad D_+(f) = \{[\mathfrak{p}] \in \operatorname{Proj} S \mid f \notin \mathfrak{p}\}$$

The sets $D_+(f)$ cover $\operatorname{Proj} S$: for $[\mathfrak{p}] \in \operatorname{Proj} S$ we have $S_+ \not\subseteq \mathfrak{p}$, and S_+ is generated by homogeneous elements of positive degree, so one of them lies outside $\mathfrak{p}$. Since $\mathfrak{p}$ is prime, $D_+(f) \cap D_+(g) = D_+(fg)$. So the sets $D_+(f)$ form a basis of a topology on $\operatorname{Proj} S$, the *Zariski topology*, whose closed sets are the sets $V_+(T)$, as $\operatorname{Proj} S \setminus V_+(T) = \bigcup_{t \in T} D_+(t)$.

If $f = x_i$, we would like this to correspond to the usual affine space $\mathbb{A}^n_R$. In the classical case, we got this isomorphism by doing:

$$\begin{aligned}
\{[x_0 : \cdots : x_n] \mid x_i \neq 0\} &= \left\{ \left[\frac{x_0}{x_i} : \cdots : 1 : \cdots : \frac{x_n}{x_i} \right] \;\middle|\; x_i \neq 0 \right\} \\
&\cong k\left[\frac{x_0}{x_i}, \ldots, \frac{x_n}{x_i} \right] \\
&= k[x_{0,i}, \ldots, x_{n,i}] && \text{relabeling} \\
&= \mathbb{A}^n_k
\end{aligned}$$

Notice how these are all degree 0 in the grading of S. These sets are not $\operatorname{Spec} S_f$ itself: the chart is the spectrum of the degree-0 part $S_{(f)} = (S_f)_0$ of the localization.

[21] More general gradings are sometimes useful, however we shall not need this either in this book or for a while

Lemma 2.3.2: Characterizing Basic Open Sets For Projective Space

Let S be an $\mathbb{N}$-graded ring, let $f \in S$ be a homogeneous element of degree $e > 0$, and define $S_{(f)} := (S_f)_0$ to be the degree-0 part of the localization. Then the map

$$\Phi_f \colon D_+(f) \to \operatorname{Spec}\left(S_{(f)}\right), \qquad \Phi_f(\mathfrak{p}) = \mathfrak{p}S_f \cap S_{(f)}$$

is a homeomorphism, $D_+(f) \cong \operatorname{Spec}(S_{(f)})$. For g homogeneous of degree $d > 0$ it carries $D_+(fg)$ onto $D(g^e/f^d)$.

***Proof*:**
Step 1: homogeneous primes. A homogeneous ideal is the sum of its homogeneous components, so two homogeneous ideals with the same homogeneous elements are equal. Let $P \neq S$ be a homogeneous ideal such that $ab \in P$ for homogeneous a, b implies $a \in P$ or $b \in P$. Then P is prime: if $a, b \notin P$, let a_i be the homogeneous component of a of least degree i with $a_i \notin P$, and b_j likewise. The component of ab of degree $i + j$ is $\sum_{k+l=i+j} a_k b_l$, and every term other than $a_i b_j$ has $k < i$ or $l < j$, hence lies in P. So if $ab \in P$, which is homogeneous, then $a_i b_j \in P$, contradicting the hypothesis.

Step 2: Φ_f is well defined. Let $[\mathfrak{p}] \in D_+(f)$ and $a \in S_{ke}$. If $a/f^k \in \mathfrak{p}S_f$, say $a/f^k = p/f^m$ with $p \in \mathfrak{p}$, then $f^N(af^m - pf^k) = 0$ for some N, so $f^{N+m}a \in \mathfrak{p}$ and $a \in \mathfrak{p}$, as $f \notin \mathfrak{p}$. Hence $\Phi_f(\mathfrak{p}) = \left\{a/f^k \;\middle|\; k \geq 0,\ a \in \mathfrak{p} \cap S_{ke}\right\}$. It is an ideal of $S_{(f)}$, being $\mathfrak{p}S_f \cap S_{(f)}$, it does not contain 1, and if $ab/f^{k+l} \in \Phi_f(\mathfrak{p})$ with $a \in S_{ke}$ and $b \in S_{le}$, then $ab \in \mathfrak{p}$, so a or b lies in $\mathfrak{p}$. So $\Phi_f(\mathfrak{p})$ is prime.

Step 3: the inverse map. For a prime $\mathfrak{q}$ of $S_{(f)}$ put

$$\Psi_f(\mathfrak{q}) = \bigoplus_{d \geq 0} \left\{a \in S_d \;\middle|\; a^e/f^d \in \mathfrak{q}\right\}$$

noting that $a^e/f^d \in S_{(f)}$ for $a \in S_d$. Each summand contains 0 and is closed under negation, as $(-a)^e = \pm a^e$. For $a, b \in S_d$ in it, each $a^j b^{e-j}/f^d$ satisfies $(a^j b^{e-j}/f^d)^e = (a^e/f^d)^j (b^e/f^d)^{e-j} \in \mathfrak{q}$, so it lies in the prime $\mathfrak{q}$, and summing the binomial expansion gives $(a + b)^e/f^d \in \mathfrak{q}$. For $c \in S_k$, $(ca)^e/f^{d+k} = (c^e/f^k)(a^e/f^d) \in \mathfrak{q}$. So $\Psi_f(\mathfrak{q})$ is a homogeneous ideal. It omits f, since $f^e/f^e = 1 \notin \mathfrak{q}$, and if $a \in S_d$ and $b \in S_k$ satisfy $ab \in \Psi_f(\mathfrak{q})$, then $(a^e/f^d)(b^e/f^k) \in \mathfrak{q}$ gives a or b in $\Psi_f(\mathfrak{q})$. By Step 1, $\Psi_f(\mathfrak{q})$ is prime, and it does not contain $f \in S_+$, so $[\Psi_f(\mathfrak{q})] \in D_+(f)$.

Step 4: Φ_f and Ψ_f are mutually inverse. For $a \in S_d$, Step 2 gives $a \in \Psi_f(\Phi_f(\mathfrak{p}))$ if and only if $a^e \in \mathfrak{p}$, if and only if $a \in \mathfrak{p}$. By Step 1, $\Psi_f(\Phi_f(\mathfrak{p})) = \mathfrak{p}$. For $a \in S_{ke}$, Step 2 applied to $\Psi_f(\mathfrak{q})$ gives $a/f^k \in \Phi_f(\Psi_f(\mathfrak{q}))$ if and only if $a \in \Psi_f(\mathfrak{q})$, if and only if $(a/f^k)^e = a^e/f^{ke} \in \mathfrak{q}$, if and only if $a/f^k \in \mathfrak{q}$. So $\Phi_f(\Psi_f(\mathfrak{q})) = \mathfrak{q}$.

Step 5: Φ_f is a homeomorphism. For $[\mathfrak{p}] \in D_+(f)$ and g homogeneous of degree $d > 0$, Step 2 gives $g^e/f^d \notin \Phi_f(\mathfrak{p})$ if and only if $g \notin \mathfrak{p}$, so $\Phi_f(D_+(fg)) = \Phi_f(D_+(f) \cap D_+(g)) = D(g^e/f^d)$. Every open subset of $D_+(f)$ is a union of sets $D_+(h) \cap D_+(f) = D_+(fh)$, so Φ_f is open. Every element of $S_{(f)}$ has the form a/f^k with $a \in S_{ke}$, and $a/f^k \notin \Phi_f(\mathfrak{p})$ if and only if $a \notin \mathfrak{p}$, so

$\Phi_f^{-1}(D(a/f^k))$ is the open set $D_+(fa)$, fa being homogeneous of positive degree. The sets $D(a/f^k)$ form a basis of $\operatorname{Spec} S_{(f)}$ (corollary 2.1.13), so Φ_f is continuous, and a continuous open bijection is a homeomorphism, completing the proof.

Let us apply this to $S = R[x_0, \ldots, x_n]$ and $f = x_i$:

$$\begin{aligned} D_+(x_i) &= \{[\mathfrak{p}] \in \operatorname{Proj} S \mid x_i \notin \mathfrak{p}\} \\ &\cong \operatorname{Spec}\left(S_{(x_i)}\right) \\ &= \operatorname{Spec}\ (\text{degree-0 elements of } R[x_0, \ldots, x_n, 1/x_i]) \\ &\overset{!}{=} \operatorname{Spec} R\left[\frac{x_0}{x_i}, \ldots, \frac{x_n}{x_i}\right] \\ &\cong \operatorname{Spec} \frac{R[x_{0,i}, \ldots, x_{n,i}]}{(x_{i,i} - 1)} \\ &\cong \mathbb{A}_R^n \end{aligned}$$

where $\overset{!}{=}$ comes from the fact that $1/x_i$ is now a unit of degree -1, hence each x_j/x_i is of degree 0, and these elements generate $S_{(x_i)}$ over R. Relabeling x_j/x_i as $x_{j,i}$ and "omitting" $x_{i,i}$ (which equals one, equivalently we quotient by $(x_{i,i} - 1)$, a form of dehomogenizing), we see that it is isomorphic to the usual $\mathbb{A}_R^n$, and hence we recover the usual affine cover! Note the importance of taking the degree-0 component, for

$$\operatorname{Spec} R[x_0, \ldots, x_n, 1/x_i] \cong \mathbb{A}_R^{n+1} \setminus \{x_i = 0\}$$

Let us now construct the structure sheaf on $\operatorname{Proj} S$. We shall see that $\mathbb{P}_R^n$ is not affine for $n \geq 1$ (we hope not, as the classical projective variety is not affine!), and so it is not determined by its global sections in the way that affine schemes are. Nonetheless, the construction below endows $\operatorname{Proj} S$ with a natural scheme structure, glued from the affine schemes $\operatorname{Spec} S_{(f)}$ along the overlaps $D_+(f) \cap D_+(g) = D_+(fg)$.

Theorem 2.3.3: Structure Sheaf On Proj

Let S be an $\mathbb{N}$-graded ring. There is a sheaf of rings $\mathcal{O}$ on $\operatorname{Proj} S$, with its Zariski topology, such that $(\operatorname{Proj} S, \mathcal{O})$ is a scheme and, for every homogeneous $f \in S$ of positive degree, $(D_+(f), \mathcal{O}|_{D_+(f)}) \cong (\operatorname{Spec} S_{(f)}, \mathcal{O}_{\operatorname{Spec} S_{(f)}})$ with underlying homeomorphism Φ_f (lemma 2.3.2). Under these isomorphisms $\mathcal{O}(D_+(f)) = S_{(f)}$, and for g homogeneous of positive degree the restriction $\mathcal{O}(D_+(f)) \to \mathcal{O}(D_+(fg))$ is the ring map $S_{(f)} \to S_{(fg)}$, $a/f^k \mapsto ag^k/(fg)^k$.

***Proof*:**
Write e_q for the degree of a homogeneous element q of positive degree. For such q, r let $\rho_{q,qr} \colon S_{(q)} \to S_{(qr)}$, $a/q^k \mapsto ar^k/(qr)^k$, be the restriction to degree 0 of the localization map $S_q \to S_{qr}$. These maps compose, $\rho_{qr,qrs} \circ \rho_{q,qr} = \rho_{q,qrs}$.

Step 1: $\rho_{q,qr}$ is the localization at $t = r^{e_q}/q^{e_r}$. We have $\rho_{q,qr}(t) = r^{e_q+e_r}/(qr)^{e_r}$, a unit with inverse $q^{e_q+e_r}/(qr)^{e_q}$, so [10, Universal Property of Localization] gives a ring map $\lambda_{q,r} \colon (S_{(q)})_t \to$

$S_{(qr)}$ extending $\rho_{q,qr}$. It is injective: if $\rho_{q,qr}(a/q^k) = ar^k/(qr)^k = 0$, then $q^N r^{N+k} a = 0$ for some N, and for $e_q M \geq N + k$ we get $t^M \cdot a/q^k = r^{e_q M} a/q^{e_r M + k} = 0$, as $q^N r^{e_q M} a = 0$. It is surjective: for $b \in S_{k(e_q+e_r)}$ put $c = br^{k(e_q-1)}$, of degree $ke_q(1+e_r)$. A computation in S_{qr} gives $\rho_{q,qr}(c/q^{k(1+e_r)}) = br^{k(e_q-1)}/q^{k(1+e_r)} = (b/(qr)^k)\,\rho_{q,qr}(t)^k$, so $b/(qr)^k = \lambda_{q,r}\big((c/q^{k(1+e_r)})/t^k\big)$.

Step 2: the maps $c_{q,r}$. Let $D_+(r) \subseteq D_+(q)$. Then $D_+(rq) = D_+(r)$, so $\Phi_r(D_+(rq)) = \operatorname{Spec} S_{(r)}$, while lemma 2.3.2 gives $\Phi_r(D_+(rq)) = D(q^{e_r}/r^{e_q})$. So q^{e_r}/r^{e_q} lies in no prime of $S_{(r)}$, the radical of the ideal it generates is $I(\emptyset) = S_{(r)}$ (proposition 2.1.6(5)), and it is a unit. Localizing at a unit is an isomorphism, so by Step 1 (with q and r exchanged) $\rho_{r,rq}$ is an isomorphism. Put $c_{q,r} = \rho_{r,qr}^{-1} \circ \rho_{q,qr} \colon S_{(q)} \to S_{(r)}$. Then $c_{q,q} = \mathrm{id}$ and $c_{r,s} \circ c_{q,r} = c_{q,s}$ when $D_+(s) \subseteq D_+(r) \subseteq D_+(q)$: using that the ρ compose, both sides followed by the injective map $\rho_{s,qrs}$ equal $\rho_{q,qrs}$. Also $c_{q,qr} = \rho_{q,qr}$, since $\rho_{qr,q^2r} \circ \rho_{q,qr} = \rho_{q,q^2r}$.

Step 3: the sheaves $\mathcal{O}_f$. For f homogeneous of positive degree let $\mathcal{O}_f$ be the sheaf on $D_+(f)$ with $\mathcal{O}_f(W) = \mathcal{O}_{\operatorname{Spec} S_{(f)}}(\Phi_f(W))$, so that Φ_f is an isomorphism of locally ringed spaces $(D_+(f), \mathcal{O}_f) \cong \operatorname{Spec} S_{(f)}$. If $D_+(q) \subseteq D_+(f)$, then $D_+(q) = D_+(fq)$ and $\Phi_f(D_+(q)) = D(t_q)$ with $t_q = q^{e_f}/f^{e_q}$ (lemma 2.3.2), so $\mathcal{O}_f(D_+(q)) = (S_{(f)})_{t_q}$, the restriction from $D_+(f)$ being the localization map and the restrictions between such sets being the canonical $S_{(f)}$-algebra maps, by the construction of the structure sheaf (section 2.2.1, proposition 2.2.1). The ring isomorphism $\varepsilon_q^f = \rho_{q,fq}^{-1} \circ \lambda_{f,q} \colon \mathcal{O}_f(D_+(q)) \to S_{(q)}$ restricts on $S_{(f)}$ to $c_{f,q}$. A ring map out of the localization $\mathcal{O}_f(D_+(q))$ is determined by its restriction to $S_{(f)}$, so Step 2 gives $\varepsilon_{q'}^f \circ \mathrm{res}_{D_+(q),D_+(q')} = c_{q,q'} \circ \varepsilon_q^f$ whenever $D_+(q') \subseteq D_+(q) \subseteq D_+(f)$, both sides restricting on $S_{(f)}$ to $c_{f,q'}$.

Step 4: gluing data. Fix f, g. The sets $D_+(q) \subseteq D_+(fg)$ form a basis of $D_+(fg)$, since $D_+(h) \cap D_+(fg) = D_+(fgh)$. On such a set put $\varphi_{D_+(q)} = (\varepsilon_q^g)^{-1} \circ \varepsilon_q^f \colon \mathcal{O}_f(D_+(q)) \to \mathcal{O}_g(D_+(q))$, a ring isomorphism. It does not depend on the choice of q: if $D_+(q'') = D_+(q)$, the identity of Step 3 gives $\varepsilon_{q''}^f = c_{q,q''} \circ \varepsilon_q^f$ and $\varepsilon_{q''}^g = c_{q,q''} \circ \varepsilon_q^g$. For $D_+(q') \subseteq D_+(q)$, the same identity for f and for g gives $\varphi_{D_+(q')} \circ \mathrm{res} = (\varepsilon_{q'}^g)^{-1} \circ c_{q,q'} \circ \varepsilon_q^f = \mathrm{res} \circ \varphi_{D_+(q)}$. By corollary 1.3.4 these maps extend uniquely to a sheaf isomorphism $\varphi_{fg} \colon \mathcal{O}_f|_{D_+(fg)} \to \mathcal{O}_g|_{D_+(fg)}$, and $\varphi_{ff} = \mathrm{id}$. On $D_+(fgh)$ the isomorphisms $\varphi_{gh} \circ \varphi_{fg}$ and φ_{fh} agree on every basic set $D_+(q)$, both being $(\varepsilon_q^h)^{-1} \circ \varepsilon_q^f$, so they are equal by the uniqueness in proposition 1.3.3.

Step 5: the scheme. By theorem 1.1.23, applied to the cover $\{D_+(f)\}$, there is a sheaf $\mathcal{O}$ on $\operatorname{Proj} S$ with isomorphisms $\psi_f \colon \mathcal{O}|_{D_+(f)} \to \mathcal{O}_f$ satisfying $\varphi_{fg} \circ \psi_f = \psi_g$ on $D_+(fg)$. The sections of $\mathcal{O}$ constructed there are compatible tuples of sections of the $\mathcal{O}_f$, and the φ_{fg} are ring isomorphisms, so these tuples form rings under componentwise operations and the ψ_f are ring isomorphisms. Composing ψ_f with Step 3 gives $(D_+(f), \mathcal{O}|_{D_+(f)}) \cong \operatorname{Spec} S_{(f)}$ with underlying map Φ_f. Each stalk of $\mathcal{O}$ is therefore a stalk of the structure sheaf of some $\operatorname{Spec} S_{(f)}$ (proposition 1.1.5), a local ring by proposition 2.2.22, so $(\operatorname{Proj} S, \mathcal{O})$ is a locally ringed space covered by affine schemes, a scheme (definition 2.2.9).

Step 6: sections and restrictions. Under ψ_f, $\mathcal{O}(D_+(f)) = \mathcal{O}_f(D_+(f)) = S_{(f)}$. By Step 3 with $q = fg$, the restriction $S_{(f)} \to \mathcal{O}_f(D_+(fg))$ followed by ε_{fg}^f is $c_{f,fg} = \rho_{f,fg}$, and $\varepsilon_{fg}^f \circ \psi_f$ agrees with the identification $\mathcal{O}(D_+(fg)) = S_{(fg)}$ given by ψ_{fg}, because $\psi_{fg} = \varphi_{f,fg} \circ \psi_f$ and $\varepsilon_{fg}^{fg} = c_{fg,fg} = \mathrm{id}$. So the restriction is $a/f^k \mapsto ag^k/(fg)^k$, completing the proof.

Definition 2.3.4: Projective Scheme

Let $S_\bullet$ be an $\mathbb{N}$-graded ring. The scheme of theorem 2.3.3 is denoted $\operatorname{Proj} S_\bullet$. For $a \in S_0$ the sections $a/1 \in S_{(f)} = \mathcal{O}(D_+(f))$ agree on the overlaps $D_+(fg)$ by the restriction formula of theorem 2.3.3, and the sets $D_+(f)$ cover $\operatorname{Proj} S_\bullet$, so they glue to a global section. This gives a ring map $S_0 \to \mathcal{O}(\operatorname{Proj} S_\bullet)$, which makes every $\mathcal{O}(W)$ an S_0-algebra with S_0-algebra restriction maps, and so makes $\operatorname{Proj} S_\bullet$ an S_0-scheme (definition 2.2.18). A *projective R-scheme* is an R-scheme isomorphic over $\operatorname{Spec} R$ to $\operatorname{Proj} S_\bullet$ for an $\mathbb{N}$-graded ring $S_\bullet$ with $S_0 = R$ that is generated as an R-algebra by finitely many elements of S_1. If $S = R[x_0, \dots, x_n]$ with the standard grading, then $\operatorname{Proj} S$, with this R-scheme structure, is the classical projective n-scheme or prototypical projective scheme over R, and is denoted $\mathbb{P}^n_R$ or $\mathbb{P}^n$ if unambiguous.

Finite generation enters only through the notion of a projective R-scheme: lemma 2.3.2 and theorem 2.3.3 hold for every $\mathbb{N}$-graded ring. By corollary 5.3.15(3), a scheme over $\operatorname{Spec} R$ is a projective R-scheme if and only if it admits a closed immersion into some $\mathbb{P}^n_R$ over $\operatorname{Spec} R$. The structure over $\operatorname{Spec} R$ is part of the data. With $\deg x_0 = 1$, the scheme $\operatorname{Proj} k[t][x_0] = D_+(x_0)$ is $\operatorname{Spec} k[t]$, and its structure morphism to $\operatorname{Spec} k[t]$ is an isomorphism. The same scheme $\mathbb{A}^1_k = \operatorname{Spec} k[t]$, made a $k[t]$-scheme through $k[t] \to k \to k[t]$, $t \mapsto 0$, has a structure morphism with image the origin, so the two $k[t]$-schemes are not isomorphic over $\operatorname{Spec} k[t]$.

We can naturally recover the notion of quasiprojective as open subschemes of projective schemes:

Definition 2.3.5: Quasiprojective Scheme

A *quasiprojective R-scheme* is an open subscheme of a projective R-scheme.

Notice that the scalars can be defined for any ring, which generalizes the result from classical algebraic geometry. If $R = k$ is an algebraically closed field, the closed points of $\mathbb{P}^n_k$ will behave like the traditional points of projective space defined in classical algebraic geometry.

Proposition 2.3.6: Global Sections of $\operatorname{Proj} S_\bullet$

Let $S_\bullet = R[x_0, \dots, x_n]$ with the standard grading, and let $X = \operatorname{Proj} S_\bullet$. Then:

$$\mathcal{O}_X(X) = S_0 = R$$

Hence $\mathbb{P}^n_R$ is not an affine scheme for $n \geq 1$: affineness would force $\mathbb{P}^n_R \cong \operatorname{Spec} R$, and the case $n = 1$ was refuted in example 2.2.57.

***Proof*:**

If $n = 0$, then $X = D_+(x_0) = \operatorname{Spec} S_{(x_0)}$ and a degree-0 fraction h/x_0^d with $h \in S_d = Rx_0^d$ is a scalar, so $\mathcal{O}_X(X) = S_{(x_0)} = R$. Assume $n \geq 1$.

The charts $U_i = D_+(x_i) \cong \operatorname{Spec} S_{(x_i)}$ for $i = 0, \dots, n$ cover X, with $\mathcal{O}_X(U_i) = S_{(x_i)}$ (theorem 2.3.3, lemma 2.3.2, proposition 2.2.1), and $U_i \cap U_j = D_+(x_i x_j) \cong \operatorname{Spec} S_{(x_i x_j)}$, since a prime avoids $x_i x_j$ exactly when it avoids both factors. By the sheaf axioms (proposition 1.1.9) a global section is a tuple $(t_i)_i$ with $t_i \in S_{(x_i)}$ such that t_i and t_j have the same image in $S_{(x_i x_j)}$ under the restriction maps, which are the localization maps constructed in the proof of theorem 2.3.3.

All these rings embed in the ring of Laurent polynomials. Localizing at $x_0 \cdots x_n$ inverts every

x_i, so

$$S_{x_0\cdots x_n} = R[x_0^{\pm 1}, \ldots, x_n^{\pm 1}]$$

a free R-module with basis the Laurent monomials $x^a = x_0^{a_0} \cdots x_n^{a_n}$, $a \in \mathbb{Z}^{n+1}$. The localization map $S_{x_i} \to S_{x_0\cdots x_n}$ is injective, since multiplication by a monomial is injective on S (compare coefficients), and its image is the R-span of the monomials x^a with $a_j \geq 0$ for all $j \neq i$. Likewise the image of $S_{x_i x_j}$ is the span of the monomials with $a_\ell \geq 0$ for all $\ell \neq i, j$, and under these identifications the restriction maps become inclusions of R-submodules. Taking degree-0 parts, a compatible tuple $(t_i)_i$ is a single Laurent polynomial

$$t \in \bigcap_{i=0}^{n} (S_{x_i})_0$$

If the monomial x^a occurs in t, then membership in S_{x_i} forces $a_j \geq 0$ for all $j \neq i$; since $n \geq 1$, every index j differs from some i, so $a_j \geq 0$ for every j. Hence $t \in S$ and t has degree 0, so $t \in S_0 = R$. Conversely every scalar in R restricts to a compatible tuple. Therefore $\mathcal{O}_X(X) = R$.

Recall from [22, Zariski Topology on Projective Space] that homogeneous polynomials of $k[x_0, \ldots, x_n]$ cut out closed sets from $\mathbb{P}^n_k$. This is because if $a \in k^{n+1}$ is a root of a homogeneous polynomial, so are all scalar multiples of a. In modern algebraic geometry, the global sections $\mathcal{O}_X(X)$ are regarded as the "functions" on a scheme X. However, $\mathcal{O}_{\mathbb{P}^n_R}(\mathbb{P}^n_R) = R$ by proposition 2.3.6, so the graded ring S that *generates* $\operatorname{Proj} S$ does not manifest as the structure sheaf.

How, then, should we interpret S? The answer is that each graded piece S_d gives rise to a sheaf $\mathcal{O}(d)$ on $\mathbb{P}^n_R$ called a *twisting sheaf* (or *Serre twist*). These are examples of *line bundles* (invertible sheaves), and the structure sheaf $\mathcal{O}_{\mathbb{P}^n_R}$ is the special case $\mathcal{O}(0)$. We shall show in proposition 5.3.10 that, for $n \geq 1$, the graded ring S can be recovered from these sheaves:

$$S_d \cong \Gamma(\mathbb{P}^n_R, \mathcal{O}(d))$$

This is significant because line bundles and their global sections control the geometry of morphisms *from* projective space: a morphism $\mathbb{P}^n_R \to \mathbb{P}^m_R$ is determined by a choice of $m+1$ global sections of a line bundle on $\mathbb{P}^n_R$ (with no common vanishing), a correspondence that will be made precise in section 5.3.3. In this way, the graded ring S – despite not being the ring of global functions – encodes the "embedding data" of projective space.

The reader ought to remember that if $f(x_{0,i}, \ldots, x_{n,i})$ is a homogeneous polynomial of degree d on $D(x_i)$, then when converting coordinates to $D(x_j)$ we get:

$$f(x_{0,i}, \ldots, x_{n,i}) = x_{j,i}^{\deg f}\, f(x_{0,j}, \ldots, x_{n,j})$$

Example 2.3.7: Projective Scheme Homogeneous Polynomial Cut

There are many examples in [22], and hence we shall just put a few as a refresher.

1. Take $C = V(x^2 + y^2 - z^2) \subseteq \mathbb{P}^2_k$ for an algebraically closed field k. Then if $z = 0$, we would get $[(0,0,0)] \in C$, however this point does not exist in $\mathbb{P}^n_k$. Hence consider $z \neq 0$. Then for each solution $[(a,b,c)] = [(\lambda a, \lambda b, \lambda c)] \in C$. In k^3, the solutions form a cone of points. Loosely visualizing $\mathbb{P}^2_k$ as a sphere (we should glue the antipodal points), we see that this resembles the image:

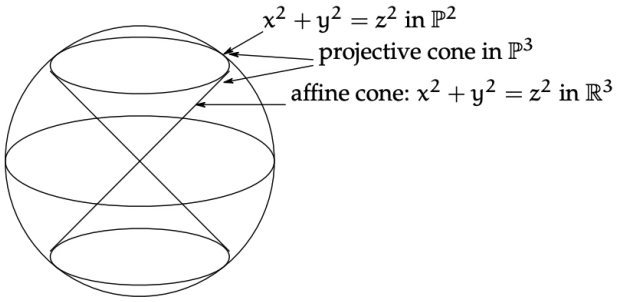


Figure 2.5: Visualization of $V(x^2 + y^2 - z^2)$

Such vanishing sets are called *vanishing cones*.

2. $\mathbb{P}^0_R = \operatorname{Proj} R[x_0] \cong \operatorname{Spec} R$, hence all 0-dimensional projective schemes are affine. Note then that $\mathbb{P}^0_k = \{\overline{0}\}$ since $\operatorname{Spec} k$ is a singleton; or this can be verified directly by noticing that the only non-irrelevant homogeneous prime ideal of $k[x]$ is (0), for all other homogeneous prime ideals are of the form (x^n) and hence contain S_+.

3. For an example that is not a hypersurface, take $V(wz - xy,\, x^2 - wy,\, y^2 - xz)$. This is called the *twisted cubic curve*. This ideal is prime, and the variety is isomorphic to $\mathbb{P}^1_k$ (hint: consider $k[w,x,y,z] \to k[s,t]$, $(w,x,y,z) \mapsto (s^3, s^2t, st^2, t^3)$).

4. When $R = k$ is a field, the definition of $\mathbb{P}^n_k$ as a scheme matches the classical definition of projective space as a quotient of $k^{n+1} \setminus \{0\}$ by $k^\times$. The points in the scheme correspond exactly to lines through the origin (plus the generic point). However, when R is not a field, $\mathbb{P}^n_R$ behaves like a "compact" object in a very powerful way. Consider the case where R is a Discrete Valuation Ring (DVR), for example $R = k[[t]]$ or $\mathbb{Z}_{(p)}$, with fraction field K. Consider a point $P \in \mathbb{P}^n_K$ (i.e., a point in projective space over the fraction field). We can write this point in coordinates $[f_0 : \cdots : f_n]$ where $f_i \in K$. We might ask: does this point "extend" to a point in $\mathbb{P}^n_R$? In other words, if we have a curve defined for $t \neq 0$, does it have a unique limit as $t \to 0$?

   The answer is yes. Since we are in projective space, we can scale coordinates by any non-zero element.

   (a) Let $v(f_i)$ be the valuation of each coordinate.

   (b) Multiply the entire tuple by t^{-m}, where $m = \min_i(v(f_i))$.

   (c) Now all coordinates are in R, and at least one coordinate is a unit (valuation 0).

   (d) We can now safely set $t = 0$ to find the unique limit point in the special fiber $\mathbb{P}^n_k \subset \mathbb{P}^n_R$.

   For example, if $R = k[[t]]$ and $P = [1/t^2 : 1/t] \in \mathbb{P}^1_K$, we scale by t^2 to get $[1 : t]$. Setting $t = 0$ gives the limit point $[1 : 0]$. This property – that any map from the punctured curve $\operatorname{Spec} K$ extends uniquely to the whole curve $\operatorname{Spec} R$ – is called the *Valuative Criterion for Properness*. It essentially states that projective schemes are "compact" (curves do not run off to infinity without hitting a limit). This implies that $\mathbb{P}^n_R$ is connected in a way affine schemes are not (an affine line with a missing origin cannot be patched!). See section 3.5.4 for more details.

5. In an affine scheme, two polynomials over a field cut out the same vanishing set if they differ by a scalar. This is not the case for homogeneous polynomials cutting out parts of $\mathbb{P}^n$: consider $x^2y = 0$ and $xy^2 = 0$. The problem is that these two are homogenizations of different polynomials that produce the same set. However, as we know from section 2.2.2, schemes remember multiplicity. This behaviour shall later be properly taken into account by saying that two polynomials shall cut out different *closed subschemes* (rather than the same *closed subset*). With this addendum, it shall once again be the case that two polynomials cut out the same closed subscheme if and only if they differ by a scalar multiple, as closed subschemes shall essentially correspond to polynomial cut-outs (as sets) that have sheaves remembering multiplicity information.

Proposition 2.3.8: Irreducibility of classical projective scheme

Let R be a domain. Then the scheme $\mathbb{P}^n_R$ is irreducible and is of finite type over R.

As irreducible spaces are connected, $\mathbb{P}^n_k$ is connected.

***Proof*:**
Since R is a domain, $S = R[x_0, \dots, x_n]$ is a domain, so the zero ideal (0) is a homogeneous prime not containing S_+. Hence $(0) \in \operatorname{Proj} S$ is a point. For each i, $x_i \notin (0)$, so $(0) \in D_+(x_i)$ for every i. Since $\{D_+(x_i)\}_{i=0}^n$ covers $\operatorname{Proj} S$, the generic point (0) lies in every basic open of the standard cover. As any nonempty open set contains some $D_+(x_i)$, it contains (0), so $\{(0)\}$ is dense and $\mathbb{P}^n_R$ is irreducible. Finite type over R follows from each chart $D_+(x_i) \cong \mathbb{A}^n_R = \operatorname{Spec} R[x_1, \dots, x_n]$ being of finite type over R, with finitely many charts.

To conclude this section, let us formally define the affine and projective cone:

Definition 2.3.9: Affine Cone

Let $S_\bullet$ be a graded ring. Then the affine cone of $\operatorname{Proj} S_\bullet$ is $\operatorname{Spec}(S_\bullet)$.

Note that $\operatorname{Spec}(S_\bullet)$ depends on $S_\bullet$, not just $\operatorname{Proj}(S_\bullet)$. Think of the affine cone as eliminating the points at infinity and adding the origin. In fact, given a field k, there is a natural map $\operatorname{Spec} S_\bullet \setminus V(S_+) \to \operatorname{Proj} S_\bullet$ (more generally, if $S_\bullet$ is a finitely generated graded ring over a base ring R, there is a natural morphism).

Definition 2.3.10: Projective Cone

Take $\operatorname{Spec} S_\bullet$. Then the projective cone or projective completion is $\operatorname{Proj}(S_\bullet[t])$ for some indeterminate t of degree 1.

Take for example $\operatorname{Proj}(k[x,y,z]/(z^2-x^2-y^2))$. Then the projective cone is $\operatorname{Proj}(k[x,y,z,t]/(z^2-x^2-y^2))$. The vanishing locus $V_+(t)$ returns the original projective scheme, and taking the complement (the distinguished open set $D_+(t)$) gives $\operatorname{Spec}(S_\bullet)$. This construction can be thought of as the opposite of the affine cone, where we attach the points at infinity of a conic curve.

The Non-Uniqueness of Graded Rings

Before moving on, we must highlight a fundamental difference between the Spec and Proj constructions. The spectrum functor provides a perfect anti-equivalence between commutative rings and affine schemes; if $\operatorname{Spec} A \cong \operatorname{Spec} B$ as schemes, then $A \cong B$ as rings.

The Proj construction completely fails this property. Two wildly different graded rings can produce the exact same projective scheme. For example, let $S_\bullet = k[x, y]$ and let $T_\bullet = k[x^2, xy, y^2] \cong k[u, v, w]/(uw - v^2)$. As standard rings, these are not isomorphic: $S_\bullet$ is a Unique Factorization Domain, while $T_\bullet$ is not (since $uw = v^2$ represents two distinct factorizations into irreducibles). Geometrically, their affine cones are drastically different: $\operatorname{Spec} S_\bullet$ is the smooth plane $\mathbb{A}^2_k$, whereas $\operatorname{Spec} T_\bullet$ is a quadric cone which is singular at the origin.

However, because Proj effectively removes the origin (by restricting to homogeneous prime ideals not containing S_+) and takes a quotient by the grading, these differences vanish: $\operatorname{Proj} S_\bullet \cong \operatorname{Proj} T_\bullet \cong \mathbb{P}^1_k$ (proven in theorem 3.1.37).

In general, the Proj construction only determines a graded ring "up to truncation at finite degrees". If two graded rings $S_\bullet$ and $T_\bullet$ share the exact same graded pieces for all degrees $d \geq N$ for some large integer N, they will yield isomorphic projective schemes. We will formalize exactly how to recover the "canonical" coordinate ring of a projective scheme using twisted sheaves in section 5.3.

Proj-like Functors for other schemes?

here

2.4 Properties of Schemes

In this section, we shall cover various different properties a scheme may have. In the process, we shall see if these properties can be checked affine-locally or stalk-locally Many of the topological properties inquired about affine schemes can be asked about for general schemes. For example, general schemes need not be quasicompact, while all affine schemes are quasicompact. Properties that have been explored already are:

1. (dis)connected: In the affine case, this corresponds to the existence of non-trivial idempotent elements, for if $e \in R$ is such an element then $R \cong eR \times (1 - e)R$, and so the spectrum is a disjoint union. For the general case of schemes, it is usually given by the disjoint union of known schemes.

2. (ir)reducible: In the affine case, this corresponds to the ring having *no zero divisors*, i.e. an integral domain. Proposition 2.4.20 shall characterize irreducible schemes[22].

   Recall that an equivalent conditions to irreducibility is that every non-empty open subset is dense (as the reader should verify).

3. quasicompact: Essentially all schemes we shall work with will be quasicompact (with the "trivial" exception being the infinite disjoint union of schemes). Most of the time, non-quasicompact schemes will come from some infinite gluing.

[22] with the reducedness condition insuring no "fuzz" in the sheaf

These are global topological properties, hence they cannot be checked affine- or stalk-locally (as the reader should verify with some counter-examples).

Finiteness Conditions

The most common types of rings in commutative algebra are Noetherian rings. The geometric equivalent is the following:

Definition 2.4.1: Noetherian Scheme

Let X be a scheme. If X can be covered by affine open sets where each ring R_i of $\operatorname{Spec}(R_i)$ is Noetherian, then X is said to be a *locally Noetherian Scheme*. If X is also quasicompact, then X is said to be a *Noetherian Scheme*.

Lemma 2.4.2: Noetherian Schemes are Noetherian Spaces

Let X be a scheme that is a finite union of affine opens $U_1, \dots, U_n$ with $U_i \cong \operatorname{Spec} R_i$ and each R_i Noetherian. Then every descending chain of closed subsets of X is eventually constant, that is, X is a Noetherian topological space. This applies to every Noetherian scheme and to $\operatorname{Spec} R$ for every Noetherian ring R. Every open subset of a Noetherian topological space is quasicompact.

***Proof*:**
Let $Z_1 \supseteq Z_2 \supseteq \cdots$ be closed subsets of X and fix i. The sets $Z_k \cap U_i$ form a descending chain of closed subsets of $U_i \cong \operatorname{Spec} R_i$, so the ideals $I(Z_k \cap U_i) \subseteq R_i$ form an ascending chain (proposition 2.1.6(2)), which is eventually constant because R_i is Noetherian. Since $V(I(W)) = W$ for closed W (proposition 2.1.6(6)), the chain $Z_k \cap U_i$ is eventually constant. There are finitely many i, so some N has $Z_k \cap U_i = Z_N \cap U_i$ for all i and all $k \geq N$, whence $Z_k = Z_N$ for $k \geq N$. A Noetherian scheme is covered by affine opens with Noetherian rings and is quasicompact (definition 2.4.1), so finitely many of them cover it, and $\operatorname{Spec} R$ is covered by itself.

Let W be an open subset of a Noetherian topological space X, covered by open sets W_i with $i \in I$. If no finitely many of them cover W, choose $i_1 \in I$ and inductively $i_{k+1} \in I$ with $W_{i_{k+1}} \not\subseteq W_{i_1} \cup \cdots \cup W_{i_k}$, which is possible because this finite union is not all of W and every point of W lies in some W_i. The complements of the sets $W_{i_1} \cup \cdots \cup W_{i_k}$ form a descending chain of closed subsets of X that is never constant, a contradiction. So W is quasicompact (definition 2.1.19), as claimed.

All coordinate rings are Noetherian, and hence we may think of varieties as examples of Noetherian schemes[23]. We can think of the Noetherian condition as insuring that the open affine do not have "difficult" infinite behavior. Most examples which involve non-Noetherian rings involve infinitely many variables (ex. $k[x_1, x_2, ...]$, $k[x, y, y/x, y/x^2, ...]$ and so forth). Examples of non-Noetherian topological spaces we shall encounter will be the fibered product of Noetherian rings (see section 2.2.10), and the special product topology put on the ring of adèles. In general, we shall see that there are many

[23] Once we introduce theorem 3.7.1 this will be made precise

results that do not require finite generation, and hence it is worth keeping track of it when this condition is necessary.

One important consequence of being Noetherian is the existence of finitely many irreducible components:

Lemma 2.4.3: Finitely Many Components

Let X be a Noetherian scheme. Then X has finitely many irreducible components.

***Proof*:**
By lemma 2.4.2, X is a Noetherian topological space. If $X = \emptyset$ there is nothing to prove. Otherwise proposition 2.1.25, applied to the closed set X, writes $X = Z_1 \cup \cdots \cup Z_n$ with each Z_i closed and irreducible. Let $W \subseteq X$ be irreducible. Then W is the union of its closed subsets $W \cap Z_i$, and by induction on n it lies in one of the Z_i: if $W \not\subseteq Z_1$, irreducibility of $W = (W \cap Z_1) \cup (W \cap (Z_2 \cup \cdots \cup Z_n))$ gives $W \subseteq Z_2 \cup \cdots \cup Z_n$. So every maximal irreducible subset of X, that is, every irreducible component, is one of $Z_1, \ldots, Z_n$, completing the proof.

This is not true if the ring is not Noetherian: the spectrum of the ring

$$\varinjlim_{n \to \infty} \frac{k[x_i \mid 1 \leq i \leq n]}{\prod_i^n x_i}$$

has infinitely many components, but it is a quasicompact scheme (as it is affine).

Let us next turn to an important property of quasicompact schemes. When working with affine schemes, we always have that there exists closed points corresponding to the maximal ideals of the ring. When working over general schemes, there may be no closed points[24]. Fortunately, all quasicompact scheme have closed points, and even better:

Proposition 2.4.4: Quasicompact Schemes and Closed Points

Let X be a quasicompact scheme. Then X contains closed points. Furthermore, the closure of any non-closed point contains closed points.

***Proof*:**
First, if X is affine, it certainly contains closed points as $\mathcal{O}_X(X) = R$ contain at least one maximal ideal. Furthermore, the closure of any non-closed point would contain maximal ideal as any prime ideal is contained in a maximal ideal.

Now, let X be some quasicompact scheme. Let $X = \bigcup_i^n U_i$ be a finite covering of affine open subsets. Take any closed point $\mathfrak{p}_1 \in U_1$. If it is closed in X we are done. If not, take its closure $\overline{\mathfrak{p}_1}$ in X. As $\mathfrak{p}_1$ is not closed in X, its closure contains more than $\mathfrak{p}_1$, call this point $\mathfrak{p}_2$, and without loss of generality say it is in U_2. If $\mathfrak{p}_2$ is closed in X, we are done. If not, take $\overline{\mathfrak{p}_2}$ in X. Its closure must contain a point $\mathfrak{p}_3$ that is not in U_2 or U_1 (as $\mathfrak{p}_3$ would be in the closure of $\mathfrak{p}_1$ or $\mathfrak{p}_2$ in U_1 and U_2 respectively). Continue this process until a point is closed, or if we get to $\mathfrak{p}_n \in U_n$, then the closure of this point cannot contain any other point in $U_1, \ldots, U_n$, making $\mathfrak{p}_n$ closed and completing the proof.

[24] see https://math.stanford.edu/~vakil/files/schwede03.pdf

Note that $\mathfrak{p}_n$ is in the closure of $\mathfrak{p}_1$ in X : by a topological argument it is easy to show that:

$$\overline{\mathfrak{p}_1} \supseteq \overline{\mathfrak{p}_2} \supseteq \cdots \supseteq \overline{\mathfrak{p}_n} = \mathfrak{p}_n$$

and hence the closure of any non-closed point contains a closed point.

Note that this does not mean that the closed points are dense, that is it does not imply that the closure of the collection of closed point is the entire space. Such schemes are called Jacobson schemes and are related to Jacobson rings and Jacobson radicals (see [10]). Once we introduce (abstract) varieties, we shall see that the closed points are dense for varieties. The following gives a good set of (affine) schemes whose closed points are dense:

Lemma 2.4.5: Dense Closed Points

Let A be a finitely generated k-algebra. Then the closed points of $\operatorname{Spec} A$ are dense

***Proof*:**
It suffices show that for each $f \in A$ where $D(f) \neq \emptyset$ (i.e. f is not nilpotent, see lemma 2.2.6), $\operatorname{Spec}(A_f) \cong D(f)$, and A_f certainly contains maximal ideals, there is a maximal ideal $\mathfrak{p}_f \subseteq A_f$ which corresponds to a (at least) prime ideal $\mathfrak{p} \subseteq A$ not containing f. If we show that $A/\mathfrak{p}$ is a field, we are done. By Zariski's lemma, F is a finitely generated k-algebra if it is a finitely generated k-module, hence it suffice to show that $A/\mathfrak{p}$ is a finitely generated k-vector space. As $(A/\mathfrak{p})_f \cong A_f/\mathfrak{p}_f$, if we show that $A_f/\mathfrak{p}_f$ is a finite dimensional k-vector space, we can lift to a finite basis for $A/\mathfrak{p}$.

Now, as $A_f = A[1/f]$, it is a finitely generated k-algebra and $\mathfrak{p}_f$ is maximal, $A_f/\mathfrak{p}_f$ is a finite dimensional k-vector space by Zariski. But then we get the desired conclusion

We can also show it in the following more "straightforward" with a more subtle application of Nullstellensatz. For the sake of contradiction, suppose $D(f)$ has no maximal ideals. Then for each maximal ideal $\mathfrak{m} \subseteq A$, $f \in \mathfrak{m}$. Thus:

$$f \in \bigcap_{\mathfrak{m} \subseteq A} \mathfrak{m} \overset{!}{=} \sqrt{(0)}$$

where $\overset{!}{=}$ comes as a consequence of the Nullstellensatz (key here is the finitely generation as a k-algebra). But then f is nilpotent, and so $D(f) = \emptyset$. Hence, any nonempty set contains a closed point, completing the proof.

Recall in proposition 2.3.18 it was mentioned that invertibility can be determined if a function is non-zero at every point, however a function being zero at all points requires more care to study. Part of this is because this is true for quasicompact schemes

Proposition 2.4.6: Quasicompact Schemes and Vanishing Function

Let X be a quasicompact scheme. Then if $f \in \mathcal{O}_X(X)$ vanishes at all points in X, then there exists an n such that $f^n = 0$.

***Proof*:**
Say $f([\mathfrak{p}]) = 0$ so that $f \in \mathfrak{p}$ for all $\mathfrak{p}$. Then on $U_i \cong \operatorname{Spec} R_i \subseteq X, f \in \sqrt{R_i}$, implying $f^{n_i} = 0$ for some n. Cover X with these open sets, and by Quasicompactness we only require finitely many such sets, where in each set we have an f_i and n_i such that $f_i^{n_i} = 0$. As X is a scheme, on intersection $U_i \cap U_j$ we have $g_{i,j} = f_i|_{U_i \cap U_j} = f_j|_{U_i \cap U_j}$ and as ring homomorphisms must map 0 to 0, $g_{i,j}^{\max n_i, n_j} = 0$. Take $n = \max_i n_i$ so that for each intersection $U_i \cap U_j$:

$$f_i|_{U_i \cap U_j} = f_j|_{U_i \cap U_j} = g_{i,j} \quad g_{i,j}^n = 0$$

Now, as these agree on intersections and cover X, by definition there must exist a function $f \in \mathcal{O}_X(X)$ such that $f|_{U_i} = f_i$. Then as $\operatorname{res}_{X,U_i}(f^n) = \operatorname{res}_{X,U_i}(f)^n = f_i^n f_i^{n_i n'} = 0 f^{n'} = 0$, we have:

$$(f^n)|_{U_i} = \left(f|_{U_i}\right)^n = 0$$

which implies $(f^n)|_{U_i} = 0 \in R_i$. Then by guability, it must be that $f^n = 0$, completing the proof.

This is not necessarily true for a general scheme, namely since we can have nilpotence of every degree:

Example 2.4.7: Non-vanishing Function On Scheme
Take

$$X = \coprod_{n \geq 1} \operatorname{Spec}\left(k[x]/\left(x^n\right)\right)$$

Then the global function ϵ is zero every-where, namely there is only one point to evaluate to, but essentially by definition there cannot exist an n such that $\epsilon^n = 0$.

Lemma 2.4.8: Projective Scheme is Noetherian

Let $X = \mathbb{P}^n_{\mathbb{Z}}$ or $\mathbb{P}^n_k$. Then X is Noetherian.

***Proof*:**
Let $R = \mathbb{Z}$ or $R = k$, $S = R[x_0, \ldots, x_n]$, and $X = \operatorname{Proj} S$. The charts $D_+(x_i) \cong \operatorname{Spec} S_{(x_i)}$ for $i = 0, \ldots, n$ cover X (theorem 2.3.3, lemma 2.3.2): a point of X is a homogeneous prime $\mathfrak{p}$ with $S_+ = (x_0, \ldots, x_n) \not\subseteq \mathfrak{p}$, so some $x_i \notin \mathfrak{p}$. Each chart ring is a polynomial ring in n variables over R:

$$S_{(x_i)} = R\left[\frac{x_0}{x_i}, \ldots, \frac{x_n}{x_i}\right] \cong R[y_1, \ldots, y_n]$$

The equality holds because a degree-0 fraction h/x_i^d with h a degree-d monomial is a product of d of the fractions x_j/x_i. The isomorphism holds because the fractions x_j/x_i for $j \neq i$ are algebraically independent over R: a polynomial relation among them, multiplied by a large power of x_i, becomes a vanishing R-linear combination of distinct monomials in S, whose coefficients must vanish. The base rings $\mathbb{Z}$ (a principal ideal domain) and k (a field) are Noetherian, so each

$S_{(x_i)}$ is Noetherian by [10, HBT on multi-variable Polynomials], and X is locally Noetherian (definition 2.4.1).

Each chart is affine, hence quasicompact (section 2.1.3), and a finite union of quasicompact open sets is quasicompact: an open cover of the union admits a finite subcover of each piece, and the union of these finitely many subcovers is a finite subcover. So X is quasicompact, and by definition 2.4.1 it is Noetherian.

To conclude this section, we shall define one more type of scheme here due to the fact that all affine and projective schemes have this property, however it shall not be of great use to us until chapter 6. The following establishes some finiteness conditions on the intersection of important open subsets:

Definition 2.4.9: Quasi-separated

Let X be a topological space. Then X is *quasiseparated* if the intersection of any two quasicompact open subsets is quasicompact.

Proposition 2.4.10: Characterizing Quasi-separated Schemes

Let X be scheme. Then X is quasi-separated if and only if the intersection of any two affine open subsets is a finite union of affine open subsets

***Proof*:**

Step 1: three facts. First, an affine scheme $\operatorname{Spec}(T)$ is quasicompact (lemma 2.1.20). Second, every open subset of a scheme X is a union of affine open subsets: the distinguished opens of the members of an affine open cover form a basis (proposition 2.2.17), and each of them is affine by proposition 2.2.13. Third, a finite union of quasicompact subsets is quasicompact (lemma 2.1.20).

Step 2: the forward direction. Let X be quasi-separated and $U, V \subseteq X$ affine open subsets. Both are quasicompact by Step 1, so $U \cap V$ is quasicompact by definition 2.4.9. It is open, hence a union of affine open subsets, and finitely many of them cover it.

Step 3: the reverse direction. Suppose the intersection of any two affine open subsets is a finite union of affine open subsets, and let $A, B \subseteq X$ be quasicompact open subsets. Each is a union of affine open subsets, so $A = A_1 \cup \cdots \cup A_p$ and $B = B_1 \cup \cdots \cup B_q$ with all A_k, B_l affine open. Then

$$A \cap B = \bigcup_{k=1}^{p} \bigcup_{l=1}^{q} A_k \cap B_l$$

and each $A_k \cap B_l$ is a finite union of affine, hence quasicompact, open subsets. So $A \cap B$ is a finite union of quasicompact subsets, hence quasicompact, and X is quasi-separated, as we sought to show.

Proposition 2.4.11: Affine Scheme and Quasiseparatedned

Let $X = \operatorname{Spec} R$ be an affine scheme. Then X is quasiseparated.

***Proof*:**
Let $A, B \subseteq X$ be quasicompact open subsets. The distinguished opens form a basis (corollary 2.1.13), so A and B are unions of distinguished opens, and by quasicompactness $A = D(f_1) \cup \cdots \cup D(f_p)$ and $B = D(g_1) \cup \cdots \cup D(g_q)$. Then

$$A \cap B = \bigcup_{k=1}^{p} \bigcup_{l=1}^{q} D(f_k g_l)$$

by proposition 2.1.12. Each $D(f_k g_l)$ is homeomorphic to $\operatorname{Spec}(R_{f_k g_l})$, hence quasicompact, and a finite union of quasicompact sets is quasicompact (lemma 2.1.20). So $A \cap B$ is quasicompact and X is quasiseparated (definition 2.4.9), as claimed.

A large family of schemes that are quasi-separated and quasicompact are projective R-schemes, and as we shall see many schemes we shall be working with can be embedded into a projective R-scheme, and hence these two finiteness properties will be very common. There do exist non quasi-separated schemes:

Example 2.4.12: Non-quasiseparated Scheme
Let $X = \operatorname{Spec}(k[x_1, x_2, \dots])$ and let $U = X \backslash [\mathfrak{m}]$ where $\mathfrak{m}$ is the maximal ideal $(x_1, x_2, \dots)$. Take two copies of X and glue along U to create a double-origin scheme, namely the infinite version of the double-origin scheme seen in example 2.15. Show this is not quasiseparated.

Lemma 2.4.13: Projective Scheme Quasicompact And Quasiseperated

Let X be a projective R-scheme (definition 2.3.4). Then X is quasicompact and quasiseparated

Note that R need not be Noetherian, hence any projective $\mathbb{P}^n_R$ scheme is quasicompact.

***Proof*:**
Both properties are preserved by homeomorphisms, so we may take $X = \operatorname{Proj} S_\bullet$ with $S_0 = R$ and $S_\bullet$ generated as an R-algebra by finitely many elements $f_1, \dots, f_m \in S_1$, which are homogeneous of positive degree.

Step 1: $X = D_+(f_1) \cup \cdots \cup D_+(f_m)$. A homogeneous element of degree $d > 0$ is the degree-d part of a polynomial expression in the f_i with coefficients in R, hence an R-linear combination of monomials $f_1^{c_1} \cdots f_m^{c_m}$ of degree d, and each such monomial has some $c_i \geq 1$. So every homogeneous element of positive degree lies in the ideal $(f_1, \dots, f_m)$, and a homogeneous prime containing every f_i contains S_+. Hence each $[\mathfrak{p}] \in X$ has $f_i \notin \mathfrak{p}$ for some i.

Step 2: basic opens. For homogeneous g, h of positive degree, primality gives $D_+(g) \cap D_+(h) = D_+(gh)$. Each $D_+(g)$ is homeomorphic to $\operatorname{Spec}(S_{(g)})$ by lemma 2.3.2, hence quasicompact, and a finite union of quasicompact subsets is quasicompact (lemma 2.1.20).

Step 3: conclusion. By Steps 1 and 2, X is quasicompact. The sets $D_+(g)$ form a basis of the topology of X, by the definition of the Zariski topology on $\operatorname{Proj} S$. If $A, B \subseteq X$ are quasicompact open subsets, then $A = D_+(g_1) \cup \cdots \cup D_+(g_p)$ and $B = D_+(h_1) \cup \cdots \cup D_+(h_q)$ for homogeneous

g_k, h_l of positive degree, so

$$A \cap B = \bigcup_{k=1}^{p} \bigcup_{l=1}^{q} D_+(g_k h_l)$$

is a finite union of quasicompact sets by Step 2, hence quasicompact. So X is quasiseparated (definition 2.4.9), completing the proof.

Exercise 2.4.1

1. Let X be a scheme. Show that X is a locally Noetherian scheme if and only if every open affine subset corresponds to a Noetherian ring.
2. Show that $\prod_i^\infty \operatorname{Spec} R_i$ properly embeds in $\operatorname{Spec}(\prod_i^\infty R_i)$. More precisely, the right hand side is (significantly) larger than the left hand side.

2.4.1 Reducedness and Integrality

The next important property that general rings have that coordinate rings for varieties do not have is the presence of zero-divisors and nilpotence. A ring with zero-divisors can be decomposed into a product to two rings and corresponds to a function vanishing on separate irreducible components (there must be more than 1), while a ring with nilpotence have effect on the sheaf of a ring as discussed in section 2.2.2. A ring with no nilpotent elements is said to be *reduced* (note that it may have zero-divisors[25]). A scheme that has this property of being reduced for every ring of sections is given a name:

Definition 2.4.14: Reduced Scheme

Let X be a scheme. Then X is a *reduced scheme* if $\mathcal{O}_X(U)$ is reduced for every open set U of X

Naturally, on a reduced scheme by proposition 2.4.6 a global section that evaluates to zero at every point will be the zero-section.

Proposition 2.4.15: Reducedness Is Stalk-local

Let X be a scheme. Then X is reduced if and only if non of the stalks have nonzero nilpotents. As a consequence, if $f, g \in \mathcal{O}_X(U)$ are two functions and agree on all points, then $f = g$

***Proof*:**
if X is a reduced scheme, then the localization will preserve reducedness, and hence each stalk is reduced. For the converse, we can show the contrapositive and assume there exists a $f \in \mathcal{O}_X(U)$ such that $f^n = 0$. Note that f cannot be a unit (as the reader could verify). Then some affine open subset $\operatorname{Spec} R \subseteq U$ we have $f|_U^n = f^n|_U = 0|_U$, showing that f is still nilpotent in $\mathcal{O}_X(\operatorname{Spec} R)$. Then it is a non-unit element in $\operatorname{Spec} R$, and hence belongs in some maximal ideal $f \in \mathfrak{m}$. Then localizing at $[\mathfrak{m}]$, f remains to be a nilpotent element, showing that not all stalks are reduced.

[25] if D is the set of all zero-divisors of a reduced ring, then it is the union of all minimal ideals

For the next assertion, this will be a simpler version of proposition 2.4.6 as we do not need to keep track of any exponents, and hence the Quasicompactness condition is not needed! Assume $f, g \in \mathcal{O}_X(U)$ such that $f(x) = g(x)$ for all $x \in U$. We want to show that $f = g$. Take $h = f - g$. Now, by definition $h(x) = f(x) - g(x) = 0$, so h must be contained in every prime ideal on every affine open subset of U. As $\mathcal{O}_X(U)$ is reduced, $\sqrt{(0)} = (0)$ on each affine open, and so $h|_{\operatorname{Spec} R} = 0$ for each element in the affine open covering, which by the gluging gives us $h = 0$, completing the proof.

Lemma 2.4.16: Reducedness For Affine And Projective Schemes

Let R be a reduced ring. Then $\operatorname{Spec}(R)$ is reduced. Hence, $\mathbb{A}^n$ and $\mathbb{P}^n$ are reduced.

***Proof*:**
The stalks of $\operatorname{Spec}(R)$ are the localizations $R_{\mathfrak{p}}$, and a localization of a reduced ring is reduced: if $(r/s)^n = 0$ in $R_{\mathfrak{p}}$, then $tr^n = 0$ for some $t \notin \mathfrak{p}$, so $(tr)^n = t^{n-1}(tr^n) = 0$, so $tr = 0$ since R is reduced, so $r/s = 0$ in $R_{\mathfrak{p}}$. By proposition 2.4.15, $\operatorname{Spec}(R)$ is reduced.

A polynomial ring over a reduced ring is reduced. Let $0 \neq f \in R[x]$ with top-degree term rx^d, $r \neq 0$. The coefficient of x^{nd} in f^n is r^n, which is nonzero since R is reduced, so $f^n \neq 0$ and f is not nilpotent. Iterating over the variables, $R[x_1, \dots, x_n]$ is reduced, so $\mathbb{A}^n_R = \operatorname{Spec} R[x_1, \dots, x_n]$ is reduced whenever R is; in particular this covers $\mathbb{A}^n_k$ and $\mathbb{A}^n_{\mathbb{Z}}$.

For $\mathbb{P}^n_R$ with R reduced: it is covered by the charts $D_+(x_i) \cong \operatorname{Spec} S_{(x_i)}$ for $S = R[x_0, \dots, x_n]$ (theorem 2.3.3, lemma 2.3.2), and $S_{(x_i)}$ is a subring of the localization S_{x_i} of the reduced ring S, hence reduced, since subrings and localizations of reduced rings are reduced by the arguments above. Every stalk of $\mathbb{P}^n_R$ is a stalk of one of these charts, hence has no nonzero nilpotents, so $\mathbb{P}^n_R$ is reduced by proposition 2.4.15.

Proposition 2.4.17: Reducedness On Quasicompact Schemes

Let X be a quasicompact scheme. Then X is reduced if and only if it is reduced at closed points

***Proof*:**
Assume it is reduced at all closed points. As X is quasi-compact, the closure of each non-closed point contains a closed point; say $Y \in X$ is a non-closed and $x \in X$ is a closed point contained in its closure. Then $\mathcal{O}_{X,Y}$ is the localization of $\mathcal{O}_{X,x}$. As localization preserves reducedness, $\mathcal{O}_{X,Y}$ must be reduced, completing the proof.

The above showed that reducedness for quasicompact schemes is a "closed-point" local condition. It is not an open set local condition. Furthermore, it is not enough to check the global sections:

Example 2.4.18: Reducedness Edge Case

1. The scheme given by:

$$X = \operatorname{Spec} \frac{\mathbb{C}[x, y_1, y_2, \dots]}{(y_1^2, y_2^2, \dots, (x-1)y - 1, (x-2)y_2, \dots)}$$

is nonreduced at the closed points corresponding to the positive integers. The complement of this set is not Zariski open.
The key is that if X is also locally Noetherian, then reducedness is a locally open condition.

2. Another mistake that is easy to make: let X be scheme where the global sections is reduced. This does not imply X is reduced (take the scheme cut out by $x^2 = 0$ in $\mathbb{P}^2_k$; note that the global section will be k, but X will be non-reduced).

One of the most important type of reduced ring is an integral domain. If a scheme is an integral domain for each section, it is given a name:

Definition 2.4.19: Integral Scheme

Let X be a scheme. Then X is *integral* if it is nonempty and $\mathcal{O}_X(U)$ is an integral domain for every nonempty open subset $U \subseteq X$.

Recall that the product of integral domain is not an integral domain, or geometrically, the disjoint union of integral scheme need not be integral. This global behavior cannot be captured through stalks, as the reader should verify, and hence integrality is not stalk-local. Integral schemes can be thought of as schemes that are in "one piece" and have "no fuzz":

Proposition 2.4.20: Integral, then Irreducible and Reduced

Let X be a scheme. Then if X is an integral scheme, it is irreducible and reduced

It is in fact an if and only if, which we shall show in proposition 2.4.26.

***Proof*:**
If X is integral, it is certainly reduced as each ring of local sections is integral (and hence reduced). To show it is irreducible, assume $X = X_1 \cup X_2$, X_1, X_2 closed. For the sake of contradiction, let's say these sets do not contain each other. Then there exists open subsets $U \subseteq X_1, V \subseteq X_2$ such that $U \cap V = \emptyset$. Indeed, given $X_1 \not\subseteq X_2$, then here exists a point $p \in X_1 \setminus X_2$. Let

$$U = (X \setminus X_2) \cap X_1$$

Then U is open in X_1, and hence by construction in X. Similarly, define $V = (X \setminus X_1) \cap X_2$. Then $U \cap V = \emptyset$.

Then, since U and V are disjoint opens, the sheaf axioms give:

$$\mathcal{O}_X(U \coprod V) \cong \mathcal{O}_X(U) \times \mathcal{O}_X(V)$$

But then $\mathcal{O}_X(U \coprod V)$ certainly contains zero-divisors, and hence cannot be integral.

An important property that integral domains is the ability to localize to obtain a fractional field. Localizing at an integral schemes generic point shall give us a "global" function field in which all local sections can embed, similarity to varieties! We first prove that an integral scheme has a unique generic point. Note the proof can be done if we replace integrality with irreducibility, that is we do not need every ring of local sections to be integral.

Lemma 2.4.21: Irreducible Schemes and Generic Points

Let X be an integral scheme. Then for every irreducible closed subset $Z \subseteq X$, there exists a unique generic point η as $\overline{\eta} = Z$. In particular, as X is irreducible, then there exists a point η such that $\overline{\eta} = X$.

***Proof*:**

Step 1: existence. The set Z is nonempty (definition 2.1.22), so some affine open $U \subseteq X$ meets it. Fix a homeomorphism of U with $\operatorname{Spec}(R)$ underlying an isomorphism of definition 2.2.9, and put $W = Z \cap U$. By lemma 2.1.24(3), W is dense in Z, and W is irreducible: a nonempty relatively open subset of W is relatively open in Z, hence dense in Z and so in W, and lemma 2.1.24(3) applies to W. Regard W as an irreducible closed subset of $\operatorname{Spec}(R)$. By proposition 2.1.23, $\mathfrak{p} = I(W)$ is prime and the closure of $[\mathfrak{p}]$ in U is $V(\mathfrak{p}) = W$. Let $\eta \in W$ correspond to $[\mathfrak{p}]$. So the closure $\overline{\{\eta\}}$ in X meets U in W. It is contained in the closed set Z and contains W, hence contains the closure of W in X, which is Z because W is dense in the closed set Z. Hence $\overline{\{\eta\}} = Z$.

Step 2: uniqueness. Let $\eta_1, \eta_2 \in X$ with $\overline{\{\eta_1\}} = \overline{\{\eta_2\}} = Z$, and let $U \cong \operatorname{Spec}(R)$ be an affine open meeting Z. If $\eta_k \notin U$, then η_k lies in the closed set $X \setminus U$, so $Z = \overline{\{\eta_k\}} \subseteq X \setminus U$, which contradicts $U \cap Z \neq \emptyset$. So $\eta_1, \eta_2 \in U$ correspond to primes $\mathfrak{p}_1, \mathfrak{p}_2$ of R, and the closure of each $\{\eta_k\}$ in U is $Z \cap U$. The closure of $\{[\mathfrak{p}_k]\}$ in $\operatorname{Spec}(R)$ is $V(\mathfrak{p}_k)$, so $V(\mathfrak{p}_1) = V(\mathfrak{p}_2)$, and proposition 2.1.6(6) gives $\mathfrak{p}_1 = I(V(\mathfrak{p}_1)) = I(V(\mathfrak{p}_2)) = \mathfrak{p}_2$, primes being radical. Hence $\eta_1 = \eta_2$.

Step 3: the case $Z = X$. An integral scheme is irreducible by proposition 2.4.20, and X is closed in itself, so Steps 1 and 2 give a unique η with $\overline{\{\eta\}} = X$, as we sought to show.

This is a purely topological fact, and hence can also be proven using only irreducibility.

The existence was given by reducing to an affine open and "using" its generic point. Using this, we can quickly calculate the stalk at the generic point of an integral scheme. We give this stalk a name:

Definition 2.4.22: Rational Field

Let X be an integral scheme. Then the function field is the collection

$$\mathcal{O}_{X,\eta} = R_{(0)} = \operatorname{Frac}(R)$$

for any R given by $\operatorname{Spec}(R) \subseteq X$, and it is often denoted $\kappa(X)$. If X is a k-scheme, then it is denoted $k(X)$.

Thus, integral schemes have a similar property to that of classical affine and projective varieties, namely that their sections over different open sets can be considered as subsets of a single ring! Gluing is easy as we can see that two elements are glued if and only if they are the same element of $\kappa(X)$.

Proposition 2.4.23: Integral Scheme: Restriction Maps and Function Field

Let X be a scheme that is irreducible and reduced. If $U \subseteq V$ where $V \neq \emptyset$, then

$$\text{res}_{V,U} : \mathcal{O}_X(V) \hookrightarrow \mathcal{O}_X(U)$$

is an inclusion.

***Proof*:**
As X is reduced, its sections are all determined by their value on the points, and hence can be treated as function (see example 2.4.24 on why this proof fails otherwise). Take $U \subseteq V \subseteq X$, $f, g \in \mathcal{O}_X(U)$ and assume $f|_U = g|_U$. We want to show that $f|_V = g|_V$. Define:

$$Z = \{x \in V \mid f(x) = g(x)\}$$

Z is closed in V as it is the zero set of $f - g$. By definition $U \subseteq Z$, $Z \subseteq V$, and U is open in V. As X is integral, the open subset V is irreducible. But then U is *dense* in V. As Z is closed in V and contains U, it must be that $Z = V$, but then

$$f|_V = g|_V$$

as we sought to show

Note that we require *both* the irreducibility and the reducedness, having *only* irreducibility is not enough

Example 2.4.24: Non-injective Non-reduced Scheme

Let $R = \frac{k[x,y]}{(xy,y^2)}$ so that $\operatorname{Spec} R$ is the x axis there is a bit of fluff in direction of the y-axis at the point 0. Then given $D(x) \subseteq \operatorname{Spec} R$, the map $\text{res} : \mathcal{O}_{\operatorname{Spec} R}(\operatorname{Spec} R) \to \mathcal{O}_{\operatorname{Spec} R}(D(x))$ maps $y \mapsto 0$ as

$$y \mapsto \frac{y}{1} = \frac{xy}{x} = \frac{0}{x} = 0$$

Corollary 2.4.25: Integral Scheme: Generic Point

Let X be an integral scheme. Then there exists a generic point η such that given any choice of $\operatorname{Spec}(R) \subseteq X$, there is always an inclusion:

$$\mathcal{O}_X(U) \hookrightarrow \mathcal{O}_{X,\eta} \cong \kappa(R)$$

***Proof*:**
By definition:

$$\mathcal{O}_{X,\eta} = \varinjlim_{\eta \in U} \mathcal{O}_X(U)$$

and as we saw lemma 2.4.21 the generic point is in each open set. By proposition 2.4.23, each map in the commutative diagram are injective, hence the map $\mathcal{O}_X(U)$ is injective into $\mathcal{O}_{X,\eta}$.

Proposition 2.4.26: Irreducible and Reduced, then Integral

Let X be a scheme. If X is irreducible and reduced, it is integral

***Proof*:**
Let X be irreducible and reduced. Then any open subscheme is irreducible and reduced (the irreducibility is proved topologically, and the reducedness can be deduced by the stalks). In particular, any affine open subscheme $\operatorname{Spec} R$ must be irreducible and reduced. Certainly R must then be reduced, hence to show that $\operatorname{Spec} R$ is integral consider $xy \in R$ such that $xy = 0$. Then

$$\operatorname{Spec} R = V(0) = V(xy) = V(x) \cup V(y)$$

But that contradicts irreducedness. Hence, all ring of sections of affine open subsets are integral. For general open $U \subseteq X$, we must have an affine open $\operatorname{Spec} R \subseteq U$, and hence a restriction $\mathcal{O}_X(U) \to \mathcal{O}_X(\operatorname{Spec} R)$. By proposition 2.4.23, this map is injective, and hence $\mathcal{O}_X(U)$ is a subring of an integral domain, making it an integral domain. Hence all rings of local sections are integral domains, completing the proof.

If we have the added condition of being Noetherian, then we only need to check that it is connected and that all its stalks are integral domains:

Proposition 2.4.27: Characterizing Integrality in Noetherian Schemes

Let X be a Noetherian Scheme. Then X is integral if and only if X is nonempty, connected, and all stalks $\mathcal{O}_{X,\mathfrak{p}}$ are integral domains.

Intuitively, the stalk at the intersection of irreducible components will "remember" the generic points of the respective irreducible component.

***Proof*:**
By proposition 2.4.20 and proposition 2.4.26, X is integral if and only if it is reduced and irreducible. Then X is certainly non-empty, connected, and the stalks are integral (the localization of integral domains is integral).

Conversely, assume X is nonempty, connected, with integral stalks. First of all, as the stalks are integral, they are reduced. We must show that X is irreducible. As X is Noetherian, by lemma 2.4.3 X can be decomposed into finitely many irreducible components:

$$X = X_1 \cup X_2 \cup \cdots \cup X_n$$

As X is connected, for each i, j we must have $X_i \cap X_j \neq \emptyset$. Take an affine open subset $\operatorname{Spec} R \subseteq X$ and:

$$\operatorname{Spec} R = \bigcup_i^n (X_i \cap \operatorname{Spec} R)$$

Then as each X_i is irreducible, it is a topological argument to see that the sets $(X_i \cap \operatorname{Spec} R)$ form an irreducible decomposition of $\operatorname{Spec} R$. Hence, they must each have a generic point that corresponds to a minimal prime ideal. Hence, if the X_i are distinct, R_i must contains at least two distinct minimal prime ideals. Then if $[\mathfrak{p}] \in X_i \cap X_j$, $\mathcal{O}_{X,[\mathfrak{p}]} = (R_i)_{\mathfrak{p}}$ contains two distinct

minimal prime ideals. But integral domains could only contain a single minimal prime ideal (namely the one associated to the generic point $[(0)]$), and so this stalk cannot be integral. Thus, it must be that $X_i = X_j$ for all i, j, showing that X is irreducible and completing the proof.

Exercise 2.4.2

1. Show that every point $x \in X$ in a topological space must be contains in an irreducible component (you will use Zorn's Lemma).
2. Show that connected components (maximal connected subset) are closed. Show that the connected components of Spec $(\prod_i^\infty \mathbb{F}_2)$ are not open.
3. Show that $D(f)$ is empty if and only if f is nilpotent. Show that if X is an integral scheme, then $D(f)$ is always nonempty and contains the generic point.
4. If $f \in R$ is nilpotent, then R_f is the zero-ring.
5. Let X be an irreducible scheme. Show that there is a unique generic point η such that $\overline{\eta} = X$ (note that lack of reducedness). More generally, show that if $Z \subseteq X$ is a (nonempty) irreducible subset, there is a unique generic point η such that $\overline{\eta} = Z$
6. Let X be a reduced scheme and $Y \hookrightarrow X$ a closed immersion where $\iota(Y) = X$. Then ι is an isomorphism.

2.4.2 Normality and Factoriality

Recall that a normal ring R is an integral domain that is integrally closed in its field of fractions, $\overline{R}^{\mathrm{Frac}(R)} = R$. Due to their importance in number theory, we shall take the time to develop the theory of Schemes for which all the rings of sections are normal. These shall correspond to schemes that have some nicer behaviour for their singularities. It can be thought of as a condition weaker than smoothness (smoothness shall be called regular in section 2.7) as normality shall still allow for singularities, however these singularities shall be more tame. In particular, the reader can intuitively have the following geometric picture in mind for subschemes of Normal Schemes:

1. In codimension 1, all singularities are resolved, meaning it is "smooth" in codim 1
2. In codimension ≥ 2, being normal means being nice enough so that rational functions defined on the compliment of the singularities can be extended to the singularity (i.e. Algebraic Hartogs's lemma applies)

Definition 2.4.28: Normal Scheme

Let X be a scheme. Then X is said to be *normal* if every stalk $\mathcal{O}_{X,\mathfrak{p}}$ is a normal domain.

Proposition 2.4.29: Normal Scheme Stalk Local

Let X be a scheme. Then X is a normal scheme if and only if $\mathcal{O}_{X,\mathfrak{p}}$ is normal for each $\mathfrak{p} \in X$. If X is integral, this holds if and only if $\mathcal{O}_X(U)$ is integrally closed for every nonempty affine open $U \subseteq X$. In particular, for an integral domain A, the scheme $\operatorname{Spec}(A)$ is normal if and only if A is integrally closed.

***Proof*:**
The first assertion is definition 2.4.28, a normal ring being an integrally closed domain. Let X be integral and $U \subseteq X$ a nonempty affine open, with an isomorphism of locally ringed spaces $U \cong \operatorname{Spec}(A)$. Then $A \cong \mathcal{O}_X(U)$ is a domain (definition 2.4.19), and for $x \in U$ corresponding to $\mathfrak{p} \in \operatorname{Spec}(A)$ we have $\mathcal{O}_{X,x} \cong A_{\mathfrak{p}}$ by proposition 2.2.22. As x runs over U, the prime $\mathfrak{p}$ runs over $\operatorname{Spec}(A)$. So the stalks at all points of U are normal if and only if $A_{\mathfrak{p}}$ is integrally closed for every prime $\mathfrak{p}$, which by [10, Integral Closure Is A Local Property] holds if and only if A, equivalently $\mathcal{O}_X(U)$, is integrally closed. Every point of X lies in a nonempty affine open (definition 2.2.9), so X is normal if and only if $\mathcal{O}_X(U)$ is integrally closed for every nonempty affine open U. For a domain A the stalks of $\operatorname{Spec}(A)$ are the rings $A_{\mathfrak{p}}$ (proposition 2.2.22), so the same citation shows that $\operatorname{Spec}(A)$ is normal if and only if A is integrally closed, completing the proof.

Note that reducedness was stalk-local, and each stalk of a normal scheme is a domain and hence reduced, so normal schemes are reduced. By proposition 2.4.29, if R is an integrally closed domain then $\operatorname{Spec}(R)$ is a normal scheme. A UFD is integrally closed ([10, A Unique Factorization Domain Is Integrally Closed]), so a scheme whose stalks are all UFDs is normal.

Definition 2.4.30: Factorial Scheme

Let X be a scheme. Then X is said to be *factorial*, or *locally factorial*, if every stalk $\mathcal{O}_{X,x}$ is a UFD.

If R is a Noetherian UFD, then $\operatorname{Spec} R$ is factorial. Indeed, by [10, Ideals of Localization] a height-one prime of $R_{\mathfrak{p}}$ has the form $\mathfrak{q}R_{\mathfrak{p}}$ for a height-one prime $\mathfrak{q} \subseteq \mathfrak{p}$ of R. That $\mathfrak{q}$ is principal by [10, Factoriality by Height-One Primes], so every height-one prime of the Noetherian domain $R_{\mathfrak{p}}$ is principal, and the same lemma makes $R_{\mathfrak{p}}$ a UFD. The converse fails, since every stalk can be a UFD while the ring of global sections is not one. The ring $R = \mathbb{Z}[\sqrt{-5}]$ is the ring of integers of $\mathbb{Q}(\sqrt{-5})$, hence a Dedekind domain ([7, Ring Of Integers Are Dedekind Domains]). Its local ring at each nonzero prime is therefore a DVR ([7, Dedekind Domain As DVR]), hence a UFD, and its stalk at the generic point is the field $\mathbb{Q}(\sqrt{-5})$, so $\operatorname{Spec} R$ is factorial. Yet the height-one prime $(2, 1+\sqrt{-5})$ is not principal ([7, Class Groups]), so R is not a UFD by [10, Factoriality by Height-One Primes].

Example 2.4.31: Normal Schemes

1. The schemes $\mathbb{A}^n_k$, $\mathbb{P}^n_k$ and $\operatorname{Spec}\mathbb{Z}$ are normal. If R is normal, by definition $\operatorname{Spec} R$ is normal.
2. Starting on p.164 section 5.4.8, lots and lots of good examples

Exercise 2.4.3

1. Find the structure for the following rings:

a) $\operatorname{Spec} \mathbb{C}[x]/(x^3 - x^2)$

b) $\operatorname{Spec} \mathbb{C}[x]/(x^2 - d)$

c) $\operatorname{Spec} \mathbb{C}[x]/(x^2 + 1)$

2. Let X be a quasicompact scheme. Show that the closure of each point contains a closed point (this is not true for non quasicompact schemes in general)

3. Let X be a quasicompact scheme. Suppose f vanishes at all point of X. Then there exists some N st $f^n = 0$. Show that this fails when X is not quasicompact (hint: take infinite disjoint unions of $\operatorname{Spec}(R_n)$ where $R_n := k[x]/(x^n)$)

4. Let X be a quasicompact scheme. Then it suffices to cheek reducedness at closed points (show that any nonreduced point has a nonreduced closed point in its closure)

5. Show that reducedness is not in general an open condition by considering:

$$X = \operatorname{Spec}\left(\frac{\mathbb{C}[x, y_1, y_2, ...,}{(y_1^2, y_2^2, y_3^3, ..., (x-1)y_1, (x-2)y_2, (x-3)y_3), ...}\right)$$

and $Y = \operatorname{Spec}[x]$. Show the nonreduced points of X are the closed points corresponding to the positive integers. The compliment of this set is *not* Zariski open.

6. Let X be an integral scheme. Show that if $\emptyset \neq V \subseteq U$, then the map $\operatorname{res}_{U,V} : \mathcal{O}_X(U) \to \mathcal{O}_X(V)$ is an inclusion. very important, see p.156 under exercise

7. Let X be an integral scheme, γ the generic point, and $\operatorname{Spec}(R) \cong U \subseteq X$ is any nonempty affine open subset of X. Then $\mathcal{O}_X(U) \to \mathcal{O}_{X,\gamma} = K(A)$ is an inclusion. Conclude that gluing is easy: every f_i in U_i for an open cover for U glue together if they are same element in $\operatorname{Frac}(X)$ (recall that $\operatorname{Frac}(X)$ is defined to be $\operatorname{Frac}(R)$ where R is given by $\mathcal{O}_X(\operatorname{Spec}(R))$

8. Show that integrality is not stalk local via the example $\operatorname{Spec}(A) \coprod \operatorname{Spec}(B) = \operatorname{Spec}(A \times B)$.

9. Show that reducedness is an affine local property.

10. Let X be a locally Noetherian Scheme. Then X is quasiseparated.

11. Let X be a Noetherian Scheme. Then it has a finite number of connected components, each a finite union of irreducible components.

12. Show that the ring of sections of a Noetherian Scheme need not be Noetherian.

13. Let X be a scheme of locally finite type over $\mathbb{Z}$ or k. Then X is locally Noetherian.

14. A scheme that is of finite type over any Noetherian ring is Neotherian.

15. Let X be a quasiprojective R-scheme. Then if R is Noetherian, X is a Noetherian scheme, and hence has a finite number of irreducible components.

16. Let X be a locally Noetherian scheme. Then the reduced locus is open. (Vakil p.156)

17. Let $U \subseteq X$ be an open subscheme of a projective R-scheme X. Show that U is locally of finite type R

18. Find a counter-example showing that normal schemes are not in general integral schemes (hint: think of the disjoint union of two normal scheme. For more hint, Vakil p. 162).

19. Let X be a Noetherian scheme. Then X is normal if and only if if is a finite disjoint union of integral normal schemes.

2.5 Associated Points

Associated points capture the "component structure" of a scheme, including both the obvious irreducible components and more subtle "embedded" components arising from nilpotent structure. For a reduced scheme, the associated points are simply the generic points of the irreducible components. For non-reduced schemes, there can be additional associated points that "remember" how components come together or where nilpotent thickening occurs.

The algebraic foundation is the theory of associated primes of modules, which we briefly recall before giving the scheme-theoretic definition.

2.5.1 Associated Primes of Modules

Definition 2.5.1: Associated Prime

Let R be a ring and M an R-module. A prime ideal $\mathfrak{p} \in \operatorname{Spec} R$ is an *associated prime* of M if $\mathfrak{p} = \operatorname{Ann}(m)$ for some $m \in M$. The set of associated primes of M is denoted $\operatorname{Ass}_R(M)$ or simply $\operatorname{Ass}(M)$.

Equivalently, $\mathfrak{p} \in \operatorname{Ass}(M)$ if and only if $R/\mathfrak{p}$ embeds into M as an R-submodule (via $1 \mapsto m$). The condition $\mathfrak{p} = \operatorname{Ann}(m)$ is quite stringent: it is not enough for $\mathfrak{p}$ to annihilate some element; it must be the *full* annihilator.

Example 2.5.2: Associated Primes

1. Let $R = k[x]$ and $M = R$. The only associated prime is (0): if $\operatorname{Ann}(f) = \mathfrak{p}$ for some $f \neq 0$, then $\mathfrak{p} \cdot f = 0$, but $k[x]$ is a domain so $\mathfrak{p} = (0)$.
2. Let $R = k[x, y]$ and $M = R/(xy)$. The associated primes are (x) and (y): we have $\operatorname{Ann}(\bar{x}) = (y)$ and $\operatorname{Ann}(\bar{y}) = (x)$ in the quotient.
3. Let $R = k[x, y]$ and $M = R/(y^2, xy)$. The associated primes are (y) and (x, y): we have $\operatorname{Ann}(\bar{x}) = (y)$, and $\operatorname{Ann}(\bar{y}) = (x, y)$. The prime (x, y) is an *embedded* associated prime (see below).
4. Let $R = \mathbb{Z}$ and $M = \mathbb{Z}/12\mathbb{Z}$. Then $\operatorname{Ass}(M) = \{(2), (3)\}$: we have $\operatorname{Ann}(\bar{4}) = (3)$ and $\operatorname{Ann}(\bar{3}) = (4) \supsetneq (2)$ ... but $\operatorname{Ann}(\bar{6}) = (2)$. Note (2) and (3) are the primes appearing in the primary decomposition $12 = 4 \cdot 3 = 2^2 \cdot 3$.

The key properties of associated primes in the Noetherian setting are:

Proposition 2.5.3: Properties of Associated Primes

Let R be a Noetherian ring and M a finitely generated R-module. Then:

1. $\mathrm{Ass}(M) \neq \emptyset$ if and only if $M \neq 0$.
2. $\mathrm{Ass}(M)$ is finite.
3. the set of zero divisors on M is given by:
$$\bigcup_{\mathfrak{p} \in \mathrm{Ass}(M)} \mathfrak{p} = \{r \in R \mid rm = 0 \text{ for some } m \neq 0\}$$
,
4. The minimal elements of $\mathrm{Ass}(M)$ are exactly the minimal primes of $\mathrm{Supp}(M)$ (equivalently, the minimal primes of $\mathrm{Ann}(M)$).

***Proof*:**
See [10, Zero-Divisors and Annihilators, Finiteness of Associated Primes, The Zero-Divisors Are the Union of the Associated Primes, Associated Primes Sit in the Support] for the full proof. We sketch the key ideas:

1. If $M \neq 0$, choose $m \neq 0$. The set of ideals $\{\mathrm{Ann}(m') : m' \in Rm,\ m' \neq 0\}$ is nonempty and has a maximal element $\mathrm{Ann}(m_0)$ by the Noetherian condition. One checks this maximal annihilator is prime.
2. Primary decomposition: $0 = \bigcap_{i=1}^{n} Q_i$ in M gives $\mathrm{Ass}(M) \subseteq \{\sqrt{Q_1}, \ldots, \sqrt{Q_n}\}$. Equality holds for the minimal primes; the others may or may not be associated.
3. If $r \in \mathfrak{p} \in \mathrm{Ass}(M)$, write $\mathfrak{p} = \mathrm{Ann}(m)$; then $rm = 0$ with $m \neq 0$. Conversely, if $rm = 0$ for $m \neq 0$, then $r \in \mathrm{Ann}(m) \subseteq \mathfrak{p}$ for some associated prime $\mathfrak{p}$ containing $\mathrm{Ann}(m)$.
4. Follows from the relationship between associated primes and primary decomposition.

Property (3) is particularly important: it says that in a Noetherian ring, the zero divisors are precisely the elements lying in some associated prime. This will translate to the geometric statement that a function is a zero divisor if and only if it vanishes at an associated point.

Definition 2.5.4: Minimal and Embedded Primes

Let R be a Noetherian ring and M a finitely generated R-module. An associated prime $\mathfrak{p} \in \mathrm{Ass}(M)$ is:

- *minimal* (or *isolated*) if it is minimal among associated primes, i.e., there is no $\mathfrak{q} \in \mathrm{Ass}(M)$ with $\mathfrak{q} \subsetneq \mathfrak{p}$;
- *embedded* if it is not minimal, i.e., there exists $\mathfrak{q} \in \mathrm{Ass}(M)$ with $\mathfrak{q} \subsetneq \mathfrak{p}$.

Minimal associated primes correspond to the irreducible components of $\mathrm{Supp}(M)$. Embedded associated primes correspond to "hidden" structure not visible in the support as a set, but detected

by the module structure. Geometrically, embedded points arise from nilpotent thickening or non-transverse intersections.

2.5.2 Associated Points of Schemes

Definition 2.5.5: Associated Point of a Scheme

Let X be a scheme. A point $x \in X$ is an *associated point* of X if, for some (equivalently, every) affine open $U = \operatorname{Spec} R$ containing x with x corresponding to $\mathfrak{p} \in \operatorname{Spec} R$, we have:

$$\mathfrak{p} \in \operatorname{Ass}_A(A)$$

The set of associated points of X is denoted $\operatorname{Ass}(X)$.

The "some/every" equivalence follows from the fact that associated primes behave well under localization and thus by the affine communication lemma: if $\mathfrak{p} \in \operatorname{Ass}(R)$ and $S \subseteq A$ is a multiplicative set with $S \cap \mathfrak{p} = \emptyset$, then $S^{-1}\mathfrak{p} \in \operatorname{Ass}(S^{-1}R)$.

Proposition 2.5.6: Characterization of Associated Points

Let X be a locally Noetherian scheme. A point $x \in X$ is an associated point if and only if the maximal ideal $\mathfrak{m}_x \subseteq \mathcal{O}_{X,x}$ is an associated prime of $\mathcal{O}_{X,x}$ (as a module over itself). Equivalently, $x \in \operatorname{Ass}(X)$ if and only if there exists $f \in \mathcal{O}_{X,x}$ with $\operatorname{Ann}(f) = \mathfrak{m}_x$.

***Proof*:**
Take an affine open $U = \operatorname{Spec} A$ containing x, with x corresponding to $\mathfrak{p}$. Then $\mathcal{O}_{X,x} = A_{\mathfrak{p}}$ and $\mathfrak{m}_x = \mathfrak{p}A_{\mathfrak{p}}$. We have $\mathfrak{p} \in \operatorname{Ass}_A(A)$ if and only if $\mathfrak{p}A_{\mathfrak{p}} \in \operatorname{Ass}_{A_{\mathfrak{p}}}(A_{\mathfrak{p}})$ by the localization property of associated primes.

Proposition 2.5.7: Associated Points of Locally Noetherian Schemes

Let X be a locally Noetherian scheme. Then:

1. $\operatorname{Ass}(X)$ is a discrete subset of X: every associated point has an open neighborhood containing no other associated points.
2. If X is quasi-compact (e.g., Noetherian), then $\operatorname{Ass}(X)$ is finite.
3. If X is reduced, then $\operatorname{Ass}(X)$ equals the set of generic points of the irreducible components of X.
4. The minimal associated points are exactly the generic points of the irreducible components of X.
5. X has embedded points if and only if X is not reduced "at" those points (more precisely: X has no embedded points if and only if X is reduced).

***Proof*:**

1. Each associated point x lies in an affine open $\operatorname{Spec} A$ where it corresponds to $\mathfrak{p} \in \operatorname{Ass}(A)$. Shrinking to $\operatorname{Spec} A_{\mathfrak{p}}$, the only associated prime is $\mathfrak{p}A_{\mathfrak{p}}$ by the localization property.
2. Cover X by finitely many affine opens $\operatorname{Spec} A_i$; each has finitely many associated primes, and associated points of X lying in $\operatorname{Spec} A_i$ correspond to associated primes of A_i.
3. If X is reduced, then for any affine open $\operatorname{Spec} A$, we have A reduced (no nilpotents). A reduced Noetherian ring has $\operatorname{Ass}(A) = \operatorname{Min}(A)$ (the minimal primes), and these correspond to the generic points of irreducible components.
4. The minimal primes of A are both the minimal elements of $\operatorname{Ass}(A)$ and the generic points of the irreducible components of $\operatorname{Spec} A$.
5. If X is reduced, (3) shows there are no embedded points. Conversely, if X has no embedded points and $\mathfrak{p}$ is a minimal prime, then $A_{\mathfrak{p}}$ has unique associated prime $\mathfrak{p}A_{\mathfrak{p}}$, which forces $A_{\mathfrak{p}}$ to be a field (it's a local ring with no zero divisors of depth 0), so A is generically reduced. A bit more work shows A is reduced.

Example 2.5.8: Associated Points of Integral Schemes

1. Let R be an integral domain. Then the generic point (0) is the unique associated point of $\operatorname{Spec} R$. Indeed, $\operatorname{Ass}(R) = \{(0)\}$ since R has no zero divisors: if $\operatorname{Ann}(r) = \mathfrak{p}$ for some $r \neq 0$, then $\mathfrak{p} \cdot r = 0$ forces $\mathfrak{p} = (0)$.

   More generally, if X is an integral scheme, then $\operatorname{Ass}(X) = \{\eta\}$ where η is the generic point.

2. Let $X = \operatorname{Spec}(k[x,y]/(xy))$. This is two lines (the x-axis and y-axis) meeting at the origin. The associated primes of $k[x,y]/(xy)$ are (x) and (y):

   $$\operatorname{Ann}(\bar{x}) = (y), \qquad \operatorname{Ann}(\bar{y}) = (x)$$

   These are both minimal, corresponding to the generic points of the two irreducible components. The origin (x,y) is *not* an associated point: one can check that $\operatorname{Ann}(\bar{f}) \neq (x,y)$ for any $\bar{f} \in k[x,y]/(xy)$.

   Geometrically: the two lines meet transversally at the origin, and transverse intersections do not create embedded points.

3. Let $X = \operatorname{Spec}(k[x,y]/(y^2, xy))$. This scheme has the same underlying topological space as $\operatorname{Spec}(k[x,y]/(y)) = \mathbb{A}^1_k$ (the x-axis), but with nilpotent "fuzz" at the origin.

   The associated primes are (y) and (x,y):

   - $\operatorname{Ann}(\bar{x}) = (y)$: if $f \cdot \bar{x} = 0$ in $k[x,y]/(y^2, xy)$, then $fx \in (y^2, xy)$, forcing $f \in (y)$.
   - $\operatorname{Ann}(\bar{y}) = (x,y)$: we have $x \cdot \bar{y} = \overline{xy} = 0$ and $y \cdot \bar{y} = \overline{y^2} = 0$, so $(x,y) \subseteq \operatorname{Ann}(\bar{y})$. Maximality is clear since $\bar{y} \neq 0$.

   Thus $\operatorname{Ass}(X) = \{[(y)], [(x,y)]\}$: the generic point of the x-axis (minimal) and the origin (embedded). The embedded point at the origin "remembers" that X is non-reduced there, even though the underlying set is just a line.

4. Continuing the previous example, let $X = \operatorname{Spec}(k[x,y]/(y^2,xy))$ and consider the function $\bar{y} \in \mathcal{O}_X(X)$.

   Where does $\bar{y}$ vanish? At a point $\mathfrak{p} \in X$, the function $\bar{y}$ vanishes if and only if its image in the residue field $\kappa(\mathfrak{p})$ is zero, which happens if and only if $y \in \mathfrak{p}$. Thus $\bar{y}$ vanishes at every point of X (since every prime of $k[x,y]/(y^2,xy)$ contains y).

   What is $\operatorname{Supp}(\bar{y})$*?* The support of $\bar{y}$ as a section is the set of points where $\bar{y}$ is not zero in the stalk:

   $$\operatorname{Supp}(\bar{y}) = \{\mathfrak{p} \in X \mid \bar{y} \neq 0 \in \mathcal{O}_{X,\mathfrak{p}}\}$$

   At the generic point $[(y)]$: the stalk is $(k[x,y]/(y^2,xy))_{(y)} = k(x)[y]/(y^2)$ (localize away from (y), but y remains nilpotent). Here $\bar{y} \neq 0$.

   At the origin $[(x,y)]$: the stalk is $(k[x,y]/(y^2,xy))_{(x,y)}$, and $\bar{y} \neq 0$ there as well.

   At any other closed point $[(x-a,y)]$ for $a \neq 0$: the stalk is $k[x,y]_{(x-a,y)}/(y^2,xy)$. Writing $x = (x-a)+a$ in the relation $xy = 0$ gives $(x-a)y = -ay$, and multiplying by a^{-1} gives $y\left(1 + a^{-1}(x-a)\right) = 0$. Since $x - a$ lies in the maximal ideal, $1 + a^{-1}(x-a)$ is a unit, so $y = 0$ in this stalk.

   Thus $\operatorname{Supp}(\bar{y}) = \{[(y)], [(x,y)]\}$, which is exactly $\operatorname{Ass}(X)$. The function $\bar{y}$ is a *zero divisor* (since $(x) \cdot \bar{y} = 0$), and its support is concentrated at the associated points.

Proposition 2.5.9: Zero Divisors and Associated Points

Let X be a locally Noetherian scheme and $f \in \mathcal{O}_X(X)$ a global section. Then f is a zero divisor if and only if f vanishes at some associated point of X. Equivalently:

$$f \text{ is a non-zero-divisor} \iff f|_\xi \neq 0 \text{ for all } \xi \in \operatorname{Ass}(X)$$

***Proof*:**
The question is local, so we may assume $X = \operatorname{Spec} A$ for a Noetherian ring A. Then f is a zero divisor in A if and only if $f \in \bigcup_{\mathfrak{p} \in \operatorname{Ass}(A)} \mathfrak{p}$ by proposition 2.5.3(3), which happens if and only if $f \in \mathfrak{p}$ for some associated prime $\mathfrak{p}$, i.e., f vanishes at the corresponding associated point.

This proposition is extremely useful: to check whether a function is a non-zero-divisor, it suffices to check non-vanishing at the finitely many associated points, rather than at every point.

Example 2.5.10: Double Line

Let $X = \operatorname{Spec}(k[x,y]/(y^2))$, the "double line" or x-axis with nilpotent thickening. The only prime ideal of $k[x,y]/(y^2)$ containing (y^2) is (y) (since (y) is radical and $(y^2) \subseteq (y)$, and any prime containing y^2 must contain y).

Is (y) an associated prime? We need $\operatorname{Ann}(f) = (y)$ for some f. Take $f = \bar{y}$: then $g \cdot \bar{y} = 0$ iff $gy \in (y^2)$ iff $g \in (y)$. So $\operatorname{Ann}(\bar{y}) = (y)$.

Thus $\operatorname{Ass}(X) = \{[(y)]\}$: just the generic point. There is no embedded point! Even though X is non-reduced, the nilpotent thickening is "uniform" along the entire line, not concentrated at any particular point.

Compare this to $k[x,y]/(y^2,xy)$, where the xy relation creates an embedded point at the origin by making y "more nilpotent" there.

Example 2.5.11: Associated Points of $\operatorname{Spec}\mathbb{Z}$
Let $X = \operatorname{Spec}\mathbb{Z}$. Since $\mathbb{Z}$ is an integral domain, the unique associated point is the generic point (0). The closed points (p) for primes p are *not* associated.

Now let $M = \mathbb{Z}/n\mathbb{Z}$ for some $n > 1$, and consider $\widetilde{M}$ as a quasi-coherent sheaf on $\operatorname{Spec}\mathbb{Z}$. The associated primes of M are the primes dividing n. For $n = 12 = 2^2 \cdot 3$, we have $\operatorname{Ass}(M) = \{(2), (3)\}$, corresponding to the points where the "support" of the module is concentrated.

Associated points play a key role in many constructions:

Proposition 2.5.12: Sections Determined by Associated Points

Let X be a locally Noetherian scheme and $\mathscr{F}$ a quasi-coherent sheaf on X. A section $s \in \mathscr{F}(X)$ is zero if and only if $s_\xi = 0$ in the stalk $\mathscr{F}_\xi$ for every associated point $\xi \in \operatorname{Ass}(X)$.

***Proof*:**
This reduces to the affine case: if $X = \operatorname{Spec} A$ and $\mathscr{F} = \widetilde{M}$ for an A-module M, then $s \in M$ is zero iff $s = 0$ in $M_\mathfrak{p}$ for all $\mathfrak{p} \in \operatorname{Ass}(M)$. This follows from the fact that the natural map $M \to \prod_{\mathfrak{p} \in \operatorname{Ass}(M)} M_\mathfrak{p}$ is injective for finitely generated modules over Noetherian rings.

2.5.13 Associated Points and Depth For a locally Noetherian scheme X, the associated points are exactly the points where the depth of the local ring is zero:

$$\operatorname{Ass}(X) = \{x \in X \mid \operatorname{depth}(\mathcal{O}_{X,x}) = 0\}$$

Recall that the depth of a local ring $(R, \mathfrak{m})$ is the length of a maximal regular sequence in $\mathfrak{m}$. A point x has depth 0 if and only if $\mathfrak{m}_x$ consists entirely of zero divisors (plus nilpotents), which happens precisely when $\mathfrak{m}_x \in \operatorname{Ass}(\mathcal{O}_{X,x})$.
This perspective connects associated points to Cohen-Macaulay and Gorenstein conditions: a Noetherian scheme is Cohen-Macaulay if and only if at every point, the depth equals the dimension, which imposes strong restrictions on the associated points.

Exercise 2.5.1

1. Let $A = k[x, y, z]/(xy, xz)$. Find $\operatorname{Ass}(A)$ and describe the associated points of $\operatorname{Spec} A$ geometrically. Which are minimal and which are embedded?
2. Let $A = k[x, y]/(x^2, xy)$. Show that $\operatorname{Ass}(A) = \{(x), (x, y)\}$. Interpret this geometrically: what is the underlying space, and where is the embedded point?
3. Prove that a Noetherian scheme X is reduced if and only if it has no embedded associated points.
4. Let X be a Noetherian scheme and $U \subseteq X$ an open subset. Show that $\operatorname{Ass}(U) = \operatorname{Ass}(X) \cap U$. (Hint: use the localization property of associated primes.)
5. Let $f : X \to Y$ be a closed immersion of Noetherian schemes. Show that $\operatorname{Ass}(X) \subseteq f^{-1}(\operatorname{Ass}(Y))$ does NOT hold in general. Give a counterexample. (Hint: consider $\operatorname{Spec}(k) \hookrightarrow$

$\operatorname{Spec}(k[x]/(x^2))$.)

6. Let X be an integral scheme with generic point η. Show that a rational function $f \in k(X)$ is regular (i.e., lies in $\mathcal{O}_X(U)$ for some open U) if and only if it is regular at η. (This is a version of the "associated points control regularity" principle for integral schemes.)

7. (*Primary decomposition and associated points*) Let $I \subseteq R$ be an ideal in a Noetherian ring with primary decomposition $I = Q_1 \cap \cdots \cap Q_n$. Show that $\operatorname{Ass}(R/I) = \{\sqrt{Q_1}, \ldots, \sqrt{Q_n}\}$, and identify which are minimal vs. embedded.

2.6 Dimension

The reader may want to review [10] for an in-depth look at the algebraic counterpart. Recall that the *Krull dimension* of a commutative ring R is the supremum of the lengths of proper chains of prime ideals:

$$\mathfrak{p}_0 \subsetneq \mathfrak{p}_1 \subsetneq \cdots \subsetneq \mathfrak{p}_n$$

with indexing starting at 0. Geometrically, for $\operatorname{Spec} R$ and topological spaces more generally (including schemes), the *dimension* is the supremum of lengths of proper chains of closed irreducible subsets with indexing starting at 0. Since the closed irreducible subsets of $\operatorname{Spec} R$ are exactly the sets $V(\mathfrak{p})$ for $\mathfrak{p}$ prime, we have $\dim \operatorname{Spec} R = \operatorname{kdim} R$. Furthermore:

$$\dim \operatorname{Spec} R = \dim \operatorname{Spec}(R/\sqrt{0})$$

since $\operatorname{Spec} R$ and $\operatorname{Spec}(R/\sqrt{0})$ are homeomorphic (quotienting by the nilradical does not change the prime ideals).

If the space is not irreducible, then the dimension per component can vary (for example $\mathbb{A}^1 \cup \mathbb{A}^2 \subseteq \mathbb{A}^3$, interpreted as the x-axis union the yz-plane). Hence we usually work with irreducible schemes. If necessary, we say a scheme has *pure dimension* n (or is *equidimensional*) if all of its irreducible components have the same dimension n. Dimension respects inclusions: if $X \subseteq Y$ then $\dim X \leq \dim Y$.

Dimension	Name
0	point (or collection of points)
1	curve
2	surface
$n \geq 3$	n-fold

Lemma 2.6.1: Dimension of Open Subset

Let $U \subseteq X$ be an open subset of a topological space. Then there is a bijection between irreducible closed subsets of U and the irreducible closed subsets of X that meet U, given by $Z \mapsto \overline{Z}$ (closure in X) with inverse $W \mapsto W \cap U$.

***Proof*:**
If $Z \subseteq U$ is irreducible and closed in U, then $\overline{Z}$ is irreducible and closed in X with $\overline{Z} \cap U = Z$ (since Z is closed in U, it equals the intersection of its closure with U). Conversely, if $W \subseteq X$ is irreducible closed with $W \cap U \neq \emptyset$, then $W \cap U$ is irreducible (an open subset of an irreducible set is irreducible) and closed in U, with $\overline{W \cap U} = W$ (since $W \cap U$ is dense in W). These operations are mutually inverse.

Proposition 2.6.2: Dimension of Scheme

Let X be a scheme. Then $\dim X = n$ if and only if X admits an open cover by affine open subsets of dimension at most n, where equality is achieved by at least one affine open.

***Proof*:**
If $\dim X = n$, then each open subset in an affine cover has dimension $\leq n$ (since irreducible closed subsets of an open set correspond to those of X meeting it, by lemma 2.6.1).

Conversely, suppose every affine open in some cover has dimension $\leq n$ and at least one has dimension exactly n. We must show $\dim X = n$. The dimension is at least n since some affine open has dimension n, and a chain in U_i lifts to a chain in X via lemma 2.6.1. For the reverse inequality, suppose $\dim X > n$. Then there exists a chain of irreducible closed subsets in X:

$$Z_0 \subsetneq Z_1 \subsetneq \cdots \subsetneq Z_{n+1}$$

Since Z_0 is nonempty, it meets some affine open U_i in the cover. Then each Z_j also meets U_i (since $Z_0 \subseteq Z_j$ and $Z_0 \cap U_i \neq \emptyset$). By lemma 2.6.1, the chain:

$$Z_0 \cap U_i \subsetneq Z_1 \cap U_i \subsetneq \cdots \subsetneq Z_{n+1} \cap U_i$$

consists of distinct irreducible closed subsets of U_i (since the closure map is a bijection). This gives $\dim U_i \geq n + 1 > n$, contradicting our assumption.

Lemma 2.6.3: Integral Extensions and Dimension

Let $f : \operatorname{Spec} R \to \operatorname{Spec} S$ correspond to an integral extension $S \hookrightarrow R$. Then:

$$\dim \operatorname{Spec} R = \dim \operatorname{Spec} S$$

***Proof*:**
The going-up theorem gives $\dim \operatorname{Spec} R \geq \dim \operatorname{Spec} S$ (any chain in S lifts to a chain in R), and the incomparability theorem gives $\dim \operatorname{Spec} R \leq \dim \operatorname{Spec} S$ (primes lying over the same prime cannot be comparable). See [10] for the details.

2.6.4 A word on Noetherian Schemes It is *not* the case that a Noetherian scheme must be finite-dimensional: Nagata constructed a Noetherian ring of infinite Krull dimension (see Vakil, chapter 11).
However, Noetherian *local* rings are always finite-dimensional (by [10]), and so points of locally Noetherian schemes always have finite codimension.

2.6.1 Codimension

We are often concerned with subspaces that are a few dimensions lower than the ambient scheme X. Since schemes can be unions of irreducible components of different dimension, we define codimension only for irreducible subschemes:

Definition 2.6.5: Codimension

Let X be a topological space and $Y \subseteq X$ an irreducible closed subset. Then:

$$\begin{aligned}&\operatorname{codim}_X Y \\ &= \sup\{n : \text{there exists a chain } Y = Z_0 \subsetneq Z_1 \subsetneq \cdots \subsetneq Z_n \text{ of irreducible closed subsets of } X\}\end{aligned}$$

For a scheme X and a point $x \in X$, the codimension of $\overline{\{x\}}$ in X equals the Krull dimension of the local ring $\mathcal{O}_{X,x}$.

For the affine case: if $\mathfrak{p} \in \operatorname{Spec} R$, then $\operatorname{codim}_{\operatorname{Spec} R} V(\mathfrak{p})$ equals the *height* of $\mathfrak{p}$, i.e. the supremum of lengths of chains:

$$\mathfrak{q}_0 \subsetneq \mathfrak{q}_1 \subsetneq \cdots \subsetneq \mathfrak{q}_h = \mathfrak{p}$$

and this equals $\operatorname{kdim} R_{\mathfrak{p}}$. For example, in $\operatorname{Spec} \mathbb{Z}$, the generic point $\eta = [(0)]$ has codimension 0, while any closed point $[(p)]$ has codimension 1.

If Y has codimension 0 in X, then Y must be an irreducible component of X.

Definition 2.6.6: Hypersurface

Let $Y \subseteq X$ be an irreducible closed subscheme. Then Y is called a *hypersurface* if $\operatorname{codim}_X Y = 1$.

Lemma 2.6.7: Sum of Dimension and Codimension

Let X be a topological space and $Y \subseteq X$ an irreducible closed subset. Then:

$$\operatorname{codim}_X Y + \dim Y \leq \dim X$$

***Proof*:**
A chain of length c descending from Y and a chain of length d ascending from Y can be concatenated to give a chain of length $c + d$ in X. Taking suprema gives $\operatorname{codim}_X Y + \dim Y \leq \dim X$.

Equality holds for irreducible varieties of finite type over a field, but *not* in general:

Example 2.6.8: Codimension Pathology

Consider the scheme $X = \operatorname{Spec}(k[x]_{(x)}[t])$, which can be thought of as a surface with a line attached at one point. Then $\dim X = 2$, but the closed point $p = (x, t)$ has dimension 0 and codimension 1 (not 2!), since the maximal chain from p is:

$$(0) \subsetneq (x) \subsetneq (x, t)$$

but (x) already has codimension 1 and dimension 1 (the "line" component), while the "surface" component (0) has dimension 2. Note too that this scheme has open subsets of *different dimension* – a phenomenon that does *not* occur for open subsets of n-manifolds.

The pathology above is excluded by the following class of rings:

Definition 2.6.9: Catenary Ring

A commutative ring R is *catenary* if for every pair of nested prime ideals $\mathfrak{q} \subseteq \mathfrak{p}$, all maximal chains of prime ideals between $\mathfrak{q}$ and $\mathfrak{p}$ have the same length.

Many rings are catenary: localizations of finitely generated $\mathbb{Z}$-algebras, complete Noetherian local rings, Dedekind domains, Cohen-Macaulay rings, and in particular all localizations of finitely generated k-algebras (see exercise 2.6.1).

2.6.2 Transcendence Dimension

Working with Krull dimension directly can be difficult. For finitely generated algebras over a field, we have a more computable invariant:

Definition 2.6.10: Transcendence Dimension

Let A be an integral domain that is a k-algebra. Then the *transcendence dimension* of A over k is:

$$\operatorname{trdeg}_k A := \operatorname{trdeg}_k \operatorname{Frac}(A)$$

i.e. the transcendence degree of the fraction field of A over k. Transcendence bases, the transcendence degree of a field extension, and its additivity in towers are developed in *Core Algebra* [12]; only the translation to algebras is made here.

Theorem 2.6.11: Transcendence and Krull Dimension

Let A be a finitely generated integral domain over a field k. Then:

$$\dim \operatorname{Spec} A = \operatorname{trdeg}_k \operatorname{Frac}(A)$$

Thus, if X is an irreducible k-variety with function field $k(X)$, then $\dim X = \operatorname{trdeg}_k k(X)$.

***Proof*:**
See [10, Dimension of Finitely Generated Algebras] for the algebraic proof. The key steps are: Noether normalization gives a finite (hence integral) extension $k[y_1, \dots, y_d] \hookrightarrow A$ with $d = \operatorname{trdeg}_k \operatorname{Frac}(A)$; by lemma 2.6.3, $\dim \operatorname{Spec} A = \dim \mathbb{A}^d_k = d$.

For catenary rings, dimension and codimension add up:

Theorem 2.6.12: Dimension Formula for Finite Type k-Schemes

Let X be an irreducible scheme locally of finite type over a field k, $Y \subseteq X$ an irreducible closed subset, and η the generic point of Y. Then:

$$\operatorname{codim}_X Y + \dim Y = \dim X$$

equivalently:

$$\dim \mathcal{O}_{X,\eta} + \dim Y = \dim X$$

***Proof*:**
The question is local, so we may reduce to $X = \operatorname{Spec} A$ for a finitely generated k-domain A. Let $\mathfrak{p}$ be the prime corresponding to η. Then $\operatorname{codim}_X Y = \operatorname{kdim} A_{\mathfrak{p}}$ and $\dim Y = \operatorname{kdim} A/\mathfrak{p}$, so we must show $\operatorname{kdim} A_{\mathfrak{p}} + \operatorname{kdim} A/\mathfrak{p} = \operatorname{kdim} A$. This follows from the fact that finitely generated k-algebras are catenary and equidimensional (see [10] or Vakil, chapter 11 for a guided exercise).

2.6.3 Krull's Height Theorem and Algebraic Hartogs' Lemma

We now turn to two fundamental results about codimension-1 phenomena. Krull's theorem gives an upper bound on codimension in terms of the number of defining equations, while the algebraic Hartogs' lemma characterizes regular functions on normal schemes in terms of codimension-1 behavior. Both will be used extensively in the theory of divisors (section 5.4).

Krull's Height Theorem

The following theorem, proved in [10], is one of the most important tools for computing codimension:

Theorem 2.6.13: Krull's Height Theorem (Hauptidealsatz)

Let R be a Noetherian ring. If $\mathfrak{a} = (f_1, \dots, f_n)$ is a proper ideal generated by n elements, then every minimal prime $\mathfrak{p}$ over $\mathfrak{a}$ has:

$$\operatorname{height}(\mathfrak{p}) \leq n$$

In particular, if $f \in R$ is neither a zero-divisor nor a unit, then every minimal prime over (f) has height exactly 1.

***Proof*:**
See [10, Krull's Height Theorem]. The proof uses the dimension theorem for Noetherian local rings: localizing at $\mathfrak{p}$, the ideal $\mathfrak{a}_\mathfrak{p}$ is $\mathfrak{p}_\mathfrak{p}$-primary and generated by n elements, so $\operatorname{kdim} R_\mathfrak{p} \leq n$ by the characterization of dimension as the minimal number of generators of a parameter ideal.

Geometrically, Krull's theorem says: in a Noetherian scheme, the vanishing locus of n functions has all irreducible components of codimension $\leq n$. In particular:

Corollary 2.6.14: Principal Ideal Theorem – Geometric Form

Let X be a locally Noetherian scheme and $f \in \mathcal{O}_X(X)$ a regular function that is not a zero-divisor on any irreducible component. Then every irreducible component of $V(f)$ has codimension exactly 1 in X.

Corollary 2.6.15: Codimension Bound for Complete Intersections

Let X be a locally Noetherian scheme and $Z = V(f_1, \ldots, f_n) \subseteq X$. Then every irreducible component of Z has codimension $\leq n$. If equality holds for every component, Z is called a *(set-theoretic) complete intersection.*

Example 2.6.16: Krull's Theorem in Practice

1. In $\mathbb{A}^3_k = \operatorname{Spec} k[x, y, z]$, the ideal (xy, xz) defines the vanishing locus $V(x) \cup V(y, z)$, which is the yz-plane $\cup$ the x-axis. The yz-plane has codimension 1, while the x-axis has codimension 2. Krull's theorem guarantees codimension ≤ 2, and both bounds are achieved.
2. The ideal $(x^2 + y^2 + z^2, x + y + z) \subseteq k[x, y, z]$ defines a curve in $\mathbb{A}^3_k$. By Krull's theorem, each component has codimension ≤ 2. In this case, the curve has codimension exactly 2, so it is a complete intersection.
3. The twisted cubic curve $C \subseteq \mathbb{A}^3_k$ (the image of $t \mapsto (t, t^2, t^3)$) has codimension 2 but ideal $I(C) = (y - x^2, z - x^3)$, which requires 2 generators. However, in $\mathbb{P}^3_k$ the twisted cubic requires 3 generators of its homogeneous ideal (it is *not* a complete intersection).

The converse direction of Krull's theorem also holds and is equally useful:

Proposition 2.6.17: Converse to Krull's Theorem

Let R be a Noetherian ring and $\mathfrak{p}$ a prime of height n. Then there exist $f_1, \ldots, f_n \in \mathfrak{p}$ such that $\mathfrak{p}$ is a minimal prime over $(f_1, \ldots, f_n)$.

***Proof*:**
See [10]. One constructs the f_i inductively using prime avoidance: at each step, choose f_i in $\mathfrak{p}$ but outside all minimal primes of $(f_1, \ldots, f_{i-1})$ of height $< i$.

Algebraic Hartogs' Lemma

In complex analysis, Hartogs' extension theorem states that a holomorphic function defined on the complement of a codimension-≥ 2 set extends uniquely to the ambient space. Amazingly, there is an algebraic analogue for normal schemes:

Theorem 2.6.18: Algebraic Hartogs' Lemma

Let A be a Noetherian normal domain. Then:

$$A = \bigcap_{\substack{\mathfrak{p} \in \operatorname{Spec} A \\ \operatorname{height}(\mathfrak{p}) = 1}} A_{\mathfrak{p}}$$

where the intersection takes place inside $\operatorname{Frac}(A)$.

In geometric language: a rational function on a Noetherian normal scheme with no poles along any codimension-1 subvariety is in fact a regular function. Equivalently, regular functions on a normal scheme are determined by their behavior at codimension-1 points – you cannot have a function that is "only irregular in codimension ≥ 2."

***Proof*:**
The inclusion $A \subseteq \bigcap_{\operatorname{ht}(\mathfrak{p})=1} A_{\mathfrak{p}}$ is clear (any element of A lies in every localization). For the reverse, suppose $x \in \operatorname{Frac}(A)$ lies in $A_{\mathfrak{p}}$ for every height-1 prime $\mathfrak{p}$. Let $I = \{a \in A \mid ax \in A\}$ be the ideal of "denominators" of x. If $x \notin A$, then $I \neq A$. Let $\mathfrak{q}$ be a minimal prime over I.

Claim: $\operatorname{height}(\mathfrak{q}) \leq 1$.

If $\operatorname{height}(\mathfrak{q}) = 0$, this is fine. If $\operatorname{height}(\mathfrak{q}) \geq 2$, then $x \in A_{\mathfrak{p}}$ for all height-1 primes $\mathfrak{p} \subseteq \mathfrak{q}$, and since $A_{\mathfrak{q}}$ is a normal local domain, $A_{\mathfrak{q}} = \bigcap_{\operatorname{ht}(\mathfrak{p})=1,\, \mathfrak{p} \subseteq \mathfrak{q}} (A_{\mathfrak{q}})_{\mathfrak{p}A_{\mathfrak{q}}}$ (applying the result in the local case, where it holds by the characterization of normal domains as S_2 and R_1). This gives $x \in A_{\mathfrak{q}}$, contradicting $I \subseteq \mathfrak{q}$. So $\operatorname{height}(\mathfrak{q}) = 1$.

But then $x \in A_{\mathfrak{q}}$ by hypothesis, so there exist $a \in A$, $s \notin \mathfrak{q}$ with $x = a/s$, giving $sx = a \in A$ and hence $s \in I$. But $I \subseteq \mathfrak{q}$ and $s \notin \mathfrak{q}$, a contradiction. Therefore $I = A$ and $x \in A$.

Example 2.6.19: Algebraic Hartogs in Practice

1. Consider $\mathbb{A}^2_k = \operatorname{Spec} k[x, y]$ and the open set $U = D(x) \cup D(y) = \mathbb{A}^2 \setminus \{(0, 0)\}$. We showed in example 2.2.2 that $\mathcal{O}_{\mathbb{A}^2}(U) = k[x, y]$. This is now immediate from algebraic Hartogs: the complement of U is $\{(0, 0)\}$, which has codimension 2, and $k[x, y]$ is a normal domain. Hence any regular function on U extends to all of $\mathbb{A}^2$.

2. The same argument fails for $\mathbb{A}^1$: removing a point from $\mathbb{A}^1$ removes a codimension-1 subset, and indeed $\mathcal{O}_{\mathbb{A}^1}(D(x)) = k[x, 1/x] \neq k[x]$.

3. Let $A = k[x, y, z]/(x^2 + y^2 - z^2)$, the quadric cone, which is a normal domain (exercise). The singular point at the origin has codimension 2, so algebraic Hartogs implies that regular functions on the smooth locus extend to all of $\operatorname{Spec} A$.

4. The algebraic Hartogs' lemma *fails* without normality. Consider the nodal cubic $A =$

$k[x,y]/(y^2 - x^2(x+1))$, which is not normal at the origin. The element y/x in the fraction field is regular at all codimension-1 points away from the node, yet $y/x \notin A$.

The connection between these two results is fascinating: Krull's theorem says that the zero locus of a single function has codimension ≤ 1, while Hartogs' lemma says that on normal schemes, a rational function is determined by its poles, which also occur in codimension 1. Both point toward the centrality of codimension-1 phenomena, which will be formalized in the theory of *Weil and Cartier divisors* in section 5.4.

Exercise 2.6.1

1. Show that a Noetherian scheme of dimension 0 has finitely many points.
2. Let $f : X \to Y$ be an integral morphism. Show that $\dim f^{-1}(y) = 0$ for all $y \in Y$ (i.e., the fibers are zero-dimensional).
3. If $\nu : \widetilde{X} \to X$ is the normalization of a scheme, show that $\dim \widetilde{X} = \dim X$.
4. If K/k is an algebraic field extension and X is a k-scheme of finite type, show that $\dim X = \dim X_K$ where $X_K = X \times_k K$ is the base change.
5. Generalize the above result: if K/k is an arbitrary field extension and X is an irreducible k-scheme of finite type, show that $\dim X_K = \dim X$.
6. Show the following rings are catenary: localizations of finitely generated $\mathbb{Z}$-algebras, complete Noetherian local rings, Dedekind domains.
7. Let X be an integral scheme locally of finite type over a field k, and let $U \subseteq X$ be a nonempty open subset. Show that $\dim U = \dim X$.
8. (*Hartogs' lemma for schemes*) Let X be a normal Noetherian integral scheme, and let $U \subseteq X$ be an open subset whose complement has codimension ≥ 2. Show that the restriction map $\mathcal{O}_X(X) \to \mathcal{O}_X(U)$ is an isomorphism. (Hint: reduce to the affine case and apply theorem 2.6.18.)
9. Show that the quadric cone $\operatorname{Spec} k[x,y,z]/(x^2 + y^2 - z^2)$ is a normal domain ($\operatorname{char} k \neq 2$). (Hint: show it is R_1 and S_2, or use the Jacobian criterion to show regularity in codimension 1 and then invoke Serre's criterion.)

2.7 Regular and Smooth Schemes

The notions of regularity and smoothness are central to both differential and algebraic geometry. In differential geometry [59], a smooth manifold is a topological manifold whose transition maps of its atlas are smooth. A coordinate chart $f\colon U \to V \subseteq \mathbb{R}^n$ induces linear isomorphisms

$$T_p f : T_p M \longrightarrow T_{f(p)}\mathbb{R}^n \cong \mathbb{R}^n$$

Let $C^\infty_p(M)$ denote the local ring of germs of smooth real-valued functions at p, with maximal ideal

$$\mathfrak{m}_p = \left\{ [h] \in C^\infty_p(M) \;\middle|\; h(p) = 0 \right\}$$

Then there are canonical identifications

$$T_p^* M \cong \mathfrak{m}_p/\mathfrak{m}_p^2, \qquad T_p M \cong \operatorname{Hom}_{\mathbb{R}}(\mathfrak{m}_p/\mathfrak{m}_p^2, \mathbb{R})$$

The first identification sends the class of h to dh_p, so the quotient $\mathfrak{m}_p/\mathfrak{m}_p^2$ records the first-order parts of functions vanishing at p. A chart $f\colon U \to V \subseteq \mathbb{R}^n$ at p identifies $\mathfrak{m}_p/\mathfrak{m}_p^2$ with $(\mathbb{R}^n)^*$, the classes of $f_1 - f_1(p), \ldots, f_n - f_n(p)$ forming a basis, so

$$\dim_{\mathbb{R}}(\mathfrak{m}_p/\mathfrak{m}_p^2) = n = \dim M$$

In algebraic geometry, the stalk $\mathcal{O}_{X,x}$ is likewise a local ring, with maximal ideal $\mathfrak{m}_x$ and residue field $\kappa(x)$, and $\mathfrak{m}_x/\mathfrak{m}_x^2$ is a vector space over $\kappa(x)$, the *Zariski cotangent space*. If X is locally Noetherian, then [10, Dimension and Embedding Dimension] gives

$$\dim \mathcal{O}_{X,x} \leq \dim_{\kappa(x)}(\mathfrak{m}_x/\mathfrak{m}_x^2)$$

Equality need not hold. At the origin of

$$X = \operatorname{Spec} \frac{k[x,y]}{(xy)}$$

the two sides are

$$\dim \mathcal{O}_{X,0} = 1, \qquad \dim_k(\mathfrak{m}_0/\mathfrak{m}_0^2) = 2$$

The relation $xy = 0$ is quadratic and imposes no linear relation between the classes of x and y in $\mathfrak{m}_0/\mathfrak{m}_0^2$, so the Zariski cotangent space at the origin has dimension 2, one more than the dimension of X there. A Noetherian local ring $(A, \mathfrak{m})$ is called *regular* when $\dim A = \dim_{A/\mathfrak{m}}(\mathfrak{m}/\mathfrak{m}^2)$.

Smoothness is a property of a morphism $f\colon X \to S$ (definition 2.7.10).

2.7.1 Smooth Morphisms A morphism is smooth exactly when it is locally of finite presentation, flat, and has geometrically regular fibres. A scheme is *smooth over k* when its structure morphism $X \to \operatorname{Spec} k$ is smooth. For schemes locally of finite type over a perfect field, smoothness over the field is equivalent to regularity. Over an imperfect field the two conditions can differ. Smooth morphisms are the algebraic analogue of smooth submersions of manifolds.

The algebraic foundations for this section (Hilbert functions, the dimension theorem, regular local rings, and their characterization via homological methods) were developed in [10, 18]. We recall the key definitions and build on them:

Definition 2.7.2: Tangent Space

Let X be a scheme and $x \in X$ a point. Then the *(Zariski) cotangent space* at x is the $\kappa(x)$-vector space:

$$\mathfrak{m}_x/\mathfrak{m}_x^2$$

where $\mathfrak{m}_x \subseteq \mathcal{O}_{X,x}$ is the maximal ideal of the stalk at x, and $\kappa(x) = \mathcal{O}_{X,x}/\mathfrak{m}_x$ is the residue field. The *(Zariski) tangent space* is its dual:

$$T_x X := (\mathfrak{m}_x/\mathfrak{m}_x^2)^\vee = \operatorname{Hom}_{\kappa(x)}(\mathfrak{m}_x/\mathfrak{m}_x^2,\, \kappa(x))$$

Definition 2.7.3: Regular Local Ring

Let $(R, \mathfrak{m})$ be a Noetherian local ring with residue field $k = R/\mathfrak{m}$. Then R is *regular* if:

$$\mathrm{kdim} R = \dim_k(\mathfrak{m}/\mathfrak{m}^2)$$

By [10], we always have $\mathrm{kdim} R \leq \dim_k(\mathfrak{m}/\mathfrak{m}^2)$, so regularity asks for equality.

Proposition 2.7.4: Regular Local Rings are Domains

Every regular local ring is an integral domain.

***Proof*:**
The associated graded ring is a polynomial ring, $G_{\mathfrak{m}}(R) \cong k[t_1, \ldots, t_d]$, by [10, Characterizations of a Regular Local Ring], hence a domain; and $\bigcap_n \mathfrak{m}^n = 0$ by the Krull intersection theorem for Noetherian local rings. So [10, A Graded Domain Lifts] applies and R is a domain.

Definition 2.7.5: Regular Scheme

A scheme X is *regular* if $\mathcal{O}_{X,x}$ is a regular local ring for every point $x \in X$. A scheme is *singular* at a point x if $\mathcal{O}_{X,x}$ is not regular.

Example 2.7.6: Regular and Singular Schemes

1. $\mathbb{A}^2_k$ *is regular.* For a closed point $\mathfrak{m} = (x - a, y - b)$, first observe that $\kappa(\mathfrak{m}) = k$ (since k is algebraically closed). The cotangent space $\mathfrak{m}/\mathfrak{m}^2$ is spanned by the images of $x - a$ and $y - b$, so $\dim_k(\mathfrak{m}/\mathfrak{m}^2) = 2$. The local ring $k[x, y]_{\mathfrak{m}}$ has a maximal chain $0 \subsetneq (x - a) \subsetneq (x - a, y - b)$, giving kdim $= 2$.

   Since we are working over schemes, we also have higher-dimensional points. Consider $[\mathfrak{p}] = [(x^2 - y)] \in \mathbb{A}^2_k$. The local ring $k[x, y]_{\mathfrak{p}}$ has dimension 1 (one can show the chain $0 \subsetneq (x^2 - y)$ is maximal), and $\mathfrak{m}_{\mathfrak{p}}/\mathfrak{m}^2_{\mathfrak{p}}$ is a 1-dimensional $\kappa(\mathfrak{p})$-vector space. Hence this point is regular.

   An important observation: $[(x^3 - y^2)] \in \mathbb{A}^2_k$ is *also* a regular point of $\mathbb{A}^2_k$, even though $x^3 - y^2$ defines a singular curve! This is because regularity at a generic point of $\mathbb{A}^2_k$ concerns the local ring of the *ambient* scheme, not the subscheme $V(x^3 - y^2)$. The point $[(x^3 - y^2)]$ is "generically regular" in $\mathbb{A}^2_k$ – further motivating the name *generic* point.

2. *The cuspidal cubic is singular.* Let $X = \operatorname{Spec} k[x, y]/(x^3 - y^2)$. Then X is not regular at the origin: the local ring $\mathcal{O}_{X,(x,y)}$ has kdim $= 1$ (since X is an irreducible curve), but the cotangent space $\mathfrak{m}/\mathfrak{m}^2$ at the origin has $\dim_k = 2$, since neither $\overline{x}$ nor $\overline{y}$ is redundant modulo $\mathfrak{m}^2$ (the relation $x^3 = y^2$ lies in $\mathfrak{m}^3$, not $\mathfrak{m}^2$). On the other hand, X is regular at every other closed point (exercise).

3. *Normal does not imply regular.* If X is a normal scheme, then its stalks are integrally closed, and all codimension-1 points are regular: the local ring at such a point is a one-dimensional normal local domain, hence a discrete valuation ring ([10, Characterizations of a DVR])

and so regular. However, normality does *not* imply regularity in dimension ≥ 2. For example, $\operatorname{Spec} k[x,y,z]/(x^2+y^2-z^2)$ is normal but not regular at the origin (exercise: show $\dim_k \mathfrak{m}/\mathfrak{m}^2 = 3$ while kdim $= 2$).

However, for a 1-dimensional connected Noetherian scheme, normality *does* imply regularity: the codimension-1 points are exactly the closed points, and the generic point is always regular. In fact, a 1-dimensional normal local domain is a DVR by [10, Characterizations of a DVR], which is a regular local ring of dimension 1.

The singular locus of a scheme is typically "small." For varieties over algebraically closed fields, this was shown in [22, Singular Points of Variety]: the set of singular points forms a proper closed subset.

For schemes of finite type over a field, regularity can be checked via partial derivatives, generalizing the classical criterion from [22, Jacobian Criterion and Regular Local Rings]:

Proposition 2.7.7: Jacobian Criterion for Schemes

Let k be a field and $X = \operatorname{Spec}(k[x_1,\dots,x_n]/(f_1,\dots,f_m))$ a closed subscheme of $\mathbb{A}^n_k$. Let $x \in X$ be a closed point whose residue field $\kappa(x)$ is separable over k, as holds automatically when k is perfect and in particular when k is algebraically closed. Then X is regular at x if and only if:

$$\operatorname{rank}\left(\frac{\partial f_i}{\partial x_j}(x)\right) = n - \dim \mathcal{O}_{X,x}$$

where the partial derivatives are evaluated in $\kappa(x)$.

***Proof*:**
This is the scheme-theoretic translation of [22, Jacobian Criterion and Regular Local Rings]. The key computation is that the cotangent space satisfies:

$$\mathfrak{m}_x/\mathfrak{m}_x^2 \cong \frac{\mathfrak{m}_{\mathbb{A}^n,x}}{(\mathfrak{m}^2_{\mathbb{A}^n,x} + (f_1,\dots,f_m))}$$

That display is valid over any field. Separability enters in the next step, the count $\dim_{\kappa(x)}(\mathfrak{m}_x/\mathfrak{m}_x^2) = n - \operatorname{rank}(J_x)$, which needs $\mathfrak{m}_{\mathbb{A}^n,x}/\mathfrak{m}^2_{\mathbb{A}^n,x} \to \Omega_{\mathbb{A}^n/k} \otimes \kappa(x)$ to be an isomorphism. Here no external input is required. The local ring $\mathcal{O}_{\mathbb{A}^n,x}$ is a localization of $k[x_1,\dots,x_n]$, which has global dimension n ([18, Hilbert's Syzygy Theorem]), so it has finite global dimension and is regular by [18, Auslander-Buchsbaum-Serre Theorem]; at a closed point its dimension is n by theorem 2.6.12. Hence both sides have dimension n, and the map is surjective because $\Omega_{\kappa(x)/k} = 0$ for the finite separable extension $\kappa(x)/k$ (proposition 5.5.9); a surjection between $\kappa(x)$-spaces of equal finite dimension is an isomorphism. So the span of the images of $f_1,\dots,f_m$ has dimension exactly $\operatorname{rank}(J_x)$. Regularity is then the condition that $\dim_{\kappa(x)}(\mathfrak{m}_x/\mathfrak{m}_x^2)$ equals $\dim \mathcal{O}_{X,x}$, which is equivalent to $\operatorname{rank}(J_x) = n - \dim \mathcal{O}_{X,x}$.

2.7.8 Separability Is Not Removable Without the hypothesis the criterion fails. Over $k = \mathbb{F}_p(t)$ the scheme $Y = \operatorname{Spec} k[w]/(w^p - t)$ is the spectrum of the field $\mathbb{F}_p(t^{1/p})$, hence regular of dimension 0, so the criterion would require Jacobian rank $1 - 0 = 1$; but $\partial(w^p - t)/\partial w = 0$ and the rank is 0. The residue field $k(t^{1/p})$ is purely inseparable over k, which is exactly what fails.

Example 2.7.9: Jacobian Criterion in Practice

1. The cuspidal cubic $X = V(y^2 - x^3) \subseteq \mathbb{A}^2_k$. The Jacobian is $(-3x^2,\, 2y)$. At the origin, this is $(0, 0)$, which has rank $0 < 2 - 1 = 1$. At any other closed point (a, b) with $b \neq 0$, the rank is $1 = n - \dim X$. Hence X is singular only at the origin.
2. The nodal cubic $X = V(y^2 - x^2(x + 1)) \subseteq \mathbb{A}^2_k$. The Jacobian is $(-3x^2 - 2x,\, 2y)$. At the origin, this is $(0, 0)$, so the origin is again singular. At any other point on X where $y \neq 0$, the rank is 1.
3. The quadric surface $X = V(x^2 + y^2 - z^2) \subseteq \mathbb{A}^3_k$ ($\operatorname{char} k \neq 2$). The Jacobian is $(2x,\, 2y,\, -2z)$, which has rank 0 only at the origin. Hence X is regular away from the origin.

Smooth Morphisms and Smooth Schemes

Regularity is a property of a scheme in isolation, while *smoothness* is a relative notion: it concerns a morphism $X \to S$ and asks whether the fibers are "non-singular" in a way that varies well over the base. The two notions coincide over perfect fields.

Definition 2.7.10: Smooth Morphism (Preliminary)

A morphism $f : X \to S$ of schemes locally of finite type is *smooth of relative dimension* n at a point $x \in X$ if there exists an open neighborhood U of x and a factorization:

$$\begin{array}{ccc} U & \xrightarrow{\text{open}} & \operatorname{Spec} \dfrac{\mathcal{O}_S(V)[x_1, \ldots, x_{n+r}]}{(f_1, \ldots, f_r)} \\ & \searrow^{f|_U} & \downarrow \\ & & V \end{array}$$

where V is an open neighborhood of $f(x)$ in S, and the Jacobian matrix $(\partial f_i/\partial x_j)$ has rank r at x.

The morphism f is *smooth* if it is smooth at every point. It is *étale* if it is smooth of relative dimension 0 — equivalently, flat and unramified (definition 2.7.19), the two descriptions reconciled by theorem 2.7.20.

The definition is local and built from a factorization, so the two stability properties one wants follow by manipulating factorizations rather than by any new geometry.

Proposition 2.7.11: Stability of Smooth Morphisms

Let $f\colon X \to S$ be smooth of relative dimension n.

1. *Base change.* For any morphism $S' \to S$, the projection $X \times_S S' \to S'$ is smooth of relative dimension n.
2. *Composition.* If $g\colon S \to T$ is smooth of relative dimension m, then $g \circ f$ is smooth of relative dimension $m+n$.
3. *Products.* If X and Y are smooth over a field k of dimensions m and n, then $X \times_k Y$ is smooth over k of dimension $m+n$.

***Proof*:**
(1). Smoothness is local on source and target, so work with a factorization as in definition 2.7.10: U open in X mapping to $\operatorname{Spec} \mathcal{O}_S(V)[x_1,\dots,x_{n+r}]/(f_1,\dots,f_r)$ over V, with the Jacobian of rank r. Base change along $S' \to S$ replaces $\mathcal{O}_S(V)$ by $\mathcal{O}_{S'}(V')$ and leaves the polynomials f_i and hence the Jacobian matrix formally unchanged, so its rank at a point of the fibre product is again r. The relative dimension $(n+r)-r=n$ is unchanged.

(2). Compose the two factorizations: over a suitable open, X is cut out in an affine space of relative dimension $n+r$ over S by r equations, and S in one of relative dimension $m+s$ over T by s equations. Pulling the first set of equations up and adjoining the second exhibits X inside an affine space of relative dimension $(m+s)+(n+r)$ over T cut out by $r+s$ equations, and the Jacobian is block triangular with the two original blocks on the diagonal, hence of rank $r+s$. The relative dimension is $(m+s+n+r)-(r+s)=m+n$.

(3). The projection $X \times_k Y \to Y$ is the base change of $X \to \operatorname{Spec} k$ along $Y \to \operatorname{Spec} k$, so it is smooth of relative dimension m by (1). Composing with $Y \to \operatorname{Spec} k$, smooth of relative dimension n, gives $X \times_k Y \to \operatorname{Spec} k$ smooth of relative dimension $m+n$ by (2), completing the proof.

Smoothness is a condition on a Jacobian and flatness is a condition on modules. The implication from the first to the second is proved below from the local criterion for flatness, in the homological form of [18].

Proposition 2.7.12: Smooth Morphisms Are Flat

A morphism $f : X \to S$ smooth of relative dimension n (definition 2.7.10) with S locally Noetherian is flat, and all of its nonempty fibres have pure dimension n.

***Proof*:**
Both conclusions are local on source and target, so fix $x \in X$ and take the presentation of definition 2.7.10: an affine open $V = \operatorname{Spec} A$ of S around $f(x)$, with A Noetherian, and an open neighbourhood of x realized as $\operatorname{Spec} B$ for

$$B \;=\; C/(f_1,\dots,f_r), \qquad C \;=\; A[x_1,\dots,x_N], \qquad N = n+r$$

the Jacobian $(\partial f_i/\partial x_j)$ having rank r at x. Write $\mathfrak{p} \subseteq A$ for the image of x, $\kappa = \kappa(\mathfrak{p})$, and $\mathfrak{q} \subseteq B$ for x itself. Reduction modulo $\mathfrak{p}$ identifies $C \otimes_A \kappa$ with the polynomial ring $\kappa[x_1,\dots,x_N]$, inside

which the images $\bar{f}_1, \dots, \bar{f}_r$ cut out the fibre through x. Let $\mathfrak{n}$ be the prime of $\kappa[x]$ corresponding to x and set $R = \kappa[x_1, \dots, x_N]_{\mathfrak{n}}$.

Step 1: R is regular. Iterating the formula $\operatorname{gldim} R'[t] = \operatorname{gldim} R' + 1$ of [18, Global Dimension of Polynomial Rings] down to the field κ, whose global dimension is 0, gives $\operatorname{gldim} \kappa[x_1, \dots, x_N] = N$. Localization is exact, so localizing a projective resolution of length $\le N$ resolves the localized module, and every R-module is the localization of itself viewed over $\kappa[x]$; hence $\operatorname{gldim} R \le N$ is finite. The Auslander–Buchsbaum–Serre theorem of [18, Auslander-Buchsbaum-Serre Theorem] then says exactly that R is a regular local ring.

Step 2: $\bar{f}_1, \dots, \bar{f}_r$ is a regular sequence on R. First, their classes in $\mathfrak{n}R/\mathfrak{n}^2R$ are linearly independent over $\kappa(x)$. Indeed the universal κ-derivation sends $g \mapsto dg = \sum_j (\partial g/\partial x_j)\, dx_j$ into the free $\kappa(x)$-module on $dx_1, \dots, dx_N$, and it kills $\mathfrak{n}^2$ after reduction because $d(gh) = g\, dh + h\, dg$ vanishes at x when both factors do. So a relation $\sum_i a_i \bar{f}_i \in \mathfrak{n}^2 R$ with $a_i \in R$ yields, using $\bar{f}_i \in \mathfrak{n}R$, the relation $\sum_i a_i(x)\, d\bar{f}_i = 0$; since the Jacobian has rank r at x the vectors $d\bar{f}_i$ are independent, so every $a_i(x) = 0$, that is $a_i \in \mathfrak{n}R$. This is the asserted independence.

Now induct, writing $d = \dim R$: suppose $R_m = R/(\bar{f}_1, \dots, \bar{f}_m)$ is regular of dimension $d - m$ for some $m < r$, the case $m = 0$ being Step 1. Independence in the cotangent space says $\bar{f}_{m+1}$ lies in the maximal ideal of R_m but not its square, so it is nonzero; and R_m is a domain by proposition 2.7.4, so $\bar{f}_{m+1}$ is a non-zero divisor on R_m. Killing a non-zero divisor drops the dimension by exactly one [10, One Equation Cuts Dimension by One], while killing an element outside the square of the maximal ideal drops the cotangent dimension by exactly one; both were $d - m$, so R_{m+1} is regular of dimension $d - m - 1$. The induction runs to $m = r$, proving both that the sequence is regular and that the local ring R_r of the fibre at x is regular of dimension $d - r$.

Step 3: flatness. Write $B_m = C/(f_1, \dots, f_m)$ and localize everything at x; each $(B_m)_{\mathfrak{q}}$ is a Noetherian local ring receiving a local homomorphism from $A_{\mathfrak{p}}$, and is finitely generated as a module over itself. We show by induction that $(B_m)_{\mathfrak{q}}$ is flat over $A_{\mathfrak{p}}$. For $m = 0$, C is a polynomial ring over A, hence free, hence flat, and flatness survives localization. Given flatness at stage m, note that $(B_m)_{\mathfrak{q}} \otimes_{A_{\mathfrak{p}}} \kappa = R_m$, on which $\bar{f}_{m+1}$ is a non-zero divisor by Step 2. The Fibrewise Nonzerodivisor Criterion of [18, Fibrewise Nonzerodivisor Criterion for Flatness] therefore makes f_{m+1} a non-zero divisor on $(B_m)_{\mathfrak{q}}$ and $(B_{m+1})_{\mathfrak{q}} = (B_m)_{\mathfrak{q}}/f_{m+1}(B_m)_{\mathfrak{q}}$ flat over $A_{\mathfrak{p}}$. At $m = r$ this says $B_{\mathfrak{q}}$ is flat over $A_{\mathfrak{p}}$, and x was arbitrary, so f is flat.

Step 4: fibre dimension. Steps 1 and 2 apply at every point of every nonempty fibre, so each fibre is a regular scheme of finite type over the residue field of its image, and is therefore a disjoint union of integral components (proposition 2.7.28). Take a closed point x of such a component Z. Then $\mathfrak{n}$ is a maximal ideal of $\kappa[x_1, \dots, x_N]$, so $d = \dim R = N$, and Step 2 gives $\dim \mathcal{O}_{Z,x} = N - r = n$. For an integral scheme of finite type over a field the local ring at a closed point has dimension equal to that of the whole [10], so $\dim Z = n$. Every component has dimension n, which is pure dimension n.

Definition 2.7.13: Smooth Scheme Over a Field

Let k be a field. A k-scheme X is *smooth over k* (or *k-smooth*) if the structure morphism $X \to \operatorname{Spec} k$ is smooth.

The relationship between smoothness and regularity is as follows:

Proposition 2.7.14: Smooth Implies Regular

Let k be a field and X a k-scheme locally of finite type. If X is smooth over k, then X is regular.

***Proof*:**
Smoothness over k implies that the cotangent space $\mathfrak{m}_x/\mathfrak{m}_x^2$ has dimension equal to the local dimension of X at x for every point x, which is exactly regularity. The key point is that smoothness gives the Jacobian rank condition over every residue field, not just over k itself.

The converse holds over perfect fields:

Theorem 2.7.15: Regularity and Smoothness Over Perfect Fields

Let k be a perfect field (e.g., any field of characteristic 0, or any finite field) and X a k-scheme locally of finite type. Then X is smooth over k if and only if X is regular.

***Proof*:**
The forward direction is proposition 2.7.14. For the converse, regularity gives the Jacobian rank condition at all k-rational points. The perfection of k ensures that this extends to all points: if k is perfect, then every finitely generated field extension $\kappa(x)/k$ is separably generated, and the cotangent space computation over $\kappa(x)$ agrees with the one over k. The details require the theory of Kähler differentials, which will be developed in section 5.5.

Example 2.7.16: Regularity $\neq$ Smoothness Over Imperfect Fields
Let $k = \mathbb{F}_p(t)$ (an imperfect field) and consider $X = \operatorname{Spec} k[x]/(x^p - t)$. Then X is a single closed point with residue field $k(\alpha)$ where $\alpha^p = t$. The local ring is $k(\alpha)$, which is a field and hence a regular local ring of dimension 0.

However, X is *not* smooth over k: the polynomial $f(x) = x^p - t$ has $f'(x) = px^{p-1} = 0$, so the Jacobian criterion fails. Geometrically, the extension $k(\alpha)/k$ is purely inseparable, and the fiber over $\operatorname{Spec} k$ is "infinitesimally thickened" from the perspective of k. This is the prototypical example of a regular but non-smooth scheme.

2.7.1 Unramified and Étale Morphisms

Among the smooth morphisms, those of relative dimension 0 are the relative analogue of a covering map: their fibres are finite and reduced, and cannot jump in dimension. Isolating this class calls for a companion to smoothness that sees the fibre direction alone, and that companion is unramifiedness, the vanishing of the relative cotangent sheaf. The two notions together pin down the étale morphisms, on whose theory the cohomological developments of the series rest.

The fibre of an unramified morphism is controlled by one fact about schemes over a field, which is recorded first.

Lemma 2.7.17: Vanishing Differentials over a Field

Let k be a field and Z a k-scheme locally of finite type with $\Omega_{Z/k} = 0$. Then Z has dimension 0, and near each point the scheme Z is the spectrum of a finite separable field extension of k. In particular Z is a disjoint union of spectra of finite separable extensions of k.

***Proof*:**
The claim is local, so take $Z = \operatorname{Spec} C$ with C a finitely generated k-algebra and $\Omega_{C/k} = 0$.

Dimension zero. Suppose $\dim C \geq 1$, and choose a minimal prime $\mathfrak{p}$ with $\dim C/\mathfrak{p} \geq 1$. The second exact sequence (proposition 5.5.6) applied to $C \twoheadrightarrow C/\mathfrak{p}$ exhibits $\Omega_{(C/\mathfrak{p})/k}$ as a quotient of $\Omega_{C/k} \otimes_C (C/\mathfrak{p}) = 0$, so $\Omega_{(C/\mathfrak{p})/k} = 0$. Writing $K = \operatorname{Frac}(C/\mathfrak{p})$, the module of differentials commutes with localization of the top ring, $\Omega_{K/k} \cong K \otimes_{C/\mathfrak{p}} \Omega_{(C/\mathfrak{p})/k}$, which is 0. But $C/\mathfrak{p}$ is a finitely generated domain of dimension ≥ 1, so $\operatorname{trdeg}(K/k) \geq 1$, and proposition 5.5.9 forces $\dim_K \Omega_{K/k} \geq 1$, a contradiction. Hence $\dim C = 0$.

Structure. A finitely generated k-algebra of dimension 0 is Artinian, hence a finite product $C \cong \prod_i C_i$ of local Artinian k-algebras, and each $\Omega_{C_i/k}$ is a direct factor of $\Omega_{C/k} = 0$. Fix one factor $(C_i, \mathfrak{m}_i, \ell_i)$. Applying proposition 5.5.6 to $C_i \twoheadrightarrow \ell_i$ exhibits $\Omega_{\ell_i/k}$ as a quotient of $\Omega_{C_i/k} \otimes_{C_i} \ell_i = 0$, so $\Omega_{\ell_i/k} = 0$, and since ℓ_i/k is finite by the Nullstellensatz this makes ℓ_i/k separable (proposition 5.5.9). With ℓ_i/k separable algebraic, proposition 5.5.10 gives $\mathfrak{m}_i/\mathfrak{m}_i^2 \cong \Omega_{C_i/k} \otimes_{C_i} \ell_i = 0$, so $\mathfrak{m}_i = 0$ by Nakayama's lemma and $C_i = \ell_i$ is a finite separable extension of k. Thus C is a finite product of finite separable field extensions of k, which is the assertion.

Definition 2.7.18: Unramified Morphism

A morphism $f\colon X \to S$ locally of finite type, with S locally Noetherian, is *unramified* if its sheaf of relative differentials vanishes,

$$\Omega_{X/S} = 0$$

The formation of $\Omega_{X/S}$ commutes with base change (proposition 5.5.16), so unramifiedness passes to every fibre: if f is unramified then $\Omega_{X_s/\kappa(s)} = 0$ for each $s \in S$, and lemma 2.7.17 identifies the fibre X_s as a disjoint union of spectra of finite separable extensions of $\kappa(s)$. This is the sense in which an unramified morphism neither thickens nor folds a fibre, since the maximal ideal upstairs is generated from downstairs and the residue extension carries no inseparability. The name is inherited from the arithmetic of Dedekind domains, where $\Omega_{X/S}$ is supported exactly on the ramification locus. It should not be confused with the ramification of a finite morphism of curves studied in section 7.4, nor with the unramified diagonal condition met in the theory of stacks.

Definition 2.7.19: Étale Morphism

A morphism $f\colon X \to S$ with S locally Noetherian is *étale* if it is flat and unramified, that is, flat, locally of finite type, and with $\Omega_{X/S} = 0$.

The definition packages étale as flat plus unramified, but in practice the class is recognized as the relative-dimension-zero case of smoothness. That the two descriptions agree is the substantive

equivalence.

Theorem 2.7.20: Étale Equals Smooth of Relative Dimension Zero

Let $f\colon X \to S$ be locally of finite type with S locally Noetherian. The following conditions are equivalent.

1. f is étale, that is, flat with $\Omega_{X/S} = 0$.
2. f is smooth of relative dimension 0 in the sense of definition 2.7.10.

When they hold, every fibre X_s is a disjoint union of spectra of finite separable extensions of $\kappa(s)$.

***Proof*:**

(2) implies (1). Suppose f is smooth of relative dimension 0. Flatness is proposition 2.7.12, which applies because S is locally Noetherian. For the differentials, work in a presentation as in definition 2.7.10: an open $U \subseteq X$ is realized as $\operatorname{Spec} \mathcal{O}_S(V)[x_1, \dots, x_r]/(g_1, \dots, g_r)$ over an affine open $V \subseteq S$, with the Jacobian $(\partial g_i/\partial x_j)$ of rank r at the point in question, relative dimension 0 making the number of variables equal to the number r of equations. The conormal sequence (proposition 5.5.18) for this closed immersion into affine space over V reads

$$\bigoplus_{i=1}^{r} \mathcal{O}_U \, \overline{g_i} \xrightarrow{J} \bigoplus_{j=1}^{r} \mathcal{O}_U \, dx_j \longrightarrow \Omega_{U/V} \longrightarrow 0,$$

with J the Jacobian. At the given point J has rank r, so $J \otimes \kappa$ is surjective onto the r-dimensional target, and Nakayama's lemma makes J surjective in a neighbourhood. Hence $\Omega_{X/S} = 0$ there, and as the point was arbitrary $\Omega_{X/S} = 0$.

(1) implies (2). Suppose f is flat with $\Omega_{X/S} = 0$, and fix $x \in X$ with image $s = f(x)$. Locally $X = \operatorname{Spec} B$ over an affine open $\operatorname{Spec} A \subseteq S$ with A Noetherian, and $B = A[x_1, \dots, x_N]/I$ is finitely generated. The conormal sequence (proposition 5.5.6) for $\operatorname{Spec} B \hookrightarrow \mathbb{A}^N_A$ is

$$I/I^2 \xrightarrow{\delta} \bigoplus_{j=1}^{N} B \, dx_j \longrightarrow \Omega_{B/A} = 0,$$

so δ is surjective. Passing to the residue field $\kappa = \kappa(x)$, the differentials of suitable elements $f_1, \dots, f_N \in I$ span the N-dimensional space $\bigoplus_j \kappa \, dx_j$, that is, the Jacobian $(\partial f_i/\partial x_j)(x)$ is invertible. Set

$$B' = A[x_1, \dots, x_N]/(f_1, \dots, f_N),$$

a quotient of $\mathbb{A}^N_A$ by N equations of full Jacobian rank at x, hence smooth over A of relative dimension 0 at x by definition 2.7.10. By the implication already proved, B' is flat over A with $\Omega_{B'/A} = 0$ near x.

The inclusion $(f_1, \dots, f_N) \subseteq I$ gives a surjection $\varphi\colon B' \twoheadrightarrow B$. Both fibres over s are, near x, finite separable field extensions of $\kappa(s)$: for B' this is lemma 2.7.17 applied to $\Omega_{X'_s/\kappa(s)} = 0$, and for B it is the same lemma applied to $\Omega_{X_s/\kappa(s)} = 0$, the latter being the base change (proposition 5.5.16) of $\Omega_{X/S} = 0$. Reducing φ modulo the maximal ideal $\mathfrak{m}_s$ of A at s gives a surjection of these fibre local rings at x, a surjection of fields, which is an isomorphism. Write $J' = \ker \varphi$ and

tensor $0 \to J' \to B' \to B \to 0$ with $\kappa(s)$. Flatness of B over A gives $\operatorname{Tor}_1^A(B, \kappa(s)) = 0$, so $J' \otimes_A \kappa(s)$ injects into $B' \otimes_A \kappa(s)$ with image the kernel of the fibre surjection, which vanishes at x. Nakayama's lemma applied to the finitely generated module J' over the local ring of B' at x then forces $J' = 0$ near x, so φ is an isomorphism there and f is smooth of relative dimension 0 at x. As x was arbitrary, (2) holds.

The fibre statement is lemma 2.7.17 applied to each $\Omega_{X_s/\kappa(s)} = 0$, which is the base change of $\Omega_{X/S} = 0$.

The theorem upgrades étaleness from the abbreviation "smooth of relative dimension 0" to a criterion in terms of flatness and differentials, and it is the equivalence on which the study of the étale topology rests when it treats étale maps that way. The hypothesis that S be locally Noetherian is inherited from proposition 2.7.12 and from the finite-type presentations above. Over a base that is not locally Noetherian the correct blanket condition is local finite presentation rather than finite type, and the flatness hypothesis must be read in that form. A statement at that generality is not proved here.

Example 2.7.21: Étale but Not an Open Immersion

Over a field k with $\operatorname{char} k \neq 2$, the squaring map $\operatorname{Spec} k[t, t^{-1}] \to \operatorname{Spec} k[u, u^{-1}]$ sending $u \mapsto t^2$ is étale. It is finite and flat, free of rank 2 with basis $1, t$, and $\Omega_{k[t,t^{-1}]/k[u,u^{-1}]}$ is generated by dt subject to $0 = du = 2t\,dt$, which forces $dt = 0$ since $2t$ is a unit, so the relative differentials vanish. Over a rational point $u = a$ the fiber is $\operatorname{Spec} k[t]/(t^2 - a)$, two rational points when a is a square in k and one point with residue field $k(\sqrt{a})$ otherwise, so the map is not injective over the squares and in no case an open immersion. It has no section over any nonempty open subset of $\operatorname{Spec} k[u, u^{-1}]$, since a section would give $t \in k(u)$ with $t^2 = u$. This gap between étale maps and open immersions is exactly what the étale topology is built to close.

The Singular Locus

Definition 2.7.22: Singular Locus

Let X be a locally Noetherian scheme. The *singular locus* of X is the set:

$$\operatorname{Sing}(X) = \{x \in X \mid \mathcal{O}_{X,x} \text{ is not a regular local ring}\}$$

Proposition 2.7.23: Singular Locus is Closed

Let X be a scheme locally of finite type over a field k. Then $\operatorname{Sing}(X)$ is a closed subset of X.

***Proof*:**

The question is local, so we may assume $X = \operatorname{Spec}(k[x_1, \ldots, x_n]/(f_1, \ldots, f_m))$. By the Jacobian criterion, the singular locus is cut out by the vanishing of all $(n - \dim X) \times (n - \dim X)$ minors of the Jacobian matrix, which defines a closed set.

For irreducible varieties, the singular locus is not just closed but *proper*:

Theorem 2.7.24: Generic Smoothness

Let X be an irreducible variety over an algebraically closed field k. Then $\mathrm{Sing}(X)$ is a proper closed subset of X. In particular, the smooth locus is a dense open subset.

***Proof*:**
See [22, Singular Points of Variety]. The key idea is that if X is a hypersurface $V(f)$ and $\mathrm{Sing}(X) = X$, then all partial derivatives of f vanish identically on X. In characteristic 0, this forces f to be constant. In characteristic $p > 0$, it forces $f = g^p$ for some polynomial g (using algebraic closure to take p-th roots), contradicting the irreducibility of f. The general case reduces to the hypersurface case via birational equivalence.

Over non-algebraically-closed fields, the correct generalization replaces "regular" with "smooth":

Corollary 2.7.25: Generic Smoothness Over Perfect Fields

Let X be a geometrically integral[a] k-scheme of finite type over a perfect field k. Then the smooth locus of $X \to \operatorname{Spec} k$ is a dense open subset.

[a] geometrically integral means change the base ring k to $\overline{k}$, in the sense of section 3.2

2.7.2 Properties of Regular Schemes

Beyond being domains, regular local rings enjoy two stronger finiteness properties:

Proposition 2.7.26: Regular Local Rings are Normal

Every regular local ring is integrally closed (normal).

***Proof*:**
This is [10, A Regular Local Ring Is Normal], where it is deduced from factoriality (proposition 2.7.27) together with the fact that a unique factorization domain is integrally closed.

Proposition 2.7.27: Regular Local Rings are UFDs

Every regular local ring is a unique factorization domain.

***Proof*:**
This is the theorem of Auslander and Buchsbaum, proved as [10, Regular Local Rings Are Factorial]. The argument is an induction on dimension: a parameter $x \notin \mathfrak{m}^2$ is a prime element because $R/(x)$ is again regular, the localizations $R_{\mathfrak{p}}$ at $\mathfrak{p} \neq \mathfrak{m}$ are regular of smaller dimension by the Auslander–Buchsbaum–Serre characterization via finite global dimension ([18, Auslander-Buchsbaum-Serre Theorem]), and factoriality descends from $R[1/x]$ by Nagata's criterion.

The chain of implications for Noetherian local domains is:

$$\text{regular} \implies \text{UFD} \implies \text{normal} \implies \text{regular in codimension 1}$$

None of these implications reverse in general (except in dimension 1, where regular, normal, and DVR are equivalent by [10, Characterizations of a DVR]).

Proposition 2.7.28: Regular Scheme and Irreducible Components

A regular scheme is a disjoint union of its irreducible components (each of which is integral).

***Proof*:**
Since each stalk is a domain, every point lies on a unique irreducible component. Hence distinct components cannot meet, and the scheme decomposes as a disjoint union. Each component is integral (irreducible and reduced) since regularity implies the stalks are domains.

2.7.3 Dimension and Regularity for Schemes Over $\mathbb{Z}$

An important feature of the scheme-theoretic framework is that the notion of regularity applies uniformly to schemes over any base, including $\operatorname{Spec}\mathbb{Z}$. This allows number-theoretic objects to be treated geometrically:

Example 2.7.29: Regularity of Arithmetic Schemes

1. $\operatorname{Spec}\mathbb{Z}$ is a regular scheme of dimension 1: for each prime (p), the local ring $\mathbb{Z}_{(p)}$ is a DVR (hence a regular local ring of dimension 1), and the generic point has local ring $\mathbb{Q}$, a field.
2. $\operatorname{Spec}\mathbb{Z}[i]$ is a regular scheme. At each maximal ideal $\mathfrak{p}$, the local ring $\mathbb{Z}[i]_{\mathfrak{p}}$ is a DVR (since $\mathbb{Z}[i]$ is a Dedekind domain).
3. $\operatorname{Spec}\mathbb{Z}[\sqrt{-3}]$ is *not* regular at the prime $\mathfrak{p} = (2, 1+\sqrt{-3})$: the ring $\mathbb{Z}[\sqrt{-3}]$ is a non-maximal order in $\mathbb{Q}(\sqrt{-3})$ whose conductor is exactly $\mathfrak{p}$, so the local ring $\mathbb{Z}[\sqrt{-3}]_{\mathfrak{p}}$ is a 1-dimensional local domain that is not integrally closed and hence not a DVR. The normalization (integral closure in the fraction field) is the ring of Eisenstein integers $\mathbb{Z}\left[\frac{1+\sqrt{-3}}{2}\right]$ (see [7, Class Groups]), which *is* a Dedekind domain and hence regular.

Exercise 2.7.1

1. Show that $\mathbb{A}^n_k$ is a regular scheme for any field k.
2. Let $X = \operatorname{Spec} k[x,y]/(y^2 - x^3 + x)$. Show that X is a regular scheme (assuming $\operatorname{char} k \neq 2$).
3. Show that the generic point of any integral scheme is a regular point.
4. Let $X = \operatorname{Spec} k[x,y,z,w]/(xw - yz)$ be the quadric cone (from example 2.2.2). Find $\operatorname{Sing}(X)$ and show it consists of a single point.
5. Let k be a perfect field and $f \in k[x_1, \ldots, x_n]$ irreducible. Show that $V(f)$ is smooth over k if and only if f and its partial derivatives have no common zero.

6. Show that the product of regular schemes is regular (hint: use that regularity is detected at stalks, and compute the stalks of a product).

7. Let R be a regular local ring and $x \in \mathfrak{m} \setminus \mathfrak{m}^2$. Show that $R/(x)$ is again a regular local ring (of dimension one less).

2.8⋆ *How Much Does the Topology Know?

⋆ *Optional. This section is an aside; nothing later in the book depends on it.*

Throughout this chapter, we have constructed schemes as locally ringed spaces: a topological space X together with a structure sheaf $\mathcal{O}_X$. The topological space carries the Zariski topology, which for affine schemes $\operatorname{Spec}(R)$ is entirely determined by the lattice of prime ideals of R. A natural question arises: how much of the scheme-theoretic structure is already determined at the level of topology?

One might initially expect a clean dichotomy: that the topological spaces arising from affine schemes form a special class, and that the topology of a non-affine scheme would look fundamentally "different"—perhaps failing some property that all affine topologies share. This turns out to be false, and understanding *why* it is false sharpens our appreciation for the role of the structure sheaf.

Spectral spaces

To state the result precisely, we need to isolate the topological properties that characterize the spaces $|\operatorname{Spec}(R)|$.

Definition 2.8.1: Spectral Space

A topological space X is a *spectral space* (or *coherent space*) if it satisfies the following four conditions:

1. X is quasi-compact.
2. X is sober: every irreducible closed subset has a unique generic point.
3. The quasi-compact open subsets of X are closed under finite intersection (i.e. X is quasi-separated, see definition 3.5.6) and form a basis for the topology.
4. Every non-empty irreducible closed subset of X has a generic point.

Condition (2) already implies (4), so the definition can be stated with just (1)–(3), but we include (4) for emphasis as it is the condition most often violated by "pathological" topological spaces. Note that condition (3) asks for two things at once: the quasi-compact opens form a basis (so there are enough of them), and a finite intersection of quasi-compact opens is again quasi-compact (a condition that can fail even for well-behaved spaces—it fails, for instance, for the Zariski topology on an infinite product of affine lines, though not for any Noetherian space).

The reader should recognize that every affine scheme satisfies these conditions. In $\operatorname{Spec}(R)$: the space is quasi-compact (section 2.1.3); it is sober by construction (generic points of irreducible closed sets correspond to prime ideals); and the distinguished open sets $D(f)$, which are quasi-compact, form a basis that is closed under finite intersection ($D(f) \cap D(g) = D(fg)$).

The fact, proved by Hochster in 1969, is that these conditions are not just necessary but *sufficient*.

Theorem 2.8.2: Hochster's Theorem

A topological space X is homeomorphic to $\mathrm{Spec}\,(R)$ for some commutative ring R if and only if X is a spectral space.

***Proof*:**
see [50]

The "only if" direction is the easy half: we have just verified that every $\mathrm{Spec}\,(R)$ is spectral. The "if" direction is: given any spectral space X, Hochster explicitly constructs a commutative ring R such that $|\mathrm{Spec}\,(R)| \cong X$. The construction uses the lattice of quasi-compact open subsets of X to build R as a quotient of a polynomial ring, with generators corresponding to the quasi-compact opens and relations encoding their intersection and covering behavior.

We do not reproduce Hochster's construction here, but the theorem has an immediate and perhaps surprising consequence.

Non-affine schemes with affine topology

Consider projective space $\mathbb{P}^n_k$ over a field k, with $n \geq 1$. This is emphatically not an affine scheme: its ring of global sections is just k, and $\mathrm{Spec}\,(k)$ is a single point, while $\mathbb{P}^n_k$ has many points. Yet the underlying topological space $|\mathbb{P}^n_k|$ is:

1. quasi-compact (it is covered by finitely many affine charts $U_i \cong \mathbb{A}^n$),
2. sober (it is a scheme, and every scheme is sober),
3. equipped with a basis of quasi-compact opens closed under finite intersection (the distinguished opens in each affine chart, together with their finite intersections, which are again quasi-compact by proposition 2.1.15).

Hence $|\mathbb{P}^n_k|$ is a spectral space, and by Hochster's theorem, there exists a commutative ring R with $|\mathbb{P}^n_k| \cong |\mathrm{Spec}\,(R)|$ as topological spaces. The topology of projective space is indistinguishable from that of an affine scheme.

This is not an isolated phenomenon. Any quasi-compact and quasi-separated scheme has an underlying spectral space, and hence is homeomorphic to $\mathrm{Spec}\,(R)$ for some ring R. This includes virtually all schemes encountered in practice: all Noetherian schemes, all schemes of finite type over a field or over $\mathbb{Z}$, all separated schemes.

Proposition 2.8.3: Topology of Quasi-Compact Quasi-Separated Schemes

Let X be a quasi-compact and quasi-separated scheme. Then $|X|$ is a spectral space, and hence there exists a commutative ring R such that $|X| \cong |\mathrm{Spec}\,(R)|$ as topological spaces.

***Proof*:**
Quasi-compactness of X gives condition (1). Every scheme is sober (a point $x \in X$ has closure $\overline{\{x\}}$ which is irreducible, and if Z is irreducible closed then its generic point is the unique point η with $\overline{\{\eta\}} = Z$; this follows from the corresponding fact for affine schemes, which glues). For condition (3): the quasi-compact opens of X include all affine opens and their finite unions, which form a basis. The quasi-separated hypothesis says exactly that the intersection of two quasi-compact opens is quasi-compact, giving closure under finite intersection.

Here are some examples.

Example 2.8.4: Non-Affine Schemes with Affine Topology

1. *Projective space.* As discussed above, $|\mathbb{P}^n_k|$ is spectral. Over $k = \overline{k}$ algebraically closed, the closed points of $\mathbb{P}^1_k$ are in bijection with $\overline{k} \cup \{\infty\}$, and the topology is the cofinite topology on this set together with a generic point whose closure is the whole space. This is homeomorphic to $|\operatorname{Spec}(k[x])|$ (which also has the cofinite topology on its closed points, plus a generic point). In particular, $|\mathbb{P}^1_k| \cong |\mathbb{A}^1_k|$ as topological spaces, even though $\mathbb{P}^1_k$ and $\mathbb{A}^1_k$ are not isomorphic as schemes.
2. *The line with doubled origin.* The non-separated scheme obtained by gluing two copies of $\mathbb{A}^1_k$ along $\mathbb{A}^1_k \setminus \{0\}$ is quasi-compact and quasi-separated, so its topology is spectral. This scheme is not affine (it is not even separated), yet topologically it is homeomorphic to $\operatorname{Spec}(R)$ for some ring R.
3. *The punctured plane.* The scheme $U = \mathbb{A}^2_k \setminus \{(x, y)\}$ from Example 2.2.53 is quasi-compact and quasi-separated (it is separated, being an open subscheme of an affine scheme), hence spectral. It is not affine, yet its topology is that of some $\operatorname{Spec}(R)$.

The constructible topology and Stone spaces

Hochster's theorem shows that a topological space is the spectrum of a ring if and only if it is spectral. But the definition of a spectral space involves a somewhat asymmetric collection of conditions. There is a cleaner perspective that explains *what structure a spectral space carries*, and it comes from a classical duality in topology and logic.

Given a spectral space X, define the *constructible topology* (also called the *patch topology*) on the underlying set of X by declaring both the quasi-compact open subsets *and their complements* to be open. Equivalently, the constructible topology is the coarsest topology in which every quasi-compact open of X is clopen (both open and closed).

Proposition 2.8.5: The Constructible Topology is a Stone Space

Let X be a spectral space. Then X equipped with the constructible topology is a *Stone space*: a topological space that is compact, Hausdorff, and totally disconnected.

***Proof*:**
(Sketch) The constructible topology is Hausdorff because any two distinct points of a spectral space can be separated by a quasi-compact open (one point lies in it, the other does not), which is clopen in the constructible topology. It is compact because the constructible topology is finer than the original spectral topology (which is already quasi-compact), and one can verify that any cover by constructible opens has a finite subcover using the quasi-compactness of the spectral topology together with the finiteness of intersections. Total disconnectedness follows because the clopen sets (which include all quasi-compact opens and their complements) separate points.

We denote the set X equipped with the constructible topology by X^{cons}. The identity map $X^{\mathrm{cons}} \to X$ is continuous (the constructible topology is finer), but not a homeomorphism: the constructible topology is strictly finer than the spectral topology whenever X has a non-closed point (i.e., whenever X is not discrete).

The name "constructible topology" is literal. A subset $C \subseteq X$ of a topological space is *constructible* if it belongs to the Boolean algebra generated by the open sets—equivalently, if C is a finite union of locally closed subsets ($U \cap Z$ where U is open and Z is closed). In a spectral space, the constructible subsets have a clean characterization:

Proposition 2.8.6: Constructible Sets are Clopens

Let X be a spectral space. A subset $C \subseteq X$ is constructible (in the classical sense: a finite Boolean combination of quasi-compact opens) if and only if C is clopen in the constructible topology.

This connects directly to Chevalley's theorem (definition 5.2.2): if $f : X \to Y$ is a morphism of finite type between Noetherian schemes, then the image of a constructible set is constructible. In the language of the constructible topology, this says that f induces a continuous map $X^{\mathrm{cons}} \to Y^{\mathrm{cons}}$ between the associated Stone spaces, and continuous maps between Stone spaces preserve clopens. From this perspective, Chevalley's theorem is a statement about the *continuity of scheme morphisms with respect to the constructible topology*—a viewpoint that is both conceptually cleaner and technically powerful.

Stone duality and the lattice of opens

The appearance of Stone spaces is not accidental; it reflects a duality from mathematical logic and lattice theory.

Stone duality (1936) establishes an anti-equivalence of categories between Boolean algebras and Stone spaces: every Boolean algebra B is the algebra of clopen subsets of a unique Stone space $S(B)$, and conversely, every Stone space arises in this way. Restricting to distributive lattices (which need not be Boolean), one obtains *spectral spaces*: a spectral space is precisely a Stone space together with a distinguished sublattice of its Boolean algebra of clopens.

More precisely, let X^{cons} be a Stone space. A sublattice $L \subseteq \mathrm{Clopen}(X^{\mathrm{cons}})$ that forms a basis for some topology on the underlying set, and is closed under finite intersection and finite union, determines a spectral topology on X: the topology generated by L. Different choices of L give different spectral topologies on the same underlying Stone space.

This gives a clean factorization of the data in a spectral space:

$$\text{Spectral space} \;=\; \text{Stone space} \;+\; \text{Choice of lattice.}$$

The Stone space is the "combinatorial skeleton"—it knows which subsets are constructible (i.e., definable by first-order formulas in model-theoretic language). The lattice records which constructible sets are "open," i.e., which direction the specialization order runs.

Example 2.8.7: Different Spectral Structures on the Same Stone Space

Consider the Cantor set $C \subseteq [0,1]$, which is a Stone space (compact, Hausdorff, totally disconnected). Every countable Boolean algebra has C as its Stone space. In particular, the spaces $|\mathrm{Spec}\,(\mathbb{Z})|$ and $|\mathrm{Spec}\,(k[x])|$ (for k algebraically closed) both have the same constructible topology—they are both countable spectral spaces whose Stone space is the Cantor set—but they carry different spectral topologies (different lattices of opens) corresponding to the different lattice structures of their prime ideals.

More concretely, both spaces consist of a generic point η together with countably many closed points, and the topology in both cases has $\{\eta\}$ dense and the closed points discrete in the complement. But the "arithmetic" of which opens contain which closed points differs: in $\mathrm{Spec}\,(\mathbb{Z})$, the closed point (p) lies in $D(q)$ for $q \neq p$, while in $\mathrm{Spec}\,(k[x])$, the closed point $(x-a)$ lies in $D(x-b)$ for $b \neq a$. These are different lattices on the same Stone space.

The three layers in the data of a scheme

The data of a scheme $(X, \mathcal{O}_X)$ decomposes into three layers:

1. *The Stone space* X^{cons}: the combinatorial skeleton. It knows the set of points, the constructible subsets, and (via Stone duality) the Boolean algebra they generate. This is the coarsest invariant—it is shared by many different schemes.
2. *The spectral topology*: a choice of sublattice L of the clopens of X^{cons}, determining which constructible sets are "open." This encodes the specialization order (which points are limits of which others) and hence the notion of "generic" and "closed" points. Different spectral topologies on the same Stone space give genuinely different spaces (e.g., $\mathrm{Spec}\,(\mathbb{Z})$ vs. $\mathrm{Spec}\,(k[x])$), but by Hochster's theorem, both are realizable as $\mathrm{Spec}\,(R)$ for some ring.
3. *The structure sheaf* $\mathcal{O}_X$: the algebraic data. It determines which continuous maps are "algebraic," what the global sections and cohomology are, and whether the scheme is affine, projective, smooth, etc. This is the finest invariant, and it carries the geometry.

The passage from (3) to (2) forgets the algebra; the passage from (2) to (1) forgets the specialization order. Neither passage is reversible: many non-isomorphic schemes share the same topology (as we saw with $\mathbb{P}^1_k$ and $\mathbb{A}^1_k$), and many non-homeomorphic spectral spaces share the same Stone space (as with $\mathrm{Spec}\,(\mathbb{Z})$ and $\mathrm{Spec}\,(k[x])$).

The conclusion is that the Zariski topology, while important for defining the notion of "local" and for the combinatorics of specialization and generization, carries far less information than one might expect. The structure sheaf is the essential datum: it determines which continuous maps are "algebraic," which open subsets are "distinguished," and what "functions" look like. Two schemes can share the same topological space—or even the same constructible topology—while having completely different algebraic geometry.

3

Schemes II: Morphisms and Constructions

Scheme morphisms are inherently more *algebraic* than *smooth* in nature: they are in some ways more stringent than holomorphic maps (only "polynomial-like" functions are allowed), but they also permit singularities. In this chapter, we shall properly explore morphisms of schemes, classify many important types, and interpret them geometrically. In the process, we will be able to "recover" the correct analogues of Hausdorffness and compactness for schemes in the form of *separatedness* and *properness*. The condition of being proper will turn out to be closely related to being projective: by Chow's lemma, every proper scheme over a Noetherian base admits a surjective birational morphism from a projective scheme, showing that projective schemes are the natural "compact schemes" to consider. The chapter concludes by defining an abstract variety and showing that $\mathbf{Var}_k$ is a fully faithful subcategory of $\mathbf{Sch}_k$.

Algebraically, smooth and holomorphic morphisms share an important feature with scheme morphisms: a continuous map $f : M \to N$ between smooth manifolds is a *smooth map* if and only if $f^*(C^\infty(N)) \subseteq C^\infty(M)$ where

$$f^* : C^\infty(N) \to C^\infty(M) \qquad f^*(\varphi) = \varphi \circ f$$

and in particular if f is a homeomorphism, it is a diffeomorphism if and only if f^* is a ring isomorphism. In essence, f "preserves" and is "compatible with" all the relations on N given by smooth functions. As schemes are defined by their algebraic relations, this pullback approach generalizes naturally[1].

With this in mind, we shall require the following properties from scheme morphisms:

1. A scheme morphism $f : X \to Y$ must preserve the structure sheaf (which encodes all algebraic

[1] In the smooth case, the "if and only if" makes this an equivalent definition of smooth map; since there is only one type of "affine patch" (open subsets of $\mathbb{R}^n$), one can then recover the usual definition. Proposition 3.1.10 establishes the analogous result for schemes.

information). Concretely, for every open set $V \subseteq Y$, there should be a ring homomorphism $f^{\#}(V) : \mathcal{O}_Y(V) \to \mathcal{O}_X(f^{-1}(V))$ that pulls back local sections on Y to local sections on X. The intuition is that we treat sections as genuine (scheme) functions, and we require their composition with f to remain a scheme function. Thus, a scheme morphism must be a *morphism of ringed spaces* (a continuous map together with a compatible sheaf map).

This additional structure is genuinely necessary: as we saw in example 2.1.29, not all continuous maps between spectra are induced by ring homomorphisms, and there is no natural way to recover a ring homomorphism from a mere continuous map. A simple instance is that $\operatorname{Spec} k$ and $\operatorname{Spec} k'$ for two non-isomorphic fields k, k' have homeomorphic underlying spaces (both are single points) yet are not isomorphic as schemes.

2. The pullback of functions must be compatible with evaluation: if $s \in \mathcal{O}_Y(V)$ vanishes at $f(x) \in Y$, then $f^{\#}(s)$ must vanish at $x \in X$. This means the pullback must respect the local ring structure at each point, making f a morphism of *locally* ringed spaces.

3. Ring homomorphisms between global sections and scheme morphisms between affine schemes must be dual to each other (generalizing the fact that a map between k-varieties is regular if and only if the contravariant map between coordinate rings is a k-algebra homomorphism).

This final condition means we want an equivalence of categories:

$$\mathbf{Sch}^{\mathbf{Aff}} \underset{\operatorname{Spec}(-)}{\overset{\Gamma(-)}{\rightleftarrows}} \mathbf{CRing}^{\mathrm{op}}$$

In section 3.3, this equivalence is generalized from affine schemes to all schemes by replacing $\mathbf{CRing}^{\mathrm{op}}$ with the appropriate category of sheaves of rings on a site (the functor-of-points perspective).

For those familiar with differential geometry [59], let us take a moment to compare and contrast smooth maps with scheme morphisms.

In differential geometry, continuous maps between manifolds can be arbitrarily well approximated by smooth maps (as a consequence of *Whitney's Approximation Theorem*). Hence, smooth maps can be thought of as "smoothing out" continuous maps – the smooth and continuous worlds are not far apart. This is *not at all* the case in the Zariski topology. Consider, for example, any permutation $\sigma : \overline{k} \to \overline{k}$ that preserves finite subsets (where $\overline{k}$ is an algebraically closed field). Such a σ induces a homeomorphism of $\mathbb{A}^1_{\overline{k}}$ in the Zariski topology – since the closed sets are exactly the finite sets and the whole space, any bijection preserving finiteness is continuous. Yet a "wild" permutation (one not given by a polynomial) will not be a scheme morphism[2]. Thus, there is no hope of approximating arbitrary continuous maps by regular maps: scheme morphisms are fundamentally more rigid than smooth maps.

In fact, regular maps behave much more like *holomorphic* maps. Both are determined by their local algebraic/analytic expressions, both satisfy analogues of the identity theorem (a regular map vanishing on a dense open must vanish everywhere), and both have fibers of controlled dimension. This similarity becomes an exact correspondence for the right categories: the GAGA theorems of Serre establish an equivalence between algebraic and analytic coherent sheaves on projective varieties over $\mathbb{C}$ (see section 6.9). Hence, when building geometric intuition for scheme morphisms via [59]

[2] Over $\mathbb{C}$, any discontinuous field automorphism $\sigma \in \operatorname{Aut}(\mathbb{C}/\mathbb{Q})$ gives such an example; these exist in abundance (assuming the axiom of choice) but none are $\mathbb{C}$-algebra maps.

and [58], it is holomorphic maps, not smooth maps, that provide the better mental model. A key difference is that scheme morphisms freely allow singularities and have unique algebraic-geometric features absent from the complex-analytic or smooth settings (for example, the Frobenius morphism from example 3.1.11, or the behavior of morphisms between schemes over $\operatorname{Spec}\mathbb{Z}$).

3.1 Elementary Concepts

Some of the discussion here shall overlap with the content in section 1.4; this was done as the self-completeness of the material in this section is a priority.

Recall (definition 1.4.4) that a morphism of ringed spaces $\pi : X \to Y$ is a continuous map of topological spaces together with a map $\pi^{\#} : \mathcal{O}_Y \to \pi_*\mathcal{O}_X$ that replicates composing any function in $\mathcal{O}_Y$ with π[3]. However, this notion is not restrictive enough for scheme theory for two reasons:

(i) Not all *continuous* maps between spectra are induced by ring homomorphisms (see Example 2.1.29), so specifying a sheaf map is genuinely additional data beyond the continuous map. Again, to abuse the same example, $\operatorname{Spec}(k)$ and $\operatorname{Spec}(k')$ for non-isomorphic fields k, k' are homeomorphic (both are single points) but carry different structure sheaves.

(ii) Even among ringed space morphisms, the global sections functor $\Gamma(-)$ is *not injective* on morphism sets: distinct ringed space morphisms can induce the same ring homomorphism on global sections. The following example demonstrates this.

Example 3.1.1: Two Ringed Space Morphisms for One Ring Homomorphism

Take $R = \mathbb{Z}_{(p)}$ and $S = \mathbb{Q}$ so that $\operatorname{Spec}(R) = \{[(p)], [(0)]\}$ and $\operatorname{Spec}(S) = \{[(0)]\}$. Let us drop the parentheses and square brackets to declutter notation. The open sets of $\operatorname{Spec}(R)$ are:

$$\emptyset, \quad \{0\}, \quad \{0, p\} = \operatorname{Spec}(R)$$

The stalks of $\operatorname{Spec}(R)$ are $\mathbb{Z}_{(p)}$ at p and $\mathbb{Q}$ at 0. The stalk of $\operatorname{Spec}(S)$ is $\mathbb{Q}$ at 0.

There is only one ring homomorphism $\psi : R \to S$, namely the inclusion $\mathbb{Z}_{(p)} \hookrightarrow \mathbb{Q}$.

In the category **Sch**, this should correspond to a unique map $\iota^a : \operatorname{Spec}(S) \to \operatorname{Spec}(R)$ defined by $\psi^{-1}((0)) = (0)$. However, in the category **RingedSp** we can define two morphisms between these affine schemes:

1. *Map 1 (The "correct" map):* $\pi_1 : \operatorname{Spec}(S) \to \operatorname{Spec}(R)$ defined by $0 \mapsto 0$. The sheaf map is determined by the stalk map at the unique point $0 \in \operatorname{Spec}(S)$:

$$(\pi_1)_0^{\#} : \mathcal{O}_{R,0} \to \mathcal{O}_{S,0} \qquad \Longrightarrow \qquad \mathbb{Q} \xrightarrow{\text{id}} \mathbb{Q}$$

   This is a local ring homomorphism (both sides are fields, with maximal ideal (0) mapping to (0)). On global sections, it induces the inclusion $\mathbb{Z}_{(p)} \hookrightarrow \mathbb{Q}$.

2. *Map 2 (The "phantom" map):* $\pi_2 : \operatorname{Spec}(S) \to \operatorname{Spec}(R)$ defined by $0 \mapsto p$. Since $\{p\}$ is closed, this is a continuous map. The sheaf map is determined by the stalk map at $0 \in \operatorname{Spec}(S)$:

$$(\pi_2)_0^{\#} : \mathcal{O}_{R,p} \to \mathcal{O}_{S,0} \qquad \Longrightarrow \qquad \mathbb{Z}_{(p)} \xrightarrow{\text{inclusion}} \mathbb{Q}$$

[3] Or equivalently by adjointness, the map $\pi^{-1}(\mathcal{O}_Y) \to \mathcal{O}_X$.

This is a valid ring homomorphism, so π_2 is a valid morphism of ringed spaces. On global sections, this *also* induces the inclusion $\mathbb{Z}_{(p)} \hookrightarrow \mathbb{Q}$.

Thus, we have found two distinct ringed space morphisms $\pi_1 \neq \pi_2$ that induce the *same* ring homomorphism on global sections. We reject π_2 because the stalk map $\mathbb{Z}_{(p)} \to \mathbb{Q}$ is *not* a local ring homomorphism: the maximal ideal $(p) \subseteq \mathbb{Z}_{(p)}$ maps to the unit ideal in $\mathbb{Q}$ (since p is invertible in $\mathbb{Q}$), violating the condition $\varphi(\mathfrak{m}_R) \subseteq \mathfrak{m}_S$. The definition of locally ringed space morphism correctly filters out this non-geometric map.

Thus, if $\mathbf{RingedSp}^{\mathbf{Aff}}$ denotes the category whose objects are $(\operatorname{Spec}(R), \mathcal{O}_{\operatorname{Spec}(R)})$ and whose morphisms are ringed space morphisms, the functor:

$$\Gamma(-) : \mathbf{RingedSp}^{\mathbf{Aff}} \to \mathbf{CRing}^{\mathrm{op}}$$

is surjective on hom-sets (every ring homomorphism $R \to S$ induces at least one ringed space morphism via the standard contraction-of-primes construction), but not injective (as the example above demonstrates).

The key realization is that ring homomorphisms induce local structure that must be preserved. By proposition 2.2.22, the stalks of schemes are all *local rings*. A map between local rings need not preserve maximal ideals (for instance, the inclusion $\mathbb{Z}_{(p)} \hookrightarrow \mathbb{Q}$ sends the maximal ideal (p) to the unit ideal). We thus single out maps that do preserve this structure:

Definition 3.1.2: Local Ring Homomorphism

Let $\varphi : (R, \mathfrak{m}_R) \to (S, \mathfrak{m}_S)$ be a ring homomorphism between local rings with maximal ideals $\mathfrak{m}_R$ and $\mathfrak{m}_S$ respectively. Then φ is a *local ring homomorphism* if $\varphi(\mathfrak{m}_R) \subseteq \mathfrak{m}_S$, or equivalently $\varphi^{-1}(\mathfrak{m}_S) = \mathfrak{m}_R$.

A ringed space morphism that is a local ring homomorphism at each stalk is given a name:

Definition 3.1.3: Locally Ringed Space Morphism

Let X, Y be locally ringed spaces. A *morphism of locally ringed spaces* $(\pi, \pi^{\#}) : X \to Y$ is a morphism of ringed spaces such that the induced stalk map $\pi_p^{\#} : \mathcal{O}_{Y,\pi(p)} \to \mathcal{O}_{X,p}$ is a local ring homomorphism for every $p \in X$.

Observe that ring homomorphisms *automatically* induce local maps on localizations: if $\varphi : R \to S$ is a ring homomorphism and $\mathfrak{p} \subseteq S$ is prime, then the induced map $\varphi_{\mathfrak{p}} : R_{\varphi^{-1}(\mathfrak{p})} \to S_{\mathfrak{p}}$ given by $r/t \mapsto \varphi(r)/\varphi(t)$ satisfies $\varphi_{\mathfrak{p}}(\mathfrak{m}_R) \subseteq \mathfrak{m}_S$. Indeed, if $r/t \in \mathfrak{m}_R$ then $r \in \varphi^{-1}(\mathfrak{p})$, so $\varphi(r) \in \mathfrak{p}$, giving $\varphi(r)/\varphi(t) \in \mathfrak{p}S_{\mathfrak{p}} = \mathfrak{m}_S$. Equivalently, $\mathfrak{m}_R = \varphi_{\mathfrak{p}}^{-1}(\mathfrak{m}_S)$[4]. Example 3.1.1 shows that this locality condition need *not* hold for arbitrary ringed space morphisms, even between locally ringed spaces. The local ring condition forces the morphism to "respect" not just the lattice structure of opens but also the evaluation of functions at each point.

With this added structure, we recover a bijection:

[4] Note this does not imply $\varphi(\mathfrak{m}_R) = \mathfrak{m}_S$: the inclusion $\mathbb{Z}_{(p)} \hookrightarrow \mathbb{Z}_{(p)}[1/q]$ for primes $p \neq q$ is a local ring homomorphism where $\varphi(\mathfrak{m}_R) \subsetneq \mathfrak{m}_S$.

Theorem 3.1.4: Ring Homomorphisms and Locally Ringed Space Morphisms

Let R, S be commutative rings. Then:

1. Every ring homomorphism $\varphi : R \to S$ induces a natural morphism of locally ringed spaces:
$$(f, f^{\#}) : (\operatorname{Spec}(S), \mathcal{O}_{\operatorname{Spec}(S)}) \to (\operatorname{Spec}(R), \mathcal{O}_{\operatorname{Spec}(R)})$$
2. Every morphism of locally ringed spaces $(\operatorname{Spec}(S), \mathcal{O}_{\operatorname{Spec}(S)}) \to (\operatorname{Spec}(R), \mathcal{O}_{\operatorname{Spec}(R)})$ induces a ring homomorphism $\varphi : R \to S$ (via global sections).
3. These operations are inverse to each other. In particular, the locally ringed space morphism is uniquely determined by the ring homomorphism on global sections:
$$\operatorname{Hom}_{\mathbf{LRS}}(\operatorname{Spec}(S), \operatorname{Spec}(R)) \cong \operatorname{Hom}_{\mathbf{CRing}}(R, S)$$

***Proof*:**
(1) Ring homomorphism ⇒ locally ringed space morphism. Let $\varphi : R \to S$ be a ring homomorphism. Define the continuous map $f : \operatorname{Spec}(S) \to \operatorname{Spec}(R)$ by the standard contraction $f(\mathfrak{p}) = \varphi^{-1}(\mathfrak{p})$.

We construct the sheaf morphism $f^{\#} : \mathcal{O}_{\operatorname{Spec}(R)} \to f_*\mathcal{O}_{\operatorname{Spec}(S)}$ on the basis of distinguished open sets, using proposition 1.3.3. For each $g \in R$, we must define a map

$$f^{\#}(D(g)) : \mathcal{O}_{\operatorname{Spec}(R)}(D(g)) \longrightarrow (f_*\mathcal{O}_{\operatorname{Spec}(S)})(D(g)) = \mathcal{O}_{\operatorname{Spec}(S)}(f^{-1}(D(g))).$$

The preimage of a distinguished open under f is again distinguished:

$$\begin{aligned} f^{-1}(D(g)) &= \left\{[\mathfrak{p}] \in \operatorname{Spec}(S) \;\middle|\; \varphi^{-1}(\mathfrak{p}) \in D(g)\right\} \\ &= \left\{[\mathfrak{p}] \in \operatorname{Spec}(S) \;\middle|\; g \notin \varphi^{-1}(\mathfrak{p})\right\} \\ &= \{[\mathfrak{p}] \in \operatorname{Spec}(S) \mid \varphi(g) \notin \mathfrak{p}\} \\ &= D(\varphi(g)) \end{aligned}$$

This is a special property of the affine setting: the pullback of a "basic" geometric object is again basic. Using the identifications $\mathcal{O}_{\operatorname{Spec}(R)}(D(g)) \cong R_g$ and $\mathcal{O}_{\operatorname{Spec}(S)}(D(\varphi(g))) \cong S_{\varphi(g)}$, the global map φ induces a map on localizations. Since $\varphi(g)$ is a unit in $S_{\varphi(g)}$, the universal property of localization provides a unique ring homomorphism:

$$\Phi_g : R_g \longrightarrow S_{\varphi(g)}, \qquad \frac{a}{g^n} \longmapsto \frac{\varphi(a)}{\varphi(g)^n}.$$

These maps are compatible with restriction: if $D(h) \subseteq D(g)$ (so $h \in \sqrt{(g)}$), the diagram

$$\begin{array}{ccc} R_g & \xrightarrow{\Phi_g} & S_{\varphi(g)} \\ \downarrow{\scriptstyle \mathrm{loc}} & & \downarrow{\scriptstyle \mathrm{loc}} \\ R_h & \xrightarrow{\Phi_h} & S_{\varphi(h)} \end{array}$$

commutes because both paths send a/g^n to $\varphi(a)/\varphi(g)^n$ viewed in $S_{\varphi(h)}$. By proposition 1.3.3, the maps Φ_g glue to a unique sheaf morphism $f^{\#}$.

At each stalk, the induced map $f_{\mathfrak{p}}^{\#} : R_{\varphi^{-1}(\mathfrak{p})} \to S_{\mathfrak{p}}$ is the localization of φ at $\mathfrak{p}$, which is a local ring homomorphism as shown above. Hence $(f, f^{\#})$ is a morphism of locally ringed spaces.

(2) Locally ringed space morphism $\Rightarrow$ ring homomorphism. Given a morphism of locally ringed spaces $(f, f^{\#}) : \operatorname{Spec}(S) \to \operatorname{Spec}(R)$, evaluate the sheaf map on global sections:

$$f^{\#}(\operatorname{Spec}(R)) : \mathcal{O}_{\operatorname{Spec}(R)}(\operatorname{Spec}(R)) \to \mathcal{O}_{\operatorname{Spec}(S)}(f^{-1}(\operatorname{Spec}(R)))$$

Since $\mathcal{O}_{\operatorname{Spec}(R)}(\operatorname{Spec}(R)) \cong R$ and $\mathcal{O}_{\operatorname{Spec}(S)}(\operatorname{Spec}(S)) \cong S$, this yields a ring homomorphism $\varphi : R \to S$.

(3) These operations are inverse. Starting with $\varphi : R \to S$, constructing $(f, f^{\#})$, and taking global sections recovers φ by construction.

Conversely, starting with a locally ringed space morphism $(f, f^{\#})$ and letting $\varphi = f^{\#}(\operatorname{Spec}(R)) : R \to S$, and g be the morphism defined in step (1) given by $g(\mathfrak{p}) = \varphi^{-1}(\mathfrak{p})$, we must show that $(f, f^{\#})$ equals the morphism $(g, g^{\#})$ constructed in step (1).

The continuous maps agree: Let $\mathfrak{p} \in \operatorname{Spec}(S)$. Consider the commutative diagram relating global sections to stalks:

$$\begin{array}{ccc} R & \xrightarrow{\varphi} & S \\ \downarrow{\scriptstyle \text{loc}} & & \downarrow{\scriptstyle \text{loc}} \\ R_{f(\mathfrak{p})} & \xrightarrow{f_{\mathfrak{p}}^{\#}} & S_{\mathfrak{p}} \end{array}$$

We compute $\varphi^{-1}(\mathfrak{p}) \subseteq R$ by tracing the preimage of $\mathfrak{m}_{S_{\mathfrak{p}}} \subseteq S_{\mathfrak{p}}$ along two paths:

- *Algebraic path (right then down):* The localization $S \to S_{\mathfrak{p}}$ pulls $\mathfrak{m}_{S_{\mathfrak{p}}}$ back to $\mathfrak{p}$. Then φ pulls $\mathfrak{p}$ back to $\varphi^{-1}(\mathfrak{p})$.
- *Geometric path (down then right):* Since $(f, f^{\#})$ is a morphism of *locally* ringed spaces, $f_{\mathfrak{p}}^{\#}$ is a local ring homomorphism, so $(f_{\mathfrak{p}}^{\#})^{-1}(\mathfrak{m}_{S_{\mathfrak{p}}}) = \mathfrak{m}_{R_{f(\mathfrak{p})}}$. The localization $R \to R_{f(\mathfrak{p})}$ pulls this maximal ideal back to $f(\mathfrak{p})$.

By commutativity, both paths yield the same ideal: $f(\mathfrak{p}) = \varphi^{-1}(\mathfrak{p}) = g(\mathfrak{p})$. Hence $f = g$ as continuous maps.

The sheaf maps agree: Since $f = g$, both $(f, f^{\#})$ and $(g, g^{\#})$ are sheaf morphisms over the same continuous map. To show $f^{\#} = g^{\#}$, it suffices to check equality on stalks ($\mathcal{O}_{\operatorname{Spec}(R)}$ is a sheaf, so morphisms out of it are determined by stalks). At each $\mathfrak{p} \in \operatorname{Spec}(S)$, both stalk maps are local ring homomorphisms $R_{\varphi^{-1}(\mathfrak{p})} \to S_{\mathfrak{p}}$ making the above diagram commute. Since the localization map $R \to R_{\varphi^{-1}(\mathfrak{p})}$ is an epimorphism in the category of rings (every element of $R_{\varphi^{-1}(\mathfrak{p})}$ is a fraction r/s with $r, s \in R$), any ring homomorphism out of $R_{\varphi^{-1}(\mathfrak{p})}$ that agrees with φ when precomposed with the localization map must be the unique localization $\varphi_{\mathfrak{p}}$. Hence $f_{\mathfrak{p}}^{\#} = g_{\mathfrak{p}}^{\#}$ for all $\mathfrak{p}$, giving $f^{\#} = g^{\#}$.

Affine schemes form the correct category to be the opposite category of commutative rings. We can rephrase what we just proved as:

Corollary 3.1.5: Equivalence of Rings and Affine Schemes

The category of affine schemes with locally ringed space morphisms is equivalent to the opposite category of commutative rings:

$$\mathbf{Sch}^{\mathbf{Aff}} \underset{\operatorname{Spec}(-)}{\overset{\Gamma(-)}{\rightleftarrows}} \mathbf{CRing}^{\mathrm{op}}$$

where Spec and Γ are inverse equivalences. More generally, for any locally ringed space X and any ring R, this equivalence extends to an adjunction on all of **LRS**:

$$\mathrm{Hom}_{\mathbf{CRing}}(R, \Gamma(X)) \cong \mathrm{Hom}_{\mathbf{LRS}}(X, \operatorname{Spec}(R))$$

where $\Gamma(-)$ is the left adjoint and $\operatorname{Spec}(-)$ is the right adjoint. On the subcategory of affine schemes, this adjunction restricts to an equivalence because both the unit $(X \to \operatorname{Spec}\Gamma(X))$ and the counit $(\Gamma(\operatorname{Spec}(R)) \to R)$ are isomorphisms.

Having shown that locally ringed morphisms are the right choice, let us take a moment to understand better how evaluation at a point is "preserved" by locally ringed space morphisms. Given a morphism of locally ringed spaces $\pi : X \to Y$, a point $q \in Y$ and a point $p \in X$ with $\pi(p) = q$, how do we interpret the local ring homomorphism between stalks? Say there is a section $g \in \mathcal{O}_Y(V)$ on an open subset $V \subseteq Y$ that vanishes at q, i.e., $g(q) = 0$ (equivalently, $g \in \mathfrak{m}_q$ in the stalk $\mathcal{O}_{Y,q}$). We want the pullback $\pi^{\#}(g)$ to vanish at p:

$$g(\pi(p)) = 0 \quad \Rightarrow \quad (\pi^{\#}g)(p) = 0.$$

The local ring condition ensures exactly this: since $\pi_p^{\#} : \mathcal{O}_{Y,q} \to \mathcal{O}_{X,p}$ satisfies $\pi_p^{\#}(\mathfrak{m}_q) \subseteq \mathfrak{m}_p$, the vanishing $g \in \mathfrak{m}_q$ implies $\pi^{\#}(g) \in \mathfrak{m}_p$, which means $\pi^{\#}(g)$ vanishes at p.

$$\begin{array}{ccc} \mathcal{O}_{Y,q} & \xrightarrow{\pi_p^{\#}} & \mathcal{O}_{X,p} \\ g \in \mathfrak{m}_q & \longmapsto & \pi^{\#}(g) \in \mathfrak{m}_p \end{array}$$

$$\begin{array}{ccc} Y \supseteq V & & X \supseteq \pi^{-1}(V) \\ \in & & \in \\ q = \pi(p) & \xleftarrow[\pi]{} & p \end{array}$$

The top row encodes the algebraic condition (the stalk map sends the maximal ideal into the maximal ideal); the bottom row encodes the geometric picture (the point p maps to q under π). The locally ringed space condition is the requirement that these two rows are compatible: vanishing is preserved under pullback. π is not injective, for each $p \in \pi^{-1}(q)$, there would be a local map.

This is "obvious" when g and π are our usual functions since $g(\pi(p)) = (g \circ \pi)(p)$. However, the phantom map from Example 3.1.1 shows that this can fail for general ringed space morphisms.

Consider for the last time:

$$\pi_2 : \operatorname{Spec}(\mathbb{Q}) \to \operatorname{Spec}(\mathbb{Z}_{(p)}), \qquad \pi_2([(0)]) = [(p)].$$

The function $p \in \mathcal{O}_Y(Y) = \mathbb{Z}_{(p)}$ vanishes at the closed point $[(p)]$: its value in the residue field $\kappa([(p)]) = \mathbb{Z}_{(p)}/(p) \cong \mathbb{F}_p$ is $p \bmod p = 0$. However, the pullback $\pi_2^{\#}(p)$ is $p \in \mathbb{Q}$ (via the inclusion $\mathbb{Z}_{(p)} \hookrightarrow \mathbb{Q}$), and the value of p at the unique point $[(0)] \in \operatorname{Spec}(\mathbb{Q})$ is $p \bmod (0) = p \neq 0$ in $\kappa([(0)]) = \mathbb{Q}$. The vanishing was not preserved.

3.1.6 Remark: the generic point of $\operatorname{Spec}(\mathbb{Z}_{(p)})$ The reader may find it surprising that $\{[(0)]\}$ is an open subset of $\operatorname{Spec}(\mathbb{Z}_{(p)})$: it is the distinguished open $D(p)$, the complement of the closed point $V(p) = \{[(p)]\}$. This is *not* typical behaviour for generic points. In $\operatorname{Spec}(\mathbb{Z})$ or $\operatorname{Spec}(k[x, y])$, the generic point is *not* open—its closure is the entire space. The peculiarity here is that $\mathbb{Z}_{(p)}$ is a local domain with exactly two prime ideals, so the space has only two points, and removing a closed point from a two-point space leaves an open point. This makes $\operatorname{Spec}(\mathbb{Z}_{(p)})$ the simplest non-trivial example for testing morphism conditions, precisely because the topology is so small that pathological maps are easy to construct explicitly.

3.1.7 Sections as Functions: Three Perspectives The reader may be troubled by a philosophical point: in differential geometry, a section $f \in C^\infty(U)$ *is* a function $f : U \to \mathbb{R}$, and evaluation at a point x is simply $f(x)$. For schemes, a section $f \in \mathcal{O}_X(U)$ is an element of a ring, and evaluation at x is the composite

$$\mathcal{O}_X(U) \longrightarrow \mathcal{O}_{X,x} \longrightarrow \kappa(x) = \mathcal{O}_{X,x}/\mathfrak{m}_x,$$

which is an additional construction. Is the datum of evaluation "missing" from the definition? The answer is no—it is encoded in the locally ringed space condition, but implicitly. There are three equivalent ways to see this.

(a) Hartshorne's approach. Define a section over U to be a function $f : U \to \coprod_{x \in U} \mathcal{O}_{X,x}$ such that $f(x) \in \mathcal{O}_{X,x}$ for each x, and such that f is "locally a quotient of ring elements." This makes sections into genuine functions and evaluation into literal function evaluation. The cost is that the codomain $\coprod \mathcal{O}_{X,x}$ varies from point to point and the definition requires a local coherence condition.

(b) The Modern approach. Keep sections as abstract ring elements. The locally ringed space axiom guarantees that each stalk $\mathcal{O}_{X,x}$ is local with a unique maximal ideal $\mathfrak{m}_x$, hence a unique residue field $\kappa(x)$, hence a unique evaluation map. The datum of evaluation is not "added"—it is a *consequence* of the locally ringed space structure. If the stalks were not local (i.e., if there were multiple maximal ideals), there would be multiple residue fields and no canonical notion of "the value of f at x." The LRS axiom is precisely the condition that evaluation is well-defined.

(c) The functor-of-points approach. By the representability of the global section functor (Proposition 3.3.9), a section $f \in \mathcal{O}_X(U)$ corresponds to a morphism $U \to \mathbb{A}^1_k$. In this picture, sections *are* morphisms, and evaluation at a point x is the composite $\operatorname{Spec}(\kappa(x)) \to U \to \mathbb{A}^1_k$, which is genuinely a function value.

All three perspectives carry exactly the same information. Approach (a) is most concrete; (b) is most algebraically efficient; (c) is most categorically natural. This text primarily uses (b), with (c) developed in section 3.3.

3.1.1 Scheme Morphisms

Inspired by the algebraic-geometric properties of locally ringed space morphisms, we define morphisms between schemes as locally ringed space morphisms:

Definition 3.1.8: Morphism of Schemes

Let X, Y be schemes. A morphism of locally ringed spaces $\pi : X \to Y$ is called a *morphism of schemes*. Schemes together with scheme morphisms form the category **Sch**.

One of the most important ways to control scheme morphisms is to reduce them to an affine cover. The following result is slightly more general but implies the desired result:

Lemma 3.1.9: Gluing Morphisms of Ringed Spaces

Let X, Y be ringed spaces and $X = \bigcup_i U_i$ an open cover. Suppose there exist ringed space morphisms $f_i : U_i \to Y$ that agree on overlaps:

$$f_i\big|_{U_i \cap U_j} = f_j\big|_{U_i \cap U_j}$$

Then there is a unique morphism of ringed spaces $f : X \to Y$ such that $f\big|_{U_i} = f_i$ for all i. Furthermore, if each f_i is a morphism of locally ringed spaces, then so is f.

***Proof*:**

Continuity. By the topological gluing lemma, the maps f_i glue to a unique continuous map $f : X \to Y$ with $f|_{U_i} = f_i$.

Sheaf morphism. We must construct $f^{\#} : \mathcal{O}_Y \to f_*\mathcal{O}_X$. For any open $V \subseteq Y$, we need a ring homomorphism $f^{\#}(V) : \mathcal{O}_Y(V) \to \mathcal{O}_X(f^{-1}(V))$.

The open set $f^{-1}(V) \subseteq X$ is covered by $\{f^{-1}(V) \cap U_i\}_i$. Since $f|_{U_i} = f_i$, we have $f_i^{-1}(V) = f^{-1}(V) \cap U_i$. The sheaf morphism $f_i^{\#} : \mathcal{O}_Y \to (f_i)_*\mathcal{O}_{U_i}$ gives, for each i, a ring homomorphism:

$$f_i^{\#}(V) : \mathcal{O}_Y(V) \longrightarrow \mathcal{O}_{U_i}(f^{-1}(V) \cap U_i)$$

On overlaps $f^{-1}(V) \cap U_i \cap U_j$, the agreement condition $f_i|_{U_i \cap U_j} = f_j|_{U_i \cap U_j}$ implies $f_i^{\#}|_{U_i \cap U_j} = f_j^{\#}|_{U_i \cap U_j}$, so:

$$f_i^{\#}(V)(s)\big|_{f^{-1}(V) \cap U_i \cap U_j} = f_j^{\#}(V)(s)\big|_{f^{-1}(V) \cap U_i \cap U_j} \qquad \forall s \in \mathcal{O}_Y(V)$$

The sets $\{f^{-1}(V) \cap U_i\}_i$ cover $f^{-1}(V)$, and the local images $f_i^{\#}(V)(s)$ are sections of $\mathcal{O}_X$ that agree on overlaps. By the gluability axiom of $\mathcal{O}_X$, there exists a unique section $t \in \mathcal{O}_X(f^{-1}(V))$ restricting to $f_i^{\#}(V)(s)$ on each $f^{-1}(V) \cap U_i$. Define $f^{\#}(V)(s) := t$. This is a ring homomorphism since each $f_i^{\#}(V)$ is, and it commutes with restriction by construction. Uniqueness of $f^{\#}$ follows from the locality axiom of $\mathcal{O}_X$.

Locally ringed space condition. If each f_i is an LRS morphism, then for any $p \in X$, p lies in some U_i and the stalk map $(f^{\#})_p = (f_i^{\#})_p$ is a local ring homomorphism by assumption. Hence f is an LRS morphism.

By lemma 3.1.9, we can control morphisms of schemes by looking at ring homomorphisms. We can even ensure the codomain is an affine scheme:

Proposition 3.1.10: Scheme Morphisms are Locally Affine

A morphism $\pi : X \to Y$ between schemes is a morphism of schemes if and only if it is *locally affine*: every point $x \in X$ has affine open neighborhoods $U = \operatorname{Spec} R \ni x$ and $V = \operatorname{Spec} S \ni \pi(x)$ with $\pi(U) \subseteq V$, such that $\pi|_U : U \to V$ is induced by a ring homomorphism $\varphi : S \to R$.

***Proof*:**

($\Rightarrow$) Scheme morphisms are locally affine. Let $\pi : X \to Y$ be a morphism of schemes and $x \in X$. Since Y is a scheme, there exists an affine open $V = \operatorname{Spec} S$ containing $\pi(x)$. By continuity, $\pi^{-1}(V)$ is an open neighborhood of x in X; since X is a scheme, it can be covered by affine opens, so there exists an affine open $U = \operatorname{Spec} R \ni x$ with $U \subseteq \pi^{-1}(V)$, hence $\pi(U) \subseteq V$.

The restricted map $\pi|_U : \operatorname{Spec} R \to \operatorname{Spec} S$ is a morphism of locally ringed spaces between affine schemes. By theorem 3.1.4, it is uniquely induced by the ring homomorphism on global sections:

$$\varphi := \Gamma(\pi|_U) : S \cong \mathcal{O}_Y(V) \xrightarrow{\pi|_U^{\#}(V)} \mathcal{O}_X(U) \cong R$$

($\Leftarrow$) Locally affine morphisms are scheme morphisms. Suppose $\pi : X \to Y$ is an LRS morphism and every $x \in X$ admits affine neighborhoods as above. Let $\{U_i = \operatorname{Spec} R_i\}$ be an open cover of X with corresponding affine opens $\{V_i = \operatorname{Spec} S_i\}$ in Y such that $\pi(U_i) \subseteq V_i$ and $\pi|_{U_i}$ is induced by a ring homomorphism $\varphi_i : S_i \to R_i$.

Each $\pi|_{U_i}$ is an LRS morphism (by theorem 3.1.4), so the stalk map at every $p \in U_i$ is a local ring homomorphism. Since $\{U_i\}$ covers X, the stalk map at every $p \in X$ is local. Hence π is an LRS morphism between schemes, i.e., a morphism of schemes.

Example 3.1.11: Morphisms Of Schemes

1. Recall the ring map $\varphi : \mathbb{C}[y] \to \mathbb{C}[x]$ given by $y \mapsto x^2$. Geometrically, this corresponds to the map $f : \mathbb{A}^1_x \to \mathbb{A}^1_y$ sending $a \mapsto a^2$. Let us verify the sheaf map on a specific open set. Consider the section $s = 3/(y-4) \in \mathcal{O}_{\mathbb{A}^1_y}(D(y-4))$, defined on the open set $V = D(y-4) = \mathbb{A}^1 \setminus \{4\}$. The preimage of this set is:

   $$f^{-1}(V) = \left\{a \in \mathbb{C} \;\middle|\; a^2 \neq 4\right\} = \mathbb{A}^1 \setminus \{-2, 2\}$$

   The induced map on sections $f_V^{\#} : \mathcal{O}_{\mathbb{A}^1_y}(V) \to \mathcal{O}_{\mathbb{A}^1_x}(f^{-1}(V))$ sends s to:

   $$f_V^{\#}(s) = \frac{3}{x^2 - 4}$$

   which is indeed a regular function on $f^{-1}(V) = D(x-2) \cap D(x+2)$.

2. Let $U \subseteq Y$ be an open subset of a scheme Y. There is a natural morphism of ringed spaces:

   $$\iota : (U, \mathcal{O}_Y|_U) \to (Y, \mathcal{O}_Y)$$

This is the open embedding as seen in definition 2.2.49. It is important to note the behavior of the stalks. For any point $p \in U$, the stalk map $\iota_p^\# : \mathcal{O}_{Y,\iota(p)} \to \mathcal{O}_{U,p}$ is an *isomorphism* (specifically, the identity map). This distinguishes open embeddings from other types of immersions.

3. *Tautological Projection (Affine Cone):* Consider the projection from the punctured affine space to projective space:

$$\pi : \mathbb{A}_k^{n+1} \setminus \{0\} \to \mathbb{P}_k^n \qquad (x_0, \dots, x_n) \mapsto [x_0 : \cdots : x_n]$$

This is a morphism of schemes. To see the sheaf map, we check it on the standard affine cover of $\mathbb{P}^n$. Let $U_i = D_+(x_i) \cong \mathbb{A}_k^n$ be the standard open charts of projective space. The preimage is the distinguished open set $\pi^{-1}(U_i) = D(x_i) \subseteq \mathbb{A}^{n+1} \setminus \{0\}$. The rings of sections are:

$$\mathcal{O}_{\mathbb{P}^n}(U_i) \cong k\left[\frac{x_0}{x_i}, \dots, \frac{x_n}{x_i}\right] \qquad \text{and} \qquad \mathcal{O}_{\mathbb{A}^{n+1}\setminus\{0\}}(D(x_i)) \cong k[x_0, \dots, x_n]_{x_i}$$

The sheaf map $\pi^\#(U_i)$ is the natural inclusion of the degree-0 subring into the full localization:

$$k\left[\frac{x_0}{x_i}, \dots, \frac{x_n}{x_i}\right] \hookrightarrow k[x_0, \dots, x_n]_{x_i}$$

Since these inclusions commute with further localization (restriction), they glue to form a valid sheaf morphism. Furthermore, the topological map π restricted to $D(x_i) \to U_i$ is exactly the map of spectra induced by this ring homomorphism (given by pulling back prime ideals $\mathfrak{p} \mapsto \mathfrak{p} \cap \mathcal{O}_{\mathbb{P}^n}(U_i)$), confirming this matched pair forms a valid morphism of schemes.

4. *Structure Map of Proj:* Let $S_\bullet$ be a finitely generated graded R-algebra (where $S_0 = R$). There is a natural structure morphism:

$$\pi : \operatorname{Proj} S_\bullet \to \operatorname{Spec}(R)$$

This is induced locally. On the basic open set $D_+(f) \subseteq \operatorname{Proj} S_\bullet$ (for homogeneous f), the ring of sections is $S_{(f)}$. The map $R \to S_\bullet$ induces a map $R \to S_{(f)}$ via $r \mapsto r/1$. These local maps glue to give the global morphism π.

5. Let $\varphi : S_\bullet \to R_\bullet$ be a *surjective* homomorphism of graded rings. This induces a *closed embedding* of schemes in the opposite direction:

$$f : \operatorname{Proj} R_\bullet \hookrightarrow \operatorname{Proj} S_\bullet$$

For example, the projection $k[x, y, z] \to k[x, y, z]/(x^2 + y^2 - z^2)$ induces the inclusion of the conic into $\mathbb{P}^2$.

6. Consider the projection $\pi : \operatorname{Spec}(\mathbb{C}[x, y]) \to \operatorname{Spec}(\mathbb{C}[x])$ induced by the inclusion $\mathbb{C}[x] \hookrightarrow \mathbb{C}[x, y]$.

   - *Closed Points:* A point $(x - 3, y - a)$ maps to the contraction $(x - 3)$. Geometrically, $(3, a) \mapsto 3$.
   - *Generic Points:* Consider the generic point of the parabola, $\eta = [(y - x^2)]$. The image is the contraction $(y - x^2) \cap \mathbb{C}[x]$. Since no nonzero polynomial in $\mathbb{C}[x]$ alone lies in $(y - x^2)$, the intersection is (0). Thus, the parabola maps to the generic point of the affine line!

7. Let $p \in X$ be a point of a scheme. We can form a scheme $\operatorname{Spec}(\mathcal{O}_{X,p})$. There is a canonical morphism $\operatorname{Spec}(\mathcal{O}_{X,p}) \to X$ induced by the localization maps $\mathcal{O}_X(U) \to \mathcal{O}_{X,p}$ for open neighborhoods U of p. For $X = \mathbb{A}^1_k$ and $p = (x)$, this is the map $\operatorname{Spec}(k[x]_{(x)}) \to \operatorname{Spec}(k[x])$.

   The scheme $\operatorname{Spec}(\mathcal{O}_{X,p})$ can be thought of as the "intersection of all open neighborhoods of p," but it is *not* an open embedding: $\operatorname{Spec}(\mathcal{O}_{X,p})$ has only the points of X that specialize to p (i.e., the prime ideals contained in $\mathfrak{m}_p$), which typically includes the generic point but not other closed points. The key property is that the stalk map at p is an isomorphism $\mathcal{O}_{X,p} \xrightarrow{\sim} \mathcal{O}_{\operatorname{Spec}(\mathcal{O}_{X,p}),\mathfrak{m}_p}$, since localizing a local ring at its maximal ideal has no effect.

8. *The Frobenius Morphism (Non-isomorphism bijection):* Let $X = \operatorname{Spec}(\mathbb{F}_p[t])$. Consider the ring homomorphism $\varphi : \mathbb{F}_p[t] \to \mathbb{F}_p[t]$ given by $t \mapsto t^p$. This induces a scheme morphism $F : X \to X$.

   - *Topologically:* The map on points is $\mathfrak{p} \mapsto \varphi^{-1}(\mathfrak{p})$. For a closed point $(t-a)$ where $a \in \overline{\mathbb{F}_p}$, we compute:

     $$\varphi^{-1}((t-a)) = \{f \in \mathbb{F}_p[t] \mid f(t^p) \in (t-a)\} = \{f \mid f(a^p) = 0\} = (t-a^p)$$

     Since $a^p = a$ in $\mathbb{F}_p$ (by Fermat's little theorem), we get $\varphi^{-1}((t-a)) = (t-a)$. Similarly, $\varphi^{-1}((0)) = (0)$. Thus, F is the *identity* on the underlying topological space.
   - *Sheaf-theoretically:* The map on sections is $t \mapsto t^p$. This is *not* an isomorphism: it is not surjective, as t is not in the image.

   Thus, F is a homeomorphism that is not an isomorphism of schemes. This distinction is invisible to the Zariski topology but detected by the structure sheaf.

   To give some insight on how we can extract information from this geometric perspective: once we define differentials in section 5.5, we shall see that $d(t^p) = pt^{p-1}\,dt = 0$, and hence the tangent map between tangent spaces will be the *zero map* at each point. We can think of this as "infinitesimal shrinking" at each point, while globally it is the identity!

If the codomain of a scheme morphism is an affine scheme, then we can completely determine the behaviour of the scheme morphisms by looking at the ring morphism between the global sections:

Proposition 3.1.12: Adjunction of Affine Schemes

Let X be a scheme and R a ring. Sending a morphism $\pi\colon X \to \operatorname{Spec}(R)$ to its ring map on global sections $\Gamma(\pi) = \pi^{\#}(\operatorname{Spec}(R))\colon R = \mathcal{O}_{\operatorname{Spec}(R)}(\operatorname{Spec}(R)) \to \mathcal{O}_X(X)$ is a bijection

$$\operatorname{Hom}_{\mathbf{Sch}}(X, \operatorname{Spec}(R)) \cong \operatorname{Hom}_{\mathbf{CRing}}(R, \mathcal{O}_X(X))$$

natural in X and in R: for a morphism $g\colon X' \to X$, $\Gamma(\pi \circ g) = g^{\#}(X) \circ \Gamma(\pi)$, and for a ring map $\psi\colon R' \to R$ with induced morphism $\operatorname{Spec}(\psi)\colon \operatorname{Spec}(R) \to \operatorname{Spec}(R')$ (theorem 3.1.4), $\Gamma(\operatorname{Spec}(\psi) \circ \pi) = \Gamma(\pi) \circ \psi$. The identity of $\mathcal{O}_X(X)$ corresponds to a morphism $\theta\colon X \to \operatorname{Spec}(\Gamma(X, \mathcal{O}_X))$, and θ is an isomorphism if and only if X is an affine scheme.

Note that the ring $\mathcal{O}_X(X)$ can be very complex, even in nice cases where X is a finite-type k-scheme, for example it need not be finitely generated.

***Proof*:**
The sheaf map of $\pi \circ g$ at $\operatorname{Spec}(R)$ is $g^{\#}(\pi^{-1}(\operatorname{Spec}(R))) \circ \pi^{\#}(\operatorname{Spec}(R)) = g^{\#}(X) \circ \Gamma(\pi)$, which is the naturality in X. Likewise the sheaf map of $\operatorname{Spec}(\psi) \circ \pi$ at $\operatorname{Spec}(R')$ is $\pi^{\#}(\operatorname{Spec}(R)) \circ \operatorname{Spec}(\psi)^{\#}(\operatorname{Spec}(R')) = \Gamma(\pi) \circ \psi$, as the map of $\operatorname{Spec}(\psi)$ on global sections is ψ (theorem 3.1.4), which is the naturality in R.

Injectivity. Let $\Gamma(\pi) = \Gamma(\pi')$. Let $U \subseteq X$ be an affine open with inclusion j_U and $\alpha \colon \operatorname{Spec}(S) \to U$ an isomorphism. By naturality the morphisms $\pi \circ j_U \circ \alpha$ and $\pi' \circ j_U \circ \alpha$ from $\operatorname{Spec}(S)$ to $\operatorname{Spec}(R)$ have the same ring map on global sections, so they are equal (theorem 3.1.4), and $\pi \circ j_U = \pi' \circ j_U$. The affine opens cover X, so $\pi = \pi'$ by the uniqueness in lemma 3.1.9.

Surjectivity. Let $\varphi \colon R \to \mathcal{O}_X(X)$. For each affine open U fix an isomorphism $\alpha_U : \operatorname{Spec}(S_U) \to U$, so that $\alpha_U^{\#}(U) \colon \mathcal{O}_X(U) \to S_U$ is a ring isomorphism. By theorem 3.1.4 there is a morphism $p_U : \operatorname{Spec}(S_U) \to \operatorname{Spec}(R)$ with $\Gamma(p_U) = \alpha_U^{\#}(U) \circ \operatorname{res}_{X,U} \circ \varphi$. Put $\pi_U = p_U \circ \alpha_U^{-1} \colon U \to \operatorname{Spec}(R)$. By naturality $\Gamma(\pi_U) = \operatorname{res}_{X,U} \circ \varphi$. For affine opens U, U' and an affine open $W \subseteq U \cap U'$, the restrictions of π_U and $\pi_{U'}$ to W both have ring map $\operatorname{res}_{X,W} \circ \varphi$, so they agree by injectivity applied to the scheme W. Such sets W cover $U \cap U'$, so π_U and $\pi_{U'}$ agree on $U \cap U'$, and lemma 3.1.9 glues the π_U to a morphism $\pi \colon X \to \operatorname{Spec}(R)$. For $r \in R$ the section $\Gamma(\pi)(r)$ restricts on each affine open U to $\Gamma(\pi_U)(r) = \varphi(r)|_U$, so $\Gamma(\pi) = \varphi$ by locality.

The morphism θ. If θ is an isomorphism, X is affine. Conversely, let $\alpha \colon \operatorname{Spec}(S) \to X$ be an isomorphism. By naturality, $\Gamma(\theta \circ \alpha) = \alpha^{\#}(X) \circ \Gamma(\theta) = \alpha^{\#}(X)$, a ring isomorphism $\mathcal{O}_X(X) \to S$. By theorem 3.1.4, $\theta \circ \alpha$ is the morphism induced by this ring isomorphism, an isomorphism by theorem 2.2.16. Hence $\theta = (\theta \circ \alpha) \circ \alpha^{-1}$ is an isomorphism, completing the proof.

The converse is not true:

Example 3.1.13: Maps From Affine To Projective
Take $\operatorname{Hom}_{\mathrm{Sch}}\left(\operatorname{Spec}\mathbb{Q}[x], \mathbb{P}^1_{\mathbb{Q}}\right)$ and $\operatorname{Hom}_{\mathbf{Ring}}(\mathbb{Q}, \mathbb{Q}[x])$ (recall that $\mathcal{O}_X\left(\mathbb{P}^1_{\mathbb{Q}}\right) \cong \mathbb{Q}$). Then

$$\operatorname{Hom}_{\mathrm{Ring}}(\mathbb{Q}, \mathbb{Q}[x]) = \{\mathrm{id}\}$$

namely since since the units of $\mathbb{Q}$ must map to the units of $\mathbb{Q}[x]$, which are all contains in $\mathbb{Q}$, and as it must be that $1 \mapsto 1$, we are forced to have the identity. However, there are multiple maps from $\operatorname{Spec}\mathbb{Q}[x]$ to $\mathbb{P}^1_k$; simply map $\operatorname{Spec}\mathbb{Q}[x]$ into the open subset $U_0 \cong k[y] \subseteq \mathbb{P}^1_k$ by taking $x \mapsto p(y)$ for some polynomial $p(y) \in k[y]$. More generally, we can let choose a field k[a]

The intuition is that $\mathcal{O}_X(X)$ in general contains less information than X as the global sections on a general scheme need to restrict to local sections on affine open subsets, and hence there may be in general less of them.

Show that the canonical morphism $X \to \operatorname{Spec}(\mathcal{O}_X(X))$ is an isomorphism if and only if X is a affine.

[a] Note that an [injective] ring homomorphism $k \to k$ need not be surjective, consider $F(x) \to F(x)$ given by $x \mapsto x^2$, hence the argument requires a bit more work

Using proposition 3.1.12, we can get a few more interesting results:

Example 3.1.14: Morphisms to Affine and Projective Space

1. *Maps to Affine Space are Tuples of Functions:* Let X be *any* scheme (not necessarily affine). What is a morphism $X \to \mathbb{A}^n_k$? Using the Adjunction property (proposition 3.1.12), we have:

$$\begin{aligned}\mathrm{Hom}_{\mathbf{Sch}}(X, \mathbb{A}^n_k) &\cong \mathrm{Hom}_{\mathbf{Sch}}(X, \operatorname{Spec} k[t_1, \ldots, t_n]) \\ &\cong \mathrm{Hom}_{\mathbf{Ring}}(k[t_1, \ldots, t_n], \mathcal{O}_X(X))\end{aligned}$$

   A ring homomorphism from a polynomial ring is uniquely determined by where it sends the variables t_i. Let f_i be the image of t_i in $\Gamma(X, \mathcal{O}_X)$. Thus, a morphism to affine space is exactly equivalent to choosing n global functions $(f_1, \ldots, f_n)$ on X. This generalizes the classical fact that a map to $\mathbb{R}^n$ is given by n coordinate functions.

   Notice how this also shows that there is no non-trivial map $\mathbb{P}^n_k \to \mathbb{A}^m_k$ (exercise 3.1.2(3.1.2 (4))).

   s

2. *Maps to Projective Space (The need for Line Bundles):* One might hope to define a map $X \to \mathbb{P}^n_k$ similarly by choosing $n+1$ global functions $f_0, \ldots, f_n$ to serve as homogeneous coordinates $[f_0, \ldots, f_n]$. This naturally fails when using the structure sheaf since global functions on projective space are just constants, which would make such maps trivial (as illustrated in example 3.1.13). Instead, the coordinate functions x_i on $\mathbb{P}^n$ are sections of a twisted sheaf, not the structure sheaf. Consequently, to define a scheme morphism to projective space, one requires a *Line Bundle* (an invertible sheaf $\mathcal{L}$) on X and $n+1$ global sections $s_i \in \Gamma(X, \mathcal{L})$ that do not vanish simultaneously. This distinction – that maps to affine space use functions, while maps to projective space use bundles – is the primary motivation for the study of *Invertible Sheaves* in section 5.3.3.

Our last result in this section is a special case where gluing affine schemes *does* produce an affine scheme:

Corollary 3.1.15: Characterizing Affine Schemes Locally

Let X be a scheme. Then X is an affine scheme if and only if there is a finite subset of elements $f_1, \ldots, f_n \in \mathcal{O}_X(X)$ such that X_f is affine and $(f_1, \ldots, f_n) = \mathcal{O}_X(X)$

***Proof*:**
If X is affine, take $f_1 = 1$. Conversely, assume there exist $f_1, \ldots, f_n \in \mathcal{O}_X(X)$ such that each X_{f_i} is affine and $(f_1, \ldots, f_n) = \mathcal{O}_X(X)$. This ideal condition implies there exist $a_i \in \mathcal{O}_X(X)$ such that $\sum_{i=1}^n a_i f_i = 1$. Let $R = \mathcal{O}_X(X)$. We claim that $X \cong \operatorname{Spec} R$.

By proposition 3.1.12, the identity homomorphism on R induces a canonical morphism of schemes:

$$\theta : X \longrightarrow \operatorname{Spec} R$$

To show θ is an isomorphism, it suffices to verify it locally on an open cover of the target $\operatorname{Spec} R$. The condition $\sum_{i=1}^n a_i f_i = 1$ ensures that the distinguished open sets $D(f_1), \ldots, D(f_n)$ form an open cover of $\operatorname{Spec} R$.

For each i, the preimage of $D(f_i)$ under θ is precisely the locus of points in X where the global

section f_i does not vanish, which is by definition X_{f_i}. Since the $D(f_i)$ cover $\operatorname{Spec} R$, their preimages X_{f_i} necessarily cover X.

Consider the restriction of θ to one of these patches:

$$\theta|_{X_{f_i}} : X_{f_i} \longrightarrow D(f_i) \cong \operatorname{Spec}(R_{f_i})$$

By our hypothesis, X_{f_i} is an affine scheme. Because the global sections of a scheme on a distinguished open set defined by a global function f_i are given by the localization of the global sections ring, we have

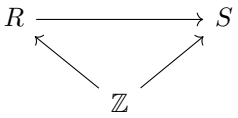
$$\mathcal{O}_X(X_{f_i}) \cong R_{f_i}$$

Thus, by the Adjunction property (corollary 3.1.5), the morphism $\theta|_{X_{f_i}}$ corresponds exactly to the identity ring isomorphism $R_{f_i} \xrightarrow{\sim} R_{f_i}$. Hence, each restriction $\theta|_{X_{f_i}}$ is an isomorphism of schemes.

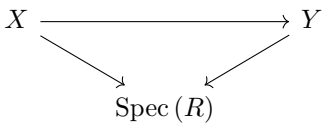
Since θ is a morphism of schemes that restricts to an isomorphism on the sets of an open cover $\{X_{f_i}\}$ mapping to an open cover $\{D(f_i)\}$, we can apply the gluing lemma for morphisms (lemma 3.1.9). The local inverses glue uniquely to form a global inverse, proving that θ is an isomorphism and $X \cong \operatorname{Spec} R$.

3.1.2 S-Schemes

Recall that all abelian rings can be seen as $\mathbb{Z}$-algebras, and that $\mathbb{Z}$ is initial in the category of commutative rings. From this perspective, all ring homomorphism $\varphi : R \to S$ are in bijective correspondence with morphisms that make the following diagram commute:



Then we may generalize the notion of abelian groups to R-algebras by replacing $\mathbb{Z}$ with a ring R. We may further augment the restriction morphisms to R-algebra homomorphisms as we are working with commutative rings; this was done in definition definition 2.2.18. Now, this structure was given by a scheme morphism $X \to \operatorname{Spec} R$, and an R-scheme morphism is a scheme morphism between R-schemes $X \to Y$ where the following diagram commutes:



This is the commutative diagram of fiber-preserving morphism, the same requirement that is imposed of fiber-bundle morphisms. And indeed, R-schemes behave very much like bundles over $\operatorname{Spec} R$ (as the examples bellow will demonstrates), and when the morphism $X \to \operatorname{Spec} R$ is nice enough (étale, in the sense developed in *Étale Cohomology* [16]), then it shall even be locally trivial.

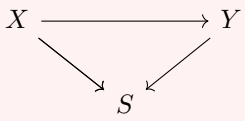
Thus, it is natural to consider the base space to be an arbitrary scheme:

Definition 3.1.16: S-Scheme

Let $\mathbf{Sch}_S$ be the category who objects are scheme morphisms:

$$X \to S$$

and whose morphisms are commutative diagrams:



As compositions, $f \circ \pi_X = \pi_Y$. The objects of this category or called *S-scheme*. The morphism of schemes $X \to S$ is called the *structure morphism*.

To see what extra structure is added, let us start the case of $S = \operatorname{Spec}(R)$. Then each open subset of X has the structure of an R-algebra, and these generalize definition 2.2.18. If $S = \operatorname{Spec}(R)$, then we would often say that the S-scheme (i.e. $\operatorname{Spec}(R)$-scheme) is an R-scheme

Proposition 3.1.17: Final Object Of Scheme

In **Sch**, the scheme $\operatorname{Spec}\mathbb{Z}$ is the final object. This makes the category of schemes isomorphic to the category of $\mathbb{Z}$-schemes.

If k is a field, $\operatorname{Spec} k$ is the final object in the category of k-schemes.

***Proof*:**
A ring map $\varphi\colon \mathbb{Z} \to A$ satisfies $\varphi(1) = 1$, hence $\varphi(n) = n \cdot 1_A$ for every $n \in \mathbb{Z}$ by additivity, and $n \mapsto n \cdot 1_A$ is a ring map. So $\operatorname{Hom}_{\mathbf{CRing}}(\mathbb{Z}, A)$ has exactly one element for every ring A.

Step 1: $\operatorname{Spec}\mathbb{Z}$ is final. For a scheme X, proposition 3.1.12 gives a bijection $\operatorname{Hom}_{\mathbf{Sch}}(X, \operatorname{Spec}\mathbb{Z}) \cong \operatorname{Hom}_{\mathbf{CRing}}(\mathbb{Z}, \mathcal{O}_X(X))$, whose right side has exactly one element. So there is exactly one morphism $\pi_X : X \to \operatorname{Spec}\mathbb{Z}$.

Step 2: schemes and $\mathbb{Z}$-schemes. The functor from $\mathbb{Z}$-schemes (definition 3.1.16 with $S = \operatorname{Spec}\mathbb{Z}$) to schemes that forgets the structure morphism is bijective on objects, since each scheme X has exactly one structure morphism π_X. A morphism of $\mathbb{Z}$-schemes $X \to Y$ is a morphism of schemes f with $\pi_Y \circ f = \pi_X$, and every morphism of schemes satisfies this equation, both sides being morphisms $X \to \operatorname{Spec}\mathbb{Z}$. So the functor is also bijective on morphisms, hence an isomorphism of categories, with inverse $X \mapsto (X, \pi_X)$.

Step 3: k-schemes. In the category of $\operatorname{Spec} k$-schemes of definition 3.1.16, a morphism from (X, π_X) to $(\operatorname{Spec} k, \mathrm{id})$ is a morphism $f\colon X \to \operatorname{Spec} k$ with $\mathrm{id} \circ f = \pi_X$, so $f = \pi_X$ is the only one. Hence $(\operatorname{Spec} k, \mathrm{id})$ is the final object, as claimed.

Working over S-schemes shall restrict to much "simpler" cases than over general schemes. For

example, if studying complex analysis and complex varieties, we would expect $\mathrm{Spec}(\mathbb{C})$ to have trivial automorphisms, and we may want to interpret $\mathrm{Spec}\left(\mathbb{R}[x]/\left(x^2+1\right)\right)$ without too much "gluing" done by the automorphism ring, namely we want $\mathbb{C}$ be be trivial.

Example 3.1.18: The Category of S-schemes

An *S-scheme* is a scheme X equipped with a morphism $\pi : X \to S$, called the structure morphism. A morphism of S-schemes $f : X \to Y$ is a morphism of schemes such that the triangle with the base S commutes .

1. *Schemes over k ("Geometry" Schemes)*: If $S = \mathrm{Spec}(k)$ for a field k, an S-scheme closely resembles an algebraic set (over k). By proposition 3.1.12, the structure morphism $X \to \mathrm{Spec}(k)$ is equivalent to the data of a k-algebra homomorphism $k \to \Gamma(X, \mathcal{O}_X)$. This fixes the field of scalars, ensuring that the functions on X are k-algebras, not just rings.

   When working with polynomials, we are often interested in k-linear homomorphisms $k[x] \to k[x]$ where k is fixed (e.g., coordinate changes). If we only looked at ring homomorphisms, include field automorphisms (like complex conjugation or the Frobenius). We enforce linearity by requiring the algebra diagram to commute:

   $$\begin{array}{ccc} k[x] & \xrightarrow{\varphi} & k[x] \\ & {\scriptstyle\iota}\nwarrow \quad \nearrow{\scriptstyle\iota} & \\ & k & \end{array}$$

   In the language of schemes, this corresponds to a morphism of S-schemes (where $S = \mathrm{Spec}(k)$):

   $$\begin{array}{ccc} \mathrm{Spec}(k[x]) & \xrightarrow{f} & \mathrm{Spec}(k[x]) \\ & {\scriptstyle\pi_X}\searrow \quad \swarrow{\scriptstyle\pi_Y} & \\ & \mathrm{Spec}(k) & \end{array}$$

   The condition $f \circ \pi_X = \pi_Y$ ensures that the morphism preserves the underlying field structure.

2. *Projective Schemes:* Projective schemes over a field, $\mathbb{P}^n_k = \mathrm{Proj}(k[x_0, \ldots, x_n])$, are naturally k-schemes. The structure map $\mathbb{P}^n_k \to \mathrm{Spec}(k)$ is induced by the inclusion of the degree 0 scalars $k \hookrightarrow k[x_0, \ldots, x_n]$.

3. *Field Extensions as Coverings:* Let L/K be a finite field extension. Then $\mathrm{Spec}(L)$ and $\mathrm{Spec}(K)$ are both single points topologically. However, the inclusion $K \hookrightarrow L$ induces a morphism of schemes $\pi : \mathrm{Spec}(L) \to \mathrm{Spec}(K)$. We do not want to include morphisms permuting K, hence we limit to k-scheme morphisms While topologically trivial, this morphism encodes the arithmetic complexity of the extension.

3.1.19 Foreshadowing Étale Fundamental Group Later, we shall see that $\mathrm{Spec}(L)$ acts as a "covering space" for $\mathrm{Spec}(K)$. This analogy can be made more precise K, L be perfect fields and let L/K be a finite extension. Then $L = k[\alpha]$ for a root of the appropriate irreducible polynomial $m_{\alpha,L/K}$. Then taking advantage of the fact that the category of K-field extensions is *isomorphic* (not just equivalent) to the category of rings of integers with base ring $\mathcal{O}_K$ (exercise if the reader forgot) The morphism $\mathrm{Spec}\,(L) \to \mathrm{Spec}\,(K)$ is associated to a (dense) morphism $\mathrm{Spec}\,(\mathcal{O}_L) \to \mathrm{Spec}\,(\mathcal{O}_K)$ (as $\mathcal{O}_K \to \mathcal{O}_L$ is injective). In section 3.5.1, we shall see morphism between affine schemes are uniquely determined by their value on dense sets. As $\mathcal{O}_L$ and $\mathcal{O}_K$ are both dimension 1 rings, these are *covering of curves*. This covering hints at the generalization of covering spaces from algebraic topology, leading to the *Étale Fundamental Group*, which generalizes the Galois group $\mathrm{Gal}(L/K)$ to schemes, allowing us to study loops and homotopy in an arithmetic setting.

Lemma 3.1.20: Stalks of k-schemes

Let X be a k-scheme and $x \in X$ so that $x = [\mathfrak{p}]$ for some prime $\mathfrak{p} \subseteq R_i = k\left[u_i\right]_i$. Then

$$\mathcal{O}_{X,\mathfrak{p}} \cong \left(k\left[u_i\right]_i\right)_{\mathfrak{p}}$$

In particular, we have that the quotient by $\mathfrak{m}_x$ is a k-algebra:

$$\kappa(x) = k(x)$$

***Proof*:**
As X is a k-scheme, each $\mathcal{O}_X(U)$ is a k-algebra, and the colimit of k-algebra's is a k-algebra (the stalk colimit is filtered, and a filtered colimit of k-algebras is again a k-algebra). The rest comes from proposition 1.1.5.

Note that the residue may be other fields, ex. $\mathbb{R}(x) = \mathbb{C}$ (take $\mathrm{Spec}\,\mathbb{R}[t]$ where and choose a degree > 1 point).

One of the most important first impositions for many algebraic manipulations are finiteness conditions:

Definition 3.1.21: Locally Finite Type And Finite Type

Let X, Y be S-schemes. Then a morphism $f : X \to Y$ is locally of finite type if there exists an affine open cover $V_j = \mathrm{Spec}\,(S_j)$ of Y where $f^{-1}(V_i)$ can be covered by affine open subsets $U_{ij} = \mathrm{Spec}\,(A_{ij})$ where A_{ij} is a finitely generated S_i-algebra, that is

$$f^{\#}(V_i) : S_j = \mathcal{O}_Y(V_j) \to \mathcal{O}_X\left(f^{-1}(V_i)\right) \xrightarrow{\text{res}} \mathcal{O}_Y(U_{ij}) = R_{ij}$$

induces a finitely generated S_j-algebra on each R_{ij}. If furthermore each $f^{-1}(V_j)$ can be covered by finite number of affine subsets, f is said to be of finite type.

An S-scheme X is said to be of locally finite type (resp. finite type) over S if $f : X \to S$ is of locally finite type (resp. finite type).

3.1.3 Subschemes: Closed, Locally Closed, and More

Open subschemes (section 2.2.8) are the simplest type of subscheme: given an open subset $U \subseteq X$, the restriction $(U, \mathcal{O}_X|_U)$ carries a unique scheme structure compatible with X. The situation for closed subsets is fundamentally different: a closed subset can carry *multiple* scheme structures, distinguished by the amount of "infinitesimal data" they retain. Beyond the open and closed cases, there are subschemes that are locally closed (closed inside an open subset), and there are natural "local" subobjects that do not correspond to any topological subset at all.

The reader may remark that sub-objects usually correspond to monomorphisms. While the definition is formally definable the resulting schemes for **Sch**

Closed subschemes and closed immersions

Given a surjection $R \twoheadrightarrow R/I$, by proposition 2.1.30 the image $\operatorname{Spec}(R/I) \subseteq \operatorname{Spec}(R)$ is a closed subset, and the inclusion is a morphism of locally ringed spaces. For a general scheme, this condition generalizes to requiring that the sheaf map be surjective.

Consider the closed point $\{0\} \subseteq \mathbb{A}^1_k$. As a topological subspace, $\{0\}$ is just a point. But there are infinitely many scheme structures on this point compatible with the ambient scheme: $\operatorname{Spec}(k[x]/(x))$, $\operatorname{Spec}(k[x]/(x^2))$, $\operatorname{Spec}(k[x]/(x^3))$, and so on. These differ in how much nilpotent (infinitesimal) structure they retain. The underlying reason is elementary: $0^n = 0$ for all n, so the vanishing locus of x and x^n are identical as subsets, but $k[x]/(x^n)$ remembers the first $n-1$ derivatives. In contrast, open subschemes have a *unique* compatible scheme structure: the restriction of $\mathcal{O}_X$ to the open set.

Definition 3.1.22: Closed Immersion and Closed Subscheme

Let $f : Y \to X$ be a morphism of schemes. Then f is a *closed immersion* if:

1. f induces a homeomorphism from $|Y|$ onto a closed subset of $|X|$, and
2. the sheaf map $f^\# : \mathcal{O}_X \to f_*\mathcal{O}_Y$ is surjective (equivalently, surjective on stalks).

A *closed subscheme* of X is a closed subset $Z \subseteq |X|$ together with a scheme structure on Z such that the inclusion $\iota : Z \hookrightarrow X$ is a closed immersion.

The distinction between a closed immersion and a closed subscheme is that a closed immersion $f : Y \to X$ is a morphism (which might come with an isomorphism $Y \cong Z$ for some closed subscheme Z), while a closed subscheme is always given with the literal inclusion as the map. In practice, the two are used interchangeably.

When define closed subsets of affine schemes, we used $V(-)$. This construction can also be used to construct closed subschemes of an affine scheme, but it shall forget the sheaf-theorem information (ex. nilpotence). This shall be recovered in definition 5.1.36 by inputting *ideal shaves* in $V(-)$.

Note that the complement of a closed subscheme is a unique open subscheme (with its canonical structure), while the complement of an open subscheme admits *multiple* closed subscheme structures, usually infinitely many, corresponding to different choices of nilpotent thickening. This asymmetry between open and closed is a recurring theme. There is a natural choice of closed subscheme structure once nilpotent information is removed, the *reduced induced structure* of proposition 3.1.25.

Example 3.1.23: Closed Subschemes

1. *A point with its reduced structure.* The inclusion $\{0\} \hookrightarrow \mathbb{A}^1_k$ corresponds to the surjection $k[x] \twoheadrightarrow k[x]/(x) \cong k$. The image is closed, the sheaf map is surjective (and remains so on all localizations), so this is a closed immersion.

2. *A fat point.* The inclusion $\operatorname{Spec}(k[x]/(x^2)) \hookrightarrow \mathbb{A}^1_k$ corresponds to $k[x] \twoheadrightarrow k[x]/(x^2)$. The image is the same closed point $\{0\}$, but the scheme structure is different: the coordinate ring $k[x]/(x^2) \cong k[\epsilon]/(\epsilon^2)$ remembers one infinitesimal direction. More generally, $\operatorname{Spec}(k[x]/(x^n)) \hookrightarrow \mathbb{A}^1_k$ is a closed immersion for every $n \geq 1$, giving countably many closed subscheme structures on the single point $\{0\}$.

   This example also shows that the structure sheaf of a closed subscheme is *not* a subsheaf of $\mathcal{O}_X$: the quotient $k[x]/(x^2)$ is not a subring of $k[x]$.

3. *The general affine case.* For any ideal $I \subseteq R$, the surjection $R \twoheadrightarrow R/I$ induces a closed immersion $\operatorname{Spec}(R/I) \hookrightarrow \operatorname{Spec}(R)$ with image $V(I)$.

4. *Projective subspaces.* The inclusion $\mathbb{P}^1_k \hookrightarrow \mathbb{P}^2_k$ given by $[s : t] \mapsto [s : t : 0]$ is a closed immersion. On the standard affine chart $U_0 = \{x_0 \neq 0\}$, it corresponds to the surjection $k[y_1, y_2] \twoheadrightarrow k[y_1, y_2]/(y_2) \cong k[y_1]$.

Neither of the two conditions in the definition is redundant. The surjectivity of the sheaf map is not enough. The open immersion $D(x) \cup D(y) \hookrightarrow \mathbb{A}^2_k$ has a surjective sheaf map (in fact an isomorphism on each open), but the image is open, not closed. More exotic examples arise from non-separated schemes.

Similarly, topological closedness is not enough. The normalization of the cuspidal cubic provides an example. Let $Y = \operatorname{Spec}(k[x, y]/(y^2 - x^3))$ and consider the map $\nu : \mathbb{A}^1_k \to Y$ given by $t \mapsto (t^2, t^3)$, corresponding to the ring map $k[x, y]/(y^2 - x^3) \to k[t]$ with $x \mapsto t^2$, $y \mapsto t^3$. This map is a homeomorphism onto its image (which is all of $|Y|$) and $|Y|$ is closed in itself, but the ring map is *not* surjective: the element $t \in k[t]$ is not in the image (even though $y/x = t^3/t^2 = t$ is a rational function on Y, it is not regular). The failure of surjectivity reflects the singularity of Y at the origin—or algebraically, the failure of $k[t^2, t^3]$ to be integrally closed in $k[t]$.

Proposition 3.1.24: Properties of Closed Immersions

1. A morphism $f : Y \to X$ is a closed immersion if and only if for every affine open $\operatorname{Spec}(S) \subseteq X$, the preimage $f^{-1}(\operatorname{Spec}(S)) = \operatorname{Spec}(R)$ is affine and the induced map $S \to R$ is surjective.

2. Every surjective ring homomorphism $S \twoheadrightarrow R$ induces a closed immersion $\operatorname{Spec}(R) \hookrightarrow \operatorname{Spec}(S)$.

3. The composition of two closed immersions is a closed immersion.

4. Being a closed immersion is local on the target: if $\{V_\alpha\}$ is an open cover of X and each restriction $f^{-1}(V_\alpha) \to V_\alpha$ of $f : Y \to X$ is a closed immersion, then f is a closed immersion.

***Proof*:**

1. *Sufficiency.* Suppose that every affine open $V = \operatorname{Spec}(S) \subseteq X$ has $f^{-1}(V) = \operatorname{Spec}(R)$ with $S \to R$ surjective. By (2), whose proof does not use (1), each restriction $f_V \colon f^{-1}(V) \to V$ is a closed immersion. The map f is injective, since $f(y) = f(y')$ puts y and y' in a common $f^{-1}(V)$, on which f is injective. For $Z \subseteq Y$ closed, $f(Z) \cap V = f_V(Z \cap f^{-1}(V))$ is closed in V, and the affine opens V cover X, so $f(Z)$ is closed. Hence f is a homeomorphism onto the closed set $f(Y)$. On each V the sheaf map $\mathcal{O}_X \to f_*\mathcal{O}_Y$ restricts to that of f_V, which is surjective on stalks, so f is a closed immersion (definition 3.1.22).

   Necessity (cited). Let f be a closed immersion and $V = \operatorname{Spec}(S) \subseteq X$ an affine open. The restriction $f^{-1}(V) \to V$ (lemma 2.2.52) is a closed immersion, being a homeomorphism onto the closed subset $f(Y) \cap V$ of V whose sheaf map is the restriction of $f^\#$. That a closed immersion into an affine scheme $\operatorname{Spec}(S)$ has affine source $\operatorname{Spec}(R)$ with $S \to R$ surjective is [36, Corollary II.5.10], whose proof uses the quasi-coherence of the ideal sheaf of a closed subscheme. This book does not reprove it.

2. The surjection $S \twoheadrightarrow R = S/I$ induces a homeomorphism $\operatorname{Spec}(R) \cong V(I) \subseteq \operatorname{Spec}(S)$ onto a closed subset, and the sheaf map $\mathcal{O}_{\operatorname{Spec}(S)} \to f_*\mathcal{O}_{\operatorname{Spec}(R)}$ is surjective on each distinguished open (it corresponds to $S_g \twoheadrightarrow (S/I)_{\overline{g}}$, which is surjective).

3. Let $Z \xrightarrow{g} Y \xrightarrow{f} X$ be closed immersions. The composition $f \circ g$ is a homeomorphism onto a closed subset (a closed subset of a closed subset is closed). For a homeomorphism $h \colon A \to B$ onto a closed subset and a sheaf $\mathcal{F}$ on A, the stalk of $h_*\mathcal{F}$ at $h(a)$ is $\mathcal{F}_a$, every open neighbourhood of a being h^{-1} of an open neighbourhood of $h(a)$, and the stalk at a point outside $h(A)$ is zero, $B \setminus h(A)$ being open. So at a point $f(g(z))$ the stalk map of the sheaf map $\mathcal{O}_X \to f_*\mathcal{O}_Y \to f_*g_*\mathcal{O}_Z$ of $f \circ g$ is the composite of the surjective stalk maps $\mathcal{O}_{X,f(g(z))} \to \mathcal{O}_{Y,g(z)} \to \mathcal{O}_{Z,z}$, and at the other points its target is zero. So $f \circ g$ is a closed immersion (definition 3.1.22).

4. The map f is injective, since $f(y) = f(y')$ puts y and y' in a common $f^{-1}(V_\alpha)$, on which f is injective. For $C \subseteq Y$ closed, each $f(C) \cap V_\alpha$ is the image of the closed set $C \cap f^{-1}(V_\alpha)$ under the restriction of f, hence closed in V_α, so $f(C)$ is closed. Hence f is a homeomorphism onto the closed set $f(Y)$. On each V_α the sheaf map $\mathcal{O}_X \to f_*\mathcal{O}_Y$ restricts to that of $f^{-1}(V_\alpha) \to V_\alpha$, which is surjective on stalks, and the V_α cover X. So f is a closed immersion (definition 3.1.22).

As closed subschemes are determined by ideals on each open subset, it is natural to ask whether they can be described globally by an "ideal subsheaf" of $\mathcal{O}_X$. They can, and the theory is developed in section 5.1.4 once we have the language of quasi-coherent sheaves.

Every closed subset carries a canonical reduced closed subscheme structure:

Proposition 3.1.25: Reduced Induced Closed Subscheme

Let X be a scheme and $Z \subseteq X$ a closed subset. There is a closed subscheme structure on Z, with inclusion $i\colon Z \to X$, such that:

1. Z is a reduced scheme (definition 2.4.14).
2. for every affine open $U \subseteq X$ and every isomorphism $\alpha\colon \operatorname{Spec}(A) \to U$, the open subscheme $i^{-1}(U)$ of Z is isomorphic over U to $\operatorname{Spec}(A/J)$, where $J = I(\alpha^{-1}(Z \cap U))$ and $\operatorname{Spec}(A/J) \to U$ is α composed with the closed immersion induced by $A \to A/J$ (proposition 3.1.24(2)). In particular, for $z \in Z \cap U$ with prime $\mathfrak{p} = \alpha^{-1}(z)$, the stalk map $\mathcal{O}_{X,z} \cong A_{\mathfrak{p}} \to \mathcal{O}_{Z,z}$ is surjective with kernel $JA_{\mathfrak{p}}$.
3. every morphism $g\colon T \to X$ from a reduced scheme T with $g(T) \subseteq Z$ factors uniquely as $g = i \circ h$.

If $i'\colon Z' \to X$ is a closed immersion with Z' reduced and image Z, there is a unique isomorphism $h\colon Z' \to Z$ with $i \circ h = i'$.

***Proof*:**

Step 1: the affine case. Let A be a ring, $C \subseteq \operatorname{Spec}(A)$ closed, $J = I(C)$, and $q\colon \operatorname{Spec}(A/J) \to \operatorname{Spec}(A)$ the closed immersion induced by $A \to A/J$, a homeomorphism onto $V(J) = C$ (proposition 2.1.6(6), lemma 2.1.5). The ideal J is an intersection of primes, so A/J and its localizations are reduced, and a section of $\mathcal{O}_{\operatorname{Spec}(A/J)}$ with $s^n = 0$ has nilpotent, hence zero, germs (proposition 2.2.22) and is zero (lemma 1.1.15). So $\operatorname{Spec}(A/J)$ is reduced. Let T be reduced and $g\colon T \to \operatorname{Spec}(A)$ with $g(T) \subseteq C$. By proposition 3.1.12 and its naturality, morphisms $T \to \operatorname{Spec}(A/J)$ correspond to ring maps $A/J \to \mathcal{O}_T(T)$, and composing with q corresponds to composing with the quotient map, which is $\Gamma(q)$ (theorem 3.1.4). So g factors through q, and then uniquely, if and only if $\beta = \Gamma(g)$ vanishes on J. Let $a \in J$ and $t \in T$. The prime $g(t)$ contains J, so the germ of a at $g(t)$ lies in the maximal ideal of $\mathcal{O}_{\operatorname{Spec}(A),g(t)} = A_{g(t)}$, and the stalk map of g at t, which is local, sends it to the germ of $\beta(a)$ at t, which therefore lies in $\mathfrak{m}_t$. On an affine open $V \subseteq T$, with $V \cong \operatorname{Spec}(B)$, the restriction of $\beta(a)$ is an element $b \in B$ with $b/1 \in \mathfrak{q}B_{\mathfrak{q}}$ for every prime $\mathfrak{q}$ of B, hence $b \in \mathfrak{q}$ ([10, Ideals of Localization], part (2)). So b is nilpotent ([10, The Radical as an Intersection of Primes]), hence $b = 0$, as T is reduced. By locality $\beta(a) = 0$.

Step 2: open subsets. In the setting of Step 1 let $W \subseteq \operatorname{Spec}(A)$ be open. A morphism $g\colon T \to W$ from a reduced scheme with image in C, followed by the inclusion of W, factors through q by Step 1. The factorization lands in $q^{-1}(W)$, so it factors through the open subscheme $q^{-1}(W)$, which is reduced, and the inclusions of $q^{-1}(W)$ and of W are monomorphisms (lemma 2.2.52). Hence g factors uniquely through $q^{-1}(W) \to W$.

Step 3: gluing data. For each affine open $U \subseteq X$ fix an isomorphism $\alpha_U\colon \operatorname{Spec}(A_U) \to U$, put $J_U = I(\alpha_U^{-1}(Z \cap U))$, $Z_U = \operatorname{Spec}(A_U/J_U)$, and let $\pi_U\colon Z_U \to U$ be α_U composed with the morphism q of Step 1. Then π_U is a closed immersion with image $Z \cap U$, Z_U is reduced, and by Step 2, transported along α_U, for every open $W \subseteq U$ every morphism from a reduced scheme to W with image in Z factors uniquely through $\pi_U^{-1}(W) \to W$. For affine opens U, U' put $Z_{UU'} = \pi_U^{-1}(U \cap U')$. The morphism $Z_{UU'} \to U \cap U'$ given by π_U has reduced source

and image in Z, so it factors uniquely through $Z_{U'U} \to U \cap U'$, as $\varphi_{UU'} \colon Z_{UU'} \to Z_{U'U}$ with $\pi_{U'} \circ \varphi_{UU'} = \pi_U$. The morphisms $\varphi_{U'U} \circ \varphi_{UU'}$ and the identity both factor π_U through $Z_{UU'}$, so they are equal. Hence $\varphi_{UU'}$ is an isomorphism and $\varphi_{UU} = \mathrm{id}$. As $\pi_{U'} \circ \varphi_{UU'} = \pi_U$, the map $\varphi_{UU'}$ carries $Z_{UU'} \cap Z_{UU''}$ into $Z_{U'U} \cap Z_{U'U''}$, and there $\varphi_{U'U''} \circ \varphi_{UU'}$ and $\varphi_{UU''}$ both factor π_U through $\pi_{U''}^{-1}(U \cap U' \cap U'')$, so they are equal. This is the cocycle condition.

Step 4: the closed subscheme. Theorem 2.2.56 gives a scheme $\tilde{Z}$ with open embeddings $\iota_U \colon Z_U \to \tilde{Z}$ covering it, with $\iota_U(Z_U) \cap \iota_{U'}(Z_{U'}) = \iota_U(Z_{UU'})$ and $\iota_{U'} \circ \varphi_{UU'} = \iota_U$ on $Z_{UU'}$. The morphisms $\pi_U \circ \iota_U^{-1}$ on the sets $\iota_U(Z_U)$ agree on overlaps, so lemma 3.1.9 glues them to $\tilde{\imath} \colon \tilde{Z} \to X$ with $\tilde{\imath} \circ \iota_U = \pi_U$, whose image is Z. If $\tilde{\imath}(\iota_U(a)) = \tilde{\imath}(\iota_{U'}(b))$, then $a \in Z_{UU'}$, and $\varphi_{UU'}(a)$ and b have the same image under the injective map $\pi_{U'}$, so $\iota_U(a) = \iota_{U'}(\varphi_{UU'}(a)) = \iota_{U'}(b)$. So $\tilde{\imath}$ is injective. For an affine open $V \subseteq X$, a point $\iota_U(a)$ maps into V only if $a \in Z_{UV}$, and then $\iota_U(a) = \iota_V(\varphi_{UV}(a))$. So $\tilde{\imath}^{-1}(V) = \iota_V(Z_V)$, on which $\tilde{\imath}$ is π_V, given through α_V by the surjection $A_V \to A_V/J_V$, and $\tilde{\imath}$ is a closed immersion by proposition 3.1.24(1). Transporting the structure sheaf along the homeomorphism $\tilde{Z} \to Z$ makes Z a scheme whose inclusion i is a closed immersion (definition 3.1.22).

1. A nilpotent section of $\mathcal{O}_Z$ restricts to nilpotent, hence zero, sections on the reduced open subschemes corresponding to the Z_U, so it is zero by locality.

2. The open subscheme $i^{-1}(U)$ is isomorphic to Z_U over U by construction. Another isomorphism $\alpha \colon \mathrm{Spec}\,(A) \to U$ satisfies $\alpha_U^{-1} \circ \alpha = \mathrm{Spec}\,(\gamma)$ for a ring isomorphism $\gamma \colon A_U \to A$ (theorem 2.2.16), which sends a prime $\mathfrak{p}$ of A to $\gamma^{-1}(\mathfrak{p})$. Hence $J_U = \gamma^{-1}(J)$, and γ induces an isomorphism $\mathrm{Spec}\,(A/J) \cong \mathrm{Spec}\,(A_U/J_U)$ over U. By proposition 2.2.22(3), the stalk map at the point $\mathfrak{p}/J$ of the morphism $\mathrm{Spec}\,(A/J) \to \mathrm{Spec}\,(A)$ induced by $A \to A/J$ is the map $A_{\mathfrak{p}} \to (A/J)_{\mathfrak{p}/J}$, $a/c \mapsto \bar{a}/\bar{c}$. It is surjective with kernel $JA_{\mathfrak{p}}$, as localization at $A \setminus \mathfrak{p}$ is exact ([10, Localization Is Exact]).

3. Let $g \colon T \to X$ with T reduced and $g(T) \subseteq Z$. For each affine open U, the restriction of g to the reduced open subscheme $g^{-1}(U)$ factors uniquely through π_U as h_U (Step 3). On $g^{-1}(U \cap U')$ both $\varphi_{UU'} \circ h_U$ and $h_{U'}$ factor g through $Z_{U'U} \to U \cap U'$, so they agree, and $\iota_U \circ h_U = \iota_{U'} \circ h_{U'}$ there. By lemma 3.1.9 the $\iota_U \circ h_U$ glue to $h \colon T \to \tilde{Z} \cong Z$ with $i \circ h = g$. If also $i \circ h' = g$, then h' maps $g^{-1}(U)$ into $\tilde{\imath}^{-1}(U) = \iota_U(Z_U)$, so on $g^{-1}(U)$ it is ι_U composed with a factorization of g through π_U (lemma 2.2.52), which is h_U. Hence $h' = h$.

Uniqueness. Let $i' \colon Z' \to X$ be as in the statement. By (3), $i' = i \circ h$ for a unique h, which is a homeomorphism, as i and i' are homeomorphisms onto Z. Let $z' \in Z'$ and let $\alpha \colon \mathrm{Spec}\,(A) \to U$ be an isomorphism onto an affine open containing $i'(z')$. By proposition 3.1.24(1) (the direction cited from Hartshorne), $i'^{-1}(U) \cong \mathrm{Spec}\,(A/J')$ over U with $A \to A/J'$ surjective. The ring A/J' is the ring of sections of the reduced scheme Z' over $i'^{-1}(U)$, so J' is radical, and $V(J') = \alpha^{-1}(Z \cap U)$, so $J' = I(V(J')) = J$ (proposition 2.1.6(6)). With $\mathfrak{p} = \alpha^{-1}(i'(z'))$, the stalk maps $i^{\#}_{h(z')}$ and $i'^{\#}_{z'}$, both defined on $\mathcal{O}_{X,i'(z')} \cong A_{\mathfrak{p}}$, are therefore surjective with kernel $JA_{\mathfrak{p}}$, by (2) and by proposition 2.2.22(3) applied to $A \to A/J' = A/J$, and $i'^{\#}_{z'} = h^{\#}_{z'} \circ i^{\#}_{h(z')}$, so $h^{\#}_{z'}$ is an isomorphism. By proposition 1.2.24, h is an isomorphism, completing the proof.

Definition 3.1.26: Reduced Induced Structure

The closed subscheme structure on a closed subset Z of a scheme X given by proposition 3.1.25 is the *reduced induced structure* on Z. By the uniqueness clause of that proposition, it is the only reduced closed subscheme structure on Z, up to unique isomorphism over X.

Locally closed subschemes: immersions

If $X = \operatorname{Spec} R$ is an affine scheme and $U \subseteq X$ is an open subset, for any $S \subseteq \mathcal{O}_X(U)$, $V(S)$ is usually *not* a closed subset but a *locally closed subset*. These are the natural local solutions to regular functions, and thus deserve some attention:

Definition 3.1.27: Locally Closed Immersion

A morphism $f : Y \to X$ is a *locally closed immersion* (or simply an *immersion*) if f factors as

$$Y \xrightarrow{i} U \xhookrightarrow{\text{open}} X$$

where $U \subseteq X$ is an open subscheme and $i : Y \hookrightarrow U$ is a closed immersion. Equivalently, f induces a homeomorphism from $|Y|$ onto a locally closed subset of $|X|$ (i.e., the intersection $C \cap U$ of a closed set C and an open set U), with $f^{\#}$ surjective onto the structure sheaf of Y. A *locally closed subscheme* of X is a locally closed subset with the scheme structure making the inclusion an immersion.

Proposition 3.1.28: Open and Closed Immersions are Immersions

Every open immersion and every closed immersion is a locally closed immersion.

***Proof*:**

An open immersion $U \hookrightarrow X$ factors as $U \xrightarrow{\text{id}} U \hookrightarrow X$, where id is (trivially) a closed immersion.
A closed immersion $Y \hookrightarrow X$ factors as $Y \hookrightarrow X \xrightarrow{\text{id}} X$, taking $U = X$.

Closedness of a subset $C \subseteq X$ can be verified on an open cover: C is closed if and only if $C \cap U_i$ is closed in U_i for every member of an open cover $\{U_i\}$ of X. Local closedness is a strictly weaker condition: C must be the intersection of a closed set with a *single* open set, not locally closed in each chart.

Locally closed subsets appear naturally in at least two important contexts. First, images of morphisms of varieties are typically locally closed (or finite unions thereof); this is the content of Chevalley's theorem (definition 5.2.2). Second, the diagonal morphism $\Delta : X \to X \times_S X$ is always a locally closed immersion, and the condition that Δ be a *closed* immersion is precisely the definition of separatedness (section 3.5.1).

Local schemes: beyond topological subsets

The open, closed, and locally closed subschemes above all correspond to topological subsets of X equipped with a scheme structure. There is a fourth type of "subobject" that arises naturally but does *not* correspond to any topological subset: the *local scheme* at a point.

Let X be a scheme and $x \in X$ a point. The stalk $\mathcal{O}_{X,x}$ is a local ring, and we can form the scheme $\operatorname{Spec}(\mathcal{O}_{X,x})$. There is a canonical morphism

$$\iota_x : \operatorname{Spec}(\mathcal{O}_{X,x}) \longrightarrow X$$

defined as follows. Choose any affine open $\operatorname{Spec}(R) \subseteq X$ containing x, with x corresponding to a prime $\mathfrak{p} \subseteq R$. The localization map $R \to R_{\mathfrak{p}} = \mathcal{O}_{X,x}$ induces $\operatorname{Spec}(\mathcal{O}_{X,x}) \to \operatorname{Spec}(R) \hookrightarrow X$. This morphism is independent of the choice of affine open (the stalk is a colimit over all open neighborhoods, so any two choices yield the same map).

In what sense is $\operatorname{Spec}(\mathcal{O}_{X,x})$ a "subobject" of X? The morphism ι_x is *not* an immersion in any topological sense:

Proposition 3.1.29: The Local Scheme is Not a Topological Subscheme

Let X be a scheme of finite type over a field k and let $x \in X$ be a non-isolated point. Then $\iota_x : \operatorname{Spec}(\mathcal{O}_{X,x}) \to X$ is not an open immersion, not a closed immersion, and not a locally closed immersion. In particular, the image of ι_x is not a locally closed subset of $|X|$.

***Proof*:**
The image of ι_x consists of those primes $\mathfrak{q} \subseteq R$ with $\mathfrak{q} \subseteq \mathfrak{p}$ (these are exactly the primes that remain in the localization $R \to R_{\mathfrak{p}}$). This is the set of *generizations* of x: the points $y \in X$ with $x \in \overline{\{y\}}$. In particular, the image contains the generic point of every irreducible component passing through x.

This set is neither open nor closed in general. It is not closed because it contains the generic point η of X (if X is irreducible), and $\overline{\{\eta\}} = X \neq \operatorname{im}(\iota_x)$. It is not open because removing the image from X does not yield a closed set (the closure of $\{x\}$ is not contained in $X \setminus \operatorname{im}(\iota_x)$). It is not locally closed because a locally closed set in a Noetherian space is a finite union of locally closed irreducible sets, and the set of generizations of x does not have this form when x has positive codimension.

Example 3.1.30: Local Schemes

1. *The generic point of $\mathbb{A}^1_k$.* Let $X = \mathbb{A}^1_k = \operatorname{Spec}(k[x])$ and let $\eta = [(0)]$ be the generic point. Then $\mathcal{O}_{X,\eta} = k(x)$, the function field, and $\operatorname{Spec}(\mathcal{O}_{X,\eta}) = \operatorname{Spec}(k(x))$ is a single point. The morphism $\iota_\eta : \operatorname{Spec}(k(x)) \to \mathbb{A}^1_k$ sends the unique point to η. This is the inclusion of the generic point, which is neither open nor closed in $\mathbb{A}^1_k$ (its closure is all of $\mathbb{A}^1_k$, but it is not itself closed).

   For $\mathbb{A}^1_k$ the generic point *is* a locally closed subset (it is the intersection of the open set X with the closure $\overline{\{\eta\}} = X$, so it is a dense locally closed subset). But the map ι_η is still not a locally closed immersion, because $\operatorname{Spec}(k(x))$ and $\{\eta\}$ (with its induced scheme structure as a subspace of $\mathbb{A}^1_k$) are not isomorphic as locally ringed spaces: the stalk of $\mathcal{O}_{\mathbb{A}^1}$ at η is

$k(x)$, but the induced structure sheaf on $\{\eta\}$ as a subspace is $\mathcal{O}_{\mathbb{A}^1}(\mathbb{A}^1) = k[x]$, not $k(x)$.

2. *A closed point of* $\mathbb{A}^2_k$. Let $X = \mathbb{A}^2_k$ and $x = [(x, y)]$, the origin. Then $\mathcal{O}_{X,x} = k[x, y]_{(x,y)}$, and $\operatorname{Spec}(\mathcal{O}_{X,x})$ has two points: the closed point (the maximal ideal (x, y)) and the generic point (the zero ideal). The image of ι_x is $\{[(x, y)], [(0)]\}$—the origin together with the generic point of $\mathbb{A}^2$. This is not open, not closed, and not locally closed in $\mathbb{A}^2_k$.

3. *A point on a curve in* $\mathbb{A}^2_k$. Let $C = V(y - x^2) \subseteq \mathbb{A}^2_k$ and let $x = [(x, y)]$ be the origin of C. Then $\mathcal{O}_{C,x} = (k[x, y]/(y - x^2))_{(x,y)} \cong k[x]_{(x)}$, which has two primes. The image of ι_x consists of the origin and the generic point of C—but not the generic point of $\mathbb{A}^2$ (which does not lie on C).

Despite not being a topological subscheme, the local scheme $\operatorname{Spec}(\mathcal{O}_{X,x})$ is arguably the most important "subobject" associated to a point. It captures the local algebra of X at x (the ring $\mathcal{O}_{X,x}$ encodes the germs of regular functions near x), and many properties of X at x (regularity, normality, Cohen-Macaulay-ness) are properties of this single local ring. The morphism ι_x is flat (since localization is an exact functor), which makes it well-behaved from a homological perspective even though it is topologically ill-behaved.

We can place this in context with the other types of subobjects:

Type	*Topological condition*	*Sheaf condition*	*Uniqueness*
Open subscheme	image is open	$f^\#$ is an isomorphism	unique
Closed subscheme	image is closed	$f^\#$ is surjective	not unique
Locally closed	image is loc. closed	$f^\#$ surjective onto image	not unique
Local scheme at x	image = generizations of x	$f^\#$ is localization	unique

The first three are topological subobjects: the image is a subset with a specific topological property, and the scheme structure is constrained by the sheaf map. The fourth is algebraic rather than topological: it is determined by the localization at a prime, and the image has no clean topological description. Yet it is the one that controls the local geometry.

3.1.31 Immersions as finite-type monomorphisms There is a clean characterization that explains the role of locally closed immersions. Among monomorphisms of schemes, the immersions are (essentially) the ones that are *locally of finite type*: they impose only finitely many algebraic conditions locally. All three of our topological types—open immersions, closed immersions, and locally closed immersions—are locally of finite type. The local scheme map $\operatorname{Spec}(\mathcal{O}_{X,x}) \to X$ is a monomorphism but is *not* locally of finite type (the local ring $\mathcal{O}_{X,x}$ is generally not a finitely generated algebra over the coordinate ring of any affine chart).
This gives a clean heuristic: schemes are locally affine, and the sub-objects of an affine scheme $\operatorname{Spec}(R)$ that are "of finite type" are exactly the closed subschemes $\operatorname{Spec}(R/I)$. The locally closed immersions are what you get by globalizing this: locally on X, a finite-type sub-object is a closed subscheme of an affine chart, which is a locally closed subscheme of X.

Formal local schemes: the analytic germ

One can zoom in even further than the local scheme by taking the completion. Let $\mathfrak{m}_x \subseteq \mathcal{O}_{X,x}$ be the maximal ideal. We can form the $\mathfrak{m}_x$-adic completion of the local ring, $\widehat{\mathcal{O}}_{X,x}$, and consider the affine scheme $\operatorname{Spec}(\widehat{\mathcal{O}}_{X,x})$.

The natural map $\mathcal{O}_{X,x} \to \widehat{\mathcal{O}}_{X,x}$ induces a canonical morphism

$$\widehat{\iota}_x : \operatorname{Spec}(\widehat{\mathcal{O}}_{X,x}) \longrightarrow \operatorname{Spec}(\mathcal{O}_{X,x}) \longrightarrow X.$$

For Noetherian schemes, the completion map $\mathcal{O}_{X,x} \to \widehat{\mathcal{O}}_{X,x}$ is faithfully flat. Consequently, the induced map on spectra is surjective, meaning the image of $\widehat{\iota}_x$ in X is exactly the same as the image of the local scheme: the set of generizations of x.

If the image is the same, how is this a "smaller" neighborhood? While the local scheme captures arbitrarily small *Zariski* neighborhoods, the completion captures arbitrarily small *formal* (or analytic) neighborhoods. The Zariski topology is famously coarse, and the completion detects geometric features that the local scheme cannot.

Example 3.1.32: Formal Branches of a Node

Let $X = \operatorname{Spec}(k[x,y]/(y^2 - x^2(x+1)))$ be a nodal cubic curve over an algebraically closed field of characteristic $\neq 2$, and let $p = [(x,y)]$ be the node at the origin.

The local ring $\mathcal{O}_{X,p} = (k[x,y]/(y^2 - x^2(x+1)))_{(x,y)}$ is an integral domain; in the Zariski topology, any open neighborhood of the origin is irreducible, so the local scheme $\operatorname{Spec}(\mathcal{O}_{X,p})$ is an integral scheme.

However, analytically, the curve consists of two branches crossing at the origin. If we pass to the completion $\widehat{\mathcal{O}}_{X,p} \cong k[\![x,y]\!]/(y^2 - x^2(x+1))$, the element $x+1$ has a square root (by Hensel's Lemma, or the binomial series). Thus, the defining equation factors:

$$y^2 - x^2(x+1) = (y - x\sqrt{x+1})(y + x\sqrt{x+1}).$$

The complete local ring is not a domain! The formal local scheme $\operatorname{Spec}(\widehat{\mathcal{O}}_{X,p})$ has two irreducible components, perfectly capturing the two analytic branches of the curve passing through the node.

Geometrically, passing to the formal local scheme feels like taking a smaller, more precise subspace. Categorically, however, this marks a hard boundary: *the morphism $\widehat{\iota}_x$ is not a monomorphism.* For a map of affine schemes to be a monomorphism, the underlying ring map must be an epimorphism (meaning the multiplication map $S \otimes_R S \to S$ is an isomorphism). However, there is a fundamental algebraic fact: *any faithfully flat epimorphism of rings is an isomorphism.* Because the completion of a Noetherian local ring is faithfully flat, if it were an epimorphism, the local ring would already have to be complete!

Because $\widehat{\iota}_x$ is not a monomorphism, the formal local scheme does not belong in our hierarchy of subobjects (immersions and local schemes). It is not "injective on points" functorially, and its diagonal map is very far from being an isomorphism. Instead of strictly restricting the space of functions, completion conceptually "thickens" the local ring by adding infinite limits (Taylor series). This breaks categorical injectivity, but it is exactly the mechanism required to reveal finer analytic structure. Geometrically, we can think of the fact that the two branches of a node our now two separate points as the analytic scheme "adding" more points, and thus failing the required injectivity.

In modern algebraic geometry, one often prefers to treat the completion as a *formal scheme*, denoted $\mathrm{Spf}(\widehat{\mathcal{O}}_{X,x})$. As a formal scheme, its underlying topological space is just the single point $\{x\}$, but its sheaf of topological rings remembers the "infinitesimal thickenings" of the point. See section 5.6⋆ for more details.

Quick Aside: Dominant morphisms

Closed immersions of affine schemes correspond to surjective ring maps $R \twoheadrightarrow R/I$. The opposite extreme—injective ring maps—does not yield a surjective scheme morphism, but it yields the next best thing.

Definition 3.1.33: Dominant Morphism

A scheme morphism $f : X \to Y$ is *dominant* if the image of f is dense in Y, i.e., $\overline{f(X)} = Y$.

Proposition 3.1.34: Injective Ring Maps and Dominant Morphisms

Let $\varphi : R \hookrightarrow S$ be an injective ring homomorphism. Then the corresponding scheme morphism $f : \mathrm{Spec}\,(S) \to \mathrm{Spec}\,(R)$ is dominant.

***Proof*:**
The image $f(\mathrm{Spec}\,(S))$ is the set $\{\varphi^{-1}(\mathfrak{q}) : \mathfrak{q} \in \mathrm{Spec}\,(S)\}$. Its closure in $\mathrm{Spec}\,(R)$ is $V(I)$ where $I = \bigcap_{\mathfrak{q}\in\mathrm{Spec}\,(S)} \varphi^{-1}(\mathfrak{q}) = \varphi^{-1}\left(\bigcap_{\mathfrak{q}\in\mathrm{Spec}\,(S)} \mathfrak{q}\right) = \varphi^{-1}(\mathrm{Nil}(S))$, the preimage of the nilradical of S. Since φ is injective, $\varphi^{-1}(\mathrm{Nil}(S)) \subseteq \mathrm{Nil}(R)$. Thus $V(I) \supseteq V(\mathrm{Nil}(R)) = \mathrm{Spec}\,(R)$, giving $\overline{f(\mathrm{Spec}\,(S))} = \mathrm{Spec}\,(R)$.

Note that such a map is *not* necessarily injective, and in fact often fails to be injective (ex. on curves it fails at the non-branching points). Thus, we do not consider such maps subschemes; dominant maps are recorded here as it is likely the reader would be curious to know how to think of injective ring homomorphisms at this early stage. A fuller treatment of dominant morphisms and rational maps is given in definition 5.2.2. In particular, we will see that for morphisms of finite type, dominance can be characterized by the density of the image, and Chevalley's theorem (theorem 5.2.7) gives a precise description of the image (it is a constructible set).

Exercise 3.1.1

1. Let X be a scheme and $Z \subseteq |X|$ a closed subset. Define an ideal sheaf $\mathcal{I}$ on X by

$$\mathcal{I}(U) = \{f \in \mathcal{O}_X(U) \mid f(x) = 0 \text{ for all } x \in Z \cap U\}$$

   where $f(x) = 0$ means the image of f in $\kappa(x)$ vanishes. Suppose $\mathrm{Spec}\,(R) \subseteq X$ is an affine open subset such that $Z \cap \mathrm{Spec}\,(R) = V(I)$ for some ideal $I \subseteq R$. Show that the ideal of R corresponding to $\mathcal{I}(\mathrm{Spec}\,(R))$ is $\sqrt{I}$.

2. Use the previous result to conclude that the closed subscheme Z_{red} defined by $\mathcal{I}$ is a scheme whose underlying topological space is exactly Z, and whose structure sheaf has no nilpotents. Argue why Z_{red} is the unique closed subscheme structure on Z with this property.

3. Consider the closed subset $Z = \{0\}$ in the affine line $\mathbb{A}^1_k = \operatorname{Spec}(k[x])$. Describe the reduced subscheme Z_{red} and compare it to the closed subschemes defined by the ideals (x^n) for $n \geq 2$.

4. The *reduction* of a scheme X, denoted X_{red}, is the reduced closed subscheme supported on the entire underlying space $|X|$. Show that the inclusion $X_{\text{red}} \hookrightarrow X$ satisfies the following universal property: for any reduced scheme Y and any morphism $f : Y \to X$, f factors uniquely through X_{red}.

3.1.4 More on Projective Schemes

In section 2.3, we defined the prototypical projective scheme $\mathbb{P}^n_R = \operatorname{Proj} R[x_0, \ldots, x_n]$. To do geometry, we need to understand its closed subschemes.

As we will prove rigorously in the next chapter using ideal sheaves (specifically corollary 5.3.15), closed subschemes of a projective scheme are governed by homogeneous ideals. For now, we take this as our primary geometric construction: given a graded ring $S_\bullet$ and a homogeneous ideal $I \subseteq S_\bullet$, the canonical surjective graded ring homomorphism $S_\bullet \twoheadrightarrow S_\bullet/I$ induces a closed immersion:

$$\operatorname{Proj}(S_\bullet/I) \hookrightarrow \operatorname{Proj} S_\bullet$$

Conceptually, this is the exact generalization of the affine case. If $S_\bullet$ is a finitely generated graded R-algebra generated in degree 1, we can choose $n+1$ degree 1 generators to form a surjective graded map $R[x_0, \ldots, x_n] \twoheadrightarrow S_\bullet$. This implies $S_\bullet \cong R[x_0, \ldots, x_n]/I$, meaning $\operatorname{Proj} S_\bullet$ naturally embeds as a closed subscheme of $\mathbb{P}^n_R$. Choosing these generators is geometrically equivalent to choosing projective coordinates.

We can classify the simplest closed subschemes of $\mathbb{P}^n_k$ by the generators of their ideals:

Definition 3.1.35: Projective Hypersurface and Linear Spaces

A *hypersurface* of $\mathbb{P}^n_k$ is a closed subscheme $\operatorname{Proj} k[x_0, \ldots, x_n]/(f)$ given by a single homogeneous polynomial f. The *degree* of the hypersurface is the degree of f.

A hypersurface of degree 1 is called a *hyperplane*. A hypersurface of degree $2, 3, 4, 5 \ldots$ is called a *quadric*, *cubic*, *quartic*, *quintic*, and so forth.

If $n = 2$ (so we are in $\mathbb{P}^2_k$), a hypersurface is called a *curve*. A hypersurface of degree $1, 2, 3$ is called a *line*, *conic curve* (or *conic*), and *cubic curve* respectively.

A *linear space* is a closed subscheme cut out by a collection of degree 1 homogeneous polynomials (an intersection of hyperplanes). Any injective linear transformation $V \to W$ of vector spaces induces a closed immersion $\mathbb{P}(V) \hookrightarrow \mathbb{P}(W)$ whose image is a linear space.

Recall that $\mathbb{P}^n_k$ has only constant global functions, so these subschemes are cut out by regular functions defined only on open affine patches. Let us look at some classic examples.

Example 3.1.36: Closed Projective Subschemes

1. *Twisted Cubic*: Take the homogeneous ideal $I = (xy - zw, x^2 - wy, y^2 - xz)$ and look at the closed subscheme $\operatorname{Proj} k[w, x, y, z]/I \subseteq \mathbb{P}^3_k$. This shape is called the *twisted cubic*. It is a

curve, and it is isomorphic to $\mathbb{P}^1_k$ via the map induced by the graded ring homomorphism:

$$k[w,x,y,z] \to k[s,t] \qquad (w,x,y,z) \mapsto (s^3, s^2t, st^2, t^3)$$

To see this is an isomorphism of schemes, one can take the standard affine open sets and show it gives closed immersions on the charts. For example, on the chart $D_+(w)$ where $w \neq 0$ and $s \neq 0$, the map of degree 0 localizations is:

$$k\left[\frac{x}{w}, \frac{y}{w}, \frac{z}{w}\right] \to k\left[\frac{t}{s}\right] \qquad \frac{x}{w} \mapsto \frac{t}{s},\ \frac{y}{w} \mapsto \frac{t^2}{s^2},\ \frac{z}{w} \mapsto \frac{t^3}{s^3}$$

2. *Lines on a Quadric Surface*: The following shows a way to parameterize certain projective varieties. Take the quadric surface $V_+(wz - xy) \subseteq \mathbb{P}^3_k$ (with projective coordinates w, x, y, z). For any two elements $\alpha, \beta \in k$ that are not both zero, notice that as $[c,d] \in \mathbb{P}^1_k$ varies, $[\alpha c, \beta c, \alpha d, \beta d]$ forms a line completely contained on the surface. Show there is one other parameterization of this surface via lines, and show it must be the case that these are the only two families (hint: one parameterization contains $V_+(w,x)$ and the other $V_+(w,y)$).

Another important construction arises from taking a different grading of the polynomial ring, denoted $S_{d\bullet}$. This allows us to "linearize" high-degree equations. First, we must establish that altering the grading in this specific way does not change the underlying abstract scheme.

Theorem 3.1.37: Isomorphism of Veronese Subrings

Let $S_\bullet = R[x_0, \ldots, x_n]$ be the standard graded polynomial ring. For any integer $d > 0$, let $S_{d\bullet} = \bigoplus_{m=0}^{\infty} S_{md}$ be the subring consisting of degrees that are multiples of d. Then there is a canonical isomorphism of schemes:

$$\operatorname{Proj} S_{d\bullet} \cong \operatorname{Proj} S_\bullet$$

***Proof*:**
We construct this isomorphism by comparing the standard affine open covers. By lemma 2.3.2, the scheme $\operatorname{Proj} S_\bullet$ is covered by the distinguished open sets $D_+(x_i) \cong \operatorname{Spec}((S_{x_i})_0)$ for $i = 0, \ldots, n$.

For the subring $S_{d\bullet}$, the elements x_i^d have degree d and therefore reside in $S_{d\bullet}$. The corresponding distinguished open sets $D_+(x_i^d)$ cover $\operatorname{Proj} S_{d\bullet}$, and applying lemma 2.3.2 again, each is isomorphic to $\operatorname{Spec}(((S_{d\bullet})_{x_i^d})_0)$.

We claim the rings $(S_{x_i})_0$ and $((S_{d\bullet})_{x_i^d})_0$ are identically the same. An arbitrary element in $(S_{x_i})_0$ is of the form a/x_i^k, where $a \in S_k$ is a homogeneous polynomial of degree k. We can rewrite this fraction by multiplying the numerator and denominator by x_i^{kd-k}:

$$\frac{a}{x_i^k} = \frac{a x_i^{kd-k}}{x_i^{kd}} = \frac{a x_i^{kd-k}}{(x_i^d)^k}$$

The new numerator $a x_i^{kd-k}$ has degree $k + (kd - k) = kd$, meaning it is an element of S_{kd}, which is a valid component of $S_{d\bullet}$. The denominator is a power of x_i^d. Therefore, this fraction is naturally an element of $((S_{d\bullet})_{x_i^d})_0$. Since this containment goes both ways, the rings are equal. These local isomorphisms canonically glue together to yield the global isomorphism $\operatorname{Proj} S_{d\bullet} \cong \operatorname{Proj} S_\bullet$.

Definition 3.1.38: Veronese Embedding

Let $S_\bullet = k[x_0, \dots, x_n]$ and let $S_{d\bullet}$ be the d-tuple Veronese subring. By theorem 3.1.37, $\operatorname{Proj} S_{d\bullet} \cong \mathbb{P}^n_k$. We can view $S_{d\bullet}$ as being generated in "degree 1" by the monomials of degree d from the original ring. There are $N+1 = \binom{n+d}{d}$ such monomials. Mapping a new polynomial ring $k[y_0, \dots, y_N] \twoheadrightarrow S_{d\bullet}$ by sending the variables y_i to these monomials induces a closed immersion:

$$\mathbb{P}^n_k \cong \operatorname{Proj} S_{d\bullet} \hookrightarrow \mathbb{P}^N_k$$

This map is called the *d-tuple Veronese Embedding.*

The geometric utility of the Veronese embedding is that a hypersurface of degree d in $\mathbb{P}^n_k$ gets mapped exactly to a hyperplane intersection (degree 1) in $\mathbb{P}^N_k$.

3.1.39 Foreshadowing: Sheaves and Morphisms In the next chapter, we will show that morphisms to projective space are governed by invertible sheaves. The Veronese embedding is precisely the morphism $X \to \mathbb{P}^N_k$ generated by the global sections of the twisting sheaf $\mathcal{O}_X(d)$.

As noted earlier, choosing a set of degree 1 generators is equivalent to choosing an embedding into a specific ambient projective space $\mathbb{P}^N_R$. Therefore, it is necessary to distinguish the intrinsic geometry of the abstract scheme from the extrinsic geometry of its embedding.

Consider the scheme $X = \mathbb{P}^1_k$. We can embed it in several ways:

1. *Degree 1*: Use $S_\bullet = k[x, y]$. It is generated in degree 1 by 2 elements. This gives the identity embedding $\mathbb{P}^1_k \hookrightarrow \mathbb{P}^1_k$ as a straight line.
2. *Degree 2*: Consider the 2-tuple Veronese subring $S_{2\bullet} = k[x^2, xy, y^2]$. As an abstract scheme, $\operatorname{Proj} S_{2\bullet} \cong \mathbb{P}^1_k$. To write $S_{2\bullet}$ as a ring generated in degree 1, we require 3 generators: let $u = x^2$, $v = xy$, $w = y^2$. They satisfy $uw = v^2$, yielding an embedding $\mathbb{P}^1_k \hookrightarrow \mathbb{P}^2_k$. The image is a conic curve.
3. *Degree 3*: The 3-tuple Veronese subring $S_{3\bullet} = k[x^3, x^2y, xy^2, y^3]$ requires 4 generators, yielding an embedding $\mathbb{P}^1_k \hookrightarrow \mathbb{P}^3_k$. The image is the twisted cubic.

The line in $\mathbb{P}^1_k$, the conic in $\mathbb{P}^2_k$, and the twisted cubic in $\mathbb{P}^3_k$ are isomorphic as abstract schemes. Extrinsically, however, they are embedded differently, have different geometric degrees, and are cut out by different homogeneous ideals.

One might ask why we do not always restrict to degree 1 generators. Certain equations become unnecessarily complicated if forced into a standard degree 1 embedding, but simplify if we assign different *weights* (degrees) to the variables. For example, embedding certain curves or moduli spaces into standard $\mathbb{P}^N$ via Veronese embeddings increases the ambient dimension N significantly and requires many equations to cut out. This motivates the following construction.

Definition 3.1.40: Weighted Projective Space

Let $S_\bullet = k[a_1, \dots, a_n]$ be a graded ring where each variable a_i is assigned a specific degree $d_i \in \mathbb{Z}_{>0}$. Then $\operatorname{Proj} k[a_1, \dots, a_n]$ is called a *weighted projective space* and is denoted $\mathbb{P}(d_1, \dots, d_n)$.

Geometrically, over a field, assigning different weights corresponds to taking the quotient of $\mathbb{A}^n_k \setminus \{0\}$ by the action of the multiplicative group $\mathbb{G}_m$ (i.e. $(\mathbb{A}^*_k, \cdot)$) where the scaling acts as $\lambda \cdot (x_1, \ldots, x_n) = (\lambda^{d_1} x_1, \ldots, \lambda^{d_n} x_n)$.

Example 3.1.41: Weighted Projective Space Examples

1. *Show that $\mathbb{P}(m, n) \cong \mathbb{P}^1_k$ for any positive integers m, n.*

   Let $S_\bullet = k[x, y]$ with $\deg(x) = m$ and $\deg(y) = n$. The scheme is covered by the two open sets $D_+(x)$ and $D_+(y)$. For $D_+(x) \cong \operatorname{Spec}((S_x)_0)$, the degree 0 elements are generated by $u = y^m/x^n$. Thus, $(S_x)_0 \cong k[u]$, which is isomorphic to $\mathbb{A}^1_k$. Similarly, for $D_+(y) \cong \operatorname{Spec}((S_y)_0)$, the degree 0 elements are generated by $v = x^n/y^m$. Thus, $(S_y)_0 \cong k[v]$, which is also isomorphic to $\mathbb{A}^1_k$. On the intersection $D_+(xy)$, the variable v is exactly u^{-1}. This translates to gluing two copies of $\mathbb{A}^1_k$ along $\mathbb{A}^1_k \setminus \{0\}$ via the map $u \mapsto u^{-1}$, which is the precise definition of $\mathbb{P}^1_k$.

2. *Show that $\mathbb{P}(1, 1, 2) \cong \operatorname{Proj} k[u, v, w, z]/(uw - v^2)$.*

   Let $S_\bullet = k[x, y, t]$ with $\deg(x) = 1, \deg(y) = 1$, and $\deg(t) = 2$. By theorem 3.1.37, $\operatorname{Proj} S_\bullet \cong \operatorname{Proj} S_{2\bullet}$. The subring $S_{2\bullet}$ is generated by all degree 2 monomials from $S_\bullet$. These monomials are exactly x^2, xy, y^2, and t. Let us map a standard polynomial ring into $S_{2\bullet}$ by setting $u = x^2$, $v = xy$, $w = y^2$, and $z = t$. The only algebraic relation between these generators is $uw = x^2y^2 = (xy)^2 = v^2$. The generator z is algebraically independent of the others. Therefore, $S_{2\bullet} \cong k[u, v, w, z]/(uw - v^2)$. Since the spaces are isomorphic, it follows that $\mathbb{P}(1, 1, 2)$ can be realized as the quadric cone $\operatorname{Proj} k[u, v, w, z]/(uw - v^2)$ in standard $\mathbb{P}^3_k$.

3. *Smooth Compactifications of Hyperelliptic Curves.*

   Weighted projective spaces are highly practical for compactifying curves without introducing artificial singularities. Consider the affine hyperelliptic curve $y^2 = x^5 + 1$. If we attempt to compactify this by homogenizing it into standard $\mathbb{P}^2_k$ (where all variables have degree 1), we obtain the curve $y^2z^3 = x^5 + z^5$. Calculating the Jacobian reveals that this curve has a singularity at the point at infinity $[0 : 1 : 0]$.

   However, if we embed the curve into the weighted projective space $\mathbb{P}(1, 1, 3)$ with coordinates (x, z, y) having degrees $1, 1$, and 3 respectively, we can form a homogeneous equation of total degree 6:

   $$y^2 = x^5z + z^6$$

   In this weighted space, the curve remains smooth at infinity.

3.1.5 The Category of Schemes

Having defined schemes, morphisms, subschemes, and projective schemes, we have formally constructed the category of schemes, denoted **Sch** (or $\mathbf{Sch}_S$ for S-schemes). Before moving on, it is valuable to look at the categorical properties of **Sch** as a whole.

How well-behaved is this category? Does it have limits and colimits? The short answer is that the local affine structure is well-behaved, but the global structure is much more complex.

The Affine Subcategory

Let $\mathbf{Sch}^{\mathbf{Aff}}$ be the full subcategory of affine schemes. By the definition of the spectrum functor, we have an anti-equivalence of categories:

$$\mathbf{Sch}^{\mathbf{Aff}} \simeq \mathbf{CRing}^{op}$$

Because the category of commutative rings is both complete (has all limits) and cocomplete (has all colimits), the category $\mathbf{Sch}^{\mathbf{Aff}}$ is also completely well-behaved: it has all limits and colimits. As discussed in section 2.2.10, operations here simply mirror the algebraic operations in rings (for example, the categorical coproduct of affine schemes corresponds to the direct product of their rings).

For instance, the initial object in **CRing** is $\mathbb{Z}$, which means the terminal object in $\mathbf{Sch}^{\mathbf{Aff}}$ is $\operatorname{Spec}\mathbb{Z}$. In fact, $\operatorname{Spec}\mathbb{Z}$ is the terminal object in the entirety of **Sch**. To prove this quickly, let X be any scheme, and let X be covered by affine open sets $\operatorname{Spec} A_i$. A morphism $X \to \operatorname{Spec}\mathbb{Z}$ restricts to local morphisms $\operatorname{Spec} A_i \to \operatorname{Spec}\mathbb{Z}$, which correspond exactly to ring homomorphisms $\mathbb{Z} \to A_i$. Since $\mathbb{Z}$ is the initial object in the category of rings, there exists exactly one such ring homomorphism for each A_i. Because these local morphisms are completely unique, they trivially agree on all overlaps and glue together to form a unique global morphism $X \to \operatorname{Spec}\mathbb{Z}$.

The Projective Subcategory

On the opposite end of the spectrum is the subcategory of projective schemes over a fixed base ring. While affine schemes are determined entirely by their global functions, connected projective schemes have only constant global functions.

Categorically, the projective subcategory is highly rigid and lacks the completeness of $\mathbf{Sch}^{\mathbf{Aff}}$. It does not admit infinite limits, and finite colimits (such as general pushouts) often fail to produce a scheme that remains projective. However, what the projective subcategory lacks in categorical completeness, it makes up for in geometric properness. Because projective schemes satisfy the valuative criterion for properness (as we will see later), curves do not escape to infinity without a limit. This makes the projective subcategory the ideal setting for intersection theory and global counting problems, whereas the affine category remains the setting for local algebraic calculations.

Subobjects and Immersions

In standard category theory, a "subobject" is defined as an equivalence class of monomorphisms. If we apply this strict categorical definition to **Sch**, we encounter non-geometric behavior. A morphism of schemes $X \to Y$ being a categorical monomorphism simply means it is left-cancellative.

To see why this is geometrically problematic, consider the affine subcategory $\mathbf{Sch}^{\mathbf{Aff}}$. Because of the anti-equivalence to **CRing**, a monomorphism of affine schemes corresponds exactly to an *epimorphism* of commutative rings. Ring epimorphisms certainly include surjective maps $A \to A/I$ (which yield closed immersions) and principal localizations $A \to A_f$ (which yield basic open immersions). However, ring epimorphisms also include total localizations like $\mathbb{Z} \to \mathbb{Q}$. Therefore, the map $\operatorname{Spec}\mathbb{Q} \to \operatorname{Spec}\mathbb{Z}$ is a strict categorical subobject in $\mathbf{Sch}^{\mathbf{Aff}}$, even though geometrically $\operatorname{Spec}\mathbb{Q}$ does not resemble a "piece" of $\operatorname{Spec}\mathbb{Z}$ (it is neither an open nor a closed subset).

For this reason, algebraic geometry rarely focuses on bare categorical monomorphisms. Instead, we restrict our attention to specific geometric embeddings: *open immersions* and *closed immersions*

(and their composition, locally closed immersions). These immersions ensure that the subscheme maps homeomorphically onto a topological subspace of the target and that the structure sheaves behave compatibly.

In $\mathbf{Sch}^{\mathbf{Aff}}$, these geometric subobjects are strictly generated by ideal quotients and localizations. In the projective subcategory, the geometric subobjects are almost exclusively closed immersions; because projective schemes are proper (compact), any open immersion that is also a projective scheme must be a closed set (a union of connected components). Thus, geometric "subschemes" form a much narrower and better-behaved class of morphisms than general categorical subobjects.

Limits in Sch

When we glue affine schemes together to form general schemes, we lose some of the categorical completeness found in $\mathbf{Sch}^{\mathbf{Aff}}$.

1. *Finite limits exist:* As proved above, the category **Sch** contains a terminal object $\operatorname{Spec}\mathbb{Z}$. Furthermore, as we will see in section 3.2, the category of schemes also admits pullbacks (fibered products). A standard result in category theory dictates that any category with a terminal object and pullbacks automatically has all finite limits (including equalizers and finite products).

2. *Infinite limits generally do not exist:* Unlike affine schemes, general schemes are not closed under infinite products. For example, the infinite product $\prod_{i=1}^{\infty} \mathbb{P}^1_k$ does not exist in the category of schemes. The intuitive reason for this failure is that the Zariski topology behaves poorly under infinite Cartesian products, making it impossible to find a valid affine open cover for the resulting space.

3. *Limits along affine morphisms exist:* an inverse system of schemes indexed by a directed set whose transition morphisms are affine has a limit in **Sch**. For affine schemes the limit is the spectrum of the colimit of the rings, so $\operatorname{Spec}\mathbb{Q}$ is the limit of the schemes $\operatorname{Spec}\mathbb{Z}[1/n]$, $n \geq 1$, ordered by divisibility. The general case is glued from this one (section 3.10⋆).

Colimits in Sch

1. *Arbitrary coproducts exist:* One of the simplest and most useful colimits in **Sch** is the coproduct. Any arbitrary collection of schemes $\{X_i\}$ has a coproduct given by their disjoint union $\coprod X_i$. As we established in section 2.2.9, this holds true for both finite and infinite collections.

2. *Finite colimits generally do not exist:* Outside of disjoint unions, the category of schemes does not have general pushouts or coequalizers. A standard counter-example involves trying to glue a scheme to itself along a generic point. For instance, the pushout of $\operatorname{Spec}(K) \rightrightarrows \mathbb{A}^1_K$ (where we attempt to glue the generic point to something else) fails to be a scheme. We will see this explicitly in example 3.2.7.

3. *Pushouts of open immersions exist:* There is one important exception to the failure of pushouts. If $f : X \to Z$ and $g : Y \to Z$ are both open immersions, their pushout always exists. This is exactly the categorical formulation of gluing schemes together, which we covered in section 2.2.9.

4. *Filtered colimits need not be schemes:* a directed union of schemes along open immersions is again a scheme, glued from the pieces as in theorem 2.2.56, and it is the colimit in **Sch**, since

compatible morphisms out of the pieces glue (lemma 3.1.9). Without the open-immersion hypothesis the colimit is formed as the functor $T \mapsto \varinjlim_i \mathrm{Hom}(T, X_i)$ on schemes, an *ind-scheme*. Infinite-dimensional projective space $\mathbb{P}^\infty_k = \varinjlim_n \mathbb{P}^n_k$, along the linear closed immersions, is the standard example. It is not represented by the scheme $\mathrm{Proj}\, k[x_0, x_1, \dots]$: that scheme has the k-point $(1:1:1:\cdots)$, which lies on no $\mathbb{P}^n_k$, while every k-point of $\varinjlim_n \mathbb{P}^n_k$ comes from some $\mathbb{P}^n_k(k)$. No scheme represents it either. If a scheme X did, the universal element id_X would come from some stage n, hence so would every element of $\varinjlim_i \mathrm{Hom}(T, \mathbb{P}^i_k)$ for every scheme T (each element is the pullback of id_X along some $T \to X$). For $T = \mathbb{P}^{n+1}_k$ and the element given by the identity, the linear closed immersion $\mathbb{P}^{n+1}_k \to \mathbb{P}^m_k$ would then factor through $\mathbb{P}^n_k$ for some m. That is impossible: a closed immersion induces surjections of local rings, hence injections of Zariski tangent spaces, and at a k-rational point p of $\mathbb{P}^{n+1}_k$ these have dimension $n+1$ at p and n at its image in $\mathbb{P}^n_k$.

Schemes as Colimits of Affine Schemes

We have seen that the category **Sch** contains $\mathbf{Aff} = \mathbf{CRing}^{\mathrm{op}}$ as a full subcategory, and that general schemes are obtained by "gluing" affine pieces. We can now make the gluing process categorically precise: every scheme is a colimit of affine schemes, and the colimit has a particularly clean form.

The coequalizer picture. Let X be a scheme admitting a finite affine open cover $X = U_1 \cup \cdots \cup U_n$. Form the disjoint union

$$U = U_1 \sqcup \cdots \sqcup U_n,$$

which is itself an affine scheme: if $U_i = \mathrm{Spec}\,(A_i)$, then $U = \mathrm{Spec}\,(A_1 \times \cdots \times A_n)$. The natural map $\pi : U \to X$ (the inclusion of each U_i into X) is surjective but not injective: a point lying in $U_i \cap U_j$ appears twice in U, once in the U_i-copy and once in the U_j-copy.

To recover X from U, we must identify these redundant copies. Define the "overlap scheme"

$$R = U \times_X U = \coprod_{i,j} (U_i \cap U_j).$$

By the Build-up Lemma (lemma 2.2.30), each intersection $U_i \cap U_j$ can be covered by open sets that are simultaneously distinguished in U_i and U_j. If X is separated, the intersections $U_i \cap U_j$ are themselves affine (section 3.5.1), and R is affine as well; in general, R can be covered by affines, and the construction still works by refining the cover.

The scheme R comes equipped with two natural morphisms to U: the projections

$$s, t : R \rightrightarrows U$$

where s includes $U_i \cap U_j$ into its U_i-copy and t includes it into its U_j-copy. The scheme X is then recovered as the *coequalizer* of s and t in the category of locally ringed spaces:

$$R \underset{t}{\overset{s}{\rightrightarrows}} U \xrightarrow{\pi} X.$$

That is, X is the universal locally ringed space receiving a map from U that identifies $s(r)$ with $t(r)$ for every point $r \in R$.

Let us form an equivalence relation between the glued points categorically. We shall look at the *groupoid object* in the category of schemes (or, when everything is affine, in **Aff**). The additional structure maps are:

1. *Identity:* $e : U \to R$, the diagonal, sending a point of U_i to the "trivial overlap" of U_i with itself.
2. *Composition:* $c : R \times_{s,U,t} R \to R$, expressing transitivity: if a point of $U_i \cap U_j$ and a point of $U_j \cap U_k$ agree on U_j, they compose to a point of $U_i \cap U_k$.
3. *Inverse:* $\iota : R \to R$, the swap: a point of $U_i \cap U_j$ is also a point of $U_j \cap U_i$.

These satisfy the usual groupoid axioms (associativity, unit, inverse), which encode the fact that the relation "x and y represent the same point of X" is an equivalence relation. On the ring side, dualizing: if $U = \operatorname{Spec}(A)$ and $R = \operatorname{Spec}(B)$, the two ring homomorphisms $s^\#, t^\# : A \rightrightarrows B$ together with the identity, composition, and inverse maps make (A, B) into a *cogroupoid object* in **CRing**.

Conversely, one can *start* with affine groupoid data and produce a scheme. Given rings A and B with homomorphisms $s^\#, t^\# : A \rightrightarrows B$ and maps $e^\# : B \to A$, $c^\# : B \to B \otimes_A B$, $\iota^\# : B \to B$ satisfying the cogroupoid axioms, together with the condition that s and t are open immersions (so that the "gluing maps" identify open subsets), the coequalizer of $\operatorname{Spec}(B) \rightrightarrows \operatorname{Spec}(A)$ in the category of locally ringed spaces is a scheme.

This gives a precise sense in which the category of schemes is built from the category of affine schemes: **Sch** is obtained from **Aff** by freely adjoining coequalizers of affine groupoid presentations whose structure maps are open immersions. Every scheme admits such a presentation (by choosing an affine cover), and the result is independent of the choice (by the Affine Communication Lemma and the uniqueness of colimits).

The reader should keep this in mind when going through the functorial perspective developed in section 3.3.7, where schemes are characterized as Zariski sheaves with affine open covers. The two viewpoints are dual: the functorial approach defines schemes as certain *limits* (sheaves are defined by equalizer conditions), while the groupoid approach constructs schemes as certain *colimits* (coequalizers of equivalence relations). The equivalence of these two perspectives is a manifestation of a general principle: in algebraic geometry, limit-based and colimit-based descriptions of the same object coexist, and switching between them is sometimes the key to a proof.

Exercise 3.1.2

1. Show that the map $\operatorname{Spec} k(x) \to \operatorname{Spec} k[y]_{(y)}$ sending the unique (generic) point η to the closed point of the codomain is a valid morphism of ringed spaces, but *not* a morphism of locally ringed spaces. (This highlights why the local ring condition is necessary for scheme morphisms).
2. Let $p \in X$ be a point in a scheme. Show there is a canonical morphism of schemes $\operatorname{Spec}(\mathcal{O}_{X,p}) \to X$.
3. Using the above, define a canonical morphism $\operatorname{Spec}(\kappa(p)) \to X$. This is the formal scheme-theoretic morphism corresponding to the inclusion of a point.
4. Show there is no non-trivial map $\mathbb{P}^n_k \to \mathbb{A}^m_k$.

3.2 (Fiber) Product and Fibers

https://stacks.math.columbia.edu/tag/01JW

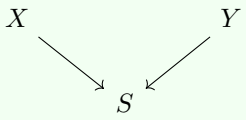
One of the most common tools in constructing new algebraic and geometric objects are limits and colimits. Given a category that is complete and cocomplete (having all limits and colimits), then:

1. limits are a combination of products and equalizers (i.e. cutting)
2. colimits are a combination of coproducts and coequalizers (i.e. gluing)

The category $\mathbf{Sch}_S$ is closed under coproducts (disjoint union), but not necessarily colimits as we require non-trivial gluing conditions (recall theorem 2.2.56). However, it *is* closed under products and equalizes (equalizer of an affine scheme R usually becomes a closed subscheme). To do this, we must show it is closed under products. Starting with the affine case, by corollary 3.1.5 it is equivalence for R-algebra to be closed under coproducts, and indeed the coproduct of A, B is $A \otimes_R B$ (see [12, Universal Property of the Coproduct in R-Alg]).

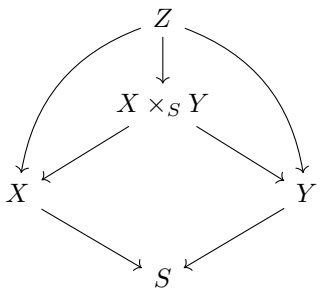
Theorem 3.2.1: Fibered Product Of Schemes

Let X, Y be schemes over S. Then then the fibered product of X and Y over S exists, that is the limit of the diagram exists:



The fibered product is denoted $X \times_S Y$

More commonly, we would say that the fibered product satisfies the universal property that if $Z \to X$ and $Z \to Y$ commutes in the diagram, then there exists a unique morphism $Z \to X \times_S Y$ such that



commutes. If S is the final object, or more concisely it is empty or "does not matter", then we have the usual product. If S is not specified, the product shall be defined as $X \times Y := X \times_{\mathrm{Spec}\,(\mathbb{Z})} Y$

***Proof*:**
We shall construct the products of s, and then glue them together.

Let $X = \mathrm{Spec}\,(A)$, $Y = \mathrm{Spec}\,(B)$, and $S = \mathrm{Spec}\,(R)$ be affine. Then A, B are R-algebras, and the product of X and Y over S is:

$$\mathrm{Spec}\,(A \otimes_R B)$$

This comes form the universal property of tensor products along with the property of adjoints. Indeed, if $Z \to \mathrm{Spec}\,(A \otimes_R B)$ is a scheme morphism, then this gives a ring homomorphism

$A \otimes_R B$ into $\mathcal{O}_Z(Z)$. By the universal property of tensor products, a morphism of $A \otimes_R B$ into is the same as a morphism from A and B to that ring. Then by proposition 3.1.12, this gives a morphism of Z into $\operatorname{Spec}(A \otimes_R B)$, and this morphism will satisfy the universal property of fibered products given by the universal property of tensor products.

Let us now construct the more general case via gluing. Let X, Y be schemes. To do this, we shall reduce to a covering of $X \times_S Y$. For any open affine $U \subseteq X$, if the product existed then $\pi_1^{-1}(U) \cong U \times_S Y$ (check this[a]). Now if $\{X_i\}$ is an open covering for x, then I claim that if $X_i \times_S X$ exists, so does $X \times_S Y$. Indeed, for each $U_{ij} = \pi_1^{-1}(X_i) \subseteq X_i \times_S Y$, by the uniqueness of the product there are unique isomorphisms: $\varphi_{ij} : U_{ij} \to U_{ji}$ for each i, j for compatible projections. Furthermore, these isomorphisms are compatible with the gluing condition (as you should check if you are not convinced!). As a remind, the condition was:

$$\varphi_{ik}|_{X_{ij} \cap X_{ik}} = \varphi_{jk} \circ \varphi_{ij}|_{X_{ij} \cap X_{ik}}$$

Thus, we get that we may glue them together by lemma 3.1.9 we have that this glued scheme is isomorphic to $X \times_S Y$, and through this isomorphism we can check that the universal property is satisfied.

By interchanging X and Y, we can now conclude from the proof on affine schemes that if X and Y are any schemes, and S is an affine, then $X \times_S Y$ exists. To show that S can be an arbitrary scheme, let $\{S_i\}$ be an open affine cover of S. Let $\alpha : X \to S$ and $\beta : Y \to S$ be two scheme morphisms, and let $X_i = \alpha^{-1}(S_i)$, $Y_i = \beta^{-1}(S_i)$. Then $X_i \times_{S_i} Y_i$ is well-defined.

Now, what's key is that this scheme is the same as the scheme $X_i \times_S Y$. To see this, we must invoke the uniqueness of the universal property, and indeed if $f : Z \to X_i$ and $g : Z \to Y$ are scheme morphisms over S, by the properties of the commutative diagram we must have that the image of v must land inside Y_i. But by universality they must be equal. Thus $X_i \times_S Y$ is well-defined for each i, and so by the above argument it glues to create $X \times_S Y$, as we sought to show.

[a] If $f : Z \to U$, compose with the inclusion map and invoke the appropriate universal properties

Example 3.2.2: Fibered Products

1. Show that $\mathbb{A}^n_k \times_k \mathbb{A}^m_k \cong \mathbb{A}^{n+m}_k$, which is exactly what we would hope from a product. Notably, prove $k[x] \otimes_k k[y] \cong k[x, y]$.
2. Similarly, if k is a characteristic 0 field show that $\mathbb{Z}[x] \otimes_{\mathbb{Z}} k[y] \cong k[x, y]$. Hence $\mathbb{A}^1_{\mathbb{Z}} \times \mathbb{A}^1_k \cong \mathbb{A}^2_k$.
3. Let $X = \operatorname{Spec} \mathbb{C}[x, y]/(x^2 - y)$ and $Y = \mathbb{C}[z, w]/(z^3 - w^4)$. Then:

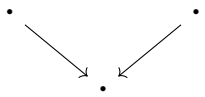
$$X \times_{\mathbb{C}} Y \cong \operatorname{Spec} \frac{\mathbb{C}[x, y, z, w]}{(x^2 - y, z^3 - w^3)}$$

   The reader should stare at this until they understand why it is called a fibered product.

4. There is new behaviour when considering schemes being tensored over non-algebraically closed fields. Take $\operatorname{Spec} \mathbb{C} \times_{\operatorname{Spec} \mathbb{R}} \operatorname{Spec} \mathbb{C}$. As these are affine, this is $\operatorname{Spec}(\mathbb{C} \otimes_{\mathbb{R}} \mathbb{C})$. As $\mathbb{C} \otimes_{\mathbb{R}} \mathbb{C} \cong \mathbb{C} \times \mathbb{C}$ (exercise),

$$\operatorname{Spec} \mathbb{C} \times_{\operatorname{Spec} \mathbb{R}} \operatorname{Spec} \mathbb{C} \cong \operatorname{Spec}(\mathbb{C}) \amalg \operatorname{Spec}(\mathbb{C})$$

Notice here how the choice of sheaf completely changed the answer, notably $\operatorname{Spec}\mathbb{C}$ and $\operatorname{Spec}\mathbb{R}$ are schemes over a single point, hence the fibered product diagram when only considering points is an identity map on points:



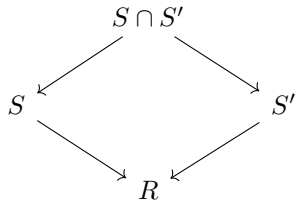
and in **Top** this results is a single point. However, the resulting scheme is *two* points. Thus, the fibered product here somehow "unravelled" the fact that $\operatorname{Spec}\mathbb{R}$ should be thought of as "gluing" conjugates together (think of the intuition for $\operatorname{Spec}\mathbb{R}[x]$ as a $\operatorname{Spec}\mathbb{C}[x]/\operatorname{Gal}_{\mathbb{R}}(\mathbb{C})$, the real line then do not glue to another points, but their residue field is $\mathbb{R}$ instead of $\mathbb{C}$, and hence it codimension 2 with respect to $\mathbb{C}$).

More generally show that if L/K is a finite galois extension, then

$$L \otimes_K L \cong \prod_{|\operatorname{Aut}(L/K)|} K$$

is isomorphic to $L^{|G|}$($|G|$ copies of L), hint: this is the linear independence of characters proof in geometric form. Hence, over fields fibered products capture some idea properties of galois groups.

5. Recall that the intersection of rings is a ring. This is a special case of the fibered product in the category of commutative rings, namely that if $S \to R$ and $S' \to R$ exists, then:



Let us see how this translates to schemes and affine schemes. Recall that the product and coproduct of affine schemes is an affine scheme:

$$\operatorname{Spec} S \times \operatorname{Spec} S' = \operatorname{Spec} S \otimes_{\mathbb{Z}} S'$$
$$\operatorname{Spec} S \amalg \operatorname{Spec} S' = \operatorname{Spec} S \times S'$$

If $U, U' \subseteq \operatorname{Spec} R$ are open affine, $U = \operatorname{Spec} S$ and $U' = \operatorname{Spec} S'$, Then as $\operatorname{Spec}(-)$ is contravariant we need that the intersection $(\operatorname{Spec}(S)) \cap (\operatorname{Spec}(S'))$ becomes the "union" of rings. Naturally, the union of rings need not be a ring, and as both S, S' contain units then $\langle S, S' \rangle = R$. Instead, the natural dual is:

$$\operatorname{Spec} S \cap \operatorname{Spec} S' = \operatorname{Spec} S \otimes_R S'$$

As we saw, the union of affine need not be affine (see example 2.2.57).

One of the most important uses of the fibered is to change the base scheme:

Definition 3.2.3: Base Change (Scheme-theoretic Pullback)

Let X be an S-scheme, and $T \to S$ a scheme morphism. Then the *base change* of X to T is the T-scheme:

$$X_T = T \times_S X$$

If $f : X \to Y$ is an S-scheme morphism, and $T \to S$ a scheme morphism, then the *base change* of f is the map $f' : X_T \to Y_T$ where $f' = \mathrm{id}_T \times_{\mathrm{id}_S} f$.

Example 3.2.4: Base Change

1. For any R-scheme X and morphism $\operatorname{Spec} S \to \operatorname{Spec} R$ (i.e. ring homomorphism $R \to S$), we get the S-scheme:
$$X' = S \times_R X$$
The category of R-schemes maps to the category of S schemes via base change.
2. If $X = R[x_1, ..., x_m]/I$, is an R-scheme and $S \to R$ is a ring homomorphism, then:
$$X_S = \frac{S[x_1, ..., x_m]}{IS}$$
that is, the set of constant change.
3. As the map $S \to R$ need not be surjective, we can get something like $X = \operatorname{Spec} \mathbb{C}$ as a $\mathbb{C}$-scheme and base change $\mathbb{R} \hookrightarrow \mathbb{C}$.
4.

As the tensor product is a very versatile operation, the fibered product is as well. Let us demonstrate by translating some important results:

Proposition 3.2.5: Base Change Properties

The following properties hold for fibered products:

1. Let $\iota : U \to Z$ be an open immersion and $\varphi : Y \to Z$ be any scheme morphism. Then $U \times_Z Y$ is isomorphic to the open subscheme of Y given by the preimage of U:
$$U \times_Z Y \cong \left(\varphi^{-1}(U), \mathcal{O}_Y|_{\varphi^{-1}(U)}\right)$$
2. *Stability of Closed Immersions:* If $X \to Z$ is a closed immersion, then for any map $Y \to Z$, the base change $X \times_Z Y \to Y$ is a closed immersion.
3. *Stability of Open Immersions:* If $X \to Z$ is an open immersion, then for any map $Y \to Z$, the base change $X \times_Z Y \to Y$ is an open immersion.

***Proof*:**

1. An open immersion is an isomorphism onto an open subscheme followed by its inclusion (definition 2.2.49), so we may take $U \subseteq Z$ open with inclusion ι. Let $W = \varphi^{-1}(U)$, with inclusion $j\colon W \to Y$ and the restriction $\varphi_W\colon W \to U$ of φ (lemma 2.2.52), so that $\iota \circ \varphi_W = \varphi \circ j$. Let $a\colon T \to U$ and $b\colon T \to Y$ satisfy $\iota \circ a = \varphi \circ b$. Then $\varphi(b(T)) \subseteq U$, so $b(T) \subseteq W$ and $b = j \circ b_W$ for a unique $b_W\colon T \to W$ (lemma 2.2.52). Now $\iota \circ \varphi_W \circ b_W = \varphi \circ b = \iota \circ a$, and ι is a monomorphism (lemma 2.2.52), so $\varphi_W \circ b_W = a$. A morphism $c\colon T \to W$ with $j \circ c = b$ equals b_W, as j is a monomorphism. So W, with φ_W and j, satisfies the universal property of $U \times_Z Y$, and $U \times_Z Y \cong W$ by the uniqueness of fibered products.
2. Let $i\colon X \to Z$ be a closed immersion and $p\colon X' = X \times_Z Y \to Y$ the projection. Cover Y by affine opens $V \cong \operatorname{Spec}(S)$, each mapping into an affine open $W \cong \operatorname{Spec}(R)$ of Z. By proposition 3.1.24(1) (the direction cited from Hartshorne), $i^{-1}(W) \cong \operatorname{Spec}(R/I)$ over W for an ideal I. By (1), applied twice, and the universal property of theorem 3.2.1, the open subscheme $p^{-1}(V)$ is $X \times_Z V \cong i^{-1}(W) \times_W V$, which is $\operatorname{Spec}((R/I) \otimes_R S) = \operatorname{Spec}(S/IS)$ by the affine case of theorem 3.2.1, with $p^{-1}(V) \to V$ induced by the surjection $S \to S/IS$. So $p^{-1}(V) \to V$ is a closed immersion (proposition 3.1.24(2)), and p is a closed immersion by proposition 3.1.24(4).
3. If $X \to Z$ is an open immersion with image the open subscheme U, then by (1) the base change $X \times_Z Y \to Y$ is, up to isomorphism, the inclusion of the open subscheme $\varphi^{-1}(U)$, an open immersion, completing the proof.

This motivates an important definition generalizing affine and projective schemes (idk if this should be here)

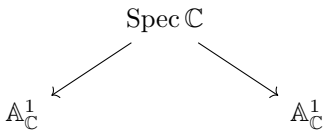
Definition 3.2.6: Affine Schemes over a Scheme

Let X be a scheme. Then:

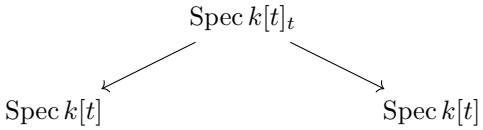
$$\mathbb{A}^n_X = X \times_{\mathbb{Z}} \mathbb{Z}[x_1, \ldots, x_n]$$

Example 3.2.7: Pushout Need Not Exist

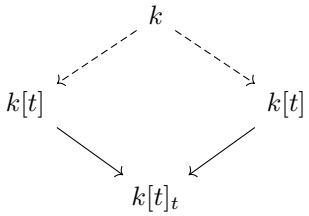
Consider two copies of $\mathbb{A}^1_{\mathbb{C}}$. Consider:



where $[(0)] \mapsto [(0)]$ in both cases. This is like the double-origin example, except every point is a double origin. This shall be shown to not be a scheme as it is "too non-separated".
We cannot fix this by limiting to the category of affine schemes. Consider for example:



where $\operatorname{Spec} k[t]_t = \mathbb{A}^1_k \backslash \{0\}$, and the maps are the maps given by $t \mapsto t$ and $t \mapsto 1/t$. Then we saw in example 2.15 that the colimit of this diagram is projective space, $\mathbb{P}^1_k$. Hence affine schemes are not in general closed under pushout. Note that the corresponding fibered product does exist in CRing, namely it will have to be:



However, applying Spec to this will not return the pushout, namely the pair of adjoint functors (Spec, Γ) does not preserve pushouts.

Note however that colimits do exist in the category of locally ringed spaces. See this link (which link??) for more information.

(words here)

Definition 3.2.8: Geometrically Integral

Let S be a k-scheme. Then S is *geometrically integral* if it is integral over the algebraic closure$\overline{k}$.

The term "geometrically PROPERTY" is a common term to specify that a PROPERTY occurs for the k-scheme when taking the algebraic closure of k.

(want to include this somewhere: *Tensor Product Interpretation (Base Change):* The map constructed above is canonically isomorphic to the base change of the ring R_r to S. Using the tensor product:

$$S \otimes_R R_r \cong S \otimes_R (R[x]/\langle xr - 1\rangle) \cong S[x]/\langle x\varphi(r) - 1\rangle \cong S_{\varphi(r)}$$

This interpretation is powerful because it links the *geometric pullback* of sheaves to the *algebraic tensor product.*)

3.2.1 Fibers and family of Schemes

This gives a huge class of new object's that we may define. Let us start with defining fibers:

Definition 3.2.9: Fibers

Let $f : X \to Y$ be a morphism of schemes. Let $y \in Y$, and $k(y)$ be the residue field of y. Let $\operatorname{Spec}(k(y)) \to Y$ be the natural morphism[a]. Then the *fiber* of the morphism f over the point y is the scheme

$$X_y = X \times_Y \operatorname{Spec}(\kappa(y))$$

In the case where $X = \operatorname{Spec}(R), Y = \operatorname{Spec}(S)$ are affine, then for a point $\mathfrak{p} \in Y$:

$$f^{-1}(\mathfrak{p}) = \operatorname{Spec} R \times_S \operatorname{Spec} \kappa(\mathfrak{p}) = \operatorname{Spec}\ (R \otimes_S S_{\mathfrak{p}}/\mathfrak{p}S_{\mathfrak{p}})$$

[a] It is the composite of $\operatorname{Spec} k(y) \to \operatorname{Spec} \mathcal{O}_{Y,y}$, dual to the quotient by the maximal ideal, with the canonical $\operatorname{Spec} \mathcal{O}_{Y,y} \to Y$.

Proposition 3.2.10: Topological Space of Fibers

Let $f : X \to Y$ be a morphism of schemes, and let $y \in Y$. The natural map $|X_y| \to f^{-1}(y)$ is a homeomorphism.

Furthermore, $X_y \to X$ is a *closed immersion* identified with the ideal $f^{\#}(\mathfrak{m}_y)$.

***Proof*:**
The question is affine local on source and target (proving bijectivity can be reduced to compatible open covers), so let's assume $X = \operatorname{Spec}(R)$ and $Y = \operatorname{Spec}(S)$ are affine schemes. Then $f : \operatorname{Spec} R \to \operatorname{Spec} S$ corresponds to a ring homomorphism $f^{\#} : S \to R$. Let y correspond to the prime ideal $\mathfrak{p}$ in S. Then:

$$X_y = X \times_Y \operatorname{Spec}(\kappa(y)) = \operatorname{Spec}(R \otimes_S S_{\mathfrak{p}}/\mathfrak{p}S_{\mathfrak{p}})$$

We shall take advantage of the description of points as maps to residue fields, taken up in definition 3.3.1. Suppose $(f^{\#})^{-1}(\mathfrak{q}) = \mathfrak{p}$ where $\mathfrak{q} \in \operatorname{Spec}(R)$ (i.e. $f(\mathfrak{q}) = \mathfrak{p}$). Then Let g represent the composition $S \to R \to R_{\mathfrak{q}}$. Then $g : S \to R_{\mathfrak{q}}$ factors uniquely through $\overline{g} : S_{\mathfrak{p}} \to R_{\mathfrak{q}}$. We can extend this map by tensoring with $R \otimes_S (-)$ so that:

$$\Phi : R \otimes_S S_{\mathfrak{p}} \to R_{\mathfrak{q}} \qquad (r, s/t) = r\overline{g}(s/t)$$

Note that any element in $\mathfrak{p}S_{\mathfrak{p}}$ maps to $\mathfrak{q}R_{\mathfrak{q}}$, hence have have a unique map:

$$R \otimes_S S_{\mathfrak{p}}/\mathfrak{p}S_{\mathfrak{p}} \twoheadrightarrow R_{\mathfrak{q}}/\mathfrak{q}R_{\mathfrak{q}} = \kappa(\mathfrak{q})$$

As this map is unique, the correspondence between $k(\mathfrak{q})$ and the points of the fiber is unique, and certainly it surjective. Thus, for each point $\operatorname{Spec}(R_{\mathfrak{q}}/\mathfrak{q}R_{\mathfrak{q}})$, it is associated to a unique point in $f^{-1}(\mathfrak{p})$.

Finally, $f^{-1}(\{y\})$ satisfies the (topological) pullback

$$\begin{array}{ccc} f^{-1}(y) & \longrightarrow & X \\ \downarrow & & \downarrow f \\ \operatorname{Spec}(k(y)) & \longrightarrow & Y \end{array}$$

Taking:

$$\begin{array}{ccc} X_y & \longrightarrow & X \\ \downarrow & & \downarrow f \\ \operatorname{Spec}(k(y)) & \longrightarrow & Y \end{array}$$

The maps are all continuous and follow from properties of fiber product, and the algebra above shows it is a bijection. It is a homeomorphism because $R \otimes_S S_{\mathfrak{p}}/\mathfrak{p}S_{\mathfrak{p}} \cong (S \setminus \mathfrak{p})^{-1}(R/\mathfrak{p}R)$, so the map to $\operatorname{Spec} R$ factors as a localization of $\operatorname{Spec}(R/\mathfrak{p}R)$ followed by the closed immersion $\operatorname{Spec}(R/\mathfrak{p}R) \hookrightarrow \operatorname{Spec} R$, and each of these is a homeomorphism onto its image.

In some ways, fibers of schemes are more "regular" than fibers of sets, as the extra information of schemes will provide some more possible pre-images:

Example 3.2.11: Fibers

1. Take the familiar projection of the parabola onto the x axis induced by the ring map $\mathbb{Q}[x] \to \mathbb{Q}[y]$ mapping $x \mapsto y^2$. We can think of this as the map $\mathbb{Q}[x] \to \mathbb{Q}[y^2]$ where the codomain is isomorphic to:
$$\mathbb{Q}[y^2] \cong \frac{\mathbb{Q}[x,y]}{x-y^2}$$
This is sometimes called the parameterization of the map. Then the pre-image of 1 is:
$$\begin{aligned} \operatorname{Spec} \mathbb{Q}[x,y]/\left(y^2-x\right) \otimes_{\mathbb{Q}[x]} \mathbb{Q}[x]/(x-1) &\cong \operatorname{Spec} \mathbb{Q}[x,y]/\left(y^2-x, x-1\right) \\ &\cong \operatorname{Spec}\left(\mathbb{Q}[y]/\left(y^2-1\right)\right) \\ &\cong \operatorname{Spec}(\mathbb{Q}[y]/(y-1)(y+1)) \\ &\overset{\text{C.R.T}}{\cong} \operatorname{Spec}(\mathbb{Q}[y]/(y-1) \times \mathbb{Q}[y]/(y+1)) \\ &\cong \operatorname{Spec}(\mathbb{Q}[y]/(y-1)) \amalg \operatorname{Spec}(\mathbb{Q}[y]/(y+1)) \end{aligned}$$
That is, it is the points ± 1. The pre-image of 0 is:
$$\operatorname{Spec} \mathbb{Q}[x,y]/(y^2-x,x) \cong \operatorname{Spec} \mathbb{Q}[y]/(y^2)$$
Notice that the pre-image remembers multiplicity information!
As the spectrum of a polynomial ring over rationals is the quotient by the galois action on the polynomial ring over $\bar{k}$, the pre-image of -1 exists! It is:
$$\operatorname{Spec} \mathbb{Q}[x,y]/(y^2-x,x+1) \cong \operatorname{Spec} \mathbb{Q}[y]/(y^2+1) \cong \operatorname{Spec} \mathbb{Q}[i] = \operatorname{Spec} \mathbb{Q}(i)$$
This can be thought of as "size 2" point over the base field. The pre-image of the generic point is a reduced point, and can be thought of as "size 2" over the residue field:
$$\operatorname{Spec} \mathbb{Q}[x,y]/(y^2-x) \otimes_{\mathbb{Q}[x]} \mathbb{Q}(x) \cong \operatorname{Spec} \mathbb{Q}[y] \otimes_{\mathbb{Q}[y]} \mathbb{Q}(y^2)$$
Try working through the pre-image of other points that are not commonly associated to $\mathbb{Q}$. Notice that the pre-image is the two points, or you get nonreduced behaviour that is "of degree 2" or you get a field extension of degree 2.

2. Let $X = \text{Spec}(\mathbb{Z}[i])$ (the Gaussian integers) and $Y = \text{Spec}(\mathbb{Z})$. The inclusion $\mathbb{Z} \hookrightarrow \mathbb{Z}[i]$ induces a morphism $\pi : X \to Y$. Let us analyze the fibers of this map (the pre-images of points).

   - *Ramification:* Consider the point $(2) \in \text{Spec}(\mathbb{Z})$. The fiber is $\text{Spec}(\mathbb{Z}[i] \otimes_{\mathbb{Z}} \mathbb{Z}/(2)) \cong \text{Spec}(\mathbb{F}_2[x]/(x^2+1)) = \text{Spec}(\mathbb{F}_2[x]/(x+1)^2)$. Topologically, there is one point, but the ring has a nilpotent element. The prime 2 "ramifies."
   - *Splitting:* Consider $(5) \in \text{Spec}(\mathbb{Z})$. In $\mathbb{Z}[i]$, $5 = (2+i)(2-i)$. The fiber corresponds to $\mathbb{F}_5[x]/(x^2+1)$. Since x^2+1 has roots $2, 3$ in $\mathbb{F}_5$, this ring is $\mathbb{F}_5 \times \mathbb{F}_5$. Geometrically, the point (5) pulls back to *two* distinct points.
   - *Inert:* Consider $(3) \in \text{Spec}(\mathbb{Z})$. x^2+1 is irreducible in $\mathbb{F}_3$. The fiber is $\text{Spec}(\mathbb{F}_9)$, a single point with a larger residue field.

   This example demonstrates that scheme morphisms provide the correct geometric framework for Algebraic Number Theory.

3. As $X \to \text{Spec}\,(\mathbb{Z})$ is always well-defined and unique, there is always fibers over $\text{Spec}\,(\mathbb{Z})$ and hence they are usually given a name: The fiber over (0) is often labeled $X_{\mathbb{Q}}$ (over $\mathbb{Q}$), and the fibers for all the primes are X_p (over $\mathbb{F}_p$). A common terminology is that X_p *arises by reduction* mod p of the scheme X.

4. One way of characterizing surjectivity $\text{Spec}\,(B) \to \text{Spec}\,(A)$ ring theoretically is to require that for each $\mathfrak{p} \in A$, $B \otimes_A \kappa(\mathfrak{p}) \neq 0$, that is that every fiber is nonempty; the fiber itself is definition 3.2.9.

Proposition 3.2.12: Pre-image of Generic (Spreading out)

Let $f : X \to Y$ be a morphism of finite type and Y Noetherian. Then if $f^{-1}(\eta) = \emptyset$ for a generic point $\eta \in Y$, there exists an open subset $U \subseteq Y$ containing η such that $f^{-1}(U) = \emptyset$.

***Proof*:**
Since the problem is local on Y around η, we may assume $Y = \text{Spec}(A)$ is affine. Since Y has a generic point η, Y is irreducible. As Y is Noetherian, we can treat A as an integral domain (modulo nilpotents, which do not affect the topology or emptiness of schemes). Let $K = \text{Frac}(A)$ be the residue field of η.

Since f is of finite type, X is quasi-compact. We can cover X by a finite number of affine open subsets $X = \bigcup_{i=1}^{n} \text{Spec}(B_i)$, where each B_i is a finitely generated A-algebra.

The hypothesis $f^{-1}(\eta) = \emptyset$ means the fiber product $X \times_Y \text{Spec}(K)$ is the empty scheme. For each affine patch $\text{Spec}(B_i)$, this corresponds to the ring:

$$B_i \otimes_A K = 0$$

Note that $B_i \otimes_A K$ is isomorphic to the localization of B_i by the multiplicative set $S = A \setminus \{0\}$. If a localization is the zero ring, then $1 = 0$ in that ring. By the definition of localization, this implies there exists an element $s_i \in S$ (i.e., a non-zero element $s_i \in A$) such that:

$$s_i \cdot 1_{B_i} = 0 \quad \text{in } B_i$$

Now, consider the basic open set $U_i = D(s_i) \subseteq Y$. Since $s_i \neq 0$, $\eta \in U_i$. The preimage of this

open set in the chart $\operatorname{Spec}(B_i)$ is $\operatorname{Spec}((B_i)_{s_i})$. However, since s_i annihilates the unity in B_i, the localized ring is zero:

$$1 = \frac{s_i}{s_i} = \frac{1}{s_i} \cdot (s_i \cdot 1_{B_i}) = 0$$

Thus, $\operatorname{Spec}((B_i)_{s_i}) = \emptyset$.

To handle the entire scheme X, let $s = s_1 s_2 \cdots s_n$. Since A is a domain, $s \neq 0$, so $U = D(s)$ is a non-empty open set containing η. Furthermore, $U \subseteq U_i$ for all i. Consequently, the preimage $f^{-1}(U)$ is the union of the preimages restricted to each chart. Since s is a multiple of s_i, s acts as zero on every B_i, making the localization $(B_i)_s = 0$ for all i. Therefore, $f^{-1}(U) = \emptyset$.

On p. 260 of Vakil he gives more on general fibers, generic fibers, and generically finite morphisms

do not know where to put these examples yet

Example 3.2.13: Parameterization of Curves using Schemes
Take

$$X = \operatorname{Spec} \frac{k[x,y,t]}{(ty - x^2)}$$
$$Y = \operatorname{Spec} k[t]$$

Let $f : X \to Y$ be given by the natural ring homomorphism $k[t] \to \frac{k[x,y,t]}{(ty-x^2)}$. Note both these schemes are integral schemes of finite type over k.

Now, identify the closed points of Y with k. For $a \neq 0 \in k$, we get the fibers correspond to the parabola's $ay = x^2$ or $y = x^2/a$. All of these are irreducible, reduced curves. At $a = 0$, we get $x^2 = 0$, which is a reduced scheme.

More generally, S-schemes can be geometrically interpreted as schemes paramaterized by the points of S whose maps are fiber-bundle morphisms. Hence, the scheme $\operatorname{Spec} k[t]$ *paramaterized* the curve.

A quick note must be done about general products and why it may be preferable to take the fibered product over a field:

Example 3.2.14: Strange Fibered Products

1. Show that $\operatorname{Spec}(\mathbb{Z}/m\mathbb{Z}) \times_{\operatorname{Spec}(\mathbb{Z})} \operatorname{Spec}(\mathbb{Z}/n\mathbb{Z}) = \emptyset$.
2. Another product of taking the "absolute" fibered product is that if $X = \operatorname{Spec} \mathbb{Z}[x]$ and $Y = \operatorname{Spec} \mathbb{Z}[y]$ then:

$$\dim X \times Y = \dim(X) + \dim(Y) - \dim(\operatorname{Spec} Z) = \dim X + \dim Y - 1$$

If we restrict to k-schemes for a field k, these two pathologies do not occur.

Put an appropriate definition here for the following

Hence, given a surjective map $Y \to X$, Y can be regarded as a collection of schemes paramaterized by Y. Conversely, we may want to study a scheme X_0, and see if there are some deformations we

may want to do. Then a common way of defining a *family of algebraic deformation* is by a scheme morphisms $f : X \to Y$ where one of the fibers of f is isomorphic to X_0.

Some more here on representable Functors

Exercise 3.2.1

1. Let $\varphi : X \to Y$ be a morphism of S-scheme of finite type. Show that if $S \to S'$ is any base extension, $\varphi' : X' \to Y'$ is an S'-scheme morphism preserves finite type.
2. Let $\varphi : X \to Y$ be a scheme morphism of integral schemes. Show the fibers need not be irreducible or reduced.
3. Let $\mathfrak{p} \in X$ be a point in a scheme. Show there is a canonical map
$$\operatorname{Spec}(\mathcal{O}_{X,\mathfrak{p}}) \to X$$
4. Using the above, define a canonical morphism $\operatorname{Spec}(\kappa(\mathfrak{p})) \to X$.
5. Perhaps put here some base-change exercises relating the tensor product exercises you had in [22] and now give them their geometric counter-part.
6. Show if $U, V \subseteq X$ are open subschemes then $U \times_X V \cong U \cap V$.
7. Let $\varphi : B \to A$ be a ring morphism, $S \subseteq B$ a multiplicative subset, so $\varphi(S)$ is a multiplicative subset of A. Show that $\varphi(S)^{-1}A \cong A \otimes_B (S^{-1}B)$. What does this imply for the relation between localization and the fibered product?
8. Let X be a k-scheme. Show that if L/k is a field extension, $X \times_{\operatorname{Spec}(k)} \operatorname{Spec}(L)$ is an L-scheme. Show that this in fact gives a functor (define the appropriate maps, and verify).

3.2.2 *Monomorphisms in Sch

Using fiber products, we can better characterize subschemes categorically. It is natural to ask: what are the *categorical* sub-objects of a scheme? In any category, the correct notion of "sub-object" is a monomorphism: a morphism $f : Y \to X$ such that for any two morphisms $g, h : Z \rightrightarrows Y$, the equality $f \circ g = f \circ h$ implies $g = h$. In the category of sets, monomorphisms are injective functions. In the category of groups, they are injective homomorphisms. In the category of schemes, the answer is more interesting.

Proposition 3.2.15: Monomorphisms of Schemes

A morphism $f : Y \to X$ of schemes is a monomorphism if and only if the diagonal morphism $\Delta : Y \to Y \times_X Y$ is an isomorphism.

***Proof*:**
By definition, f is a monomorphism if and only if for every scheme Z, the map $f_* : \operatorname{Hom}(Z, Y) \to \operatorname{Hom}(Z, X)$ is injective. By the universal property of the fiber product, $\operatorname{Hom}(Z, Y \times_X Y) \cong \operatorname{Hom}(Z, Y) \times_{\operatorname{Hom}(Z,X)} \operatorname{Hom}(Z, Y)$, so f_* is injective for all Z if and only if the two projections

$Y \times_X Y \rightrightarrows Y$ are equal, which holds if and only if Δ is an isomorphism.

For affine schemes, the characterization is purely algebraic. A morphism $\operatorname{Spec}(S) \to \operatorname{Spec}(R)$ is a monomorphism if and only if the corresponding ring map $\varphi : R \to S$ is an *epimorphism of commutative rings*: a map such that for any two ring maps $\alpha, \beta : S \rightrightarrows T$, the equality $\alpha \circ \varphi = \beta \circ \varphi$ implies $\alpha = \beta$. Equivalently, φ is an epimorphism if and only if the multiplication map $S \otimes_R S \to S$ is an isomorphism.

Note that monomorphissm of schemes are at least always injective. It is easy to see this reduces to affine patches. THen for a ring map $f : R \to S$ to be an epimorphism in the category of commutative rings, it must satisfy a strict tensor condition: the multiplication map $S \otimes_R S \to S$ must be an isomorphism. Geometrically, this injective on points, and it does not ramify or fold back on itself; the localizations $R \to R_f$ are the standard example.

The crucial observation is that epimorphisms of rings are *much more general* than surjections. The three main classes are:

Example 3.2.16: Epimorphisms in CRing

1. *Surjections* $R \twoheadrightarrow R/I$. These are epimorphisms because any map out of R/I is determined by its values on the images of elements of R. The corresponding monomorphisms of affine schemes are the *closed immersions*.

2. *Localizations* $R \to S^{-1}R$. These are epimorphisms because any map out of $S^{-1}R$ is determined by its values on elements of R (the images of elements of S must be invertible, which leaves no freedom). In general these maps are neither open nor closed (recall proposition 2.1.30). This includes both $R \to R_f$ (giving *open immersions* to distinguished opens) and $R \to R_{\mathfrak{p}}$ (giving the *local scheme* morphisms $\operatorname{Spec}(\mathcal{O}_{X,x}) \to X$).

   The most famous such example is $\mathbb{Z} \to \mathbb{Q}$ which is the localization away from the prime (0).

3. *Gabriel Localizations.* Most localization is done at either a single element or the compliment of a prime ideal. However, we can localize ideals. More generally, we can localize on *filters of ideals* (Gabriel localization) corresponding to an arbitrary intersection of open sets. Naturally, open sets are not closed under arbitrary intersection! These can result in pathological "fractal" sets that are still categorically a subscheme.

All four types of subobjects from the summary table are monomorphisms of schemes, but the converse is false: there exist monomorphisms that do not fall into any of the four classes. For instance, the map $\operatorname{Spec}(\mathbb{Q}) \to \operatorname{Spec}(\mathbb{Z})$ is a monomorphism (the inclusion $\mathbb{Z} \hookrightarrow \mathbb{Q}$ is an epimorphism of rings—it is the localization at the multiplicative set $\mathbb{Z} \setminus \{0\}$), but the image is the single point $\{[(0)]\}$, which is the generic point of $\operatorname{Spec}(\mathbb{Z})$. This is neither open, nor closed, nor locally closed: it is a local scheme map.

3.3 The Functor of Points

Throughout the development of scheme theory, we have built our geometric objects from locally ringed spaces: a topological space equipped with a structure sheaf, glued from affine pieces. This is a powerful framework, but it carries a certain asymmetry: the points of a scheme and the morphisms between schemes are defined by quite different-looking data. The goal of this section is to

develop a perspective—the *functor of points*—in which schemes are understood entirely through their morphisms. The result is not a rephrasing; it reveals structure that the locally ringed space definition obscures, and it provides the language in which most modern moduli problems are formulated.

The idea has a simple origin. Recall from section 2.2.5 that a global section $f \in \Gamma(X, \mathcal{O}_X)$ of a scheme X assigns to each point $x \in X$ an element $f(x)$ in the residue field $\kappa(x)$, but that these residue fields vary from point to point. A global section is therefore *not* a function in the classical sense. Let us trace how this observation leads, step by step, to a functorial description of schemes.

3.3.1 From Evaluation to Morphisms

When the base field is algebraically closed, the situation is familiar. Over an algebraically closed field $k = \overline{k}$, a polynomial $f \in \overline{k}[x_1, \dots, x_n]$ determines a genuine set-theoretic function:

$$\overline{k}[x_1, \dots, x_n] \longrightarrow \mathrm{Hom}_{\mathbf{Set}}(\overline{k}^n, \overline{k}), \qquad f \longmapsto \big((a_1, \dots, a_n) \mapsto f(a_1, \dots, a_n)\big). \tag{3.1}$$

Here $\overline{k}^n$ can be identified with $\mathrm{mSpec}\,(\overline{k}[x_1, \dots, x_n])$ via the Nullstellensatz, and each evaluation $f(a_1, \dots, a_n)$ lands in $\overline{k} \cong \kappa(\mathfrak{m}_{(a_1,\dots,a_n)})$. The map in equation (3.1) is injective precisely because an algebraically closed field has "enough points" to separate polynomials.

Over a general field k, this breaks down: the maximal ideals of $k[x_1, \dots, x_n]$ no longer correspond to k-rational tuples. As discussed in example 2.1.3, $\mathrm{mSpec}\,(k[x_1, \dots, x_n])$ contains orbits of geometric points under the absolute Galois group. One can partially recover the situation by mapping not into k, but into the affine line $\mathbb{A}^1_k$, which itself carries these additional closed points. Concretely, a polynomial $f \in k[x_1, \dots, x_n]$ induces a morphism of k-algebras

$$k[t] \longrightarrow k[x_1, \dots, x_n], \qquad t \longmapsto f,$$

which, by the equivalence of affine schemes with rings, corresponds to a scheme morphism $\mathbb{A}^n_k \to \mathbb{A}^1_k$. Evaluating at a closed point $\mathfrak{m}$ of $\mathbb{A}^n_k$ yields the image of f in $\kappa(\mathfrak{m})$, and this image determines a closed point of $\mathbb{A}^1_k$. Thus, we arrive at a map

$$k[x_1, \dots, x_n] \longrightarrow \mathrm{Hom}_{\mathbf{Sch}_k}(\mathrm{mSpec}\, k[x_1, ..., x_n], \mathrm{mSpec}\, k[t]), \qquad f \longmapsto \text{the morphism induced by } t \mapsto f. \tag{3.2}$$

This is already an improvement: it works over any field, and the target is a set of *scheme morphisms* rather than set-theoretic functions. However, it still only detects the behavior of f at closed points. Recall that such schemes are called *Jacobson schemes* (see definition 3.7.4). For general schemes this is insufficient. For instance, on $\mathrm{Spec}\,(k[[x]])$ or $\mathrm{Spec}\,(\mathbb{Z}_{(p)})$, a function is not determined by its values at closed points alone.

The natural fix is to allow *all* prime ideals as test points. For any k-algebra A, a scheme morphism $\mathrm{Spec}\,(A) \to \mathbb{A}^1_k$ is the same data as a k-algebra map $k[t] \to A$, which since this is a k-algebra homomorphism is the same as choosing an element of A for t to map to. This gives:

$$\Gamma(\mathrm{Spec}\,(A), \mathcal{O}_{\mathrm{Spec}\,(A)}) = A \;\cong\; \mathrm{Hom}_{\mathbf{Sch}_k}(\mathrm{Spec}\,(A),\, \mathbb{A}^1_k). \tag{3.3}$$

The left-hand side is the ring of global sections; the right-hand side is the set of scheme morphisms to the affine line. Each element $f \in A$ corresponds to the unique morphism $\pi_f : \mathrm{Spec}\,(A) \to \mathbb{A}^1_k$ for which $\pi_f^{\#}(t) = f$. The "value" of f at a prime $\mathfrak{p}$ is recovered by following $t \mapsto f \mapsto f \bmod \mathfrak{p} \in \kappa(\mathfrak{p})$, which is precisely the scheme-theoretic notion of evaluation from section 2.2.5.

Two observations are now in order. First, the construction (3.3) is not specific to affine schemes: it will extend to arbitrary k-schemes X once we establish the representability of the global section functor (Proposition 3.3.9 below). Second, and more importantly, there is nothing special about $\mathbb{A}^1_k$ as a target apart from the fact that it *represents* the functor of global sections. One can ask: what happens if we replace $\mathbb{A}^1_k$ by another scheme T?

The set $\mathrm{Hom}_{\mathbf{Sch}_k}(X, T)$ makes perfect sense for any T, and different choices of T probe different aspects of X. Taking $T = \mathbb{A}^1_k$ recovers global sections; taking $T = \mathbb{G}_m$ recovers invertible global sections; taking $T = \mathbb{P}^n_k$ recovers line bundles generated by $n+1$ global sections (cf. section 5.3.3). The test object T acts as a "measuring device" that detects a particular piece of the geometry of X. The fact—a consequence of the Yoneda Lemma applied to schemes—is that if we keep track of $\mathrm{Hom}(-, X)$ for *all* possible test objects simultaneously, we recover X entirely.

3.3.2 Z-valued Points

Consider the scheme

$$X = \mathrm{Spec}\ \frac{\mathbb{R}[x,y]}{(x^2+y^2-1)},$$

whose underlying topological space consists of $\mathrm{Gal}(\mathbb{C}/\mathbb{R})$-conjugate pairs of maximal ideals together with a generic point. The closed points do not individually look like points on a circle—many have residue field $\mathbb{C}$, not $\mathbb{R}$—and so the "usual" picture of the unit circle is not immediately visible.

If we want to recover that picture, we should look for $\mathbb{R}$-valued points: ring homomorphisms

$$\mathrm{ev}_{a,b} : \frac{\mathbb{R}[x,y]}{(x^2+y^2-1)} \longrightarrow \mathbb{R}$$

sending $x \mapsto a$ and $y \mapsto b$. The quotient relation forces $a^2 + b^2 = 1$, so the collection of all such homomorphisms is in bijection with

$$X(\mathbb{R}) \ \simeq\ \left\{(a,b) \in \mathbb{R}^2 \ \middle|\ a^2+b^2=1\right\},$$

which is exactly the unit circle. Equivalently, each such homomorphism factors through the evaluation map $\mathbb{R}[x,y] \to \mathbb{R}$ as a map that kills the ideal (x^2+y^2-1). Via the contravariant equivalence of affine schemes and rings, these ring homomorphisms correspond to scheme morphisms $\mathrm{Spec}\,(\mathbb{R}) \to X$. The sheaf morphism on global sections is:

$$\mathrm{ev}^{\#}_{a,b}(X) : \frac{\mathbb{R}[x,y]}{(x^2+y^2-1)} \longrightarrow \mathbb{R}.$$

Now suppose we change the target ring. Over $\mathbb{Z}$, the equation $a^2+b^2=1$ has only the four solutions $(\pm 1, 0)$ and $(0, \pm 1)$. Over $\mathbb{Q}$, a Pythagorean-triple parametrization yields infinitely many solutions such as $\left(\frac{3}{5}, \frac{4}{5}\right)$. Over $\mathbb{C}$, the circle becomes a rational curve, and the solution set is much larger still. Each choice of "coefficient ring" yields a different collection of evaluation maps, probing different arithmetic and geometric features of the same scheme.

There is one subtlety to address regarding the base. For the evaluation map $\mathrm{ev}_{a,b}$ to exist as a map of $\mathbb{R}$-algebras, the target ring must receive a structure map from $\mathbb{R}$. There is no nontrivial ring homomorphism $\mathbb{R} \to \mathbb{Q}$, so asking for $\mathbb{Q}$-valued points of the $\mathbb{R}$-scheme X is not well-posed. To allow all these evaluations simultaneously, it is natural to work over the universal base $\mathbb{Z}$. Set

$X_{\mathbb{Z}} = \operatorname{Spec} \mathbb{Z}[x,y]/(x^2+y^2-1)$. Then for any commutative ring R, the R-valued points of $X_{\mathbb{Z}}$ are the ring homomorphisms $\mathbb{Z}[x,y]/(x^2+y^2-1) \to R$, or equivalently the solutions of $a^2+b^2=1$ in R. If $A \to B$ is a ring homomorphism and X_A, X_B denote the base changes, then $X_A(B) = X_B(B)$: the B-valued points see only the base-changed scheme.

It is worth pausing to understand the $\mathbb{Z}$-points of $X_{\mathbb{Z}}$ scheme-theoretically.

A $\mathbb{Z}$-point of $X_{\mathbb{Z}}$ is a morphism $\operatorname{Spec}(\mathbb{Z}) \to X_{\mathbb{Z}}$, which is a section of the structure map $X_{\mathbb{Z}} \to \operatorname{Spec}(\mathbb{Z})$. Geometrically, one can visualize $X_{\mathbb{Z}}$ as an arithmetic surface fibered over $\operatorname{Spec}(\mathbb{Z})$, with each fiber X_p being the circle $x^2+y^2=1$ reduced modulo the prime p. A $\mathbb{Z}$-point is then a "horizontal" section of this fibration: it passes through every fiber simultaneously.

These $\mathbb{Z}$-points are *not* closed points of $X_{\mathbb{Z}}$. Consider the point $P=(1,0)$, corresponding to the homomorphism $\varphi : \mathbb{Z}[x,y]/(x^2+y^2-1) \to \mathbb{Z}$ defined by $x \mapsto 1$, $y \mapsto 0$. The kernel is the prime ideal $\mathfrak{p} = (x-1, y)$, and the quotient $\mathbb{Z}[x,y]/\mathfrak{p} \cong \mathbb{Z}$. Since $\mathbb{Z}$ is not a field, $\mathfrak{p}$ is prime but not maximal, and the image of P is a closed subscheme isomorphic to $\operatorname{Spec}(\mathbb{Z})$—it has dimension 1 and residue field $\mathbb{Q}$ at its generic point. In contrast, a closed point of $X_{\mathbb{Z}}$ has dimension 0 and a finite residue field $\mathbb{F}_q$.

The four $\mathbb{Z}$-points interact with the fibers in instructive ways. Modulo 2, since $1 \equiv -1$ in $\mathbb{F}_2$, the points $(1,0)$ and $(-1,0)$ collapse to the same point of X_2, as do $(0,1)$ and $(0,-1)$. Modulo an odd prime $p \geq 3$, all four points remain distinct in $X_p(\mathbb{F}_p)$. These four "horizontal sections" are the only integer solutions on the circle; any other rational point, such as $\left(\frac{3}{5}, \frac{4}{5}\right)$, gives a $\mathbb{Q}$-point but not a $\mathbb{Z}$-point.

More generally, there is no reason to restrict the "value scheme" to be affine. One might wish to evaluate points on a torus with values in a curve, or to study morphisms between projective schemes. The evaluation interpretation as a global ring homomorphism is then lost, but this is precisely what the sheaf was built to handle: evaluation can be performed locally. Geometrically, a morphism $Z \to X$ between arbitrary schemes is the datum of compatible local evaluations. This motivates the following definition.

Definition 3.3.1: Z-valued Points

Let X be a scheme. The *Z-valued points* of X are the morphisms $Z \to X$ in the category of schemes. The set of all such morphisms is denoted

$$X(Z) := \operatorname{Hom}_{\mathbf{Sch}}(Z, X).$$

When $Z = \operatorname{Spec}(R)$ is affine, this is written $X(R)$.

The reader should note a potential source of confusion: a "Z-valued point" is the *morphism* $Z \to X$ itself, not any particular point in Z or in the image of Z. This terminology is justified by the fact that a morphism of schemes is not determined by the underlying continuous map of topological spaces—the sheaf map $\varphi^{\#}$ carries additional information. The full morphism is the "point".

the fibered product of two schemes need not be the schemes of these two fibered products. In a concrete sense, this is because we must be more specific with what types of points we are dealing with. If we specify the points, we in fact recover the usual "sets"[5].

[5]The topological product can also be thought of as being over the right valued points

Proposition 3.3.2: Fibered Product With Z-valued Points

Let X, Y be schemes over a scheme W and take Z-valued points on each scheme i.e. morphisms in $\mathrm{Hom}(Z, X)$ and $\mathrm{Hom}(Z, Y)$. Then:

$$\mathrm{Hom}(Z, X \times_W Y) = \mathrm{Hom}(Z, X) \times_{\mathrm{Hom}(Z,W)} \mathrm{Hom}(Z, Y)$$

***Proof*:**
Write $\alpha : X \to W$ and $\beta : Y \to W$ for the structure morphisms and $p : X \times_W Y \to X$, $q : X \times_W Y \to Y$ for the projections, so that $\alpha \circ p = \beta \circ q$. The right side is the fibered product in **Set** (example 0.4.16): the set of pairs $(f, g) \in \mathrm{Hom}(Z, X) \times \mathrm{Hom}(Z, Y)$ with $\alpha \circ f = \beta \circ g$ in $\mathrm{Hom}(Z, W)$.

Define

$$\Theta : \mathrm{Hom}(Z, X \times_W Y) \to \mathrm{Hom}(Z, X) \times_{\mathrm{Hom}(Z,W)} \mathrm{Hom}(Z, Y) \qquad \varphi \mapsto (p \circ \varphi,\ q \circ \varphi)$$

The pair on the right satisfies $\alpha \circ p \circ \varphi = \beta \circ q \circ \varphi$, so Θ is well-defined. Given a pair (f, g) with $\alpha \circ f = \beta \circ g$, the universal property of the fibered product (theorem 3.2.1) produces a morphism $\varphi : Z \to X \times_W Y$ with $p \circ \varphi = f$ and $q \circ \varphi = g$, and states that this φ is unique. Existence makes Θ surjective and uniqueness makes it injective, so Θ is a bijection.

Example 3.3.3: Non-trivial Projective Line: Severi–Brauer Variety
A Severi–Brauer Variety over k is a variety P that becomes isomorphic to $\mathbb{P}^n$ for appropriate n after base-changing to an algebraic closure[a]. Take the conic:

$$C : x^2 + y^2 + z^2 = 0 \subseteq \mathbb{P}^2_{\mathbb{R}}$$

This connect is simply $\mathrm{Proj}\, \frac{\mathbb{R}[x,y,z]}{(x^2+y^2+z^2)}$. Taking the standard chart $D_+(x)$ and letting $u = y/x$, $v = z/x$ we see that:

$$C \cap D_+(x) = \mathrm{Spec}\left(\frac{R[u,v]}{(1+u^2+v^2)}\right)$$

In this chart that (v, u^2+1) is a closed point, since:

$$\frac{\mathbb{R}[u,v]/(1+u^2+v^2)}{(v,u^2+1)} \cong \frac{\mathbb{R}[u]}{(u^2+1)} \cong \mathbb{C}$$

In C itself, this becomes the homogeneous principle ideal (z, x^2+y^2). When base-changing to $\mathbb{C}$, this becomes the two point $[1 : i : 0], [1 : -i : 0] \in C_{\mathbb{C}}$.

However, there are no $\mathbb{R}$-points: there is no triple $a, b, c \in \mathbb{R} \setminus \{0\}$ such that $a^2 + b^2 + c^2 = 0$, hence $C(\mathbb{R}) = \emptyset$.

In cite:HERE, we shall see this corresponds to its Brauer class being the quaternion algebra $\left(\frac{-1,-1}{\mathbb{R}}\right)$, and that $\mathrm{Br}(\mathbb{R}) \cong \mathbb{Z}/2\mathbb{Z}$.

[a] For those who know projective bundles, it's the function-field analogue of a projective bundle with no vector-bundle lift

3.3.3 The Yoneda Embedding for Schemes

Having defined Z-valued points, we can now ask: how much of a scheme X is captured by the totality of its valued points? The answer, given by the Yoneda Lemma, is: *everything.*

Proposition 3.3.4: Scheme Morphisms from Valued Points

Let X and Y be schemes. A morphism of schemes $\varphi : X \to Y$ is equivalent to a natural transformation $\eta : h_X \to h_Y$ between the associated functors of points. That is, giving a collection of maps of sets

$$\eta_Z : X(Z) \longrightarrow Y(Z)$$

for every scheme Z, natural in Z, uniquely determines a scheme morphism $\varphi : X \to Y$, and conversely.

***Proof*:**
Every scheme X defines a contravariant functor $h_X : \mathbf{Sch}^{\mathrm{op}} \to \mathbf{Set}$ by

$$h_X(Z) = \mathrm{Hom}_{\mathbf{Sch}}(Z, X) = X(Z).$$

A morphism $\varphi : X \to Y$ induces a natural transformation $\varphi_* : h_X \to h_Y$ by post-composition: for each Z and each $f \in X(Z)$, set $(\varphi_*)_Z(f) = \varphi \circ f$. Naturality follows from the associativity of composition.

Conversely, given a natural transformation $\eta : h_X \to h_Y$, we evaluate at $Z = X$. The set $X(X) = \mathrm{Hom}(X, X)$ contains the identity id_X, and we define

$$\varphi := \eta_X(\mathrm{id}_X) \in Y(X) = \mathrm{Hom}(X, Y).$$

By the Yoneda Lemma (section 0.4),

$$\mathrm{Nat}(h_X, h_Y) \;\cong\; h_Y(X) = \mathrm{Hom}(X, Y),$$

and this correspondence is bijective. Thus, the assignment $X \mapsto h_X$ defines a fully faithful embedding $\mathbf{Sch} \hookrightarrow \mathrm{Fun}(\mathbf{Sch}^{\mathrm{op}}, \mathbf{Set})$: the category of schemes embeds into the category of presheaves on $\mathbf{Sch}$.

The proof is short, but the content is not trivial. It says that a scheme is *completely determined* by how all other schemes map into it. No information is lost by passing from the locally ringed space X to the functor h_X.

A useful analogy organizes this perspective. In linear algebra, a vector $v \in V$ is determined by the collection of values $\ell(v)$ for all linear functionals $\ell \in V^*$: the dual space "separates points." Choosing a basis for V^* gives coordinates on V. The Yoneda embedding plays the same role for schemes. The functor h_X is the "dual description" of X: the scheme is determined by the collection of sets $X(Z)$ for all test schemes Z. Each choice of test scheme is analogous to choosing a basis vector in the dual—it extracts one piece of information about X.

Different test objects give genuinely different perspectives on the same scheme:

Test object Z	$X(Z)$	*What it detects*
$\mathrm{Spec}\,(\overline{k})$	geometric points	the classical variety
$\mathrm{Spec}\,(k)$	rational points	arithmetic information
$\mathrm{Spec}\,(k[\epsilon]/(\epsilon^2))$	tangent vectors	infinitesimal geometry (order 1)
$\mathrm{Spec}\,(k[t]/(t^n))$	$(n-1)$-jets	higher infinitesimal geometry
$\mathrm{Spec}\,(k[[t]])$	formal arcs	formal local behavior
$\mathrm{Spec}\,(K),\ K = k(C)$	rational curves	birational geometry
$\mathrm{Spec}\,(\mathcal{O}_{X,x})^\dagger$	the germ at x	local scheme structure
$\mathrm{Spec}\,(\hat{\mathcal{O}}_{X,x})^\dagger$	formal neighborhood	formal geometry at x

† The last two entries differ in character from the others: they depend on the scheme X itself, rather than being universal test objects. Their role is not to "probe" X from outside, but to isolate the local structure at a point $x \in X$.

The local ring $\mathcal{O}_{X,x}$ is the stalk of the structure sheaf, and $\mathrm{Spec}\,(\mathcal{O}_{X,x})$ is the "germ" of X at x: the limit of all open neighborhoods of x, retaining only the information visible in an arbitrarily small neighborhood. There is a canonical morphism $\mathrm{Spec}\,(\mathcal{O}_{X,x}) \to X$ (induced by the localization maps $\mathcal{O}_X(U) \to \mathcal{O}_{X,x}$), and any morphism $f : Y \to X$ sending a point $y \in Y$ to x factors through $\mathrm{Spec}\,(\mathcal{O}_{X,x})$ at the level of germs:

$$\mathrm{Spec}\,(\mathcal{O}_{Y,y}) \longrightarrow \mathrm{Spec}\,(\mathcal{O}_{X,x}) \longrightarrow X.$$

In this sense, the local ring "sees" every morphism that passes through x. This is why properties of morphisms (flatness, smoothness, étaleness) can often be checked at the level of local rings.

The completion $\hat{\mathcal{O}}_{X,x}$ captures even finer information: the formal power series behavior at x, analogous to the Taylor expansion of a smooth function. The formal arc entry $\mathrm{Spec}\,(k[[t]])$ in the table is in fact the special case $\mathrm{Spec}\,(\hat{\mathcal{O}}_{\mathbb{A}^1,0})$—the formal neighborhood of the origin in the affine line. For a general point x on a smooth n-dimensional variety, the completion is $\hat{\mathcal{O}}_{X,x} \cong k[[t_1, \ldots, t_n]]$, and the formal neighborhood is a "formal polydisk" of dimension n.

The passage from one test object to another is analogous to a change of basis. A morphism $Z' \to Z$ induces, by precomposition, a map $X(Z) \to X(Z')$ that relates the two perspectives. For instance, the closed immersion $\mathrm{Spec}\,(k) \hookrightarrow \mathrm{Spec}\,(k[\epsilon]/(\epsilon^2))$, dual to the quotient map $k[\epsilon]/(\epsilon^2) \twoheadrightarrow k$, sends a tangent vector to its basepoint. This is the "forgetful" map from infinitesimal data to point data. The inclusions among the infinitesimal test objects form a tower:

$$\mathrm{Spec}\,(k) \hookrightarrow \mathrm{Spec}\,(k[\epsilon]/(\epsilon^2)) \hookrightarrow \mathrm{Spec}\,(k[t]/(t^3)) \hookrightarrow \cdots \hookrightarrow \mathrm{Spec}\,(k[[t]])$$

where each step remembers one more order of infinitesimal information, and the formal arc $\mathrm{Spec}\,(k[[t]])$ is the limit, retaining all orders simultaneously.

3.3.4 Key Examples

The definition of Z-valued points subsumes many classical constructions. We now develop the most important instances in detail.

Example 3.3.5: Geometric Points

Let X be a scheme locally of finite type over a field k, and let $\overline{k}$ be an algebraic closure. A $\overline{k}$-valued point is a k-morphism

$$\varphi : \operatorname{Spec}(\overline{k}) \longrightarrow X.$$

Since $\operatorname{Spec}(\overline{k})$ has a unique point η, the morphism φ is determined by the image $y = \varphi(\eta) \in X$ together with the stalk map $\varphi_y^{\#} : \mathcal{O}_{X,y} \to \overline{k}$. Passing to the residue field, we obtain a k-embedding $\kappa(y) \hookrightarrow \overline{k}$.

We claim that y must be a closed point. Indeed, if y were not closed, then $\kappa(y)$ would contain an element transcendental over k; say t. In particular, as $\varphi_y(t) \in \overline{k}$ must be algebraic, there exists a k-polynomial p such that $\varphi_y(p(t)) = p(\varphi_y(t)) = 0$. As $\varphi_y(0) = 0$ and t is transcendental, $p(t) \neq 0$, demonstrating the failure of injectivity—a contradiction, since field homomorphisms are injective. Conversely, if y is closed, then $\kappa(y)/k$ is a finite extension (by Zariski's Lemma, since X is locally of finite type), and any finite extension of k embeds into $\overline{k}$. Thus, y must be closed, and every closed point arises in this way.

However, there may be *more* $\overline{k}$-valued points than closed points. For example, the $\mathbb{R}$-scheme $\operatorname{Spec}(\mathbb{R}[x]/(x^2+1)) \cong \operatorname{Spec}(\mathbb{C})$ has a single closed point, but two $\mathbb{C}$-valued points: these correspond to the two $\mathbb{R}$-algebra homomorphisms $\mathbb{C} \to \mathbb{C}$ (namely the identity and complex conjugation). In general, the number of $\overline{k}$-valued points lying over a closed point y equals the number of k-embeddings $\kappa(y) \hookrightarrow \overline{k}$, which is $[\kappa(y) : k]$ when $\kappa(y)/k$ is separable.

To obtain a bijection, one can either base-change to $\overline{k}$ (replacing X by $X_{\overline{k}} = X \times_k \operatorname{Spec}(\overline{k})$, which identifies all the Galois conjugates; see section 3.2), or work directly over an algebraically closed field. In the latter case, $\kappa(y) = \overline{k}$ for every closed point, and the correspondence

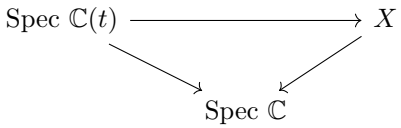
$$\left\{\operatorname{Spec}(\overline{k}) \to X\right\} \xrightarrow{\sim} \{y \in X : y \text{ is closed}\}$$

is bijective. This recovers the classical variety associated to X: the "geometric points" are exactly the closed points over an algebraically closed field.

Example 3.3.6: Generic Points and Rational Maps

Let C be a curve over $\mathbb{C}$ with function field $K = \mathbb{C}(C)$. A K-valued point of a $\mathbb{C}$-scheme X is a $\mathbb{C}$-morphism $\operatorname{Spec}(K) \to X$. Since $\operatorname{Spec}(K)$ is the generic point of C, such a morphism amounts to a rational map $C \dashrightarrow X$: it is defined on some dense open subset of C, but may not extend to all of C.

More concretely, taking $K = \mathbb{C}(t)$ (the function field of $\mathbb{A}^1_{\mathbb{C}}$), any $\mathbb{C}(t)$-valued point fits into the commutative diagram:



This is the algebro-geometric analogue of a "generic curve" in X: rather than picking out a single closed point, it describes a one-parameter family of points parametrized by the generic point of the affine line. Whether such a rational map extends to a regular morphism $\mathbb{A}^1_{\mathbb{C}} \to X$ (or $C \to X$ in general) is precisely the content of the valuative criteria discussed in section 3.5.1 and section 3.5.4.

Example 3.3.7: Tangent Vectors and the Dual Numbers

Let $D = k[\epsilon]/(\epsilon^2)$ be the ring of dual numbers. The scheme $\operatorname{Spec}(D)$ is a "fat point": it has a unique closed point $\overline{o}$ (corresponding to the maximal ideal (ϵ)), but its structure sheaf remembers one infinitesimal direction. The canonical surjection $D \twoheadrightarrow k$ corresponds to a closed immersion $\iota : \operatorname{Spec}(k) \hookrightarrow \operatorname{Spec}(D)$, and we write $o = \iota^{-1}(\overline{o})$.

Let X be a k-scheme of finite type and let $x \in X$ be a closed point (so $\kappa(x) = k$). We classify all D-valued points of X based at x, namely all morphisms $\varphi : \operatorname{Spec}(D) \to X$ such that $\varphi \circ \iota$ sends o to x. Choose an affine open neighborhood $x \in \operatorname{Spec}(R) \subseteq X$; then $\operatorname{Hom}_x(\operatorname{Spec}(D), X) \cong \operatorname{Hom}_x(\operatorname{Spec}(D), \operatorname{Spec}(R))$, where the subscript x indicates that the closed point maps to x. By the equivalence of affine schemes and rings, these are in bijection with k-algebra homomorphisms $f : R \to D$ satisfying $f(\mathfrak{m}_x) \subseteq (\epsilon)$.

Now R decomposes as a k-vector space as $R = k \oplus \mathfrak{m}_x$. Indeed, as x is closed,

$$0 \to \mathfrak{m}_x \hookrightarrow R \twoheadrightarrow R/\mathfrak{m}_x \cong k \to 0$$

and as k is free this splits. Then f is the identity on k, so f is determined by its restriction to $\mathfrak{m}_x$. Since $\epsilon^2 = 0$ in D, we have $f(\mathfrak{m}_x^2) = 0$, and so f descends to a k-linear functional on $\mathfrak{m}_x/\mathfrak{m}_x^2$. Conversely, any linear functional $\ell : \mathfrak{m}_x/\mathfrak{m}_x^2 \to k$, extended by zero on $\mathfrak{m}_x^2$ and by the identity on k, determines a k-algebra map $R \to D$ sending $\mathfrak{m}_x$ into (ϵ). We conclude:

$$\operatorname{Hom}_x(\operatorname{Spec}(D),\, X) \;\cong\; (\mathfrak{m}_x/\mathfrak{m}_x^2)^\vee = \Theta_{X,x},$$

the Zariski tangent space at x. This is a satisfying realization: the tangent space, originally defined via the maximal ideal and its square (cf. section 2.7), is recovered purely by probing X with the fat point $\operatorname{Spec}(D)$.

Moreover, if $f : X \to Y$ is a morphism of k-schemes, then post-composition with f sends a D-valued point based at x to one based at $f(x)$:

$$\mathrm{d}_x f :\; \Theta_{X,x} \longrightarrow \Theta_{Y,f(x)}, \qquad v \longmapsto f \circ v.$$

This is the differential of f, now defined without any reference to local rings or Kähler differentials—it is simply the functoriality of D-valued points.

Example 3.3.8: Group Schemes via Valued Points

By Proposition 3.3.4, a group scheme G over k can equivalently be described as a scheme such that for every k-scheme Z, the set $G(Z)$ carries a group structure, and all maps $G(Z) \to G(Z')$ induced by morphisms $Z' \to Z$ are group homomorphisms. In other words, G is a "group object" in the functor-of-points sense: the functor $h_G : \mathbf{Sch}_k^{\mathrm{op}} \to \mathbf{Set}$ factors through the forgetful functor $\mathbf{Grp} \to \mathbf{Set}$.

For a concrete instance, consider the multiplicative group scheme $\mathbb{G}_m = \operatorname{Spec}(k[t, t^{-1}])$. For any k-algebra A:

$$\mathbb{G}_m(A) = \operatorname{Hom}_{k\text{-}\mathbf{Alg}}(k[t, t^{-1}],\, A) \;\cong\; A^\times.$$

A k-algebra homomorphism out of $k[t, t^{-1}]$ is determined by the image of t, and invertibility of t in the source forces the image to be a unit in A. Since the units of any ring form a group under multiplication, and ring homomorphisms preserve units, $\mathbb{G}_m(A)$ is a group for every A, functorially in A.

Similarly, the additive group scheme $\mathbb{G}_a = \operatorname{Spec}(k[t]) = \mathbb{A}^1_k$ satisfies $\mathbb{G}_a(A) = A$ with its additive group structure; and the general linear group $\mathrm{GL}_{n,k} = \operatorname{Spec}(k[x_{ij}, \det^{-1}])$ satisfies $\mathrm{GL}_{n,k}(A) \cong \mathrm{GL}_n(A)$ for any k-algebra A. These examples demonstrate a recurring theme: when working over a non-algebraically-closed base, or with non-reduced schemes, the functor of points often gives a cleaner description than the underlying topological space. One *defines* the algebraic group by specifying what it does on all test rings, and the scheme structure is a consequence.

3.3.5 Representable Functors in Algebraic Geometry

The examples of the previous subsection probe a fixed scheme X by varying the test object Z. We now reverse the perspective: fix the type of geometric data we wish to extract, and ask whether there exists a "universal" scheme that classifies it. Formally, given a contravariant functor $\mathcal{F} : \mathbf{Sch}_k^{\mathrm{op}} \to \mathbf{Set}$, we say $\mathcal{F}$ is *representable* if there exists a scheme M and a natural isomorphism $\mathcal{F} \cong h_M$. The scheme M is then called a *fine moduli space* for the problem encoded by $\mathcal{F}$, and by the Yoneda Lemma it is unique up to unique isomorphism.

We have already encountered the simplest instance of this idea.

Proposition 3.3.9: Global Sections and the Affine Line

Let X be a k-scheme. There is a natural bijection

$$\Gamma(X, \mathcal{O}_X) \;\cong\; \mathrm{Hom}_{\mathbf{Sch}_k}(X,\, \mathbb{A}^1_k).$$

Concretely, a global section $f \in \Gamma(X, \mathcal{O}_X)$ corresponds to the unique morphism $\pi_f : X \to \mathbb{A}^1_k$ such that $\pi_f^{\#}(t) = f$, where t is the coordinate on $\mathbb{A}^1_k = \operatorname{Spec}(k[t])$. In categorical language, the global section functor $\Gamma(-, \mathcal{O})$ is representable by $\mathbb{A}^1_k$.

***Proof*:**
By the adjunction between Spec and Γ (proposition 3.1.12):

$$\mathrm{Hom}_{\mathbf{Sch}_k}(X,\, \operatorname{Spec} k[t]) \;\cong\; \mathrm{Hom}_{k\text{-}\mathbf{Alg}}(k[t],\, \Gamma(X, \mathcal{O}_X)).$$

The polynomial ring $k[t]$ is the free k-algebra on one generator: a k-algebra homomorphism $k[t] \to R$ is uniquely determined by the image of t. Thus:

$$\mathrm{Hom}_{k\text{-}\mathbf{Alg}}(k[t],\, \Gamma(X, \mathcal{O}_X)) \;\cong\; \Gamma(X, \mathcal{O}_X).$$

Let us verify that the bijection matches our intuition of evaluation. Let $x \in X$ and let $\pi_f : X \to \mathbb{A}^1_k$ be the morphism corresponding to $f \in \Gamma(X, \mathcal{O}_X)$. The definition of a scheme morphism gives a commutative diagram of local rings:

$$\begin{array}{ccccc} k[t] & \xrightarrow{\;\psi\;} & \Gamma(X, \mathcal{O}_X) & \longrightarrow & \mathcal{O}_{X,x} \\ \downarrow & & & & \downarrow{\scriptstyle \text{residue}} \\ \mathcal{O}_{\mathbb{A}^1, \pi_f(x)} & & \longrightarrow & & \kappa(x) \end{array}$$

The composition across the top sends $t \mapsto f \mapsto f(x) \in \kappa(x)$, and the image point $\pi_f(x)$ is the prime ideal $\mathfrak{q} = \{P(t) \in k[t] : P(f(x)) = 0 \text{ in } \kappa(x)\}$. This is exactly the scheme-theoretic "value" of f at x.

3.3.10 Remark: no field hypothesis is needed The above proof uses only that $k[t]$ is free on one generator; it did not use that k is a field. For any base scheme S, the relative affine line $\mathbb{A}^1_S$ represents the global section functor on $\mathbf{Sch}_S$. The same remark applies to several of the examples that follow.

The representability of $\Gamma(-, \mathcal{O})$ by $\mathbb{A}^1_k$ is the first instance of a broad pattern. Let us now survey the most important representable functors in algebraic geometry. In each case, a piece of geometric data attached to a scheme X is captured by morphisms from X into a single "classifying" scheme.

The multiplicative group. Replacing the additive structure of $\mathbb{A}^1_k$ with the multiplicative structure of $\mathbb{G}_m = \operatorname{Spec}(k[t, t^{-1}])$ gives:

$$\operatorname{Hom}_{\mathbf{Sch}_k}(X, \mathbb{G}_m) \;\cong\; \Gamma(X, \mathcal{O}_X)^\times.$$

The proof follows the same pattern: a k-algebra map $k[t, t^{-1}] \to R$ is determined by the image of t, and invertibility of t in the source forces the image to be a unit. The $\mathbb{G}_m$-valued points of X are precisely the invertible global sections.

Projective space and line bundles. The representability of projective space is quite interesting. Define the functor

$$\mathcal{F}(X) = \{(\mathcal{L},\, s_0, \ldots, s_n) \mid \mathcal{L} \in \operatorname{Pic}(X),\ s_i \in \Gamma(X, \mathcal{L}),\ \text{the } s_i \text{ generate } \mathcal{L}\} \,/\!\cong$$

where two tuples $(\mathcal{L}, s_0, \ldots, s_n)$ and $(\mathcal{L}', s_0', \ldots, s_n')$ are identified if there is an isomorphism $\mathcal{L} \xrightarrow{\sim} \mathcal{L}'$ carrying s_i to s_i' for each i. Then there is a natural bijection

$$\operatorname{Hom}_{\mathbf{Sch}_k}(X, \mathbb{P}^n_k) \;\cong\; \mathcal{F}(X).$$

A morphism to $\mathbb{P}^n$ is the same data as a line bundle equipped with $n+1$ globally generating sections; these sections serve as "homogeneous coordinates" for the map. This is the content of the theory developed in section 5.3.3. When $n = 1$, a map $X \to \mathbb{P}^1$ corresponds to a line bundle with two generating sections—the algebro-geometric analogue of a meromorphic function.

Grassmannians. The Grassmannian $\operatorname{Gr}(r, n)$ generalizes projective space from lines to r-dimensional subspaces. It represents the functor

$$\mathcal{F}(X) = \left\{\mathcal{O}_X^{\oplus n} \twoheadrightarrow \mathcal{E} \;\middle|\; \mathcal{E} \text{ is a locally free sheaf of rank } r\right\} /\!\cong.$$

A morphism $X \to \operatorname{Gr}(r, n)$ classifies a rank-r quotient bundle of the trivial bundle $\mathcal{O}_X^{\oplus n}$, or equivalently, a sub-bundle of rank $n - r$. When $r = 1$, this recovers the description of projective space above. The Grassmannian is itself a smooth projective variety; its functor-of-points description gives a uniform construction that works over any base ring, without the need to choose coordinates or Plücker embeddings.

The general linear group. The group scheme $\mathrm{GL}_{n,k} = \operatorname{Spec}(k[x_{11}, \ldots, x_{nn}, \det^{-1}])$ represents the functor

$$\operatorname{Hom}_{\mathbf{Sch}_k}(\operatorname{Spec}(A), \mathrm{GL}_{n,k}) \;\cong\; \mathrm{GL}_n(A)$$

for any k-algebra A. A map into $\mathrm{GL}_{n,k}$ is the same as giving an invertible $n \times n$ matrix with entries in A. Combined with the group scheme structure from Example 3.3.8, this gives a definition of the general linear group that works uniformly over all base rings, including rings with nilpotents or mixed characteristic.

Hilbert schemes. Perhaps the most application of representability is given by the Hilbert scheme, which we develop in the next subsection.

The common thread in each of the above examples is the Yoneda philosophy: a geometric invariant of X—global sections, line bundles, quotient bundles, invertible matrices—is captured by morphisms from X into a single classifying scheme. The Yoneda Lemma ensures that the classifying scheme, when it exists, is unique up to unique isomorphism.

3.3.6 Hilbert Schemes

The representable functors considered so far classify data that is "linear" in nature: sections of a sheaf, quotients of a free module, invertible matrices. The Hilbert scheme is of a fundamentally different character: it parametrizes *subschemes* of a given projective scheme. That such a moduli problem admits a scheme as its solution is one of the most important results in the functorial approach to algebraic geometry, and it provides a rich source of examples and applications.

Flatness as the continuity condition

Before stating the result, we must address a foundational question: what does it mean for a collection of subschemes to "vary continuously" over a parameter space? Consider a family of closed subschemes $\{Z_s\}_{s \in S}$ of a fixed projective scheme $Y \subseteq \mathbb{P}^n_k$, parametrized by a scheme S. Algebraically, this family is encoded by a closed subscheme $Z \subseteq Y \times_k S$ such that each fiber $Z_s = Z \times_S \{s\}$ is the corresponding member of the family.

Not every such Z should be considered a "nice" family. For instance, one could construct a family where a curve degenerates to a union of a curve and an isolated point, causing the Hilbert polynomial to jump. The condition that prevents such pathologies is *flatness*: we require that Z be flat over S. Flatness ensures that the Hilbert polynomial $P_{Z_s}(m) = \chi(Z_s, \mathcal{O}_{Z_s}(m))$ is constant as s varies over connected components of S. This is the correct algebraic analogue of "continuous variation": the members of the family all share the same discrete numerical invariants.

The reader who has not yet encountered flatness in detail should think of it as a homological condition: a morphism $\pi : Z \to S$ is flat if the fibers Z_s do not "jump" in any cohomological sense. A thorough treatment will be given in section 6.8.

The Hilbert functor and Grothendieck's theorem

Fix a projective scheme $Y \subseteq \mathbb{P}^n_k$ and a numerical polynomial $p \in \mathbb{Q}[t]$ (recall that the Hilbert polynomial of any coherent sheaf on projective space is a polynomial with rational coefficients). The *Hilbert functor* is the contravariant functor $\mathcal{H}ilb^p_Y : \mathbf{Sch}_k^{\mathrm{op}} \to \mathbf{Set}$ defined by

$$\mathcal{H}ilb^p_Y(S) = \left\{ Z \subseteq Y \times_k S \;\middle|\; \begin{array}{l} Z \text{ is a closed subscheme, flat over } S, \\ \text{with Hilbert polynomial } p \text{ on each fiber} \end{array} \right\}.$$

For a morphism $f : S' \to S$, the induced map $\mathcal{H}ilb^p_Y(S) \to \mathcal{H}ilb^p_Y(S')$ sends a family $Z \subseteq Y \times S$ to its pullback $Z \times_S S' \subseteq Y \times S'$.

Theorem 3.3.11: Grothendieck's Existence Theorem

Let $Y \subseteq \mathbb{P}^n_k$ be a projective scheme over a Noetherian ring k, and let $p \in \mathbb{Q}[t]$ be a numerical polynomial. Then the Hilbert functor $\mathcal{H}ilb^p_Y$ is representable by a projective scheme Hilb^p_Y over k:

$$\mathrm{Hom}_{\mathbf{Sch}_k}(S, \mathrm{Hilb}^p_Y) \;\cong\; \mathcal{H}ilb^p_Y(S)$$

naturally in S.

The proof is beyond our scope (it relies on a careful analysis of Castelnuovo–Mumford regularity and the Grassmannian embedding), but the statement is worth absorbing. It says that the totality of all subschemes of Y with a given Hilbert polynomial is itself a geometric object—a projective scheme—on which one can do geometry. There is a *universal family*: a closed subscheme $\mathcal{Z} \subseteq Y \times \mathrm{Hilb}^p_Y$, flat over Hilb^p_Y, such that every flat family $Z \subseteq Y \times S$ with Hilbert polynomial p is the pullback of $\mathcal{Z}$ along a unique morphism $S \to \mathrm{Hilb}^p_Y$.

Example: the Hilbert scheme of points

The simplest interesting case arises when p is the constant polynomial $p(m) = n$ for a positive integer n. Then a closed subscheme $Z \subseteq Y$ with Hilbert polynomial $p(m) = n$ is a zero-dimensional subscheme of length n: it is supported on at most n points of Y, possibly with multiplicity (i.e., with nilpotent structure). The corresponding Hilbert scheme is denoted $\mathrm{Hilb}^n(Y)$ and is called the *Hilbert scheme of n points* on Y.

When Y is a smooth surface, $\mathrm{Hilb}^n(Y)$ turns out to be a smooth variety of dimension $2n$. This is a nontrivial theorem (due to Fogarty) and stands in contrast to the naive expectation: the "symmetric product" $\mathrm{Sym}^n(Y) = Y^n/S_n$, which parametrizes unordered n-tuples of points, is singular along the diagonals where points collide. The Hilbert scheme resolves these singularities by remembering the infinitesimal directions in which colliding points approach each other.

To see this concretely, consider $Y = \mathbb{A}^2_k$ and $n = 2$. A length-2 subscheme $Z \subseteq \mathbb{A}^2$ is one of two types:

1. Two distinct points $\{p, q\}$, corresponding to the ideal $I_p \cap I_q \subseteq k[x, y]$ where I_p and I_q are the maximal ideals of p and q. The quotient $k[x, y]/I$ has dimension 2 as a k-vector space.
2. A single point p with a tangent direction $v \in T_p\mathbb{A}^2$. If $p = (0, 0)$ and v points in the x-direction, the corresponding ideal is (y, x^2), and $k[x, y]/(y, x^2) \cong k[x]/(x^2)$ has dimension 2. The extra nilpotent direction records "how the two points came together."

Thus $\mathrm{Hilb}^2(\mathbb{A}^2)$ parametrizes pairs of distinct points together with a "blow-up" of the diagonal: at each point of the diagonal, the fiber is a $\mathbb{P}^1$ recording the tangent direction of approach. This is precisely the blow-up of $\mathrm{Sym}^2(\mathbb{A}^2)$ along its singular locus.

Tangent spaces to the Hilbert scheme

The functor-of-points perspective makes the computation of tangent spaces to moduli spaces almost mechanical. By Example 3.3.7, a tangent vector to Hilb^p_Y at a point $[Z]$ (corresponding to a subscheme

$Z \subseteq Y$) is a morphism $\operatorname{Spec}(k[\epsilon]/(\epsilon^2)) \to \operatorname{Hilb}^p_Y$ sending the closed point to $[Z]$. By the universal property, this is the same as a flat family of subschemes of Y over $\operatorname{Spec}(k[\epsilon]/(\epsilon^2))$ whose special fiber is Z. Such a family is a first-order deformation of Z inside Y, and these deformations are in bijection with the $\mathcal{O}_Y$-homomorphisms $\mathcal{I}_Z \to \mathcal{O}_Z$:

$$\Theta_{\operatorname{Hilb}^p_Y,\,[Z]} \;\cong\; \operatorname{Hom}_{\mathcal{O}_Y}(\mathcal{I}_Z,\, \mathcal{O}_Z)$$

where $\mathcal{I}_Z$ is the ideal sheaf of Z in Y. This identification—tangent vectors to the Hilbert scheme are the same as homomorphisms from the ideal sheaf to the structure sheaf of the subscheme—is a prototype for the general principle that tangent spaces to moduli spaces are computed by Hom or Ext groups.

3.3.7 Schemes as Functors

The reader may have noticed an apparent circularity in the functor-of-points perspective. We define the functor $h_X : \mathbf{Sch}^{\mathrm{op}} \to \mathbf{Set}$ using the category $\mathbf{Sch}$ of schemes, but if the functor of points is to serve as a *definition* of schemes, we seem to need the category $\mathbf{Sch}$ already in hand.

The resolution, due to Grothendieck and developed systematically by Demazure and Gabriel, proceeds in stages and is worth spelling out.

Step 1: Start with rings. The category $\mathbf{CRing}$ of commutative rings is purely algebraic. We define the category of *affine schemes* as $\mathbf{Aff} = \mathbf{CRing}^{\mathrm{op}}$, the formal opposite category. No geometry has been invoked.

Step 2: Embed into functors. Any functor $F : \mathbf{CRing} \to \mathbf{Set}$ (equivalently, a presheaf on $\mathbf{Aff}$) can be thought of as a "generalized space." An affine scheme $\operatorname{Spec}(A)$ defines such a functor by $h_{\operatorname{Spec}(A)}(R) = \operatorname{Hom}_{\mathbf{CRing}}(A, R)$. By the Yoneda Lemma, the assignment $\operatorname{Spec}(A) \mapsto h_{\operatorname{Spec}(A)}$ is a fully faithful embedding $\mathbf{Aff} \hookrightarrow \operatorname{Fun}(\mathbf{CRing}, \mathbf{Set})$.

Step 3: Impose a sheaf condition. Not every functor $\mathbf{CRing} \to \mathbf{Set}$ deserves to be called geometric. We require that F be a *sheaf* for the Zariski topology. Concretely, if $f_1, \dots, f_n$ generate the unit ideal in a ring A (so that $\operatorname{Spec}(A) = \bigcup_i D(f_i)$ is a cover by distinguished open sets), then the diagram

$$F(A) \longrightarrow \prod_i F(A_{f_i}) \rightrightarrows \prod_{i,j} F(A_{f_i f_j})$$

should be an equalizer. This is the local-to-global property that geometric objects possess: compatible data on an open cover should glue uniquely. A sheaf satisfying this property is called a *Zariski sheaf*.

Step 4: Require an open cover by affines. A *scheme* is a Zariski sheaf F that admits an open cover by representable subfunctors. That is, there exist rings A_i and natural transformations $h_{\operatorname{Spec}(A_i)} \hookrightarrow F$ that are "open immersions" (in the sense defined below) and that jointly cover F.

This definition recovers the usual category of schemes without circularity: we begin with rings, pass to representable functors (affine schemes), impose a sheaf condition (gluing), and characterize schemes as those sheaves admitting an affine open cover. The result is a full subcategory of $\operatorname{Fun}(\mathbf{CRing}, \mathbf{Set})$ defined by intrinsic conditions.

The practical payoff is significant. To define a scheme, it suffices to specify a functor $\mathcal{F} : \mathbf{CRing} \to \mathbf{Set}$ and verify the Zariski sheaf condition and the existence of an affine open cover. This is often considerably easier than constructing a locally ringed space by hand, especially for moduli problems: the Hilbert scheme was originally constructed by Grothendieck precisely by first defining the functor

$\mathcal{H}ilb^p_Y$ and then proving it satisfies the above conditions. The same strategy applies to Picard schemes, moduli of curves, and many other classifying spaces that arise in algebraic geometry.

Open subfunctors

Step 4 above relies on a notion of "open immersion" for functors, which we now make precise. Let $F : \mathbf{CRing} \to \mathbf{Set}$ be a Zariski sheaf. A subfunctor $G \subseteq F$ (meaning $G(A) \subseteq F(A)$ for every ring A, compatibly with restriction maps) is called an *open subfunctor* if the following condition holds: for every ring A and every element $\xi \in F(A)$, there exist elements $f_1, \dots, f_n \in A$ generating the unit ideal such that, for each i, the image of ξ in $F(A_{f_i})$ lies in $G(A_{f_i})$.

Informally, G is open in F if membership in G can be "checked locally." To see why this is the right definition, consider the representable case. If $F = h_{\operatorname{Spec}(B)}$ and G corresponds to an open subscheme $U \subseteq \operatorname{Spec}(B)$, then a morphism $\xi : \operatorname{Spec}(A) \to \operatorname{Spec}(B)$ factors through U if and only if its image lands in U. The preimage $\xi^{-1}(U)$ is an open subset of $\operatorname{Spec}(A)$, and the condition that $f_1, \dots, f_n$ generate the unit ideal with $\xi|_{D(f_i)}$ landing in U says exactly that $\xi^{-1}(U)$ covers $\operatorname{Spec}(A)$—i.e., ξ factors through U.

A collection of open subfunctors $\{G_i \subseteq F\}$ is said to *cover* F if for every field K, the sets $G_i(K)$ jointly cover $F(K)$. The condition in Step 4 can now be restated: a Zariski sheaf F is a scheme if it admits a cover by open subfunctors, each of which is representable by an affine scheme.

Tangent spaces to functors

One of the practical advantages of the functorial viewpoint is that many geometric constructions can be performed directly on functors, without first knowing that the functor is representable. We have already seen in Example 3.3.7 that the tangent space to a scheme X at a k-rational point x is recovered by probing with the dual numbers $D = k[\epsilon]/(\epsilon^2)$. This construction generalizes immediately to arbitrary functors.

Definition 3.3.12: Tangent Space of a Functor

Let $F : \mathbf{Sch}_k^{\mathrm{op}} \to \mathbf{Set}$ be a functor and let $\xi_0 \in F(\operatorname{Spec}(k))$ be a k-rational point of F. The *tangent space* of F at ξ_0 is

$$T_{\xi_0} F := \left\{ \xi \in F(\operatorname{Spec}(k[\epsilon]/(\epsilon^2))) \;\middle|\; \xi \text{ maps to } \xi_0 \text{ under } F(\operatorname{Spec}(k) \hookrightarrow \operatorname{Spec}(D)) \right\}.$$

When $F = h_X$ for a scheme X and ξ_0 corresponds to a closed point $x \in X$ with $\kappa(x) = k$, this recovers the Zariski tangent space $\Theta_{X,x}$ by Example 3.3.7. The key point is that the definition makes sense even when F is not representable.

For the tangent space to carry a k-vector space structure, one needs the functor F to satisfy a mild condition: $T_{\xi_0} F$ should be compatible with the k-module structure on square-zero extensions of k. This is automatic for representable functors, and holds for most functors arising in practice (including the Hilbert functor and the Picard functor).

As an application, we computed the tangent space of the Hilbert functor at the end of section 3.3.6: for a subscheme $Z \subseteq Y$ with ideal sheaf $\mathcal{I}_Z$, the tangent space $T_{[Z]} \operatorname{Hilb}^p_Y \cong \operatorname{Hom}_{\mathcal{O}_Y}(\mathcal{I}_Z, \mathcal{O}_Z)$. This computation was carried out purely at the level of the functor $\mathcal{H}ilb^p_Y$; the fact that Hilb^p_Y is actually

a scheme (Theorem 3.3.11) was not needed for the computation itself, only for the conclusion that this tangent space has geometric meaning.

3.3.8 Moduli Problems and the Limits of Representability

The Hilbert scheme demonstrates the power of the functorial approach: define a moduli functor, prove it is representable, and take the representing scheme as the parameter space. A natural question is how far this method extends. Can every reasonable classification problem in algebraic geometry be solved by a representing scheme?

The answer is no, and understanding *why* it fails is as instructive as the successes.

The moduli problem for elliptic curves

An *elliptic curve* over a field k is a smooth projective curve of genus 1 equipped with a distinguished rational point (the reader unfamiliar with this notion may consult section 7.6 and return here afterward). Two elliptic curves are isomorphic if and only if they have the same j-invariant, an element of k.

We would like a "moduli space of elliptic curves": a scheme M whose points correspond to isomorphism classes of elliptic curves, and such that families of elliptic curves over a base S correspond to morphisms $S \to M$. The functor to be represented is:

$$\mathcal{M}_{1,1}(S) = \left\{ (E \to S,\, \sigma : S \to E) \;\middle|\; \begin{array}{l} E \to S \text{ is a smooth proper morphism} \\ \text{with genus-1 geometric fibers,} \\ \sigma \text{ is a section} \end{array} \right\} / \cong_S$$

where $\cong_S$ denotes isomorphism of families over S. If this functor were representable by a scheme M, there would exist a *universal family*: an elliptic curve $\mathcal{E} \to M$ such that every family $E \to S$ is the pullback of $\mathcal{E}$ along a unique morphism $S \to M$.

Over an algebraically closed field, the isomorphism classes of elliptic curves are classified by the j-invariant, so the underlying set of M should be $\mathbb{A}^1$. Indeed, there is a scheme—the j-line $\mathbb{A}^1_k$—that serves as a *coarse moduli space*: its closed points are in bijection with isomorphism classes, and for any family $E \to S$ there exists a unique morphism $S \to \mathbb{A}^1$ whose fiber over a point s has j-invariant equal to the j-invariant of E_s. But the j-line is *not* a fine moduli space. The functor $\mathcal{M}_{1,1}$ is not representable.

The failure is easy to spot: every elliptic curve E over a field of characteristic $\neq 2$ has a nontrivial automorphism, namely it has at least the inversion map $\iota : E \to E$ sending $P \mapsto -P$ (with respect to the group law). For the curves with $j = 0$ and $j = 1728$, the automorphism group is even larger ($\mathbb{Z}/6\mathbb{Z}$ and $\mathbb{Z}/4\mathbb{Z}$, respectively). These automorphisms are the obstruction to representability.

To see why concretely, suppose a fine moduli space M existed with universal family $\mathcal{E} \to M$. Consider the trivial family $E \times S \to S$ for a fixed elliptic curve E with nontrivial automorphism α. The identity map $\mathrm{id} : E \times S \to E \times S$ and the automorphism $\alpha \times \mathrm{id}_S : E \times S \to E \times S$ are *different* isomorphisms of the family, but they give rise to the *same* classifying morphism $S \to M$ (since they classify the same family up to isomorphism). This violates the requirement that the classifying morphism be unique.

More precisely, the issue is that the functor $\mathcal{M}_{1,1}$ does not satisfy the sheaf condition in sufficient generality: it fails descent for étale covers. A family that is locally trivial in the étale topology need

not be globally trivial, and the automorphisms of the fibers obstruct the gluing.

Coarse moduli spaces

The j-line is an example of a *coarse moduli space*: a scheme M equipped with a natural transformation $\mathcal{M}_{1,1} \to h_M$ that is universal among maps to representable functors, and that induces a bijection $\mathcal{M}_{1,1}(\operatorname{Spec}(\overline{k})) \xrightarrow{\sim} M(\overline{k})$ on geometric points. A coarse moduli space classifies objects up to isomorphism at the level of points, but does not carry a universal family and does not faithfully track families over general bases.

Beyond schemes: stacks

The correct solution is to enlarge the category of geometric objects. Rather than forcing $\mathcal{M}_{1,1}(S)$ to be a *set* (of isomorphism classes), one retains it as a *groupoid*: the category whose objects are families $E \to S$ and whose morphisms are isomorphisms of families. The resulting structure is an *algebraic stack*, often denoted $\mathcal{M}_{1,1}$ as well. The stack remembers the automorphisms of each object, and this additional data resolves the descent obstruction.

An intermediate notion is that of an *algebraic space*. These arise when the moduli problem has no nontrivial automorphisms but the functor still fails to be representable by a scheme—for instance, because the "correct" topology for gluing is the étale topology rather than the Zariski topology. Algebraic spaces are Zariski-locally schemes but may not be globally so; they are to schemes what manifolds are to open subsets of $\mathbb{R}^n$.

The hierarchy, in increasing generality, is:

$$\text{Schemes} \subset \text{Algebraic Spaces} \subset \text{Algebraic Stacks} \subset \text{Higher Stacks},$$

where each step relaxes a condition on the moduli functor. Schemes require representability by a Zariski sheaf with an affine cover. Algebraic spaces replace the Zariski topology with the étale topology. Algebraic stacks allow the functor to take values in groupoids rather than sets. The development of this hierarchy is beyond our scope, but the reader should be aware that the functor-of-points perspective developed in this section is the entry point to this hierarchy: every step in it is formulated in terms of functors and their properties.

3.3.9 *The Fourier–Mukai Transform

The functor-of-points philosophy associates to a scheme X the collection of all morphisms into X. One can push this idea further by asking: what information is carried not by morphisms, but by *sheaves* on products? This line of thought leads to the Fourier–Mukai transform, which can be understood as a "categorified" version of the change-of-basis perspective developed above. This subsection is intended as a brief orientation for the interested reader; the full theory belongs to the realm of derived categories and will not be developed in detail here.

From graphs to correspondences

Given a morphism $f : X \to Y$, its *graph* $\Gamma_f = \{(x, f(x)) \mid x \in X\}$ is a closed subscheme of $X \times Y$. The morphism f is entirely recoverable from Γ_f: project to X (an isomorphism) and then to Y.

A natural generalization is to drop the requirement that the projection to X be an isomorphism. Any closed subscheme $Z \subseteq X \times Y$—or, more generally, any sheaf $\mathcal{P}$ on $X \times Y$—defines a "correspondence" between X and Y. For a point $x \in X$, the fiber $Z_x = Z \cap (\{x\} \times Y)$ is a subscheme of Y, and as x varies, one obtains a family of subschemes of Y parametrized by X. This is precisely the kind of data that Hilbert schemes classify.

A word on derived categories

To formulate the Fourier–Mukai transform precisely, one works not with the abelian category $\mathbf{Coh}(X)$ of coherent sheaves, but with its *bounded derived category* $D^b(X)$. The motivation is that many natural operations on sheaves—tensor products, pushforwards, global sections—are not exact, and their "corrections" (the higher derived functors Tor, $R^i f_*$, H^i) carry essential geometric information. The derived category is a framework in which a sheaf is replaced by a *complex* of sheaves, and these higher derived functors become part of a single, coherent operation.

An object of $D^b(X)$ is a bounded complex

$$\cdots \to \mathcal{F}^{i-1} \to \mathcal{F}^{i} \to \mathcal{F}^{i+1} \to \cdots$$

of coherent sheaves, considered up to *quasi-isomorphism*: two complexes are identified if they have isomorphic cohomology sheaves. A single sheaf $\mathcal{F}$ is regarded as a complex concentrated in degree zero. The derived category retains the information of the abelian category (via the cohomology functor H^i) but also remembers the "process" of resolution, which is essential for the functoriality that follows.

The reader familiar with the construction of Ext groups via projective or injective resolutions will recognize the underlying idea: in the derived category, one works with the resolutions themselves rather than discarding them after taking cohomology.

The transform

Let X and Y be smooth projective varieties over a field k, and let $\pi_X : X \times Y \to X$ and $\pi_Y : X \times Y \to Y$ denote the projection morphisms. Given a "kernel" $\mathcal{P} \in D^b(X \times Y)$, the *Fourier–Mukai transform* with kernel $\mathcal{P}$ is the functor

$$\Phi_{\mathcal{P}} : D^b(X) \longrightarrow D^b(Y), \qquad \mathcal{F} \longmapsto R\pi_{Y,*}(\mathcal{P} \otimes^L \pi_X^* \mathcal{F}),$$

where π_X^* is the (exact) pullback, $\otimes^L$ is the derived tensor product, and $R\pi_{Y,*}$ is the derived pushforward. In words: pull the sheaf $\mathcal{F}$ back from X to the product, "multiply" by the kernel $\mathcal{P}$, and "integrate" (push forward) along Y.

The analogy with the classical Fourier transform on $\mathbb{R}^n$ is worth making explicit. Recall that the classical transform sends a function f on $\mathbb{R}^n$ to a function $\hat{f}$ on the dual space $(\mathbb{R}^n)^*$ via

$$\hat{f}(\xi) = \int_{\mathbb{R}^n} f(x)\, e^{2\pi i \langle x, \xi \rangle}\, dx.$$

The exponential $e^{2\pi i\langle x,\xi\rangle}$ is a "kernel" on the product $\mathbb{R}^n \times (\mathbb{R}^n)^*$; the multiplication by f is the analogue of tensoring with $\pi_X^*\mathcal{F}$; and the integration over $\mathbb{R}^n$ is the analogue of the derived pushforward $R\pi_{Y,*}$. The Fourier–Mukai transform is, in this precise sense, the algebraic geometer's Fourier transform, with coherent sheaves playing the role of functions and the derived pushforward replacing integration.

From delta-points to character-points

The classical Fourier transform admits a particularly illuminating description in terms of two distinguished "bases" for the space of functions on a locally compact abelian group G. The first basis consists of the *delta functions* δ_x for $x \in G$: any function can be written as $f = \int_G f(x)\,\delta_x\,dx$, expressing f in terms of where it is "supported." The second basis consists of the *characters* χ_ξ for $\xi \in \hat{G}$ (the Pontryagin dual group): the Fourier transform rewrites f as $\hat{f} = \int_G f(x)\,\chi_\xi(x)\,dx$, expressing f in terms of its "frequency content." The Fourier transform is, at its core, a change of basis from delta-points to character-points:

$$\widehat{\delta_x} = \chi_x, \qquad \widehat{\chi_\xi} = \delta_\xi.$$

The Fourier–Mukai transform makes this change of basis precise in algebraic geometry, and the natural setting is that of *abelian varieties*. Let A be an abelian variety of dimension g over k (a smooth projective group variety—the higher-dimensional generalization of an elliptic curve). Its *dual abelian variety* $\hat{A} = \mathrm{Pic}^0(A)$ parametrizes the degree-zero line bundles on A: a point $\xi \in \hat{A}$ corresponds to a line bundle L_ξ on A that is algebraically equivalent to zero. The dual $\hat{A}$ is the algebro-geometric analogue of the Pontryagin dual $\hat{G}$.

On the product $A \times \hat{A}$ there is a canonical line bundle $\mathcal{P}$, the *Poincaré bundle*, characterized by the universal property: for each $\xi \in \hat{A}$, the restriction $\mathcal{P}|_{A\times\{\xi\}} \cong L_\xi$. The Poincaré bundle is the algebro-geometric analogue of the exponential kernel $e^{2\pi i\langle x,\xi\rangle}$: it "pairs" a point of A with a character from $\hat{A}$.

The Fourier–Mukai transform $\Phi_{\mathcal{P}} : D^b(A) \to D^b(\hat{A})$ with kernel $\mathcal{P}$ now enacts the change of basis between delta-points and character-points. The two fundamental computations are:

Deltas map to characters. Let $a \in A$ be a closed point and let $k(a)$ denote the skyscraper sheaf at a (the sheaf supported at the single point a with stalk k). The skyscraper $k(a)$ is the algebro-geometric δ-function: it is "concentrated at a" and detects nothing elsewhere. Applying the transform:

$$\Phi_{\mathcal{P}}(k(a)) = R\pi_{\hat{A},*}(\mathcal{P} \otimes^L \pi_A^* k(a)) = R\pi_{\hat{A},*}(\mathcal{P}|_{\{a\}\times\hat{A}}) = \mathcal{P}|_{\{a\}\times\hat{A}}.$$

The restriction $\mathcal{P}|_{\{a\}\times\hat{A}}$ is a line bundle on $\hat{A}$. Under the double-duality identification $A \cong \hat{\hat{A}}$, this is precisely the "character" on $\hat{A}$ associated to the point a—just as the classical Fourier transform sends δ_x to the character χ_x.

Characters map to deltas. Let $\xi \in \hat{A}$ and let $L_\xi = \mathcal{P}|_{A\times\{\xi\}}$ be the corresponding degree-zero line bundle on A. The line bundle L_ξ is the algebro-geometric "character": tensoring with L_ξ defines a "twist" of sheaves on A, analogous to multiplication by χ_ξ. One computes:

$$\Phi_{\mathcal{P}}(L_\xi) \cong k(\xi)[-g]$$

where $k(\xi)$ is the skyscraper sheaf at the point $\xi \in \hat{A}$ and $[-g]$ denotes a cohomological shift by $g = \dim A$. Up to this shift (which accounts for the difference between underived and derived pushforward on a g-dimensional variety), the character L_ξ is sent to the delta-function $k(\xi)$ on the dual—exactly as $\widehat{\chi_\xi} = \delta_\xi$ in the classical setting.

The cohomological shift $[-g]$ is the price of working in the derived category. It appears because the higher direct images $R^i\pi_{\hat{A},*}$ of L_ξ vanish for $i \neq g$ (a consequence of the fact that L_ξ has no global sections for $\xi \neq 0$, and its cohomology is concentrated in degree g). The shift is harmless: it is an

automorphism of $D^b(\hat{A})$, and Mukai's theorem accounts for it by asserting that the *composition*

$$D^b(A) \xrightarrow{\Phi_{\mathcal{P}}} D^b(\hat{A}) \xrightarrow{\Phi_{\mathcal{P}^\vee}} D^b(A)$$

is isomorphic to the shift functor $(-1)_A^*[-g]$, where $(-1)_A$ is the inversion on A. This is the precise analogue of the Fourier inversion formula $\hat{\hat{f}}(x) = f(-x)$.

Mukai's theorem and Orlov representability

The computations above are instances of a fundamental result due to Mukai: the Fourier–Mukai transform $\Phi_{\mathcal{P}} : D^b(A) \to D^b(\hat{A})$ is an *equivalence of categories*. This is a strong statement: it says that the derived category of coherent sheaves on an abelian variety is equivalent to that of its dual. Two geometrically distinct varieties—A and $\hat{A}$ need not be isomorphic—are "the same" from the perspective of their sheaf theory.

More broadly, Orlov's representability theorem asserts that for smooth projective varieties, every equivalence $D^b(X) \xrightarrow{\sim} D^b(Y)$ is isomorphic to a Fourier–Mukai transform $\Phi_{\mathcal{P}}$ for some kernel $\mathcal{P} \in D^b(X \times Y)$. This is the categorified Yoneda Lemma: just as a natural transformation $h_X \to h_Y$ is determined by a single element of $\mathrm{Hom}(X, Y)$, a sufficiently nice functor $D^b(X) \to D^b(Y)$ is determined by a single object of $D^b(X \times Y)$.

The analogy with the change-of-basis perspective from section 3.3.3 is clean. The Yoneda embedding probes X by mapping test objects *into* X; the Fourier–Mukai transform probes X by its category of sheaves. The kernel $\mathcal{P}$ on $X \times Y$ mediates between these two descriptions, playing the role of a "change-of-basis matrix" between two categorical coordinate systems. On abelian varieties, this change of basis is literally the passage from delta-points to character-points—the oldest and most fundamental duality in harmonic analysis, now transplanted into algebraic geometry.

3.4 Finiteness and Affine Conditions

As schemes are rather general notions, so to scheme morphisms can be very general. The following notions are guiding principles presented by Vakil on what a good type of morphism should have. This terminology will be very useful in labeling many of the following lemmas an propositions. Let's say that C is a collection of morphisms between schemes. Then:

1. it is *local on target* if
   (a) $\pi : X \to Y$ is in the class, then for any open subset $V \subseteq Y$, the restricted homomorphism $\pi^{-1}(V) \to V$ is in the class
   (b) $\pi : X \to Y$ is a morphism, $\cup_i V_i = Y$ is an open cover where each $\pi^{-1}(V_i) \to V_i$ is in this class, then π is in this class

   Usually we consider affine opens, but it need not be the case.

2. is *local on source* if there is
   (a) $\pi : X \to Y$ is in the class, then for any open subset $U \subseteq X$, the restricted homomorphism $U \to \pi(U)$ is in the class

(b) $\pi : X \to Y$ is a morphism, $\cup_i U_i = X$ is an open cover where each $U_i \to \pi(U_i)$ is in this class, then π is in this class

3. it is closed under composition
4. it is closed under fibered products. As a special case, is closed under change of basis from an S-scheme to a T-scheme

Finiteness Conditions

Definition 3.4.1: Quasicompact Morphism

Let $\pi : X \to Y$ be a morphism of schemes. Then π is *quasicompact* if every open affine subset $U \subseteq Y$ has quasicompact pre-image, that is $\pi^{-1}(U)$ is quasicompact.

Naturally, Quasicompactness is is closed under composition, and is local on target.

Example 3.4.2: Quasicompact Morphism

We shall in this section show that other types of morphism shall be quasicompact (finite, embeddings, proper, etc). At the moment, we shall give concrete examples:

1. Any scheme morphism of affine schemes is quasicompact, this relies on the use of the distinguished open base: Let $f : \operatorname{Spec} S \to \operatorname{Spec} R$ and $U \subseteq \operatorname{Spec} R$ be quasi-compact. Then $U = \bigcup_i^n D(g_i)$ for appropriate distinguished open sets. Then since $f^{-1}(D(g_i)) = D(f^{\#}(g_i))$, we have that
$$f^{-1}(U) = \bigcup_i^n D(f^{\#}(g_i))$$
showing the pre-image is quasi-compact.
2. Let $X = \coprod_i^{\infty} \operatorname{Spec} k$ and $Y = \operatorname{Spec} k$. Then the natural projection $\pi : X \to Y$ is certainly not quasicompact, the pre-image of $\operatorname{Spec} k$ not being quasicompact.
3. Let X be Noetherian schemes and $f : X \to Y$ a scheme morphism. Then f is quasicompact. This follows from the following topological property: If X is a Noetherian topological space, every subset is Noetherian.

   Hence, any non-quasicompact map *must* have X as a non-Noetherian scheme, and so in practice we shall not often see non-quasicompact schemes.
4. Let $X = \operatorname{Spec} k[x]$ and $\tilde{X}$ be the blow up at every point. Then the natural map $\varphi : \tilde{X} \to X$ is not quasicompact.

For future reference, the following definition is put down; it is used throughout chapter 6:

Definition 3.4.3: Quasiseparated Morphism

Let $\pi : X \to Y$ be a morphism of schemes. Then π is *quasiseparated* if for every open affine subset $U \subseteq Y$ the pre-image $\pi^{-1}(U)$ is quasi-separated (definition 2.4.9).

Example 3.4.4: Quasiseparated Morphism
We shall see examples of such morphisms later.

The next important type of morphism that will always have affine open pre-images:

Definition 3.4.5: Affine Morphism

Let $\pi : X \to Y$ be a morphism of schemes. Then π is *affine* if for every affine open subset $U \subseteq Y$, $\pi^{-1}(U)$ is an affine scheme (interpreted as an open subscheme of X)

Affine morphisms can be thought of as being the closest types of morphisms between schemes that "behave" like morphisms of affine schemes, namely since the pre-image of every affine is affine we can just cover in affine sets and study those without needing to "think" about the gluing structure. Proposition 3.4.10 shall show we only need to check a morphism is affine on a single affine cover, which is not immediately obvious as it is not *a priori* clear that we cannot "gerrymander" a partition to force affine pre-images on the open subsets of the cover.

3.4.6 Advanced Intuition This is for the reader already familiar with all this material. Let $f : X \to Y$ is a morphism. Then f is an affine morphisms if and only if there exists a quasi-coherent sheaf of $\mathcal{O}_Y$-algebras $\mathcal{A}$ such that $X \cong \operatorname{Spec}_Y(\mathcal{A})$ where $\operatorname{Spec}_Y(\mathcal{A})$ is the relative spectrum with respect to S (this is a generalization of X as an S-algebra)

We shall give examples of affine morphisms after we prove some nice results, one important such example shall be that when the domain and codomain are affine, the morphism must be affine. Let us start by showing that if the domain/codomain are not affine, the morphism may not be affine:

Example 3.4.7: Non Affine Morphisms
The prototypical example of a non affine morphism is $\mathbb{P}^1_k \to \operatorname{Spec} k$ (i.e. the structure morphism). Then the pre-image is *not* an (as shown in proposition 2.3.6). Example of a non-affine scheme where the codomain is not affine is the embedding $\mathbb{A}^n_k \to \mathbb{P}^n_k$ for $\mathbb{A}^n_k$ embeds in a choice of U_i ($n = 1$ it is enough to just do $n = 1$ to see why).

Being affine can be tested on a single affine open cover of the target (proposition 3.4.10), and from this affine morphisms are stable under composition and base change (proposition 3.4.12).

Lemma 3.4.8: Affine Morphisms and Qcqs

Let $\pi : X \to Y$ be an affine morphism. Then it is quasicompact and quasiseparated.

***Proof*:**
The key is that *affine schemes* are quasi-separated (and quasicompact). By the definition of an affine morphism, for every affine open subset $V \subseteq Y$, the preimage $U = \pi^{-1}(V)$ is an affine scheme, say $U \cong \operatorname{Spec} A$.

1. *Quasicompactness:* Every affine scheme is quasicompact. Since the preimage of every affine open $V \subseteq Y$ is quasicompact, the morphism π is quasicompact.
2. *Quasiseparatedness:* A morphism is quasiseparated if the preimage of every affine open

in Y is a quasiseparated scheme. Since $U = \operatorname{Spec} A$ is an affine scheme, it is separated (the diagonal $\Delta : U \to U \times_{\mathbb{Z}} U$ is a closed immersion), and every separated scheme is quasiseparated. Therefore, π is quasiseparated.

Lemma 3.4.9: Qcqs Lemma

Let X be a quasicompact and quasiseparated scheme and $s \in \mathcal{O}_X(X)$. Then

$$\mathcal{O}_X(X)_s \cong \mathcal{O}_X(X_s)$$

This theorem can be thought of as the generalization of the result where $X = \operatorname{Spec}(R)$, we we already know that:

$$\mathcal{O}_{\operatorname{Spec}(R)}(\operatorname{Spec}(R))_s \cong, \mathcal{O}_{\operatorname{Spec}(R)}(\operatorname{Spec}(R)_s) = \mathcal{O}_{\operatorname{Spec}(R)}(D(s))$$

***Proof*:**
As X is quasicompact, cover it with finitely many affine open sets $U_i = \operatorname{Spec} R_i$. take $U_{ij} = U_i \cap U_j$. Then the following is exact

$$0 \to \mathcal{O}_X(X) \to \prod_i R_i \to \prod_{i,j} \mathcal{O}_X(U_{ij})$$

As X is quasiseparated, each U_{ij} is covered by finitely many affine open sets $U_{ijk} = \operatorname{Spec}(R_{ij})$ so that the following is exact:

$$0 \to \mathcal{O}_X(X) \to \prod_i R_i \to \prod_{(i,j),k} R_{ijk}$$

Now, localization (being an exact functor) gives:

$$0 \to \mathcal{O}_X(X)_s \to \left(\prod_i R_i\right)_s \to \left(\prod_{(i,j),k} R_{ijk}\right)_s$$

which is still exact. As localization commutes with finite products (importance on finiteness), we get for appropriates values:

$$0 \to \mathcal{O}_X(X)_s \to \prod_i (R_i)_{s_i} \to \prod_{i,j,k} (R_{ijk})_{s_{ijk}}$$

which again is still exact, where the s_i and s_{ijk} is is the restriction to the rings R_i and R_{ijk}. Now, the scheme X_s can be covered by the affine open sets $\operatorname{Spec}(R_i)_{s_i}$ and $\operatorname{Spec}(R_i)_{s_i} \cap \operatorname{Spec}(R_{ijk})_{s_{ijk}}$ giving almost the same exact sequence:

$$0 \to \mathcal{O}_X(X_s) \to \prod_i (R_i)_{s_i} \to \prod_{i,j,k} (R_{ijk})_{s_{ijk}}$$

Hence, the kernel of the map

$$\mathcal{O}_X(X)_s \to \mathcal{O}_X(X_s)$$

is the same, and we get an isomorphism.

Proposition 3.4.10: Affine Morphisms Are Affine-local On Target

Let $\pi : X \to Y$ be a scheme morphism. Then π is an affine morphism if and only if there is a affine open cover $\cup_i U_i = Y$ such that $\pi^{-1}(U_i)$ is affine.

***Proof*:**
We shall want to apply the affine communication lemma. For that, we must show the conditions are satisfied. First, suppose $\pi : X \to Y$ is affine over $\operatorname{Spec}(S)$ so that $\pi^{-1}(\operatorname{Spec}(S)) = \operatorname{Spec}(R)$. Then notice that for any $s \in S$:

$$\pi^{-1}(\operatorname{Spec}(S_s)) = \operatorname{Spec}(R)_{\pi^\# s}$$

showing the first condition. For the next, suppose that $\pi : X \to \operatorname{Spec}(S)$ and $(s_1, ..., s_n)$ with $X_{\pi^\# s_i} = \operatorname{Spec}(R_i)$. We will show that X is affine as well. To that end, let $R = \mathcal{O}_X(X)$. Then $X \to \operatorname{Spec}(S)$ factors through $\alpha : X \to \operatorname{Spec}(R)$ (given by the isomorphism $R \to \mathcal{O}_X(X)$), giving us:

$$\begin{array}{ccc} \bigcup_i X_{\pi^\# s_i} & \xrightarrow{\quad\alpha\quad} & \operatorname{Spec}(R) \\ & {\scriptstyle\pi}\searrow \quad \swarrow{\scriptstyle\beta} & \\ & \bigcup_i D(s_i) = \operatorname{Spec}(S) & \end{array}$$

Then, as we are working with scheme morphisms, the pre-image of a distinguished open set is a distinguished open set and so

$$\beta^{-1}(D(s_i)) = D(\beta^\#(s_i)) \cong \operatorname{Spec} R_{\beta^\# s_i}$$

and $\pi^{-1}(D(s_i)) = \operatorname{Spec}(R_i)$. Now, as X is quasicompact and quasiseparated, by the Qcqs lemma we have the isomorphism:

$$A_{\beta^\# s_i} = \mathcal{O}_X(X)_{\beta^\# s_i} = \mathcal{O}_X(X_{\beta^\# s_i}) = R_i$$

Hence, α induces an isomorphism $\operatorname{Spec}(R_i) \cong \operatorname{Spec}(R_{\beta^\# s_i})$, and so α is an isomorphism over each $\operatorname{Spec}(R_{\beta^\# s_i})$, which covers $\operatorname{Spec}(R)$, and hence α is an isomorphism, and so $X \cong \operatorname{Spec}(R)$, completing the proof.

From this, we can conclude that if the domain/codomain are affine, the morphisms must be affine

Corollary 3.4.11: Morphism Between Affine is Affine

Let $f : \operatorname{Spec} R \to \operatorname{Spec} S$ be a morphism of schemes. Then f is an affine morphism

***Proof*:**
By proposition 3.4.10, it suffices to prove it on an affine open cover of the codomain. Then naturally, take an open cover by distinguished open sets, $D(g_i)$ whose pre-image $D(f^\#(g_i))$ which is an affine scheme.

Proposition 3.4.12: Affine Morphisms Are Stable under Composition and Base Change

1. If $f\colon X \to Y$ and $g\colon Y \to Z$ are affine, so is $g \circ f$.
2. If $f\colon X \to Y$ is affine and $Y' \to Y$ is any morphism, the base change $X \times_Y Y' \to Y'$ is affine.

***Proof*:**

1. For an affine open $W \subseteq Z$ the preimage $g^{-1}(W)$ is affine, and hence so is $f^{-1}(g^{-1}(W)) = (g \circ f)^{-1}(W)$ (definition 3.4.5).
2. Let $p\colon X \times_Y Y' \to Y'$ be the projection, and cover Y' by affine opens $\operatorname{Spec}(S')$ each mapping into an affine open $\operatorname{Spec}(S) \subseteq Y$, as in Step 2 of the proof of proposition 3.4.25. That step identifies $p^{-1}(\operatorname{Spec}(S'))$ with $\operatorname{Spec}(R) \times_{\operatorname{Spec}(S)} \operatorname{Spec}(S') \cong \operatorname{Spec}(R \otimes_S S')$, where $\operatorname{Spec}(R) = f^{-1}(\operatorname{Spec}(S))$, and uses only that $f^{-1}(\operatorname{Spec}(S))$ is affine. So p has affine preimages over an affine open cover of Y', and p is affine by proposition 3.4.10, as we sought to show.

Corollary 3.4.13: Hypersurfaces and Affine Morphissm

Let $Z \subseteq \operatorname{Spec}(R)$ be a closed subset which is locally cut out by one equation (Z is covered by open subsets of $\operatorname{Spec}(R)$ on each of which Z is the vanishing set of a single section). Then the complement of Z, Y, is affine

Note that in the global case, $Y = \operatorname{Spec}(R_f)$, but Y need not be of this form for the local case.

***Proof*:**
Let $\iota : Y \to \operatorname{Spec}(R)$ be the inclusion of the open subscheme $Y = \operatorname{Spec}(R) \backslash Z$ (proposition 2.2.47). We produce an affine open cover of $\operatorname{Spec}(R)$ each of whose members has affine preimage under ι, the preimage being the intersection with Y.

Let $x \in \operatorname{Spec}(R)$. If $x \notin Z$, then since the distinguished opens form a basis (corollary 2.1.13) there is a $D(g)$ with $x \in D(g) \subseteq Y$, and $D(g) \cap Y = D(g)$ is affine. If $x \in Z$, choose an open U from the hypothesis with $x \in U$ and $Z \cap U$ the vanishing set of a single $f_U \in \mathcal{O}_{\operatorname{Spec} R}(U)$, and then a distinguished open $D(g)$ with $x \in D(g) \subseteq U$. Write $f \in R_g = \mathcal{O}_{\operatorname{Spec} R}(D(g))$ for the restriction of f_U; restriction does not change the value of a section at a point of $D(g)$, so under $D(g) \cong \operatorname{Spec}(R_g)$ (proposition 2.1.12, proposition 2.2.1) the set $Z \cap D(g)$ is the vanishing set $V(f)$. Hence

$$D(g) \cap Y = D(g) \setminus V(f) = D(f) \cong \operatorname{Spec}\left((R_g)_f\right)$$

is affine.

These distinguished opens cover $\operatorname{Spec}(R)$, so ι is an affine morphism by proposition 3.4.10. The whole space $\operatorname{Spec}(R)$ is an affine open subset of itself, so $Y = \iota^{-1}(\operatorname{Spec}(R))$ is affine (definition 3.4.5).

Example 3.4.14: Affine Morphisms

1. Take $f\colon \mathbb{A}^2_k \setminus \{0\} \to \mathbb{P}^1_k$ given by $(a,b) \mapsto [a:b]$. The origin must be removed: it has no image, the point $[0:0]$ not existing, so the assignment defines a morphism on the complement of the origin and on no larger open set. This is an affine morphism: take the usual open cover U_0, U_1. Then:
$$f^{-1}(U_0) \cong \operatorname{Spec} k[x,y]_x \qquad \text{and} \qquad f^{-1}(U_1) \cong \operatorname{Spec} k[x,y]_y$$
hence by proposition 3.4.10 it is affine.
2. More generally let $\iota : X \to Y$ be either an open or closed immersion. Show that ι is an affine morphism.

The next two types of morphisms are affine morphisms where the pre-image has some extra properties. The next one can be thought of as the generalization of finite integral extensions (or more simply a finite extension or finite ring homomorphism):

Definition 3.4.15: Finite Morphism

Let $\pi : X \to Y$ be a scheme morphism. Then π is said to be *finite* if for every affine open set $\operatorname{Spec}(S) \subseteq Y$, $\pi^{-1}(\operatorname{Spec}(S))$ is the spectrum of a S-algebra that is finitely generated as a S-module, i.e. it is a finite S-algebra.

Note how being a finitely generated S-module is a rather stronger condition than being a finitely generated S-algebra: it forces each element to be integral and "globally bounded" (since R is a finitely generated S-module, there cannot be elements of arbitrary order). The composition of two finite morphisms is finite, module-finiteness being transitive (proposition 3.4.25).

Lemma 3.4.16: Algebra Build-up For Local On Target

Let $R \to S$ be a ring map, and $(f_1, ..., f_n) = R$ for appropriate $f_i \in R$. Then:

1. If $R_{f_i} \to S_{f_i}$ is integral, $R \to S$ is integral
2. If $R_{f_i} \to S_{f_i}$ is finite, $R \to S$ is finite

***Proof*:**

1. Take $s \in S$, and consider the ideal
$$(p \in R[x] \mid p(s) = 0) \subseteq R[x]$$
Then take the subset $J \subseteq R$ of the leading coefficients of the elements in I. This is in fact an ideal, as the reader could verify or see [10]. By clearing denominators, we get that:
$$f_i^{n_i} \in J$$
Hence, as shown in [12], $1 \in J$
2. The above shows that $R \to S$ is integral, for the finiteness note that we may choose a maximal N_i as the degree's of the generators are globally bounded.

Proposition 3.4.17: Finite Morphism Affine Local On Target

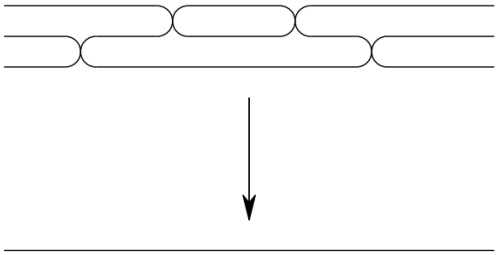
Let $f : X \to Y$ be a scheme morphism. Then it is finite if there is an affine cover $\bigcup_i \operatorname{Spec}(S_i) = Y$ such that $f^{-1}(\operatorname{Spec}(S_i))$ is the spectrum of a finite S_i-algebra.

***Proof*:**

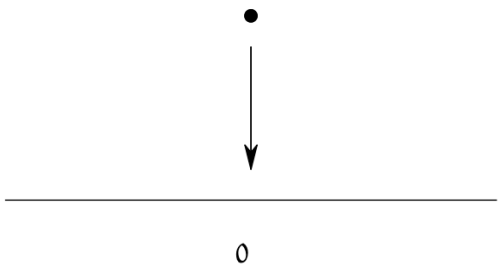
If f is finite, the result is *a fortiori* true, for the converse take $\operatorname{Spec} S_i \subseteq Y$, cover it in distinguished open sets, and apply corollary 3.1.15 and lemma 3.4.16.

Example 3.4.18: Finite Morphism

1. If L/K is a finite extension, $\operatorname{Spec}(K) \to \operatorname{Spec}(L)$ is a finite morphism.
2. Take the map $\operatorname{Spec} k[t] \to \operatorname{Spec} k[u]$ given by $u \mapsto p(t)$ for some degree n polynomial $p(t) \in k[t]$. Then this map is finite: notice that $k[t]$ is generated as a $k[u]$-module by $1, t, ..., t^{n-1}$. This can be visualized as:



3. Let $I \subseteq R$ be an ideal and take the natural schemes and map $\operatorname{Spec}(R/I) \to \operatorname{Spec}(R)$. Then this is a finite morphism, as R/I is generated as an R-module by $1 \in R/I$. This can be visualized as:



4. More generally, all closed immersions are finite morphisms.

5. Let $f \in R[x]$ be a polynomial whose leading coefficient is *not* a unit. Show that

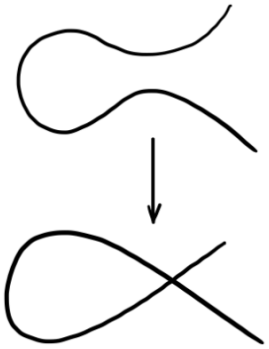
$$\operatorname{Spec} \frac{R[x]}{(f)} \to \operatorname{Spec} R$$

is *not* a finite morphism. Show that it *is* a finite morphism if the leading coefficient is a unit, making this an iff.

6. Take $\operatorname{Spec} k[t] \to \operatorname{Spec} \frac{k[x,y]}{(y^2-x^2-x^3)}$ corresponding to the ring map given by:

$$x \mapsto t^2 - 1 \qquad y \to t^3 - t$$

Namely, this shows this curve is rational. This is a finite morphism, since $k[t]$ is generated as a $\frac{k[x,y]}{(y^2-x^2-x^3)}$-module by 1 and t. This can be visualized as:



As the reader may have noticed, there is only one point that is 2-to-1. The reader should check that this will induce an isomorphism between $D(t^2-1)$ and $D(x)$, i.e. if the node is removed there is an isomorphism. This shall come up many times as an example of normalizing a curve.

7. Let $X \to \operatorname{Spec} k$ be a scheme morphism. Show that if the morphism if finite, X is the finite union of points with the discrete topology, each points having as residue field a finite extension of k. This can be visualized as:



Notice that all the fibers were finite. This is no happenstance of the chosen examples:

Proposition 3.4.19: Finite Morphism is Finite-to-one

Let $\pi : X \to Y$ be a finite morphism. Then it's fibers are finite

***Proof*:**
Since the property of the fiber is local on the target, we may assume $Y = \operatorname{Spec} B$ is affine. By the definition of a finite morphism, X is affine, say $X = \operatorname{Spec} A$, and the induced ring map $B \to A$ makes A a *finite* B-module (finitely generated as a module).

Let $y \in Y$ be a point corresponding to the prime ideal $\mathfrak{p} \subseteq B$. The fiber over y, denoted X_y, is given by the spectrum of the fiber ring:

$$A_y = A \otimes_B \kappa(y)$$

where $\kappa(y)$ is the residue field at $\mathfrak{p}$. Since A is generated by finitely many elements as a B-module, the tensor product A_y is generated by the images of those elements as a vector space over $\kappa(y)$. Thus, A_y is a finite-dimensional $\kappa(y)$-algebra.

Any finite-dimensional algebra over a field is an Artinian ring. A basic property of Artinian rings is that they have finitely many prime ideals (in fact, finitely many points, all of which are closed). Therefore, $\operatorname{Spec}(A_y)$ consists of finitely many points.

The converse is not the case

Example 3.4.20: Finite Fibers But Not Finite Morphisms
The open embedding $\mathbb{A}^2_k \setminus \{(0,0)\} \hookrightarrow \mathbb{A}^2_k$ has finite fibered (they are one-to-one), but it is not finite (as $\mathbb{A}^2_k \setminus \{(0,0)\}$ is not affine).

It also possible to have finite fibers and be affine, but not finite: Take $\mathbb{A}^1_{\mathbb{C}} \setminus \{0\} \hookrightarrow \mathbb{A}^1_{\mathbb{C}}$. Then $\mathcal{O}(f(\mathbb{A}_{\mathbb{C}})^{-1}) = \mathbb{C}[x, x^{-1}]$, which is not a finitely generated $\mathbb{C}[x]$-module.

It looks important to add 7.3.I to get a grasp on finite morphisms to affine spectrums relation to projective scheme

Definition 3.4.21: Integral Morphism

Let $\pi : X \to Y$ be a scheme morphism. Then it is an *integral morphism* if π is affine, and for every affine open subset $\operatorname{Spec}(S) \subseteq Y$, where $\pi^{-1}(\operatorname{Spec}(S)) = \operatorname{Spec}(R)$, every element of R is integral over the image of S.

A finite morphism is an integral morphism, module generators bounding the degrees of integral equations ([10, Integral Element Equivalences]), however an integral morphism need not be finite (think $\operatorname{Spec}\overline{\mathbb{Q}} \to \operatorname{Spec}\mathbb{Q}$). A closed immersion is integral, the induced ring maps being surjective. The composition of integral morphisms is integral (proposition 3.4.25).

Proposition 3.4.22: Integral Morphisms Affine-local On Target

Let $\pi : X \to Y$ be a scheme morphism. Then π is an integral morphism if and only if there is an affine open cover $\cup_i U_i = Y$ such that for each i the preimage $\pi^{-1}(U_i)$ is affine and every element of $\mathcal{O}_X(\pi^{-1}(U_i))$ is integral over the image of $\mathcal{O}_Y(U_i)$.

***Proof*:**
If π is integral, any affine open cover of Y has the stated properties, by definition 3.4.21. Conversely, let Q be the property of an affine open $\operatorname{Spec}(S) \subseteq Y$ that $\pi^{-1}(\operatorname{Spec}(S))$ is affine, say $\operatorname{Spec}(R)$, with every element of R integral over the image of S. By hypothesis Q holds on an affine open cover of Y. We verify the two conditions of theorem 2.2.31.

Localization. Let Q hold for $\operatorname{Spec}(S)$, say $\pi^{-1}(\operatorname{Spec}(S)) = \operatorname{Spec}(R)$, and let $s \in S$ with image $\bar{s} \in R$. The preimage of $D(s) = \operatorname{Spec}(S_s)$ is $D(\bar{s}) \cong \operatorname{Spec}(R_{\bar{s}})$ (proposition 3.1.10, lemma 2.1.28 and the display following it), which is affine. An element of $R_{\bar{s}}$ has the form $(r/1)(1/\bar{s})^k$. The element $r/1$ is integral over the image of S_s, an integral equation for r over the image of S localizing to one for $r/1$. The element $1/\bar{s}$ is the image of $1/s \in S_s$. Sums and products of elements integral over a subring are integral over it ([10, Integral Elements Form Rings]), so every element of $R_{\bar{s}}$ is integral over the image of S_s, and Q holds for $\operatorname{Spec}(S_s)$.

Gluing. Let $(s_1, \ldots, s_n) = S$ with Q holding for each $\operatorname{Spec}(S_{s_i})$. The $D(s_i)$ form an affine open cover of $\operatorname{Spec}(S)$ with affine preimages, so the restriction of π over $\operatorname{Spec}(S)$ is an affine morphism by proposition 3.4.10, and $\pi^{-1}(\operatorname{Spec}(S)) = \operatorname{Spec}(R)$ is affine. Writing $\bar{s}_i$ for the image of s_i in R, the preimage of $D(s_i)$ is $\operatorname{Spec}(R_{\bar{s}_i})$, so the hypothesis says each map $S_{s_i} \to R_{\bar{s}_i}$ is integral. By lemma 3.4.16(1), applied to the map $S \to R$ and the elements $s_1, \ldots, s_n$, the map $S \to R$ is integral, and Q holds for $\operatorname{Spec}(S)$.

Theorem 2.2.31 now gives Q for every affine open subset of Y, which is precisely definition 3.4.21.

Proposition 3.4.23: Integral Morphisms Are Closed Maps

Let $\pi : X \to Y$ be an integral morphism. Then it is a closed map: it maps closed subsets to closed subsets.

***Proof*:**
Step 1: reduction to the affine case. Closedness is local on the target: if $\{V_i\}$ is an open cover of Y and $C \subseteq X$ is closed, then $\pi(C)$ is closed as soon as each $\pi(C) \cap V_i$ is closed in V_i, and $\pi(C) \cap V_i$ is the image of the closed set $C \cap \pi^{-1}(V_i)$ under the restriction of π. Cover Y by affine open sets $\operatorname{Spec}(S)$. By definition 3.4.21 the preimage of each is affine, say $\operatorname{Spec}(R)$, with every element of R integral over the image of S. So it suffices to show that each induced map $\operatorname{Spec}(R) \to \operatorname{Spec}(S)$ is closed.

Step 2: the image of $V(J)$ is $V(I)$ for the contracted ideal I. Let $V(J) \subseteq \operatorname{Spec}(R)$ be closed, with $J \subseteq R$ an ideal, and let $I = (\pi^{\#})^{-1}(J) \subseteq S$. Contraction carries primes containing J to primes containing I, so $\pi(V(J)) \subseteq V(I)$. For the reverse inclusion, write $B := R/J$ and let $T \subseteq B$ be the image of S in B, so that $T \cong S/I$ by the choice of I and the primes of S containing I correspond to the primes of T. The ring B is integral over T: reducing modulo J an integral equation over the image of S gives one over T. Given a prime $\mathfrak{p} \in V(I)$ with corresponding prime $\overline{\mathfrak{p}} \subseteq T$, Step 3 produces a prime $\overline{\mathfrak{q}} \subseteq B$ with $\overline{\mathfrak{q}} \cap T = \overline{\mathfrak{p}}$. The preimage $\mathfrak{q} \subseteq R$ of $\overline{\mathfrak{q}}$ is then a prime containing J whose contraction to S is $\mathfrak{p}$, showing $V(I) \subseteq \pi(V(J))$.

Step 3: lying over. Let $T \subseteq B$ be a subring with B integral over T, and let $\overline{\mathfrak{p}} \subseteq T$ be prime. We claim there is a prime $\overline{\mathfrak{q}} \subseteq B$ with $\overline{\mathfrak{q}} \cap T = \overline{\mathfrak{p}}$. Localize at $D = T \setminus \overline{\mathfrak{p}}$. The map $T_{\overline{\mathfrak{p}}} \to D^{-1}B$ is integral, since dividing an integral equation $b^n + t_{n-1}b^{n-1} + \cdots + t_0 = 0$ by d^n exhibits b/d as a root of a monic polynomial with coefficients $t_{n-i}/d^i \in T_{\overline{\mathfrak{p}}}$. It is also injective: if t/d maps to 0, then $et = 0$ in B for some $e \in D$, and $t, e \in T$, so $t/d = 0$ already in $T_{\overline{\mathfrak{p}}}$. In particular $D^{-1}B \neq 0$, so it contains a maximal ideal $\mathfrak{m}$. The quotient $D^{-1}B/\mathfrak{m}$ is a field, integral over the image of $T_{\overline{\mathfrak{p}}}$, which is an integral domain. By [10, Integral Extension over Fields] that domain is a field, so $\mathfrak{m} \cap T_{\overline{\mathfrak{p}}}$ is the unique maximal ideal $\overline{\mathfrak{p}}\,T_{\overline{\mathfrak{p}}}$ of the local ring $T_{\overline{\mathfrak{p}}}$. Contracting $\mathfrak{m}$ to B gives a prime $\overline{\mathfrak{q}}$ with $\overline{\mathfrak{q}} \cap T = \overline{\mathfrak{p}}$, since contraction to T can be computed through $T_{\overline{\mathfrak{p}}}$, completing the proof.

As a consequence, finite morphisms are closed.

Corollary 3.4.24: Finite Morphisms Are Closed Maps

Let $\pi : X \to Y$ be a finite morphism. Then π is a closed map.

***Proof*:**
For every affine open $\operatorname{Spec}(S) \subseteq Y$ the preimage is $\operatorname{Spec}(R)$ with R finitely generated as an S-module (definition 3.4.15), so every element of R is integral over the image of S by [10, Integral Element Equivalences], the ring R being a module-finite ring extension of that image. Hence π is an integral morphism (definition 3.4.21) and proposition 3.4.23 applies. Compare proposition 3.4.32: a morphism that is integral and of finite type is finite.

Proposition 3.4.25: Finite Morphisms Are Stable under Composition and Base Change

1. The composition of two finite morphisms is finite.
2. If $f : X \to Y$ is finite and $g : Y' \to Y$ is any morphism, then the base change $X \times_Y Y' \to Y'$ is finite.
3. The composition of two integral morphisms is integral, and the base change of an integral morphism is integral.

***Proof*:**
Step 1: compositions. Let $f : X \to Y$ and $h : Y \to Z$ be finite and let $\operatorname{Spec}(S) \subseteq Z$ be any affine open. By definition 3.4.15, $h^{-1}(\operatorname{Spec}(S)) = \operatorname{Spec}(R)$ with R a finite S-module, and $f^{-1}(\operatorname{Spec}(R)) = \operatorname{Spec}(R')$ with R' a finite R-module. If $b_1, \dots, b_n$ generate R over S and $c_1, \dots, c_m$ generate R' over R, then the products $b_i c_j$ generate R' over S. So $(h \circ f)^{-1}(\operatorname{Spec}(S))$ is the spectrum of a finite S-algebra, for every affine open, and $h \circ f$ is finite. For two integral morphisms, the same reduction produces ring maps $S \to R \to R'$ with R integral over the image of S and R' integral over the image of R. The image of R in R' is integral over the image of S in R', integral equations mapping to integral equations, and [10, Transitivity of Integrality] applied to the tower of subrings of R' shows R' is integral over the image of S.

Step 2: base change, reduction to rings. Let $f : X \to Y$ be finite (or integral), $g : Y' \to Y$ any morphism, and $p : X \times_Y Y' \to Y'$ the projection. Cover Y' by affine opens $\operatorname{Spec}(S')$, each mapping into some affine open $\operatorname{Spec}(S) \subseteq Y$: for $y' \in Y'$, choose an affine open $\operatorname{Spec}(S) \ni g(y')$ and then an affine open neighborhood of y' inside $g^{-1}(\operatorname{Spec}(S))$, affine opens forming a basis. By proposition 3.4.17 (resp. proposition 3.4.22) it suffices to identify $p^{-1}(\operatorname{Spec}(S'))$ as the spectrum of a finite (resp. integral) S'-algebra. Proposition 3.2.5(1) identifies $p^{-1}(\operatorname{Spec}(S'))$ with $\operatorname{Spec}(S') \times_{Y'} (X \times_Y Y')$, which the universal property identifies with $X \times_Y \operatorname{Spec}(S')$, and since $\operatorname{Spec}(S') \to Y$ factors through $\operatorname{Spec}(S)$, the universal property gives

$$X \times_Y \operatorname{Spec}(S') \cong (X \times_Y \operatorname{Spec}(S)) \times_{\operatorname{Spec}(S)} \operatorname{Spec}(S') \cong \operatorname{Spec}(R) \times_{\operatorname{Spec}(S)} \operatorname{Spec}(S')$$

where $f^{-1}(\operatorname{Spec}(S)) = \operatorname{Spec}(R)$, using proposition 3.2.5(1) once more. The fibered product of affine schemes is the spectrum of the tensor product, by the construction in theorem 3.2.1, so $p^{-1}(\operatorname{Spec}(S')) \cong \operatorname{Spec}(R \otimes_S S')$.

Step 3: base change, the algebra. If $b_1, \dots, b_n$ generate R as an S-module, then the elements $b_i \otimes 1$ generate $R \otimes_S S'$ as an S'-module: every element is a finite sum of simple tensors $b \otimes s'$, and writing $b = \sum_i s_i b_i$ gives $b \otimes s' = \sum_i b_i \otimes s_i s'$. So the base change of a finite morphism is finite. If instead R is integral over the image of S, then each $b \otimes 1$ is integral over the image of S': an integral equation for b with coefficients from the image of S maps to one for $b \otimes 1$ whose coefficients, of the form $s \cdot (1 \otimes 1) = 1 \otimes (s \cdot 1_{S'})$ for $s \in S$, lie in the image of S'. The ring $R \otimes_S S'$ is generated as an algebra over the image of S' by the elements $b \otimes 1$, by [12, Generation of $A \otimes_R B$ as an Algebra] together with the fact that the elements $1 \otimes s'$ already lie in the image of S'. Sums and products of integral elements are integral ([10, Integral Elements Form Rings], applied to the image of S' as the base subring), so every element of $R \otimes_S S'$ is integral over the image of S', as we sought to show.

The definition above invites the question of which properties are preserved in the passage to $\overline{k}$, and connectedness is the first case worth settling. It can fail: $\operatorname{Spec}\mathbb{C}$ is a single point over $\mathbb{R}$, while its base change

$$\operatorname{Spec}(\mathbb{C}\otimes_{\mathbb{R}}\mathbb{C})\cong\operatorname{Spec}(\mathbb{C}\times\mathbb{C})$$

is a pair of points. The failure is visible in the residue field, which is $\mathbb{C}$, strictly larger than $\mathbb{R}$, and that extra room is what splits. A point whose residue field is exactly k leaves no such room. Call a morphism $\operatorname{Spec}k\to X$ of k-schemes a *k-rational point* of X; its image is a point $x_0\in X$ with $\kappa(x_0)=k$, since the composite $\operatorname{Spec}k\to X\to\operatorname{Spec}k$ is the identity and so the induced k-algebra map $\kappa(x_0)\to k$ is an isomorphism. Write $X(k)$ for the set of k-rational points.

The topological input is recorded first, and it is here that the finiteness hypothesis on X gets spent.

Lemma 3.4.26: Base Change to the Algebraic Closure is Open and Closed

Let k be a field and let X be a scheme locally of finite type over k. Then the projection $\pi\colon X_{\overline{k}}\to X$ is surjective, closed, and open.

***Proof*:**

Step 1: π is integral, hence surjective and closed. By definition 3.1.21 we may cover X by affine opens $\operatorname{Spec}A$ with A a finitely generated k-algebra, and by proposition 3.2.5(1) together with example 3.2.4 we have $\pi^{-1}(\operatorname{Spec}A)=\operatorname{Spec}(A\otimes_k\overline{k})$. As an A-algebra, $A\otimes_k\overline{k}$ is generated by the elements $1\otimes\lambda$ for $\lambda\in\overline{k}$, because $a\otimes\lambda=(a\otimes1)(1\otimes\lambda)$. Each λ is algebraic over k, so $1\otimes\lambda$ is a root of the monic minimal polynomial of λ, read in $A[T]$. An algebra generated by integral elements is integral over its base [10, Integral Elements Form Rings], so π is an integral morphism (definition 3.4.21) and hence a closed map by proposition 3.4.23.

For surjectivity, fix $\mathfrak{p}\in\operatorname{Spec}A$. Localizing at $\mathfrak{p}$ and killing $\mathfrak{p}$ shows that the primes of $A\otimes_k\overline{k}$ contracting to $\mathfrak{p}$ are the primes of

$$(A\otimes_k\overline{k})\otimes_A\kappa(\mathfrak{p})\cong\kappa(\mathfrak{p})\otimes_k\overline{k}$$

This ring is a tensor product of nonzero k-vector spaces, hence nonzero, hence has a prime.

Step 2: openness reduces to a finite subextension. Openness may be tested on the cover: for $U\subseteq X_{\overline{k}}$ open one has $\pi(U)\cap\operatorname{Spec}A=\pi(U\cap\pi^{-1}(\operatorname{Spec}A))$, and every open of $\operatorname{Spec}(A\otimes_k\overline{k})$ is a union of distinguished opens, so it is enough to show that $\pi(D(f))$ is open for $f\in A\otimes_k\overline{k}$. Write $f=\sum_{i=1}^{n}a_i\otimes\lambda_i$ and set $k'=k(\lambda_1,\dots,\lambda_n)$, a finite extension of k since each λ_i is algebraic. Put $B=A\otimes_k k'$ and $C=A\otimes_k\overline{k}\cong B\otimes_{k'}\overline{k}$, and let $f'\in B$ be the element $\sum_i a_i\otimes\lambda_i$, which maps to f. Factor π as $\pi'\circ\rho$ with $\rho\colon\operatorname{Spec}C\to\operatorname{Spec}B$ and $\pi'\colon\operatorname{Spec}B\to\operatorname{Spec}A$. Since $\overline{k}$ is an algebraic closure of k' and B is a finitely generated k'-algebra, Step 1 applies to ρ and makes it surjective; as $D(f)=\rho^{-1}(D(f'))$ this gives $\rho(D(f))=D(f')$ and therefore $\pi(D(f))=\pi'(D(f'))$.

Step 3: the finite case. Choose a k-basis $\lambda_1,\dots,\lambda_d$ of k', where $d=[k':k]$. Then $1\otimes\lambda_1,\dots,1\otimes\lambda_d$ is an A-basis of B, so B is a free A-module of rank d. For $b\in B$ let

$$\chi_b(T)=T^d+c_{d-1}(b)T^{d-1}+\cdots+c_0(b)\in A[T]$$

be the characteristic polynomial of multiplication by b on B. We claim $\pi'(D(b))=\bigcup_{i=0}^{d-1}D(c_i(b))$, which is open.

Fix $\mathfrak{p}\in\operatorname{Spec}A$. By the correspondence of Step 1, the primes of B over $\mathfrak{p}$ are the primes of $\overline{B}=B\otimes_A\kappa(\mathfrak{p})$, a $\kappa(\mathfrak{p})$-algebra free of rank d on the reduced basis, and the matrix of

multiplication by the image $\overline{b}$ is the reduction of the matrix over A; so $\chi_{\overline{b}} = \chi_b \bmod \mathfrak{p}$. Now $\mathfrak{p} \notin \pi'(D(b))$ says that $\overline{b}$ lies in every prime of $\overline{B}$, that is, $\overline{b}$ is nilpotent. If $\overline{b}$ is nilpotent then multiplication by $\overline{b}$ is a nilpotent endomorphism of a finite dimensional vector space, so $\chi_{\overline{b}} = T^d$ and every $c_i(b)$ lies in $\mathfrak{p}$. Conversely, if every $c_i(b)$ lies in $\mathfrak{p}$ then $\chi_{\overline{b}} = T^d$, so Cayley-Hamilton makes multiplication by $\overline{b}$ nilpotent of order d, whence $\overline{b}^d = 0$. The two conditions agree, which proves the claim and shows π' is open. Combining with Step 2, π carries distinguished opens to opens and is therefore open, completing the proof.

The finiteness hypothesis on X enters through Step 2, where a single element of $A \otimes_k \overline{k}$ is seen to involve only finitely many elements of $\overline{k}$; this is what keeps the infinite extension $\overline{k}/k$ out of the topological argument below.

Proposition 3.4.27: Connected Plus a Rational Point Gives Geometric Connectedness

Let k be a field and let X be a connected scheme locally of finite type over k with $X(k) \neq \emptyset$. Then X is geometrically connected, that is, $X_{\overline{k}}$ is connected.

***Proof*:**
Step 1: the fiber over a rational point is a single point. Fix a k-rational point with image $x_0 \in X$, so $\kappa(x_0) = k$. Choose an affine open $\operatorname{Spec} A \subseteq X$ containing x_0 with A finitely generated over k, and let $\mathfrak{p} \subseteq A$ be the prime of x_0. By the correspondence in Step 1 of lemma 3.4.26, the points of $X_{\overline{k}}$ over x_0 are the primes of

$$\kappa(\mathfrak{p}) \otimes_k \overline{k} = k \otimes_k \overline{k} \cong \overline{k}$$

a field, so there is exactly one. Write $\tilde{x}_0$ for it.
Step 2: every connected component of $X_{\overline{k}}$ contains $\tilde{x}_0$. The scheme $X_{\overline{k}}$ is covered by spectra of finitely generated $\overline{k}$-algebras, which are Noetherian, so each such affine has finitely many irreducible components by lemma 2.4.3 and hence finitely many connected components, each of them open. So $X_{\overline{k}}$ is locally connected and its connected components are open as well as closed. Let Z be one of them. By lemma 3.4.26 the image $\pi(Z)$ is then open and closed in X, and it is nonempty, so $\pi(Z) = X$ because X is connected. In particular $x_0 \in \pi(Z)$, so Z meets the fiber over x_0, which by Step 1 forces $\tilde{x}_0 \in Z$.
Step 3: conclusion. The scheme $X_{\overline{k}}$ is nonempty, since it contains $\tilde{x}_0$. Distinct connected components are disjoint, so Step 2 leaves exactly one component, and $X_{\overline{k}}$ is connected, as we sought to show.

The classical phrasing of the same argument uses the action of $\operatorname{Aut}(\overline{k}/k)$ on $X_{\overline{k}}$ over X: the group permutes the connected components and fixes $\tilde{x}_0$, since that point is alone in its fiber, so the component through $\tilde{x}_0$ is stable. Step 2 above trades the transitivity of that action on fibers for the cheaper statement that π is open and closed. Dropping the rational point returns the failure that opened the discussion: $(\operatorname{Spec}\mathbb{C})(\mathbb{R}) = \emptyset$, and $\operatorname{Spec}\mathbb{C}$ is a connected $\mathbb{R}$-scheme of finite type that is not geometrically connected.

Definition 3.4.28: Locally Finite Type Morphism

Let $\pi : X \to Y$ be a scheme morphism. Then π is said to be *locally of finite type* if for every affine open subset $\mathrm{Spec}\,(S) \subseteq Y$, and every affine open subset $\mathrm{Spec}\,(R) \subseteq \pi^{-1}(\mathrm{Spec}\,(S))$, the induced morphism $S \to R$ makes R a finitely generated S-algebra. Equivalently, $\pi^{-1}(\mathrm{Spec}\,(S))$ can be covered by affine open subsets $\mathrm{Spec}\,(R_i)$ where each R_i is a finitely generated S-algebra.

Definition 3.4.29: Finite Type Morphism

Let $\pi : X \to Y$ be a scheme morphism. Then π is said to be *of finite type* if it is locally of finite type and quasicompact.

Proposition 3.4.30: Locally Finite Type Morphism Affine-local On Target

Let $\pi : X \to Y$ be an scheme morphism. Then π is locally of finite type (resp. of finite type) if there is a cover of Y by affine open sets $\mathrm{Spec}\,(S_i)$ such that $\pi^{-1}(\mathrm{Spec}\,(S_i))$ is locally finite type (resp. finite type) over S_i.

***Proof*:**

Let Q be the following property of an affine open subscheme $\mathrm{Spec}\,(S) \subseteq Y$: the preimage $\pi^{-1}(\mathrm{Spec}\,(S))$ can be covered by affine open subsets $\mathrm{Spec}\,(R_j)$ with each R_j a finitely generated S-algebra. We verify the two conditions of theorem 2.2.31.

Localization. Suppose Q holds for $\mathrm{Spec}\,(S)$ with cover $\{\mathrm{Spec}\,(R_j)\}_j$, and let $s \in S$ with image $s_j \in R_j$. The restriction of π to each $\mathrm{Spec}\,(R_j)$ is induced by the ring map $S \to R_j$ (proposition 3.1.10), under which the preimage of $D(s)$ is $D(s_j) \cong \mathrm{Spec}\,((R_j)_{s_j})$ (lemma 2.1.28 and the display following it). So the $\mathrm{Spec}\,((R_j)_{s_j})$ cover $\pi^{-1}(\mathrm{Spec}\,(S_s))$. If $a_1, \ldots, a_m$ generate R_j as an S-algebra, their images generate $(R_j)_{s_j}$ as an S_s-algebra, since r/s_j^k is the image of r times the image of $1/s^k \in S_s$. Hence Q holds for $\mathrm{Spec}\,(S_s)$.

Gluing. Suppose $(s_1, \ldots, s_n) = S$ and Q holds for each $\mathrm{Spec}\,(S_{s_i})$, with covers $\{\mathrm{Spec}\,(R_{ij})\}_j$. Each $S_{s_i} = S[1/s_i]$ is a finitely generated S-algebra, and a finitely generated algebra over a finitely generated algebra is finitely generated (concatenate the generators), so each R_{ij} is a finitely generated S-algebra. The sets $D(s_i) = \mathrm{Spec}\,(S_{s_i})$ cover $\mathrm{Spec}\,(S)$, so the $\mathrm{Spec}\,(R_{ij})$ cover $\pi^{-1}(\mathrm{Spec}\,(S))$, and Q holds for $\mathrm{Spec}\,(S)$.

By hypothesis Q holds for every member of the affine open cover $\{\mathrm{Spec}\,(S_i)\}_i$ of Y, so theorem 2.2.31 gives Q for every affine open subset of Y. This is the covering form of definition 3.4.28, so π is locally of finite type.

For the finite type statement it remains, by definition 3.4.29, to show that π is quasicompact when each $\pi^{-1}(\mathrm{Spec}\,(S_i))$ is additionally quasicompact. Let Q' be the property of an affine open $\mathrm{Spec}\,(S) \subseteq Y$ that $\pi^{-1}(\mathrm{Spec}\,(S))$ is quasicompact, equivalently that it is a finite union of affine open subsets of X: a quasicompact scheme has a finite subcover of any affine open cover, and conversely affine schemes are quasicompact (section 2.1.3) and a finite union of quasicompact open sets is quasicompact, as an open cover admits a finite subcover of each piece. Q' satisfies the localization condition: if $\pi^{-1}(\mathrm{Spec}\,(S)) = \bigcup_{m=1}^{N} \mathrm{Spec}\,(A_m)$, then as above $\pi^{-1}(D(s))$ is the union of the finitely many affine sets $\mathrm{Spec}\,((A_m)_{s_m})$, with s_m the image of s. It satisfies

the gluing condition: $\pi^{-1}(\operatorname{Spec}(S)) = \bigcup_{i=1}^{n} \pi^{-1}(D(s_i))$ is a finite union of quasicompact open sets. By theorem 2.2.31, every affine open subset of Y has quasicompact preimage, which is definition 3.4.1.

Example 3.4.31: Finite Type Morphisms

1. All Schemes that are finite type over k are example of finite type scheme morphisms $X \to \operatorname{Spec} k$
2. The map $\mathbb{P}^n_R \to \operatorname{Spec} R$ is of finite type ($\mathbb{P}^n_R$ is covered by $n+1$ open sets)

Proposition 3.4.32: Integral and Finite Type iff Finite

1. finite morphisms are of finite type
2. A morphism is finite if it is integral and of finite type

***Proof*:**

1. Let $\pi : X \to Y$ be finite and $\operatorname{Spec}(S) \subseteq Y$ an affine open subset, so $\pi^{-1}(\operatorname{Spec}(S)) = \operatorname{Spec}(R)$ with R finitely generated as an S-module (definition 3.4.15), say $R = Sr_1 + \cdots + Sr_m$. Then the r_i already generate R as an S-algebra, so the preimage of every affine open of Y is covered by (indeed equal to) the spectrum of a finitely generated algebra, and π is locally of finite type (definition 3.4.28). Since the preimage of every affine open is affine, π is an affine morphism (definition 3.4.5), hence quasicompact (lemma 3.4.8). By definition 3.4.29, π is of finite type.
2. Let π be integral and of finite type, and let $\operatorname{Spec}(S) \subseteq Y$ be an affine open subset. Since π is integral it is affine, so $\pi^{-1}(\operatorname{Spec}(S)) = \operatorname{Spec}(R)$, and every element of R is integral over the image of S (definition 3.4.21). Since π is locally of finite type and $\operatorname{Spec}(R)$ is an affine open subset of $\pi^{-1}(\operatorname{Spec}(S))$, the ring R is a finitely generated S-algebra (definition 3.4.28). An algebra that is finitely generated and integral is finitely generated as a module ([10, Module-Finite versus Finite-Type-and-Integral]). Hence R is a finite S-algebra, and π is finite (definition 3.4.15).

Finite type bounds the generators of the coordinate algebras. Finite presentation bounds the relations as well. Over a locally Noetherian base the two conditions agree (proposition 3.4.34), and the distinction matters only over general bases, where the theory of étale morphisms takes the stronger condition as its standing hypothesis ([16]).

Definition 3.4.33: Locally of Finite Presentation

Let $\pi : X \to Y$ be a scheme morphism. Then π is *locally of finite presentation* if Y can be covered by affine open sets $\operatorname{Spec}(S_j)$ such that each $\pi^{-1}(\operatorname{Spec}(S_j))$ can be covered by affine open sets $\operatorname{Spec}(R_{ij})$, where each R_{ij} is a *finitely presented* S_j-algebra, meaning

$$R_{ij} \cong \frac{S_j[x_1, \ldots, x_n]}{(g_1, \ldots, g_m)}$$

for some $n, m \geq 0$ and polynomials $g_1, \ldots, g_m \in S_j[x_1, \ldots, x_n]$.

Definition 3.4.28 quantifies over every affine open of the target and of the preimage. For the target, the covering form of definition 3.4.33 implies the quantified form by the Affine Communication Lemma, and this is Step 2 of the proof of proposition 3.4.35. For affine opens of the preimage the quantified form is also true: it is *Finite Presentation on Every Pair of Affine Opens* in section 3.11⋆, proved by a limit argument (compare [70, Tag 01TQ]).

Proposition 3.4.34: Finite Presentation versus Finite Type

1. A morphism locally of finite presentation is locally of finite type.
2. If Y is locally Noetherian (definition 2.4.1) and $\pi : X \to Y$ is locally of finite type, then π is locally of finite presentation.

***Proof*:**
(1) A finitely presented S-algebra is a finitely generated S-algebra, and definition 3.4.28 accepts covers of this kind.

(2) Cover Y by affine opens $\operatorname{Spec}(S)$ with S Noetherian (definition 2.4.1). Each $\pi^{-1}(\operatorname{Spec}(S))$ is covered by affine opens $\operatorname{Spec}(R)$ with R a finitely generated S-algebra (definition 3.4.28), so $R \cong S[x_1, \ldots, x_n]/I$ for some ideal I. The ring $S[x_1, \ldots, x_n]$ is Noetherian by [10, HBT on multi-variable Polynomials], so I is finitely generated and R is finitely presented over S.

Proposition 3.4.35: Finite Presentation Is Stable under Composition and Base Change

1. If $\pi : X \to Y$ and $\rho : Y \to Z$ are locally of finite type (resp. locally of finite presentation), then so is $\rho \circ \pi$.
2. If $\pi : X \to Y$ is locally of finite type (resp. locally of finite presentation) and $Y' \to Y$ is any morphism, then the base change $X \times_Y Y' \to Y'$ is locally of finite type (resp. locally of finite presentation).
3. Both statements hold with locally of finite type replaced by of finite type (definition 3.4.29).

***Proof*:**
Step 1: the ring-level facts. First, compositions. If B is generated over A by finitely many elements and C over B by finitely many elements, then C is generated over A by the union of the two finite sets. For finite presentation, write $B = A[x_1, \dots, x_r]/(f_1, \dots, f_s)$ and $C = B[y_1, \dots, y_t]/(g_1, \dots, g_u)$. The coefficientwise surjection $A[x, y] = (A[x])[y] \to B[y]$ has kernel the polynomials in y with all coefficients in the ideal (f), which is the ideal generated by $f_1, \dots, f_s$ in $A[x, y]$: one inclusion is clear, and a polynomial whose coefficients are $A[x]$-combinations of the f_k is an $A[x, y]$-combination of the f_k. Choosing lifts $\tilde{g}_l \in A[x, y]$ of the g_l, the correspondence of ideals under this surjection gives

$$C \cong \frac{A[x_1, \dots, x_r, y_1, \dots, y_t]}{(f_1, \dots, f_s, \tilde{g}_1, \dots, \tilde{g}_u)}$$

so C is finitely presented over A. Second, base changes. Let $S \to S'$ be any ring map. The S'-algebra $S[x_1, \dots, x_n] \otimes_S S'$ is the polynomial algebra $S'[x_1, \dots, x_n]$: both are free S'-modules on the monomials, and the multiplication of [12, Tensor Product of R-Algebras] agrees with that of the polynomial ring on them. For $R = S[x]/I$, tensoring the exact sequence $I \to S[x] \to R \to 0$ with S' over S and applying [12, $D \otimes_R -$ Is Right Exact] identifies $R \otimes_S S'$ with the quotient of $S'[x]$ by the ideal generated by the images of elements of I. If I is generated by $f_1, \dots, f_s$, that ideal is generated by their images. So $R \otimes_S S'$ is finitely generated (resp. finitely presented) over S' whenever R is so over S.

Step 2: locally of finite presentation is affine-local on the target. We claim that if $\pi : X \to Y$ is locally of finite presentation, then for *every* affine open $\operatorname{Spec}(B) \subseteq Y$ the preimage $\pi^{-1}(\operatorname{Spec}(B))$ can be covered by affine opens whose rings are finitely presented B-algebras. Let Q be this property of affine open subschemes of Y, and verify the two conditions of theorem 2.2.31. Localization: suppose Q holds for $\operatorname{Spec}(B)$ and $g \in B$, with charts $\operatorname{Spec}(R)$ covering $\pi^{-1}(\operatorname{Spec}(B))$. The preimage of the distinguished open $\operatorname{Spec}(B_g)$ meets each chart in $\operatorname{Spec}(R_{\tilde{g}})$, with $\tilde{g}$ the image of g in R, and $R_{\tilde{g}} \cong R \otimes_B B_g$, both being the localization of R at $\tilde{g}$, so Step 1 makes it finitely presented over B_g. Gluing: suppose $g_1, \dots, g_n$ generate the unit ideal of B and Q holds for each $\operatorname{Spec}(B_{g_k})$. Then $\pi^{-1}(\operatorname{Spec}(B))$ is the union of the sets $\pi^{-1}(\operatorname{Spec}(B_{g_k}))$, covered by affines finitely presented over B_{g_k}, and $B_{g_k} \cong B[t]/(tg_k - 1)$ is finitely presented over B, so Step 1 makes those affines finitely presented over B. The property Q holds on the cover provided by definition 3.4.33, so theorem 2.2.31 gives it for every affine open of Y. The corresponding statement for locally of finite type is part of definition 3.4.28.

Step 3: compositions. Let $\operatorname{Spec}(A) \subseteq Z$ be any affine open. By definition 3.4.28 (resp. Step 2 applied to ρ), $\rho^{-1}(\operatorname{Spec}(A))$ is covered by affine opens $\operatorname{Spec}(B)$ with B finitely generated (resp. finitely presented) over A. By the same statements applied to π, each $\pi^{-1}(\operatorname{Spec}(B))$ is covered by affine opens $\operatorname{Spec}(C)$ with C finitely generated (resp. finitely presented) over B. These $\operatorname{Spec}(C)$ together cover $(\rho \circ \pi)^{-1}(\operatorname{Spec}(A))$, and Step 1 makes each C finitely generated (resp. finitely presented) over A.

Step 4: base changes. Cover Y' by affine opens $\operatorname{Spec}(S')$, each mapping into an affine open $\operatorname{Spec}(S) \subseteq Y$, shrinking around each point of Y' as in the proof of proposition 3.4.25. As there, proposition 3.2.5(1) and the affine construction of theorem 3.2.1 identify the preimage of $\operatorname{Spec}(S')$ in $X \times_Y Y'$ with $\pi^{-1}(\operatorname{Spec}(S)) \times_{\operatorname{Spec}(S)} \operatorname{Spec}(S')$, and covering $\pi^{-1}(\operatorname{Spec}(S))$ by affine opens $\operatorname{Spec}(R)$ with R finitely generated (resp. finitely presented) over S, which definition 3.4.28

(resp. Step 2) gives, covers the preimage by the affine opens $\operatorname{Spec}(R \otimes_S S')$. Step 1 makes each $R \otimes_S S'$ finitely generated (resp. finitely presented) over S'. These covers sit over an affine open cover of Y', which suffices by proposition 3.4.30 (resp. directly by definition 3.4.33, which is the covering form).

Step 5: quasicompactness, for the finite type statements. By definition 3.4.29 it remains to check that quasicompactness of morphisms (definition 3.4.1) is preserved. Composition: for an affine open $\operatorname{Spec}(A) \subseteq Z$, the set $\rho^{-1}(\operatorname{Spec}(A))$ is quasicompact, hence a finite union of affine opens, each with quasicompact π-preimage, and a finite union of quasicompact sets is quasicompact. Base change: let $V' \subseteq Y'$ be an affine open. Being quasicompact, V' is a finite union of affine opens $\operatorname{Spec}(S')$ each mapping into an affine open $\operatorname{Spec}(S) \subseteq Y$, and as in Step 4 the preimage of each $\operatorname{Spec}(S')$ is $\pi^{-1}(\operatorname{Spec}(S)) \times_{\operatorname{Spec}(S)} \operatorname{Spec}(S')$. Writing the quasicompact set $\pi^{-1}(\operatorname{Spec}(S))$ as a finite union of affine opens $\operatorname{Spec}(R)$ exhibits that preimage as a finite union of the affine, hence quasicompact, sets $\operatorname{Spec}(R \otimes_S S')$. So the preimage of V' is a finite union of quasicompact sets, as we sought to show.

Definition 3.4.36: Quasifinite Morphism

Let $\pi : X \to Y$ be a scheme morphism. Then π is said to be *quasifinite* if it is of finite type, and for each $\mathfrak{q} \in Y$, $\pi^{-1}(\mathfrak{q})$ is a finite set.

By proposition proposition 3.4.19 and proposition 3.4.32(1), finite morphisms are quasifinite. The converse is not true (think of $\mathbb{A}^2_k \setminus \{(0,0)\} \to \mathbb{A}^2_k$. The finite type condition is necessary, there exists morphisms with finite fibers that are not quasifinite

Example 3.4.37: Finite Fibers, But Not Quasifinite

The morphism $\operatorname{Spec}(\mathbb{C}(t)) \to \operatorname{Spec}(\mathbb{C})$ has finite fibers, but is not quasifinite. Similarly to $\operatorname{Spec}(\overline{\mathbb{Q}}) \to \operatorname{Spec}(\mathbb{Q})$.

The failure of the converse in the example above is the model for the general situation: the morphism $\mathbb{A}^2_k \setminus \{(0,0)\} \to \mathbb{A}^2_k$ is quasifinite but not finite, and the only thing standing between it and finiteness is that it has been restricted to an open subset. Zariski's Main Theorem says this is always the obstruction, and it is what makes quasifiniteness a useful hypothesis.

Theorem 3.4.38: Zariski's Main Theorem

Let Y be a Noetherian scheme and let $\pi\colon X \to Y$ be a separated, quasifinite morphism of finite type. Then π factors as

$$X \xrightarrow{j} Z \xrightarrow{g} Y$$

where j is an open immersion and g is finite.

***Proof Key ideas*:**
The full proof is outside the scope of this book. It rests either on the theorem on formal functions together with Stein factorization, neither of which is developed here, or on a dévissage reducing the general case to that of a finitely generated algebra over a Noetherian ring; the argument is long and its details are not illuminating at this stage. See EGA IV_3, 8.12.6 [32], or [70, Tag

05K0] for a self-contained modern account.

The shape of the argument is worth recording, because the substance of the theorem is a statement about rings. Reduce to the case $Y = \operatorname{Spec} A$ and $X = \operatorname{Spec} B$ affine, with B a finitely generated A-algebra. The local assertion is that if $\mathfrak{q} \in \operatorname{Spec} B$ is isolated in its fibre over A, then there are an element $b \in B \setminus \mathfrak{q}$ and a finite A-subalgebra $A' \subseteq B$ with $B_b = A'_b$; that is, after inverting a single element the algebra becomes finite over A. The distinguished open $D(b)$ is the piece of X on which π already looks finite, and Z is assembled by gluing the finite algebras A' obtained this way over a cover of Y.

The theorem separates the two mechanisms by which a morphism can have finite fibres. A finite morphism has them because it is proper and affine; a quasifinite morphism has them because its fibres are discrete, and it may have been cut down to an open subset along the way. Theorem 3.4.38 says that these are the only two mechanisms, and that the second sits inside the first as an open immersion.

The form in which the theorem is usually applied adds a normality hypothesis on the target, which collapses the finite part to an isomorphism and leaves only the open immersion.

Corollary 3.4.39: Birational Form of Zariski's Main Theorem

Let $\pi\colon X \to Y$ be a separated, quasifinite morphism of finite type between integral Noetherian schemes, and suppose that Y is normal and that π is dominant and induces an isomorphism $K(Y) \to K(X)$ of function fields (such a morphism is called *birational*; see definition 3.6.2). Then π is an open immersion onto an open subscheme of Y.

***Proof*:**
Step 1: factor, and cut the middle term down to an integral scheme. By theorem 3.4.38 there is a factorization $\pi = g \circ j$ with $j\colon X \to Z$ an open immersion and $g\colon Z \to Y$ finite. Let Z' be the closure of $j(X)$ in Z, taken with its reduced induced structure (definition 3.1.26). Then Z' is integral, being the closure of the integral scheme $j(X) \cong X$, and $j(X)$ is an open subscheme of Z': it is open in Z and contained in Z', and its two induced structures agree because both are reduced with the same underlying space (proposition 3.1.25, applied in the open subscheme $j(X)$ of Z). The restriction $g' = g|_{Z'}\colon Z' \to Y$ is finite, since a closed subscheme of a scheme finite over Y is again finite over Y.

Step 2: g' is surjective and changes no function field. A finite morphism is proper (proposition 3.5.24) and hence closed, so $g'(Z')$ is closed in Y; it contains $\pi(X)$, which is dense because π is dominant, so g' is surjective. The generic point of Z' is the image of the generic point of X, so $K(Z') = K(X)$, and the hypothesis on π gives $K(Z') = K(Y)$.

Step 3: normality forces g' to be an isomorphism. The claim is local on Y, so let $\operatorname{Spec} A \subseteq Y$ be affine, with A a normal domain with fraction field $K(Y)$. A finite morphism is affine, so $g'^{-1}(\operatorname{Spec} A) = \operatorname{Spec} B$ with B a finite A-module, and B is a domain with fraction field $K(Z') = K(Y)$. The map $A \to B$ is injective because g' is dominant between integral schemes, so $A \subseteq B$ is an inclusion of subrings of $K(Y)$. Every element of B is integral over A, being an element of a finite A-module, and A is integrally closed in $K(Y)$ by normality; hence $B \subseteq A$

and so $B = A$. Therefore g' is an isomorphism.

Step 4: conclusion. The morphism π is the composite of the open immersion $X \to Z'$ with the isomorphism g', hence is an open immersion, as we sought to show.

Normality of the target is not a technical convenience; without it the corollary is false, and the standard counterexample is the one every treatment of normalization begins with.

Example 3.4.40: Normalization of the Cuspidal Cubic

Let k be algebraically closed and let $C = \operatorname{Spec} k[u,v]/(v^2 - u^3)$ be the cuspidal cubic. The morphism $\nu\colon \mathbb{A}^1_k \to C$ determined by $u \mapsto t^2$ and $v \mapsto t^3$ is finite, hence separated and quasifinite, and it is bijective on points; it is birational, being an isomorphism over $C \setminus \{(0,0)\}$.

It is not an open immersion. Being surjective, an open immersion would make it an isomorphism, and it is not: at the origin the maximal ideal of $\mathcal{O}_{C,(0,0)}$ is generated by the images of u and v, and neither lies in its square, since the only relation $v^2 = u^3$ has all its terms of degree at least 2. So $\dim_k \mathfrak{m}/\mathfrak{m}^2 = 2$ while $\dim \mathcal{O}_{C,(0,0)} = 1$, and the local ring is not regular, whereas $\mathcal{O}_{\mathbb{A}^1_k,0}$ is. The hypothesis of corollary 3.4.39 that fails is normality of C; the morphism ν is precisely its normalization.

The example names a construction that has not yet been made in general. It exists for every variety, and it is finite, which is what makes it usable: normalizing does not leave the category of schemes of finite type, and it does not disturb properness.

Theorem 3.4.41: Normalization of a Variety

Let k be a field and let X be an integral scheme of finite type over k. Then there exist an integral normal scheme $\widetilde{X}$ of finite type over k and a morphism $\nu\colon \widetilde{X} \to X$ such that

1. ν is finite and surjective;
2. ν is birational, inducing an isomorphism $K(X) \to K(\widetilde{X})$ on function fields, and an isomorphism over the normal locus of X;
3. $(\widetilde{X}, \nu)$ is universal: every dominant morphism $Z \to X$ from an integral normal scheme factors uniquely through ν.

The pair $(\widetilde{X}, \nu)$ is unique up to unique isomorphism over X, and is called the *normalization* of X. If X is proper over k, so is $\widetilde{X}$.

***Proof*:**

Step 1: the affine case. Let $X = \operatorname{Spec} A$ with A a domain finitely generated over k, and let $\overline{A}$ be the integral closure of A in $K = \operatorname{Frac}(A)$. By the finiteness of integral closure for finitely generated algebras over a field, proved in *Commutative Algebra* [10, Finiteness of Integral Closure], the ring $\overline{A}$ is a finite A-module, so it is again a finitely generated k-algebra, and it is a normal domain with fraction field K. Set $\widetilde{X} = \operatorname{Spec} \overline{A}$. The inclusion $A \subseteq \overline{A}$ is an integral ring extension with $\overline{A}$ finite over A, so $\nu\colon \widetilde{X} \to X$ is finite by proposition 3.4.32, and it is surjective because an integral extension induces a surjection on spectra. It is birational since the two rings

share the fraction field K, and it is an isomorphism over the locus where A is already integrally closed.

Step 2: gluing. For a distinguished open $\operatorname{Spec} A_f \subseteq X$, integral closure commutes with localization (*Commutative Algebra* [10, Normalization Commutes with Localization]), giving $\overline{A_f} = (\overline{A})_f$, so the affine normalizations agree on overlaps and glue: covering a general X by affine opens $\operatorname{Spec} A_i$ and normalizing each, the resulting schemes patch to a scheme $\widetilde{X}$ with a finite morphism to X, since finiteness is affine-local on the target (proposition 3.4.17). The glued $\widetilde{X}$ is integral with function field $K(X)$ and is normal, normality being a stalk condition.

Step 3: the universal property and uniqueness. Let $g\colon Z \to X$ be dominant with Z integral and normal. Dominance gives an inclusion of function fields $K(X) \subseteq K(Z)$. Working over an affine $\operatorname{Spec} A \subseteq X$ with preimage covered by affines $\operatorname{Spec} C \subseteq Z$, every element of $\overline{A}$ is integral over A, hence over C, and lies in $K(Z)$; as C is integrally closed in $K(Z)$, it lies in C. So $A \to C$ extends uniquely to $\overline{A} \to C$, and these glue to the required factorization. Uniqueness up to unique isomorphism is the usual consequence of a universal property.

Step 4: properness. A finite morphism is proper (proposition 3.5.24), so if $X \to \operatorname{Spec} k$ is proper then the composite $\widetilde{X} \to X \to \operatorname{Spec} k$ is proper, as we sought to show.

Two features of the normalization matter for what follows. It changes nothing in codimension zero, since it is birational; and by corollary 3.4.39 it is an isomorphism exactly where X was already normal, so it is a repair restricted to the non-normal locus. What it gives is that $\widetilde{X}$ is regular in codimension 1, because a normal Noetherian scheme is, and therefore that the divisor theory of chapter 5 applies to it.

Vakil promises exercise 9.3.H shows why this would naturally arise

Exercise 3.4.1

1. Let $\pi : X \to Y$ be an open embedding. Then if Y is locally Noetherian, X is locally Noetherian. If Y is Noetherian, X is Noetherian.
2. Let $\pi : X \to Y$ be an open embedding. Show that if Y is quasicompact, X need not be (there is an affine counter-example)
3. Show that the composition of two finite morphism is a finite morphism.
4. Show that being an open embedding is not local on the source.
5. Let B/A be an integral extension. Then show that any ring homomorphism $B \to C$ induces an integral extensions $C \otimes_B A/C$. Conclude that integrality is preserved by base change (see [12, Extension of Scalars] if you are unsure what it means to change base).
6. Show that every open embedding is locally of finite type. Hence, every quasicompact open embedding is of finite type.
7. Every open embedding into a locally Noetherian scheme is of finite type
8. The composition of two morphisms locally of finite type is locally of finite type.

9. Let $\pi : X \to Y$ be locally of finite type and Y locally Noetherian. Then X is locally Noetherian. IF $\pi : X \to Y$ is of finite type and Y is Noetherian, then X is Noetherian,
10. Show that quasifinite morphisms to Spec (k) are finite)
11. Let $k = \overline{\mathbb{F}_p}$. Show that the morphism $F : \mathbb{A}^n_k \to \mathbb{A}^n_k$ of k-schemes corresponding to the bijection $x_i \mapsto x_i^p$ is a bijection, and no power of F is the identity (as sets)
12. (Generalizing the above) Let X be an $\mathbb{F}_p$-scheme. Define an endomorphism $F : X \to X$ such that for each affine open subset Spec $(R) \subseteq R< F$ corresponds to the map $R \to R$ given by $f \mapsto f^p$. The morphism F is called the *absolute Frobenius morphism.*

3.5 Separated and Proper Schemes

The last two major types of schemes we must cover are *separated schemes* that recover the notion of Hausdroffness for schemes, and *proper schemes* which recover a notion of compactness for schemes. As is the theme in this section, these two properties can be defined purely in terms of scheme morphisms.

From an algebraic-geometric perspective, these conditions allow for better behaviour of closed sets, whether the intersection of two closed sets is closed or whether the image of closed sets is closed is governed by separatedness and properness respectively. Both separatedness and properness can be defined in terms of morphisms, and this is in fact the more practical approach. As closed sets are locally closed set of affine scheme (i.e. vanishing sets of global sections of Spec R_i), the preservation of closedness of these sets under various kinds of mappings is of key interest in scheme theory.

3.5.1 Separated Morphisms

Recall that if $f : X \to Y$ is a continuous map and $K \subseteq X$ is compact, then $f(K)$ is compact. In euclidean topology, this implies closed and bounded maps to closed and bounded, importantly this is a special case of when closed sets maps to closed sets. As affine schemes are quasi-compact (recall we use the term quasi-compact instead of compact due to the differing behaviour without Hausdorffness), if $K \subseteq X$ is closed it is quasi-compact, and hence its image is quasi-compact if $f : X \to Y$ is a scheme morphism (which is continuous). However, such a map *need not be closed*, namely it need not map closed sets to closed sets: the projection $\mathcal{Z}(xy - 1) \to \mathbb{A}^1_k$ has closed source and image $\mathbb{A}^1_k \setminus \{0\}$. In particular a quasicompact set need not be closed (see [30, section 3.2]).

Another example that shows the failure of separatedness is the following: Take X to be the line with double origin (recall example 2.2.57). Then the two natural inclusions $\iota_1, \iota_2 : \mathbb{A}^1_k \to X$ do not intersect on a closet set!

To recover the good properties of Hausdroffness, the notion of *separatedness* is defined. The idea is that we want to avoid cases where there can be "double points" or even more points that are intuitively arbitrarily close to each other; such points cannot be distinguished, or *separated* by (regular) functions.

In the topological setting, we saw X is Hausdorff if and only if $\Delta \subseteq X \times X$ is closed. In the case of scheme every "diagonal" (when working with fibered products) is a locally closed immersion:

Proposition 3.5.1: Diagonal Locally closed immersion

Let $f : X \to S$ be a morphism of schemes. Then the diagonal morphism $\delta : X \to X \times_S X$ is a locally closed immersion (definition 3.1.27): there is an open subset $W \subseteq X \times_S X$ containing $\delta(X)$ such that $\delta\colon X \to W$ is a closed immersion. The diagonal δ is a closed immersion if and only if $\delta(X)$ is closed in $X \times_S X$. If f is affine (definition 3.4.5), then W can be taken to be $X \times_S X$, so δ is a closed immersion.

***Proof*:**
Let $p_1, p_2\colon X \times_S X \to X$ be the projections, so $p_1 \circ \delta = p_2 \circ \delta = \mathrm{id}_X$. Cover S by affine opens $V_i \cong \operatorname{Spec} B_i$ and each $f^{-1}(V_i)$ by affine opens $U_{ij} \cong \operatorname{Spec} A_{ij}$, and when f is affine take the single affine open $f^{-1}(V_i)$ for each i. Put $W_{ij} = p_1^{-1}(U_{ij}) \cap p_2^{-1}(U_{ij})$. By proposition 3.2.5(1), applied twice, the universal property of theorem 3.2.1, and lemma 2.2.52 (morphisms into V_i are morphisms into S with image in V_i), the open subscheme W_{ij} is $U_{ij} \times_{V_i} U_{ij}$, which is $\operatorname{Spec}(A_{ij} \otimes_{B_i} A_{ij})$ by the affine case of theorem 3.2.1. Let W be the union of the W_{ij}. Since $p_1 \circ \delta = p_2 \circ \delta = \mathrm{id}_X$, we have $\delta^{-1}(W_{ij}) = U_{ij}$, so $\delta(X) \subseteq W$.

Step 1: each piece. The restriction $\delta_{ij}\colon U_{ij} \to W_{ij}$ of δ (lemma 2.2.52) has both components $\mathrm{id}_{U_{ij}}$. The projections of $\operatorname{Spec}(A_{ij} \otimes_{B_i} A_{ij})$ correspond to $a \mapsto a \otimes 1$ and $a \mapsto 1 \otimes a$, so δ_{ij} corresponds to a ring map sending $a \otimes 1$ and $1 \otimes a$ to a (theorem 3.1.4), that is, to $a \otimes a' \mapsto aa'$. This map is surjective, so δ_{ij} is a closed immersion (proposition 3.1.24(2)).

Step 2: gluing the pieces. The map δ is injective, as $p_1 \circ \delta = \mathrm{id}_X$. For $Z \subseteq X$ closed, $\delta(Z) \cap W_{ij} = \delta_{ij}(Z \cap U_{ij})$ is closed in W_{ij} by Step 1, so $W \setminus \delta(Z)$ is the union of the open sets $W_{ij} \setminus \delta(Z)$ and $\delta(Z)$ is closed in W. Hence δ is a homeomorphism of X onto the closed subset $\delta(X)$ of W. As $\delta^{-1}(W_{ij}) = U_{ij}$, the restriction of $\mathcal{O}_W \to \delta_*\mathcal{O}_X$ to W_{ij} is the sheaf map of δ_{ij}, which is surjective on stalks. So $\delta\colon X \to W$ is a closed immersion (definition 3.1.22).

Step 3: closed image. A closed immersion has closed image. Conversely, if $\delta(X)$ is closed, then δ is a homeomorphism onto a closed set by Step 2, and the sheaf map $\mathcal{O}_{X \times_S X} \to \delta_*\mathcal{O}_X$ is surjective on stalks at the points of W, where it restricts to the sheaf map of $\delta\colon X \to W$, and at the points of the open set $(X \times_S X) \setminus \delta(X)$, where $\delta_*\mathcal{O}_X$ has zero stalks. These two open sets cover $X \times_S X$, so δ is a closed immersion (definition 3.1.22).

Step 4: the affine case. Let f be affine, with $U_i = f^{-1}(V_i)$, and put $q = f \circ p_1 = f \circ p_2$. Then $W_i = p_1^{-1}(f^{-1}(V_i)) \cap p_2^{-1}(f^{-1}(V_i)) = q^{-1}(V_i)$, and these sets cover $X \times_S X$. So $W = X \times_S X$, completing the proof.

The key is that these open subsets need not cover $X \times_S X$, hence this does not imply the embedding is closed.

In our case, a subset is closed if it is locally closed in an affine covering, this means that the diagonal can be expressed (locally) as vanishing sets. Furthermore, the product is not well-defined for schemes, as it must always be fibered over some other scheme. This leads to the following:

Definition 3.5.2: Separated Scheme

Let X be a scheme. Then X is *separated* if the map $\delta : X \to X \times_{\mathbb{Z}} X$ is a closed immersion.

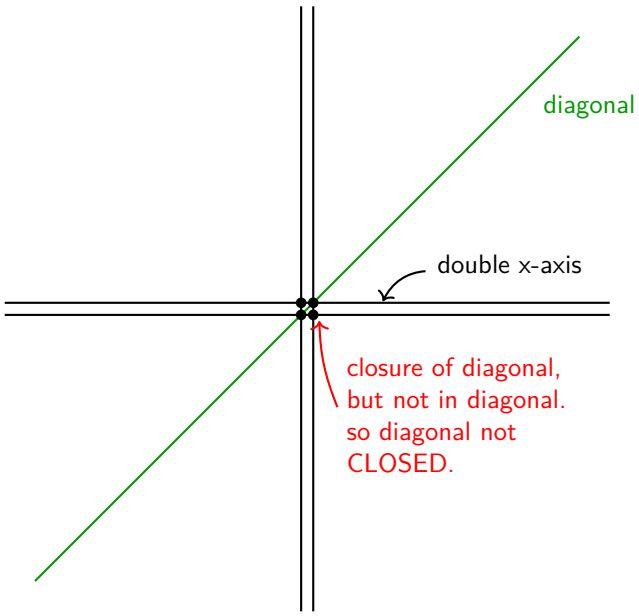
More generally, a map $X \to S$ between schemes is *separated* if the map $\delta : X \to X \times_S X$ is a closed immersion. Denote the image of δ as $\Delta(X)$ or Δ_X.

As a historical footnote, the term scheme used to mean separated scheme back before the 1960s, and the term pre-scheme was used for the modern term scheme.

Note that the first condition is equivalent to saying that the morphism $X \to \operatorname{Spec}(\mathbb{Z})$ is separated. The fact that separatedness is given by the purely topological condition of $\Delta(X)$ being closed shows its closeness to the original topological condition of Hausdorffness.

Example 3.5.3: Separated Morphisms

1. Take the line with two origins over k, label it X. Then when considering $X \times_{\operatorname{Spec} k} X$, we can think to it as in image bellow. The double origin will correspond to 4 points, but the diagonal will only have 2 points. Taking the closure of the diagonal will include all 4 points (show that any open set containing either $(0_1, 0_2)$ or $(0_2, 0_1)$ must intersect $\Delta(X)$, and hence is in the closure; see [30, section 1.5]), showing that it is *not* separated.



Note that $X \to X \times_k X$ is still a locally closed immersion. To show that it is not closed, we can take an open cover that covers *all* of $X \times_k X$ and show that for the affine open W_i containing $P_1 = (0_1, 0_2)$, $\Delta(X) \cap W_i$ is not closed. Indeed, As $P_1 \notin \Delta(X)$, $P_1 \notin \Delta(X) \cap W_i$. Taking W_i to be the product of the first copy of $\mathbb{A}^1_k$ with the second, $W_i \cong \mathbb{A}^2_k$ and $\Delta(X) \cap W_i$ is the punctured diagonal $\{(x, x) : x \neq 0\}$, whose closure is the whole line $t_1 = t_2$. Since $P_1 \in \overline{\Delta(X) \cap W_i}$, we see that $\Delta(X) \cap W_i$ is *not* closed.

Note that if we take $X \to X$ to be the identity map, then this *is* a separated morphism, since:

$$X \cong X \times_X X$$

showing the relativeness of separatedness when picking different base schemes.

2. Let $X = \operatorname{Spec}(A) \to \operatorname{Spec}(B) = Y$ be any morphism of schemes between affine schemes. Then this is *always* a separated morphism. Unwind the definitions: when converting into rings we get the map:

$$A \otimes_B A \to A$$

which is surjective, and hence we get:

$$A \cong A \otimes_B A/\mathfrak{a}$$

that is, we are taking the quotient by an ideal, which we know will give us a closed subscheme of $X \times_Y X$ isomorphic to X.

3. More generally, if $X \to Y$ is an affine morphism, it is separated (follows from proposition 3.5.1).

4. The following scheme is a famous example of non-separatedness appearing naturally. Consider $\mathbb{C}^* \curvearrowright \mathbb{A}^2_{\mathbb{C}}$ via $t \cdot (x, y) = (tx, t^{-1}x)$; this is usually sated as "the algebraic torus acts on the affine plane with weights $(1, -1)$". Then taking $(\mathbb{A}^2_{\mathbb{C}} \setminus \{0\})/\mathbb{C}^*$ where the origin is removed as the action is non-free, the resulting space is a non-separated scheme.

   In many cases, non-separated scheme appear naturally in moduli theory.

An important example worth singling out is that of projective scheme:

Lemma 3.5.4: Projective Scheme is Separated

The scheme $\mathbb{P}^1_{\mathbb{Z}}$ is separated.

***Proof*:**
Let $X = \mathbb{P}^1_{\mathbb{Z}}$ and take a standard affine cover $U_0 = \operatorname{Spec} \mathbb{Z}[t]$ and $U_1 = \operatorname{Spec} \mathbb{Z}[u]$, glued via $u = 1/t$. To prove separatedness, we must show $\Delta(X)$ is closed in $X \times_{\mathbb{Z}} X$. It suffices to show the intersection of $\Delta(X)$ with the members of an open cover of the *product* is closed. The product is covered by the four affine patches $U_{ij} = U_i \times U_j$.

We analyze the corresponding ring maps (dual maps) $\Delta^\#$. Recall that a morphism of affine schemes is a closed immersion if and only if its ring dual map is surjective. If surjective, the image corresponds to $V(\ker(\Delta^\#))$.

- *Case 1: The standard charts* (e.g., $W_{00} = U_0 \times U_0$).
  Algebraically, $W_{00} \cong \operatorname{Spec}(\mathbb{Z}[t] \otimes_{\mathbb{Z}} \mathbb{Z}[t]) \cong \operatorname{Spec} \mathbb{Z}[t_1, t_2]$. The diagonal restricted to this chart maps U_0 into $U_0 \times U_0$. The corresponding ring dual map $\Delta^\#$ is:

  $$\varphi : \mathbb{Z}[t_1, t_2] \to \mathbb{Z}[t], \quad \text{defined by } t_1 \mapsto t,\ t_2 \mapsto t.$$

  This map is surjective, since t lies in its image and $\mathbb{Z}[t]$ is generated by t as a $\mathbb{Z}$-algebra. By the First Isomorphism Theorem, $\mathbb{Z}[t] \cong \mathbb{Z}[t_1, t_2]/\ker(\varphi)$. The kernel is the ideal generated

by $(t_1 - t_2)$. Consequently, the diagonal corresponds to the closed subscheme defined by this ideal, $V(t_1 - t_2)$, representing the line $y = x$.

- *Case 2: The mixed charts* (e.g., $W_{01} = U_0 \times U_1$).
  Algebraically, the target is $W_{01} \cong \operatorname{Spec} \mathbb{Z}[t] \times \operatorname{Spec} \mathbb{Z}[u] \cong \operatorname{Spec} \mathbb{Z}[t, u]$. The source (the part of the diagonal mapping into this chart) is the intersection $U_0 \cap U_1$. This intersection is the locus where $t \neq 0$, which is affine: $\operatorname{Spec} \mathbb{Z}[t, t^{-1}]$. The ring dual map $\Delta^{\#}$ maps the coordinates of the product to their values on the intersection (determined by the gluing $u = 1/t$):
  $$\psi : \mathbb{Z}[t, u] \to \mathbb{Z}[t, t^{-1}].$$
  Defined by $t \mapsto t$ and $u \mapsto t^{-1}$. Since t and t^{-1} are in the image, ψ is surjective. This implies the geometric map is a closed immersion. The algebraic geometry dictionary gives the image is defined by the kernel of ψ:
  $$\operatorname{Spec} \mathbb{Z}[t, t^{-1}] \cong \operatorname{Spec} \left(\mathbb{Z}[t, u] / \ker(\psi) \right).$$
  Since $t(t^{-1}) - 1 = 0$ in the codomain, the element $(tu - 1)$ is in the kernel. In fact, $\ker(\psi) = (tu - 1)$. Thus, the image of the diagonal in this chart is the closed subset $V(tu - 1) \subseteq \mathbb{A}^2_{\mathbb{Z}}$, which describes the hyperbola $u = 1/t$.

The charts $U_1 \times U_1$ and $U_1 \times U_0$ are handled in the same way, with the roles of t and u exchanged. So the image of the diagonal is closed in every chart of an open cover of $X \times_{\mathbb{Z}} X$, hence closed, and the diagonal is a closed immersion by proposition 3.5.1.

The same argument works for every $\mathbb{P}^n_R$, with the structure morphism to $\operatorname{Spec}(R)$:

Corollary 3.5.5: Projective Space is Separated

For any ring R and $n \geq 0$, the structure morphism $\mathbb{P}^n_R \to \operatorname{Spec}(R)$ (definition 2.3.4) is separated. In particular $\mathbb{P}^n_{\mathbb{Z}}$ is separated.

***Proof*:**
Let $X = \mathbb{P}^n_R = \operatorname{Proj} S$ with $S = R[x_0, \ldots, x_n]$, let $U_i = D_+(x_i) \cong \operatorname{Spec} A_i$ with $A_i = S_{(x_i)}$, and let $\delta \colon X \to X \times_R X$ be the diagonal, where $X \times_R X = X \times_{\operatorname{Spec}(R)} X$, with projections p_1, p_2. By proposition 3.2.5(1), applied twice, and theorem 3.2.1, the sets $p_1^{-1}(U_i) \cap p_2^{-1}(U_j)$ are open subschemes isomorphic to $U_i \times_R U_j \cong \operatorname{Spec}(A_i \otimes_R A_j)$, and they cover $X \times_R X$, since every point has its two projections in some U_i and U_j. As $p_1 \circ \delta = p_2 \circ \delta = \mathrm{id}$, the preimage of $U_i \times_R U_j$ under δ is $U_i \cap U_j = D_+(x_i x_j) \cong \operatorname{Spec} S_{(x_i x_j)}$ (lemma 2.3.2, theorem 2.3.3). The restriction δ_{ij} of δ has the two inclusions of $U_i \cap U_j$ as components, so it corresponds to the ring map $A_i \otimes_R A_j \to S_{(x_i x_j)}$ sending $a \otimes b$ to the product of the restrictions of a and b, which are given by theorem 2.3.3 (theorem 3.1.4). This map is surjective: an element of $S_{(x_i x_j)}$ is an R-linear combination of fractions $m/(x_i x_j)^d$ with m a monomial of degree $2d$, and writing $m = m_1 m_2$ with monomials m_1, m_2 of degree d exhibits the fraction as the image of $(m_1/x_i^d) \otimes (m_2/x_j^d)$. So δ_{ij} is a closed immersion (proposition 3.1.24(2)), and $\delta(X) \cap (U_i \times_R U_j) = \delta_{ij}(U_i \cap U_j)$ is closed in $U_i \times_R U_j$. Being closed in each member of an open cover, $\delta(X)$ is closed, and δ is a closed immersion by proposition 3.5.1. Hence $X \to \operatorname{Spec}(R)$ is separated (definition 3.5.2), and for $R = \mathbb{Z}$ this is the absolute statement, completing the proof.

The next concept will not be too useful to us for awhile, and shall come back up when we see algebraic stacks and algebraic spaces, but is useful to define:

Definition 3.5.6: Quasi-separated

Let X be a scheme. Then X is quasi-separated if the image of $X \to X \times X$ is quasi-compact. More generally, a map $X \to S$ between schemes is quasi-separated if the image map of $X \to X \times_S X$ is quasi-compact.

Proposition 3.5.7: Separated Implies Quasi-separated

Let X be a separated scheme. Then X is quasi-separated.

***Proof*:**
By definition, X is separated if the diagonal morphism $\Delta : X \to X \times X$ is a closed immersion. By definition, X is quasi-separated if Δ is a quasi-compact morphism. Thus, it suffices to prove that any closed immersion is quasi-compact.

Let $f : Z \to Y$ be a closed immersion. To prove f is quasi-compact, we check the condition on an affine open cover. Let $V \subseteq Y$ be an affine open subset, say $V = \operatorname{Spec}(B)$. Because f is a closed immersion, the restriction $f|_V : f^{-1}(V) \to V$ corresponds to a surjective ring homomorphism $B \to A$ (specifically $A \cong B/I$ for the ideal I cutting out Z). Therefore, the preimage $f^{-1}(V)$ is isomorphic to $\operatorname{Spec}(A)$. Since every affine scheme is quasi-compact, $f^{-1}(V)$ is quasi-compact.

As the preimage of any affine open is quasi-compact, the map f is quasi-compact. Applying this to Δ, we conclude X is quasi-separated.

Separatedness gives us nicer behaviour on closed sets. The following results outline some important examples:

Proposition 3.5.8: Intersection of Separated affine schemes

Let $X \to S$ be a separated scheme morphism where S is affine. Then if $U, V \subseteq X$ are open affine subsets, their intersection

$$U \cap V$$

is an affine open subset.

***Proof*:**
Let $S = \operatorname{Spec} R$, $U = \operatorname{Spec} A$, and $V = \operatorname{Spec} B$.

1. **The product of affines is affine.** Recall that the product $X \times_S X$ is constructed by gluing affine charts. Specifically, the open subset $U \times_S V \subseteq X \times_S X$ is isomorphic to the spectrum of the tensor product:

   $$U \times_S V \cong \operatorname{Spec}(A \otimes_R B).$$

   Thus, $U \times_S V$ is an affine open subscheme of the product.

2. **The intersection is a preimage.** Consider the diagonal morphism $\Delta : X \to X \times_S X$.

We observe that the intersection of the open sets is exactly the pullback of their product by the diagonal:

$$\Delta^{-1}(U \times_S V) = \{x \in X \big| \Delta(x) \in U \times_S V\} = \{x \in X \big| x \in U \text{ and } x \in V\} = U \cap V.$$

3. **Separatedness implies affine.** By the definition of separatedness, Δ is a closed immersion. This means that Δ induces a closed immersion of $U \cap V$ into the open subset $U \times_S V$:

$$U \cap V \hookrightarrow U \times_S V.$$

A closed immersion into an affine scheme corresponds to a surjective ring map. Since $U \times_S V$ is affine ($\operatorname{Spec}(A \otimes_R B)$), $U \cap V$ must be isomorphic to $\operatorname{Spec}((A \otimes_R B)/I)$ for some ideal I. Since the spectrum of a quotient ring is an affine scheme, $U \cap V$ is affine.

For a counter-example, taking affine *plane* with double origin we get that taking two affine lines each containing a different origin will intersect to produce the plane without the origin, which in example 2.2.53 is not affine.

Proposition 3.5.9: Graphs Are Closed

Let X be a separated S-scheme, Y any S-scheme and $f\colon Y \to X$ an S-morphism. Then the graph morphism $\Gamma = (\mathrm{id}_Y, f)\colon Y \to Y \times_S X$ is a closed immersion, so the graph

$$\Gamma_f = \operatorname{im}((\mathrm{id}_Y, f)\colon Y \to Y \times_S X)$$

is closed.

***Proof*:**
Let $\delta\colon X \to X \times_S X$ be the diagonal, a closed immersion as X is separated (definition 3.5.2), and let $\varphi\colon Y \times_S X \to X \times_S X$ have components $f \circ p_Y$ and p_X (theorem 3.2.1). Both $\varphi \circ \Gamma$ and $\delta \circ f$ have components f and f, so they are equal. Let $a\colon T \to Y \times_S X$ and $c\colon T \to X$ satisfy $\varphi \circ a = \delta \circ c$, that is, $f \circ p_Y \circ a = c = p_X \circ a$, and put $b = p_Y \circ a$. Then $\Gamma \circ b$ has components b and $f \circ b = p_X \circ a$, so $\Gamma \circ b = a$, and $f \circ b = c$. Any b' with $\Gamma \circ b' = a$ satisfies $b' = p_Y \circ \Gamma \circ b' = b$. So Y, with Γ and f, is a fibered product of φ and δ, and Γ is, up to isomorphism, the base change of δ along φ. By proposition 3.2.5(2), Γ is a closed immersion, and its image Γ_f is closed, completing the proof.

For the case where it is not separated as usual let X be the line with two origins (not separated). Let Y be the affine line $\mathbb{A}^1$. Consider the morphism $g : Y \setminus \{0\} \to X$ which is the natural inclusion into the "un-doubled" part of X. What is the graph of this morphism? It's a subset of $(Y \setminus \{0\}) \times_k X$. The key is that this map cannot be extended to the origin of Y in a unique way. Indeed, $0 \in Y$ could map to either $0_1 \in X$ or $0_2 \in X$. This ambiguity means the graph of any potential extension is *not* closed; the point $(0, 0_1)$ and $(0, 0_2)$ are "limit points" of the graph but there is no natural choice of which one is in the graph. The graph is not well-behaved at the boundary. This intuition will in fact be a central way of interpreting separatedness, as shall be explored in the next section.

Proposition 3.5.10: Morphism Determined By Dense Open Subset

Let X be a *separated* S-scheme, let Z be a *reduced* S-scheme, and let $U \subseteq Z$ be a dense open subset. Then if two S-morphisms $f, g : Z \to X$ agree on U, they agree everywhere:

$$f|_U = g|_U \qquad \Rightarrow \qquad f = g$$

***Proof*:**
Let $\delta\colon X \to X \times_S X$ be the diagonal, with projections p_1, p_2, and $h = (f, g)\colon Z \to X \times_S X$. As X is separated over S, $\delta(X)$ is closed (definition 3.5.2), so $h^{-1}(\delta(X))$ is closed. It contains U, where $h = \delta \circ f$, and U is dense, so $h(Z) \subseteq \delta(X)$. Since $p_1 \circ \delta = p_2 \circ \delta$, this gives $f(z) = p_1(h(z)) = p_2(h(z)) = g(z)$ for every $z \in Z$, so $f = g$ as maps.

It remains to compare the sheaf maps. Let $z \in Z$ and $x = f(z) = g(z)$. Choose affine opens $W \cong \operatorname{Spec}(R)$ of S containing the image of x, $V \cong \operatorname{Spec}(C)$ of X over W containing x, and $Z' \cong \operatorname{Spec}(D)$ of Z inside $f^{-1}(V)$ containing z. By lemma 2.2.52 and theorem 3.1.4, f and g restrict to morphisms $Z' \to V$ over W given by R-algebra maps $\alpha, \beta\colon C \to D$. By proposition 3.2.5(1), applied twice, and theorem 3.2.1, $V \times_W V \cong \operatorname{Spec}(C \otimes_R C)$ is an open subscheme of $X \times_S X$ meeting $\delta(X)$ in $\delta(V)$, and $\delta|_V$ is the closed immersion given by the multiplication $C \otimes_R C \to C$, whose kernel contains $c \otimes 1 - 1 \otimes c$ for $c \in C$. The restriction of h to Z' is given by $c \otimes c' \mapsto \alpha(c)\beta(c')$, and its image lies in $\delta(V)$, the set of primes containing that kernel. So for every prime $\mathfrak{q}$ of D and $c \in C$, the image $\alpha(c) - \beta(c)$ of $c \otimes 1 - 1 \otimes c$ lies in $\mathfrak{q}$, hence is nilpotent ([10, The Radical as an Intersection of Primes]), hence zero, as $D = \mathcal{O}_Z(Z')$ is reduced. So $\alpha = \beta$, and f and g agree on Z' (theorem 3.1.4). Such Z' cover Z, so $f = g$ by the uniqueness in lemma 3.1.9, as we sought to show.

Both hypotheses are needed, and each fails in a way worth seeing once. Dropping separatedness, let X be the line with two origins of the discussion above, let $Z = \mathbb{A}^1$ and $U = \mathbb{A}^1 \setminus \{0\}$, and let f and g be the two evident inclusions: they agree on the dense open U and differ at the origin. Dropping reducedness, let $Z = \operatorname{Spec} k[x, \epsilon]/(\epsilon^2, x\epsilon)$ with $U = D(x)$, and let $f, g : Z \to \mathbb{A}^1$ be given by $y \mapsto 0$ and $y \mapsto \epsilon$. Since $x\epsilon = 0$, the function ϵ restricts to 0 on U, so f and g agree there even though they differ on Z; here the target is as well behaved as one could ask, being both reduced and separated, and it is the nilpotent on the source that defeats the conclusion.

Lemma 3.5.11: Monomorphisms and Separatedness

1. Closed immersions (definition 3.1.22) are monomorphisms of schemes.
2. If $i\colon A \to B$ is a monomorphism of schemes, then the diagonal $A \to A \times_B A$ is an isomorphism, and i is separated (definition 3.5.2). In particular open embeddings and closed immersions are separated.

***Proof*:**

1. Let $i\colon A \to B$ be a closed immersion and $a, b\colon T \to A$ morphisms with $i \circ a = i \circ b$. As i is injective, $a = b$ as maps. The sheaf map of a composite is $(i \circ a)^{\#} = i_*(a^{\#}) \circ i^{\#}$, so $i_*(a^{\#}) \circ i^{\#} = i_*(b^{\#}) \circ i^{\#}$. Since $i^{\#}$ is surjective on stalks, the morphisms $i_*(a^{\#})$ and $i_*(b^{\#})$

from $i_*\mathcal{O}_A$ to the sheaf $i_*a_*\mathcal{O}_T$ agree on every stalk, so they are equal (proposition 1.2.23). Every open $W \subseteq A$ has the form $i^{-1}(W')$ for an open $W' \subseteq B$, i being a homeomorphism onto its image, so $a^\#(W) = i_*(a^\#)(W') = i_*(b^\#)(W') = b^\#(W)$.

2. Let $p_1, p_2 \colon A \times_B A \to A$ be the projections and δ the diagonal. Since $i \circ p_1 = i \circ p_2$ and i is a monomorphism, $p_1 = p_2$. Then $p_k \circ (\delta \circ p_1) = p_1 = p_k$ for $k = 1, 2$, so $\delta \circ p_1 = \mathrm{id}$ by the uniqueness in theorem 3.2.1, while $p_1 \circ \delta = \mathrm{id}_A$. So δ is an isomorphism, in particular a homeomorphism onto the closed set $A \times_B A$ with bijective sheaf map, hence a closed immersion, and i is separated. Open embeddings are monomorphisms by lemma 2.2.52, and closed immersions by (1), completing the proof.

3.5.2 Valuation Criterion for Separatedness

We shall next characterize separated schemes and morphisms in using valuation rings. This will be analogous to the fact that in a Hausdorff space, the limit of a sequence has at most 1 point. From this result, think of a continuous map $(0,1) \to X$. If we want to extend this to a map $[0,1] \to X$, if X is Hausdorff, there is only one possible extension. If X was not, say it had two origins, then there can be more than one map that can extend (the new 0 point will have two possible places to map to).

Let us now extend this idea to a map $X \to Y$, and we have an embedding $(0,1) \to X$ an identity map $(0,1) \to [0,1]$ and an embedding $(0,1] \to Y$ where we are only adding one point. Then there should only be one lift. Note that this lift is not dependent on X and Y. Let's say Y was the double-origin scheme and X was the double y-axis scheme. Then the embedding $(0,1)$ can be extended to two way into Y, but that will correspond to a unique lift.

With this intuition in mind, let us now look at what is the equivalent of the open interval for schemes. This would be a Discrete valuation ring (DVR) R. It has only two points. We can think of as $\mathrm{Spec}\,(\mathrm{Frac}(R))$ as the open interval and $\mathrm{Spec}\,(R)$ as adding the closed 0 (notice that the new point in $\mathrm{Spec}\,(R)$ is also a closed point, while the other point is "open" and is the generic point that can be thought of was closed to everywhere else). Then we have the following situation:

$$\begin{array}{ccc} \mathrm{Spec}\,(\mathrm{Frac}(R)) & \longrightarrow & Y \\ \downarrow & \nearrow & \downarrow \\ \mathrm{Spec}\,(R) & \longrightarrow & X \end{array}$$

With the hope that having at most one lift corresponding to the map being separable. This is *sometimes* the case. To prove this, we require two technical lemmas regarding valuation rings and specialization.

Lemma 3.5.12: Lifts from Valuation Rings

Let R be a valuation ring of a field K, let $T = \operatorname{Spec} R$ and $U = \operatorname{Spec} K$, with closed point $t_0 = \mathfrak{m}_R$ and generic point $t_1 = (0)$ of T, and let $\iota\colon U \to T$ be induced by $R \subseteq K$. Let X be a scheme. For points $x_0, x_1 \in X$ with $x_0 \in \overline{\{x_1\}}$, every open neighbourhood of x_0 contains x_1, so germs at x_0 define germs at x_1. Let $\mathcal{O}_{x_0,x_1} \subseteq \kappa(x_1)$ be the image of $\mathcal{O}_{X,x_0}$ under this map $\mathcal{O}_{X,x_0} \to \mathcal{O}_{X,x_1}$ followed by the residue map.

1. A morphism $U \to X$ is equivalent to giving a point $x_1 \in X$ and an inclusion of fields $\kappa(x_1) \subseteq K$.
2. A morphism $g\colon T \to X$ is equivalent to giving
   - two points $x_0, x_1 \in X$, with x_0 a specialization of x_1 (i.e., $x_0 \in \overline{\{x_1\}}$), and
   - an inclusion $\kappa(x_1) \subseteq K$ such that R dominates the local ring $\mathcal{O}_{x_0,x_1} \subseteq \kappa(x_1) \subseteq K$.

   Here $x_0 = g(t_0)$, $x_1 = g(t_1)$, and the inclusion is the one that (1) attaches to $g \circ \iota$.

When $\overline{\{x_1\}}$ carries its reduced induced structure (definition 3.1.26), $\mathcal{O}_{x_0,x_1}$ is its local ring at x_0 by proposition 3.1.25(2), which is the form of (2) in [36, Lemma II.4.4].

***Proof*:**
For $x \in X$ and an affine open $V \cong \operatorname{Spec} B$ containing x, write $\mathfrak{p}_x \subseteq B$ for the prime of x. By proposition 2.2.22(1), $\mathcal{O}_{X,x} = B_{\mathfrak{p}_x}$, the germ of $b \in B$ being $b/1$. If a morphism $\operatorname{Spec} A \to V$ is induced by a ring map $\gamma\colon B \to A$ (theorem 3.1.4) and $\mathfrak{q}$ is a prime of A, the stalk map at $[\mathfrak{q}]$ is the map $B_{\gamma^{-1}(\mathfrak{q})} \to A_{\mathfrak{q}}$, $b/c \mapsto \gamma(b)/\gamma(c)$ (proposition 2.2.22(3)).

Morphisms out of a local ring. Let A be a local ring with maximal ideal $\mathfrak{m}$ and $a = [\mathfrak{m}]$. An open subset $\operatorname{Spec} A \setminus V(J)$ containing a has $J \not\subseteq \mathfrak{m}$, so $J = A$ and the open subset is $\operatorname{Spec} A$. The assignment $g \mapsto (g(a), g_a^\#)$, with $g_a^\#\colon \mathcal{O}_{X,g(a)} \to \mathcal{O}_{\operatorname{Spec} A,a} = A$ the stalk map, is a bijection from morphisms $g\colon \operatorname{Spec} A \to X$ to pairs (x, φ) of a point $x \in X$ and a local homomorphism $\varphi\colon \mathcal{O}_{X,x} \to A$. Let $x \in X$ and let $V \cong \operatorname{Spec} B$ be an affine open containing x. A morphism g with $g(a) = x$ has $g^{-1}(V) = \operatorname{Spec} A$, so it factors through V (lemma 2.2.52), and by theorem 3.1.4 the factored morphism is induced by a ring map $\beta\colon B \to A$, with $\beta^{-1}(\mathfrak{m}) = \mathfrak{p}_x$. By the opening paragraph its stalk map at a is the map $B_{\mathfrak{p}_x} \to A_{\mathfrak{m}} = A$ induced by β, so β is $g_a^\#$ composed with $B \to B_{\mathfrak{p}_x}$, and g is determined by $(x, g_a^\#)$. Conversely, for a local $\varphi\colon \mathcal{O}_{X,x} \to A$, the composite $\beta = \varphi \circ (B \to B_{\mathfrak{p}_x})$ has $\beta^{-1}(\mathfrak{m}) = \mathfrak{p}_x$, so the morphism $g_V\colon \operatorname{Spec} A \to V \subseteq X$ it induces sends a to x, and by the opening paragraph its stalk map at a is the map induced by β, which is φ. If $V' \subseteq V$ is an affine open containing x, with $V' \cong \operatorname{Spec} B'$, then the restriction $B \to B'$ followed by the germ map $B' \to \mathcal{O}_{X,x}$ is the germ map of B, so $g_{V'}$, viewed as a morphism into V, is induced by the same ring map $B \to A$ as g_V, and $g_{V'} = g_V$ (theorem 3.1.4). Two affine opens containing x contain a third, as their intersection is a scheme (proposition 2.2.47), so g_V does not depend on V, and the two constructions are mutually inverse.

1. With $A = K$, whose spectrum is the single point of U, a morphism $U \to X$ amounts to a point x_1 and a local homomorphism $\varphi\colon \mathcal{O}_{X,x_1} \to K$. Being local, φ has kernel $\varphi^{-1}((0)) = \mathfrak{m}_{x_1}$, so it factors through an injection $\kappa(x_1) \to K$, and every field embedding $\kappa(x_1) \to K$ composed with the residue map is a local homomorphism. The residue map is

surjective, so these correspondences are mutually inverse.

2. *From a morphism to the data.* Let $g\colon T \to X$ and $x_0 = g(t_0)$, and fix an affine open $V \cong \operatorname{Spec} B$ containing x_0, through which g factors via a ring map $\beta\colon B \to R$ as above. Then $\mathfrak{p}_0 = \mathfrak{p}_{x_0} = \beta^{-1}(\mathfrak{m}_R)$, and $\mathfrak{p}_1 = \beta^{-1}((0))$ is a prime contained in $\mathfrak{p}_0$, so $x_1 = [\mathfrak{p}_1] = g(t_1)$ satisfies $x_0 \in V(\mathfrak{p}_1) \subseteq \overline{\{x_1\}}$. The morphism $g \circ \iota$ is induced by $B \to R \subseteq K$, whose kernel is $\mathfrak{p}_1$, and by the opening paragraph its stalk map $B_{\mathfrak{p}_1} \to K$ sends b/c to $\beta(b)/\beta(c)$, so the inclusion $\kappa(x_1) \subseteq K$ that (1) attaches to it sends the class of b/c to $\beta(b)/\beta(c)$. The map $\mathcal{O}_{X,x_0} \to \mathcal{O}_{X,x_1}$ of germs is the localization map $B_{\mathfrak{p}_0} \to B_{\mathfrak{p}_1}$, so $\mathcal{O}_{X,x_0} \to \mathcal{O}_{x_0,x_1} \subseteq \kappa(x_1) \subseteq K$ sends b/c, with $c \notin \mathfrak{p}_0$, to $\beta(b)/\beta(c) = \varphi(b/c)$, where $\varphi = g_{t_0}^{\#}$ is the map induced by β. So $\mathcal{O}_{x_0,x_1}$, viewed in K, is $\varphi(B_{\mathfrak{p}_0}) \subseteq R$, a quotient of the local ring $B_{\mathfrak{p}_0}$ and hence local with maximal ideal $\varphi(\mathfrak{p}_0 B_{\mathfrak{p}_0})$. As φ is local, $\mathfrak{m}_R \cap \varphi(B_{\mathfrak{p}_0}) = \varphi(\varphi^{-1}(\mathfrak{m}_R)) = \varphi(\mathfrak{p}_0 B_{\mathfrak{p}_0})$, so R dominates $\mathcal{O}_{x_0,x_1}$.

   From the data to a morphism. Let $x_0 \in \overline{\{x_1\}}$ and $\kappa(x_1) \subseteq K$ with R dominating $\mathcal{O}_{x_0,x_1}$, and let $\varphi\colon \mathcal{O}_{X,x_0} \to R$ be the map onto $\mathcal{O}_{x_0,x_1} \subseteq R$. A surjection from a local ring onto a nonzero ring maps the maximal ideal onto the unique maximal ideal, so domination gives $\varphi^{-1}(\mathfrak{m}_R) = \mathfrak{m}_{x_0}$, and φ is local. The bijection above gives $g\colon T \to X$ with $g(t_0) = x_0$ and $g_{t_0}^{\#} = \varphi$. Take V and $\beta = \varphi \circ (B \to B_{\mathfrak{p}_0})$ as before. The open set V contains x_0, hence x_1, and $x_0 \in \overline{\{x_1\}} \cap V = V(\mathfrak{p}_1)$ for the prime $\mathfrak{p}_1$ of x_1, so $\mathfrak{p}_1 \subseteq \mathfrak{p}_0$. The kernel of φ is the kernel of $B_{\mathfrak{p}_0} \to B_{\mathfrak{p}_1} \to \kappa(x_1)$, which is $\mathfrak{p}_1 B_{\mathfrak{p}_0}$, and its contraction to B is $\mathfrak{p}_1$ ([10, Ideals of Localization], part (2)). Hence $g(t_1) = [\beta^{-1}((0))] = x_1$, and by the first half of this part the inclusion attached to $g \circ \iota$ sends the class of b/c to $\beta(b)/\beta(c)$, which is the given inclusion applied to that class. So the first half, applied to g, returns the data we started with. Conversely, the first half shows that for any g the stalk map $g_{t_0}^{\#}$ is the map $\mathcal{O}_{X,x_0} \to \mathcal{O}_{x_0,x_1} \subseteq R$ built from the data of g, so by the bijection above the morphism built from the data of g is g, completing the proof.

To make the conditions of lemma 3.5.12 concrete, let us look at a geometric example where T is a curve germ and X is the affine plane.

Example 3.5.13: Visualizing Valuation Lifts

1. Let $X = \mathbb{A}_k^2 = \operatorname{Spec} k[x,y]$. Let $R = k[t]_{(t)}$ be the discrete valuation ring at the origin of the affine line. Then $K = k(t)$. Let $T = \operatorname{Spec} R = \{t_0, t_1\}$ where $t_0 = (t)$ is the closed point and $t_1 = (0)$ is the generic point.

   Consider the morphism $T \to X$ defined by the ring homomorphism:

   $$\varphi^{\#} : k[x,y] \to k[t]_{(t)}, \quad x \mapsto t, \quad y \mapsto t^2$$

   Geometrically, this represents a curve parameterized by t approaching the origin along the parabola $y = x^2$.

   (a) *Where the generic point maps ($U \to X$):* The generic point t_1 corresponds to the prime ideal $(0) \subset R$. Its image $x_1 \in X$ is the prime ideal $\mathfrak{p} = (\varphi^{\#})^{-1}((0))$. Since $y - x^2 \mapsto t^2 - t^2 = 0$, the ideal $(y - x^2)$ is in the kernel. In fact, the kernel is exactly $(y - x^2)$. Thus, x_1 is the *generic point of the parabola* $V(y - x^2)$. The inclusion of fields

is the map on function fields:

$$\kappa(x_1) = \operatorname{Frac}\left(\frac{k[x,y]}{(y-x^2)}\right) \xrightarrow{\cong} k(t) = K$$

(b) *Where the closed point maps:* The closed point t_0 corresponds to the maximal ideal $(t) \subset R$. Its image $x_0 \in X$ is $(\varphi^\#)^{-1}((t)) = (x,y)$, which is the origin.

(c) *Specialization:* Since $(y-x^2) \subseteq (x,y)$, the origin x_0 lies in $V(y-x^2) = \overline{\{x_1\}}$, the parabola, which is the required specialization condition $x_0 \in \overline{\{x_1\}}$.

(d) *Domination:* The local ring $\mathcal{O}_{x_0,x_1}$ of lemma 3.5.12, the image of $\mathcal{O}_{X,x_0} = k[x,y]_{(x,y)}$ in $\kappa(x_1)$, is the ring of functions on the parabola regular at the origin, which is isomorphic to $k[t]_{(t)}$. The map induced by $\varphi^\#$ is the identity $k[t]_{(t)} \to k[t]_{(t)}$, which is a local homomorphism (maps maximal ideal to maximal ideal), satisfying the domination condition.

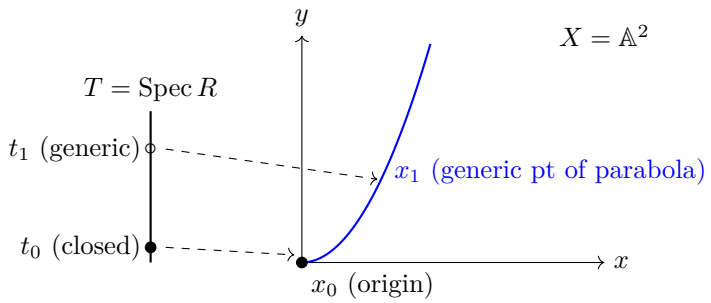


Figure 3.1: The generic point of the DVR maps to the generic point of the curve, while the closed point of the DVR maps to the limit point on the curve.

2. Let us look at a case where the generic point of T maps to the *generic point* of $X = \mathbb{A}^2_k$, rather than a curve inside it. This requires constructing a valuation on the function field $k(x,y)$.

   Let $K = k(x,y)$. We define a valuation $v : K^\times \to \mathbb{Z}$ by the "order of vanishing along the y-axis". Any rational function $f \in k(x,y)$ can be written as $f = x^n \frac{g}{h}$ where x does not divide g or h. We define $v(f) = n$.

   Let R be the valuation ring for v:

   $$R = \{f \in k(x,y) \,|\, v(f) \geq 0\} = k(x,y)_{(x)}$$

   This is the local ring of the affine plane localized at the prime ideal (x) (the generic point of the y-axis).

   Let us analyze the map $T = \operatorname{Spec} R \to X = \operatorname{Spec} k[x,y]$ induced by the natural inclusion $k[x,y] \subset R$.

   (a) *Generic Point:* The generic point t_1 of T (ideal (0) in R) maps to the ideal (0) in $k[x,y]$. Thus, $t_1 \mapsto \eta$, the generic point of the plane.

   (b) *Closed Point:* The closed point t_0 of T (maximal ideal xR) maps to the ideal $(x) \subset k[x,y]$. This is the *generic point of the line* $x = 0$.

(c) *Visualization:*

$$\begin{array}{ccc} U = \operatorname{Spec} k(x,y) & \longrightarrow & \eta \text{ (generic pt of } \mathbb{A}^2) \\ \downarrow & & \vdots \text{ specializes} \\ T = \operatorname{Spec} k(x,y)_{(x)} & \longrightarrow & \xi \text{ (generic pt of } y\text{-axis)} \end{array}$$

Even though t_0 is a "closed point" in the topological space T, it maps to a non-closed point in X.

(d) *Domination:* The stalk of X at the image point ξ is $\mathcal{O}_{X,\xi} = k[x,y]_{(x)}$. In this specific case, the valuation ring R is *isomorphic* to the stalk. This happens because we approached a codimension 1 point. If we approached the origin (codimension 2), R would be a further localization/quotient of the stalk.

Lemma 3.5.14: Closure and Specialization

Let $f : X \to Y$ be a quasicompact morphism of schemes and $Z \subseteq X$ a closed subset. Then $f(Z) \subseteq Y$ is closed if and only if it is stable under specialization. In particular this applies to $f(X)$.

***Proof*:**

A closed set C is stable under specialization: if $y \in C$ then $\overline{\{y\}} \subseteq C$. Conversely, assume $f(Z)$ is stable under specialization and let $y \in \overline{f(Z)}$. Choose an affine open $V \ni y$, with $V \cong \operatorname{Spec} B$, and let $\mathfrak{p}_y \subseteq B$ be the prime of y.

Step 1: reduction to one affine piece. Since f is quasicompact, $f^{-1}(V)$ is a finite union of affine opens $X_1, \dots, X_m$ with $X_i \cong \operatorname{Spec} A_i$, and $f(Z) \cap V = \bigcup_i f(Z \cap X_i)$. Every open neighbourhood of y inside V meets $f(Z)$, hence meets $f(Z) \cap V$, so y lies in the closure of $f(Z) \cap V$ in V. The closure of a finite union is the union of the closures, so y lies in the closure in V of $f(Z \cap X_i)$ for some i. The morphism $f|_{X_i} \colon X_i \to V$ (lemma 2.2.52) is induced by a ring homomorphism $\varphi \colon B \to A_i$ (theorem 3.1.4), and the closed subset $Z \cap X_i$ of X_i is $V(\mathfrak{a})$ for an ideal $\mathfrak{a} \subseteq A_i$. Let $A' = A_i/\mathfrak{a}$ and $\psi \colon B \to A'$ the composite of φ with the quotient map. By lemma 2.1.5 and lemma 2.1.28, $f(Z \cap X_i) = \psi^a(\operatorname{Spec} A')$.

Step 2: the closure of the image. If $A' = 0$, the image is empty, which is impossible. Otherwise $I(\psi^a(\operatorname{Spec} A')) = \bigcap_{\mathfrak{q}} \psi^{-1}(\mathfrak{q}) = \psi^{-1}(\operatorname{Nil}(A'))$ by [10, The Radical as an Intersection of Primes], and $\psi^{-1}(\operatorname{Nil}(A')) = \sqrt{\ker \psi}$. By proposition 2.1.6(5), the closure of $\psi^a(\operatorname{Spec} A')$ is $V(\sqrt{\ker \psi}) = V(\ker \psi)$. So $\ker \psi \subseteq \mathfrak{p}_y$.

Step 3: a point of the image specializing to y. By Zorn's lemma there is a prime $\mathfrak{p}'$ with $\ker \psi \subseteq \mathfrak{p}' \subseteq \mathfrak{p}_y$, minimal among primes containing $\ker \psi$: the intersection of a chain of such primes is again one. Let $C = B/\ker \psi$, so that ψ induces an injection $\overline{\psi} \colon C \to A'$, and let $\mathfrak{p}'' = \mathfrak{p}'/\ker \psi$, a minimal prime of C. By [10, Ideals of Localization], part (3), the primes of $C_{\mathfrak{p}''}$ correspond to the primes of C contained in $\mathfrak{p}''$, so $C_{\mathfrak{p}''}$ has the single prime $\mathfrak{p}'' C_{\mathfrak{p}''}$. Let $D = \overline{\psi}(C \setminus \mathfrak{p}'')$. Localization is exact ([10, Localization Is Exact]), so $\overline{\psi}$ induces an injection $C_{\mathfrak{p}''} \to D^{-1}A'$. As

$C_{\mathfrak{p}''} \neq 0$, the ring $D^{-1}A'$ is nonzero and has a prime $\mathfrak{Q}$ (proposition 2.1.6(5) with the proper ideal (0)). Its preimage in $C_{\mathfrak{p}''}$ is prime, hence equals $\mathfrak{p}''C_{\mathfrak{p}''}$. Let $\mathfrak{q}$ be the preimage of $\mathfrak{Q}$ in A'. The two composites $C \to C_{\mathfrak{p}''} \to D^{-1}A'$ and $C \to A' \to D^{-1}A'$ agree, so $\overline{\psi}^{-1}(\mathfrak{q})$ is the preimage of $\mathfrak{p}''C_{\mathfrak{p}''}$ in C, which is $\mathfrak{p}''$. Hence $\psi^{-1}(\mathfrak{q}) = \mathfrak{p}'$, and $[\mathfrak{p}'] = \psi^a(\mathfrak{q}) \in f(Z \cap X_i) \subseteq f(Z)$. Since $\mathfrak{p}' \subseteq \mathfrak{p}_y$, the point y lies in $V(\mathfrak{p}')$, the closure of $[\mathfrak{p}']$ in V, hence in its closure in Y. Stability under specialization gives $y \in f(Z)$, so $f(Z)$ is closed, as we sought to show.

Theorem 3.5.15: Valuation Criterion for Separatedness

Let $f : X \to Y$ be a morphism of schemes, and assume that X is Noetherian. Then f is separated if and only if for any field K, and for any valuation ring R with quotient field K, $T = \operatorname{Spec} R, U = \operatorname{Spec} K$, and the morphism $\iota : U \to T$ induced by the inclusion $R \subseteq K$, given a morphism of T to Y and a morphism of U to X making the following diagram commute:

$$\begin{array}{ccc} U & \longrightarrow & X \\ \downarrow{\scriptstyle \iota} & \nearrow & \downarrow{\scriptstyle f} \\ T & \longrightarrow & Y \end{array}$$

there is *at most one* morphism of T to X making the whole diagram commutative.

***Proof*:**
Let $P = X \times_Y X$ with projections p_1, p_2, let $\delta\colon X \to P$ be the diagonal, and let $t_0 = \mathfrak{m}_R$ and $t_1 = (0)$ be the closed and generic points of T. Since $p_1 \circ \delta = p_2 \circ \delta = \mathrm{id}_X$, every point $z \in \delta(X)$ has $p_1(z) = p_2(z)$.

Separated implies uniqueness. Let f be separated, and let $h, h'\colon T \to X$ both make the diagram commute, with $u\colon U \to X$ the top arrow. As $f \circ h = f \circ h'$, theorem 3.2.1 gives $L\colon T \to P$ with $p_1 \circ L = h$ and $p_2 \circ L = h'$, and $L \circ \iota = \delta \circ u$, both having components u. So $L(t_1) \in \delta(X)$, which is closed as δ is a closed immersion. Every prime of R contains (0), so $T = \overline{\{t_1\}}$ and $L(T) \subseteq \overline{\{L(t_1)\}} \subseteq \delta(X)$. Hence $h(t) = p_1(L(t)) = p_2(L(t)) = h'(t)$ for every $t \in T$. The restrictions $h \circ \iota = h' \circ \iota = u$ attach the same inclusion $\kappa(h(t_1)) \subseteq K$, so $h = h'$ by lemma 3.5.12(2).

Uniqueness implies that $\delta(X)$ is closed. Assume every diagram has at most one lift. Every open subset of the Noetherian scheme X is quasicompact (lemma 2.4.2), so δ is quasicompact, and by lemma 3.5.14 it suffices to show that $\delta(X)$ is stable under specialization. Let $\xi_1 = \delta(x)$ and $\xi_0 \in \overline{\{\xi_1\}}$, put $K = \kappa(\xi_1)$, and let $\mathcal{O}_{\xi_0,\xi_1} \subseteq K$ be the local ring of lemma 3.5.12(2). By [10, Domination and Existence of Valuation Rings] some valuation ring R of K dominates it, and lemma 3.5.12(2) gives $\varphi\colon T \to P$ with $\varphi(t_0) = \xi_0$, $\varphi(t_1) = \xi_1$ and attached inclusion id_K. Put $u_k = p_k \circ \varphi$ for $k = 1, 2$, so $f \circ u_1 = f \circ u_2$. By lemma 3.5.12(1), $u_k \circ \iota$ is determined by the point $p_k(\xi_1) = x$ and the residue field map $p_k^{\#}\colon \kappa(x) \to \kappa(\xi_1) = K$. Since $p_k \circ \delta = \mathrm{id}_X$, the composite $\kappa(x) \xrightarrow{p_k^{\#}} \kappa(\xi_1) \xrightarrow{\delta^{\#}} \kappa(x)$ is the identity, so the field map $\delta^{\#}$ is surjective, hence an isomorphism, and $p_k^{\#} = (\delta^{\#})^{-1}$ for $k = 1, 2$. Hence $u_1 \circ \iota = u_2 \circ \iota$, and u_1 and u_2 both lift the diagram with top arrow $u_1 \circ \iota$ and bottom arrow $f \circ u_1$. By hypothesis $u_1 = u_2 =: v$. The morphisms φ and $\delta \circ v$ have the same components $p_k \circ \varphi = v = p_k \circ \delta \circ v$, so $\varphi = \delta \circ v$ by the

uniqueness in theorem 3.2.1, and $\xi_0 = \varphi(t_0) \in \delta(X)$.

Closed image implies closed immersion. By proposition 3.5.1 there is an open $W \subseteq P$ containing $\delta(X)$ such that $\delta\colon X \to W$ is a closed immersion. So δ is a homeomorphism onto $\delta(X)$, which is closed in P. The sheaf map $\mathcal{O}_P \to \delta_*\mathcal{O}_X$ is surjective on stalks at the points of W, where it restricts to the sheaf map of $\delta\colon X \to W$, and at the points of the open set $P \setminus \delta(X)$, where $\delta_*\mathcal{O}_X$ has zero stalks. These two open sets cover P, so δ is a closed immersion (definition 3.1.22) and f is separated (definition 3.5.2), completing the proof.

Example 3.5.16: Valuation Criterion on $\mathbb{A}^1$

Let us demonstrate the mechanics of the valuation criterion by proving that the affine line $X = \mathbb{A}^1_k = \operatorname{Spec} k[t]$ is separated over $Y = \operatorname{Spec} k$.

Suppose we are given a valuation ring R with fraction field K, and the standard diagram:

$$\begin{array}{ccc} U = \operatorname{Spec} K & \xrightarrow{\alpha} & \operatorname{Spec} k[t] \\ {\scriptstyle\iota}\downarrow & \overset{\exists ?\tilde{\alpha}}{\nearrow} & \downarrow \\ T = \operatorname{Spec} R & \longrightarrow & \operatorname{Spec} k \end{array}$$

We translate this geometric data into algebraic data (reversing arrows):

1. The morphism $\alpha : \operatorname{Spec} K \to \operatorname{Spec} k[t]$ corresponds to a k-algebra homomorphism $\alpha^\# : k[t] \to K$. This homomorphism is completely determined by the image of the generator t. Let $\alpha^\#(t) = \lambda \in K$.
2. The inclusion $\iota : \operatorname{Spec} K \to \operatorname{Spec} R$ corresponds to the inclusion of the valuation ring into its fraction field $R \hookrightarrow K$.
3. A lift $\tilde{\alpha} : \operatorname{Spec} R \to \operatorname{Spec} k[t]$ corresponds to a homomorphism $\tilde{\alpha}^\# : k[t] \to R$. Let $\tilde{\alpha}^\#(t) = r \in R$.

For the diagram to commute, we require that the composition $k[t] \xrightarrow{\tilde{\alpha}^\#} R \hookrightarrow K$ equals the original map $\alpha^\#$. Evaluating at t, this means we must have $r = \lambda$ inside K.

Now we analyze the possibilities based on the element $\lambda \in K$:

- *Case 1:* $\lambda \in R$. Then we must define $\tilde{\alpha}^\#(t) = \lambda$. Since the map is determined on generators, this is the *unique* homomorphism making the diagram commute.
- *Case 2:* $\lambda \notin R$. Then there is no element $r \in R$ such that $r = \lambda$. Thus, no lift exists.

In either case, the number of lifts is *at most one* (it is either 1 or 0). Therefore, $\mathbb{A}^1_k$ is separated.

Contrast this with the "Line with two origins" (X). If we map $\operatorname{Spec} K$ to the generic point of X, and R is a DVR, we essentially have a map sending t to 0 in the residue field. In the algebraic formulation, the gluing data allows for two distinct maps $T \to X$ (one to each origin) that become identical when restricted to the generic point U. This violates the uniqueness condition.

Corollary 3.5.17: Properties of Separated Morphisms

Assume that all schemes are Noetherian in the following statements.

1. Open and closed immersions are separated.
2. A composition of two separated morphisms is separated.
3. Separated morphisms are stable under base extension.
4. If $f : X \to Y$ and $f' : X' \to Y'$ are separated morphisms of schemes over a base scheme S, then the product morphism $f \times f' : X \times_S X' \to Y \times_S Y'$ is also separated.
5. If $f : X \to Y$ and $g : Y \to Z$ are two morphisms and if $g \circ f$ is separated, then f is separated.
6. A morphism $f : X \to Y$ is separated if and only if Y can be covered by open subsets V_i such that $f^{-1}(V_i) \to V_i$ is separated for each i.

***Proof*:**
Every scheme in sight is Noetherian, as the statement assumes, including the fibered products in (3) and (4) and the open subschemes in (6), so theorem 3.5.15 applies to each morphism below. We call the commutative squares of that theorem *valuative diagrams* and write $t_0 = \mathfrak{m}_R$ for the closed point of T.

1. This is lemma 3.5.11(2).
2. Let $f\colon X \to Y$ and $g\colon Y \to Z$ be separated, and let $h, h'\colon T \to X$ lift a valuative diagram for $g \circ f$ with top arrow $u\colon U \to X$ and bottom arrow $w\colon T \to Z$. Then $f \circ h$ and $f \circ h'$ lift the valuative diagram for g with top arrow $f \circ u$ and bottom arrow w, so $f \circ h = f \circ h'$ by theorem 3.5.15. Now h and h' lift the valuative diagram for f with top arrow u and bottom arrow $f \circ h$, so $h = h'$. By theorem 3.5.15, $g \circ f$ is separated.
3. Let $f\colon X \to Y$ be separated, $Y' \to Y$ a morphism, $X' = X \times_Y Y'$ with projection $p\colon X' \to X$, and $f'\colon X' \to Y'$ the base change. Let $h, h'\colon T \to X'$ lift a valuative diagram for f' with top arrow u' and bottom arrow v. Then $p \circ h$ and $p \circ h'$ lift the valuative diagram for f with top arrow $p \circ u'$ and bottom arrow v followed by $Y' \to Y$, so $p \circ h = p \circ h'$ by theorem 3.5.15. Also $f' \circ h = v = f' \circ h'$, so $h = h'$ by the uniqueness in theorem 3.2.1. By theorem 3.5.15, f' is separated.
4. A morphism into $X \times_Y (Y \times_S X')$ amounts to morphisms into X and into X' that agree over S, the component into Y being forced, so by theorem 3.2.1 there is a canonical isomorphism $X \times_Y (Y \times_S X') \cong X \times_S X'$ under which the base change of f along the projection $Y \times_S X' \to Y$ is $f \times \mathrm{id}_{X'}$. In the same way $\mathrm{id}_Y \times f'\colon Y \times_S X' \to Y \times_S Y'$ is the base change of f' along $Y \times_S Y' \to Y'$. Both are separated by (3), and $f \times f' = (\mathrm{id}_Y \times f') \circ (f \times \mathrm{id}_{X'})$ is separated by (2).
5. Let $g \circ f$ be separated, and let $h, h'\colon T \to X$ lift a valuative diagram for f with top arrow u and bottom arrow w. They also lift the valuative diagram for $g \circ f$ with top arrow u and bottom arrow $g \circ w$, so $h = h'$ by theorem 3.5.15, and f is separated by the same theorem.

6. Let $f_i\colon f^{-1}(V_i) \to V_i$ be the restrictions. If f is separated, each f_i is the base change of f along $V_i \to Y$ (proposition 3.2.5(1)), hence separated by (3). Conversely, let every f_i be separated, and let $h, h'\colon T \to X$ lift a valuative diagram for f with top arrow u and bottom arrow w. Choose i with $w(t_0) \in V_i$. The only open subset of T containing t_0 is T, since an ideal not contained in $\mathfrak{m}_R$ is R, so $w(T) \subseteq V_i$, and h, h' and u have image in $f^{-1}(V_i)$ because $f \circ h = f \circ h' = w$ and $f \circ u = w \circ \iota$. By lemma 2.2.52 these four morphisms factor through V_i and $f^{-1}(V_i)$, and the factored morphisms form a valuative diagram for f_i with two lifts, the required equalities following from those for f because the open embeddings $V_i \to Y$ and $f^{-1}(V_i) \to X$ are monomorphisms. So the factored h and h' are equal by theorem 3.5.15, and $h = h'$. By the same theorem, f is separated, completing the proof.

3.5.18 Note on Noetherian Condition You have probably noticed that in order to apply the theorem, it is not necessary to assume that all the schemes mentioned in the corollary are Noetherian. In fact, even in the theorem itself, you can get by with assuming something less than X Noetherian (see EGA I, new ed., 5.5.4).
What Hartshorne says is that if a Noetherian hypothesis will make statements and proofs substantially simpler, then he takes the hypothesis, even though it may not be necessary. This is reasonable as most cases where the valuation citation is useful is when working with stacks which will satisfy these conditions.

Rational maps to Separated Schemes

here

Exercise 3.5.1

1. add exercise 4.5 of Hartshorne: important for Weil divisors to show characterization of irred. closed subscheme of (separated noethean integral) reg. in codim 1 scheme
2. Show that $X \cong \Delta(X)$. Show that in general, $\Delta(X)$ is a locally closed immersion.
3. Show that a scheme X over S is separated if and only if it is separated over $\mathbb{Z}$.
4. If $\varphi : A \to B$ is a ring homomorphism, then the corresponding morphism $\operatorname{Spec}(B) \to \operatorname{Spec}(A)$ is dominant if and only if $\ker \varphi$ is contained in the nilradical of A.

3.5.3 Universally Closed Morphisms

We now build up to the notion of properness. Proper schemes and morphisms capture the idea of compactness and "compact-preserving" maps. In topology, a map is often called *proper* if the preimage of any compact set is compact. In algebraic geometry, because the Zariski topology is so coarse (and affine schemes are almost always quasicompact), this definition is not strong enough.

Instead, we look for a property that ensures the *image* of a closed set is closed, and remains closed after any base change.

Definition 3.5.19: Universally Closed

Let $f : X \to Y$ be a morphism of schemes. Then f is *universally closed* if it is a closed map (images of closed sets are closed), and for any morphism $Y' \to Y$, the base change $f' : X \times_Y Y' \to Y'$ is also a closed map.

The intuition is that we want to avoid "missing points" at the boundary. Recall the example of $\mathbb{A}^1_k \to \operatorname{Spec} k$. This is closed, but base changing to $\mathbb{A}^1_k$ (projection of the plane to the line) maps the hyperbola $xy = 1$ to the open set $\mathbb{A}^1 \setminus \{0\}$. The "missing point" at infinity prevents the map from being universally closed.

Just as Separatedness was characterized by the *uniqueness* of lifts in the valuation criterion, Universally Closed morphisms are characterized by the *existence* of lifts.

Theorem 3.5.20: Valuation Criterion for Universally Closed Morphisms

Let $f : X \to Y$ be a quasicompact morphism of schemes. Then f is universally closed if and only if for any valuation ring R with fraction field K, and any diagram:

$$\begin{array}{ccc} U = \operatorname{Spec} K & \longrightarrow & X \\ \downarrow{\scriptstyle \iota} & \overset{\exists}{\nearrow} & \downarrow{\scriptstyle f} \\ T = \operatorname{Spec} R & \longrightarrow & Y \end{array}$$

there exists **at least one** morphism $T \to X$ making the diagram commute.

From the perspective of functors of points, the valuative criterion is naturally a statement about extending $\operatorname{Spec}(K)$-valued points to $\operatorname{Spec}(R)$-valued points.

***Proof*:**
Write $t_1 = (0)$ and $t_0 = \mathfrak{m}_R$ for the generic and closed points of $T = \operatorname{Spec} R$. By lemma 3.5.12, a morphism $T \to X$ is determined by the images x_0, x_1 of t_0, t_1 together with the inclusion $\kappa(x_1) \subseteq K$ that part (1) attaches to its restriction to U, and every admissible datum arises. For $x_0 \in \overline{\{x_1\}}$ we use the local ring $\mathcal{O}_{x_0,x_1} \subseteq \kappa(x_1)$ of that lemma, the image of $\mathcal{O}_{X,x_0}$ under the map of germs $\mathcal{O}_{X,x_0} \to \mathcal{O}_{X,x_1}$ followed by the residue map.

- *Valuation rings are maximal for domination.* If $\mathcal{O} \subseteq K$ is a local ring with $R \subseteq \mathcal{O}$ and $\mathfrak{m}_{\mathcal{O}} \cap R = \mathfrak{m}_R$, then $\mathcal{O} = R$. For $a \in \mathcal{O} \setminus R$ we would have $a^{-1} \in R$ ([10, Characterizing Valuation Ring]) with a^{-1} not a unit of R, so $a^{-1} \in \mathfrak{m}_R \subseteq \mathfrak{m}_{\mathcal{O}}$ ([10, Valuation Ring Properties]), and $1 = a \cdot a^{-1} \in \mathfrak{m}_{\mathcal{O}}$, a contradiction.
- *Stalk maps respect generization.* If $g\colon A \to B$ is a morphism of ringed spaces and $a_0 \in \overline{\{a_1\}}$, every open neighbourhood of a_0 contains a_1, so germs at a_0 map to germs at a_1, and the stalk maps of g at a_0 and a_1 commute with these generization maps, both composites sending the germ of s to the germ of $g^{\#}(s)$.

Existence of lifts implies universally closed. Suppose every diagram admits a lift. Let $Y' \to Y$ be a morphism, $X' = X \times_Y Y'$, with projections $f'\colon X' \to Y'$ and $p\colon X' \to X$. First, f' is quasicompact. An affine open $V' \subseteq Y'$ is quasicompact (lemma 2.1.20), so it is a finite union

of affine opens V_k' each mapping into an affine open $V_k \subseteq Y$. By proposition 3.2.5(1), applied twice, and the universal property of theorem 3.2.1, $f'^{-1}(V_k') = f^{-1}(V_k) \times_{V_k} V_k'$. As $f^{-1}(V_k)$ is a finite union of affine opens W, this is covered by the finitely many affine opens $W \times_{V_k} V_k'$ (the affine case of theorem 3.2.1), so $f'^{-1}(V')$ is quasicompact.

Now let $Z' \subseteq X'$ be closed. By lemma 3.5.14 it suffices to show that $f'(Z')$ is stable under specialization. Let $z_1 \in Z'$, $y_1 = f'(z_1)$ and $y_0 \in \overline{\{y_1\}}$. Put $K = \kappa(z_1)$ and $\mathcal{O} = \mathcal{O}_{y_0,y_1} \subseteq \kappa(y_1)$, a local ring (lemma 3.5.12). The residue field map of f' at z_1 embeds $\kappa(y_1)$ in $\kappa(z_1) = K$. By [10, Domination and Existence of Valuation Rings] some valuation ring R of K dominates $\mathcal{O}$. By lemma 3.5.12(2) there is $v\colon T \to Y'$ with $v(t_0) = y_0$, $v(t_1) = y_1$ and attached inclusion $\kappa(y_1) \subseteq K$. By lemma 3.5.12(1) there is $u'\colon U \to X'$ with image z_1 and attached map id_K. Then $f' \circ u'$ and $v \circ \iota$ both correspond to y_1 with the inclusion $\kappa(y_1) \subseteq K$, so they are equal. Put $u = p \circ u'$ and let w be v followed by $Y' \to Y$. Then $f \circ u = w \circ \iota$, and the hypothesis gives $h\colon T \to X$ with $h \circ \iota = u$ and $f \circ h = w$. By theorem 3.2.1, (h, v) defines $h'\colon T \to X'$ with $p \circ h' = h$ and $f' \circ h' = v$, and $h' \circ \iota = u'$, both having components u and $v \circ \iota$. So $h'(t_1) = z_1 \in Z'$, and $t_0 \in \overline{\{t_1\}}$ gives $h'(t_0) \in \overline{\{z_1\}} \subseteq Z'$, while $f'(h'(t_0)) = v(t_0) = y_0$. Hence $y_0 \in f'(Z')$, so $f'(Z')$ is closed and f' is a closed map. Taking $Y' = Y$, for which $X' \cong X$ with f' corresponding to f, shows that f is closed, so f is universally closed (definition 3.5.19).

Universally closed implies existence of lifts. Suppose f is universally closed and let $u\colon U \to X$ and $w\colon T \to Y$ satisfy $f \circ u = w \circ \iota$. Let $X_T = X \times_Y T$ with projections $p\colon X_T \to X$ and $q\colon X_T \to T$, a closed map by definition 3.5.19. The pair (u, ι) defines $u'\colon U \to X_T$. Let ξ_1 be its image, with attached inclusion $\kappa(\xi_1) \subseteq K$, and let $Z = \overline{\{\xi_1\}}$. The set $q(Z)$ is closed and contains $q(\xi_1) = t_1$, hence contains $\overline{\{t_1\}} = T$, so some $\xi_0 \in Z$ has $q(\xi_0) = t_0$. Let $\mathcal{O} = \mathcal{O}_{\xi_0,\xi_1} \subseteq \kappa(\xi_1) \subseteq K$. The stalk map of q at ξ_0 is a local homomorphism $R = \mathcal{O}_{T,t_0} \to \mathcal{O}_{X_T,\xi_0}$, and composing it with the surjection $\mathcal{O}_{X_T,\xi_0} \to \mathcal{O}$, which is local, gives a local homomorphism $R \to \mathcal{O}$. The stalk map of q at ξ_1 is a map $K = \mathcal{O}_{T,t_1} \to \mathcal{O}_{X_T,\xi_1}$ whose composite with $\mathcal{O}_{X_T,\xi_1} \to \kappa(\xi_1) \subseteq K$ is the stalk map of $q \circ u' = \iota$, which is id_K. By the second fact, applied to q with $t_0 \in \overline{\{t_1\}}$, the map $R \to \mathcal{O} \subseteq K$ is therefore the inclusion of R. So $\mathcal{O}$ is a local subring of K containing R with $\mathfrak{m}_{\mathcal{O}} \cap R = \mathfrak{m}_R$, and the first fact gives $\mathcal{O} = R$. In particular R dominates $\mathcal{O}$, and lemma 3.5.12(2) gives $h'\colon T \to X_T$ with $h'(t_0) = \xi_0$, $h'(t_1) = \xi_1$ and attached inclusion $\kappa(\xi_1) \subseteq K$, so $h' \circ \iota = u'$ by part (1). Put $h = p \circ h'$. Then $h \circ \iota = p \circ u' = u$. The morphisms $q \circ h'$ and id_T fix t_0 and t_1 and restrict on U to $q \circ u' = \iota$, so they are equal by lemma 3.5.12(2). Hence $f \circ h = f \circ p \circ h' = w \circ q \circ h' = w$, and h is the required lift, completing the proof.

3.5.4 Proper Morphisms

We now combine the concepts. We want a scheme (or morphism) that behaves like a compact Hausdorff space.

- Hausdorff iff Separated iff Uniqueness of limits.
- Compact iff Universally Closed + Finite Type iff Existence of limits.

Definition 3.5.21: Proper Morphism

A morphism $f : X \to Y$ is called *proper* if it satisfies three conditions:

1. Separated (Hausdorff-like).
2. Universally Closed (Image of closed is closed).
3. Finite Type (Finite covering / geometrically bounded).

A scheme X is said to be *proper* if the morphism $X \to \operatorname{Spec}(\mathbb{Z})$ is proper, and an R-scheme is proper if the map $X \to \operatorname{Spec}(R)$ is proper.

Theorem 3.5.22: Valuation Criterion for Properness

Let $f : X \to Y$ be a morphism of finite type, with X Noetherian. Then f is proper if and only if for every valuation ring R with fraction field K, and every diagram:

$$\begin{array}{ccc} U & \longrightarrow & X \\ {\scriptstyle \iota}\downarrow & \nearrow & \downarrow{\scriptstyle f} \\ T & \longrightarrow & Y \end{array}$$

there exists a *unique* lifting $T \to X$ making the diagram commute.

***Proof*:**
A morphism of finite type is quasicompact (definition 3.4.29). Since f is of finite type, definition 3.5.21 says that f is proper if and only if it is separated and universally closed. As X is Noetherian, theorem 3.5.15 shows that f is separated if and only if every diagram as in the statement admits at most one lifting $T \to X$. As f is quasicompact, theorem 3.5.20 shows that f is universally closed if and only if every such diagram admits at least one lifting. Both criteria quantify over the same diagrams, with R a valuation ring, K its fraction field, $T = \operatorname{Spec} R$, $U = \operatorname{Spec} K$ and ι induced by $R \subseteq K$. Hence f is proper if and only if every such diagram admits exactly one lifting, as claimed.

Proposition 3.5.23: Properties of Proper Morphisms

Assume all schemes are Noetherian.

1. A closed immersion is proper.
2. A composition of proper morphisms is proper.
3. Proper morphisms are stable under base change.
4. Products of proper morphisms are proper.
5. If $f : X \to Y$ and $g : Y \to Z$ are morphisms, g is separated, and $g \circ f$ is proper, then f is proper.
6. Properness is local on the base (target).

***Proof*:**
Every scheme in sight is Noetherian, as the statement assumes, including the fibered products in (3) and (4) and the open subschemes in (6). We call the diagrams of theorem 3.5.22 *valuative diagrams*.

- *(a) Finite type.* Closed immersions are of finite type: by proposition 3.1.24(1) the preimage of an affine open $\operatorname{Spec} C$ is an affine $\operatorname{Spec} C'$ with $C \to C'$ surjective, so C' is a finitely generated C-algebra and the preimage is quasicompact (lemma 2.1.20). If $f\colon X \to Y$ and $g\colon Y \to Z$ are of finite type, so is $g \circ f$: for an affine $\operatorname{Spec} C \subseteq Z$, $g^{-1}(\operatorname{Spec} C)$ is covered by finitely many affines $\operatorname{Spec} B$ with B finitely generated over C, each $f^{-1}(\operatorname{Spec} B)$ by finitely many affines $\operatorname{Spec} A$ with A finitely generated over B, hence over C (generators of B together with those of A), so $(g \circ f)^{-1}(\operatorname{Spec} C)$ is covered by finitely many such $\operatorname{Spec} A$ and definition 3.4.28, definition 3.4.29 apply. If f is of finite type and $Y' \to Y$ is a morphism, the base change $f'\colon X' = X \times_Y Y' \to Y'$ is of finite type. Cover Y' by affines $\operatorname{Spec} B$ each mapping into an affine $\operatorname{Spec} A \subseteq Y$. By proposition 3.2.5(1), applied twice, and the universal property of theorem 3.2.1, $f'^{-1}(\operatorname{Spec} B) = f^{-1}(\operatorname{Spec} A) \times_{\operatorname{Spec} A} \operatorname{Spec} B$, covered by the finitely many affines $\operatorname{Spec}(C_j \otimes_A B)$ where $f^{-1}(\operatorname{Spec} A) = \bigcup_j \operatorname{Spec} C_j$ with C_j finitely generated over A, and $C_j \otimes_A B$ is generated over B by the elements $c \otimes 1$ for generators c of C_j. So f' is of finite type by proposition 3.4.30.
- *(b) Open subsets.* Every open subset of a Noetherian scheme is quasicompact (lemma 2.4.2).

1. Let $i\colon Z \to X$ be a closed immersion. It is of finite type by (a). Every base change of i is a closed immersion (proposition 3.2.5(2)), hence a homeomorphism onto a closed subset and a closed map, so i is universally closed. It is separated by lemma 3.5.11(2). Hence i is proper (definition 3.5.21).
2. Let $f\colon X \to Y$ and $g\colon Y \to Z$ be proper. Then $g \circ f$ is of finite type by (a). Given a valuative diagram with $u\colon U \to X$ and $w\colon T \to Z$, the pair $(f \circ u, w)$ is a valuative diagram for g, with a unique lift $h_1\colon T \to Y$, and (u, h_1) is one for f, with a unique lift $h\colon T \to X$. Then h lifts the original diagram. If h' is another lift, $f \circ h'$ lifts the diagram for g, so $f \circ h' = h_1$, and then h' lifts the diagram for f, so $h' = h$. By theorem 3.5.22, $g \circ f$ is proper.

3. Let $f\colon X \to Y$ be proper, $Y' \to Y$ a morphism, and $f'\colon X' = X \times_Y Y' \to Y'$ with projection $p\colon X' \to X$. Then f' is of finite type by (a). Let $u'\colon U \to X'$ and $v\colon T \to Y'$ form a valuative diagram for f'. Then $p \circ u'$ and v followed by $Y' \to Y$ form one for f, with a unique lift $h\colon T \to X$. The pair (h, v) defines $h'\colon T \to X'$ (theorem 3.2.1) with $p \circ h' = h$ and $f' \circ h' = v$, and $h' \circ \iota = u'$ since both have components $p \circ u'$ and $v \circ \iota$. If h'' is another lift for f', then $p \circ h''$ lifts the diagram for f, so $p \circ h'' = h$, and $f' \circ h'' = v$, so $h'' = h'$ by the uniqueness in theorem 3.2.1. By theorem 3.5.22, f' is proper.

4. Let $f\colon X \to Y$ and $f'\colon X' \to Y'$ be proper morphisms of S-schemes. A morphism into $X \times_Y (Y \times_S X')$ amounts to morphisms into X and into X' that agree over S, the component into Y being forced, so by theorem 3.2.1 there is a canonical isomorphism $X \times_Y (Y \times_S X') \cong X \times_S X'$ under which the base change of f along the projection $Y \times_S X' \to Y$ is $f \times \mathrm{id}_{X'}$. In the same way $\mathrm{id}_Y \times f'\colon Y \times_S X' \to Y \times_S Y'$ is the base change of f' along $Y \times_S Y' \to Y'$. Both are proper by (3), and $f \times f' = (\mathrm{id}_Y \times f') \circ (f \times \mathrm{id}_{X'})$ is proper by (2).

5. Let g be separated and $g \circ f$ proper. First, f is of finite type. Cover Y by affines $\operatorname{Spec} B$ each mapping into an affine $\operatorname{Spec} C \subseteq Z$. For an affine $\operatorname{Spec} A \subseteq f^{-1}(\operatorname{Spec} B)$, the ring A is finitely generated over C (definition 3.4.28 for $g \circ f$), and the map $C \to A$ factors through B, so the same generators generate A over B. Hence f is locally of finite type, $f^{-1}(\operatorname{Spec} B)$ is quasicompact by (b), and f is of finite type by proposition 3.4.30. Now let $u\colon U \to X$ and $w\colon T \to Y$ form a valuative diagram for f. Then u and $g \circ w$ form one for $g \circ f$, with a unique lift h. The morphisms $f \circ h$ and w both lift the valuative diagram for g given by $f \circ u$ and $g \circ w$, so $f \circ h = w$ by theorem 3.5.15, g being separated and Y Noetherian. So h lifts the diagram for f, and any lift for f lifts the diagram for $g \circ f$, hence equals h. By theorem 3.5.22, f is proper.

6. Let $\{V_i\}$ be an open cover of Y and $f_i\colon f^{-1}(V_i) \to V_i$ the restrictions. If f is proper, each f_i is the base change of f along $V_i \to Y$ (proposition 3.2.5(1)), hence proper by (3). Conversely, let every f_i be proper. The affine opens of Y contained in some V_i cover Y, and f is of finite type over each of them because f_i is, so f is of finite type by proposition 3.4.30. Let $u\colon U \to X$ and $w\colon T \to Y$ form a valuative diagram for f, and choose i with $w(\mathfrak{m}_R) \in V_i$. The only open subset of T containing the closed point $\mathfrak{m}_R$ is T, since an ideal not contained in $\mathfrak{m}_R$ is R, so $w(T) \subseteq V_i$, and u has image in $f^{-1}(V_i)$ because $f \circ u = w \circ \iota$. By lemma 2.2.52, w and u factor through V_i and $f^{-1}(V_i)$, and the factored maps form a valuative diagram for f_i, the open embedding $V_i \to Y$ being a monomorphism. Its unique lift is a lift for f. Conversely, every lift for f has image in $f^{-1}(V_i)$, so it factors through $f^{-1}(V_i)$ by the same lemma and, the open embeddings being monomorphisms, gives a lift for f_i, hence equals the one already found. By theorem 3.5.22, f is proper, completing the proof.

The following two propositions give a large class of useful proper morphisms, and consequence proper schemes over Y:

Proposition 3.5.24: Finite Morphisms are Proper

Let $f : X \to Y$ be a finite morphism. Then f is proper.

***Proof*:**
The preimage of every affine open under f is affine (definition 3.4.15), so f is affine (definition 3.4.5), its diagonal is a closed immersion by proposition 3.5.1, and f is separated (definition 3.5.2). It is of finite type by proposition 3.4.32(1). The morphism f and all its base changes are finite (proposition 3.4.25(2)), hence closed maps (corollary 3.4.24), so f is universally closed (definition 3.5.19) and proper (definition 3.5.21), as claimed.

If the focus is put on morphisms between affine schemes (or slightly more generally affine morphisms), then properness reduces to finiteness:

Proposition 3.5.25: Affine Proper Morphisms are Finite

Let $f : X \to Y$ be an affine morphism. Then f is proper if and only if f is finite.

***Proof*:**
($\Leftarrow$) This follows immediately from proposition 3.5.24. A finite morphism is always affine and proper.

($\Rightarrow$) Suppose f is affine and proper. We must show it is finite. Since finiteness is local on the base, we may assume $Y = \operatorname{Spec} A$ is affine. Because f is affine, $X = \operatorname{Spec} B$ can also be assumed to be affine, and so f corresponds to a ring homomorphism $\varphi : A \to B$.

Since f is proper, it is of *finite type*. This means B is a finitely generated A-algebra.

To show finiteness, we must show that φ is *integral*. We use the Valuative Criterion for Universally Closed Morphisms (theorem 3.5.20). Let K be the fraction field of B (or any field containing B) and let $R \subset K$ be any valuation ring containing A (more precisely, containing $\varphi(A)$). Consider the diagram:

$$\begin{array}{ccc} \operatorname{Spec} K & \longrightarrow & \operatorname{Spec} B \\ \downarrow & & \downarrow \\ \operatorname{Spec} R & \longrightarrow & \operatorname{Spec} A \end{array}$$

Since f is universally closed, a lift $\operatorname{Spec} R \to \operatorname{Spec} B$ exists. Algebraically, this means the map $B \to K$ factors through R, i.e., $B \subseteq R$.

Thus, B is contained in every valuation ring of K that contains A. A standard theorem in commutative algebra states that the integral closure of A in K is the intersection of all valuation rings of K containing A, see lemma 5.4.13 for a slightly stronger proof. Therefore, B is integral over A.

We now have that B is a finitely generated A-algebra that is integral over A. A standard result in commutative algebra states that such a ring is a finitely generated A-module (see [10, Module-Finite versus Finite-Type-and-Integral]). Since B is a finite A-module, f is a finite morphism, completing the proof.

Example 3.5.26: Non-Proper Morphisms

1. $\mathbb{A}^1_k \to \operatorname{Spec} k$ is not proper. It is not universally closed (base change to $\mathbb{A}^1$ maps the hyperbola to an open set).

2. $\mathbb{A}^1_k \setminus \{0\} \to \mathbb{A}^1_k$ (open immersion). This is separated and finite type, but not closed (image is open).

3. The "Line with two origins" is not proper because it is not separated.

The Criteria over Discrete Valuation Rings

The criteria theorem 3.5.15, theorem 3.5.20 and theorem 3.5.22 quantify over every valuation ring of every field. For a morphism of finite type over a Noetherian base the discrete valuation rings suffice, and this is the form in which the criteria are applied: the properness of the Picard scheme in [5] is proved by extending K-points to R-points for discrete valuation rings R only. The algebraic input is [10, Discrete Valuation Rings Dominating a Noetherian Local Domain]: a Noetherian local domain that is not a field is dominated by a discrete valuation ring whose fraction field is any prescribed finitely generated extension of its own. Geometrically, every nontrivial specialization in a Noetherian scheme is traced out by the spectrum of a discrete valuation ring, and this replaces the domination argument for arbitrary valuation rings in the proofs of the criteria. The reduction for universal closedness needs one further ingredient, a test for universal closedness on the base changes along affine spaces alone, which holds for every quasicompact morphism.

Lemma 3.5.27: Finite Type over a Noetherian Base

Let Y be a Noetherian scheme and let $f : X \to Y$ and $f' : X' \to Y$ be morphisms of finite type.

1. X is Noetherian, every open subset of X is quasicompact, and every morphism from X to a scheme is quasicompact.

2. $X \times_Y X'$ is Noetherian and the morphism $X \times_Y X' \to Y$ is of finite type.

3. For every $n \geq 0$ the scheme $Y \times \mathbb{A}^n$ is Noetherian and $f \times \mathrm{id} : X \times \mathbb{A}^n \to Y \times \mathbb{A}^n$ is of finite type, where $\mathbb{A}^n = \mathbb{A}^n_{\mathbb{Z}}$ and the products are over $\operatorname{Spec} \mathbb{Z}$.

***Proof*:**
Step 1: An affine cover. By definition 2.4.1, Y is quasicompact and covered by affine opens with Noetherian rings, so $Y = \bigcup_{\alpha=1}^{r} \operatorname{Spec} A_\alpha$ with each A_α Noetherian. Since f is quasicompact (definition 3.4.29), $f^{-1}(\operatorname{Spec} A_\alpha)$ is quasicompact, and by definition 3.4.28 it is covered by affine opens $\operatorname{Spec} C$ with C a finitely generated A_α-algebra, so finitely many of them cover it:

$$f^{-1}(\operatorname{Spec} A_\alpha) = \bigcup_{\beta=1}^{s_\alpha} \operatorname{Spec} C_{\alpha\beta}$$

with each $C_{\alpha\beta}$ a finitely generated A_α-algebra, hence Noetherian by [10, Finitely Generated Algebras Are Noetherian]. In the same way $f'^{-1}(\operatorname{Spec} A_\alpha) = \bigcup_\gamma \operatorname{Spec} C'_{\alpha\gamma}$ with finitely many finitely generated A_α-algebras $C'_{\alpha\gamma}$.

Step 2: Finite unions of Noetherian affines. Let W be a scheme covered by finitely many affine opens $\operatorname{Spec} D_1, \dots, \operatorname{Spec} D_N$ with each D_k Noetherian. Then W is quasicompact, being a finite union of quasicompact opens, hence Noetherian by definition 2.4.1, and every open $V \subseteq W$ is quasicompact: V is the finite union of the opens $V \cap \operatorname{Spec} D_k$, and an open subset of $\operatorname{Spec} D_k$

has the form $\operatorname{Spec} D_k \setminus V(\mathfrak{a})$ for an ideal $\mathfrak{a} = (h_1, \dots, h_m)$, which is finitely generated because D_k is Noetherian, so that

$$\operatorname{Spec} D_k \setminus V(\mathfrak{a}) = D(h_1) \cup \cdots \cup D(h_m)$$

is a finite union of the affine, hence quasicompact, opens $D(h_\ell) \cong \operatorname{Spec}(D_k)_{h_\ell}$. Consequently every morphism $g : W \to S$ to a scheme is quasicompact, since $g^{-1}(V)$ is an open subset of W for every affine open $V \subseteq S$ (definition 3.4.1). Applied to $W = X$ with the cover of Step 1, this proves (1).

Step 3: Proof of (2). Write $P = X \times_Y X'$ with projections $p : P \to X$ and $p' : P \to X'$, and let $h : P \to Y$ be the composite. Fix α. By proposition 3.2.5(1) and the universal property of theorem 3.2.1, the open $h^{-1}(\operatorname{Spec} A_\alpha) = p^{-1}(f^{-1}(\operatorname{Spec} A_\alpha))$ is the fibered product $f^{-1}(\operatorname{Spec} A_\alpha) \times_{\operatorname{Spec} A_\alpha} f'^{-1}(\operatorname{Spec} A_\alpha)$, and inside it the opens $p^{-1}(\operatorname{Spec} C_{\alpha\beta}) \cap p'^{-1}(\operatorname{Spec} C'_{\alpha\gamma})$ are the fibered products

$$\operatorname{Spec} C_{\alpha\beta} \times_{\operatorname{Spec} A_\alpha} \operatorname{Spec} C'_{\alpha\gamma} = \operatorname{Spec}(C_{\alpha\beta} \otimes_{A_\alpha} C'_{\alpha\gamma})$$

(the affine case in the proof of theorem 3.2.1). These finitely many affine opens cover $h^{-1}(\operatorname{Spec} A_\alpha)$, because for z in it the point $p(z)$ lies in some $\operatorname{Spec} C_{\alpha\beta}$ and $p'(z)$ in some $\operatorname{Spec} C'_{\alpha\gamma}$. If $c_1, \dots, c_a$ generate $C_{\alpha\beta}$ and $c'_1, \dots, c'_b$ generate $C'_{\alpha\gamma}$ as A_α-algebras, then $c_i \otimes 1$ and $1 \otimes c'_j$ generate $C_{\alpha\beta} \otimes_{A_\alpha} C'_{\alpha\gamma}$, so these rings are finitely generated A_α-algebras and Noetherian by [10, Finitely Generated Algebras Are Noetherian]. Hence h is locally of finite type by definition 3.4.28 and proposition 3.4.30, and P is covered by finitely many affine opens with Noetherian rings, so by Step 2 it is Noetherian and h is quasicompact. Thus h is of finite type.

Step 4: Proof of (3). The preimage of $\operatorname{Spec} A_\alpha$ under the projection $Y \times \mathbb{A}^n \to Y$ is $\operatorname{Spec} A_\alpha \times \mathbb{A}^n = \operatorname{Spec} A_\alpha[x_1, \dots, x_n]$ (proposition 3.2.5(1) and the affine case of theorem 3.2.1), a finitely generated A_α-algebra and so Noetherian by [10, Finitely Generated Algebras Are Noetherian]. Hence $Y \times \mathbb{A}^n$ is Noetherian by Step 2. The preimage of $\operatorname{Spec} A_\alpha \times \mathbb{A}^n$ under $f \times \mathrm{id}$ is the preimage of $\operatorname{Spec} A_\alpha$ under $f \circ p_X = p_Y \circ (f \times \mathrm{id})$, namely $f^{-1}(\operatorname{Spec} A_\alpha) \times \mathbb{A}^n$, which is covered by the affine opens $\operatorname{Spec} C_{\alpha\beta} \times \mathbb{A}^n = \operatorname{Spec} C_{\alpha\beta}[x_1, \dots, x_n]$, and $C_{\alpha\beta}[x_1, \dots, x_n]$ is generated over $A_\alpha[x_1, \dots, x_n]$ by the generators of $C_{\alpha\beta}$ over A_α. Hence $f \times \mathrm{id}$ is locally of finite type by definition 3.4.28 and proposition 3.4.30, and it is quasicompact by Step 2 applied to $X \times \mathbb{A}^n$, which is covered by the finitely many Noetherian affines $\operatorname{Spec} C_{\alpha\beta}[x_1, \dots, x_n]$.

Lemma 3.5.28: Specializations Are Realized by Discrete Valuation Rings

Let Y be a Noetherian scheme, let $y \in Y$, and let $y_0 \in \overline{\{y\}}$ be a point different from y.

1. Let $\kappa(y) \subseteq L$ be a finitely generated extension of fields. Then there are a discrete valuation ring R with fraction field L and a morphism $\operatorname{Spec} R \to Y$ sending the generic point to y and the closed point to y_0, whose restriction to $\operatorname{Spec} L$ is the morphism $\operatorname{Spec} L \to Y$ determined by y and $\kappa(y) \subseteq L$ as in lemma 3.5.12(1).
2. Let $g : X \to Y$ be a morphism of finite type and let $x \in X$ with $g(x) = y$. Then $\kappa(x)$ is a finitely generated extension of $\kappa(y)$, so (1) applies with $L = \kappa(x)$, and the morphism $\operatorname{Spec} R \to Y$ it produces restricts on $\operatorname{Spec} \kappa(x)$ to the composite $\operatorname{Spec} \kappa(x) \to X \xrightarrow{g} Y$ of the natural morphism (definition 3.2.9) with g.

***Proof*:**

Step 1: An affine model of the specialization. By definition 2.4.1 choose an affine open $\operatorname{Spec} A' \subseteq Y$ containing y_0 with A' Noetherian. Since $y_0 \in \overline{\{y\}}$, every open neighborhood of y_0 contains y, so $y \in \operatorname{Spec} A'$. Let $\mathfrak{p}$ and $\mathfrak{q}$ be the primes of A' corresponding to y and y_0. Then $\mathfrak{p} \subseteq \mathfrak{q}$, because y_0 lies in the closure $V(\mathfrak{p})$ of $\{y\}$, and $\mathfrak{p} \neq \mathfrak{q}$. The ring $B = A'/\mathfrak{p}$ is a Noetherian domain with $\operatorname{Frac} B = \kappa(\mathfrak{p}) = \kappa(y)$, and $\mathfrak{q}/\mathfrak{p}$ is a nonzero prime of B. Hence $R_0 = B_{\mathfrak{q}/\mathfrak{p}}$ is a Noetherian local domain with fraction field $\kappa(y)$ whose maximal ideal $(\mathfrak{q}/\mathfrak{p})R_0$ is nonzero, so R_0 is not a field.

Step 2: The discrete valuation ring. Regard $R_0 \subseteq \kappa(y) \subseteq L$. Since L is a finitely generated extension of $\operatorname{Frac} R_0 = \kappa(y)$, [10, Discrete Valuation Rings Dominating a Noetherian Local Domain] provides a discrete valuation ring $R \subseteq L$ with fraction field L dominating R_0, that is, $R_0 \subseteq R$ and $\mathfrak{m}_R \cap R_0 = (\mathfrak{q}/\mathfrak{p})R_0$. Let $t_1 = (0)$ and $t_0 = \mathfrak{m}_R$ be the generic and closed points of $\operatorname{Spec} R$, and let $\operatorname{Spec} R \to Y$ be the composite

$$\operatorname{Spec} R \to \operatorname{Spec} R_0 \to \operatorname{Spec} B \to \operatorname{Spec} A' \subseteq Y$$

of the morphisms induced by $A' \to B \to R_0 \to R$. Since $R_0 \to R$ is an injection of domains, t_1 maps to $(0) \in \operatorname{Spec} R_0$, then to $(0) \in \operatorname{Spec} B$, then to $\mathfrak{p}$, so $t_1 \mapsto y$. By domination, $t_0 = \mathfrak{m}_R$ maps to $\mathfrak{m}_R \cap R_0 = (\mathfrak{q}/\mathfrak{p})R_0$, then to $\mathfrak{q}/\mathfrak{p}$, then to $\mathfrak{q}$, so $t_0 \mapsto y_0$. The restriction to $\operatorname{Spec} L$ sends the unique point to y and induces on residue fields the composite $\kappa(y) = \operatorname{Frac} B \to \operatorname{Frac} R_0 \to \operatorname{Frac} R = L$, which is the given inclusion $\kappa(y) \subseteq L$, so by lemma 3.5.12(1) this restriction is the morphism determined by y and $\kappa(y) \subseteq L$. This proves (1).

Step 3: Residue fields along a morphism of finite type. Let g and x be as in (2). Choose an affine open $\operatorname{Spec} A' \subseteq Y$ containing y and an affine open $\operatorname{Spec} C \subseteq g^{-1}(\operatorname{Spec} A')$ containing x. By definition 3.4.28, C is a finitely generated A'-algebra, say generated by $c_1, \ldots, c_n$. Let $\mathfrak{P} \subseteq C$ be the prime of x, which lies over the prime $\mathfrak{p}$ of y. Then $\kappa(x) = \operatorname{Frac}(C/\mathfrak{P})$, and $C/\mathfrak{P}$ is generated as an $A'/\mathfrak{p}$-algebra by the classes $\bar{c}_1, \ldots, \bar{c}_n$, so every element of $\kappa(x)$ is a quotient of two polynomial expressions in the $\bar{c}_i$ with coefficients in $A'/\mathfrak{p}$. Hence $\kappa(x)$ is generated as a field over $\operatorname{Frac}(A'/\mathfrak{p}) = \kappa(y)$ by $\bar{c}_1, \ldots, \bar{c}_n$, where $\kappa(y) \to \kappa(x)$ is the map induced by g. Apply (1) with $L = \kappa(x)$ and this inclusion. The composite $\operatorname{Spec} \kappa(x) \to X \to Y$ sends the unique point to y and induces on residue fields the same map $\kappa(y) \to \kappa(x)$, so by lemma 3.5.12(1) it is the restriction to $\operatorname{Spec} \kappa(x)$ of the morphism $\operatorname{Spec} R \to Y$ constructed in (1).

Theorem 3.5.29: Separatedness over Discrete Valuation Rings

Let Y be a Noetherian scheme and let $f : X \to Y$ be a morphism of finite type. Then f is separated if and only if for every discrete valuation ring R with fraction field K, with $T = \operatorname{Spec} R$, $U = \operatorname{Spec} K$ and $\iota : U \to T$ induced by $R \subseteq K$, and every commutative diagram

$$\begin{array}{ccc} U & \longrightarrow & X \\ {\scriptstyle \iota}\downarrow & \nearrow & \downarrow{\scriptstyle f} \\ T & \longrightarrow & Y \end{array}$$

there is at most one morphism $T \to X$ making the whole diagram commute.

***Proof*:**
Step 1: Separated implies uniqueness. By lemma 3.5.27(1), X is Noetherian, so theorem 3.5.15 applies to f and gives uniqueness of the lift for every valuation ring, in particular for every discrete valuation ring.

Step 2: Uniqueness implies that the diagonal has closed image. Assume lifts from discrete valuation rings are unique. Let $P = X \times_Y X$ with projections $p_1, p_2 : P \to X$ and structure morphism $h = f \circ p_1 = f \circ p_2 : P \to Y$, and let $\Delta : X \to P$ be the diagonal. By lemma 3.5.27(2), P is Noetherian, and by lemma 3.5.27(1) the morphism Δ is quasicompact, so by lemma 3.5.14 it suffices to show that $\Delta(X)$ is stable under specialization. Let $z = \Delta(x)$ with $x \in X$ and let $z_0 \in \overline{\{z\}}$. If $z_0 = z$ there is nothing to show, so assume $z_0 \neq z$. Lemma 3.5.28(1), applied to the Noetherian scheme P, the points z, z_0 and $L = \kappa(z)$, gives a discrete valuation ring R with fraction field $K = \kappa(z)$ and a morphism $\varphi : T = \operatorname{Spec} R \to P$ with $\varphi(t_1) = z$ and $\varphi(t_0) = z_0$, whose restriction $\varphi \circ \iota$ is the natural morphism $j : \operatorname{Spec} \kappa(z) \to P$. Put $u_i = p_i \circ \varphi : T \to X$ for $i = 1, 2$. Both satisfy $f \circ u_i = h \circ \varphi$. Their restrictions to U are $p_i \circ j$, which by lemma 3.5.12(1) is determined by the point $p_i(z) = p_i(\Delta(x)) = x$ and the map $p_i^{\#} : \kappa(x) \to \kappa(z)$ induced on residue fields. Since $p_i \circ \Delta = \mathrm{id}_X$, the composite $\kappa(x) \xrightarrow{p_i^{\#}} \kappa(z) \xrightarrow{\Delta^{\#}} \kappa(x)$ is the identity, so $\Delta^{\#}$ is surjective, and being a homomorphism of fields it is an isomorphism with $p_i^{\#} = (\Delta^{\#})^{-1}$ for both i. Hence $u_1 \circ \iota = u_2 \circ \iota$, and u_1, u_2 are two lifts in the diagram with top arrow $u_1 \circ \iota : U \to X$ and bottom arrow $h \circ \varphi : T \to Y$. By hypothesis $u_1 = u_2 =: u$. The morphisms φ and $\Delta \circ u$ into P have the same components $p_i \circ \varphi = u = p_i \circ \Delta \circ u$, so $\varphi = \Delta \circ u$ by the uniqueness in theorem 3.2.1, and $z_0 = \varphi(t_0) = \Delta(u(t_0)) \in \Delta(X)$.

Step 3: Closed image implies closed immersion. By proposition 3.5.1 and its proof there is an open $W \subseteq P$ containing $\Delta(X)$ such that $\Delta : X \to W$ is a closed immersion (the proof constructs W as $\bigcup_{i,j} U_{ij} \times_{V_i} U_{ij}$). Since $\Delta(X)$ is closed in P by Step 2, Δ is a homeomorphism of X onto a closed subset of P, and $\mathcal{O}_P \to \Delta_* \mathcal{O}_X$ is surjective on stalks: at the points of W because $\Delta : X \to W$ is a closed immersion, and at the points of the open set $P \setminus \Delta(X)$ because $\Delta_* \mathcal{O}_X$ vanishes there. Hence Δ is a closed immersion (definition 3.1.22) and f is separated (definition 3.5.2).

Lemma 3.5.30: Universal Closedness Is Detected by Affine Spaces

Let $f : X \to Y$ be a quasicompact morphism of schemes. Then f is universally closed if and only if for every $n \geq 0$ the morphism

$$f \times \mathrm{id} : X \times \mathbb{A}^n \to Y \times \mathbb{A}^n$$

is a closed map, where $\mathbb{A}^n = \mathbb{A}^n_{\mathbb{Z}}$ and the products are over $\operatorname{Spec} \mathbb{Z}$.

***Proof*:**
Step 1: The base change along an affine space. A morphism $Z \to X \times_Y (Y \times \mathbb{A}^n)$ amounts to morphisms $Z \to X$ and $Z \to Y \times \mathbb{A}^n$ with the same composite to Y, that is, to morphisms $Z \to X$ and $Z \to \mathbb{A}^n$, the component $Z \to Y$ being forced. By the universal property of theorem 3.2.1 there is a canonical isomorphism $X \times_Y (Y \times \mathbb{A}^n) \cong X \times \mathbb{A}^n$ under which the base change of f along the projection $Y \times \mathbb{A}^n \to Y$ becomes $f \times \mathrm{id}$. In particular, if f is universally closed then every $f \times \mathrm{id}$ is closed (definition 3.5.19).

Step 2: Reduction to an affine base and an affine base change. Conversely, assume every $f \times \mathrm{id}$ is closed, let $Y' \to Y$ be a morphism, $X' = X \times_Y Y'$, and $f' : X' \to Y'$ the projection. Closedness is local on the target: if $\{V_\lambda\}$ is an open cover of Y' such that each restriction $f'^{-1}(V_\lambda) \to V_\lambda$ is closed, then for $Z \subseteq X'$ closed the set $f'(Z) \cap V_\lambda = f'(Z \cap f'^{-1}(V_\lambda))$ is closed in V_λ for every λ, so $f'(Z)$ is closed. Cover Y by affine opens $\operatorname{Spec} A$ and the preimage of each $\operatorname{Spec} A$ in Y' by affine opens $\operatorname{Spec} B$. For such a pair let $X_A = f^{-1}(\operatorname{Spec} A)$ and let $f_A : X_A \to \operatorname{Spec} A$ be the restriction of f. By proposition 3.2.5(1), applied twice, and the universal property of theorem 3.2.1, the open $f'^{-1}(\operatorname{Spec} B)$ is $X \times_Y \operatorname{Spec} B = X_A \times_{\operatorname{Spec} A} \operatorname{Spec} B$, and the restriction of f' to it is the base change of f_A along $\operatorname{Spec} B \to \operatorname{Spec} A$. The morphism f_A is quasicompact, and $f_A \times \mathrm{id} : X_A \times \mathbb{A}^n \to \operatorname{Spec} A \times \mathbb{A}^n$ is the restriction of $f \times \mathrm{id}$ to the preimage $X_A \times \mathbb{A}^n$ of the open $\operatorname{Spec} A \times \mathbb{A}^n$ (proposition 3.2.5(1)), which is again a closed map: if Z is closed in $(f \times \mathrm{id})^{-1}(V)$ for an open V, then $Z = Z' \cap (f \times \mathrm{id})^{-1}(V)$ for a closed $Z' \subseteq X \times \mathbb{A}^n$, and $(f \times \mathrm{id})(Z) = (f \times \mathrm{id})(Z') \cap V$. Replacing f by f_A and $Y' \to Y$ by $\operatorname{Spec} B \to \operatorname{Spec} A$, we may assume $Y = \operatorname{Spec} A$ and $Y' = \operatorname{Spec} B$. Then X is quasicompact, so $X = \operatorname{Spec} C_1 \cup \cdots \cup \operatorname{Spec} C_m$ for finitely many affine opens, and $X \times \mathbb{A}^n \to \operatorname{Spec} A \times \mathbb{A}^n = \mathbb{A}^n_A$ is closed for every n.

Step 3: Reduction to an affine space of arbitrary dimension. Choose a set I and a surjection $A[x_i : i \in I] \twoheadrightarrow B$ of A-algebras, for instance $I = B$ with $x_b \mapsto b$. Write $\mathbb{A}^I_A = \operatorname{Spec} A[x_i : i \in I]$, and for a finite subset $J \subseteq I$ write $\mathbb{A}^J_A = \operatorname{Spec} A[x_j : j \in J]$, with the morphisms $p_J : \mathbb{A}^I_A \to \mathbb{A}^J_A$ and, for $J' \subseteq J$, $p_{J,J'} : \mathbb{A}^J_A \to \mathbb{A}^{J'}_A$ induced by the inclusions of polynomial rings, so that $p_{J'} = p_{J,J'} \circ p_J$. Set

$$X_I = X \times_{\operatorname{Spec} A} \mathbb{A}^I_A, \qquad X_J = X \times_{\operatorname{Spec} A} \mathbb{A}^J_A$$

with projections $\pi : X_I \to \mathbb{A}^I_A$ and $\pi_J : X_J \to \mathbb{A}^J_A$, and let $q_J = \mathrm{id} \times p_J : X_I \to X_J$ and $q_{J,J'} = \mathrm{id} \times p_{J,J'} : X_J \to X_{J'}$, so that $\pi_J \circ q_J = p_J \circ \pi$ and $q_{J'} = q_{J,J'} \circ q_J$. Each π_J is a closed map: $\mathbb{A}^J_A = \operatorname{Spec} A \times \mathbb{A}^{|J|}$ and, by Step 1 with $Y = \operatorname{Spec} A$, $X_J \cong X \times \mathbb{A}^{|J|}$ compatibly, so π_J is one of the maps $f \times \mathrm{id}$. The closed immersion $\iota : \operatorname{Spec} B \to \mathbb{A}^I_A$ (proposition 3.1.24(2)) has base change $\mathrm{id} \times \iota : X' = X \times_{\operatorname{Spec} A} \operatorname{Spec} B \to X_I$, a closed immersion by proposition 3.2.5(2), and $\pi \circ (\mathrm{id} \times \iota) = \iota \circ f'$. Let $Z \subseteq X'$ be closed. Since a closed immersion is a homeomorphism onto a closed subset (definition 3.1.22), $Z_I = (\mathrm{id} \times \iota)(Z)$ is closed in X_I, and $f'(Z) = \iota^{-1}(\iota(f'(Z))) = \iota^{-1}(\pi(Z_I))$ is closed in $\operatorname{Spec} B$ as soon as $\pi(Z_I)$ is closed in $\mathbb{A}^I_A$. It therefore suffices to prove that $\pi : X_I \to \mathbb{A}^I_A$ is a closed map.

Step 4: Open subsets of X_I come from finite stages. By proposition 3.2.5(1) and the affine case of theorem 3.2.1, X_I is covered by the affine opens $\operatorname{Spec} C_k[x_i : i \in I]$ for $k = 1, \ldots, m$, the preimages of the $\operatorname{Spec} C_k$ under the projection to X, and on them π and q_J are induced by the ring maps $A[x_i : i \in I] \to C_k[x_i : i \in I]$ and $C_k[x_j : j \in J] \to C_k[x_i : i \in I]$. Likewise X_J is covered by the $\operatorname{Spec} C_k[x_j : j \in J]$, and $q_J^{-1}(\operatorname{Spec} C_k[x_j : j \in J]) = \operatorname{Spec} C_k[x_i : i \in I]$ since both are the preimage of $\operatorname{Spec} C_k$ under the projection to X. Every open subset of X_I is a union of basic opens $D(h)$ with $h \in C_k[x_i : i \in I]$ for some k. Such an h involves only finitely many variables, so $h \in C_k[x_j : j \in J]$ for a finite $J \subseteq I$, and then $D(h) = q_J^{-1}(D_J(h))$, where $D_J(h) \subseteq \operatorname{Spec} C_k[x_j : j \in J] \subseteq X_J$ is the basic open of h, because the preimage of $D(h)$ under the map of spectra induced by a ring homomorphism φ is $D(\varphi(h))$. Consequently every open subset of X_I is a union of sets $q_J^{-1}(V)$ with $J \subseteq I$ finite and $V \subseteq X_J$ open. Moreover, finitely many such sets $q_{J_1}^{-1}(V_1), \ldots, q_{J_r}^{-1}(V_r)$ have union $q_J^{-1}(V)$ for $J = J_1 \cup \cdots \cup J_r$ and $V = \bigcup_\ell q_{J,J_\ell}^{-1}(V_\ell)$, because $q_{J_\ell} = q_{J,J_\ell} \circ q_J$.

Step 5: Fibers of π are quasicompact. Let $t \in \mathbb{A}^I_A$ and $F_t = \pi^{-1}(t)$. The intersection $F_t \cap \operatorname{Spec} C_k[x_i : i \in I]$ is the fiber over t of the morphism $\operatorname{Spec} C_k[x_i : i \in I] \to \mathbb{A}^I_A$, which by

definition 3.2.9 and proposition 3.2.10 is homeomorphic to

$$\operatorname{Spec}\left(C_k[x_i : i \in I] \otimes_{A[x_i : i \in I]} \kappa(t)\right) = \operatorname{Spec}\left(C_k \otimes_A \kappa(t)\right)$$

and hence is quasicompact. Thus F_t is a finite union of quasicompact sets and is quasicompact.

Step 6: π is closed. Let $Z \subseteq X_I$ be closed. For each finite $J \subseteq I$ the set $\pi_J(\overline{q_J(Z)})$ is closed in $\mathbb{A}_A^J$ because π_J is closed, so

$$\Phi = \bigcap_{J \subseteq I \text{ finite}} p_J^{-1}\left(\pi_J(\overline{q_J(Z)})\right)$$

is closed in $\mathbb{A}_A^I$, and we claim $\pi(Z) = \Phi$. If $t = \pi(z)$ with $z \in Z$, then $p_J(t) = \pi_J(q_J(z)) \in \pi_J(\overline{q_J(Z)})$ for every J, so $\pi(Z) \subseteq \Phi$. Conversely, let $t \in \Phi$ and suppose $F_t \cap Z = \emptyset$. Then F_t is contained in the open set $X_I \setminus Z$, which by Step 4 is a union of sets $q_J^{-1}(V)$. By the quasicompactness of F_t (Step 5) finitely many of them cover F_t, and by the last assertion of Step 4 their union is $q_J^{-1}(V)$ for a single finite $J \subseteq I$ and open $V \subseteq X_J$, so

$$F_t \subseteq q_J^{-1}(V) \subseteq X_I \setminus Z$$

holds. The second inclusion says $q_J(Z) \cap V = \emptyset$, hence $\overline{q_J(Z)} \cap V = \emptyset$ as V is open. Since $t \in \Phi$, there is $w \in \overline{q_J(Z)}$ with $\pi_J(w) = p_J(t) =: s$. A morphism into X_I over $\operatorname{Spec} A$ amounts to a morphism into X and one into $\mathbb{A}_A^I$, and the latter determines the one into $\mathbb{A}_A^J$, so $X_I = X_J \times_{\mathbb{A}_A^J} \mathbb{A}_A^I$ with projections q_J and π (theorem 3.2.1), and the points $w \in X_J$ and $t \in \mathbb{A}_A^I$ lie over the same point $s \in \mathbb{A}_A^J$. The ring $\kappa(w) \otimes_{\kappa(s)} \kappa(t)$ is nonzero, being a $\kappa(w)$-vector space with basis any $\kappa(s)$-basis of $\kappa(t)$, so it has a prime ideal $\mathfrak{r}$, with residue field E. The morphisms $\operatorname{Spec} E \to \operatorname{Spec} \kappa(w) \to X_J$ and $\operatorname{Spec} E \to \operatorname{Spec} \kappa(t) \to \mathbb{A}_A^I$ have the same composite to $\mathbb{A}_A^J$: by lemma 3.5.12(1) each is determined by the point s and a field map $\kappa(s) \to E$, and these are $a \mapsto \overline{a \otimes 1}$ and $a \mapsto \overline{1 \otimes a}$, which agree. By theorem 3.2.1 they induce a morphism $\operatorname{Spec} E \to X_I$, whose image point z' satisfies $q_J(z') = w$ and $\pi(z') = t$. Then $z' \in F_t \subseteq q_J^{-1}(V)$ gives $w = q_J(z') \in V$, contradicting $\overline{q_J(Z)} \cap V = \emptyset$. Hence $F_t \cap Z \neq \emptyset$, that is, $t \in \pi(Z)$. Therefore $\pi(Z) = \Phi$ is closed.

Theorem 3.5.31: Properness over Discrete Valuation Rings

Let Y be a Noetherian scheme and let $f : X \to Y$ be a morphism of finite type. Consider commutative diagrams

$$\begin{array}{ccc} U & \longrightarrow & X \\ {\scriptstyle\iota}\downarrow & \nearrow & \downarrow{\scriptstyle f} \\ T & \longrightarrow & Y \end{array}$$

in which R is a discrete valuation ring with fraction field K, $T = \operatorname{Spec} R$, $U = \operatorname{Spec} K$, and ι is induced by $R \subseteq K$.

1. f is universally closed if and only if every such diagram admits at least one morphism $T \to X$ making the whole diagram commute.
2. f is proper if and only if every such diagram admits exactly one morphism $T \to X$ making the whole diagram commute.

In particular, if f is separated, then f is proper if and only if every such diagram admits at least one such morphism.

***Proof*:**

Step 1: The directions that follow from the criteria for valuation rings. The morphism f is quasicompact (definition 3.4.29) and X is Noetherian (lemma 3.5.27(1)). If f is universally closed, theorem 3.5.20 gives at least one lift for every valuation ring, in particular for every discrete valuation ring. If f is proper, theorem 3.5.22 gives exactly one lift for every valuation ring. Conversely, if exactly one lift exists for every discrete valuation ring, then f is separated by theorem 3.5.29, and once (1) is proved it is universally closed, hence proper by definition 3.5.21. The final assertion is (1) together with definition 3.5.21. It remains to show that the existence of lifts for discrete valuation rings implies that f is universally closed.

Step 2: Existence passes to base changes. Assume that every diagram for f admits a lift. Let $Y' \to Y$ be any morphism, $X' = X \times_Y Y'$, with projections $f' : X' \to Y'$ and $q : X' \to X$. Let $u : U \to X'$ and $v : T \to Y'$ satisfy $f' \circ u = v \circ \iota$. Then $q \circ u : U \to X$ and the composite $T \xrightarrow{v} Y' \to Y$ form a diagram for f, because $f \circ q \circ u$ equals $f' \circ u$ followed by $Y' \to Y$, which is $v \circ \iota$ followed by $Y' \to Y$. By hypothesis there is $h : T \to X$ with $h \circ \iota = q \circ u$ and $f \circ h$ equal to v followed by $Y' \to Y$. The pair (h, v) defines $\tilde{h} : T \to X'$ with $q \circ \tilde{h} = h$ and $f' \circ \tilde{h} = v$, and $\tilde{h} \circ \iota = u$ because both have components $q \circ u$ and $f' \circ u$ (uniqueness in theorem 3.2.1). Hence every diagram for f' admits a lift.

Step 3: Existence implies closedness over a Noetherian base. Let Y' be a Noetherian scheme and let $g : X' \to Y'$ be a morphism of finite type such that every diagram for g with R a discrete valuation ring admits a lift. Let $Z \subseteq X'$ be closed. By lemma 3.5.27(1), g is quasicompact, so by lemma 3.5.14 it suffices to show that $g(Z)$ is stable under specialization. Let $x \in Z$, $y = g(x)$, and $y_0 \in \overline{\{y\}}$, where we may assume $y_0 \neq y$. By lemma 3.5.28(2) there are a discrete valuation ring R with fraction field $K = \kappa(x)$ and $v : T = \operatorname{Spec} R \to Y'$ with $v(t_1) = y$ and $v(t_0) = y_0$ for the generic point t_1 and the closed point t_0, such that $v \circ \iota = g \circ u$ for the natural morphism $u : U = \operatorname{Spec} \kappa(x) \to X'$, whose image is x. By hypothesis there is $h : T \to X'$ with $g \circ h = v$ and $h \circ \iota = u$. Then $h(t_1) = x$, and $x_0 = h(t_0)$ lies in $\overline{\{x\}} \subseteq Z$ by continuity, since $t_0 \in \overline{\{t_1\}}$, while $g(x_0) = v(t_0) = y_0$. Hence $y_0 \in g(Z)$, so $g(Z)$ is closed, and since Z was an arbitrary

closed subset, g is a closed map.

Step 4: Conclusion. Assume that every diagram for f admits a lift and fix $n \geq 0$. By lemma 3.5.27(3), $Y \times \mathbb{A}^n$ is Noetherian and $f \times \mathrm{id} : X \times \mathbb{A}^n \to Y \times \mathbb{A}^n$ is of finite type. A morphism into $X \times_Y (Y \times \mathbb{A}^n)$ amounts to a morphism into X and one into $\mathbb{A}^n$, so $X \times_Y (Y \times \mathbb{A}^n) = X \times \mathbb{A}^n$ with $f \times \mathrm{id}$ the base change of f along $Y \times \mathbb{A}^n \to Y$ (theorem 3.2.1), and by Step 2 every diagram for $f \times \mathrm{id}$ admits a lift. Step 3 shows that $f \times \mathrm{id}$ is a closed map. As this holds for every n and f is quasicompact, f is universally closed by lemma 3.5.30.

Over a base that is merely locally Noetherian the same statements hold: separatedness and universal closedness are local on the target, and a diagram with R a discrete valuation ring lives over any open of Y containing the image of the closed point of $\operatorname{Spec} R$, the generic point specializing to the closed point ([70, Tag 0207], [70, Tag 0208], [70, Tag 05JY]). Without a Noetherian hypothesis the input from [10, Discrete Valuation Rings Dominating a Noetherian Local Domain] is unavailable and the proofs above break down. Theorem 3.5.20 quantifies over arbitrary valuation rings for exactly this reason.

Projective Schemes and Consequences

Theorem 3.5.32: Projective Morphisms are Proper

A projective morphism between Noetherian schemes is proper. Here $f\colon X \to Y$ is *projective* if it factors as a closed immersion $X \to P$ followed by the projection $P \to Y$, where P is either $\mathbb{P}^n_Y = \mathbb{P}^n_{\mathbb{Z}} \times_{\operatorname{Spec}\mathbb{Z}} Y$ or, when $Y = \operatorname{Spec} A$ for a Noetherian ring A, the scheme $\mathbb{P}^n_A = \operatorname{Proj} A[x_0, \ldots, x_n]$ (definition 2.3.4) with its structure morphism to $\operatorname{Spec} A$. The theorem holds for both choices.

***Proof*:**

Step 1: $\mathbb{P}^n_A \to \operatorname{Spec} A$ is of finite type and separated. Let A be a Noetherian ring, $S = A[x_0, \ldots, x_n]$, and $q_A\colon \mathbb{P}^n_A \to \operatorname{Spec} A$ the structure morphism of definition 2.3.4. A homogeneous prime not containing $S_+ = (x_0, \ldots, x_n)$ misses some x_i, so the sets $U_i = D_+(x_i)$ cover $\mathbb{P}^n_A$, and $U_i \cong \operatorname{Spec} A_i$ with $A_i = S_{(x_i)}$ (theorem 2.3.3). By definition 2.3.4, the ring map of q_A on global sections followed by restriction to U_i is the inclusion $A \to A_i$ in degree 0, so the restriction of q_A to U_i is the morphism induced by $A \to A_i$ (proposition 3.1.12). The A-algebra A_i is generated by $x_0/x_i, \ldots, x_n/x_i$, since a fraction h/x_i^d with h a monomial of degree d is a product of d of them. Each U_i is quasicompact (lemma 2.1.20). So $\mathbb{P}^n_A$ is quasicompact and of finite type over A (definition 2.2.18), q_A is of finite type by proposition 3.4.30 applied to the cover of $\operatorname{Spec} A$ by itself, and q_A is separated by corollary 3.5.5. The rings A_i are Noetherian by [10, Finitely Generated Algebras Are Noetherian], so $\mathbb{P}^n_A$ is Noetherian (definition 2.4.1).

Step 2: lifts from valuation rings. Let R be a valuation ring with fraction field K, let $\iota\colon \operatorname{Spec} K \to \operatorname{Spec} R$ be induced by $R \subseteq K$, and let $u\colon \operatorname{Spec} K \to \mathbb{P}^n_A$ and $w\colon \operatorname{Spec} R \to \operatorname{Spec} A$ satisfy $q_A \circ u = w \circ \iota$. By proposition 3.1.12, w corresponds to a ring map $\omega\colon A \to R$. The image of the point of $\operatorname{Spec} K$ lies in some U_j, so u factors through the open subscheme $U_j \cong \operatorname{Spec} A_j$ (lemma 2.2.52) and corresponds to a ring map $\varphi_j\colon A_j \to K$, whose restriction to A is ω followed by $R \subseteq K$, because $q_A \circ u = w \circ \iota$. Put $b_k = \varphi_j(x_k/x_j)$, so $b_j = 1$. For nonzero $a, b \in K$ at least one of a/b and b/a lies in R ([10, Characterizing Valuation Ring]), and if b/a and c/b lie in R

then so does c/a. So among the finitely many nonzero b_k, a set containing b_j, there is some b_i with $b_k/b_i \in R$ for every nonzero b_k. The point $u(\operatorname{Spec} K)$ is the prime $\ker \varphi_j$ of A_j, which does not contain x_i/x_j, so it lies in $D(x_i/x_j) \subseteq U_j$, which is $U_i \cap U_j = D_+(x_ix_j) \cong \operatorname{Spec} S_{(x_ix_j)}$ (lemma 2.3.2). The restriction maps from A_i and from A_j to $S_{(x_ix_j)}$ send each degree-0 fraction to the same fraction (theorem 2.3.3). So u factors through $\operatorname{Spec} S_{(x_ix_j)}$ via a ring map ψ with $\varphi_j = \psi \circ (A_j \to S_{(x_ix_j)})$, and since $x_k/x_i = (x_k/x_j)(x_i/x_j)^{-1}$ in $S_{(x_ix_j)}$, we get $\psi(x_k/x_i) = b_k/b_i$. The ring map $\varphi_i \colon A_i \to K$ of u, viewed as a morphism into U_i, is ψ composed with $A_i \to S_{(x_ix_j)}$. Hence $\varphi_i(x_k/x_i) = b_k/b_i$, which lies in R by the choice of i (it is 0 when $b_k = 0$), and φ_i agrees with ω on A. As A_i is generated over A by the x_k/x_i, the map φ_i takes values in R, and the resulting ring map $\varphi_i' \colon A_i \to R$ defines $h \colon \operatorname{Spec} R \to U_i \subseteq \mathbb{P}^n_A$. By proposition 3.1.12, $h \circ \iota = u$, and $q_A \circ h = w$ because φ_i' restricts to ω on A.

Step 3: q_A is proper. The morphism q_A is of finite type, hence quasicompact (definition 3.4.29), and by Step 2 every diagram as in theorem 3.5.20 admits a lift, so q_A is universally closed. With Step 1, q_A is proper (definition 3.5.21). In particular $q_{\mathbb{Z}} \colon \mathbb{P}^n_{\mathbb{Z}} \to \operatorname{Spec} \mathbb{Z}$ is proper.

Step 4: the projection $p \colon P \to Y$ is proper. If $P = \mathbb{P}^n_A$ with $Y = \operatorname{Spec} A$, this is Step 3. Otherwise $P = \mathbb{P}^n_Y = \mathbb{P}^n_{\mathbb{Z}} \times_{\operatorname{Spec} \mathbb{Z}} Y$ and p is the base change of $q_{\mathbb{Z}}$ along $Y \to \operatorname{Spec} \mathbb{Z}$. Write $Y = \bigcup_{\beta=1}^{r} \operatorname{Spec} B_\beta$ with each B_β Noetherian (definition 2.4.1). By proposition 3.2.5(1) and the affine case of theorem 3.2.1, $\mathbb{P}^n_Y$ is covered by the finitely many affine opens $U_i \times_{\operatorname{Spec} \mathbb{Z}} \operatorname{Spec} B_\beta \cong \operatorname{Spec} (S^{\mathbb{Z}}_{(x_i)} \otimes_{\mathbb{Z}} B_\beta)$, where $S^{\mathbb{Z}} = \mathbb{Z}[x_0, \dots, x_n]$, and these rings are finitely generated B_β-algebras, hence Noetherian by [10, Finitely Generated Algebras Are Noetherian]. Each of these affine opens is quasicompact (lemma 2.1.20), so $\mathbb{P}^n_Y$ is Noetherian (definition 2.4.1). All of $\mathbb{P}^n_{\mathbb{Z}}$, $\operatorname{Spec} \mathbb{Z}$, Y and $\mathbb{P}^n_Y$ are Noetherian, so p is proper by proposition 3.5.23(3).

Step 5: conclusion. Let $f = p \circ i$ with $i \colon X \to P$ a closed immersion and $p \colon P \to Y$ the projection, for either choice of P. The schemes X and P are Noetherian, so i is proper by proposition 3.5.23(1), p is proper by Steps 3 and 4, and f is proper by proposition 3.5.23(2), completing the proof.

Over an affine Noetherian base $\operatorname{Spec} A$ the two choices of P are isomorphic over $\operatorname{Spec} A$: this is the case $\mathcal{S} = \mathbb{Z}[x_0, \dots, x_n]$ on $\operatorname{Spec} \mathbb{Z}$ of proposition 5.3.50. The proof above does not use this.

One of the most powerful consequences of properness is the behavior of global functions. For compact complex manifolds, Liouville's theorem states that holomorphic functions on a compact connected manifold are constant. We have the algebraic analogue:

Theorem 3.5.33: Global Sections of Proper Schemes

Let k be a field. Let X be a proper k-scheme that is *geometrically* reduced and *geometrically* connected. Then:

$$\Gamma(X, \mathcal{O}_X) = k$$

(i.e., the only global regular functions are constants). Over an algebraically closed field the two hypotheses are just reducedness and connectedness.

The adverb is not decoration. Take $k = \mathbb{R}$ and $X = \operatorname{Spec} \mathbb{C}$: this is proper over $\mathbb{R}$ (it is finite), reduced, and connected, yet $\Gamma(X, \mathcal{O}_X) = \mathbb{C} \neq \mathbb{R}$. What fails is that X has a point whose residue

field is larger than k, and $X_{\mathbb{C}} = \operatorname{Spec}(\mathbb{C} \otimes_{\mathbb{R}} \mathbb{C})$ is disconnected. Geometric connectedness is exactly the hypothesis that rules this out, by proposition 3.4.27 and the discussion around it.

***Proof*:**
Let $f \in \Gamma(X, \mathcal{O}_X)$ be a global section. By proposition 3.3.9, we get a scheme morphism (that we shall also denote as f)

$$f : X \to \mathbb{A}^1_k = \operatorname{Spec} k[t]$$

Consider the open immersion $\iota : \mathbb{A}^1_k \hookrightarrow \mathbb{P}^1_k$. Compose them to get $\varphi : X \to \mathbb{P}^1_k$. Since X is proper over k and $\mathbb{P}^1_k$ is separated over k, the image $\varphi(X)$ must be a *closed* subset of $\mathbb{P}^1_k$. However, f maps into the affine line, so the image $\varphi(X)$ does not contain the point at infinity, ∞. Thus, $\varphi(X)$ is a closed subset of $\mathbb{P}^1_k$ contained entirely in $\mathbb{A}^1_k$. The closed subsets of $\mathbb{P}^1_k$ are finite sets of points or the whole space. Since it misses ∞, it must be a finite set of points. Since X is connected, the image is a single closed point $\mathfrak{p} \in \mathbb{A}^1_k$.

It remains to see that the residue field at $\mathfrak{p}$ is k, and this is where the geometric hypotheses enter; without them $\mathfrak{p}$ may be a closed point of positive degree, as the example after the statement shows. Base change to $\overline{k}$. Formation of global sections commutes with the flat base change $\operatorname{Spec} \overline{k} \to \operatorname{Spec} k$, so $\Gamma(X, \mathcal{O}_X) \otimes_k \overline{k} = \Gamma(X_{\overline{k}}, \mathcal{O}_{X_{\overline{k}}})$, and it suffices to prove the claim for $X_{\overline{k}}$. That scheme is proper over $\overline{k}$, and reduced and connected because X was geometrically so. Running the argument above over $\overline{k}$, the image of f is a single closed point of $\mathbb{A}^1_{\overline{k}}$, whose residue field is $\overline{k}$ because $\overline{k}$ is algebraically closed. So f is the constant $\lambda \in \overline{k}$, giving $\Gamma(X_{\overline{k}}, \mathcal{O}) = \overline{k}$ and therefore $\dim_k \Gamma(X, \mathcal{O}_X) = 1$. As $k \subseteq \Gamma(X, \mathcal{O}_X)$, equality holds, completing the proof.

While Projective implies Proper, the converse is not true (though examples are pathological, like Hironaka's example). However, Chow's Lemma states that proper schemes are "birationally" projective.

Theorem 3.5.34: Chow's Lemma

Let $f : X \to S$ be a proper morphism of finite type over a Noetherian base S. Then there exists a scheme X' and a morphism $\pi : X' \to X$ such that:

1. π is a surjective projective morphism.
2. The composite $X' \to S$ is a projective morphism.
3. There exists a dense open $U \subseteq X$ such that $\pi^{-1}(U) \to U$ is an isomorphism.

***Proof*:**
(Sketch). The proof involves taking a finite cover of X by quasi-projective schemes U_i. We form the disjoint union $U = \coprod U_i$ and a map $U \to X$. We then take the closure of the graph of this map inside $X \times_S \mathbb{P}^N$ (for sufficiently large N). The resulting scheme X' dominates X and sits inside a projective space over S, making it projective. The dense open isomorphism comes from the fact that we started with a cover. See [36, Hartshorne II.4.10] for the complete derivation.

Exercise 3.5.2

1. Show that two copies of $\operatorname{Spec}(k[x_1, x_2, ...])$ minus the origin, glued at the origin, is not quasi-separated.
2. Prove that the intersection of two proper subschemes is proper.

3.6 Rational Maps

When working with [quasiprojective] varieties, we had the weaker notion of a rational function that was defined almost everywhere. A rational function was more flexible, allowing for some important classification results, the definition of Kodaira dimension, study singularities, more invariances by looking at the birational automorphisms, and some "local to global" principles. The following brings these concepts over to schemes

Definition 3.6.1: Rational Map

Let X, Y be schemes. Then a *rational map* $\pi : X \dashrightarrow Y$, is an equivalence relation:

$$(\alpha U \to Y) \sim (\beta : V \to Y) \qquad \text{if and only if} \qquad Z \subseteq U \cap V \text{ s.t. } \alpha|_Z = \beta|_Z$$

where U, V, Z are all dense open subsets[a].

If X, Y are S-schemes, then rational maps of S-schemes are defined similarly.

If the image of one of the representatives is dense, then the rational map is said to be *dominant* or *dominating*.

[a] When Y is separated, we will let $Z = U \cap V$

Note that a rational map is is an equivalence relation, not necessarily defined on a single domain.

Definition 3.6.2: Birational Map

A rational map $\pi : X \dashrightarrow Y$ is *birational* if it is dominant, and it has a dominant rational inverse. Two schemes X, Y are *birationally equivalent* if there is a birational map $\pi : X \dashrightarrow Y$.

Example 3.6.3: Rational Map

1. A scheme morphisms always gives a rational map, similarily to how all regular functions are special rational functions
2. The projection $\mathbb{P}^n_R \dashrightarrow \mathbb{P}^{n-1}_R$ projecting down a dimension is a rational map (recall the same reasoning for rational functions in [22, Rational Functions]

Lemma 3.6.4: Dominating Rational Map on Irreducible Schemes

Let $\pi : X \dashrightarrow Y$ be a rational map between irreducible schemes. Then π is dominating iff it sends the generic point of X to the generic point of Y.

***Proof*:**
Let (U, φ) represent π, so $U \subseteq X$ is a dense open subscheme and $\varphi : U \to Y$ is a morphism. Since X is irreducible, the dense open U contains the generic point η_X and is itself irreducible with that same generic point, so $U = \overline{\{\eta_X\}} \cap U$ and therefore $\varphi(U) \subseteq \overline{\{\varphi(\eta_X)\}}$. Taking closures in Y,

$$\overline{\varphi(U)} = \overline{\{\varphi(\eta_X)\}}$$

the inclusion $\supseteq$ holding because $\eta_X \in U$. Now π is dominating precisely when $\overline{\varphi(U)} = Y$, and by the display that happens precisely when $\overline{\{\varphi(\eta_X)\}} = Y$, which for the irreducible Y says exactly that $\varphi(\eta_X)$ is its generic point. The condition does not depend on the representative, since any two agree on a dense open subscheme, which again contains η_X, as we sought to show.

The correspondence between rational maps and field extensions of [22, Properties of Rational Maps, Birational Maps and Isomorphisms] has a scheme-theoretic half that holds for all integral schemes.

Proposition 3.6.5: Rational Maps of Integral Schemes Induce Function Field Maps

Let X and Y be integral schemes and let $\pi\colon X \dashrightarrow Y$ be a dominant rational map. Every representative of π carries η_X to η_Y, and its stalk map at η_X is a homomorphism of fields $\pi^\sharp\colon K(Y) \to K(X)$ that does not depend on the representative. If X and Y are k-schemes and π is a rational map of k-schemes, then $\pi^\sharp$ is a k-algebra homomorphism. For a dominant rational map $\rho\colon Y \dashrightarrow Z$ to an integral scheme Z, the composite $\rho \circ \pi$ is defined and dominant, and $(\rho \circ \pi)^\sharp = \pi^\sharp \circ \rho^\sharp$. The identity of X induces the identity of $K(X)$.

***Proof*:**
Let X, Y be integral and let $\pi : X \dashrightarrow Y$ be a dominating rational map, represented by $\varphi : U \to Y$. For an integral scheme the function field is the local ring at the generic point, so $K(X) = \mathcal{O}_{X,\eta_X} = \mathcal{O}_{U,\eta_X}$ and $K(Y) = \mathcal{O}_{Y,\eta_Y}$. By lemma 3.6.4 the morphism φ carries η_X to η_Y, so it induces a local homomorphism on stalks

$$\varphi^\sharp_{\eta_X} : \mathcal{O}_{Y,\eta_Y} \longrightarrow \mathcal{O}_{U,\eta_X}$$

that is, a ring homomorphism $K(Y) \to K(X)$, which is a k-algebra homomorphism when φ is a k-morphism, the stalk maps of a k-morphism being k-algebra homomorphisms. A homomorphism out of a field with $1 \mapsto 1$ is injective, so this is a field extension.

The construction does not depend on the representative: two representatives of π agree on a dense open subscheme, which contains η_X, so they induce the same map on the stalk there. It is contravariantly functorial: if $\rho : Y \dashrightarrow Z$ is a further dominating rational map, then $\rho \circ \pi$ is defined and dominating because the composite carries η_X to η_Y to η_Z, and the stalk maps compose in the opposite order. Identities go to identities, which is the assertion, as we sought to show.

Corollary 3.6.6: Birational Map on Irreducible Schemes

Let $\pi : X \dashrightarrow Y$ be a birational map between integral schemes. Then the induced map of function fields is an isomorphism.

***Proof*:**
By definition a birational map admits a rational inverse: there is $\psi : Y \dashrightarrow X$ with $\psi \circ \pi = \mathrm{id}_X$ and $\pi \circ \psi = \mathrm{id}_Y$ as rational maps. Both π and ψ are then dominating, since a composite that carries η_X to itself cannot lose the generic point at the middle stage, and lemma 3.6.4 applies. So proposition 3.6.5 gives field maps $\pi^\sharp : K(Y) \to K(X)$ and $\psi^\sharp : K(X) \to K(Y)$, and the functoriality recorded there turns the two identities of rational maps into

$$\pi^\sharp \circ \psi^\sharp = \mathrm{id}_{K(X)} \qquad \psi^\sharp \circ \pi^\sharp = \mathrm{id}_{K(Y)}$$

so each is an isomorphism inverse to the other, as we sought to show.

Every dominant rational map of integral schemes induces a homomorphism of function fields in the opposite direction (proposition 3.6.5). The converse, that every k-algebra homomorphism $K(Y) \to K(X)$ arises in this way, can fail when Y is not locally of finite type over k.

Example 3.6.7: Function Field Not Inducing Rational Map
Let k be a field. The integral schemes $\operatorname{Spec} k[x]$ and $\operatorname{Spec} k(x)$ have the same function field $k(x)$, and no rational map $\operatorname{Spec} k[x] \dashrightarrow \operatorname{Spec} k(x)$ of k-schemes induces the identity of $k(x)$. Every nonempty open subset of $\operatorname{Spec} k[x]$ contains a distinguished open $D(f) = \operatorname{Spec} k[x, 1/f]$ with $f \neq 0$ (corollary 2.1.13, proposition 2.2.13), so a representative of such a map restricts to a k-morphism $\operatorname{Spec} R \to \operatorname{Spec} k(x)$ with $R = k[x, 1/f]$, that is, a k-algebra homomorphism $\theta\colon k(x) \to R$ (theorem 3.1.4). Let $\mathfrak{m}$ be a maximal ideal of R. The field $R/\mathfrak{m}$ is a finitely generated k-algebra of Krull dimension 0, so it is algebraic over k (theorem 2.6.11). The composite $k(x) \to R \to R/\mathfrak{m}$ is injective, $k(x)$ being a field, so x would be algebraic over k, which it is not.

For a target locally of finite type the converse holds: a homomorphism out of a finitely generated k-algebra is determined by the images of finitely many generators, and finitely many elements of the function field of an integral scheme are regular on a common dense open subset.

Lemma 3.6.8: Homomorphisms of Function Fields Give Rational Maps

Let k be a field, and let X and Y be integral k-schemes with Y locally of finite type over k. For a dominant rational map $\pi\colon X \dashrightarrow Y$ of k-schemes, let $\pi^\sharp\colon K(Y) \to K(X)$ be the induced k-algebra homomorphism (proposition 3.6.5). Then $\pi \mapsto \pi^\sharp$ is a bijection from the set of dominant rational maps $X \dashrightarrow Y$ of k-schemes to the set of k-algebra homomorphisms $K(Y) \to K(X)$.

***Proof*:**
Write η_X and η_Y for the generic points. As Y is locally of finite type over k, it has a nonempty affine open $\operatorname{Spec} B$ with $B = k[b_1, \ldots, b_n]$ a finitely generated k-algebra (definition 2.2.18). Since Y is integral, B is a domain, the open set $\operatorname{Spec} B$ contains η_Y as its generic point, the prime (0), and $K(Y) = \mathcal{O}_{Y,\eta_Y} = B_{(0)} = \operatorname{Frac} B$ (corollary 2.4.25). Likewise, for every nonempty affine open $\operatorname{Spec} A \subseteq X$ the ring A is a domain, η_X is the prime (0) of A, and $A \subseteq \operatorname{Frac} A = K(X)$. If a morphism $\varphi\colon \operatorname{Spec} A \to \operatorname{Spec} B$ is given by a ring homomorphism $\alpha\colon B \to A$ (theorem 3.1.4) and carries η_X to η_Y, then its stalk map at η_X is the localization $B_{(0)} \to A_{(0)}$, $b/t \mapsto \alpha(b)/\alpha(t)$, whose restriction to B is α followed by the inclusion $A \subseteq K(X)$.

Step 1: injectivity. Let π and π' be dominant rational maps with $\pi^\sharp = \pi'^\sharp$, represented by $\varphi\colon U \to Y$ and $\varphi'\colon U' \to Y$. Both carry η_X to η_Y (lemma 3.6.4), so $\varphi^{-1}(\operatorname{Spec} B) \cap \varphi'^{-1}(\operatorname{Spec} B)$ is a nonempty open subset of $U \cap U'$, and it contains a nonempty affine open $W = \operatorname{Spec} A$. By lemma 2.2.52 and theorem 3.1.4, the restrictions of φ and φ' to W are morphisms $W \to \operatorname{Spec} B$ given by ring homomorphisms $\alpha, \alpha'\colon B \to A$. Their stalk maps at η_X are $\pi^\sharp$ and $\pi'^\sharp$ (proposition 3.6.5), so by the previous paragraph $\alpha(b) = \pi^\sharp(b) = \pi'^\sharp(b) = \alpha'(b)$ in $K(X)$ for every $b \in B$. Hence $\alpha = \alpha'$, and φ and φ' agree on W. The nonempty open subset W of the irreducible space X is dense (lemma 2.1.24(3)), so $\pi = \pi'$ (definition 3.6.1).

Step 2: surjectivity. Let $\sigma\colon K(Y) \to K(X)$ be a k-algebra homomorphism. It is injective, being a homomorphism of fields. Choose a nonempty affine open $\operatorname{Spec} A_0 \subseteq X$, so that $K(X) = \operatorname{Frac} A_0$, and write $\sigma(b_i) = a_i/s$ with $a_1, \ldots, a_n, s \in A_0$ and $s \neq 0$. The distinguished open $U = D(s) \subseteq \operatorname{Spec} A_0$ is the affine scheme $\operatorname{Spec} A$ with $A = (A_0)_s$ (proposition 2.2.13), and it is nonempty because it contains the prime (0). The k-subalgebra $A \subseteq K(X)$ contains $\sigma(b_1), \ldots, \sigma(b_n)$, hence contains $\sigma(B) = k[\sigma(b_1), \ldots, \sigma(b_n)]$, so σ restricts to an injective k-algebra homomorphism $\beta\colon B \to A$. Let $\varphi\colon U \to \operatorname{Spec} B \subseteq Y$ be the corresponding k-morphism (theorem 3.1.4). It is defined on the dense open subset U, so it represents a rational map $\pi\colon X \dashrightarrow Y$ of k-schemes. Since β is injective, $\varphi(U)$ is dense in $\operatorname{Spec} B$ (proposition 3.1.34), and $\operatorname{Spec} B$ is dense in Y, so π is dominant. The point $\varphi(\eta_X)$ is the prime $\beta^{-1}((0)) = (0)$ of B, that is, η_Y, so by the first paragraph $\pi^\sharp$ agrees with σ on B. Both are homomorphisms of fields out of $K(Y) = \operatorname{Frac} B$, so $\pi^\sharp(b/t) = \sigma(b)/\sigma(t) = \sigma(b/t)$ for $b, t \in B$ with $t \neq 0$, and $\pi^\sharp = \sigma$, completing the proof.

Proposition 3.6.9: Reduced Schemes and Birational Maps

Let X, Y be reduced schemes. Then X and Y are birational if and only if there exists a dense open subscheme $U \subseteq X$ and $V \subseteq Y$ such that

$$U \cong V$$

***Proof*:**
One direction is immediate from definition 3.6.1. If $U \subseteq X$ and $V \subseteq Y$ are dense open subschemes and $\theta : U \to V$ is an isomorphism, then $U \to V \hookrightarrow Y$ is a morphism defined on a dense open subscheme of X and so represents a rational map $\pi : X \dashrightarrow Y$. Its image is V, which is dense in Y, so π is dominant. The inverse isomorphism represents a dominant rational map $\psi : Y \dashrightarrow X$, and on U the composite $\psi \circ \pi$ is the inclusion $U \hookrightarrow X$ while on V the composite $\pi \circ \psi$ is the inclusion $V \hookrightarrow Y$. So ψ is a rational inverse for π and the two schemes are birational in the sense of definition 3.6.2.

For the converse let $\pi : X \dashrightarrow Y$ be birational with dominant rational inverse ψ, and fix representatives $F : X_1 \to Y$ of π and $G : Y_1 \to X$ of ψ, with $X_1 \subseteq X$ and $Y_1 \subseteq Y$ dense open. That $\psi \circ \pi$ is the identity rational map of X gives a dense open $W \subseteq X$ on which the composite is defined and agrees with the identity: $W \subseteq X_1$, $F(W) \subseteq Y_1$, and $G \circ F|_W$ is the inclusion $W \hookrightarrow X$. Symmetrically there is a dense open $W' \subseteq Y$ with $W' \subseteq Y_1$, $G(W') \subseteq X_1$, and $F \circ G|_{W'}$ the inclusion $W' \hookrightarrow Y$. Put

$$U = (F|_W)^{-1}(W') \qquad V = (G|_{W'})^{-1}(W)$$

which are open in X and in Y respectively.

Each of these two open subschemes is carried into the other. If $x \in U$ then $F(x) \in W'$ by the definition of U, and $G(F(x)) = x$ lies in W, so $F(x) \in V$. Hence $F(U) \subseteq V$, and exchanging the roles of the two maps gives $G(V) \subseteq U$. The morphisms $F|_U : U \to V$ and $G|_V : V \to U$ obtained this way satisfy

$$G|_V \circ F|_U = \mathrm{id}_U \qquad F|_U \circ G|_V = \mathrm{id}_V$$

since each is a restriction of one of the two identities of the previous paragraph. So $F|_U$ is an isomorphism of schemes onto V.

What remains is the density of U in X and of V in Y. Suppose $\alpha : U_0 \to Y$ and $\beta : V_0 \to Y$ represent the same rational map, agreeing on a dense open $Z \subseteq U_0 \cap V_0$. The closure of Z in U_0 is U_0, so continuity gives $\alpha(U_0) \subseteq \overline{\alpha(Z)} = \overline{\beta(Z)} \subseteq \overline{\beta(V_0)}$, and by symmetry $\overline{\alpha(U_0)} = \overline{\beta(V_0)}$. The closure of the image is therefore the same for every representative, and since π is dominant and $F|_W$ represents π, the set $F(W)$ is dense in Y. Moreover G restricts on $F(W) \subseteq Y_1$ to a continuous two-sided inverse of $F|_W$, so $F|_W$ is a homeomorphism of W onto $F(W)$ with its subspace topology. Now $F(W) \cap W'$ is the intersection of a dense subset of Y with a dense open subset of Y, hence is dense in Y, hence is dense in the subspace $F(W)$ that contains it. Its preimage under the homeomorphism is U, which is accordingly dense in W and so dense in X. The symmetric argument makes V dense in Y, as we sought to show.

The hypothesis of reducedness is not called on anywhere above. It is inherited from the setting in which rational maps are taken, since it is reducedness of the source that makes a morphism on a dense open subscheme determined by its restriction to a smaller one when the target is separated (proposition 3.5.10).

Vakil looks here at rational maps of irreducible varieties that look like translating the results over from varieties, so I will omit for now

I like his translation of the Pythagorean problem into a scheme problem

non-rational complex curve, cool presentation

3.7 Abstract Varieties

Let us use the tools we have developed to our usual notion of variety. We shall see that defining varieties abstractly (other common adverbs being inherently, integrally, intrinsically) shall give us many of the same benefits as defining schemes as spectrum. Most results about quasi-projective varieties from [22] readily translate to the abstract setting with a few exceptions that we shall go over.

There is a natural way in which we can embed (quasi-projective) varieties into the category $\mathbf{Sch}_{\bar{k}}$:

Theorem 3.7.1: Varieties and Schemes: Fully Faithful Functor

Let $\bar{k}$ be an algebraically closed field. Then there exists a fully faithful functor $\mathbf{Var}_{\bar{k}} \to \mathbf{Sch}_{\bar{k}}$ where $\mathbf{Var}_{\bar{k}}$ is the usual category of varieties (over $\bar{k}$) as defined in [22, Quasi-projective Variety].

Recall that a fully faithful means the hom-sets are in bijection, and it implies injectivity on objects (but not necessarily surjective, for example Ab $\to$ Grp is a fully faithful functor). We shall also stick with Hartshorne's notation for the functor, which is the letter t, it shall come up a few times later in the book.

***Proof*:**
For any (quasi-projective) variety, we shall first find the "points" that shall be the points of the scheme by working with the irreducible closed subsets. We shall then find the structure sheaf to put on it by finding all the regular functions, which shall be enough to define the functor.

First off, for any topological space X, and let $t(X)$ be the set of (nonempty) irreducible closed subsets of X. If Y is a closed subset of X, then $t(Y) \subseteq t(X), t(Y_1 \cup Y_2) = t(Y_1) \cup t(Y_2)$, and $t(\bigcap Y_i) = \bigcap t(Y_i)$. Hence, we can define a topology $t(X)$ by taking as closed sets the subsets of the form $t(Y)$, where Y is a closed subset of X. If $f : X_1 \to X_2$ is a continuous map, then we obtain a map $t(f) : t(X_1) \to t(X_2)$ by sending an irreducible closed subset to the closure of its image. Thus t is a functor on topological spaces. Furthermore, define a continuous map $\alpha : X \to t(X)$ by $\alpha(P) = \{P\}^-$. Note that α induces a bijection between the set of open subsets of X and the set of open subsets of $t(X)$.

Next, let k be an algebraically closed field, V be a variety over k, and let $\mathcal{O}_V$ be its sheaf of regular functions. I claim that $(t(V), \alpha_*(\mathcal{O}_V))$ is a scheme over k. Since any variety can be covered by open affine sub-varieties, it will be sufficient to show that if V is affine, then $(t(V), \alpha_*(\mathcal{O}_V))$ is a scheme. Hence, let V be an affine variety with affine coordinate ring A. Define a morphism of locally ringed spaces

$$\beta : (V, \mathcal{O}_V) \to X = \operatorname{Spec} A$$

as follows: for each point $P \in V$, let $\beta(P) = \mathfrak{m}_P$, the ideal of A consisting of all regular functions which vanish at P. Then β is a bijection of V onto the set of closed points of X. Since β carries the zero set of an ideal $\mathfrak{a} \subseteq A$ onto $V(\mathfrak{a})$ intersected with the closed points of X, and every closed subset on either side has this form, β is a homeomorphism onto its image. Now for any open set $U \subseteq X$, we will define a homomorphism of rings $\mathcal{O}_X(U) \to \beta_*(\mathcal{O}_V)(U) = \mathcal{O}_V\left(\beta^{-1}U\right)$. Given a section $s \in \mathcal{O}_X(U)$, and given a point $P \in \beta^{-1}(U)$, we define $s(P)$ by taking the image of s in the stalk $\mathcal{O}_{X,\beta(P)}$, which is isomorphic to the local ring $A_{\mathfrak{m}_P}$, and then passing to the quotient ring $A_{\mathfrak{m}_P}/\mathfrak{m}_P$ which is isomorphic to the field k. Thus s gives a function from $\beta^{-1}(U)$ to k. This is a regular function, and that this map gives an isomorphism $\mathcal{O}_X(U) \cong \mathcal{O}_V\left(\beta^{-1}U\right)$. Finally, since the prime ideals of A are in 1-1 correspondence with the irreducible closed subsets of V, these remarks show that $(X, \mathcal{O}_X)$ is isomorphic to $(t(V), \alpha_*(\mathcal{O}_V))$, so the latter is indeed an affine scheme, completing the proof.

We have defined upon enough properties to characterize the image of this embedding:

Definition 3.7.2: k-Variety

Let X be a k-scheme. Then if X is reduced, irreducible, separated, and of finite type over k, then X is said to be a *variety over k* or a *k-variety*. If X is affine/(quasi)projective, then it is an affine/(quasi)projective variety over k respectively.

It is important to note that in [22, [Affine] Variety], a variety was an affine algebraic set that was irreducible, which motivates adding irreducible in the definition. However some authors drop the condition and "variety" would be "algebraic set". Note too that we have defined k-varieties and not varieties in general. Furthermore, some authors require the field $k = \overline{k}$ be algebraically closed as part of the definition to more closely reflect the geometric results of the Nullstellensatz, and also to avoid the following problem:

Example 3.7.3: Variety Not Closed Under Base Change

Take the $\mathbb{R}$-variety $\mathbb{C} \cong \mathbb{R}[x]/(x^2+1) = V$ which is a single point, hence irreducible. Then $C \times_{\mathbb{R}} V = \operatorname{Spec} \mathbb{C}[x](x^2+1)$ which now has two points, hence it is not irreducible and so *not* a variety as we have defined it.

While the abstract definition of a scheme allows for elegant categorical constructions, it introduces points that are not closed (the generic points corresponding to non-maximal prime ideals). Notice in the proof of theorem 3.7.1 that the map β is a bijection of the classical variety V onto *only* the closed points of $X = \operatorname{Spec} A$. One might naturally worry: do we lose algebraic information by only evaluating functions at the closed points?

This is where the concept of a *Jacobson scheme* is useful, serving as the exact geometric justification for why our classical intuition of evaluating polynomials at points perfectly captures the algebraic structure. Indeed, it is fair to say that Jacobson schemes are the "most general" setting in which the Nullstellensatz holds.

Definition 3.7.4: Jacobson Scheme

A topological space X is *Jacobson* if its closed points are dense in every closed subset. A scheme X is a *Jacobson scheme* if its Zariski topology is Jacobson.

Recall ([10]) that a *Jacobson ring* is a ring where the intersection of every maximal ideal of R containing $I \subseteq R$ is equal to the radical $\sqrt{I}$.

Proposition 3.7.5: Characterizing Jacobson Affine Schemes

An affine scheme $X = \operatorname{Spec} R$ is Jacobson if and only if R is *Jacobson*.

***Proof*:**
Let $X = \operatorname{Spec} R$. Any closed subset of X is of the form $V(I)$ for some ideal $I \subseteq R$. The closed points of X are exactly the maximal ideals $\operatorname{mSpec}(R)$. Therefore, the closed points residing inside $V(I)$ are exactly the maximal ideals containing I.

The closure of these points inside X is given by taking the vanishing locus of their intersection:

$$\overline{V(I) \cap \mathrm{mSpec}(R)} = V\left(\bigcap_{\substack{\mathfrak{m} \in \mathrm{mSpec}(R) \\ I \subseteq \mathfrak{m}}} \mathfrak{m} \right)$$

If R is a Jacobson ring, this intersection of maximal ideals is exactly $\sqrt{I}$. Thus, the closure becomes $V(\sqrt{I})$, which is equal to $V(I)$. This shows the closed points in $V(I)$ are dense in $V(I)$, making X a Jacobson space.

Conversely, if X is a Jacobson space, the closed points are dense in $V(I)$, so $V\left(\bigcap_{\mathfrak{m} \supset I} \mathfrak{m}\right) = V(I)$. Taking the radical of the defining ideals gives $\bigcap_{\mathfrak{m} \supset I} \mathfrak{m} = \sqrt{I}$, proving R is a Jacobson ring.

Let us formally prove that our geometric intuition of "density" translates exactly to this algebraic property of the nilradical, which ensures we do not lose information.

Corollary 3.7.6: Density of Closed Points and the Nilradical

Let R be a commutative ring and $X = \operatorname{Spec} R$. If the closed points of X are dense in X, then the intersection of all maximal ideals in R is exactly the nilradical $\mathrm{Nil}(R)$.

***Proof*:**

This is a direct application of the logic from proposition 3.7.5 using the zero ideal $I = (0)$.

The entire space X is the closed subset $V(0)$. The condition that the closed points are dense in X means that the closure of the maximal ideals is the whole space:

$$V\left(\bigcap_{\mathfrak{m} \in \mathrm{mSpec}(R)} \mathfrak{m} \right) = V(0)$$

Taking the radical of the defining ideals on both sides yields:

$$\bigcap_{\mathfrak{m} \in \mathrm{mSpec}(R)} \mathfrak{m} = \sqrt{(0)} = \mathrm{Nil}(R)$$

(Note that the left side is its own radical because an intersection of prime ideals is already a radical ideal).

Because the nilradical is zero, the geometry of the maximal ideals is dense enough to completely determine the algebra. This classically justifies viewing polynomials as functions to the base field k. Furthermore, as we saw early in the chapter in section 3.3 (Functor of Points), this classical intuition is perfectly mirrored in scheme theory, where a global section $f \in \Gamma(X, \mathcal{O}_X)$ is canonically equivalent to a scheme morphism $X \to \mathbb{A}^1_k$.

Example 3.7.7: Non-Jacobson Schemes

A scheme that is not Jacobson must have a prime ideal that is not the intersection of maximal ideals. This represents a higher dimensional point that is "not covered" by closed points. Schemes

likes $\operatorname{Spec} k[[x]]$ and $\operatorname{Spec} \mathbb{Z}_{(p)}$. The scheme $\operatorname{Spec} k[x]/(x^n)$ *is* Jacobson, having only (x) as its prime and maximal ideal.

Note that:

$$k[[x]] \cong \varprojlim_n \frac{k[x]}{(x^n)}$$

giving an example of how different the geometry and the algebra can behave under limits. We shall address this in section 5.6⋆.

Having reconciled the abstract algebraic structure with our geometric intuition of points, let us look at the standard ways to construct these varieties.

Proposition 3.7.8: Affine and Projective Varieties

1. $\operatorname{Spec}(k[x_1, ..., x_n]/I)$ is an affine k-variety if and only if $I \subseteq k[x_1, ..., x_n]$ is a prime ideal
2. Let $I \subseteq k[x_0, ..., x_n]$ be a graded prime ideal with $I \neq (x_0, ..., x_n)$. Then $\operatorname{Proj} k[x_0, ..., x_n]/I$ is a projective k-variety [a]

[a] I need not be prime, ex. $I = (x_0^2, x_0x_1, \ldots, x_0x_n)$

***Proof*:**

1. Write $A = k[x_1, \ldots, x_n]/I$ and $X = \operatorname{Spec}(A)$, so $\mathcal{O}_X(X) = A$ (proposition 2.2.1).

   The scheme X is of finite type over k for every I. The only affine open subsets of $\operatorname{Spec}(k)$ are $\emptyset$ and $\operatorname{Spec}(k)$, whose preimages under the structure morphism are $\emptyset$ and X. The latter is the spectrum of the finitely generated k-algebra A, so the structure morphism is locally of finite type (definition 3.4.28), and it is quasicompact (definition 3.4.1) because affine schemes are quasicompact (section 2.1.3).

   The scheme X is separated for every I. The product is affine, $X \times_{\mathbb{Z}} X = \operatorname{Spec}(A \otimes_{\mathbb{Z}} A)$ (theorem 3.2.1), and under theorem 3.1.4 the diagonal $\delta : X \to X \times_{\mathbb{Z}} X$ corresponds to the ring map $A \otimes_{\mathbb{Z}} A \to A$, $a \otimes b \mapsto ab$, since composing δ with either projection is the identity. This ring map is surjective, so δ is a closed immersion (proposition 3.1.24(2)) and X is separated (definition 3.5.2).

   Suppose I is prime, so that A is an integral domain. Then X is reduced (lemma 2.4.16). For irreducibility, suppose $X = V(\mathfrak{a}) \cup V(\mathfrak{b}) = V(\mathfrak{ab})$ with both closed sets proper. Then $\mathfrak{ab}$ lies in every prime of A, hence in $\sqrt{(0)} = (0)$ ([10, The Radical as an Intersection of Primes], A being a domain). If both $\mathfrak{a} \neq 0$ and $\mathfrak{b} \neq 0$, choosing nonzero $f \in \mathfrak{a}$ and $g \in \mathfrak{b}$ gives $fg = 0$, contradicting that A is a domain. So $\mathfrak{a} = 0$ or $\mathfrak{b} = 0$, making $V(\mathfrak{a})$ or $V(\mathfrak{b})$ all of X, a contradiction. Since $A \neq 0$, X is nonempty, hence irreducible (definition 2.1.22). Thus X is a k-variety (definition 3.7.2), and it is affine.

   Conversely, suppose X is an affine k-variety. Being reduced and irreducible, it is integral (proposition 2.4.26), so $A = \mathcal{O}_X(X)$ is an integral domain (definition 2.4.19). Hence I, the kernel of $k[x_1, \ldots, x_n] \to A$, is prime.

2. Write $S = k[x_0, \ldots, x_n]/I$ and $X = \operatorname{Proj} S$, a scheme covered by the charts $U_i = D_+(\overline{x}_i) \cong \operatorname{Spec}(S_{(\overline{x}_i)})$ (theorem 2.3.3, lemma 2.3.2), where $\overline{x}_i$ denotes the image of x_i. Since I is prime, S is a domain, and since I is proper and $I \neq (x_0, \ldots, x_n)$, some $\overline{x}_i \neq 0$.

Finite type. Each chart ring is a finitely generated k-algebra: an element of $S_{(\overline{x}_i)}$ is a k-linear combination of degree-0 fractions $\overline{m}/\overline{x}_i^d$ with m a degree-d monomial, and each such fraction is a product of d of the finitely many fractions $\overline{x}_j/\overline{x}_i$, which therefore generate. As in item 1, the structure morphism $X \to \operatorname{Spec}(k)$ is locally of finite type (definition 3.4.28); it is quasicompact because X is a finite union of affine, hence quasicompact, charts. So X is of finite type over k (definition 3.4.29).

Reduced. If $\overline{x}_i \neq 0$, then $S_{\overline{x}_i}$ is a subring of the fraction field of S, so the chart ring $S_{(\overline{x}_i)} \subseteq S_{\overline{x}_i}$ is a domain, in particular reduced; if $\overline{x}_i = 0$ the chart is empty. Each nonempty chart is then a reduced scheme (lemma 2.4.16) with reduced stalks (proposition 2.4.15). Every stalk of X is a stalk of a chart, so X is reduced (proposition 2.4.15).

Irreducible. The zero ideal of S is a homogeneous prime, and $S_+ \not\subseteq (0)$ because $\overline{x}_i \in S_+$ is nonzero, so $[(0)] \in \operatorname{Proj} S$ (definition 2.3.1). A closed subset $V_+(E)$ contains $[(0)]$ only if $E \subseteq (0)$, in which case $V_+(E) = X$. So the only closed subset of X containing $[(0)]$ is X itself. If $X = Y \cup Z$ with Y, Z closed and proper, then $[(0)]$ lies in one of them, which being closed must be all of X, a contradiction. Hence X is irreducible (definition 2.1.22). It is nonempty as it contains $[(0)]$.

Separated. The products $U_i \times_{\mathbb{Z}} U_j \cong \operatorname{Spec}(S_{(\overline{x}_i)} \otimes_{\mathbb{Z}} S_{(\overline{x}_j)})$ are affine open subschemes covering $X \times_{\mathbb{Z}} X$, as in the gluing construction proving theorem 3.2.1. The diagonal δ satisfies $\delta^{-1}(U_i \times_{\mathbb{Z}} U_j) = U_i \cap U_j = D_+(\overline{x}_i\overline{x}_j) \cong \operatorname{Spec}(S_{(\overline{x}_i\overline{x}_j)})$, since a prime avoids $\overline{x}_i\overline{x}_j$ exactly when it avoids both factors. Restricted to these charts, δ corresponds under theorem 3.1.4 to the ring map

$$S_{(\overline{x}_i)} \otimes_{\mathbb{Z}} S_{(\overline{x}_j)} \to S_{(\overline{x}_i\overline{x}_j)} \qquad \frac{a}{\overline{x}_i^d} \otimes \frac{b}{\overline{x}_j^e} \mapsto \frac{a\overline{x}_j^d}{(\overline{x}_i\overline{x}_j)^d} \cdot \frac{b\overline{x}_i^e}{(\overline{x}_i\overline{x}_j)^e}$$

whose two components are the restriction maps, because composing δ with the projections gives the inclusions of $U_i \cap U_j$ into U_i and U_j. This ring map is surjective: an element of $S_{(\overline{x}_i\overline{x}_j)}$ is a k-linear combination of fractions $\overline{m}/(\overline{x}_i\overline{x}_j)^d$ with m a monomial of degree $2d$, and splitting $m = m_1m_2$ into monomials of degree d exhibits the fraction as the image of $(\overline{m}_1/\overline{x}_i^d) \otimes (\overline{m}_2/\overline{x}_j^d)$. By proposition 3.1.24(2), the restriction of δ over each chart is a closed immersion. Now δ is a homeomorphism onto its image, since composing with the first projection retracts it; its image meets each member of the affine cover $\{U_i \times_{\mathbb{Z}} U_j\}$ in a closed subset, so the image is closed; and the sheaf map of δ is surjective on every stalk, each point of the image lying in a chart where δ is induced by a surjection of rings. By definition 3.1.22, δ is a closed immersion, so X is separated (definition 3.5.2).

Thus X is a k-variety, and it is a projective scheme (definition 2.3.4), hence a projective k-variety (definition 3.7.2).

An important class of varieties are *proper varieties*. For historical reasons, namely the 19th century Italian school, the following is a common name for proper varieties;

Definition 3.7.9: Complete Variety

Let X be a k-variety. Then it is said to be *complete* if it is proper.

The adjective "complete" for a proper variety comes from the fact that we are in some way thinking

of a complete variety as an extension into projective space. It can be shown (see theorem 3.5.34[6]) that if C is complete, then there exists a morphism $C \to \mathbb{P}^n$ to a projective scheme that is isomorphic to an open dense subset. There do exist complete non-projective varieties, but they are hard to come by, partly due to how bad the singularity must be[7].

Before concluding this section, it is crucial to see how finite type schemes (and thus varieties) behave under categorical products, as standard geometric products naturally map to fibered products in the category of schemes.

Proposition 3.7.10: Fibered Products of Finite Type Schemes

Let S be a k-scheme of finite type. If X and Y are k-schemes of finite type over S, then their fibered product $X \times_S Y$ is a finite type k-scheme. Consequently, the product $X \times_k Y$ of two k-varieties over an algebraically closed field is a k-variety.

***Proof*:**
Because $X \to S$ and $Y \to S$ are morphisms of finite type, their base changes are also of finite type (proposition 3.4.35). Specifically, the projection $X \times_S Y \to Y$ is a base change of $X \to S$, so it is of finite type. Composing this with the finite type structure morphism $Y \to \operatorname{Spec} k$, we find that $X \times_S Y \to \operatorname{Spec} k$ is of finite type.

In the specific case where $S = \operatorname{Spec} k$ and X, Y are k-varieties over an algebraically closed field k, we know $X \times_k Y$ is of finite type. Furthermore, varieties are separated by definition, and the product of separated schemes is separated. Since k is algebraically closed, the tensor product of two integral k-algebras over k is again an integral domain. Because this property holds locally on affine charts, $X \times_k Y$ is reduced and irreducible. Having satisfied all conditions, $X \times_k Y$ is a k-variety.

One question about varieties has gone untouched so far, and it comes up every time a statement about an arbitrary variety is to be reduced to a statement about curves: given two closed points of a variety of positive dimension, is there an irreducible curve inside the variety through both of them? The strategy for producing one is the strategy classical geometry suggests. Embed the variety in projective space, cut it by a hypersurface through the two points, and repeat until the dimension has come down to one. Each cut must do two things: drop the dimension by exactly one, and leave behind something irreducible. The first is dimension theory and is settled by Krull's theorem; the second is Bertini's theorem, and it is the one ingredient below that this book quotes rather than proves.

Lemma 3.7.11: Hypersurface Sections Drop the Dimension by One

Let k be a field, let $X \subseteq \mathbb{P}^n_k$ be an irreducible closed subvariety with $\dim X = d \geq 1$, and let $F \in k[x_0, \ldots, x_n]$ be homogeneous of degree $m \geq 1$ with $X \not\subseteq V_+(F)$. Then every irreducible component of $X \cap V_+(F)$ has dimension $d - 1$.

[6] see Borcherds scheme video abstract and projective varieties

[7] again see the same Borcherds video for an overview. The example he gave also allows for producing algebraic spaces that are not schemes

***Proof*:**
Let W be an irreducible component of $X \cap V_+(F)$, so W is nonempty, and choose an index j with $W \cap D_+(x_j) \neq \emptyset$. Write $U = X \cap D_+(x_j)$, a nonempty affine open subscheme of X. Being irreducible and reduced, X is integral by proposition 2.4.26, so $\mathcal{O}_X(U)$ is a domain by definition 2.4.19.

The quotient $f = F/x_j^m$ is a regular function on $D_+(x_j)$, since $\mathcal{O}_{\mathbb{P}^n_k}(D_+(x_j)) = k[x_0/x_j, \ldots, x_n/x_j]$ and F is homogeneous of degree m; let f also denote its restriction to U. If f vanished on U then U would lie in the closed set $V_+(F)$, and U is dense in the irreducible X, so $X \subseteq V_+(F)$, against the hypothesis. Hence f is a nonzero element of a domain, so it is a zero divisor on no irreducible component of U, and U has exactly one such component. Corollary 2.6.14 applies and gives that every irreducible component of

$$V(f) = (X \cap V_+(F)) \cap U$$

has codimension 1 in U.

By lemma 2.6.1 the assignment $Z \mapsto \overline{Z}$ is an inclusion-preserving bijection from the irreducible closed subsets of an open subset onto those of the ambient space meeting it, with inverse $W \mapsto W \cap U$. Applied to the open subset $(X \cap V_+(F)) \cap U$ of $X \cap V_+(F)$, it shows that $W \cap U$ is an irreducible component of $V(f)$, so $\operatorname{codim}_U(W \cap U) = 1$; applied to the open subset U of X, it turns chains descending from W in X into chains descending from $W \cap U$ in U and back, so $\operatorname{codim}_X W = 1$ as well. Since X is irreducible and of finite type over k, theorem 2.6.12 gives

$$\dim W = \dim X - \operatorname{codim}_X W = d - 1$$

as we sought to show.

The lemma says nothing about irreducibility, and for good reason: the section it produces is usually reducible. What is worse for the problem at hand is that the reducibility can be forced by the two points one wants to keep, so that no amount of generality in the choice of hypersurface removes it. The following is the standard warning example, and it is the reason the sections taken below have degree larger than one.

Example 3.7.12: Hyperplanes Through Two Points Need Not Cut Irreducibly
Let $Q = V_+(wz - xy) \subseteq \mathbb{P}^3_k$ be the quadric surface, which carries two families of lines by example 3.1.36, and let

$$q_0 = [0, 0, 1, 0], \qquad q = [0, 0, 0, 1]$$

be two closed points of the line $\ell = V_+(w, x) \subseteq Q$. A hyperplane $V_+(aw + bx + cy + dz)$ containing q_0 and q has $c = d = 0$, so it is $H = V_+(aw + bx)$ and contains ℓ. For $b \neq 0$ one may substitute $x = -(a/b)w$ into the equation of Q, which becomes $w(bz + ay)/b$ on H, so

$$Q \cap H = V_+(w, x) \cup V_+(aw + bx,\ ay + bz)$$

and for $b = 0$ one gets $Q \cap V_+(w) = V_+(w, x) \cup V_+(w, y)$. In every case the section is a union of two distinct lines, one of them ℓ. So the hyperplanes through q_0 and q cut Q reducibly, all of them, no matter how they are chosen. The lemma this section is aiming at is untouched by the

example, since ℓ itself is an irreducible curve through q_0 and q; what fails is the naive induction by hyperplane sections.

The obstruction in the example is that the two points span a line lying on the surface, and every hyperplane through two points of a line contains that line. Sections of degree larger than the number of points to be joined are not subject to it, because forms of degree m vanishing at a finite set S still separate the points outside S once m exceeds the number of points of S. By the Veronese embedding (definition 3.1.38) such sections are hyperplane sections in a re-embedding, so nothing is lost by taking them, and the curve produced at the end of the induction still sits inside the original variety. With that adjustment the classical Bertini theorem applies. For a finite set S of closed points of $\mathbb{P}^n_k$ and an integer $m \geq 1$ write

$$W_m(S) = \{F \in k[x_0, \dots, x_n]_m \mid F(s) = 0 \text{ for every } s \in S\}$$

for the space of degree-m forms vanishing on S, and let $\Lambda_m(S)$ denote the projective space of hypersurfaces (definition 3.1.35) whose defining form lies in $W_m(S)$.

Theorem 3.7.13: Bertini Irreducibility for Sections Through a Finite Set

Let k be algebraically closed, let $X \subseteq \mathbb{P}^n_k$ be an irreducible closed subvariety with $\dim X \geq 2$, let $S \subseteq X$ be a nonempty finite set of closed points, and let $m > |S|$. Then there is a dense open subset $\Omega \subseteq \Lambda_m(S)$ such that for every $H \in \Omega$ the closed set $X \cap H$ is irreducible of dimension $\dim X - 1$. Every such H contains S, so every such section does too.

***Proof Key ideas*:**
The dimension statement and the openness are elementary and are proved here; the irreducibility is the classical Bertini theorem for linear systems and is quoted.

Step 1: the members that swallow X form a proper closed subset. For $s \in S$ the linear forms vanishing at s cut out $\{s\}$, so if all of them vanished on X then $X \subseteq \{s\}$ and $\dim X = 0$, which is excluded; choose therefore a linear form L_s with $L_s(s) = 0$ and $L_s \notin I(X)$, and a linear form $L_1 \notin I(X)$, which exists because X is nonempty. The homogeneous ideal $I(X)$ is prime because X is irreducible, so

$$F_0 = \Big(\prod_{s \in S} L_s\Big) \cdot L_1^{\,m-|S|}$$

lies in $W_m(S) \setminus I(X)$, and $X \not\subseteq V_+(F_0)$. Consequently $W_m(S) \cap I(X)$ is a proper linear subspace of $W_m(S)$, so the members of $\Lambda_m(S)$ containing X form a proper closed subset. On its complement the section $X \cap H$ is nonempty, since it contains S, and every irreducible component of it has dimension $\dim X - 1$ by lemma 3.7.11.

Step 2: what the system does away from S. Let $p \neq q$ be closed points of $\mathbb{P}^n_k$ outside S. Choosing linear forms L_s with $L_s(s) = 0 \neq L_s(q)$, then L' with $L'(p) = 0 \neq L'(q)$, and L'' with $L''(q) \neq 0$, the form

$$\Big(\prod_{s \in S} L_s\Big) \cdot L' \cdot L''^{\,m-|S|-1}$$

lies in $W_m(S)$, vanishes at p and not at q; here $m > |S|$ is what makes the exponent nonnegative. Dropping the factor L' from the same product gives a member of $W_m(S)$ not vanishing at q, so

the common zero locus of $W_m(S)$ is exactly S. The construction with L' chosen to have nonzero differential along a prescribed tangent direction at p separates tangent vectors as well. The base locus S is finite and so of codimension at least 2 in X, and the rational map $\varphi\colon X \dashrightarrow \mathbb{P}(W_m(S)^\vee)$ attached to the system is injective and unramified on $X \setminus S$. It is therefore birational onto its image, and that image has dimension $\dim X \geq 2$.

Step 3: the input that is not available here. What remains is the assertion that a linear system on an irreducible projective variety whose base locus has codimension at least 2 and whose associated rational map is birational onto an image of dimension at least 2 has irreducible general member. The standard proof passes to the incidence variety

$$Z = \{(p, H) \in (X \setminus S) \times \Lambda_m(S) \mid p \in H\}$$

which is irreducible because its projection to $X \setminus S$ is a projective-space bundle, and then compares the number of components of a general fiber of $Z \to \Lambda_m(S)$ with the degree of the finite part of the Stein factorization of that projection. Stein factorization rests on the theorem on formal functions, and neither is developed in this book; it is the same input that forced theorem 3.4.38 to be quoted rather than proved. See [70] for a modern account of the Bertini theorems. The characteristic-zero route through the smoothness form of Bertini's theorem, [36, p. II.8.18], together with connectedness of the general section, is not enough for what follows, since the applications need all characteristics.

The theorem is the only step below that is imported. Everything else is dimension bookkeeping, and the payoff is the statement the section opened with. Note that the two points in it are allowed to coincide, in which case S is a single point; note also that the variety is not assumed complete, and that the curve produced is closed in the variety but generally not complete.

Lemma 3.7.14: Two Closed Points Lie on an Irreducible Curve

Let k be algebraically closed and let V be a quasiprojective k-variety with $\dim V \geq 1$. Given closed points $v_0, v \in V$, there is a closed subvariety $C \subseteq V$, irreducible of dimension 1, with $v_0, v \in C$.

***Proof*:**
Step 1: pass to the projective closure. By definition 3.7.2 and definition 2.3.5 the variety V is an open subscheme of a projective k-scheme, hence a locally closed subscheme of some $\mathbb{P}^n_k$; fix such an immersion and identify V with its image. Let X be the closure of V in $\mathbb{P}^n_k$, with its reduced induced structure (definition 3.1.26). Then X is an irreducible closed subvariety of $\mathbb{P}^n_k$, integral by proposition 2.4.26, and V is open in X: writing $V = Y \cap U$ with Y closed and U open in $\mathbb{P}^n_k$ gives $X \subseteq Y$ and $V = X \cap U$. The generic point η of X (lemma 2.4.21) lies in V, since otherwise the closed set $X \setminus V$ would contain $\overline{\{\eta\}} = X$ and V would be empty. So V and X have the same local ring at η, hence the same function field by corollary 2.4.25, and theorem 2.6.11 gives $\dim X = \dim V =: d$.

Step 2: cut the dimension down inside X. The points v_0 and v are closed in X and not just in V. Indeed, let Z be the closure of v_0 in X, so that $Z \cap V = \{v_0\}$; by lemma 2.6.1 the closure operation matches chains descending from $\{v_0\}$ in V with chains descending from Z in X, so

$\operatorname{codim}_X Z = \operatorname{codim}_V \{v_0\} = d$ by theorem 2.6.12 applied to V, and a second application to X gives $\dim Z = 0$. Since Z is irreducible with generic point v_0, a second point of Z would produce a chain of length one, so $Z = \{v_0\}$. Being closed in X, which is closed in $\mathbb{P}^n_k$, the two points are closed points of $\mathbb{P}^n_k$.

Put $S = \{v_0, v\}$, a set of one or two closed points of X, and fix $m = 3$, so that $m > |S|$. Set $X_0 = X$ and suppose an irreducible closed subvariety $X_i \subseteq \mathbb{P}^n_k$ of dimension $d - i \geq 2$ containing S has been produced. Theorem 3.7.13 applied to X_i gives a hypersurface $H \in \Lambda_3(S)$ with $X_i \cap H$ irreducible of dimension $d - i - 1$. Let X_{i+1} be this closed set with its reduced induced structure (definition 3.1.26). It contains S, since $S \subseteq X_i$ and $S \subseteq H$. After $d - 1$ steps this yields an irreducible closed subvariety $C_0 = X_{d-1} \subseteq X$ of dimension 1 containing v_0 and v. If $d = 1$ no step is performed and $C_0 = X$.

Step 3: intersect back into V. Let $C = C_0 \cap V$ with its reduced induced structure (definition 3.1.26). Since C_0 is closed in X and V is an open subscheme of X, the subscheme C is closed in V and open in C_0. It contains v_0 and v, so it is nonempty, and a nonempty open subscheme of an irreducible scheme is irreducible and dense; so C is irreducible and shares the generic point and function field of C_0 by lemma 2.4.21 and corollary 2.4.25, whence $\dim C = \dim C_0 = 1$ by theorem 2.6.11. Finally C is reduced, being open in a reduced scheme, and separated and of finite type over k, being a locally closed subscheme of $\mathbb{P}^n_k$; so C is a k-variety in the sense of definition 3.7.2, irreducible of dimension 1 and closed in V, containing both points, completing the proof.

Step 3 is where the hypothesis of quasiprojectivity earns its place. The cutting in Step 2 has to happen in the projective closure, because that is where hypersurface sections live, and the curve it produces need not lie in V; but it meets V in a nonempty open subset of itself, and a nonempty open subset of an irreducible curve is again an irreducible curve, closed in V because the ambient curve was closed in X. The same device settles the complete case, where the variety at hand need not be projective at all. There the projective model is given by Chow's lemma, and the curve is pushed forward rather than intersected back.

Corollary 3.7.15: Curves Joining Two Points of a Complete Variety

Let k be algebraically closed, let V be a complete k-variety (definition 3.7.9) with $\dim V \geq 1$, and let $v_0 \neq v$ be closed points of V. Then there is a closed subvariety $C \subseteq V$, irreducible of dimension 1, with $v_0, v \in C$.

***Proof*:**
Step 1: a projective model. The structure morphism $V \to \operatorname{Spec} k$ is proper and of finite type, and the base $\operatorname{Spec} k$ is Noetherian, so theorem 3.5.34 provides a projective k-scheme X', a surjective morphism $\pi\colon X' \to V$, and a dense open $U \subseteq V$ with $\pi^{-1}(U) \to U$ an isomorphism. Let X be the closure of $\pi^{-1}(U)$ in X' with its reduced induced structure (definition 3.1.26). As U is irreducible, so are $\pi^{-1}(U)$ and X; thus X is a projective k-variety. It contains the dense open subscheme $\pi^{-1}(U) \cong U$, so $\dim X = \dim U = \dim V \geq 1$ by lemma 2.4.21, corollary 2.4.25 and theorem 2.6.11. The restriction $\pi|_X$ factors as the closed immersion $X \hookrightarrow X'$ followed by π, so it is proper by theorem 3.5.32 and proposition 3.5.23; its image is therefore closed and contains the dense subset U, which makes $\pi|_X$ surjective onto V.

Step 2: lift the two points. The fibers $\pi|_X^{-1}(v_0)$ and $\pi|_X^{-1}(v)$ are nonempty closed subsets of X, and X is a closed subspace of some $\mathbb{P}_k^N$, which is Noetherian by lemma 2.4.8; so the closed subsets of X satisfy the descending chain condition and each fiber contains a minimal nonempty closed subset T. By minimality every point of T has closure equal to T, so T is irreducible and each of its points is a generic point of it; the generic point of an irreducible closed subset of an integral scheme is unique by lemma 2.4.21, so T is a single point, closed in X. Choose such points $x_0 \in \pi|_X^{-1}(v_0)$ and $x \in \pi|_X^{-1}(v)$.

Step 3: produce the curve upstairs and push it down. A projective scheme is an open subscheme of itself, so X is a quasiprojective k-variety and lemma 3.7.14 gives an irreducible closed subvariety $C' \subseteq X$ of dimension 1 with $x_0, x \in C'$. Let $C = \pi(C')$ with its reduced induced structure (definition 3.1.26). It is closed, since $\pi|_X$ is proper and C' is closed in X, and it is irreducible, being the continuous image of an irreducible space (lemma 2.1.24(2)); and it contains v_0 and v. It is a k-variety, being a reduced irreducible closed subscheme of the separated finite-type scheme V.

It remains to compute $\dim C$. The generic point η' of C' maps to the generic point of C, because the closure of its image contains $\pi(C') = C$; so π induces a homomorphism $k(C) \to k(C')$ of function fields (corollary 2.4.25), which is injective because its source is a field. Theorem 2.6.11 therefore gives $\dim C \leq \dim C' = 1$. The dimension is not 0: if $\dim C = 0$ then theorem 2.6.12 applied to the irreducible closed subset $\{v_0\} \subseteq C$ forces $\operatorname{codim}_C\{v_0\} = 0$ and hence $\{v_0\} = C$, contradicting $v \in C$ and $v \neq v_0$. So $\dim C = 1$, completing the proof.

The restriction to distinct points is what the argument gives and not an artifact of bookkeeping: the image of a curve under π can be a single point, and the only thing ruling that out above is that C has to contain two distinct points. The two statements together cover the varieties that arise in practice, and they are what permits the reduction of a question about an arbitrary variety to the same question about a curve, which is then normalized and compactified. That reduction is the engine behind the rigidity lemma and the theorem of the cube in *Abelian Varieties* [5]. One gap is left open by the pair of statements: a variety that is separated and of finite type but neither quasiprojective nor complete is reached only by first compactifying it, and Nagata's compactification theorem is not developed in this book.

3.8⋆ Bertini Theorems

⋆ *Optional. This section is an aside; nothing later in the book depends on it.*

This section proves two Bertini theorems from scratch: general hyperplane sections of a smooth variety are smooth, and the preimage of a general hyperplane under a map to projective space with two-dimensional image is geometrically irreducible. The second is the theorem theorem 3.7.13 was reaching for, though it is not the same statement, and the difference is the subject of box 3.8.11.

The section closes with a quantitative refinement, and that refinement is the third of its theorems. For the applications it is not enough to know that a general member of a family behaves; one wants to know how large the bad locus is inside the parameter space, because the members one is allowed to use are often confined to a subfamily. The refinement measures the bad locus by its codimension in a Grassmannian of linear systems, and the measurement is sharp enough that a subfamily of

codimension one still meets the good locus. Its proof is an induction that removes one hypersurface at a time, and the mechanism which makes a single cut suffice, a count of the linear subspaces lying inside a fixed closed set, is proved here as well.

Throughout, $(\mathbb{P}^N_k)^\vee$ denotes the dual projective space, whose points parametrize hyperplanes: a point $[a] = [a_0 : \cdots : a_N]$ corresponds to the hyperplane

$$H_a = V_+(a_0X_0 + \cdots + a_NX_N) \subseteq \mathbb{P}^N_k$$

and this is a bijection between the k-rational points of $(\mathbb{P}^N_k)^\vee$ and the hyperplanes of $\mathbb{P}^N_k$ defined over k. A property of hyperplanes holds for a *general* hyperplane if there is a dense open subset $\Omega \subseteq (\mathbb{P}^N_k)^\vee$ such that every H_a with $[a] \in \Omega$ a k-rational point has it. The same language is used for members of any family parametrized by a variety.

Over a finite field this convention can be vacuous, since a dense open subset of a projective space over $\mathbb{F}_q$ may have no $\mathbb{F}_q$-rational point at all. That is why an infinite ground field appears in the hypotheses of every irreducibility statement below. Bertini theorems over finite fields are a separate subject, opened up by Charles and Poonen [4]; nothing proved here says anything about that case.

Lemma 3.8.1: Rational Points of a Nonempty Open Set over an Infinite Field

Let k be an infinite field and $\Omega \subseteq \mathbb{P}^M_k$ a nonempty open subscheme. Then Ω has a k-rational point.

***Proof*:**
The standard charts $D_+(X_i) \cong \mathbb{A}^M_k$ cover $\mathbb{P}^M_k$, so Ω meets one of them; replacing Ω by $\Omega \cap D_+(X_i)$ reduces to $\Omega \subseteq \mathbb{A}^M_k = \operatorname{Spec} k[y_1, \ldots, y_M]$. A nonempty open subscheme of an affine scheme contains a nonempty basic open $D(f)$, and $D(f) \neq \emptyset$ forces $f \neq 0$.

It remains to produce $c \in k^M$ with $f(c) \neq 0$, since such a c is a k-rational point of $\mathbb{A}^M_k$ lying in $D(f)$. Induct on M. For $M = 1$ a nonzero polynomial in one variable has at most $\deg f$ roots and k is infinite. For $M > 1$ write $f = \sum_i f_i(y_1, \ldots, y_{M-1}) y_M^i$ and choose i_0 with $f_{i_0} \neq 0$; by induction there is $c' \in k^{M-1}$ with $f_{i_0}(c') \neq 0$, so $f(c', y_M)$ is a nonzero polynomial in one variable and the case $M = 1$ gives c_M with $f(c', c_M) \neq 0$, completing the proof.

Two further preliminaries recur. The first bounds the size of preimages under a dominant morphism and produces a dense open subset of the target that is genuinely hit. It is proved here rather than quoted because the general theory of fiber dimension is developed later in the book.

Lemma 3.8.2: Generic Fiber Dimension

Let k be an algebraically closed field and $\pi\colon U \to W$ a dominant morphism of irreducible k-varieties. Put $D = \dim U$, $w = \dim W$ and $e = D - w$. Then there are dense open subschemes $U^\circ \subseteq U$ and $W^\circ \subseteq W$ with $\pi(U^\circ) \subseteq W^\circ$ and a finite surjective morphism

$$\psi\colon U^\circ \longrightarrow W^\circ \times_k \mathbb{A}^e_k$$

whose composite with the first projection is $\pi|_{U^\circ}$. In particular $\pi(U^\circ) = W^\circ$, and for every closed subset $T \subseteq W$,

$$\dim\left(\pi^{-1}(T) \cap U^\circ\right) \leq \dim(T \cap W^\circ) + e$$

whenever the left side is nonempty.

***Proof*:**

Choose a nonempty affine open $W_1 \subseteq W$ and then a nonempty affine open $U_1 \subseteq \pi^{-1}(W_1)$; both are dense, being nonempty opens of irreducible schemes, and $\pi^{-1}(W_1)$ is nonempty because π is dominant. Write $A = \mathcal{O}_W(W_1)$ and $B = \mathcal{O}_U(U_1)$. Both are domains (proposition 2.4.26 and definition 2.4.19), and π induces $A \to B$, injective because $\pi|_{U_1}$ is dominant: a nonzero element of the kernel would confine $\pi(U_1)$ to a proper closed subset of W_1.

Let $K = \mathrm{Frac}A$ and $L = \mathrm{Frac}B$. Theorem 2.6.11 gives $\mathrm{trdeg}_k K = w$ and $\mathrm{trdeg}_k L = D$, hence $\mathrm{trdeg}_K L = e$. Choose $u_1, \ldots, u_e \in B$ forming a transcendence basis of L over K, which is possible because B generates L as a field over K and a transcendence basis may be extracted from any generating set. Then L is algebraic over $K(u_1, \ldots, u_e)$.

Write $B = A[z_1, \ldots, z_q]$. Each z_j is algebraic over $K(u_1, \ldots, u_e)$, hence satisfies a polynomial relation with coefficients in $A[u_1, \ldots, u_e]$; clearing denominators and dividing by the leading coefficient makes z_j integral over $A_{a_j}[u_1, \ldots, u_e]$ for a suitable nonzero $a_j \in A$. Put $a = a_1 \cdots a_q$. Then every z_j is integral over $R := A_a[u_1, \ldots, u_e]$, so B_a is a finitely generated R-algebra generated by integral elements, hence a finite R-module.

Set $W^\circ = W_1 \cap D(a)$ and $U^\circ = \mathrm{Spec}\, B_a$, the open subscheme $U_1 \cap \pi^{-1}(D(a))$ of U; it is nonempty because a is a nonzero element of the domain B, hence dense. The inclusion $R \hookrightarrow B_a$ is finite and injective, so $\psi\colon U^\circ \to \mathrm{Spec}\, R = W^\circ \times \mathbb{A}^e$ is finite, and it is surjective because an injective integral ring extension satisfies lying over. Composing with the projection $\mathrm{Spec}\, R \to W^\circ$ recovers $\pi|_{U^\circ}$, and surjectivity of ψ together with surjectivity of the projection gives $\pi(U^\circ) = W^\circ$.

For the dimension bound let $T \subseteq W$ be closed and put $T' = T \cap W^\circ$. Then $\pi^{-1}(T) \cap U^\circ = \psi^{-1}(T' \times \mathbb{A}^e)$, and ψ restricts to a finite morphism from this scheme onto a closed subscheme of $T' \times \mathbb{A}^e$. A finite morphism corresponds to an integral ring extension on affine pieces, so lemma 2.6.3 gives

$$\dim\left(\pi^{-1}(T) \cap U^\circ\right) \leq \dim(T' \times_k \mathbb{A}^e_k)$$

Finally $\dim(T' \times \mathbb{A}^e) = \dim T' + e$: it suffices to check this for T' irreducible and reduced, and then $T' \times \mathbb{A}^e$ is an irreducible k-variety by proposition 3.7.10 whose function field is $k(T')(t_1, \ldots, t_e)$, of transcendence degree $\dim T' + e$ by theorem 2.6.11, as we sought to show.

The second preliminary transports a statement proved over $\overline{k}$ back to k-rational parameters. It replaces a Galois-descent argument by a linear-algebra one and needs no separability hypothesis.

Lemma 3.8.3: Descent of a Dense Open Locus

Let k be a field with algebraic closure $\overline{k}$ and let $V \subseteq \mathbb{P}^M_{\overline{k}}$ be a dense open subscheme. Then there is a dense open subscheme $\Omega \subseteq \mathbb{P}^M_k$ such that every k-rational point of Ω, viewed as an $\overline{k}$-rational point of $\mathbb{P}^M_{\overline{k}}$, lies in V.

***Proof*:**
The complement $F = \mathbb{P}^M_{\overline{k}} \setminus V$ is a proper closed subset, so $F = V_+(g_1, \dots, g_s)$ for homogeneous $g_1, \dots, g_s \in \overline{k}[X_0, \dots, X_M]$, not all zero. Choose $\theta_1, \dots, \theta_h \in \overline{k}$, linearly independent over k, whose k-span contains every coefficient of every g_j, and write

$$g_j = \sum_{l=1}^{h} \theta_l \, g_{jl}, \qquad g_{jl} \in k[X_0, \dots, X_M]$$

with each g_{jl} homogeneous of the same degree as g_j.

For $c \in k^{M+1}$ the values $g_{jl}(c)$ lie in k, so linear independence of the θ_l gives $g_j(c) = 0$ for all j if and only if $g_{jl}(c) = 0$ for all j and l. Hence the k-rational points of $\mathbb{P}^M_k$ whose associated $\overline{k}$-rational points lie in F are exactly the k-rational points of the closed subscheme $F_0 = V_+(\{g_{jl}\}) \subseteq \mathbb{P}^M_k$. Some g_j is nonzero, so some g_{jl} is a nonzero polynomial and $F_0 \neq \mathbb{P}^M_k$. Take $\Omega = \mathbb{P}^M_k \setminus F_0$, a nonempty open subscheme of an irreducible scheme and therefore dense, completing the proof.

Smoothness of a general hyperplane section

The first Bertini theorem is a statement about tangent spaces. A hyperplane section of a smooth variety fails to be smooth at a point exactly when the hyperplane contains the embedded tangent space there, and the hyperplanes through a fixed d-dimensional linear subspace of $\mathbb{P}^N$ form a linear subspace of $(\mathbb{P}^N)^\vee$ of dimension $N - d - 1$. Sweeping this over the d-dimensional X produces a family of dimension $N - 1$, one short of $(\mathbb{P}^N)^\vee$, and the theorem follows. The content is that the tangent-space condition is algebraic in the point, which the Jacobian criterion gives.

Theorem 3.8.4: Bertini's Smoothness Theorem

Let k be an algebraically closed field and $X \subseteq \mathbb{P}^N_k$ a smooth quasiprojective k-variety of dimension d. Then there is a dense open subset $\Omega \subseteq (\mathbb{P}^N_k)^\vee$ such that for every $[a] \in \Omega$ the scheme $X \cap H_a$ is smooth over k, and of pure dimension $d - 1$ wherever it is nonempty.

***Proof*:**
If $d = N$ then X is a nonempty open subscheme of $\mathbb{P}^N_k$, no hyperplane contains it, and $X \cap H_a$ is an open subscheme of $H_a \cong \mathbb{P}^{N-1}_k$, hence smooth of pure dimension $N - 1$; take $\Omega = (\mathbb{P}^N_k)^\vee$.

Assume from now on that $d < N$.

Step 1: when a section is singular at a point. Fix a closed point $x \in X$ and an index i_0 with $x_{i_0} \neq 0$, and define a k-linear map

$$\epsilon_x \colon k^{N+1} \longrightarrow \mathcal{O}_{X,x}/\mathfrak{m}_x^2, \qquad a \longmapsto \Big[\sum_{i=0}^{N} a_i\, X_i/X_{i_0}\Big]$$

Replacing i_0 by another index multiplies ϵ_x by a unit of $\mathcal{O}_{X,x}$ and does not change its kernel, written W_x. The local ring $\mathcal{O}_{\mathbb{P}^N,x}$ is generated as a k-algebra by the fractions X_j/X_{i_0}, so $\mathcal{O}_{\mathbb{P}^N,x}/\mathfrak{m}^2$ is spanned over k by the classes of 1 and of the X_j/X_{i_0}, all of which lie in the image of the corresponding map for $\mathbb{P}^N$; and $\mathcal{O}_{X,x}/\mathfrak{m}_x^2$ is a quotient of $\mathcal{O}_{\mathbb{P}^N,x}/\mathfrak{m}^2$. Hence ϵ_x is surjective. Since X is smooth over the algebraically closed k it is regular (proposition 2.7.14), and $\kappa(x) = k$, so $\dim_k \mathfrak{m}_x/\mathfrak{m}_x^2 = \dim \mathcal{O}_{X,x} = d$ by definition 2.7.3 and theorem 2.6.12. Therefore

$$\dim_k W_x = (N+1) - (d+1) = N - d$$

Suppose $X \not\subseteq H_a$ and set $\ell = \sum_i a_i X_i/X_{i_0}$, a regular function on $X \cap D_+(X_{i_0})$. If $a \in W_x$ then $\ell \in \mathfrak{m}_x^2 \subseteq \mathfrak{m}_x$, so $x \in H_a$; conversely $x \in H_a$ says $\ell \in \mathfrak{m}_x$. The local ring of the section is $\mathcal{O}_{X\cap H_a,x} = \mathcal{O}_{X,x}/(\ell)$. Now $\ell \neq 0$ in $\mathcal{O}_{X,x}$: otherwise ℓ would vanish on a nonempty open subset of the irreducible scheme $X \cap D_+(X_{i_0})$, hence on all of it, hence on X by density, against $X \not\subseteq H_a$. As $\mathcal{O}_{X,x}$ is a domain (proposition 2.7.4), every minimal prime $\mathfrak{p}$ over (ℓ) has height exactly 1: at most 1 by theorem 2.6.13, and at least 1 because $\mathfrak{p} \neq 0$. Applying theorem 2.6.12 to X and to the integral closed subscheme cut out by $\mathfrak{p}$ gives $\dim \mathcal{O}_{X,x}/\mathfrak{p} = d - 1$, so $\dim \mathcal{O}_{X\cap H_a,x} = d - 1$. The cotangent space of the section is $\mathfrak{m}_x/(\mathfrak{m}_x^2 + (\ell))$, of dimension d if $\ell \in \mathfrak{m}_x^2$ and $d-1$ otherwise. Comparing the two computations,

$$X \cap H_a \text{ is regular of dimension } d-1 \text{ at } x \iff a \notin W_x$$

and when $a \in W_x$ the section is singular at x.

Step 2: the bad hyperplanes sweep out at most $N-1$ dimensions. Every point of the locally closed subscheme $X \subseteq \mathbb{P}_k^N$ has an affine open neighbourhood in X of the form

$$U = \operatorname{Spec} \frac{k[y_1, \dots, y_N, z]}{(f_1, \dots, f_s,\ zg - 1)}$$

where $y_1, \dots, y_N$ are the affine coordinates X_j/X_{i_0} $(j \neq i_0)$ of a chart $D_+(X_{i_0})$ and X is closed in the principal open $D(g)$ of that chart. Since X is Noetherian (lemma 2.4.8 and lemma 2.6.1), finitely many such U cover it. Fix one.

By the cotangent computation inside proposition 2.7.7, the cotangent space of X at a closed point $x \in U$ is the quotient of that of the ambient chart by the differentials of the f_i, so for ℓ as above with $\ell(x) = 0$,

$$\ell \in \mathfrak{m}_x^2 \iff (a_j)_{j\neq i_0} \in \operatorname{Row} J(x), \qquad J(x) = \Big(\tfrac{\partial f_i}{\partial y_j}(x)\Big)_{i,j}$$

because the linear part of ℓ at x has coefficient a_j in y_j. Smoothness of X and proposition 2.7.7 give $\operatorname{rank} J(x) = N - d$ at every closed point of U. Consequently

$$W_x = \left\{ a \in k^{N+1} \;\middle|\; a_{i_0} + \textstyle\sum_{j \neq i_0} a_j y_j(x) = 0 \;\text{ and }\; (a_j)_{j \neq i_0} \in \operatorname{Row} J(x) \right\}$$

the first condition being the statement $\ell(x) = 0$.

For each choice σ of $N - d$ rows and $N - d$ columns of J, let $U_\sigma \subseteq U$ be the open subscheme where the corresponding minor is invertible. These cover U, the rank being $N - d$ everywhere, and on U_σ the $N - d$ rows of J indexed by σ are linearly independent, hence a basis of $\operatorname{Row} J(x)$. Define

$$\Phi_\sigma \colon U_\sigma \times_k \mathbb{P}^{N-d-1}_k \longrightarrow (\mathbb{P}^N_k)^\vee, \qquad (x, [c]) \longmapsto [a]$$

by $(a_j)_{j \neq i_0} = \sum_{t=1}^{N-d} c_t \, J_{\sigma_t}(x)$ and $a_{i_0} = -\sum_{j \neq i_0} a_j y_j(x)$. Each coordinate of a is a regular function on U_σ times a linear form in c, and $a \neq 0$ when $c \neq 0$ because the chosen rows are independent, so Φ_σ is a morphism; by the description of W_x its image is exactly the set of $[a]$ for which some closed point $x \in U_\sigma$ has $a \in W_x$.

The source of Φ_σ is an irreducible k-variety (proposition 3.7.10) with function field $k(U_\sigma)(t_1, \ldots, t_{N-d-1})$, hence of dimension $d + (N-d-1) = N-1$ by theorem 2.6.11. The closure of its image is irreducible and the induced map of function fields is injective, so theorem 2.6.11 bounds $\dim \overline{\operatorname{im} \Phi_\sigma} \leq N - 1$. Let Σ be the union of the closures $\overline{\operatorname{im} \Phi_\sigma}$ over the finitely many pairs (U, σ): a closed subset of $(\mathbb{P}^N_k)^\vee$ of dimension at most $N - 1$, hence proper.

Step 3: conclusion. Let $\Lambda \subseteq (\mathbb{P}^N_k)^\vee$ be the set of $[a]$ with $X \subseteq H_a$, the projectivization of the space of linear forms vanishing on X; it is closed, and proper because a point of X lies outside some hyperplane. Put $\Omega = (\mathbb{P}^N_k)^\vee \setminus (\Sigma \cup \Lambda)$, a dense open subset.

Fix $[a] \in \Omega$ with $X \cap H_a \neq \emptyset$. Every closed point x of $X \cap H_a$ has $a \notin W_x$, since otherwise $[a]$ would lie in Σ, so by Step 1 the section is regular of dimension $d - 1$ at x. The singular locus of $X \cap H_a$ is closed (proposition 2.7.23); if it were nonempty it would contain a closed point of $X \cap H_a$, because closed points are dense in any affine open of a scheme of finite type over a field (lemma 2.4.5) and a closed point of a closed subscheme of $X \cap H_a$ is a closed point of $X \cap H_a$. Hence $X \cap H_a$ is regular, therefore smooth over the perfect field k by theorem 2.7.15, and of pure dimension $d - 1$, completing the proof.

The theorem says nothing about the section being nonempty and nothing about its irreducibility, and both fail routinely. A smooth conic in $\mathbb{P}^2$ meets a general line in two points, a smooth zero-dimensional scheme that is not irreducible, and a general line in $\mathbb{P}^3$ misses a fixed line altogether. What the theorem delivers is the inductive step for cutting a smooth variety down to a smooth curve, provided irreducibility is given separately.

Example 3.8.5: Hyperplane Sections of the Twisted Cubic

Let k be algebraically closed and let $X \subseteq \mathbb{P}^3_k$ be the twisted cubic, the image of $\mathbb{P}^1_k \to \mathbb{P}^3_k$, $[s : t] \mapsto [s^3 : s^2 t : s t^2 : t^3]$. It is a smooth curve, so $d = 1$ and $N = 3$, and the incidence family constructed in the proof has dimension $N - 1 = 2$ inside the three-dimensional $(\mathbb{P}^3)^\vee$. Concretely, the embedded tangent line at the parameter $[s : t]$ is spanned by $[s^3 : s^2 t : s t^2 : t^3]$

and $[3s^2 : 2st : t^2 : 0]$, and the planes containing that line form a line in $(\mathbb{P}^3)^\vee$; sweeping over the one-dimensional X gives a surface in $(\mathbb{P}^3)^\vee$, the dual surface. A plane H off that surface meets X in three distinct points, a smooth scheme of pure dimension 0; a plane tangent to X meets it in a length-two point together with a further point, and the section is singular there. The count $2 < 3$ is the inequality that makes the general plane work.

Irreducibility of a general hyperplane preimage

Irreducibility is not a local condition and cannot be read off tangent spaces, so the second Bertini theorem needs a different mechanism. The mechanism is a change of viewpoint: instead of examining the members of a family one at a time, one assembles them into a single morphism and asks about its generic fiber. Irreducibility of the generic fiber is a statement about a function field, and the point of the next lemmas is that it can be certified by a single rational point and that it then propagates to almost every closed fiber.

Geometric irreducibility, not just irreducibility, is what is required. The double cover $\mathbb{A}^1 \to \mathbb{A}^1$, $u \mapsto u^2$, has irreducible generic fiber $\operatorname{Spec} k(t)[u]/(u^2 - t)$ while every closed fiber away from the origin is a pair of points; the generic fiber is not geometrically irreducible, and that is the only thing separating the two behaviours.

Lemma 3.8.6: A Smooth Rational Point Forces Geometric Irreducibility

Let F be a field and Z an irreducible F-scheme of finite type having an F-rational point at which Z is smooth over F. Then Z is geometrically irreducible, that is, $Z_{\overline{F}}$ is irreducible.

***Proof*:**
Let Z^{sm} be the locus where $Z \to \operatorname{Spec} F$ is smooth. It is open, contains the given point, and is therefore a nonempty open subscheme of the irreducible Z, hence dense, irreducible and connected, with an F-rational point. By proposition 3.4.27 the base change $Z^{\mathrm{sm}}_{\overline{F}}$ is connected. Smoothness is preserved by base change, so $Z^{\mathrm{sm}}_{\overline{F}}$ is smooth over $\overline{F}$, hence regular by proposition 2.7.14, hence the disjoint union of its irreducible components by proposition 2.7.28. A connected scheme with such a decomposition has exactly one component, so $Z^{\mathrm{sm}}_{\overline{F}}$ is irreducible.

It remains to pass from the open subscheme to $Z_{\overline{F}}$. The scheme Z is integral (proposition 2.4.26); work on an affine open $\operatorname{Spec} B \subseteq Z$ meeting Z^{sm}, so B is a domain and $Z^{\mathrm{sm}} \cap \operatorname{Spec} B \supseteq D(f)$ for some nonzero $f \in B$. The ring $B \otimes_F \overline{F}$ is free, hence flat, over B, so the nonzerodivisor f remains a nonzerodivisor in it, and a nonzerodivisor lies in no associated prime, in particular in no minimal prime (proposition 2.5.3). Hence every irreducible component of $\operatorname{Spec}(B \otimes_F \overline{F})$ meets $D(f)_{\overline{F}}$. Running this over an affine cover shows that every irreducible component of $Z_{\overline{F}}$ meets $Z^{\mathrm{sm}}_{\overline{F}}$, in distinct components of it. Since $Z^{\mathrm{sm}}_{\overline{F}}$ is irreducible, $Z_{\overline{F}}$ has exactly one component, as we sought to show.

The propagation lemma needs one input from field theory which the book does not develop, isolated here so that its use is explicit.

Lemma 3.8.7: Separably Generated Function Fields from a Smooth Point

Let F be a field and Z an integral F-scheme of finite type which is smooth over F at some point. Then $F(Z)$ is separably generated over F.

***Proof Key ideas*:**
The argument is short but ends in a field-theoretic fact this book does not prove, so that fact is quoted rather than developed: for a finite extension of fields L/F_0, separability of L over F_0 is equivalent to the vanishing of Ω_{L/F_0} and equivalent to the vanishing of every F_0-derivation of L into L. The implication used below is that a finite extension admitting no nonzero derivation over its base is separable, and the equivalence is proved in *Core Algebra* [12, Derivations Detect Separability].

By definition 2.7.10 the smooth point has a neighbourhood of the form $\operatorname{Spec} F[x_1,\dots,x_n]/(f_1,\dots,f_r)$ on which the Jacobian matrix $(\partial f_i/\partial x_j)$ has rank r, with $n-r=\dim Z$. After renumbering, the minor on the first r columns is invertible at the point, hence on a neighbourhood. Write $F_0 = F(x_{r+1},\dots,x_n)$, a purely transcendental subextension of $L = F(Z)$ of transcendence degree $n-r=\dim Z$, so that L/F_0 is finite. An F_0-derivation δ of L kills $x_{r+1},\dots,x_n$ and satisfies $\sum_{j\le r}(\partial f_i/\partial x_j)\,\delta(x_j)=0$ for each i; invertibility of the matrix forces $\delta=0$. By the quoted fact L/F_0 is separable, so $x_{r+1},\dots,x_n$ is a separating transcendence basis, completing the account.

Lemma 3.8.8: Irreducibility of the General Fiber

Let k be an algebraically closed field, let Z and T be irreducible k-varieties, and let $f\colon Z\to T$ be a dominant morphism with generic fiber Z_η over the generic point η of T. Assume Z_η has a $\kappa(\eta)$-rational point at which it is smooth over $\kappa(\eta)$. Then there is a dense open subset $T^\circ\subseteq T$ such that $f^{-1}(\lambda)$ is nonempty and irreducible for every closed point $\lambda\in T^\circ$.

***Proof*:**
Write $D=\dim Z$, $t=\dim T$, $e=D-t$, $K=\kappa(\eta)=k(T)$ and $L=k(Z)$.

Step 1: the generic fiber is geometrically irreducible. The points of Z lying over η, with the subspace topology, are the underlying space of Z_η, and its irreducible closed subsets are the traces of the irreducible closed subsets of Z dominating T. Since Z is irreducible it is the only maximal such, so Z_η is irreducible with function field L. By hypothesis it has a smooth K-rational point, so lemma 3.8.6 makes it geometrically irreducible and lemma 3.8.7 makes L/K separably generated.

Step 2: shrinking Z is harmless. Shrink T to its smooth locus, dense by theorem 2.7.24. For a closed point $\lambda\in T$ the maximal ideal $\mathfrak{m}_\lambda$ is then generated by t elements, so $f^{-1}(\lambda)$ is locally cut out in Z by t functions and theorem 2.6.13 bounds the codimension of each of its components by t; with theorem 2.6.12 every component of every nonempty fiber has dimension at least e.

Now let $F\subsetneq Z$ be closed. There is a dense open $T_F\subseteq T$ with $\dim(F\cap f^{-1}(\lambda))\le e-1$ for every

closed $\lambda \in T_F$. Induct on $\dim F$, treating the irreducible components separately. For a component F' with $\overline{f(F')} \neq T$, delete $\overline{f(F')}$ from T. Otherwise $f|_{F'}$ is dominant with $\dim F' \leq D-1$, so lemma 3.8.2 gives dense opens $F'^{\circ} \subseteq F'$ and $T'^{\circ} \subseteq T$ with $\dim(F'^{\circ} \cap f^{-1}(\lambda)) \leq \dim F' - t \leq e-1$ for $\lambda \in T'^{\circ}$, while $F' \setminus F'^{\circ}$ has smaller dimension and is handled by induction. Combining, for λ in a dense open subset every irreducible component of $f^{-1}(\lambda)$ meets any prescribed dense open $Z^{\circ} \subseteq Z$, so the components of $f^{-1}(\lambda)$ correspond bijectively to those of $f^{-1}(\lambda) \cap Z^{\circ}$.

Step 3: a hypersurface model. Fix a separating transcendence basis $u_1, \dots, u_e$ of L over K, so that $L/K(u)$ is finite separable, and let θ be a primitive element, scaled to be integral over $K[u]$, with minimal polynomial $P(\Theta) \in K(u)[\Theta]$, monic of degree δ and separable. Choose a nonempty affine open $T_1 \subseteq T$ and put $A = \mathcal{O}_T(T_1)$, so that $K = \operatorname{Frac} A$. Only finitely many elements of K occur among the coefficients of P, so there is a nonzero $\alpha \in A$ with $P \in A_\alpha[u][\Theta]$. Put

$$B' = \frac{A_\alpha[u_1, \dots, u_e][\Theta]}{(P)}$$

Being monic, P is primitive over the unique factorization domain $A_\alpha[u]$ and irreducible over $K(u)$, so Gauss's lemma makes it irreducible in $A_\alpha[u][\Theta]$ and B' a domain with fraction field L.

Both B' and an affine model B'' of a dense open subscheme of Z over T_1 are finitely generated A-algebras with fraction field L, so they agree after inverting one element: writing each generator of one as a fraction of elements of the other and clearing the finitely many denominators produces nonzero h' and h'' with $(B')_{h'} \cong (B'')_{h''}$. Let $Z^{\circ} \subseteq Z$ be the corresponding dense open subscheme. By Step 2 it suffices to prove that the fiber of $\operatorname{Spec}(B')_{h'}$ over a general closed point λ is nonempty and irreducible.

Step 4: specializing the equation. For a closed point $\lambda \in T_1$ with $\alpha(\lambda) \neq 0$, write $P_\lambda \in k[u_1, \dots, u_e][\Theta]$ for the reduction of P, monic of degree δ in Θ; the fiber of $\operatorname{Spec} B'$ over λ is $\operatorname{Spec} k[u][\Theta]/(P_\lambda)$. Separability of P makes its discriminant with respect to Θ a nonzero element of $A_\alpha[u]$, so for λ outside a proper closed subset P_λ is squarefree, and then the fiber is irreducible precisely when P_λ is irreducible in $k[u][\Theta]$.

The generic fiber of $\operatorname{Spec} B'$ over T is $\operatorname{Spec} K[u][\Theta]/(P)$, an open subscheme of Z_η under the identification of Step 3 and therefore geometrically irreducible by Step 1; so P is a power of an irreducible polynomial over $\overline{K}$, and being squarefree it is irreducible over $\overline{K}$.

Fix i with $0 < i < \delta$ and let $\mathcal{M}_i$ be the affine space of polynomials in (u, Θ) monic of degree i in Θ and of total degree at most $\deg P$; a factorization of the monic P_λ into two factors of positive Θ-degree can always be normalized so that both factors are monic in Θ, and their total degrees are then bounded by $\deg P$. Let $\mathcal{Y}_i$ be the closed subscheme of $T_1 \times \mathcal{M}_i \times \mathcal{M}_{\delta-i}$ cut out by the coefficientwise equations $P_\lambda = GH$. If the projection of $\mathcal{Y}_i$ to T_1 had dense image, some irreducible component of $\mathcal{Y}_i$ would have dense image and the generic point ζ of that component would map to η; then P would factor into two factors of positive Θ-degree over the field $\kappa(\zeta) \supseteq K$. Choosing an algebraic closure Ξ of $\kappa(\zeta)$, which contains an algebraic closure of K, the polynomial P would be reducible over Ξ. But $\overline{K}[u][\Theta]/(P)$ is a domain, and a finitely generated domain over an algebraically closed field remains a domain after any extension of the base field, so $\Xi[u][\Theta]/(P)$ is a domain and P is irreducible over Ξ. Hence the projection of each

$\mathcal{Y}_i$ to T_1 has image inside a proper closed subset, and P_λ is irreducible for λ outside a proper closed subset of T.

Step 5: conclusion. Delete from T the proper closed subsets produced in Steps 2 and 4, the zero locus of α, the zero locus of the discriminant, and the complement of the dense open image given by lemma 3.8.2 applied to $\operatorname{Spec}(B')_{h'} \to T$; what remains is a dense open T°. For $\lambda \in T^\circ$ the fiber of $\operatorname{Spec}(B')_{h'}$ is a nonempty open subscheme of the irreducible $\operatorname{Spec} k[u][\Theta]/(P_\lambda)$, hence irreducible, and Step 2 transfers this to $f^{-1}(\lambda)$, completing the proof.

One case of the theorem below resists every argument of this kind. If π is inseparable, for instance the Frobenius $\mathbb{P}^n \to \mathbb{P}^n$ in characteristic p, then the differential of π vanishes identically, no preimage of a hyperplane is smooth anywhere, and no rational point of any generic fiber is a smooth point. The repair is to twist the target by a power of Frobenius until the map becomes generically smooth, which changes neither the topology of the source nor the family of hyperplanes on the target. The field theory behind the twist is isolated next.

Lemma 3.8.9: Generic Smoothness after a Frobenius Twist

Let k be an algebraically closed field of characteristic $p > 0$, let U be an irreducible k-variety, let $\pi\colon U \to \mathbb{P}^r_k$ be a morphism whose image is dense in an irreducible closed subvariety W, and for $s \geq 0$ let $F^s\colon \mathbb{P}^r_k \to \mathbb{P}^r_k$ be the morphism $[X_i] \mapsto [X_i^{p^s}]$. Write $U_{(s)}$ for the reduced closed subscheme underlying $U \times_{\mathbb{P}^r_k, F^s} \mathbb{P}^r_k$. Then $U_{(s)} \to U$ is a homeomorphism for every s, and for all sufficiently large s the induced morphism $U_{(s)} \to \mathbb{P}^r_k$ is smooth at some point.

***Proof Key ideas*:**
The homeomorphism statement is elementary. The morphism F^s is bijective on points, k being perfect, and it is closed, so it is a homeomorphism; its base change along π is again bijective and closed, and passing to the reduced subscheme does not change the underlying space.

The second statement is a field-theoretic assertion this book does not develop, and it is quoted for that reason: for a finitely generated extension L/K in characteristic p, the compositum $L \cdot K^{1/p^s}$ is separably generated over K^{1/p^s} once s is large. Here $K = k(W)$ and $L = k(U)$, and the function field of the twisted target is K^{1/p^s} because k is perfect. The proof compares the modules of Kähler differentials Ω_{L_s/K_s} as s grows: they are generated by the differentials of a fixed finite generating set, the transition maps are surjective, and over the perfect closure the dimension equals the transcendence degree, so the dimensions stabilize there. Kähler differentials are not available at this point in the book, and the limit statement is not developed in it at all; the assertion in this geometric form is [45, Sec. 5], completing the account.

Kollár states the theorem for a map drawn as a dashed arrow but called a morphism. The two are not interchangeable, and for a rational map the symbol $\pi^{-1}(H)$ has to be given a meaning; the statement below fixes a domain of definition and takes preimages there, which is what the proof manipulates. It also weakens dominance to the condition the proof uses, that the image has dimension at least two.

Theorem 3.8.10: Bertini's Irreducibility Theorem

Let k be an infinite field, let X be a geometrically irreducible k-variety, and let $\pi\colon X \dashrightarrow \mathbb{P}^r_k$ be a rational map represented by a morphism $\pi_U\colon U \to \mathbb{P}^r_k$ on a dense open subscheme $U \subseteq X$, and suppose the closure of the image of π_U has dimension at least 2. Then there is a dense open subset $\Omega \subseteq (\mathbb{P}^r_k)^\vee$ such that $\pi_U^{-1}(H_a)$ is geometrically irreducible for every k-rational point $[a] \in \Omega$. Such $[a]$ exist by lemma 3.8.1.

***Proof*:**

Step 1: reduction to an algebraically closed field. The scheme $\pi_U^{-1}(H_a)$ is geometrically irreducible exactly when its base change to $\overline{k}$ is irreducible, and that base change is $(\pi_U)_{\overline{k}}^{-1}\big((H_a)_{\overline{k}}\big)$, preimages commuting with base change. The scheme $X_{\overline{k}}$ is irreducible by hypothesis, $U_{\overline{k}}$ is a dense open subscheme of it, and the closure of the image still has dimension at least 2. Granting the theorem over $\overline{k}$, lemma 3.8.3 converts the dense open subset of $(\mathbb{P}^r_{\overline{k}})^\vee$ it produces into a dense open subset of $(\mathbb{P}^r_k)^\vee$ whose k-rational points work. So assume $k = \overline{k}$, replace X by U, and produce a dense open subset of $(\mathbb{P}^r_k)^\vee$ over whose closed points the preimage is irreducible.

Step 2: normal form. Let $W = \overline{\pi(U)}$ with the reduced induced structure (definition 3.1.26), an irreducible closed subvariety of $\mathbb{P}^r_k$ of dimension $w \geq 2$, and regard π as a dominant morphism $U \to W$; note $r \geq w \geq 2$. Put $D = \dim U$ and $e = D - w$. For a hyperplane H one has $\pi^{-1}(H) = \pi^{-1}(H \cap W)$, and $\pi^{-1}(H)$ is cut out in U by one equation which is not identically zero, so by theorem 2.6.13 and theorem 2.6.12 it is either empty or of pure dimension $D-1$.

Suppose first that π is smooth at some point of U, so that the smooth locus U^{sm} of π is a dense open subscheme; the other case is Step 6. Shrinking W to its smooth locus (theorem 2.7.24) and applying lemma 3.8.2 to $\pi|_{U^{\mathrm{sm}}}$, fix dense opens $U^\circ \subseteq U^{\mathrm{sm}}$ and $W^\circ \subseteq W$ with $\pi(U^\circ) = W^\circ$, with W° inside the smooth locus of W, and with a finite surjective $\psi\colon U^\circ \to W^\circ \times \mathbb{A}^e$ over W°.

Discarding $U \setminus U^\circ$ removes only a proper closed subset of hyperplanes. Indeed, let $F_1, \dots, F_c$ be the irreducible components of $U \setminus U^\circ$ of dimension $D-1$; the others cannot be components of a fiber, by the purity just recorded. A component F_i is contained in $\pi^{-1}(H_a)$ exactly when $\pi(F_i) \subseteq H_a$, and the $[a]$ with $\pi(F_i) \subseteq H_a$ form a proper linear subspace of $(\mathbb{P}^r_k)^\vee$, since F_i is nonempty. Deleting those finitely many subspaces, every component of $\pi^{-1}(H_a)$ meets U°, so $\pi^{-1}(H_a)$ is irreducible if and only if $\pi^{-1}(H_a) \cap U^\circ$ is irreducible and nonempty.

Step 3: the incidence variety and its diagonal section. Put

$$\mathcal{Q} = \{(x, [a]) \in U^\circ \times (\mathbb{P}^r_k)^\vee \mid \pi(x) \in H_a\}, \qquad \mathcal{J} = \{(y, x, [a]) \in U^\circ \times \mathcal{Q} \mid \pi(y) \in H_a\}$$

both closed in the ambient products. The projection $\mathcal{Q} \to U^\circ$ has all fibers equal to the hyperplane of $(\mathbb{P}^r_k)^\vee$ cut out by the single nonzero linear condition $\sum_i a_i \pi_i(x) = 0$, and on the open subscheme of U° where a fixed homogeneous coordinate of π is invertible one may solve for the corresponding a_i, exhibiting $\mathcal{Q}$ there as a product with $\mathbb{P}^{r-1}$. These opens cover U° and pairwise meet, so $\mathcal{Q}$ is irreducible of dimension $D + r - 1$. The fiber of $\mathcal{J}$ over a point $(x, [a])$ of $\mathcal{Q}$ is $\pi^{-1}(H_a) \cap U^\circ$, which is what the theorem is about, and

$$\varsigma\colon \mathcal{Q} \longrightarrow \mathcal{J}, \qquad (x, [a]) \longmapsto (x, x, [a])$$

is a section of $\mathcal{J} \to \mathcal{Q}$, because $\pi(x) \in H_a$ by the definition of $\mathcal{Q}$. This tautological section replaces the base point of a pencil in Kollár's argument, and it is available over the whole of $\mathcal{Q}$ rather than over a single line in $(\mathbb{P}^r_k)^\vee$.

Step 4: $\mathcal{J}$ has a unique component dominating $\mathcal{Q}$. Let $\mathcal{J}^{\neq}$ be the open subscheme of $\mathcal{J}$ where $\pi(y) \neq \pi(x)$ and $\mathcal{J}^{=}$ its closed complement. The projection of $\mathcal{J}^{\neq}$ to $U^\circ \times U^\circ$ lands in the open subscheme where $\pi(y) \neq \pi(x)$, which is nonempty because $\pi(U^\circ) = W^\circ$ has dimension $w \geq 1$, and which is irreducible of dimension $2D$ by proposition 3.7.10. Its fibers are the linear subspaces of $(\mathbb{P}^r_k)^\vee$ cut out by the two independent conditions $\sum_i a_i\pi_i(y) = \sum_i a_i\pi_i(x) = 0$, that is, copies of $\mathbb{P}^{r-2}$, and the same solving-for-coordinates argument trivializes them over an open cover. Hence $\mathcal{J}^{\neq}$ is irreducible of dimension $2D + r - 2$.

For $\mathcal{J}^{=}$, the locus $\{(y, x) \in U^\circ \times U^\circ \mid \pi(y) = \pi(x)\}$ is $U^\circ \times_{W^\circ} U^\circ$, which maps finitely to $(W^\circ \times \mathbb{A}^e) \times_{W^\circ} (W^\circ \times \mathbb{A}^e) = W^\circ \times \mathbb{A}^{2e}$ under $\psi \times \psi$; so lemma 2.6.3 bounds its dimension by $w + 2e = 2D - w$. Its fibers in $\mathcal{J}^{=}$ are copies of $\mathbb{P}^{r-1}$, whence

$$\dim \mathcal{J}^{=} \leq 2D - w + r - 1 \leq 2D + r - 3 < \dim \mathcal{J}^{\neq}$$

the middle inequality being where the hypothesis $w \geq 2$ is spent. For $w = 1$ the two dimensions agree and the theorem is false, a general hyperplane then meeting W in $\deg W$ points and $\pi^{-1}(H)$ being a union of that many fibers.

Let $\mathcal{J}_0 = \overline{\mathcal{J}^{\neq}}$, irreducible of dimension $2D + r - 2$. For $(x, [a])$ in $\mathcal{Q}$ outside the proper closed subset over which some component of $\mathcal{J}^{=}$ not contained in $\mathcal{J}_0$ lies, the fiber $\pi^{-1}(H_a) \cap U^\circ$ is pure of dimension $D - 1$ and meets $\pi^{-1}(\pi(x))$ in dimension at most $e \leq D - 2$ by lemma 3.8.2, so the locus $\pi(y) \neq \pi(x)$ is dense in that fiber and the whole fiber lies in $\mathcal{J}_0$. A component of $\mathcal{J}$ dominating $\mathcal{Q}$ therefore has a dense subset inside the closed set $\mathcal{J}_0$, hence is contained in it and equals it. Consequently the generic fiber of $\mathcal{J} \to \mathcal{Q}$ is irreducible and coincides with that of $\mathcal{J}_0 \to \mathcal{Q}$; and ς maps a dense subset of $\mathcal{Q}$, hence all of $\mathcal{Q}$, into the closed set $\mathcal{J}_0$.

Step 5: the section point is smooth, and conclusion. The scheme U° is smooth over k, being smooth over the smooth W°. Fix a closed point $x \in U^\circ$, an index i_0 with $\pi_{i_0}(x) \neq 0$, and repeat the computation of Step 1 of the proof of theorem 3.8.4 with the regular functions π_i/π_{i_0} in place of the coordinates X_i/X_{i_0}: the map

$$\epsilon_x \colon k^{r+1} \longrightarrow \mathcal{O}_{U^\circ,x}/\mathfrak{m}_x^2, \qquad a \longmapsto \Big[\sum_i a_i\, \pi_i/\pi_{i_0}\Big]$$

is k-linear, and $\pi^{-1}(H_a) \cap U^\circ$ is singular at x exactly when a lies in $W_x := \ker \epsilon_x$, by the same comparison of the cotangent space with the dimension. Here ϵ_x factors through $\mathcal{O}_{W,\pi(x)}/\mathfrak{m}^2_{\pi(x)}$, of dimension $w + 1$, and the induced map on that quotient is injective because π is smooth at x; so the image of ϵ_x has dimension $w + 1$ and $\dim_k W_x = r - w$. The hyperplanes through $\pi(x)$ form a linear subspace of $(\mathbb{P}^r_k)^\vee$ of dimension $r-1$, whereas $\mathbb{P}(W_x)$ has dimension $r-w-1 < r-1$ since $w \geq 2$; so for each fixed x the singular condition is proper on the fiber of $\mathcal{Q}$ over x.

The condition $a \in W_x$ is closed on $\mathcal{Q}$, being the vanishing of the minors of a Jacobian matrix augmented by the row a, exactly as in Step 2 of the proof of theorem 3.8.4 applied to a local

presentation of U° together with the functions π_i/π_{i_0}. Being closed and containing no fiber of $\mathcal{Q} \to U^\circ$, it is a proper closed subset, so its complement is a nonempty open subset of the irreducible $\mathcal{Q}$ and contains the generic point θ. Hence the generic fiber of $\mathcal{J}_0 \to \mathcal{Q}$ has the $\kappa(\theta)$-rational point $\varsigma(\theta)$ and is smooth there over $\kappa(\theta)$.

Lemma 3.8.8 now applies to $\mathcal{J}_0 \to \mathcal{Q}$ and yields a dense open $\mathcal{Q}^\circ \subseteq \mathcal{Q}$ over whose closed points the fiber is irreducible; intersecting with the dense open of Step 4 on which the fiber of $\mathcal{J}_0$ agrees with that of $\mathcal{J}$, the scheme $\pi^{-1}(H_a) \cap U^\circ$ is irreducible for every closed point $(x, [a]) \in \mathcal{Q}^\circ$. The projection $\mathcal{Q} \to (\mathbb{P}^r_k)^\vee$ is dominant: a hyperplane meets the projective variety W of positive dimension, because the affine cone over W has dimension $w + 1 \geq 3$ and meets the hyperplane $\sum_i a_i X_i = 0$ in dimension at least $w \geq 2$ by theorem 2.6.13, hence in more than the origin; and the $[a]$ for which $H_a \cap W$ avoids W° lie in the finitely many proper linear subspaces of hyperplanes containing a fixed $(w-1)$-dimensional component of $W \setminus W^\circ$. Applying lemma 3.8.2 to $\mathcal{Q}^\circ \to (\mathbb{P}^r_k)^\vee$ produces a dense open subset of $(\mathbb{P}^r_k)^\vee$ contained in the image, and by Step 2 the preimage $\pi^{-1}(H_a)$ itself is irreducible over that subset.

Step 6: the inseparable case. Suppose π is smooth at no point of U, so $\operatorname{char} k = p > 0$. Lemma 3.8.9 gives s with $\pi_{(s)} \colon U_{(s)} \to \mathbb{P}^r_k$ smooth at a point, where $U_{(s)}$ is the reduced scheme underlying $U \times_{\mathbb{P}^r_k, F^s} \mathbb{P}^r_k$; the morphism $U_{(s)} \to U$ is a homeomorphism, so $U_{(s)}$ is an irreducible k-variety and a closed subset of U is irreducible exactly when its preimage is. For a hyperplane H_a one has $(F^s)^{-1}(H_a) = H_{a'}$ as sets, with $a'_i = a_i^{p^{-s}}$, again a hyperplane, and $[a] \mapsto [a']$ is a homeomorphism of $(\mathbb{P}^r_k)^\vee$ onto itself, inverse to the Frobenius of the dual space. Under the homeomorphism $U_{(s)} \to U$ the closed set $\pi_{(s)}^{-1}(H_{a'})$ corresponds to $\pi^{-1}(H_a)$. The closure of the image of $\pi_{(s)}$ is $(F^s)^{-1}(W)$, of the same dimension $w \geq 2$. Steps 2 to 5 applied to $\pi_{(s)}$ give a dense open subset of $(\mathbb{P}^r_k)^\vee$, whose preimage under $[a] \mapsto [a']$ is again dense open and serves as Ω. Irreducibility over the algebraically closed k is geometric irreducibility, and Step 1 transports the conclusion back to the original ground field, completing the proof.

3.8.11 Warning: Two Different Bertini Irreducibility Theorems Theorem 3.7.13 and theorem 3.8.10 are different statements, and neither implies the other. The first fixes a subvariety $X \subseteq \mathbb{P}^n_k$ and a finite set S of closed points of it, and asserts that a general hypersurface of degree $m > |S|$ *through* S cuts X irreducibly; the divisor moves in the incomplete system $\Lambda_m(S)$ of degree-m forms vanishing on S, and the content is that the constraint imposed by S does not destroy irreducibility.

The second fixes a map π to a projective space and lets the hyperplane be entirely unconstrained. The variety being cut is the source of π, which need not be embedded in the target at all, and the divisor cutting it is the preimage $\pi_U^{-1}(H)$, which can be far from a hypersurface section of anything.

Neither direction of implication holds. Taking π to be a closed immersion $X \hookrightarrow \mathbb{P}^n_k$ and $m = 1$, theorem 3.8.10 gives irreducibility of $X \cap H$ for a general hyperplane, which theorem 3.7.13 cannot deliver, its hypothesis $m > |S| \geq 1$ excluding hyperplanes; and example 3.7.12 shows the exclusion is not an artifact. Conversely theorem 3.8.10 says nothing about members of a proper subsystem such as $\Lambda_m(S)$: a dense open subset of $(\mathbb{P}^n)^\vee$ can miss a fixed linear subspace of it entirely, so generality in the full system carries no information about generality in a subsystem. That failure is not a defect of the statement but the reason theorem 3.8.20 below is phrased with codimension bounds rather than with a single dense open subset. A reader who reaches for one theorem where the other is needed will find the hypotheses do not match.

Iterating theorem 3.8.10 together with theorem 3.8.4 cuts a variety down to a curve.

Corollary 3.8.12: General Linear Sections of a Smooth Projective Variety

Let k be an infinite field and $Y \subseteq \mathbb{P}^M_k$ a geometrically irreducible projective k-variety of dimension $n \geq 1$ which is smooth over k. Then for general hyperplanes $H_1, \ldots, H_{n-1}$ the scheme $Y \cap H_1 \cap \cdots \cap H_{n-1}$ is a geometrically irreducible curve, smooth over k.

***Proof*:**
Induct on n, the case $n = 1$ being the hypothesis. Let $n \geq 2$ and apply theorem 3.8.10 to the closed immersion $Y \hookrightarrow \mathbb{P}^M_k$, whose image has dimension $n \geq 2$, and theorem 3.8.4 to $Y_{\overline{k}}$. Intersecting the two dense open subsets produced, after passing the second through lemma 3.8.3, gives a dense open subset of $(\mathbb{P}^M_k)^\vee$ over whose k-rational points $Y \cap H_1$ is geometrically irreducible and smooth of dimension $n-1$. Lemma 3.8.1 gives such an H_1 over the infinite field k, and the induction hypothesis applies to $Y \cap H_1$, completing the proof.

Codimension bounds in the Grassmannian

The corollary just proved is not strong enough for the applications that motivate the section. There the linear sections are not free to be fully general: they are required to pass through prescribed points, or they arise as the fibers of a fixed linear projection, and in both cases they range over a proper subvariety of the parameter space. A statement of the form "the bad locus is contained in a proper closed subset" is then useless, since the subvariety one is confined to may lie inside the bad

locus. What is needed is a bound on the codimension of the bad locus, which survives intersection with subvarieties of small enough codimension.

The parameter space is a Grassmannian. Fix a projective variety $Y \subseteq \mathbb{P}^M_k$ of dimension $n \geq 2$ and an integer $m \geq 1$, and let

$$V_m = \frac{k[X_0, \dots, X_M]_m}{k[X_0, \dots, X_M]_m \cap I(Y)}$$

be the space of degree-m forms restricted to Y; its projectivization $|H| = \mathbb{P}(V_m)$ is the linear system of degree-m hypersurface sections of Y. Write $\mathrm{Grass}(j, V_m)$ for the Grassmannian of j-dimensional linear subspaces of the vector space V_m; its construction and projectivity are in *Classical Algebraic Geometry* [22, Grassmannian, The Grassmannian Is a Projective Variety], which also gives the standard affine charts used below. The relevant subspaces have *vector* dimension $n-1$, that is, projective dimension $n-2$ inside $|H|$: a subspace of vector dimension $n-1$ is spanned by $n-1$ hypersurface sections, whose common zero locus on the n-dimensional Y is a curve. Subspaces of projective dimension $n-1$ would cut out a finite set instead.

Definition 3.8.13: The Universal Family of Curve Sections

With Y, m and V_m as above, the *universal family of curve sections* is

$$\mathbf{C} = \{(y, L) \in Y \times \mathrm{Grass}(n-1, V_m) \mid G(y) = 0 \text{ for every } G \in L\}$$

a closed subscheme of $Y \times \mathrm{Grass}(n-1, V_m)$. Its fiber over a point L is written C_L; as a set it is the intersection of all the members of the linear system $\mathbb{P}(L) \subseteq |H|$.

One general fact about Grassmannians is isolated first, because it is what makes the codimension-two bound work in every dimension. The argument below cuts Y by one hypersurface at a time and has to control the subspaces L none of whose members is good. Those L are exactly the linear subspaces of $|H|$ lying inside a fixed proper closed subset, and the following count says that such subspaces are rare, with a codimension that grows with the dimension of the subspace. Nothing about linear systems enters.

Lemma 3.8.14: Linear Subspaces Contained in a Closed Set

Let k be an algebraically closed field, let $Z \subsetneq \mathbb{P}^M_k$ be a closed subset and let $r \geq 1$. Write $\mathrm{Grass}(r, \mathbb{P}^M_k)$ for the Grassmannian of r-dimensional linear subspaces of $\mathbb{P}^M_k$, of dimension $(r+1)(M-r)$, and put

$$\mathbb{F}_r(Z) = \left\{\Lambda \in \mathrm{Grass}(r, \mathbb{P}^M_k) \;\middle|\; \Lambda \subseteq Z\right\}$$

Then $\mathbb{F}_r(Z)$ is closed and

$$\mathrm{codim}\,\mathbb{F}_r(Z) \geq (r+1)\,\mathrm{codim}\,Z \geq r+1$$

with $\mathrm{codim}\,Z = M - \dim Z$ and the convention that the empty set has infinite codimension. The bound is attained when Z is a hyperplane.

***Proof*:**
Step 0: closedness. Let $\mathcal{A} = \{(x, \Lambda) \mid x \in \Lambda\}$ be the tautological incidence set, closed in $\mathbb{P}^M_k \times \mathrm{Grass}(r, \mathbb{P}^M_k)$. Over each standard chart of the Grassmannian [22, The Grassmannian Is a Projective Variety] the projection $\mathcal{A} \to \mathrm{Grass}(r, \mathbb{P}^M_k)$ is the projection of a trivial $\mathbb{P}^r$-bundle, hence an open map. The complement of $\mathbb{F}_r(Z)$ is the image of the open subset of $\mathcal{A}$ where $x \notin Z$, so it is open.

Step 1: reduction to Z irreducible. Let $Z_1, \dots, Z_s$ be the irreducible components of Z. A linear subspace is irreducible, so $\Lambda \subseteq Z$ forces $\Lambda \subseteq Z_i$ for some i, and $\mathbb{F}_r(Z) = \bigcup_i \mathbb{F}_r(Z_i)$ while $\operatorname{codim} Z = \min_i \operatorname{codim} Z_i$. It therefore suffices to treat Z irreducible, taken with its reduced induced structure (definition 3.1.26).

Step 2: induction on $\dim Z$. Induct on $\dim Z$. If $\dim Z < r$ then $\mathbb{F}_r(Z) = \emptyset$ and there is nothing to prove; since $r \geq 1$ this covers every Z of dimension at most 0 and starts the induction. Assume now $\dim Z = M - c$ with $1 \leq c \leq M - r$, let Z^{sm} be the smooth locus of Z, a dense open subset by theorem 2.7.24, and let $\operatorname{Sing} Z = Z \setminus Z^{\mathrm{sm}}$, closed (proposition 2.7.23) of dimension strictly less than $\dim Z$. Split

$$\mathbb{F}_r(Z) = \mathbb{F}' \cup \mathbb{F}_r(\operatorname{Sing} Z), \qquad \mathbb{F}' = \{\Lambda \subseteq Z \mid \Lambda \cap Z^{\mathrm{sm}} \neq \emptyset\}$$

The second piece is handled by the inductive hypothesis: it has codimension at least $(r+1)\big(M - \dim \operatorname{Sing} Z\big) \geq (r+1)(c+1) > (r+1)c$.

Step 3: a subspace through a smooth point lies in the tangent space. For $x \in Z^{\mathrm{sm}}$ let $\mathbb{T}_x Z \subseteq \mathbb{P}^M_k$ be the embedded tangent space: choosing affine coordinates centred at x, it is the common zero locus of the degree-one parts at x of the dehomogenizations of the forms in $I(Z)$. It is a linear subspace, and proposition 2.7.7 gives $\dim \mathbb{T}_x Z = \dim Z = M - c$ because x is a smooth point.

Let $\Lambda \in \mathbb{F}'$ and $x \in \Lambda \cap Z^{\mathrm{sm}}$. Every line $\ell \subseteq \Lambda$ through x lies in Z, so each $F \in I(Z)$ vanishes identically on ℓ; in the coordinates above the degree-one part of F at x then vanishes on the direction of ℓ, that is, $\ell \subseteq \mathbb{T}_x Z$. Since Λ is the union of the lines through x that it contains, $\Lambda \subseteq \mathbb{T}_x Z$.

Step 4: the incidence count. Put

$$\mathcal{I} = \{(x, \Lambda) \in Z^{\mathrm{sm}} \times \mathbb{F}' \mid x \in \Lambda\}$$

a locally closed subset of $\mathbb{P}^M_k \times \mathrm{Grass}(r, \mathbb{P}^M_k)$ and so a k-variety with finitely many irreducible components. By Step 3 the fiber of $\mathcal{I}$ over a point $x \in Z^{\mathrm{sm}}$ is contained in the set of r-dimensional subspaces of $\mathbb{T}_x Z \cong \mathbb{P}^{M-c}_k$ passing through x; those correspond to the $(r-1)$-dimensional subspaces of the quotient $\mathbb{P}^{M-c-1}_k$, so the fiber has dimension at most $r(M-c-r)$. Applying lemma 3.8.2 to each irreducible component of $\mathcal{I}$ over the closure of its image in Z^{sm},

$$\dim \mathcal{I} \leq (M - c) + r(M - c - r)$$

In the other direction let $\mathbb{F}'_0$ be an irreducible component of the locally closed set $\mathbb{F}'$. The

restriction of $\mathcal{A}$ to $\mathbb{F}_0'$ is a $\mathbb{P}^r$-bundle over $\mathbb{F}_0'$, hence irreducible of dimension $\dim \mathbb{F}_0' + r$, and $\mathcal{I}$ meets it in the open subset where $x \in Z^{\mathrm{sm}}$, which is nonempty by the definition of $\mathbb{F}'$ and therefore dense. Hence $\dim \mathbb{F}_0' + r \leq \dim \mathcal{I}$, and

$$\dim \mathbb{F}' \leq (M-c) + r(M-c-r) - r$$

Step 5: conclusion. Subtracting from $\dim \mathrm{Grass}(r, \mathbb{P}_k^M) = (r+1)(M-r)$ and expanding,

$$\mathrm{codim}\mathbb{F}' \geq (r+1)(M-r) - (M-c) - r(M-c-r) + r = c(r+1)$$

Combining with the bound of Step 2 for $\mathbb{F}_r(\mathrm{Sing}\, Z)$ gives $\mathrm{codim}\mathbb{F}_r(Z) \geq c(r+1) = (r+1)\mathrm{codim} Z$, and $c \geq 1$ gives the second inequality.

For sharpness take Z a hyperplane, so that $c = 1$ and $\mathbb{F}_r(Z) = \mathrm{Grass}(r, \mathbb{P}_k^{M-1})$, of dimension $(r+1)(M-1-r)$ and hence of codimension exactly $r+1$, completing the proof.

The case $r = 1$, the statement that the lines inside a proper closed subset form a family of codimension at least twice the codimension of that subset, is classical: it goes back to Segre in 1948 and is proved in this form by Rogora, *Manuscripta Mathematica* **82** (1994). The general case is proved above because the induction below needs it for $r = n-2$, and the growth of the bound with r is exactly what a one-shot reduction to surfaces cannot give.

A piece of dimension bookkeeping comes next, because it is what the applications use. Put $\nu = \dim_k V_m$, so that $\mathrm{Grass}(n-1, V_m)$ has dimension $(n-1)(\nu - n + 1)$. If p and q are distinct closed points of Y and $V_m(-p-q) \subseteq V_m$ is the subspace of forms vanishing at both, then $V_m(-p-q)$ has codimension 2 as soon as the degree-m forms separate p from q, and the subvariety

$$\mathrm{Grass}\left(n-1, V_m(-p-q)\right) \subseteq \mathrm{Grass}(n-1, V_m)$$

of subspaces all of whose members pass through p and q has codimension exactly $2(n-1)$. This is the locus the two-point application is confined to, and $2(n-1) \geq 2$ for $n \geq 2$. A codimension bound of 2, or even of 1, in the ambient Grassmannian therefore says nothing about whether the bad locus avoids this subvariety, and the two-point statement has to be obtained by applying the theorem to the constrained system instead; corollary 3.8.24 does that.

Four further inputs are needed before the theorem, and it is worth being explicit about which are proved here and which are quoted.

The first is that the bad locus is constructible. Every bound below is obtained by exhibiting a bad set as the image of an incidence variety and comparing dimensions, and that comparison is valid for constructible sets and false for arbitrary ones. The failure is not hypothetical: a subset of $\mathbb{P}^2$ meeting every line in exactly two points exists, by transfinite induction, and is Zariski dense, so knowing that a set meets every member of a covering family in a small set says nothing about its dimension. The statement below is the one place in this section where a result from the theory of constructible sets is imported, and it is imported explicitly.

Lemma 3.8.15: Constructibility of the Geometrically Irreducible Locus

Let $f\colon X \to T$ be a morphism of finite type between Noetherian schemes. Then

$$\{t \in T \mid X_t \text{ is geometrically irreducible}\}$$

is a constructible subset of T, and so are the corresponding loci for "geometrically reduced" and for "of dimension at most d" for each fixed d.

***Proof Key ideas*:**
The three statements are proved together in EGA by a limit argument that reduces to the case of a finitely presented morphism over a Noetherian base and then constructs a stratification of T over which the geometry of the fibers is constant; the reference is [32, IV_3, 9.7.7], and [70] has a modern account. The constructibility of the dimension loci is the semicontinuity of fiber dimension. None of this is developed in this book, and the statement is used only through the following consequence: if $E \subseteq T$ is constructible then $\dim \overline{E} = \dim E$, and the image of a constructible set under a morphism is constructible, so the fiber-dimension bound of lemma 3.8.2 may be applied to E by passing to the irreducible components of $\overline{E}$, on each of which E contains a dense open subset.

The second input is connectedness. A reducible member of a linear system on a surface is only useful if its two halves meet, and that is what turns reducibility into a singularity and so into a codimension count. Kollár's Corollary 24 attributes the step to Bertini's irreducibility theorem, which does not give it: Bertini controls a general member, while the claim quantifies over every reducible member. Connectedness is the input the step uses, and it is proved here from a pencil.

Lemma 3.8.16: Connectedness of the Fibers of a Family over a Curve

Let k be an algebraically closed field and let $q\colon \Gamma \to T$ be a proper surjective morphism from an irreducible k-variety onto a smooth irreducible k-curve. Suppose $q^{-1}(t)$ is irreducible for every closed point t in some dense open subset of T. Then $q^{-1}(t)$ is connected for every closed point $t \in T$.

***Proof Key ideas*:**
This is Zariski's connectedness theorem. What is quoted is the existence of the Stein factorization of q, a factorization $\Gamma \to T' \to T$ with $T' \to T$ finite and $\Gamma \to T'$ surjective with connected fibers; see [70, Tag 03H0]. It rests on the coherence of $q_*\mathcal{O}_\Gamma$ and on the theorem on formal functions, neither of which is available at this point in the book, and it is the same input that forced theorem 3.4.38 to be quoted rather than proved.

Granting it, the rest is short and is recorded so that the size of the import is explicit. The scheme T' is integral because Γ is and $\Gamma \to T'$ is surjective with connected fibers, and the number of connected components of $q^{-1}(t)$ equals the number of points of T' over t. If the general fiber of q is irreducible, then $T' \to T$ is injective over a dense open subset of T, hence purely inseparable, hence a homeomorphism; when it has degree one it is an isomorphism by corollary 3.4.39, the target T being a smooth curve and therefore normal. Either way $T' \to T$ is a bijection on points, so every fiber of q is connected, completing the account.

Lemma 3.8.17: Members of a Very Ample System Are Connected

Let k be an algebraically closed field and let $S \subseteq \mathbb{P}^N_k$ be an integral projective k-variety with $\dim S \geq 2$. Then $S \cap V_+(F)$ is connected for every homogeneous form F that does not vanish identically on S.

***Proof*:**
Write $d = \dim S$, let $m = \deg F$ and let $C = S \cap V_+(F)$, of pure dimension $d-1$ by lemma 3.7.11. Under the m-th Veronese re-embedding (example 0.2.27) the degree-m hypersurface sections of S become hyperplane sections, so theorem 3.8.10, applied to the resulting closed immersion of S, whose image has dimension $d \geq 2$, gives a dense open set of degree-m forms G with $S \cap V_+(G)$ irreducible. Shrinking it further, require also that $V_+(G)$ contain no irreducible component of C, which excludes finitely many proper linear subspaces, and that G be independent of F modulo $I(S)$. If no such G exists then F itself already has $S \cap V_+(F)$ irreducible and there is nothing to prove; fix one otherwise.

Step 1: the total space of the pencil. Put

$$\Gamma = \big\{(x, [\lambda : \mu]) \in S \times \mathbb{P}^1_k \;\big|\; \lambda F(x) + \mu G(x) = 0\big\}$$

with its reduced induced structure (definition 3.1.26), a closed subscheme of $S \times \mathbb{P}^1_k$, and let $q\colon \Gamma \to \mathbb{P}^1_k$ be the second projection. The fiber of q over $[\lambda : \mu]$ is $S \cap V_+(\lambda F + \mu G)$, so q is surjective, and q is proper because Γ is closed in $S \times \mathbb{P}^1_k$ and S is proper over k (proposition 3.5.23 and theorem 3.5.32). The base locus $B = C \cap V_+(G)$ has dimension $d-2$, since $V_+(G)$ contains no component of the pure $(d-1)$-dimensional C, and over $S \setminus B$ the projection $\Gamma \to S$ is bijective, exactly one ratio $[\lambda : \mu]$ solving the equation.

The scheme Γ is cut out in the $(d+1)$-dimensional $S \times \mathbb{P}^1_k$ by a single equation which is not identically zero, so every irreducible component of it has dimension d by theorem 2.6.13 and theorem 2.6.12. The closed subset $B \times \mathbb{P}^1_k$ has dimension $d-1$, so no component of Γ lies inside it; every component therefore meets $\Gamma \setminus (B \times \mathbb{P}^1_k) \cong S \setminus B$, which is irreducible, and Γ is irreducible.

Step 2: conclusion. The map $[\lambda : \mu] \mapsto [\lambda F + \mu G]$ identifies $\mathbb{P}^1_k$ with the line in the space of degree-m hypersurface sections spanned by F and G, and that line meets the dense open subset of irreducible members in the point $[G]$, hence in a dense open subset of $\mathbb{P}^1_k$. So the fiber of q over a general closed point is irreducible, and lemma 3.8.16 applies: every fiber of q is connected. Taking the fiber over $[1 : 0]$ gives that C is connected, completing the proof.

The third input is what makes the induction close. Cutting Y by one member D of the system produces a variety of one dimension less, and the whole argument depends on the linear system on that variety being the quotient of V_m by the line spanned by D, with nothing else collapsing. That is the content of the next lemma, and it is where regularity in codimension one is spent.

Lemma 3.8.18: One Cut Removes Exactly One Dimension

Let k be an algebraically closed field and let $Y \subseteq \mathbb{P}^M_k$ be an integral projective k-variety of dimension $n \geq 2$ which is regular in codimension one. Let F be a degree-m form not vanishing on Y and suppose the scheme $Y \cap V_+(F)$ is irreducible and generically reduced; write Y_F for it with its reduced induced structure (definition 3.1.26). Then the restriction map

$$V_m(Y) \longrightarrow V_m(Y_F)$$

is surjective with kernel the line spanned by the class of F. In particular $\dim_k V_m(Y_F) = \dim_k V_m(Y) - 1$.

***Proof*:**

Both spaces are quotients of the space of degree-m forms on $\mathbb{P}^M_k$, so the map is surjective, and F vanishes on Y_F, so the class of F lies in the kernel; it is nonzero because F does not vanish on Y.

For the converse let G be a degree-m form vanishing on Y_F. Since Y is regular in codimension one, the local ring at every codimension-one point y of Y is a discrete valuation ring, and after trivializing degree-m forms near y by a form not vanishing at y one obtains order functions ord_y. Let y_0 be the generic point of Y_F. The scheme $Y \cap V_+(F)$ is irreducible, so Y_F is the only codimension-one subvariety of Y on which F vanishes, and it is generically reduced, so $\mathcal{O}_{Y,y_0}/(F)$ is a field and F generates the maximal ideal of the discrete valuation ring $\mathcal{O}_{Y,y_0}$. Hence $\operatorname{ord}_{y_0}(F) = 1$ and $\operatorname{ord}_y(F) = 0$ for every other codimension-one point y, while $\operatorname{ord}_{y_0}(G) \geq 1$ and $\operatorname{ord}_y(G) \geq 0$ throughout. The ratio $\varphi = G/F$ is therefore a rational function on Y with $\operatorname{ord}_y(\varphi) \geq 0$ at every codimension-one point.

Let $\pi\colon \widetilde{Y} \to Y$ be the normalization (theorem 3.4.41), a finite birational morphism from an integral normal projective variety which is an isomorphism over the normal locus of Y. Regularity in codimension one puts every codimension-one point of Y in that locus, and a finite morphism preserves dimensions, so the codimension-one points of $\widetilde{Y}$ correspond bijectively to those of Y with matching order functions. Hence $\pi^*\varphi$ has nonnegative order at every codimension-one point of the normal $\widetilde{Y}$, and a normal domain is the intersection of its localizations at the height-one primes (*Commutative Algebra* [10, A Normal Domain Is the Intersection of Its Height-One Localizations]), so $\pi^*\varphi$ is a global regular function on $\widetilde{Y}$. That variety is integral and proper over the algebraically closed k, so theorem 3.5.33 gives $\pi^*\varphi = c \in k$, and π is birational, so $\varphi = c$. Then $G - cF$ vanishes at the generic point of Y, hence on Y, and the class of G in $V_m(Y)$ is c times that of F, completing the proof.

The fourth and last input is the dichotomy at a smooth point. On a surface, a member of the system that is singular at a prescribed smooth point s costs three conditions, and only a two-dimensional family of choices of s is available to pay for them; the surplus of one is what makes reducibility a codimension-two condition rather than a codimension-one one, provided a general such member is still irreducible. That proviso fails exactly when the system of members singular at s maps the surface to a curve, and the following lemma says how rarely that happens.

Lemma 3.8.19: The Second-Order Projection Has Two-Dimensional Image

Let k be an algebraically closed field, let $S \subseteq \mathbb{P}^N_k$ be an integral projective surface and let $m \geq 2$. Write V_m for the space of degree-m forms restricted to S and $\nu = \dim_k V_m$; for a smooth closed point $s \in S$ let $V_m(-2s) \subseteq V_m$ be the subspace of forms whose restriction lies in $\mathfrak{m}_s^2$, and let $\psi_s \colon S \dashrightarrow \mathbb{P}\big(V_m(-2s)^\vee\big)$ be the associated rational map. If the closure of the image of ψ_s is a curve for every s in a dense open subset of the smooth locus of S, then $\nu = 6$ and $m = 2$.

***Proof*:**
Step 1: the system of singular members. Fix a smooth closed point s. The local ring $\mathcal{O}_{S,s}$ is regular of dimension 2, so $\mathcal{O}_{S,s}/\mathfrak{m}_s^3$ has dimension $1 + 2 + 3 = 6$ over k. Restriction of linear forms surjects onto $\mathfrak{m}_s/\mathfrak{m}_s^2$, the closed immersion $S \subseteq \mathbb{P}^N_k$ making $\mathcal{O}_{\mathbb{P}^N,s} \to \mathcal{O}_{S,s}$ surjective, so products of j linear forms vanishing at s span $\mathfrak{m}_s^j/\mathfrak{m}_s^{j+1}$. Multiplying such products by a power of a linear form not vanishing at s shows that $V_m \to \mathcal{O}_{S,s}/\mathfrak{m}_s^3$ is surjective for $m \geq 2$. Hence $\nu \geq 6$, the subspace $V_m(-2s)$ has codimension 3, and the members of $\mathbb{P}(V_m(-2s))$ whose restriction lies in $\mathfrak{m}_s^3$ form a proper subspace, so a general member has multiplicity exactly 2 at s. Here the multiplicity $\operatorname{mult}_s M$ of an effective divisor M on S at s is the largest j with the local equation of M in $\mathfrak{m}_s^j$.

The base locus of $\mathbb{P}(V_m(-2s))$ is $\{s\}$: given a closed point $y \neq s$, choose linear forms σ, τ vanishing at s and not at y and a degree-$(m-2)$ form not vanishing at y; the product lies in $V_m(-2s)$ and does not vanish at y, which uses $m \geq 2$. So ψ_s is defined on $S \setminus \{s\}$ and has no fixed component.

Step 2: every fiber of ψ_s passes through s. Assume the closure Γ_s of the image of ψ_s is a curve, and let E be the closure of a fiber, an irreducible curve in S. If $s \notin E$, choose a hyperplane of $\mathbb{P}(V_m(-2s)^\vee)$ missing the point of Γ_s below E; the corresponding member M of the system satisfies $M \cap E \subseteq \{s\}$ and hence $M \cap E = \emptyset$. But M is cut on S by a degree-m form, and a projective variety of positive dimension meets every hypersurface by lemma 3.7.11. So $s \in E$.

Step 3: the degree of Γ_s is at most 2. Let $\delta = \deg \Gamma_s$ and let H be a general hyperplane, meeting the irreducible curve Γ_s in δ distinct points [22, Degree Counts Points of a Linear Section]. The member M of $\mathbb{P}(V_m(-2s))$ corresponding to H contains the closures $E_1, \ldots, E_\delta$ of the δ fibers over those points, pairwise distinct irreducible curves, each passing through s by Step 2. The local ring $\mathcal{O}_{S,s}$ is regular, hence a unique factorization domain (*Commutative Algebra* [10, Regular Local Rings Are Factorial]), so each E_i has a prime local equation $p_i \in \mathfrak{m}_s$, the p_i are pairwise non-associate, and the local equation of M is divisible by $p_1 \cdots p_\delta \in \mathfrak{m}_s^\delta$. Taking M general, $\operatorname{mult}_s M = 2$ by Step 1, so $\delta \leq 2$.

Step 4: the numerical conclusion. The image Γ_s is nondegenerate in $\mathbb{P}(V_m(-2s)^\vee)$, a projective space of dimension $\nu - 4$: a hyperplane containing the image would be a member of the system containing S, and no nonzero form vanishes on S by construction of V_m. An irreducible nondegenerate curve in $\mathbb{P}^d_k$ has degree at least d [22], so $\nu - 4 \leq \delta \leq 2$ and $\nu \leq 6$. With $\nu \geq 6$ from Step 1 this gives $\nu = 6$.

Step 5: $m \geq 3$ is excluded. Suppose $m \geq 3$. Choose a linear form ℓ with $\ell(s) = 0$ that does not vanish on S, possible because the linear forms vanishing at s cut out $\{s\}$ and S is not a point, and a linear form ℓ' not vanishing on S. The form $\ell^3 \ell'^{m-3}$ has degree m and does not vanish on S, the ideal of S being prime, so it is a nonzero element of V_m; and its restriction lies in $\mathfrak{m}_s^3$, since ℓ restricts into $\mathfrak{m}_s$. Hence $V_m \to \mathcal{O}_{S,s}/\mathfrak{m}_s^3$ has nonzero kernel and $\nu \geq 7$, contradicting Step 4. Therefore $m = 2$, completing the proof.

The route through Steps 3 and 4 is chosen deliberately.[8]

The statement below is Kollár's Corollary 24, restated. Several corrections to the source are built in. The integer n, undefined there, is $\dim Y$, as his own proof forces. The singular-locus condition is imposed on Y, not on an unrelated X. The Grassmannian parametrizes subspaces of vector dimension $n-1$, since subspaces of vector dimension n would cut out a finite set rather than a curve. The degree bound is $m \geq 3$ throughout, which removes every exception at once and matches what the two-point corollary needs. And the conclusion of the codimension-two part is irreducibility, not integrality, for a reason recorded after the proof.

Theorem 3.8.20: Bertini's Irreducibility Theorem II

Let k be an infinite field and let $Y \subseteq \mathbb{P}_k^M$ be a closed subscheme of dimension $n \geq 2$ which is geometrically irreducible and generically reduced and whose non-regular locus $\operatorname{Sing} Y$ has codimension at least 2 in Y. Let $m \geq 3$ and let V_m, $\operatorname{Grass}(n-1, V_m)$ and C_L be as above. Then:

1. there is a closed subset $Z_1 \subseteq \operatorname{Grass}(n-1, V_m)$ of codimension at least 1 such that for every k-rational $L \notin Z_1$ the scheme C_L is a geometrically integral curve, smooth over k;
2. there is a closed subset $Z_2 \subseteq \operatorname{Grass}(n-1, V_m)$ of codimension at least 2 such that for every k-rational $L \notin Z_2$ the set C_L is a geometrically irreducible curve.

There is no exceptional case.

***Proof*:**
By lemma 3.8.3, applied to the Grassmannian in a projective embedding [22, The Grassmannian Is a Projective Variety], it suffices to construct the Z_i after base change to $\overline{k}$. Assume therefore $k = \overline{k}$.

Step 0: only the reduced structure enters. The space V_m was defined by the forms vanishing on Y, which are the forms vanishing on the underlying reduced subscheme, and part (2) is a statement about the underlying set of C_L. If Y is regular at a point then its local ring there is a domain (proposition 2.7.4) and in particular reduced, so Y and Y_{red} agree near that point and Y_{red} is regular there too; hence $\operatorname{Sing} Y_{\mathrm{red}} \subseteq \operatorname{Sing} Y$ and the hypotheses pass to Y_{red}. Replacing Y by Y_{red} changes neither V_m nor any C_L as a set, so assume from now on that Y is integral. Note that irreducible and generically reduced is what the hypothesis says, not integral; the reason is recorded after the proof.

[8] The classical way to identify a surface all of whose second-order projections collapse is the Kronecker-Castelnuovo theorem on irreducible plane curve systems. That theorem has no known analogue in positive characteristic, as Ji records in Remark 2.14 of [43]; a reader who shortens the argument above by invoking it would silently restrict the whole section to characteristic zero. The count used here is characteristic-free.

Step 0': from subspaces to bases. Let $\mathcal{F} \subseteq (V_m)^{n-1}$ be the open subscheme of linearly independent tuples $(G_1, \dots, G_{n-1})$ and let $\rho\colon \mathcal{F} \to \mathrm{Grass}(n-1, V_m)$ send a tuple to its span. In each standard chart of the Grassmannian [22, The Grassmannian Is a Projective Variety] a subspace carries a canonical basis, so ρ is there the projection of a trivial bundle with fiber GL_{n-1}; it is surjective and open. A dense open subset of $\mathcal{F}$ that is a union of fibers of ρ therefore has dense open image.

Step 1: the good members. Under the m-th Veronese re-embedding (example 0.2.27) the degree-m hypersurface sections of Y become the hyperplane sections of Y inside $\mathbb{P}(V_m^\vee)$, and $\mathbb{P}(V_m)$ is the corresponding dual projective space. Let $U \subseteq \mathbb{P}(V_m)$ be the intersection of the following three dense open subsets: the set of $[F]$ with $Y \cap V_+(F)$ irreducible of dimension $n-1$, dense open by theorem 3.8.10 applied to the closed immersion of Y, whose image has dimension $n \geq 2$; the set of $[F]$ with $Y^{\mathrm{sm}} \cap V_+(F)$ smooth over k, dense open by theorem 3.8.4 applied to the smooth quasiprojective Y^{sm} in the same re-embedding; and the complement of the finitely many proper linear subspaces consisting of the $[F]$ whose $V_+(F)$ contains an irreducible component of $\operatorname{Sing} Y$.

For $[F] \in U$ write Y_F for $Y \cap V_+(F)$ with its reduced induced structure (definition 3.1.26), an integral projective variety of dimension $n-1$. Off $\operatorname{Sing} Y$ the scheme $Y \cap V_+(F)$ is smooth, hence reduced, hence equal to Y_F there, so $\operatorname{Sing} Y_F \subseteq \operatorname{Sing} Y \cap V_+(F)$, of dimension at most $n-3$ because no component of $\operatorname{Sing} Y$ lies in $V_+(F)$. Thus $\operatorname{Sing} Y_F$ has codimension at least 2 in Y_F, and Y_F satisfies the hypotheses of the theorem in dimension $n-1$. Moreover $\dim(\operatorname{Sing} Y \cap V_+(F)) < n-1$, so the generic point of $Y \cap V_+(F)$ is a smooth point and $Y \cap V_+(F)$ is generically reduced; lemma 3.8.18 therefore applies and

$$V_m/\langle F\rangle \xrightarrow{\sim} V_m(Y_F), \qquad \dim_k V_m(Y_F) = \nu - 1$$

Two consequences are recorded for use below. First, for $L \in \mathrm{Grass}(n-1, V_m)$ containing F the image $\overline{L}$ of L in $V_m(Y_F)$ has vector dimension $n-2$, and $L \mapsto \overline{L}$ is an isomorphism from the set of such L onto $\mathrm{Grass}\big(n-2, V_m(Y_F)\big)$. Second, since $F \in L$ forces $C_L \subseteq Y \cap V_+(F)$, and since two forms differing by a multiple of F have the same zero set on Y_F,

$$C_L = C_{\overline{L}} \quad \text{as subsets of } Y$$

the right-hand side computed on Y_F. This is the recursion, and it is exact.

Step 2: part (1). Let $\mathcal{F}^\circ \subseteq \mathcal{F}$ be the set of tuples $(G_1, \dots, G_{n-1})$ such that $[G_1] \in U$, then $[\overline{G_2}]$ lies in the corresponding good open subset for Y_{G_1}, and so on down to a curve. Each condition is dense open given the previous ones, so $\mathcal{F}^\circ$ is dense open, and C_L depends only on the span, so $\mathcal{F}^\circ$ may be enlarged to a union of fibers of ρ without leaving the good locus. Shrink it once more so that no cut contains an irreducible component of what is left of $\operatorname{Sing} Y$ at that stage, again finitely many proper linear conditions. After $n-1$ such cuts the curve C_L is irreducible, and it misses $\operatorname{Sing} Y$ altogether: that set has dimension at most $n-2$ and each cut drops the dimension of what remains of it by one, so after $n-1$ cuts nothing is left. Hence $C_L \subseteq Y^{\mathrm{sm}}$ and C_L is smooth over k by theorem 3.8.4, therefore reduced and integral. Take $Z_1 = \mathrm{Grass}(n-1, V_m) \setminus \rho(\mathcal{F}^\circ)$, closed of codimension at least 1 by Step 0'.

Step 3: the bad locus, and the base case $n = 2$. Let

$$\mathcal{B} = \{L \in \mathrm{Grass}(n-1, V_m) \mid C_L \text{ is not a geometrically irreducible curve}\}$$

By lemma 3.8.15, applied to the universal family $\mathbf{C} \to \mathrm{Grass}(n-1, V_m)$, the set $\mathcal{B}$ is constructible. This is the point at which the dimension counts below become legitimate.

Suppose $n = 2$, so that $\mathrm{Grass}(1, V_m) = \mathbb{P}(V_m)$ and C_L is a single member of the system, of pure dimension 1 by lemma 3.7.11. By lemma 3.8.17 every member is connected. If a member C is reducible, two distinct irreducible components of it meet at a point s; if $s \notin \operatorname{Sing} Y$ then $\mathcal{O}_{C,s}$ is not a domain, hence not regular by proposition 2.7.4, so C is singular at s. Every reducible member therefore meets $\operatorname{Sing} Y$, a finite set since it has codimension 2 in the surface Y, or is singular at a smooth point of Y.

Take first $s \in \operatorname{Sing} Y$ and let $V_m(-s) \subseteq V_m$ be the forms vanishing at s, of codimension 1 because the system separates points. The base locus of $\mathbb{P}(V_m(-s))$ is $\{s\}$ and the associated rational map is the linear projection of Y from s in the embedding by $|H|$. If the image of that projection has dimension 2, then theorem 3.8.10 makes a general member of $\mathbb{P}(V_m(-s))$ irreducible, so $\mathcal{B} \cap \mathbb{P}(V_m(-s))$ lies in a proper closed subset and has dimension at most $\nu - 3$. If instead the image is a curve, then for a general $y \in Y$ the line joining s to y meets Y in a curve and hence lies in Y, so Y is a cone with vertex s whose rulings are lines; but the hyperplane class of that embedding is m times the hyperplane class of $Y \subseteq \mathbb{P}^M_k$, which has degree at least 1 on any curve, so a ruling would have degree at least $m \geq 3$, while a line has degree 1. That case does not occur. Since $\operatorname{Sing} Y$ is finite, the union of these loci has dimension at most $\nu - 3$.

Take now s a smooth point. By Step 1 of the proof of lemma 3.8.19 the subspace $V_m(-2s)$ has codimension 3, so $\mathbb{P}(V_m(-2s))$ has dimension $\nu - 4$. Let

$$\mathcal{I} = \{(s, [C]) \in Y^{\mathrm{sm}} \times \mathcal{B} \mid \operatorname{mult}_s C \geq 2\}$$

a constructible set, since the multiplicity condition is closed, and let T be the set of $s \in Y^{\mathrm{sm}}$ for which $\mathcal{B} \cap \mathbb{P}(V_m(-2s))$ is dense in $\mathbb{P}(V_m(-2s))$; the set T is constructible, being a fiber-dimension locus of the constructible $\mathcal{I}$. For $s \notin T$ the fiber of $\mathcal{I}$ has dimension at most $\nu - 5$. For $s \in T$ the intersection $\mathcal{B} \cap \mathbb{P}(V_m(-2s))$ is constructible and dense, hence contains a dense open subset, so a general member of $\mathbb{P}(V_m(-2s))$ is reducible and theorem 3.8.10 forbids the associated map ψ_s from having two-dimensional image; its image is a curve. Were T dense in Y it would, being constructible, contain a dense open subset, and lemma 3.8.19 would force $m = 2$, against $m \geq 3$. So T is not dense and $\dim T \leq 1$. Consequently

$$\dim \mathcal{I} \leq \max\big\{2 + (\nu - 5),\ 1 + (\nu - 4)\big\} = \nu - 3$$

and the projection of $\mathcal{I}$ to $\mathbb{P}(V_m)$, which contains every reducible member singular at a smooth point, has dimension at most $\nu - 3$.

Combining the two, $\dim \mathcal{B} \leq \nu - 3 = \dim \mathbb{P}(V_m) - 2$, and $Z_2 = \overline{\mathcal{B}}$ is a closed subset of codimension at least 2. This proves part (2) for $n = 2$.

Step 4: the one-cut induction. Let $n \geq 3$ and assume part (2) for all varieties of dimension $n - 1$

satisfying the hypotheses, with the same m. Put $Z = \mathbb{P}(V_m) \setminus U$, a closed subset of codimension at least 1, and split $\mathcal{B}$ according to whether the linear subspace $\mathbb{P}(L) \subseteq \mathbb{P}(V_m)$, of projective dimension $n-2$, meets U.

If $\mathbb{P}(L) \cap U = \emptyset$ then $\mathbb{P}(L) \subseteq Z$, so L lies in $\mathbb{F}_{n-2}(Z)$ in the notation of lemma 3.8.14, applied with $M = \nu - 1$ and $r = n - 2 \geq 1$. That lemma gives

$$\operatorname{codim}\mathbb{F}_{n-2}(Z) \geq (n-1)\operatorname{codim} Z \geq n - 1 \geq 2$$

so this half of the bad locus is contained in a closed subset of codimension at least 2. This is the half that a one-shot reduction to a surface cannot see, and it needs no extra work.

For the other half consider the incidence variety

$$\mathcal{J} = \{([F], L) \in U \times \operatorname{Grass}(n-1, V_m) \mid F \in L\}$$

locally closed in $\mathbb{P}(V_m) \times \operatorname{Grass}(n-1, V_m)$, and write π_1 and π_2 for its two projections. Its fiber over $[F] \in U$ is the set of L containing F, isomorphic by Step 1 to $\operatorname{Grass}\big(n-2, V_m(Y_F)\big)$, of dimension $(n-2)(\nu - n + 1)$; hence

$$\dim \mathcal{J} = (\nu - 1) + (n-2)(\nu - n + 1)$$

Its fiber over L is $\mathbb{P}(L) \cap U$, a nonempty open subset of $\mathbb{P}(L) \cong \mathbb{P}_k^{n-2}$ precisely when L belongs to the second half, so that fiber has dimension $n-2$.

Put $\mathcal{J}_{\text{bad}} = \mathcal{J} \cap \pi_2^{-1}(\mathcal{B})$, constructible because $\mathcal{B}$ is. Fix $[F] \in U$. Under the isomorphism of Step 1 the fiber of $\mathcal{J}_{\text{bad}}$ over $[F]$ is exactly the bad locus of Y_F inside $\operatorname{Grass}(n-2, V_m(Y_F))$, because $C_L = C_{\overline{L}}$ as sets. The inductive hypothesis, applied to the $(n-1)$-dimensional Y_F, puts that locus inside a closed subset of codimension at least 2, so the fiber has dimension at most $(n-2)(\nu - n + 1) - 2$. Applying lemma 3.8.2 to the irreducible components of the closure of $\mathcal{J}_{\text{bad}}$ over U,

$$\dim \mathcal{J}_{\text{bad}} \leq (\nu - 1) + (n-2)(\nu - n + 1) - 2$$

Every fiber of $\mathcal{J}_{\text{bad}} \to \operatorname{Grass}(n-1, V_m)$ over a point of the second half has dimension $n-2$, so the same lemma, applied to the components of the closure, bounds the image by

$$\dim \pi_2(\mathcal{J}_{\text{bad}}) \leq (n-2)(\nu - n + 1) + \nu - n - 1$$

while $\dim \operatorname{Grass}(n-1, V_m) = (n-2)(\nu - n + 1) + (\nu - n + 1)$. The difference is 2, so the second half of the bad locus also has codimension at least 2.

Taking $Z_2 = \mathbb{F}_{n-2}(Z) \cup \overline{\pi_2(\mathcal{J}_{\text{bad}})}$, a closed subset of codimension at least 2 containing $\mathcal{B}$, completes the induction and the proof.

The hypothesis is that Y be geometrically irreducible and generically reduced, not that it be geometrically integral, and the conclusion of part (2) is irreducibility, not integrality. Neither is pedantry. Both are forced by the induction, and the mechanism is depth. Cutting a variety by a

general member of a very ample system drops the depth at every point by one, so a point of Y where the local ring has depth 1 becomes an embedded point of $Y \cap V_+(F)$ for every F through it. A variety that is only regular in codimension one may have a positive-dimensional locus of such points, since they are confined to $\operatorname{Sing} Y$ but nothing bounds their depth; and every hypersurface meets a positive-dimensional set. For such a Y no member of the system cuts an integral divisor at all, and an induction phrased with an integrality hypothesis would not close after a single step. The sharp form of the obstruction is due to Benoist, *Le théorème de Bertini en famille*, Bulletin de la Société Mathématique de France **139** (2011), Theorem 0.5(iii): if Y has a closed point of depth 1 then the locus of members with non-integral section has codimension exactly 1. A codimension-two claim about integrality is therefore false, while the irreducible and generically reduced version above is insensitive to depth.

3.8.21 Warning: What Part (2) Does Not Say Part (2) of theorem 3.8.20 asserts that C_L is irreducible, and says nothing about C_L being reduced. Part (1) does assert integrality, and the two are consistent because part (1) only speaks about the complement of a codimension-one set. Sharpening part (2) to integrality would be a genuine error, not a missed opportunity: by the theorem of Benoist quoted above, a variety with a closed point of depth 1 has a codimension-one family of members cutting a non-reduced divisor, and no codimension-two subset can contain that family.

The statement of part (2) is corroborated independently. It is Corollary 2.15 of Ji's *The Noether-Lefschetz theorem in arbitrary characteristic* [43], where it appears in the same form and with the same codimension. That is corroboration, not the proof used here: her argument runs on the theory of Flenner, O'Carroll and Vogel, on SGA 2 XIII 2.1 and on the Enriques-Severi-Zariski vanishing theorem, none of which this series develops. The proof above uses only the two Bertini theorems proved earlier in this section, the Grassmannian count of lemma 3.8.14, and the two quoted inputs isolated in lemma 3.8.15 and lemma 3.8.16.

3.8.22 Warning: Why the Degree Bound Is $m \geq 3$ Theorem 3.8.20 is stated with $m \geq 3$, and part (2) of it is false at $m = 2$. Take $Y = \mathbb{P}^2$ embedded as a plane and $m = 2$, so $|H|$ is the five-dimensional system of conics and $\mathrm{Grass}(1, V_2)$ is $|H|$ itself, a $\mathbb{P}^5$; a subspace of vector dimension $n - 1 = 1$ is a single conic, and C_L is that conic. A conic is irreducible unless it is a pair of lines or a double line, and those form the discriminant hypersurface, of codimension exactly 1 in $\mathbb{P}^5$. So a Z_1 exists and no Z_2 can.

The mechanism is visible in lemma 3.8.19. What makes reducibility a codimension-two rather than a codimension-one condition is that a member singular at a prescribed smooth point s is still usually irreducible: the singular point costs three conditions, and only the two-dimensional family of choices of s is available to pay for them, leaving a surplus of one. For conics the surplus is spent, because a conic with a point of multiplicity two meets every line through that point in a scheme of length 2 concentrated there, so any such line meeting the conic elsewhere is a component and the conic splits. Raising m to 3 restores the general nodal cubic, and Step 5 of the proof of lemma 3.8.19 is exactly the observation that the form $\ell^3 \ell'^{m-3}$ becomes available at $m = 3$.

The plane with its conics is the only exception. Lemma 3.8.19 forces $\nu = 6$ and $m = 2$ in the uncontrolled case, and the classical inequality of Hopf and Riemann, that finite-dimensional spaces W_1, W_2 of functions on an integral variety over an algebraically closed field satisfy $\dim(W_1 W_2) \geq \dim W_1 + \dim W_2 - 1$, then gives $\nu \geq mN + 1$ for a surface nondegenerate in $\mathbb{P}^N$; so $6 \geq 2N + 1$, forcing $N = 2$ and $Y = \mathbb{P}^2$. That inequality is quoted rather than proved, and it is used only for this classification. The theorem itself does not need it.

The bound $m \geq 3$ is what theorem 3.8.20 assumes, and it disposes of the exception in every dimension at once, so nothing below depends on the classification just given.

Two ways of reaching a subfamily

The applications never use a fully general L. They reach the Grassmannian along a constrained family, and the two families that occur behave in opposite ways, so it is worth separating them.

The first is the two-point family. As recorded before theorem 3.8.20, the subspaces all of whose members pass through two prescribed points form $\mathrm{Grass}(n - 1, V_m(-p - q))$, of codimension exactly $2(n - 1)$ in $\mathrm{Grass}(n - 1, V_m)$. This is a *special* subvariety, chosen for its geometry, and there is no reason for the bad locus to meet it in the expected codimension; a codimension-two bound in the ambient Grassmannian carries no information about it at all. The remedy is not to transfer but to re-run the argument inside the constrained system, which is what corollary 3.8.24 does.

The second is the family that a general linear projection produces: the fibers of a projection are cut by the hyperplanes of a general n-dimensional subspace $V_0 \subseteq V_m$. Here the subvariety is *general*, and generality is exactly what makes the ambient bound transfer with no loss. This is the one place where a codimension bound in the full Grassmannian is genuinely what is required, and it is why theorem 3.8.20 is proved in the ambient space rather than only in subsystems.

Lemma 3.8.23: Transfer to a General Subsystem

Let k be an algebraically closed field, let W be a k-vector space of finite dimension ν and let $n \geq 2$. For an n-dimensional subspace $W_0 \subseteq W$ let

$$\mathbb{P}(W_0^\vee) = \{L \in \mathrm{Grass}(n-1, W) \mid L \subseteq W_0\}$$

a closed subvariety of $\mathrm{Grass}(n-1, W)$ isomorphic to $\mathbb{P}_k^{n-1}$. Let $Z \subseteq \mathrm{Grass}(n-1, W)$ be closed and irreducible of codimension c. Then there is a dense open subset of $\mathrm{Grass}(n, W)$ over whose points

$$Z \cap \mathbb{P}(W_0^\vee) \text{ is empty, or of dimension exactly } n-1-c$$

and it is empty for every W_0 in a dense open subset as soon as $c \geq n$.

***Proof*:**
Put

$$\mathcal{K} = \{(L, W_0) \in Z \times \mathrm{Grass}(n, W) \mid L \subseteq W_0\}$$

a closed subset of the product of two projective varieties. Its projection to Z has fiber over L the set of n-dimensional subspaces containing L, that is, the lines of W/L, a projective space of dimension $\nu - n$; over each standard chart this is a trivial bundle, so $\mathcal{K}$ is irreducible of dimension

$$\dim \mathcal{K} = (n-1)(\nu - n + 1) - c + (\nu - n)$$

Since $\dim \mathrm{Grass}(n, W) = n(\nu - n)$, expanding gives

$$\dim \mathcal{K} - \dim \mathrm{Grass}(n, W) = n - 1 - c$$

The projection $\pi \colon \mathcal{K} \to \mathrm{Grass}(n, W)$ is a morphism of projective varieties, so its image is closed, and its fiber over W_0 is $Z \cap \mathbb{P}(W_0^\vee)$. If π is not dominant its image is a proper closed subset and every W_0 in the complement has empty intersection. If π is dominant, lemma 3.8.2 gives a dense open subset of $\mathrm{Grass}(n, W)$ over whose points the fiber has dimension exactly $\dim \mathcal{K} - \dim \mathrm{Grass}(n, W) = n - 1 - c$. Finally, if $c \geq n$ then $\dim \mathcal{K} < \dim \mathrm{Grass}(n, W)$, so π cannot be dominant and the first alternative holds, completing the proof.

Applied to a component of the bad locus of theorem 3.8.20, which has $c \geq 2$, the lemma gives $\dim(Z \cap \mathbb{P}(V_0^\vee)) \leq n - 3$ for a general n-dimensional $V_0 \subseteq V_m$, so a general hyperplane of a general V_0 is good, and for $c \geq n$ the bad locus is missed outright. The ambient bound and the constrained bound agree, which is the whole point: no re-running of the argument inside a subsystem is possible here, because the subsystem is not given in advance.

The last statement is the form the two-point application needs, and it is the opposite case. The remedy there is to run the cutting argument inside the constrained system from the start.

Corollary 3.8.24: Curves Through Two Prescribed Points

Let k be an infinite field, let $Y \subseteq \mathbb{P}^M_k$ be a geometrically irreducible and generically reduced closed subscheme of dimension $n \geq 2$, and let $p \neq q$ be k-rational points of Y. Let $m \geq 3$ and let $V_m(-p-q) \subseteq V_m$ be the subspace of degree-m forms vanishing at p and at q, of codimension 2. Then there is a closed subset

$$Z \subseteq \mathrm{Grass}\big(n-1, V_m(-p-q)\big), \qquad \mathrm{codim} Z \geq 1$$

such that for every k-rational $L \notin Z$ the set C_L is a geometrically irreducible curve containing p and q. Such L exist by lemma 3.8.1.

***Proof*:**
Every member of L vanishes at p and at q, so $p, q \in C_L$ for every L in this Grassmannian, which is the point of the constraint. It remains to run the cutting argument inside the constrained system, and by Step 0 of the proof of theorem 3.8.20 it suffices to work with Y integral.

The system $\mathbb{P}(V_m(-p-q))$ has base locus $\{p, q\}$. After base change to $\overline{k}$ it realizes every 1-jet (definition 6.3.4) at every closed point $y \notin \{p, q\}$: choosing degree-1 forms σ_p, σ_q vanishing at p respectively q and not at y, the products $\sigma_p \sigma_q G$ with G of degree $m-2 \geq 1$ lie in $V_m(-p-q)$ and already realize every 1-jet at y, since degree-1 forms surject onto $\mathcal{O}_{Y,y}/\mathfrak{m}_y^2$. So off $\{p, q\}$ the associated rational map is unramified, and its image has dimension $n \geq 2$.

Theorem 3.8.10 applied to that map makes a general member of $\mathbb{P}(V_m(-p-q))$ cut Y irreducibly. Iterating $n-1$ times as in corollary 3.8.12, and passing from tuples to spans by the bundle ρ of Step 0' of the proof of theorem 3.8.20 applied to $V_m(-p-q)$ in place of V_m, gives a dense open subset of $\mathrm{Grass}(n-1, V_m(-p-q))$ over which C_L is an irreducible curve. Its complement is the required Z, and lemma 3.8.3 together with lemma 3.8.1 produces a k-rational point outside it, completing the proof.

3.9⋆ The Category of Affine Schemes as Ringed Spaces

⋆ *Optional. This section is an aside; nothing later in the book depends on it.*

Having established that scheme morphisms must be morphisms of *locally* ringed spaces to properly reflect geometric evaluation, it is natural to ask: just how wildly does the equivalence $\mathbf{Sch}^{\mathbf{Aff}} \cong \mathbf{CRing}^{\mathrm{op}}$ fail if we drop the "locally" condition? What does the category of affine schemes look like if we restrict our morphisms to ringed space morphisms?

As we saw with the "phantom map" in example 3.1.1, multiple distinct ringed space morphisms can induce the exact same ring homomorphism on global sections. The topological map f becomes somehow decoupled from the algebraic pullback $f^{\#}$. Fortunately, this decoupling is not entire; it is classified by the topology of the Zariski spectrum, specifically:

Proposition 3.9.1: Classification of Ringed Space Morphisms

Let $X = \operatorname{Spec}(S)$ and $Y = \operatorname{Spec}(R)$ be affine schemes. A pair consisting of a continuous map $f : X \to Y$ and a ring homomorphism $\varphi : R \to S$ defines a valid morphism of ringed spaces $(f, f^\#) : X \to Y$ (where $f^\#$ on global sections is φ) if and only if:

$$\varphi^{-1}(\mathfrak{p}) \subseteq f(\mathfrak{p}) \qquad \text{for all } \mathfrak{p} \in \operatorname{Spec}(S).$$

Furthermore, when this condition holds, the sheaf map $f^\#$ is uniquely determined.

***Proof*:**

Let $g : X \to Y$ be the canonical "true" continuous map defined by $g(\mathfrak{p}) = \varphi^{-1}(\mathfrak{p})$.

For $(f, f^\#)$ to be a valid morphism of ringed spaces, the global ring homomorphism $\varphi : R \to S$ must induce a well-defined local map on the stalks for every $\mathfrak{p} \in X$. Specifically, the composite map $R \to S \to S_{\mathfrak{p}}$ must factor uniquely through the localization of R at $f(\mathfrak{p})$:

$$\begin{array}{ccc} R & \xrightarrow{\varphi} & S \\ \downarrow{\scriptstyle \text{loc}} & & \downarrow{\scriptstyle \text{loc}} \\ R_{f(\mathfrak{p})} & \overset{\exists!\, f_{\mathfrak{p}}^\#}{\dashrightarrow} & S_{\mathfrak{p}} \end{array}$$

By the universal property of localization, this factorization exists if and only if every element in the denominator of $R_{f(\mathfrak{p})}$ maps to a unit in $S_{\mathfrak{p}}$.

- The denominators of $R_{f(\mathfrak{p})}$ are exactly the elements of the set $R \setminus f(\mathfrak{p})$.
- The units of $S_{\mathfrak{p}}$ are exactly the fractions with numerators in S and denominators in $S \setminus \mathfrak{p}$, meaning the image of the element in S must lie in $S \setminus \mathfrak{p}$.

Thus, we require:

$$\varphi(R \setminus f(\mathfrak{p})) \subseteq S \setminus \mathfrak{p}$$

Taking the set-theoretic complement of this condition in R, we see that an element $r \in R$ avoids the left-hand side if and only if it avoids the right-hand side. This gives:

$$\varphi^{-1}(\mathfrak{p}) \subseteq f(\mathfrak{p})$$

Since localization maps are epimorphisms in the category of commutative rings, if this factorization exists, it is unique, uniquely determining the sheaf map $f^\#$ on all stalks, and hence on all of X.

Recall that in the Zariski topology, the containment of prime ideals $\mathfrak{q} \subseteq \mathfrak{q}'$ means exactly that $\mathfrak{q}'$ is a specialization of $\mathfrak{q}$ (i.e., $\mathfrak{q}'$ lies in the closure of the point $\mathfrak{q}$, so $\mathfrak{q}' \in \overline{\{\mathfrak{q}\}}$). The proposition reveals a geometric reality about the category of ringed spaces: the algebraic condition $\varphi^{-1}(\mathfrak{p}) \subseteq f(\mathfrak{p})$ translates to:

$$f(\mathfrak{p}) \in \overline{\{g(\mathfrak{p})\}}$$

A ringed space morphism allows the topological map f to take a point $\mathfrak{p}$ and map it to *any specialization* (any point in the closure) of the "correct" algebraic target $g(\mathfrak{p})$, provided the resulting

assignment f remains globally continuous.

Example 3.9.2: Revisiting the Phantom Map

Consider again $R = \mathbb{Z}_{(p)}$, $S = \mathbb{Q}$, and φ the standard inclusion. We have $\operatorname{Spec}(\mathbb{Q}) = \{[(0)]\}$ and $\operatorname{Spec}(\mathbb{Z}_{(p)}) = \{[(0)], [(p)]\}$.

The "true" algebraic target for the unique point in $\operatorname{Spec}(\mathbb{Q})$ is $g([(0)]) = \varphi^{-1}((0)) = [(0)]$.

What are the specializations of $[(0)]$ in $\operatorname{Spec}(\mathbb{Z}_{(p)})$? The closure of the generic point $[(0)]$ is the entire space, since the prime ideal (0) is contained in the maximal ideal (p). Therefore, $[(p)]$ is a valid specialization of $[(0)]$.

By proposition 3.9.1, mapping $f([(0)]) = [(p)]$ satisfies the condition $(0) \subseteq (p)$. Thus, it forms a valid ringed space morphism, giving us the phantom map π_2.

3.10⋆ Limits of Schemes

⋆ *Optional. This section is an aside; nothing later in the book depends on it.*

A local ring $A_{\mathfrak{p}}$ is the colimit of the localizations A_f with $f \notin \mathfrak{p}$, and the stalk $\mathcal{O}_{X,x}$ of a scheme is the colimit of the rings of the affine open neighborhoods of x. On spectra these colimits of rings become limits of schemes: $\operatorname{Spec} A_{\mathfrak{p}}$ is the limit of the affine schemes $\operatorname{Spec} A_f$.

Definition 3.10.1: Directed Set and Inverse System of Schemes

A *directed set* is a nonempty set I with a reflexive and transitive relation $\leq$ such that any $i, j \in I$ have a common upper bound, an element k with $i \leq k$ and $j \leq k$. Viewed as a category with one morphism $i \to j$ exactly when $i \leq j$, a directed set is a filtered category (definition 0.4.25).

1. A *directed system of rings* over I consists of rings A_i, $i \in I$, and ring maps $\varphi_{ij} \colon A_i \to A_j$ for $i \leq j$, with $\varphi_{ii} = \mathrm{id}$ and $\varphi_{jk} \circ \varphi_{ij} = \varphi_{ik}$ for $i \leq j \leq k$. Its colimit (definition 0.4.26) is written $A = \varinjlim_i A_i$, with maps $\varphi_i \colon A_i \to A$. Directed systems of sets, and of R-algebras for a ring R, are defined in the same way, with maps of sets and R-algebra maps.

2. An *inverse system of schemes* over I consists of schemes S_i, $i \in I$, and morphisms $f_{ij} \colon S_j \to S_i$ for $i \leq j$, with $f_{ii} = \mathrm{id}$ and $f_{ij} \circ f_{jk} = f_{ik}$ for $i \leq j \leq k$. It has *affine transition morphisms* if every f_{ij} is affine (definition 3.4.5). For a scheme S, an inverse system of S-schemes is one in which every S_i is an S-scheme and every f_{ij} is an S-morphism.

3. A *limit* of an inverse system of schemes is a scheme L with morphisms $\ell_i \colon L \to S_i$ satisfying $f_{ij} \circ \ell_j = \ell_i$ for $i \leq j$, such that for every scheme T and every family of morphisms $g_i \colon T \to S_i$ with $f_{ij} \circ g_j = g_i$ for $i \leq j$ there is a unique morphism $g \colon T \to L$ with $\ell_i \circ g = g_i$ for all i. It is written $L = \varprojlim_i S_i$, and the ℓ_i are its *projections*.

A limit is unique up to a unique isomorphism commuting with the projections [9, Uniqueness of Limits]. The limit L of an inverse system of S-schemes is an S-scheme through the composite of ℓ_i with the structure morphism $\sigma_i \colon S_i \to S$, which does not depend on i: for $i, j \leq k$ one has

$\sigma_i \circ \ell_i = \sigma_i \circ f_{ik} \circ \ell_k = \sigma_k \circ \ell_k$, and likewise for j. With this structure, the morphism g induced by a family of S-morphisms g_i is an S-morphism, since $\sigma_i \circ \ell_i \circ g = \sigma_i \circ g_i$.

3.10.2 Remark: Cofiltered Index Categories Limits over a cofiltered category reduce to limits over a directed set. For every filtered category J there are a directed set I and a functor $M\colon I \to J$ such that, for every category $\mathcal{C}$ and every diagram $D\colon J^{\mathrm{op}} \to \mathcal{C}$, a limit of the restriction of D along M is a limit of D, and dually for colimits of diagrams $J \to \mathcal{C}$ [70, Tag 0032]. The results of this section and of section 3.11⋆ therefore apply to diagrams of schemes indexed by the opposite of a filtered category whose morphisms are all affine.

Lemma 3.10.3: Filtered Colimits of Sets and Rings

Let (E_i, φ_{ij}) be a directed system of sets over a directed set I. On the disjoint union $\coprod_{i\in I} E_i$ write $(i,a) \sim (j,b)$ when $\varphi_{ik}(a) = \varphi_{jk}(b)$ for some k with $i \le k$ and $j \le k$.

1. The relation $\sim$ is an equivalence relation, and the set E of its classes $[i,a]$, with the maps $\varphi_i\colon E_i \to E$, $a \mapsto [i,a]$, is a colimit of the system in the category of sets.
2. Every element of E is $\varphi_i(a)$ for some $i \in I$ and $a \in E_i$, and for $a, a' \in E_i$ one has $\varphi_i(a) = \varphi_i(a')$ if and only if $\varphi_{ij}(a) = \varphi_{ij}(a')$ for some $j \ge i$.
3. If the $E_i = A_i$ are rings and the φ_{ij} ring maps, then $A = E$ is a ring under

$$[i,a] + [j,b] = [k, \varphi_{ik}(a) + \varphi_{jk}(b)], \qquad [i,a]\,[j,b] = [k, \varphi_{ik}(a)\,\varphi_{jk}(b)]$$

   for any k with $i, j \le k$, the maps φ_i are ring maps, and (A, φ_i) is a colimit of the system in the category of rings. If the A_i form a directed system of R-algebras, then A is an R-algebra through any of the φ_i and is the colimit in the category of R-algebras. In particular $\varphi_i(a) = 0$ if and only if $\varphi_{ij}(a) = 0$ for some $j \ge i$.
4. For rings, $A = 0$ if and only if $A_i = 0$ for some i.
5. For rings, if every A_i is reduced, then A is reduced.

***Proof*:**

1. The relation is reflexive (take $k = i$) and symmetric. Let $(i,a) \sim (j,b)$ through k and $(j,b) \sim (l,c)$ through m, and choose n with $k, m \le n$. Then
$$\varphi_{in}(a) = \varphi_{kn}\varphi_{ik}(a) = \varphi_{kn}\varphi_{jk}(b) = \varphi_{jn}(b) = \varphi_{mn}\varphi_{jm}(b) = \varphi_{mn}\varphi_{lm}(c) = \varphi_{ln}(c)$$
   so $(i,a) \sim (l,c)$. The relation $(i,a) \sim (j, \varphi_{ij}(a))$ gives $\varphi_j \circ \varphi_{ij} = \varphi_i$. Let $\psi_i\colon E_i \to B$ be maps with $\psi_j \circ \varphi_{ij} = \psi_i$. If $(i,a) \sim (j,b)$ through k, then $\psi_i(a) = \psi_k\varphi_{ik}(a) = \psi_k\varphi_{jk}(b) = \psi_j(b)$, so $\psi([i,a]) = \psi_i(a)$ is a well-defined map $E \to B$ with $\psi \circ \varphi_i = \psi_i$, and it is the only such map because the images of the φ_i cover E.
2. The first assertion holds by construction. If $\varphi_i(a) = \varphi_i(a')$, then $\varphi_{ik}(a) = \varphi_{ik}(a')$ for some $k \ge i$, and conversely.
3. If two pairs are equivalent through k, they are equivalent through every $k' \ge k$, by applying $\varphi_{kk'}$. Hence, given $(i,a) \sim (i',a')$ and $(j,b) \sim (j',b')$ and upper bounds k of i, j and k' of

i', j', there is an index l above k, k' and indices witnessing both equivalences, and then $\varphi_{il}(a) = \varphi_{i'l}(a')$ and $\varphi_{jl}(b) = \varphi_{j'l}(b')$. Applying the ring maps φ_{kl} and $\varphi_{k'l}$ shows that the two proposed sums, and the two proposed products, have the same image in A_l. So the operations are well defined. Finitely many classes have representatives at a common index, since $(i, a) \sim (k, \varphi_{ik}(a))$ for $i \leq k$, and the ring axioms for them follow from those of A_k. The φ_i are ring maps by the formulas with $k = i$. If the maps ψ_i of (1) are ring maps, then ψ is a ring map, as one checks at a common index. For R-algebras, the composites $R \to A_i \to A$ agree for all i because the φ_{ij} are R-algebra maps, and when the ψ_i are R-algebra maps so is ψ. The last assertion is (2) with $a' = 0$.

4. If $A_i = 0$ then $1_A = \varphi_i(1) = \varphi_i(0) = 0$. If $A = 0$ then $\varphi_i(1) = 0$ for any i, so by (3) $1 = \varphi_{ij}(1) = 0$ in some A_j.

5. Let $x \in A$ with $x^n = 0$, and write $x = \varphi_i(a)$. By (3) there is $j \geq i$ with $\varphi_{ij}(a)^n = \varphi_{ij}(a^n) = 0$, so $\varphi_{ij}(a) = 0$ as A_j is reduced, and $x = \varphi_j(\varphi_{ij}(a)) = 0$, completing the proof.

Lemma 3.10.4: Spectrum of a Filtered Colimit

Let (A_i, φ_{ij}) be a directed system of rings with colimit A. Then $\operatorname{Spec} A$, with the morphisms $\operatorname{Spec}(\varphi_i) \colon \operatorname{Spec} A \to \operatorname{Spec} A_i$, is a limit of the inverse system $(\operatorname{Spec} A_i, \operatorname{Spec}(\varphi_{ij}))$. Every inverse system of affine schemes is isomorphic to one of this form, by way of the directed system of its rings of global sections.

***Proof*:**
Let T be a scheme. By proposition 3.1.12, sending a morphism $g_i \colon T \to \operatorname{Spec} A_i$ to $\Gamma(g_i) \colon A_i \to \mathcal{O}_T(T)$ is a bijection, and $\Gamma(\operatorname{Spec}(\varphi_{ij}) \circ g_j) = \Gamma(g_j) \circ \varphi_{ij}$. So a family (g_i) satisfies $\operatorname{Spec}(\varphi_{ij}) \circ g_j = g_i$ for all $i \leq j$ exactly when the ring maps $\psi_i = \Gamma(g_i)$ satisfy $\psi_j \circ \varphi_{ij} = \psi_i$. By lemma 3.10.3(3) such families correspond bijectively to ring maps $\psi \colon A \to \mathcal{O}_T(T)$ with $\psi \circ \varphi_i = \psi_i$, and by proposition 3.1.12 again to morphisms $g \colon T \to \operatorname{Spec} A$ with $\Gamma(\operatorname{Spec}(\varphi_i) \circ g) = \Gamma(g) \circ \varphi_i = \psi_i$, that is, with $\operatorname{Spec}(\varphi_i) \circ g = g_i$. This is the universal property of the limit. For the second assertion, let (S_i, f_{ij}) be an inverse system of affine schemes. The rings $\mathcal{O}(S_i)$ with the maps $\Gamma(f_{ij})$ form a directed system, by functoriality of global sections, and the isomorphisms $S_i \cong \operatorname{Spec} \mathcal{O}(S_i)$ of proposition 3.1.12 carry f_{ij} to $\operatorname{Spec}(\Gamma(f_{ij}))$ by the naturality stated there, as we sought to show.

Example 3.10.5: Local Rings and Stalks as Limits

1. Let A be a ring and $\Sigma \subseteq A$ a multiplicative subset, preordered by divisibility, $s \leq t$ when $s \mid t$. This is a directed set, st being an upper bound of s and t. For $t = su$ the map $A_s \to A_t$, $a/s^n \mapsto au^n/t^n$, is the unique A-algebra map, and these maps form a directed system. The maps $A_s \to \Sigma^{-1}A$ induce a map from the colimit of the A_s to $\Sigma^{-1}A$, which is surjective because a/s is the image of $a/s \in A_s$, and injective by lemma 3.10.3(2): if $a/s^n \in A_s$ maps to zero in $\Sigma^{-1}A$, then $va = 0$ for some $v \in \Sigma$, and a/s^n maps to zero in A_{sv}, where v is a unit. By lemma 3.10.4, $\operatorname{Spec} \Sigma^{-1}A$ is the limit of the affine schemes $\operatorname{Spec} A_s$, whose transition morphisms are the inclusions $D(t) \subseteq D(s)$ of distinguished open subsets of $\operatorname{Spec} A$ (proposition 2.2.13). For $\Sigma = A \setminus \mathfrak{p}$ this exhibits $\operatorname{Spec} A_{\mathfrak{p}}$ as a limit, and for $A = \mathbb{Z}$ and $\Sigma = \mathbb{Z} \setminus \{0\}$ it exhibits $\operatorname{Spec} \mathbb{Q}$ as the limit of the schemes $\operatorname{Spec} \mathbb{Z}[1/n]$.

2. Let X be a scheme and $x \in X$. The affine open neighborhoods of x, preordered by reverse inclusion, form a directed set: the intersection of two of them contains a distinguished open subset of the first one containing x (proposition 2.2.17), which is affine (proposition 2.2.13). Every open neighborhood of x contains an affine one, so the colimit of the rings $\mathcal{O}_X(U)$ over the affine neighborhoods U is the stalk $\mathcal{O}_{X,x}$ (definition 1.1.4). By lemma 3.10.4, $\operatorname{Spec}\mathcal{O}_{X,x}$ is the limit of the affine neighborhoods of x, with the inclusions as transition morphisms, which are affine by corollary 3.4.11. When X is integral with generic point η, every nonempty affine open subset contains η, and $\operatorname{Spec}K(X)$, for the function field $K(X) = \mathcal{O}_{X,\eta}$ (definition 2.4.22), is the limit of the nonempty affine open subsets of X.

For non-affine stages the limit is glued from limits of affine schemes (EGA IV_3, 8.2.3 [32]): since the transition morphisms are affine, the preimages of an affine open subset of one stage are affine in every later stage.

Theorem 3.10.6: Limits of Schemes with Affine Transition Morphisms

Let (S_i, f_{ij}) be an inverse system of schemes over a directed set I with affine transition morphisms. For $0 \in I$ write $I_{\geq 0} = \{i \in I : 0 \leq i\}$, a directed set.

1. Let $0 \in I$. A scheme L with morphisms $\ell_i \colon L \to S_i$, $i \in I$, satisfying $f_{ij} \circ \ell_j = \ell_i$, is a limit of the system if and only if L with the ℓ_i for $i \geq 0$ is a limit of the restricted system over $I_{\geq 0}$.

2. The limit $S = \varprojlim_i S_i$ exists, and every projection $f_i \colon S \to S_i$ is affine.

3. Let $0 \in I$ and let $U \subseteq S_0$ be open. The open subscheme $f_0^{-1}(U)$ of S, with the restrictions of the f_i, is a limit of the inverse system of open subschemes $f_{0i}^{-1}(U) \subseteq S_i$, $i \geq 0$, with the restricted transition morphisms.

4. Let $0 \in I$ and let $U \subseteq S_0$ be an affine open subset. Then $f_{0i}^{-1}(U)$ is affine for $i \geq 0$, the open subset $f_0^{-1}(U) \subseteq S$ is affine, and
$$\mathcal{O}_S\big(f_0^{-1}(U)\big) = \varinjlim_{i \geq 0} \mathcal{O}_{S_i}\big(f_{0i}^{-1}(U)\big)$$
with the colimit maps induced by the projections f_i.

5. If every S_i is reduced, then S is reduced.

***Proof*:**
Step 1: statement (1). Let $(g_i)_{i\geq 0}$ be a compatible family of morphisms $g_i \colon T \to S_i$. For $j \in I$ choose $k \geq j, 0$ and put $g_j = f_{jk} \circ g_k$. This does not depend on k: for two choices k, k' take $l \geq k, k'$, so that $f_{jk}g_k = f_{jk}f_{kl}g_l = f_{jl}g_l = f_{jk'}g_{k'}$. The extended family is compatible, since for $j \leq j'$ and $k \geq j', 0$ one has $f_{jj'}g_{j'} = f_{jj'}f_{j'k}g_k = f_{jk}g_k = g_j$. A morphism $g \colon T \to L$ with $\ell_i g = g_i$ for all $i \geq 0$ satisfies $\ell_j g = f_{jk}\ell_k g = f_{jk}g_k = g_j$ for every j. So compatible families over $I_{\geq 0}$ and over I correspond bijectively, and so do the morphisms $T \to L$ inducing them, which proves (1).

Step 2: open subschemes of a limit. A morphism $h \colon T \to Z$ whose image lies in an open subset

$W \subseteq Z$ factors through the open subscheme W: by proposition 3.2.5(1), $W \times_Z T$ is the open subscheme $h^{-1}(W) = T$, and the first projection is the factorization. The factorization is unique, because two morphisms to W that agree as morphisms to Z have the same underlying map and, $\mathcal{O}_W$ being the restriction of $\mathcal{O}_Z$, the same sheaf map. Now let $0 \in I$, let (L, ℓ_i) be a limit of the system over $I_{\geq 0}$, and let $U \subseteq S_0$ be open. The argument uses only the universal property, so it applies to every inverse system over $I_{\geq 0}$, and below it is also applied to systems of open subschemes of the S_i. For $i \geq 0$ the restriction of ℓ_i maps $\ell_0^{-1}(U)$ into $f_{0i}^{-1}(U)$, since $f_{0i}\ell_i = \ell_0$. Let $g_i \colon T \to f_{0i}^{-1}(U)$ be compatible with the restricted transition morphisms. Composed with the inclusions they form a compatible family of morphisms to the S_i, hence $g_i = \ell_i \circ g$ for a unique $g \colon T \to L$. The image of g lies in $\ell_0^{-1}(U)$ because $\ell_0 g = g_0$, so g factors uniquely through the open subscheme $\ell_0^{-1}(U)$. Thus $\ell_0^{-1}(U)$ is a limit of the system $(f_{0i}^{-1}(U))_{i\geq 0}$.

Step 3: the affine pieces. Let $0 \in I$ and let $U \subseteq S_0$ be an affine open subset. For $i \geq 0$ the open subscheme $U_i = f_{0i}^{-1}(U)$ is affine, f_{0i} being affine, and the restrictions of the f_{ij} make $(U_i)_{i\geq 0}$ an inverse system of affine schemes. By lemma 3.10.4, its limit is $L_U = \operatorname{Spec}\big(\varinjlim_{i\geq 0} \mathcal{O}_{S_i}(U_i)\big)$, with projections $p_i^U \colon L_U \to U_i$ corresponding to the colimit maps.

Step 4: gluing. Fix $0 \in I$ and let $\mathcal{A}$ be the set of affine open subsets of S_0. For $U, V \in \mathcal{A}$ let $L_{U,V} = (p_0^U)^{-1}(U \cap V) \subseteq L_U$. By Step 2 applied to the system $(U_i)_{i\geq 0}$ and the open subset $U \cap V \subseteq U$, the scheme $L_{U,V}$ is a limit of the system $(f_{0i}^{-1}(U \cap V))_{i\geq 0}$. So is $L_{V,U}$, and there is a unique isomorphism $\theta_{UV} \colon L_{U,V} \to L_{V,U}$ commuting with the projections. These data satisfy the hypotheses of theorem 2.2.56. First, $L_{U,U} = L_U$ and $\theta_{UU} = \mathrm{id}$ by uniqueness. Second, θ_{UV} commutes with the projections to S_0, so it maps $L_{U,V} \cap L_{U,W} = (p_0^U)^{-1}(U \cap V \cap W)$ into $(p_0^V)^{-1}(U \cap V \cap W) = L_{V,U} \cap L_{V,W}$. Third, on $L_{U,V} \cap L_{U,W}$ the morphisms $\theta_{VW} \circ \theta_{UV}$ and θ_{UW} both map into $L_{W,U} \cap L_{W,V} = (p_0^W)^{-1}(U \cap V \cap W)$ and commute with the projections to the $f_{0i}^{-1}(U \cap V \cap W)$. That scheme is a limit of this system by Step 2, so the two morphisms agree, which is the cocycle condition. Gluing gives a scheme S with open embeddings $\iota_U \colon L_U \to S$ covering S, with $\iota_U(L_U) \cap \iota_V(L_V) = \iota_U(L_{U,V})$ and $\iota_V \circ \theta_{UV} = \iota_U$ on $L_{U,V}$. For $i \geq 0$ the morphisms $p_i^U \circ \iota_U^{-1}$ from $\iota_U(L_U)$ to S_i agree on overlaps, because $p_i^V \circ \theta_{UV} = p_i^U$, and so glue to a morphism $f_i \colon S \to S_i$ (lemma 3.1.9). The relations $f_{ij} \circ f_j = f_i$ hold on each $\iota_U(L_U)$, hence on S. Moreover $f_0^{-1}(U) = \iota_U(L_U)$: the inclusion $\supseteq$ holds as $f_0 \circ \iota_U = p_0^U$ lands in U, and a point $\iota_V(z)$ with $f_0(\iota_V(z)) = p_0^V(z) \in U$ lies in $\iota_V(L_{V,U}) = \iota_U(L_{U,V})$.

Step 5: the universal property. Let $g_i \colon T \to S_i$, $i \geq 0$, be a compatible family. For $U \in \mathcal{A}$ the open set $T_U = g_0^{-1}(U)$ is mapped by each g_i into U_i, so the restrictions of the g_i to T_U induce a unique morphism $h_U \colon T_U \to L_U$ with $p_i^U h_U = g_i$. On $T_U \cap T_V$ the morphisms $\iota_U h_U$ and $\iota_V h_V$ both map into $\iota_U(L_{U,V})$, which with the restricted f_i is a limit of the system $(f_{0i}^{-1}(U \cap V))_{i\geq 0}$ (Step 2 and the isomorphism ι_U), and both are compatible with the g_i. So they agree, and they glue to $h \colon T \to S$ (lemma 3.1.9) with $f_i h = g_i$. If $h' \colon T \to S$ also satisfies $f_i h' = g_i$, then h' maps T_U into $f_0^{-1}(U) = \iota_U(L_U)$, where it is determined by the universal property of L_U, so $h' = h$. Thus S is a limit over $I_{\geq 0}$. For $j \in I$ put $f_j = f_{jk} \circ f_k$ for any $k \geq j, 0$, which does not depend on k, as in Step 1. With these projections S is a limit over I by Step 1.

Step 6: statements (2) to (5). The limit exists by Step 5, and (3) follows from Step 2 together with Step 1. For (4), Step 3 and (3) present $f_0^{-1}(U)$ and L_U as limits of the same system, so they are isomorphic compatibly with the projections, and $f_0^{-1}(U)$ is affine with the stated

ring. The index 0 was arbitrary, so the preimage under f_i of every affine open subset of S_i is affine, and f_i is affine (definition 3.4.5), which completes (2). For (5), each $f_{0i}^{-1}(U)$ is an open subscheme of the reduced scheme S_i, so its ring is reduced (example 2.2.32(1)). By (4) and lemma 3.10.3(5) the affine open subsets $f_0^{-1}(U)$, which cover S, have reduced rings, and S is reduced by example 2.2.32(1), as we sought to show.

Lemma 3.10.7: Base Change of a Limit

Let (S_i, f_{ij}) be an inverse system with affine transition morphisms and limit S, let $0 \in I$, and let $g\colon T \to S_0$ be a morphism. For $i \geq 0$ let $T_i = T \times_{S_0} S_i$, with S_i an S_0-scheme through f_{0i}. The morphisms $\mathrm{id}_T \times f_{ij}\colon T_j \to T_i$ make $(T_i)_{i\geq 0}$ an inverse system with affine transition morphisms, and $T \times_{S_0} S$, with the morphisms $\mathrm{id}_T \times f_i$, is its limit.

***Proof*:**
A morphism $Z \to T_j$ is a pair (a, b) of morphisms $a\colon Z \to T$ and $b\colon Z \to S_j$ with $ga = f_{0j}b$. For $i \leq j$, a morphism $Z \to T_i \times_{S_i} S_j$ is a triple (a, c, b) with $ga = f_{0i}c$ and $c = f_{ij}b$. The middle entry is determined by b, and the condition becomes $ga = f_{0i}f_{ij}b = f_{0j}b$, so $T_j \cong T_i \times_{S_i} S_j$, and under this isomorphism $\mathrm{id}_T \times f_{ij}$ is the first projection. That projection is the base change of the affine morphism f_{ij}, hence affine by proposition 3.4.12(2). A compatible family of morphisms $h_i = (a_i, b_i)\colon Z \to T_i$, $i \geq 0$, satisfies $a_j = a_i$ and $f_{ij}b_j = b_i$. It amounts to one morphism $a\colon Z \to T$ and a compatible family (b_i) with $ga = f_{0i}b_i$. By theorem 3.10.6(1) and (2) the family (b_i) corresponds to a unique $b\colon Z \to S$ with $f_ib = b_i$, and the condition becomes $ga = f_0b$. So compatible families correspond to morphisms $(a, b)\colon Z \to T \times_{S_0} S$, and the correspondence is composition with the morphisms $\mathrm{id}_T \times f_i$, as we sought to show.

The remaining results assume that the stages are quasi-compact, and from section 3.11⋆ on also quasi-separated. A morphism from a quasi-compact scheme to a quasi-separated one pulls quasi-compact open subsets back to quasi-compact sets, and a scheme affine over a quasi-compact and quasi-separated scheme is again quasi-compact and quasi-separated.

Lemma 3.10.8: Quasi-Compact and Quasi-Separated Schemes

1. Let $h\colon X \to Y$ be a morphism with X quasi-compact and Y quasi-separated. Then $h^{-1}(W)$ is quasi-compact for every quasi-compact open subset $W \subseteq Y$.
2. A scheme covered by finitely many affine open subsets $X^1, \ldots, X^n$ such that every $X^r \cap X^s$ is quasi-compact is quasi-compact and quasi-separated.
3. Let $h\colon X \to Y$ be affine with Y quasi-compact and quasi-separated. Then X is quasi-compact and quasi-separated.
4. A quasi-compact open subscheme of a quasi-separated scheme is quasi-compact and quasi-separated.

***Proof*:**
Throughout, affine schemes are quasi-compact and a finite union of quasi-compact subsets is quasi-compact (lemma 2.1.20).

1. Every point of X lies in an affine open A mapped by h into an affine open of Y (proposition 3.1.10), and finitely many such pairs (A_p, W'_p) cover X. Write $W = W_1 \cup \cdots \cup W_q$ with affine W_t. Each $W_t \cap W'_p$ is a finite union of affine open subsets W_{tps} (proposition 2.4.10). The preimage of W_{tps} under the morphism $A_p \to W'_p$ of affine schemes is affine (corollary 3.4.11), and $h^{-1}(W)$ is the finite union of these preimages.

2. The scheme is quasi-compact as a finite union of affines. Let A, B be affine open subsets. The distinguished open subsets of A contained in some X^r cover A, so A is a finite union of affine opens A_a, each distinguished in A and contained in some $X^{r(a)}$, and likewise B is a finite union of affine opens $B_b \subseteq X^{s(b)}$. With $r = r(a)$ and $s = s(b)$, the sets A_a and $X^r \cap X^s$ are quasi-compact opens of the affine, hence quasi-separated, scheme X^r (proposition 2.4.11), so $A_a \cap X^s$ is quasi-compact. It is a quasi-compact open of X^s, as is B_b, so $A_a \cap B_b = (A_a \cap X^s) \cap B_b$ is quasi-compact. Hence $A \cap B$ is quasi-compact, so it is a finite union of affine opens, and the scheme is quasi-separated by proposition 2.4.10.

3. Cover Y by finitely many affine opens W_k. The sets $h^{-1}(W_k)$ are affine and cover X. Each $W_k \cap W_l$ is a finite union of affine opens W_{klt} (proposition 2.4.10), so $h^{-1}(W_k) \cap h^{-1}(W_l)$ is the finite union of the affine sets $h^{-1}(W_{klt})$, hence quasi-compact. Now apply (2).

4. The quasi-compact open subsets of an open subscheme U are quasi-compact open subsets of the ambient scheme, so their pairwise intersections are quasi-compact, completing the proof.

When one stage is quasi-compact, compatible nonempty closed subsets of the stages have a common point in the limit (compare [70, Tag 01Z3]).

Lemma 3.10.9: Compatible Closed Subsets Meet the Limit

Let (S_i, f_{ij}) be an inverse system with affine transition morphisms and limit S, and suppose S_0 is quasi-compact for some $0 \in I$. For each $i \in I$ let $Z_i \subseteq S_i$ be a nonempty closed subset, with $f_{ij}(Z_j) \subseteq Z_i$ for $i \leq j$. Then there is $s \in S$ with $f_i(s) \in Z_i$ for every i. In particular, if every S_i is nonempty, then S is nonempty.

***Proof*:**
It suffices to find s with $f_i(s) \in Z_i$ for $i \geq 0$: for $j \in I$ and $i \geq j, 0$ one then has $f_j(s) = f_{ji}(f_i(s)) \in f_{ji}(Z_i) \subseteq Z_j$. Cover S_0 by affine opens $U^1, \ldots, U^m$ and write $U_i^k = f_{0i}^{-1}(U^k)$ for $i \geq 0$, an affine scheme. Let K_i be the set of k with $Z_i \cap U_i^k \neq \emptyset$, which is nonempty. For $i \leq j$ one has $K_j \subseteq K_i$, since f_{ij} maps $Z_j \cap U_j^k$ into $Z_i \cap U_i^k$. Some k lies in every K_i: otherwise each k is missing from some K_{i_k}, and for j above all the i_k the set $K_j \subseteq K_{i_k}$ would be empty. Fix such a k and write $A_i = \mathcal{O}_{S_i}(U_i^k)$, with transition maps φ_{ij} and colimit A. By theorem 3.10.6(4), $f_0^{-1}(U^k) = \operatorname{Spec} A$, and the restriction of f_i is $\operatorname{Spec}(\varphi_i)$. Let $J_i = I(Z_i \cap U_i^k) \subseteq A_i$, the radical ideal with $V(J_i) = Z_i \cap U_i^k$ (proposition 2.1.6(6)), a proper ideal since $Z_i \cap U_i^k$ is nonempty. For $a \in J_i$ and $\mathfrak{q} \in Z_j \cap U_j^k$, the prime $\varphi_{ij}^{-1}(\mathfrak{q}) = f_{ij}(\mathfrak{q})$ lies in $Z_i \cap U_i^k$, so it contains a and $\varphi_{ij}(a) \in \mathfrak{q}$. Hence $\varphi_{ij}(J_i) \subseteq J_j$, and the rings A_i/J_i, all nonzero, form a directed system whose colimit B

is nonzero by lemma 3.10.3(4). The maps $A_i \to A_i/J_i$ induce a ring map $A \to B$. Choose a maximal ideal of B [12, Ring with $1 \neq 0$ have Maximal ideals] and let $\mathfrak{p} \subseteq A$ be its preimage, a prime ideal. For each i the composite $A_i \to A \to B$ factors through A_i/J_i, so $\varphi_i^{-1}(\mathfrak{p}) \supseteq J_i$, and $f_i(\mathfrak{p}) = \varphi_i^{-1}(\mathfrak{p}) \in V(J_i) \subseteq Z_i$. For the last assertion take $Z_i = S_i$, completing the proof.

Proposition 3.10.10: Quasi-Compact Opens of a Limit

Let (S_i, f_{ij}) be an inverse system with affine transition morphisms and limit S, and suppose S_0 is quasi-compact for some $0 \in I$.

1. The schemes S and S_i, $i \geq 0$, are quasi-compact. If S_0 is also quasi-separated, so are they.
2. For every quasi-compact open subset $V \subseteq S$ there are $i \in I$ and a quasi-compact open subset $V_i \subseteq S_i$ with $V = f_i^{-1}(V_i)$.
3. If $V_i \subseteq S_i$ and $V'_{i'} \subseteq S_{i'}$ are quasi-compact open subsets with $f_i^{-1}(V_i) = f_{i'}^{-1}(V'_{i'})$, then $f_{ij}^{-1}(V_i) = f_{i'j}^{-1}(V'_{i'})$ for some $j \geq i, i'$.
4. If $V^1, \dots, V^n \subseteq S_i$ are quasi-compact open subsets whose preimages cover S, then their preimages cover S_j for some $j \geq i$.

***Proof*:**
A quasi-compact open subset of a scheme is a finite union of affine open subsets, and its preimage under an affine morphism is therefore a finite union of affine open subsets, hence quasi-compact.

1. The morphisms f_0 and f_{0i} are affine (theorem 3.10.6(2)), so the first assertion follows from the remark just made, and the second from lemma 3.10.8(3).
2. For an affine open $U \subseteq S_0$, theorem 3.10.6(4) identifies $f_0^{-1}(U)$ with $\operatorname{Spec}(\varinjlim_{i\geq 0} A_i^U)$, where $A_i^U = \mathcal{O}_{S_i}(f_{0i}^{-1}(U))$, and the restriction of f_i to it with $\operatorname{Spec}(\varphi_i)$. As U runs over the affine open subsets of S_0, the sets $f_0^{-1}(U)$ cover S, so their distinguished open subsets form a basis of the topology of S (proposition 2.2.17). Each of them is $D(a)$ with $a = \varphi_i(a_i)$ for some $i \geq 0$ and $a_i \in A_i^U$ (lemma 3.10.3(2)), and then $D(a) = f_i^{-1}(D(a_i))$ by lemma 2.1.28, where $D(a_i) \subseteq f_{0i}^{-1}(U)$ is affine. Being quasi-compact, V is a finite union of such sets $f_{i_t}^{-1}(D_t)$. For i above all the i_t, the union V_i of the quasi-compact opens $f_{i_t i}^{-1}(D_t)$ satisfies $f_i^{-1}(V_i) = V$.
3. Choose $k \geq i, i'$. For $j \geq k$ put $P_j = f_{ij}^{-1}(V_i)$, a quasi-compact open subset of S_j. With the restricted transition morphisms, which are affine, the P_j form an inverse system whose limit is $f_i^{-1}(V_i)$, by theorem 3.10.6(1) and (3). The sets $C_j = P_j \setminus f_{i'j}^{-1}(V'_{i'})$ are closed in P_j and satisfy $f_{jj'}(C_{j'}) \subseteq C_j$ for $j \leq j'$. If all of them were nonempty, lemma 3.10.9 would give a point of $f_i^{-1}(V_i)$ outside $f_{i'}^{-1}(V'_{i'})$. So some C_j is empty, and then $C_{j'} \subseteq f_{jj'}^{-1}(C_j)$ is empty for every $j' \geq j$. The same argument applies to the other difference, and an index above both gives $f_{ij}^{-1}(V_i) = f_{i'j}^{-1}(V'_{i'})$.
4. Replacing i by an index above i and 0, and the V^l by their preimages, we may assume $i \geq 0$, so that S_i is quasi-compact by (1). Apply (3) to the quasi-compact opens $V^1 \cup \dots \cup V^n$ and S_i of S_i, whose preimages in S both equal S, as we sought to show.

Exercise 3.10.1

1. Let I be the set of positive integers ordered by divisibility. Show directly that the underlying set of $\operatorname{Spec}\mathbb{Q}$ is the limit of the underlying sets of the schemes $\operatorname{Spec}\mathbb{Z}[1/n]$, $n \in I$: a compatible family of primes $\mathfrak{p}_n \subseteq \mathbb{Z}[1/n]$ consists of the zero ideals. Compare with example 3.10.5(1).
2. For $n \geq 0$ let $S_n = \coprod_{m \geq n} \operatorname{Spec} k$, a disjoint union of copies of $\operatorname{Spec} k$ indexed by the integers $m \geq n$, and let $S_{n+1} \to S_n$ be the inclusion of the components with $m \geq n+1$. Show that these morphisms are affine, that every S_n is nonempty, and that $\varprojlim_n S_n$ is empty. Which hypothesis of lemma 3.10.9 fails?
3. Let (A_i, φ_{ij}) be a directed system of rings with colimit A. Show that A is a domain if every A_i is a domain, and a field if every A_i is a field. Show that if every A_i is local with maximal ideal $\mathfrak{m}_i$ and every φ_{ij} satisfies $\varphi_{ij}(\mathfrak{m}_i) \subseteq \mathfrak{m}_j$, then A is local with maximal ideal $\bigcup_i \varphi_i(\mathfrak{m}_i)$.
4. Let k be a field and $S = \operatorname{Spec} k[x_1, x_2, \dots]$, the limit of the schemes $S_n = \operatorname{Spec} k[x_1, \dots, x_n]$ along the inclusions of polynomial rings. Let $V = S \setminus \{\mathfrak{m}\}$ for $\mathfrak{m} = (x_1, x_2, \dots)$. Show that V is not quasi-compact and that $V \neq f_n^{-1}(V_n)$ for every n and every open $V_n \subseteq S_n$. Compare with proposition 3.10.10(2).

3.11⋆ Finite Presentation and Limits

⋆ *Optional. This section is an aside; nothing later in the book depends on it.*

Let $S = \varprojlim_i S_i$ be a limit with affine transition morphisms, and let $U_0 \subseteq S_0$ be an affine open subset. A finitely presented algebra over $\mathcal{O}_S(f_0^{-1}(U_0)) = \varinjlim_i \mathcal{O}_{S_i}(f_{0i}^{-1}(U_0))$ is given by finitely many polynomials, whose finitely many coefficients all come from a single ring $\mathcal{O}_{S_i}(f_{0i}^{-1}(U_0))$, and an identity among such data that holds over S already holds over some S_j (theorem 3.10.6(4), lemma 3.10.3).

The local scheme $\operatorname{Spec}\mathcal{O}_{X,x}$ of a smooth variety X is the limit of the affine open neighborhoods of x (example 3.10.5), which are smooth, along open immersions. Limits of quasi-compact and quasi-separated smooth schemes along affine étale morphisms are called essentially smooth (definition 3.11.12).

Lemma 3.11.1: Finitely Presented Algebras Are Compact

Let R be a ring, (A_i, φ_{ij}) a directed system of R-algebras with colimit A, and C an R-algebra.

1. Let C be finitely generated. If $u, v\colon C \to A_i$ are R-algebra maps with $\varphi_i \circ u = \varphi_i \circ v$, then $\varphi_{ij} \circ u = \varphi_{ij} \circ v$ for some $j \geq i$.
2. Let C be finitely presented. Every R-algebra map $w\colon C \to A$ equals $\varphi_i \circ u$ for some i and some R-algebra map $u\colon C \to A_i$.

Equivalently, the map $\varinjlim_i \operatorname{Hom}_R(C, A_i) \to \operatorname{Hom}_R(C, A)$ induced by the φ_i, from the colimit of sets of lemma 3.10.3(1), is injective in case (1) and bijective in case (2).

***Proof*:**

1. Let $c_1, \dots, c_n$ generate C. For each l, $\varphi_i(u(c_l) - v(c_l)) = 0$, so by lemma 3.10.3(2) there is $j_l \geq i$ with $\varphi_{ij_l}(u(c_l) - v(c_l)) = 0$. For j above all the j_l, the R-algebra maps $\varphi_{ij} \circ u$ and $\varphi_{ij} \circ v$ agree on generators, hence are equal.
2. Write $C = R[x_1, \dots, x_n]/(g_1, \dots, g_m)$, with $\overline{x}_l$ the class of x_l. By lemma 3.10.3(2) there are an index i and elements $a_1, \dots, a_n \in A_i$ with $\varphi_i(a_l) = w(\overline{x}_l)$. Then, for each t,

$$\varphi_i(g_t(a_1, \dots, a_n)) = g_t(w(\overline{x}_1), \dots, w(\overline{x}_n)) = w(g_t(\overline{x}_1, \dots, \overline{x}_n)) = 0$$

   so there is $j \geq i$ with $g_t(\varphi_{ij}(a_1), \dots, \varphi_{ij}(a_n)) = 0$ in A_j for all t. Hence $\overline{x}_l \mapsto \varphi_{ij}(a_l)$ defines an R-algebra map $u\colon C \to A_j$, and $\varphi_j \circ u$ agrees with w on the generators $\overline{x}_l$, so $\varphi_j \circ u = w$.

The reformulation follows from the description of the colimit of sets in lemma 3.10.3(1) and (2), completing the proof.

The affine pieces in the covering condition that defines a morphism locally of finite presentation (definition 3.4.33) are its finitely presented charts.

Definition 3.11.2: Finitely Presented Chart

Let $p\colon X \to S$ be a morphism of schemes. A *finitely presented chart* of p is a pair (U, V) of affine open subsets $U \subseteq X$ and $V \subseteq S$ with $p(U) \subseteq V$ such that $\mathcal{O}_X(U)$ is a finitely presented $\mathcal{O}_S(V)$-algebra.

Lemma 3.11.3: Finitely Presented Charts Exist Locally

Let $p\colon X \to S$ be a morphism of schemes.

1. If p is locally of finite presentation, then for every $x \in X$, every affine open $V \subseteq S$ containing $p(x)$, and every open $N \ni x$, there is a finitely presented chart (U, V) with $x \in U \subseteq N$.
2. If every point of X lies in U for some finitely presented chart (U, V), then p is locally of finite presentation.

In particular, if p is locally of finite presentation and $X' \subseteq X$, $S' \subseteq S$ are open with $p(X') \subseteq S'$, then $X' \to S'$ is locally of finite presentation.

***Proof*:**

1. By Step 2 of the proof of proposition 3.4.35, $p^{-1}(V)$ is covered by affine opens whose rings are finitely presented $\mathcal{O}_S(V)$-algebras. Let A be one containing x, and choose a distinguished open $D_A(e)$ with $x \in D_A(e) \subseteq A \cap N$ (corollary 2.1.13). It is affine with ring $\mathcal{O}_X(A)_e \cong \mathcal{O}_X(A)[t]/(te - 1)$ (proposition 2.2.13), a finitely presented $\mathcal{O}_X(A)$-algebra and therefore a finitely presented $\mathcal{O}_S(V)$-algebra by Step 1 of the proof of proposition 3.4.35.

2. We verify definition 3.4.33 for the cover of S by all of its affine open subsets. Let $V' \subseteq S$ be affine open and $x \in p^{-1}(V')$, and let (U, V) be a finitely presented chart with $x \in U$. By lemma 2.2.30 there is an open D with $p(x) \in D \subseteq V \cap V'$ that is distinguished in both, $D = D_V(a) = D_{V'}(b)$. The restriction $U \to V$ corresponds to the ring map $\mathcal{O}_S(V) \to \mathcal{O}_X(U)$ (proposition 3.1.12), so $U \cap p^{-1}(D) = D_U(\overline{a})$ for the image $\overline{a}$ of a (lemma 2.1.28), an affine open neighborhood of x with ring $\mathcal{O}_X(U)_{\overline{a}} \cong \mathcal{O}_X(U) \otimes_{\mathcal{O}_S(V)} \mathcal{O}_S(V)_a$. This ring is finitely presented over $\mathcal{O}_S(V)_a = \mathcal{O}_S(D) = \mathcal{O}_S(V')_b$ by Step 1 of the proof of proposition 3.4.35, and $\mathcal{O}_S(V')_b \cong \mathcal{O}_S(V')[t]/(tb-1)$ is finitely presented over $\mathcal{O}_S(V')$, so $\mathcal{O}_X(U)_{\overline{a}}$ is finitely presented over $\mathcal{O}_S(V')$ by the same Step 1. These open sets cover $p^{-1}(V')$.

For the last assertion, apply (1) with $V \subseteq S'$ and $N = X'$ to every point of X', and then (2) to $X' \to S'$, as we sought to show.

A morphism from the limit to a scheme locally of finite presentation over the base is defined at a finite stage, uniquely up to passing to a later stage (EGA IV$_3$, 8.13.1 [32], and [70, Tag 01ZC]). For affine schemes this is lemma 3.11.1.

Theorem 3.11.4: Morphisms from a Limit to a Scheme Locally of Finite Presentation

Let S be a scheme, let (T_i, f_{ij}) be an inverse system of S-schemes with affine transition morphisms and limit T, and suppose T_0 is quasi-compact and quasi-separated for some $0 \in I$. Let X be an S-scheme locally of finite presentation over S.

1. If $g, g' \colon T_i \to X$ are S-morphisms with $g \circ f_i = g' \circ f_i$, then $g \circ f_{ij} = g' \circ f_{ij}$ for some $j \geq i$.
2. Every S-morphism $T \to X$ equals $g \circ f_i$ for some $i \in I$ and some S-morphism $g \colon T_i \to X$.

Equivalently, the map $\varinjlim_i \mathrm{Hom}_S(T_i, X) \to \mathrm{Hom}_S(T, X)$, $[g] \mapsto g \circ f_i$, is bijective, the colimit being taken along the maps $g \mapsto g \circ f_{ij}$.

***Proof*:**
Write $\sigma_i \colon T_i \to S$ and $p \colon X \to S$ for the structure morphisms. By theorem 3.10.6(1) we may restrict to indices $i \geq 0$, and then T_i and T are quasi-compact and quasi-separated by proposition 3.10.10(1).

Step 1: statement (1). We may assume $i \geq 0$, replacing g and g' by their composites with a transition morphism. For $j \geq i$ let $E_j \subseteq T_j$ be the union, over all finitely presented charts (U, V) of p, of the open sets $(gf_{ij})^{-1}(U) \cap (g'f_{ij})^{-1}(U)$. Then $E_j = f_{ij}^{-1}(E_i)$, so the closed sets $T_j \setminus E_j$ satisfy $f_{jj'}(T_{j'} \setminus E_{j'}) \subseteq T_j \setminus E_j$. For $t \in T$ the point $gf_i(t) = g'f_i(t)$ lies in U for some finitely presented chart (U, V) (lemma 3.11.3(1)), so $f_i(t) \in E_i$. If every $T_j \setminus E_j$ were nonempty, lemma 3.10.9 would give $t \in T$ with $f_i(t) \notin E_i$. So some $T_j \setminus E_j$ is empty, and replacing i by j we may assume $E_i = T_i$. Then every point of T_i has an affine open neighborhood W with $g(W) \cup g'(W) \subseteq U$ for a finitely presented chart (U, V), and finitely many of these, $W^1, \ldots, W^n$ with charts (U_k, V_k), cover T_i. Fix k. For $j \geq i$ the set $W_j^k = f_{ij}^{-1}(W^k)$ is affine, $f_i^{-1}(W^k)$ is affine with ring $\varinjlim_{j \geq i} \mathcal{O}_{T_j}(W_j^k)$ (theorem 3.10.6(4)), and $\sigma_j = p \circ g \circ f_{ij}$ maps W_j^k into V_k. By proposition 3.1.12, the restrictions of gf_{ij} and $g'f_{ij}$ to W_j^k, morphisms into U_k over V_k,

correspond to $\mathcal{O}_S(V_k)$-algebra maps $\mathcal{O}_X(U_k) \to \mathcal{O}_{T_j}(W_j^k)$, namely $\varphi_{ij} \circ u$ and $\varphi_{ij} \circ u'$ for the maps u, u' at stage i, where φ_{ij} and φ_i denote the maps of the directed system $(\mathcal{O}_{T_j}(W_j^k))_{j \geq i}$ and of its colimit. The hypothesis $gf_i = g'f_i$ says $\varphi_i \circ u = \varphi_i \circ u'$, so lemma 3.11.1(1) gives j with $gf_{ij} = g'f_{ij}$ on W_j^k. An index j that works for every k gives $gf_{ij} = g'f_{ij}$ on the cover $(W_j^k)_k$ of T_j, hence on T_j (lemma 3.1.9).

Step 2: local factorizations for statement (2). Let $h\colon T \to X$ be an S-morphism. Cover T_0 by affine opens $A^1, \dots, A^m$. By theorem 3.10.6(4), $f_0^{-1}(A^l)$ is affine with ring $\varinjlim_{i \geq 0} \mathcal{O}_{T_i}(f_{0i}^{-1}(A^l))$. Every $t \in T$ lies in some $f_0^{-1}(A^l)$ and in a distinguished open $D(a)$ of it with $h(D(a)) \subseteq U$ for a finitely presented chart (U, V). Finitely many of these, $D(a^1), \dots, D(a^n)$ with charts (U_k, V_k), cover T. Choose $i \geq 0$ and $a_i^k \in \mathcal{O}_{T_i}(f_{0i}^{-1}(A^{l_k}))$ with $a^k = \varphi_i(a_i^k)$ (lemma 3.10.3(2)), and for $j \geq i$ let $W_j^k = f_{ij}^{-1}(D(a_i^k))$. These are distinguished open subsets, hence affine, $D(a^k) = f_i^{-1}(D(a_i^k))$ (lemma 2.1.28), and $\mathcal{O}_T(D(a^k)) = \varinjlim_{j \geq i} \mathcal{O}_{T_j}(W_j^k)$ by theorem 3.10.6(4) applied with i in place of 0. The closed subsets $W_j^k \setminus \sigma_j^{-1}(V_k)$ of the quasi-compact schemes W_j^k are compatible with the transition morphisms, and the limit $D(a^k)$ of the W_j^k (theorem 3.10.6(3)) maps into V_k, as $p \circ h$ does on $D(a^k)$. If all of these closed subsets were nonempty, lemma 3.10.9 would give a point of $D(a^k)$ mapping outside V_k. So one of them is empty, and for j large σ_j maps W_j^k into V_k, and the $\mathcal{O}_{T_j}(W_j^k)$ become $\mathcal{O}_S(V_k)$-algebras compatibly with the transition maps. The restriction of h to $D(a^k)$ is a morphism into U_k over V_k, so it corresponds to an $\mathcal{O}_S(V_k)$-algebra map $\mathcal{O}_X(U_k) \to \varinjlim_j \mathcal{O}_{T_j}(W_j^k)$ (proposition 3.1.12). As $\mathcal{O}_X(U_k)$ is finitely presented over $\mathcal{O}_S(V_k)$, lemma 3.11.1(2) factors this map through some $\mathcal{O}_{T_j}(W_j^k)$, which gives an S-morphism $g^k\colon W_j^k \to U_k \subseteq X$ with $g^k \circ f_j = h$ on $D(a^k)$. Fix j that works for every k.

Step 3: gluing. For k, k' the open set $W_j^k \cap W_j^{k'}$ is quasi-compact, as an intersection of two affine opens of the quasi-separated scheme T_j, and quasi-separated (lemma 3.10.8(4)). Its preimages in the $T_{j'}$, $j' \geq j$, form an inverse system with affine transition morphisms and limit $D(a^k) \cap D(a^{k'})$ (theorem 3.10.6(3)), and the restrictions of g^k and $g^{k'}$ to $W_j^k \cap W_j^{k'}$ agree after composition with the projection from that limit, both composites being h. By Step 1, applied to this inverse system, they agree on the preimage of $W_j^k \cap W_j^{k'}$ in some $T_{j'}$. Choose j' that works for all pairs k, k' and, by proposition 3.10.10(4), such that the preimages of the W_j^k cover $T_{j'}$, the $D(a^k)$ covering T. The morphisms $g^k \circ f_{jj'}$ on these preimages agree on overlaps and glue to an S-morphism $g\colon T_{j'} \to X$ (lemma 3.1.9), and $g \circ f_{j'} = h$ holds on every $D(a^k)$, hence on T. The reformulation follows from the description of the colimit of sets in lemma 3.10.3(1) and (2), as we sought to show.

Corollary 3.11.5: Morphisms between Limits

Let S be a scheme, let $(T_i)_{i \in I}$ be an inverse system of quasi-compact and quasi-separated S-schemes with affine transition morphisms, and let $(X_a)_{a \in J}$ be an inverse system of S-schemes locally of finite presentation over S with affine transition morphisms. Then an S-morphism $\varprojlim_i T_i \to \varprojlim_a X_a$ is the same as a family of elements $c_a \in \varinjlim_i \operatorname{Hom}_S(T_i, X_a)$, $a \in J$, such that c_a is the image of c_b under composition with the transition morphism $X_b \to X_a$ for all $a \leq b$.

***Proof*:**
An S-morphism into $\varprojlim_a X_a$ is a compatible family of S-morphisms into the X_a (the remark after definition 3.10.1), and by theorem 3.11.4 the S-morphisms $\varprojlim_i T_i \to X_a$ form the set $\varinjlim_i \operatorname{Hom}_S(T_i, X_a)$, compatibly with composition with the transition morphisms $X_b \to X_a$, as we sought to show.

Applied to spectra of finitely presented algebras, theorem 3.11.4 shows that a morphism locally of finite presentation is finitely presented over every pair of affine opens (compare the remark after definition 3.4.33).

Corollary 3.11.6: Finite Presentation on Every Pair of Affine Opens

Let $p\colon X \to S$ be locally of finite presentation, and let $U \subseteq X$ and $V \subseteq S$ be affine open subsets with $p(U) \subseteq V$. Then $\mathcal{O}_X(U)$ is a finitely presented $\mathcal{O}_S(V)$-algebra.

***Proof*:**
Write $B = \mathcal{O}_S(V)$ and $C = \mathcal{O}_X(U)$.

Step 1: C is a filtered colimit of finitely presented B-algebras. Let Λ be the set of pairs $\lambda = (E, F)$ with $E \subseteq C$ finite and F a finite subset of the kernel of the B-algebra map $B[x_e : e \in E] \to C$, $x_e \mapsto e$, ordered by $(E, F) \leq (E', F')$ when $E \subseteq E'$ and $F \subseteq F'$. It is a directed set. Put $C_\lambda = B[x_e : e \in E]/(F)$, with the maps $C_\lambda \to C_{\lambda'}$ induced by the inclusions of polynomial rings and the maps $\iota_\lambda\colon C_\lambda \to C$ induced by $x_e \mapsto e$. The induced map $\varinjlim_\lambda C_\lambda \to C$ is surjective, since $c = \iota_{(\{c\},\emptyset)}(\overline{x}_c)$, and injective by lemma 3.10.3(2): if $P \in B[x_e : e \in E]$ maps to 0 in C, its class vanishes in $C_{(E, F\cup\{P\})}$. So C is the colimit of the finitely presented B-algebras C_λ.

Step 2: C is a retract of some C_λ. By lemma 3.10.4 and proposition 3.1.12, $U \cong \operatorname{Spec} C$ is the limit of the affine V-schemes $\operatorname{Spec} C_\lambda$, whose transition morphisms are affine (corollary 3.4.11), with projections π_λ corresponding to the ι_λ. The restriction $U \to V$ of p is locally of finite presentation (lemma 3.11.3), so by theorem 3.11.4(2), with V as base, the identity of U equals $g \circ \pi_\lambda$ for some λ and some V-morphism $g\colon \operatorname{Spec} C_\lambda \to U$. On global sections this says $\iota_\lambda \circ u = \mathrm{id}_C$ for the B-algebra map $u = \Gamma(g)\colon C \to C_\lambda$.

Step 3: a retract of a finitely presented algebra is finitely presented. Write $C_\lambda = B[x_1, \dots, x_n]/(F)$ and let $\rho = u \circ \iota_\lambda$, a B-algebra endomorphism of C_λ with $\iota_\lambda \circ \rho = \iota_\lambda$. The map ι_λ is surjective, since $\iota_\lambda \circ u = \mathrm{id}_C$, and its kernel K is generated by the elements $\overline{x}_s - \rho(\overline{x}_s)$, which lie in K. For $y = P(\overline{x}_1, \dots, \overline{x}_n) \in K$ one has $\rho(y) = u(\iota_\lambda(y)) = 0$, so $y = P(\overline{x}_1, \dots, \overline{x}_n) - P(\rho(\overline{x}_1), \dots, \rho(\overline{x}_n))$, and every difference of monomials $a_1 \cdots a_q - b_1 \cdots b_q = \sum_l a_1 \cdots a_{l-1}(a_l - b_l)b_{l+1} \cdots b_q$ lies in the ideal generated by the $a_l - b_l$. Choosing $r_s \in B[x_1, \dots, x_n]$ mapping to $\rho(\overline{x}_s)$ gives $C \cong C_\lambda/K \cong B[x_1, \dots, x_n]/(F, x_1 - r_1, \dots, x_n - r_n)$, a finitely presented B-algebra, as we sought to show.

Throughout, for an S_0-scheme X_0 and $i \geq 0$, write $X_i = X_0 \times_{S_0} S_i$ and $X = X_0 \times_{S_0} S$. By lemma 3.10.7, X is the limit of the X_i, with affine transition morphisms, and $X_j \cong X_i \times_{S_i} S_j$. For an S_i-morphism u between such schemes, u_j and u_S denote its base changes along $S_j \to S_i$ and $S \to S_i$.

Corollary 3.11.7: Morphisms Descend to a Finite Stage

Let (S_i, f_{ij}) be an inverse system with affine transition morphisms and limit S, let $0 \in I$, and let X_0 and Y_0 be S_0-schemes with X_0 quasi-compact and quasi-separated and Y_0 locally of finite presentation over S_0.

1. If $u, v\colon X_i \to Y_i$ are S_i-morphisms with $u_S = v_S$, then $u_j = v_j$ for some $j \geq i$.
2. Every S-morphism $X \to Y$ equals u_S for some $i \geq 0$ and some S_i-morphism $u\colon X_i \to Y_i$.

***Proof*:**
Let $q_i\colon X \to X_i$ and $q_{ij}\colon X_j \to X_i$ be the projections and $\varpi_i\colon Y_i \to Y_0$ the projection. By the universal property of $Y_i = Y_0 \times_{S_0} S_i$, an S_i-morphism $u\colon X_i \to Y_i$ is determined by the S_0-morphism $\varpi_i \circ u\colon X_i \to Y_0$, every S_0-morphism $X_i \to Y_0$ arises in this way, and the same holds over S. The base change u_S satisfies $\varpi_i \circ u \circ q_i = \varpi \circ u_S$, with $\varpi\colon Y \to Y_0$ the projection, and likewise $\varpi_i \circ u \circ q_{ij} = \varpi_j \circ u_j$. The schemes X_i, $i \geq 0$, are quasi-compact and quasi-separated by proposition 3.10.10(1), and they form an inverse system of S_0-schemes with affine transition morphisms and limit X (lemma 3.10.7). So theorem 3.11.4, with S_0 as base and Y_0 as target, translates into (1) and (2), as we sought to show.

Corollary 3.11.8: Isomorphisms Descend to a Finite Stage

In the situation of corollary 3.11.7, suppose that X_0 and Y_0 are both quasi-compact, quasi-separated and locally of finite presentation over S_0.

1. If $u\colon X_i \to Y_i$ is an S_i-morphism such that u_S is an isomorphism, then u_j is an isomorphism for some $j \geq i$.
2. Every S-isomorphism $X \to Y$ equals u_S for some $i \geq 0$ and some S_i-isomorphism $u\colon X_i \to Y_i$.

***Proof*:**

1. By corollary 3.11.7(2), with the roles of X_0 and Y_0 exchanged, the inverse of u_S is v_S for an S_k-morphism $v\colon Y_k \to X_k$, and we may take $k \geq i$. Base change is compatible with composition, so $(v \circ u_k)_S = v_S \circ u_S = \mathrm{id}_X$, and corollary 3.11.7(1) gives $j \geq k$ with $v_j \circ u_j = \mathrm{id}_{X_j}$. Likewise $u_j \circ v_j = \mathrm{id}_{Y_j}$ after enlarging j, so u_j is an isomorphism.
2. Corollary 3.11.7(2) gives u with u_S the given isomorphism, and by (1) some u_j is an isomorphism, with $(u_j)_S = u_S$, completing the proof.

Theorem 3.11.9: Schemes Locally of Finite Presentation Descend to a Finite Stage

Let (S_i, f_{ij}) be an inverse system with affine transition morphisms and limit S, and suppose S_0 is quasi-compact and quasi-separated for some $0 \in I$. Let X be a quasi-compact and quasi-separated scheme locally of finite presentation over S. Then there are $i \geq 0$, a quasi-compact and quasi-separated scheme X_i locally of finite presentation over S_i, and an S-isomorphism $X_i \times_{S_i} S \cong X$. If S_0 and X are affine, X_i can be taken affine.

By corollary 3.11.8(2), the scheme X_i is unique in the sense that two such schemes, over S_i and $S_{i'}$, become isomorphic over some S_j, $j \geq i, i'$.

***Proof*:**

Step 1: algebras. Let (R_i) be a directed system of rings with colimit R, and let C be a finitely presented R-algebra, $C \cong R[x_1, \dots, x_n]/(g_1, \dots, g_m)$. The finitely many coefficients of the g_t are images of elements of a single R_i (lemma 3.10.3(2)). Lifting them gives polynomials $g_{t,i} \in R_i[x_1, \dots, x_n]$, and $C_i = R_i[x_1, \dots, x_n]/(g_{1,i}, \dots, g_{m,i})$ is a finitely presented R_i-algebra with $C_i \otimes_{R_i} R \cong C$, by the base change computation in Step 1 of the proof of proposition 3.4.35. This proves the affine case: if S_0 and X are affine, then $S = \operatorname{Spec} R$ with $R = \varinjlim_i \mathcal{O}(S_i)$ (theorem 3.10.6(4)), $\mathcal{O}_X(X)$ is finitely presented over R (corollary 3.11.6), and $X_i = \operatorname{Spec} C_i$ satisfies $X_i \times_{S_i} S \cong \operatorname{Spec}(C_i \otimes_{R_i} R) \cong X$ (theorem 3.2.1).

Step 2: affine pieces. Cover S_0 by affine opens $U^1, \dots, U^m$, and write $U_i^k = f_{0i}^{-1}(U^k)$, $R_i^k = \mathcal{O}_{S_i}(U_i^k)$ and $S^k = f_0^{-1}(U^k)$, which is affine with ring $R^k = \varinjlim_i R_i^k$. Let $p\colon X \to S$ be the structure morphism. The scheme X is covered by finitely many affine opens $X^1, \dots, X^N$ with $p(X^r) \subseteq S^{k(r)}$, and $C^r = \mathcal{O}_X(X^r)$ is finitely presented over $R^{k(r)}$ (corollary 3.11.6). By Step 1 there are i and finitely presented $R_i^{k(r)}$-algebras C_i^r with $C_i^r \otimes R^{k(r)} \cong C^r$. Let $Z_i^r = \operatorname{Spec} C_i^r$, an S_i-scheme through $U_i^{k(r)}$, and write Z_j^r and Z^r for its base changes to S_j and S. As S^k is the preimage of U_i^k, the scheme Z^r is $Z_i^r \times_{U_i^{k(r)}} S^{k(r)} \cong \operatorname{Spec}(C_i^r \otimes R^{k(r)})$, and we obtain S-isomorphisms $\gamma^r\colon Z^r \to X^r$. Likewise $Z_j^r \cong \operatorname{Spec}(C_i^r \otimes R_j^{k(r)})$ is affine, and $Z_j^r \to S_j$ is locally of finite presentation by lemma 3.11.3(2).

Step 3: overlaps. As X is quasi-separated, $X^r \cap X^s$ is a quasi-compact open of X^r, so $Z^{rs} = (\gamma^r)^{-1}(X^r \cap X^s)$ is a quasi-compact open of Z^r, the limit of the affine schemes Z_j^r (lemma 3.10.7). By proposition 3.10.10(2), after enlarging i, Z^{rs} is the preimage of a quasi-compact open $Z_i^{rs} \subseteq Z_i^r$, and we write Z_j^{rs} for its preimage in Z_j^r, which is the base change of Z_i^{rs} to S_j (proposition 3.2.5(1)). We take $Z_i^{rr} = Z_i^r$. Each Z_j^{rs} is a quasi-compact open of an affine scheme, hence quasi-compact and quasi-separated (lemma 3.10.8(4), proposition 2.4.11), and locally of finite presentation over S_j (lemma 3.11.3). The S-isomorphism $\theta^{rs} = (\gamma^s)^{-1} \circ \gamma^r$ maps Z^{rs} onto Z^{sr}, so by corollary 3.11.8(2), applied with i in place of 0, after enlarging i it is the base change of an S_i-isomorphism $\theta_i^{rs}\colon Z_i^{rs} \to Z_i^{sr}$, and we take $\theta_i^{rr} = \mathrm{id}$. Write θ_j^{rs} for the base changes.

Step 4: the gluing conditions. Fix r, s, t. The sets $(\theta_j^{rs})^{-1}(Z_j^{sr} \cap Z_j^{st})$ and $Z_j^{rs} \cap Z_j^{rt}$ are quasi-compact opens of Z_j^r, as intersections of quasi-compact opens of affine schemes, and their preimages in Z^r are $(\theta^{rs})^{-1}(Z^{sr} \cap Z^{st})$ and $Z^{rs} \cap Z^{rt}$. Both of these equal $(\gamma^r)^{-1}(X^r \cap X^s \cap X^t)$,

so by proposition 3.10.10(3) the two sets agree for j large, and then $\theta_j^{rs}(Z_j^{rs} \cap Z_j^{rt}) = Z_j^{sr} \cap Z_j^{st}$. On $Q_j = Z_j^{rs} \cap Z_j^{rt}$, the S_j-morphisms $\theta_j^{st} \circ \theta_j^{rs}$ and θ_j^{rt} into Z_j^t have the base changes $\theta^{st} \circ \theta^{rs}$ and θ^{rt}, which agree, both being $(\gamma^t)^{-1} \circ \gamma^r$ on $Q = Z^{rs} \cap Z^{rt}$. Since Q_j is quasi-compact and quasi-separated and Z_j^t is locally of finite presentation over S_j, corollary 3.11.7(1) makes them agree after a further base change. Only finitely many triples occur, and base change preserves the equalities already obtained, so for one index j the data $(Z_j^r, Z_j^{rs}, \theta_j^{rs})$ satisfy the hypotheses of theorem 2.2.56.

Step 5: gluing. Let X_j be the glued scheme, with open embeddings $\varepsilon^r \colon Z_j^r \to X_j$. The structure morphisms $Z_j^r \to S_j$ are compatible with the S_j-isomorphisms θ_j^{rs}, so they glue to a morphism $X_j \to S_j$ (lemma 3.1.9). The scheme X_j is covered by the finitely many affine opens $\varepsilon^r(Z_j^r)$, whose pairwise intersections $\varepsilon^r(Z_j^{rs})$ are quasi-compact, so it is quasi-compact and quasi-separated by lemma 3.10.8(2), and it is locally of finite presentation over S_j by lemma 3.11.3(2). The base change $X_j \times_{S_j} S$ is covered by the preimages of the $\varepsilon^r(Z_j^r)$, which are copies of the Z^r (proposition 3.2.5(1)) meeting along the Z^{rs} with transition isomorphisms θ^{rs}. The S-isomorphisms $\gamma^r \colon Z^r \to X^r \subseteq X$ satisfy $\gamma^s \circ \theta^{rs} = \gamma^r$, so they glue to an S-morphism $X_j \times_{S_j} S \to X$ (lemma 3.1.9). The inverses $(\gamma^r)^{-1}$ glue in the same way to an inverse morphism, so $X_j \times_{S_j} S \cong X$ over S, as we sought to show.

Smoothness is defined through local presentations (definition 2.7.10). For descent, the presentation is needed over an arbitrary affine open of the base.

Lemma 3.11.10: Standard Smooth Charts

Let $h \colon X \to S$ be locally of finite type, $x \in X$, and $n \geq 0$.

1. If h is smooth of relative dimension n at x, then for every affine open $W \subseteq S$ containing $h(x)$ there are an affine open $U \ni x$ with $h(U) \subseteq W$, polynomials $g_1, \dots, g_r \in \mathcal{O}_S(W)[t_1, \dots, t_{n+r}]$, and an isomorphism of $\mathcal{O}_S(W)$-algebras
$$\mathcal{O}_X(U) \cong \mathcal{O}_S(W)[t_1, \dots, t_{n+r}]/(g_1, \dots, g_r)$$
under which $\Delta = \det(\partial g_a / \partial t_b)_{1 \leq a,b \leq r}$ maps to a unit.
2. Conversely, if U, W and $g_1, \dots, g_r$ are as in (1), then h is smooth of relative dimension n at every point of U.

***Proof*:**

1. Let U', V and $g_1, \dots, g_r \in \mathcal{O}_S(V)[t_1, \dots, t_{n+r}]$ be as in definition 2.7.10, with V affine so that the vertical arrow there is defined, and let $\iota \colon U' \to \operatorname{Spec} B$ be the open immersion, $B = \mathcal{O}_S(V)[t]/(g)$. The Jacobian matrix has rank r at $\iota(x)$, so after renumbering the variables its minor Δ on the first r columns does not vanish at $\iota(x)$. By lemma 2.2.30 there is an open $D \ni h(x)$ with $D = D_V(b) = D_W(c)$. Choose $e \in B$ with $\iota(x) \in D(e) \subseteq \iota(U') \cap D(\Delta \bar{b})$, where $D(\bar{b})$ is the preimage of $D_V(b)$ (lemma 2.1.28). Every prime of B_e avoids $\Delta \bar{b}$, since the primes of B_e are those of B not containing e, so $\Delta \bar{b}$ is a unit in B_e. Let $U = \iota^{-1}(D(e))$, an affine open neighborhood of x with ring B_e (proposition 2.2.13) and with $h(U) \subseteq D \subseteq W$. Lifting e to $\tilde{e} \in \mathcal{O}_S(V)[t]$, one has $B_e \cong \mathcal{O}_S(V)[t, y]/(g_1, \dots, g_r, y\tilde{e} - 1)$, and since b becomes

a unit this is also $\mathcal{O}_S(V)_b[t,y]/(g_1,\dots,g_r,y\tilde{e}-1)$. Now $\mathcal{O}_S(V)_b=\mathcal{O}_S(D)=\mathcal{O}_S(W)_c$, and in $\mathcal{O}_S(W)_c$ every coefficient has the form w/c^m. Multiplying each of the $r+1$ relations by a power of the unit c turns them into polynomials $G_1,\dots,G_r,H\in\mathcal{O}_S(W)[t,y]$ generating the same ideal, and

$$B_e\cong\mathcal{O}_S(W)[t_1,\dots,t_{n+r},y,z]/(G_1,\dots,G_r,H,cz-1)$$

because $\mathcal{O}_S(W)_c\cong\mathcal{O}_S(W)[z]/(cz-1)$. With the variables $t_1,\dots,t_r,y,z$ in front, the matrix of partial derivatives of the $r+2$ relations with respect to them is block lower triangular. Its diagonal blocks are $(c^{m_a}\,\partial g_a/\partial t_b)$, with determinant $c^{\sum m_a}\Delta$, then $\partial H/\partial y=c^m\tilde{e}$, then c. All three are units in B_e. The presentation has $n+(r+2)$ variables and $r+2$ relations, as required.

2. The isomorphism exhibits U as an open subscheme, all of Spec of the presented algebra, over the affine W, and at every point of U the minor Δ, being a unit, has nonzero image in the residue field, so the Jacobian has rank r there. This is definition 2.7.10 at every point of U, completing the proof.

Theorem 3.11.11: Descent of Properties of Morphisms

Let (S_i,f_{ij}) be an inverse system with affine transition morphisms and limit S, let $0\in I$, and let $h_0\colon X_0\to Y_0$ be a morphism of S_0-schemes with X_0 and Y_0 quasi-compact and quasi-separated and h_0 locally of finite presentation. Let $h_i\colon X_i\to Y_i$ and $h\colon X\to Y$ be the base changes of h_0 along $S_i\to S_0$ and $S\to S_0$. If h is

1. an isomorphism,
2. an open immersion,
3. a closed immersion,
4. smooth, or smooth of relative dimension n,
5. étale, or
6. finite,

then there is $i\geq 0$ such that h_j has the same property for every $j\geq i$.

Here étale means smooth of relative dimension 0 (definition 2.7.10), which over a locally Noetherian base is the flat and unramified notion of definition 2.7.19 by theorem 2.7.20.

***Proof*:**
Step 1: reduction. Put $Y_i=Y_0\times_{S_0}S_i$ and $Y=Y_0\times_{S_0}S$. By lemma 3.10.7, Y is the limit of the Y_i, with affine transition morphisms, and the universal property of fibered products identifies $X_0\times_{S_0}S_i$ with $X_0\times_{Y_0}Y_i$, and h_i with its projection to Y_i, as in the proof of lemma 3.10.7. Replacing the system (S_i) by (Y_i), we may therefore assume $Y_0=S_0$ and that h_0 is the structure morphism of X_0. Then S_0 and X_0 are quasi-compact and quasi-separated, h_0 is locally of finite presentation, $Y_i=S_i$, $Y=S$, and the X_i and X are quasi-compact and quasi-separated, being affine over X_0 (proposition 3.4.12, lemma 3.10.8(3)). Each listed property is preserved

by base change: open and closed immersions by proposition 3.2.5(3) and (2), finite morphisms by proposition 3.4.25(2), and smooth morphisms by proposition 2.7.11(1), applied to the open subsets on which the relative dimension is constant, which cover the source by lemma 3.11.10. Since h_j is the base change of h_i, it suffices to find one i.

Step 2: isomorphisms. Apply corollary 3.11.8(1) to the S_0-schemes X_0 and S_0, both quasi-compact, quasi-separated and locally of finite presentation over S_0, and to $u = h_0$.

Step 3: open immersions. The image $V = h(X)$ is open and homeomorphic to X, hence quasi-compact, and $V = f_i^{-1}(V_i)$ for a quasi-compact open $V_i \subseteq S_i$ (proposition 3.10.10(2)). For $j \geq i$ put $V_j = f_{ij}^{-1}(V_i)$. The open sets $h_j^{-1}(V_j)$ and X_j of X_j are quasi-compact (lemma 3.10.8(1)) and have the same preimage X in X, so they agree for j large by proposition 3.10.10(3), and h_j maps X_j into V_j. The $V_{j'}$, $j' \geq j$, form an inverse system with affine transition morphisms and limit V (theorem 3.10.6(3)), V_j is quasi-compact and quasi-separated (lemma 3.10.8(4)), the morphism $X_j \to V_j$ is locally of finite presentation (lemma 3.11.3), and its base changes along $V_{j'} \to V_j$ and $V \to V_j$ are $X_{j'} \to V_{j'}$ and $X \to V$. The last of these is an isomorphism, an open immersion being an isomorphism onto its image (definition 2.2.49). By Step 2, applied to the system $(V_{j'})$, some $X_{j'} \to V_{j'}$ is an isomorphism, and $h_{j'}$ is an open immersion.

Step 4: closed immersions. Cover S_0 by finitely many affine opens W. For each of them $h_0^{-1}(W)$ is quasi-compact and quasi-separated (lemma 3.10.8(1) and (4)) and locally of finite presentation over W (lemma 3.11.3), its base changes are the restrictions of the h_i and of h over the preimages of W, and the restriction of h is a closed immersion (proposition 3.2.5(2)). Being a closed immersion is local on the target (proposition 3.1.24(4)), so we may assume S_0 affine. Then $S_i = \operatorname{Spec} B_i$ and $S = \operatorname{Spec} B$ with $B = \varinjlim_i B_i$ (theorem 3.10.6(4)), and $X = \operatorname{Spec}(B/I)$ for an ideal I, with h corresponding to $B \to B/I$ (proposition 3.1.24(1)). The morphism h is locally of finite presentation (proposition 3.4.35(2)), so B/I is a finitely presented B-algebra (corollary 3.11.6). Hence I is finitely generated: choose $B/I \cong B[x_1, \dots, x_n]/(G_1, \dots, G_q)$ and $b_l \in B$ mapping to the image of x_l, and let $\psi\colon B[x] \to B$ be the B-algebra map with $x_l \mapsto b_l$. Composed with $B \to B/I$ it is the presentation, so $\psi(G_\nu) \in I$, and for $a \in I$ one has $a = \sum c_\nu G_\nu$ in $B[x]$, whence $a = \sum \psi(c_\nu)\psi(G_\nu)$. So $I = (\psi(G_1), \dots, \psi(G_q))$. Write these generators as images of $a_1, \dots, a_q \in B_i$ and let $Z_i = \operatorname{Spec}(B_i/(a_1, \dots, a_q))$, which is closed in S_i (proposition 3.1.24(2)), affine with a finitely presented ring, and so quasi-compact, quasi-separated and locally of finite presentation over S_i (lemma 3.11.3(2)). Its base change to S is $\operatorname{Spec}(B/I)$ by Step 1 of the proof of proposition 3.4.35, that is, X as an S-scheme. By corollary 3.11.8(2), applied with i in place of 0, there are $j \geq i$ and an S_j-isomorphism $X_j \cong Z_i \times_{S_i} S_j$. So h_j is this isomorphism followed by the closed immersion $Z_i \times_{S_i} S_j \to S_j$ (proposition 3.2.5(2)), which is a closed immersion (proposition 3.1.24(3)).

Step 5: smooth and étale morphisms. The morphisms h and h_j are locally of finite presentation, hence locally of finite type (proposition 3.4.34(1)), so lemma 3.11.10 applies to them. Cover S_0 by affine opens $W^1, \dots, W^m$ and write W_i^k and W^k for their preimages in S_i and S, so that $\mathcal{O}_S(W^k) = \varinjlim_i \mathcal{O}_{S_i}(W_i^k)$. For $x \in X$ choose k with $h(x) \in W^k$ and, by lemma 3.11.10(1), an affine open $U \ni x$ with $\mathcal{O}_X(U) \cong C = \mathcal{O}_S(W^k)[t_1, \dots, t_{n+r}]/(g_1, \dots, g_r)$ and Δ a unit in C, where n is the relative dimension at x. Finitely many such charts $U_1, \dots, U_N$ cover X. Fix one, with data k, n, r, g. Write $\Delta e = 1 + \sum_a q_a g_a$ in $\mathcal{O}_S(W^k)[t]$ for polynomials e, q_a.

The finitely many coefficients of g_a, e, q_a come from $\mathcal{O}_{S_i}(W_i^k)$ for some i, and the identity holds coefficientwise after passing to some $j \geq i$ (lemma 3.10.3(2)). The resulting polynomials $g_{a,j}$ define $C_j = \mathcal{O}_{S_j}(W_j^k)[t]/(g_{1,j}, \ldots, g_{r,j})$, in which the minor Δ_j of the $g_{a,j}$ is a unit, and $C_j \otimes \mathcal{O}_S(W^k) \cong C$ (Step 1 of the proof of proposition 3.4.35). By proposition 3.10.10(2), after enlarging j, $U = U_l$ is the preimage of a quasi-compact open $U_{l,j} \subseteq X_j$. The open sets $U_{l,j} \cap h_j^{-1}(W_j^k)$ and $U_{l,j}$ are quasi-compact (lemma 3.10.8(1)) with the same preimage U_l in X, so they agree after enlarging j (proposition 3.10.10(3)), and $U_{l,j}$ maps into W_j^k. Now $U_{l,j}$ and $\operatorname{Spec} C_j$ are quasi-compact, quasi-separated W_j^k-schemes locally of finite presentation over W_j^k (lemma 3.10.8(4), lemma 3.11.3), with base changes U_l and $\operatorname{Spec} C$ to W^k, the limit of the $W_{j'}^k$ (theorem 3.10.6(3)), and these are isomorphic over W^k. By corollary 3.11.8(2), after enlarging j, $U_{l,j} \cong \operatorname{Spec} C_j$ over W_j^k, and by lemma 3.11.10(2) the morphism h_j is smooth of relative dimension n at every point of $U_{l,j}$. Do this for all l with a common j. By proposition 3.10.10(4) the $U_{l,j}$ cover X_j after enlarging j once more, so h_j is smooth, and of relative dimension n (in particular étale when $n = 0$) if h is.

Step 6: finite morphisms. The reduction at the start of Step 4 applies, with the restriction of h over the preimage of an affine open $W \subseteq S_0$ finite by proposition 3.4.25(2), and finiteness local on the target by proposition 3.4.17. So assume S_0 affine, $S_i = \operatorname{Spec} B_i$ and $S = \operatorname{Spec} B$ with $B = \varinjlim_i B_i$. As h is finite, $X = \operatorname{Spec} C$ with C a finitely generated B-module (definition 3.4.15), and C is a finitely presented B-algebra (proposition 3.4.35(2), corollary 3.11.6). By Step 1 of the proof of theorem 3.11.9, after enlarging i there is a finitely presented B_i-algebra $C_i = B_i[x_1, \ldots, x_n]/(g_1, \ldots, g_m)$ with $C_i \otimes_{B_i} B \cong C$, so that $C = B[x_1, \ldots, x_n]/(g_1, \ldots, g_m)$. Every element of C is integral over B [10, Module-Finite versus Finite-Type-and-Integral], so for each l there is a monic $P_l \in B[T]$ with $P_l(\overline{x}_l) = 0$ in C, that is, $P_l(x_l) = \sum_t q_{lt}\, g_t$ in $B[x_1, \ldots, x_n]$ for some polynomials q_{lt}. The finitely many coefficients of the P_l and the q_{lt} come from some B_j, $j \geq i$, with the P_l still monic, and the identities hold coefficientwise after enlarging j (lemma 3.10.3(2)). Then in $C_j = C_i \otimes_{B_i} B_j = B_j[x_1, \ldots, x_n]/(g_1, \ldots, g_m)$ each generator $\overline{x}_l$ satisfies a monic polynomial over the image of B_j. The elements integral over that image form a subring [10, Integral Elements Form Rings], which contains the $\overline{x}_l$ and so is all of C_j. Being of finite type and integral, C_j is a finitely generated B_j-module [10, Module-Finite versus Finite-Type-and-Integral], and $Z_j = \operatorname{Spec} C_j \to S_j$ is finite (proposition 3.4.17). It is affine with a finitely presented ring, so quasi-compact, quasi-separated and locally of finite presentation over S_j (lemma 3.11.3(2)), and its base change to S is $\operatorname{Spec}(C_j \otimes_{B_j} B) = \operatorname{Spec} C$, that is, X as an S-scheme. By corollary 3.11.8(2), applied with j in place of 0, there are $j' \geq j$ and an $S_{j'}$-isomorphism $X_{j'} \cong Z_j \times_{S_j} S_{j'}$. So $h_{j'}$ is this isomorphism followed by the finite morphism $Z_j \times_{S_j} S_{j'} \to S_{j'}$ (proposition 3.4.25(2)), and it is finite (proposition 3.4.25(1)), an isomorphism being finite, completing the proof.

The local schemes of smooth schemes are limits of smooth schemes along open immersions, and open immersions are étale. Let $j\colon U' \to U$ be an open immersion and $u \in U'$, and choose an affine open $V \ni j(u)$ of U and $e \in \mathcal{O}_U(V)$ with $j(u) \in D_V(e) \subseteq j(U')$. Then $D_V(e) \cong \operatorname{Spec} \mathcal{O}_U(V)[y]/(ey - 1)$ (proposition 2.2.13), with $\partial(ey-1)/\partial y = e$ a unit. Such charts cover U', so j is locally of finite type (lemma 3.11.3(2), proposition 3.4.34(1)), and it is smooth of relative dimension 0 at u by lemma 3.11.10(2).

Definition 3.11.12: Essentially Smooth Morphism

A morphism $T \to S$ is *essentially smooth* if there are an inverse system (T_i, f_{ij}) of S-schemes over a directed set, with every T_i quasi-compact, quasi-separated and smooth over S and every transition morphism f_{ij} affine and étale, and an S-isomorphism $T \cong \varprojlim_i T_i$. A scheme over a field k is *essentially smooth over* k if its structure morphism is essentially smooth.

No separatedness is imposed on the stages. Over a perfect field k these are Morel's essentially smooth k-schemes [55, Introduction, Conventions and notations], whose stages are smooth k-schemes of finite type, except that Morel also requires the limit to be Noetherian. The stages are quasi-compact and quasi-separated, and smooth schemes are locally of finite presentation (lemma 3.11.10(1), lemma 3.11.3(2)), so corollary 3.11.5 computes the S-morphisms between two essentially smooth S-schemes from the S-morphisms between their stages.

Example 3.11.13: Essentially Smooth Schemes

1. Let $\operatorname{Spec} A \to S$ be smooth with S affine, and let $\Sigma \subseteq A$ be multiplicative. By example 3.10.5(1), $\operatorname{Spec} \Sigma^{-1}A$ is the limit of the schemes $\operatorname{Spec} A_s$, $s \in \Sigma$, along inclusions of distinguished opens. Each $\operatorname{Spec} A_s$ is an open subscheme of $\operatorname{Spec} A$, hence smooth over S, and the transition morphisms are open immersions between affine schemes, hence affine (corollary 3.4.11) and étale. So $\operatorname{Spec} \Sigma^{-1}A$ is essentially smooth over S. This includes $\operatorname{Spec} A_{\mathfrak{p}}$ for every prime $\mathfrak{p}$.
2. Let X be smooth over S and $x \in X$. By example 3.10.5(2), $\operatorname{Spec} \mathcal{O}_{X,x}$ is the limit of the affine open neighborhoods of x, which are smooth over S, along affine open immersions. So $\operatorname{Spec} \mathcal{O}_{X,x}$ is essentially smooth over S, and if X is integral, so is $\operatorname{Spec} K(X)$.

Corollary 3.11.14: Smooth Schemes over an Essentially Smooth Scheme

Let $T \cong \varprojlim_i T_i$ be essentially smooth over S as in definition 3.11.12, and let Y be a quasi-compact and quasi-separated scheme smooth over T. Then there are an index i and a quasi-compact and quasi-separated scheme Y_i smooth over T_i with $Y \cong Y_i \times_{T_i} T$ over T, and Y is essentially smooth over S: it is the limit of the S-schemes $Y_i \times_{T_i} T_j$, $j \geq i$, each smooth over S, along affine étale transition morphisms.

***Proof*:**
The morphism $Y \to T$ is locally of finite type, as definition 2.7.10 requires, and locally of finite presentation: every point of Y lies in an affine open U over an affine open W of T with $\mathcal{O}_Y(U)$ a finitely presented $\mathcal{O}_T(W)$-algebra (lemma 3.11.10(1)), so lemma 3.11.3(2) applies. The stages are quasi-compact and quasi-separated, so by theorem 3.11.9 there are an index i and a quasi-compact and quasi-separated scheme Y_i locally of finite presentation over T_i with $Y \cong Y_i \times_{T_i} T$ over T. By theorem 3.11.11(4), applied to the structure morphism $Y_i \to T_i$ and the inverse system $(T_j)_{j \geq i}$, after enlarging i the morphism $Y_i \to T_i$ is smooth. Then so is every base change $Y_j = Y_i \times_{T_i} T_j \to T_j$, by proposition 2.7.11(1) applied to the open subsets of Y_i on which the relative dimension is constant, which cover Y_i by lemma 3.11.10. Each Y_j is smooth over S: Y_j and T_j are covered by open subsets on which $Y_j \to T_j$ and $T_j \to S$ have constant relative dimension, and proposition 2.7.11(2) applies to the restrictions. The transition

morphism $Y_{j'} \to Y_j$ is the base change of $f_{jj'}$, hence affine (proposition 3.4.12(2)) and étale (proposition 2.7.11(1)). By lemma 3.10.7 the limit of the Y_j is $Y_i \times_{T_i} T \cong Y$, as an S-scheme, as we sought to show.

Exercise 3.11.1

1. Let k be a field and $X = \operatorname{Spec} k[x_1, x_2, \dots]$, the limit over n of $T_n = \operatorname{Spec} k[x_1, \dots, x_n]$ as in exercise 3.10.1 (4). Show that the identity of X is not of the form $g \circ f_n$ for any n and any k-morphism $g\colon T_n \to X$. Which hypothesis of theorem 3.11.4 fails?

2. Let k be an infinite field, let $S_0 = \mathbb{A}^1_k = \operatorname{Spec} k[t]$, and let S be the limit of the open subschemes $D(f)$, $f \neq 0$, so that $S = \operatorname{Spec} k(t)$ (example 3.10.5(1)). Let $Y_0 = \coprod_{a \in k} \operatorname{Spec} k$, mapping the copy indexed by a to the point $t = a$, and $X_0 = \emptyset$. Show that the base change of $\emptyset \to Y_0$ to S is an isomorphism, while its base change to every $D(f)$ is not. Which hypothesis of corollary 3.11.8 fails?

3. Let A be a ring and (N_i, φ_{ij}) a directed system of A-modules with colimit N, described as in lemma 3.10.3. Show that for a finitely generated A-module M, two homomorphisms $u, v\colon M \to N_i$ with $\varphi_i \circ u = \varphi_i \circ v$ satisfy $\varphi_{ij} \circ u = \varphi_{ij} \circ v$ for some $j \geq i$, and that for a finitely presented M every homomorphism $M \to N$ factors through some N_i.

4. Let X be smooth over a field k and $x \in X$ a closed point. Show that $\operatorname{Spec} \mathcal{O}_{X,x} \setminus \{\mathfrak{m}_x\}$ is essentially smooth over k, as the limit of the schemes $U \setminus \{x\}$ for U running over the affine open neighborhoods of x. (*Hint:* use theorem 3.10.6(3) and proposition 3.5.8.)

5. Let A be a ring, $\mathfrak{p} \subseteq A$ a prime, and C a finitely presented $A_\mathfrak{p}$-algebra. Show that there are $f \in A \setminus \mathfrak{p}$ and a finitely presented A_f-algebra C' with $C' \otimes_{A_f} A_\mathfrak{p} \cong C$.

4⋆

Schemes II.5: Group Schemes

⋆ *Optional. This chapter is an aside; nothing later in the book depends on it.*

This chapter can comfortably skipped if the reader is not interested in studying algebraic groups or abelian varieties in the near future. The chapter is placed here since it is a natural follow up to all the constructions we have presented in the previous chapter, and (I think) does not require much in the way of quasi-coherent or coherent sheaves, and similar for sheaf cohomology.

In classical algebraic geometry, an *algebraic group* is a variety that is simultaneously a group, with the multiplication and inversion maps being morphisms of varieties. The theory of algebraic groups over algebraically closed fields is rich and well-developed, encompassing the classification of reductive groups, the structure theory of Borel and torus subgroups, and the representation theory that underpins much of modern mathematics.

Over a general base scheme, however, the correct notion is that of a *group scheme*: a scheme G equipped with morphisms encoding multiplication, identity, and inversion, subject to the usual group axioms expressed diagrammatically. This generalization is essential for arithmetic applications (where the base may be $\operatorname{Spec}(\mathbb{Z})$ or $\operatorname{Spec}(\mathbb{Z}_p)$), for the study of families of groups (where the group structure may degenerate or change character from fiber to fiber), and for capturing phenomena invisible over algebraically closed fields—such as infinitesimal group schemes in characteristic p.

This section develops the foundations of group schemes. The reader should be comfortable with fiber products (section 3.2) throughout.

4.1 Definition and First Examples

Recall that in any category $\mathcal{C}$ with finite products and a terminal object e, a *group object* is an object G equipped with morphisms

$$\mu : G \times G \to G, \qquad \epsilon : e \to G, \qquad \iota : G \to G$$

satisfying the usual group axioms—associativity, identity, and inverse—expressed as commutative diagrams. In the category **Set**, a group object is an ordinary group. In the category of smooth manifolds, a group object is a Lie group. In the category of schemes, a group object is a group scheme.

Definition 4.1.1: Group Scheme

Let S be a scheme. A *group scheme over* S is an S-scheme G together with S-morphisms

$$\mu : G \times_S G \longrightarrow G, \qquad \epsilon : S \longrightarrow G, \qquad \iota : G \longrightarrow G$$

called *multiplication*, *identity*, and *inversion*, such that the following diagrams commute:
Associativity:

$$\begin{array}{ccc} G \times_S G \times_S G & \xrightarrow{\mu \times \mathrm{id}} & G \times_S G \\ {\scriptstyle \mathrm{id}\times\mu}\downarrow & & \downarrow{\scriptstyle \mu} \\ G \times_S G & \xrightarrow[\mu]{} & G \end{array}$$

Identity:

$$\begin{array}{ccccc} S \times_S G & \xrightarrow{\epsilon\times\mathrm{id}} & G \times_S G & \xleftarrow{\mathrm{id}\times\epsilon} & G \times_S S \\ & {\scriptstyle\sim}\searrow & \downarrow{\scriptstyle \mu} & \swarrow{\scriptstyle\sim} & \\ & & G & & \end{array}$$

Inverse:

$$\begin{array}{ccccc} G & & \xrightarrow{\epsilon\circ\pi} & & G \\ {\scriptstyle (\mathrm{id},\iota)}\downarrow & & & \searrow{\scriptstyle \epsilon\circ\pi} & \downarrow{\scriptstyle \mu} \\ G \times_S G & \xrightarrow[\mu]{} & G & \xleftarrow{\epsilon} & S \end{array}$$

where $\pi : G \to S$ is the structure morphism.
A group scheme is *commutative* (or *abelian*) if additionally $\mu \circ \sigma = \mu$, where $\sigma : G \times_S G \to G \times_S G$ is the swap morphism.

The identity morphism $\epsilon : S \to G$ is a section of the structure map $\pi : G \to S$. Its image is a closed subscheme of G when G is separated over S, but in general it need not be a closed point or even a closed subscheme—it is an S-valued point of G in the sense of section 3.3.2. Over a field k, however, $\epsilon : \mathrm{Spec}\,(k) \to G$ picks out a k-rational point, which is closed if G is of finite type over k.

Proposition 4.1.2: Group Schemes over a Field are Separated

Let G be a group scheme of finite type over a field k. Then G is separated over k.

We shall assume a proposition from proven later, the intuition that translated directly from topology is that separatedness is the equivalent of Hausdorffness, and a space is Hausdorff is the diagonal is closed.

***Proof*:**
We must show the diagonal $\Delta : G \to G \times_k G$ is a closed immersion (section 3.5.1). Consider the morphism $\varphi : G \times_k G \to G$ defined by $\varphi = \mu \circ (\mathrm{id} \times \iota)$, i.e., $(g, h) \mapsto g \cdot h^{-1}$. The diagonal $\Delta(G)$ is the fiber $\varphi^{-1}(\epsilon(\mathrm{Spec}\,(k)))$. Since ϵ is a section of a morphism of finite type (and k is a field), the image of ϵ is a closed point of G. The fiber of a morphism over a closed point is a closed subscheme of the source. Hence $\Delta(G)$ is closed.

For a group scheme the rational point is given for free by the identity, and smoothness upgrades the conclusion from connectedness to integrality.

Corollary 4.1.3: Group Varieties Are Geometrically Integral

Let k be a field and let G be a smooth connected group scheme of finite type over k (definition 4.1.1). Then G is geometrically integral, and it is separated over k.

***Proof*:**
Step 1: geometric connectedness. The identity section $\epsilon\colon \mathrm{Spec}\,k \to G$ is a morphism of k-schemes, so $G(k) \neq \emptyset$. Being of finite type over k, the scheme G is locally of finite type, and it is connected by hypothesis, so proposition 3.4.27 makes $G_{\overline{k}}$ connected.
Step 2: $G_{\overline{k}}$ is regular. Smoothness survives extension of scalars. Locally G admits a factorization as in definition 2.7.10 through $\mathrm{Spec}\,(k[x_1, \ldots, x_{n+r}]/(f_1, \ldots, f_r))$ along an open U, and the rank condition says that U misses the closed set $V(J)$ cut out by the ideal J of $r \times r$ minors of the Jacobian. Extending scalars to $\overline{k}$ changes neither the f_i nor the minors, and $V(J\overline{k}[x_1, \ldots, x_{n+r}])$ is the preimage of $V(J)$, so $U_{\overline{k}}$ misses it too. Hence $G_{\overline{k}}$ is smooth over $\overline{k}$, and proposition 2.7.14 makes it regular.
Step 3: regular plus connected is integral. By proposition 2.7.28 a regular scheme is the disjoint union of its irreducible components, and $G_{\overline{k}}$ is Noetherian, so by lemma 2.4.3 there are finitely many of them and each is open and closed. Step 1 leaves exactly one, so $G_{\overline{k}}$ is irreducible. Its local rings are regular, hence domains by proposition 2.7.4, and in particular reduced, so $G_{\overline{k}}$ is reduced by proposition 2.4.15. Irreducible and reduced is integral by proposition 2.4.26, so $G_{\overline{k}}$ is integral and G is geometrically integral in the sense of definition 3.2.8. Separatedness over k is proposition 4.1.2, completing the proof.

Smoothness enters only through regularity after base change, and that is the exact strengthening of "regular" the geometric statement requires: over an imperfect field a regular k-scheme need not stay reduced over $\overline{k}$, so the corollary fails with "smooth" weakened to "regular". Finite type is used twice, once for proposition 3.4.27 and once for the finiteness of the component set in Step 3. Separatedness is not an extra assumption on G, since every group scheme of finite type over a field has it; it is recorded here because geometric integrality is usually invoked alongside it.

Let us now examine the fundamental examples.

Example 4.1.4: The Additive Group Scheme
The *additive group scheme* $\mathbb{G}_{a,S} = \mathbb{A}^1_S$ has underlying scheme $\mathrm{Spec}_S(\mathcal{O}_S[t])$, with multiplication given by the ring map

$$\mathcal{O}_S[t] \longrightarrow \mathcal{O}_S[t] \otimes_{\mathcal{O}_S} \mathcal{O}_S[t], \qquad t \longmapsto t \otimes 1 + 1 \otimes t,$$

identity $t \mapsto 0$, and inversion $t \mapsto -t$. These encode, respectively, addition, zero, and negation: for any S-scheme T,

$$\mathbb{G}_a(T) = \Gamma(T, \mathcal{O}_T)$$

with its additive group structure. Over a field k, this is simply the additive group of the coordinate ring.

Example 4.1.5: The Multiplicative Group Scheme
The *multiplicative group scheme* $\mathbb{G}_{m,S} = \mathrm{Spec}_S(\mathcal{O}_S[t, t^{-1}])$ has multiplication

$$t \longmapsto t \otimes t,$$

identity $t \mapsto 1$, and inversion $t \mapsto t^{-1}$. For any S-scheme T:

$$\mathbb{G}_m(T) = \Gamma(T, \mathcal{O}_T)^\times,$$

the group of invertible global sections under multiplication.

Example 4.1.6: The General Linear Group Scheme
For a positive integer n, the *general linear group scheme* is

$$\mathrm{GL}_{n,S} = \mathrm{Spec}_S\big(\mathcal{O}_S[x_{11}, x_{12}, \ldots, x_{nn},\, d]/(d \cdot \det(x_{ij}) - 1)\big)\,.$$

The variable d is the formal inverse of $\det(x_{ij})$, so the underlying scheme parametrizes $n \times n$ matrices with invertible determinant. The multiplication map is given by matrix multiplication: the ring map sends x_{ij} to $\sum_k x_{ik} \otimes x_{kj}$, and the identity sends $x_{ij} \mapsto \delta_{ij}$. For any S-scheme T:

$$\mathrm{GL}_n(T) \cong \mathrm{GL}_n(\Gamma(T, \mathcal{O}_T)),$$

the group of invertible $n \times n$ matrices with entries in $\Gamma(T, \mathcal{O}_T)$. When $n = 1$, we recover $\mathrm{GL}_1 \cong \mathbb{G}_m$.

Every example so far was produced by writing down a coordinate ring and reading off a comultiplication. A second route names a functor from k-algebras to groups and then exhibits it as the functor of points of a scheme (section 3.3), and the projective linear group is the first case where that route is the easier one. Its classical description as the quotient $\mathrm{GL}_n/\mathbb{G}_m$ is unavailable here, since quotients of group schemes are not constructed until section 4.4 and even there require faithfully flat descent rather than a pointwise construction. The way around this is to present the same object as a symmetry group: PGL_n is what acts on the matrix algebra M_n.

Definition 4.1.7: The Projective Linear Group Scheme

Let k be a commutative ring and let $n \geq 1$. For a k-algebra R let $M_n(R)$ denote the R-algebra of $n \times n$ matrices, free as an R-module on the matrix units e_{ij}. The *projective linear group* $\mathrm{PGL}_{n,k}$ is the group functor

$$\mathrm{PGL}_n \colon k\text{-}\mathbf{Alg} \longrightarrow \mathbf{Grp}, \qquad \mathrm{PGL}_n(R) := \mathrm{Aut}_{R\text{-alg}}(M_n(R))$$

assigning to R the group of R-algebra automorphisms of $M_n(R)$ under composition, and to a map $R \to R'$ the base change $\varphi \mapsto \varphi \otimes_R R'$.
Write $\mathrm{GL}(M_n)$ for the group scheme of R-linear automorphisms of the free module M_n, so that the basis $\{e_{ij}\}$ gives $\mathrm{GL}(M_n) \cong \mathrm{GL}_{n^2,k}$ as in example 4.1.6. Then $\mathrm{PGL}_{n,k}$ is the closed subscheme of $\mathrm{GL}_{n^2,k}$ cut out by the conditions

$$\varphi(e_{ij})\,\varphi(e_{kl}) = \delta_{jk}\,\varphi(e_{il}), \qquad \sum_{i=1}^{n} \varphi(e_{ii}) = 1$$

imposed for all indices, with composition of automorphisms as group law. In particular $\mathrm{PGL}_{n,k}$ is an affine group scheme and a closed subgroup scheme of $\mathrm{GL}_{n^2,k}$.

Unwinding the second display: an R-point of GL_{n^2} is an R-linear automorphism φ of $M_n(R)$, recorded by the coefficients $a_{(ij),(kl)}$ in $\varphi(e_{kl}) = \sum_{i,j} a_{(ij),(kl)} e_{ij}$. Multiplication on $M_n(R)$ is R-bilinear, so such a φ is multiplicative as soon as it is multiplicative on matrix units and unital as soon as it fixes $\sum_i e_{ii}$; expanding the two conditions in the $a_{(ij),(kl)}$ therefore yields a finite list of quadratic and linear polynomials with integer coefficients whose vanishing is equivalent to φ being an algebra map. Their zero locus is closed under composition and inversion, so $\mathrm{PGL}_{n,k} = \mathrm{Spec}\,(A/\mathfrak{a})$ is a closed subgroup scheme, where A is the coordinate ring of $\mathrm{GL}_{n^2,k}$ and $\mathfrak{a}$ is the ideal they generate; affineness is then proposition 4.1.9. The integer coefficients make the construction compatible with base change, so the same equations define $\mathrm{PGL}_{n,S}$ over an arbitrary base scheme S.

It remains to reconcile this with the classical picture. Each $g \in \mathrm{GL}_n(R)$ gives the automorphism $c_g \colon x \mapsto gxg^{-1}$ of $M_n(R)$, and $c_g = c_h$ exactly when gh^{-1} is central in $M_n(R)$, that is, scalar. Conjugation is thus a homomorphism $\mathrm{GL}_n \to \mathrm{PGL}_n$ whose kernel is the copy of $\mathbb{G}_m$ sitting in GL_n as the scalar matrices (example 4.1.5). Over a field it is surjective on points, which is the Skolem-Noether theorem, and the argument is three lines. Let k be a field and $\varphi \in \mathrm{PGL}_n(k)$; give $V = k^n$ the twisted structure $a \cdot v := \varphi(a)v$ and call the result V_φ. The ring $M_n(k)$ is simple with V as its unique simple module up to isomorphism and every module a direct sum of copies of V, so the n-dimensional module V_φ is isomorphic to V; such an isomorphism is a k-linear bijection g with $g\varphi(a) = ag$ for all a, that is, $\varphi = c_{g^{-1}}$. With the kernel computation this gives

$$\mathrm{PGL}_n(k) = \mathrm{GL}_n(k)/k^\times$$

so on points over a field the automorphism functor returns the naive quotient.

The equality concerns fields and does not survive verbatim over an arbitrary ring. For a k-algebra R the map $\mathrm{GL}_n(R)/R^\times \to \mathrm{PGL}_n(R)$ is injective but need not be surjective: an automorphism φ twists the standard module R^n, and under the Morita equivalence between $M_n(R)$-modules and R-modules the twisted module corresponds to an invertible R-module L_φ, with φ of the form c_g precisely when L_φ is free. The obstruction lies in $\mathrm{Pic}\,(R)$ and vanishes for fields and local rings [63]; see also [74] and [51]. This is also why $R \mapsto \mathrm{GL}_n(R)/R^\times$ cannot serve as the definition: it is a group functor, but

it fails to be a sheaf and is not representable.

Example 4.1.8: The Case $n = 2$
Take $n = 2$, so M_2 is free of rank 4 on $e_{11}, e_{12}, e_{21}, e_{22}$ and PGL_2 is a closed subgroup scheme of GL_4. The scalars can be watched disappearing on the diagonal torus: for $t \in R^\times$ and $g = \mathrm{diag}(t, 1)$,

$$c_g(e_{11}) = e_{11}, \qquad c_g(e_{12}) = t\, e_{12}, \qquad c_g(e_{21}) = t^{-1} e_{21}, \qquad c_g(e_{22}) = e_{22}$$

so in the ordered basis $(e_{11}, e_{12}, e_{21}, e_{22})$ the automorphism c_g is $\mathrm{diag}(1, t, t^{-1}, 1) \in \mathrm{GL}_4(R)$. A general diagonal $g = \mathrm{diag}(t_1, t_2)$ produces $\mathrm{diag}(1, t_1 t_2^{-1}, t_1^{-1} t_2, 1)$, depending only on the ratio $t_1 t_2^{-1}$. The diagonal torus $\mathbb{G}_m^2 \subseteq \mathrm{GL}_2$ therefore maps onto a one-dimensional torus in PGL_2, the scalars $t_1 = t_2$ collapsing to the identity, as the kernel computation predicts. Over a field, $\mathrm{PGL}_2(k) = \mathrm{GL}_2(k)/k^\times$ acts faithfully on $\mathbb{P}^1_k$ by fractional linear transformations.

The definition claimed affineness on the strength of PGL_n being closed inside GL_{n^2}. That step is worth isolating, since it is invoked again for every kernel and every stabilizer met later in this chapter, none of which come with a coordinate ring already in hand.

Proposition 4.1.9: Closed Subgroup Schemes of an Affine Group Scheme Are Affine

Let S be a scheme and let G be an S-group scheme (definition 4.1.1) whose structure morphism $G \to S$ is affine. If $\iota\colon H \to G$ is a closed subgroup scheme, then the structure morphism $H \to S$ is affine. In particular, if $S = \mathrm{Spec}\,(k)$ and G is an affine scheme, then H is an affine scheme.

***Proof*:**
Only the underlying schemes are at issue; the group structure plays no role. Let $U \subseteq S$ be an affine open subset and let G_U denote its preimage in G, an affine scheme because $G \to S$ is affine (definition 3.4.5). Both conditions defining a closed immersion (definition 3.1.22) are local on the target, so the restriction $\iota^{-1}(G_U) \to G_U$ is again a closed immersion. A closed subscheme of an affine scheme $\mathrm{Spec}\, A$ is $\mathrm{Spec}\,(A/\mathfrak{a})$ for an ideal $\mathfrak{a} \subseteq A$ by proposition 3.1.24(1), hence affine, and $\iota^{-1}(G_U)$ is precisely the preimage of U under $H \to S$. So every affine open of S has affine preimage in H, which is the definition of an affine morphism, completing the proof.

Short as it is, this keeps the affine theory closed under the constructions of the present chapter. Kernels (definition 4.2.2) and stabilizers (definition 4.3.3) are built as fiber products over a section and are therefore closed subgroup schemes, so an affine group scheme has affine kernels and affine stabilizers, each of which is again the spectrum of a Hopf algebra by proposition 4.1.14. The pattern fails for non-affine group schemes: an abelian variety (example 4.1.12) is proper over the base, so every closed subgroup scheme of it is proper as well, and among proper schemes the affine ones are exactly the finite ones.

Example 4.1.10: Roots of Unity
For a positive integer n, the *group scheme of n-th roots of unity* is

$$\mu_n = \mathrm{Spec}\,(\mathbb{Z}[t]/(t^n - 1)).$$

The group structure is inherited from $\mathbb{G}_m$: the comultiplication is $t \mapsto t \otimes t$ (the same as for $\mathbb{G}_m$,

but now $t^n = 1$). For any ring A:

$$\mu_n(A) = \{a \in A \mid a^n = 1\} \subseteq A^\times.$$

The behavior of μ_n depends dramatically on the characteristic of the base. Over a field k with $\mathrm{char}(k) \nmid n$, the polynomial $t^n - 1$ has n distinct roots in $\overline{k}$, and $\mu_{n,k}$ is a reduced scheme of degree n over k—an étale group scheme. Over $\overline{k}$, it is isomorphic (as a scheme) to the disjoint union of n copies of $\mathrm{Spec}\,(\overline{k})$, and $\mu_n(\overline{k}) \cong \mathbb{Z}/n\mathbb{Z}$.

In characteristic $p > 0$ with $p \mid n$, the situation changes. Taking $n = p$, we have $t^p - 1 = (t-1)^p$ in $\mathbb{F}_p[t]$, so

$$\mu_{p,\mathbb{F}_p} = \mathrm{Spec}\,(\mathbb{F}_p[t]/((t-1)^p)) = \mathrm{Spec}\,(\mathbb{F}_p[s]/(s^p))$$

where $s = t - 1$. This is a *non-reduced* scheme: a single fat point at $t = 1$ with a p-dimensional tangent space. The group $\mu_p(\mathbb{F}_p) = \{1\}$ is trivial, yet the scheme μ_p is non-trivial—it carries nilpotent structure that remembers the "ghost" of the p-th roots of unity that have collapsed to 1.

Example 4.1.11: Constant Group Schemes

Let Γ be a finite (abstract) group. The *constant group scheme* associated to Γ over a base S is the scheme

$$\Gamma_S = \coprod_{\gamma \in \Gamma} S,$$

a disjoint union of copies of S indexed by elements of Γ. The group structure is defined by declaring that the component indexed by γ times the component indexed by γ' lands in the component indexed by $\gamma\gamma'$. For any connected S-scheme T:

$$\Gamma_S(T) = \Gamma.$$

The constant group scheme $(\mathbb{Z}/n\mathbb{Z})_k$ over a field k is always étale, being a disjoint union of copies of $\mathrm{Spec}\,k$, and reduced, regardless of the characteristic. It should be carefully distinguished from μ_n: over $\mathbb{F}_p$, the scheme $(\mathbb{Z}/p\mathbb{Z})_{\mathbb{F}_p}$ consists of p disjoint points, while $\mu_{p,\mathbb{F}_p}$ is a single fat point.

Example 4.1.12: Elliptic Curves as Group Schemes

An elliptic curve E over a field k (see section 7.6) is a smooth projective curve of genus 1 equipped with a k-rational point $O \in E(k)$. The chord-and-tangent construction endows E with a group scheme structure: the multiplication is the addition law on points, $\epsilon = O$, and ι is the map $P \mapsto -P$. Unlike the previous examples, E is *not* affine: it is a projective group scheme. Smooth connected projective group schemes over a field are called *abelian varieties*. Their group law is automatically commutative, and elliptic curves are the abelian varieties of dimension 1.

4.1.1 The Hopf Algebra Perspective

The examples above were all defined by specifying ring maps encoding multiplication, identity, and inversion. For affine group schemes, this algebraic data has a clean formalization: the coordinate ring carries the structure of a *Hopf algebra*. This perspective is particularly useful for explicit computations and for connecting group schemes to representation theory.

Let $G = \operatorname{Spec}(A)$ be an affine group scheme over a field k. The group operations on G dualize to k-algebra maps on A:

$$\begin{aligned} \mu : G \times_k G \to G \quad &\rightsquigarrow \quad \Delta : A \to A \otimes_k A \qquad \text{(comultiplication)} \\ \epsilon : \operatorname{Spec}(k) \to G \quad &\rightsquigarrow \quad \varepsilon : A \to k \qquad \text{(counit)} \\ \iota : G \to G \quad &\rightsquigarrow \quad S : A \to A \qquad \text{(antipode)} \end{aligned}$$

The group axioms translate directly:

Definition 4.1.13: Hopf Algebra

A *commutative Hopf algebra* over k is a commutative k-algebra A equipped with k-algebra homomorphisms $\Delta : A \to A \otimes_k A$ (comultiplication) and $\varepsilon : A \to k$ (counit), and a k-algebra map $S : A \to A$ (antipode), satisfying:

1. *Coassociativity:* $(\Delta \otimes \mathrm{id}) \circ \Delta = (\mathrm{id} \otimes \Delta) \circ \Delta$.
2. *Counit:* $(\varepsilon \otimes \mathrm{id}) \circ \Delta = \mathrm{id} = (\mathrm{id} \otimes \varepsilon) \circ \Delta$.
3. *Antipode:* $\mu_A \circ (S \otimes \mathrm{id}) \circ \Delta = \varepsilon \cdot 1_A = \mu_A \circ (\mathrm{id} \otimes S) \circ \Delta$, where $\mu_A : A \otimes_k A \to A$ is the multiplication of A.

The coassociativity and counit axioms say that (A, Δ, ε) is a *coalgebra*; the antipode axiom says that S "undoes" the comultiplication in the same way that inversion undoes multiplication in a group. The key constraint is that Δ and ε must be *algebra* homomorphisms—this encodes the fact that multiplication $\mu : G \times G \to G$ is a morphism of schemes, not a set-theoretic map.

Proposition 4.1.14: Affine Group Schemes and Hopf Algebras

The functor $G \mapsto \mathcal{O}(G)$ is an anti-equivalence of categories:

$$\{\text{Affine group schemes over } k\}^{\mathrm{op}} \xrightarrow{\sim} \{\text{Commutative Hopf algebras over } k\}.$$

***Proof*:**
This follows from the anti-equivalence between affine k-schemes and commutative k-algebras. An affine group scheme is a group object in the category of affine k-schemes; dualizing, its coordinate ring becomes a cogroup object in k-**Alg**. The axioms of a cogroup object in a category of commutative algebras are exactly the axioms of a commutative Hopf algebra.

Let us revisit the earlier examples through this lens.

Example 4.1.15: Hopf Algebras

1. *The additive group.* The Hopf algebra of $\mathbb{G}_a = \operatorname{Spec}(k[t])$ is $k[t]$ with $\Delta(t) = t \otimes 1 + 1 \otimes t$, $\varepsilon(t) = 0$, and $S(t) = -t$. In the language of coalgebras, t is a *primitive* element: $\Delta(t) = t \otimes 1 + 1 \otimes t$.
2. *The multiplicative group.* The Hopf algebra of $\mathbb{G}_m = \operatorname{Spec}(k[t, t^{-1}])$ is $k[t, t^{-1}]$ with $\Delta(t) = t \otimes t$, $\varepsilon(t) = 1$, and $S(t) = t^{-1}$. Here t is a *group-like* element: $\Delta(t) = t \otimes t$.

3. *The general linear group.* The Hopf algebra of GL_n is $k[x_{ij}, \det^{-1}]$ with $\Delta(x_{ij}) = \sum_k x_{ik} \otimes x_{kj}$ (matrix multiplication), $\varepsilon(x_{ij}) = \delta_{ij}$ (the identity matrix), and $S(x_{ij})$ given by the (i,j)-cofactor of (x_{ij}) divided by $\det$—this is the Cramer's rule formula for the entries of the inverse matrix.

4. *Roots of unity.* The Hopf algebra of μ_n is $k[t]/(t^n - 1)$ with the same comultiplication as $\mathbb{G}_m$. The quotient by the ideal $(t^n - 1)$ is compatible with the Hopf structure because $\Delta(t^n - 1) = t^n \otimes t^n - 1 \otimes 1 = (t^n - 1) \otimes t^n + 1 \otimes (t^n - 1)$, which lies in the ideal generated by $t^n - 1$ in $A \otimes A$. In Hopf algebra language, $(t^n - 1)$ is a *Hopf ideal*.

The Hopf algebra viewpoint makes certain constructions transparent. For instance, a *homomorphism* of group schemes $\varphi : G \to H$ corresponds to a homomorphism of Hopf algebras $\varphi^* : \mathcal{O}(H) \to \mathcal{O}(G)$ (respecting Δ, ε, and S). The kernel of φ is the group scheme whose Hopf algebra is $\mathcal{O}(G)/(\ker \varepsilon_H \cdot \mathcal{O}(G))$; we will formalize this in section 4.2.

4.1.16 Remark: non-commutative Hopf algebras If we drop the requirement that A be commutative, we obtain the notion of a *non-commutative Hopf algebra*, which corresponds to a "quantum group" rather than a group scheme. The theory of quantum groups, developed by Drinfeld and Jimbo, is a rich subject in its own right but lies outside our scope. In this text, all Hopf algebras are commutative (corresponding to the fact that the coordinate ring of an affine scheme is always commutative), though the group schemes themselves need not be.

4.1.2 Finite and Infinitesimal Group Schemes

Over an algebraically closed field of characteristic zero, the theory of affine algebraic groups is largely controlled by Lie theory: connected algebraic groups are smooth, their structure is determined by their Lie algebras, and finite group schemes are just finite groups viewed as zero-dimensional varieties. In characteristic $p > 0$, the situation is fundamentally richer: there exist group schemes that are "purely infinitesimal"—non-reduced, supported at a single point, yet carrying nontrivial group structure encoded in the nilpotent elements of their coordinate ring.

Definition 4.1.17: Finite Group Scheme

A group scheme G over a field k is *finite* if it is of the form $G = \operatorname{Spec}(A)$ where A is a finite-dimensional k-algebra. The *order* of G is $\dim_k A$.

A finite group scheme of order n has at most n geometric points, but may have fewer if A has nilpotents. The extreme cases are:

Étale group schemes. A finite group scheme is *étale* if its coordinate ring A is a product of separable field extensions of k. Over an algebraically closed field, an étale group scheme of order n is simply a disjoint union of n copies of $\operatorname{Spec}(k)$, and the group structure is that of an ordinary finite group. The constant group scheme $(\mathbb{Z}/n\mathbb{Z})_k$ is the prototypical example.

Infinitesimal (or connected) group schemes. A finite group scheme $G = \operatorname{Spec}(A)$ is *infinitesimal* if A is a local ring, i.e., G is supported at a single point. The group structure is then entirely encoded in the nilpotent elements of A. These exist only in positive characteristic (or over non-reduced bases).

We have already seen the simplest examples of both types.

Example 4.1.18: μ_p in Characteristic p

As computed in Example 4.1.10, over $\mathbb{F}_p$ the group scheme $\mu_p = \operatorname{Spec}(\mathbb{F}_p[t]/(t^p - 1)) = \operatorname{Spec}(\mathbb{F}_p[s]/(s^p))$ (where $s = t - 1$) is infinitesimal of order p. Its Hopf algebra structure (expressed in terms of s) is:

$$\Delta(s) = s \otimes 1 + 1 \otimes s + s \otimes s, \qquad \varepsilon(s) = 0, \qquad S(s) = \sum_{i=1}^{p-1} (-1)^i s^i.$$

The comultiplication is neither purely primitive ($s \otimes 1 + 1 \otimes s$) nor purely group-like ($s \otimes s$)—it is a mixture, reflecting the fact that μ_p is a "twisted" infinitesimal group.

Example 4.1.19: The Frobenius Kernel α_p

The group scheme $\alpha_p = \operatorname{Spec}(k[t]/(t^p))$, defined over a field k of characteristic $p > 0$, is the kernel of the Frobenius endomorphism on $\mathbb{G}_a$. Its Hopf algebra is $k[t]/(t^p)$ with

$$\Delta(t) = t \otimes 1 + 1 \otimes t, \qquad \varepsilon(t) = 0, \qquad S(t) = -t.$$

Note that t is primitive, just as in $\mathbb{G}_a$, but the relation $t^p = 0$ (rather than no relation) makes α_p a finite group scheme of order p. For any k-algebra A:

$$\alpha_p(A) = \{a \in A \mid a^p = 0\},$$

which is a subgroup of $(A, +)$. Over a perfect field, α_p has only the trivial k-point, yet it is a non-trivial group scheme: for instance, $\alpha_p(k[\epsilon]/(\epsilon^p)) = k \cdot \epsilon$ is a p-element group.

The schemes α_p and μ_p are both infinitesimal group schemes of order p, but they are *not* isomorphic: α_p is connected and has no non-trivial characters (homomorphisms to $\mathbb{G}_m$), while μ_p is the Cartier dual of $\mathbb{Z}/p\mathbb{Z}$ and detects p-torsion in the multiplicative group.

The contrast between α_p and μ_p illustrates a general structural result. Over a perfect field k of characteristic $p > 0$, every finite group scheme G admits a canonical filtration

$$0 \subseteq G^0 \subseteq G$$

where G^0 is the *connected component of the identity* (the largest connected subgroup scheme) and G/G^0 is étale. This is the *connected-étale sequence*:

$$1 \longrightarrow G^0 \longrightarrow G \longrightarrow G^{\text{ét}} \longrightarrow 1,$$

and over a perfect field this sequence splits (non-canonically), so $G \cong G^0 \times G^{\text{ét}}$. The connected part G^0 is an infinitesimal group scheme (its coordinate ring is local), and the étale part $G^{\text{ét}}$ is determined by the action of $\operatorname{Gal}(\bar{k}/k)$ on the geometric points $G(\bar{k})$.

This decomposition has no analogue in characteristic zero, where every finite group scheme over a field is étale. It is a manifestation of the richer arithmetic of positive characteristic, and plays a fundamental role in the theory of abelian varieties (where the p-torsion $A[p]$ of an abelian variety splits into connected and étale parts whose relative sizes determine the p-rank).

The Frobenius and Verschiebung

The characteristic-p phenomena above are organized by two fundamental endomorphisms. Let k be a perfect field of characteristic p, and let G be a group scheme over k.

The *absolute Frobenius* $F_{\mathrm{abs}} : G \to G$ is the identity on the underlying topological space and acts as $a \mapsto a^p$ on the structure sheaf. This is a morphism of schemes but *not* a morphism of k-schemes (it does not commute with the structure map unless $k = \mathbb{F}_p$). To obtain a k-morphism, one passes to the *relative Frobenius* $F : G \to G^{(p)}$, where $G^{(p)}$ is the base change of G along the Frobenius $\mathrm{Spec}\,(k) \to \mathrm{Spec}\,(k)$. The relative Frobenius is a homomorphism of group schemes.

For a commutative group scheme, there is a dual morphism $V : G^{(p)} \to G$ called the *Verschiebung* (German for "shift"). It satisfies $V \circ F = [p]_G$ and $F \circ V = [p]_{G^{(p)}}$, where $[p]$ denotes multiplication by p in the group law. The kernels $\ker(F)$ and $\ker(V)$ are finite group schemes that capture, respectively, the "infinitesimal" and "étale" parts of the p-torsion.

For example, $\ker(F : \mathbb{G}_a \to \mathbb{G}_a^{(p)}) = \alpha_p$ (since F acts as $t \mapsto t^p$ on $k[t]$, and $\ker = \mathrm{Spec}\,(k[t]/(t^p))$), while $\ker(F : \mathbb{G}_m \to \mathbb{G}_m^{(p)}) = \mu_p$.

4.2 Homomorphisms, Kernels, and Exact Sequences

Definition 4.2.1: Homomorphism of Group Schemes

A *homomorphism* of S-group schemes $\varphi : G \to H$ is a morphism of S-schemes that is compatible with the group structures:

$$\varphi \circ \mu_G = \mu_H \circ (\varphi \times \varphi)$$

, $\varphi \circ \epsilon_G = \epsilon_H$, and $\varphi \circ \iota_G = \iota_H \circ \varphi$.

In the functor-of-points language, φ is a homomorphism if and only if for every S-scheme T, the induced map $\varphi(T) : G(T) \to H(T)$ is a group homomorphism.

Definition 4.2.2: Kernel of a Homomorphism

Let $\varphi : G \to H$ be a homomorphism of S-group schemes. The *kernel* of φ is the fiber product

$$\ker(\varphi) = G \times_H S$$

where the two maps to H are $\varphi : G \to H$ and $\epsilon_H : S \to H$.

By the universal property of fiber products, for any S-scheme T:

$$\ker(\varphi)(T) = \{g \in G(T) \mid \varphi(T)(g) = e_H \in H(T)\},$$

which is exactly the kernel of the group homomorphism $\varphi(T) : G(T) \to H(T)$. The kernel $\ker(\varphi)$ is a closed subgroup scheme of G (being a fiber of a morphism over a section).

Example 4.2.3: Classical Exact Sequences

1. *The n-th power map on $\mathbb{G}_m$.* The homomorphism $[n] : \mathbb{G}_m \to \mathbb{G}_m$ defined by $t \mapsto t^n$ has kernel μ_n. This gives a short exact sequence of group schemes:
$$1 \longrightarrow \mu_n \longrightarrow \mathbb{G}_m \xrightarrow{(\,)^n} \mathbb{G}_m \longrightarrow 1.$$
This is the *Kummer sequence*. Over a field k with $\mathrm{char}(k) \nmid n$, passing to $\overline{k}$-points recovers the familiar exact sequence $1 \to \mu_n(\overline{k}) \to \overline{k}^\times \xrightarrow{n} \overline{k}^\times \to 1$.

2. *The determinant.* The determinant $\det : \mathrm{GL}_n \to \mathbb{G}_m$ is a homomorphism of group schemes with kernel SL_n:
$$1 \longrightarrow \mathrm{SL}_n \longrightarrow \mathrm{GL}_n \xrightarrow{\det} \mathbb{G}_m \longrightarrow 1.$$

3. *The Frobenius on $\mathbb{G}_a$ in characteristic p.* The Frobenius $F : \mathbb{G}_a \to \mathbb{G}_a$ defined by $t \mapsto t^p$ has kernel α_p:
$$0 \longrightarrow \alpha_p \longrightarrow \mathbb{G}_a \xrightarrow{F} \mathbb{G}_a \longrightarrow 0.$$
This is the *Artin–Schreier sequence* (in its scheme-theoretic form).

A word of caution on exactness is in order. A sequence $1 \to N \xrightarrow{i} G \xrightarrow{\varphi} H \to 1$ of group schemes is said to be *exact* if: i identifies N with $\ker(\varphi)$ as a closed subgroup scheme of G, and φ is faithfully flat. The surjectivity of φ is *not* the naive statement that $\varphi(T) : G(T) \to H(T)$ is surjective for all T; this is often false even when the sequence is exact. For instance, in the Kummer sequence over $\mathbb{Q}$, the map $\mathbb{Q}^\times \xrightarrow{n} \mathbb{Q}^\times$ is not surjective (not every rational number is an n-th power), but the map of schemes $\mathbb{G}_m \xrightarrow{n} \mathbb{G}_m$ is faithfully flat. The correct notion of surjectivity for group schemes is faithfully flat descent, not surjectivity on points.

4.3 Actions of Group Schemes on Schemes

Definition 4.3.1: Action of a Group Scheme

Let G be an S-group scheme and X an S-scheme. An *action* of G on X is an S-morphism
$$\sigma : G \times_S X \longrightarrow X$$
satisfying the usual axioms: the diagram
$$\begin{array}{ccc} G \times_S G \times_S X & \xrightarrow{\mathrm{id}\times\sigma} & G \times_S X \\ {\scriptstyle \mu\times\mathrm{id}}\downarrow & & \downarrow{\scriptstyle \sigma} \\ G \times_S X & \xrightarrow[\sigma]{} & X \end{array}$$
commutes (associativity), and $\sigma \circ (\epsilon \times \mathrm{id}_X) = \mathrm{id}_X$ (the identity acts trivially).

In the functor-of-points language, this means: for every S-scheme T, the group $G(T)$ acts on the set

$X(T)$, and these actions are compatible with base change.

Example 4.3.2: Standard Actions

1. *Translation.* Any group scheme G acts on itself by left multiplication: $\sigma(g,h) = g \cdot h$. This is the *left regular action*. Each element $g \in G(T)$ defines an automorphism $\lambda_g : G_T \to G_T$ called *left translation by* g.
2. *Conjugation.* G acts on itself by conjugation: $\sigma(g,h) = ghg^{-1}$. This is the *adjoint action* of G on itself. Its derivative at the identity gives the adjoint representation of $\mathfrak{g}$ (the Lie algebra of G) on itself; see section 4.5.
3. *Linear representations.* An action of G on $\mathbb{A}^n_S$ that is $\mathcal{O}_S$-linear corresponds to a homomorphism $G \to \mathrm{GL}_{n,S}$. This is the scheme-theoretic notion of a *linear representation*.
4. $\mathbb{G}_m$*-actions and gradings.* An action of $\mathbb{G}_m$ on an affine scheme $\operatorname{Spec}(A)$ corresponds to a $\mathbb{Z}$-grading on A: $A = \bigoplus_{d \in \mathbb{Z}} A_d$. The $\mathbb{G}_m$-equivariant morphisms are the graded ring homomorphisms. This is the algebraic origin of the weight decompositions ubiquitous in representation theory.

Definition 4.3.3: Orbits and Stabilizers

Let G act on X and let $x \in X(T)$ be a T-valued point. The *orbit map* is

$$\sigma_x : G_T \longrightarrow X_T, \qquad g \longmapsto g \cdot x.$$

The *stabilizer* (or *isotropy group*) of x is the fiber product

$$G_x = G_T \times_{X_T} T$$

where the two maps to X_T are $\sigma_x : G_T \to X_T$ and $x : T \to X_T$. For any T-scheme T':

$$G_x(T') = \{g \in G(T') \mid g \cdot x = x \in X(T')\}.$$

The stabilizer is a closed subgroup scheme of G_T.

4.4 Quotients by Group Actions

Having defined actions, we now ask: given an action $\sigma : G \times_S X \to X$, does the "quotient" X/G exist as a scheme? This question is considerably more subtle than its set-theoretic counterpart, and its resolution leads to geometric invariant theory (GIT).

The categorical quotient

The most basic notion is that of a *categorical quotient*. We want a scheme Y and a G-invariant morphism $\pi : X \to Y$ (meaning $\pi \circ \sigma = \pi \circ \mathrm{pr}_2$, where $\mathrm{pr}_2 : G \times X \to X$ is the second projection) that is universal: any G-invariant morphism $X \to Z$ factors uniquely through π.

Definition 4.4.1: Categorical and Geometric Quotients

Let G act on X over S. A *categorical quotient* is an S-scheme Y with a G-invariant morphism $\pi : X \to Y$ such that:

1. (Universal property) For every G-invariant morphism $f : X \to Z$, there exists a unique morphism $\overline{f} : Y \to Z$ with $f = \overline{f} \circ \pi$.
2. (Coinvariant ring, in the affine case) If $X = \operatorname{Spec}(A)$ and G acts on A by ring automorphisms, then $Y = \operatorname{Spec}(A^G)$ where A^G denotes the subring of G-invariants.

A categorical quotient $\pi : X \to Y$ is a *geometric quotient* if additionally:

(3) π is surjective,

(4) the fibers of π are exactly the G-orbits, and

(5) π is submersive (a subset $U \subseteq Y$ is open if and only if $\pi^{-1}(U)$ is open in X).

A geometric quotient, when it exists, is the best possible outcome: the scheme Y faithfully represents the orbit space. However, geometric quotients frequently fail to exist—the simplest example being the action of $\mathbb{G}_m$ on $\mathbb{A}^2$ by scalar multiplication, where the origin is a "bad" orbit (it is the only closed orbit of dimension less than 1).

The GIT approach

Geometric invariant theory, developed by Mumford, provides a systematic method for constructing quotients by reductive group actions on projective or quasi-projective varieties. The central idea is to work not with the full quotient, but with a *good quotient* obtained by restricting to "semistable" points.

The word *reductive* is used here without being defined. It belongs to the structure theory of linear algebraic groups, which this book does not develop, and it is taken up where it is needed, in the geometric-invariant-theory chapter of *Moduli, Deformation Theory, and Stacks* [20]. For the paragraphs below it is enough to read it as naming the groups already met in section 4.1: the general and special linear groups GL_n and SL_n, the tori $\mathbb{G}_m^r$, and PGL_n. What the hypothesis gives is finite generation of the ring of invariants, without which the construction below does not produce a scheme at all. Over a field of characteristic 0 that property is equivalent to every finite-dimensional linear representation of G decomposing into irreducible subrepresentations; in characteristic p the equivalence fails, and the weaker notion of geometric reductivity is what the finite-generation theorem requires.

Let G be a reductive group acting on a projective variety $X \subseteq \mathbb{P}^n$ via a linearization (an action that lifts to the homogeneous coordinate ring). The *semistable locus* X^{ss} consists of those points $x \in X$ for which there exists a G-invariant section $f \in \Gamma(X, \mathcal{O}_X(d))^G$ (for some $d > 0$) with $f(x) \neq 0$. The *stable locus* $X^s \subseteq X^{ss}$ consists of those semistable points with finite stabilizer and closed orbit in X^{ss}.

The fundamental theorem of GIT asserts that the categorical quotient $X^{ss} \to X^{ss}/\!/G$ exists as a projective variety, and its restriction to the stable locus $X^s \to X^s/G$ is a geometric quotient. The notation "$/\!/$" (double slash) is reserved for the GIT quotient to distinguish it from the naive

set-theoretic quotient.

Moduli, Deformation Theory, and Stacks [20, The Projective GIT Quotient] constructs the quotient $X^{ss}//G$.

Proposition 4.4.2: Quotient by a Finite Group

Let G be a finite group acting on an affine scheme $X = \operatorname{Spec}(A)$ by ring automorphisms. Then the categorical quotient $X/G = \operatorname{Spec}(A^G)$ exists, and the morphism $\pi : X \to X/G$ is finite and surjective. If $|G|$ is invertible in A, then π is a geometric quotient and $X \to X/G$ is étale over the locus of points with trivial stabilizer.

***Proof*:**
The ring of invariants A^G is a subring of A, and A is integral over A^G by the theorem on integral extensions (each $a \in A$ satisfies the monic polynomial $\prod_{g \in G}(T - g \cdot a) \in A^G[T]$). Hence $\pi : \operatorname{Spec}(A) \to \operatorname{Spec}(A^G)$ is finite and surjective. The universal property of $\operatorname{Spec}(A^G)$ as a categorical quotient follows from the fact that G-invariant maps $A \to B$ factor uniquely through A^G. When $|G|$ is invertible, the Reynolds operator $R = \frac{1}{|G|}\sum_{g \in G} g$ is a projection $A \to A^G$, and the étaleness statement follows from the Jacobian criterion applied to the map π away from the ramification locus.

4.5 The Lie Algebra of a Group Scheme

In the theory of Lie groups, the tangent space at the identity carries the structure of a Lie algebra, and this infinitesimal invariant determines the local structure of the group. The same construction works for group schemes, using the Zariski tangent space from section 2.7. (In the following section, we will see that this tangent space admits a clean reformulation via dual-number valued points; see Example 3.3.7.)

Let G be a group scheme of finite type over a field k, with identity element $e \in G(k)$. The Zariski tangent space at e is:

$$T_e G = (\mathfrak{m}_e/\mathfrak{m}_e^2)^\vee$$

where $\mathfrak{m}_e$ is the maximal ideal of $\mathcal{O}_{G,e}$.

Definition 4.5.1: Lie Algebra of a Group Scheme

The *Lie algebra* of G is the tangent space $\mathfrak{g} = T_e G$, equipped with a k-linear bracket $[\,,\,] : \mathfrak{g} \otimes_k \mathfrak{g} \to \mathfrak{g}$ defined via the adjoint action (specified below). We write $\mathfrak{g} = \operatorname{Lie}(G)$.

To define the bracket, observe that the group structure on G induces a group structure on the set of dual-number points:

$$G(k[\epsilon]/(\epsilon^2))$$

is a group via the multiplication μ. The kernel of the natural map $G(k[\epsilon]/(\epsilon^2)) \to G(k)$ (induced by $\epsilon \mapsto 0$) is exactly $\mathfrak{g}$, and it sits in a split short exact sequence of groups:

$$0 \longrightarrow \mathfrak{g} \longrightarrow G(k[\epsilon]/(\epsilon^2)) \longrightarrow G(k) \longrightarrow 1.$$

In this sequence, $\mathfrak{g}$ is an *abelian* subgroup (the product of two tangent vectors $v, w \in \mathfrak{g}$ under the group law of $G(k[\epsilon]/(\epsilon^2))$ is $v + w$, since the quadratic terms in ϵ vanish). However, the *commutator* of elements in $G(k[\epsilon_1, \epsilon_2]/(\epsilon_1^2, \epsilon_2^2, \epsilon_1\epsilon_2))$ is well-defined and lands in $\mathfrak{g}$; this commutator defines the Lie bracket $[v, w]$.

More concretely, the adjoint action $\mathrm{ad} : G \to \mathrm{Aut}(\mathfrak{g})$ is defined by conjugation: for $g \in G(T)$ and $v \in \mathfrak{g} \otimes_k \mathcal{O}(T)$, set $\mathrm{ad}(g)(v) = gvg^{-1}$ (computed in $G(T[\epsilon]/(\epsilon^2))$). Differentiating ad at the identity gives a map $\mathfrak{g} \to \mathrm{End}_k(\mathfrak{g})$, and the Lie bracket is $[v, w] = (\mathrm{d}_e\mathrm{ad})(v)(w)$.

Proposition 4.5.2: Lie Algebra as Primitive Elements

Let $G = \mathrm{Spec}\,(A)$ be an affine group scheme over k with Hopf algebra (A, Δ, ε) and augmentation ideal $I = \ker(\varepsilon)$. Then:

$$\mathfrak{g} = (I/I^2)^\vee$$

as a k-vector space, and the Lie bracket corresponds to the commutator in the convolution algebra $\mathrm{Hom}_k(A, k)$. Equivalently, $\mathfrak{g}$ is the space of *primitive elements* of the dual coalgebra: those $\delta \in A^\vee$ satisfying $\delta(ab) = \varepsilon(a)\delta(b) + \delta(a)\varepsilon(b)$ for all $a, b \in A$ (i.e., δ is a derivation $A \to k$ at ε).

***Proof*:**
By the Hopf algebra structure, the augmentation ideal $I = \ker(\varepsilon)$ is the maximal ideal at the identity, so $(I/I^2)^\vee$ is the Zariski tangent space at e as expected. A k-linear functional $\delta : A \to k$ vanishing on $k \cdot 1$ corresponds to an element of $I^\vee$; it descends to $(I/I^2)^\vee$ if and only if $\delta(I^2) = 0$, which is equivalent to the derivation (Leibniz) rule $\delta(ab) = \varepsilon(a)\delta(b) + \delta(a)\varepsilon(b)$.

Example 4.5.3: Lie Algebras of Classical Group Schemes

1. $\mathbb{G}_a$*:* The augmentation ideal of $k[t]$ (with $\varepsilon(t) = 0$) is (t), and $(t)/(t^2) \cong k$. Thus $\mathrm{Lie}(\mathbb{G}_a) = k$ with the zero bracket—the Lie algebra is one-dimensional and abelian.

2. $\mathbb{G}_m$*:* The augmentation ideal of $k[t, t^{-1}]$ (with $\varepsilon(t) = 1$) is $(t-1)$. Setting $s = t - 1$, we have $(s)/(s^2) \cong k$, so $\mathrm{Lie}(\mathbb{G}_m) = k$ with the zero bracket—again one-dimensional and abelian.

3. GL_n*:* The augmentation ideal of $k[x_{ij}, \det^{-1}]$ (with $\varepsilon(x_{ij}) = \delta_{ij}$) is generated by the elements $e_{ij} := x_{ij} - \delta_{ij}$. The quotient I/I^2 is freely generated by the images of the e_{ij}, so $\mathrm{Lie}(\mathrm{GL}_n) \cong M_n(k)$, the space of all $n \times n$ matrices. The Lie bracket is the matrix commutator $[A, B] = AB - BA$, which recovers the classical Lie algebra $\mathfrak{gl}_n$.

4. SL_n*:* The additional relation $\det(x_{ij}) = 1$ imposes the linear condition $\mathrm{tr}(e_{ij}) = 0$ at the level of I/I^2. Hence $\mathrm{Lie}(\mathrm{SL}_n) = \{A \in M_n(k) : \mathrm{tr}(A) = 0\} = \mathfrak{sl}_n$.

5. μ_n *over a field with* $\mathrm{char}(k) \nmid n$*:* The augmentation ideal of $k[t]/(t^n - 1)$ (with $\varepsilon(t) = 1$) is $I = (t-1)$. Writing $t^n - 1 = (t-1)(t^{n-1} + \cdots + 1)$, and noting that $t^{n-1} + \cdots + 1$ evaluates to $n \neq 0$ at $t = 1$ (hence is coprime to $t - 1$), we find that $(t-1)^2$ and $t^n - 1$ together generate $(t-1)$, so $I/I^2 = 0$. Thus $\mathrm{Lie}(\mu_n) = 0$—the Lie algebra is trivial. This reflects the fact that μ_n is étale: it has no infinitesimal structure.

6. μ_p *in characteristic* p*:* Now $t^p - 1 = (t-1)^p$, so the coordinate ring is $k[s]/(s^p)$ with

$s = t - 1$, and $I/I^2 = (s)/(s^2) \cong k$. Hence $\mathrm{Lie}(\mu_p) = k$ is one-dimensional. The infinitesimal group scheme μ_p carries tangent information at its unique point, unlike its étale counterpart.

The contrast between the last two examples is worth emphasizing: μ_n for $\mathrm{char}(k) \nmid n$ is étale with trivial Lie algebra, while μ_p in characteristic p is infinitesimal with a one-dimensional Lie algebra. The Lie algebra "sees" the infinitesimal structure of the group scheme, and in characteristic p this structure is genuinely richer.

This observation foreshadows a general theme: in characteristic zero, the Lie algebra functor $G \mapsto \mathrm{Lie}(G)$ is an equivalence between connected algebraic groups and their Lie algebras (this is the scheme-theoretic incarnation of Lie's third theorem). In characteristic p, the Lie algebra loses information—it cannot distinguish between α_p and $\mathbb{G}_a$, for instance, since both have $\mathrm{Lie} = k$. The correct replacement is the theory of *p-Lie algebras* (or *restricted Lie algebras*), which equips $\mathfrak{g}$ with an additional "p-th power" operation $x \mapsto x^{[p]}$ that captures the Frobenius structure. This theory, while important for the classification of algebraic groups in positive characteristic, lies beyond our present scope.

4.6 Cartier Duality

For finite abelian groups (or in functional analysis, a locally compact abelian group like $\mathbb{R}$ with the euclidean topology), Pontryagin duality states that a group A is isomorphic to its double dual $\mathrm{Hom}(\mathrm{Hom}(A, S^1), S^1)$. In the realm of finite commutative group schemes over a field k, there is a perfect geometric analogue where the circle group S^1 is replaced by the multiplicative group scheme $\mathbb{G}_m$.

Definition 4.6.1: Cartier Dual

Let G be a finite commutative group scheme over a field k. The *Cartier dual* of G, denoted G^D, is the group scheme representing the functor of homomorphisms from G to $\mathbb{G}_m$. That is, for any k-algebra R:

$$G^D(R) = \mathrm{Hom}_{\text{gp-sch}/R}(G_R, \mathbb{G}_{m,R}).$$

At first glance, it is not obvious that this functor is representable by a finite commutative group scheme. However, the Hopf algebra perspective makes the construction transparent and purely algebraic.

Let $G = \mathrm{Spec}\,(A)$. Since G is finite, A is a finite-dimensional k-vector space. We can form the dual vector space $A^\vee = \mathrm{Hom}_k(A, k)$. Because G is a commutative group scheme, its Hopf algebra A is both commutative and cocommutative (meaning $\sigma \circ \Delta = \Delta$, where σ swaps the tensor factors in $A \otimes_k A$).

Dualizing the structural maps of A perfectly swaps the algebra and coalgebra structures:

- The multiplication $A \otimes_k A \to A$ dualizes to a comultiplication $A^\vee \to A^\vee \otimes_k A^\vee$.
- The comultiplication $\Delta : A \to A \otimes_k A$ dualizes to a multiplication $A^\vee \otimes_k A^\vee \to A^\vee$.
- The unit $k \to A$ dualizes to a counit $A^\vee \to k$.
- The counit $\varepsilon : A \to k$ dualizes to a unit $k \to A^\vee$.

- The antipode $S : A \to A$ dualizes to an antipode $S^\vee : A^\vee \to A^\vee$.

Since A is commutative and cocommutative, $A^\vee$ is also a commutative and cocommutative Hopf algebra. Thus, it defines a new finite commutative affine group scheme.

Proposition 4.6.2: Cartier Duality Theorem

Let $G = \operatorname{Spec}(A)$ be a finite commutative group scheme over k. Then the Cartier dual G^D is represented by the affine group scheme $\operatorname{Spec}(A^\vee)$. Furthermore, the natural evaluation map gives an isomorphism:

$$G \xrightarrow{\sim} (G^D)^D.$$

Example 4.6.3: Cartier Duals of Core Examples

1. *Roots of unity and constant groups.* The most classical pairing is between the constant group scheme $(\mathbb{Z}/n\mathbb{Z})_k$ and the group of roots of unity μ_n. A homomorphism from $\mathbb{Z}/n\mathbb{Z}$ to $\mathbb{G}_m$ is completely determined by where it sends the generator $1 \in \mathbb{Z}/n\mathbb{Z}$, and this image must be an n-th root of unity. Hence, for any k-algebra R:

$$\operatorname{Hom}((\mathbb{Z}/n\mathbb{Z})_R, \mathbb{G}_{m,R}) \cong \mu_n(R).$$

   Thus, $((\mathbb{Z}/n\mathbb{Z})_k)^D \cong \mu_n$. By the duality theorem, we also have $\mu_n^D \cong (\mathbb{Z}/n\mathbb{Z})_k$.

2. *The Frobenius kernel α_p.* Let k be a perfect field of characteristic $p > 0$, and recall $\alpha_p = \operatorname{Spec}(k[t]/(t^p))$ with $\Delta(t) = t \otimes 1 + 1 \otimes t$. To find a homomorphism from α_p to $\mathbb{G}_m = \operatorname{Spec}(k[u, u^{-1}])$, we must specify a group-like element in $k[t]/(t^p)$. The truncated exponential series:

$$E(t) = \sum_{i=0}^{p-1} \frac{t^i}{i!}$$

   is well-defined in characteristic p (since we stop before dividing by $p!$) and satisfies $E(t+s) = E(t)E(s)$ in $k[t,s]/(t^p, s^p)$. This gives a Hopf algebra map $u \mapsto E(t)$, which defines an isomorphism $\alpha_p^D \cong \alpha_p$. Thus, unlike μ_p, the infinitesimal group scheme α_p is self-dual.

4.7 Rational Maps into Group Schemes

Definition 3.6.1 introduced a rational map as an equivalence class of morphisms defined on dense open subsets, and left open how large the domain of definition of such a map is. When the target carries a group law the answer is as good as it could be: the map is defined everywhere it has any right to be. This is Weil's extension theorem, and it is the mechanism behind the construction of Néron models and behind the structure theory of algebraic groups that *Abelian Varieties* [5] takes up.

Two independent obstructions block extension in general. The target may fail to be proper: for a proper open subscheme $W \subsetneq V$, the identity of W is a rational map $V \dashrightarrow W$ with no extension, and no hypothesis on V repairs it. The source may fail to be normal: the parametrization of the cuspidal cubic is an isomorphism away from the cusp, and its inverse does not extend across it (example 4.7.5).

Removing both obstructions is still not enough for a general target, as the blow-up of the plane at a point shows (example 4.7.12). The group law is what closes the remaining gap.

Definition 4.7.1: Domain of Definition

Let X be an integral scheme, let Y be a separated scheme, and let $\varphi\colon X \dashrightarrow Y$ be a rational map. Then φ is *defined at* a point $x \in X$ if some representative (U, α) of φ has $x \in U$. The set of such points is the *domain of definition* $\operatorname{dom}(\varphi)$, and its complement $X \setminus \operatorname{dom}(\varphi)$ is the *locus of indeterminacy* of φ.

The definition is only useful once the representatives are known to fit together, so that $\operatorname{dom}(\varphi)$ carries a single morphism rather than a compatible family of them. Separatedness of the target is exactly what makes this work.

Proposition 4.7.2: The Domain of Definition Carries a Canonical Morphism

Let X be an integral scheme, Y a separated scheme, and $\varphi\colon X \dashrightarrow Y$ a rational map. Then $\operatorname{dom}(\varphi)$ is a dense open subset of X, and the representatives of φ glue to a single morphism $\operatorname{dom}(\varphi) \to Y$ representing φ.

***Proof*:**
The domain of definition is a union of the open sets underlying representatives, hence open, and it contains at least one nonempty such set, hence is dense because X is irreducible.

Let (U, α) and (V, β) be representatives. Both U and V are nonempty open subsets of an irreducible space, so $U \cap V$ is nonempty, and by definition 3.6.1 there is a dense open $Z \subseteq U \cap V$ with $\alpha|_Z = \beta|_Z$. The locus in $U \cap V$ where α and β agree is the preimage of the diagonal $\Delta_Y \subseteq Y \times Y$ under (α, β), and this is a closed subscheme of $U \cap V$ because Y is separated. Its underlying space contains Z and is closed, hence is all of $U \cap V$; and $U \cap V$ is reduced because X is integral, so a closed subscheme carried by the whole space is the whole scheme. Thus α and β agree on $U \cap V$, and the representatives glue, as we sought to show.

The gluing argument is used repeatedly below in the following form: two morphisms out of an integral scheme into a separated scheme that agree at the generic point are equal, since the locus of agreement is closed and contains a dense point. The next lemma is the other piece of routine machinery the section needs. It says that a morphism defined on the spectrum of a local ring, an object with no room in it, already comes from a genuine neighborhood.

Lemma 4.7.3: Spreading Out a Morphism from a Local Ring

Let k be a field, let X be an integral k-scheme, let $x \in X$, and let Y be a k-scheme locally of finite type. Every k-morphism $h\colon \operatorname{Spec} \mathcal{O}_{X,x} \to Y$ factors as

$$\operatorname{Spec} \mathcal{O}_{X,x} \longrightarrow V \xrightarrow{h_V} Y$$

for some open neighborhood V of x and some k-morphism h_V.

***Proof*:**
Step 1: the morphism lands in an affine chart. Let $\mathfrak{m}$ denote the closed point of $\operatorname{Spec}\mathcal{O}_{X,x}$ and put $y = h(\mathfrak{m})$. Choose an affine open $\operatorname{Spec}B \subseteq Y$ containing y, with B a finitely generated k-algebra. The preimage $h^{-1}(\operatorname{Spec}B)$ is open in $\operatorname{Spec}\mathcal{O}_{X,x}$ and contains $\mathfrak{m}$. An open subset of the spectrum of a local ring containing the closed point is the whole space: its complement is $V(\mathfrak{a})$ for some ideal $\mathfrak{a}$ with $\mathfrak{a} \not\subseteq \mathfrak{m}$, which forces $\mathfrak{a} = \mathcal{O}_{X,x}$. Hence h factors through $\operatorname{Spec}B$ and corresponds to a k-algebra homomorphism $\theta\colon B \to \mathcal{O}_{X,x}$.

Step 2: the homomorphism is defined on a neighborhood. Write $B = k[T_1,\dots,T_n]/J$ and let $\theta_i \in \mathcal{O}_{X,x}$ be the image of T_i. Each θ_i is a germ at x, so there is an open neighborhood V of x on which all n of them are regular, say $\theta_i = s_i|_x$ with $s_i \in \mathcal{O}_X(V)$. Because X is integral, both $\mathcal{O}_X(V)$ and $\mathcal{O}_{X,x}$ are subrings of the function field $K(X)$ and the restriction $\mathcal{O}_X(V) \to \mathcal{O}_{X,x}$ is injective. For $f \in J$ the section $f(s_1,\dots,s_n) \in \mathcal{O}_X(V)$ restricts to $f(\theta_1,\dots,\theta_n) = 0$, so it is itself zero. Therefore $T_i \mapsto s_i$ descends to a k-algebra homomorphism $B \to \mathcal{O}_X(V)$.

Step 3: conclusion. The homomorphism $B \to \mathcal{O}_X(V)$ is a morphism $h_V\colon V \to \operatorname{Spec}B \subseteq Y$, and composing it with $\operatorname{Spec}\mathcal{O}_{X,x} \to V$ returns the homomorphism θ, hence returns h, completing the proof.

The first extension theorem needs no group law at all. It isolates what properness of the target gives on its own: indeterminacy cannot occur along a divisor, because a divisor is seen by a discrete valuation and a proper target has nowhere for a valuation to escape to.

Theorem 4.7.4: Rational Maps to a Proper Target

Let k be a field, let X be a normal integral k-scheme of finite type, and let Y be a proper k-scheme. Then the locus of indeterminacy of any rational map $\varphi\colon X \dashrightarrow Y$ has codimension at least 2 in X.

***Proof*:**
Write $N = X \setminus \operatorname{dom}(\varphi)$. No irreducible component of N has codimension 0, since $\operatorname{dom}(\varphi)$ is dense by proposition 4.7.2. Suppose some component W has codimension 1, and let ξ be its generic point.

Step 1: the local ring at ξ is a valuation ring. The local ring $\mathcal{O}_{X,\xi}$ is Noetherian of dimension $\operatorname{codim}W = 1$, and it is normal because X is. A one-dimensional Noetherian normal local domain is a discrete valuation ring [10, Characterizations of a DVR], and its fraction field is the function field $K(X)$.

Step 2: the valuative criterion extends the map. The generic point of X lies in every nonempty open subset, so composing $\operatorname{Spec}K(X) \to \operatorname{dom}(\varphi)$ with the morphism of proposition 4.7.2 gives a k-morphism $\operatorname{Spec}K(X) \to Y$. Together with the structure morphism $\operatorname{Spec}\mathcal{O}_{X,\xi} \to \operatorname{Spec}k$, this is a lifting problem for $Y \to \operatorname{Spec}k$ against the valuation ring $\mathcal{O}_{X,\xi}$. The morphism $Y \to \operatorname{Spec}k$ is proper and of finite type and Y is Noetherian, so theorem 3.5.22 gives a k-morphism $h\colon \operatorname{Spec}\mathcal{O}_{X,\xi} \to Y$ restricting to it on the generic point.

Step 3: descending to a neighborhood. By lemma 4.7.3 there are an open neighborhood V of ξ and a morphism $h_V : V \to Y$ through which h factors. The set $V \cap \operatorname{dom}(\varphi)$ is a nonempty open subset of the irreducible scheme X, and on it h_V and the morphism of proposition 4.7.2 agree at the generic point by construction, hence agree, because Y is separated. So (V, h_V) represents φ and $\xi \in \operatorname{dom}(\varphi)$, contradicting $\xi \in N$. Hence no component of N has codimension 1, completing the proof.

Example 4.7.5: The Cuspidal Cubic

Let k be algebraically closed and let $C = \operatorname{Spec} k[u,v]/(v^2 - u^3)$ be the cuspidal cubic. The morphism $\nu : \mathbb{A}^1_k \to C$ determined by $u \mapsto t^2$ and $v \mapsto t^3$ identifies the coordinate ring of C with the subring $k[t^2, t^3] \subseteq k[t]$ of polynomials with no linear term, and restricts to an isomorphism $\mathbb{A}^1_k \setminus \{0\} \to C \setminus \{(0,0)\}$ with inverse $(u,v) \mapsto v/u$. Let $\varphi : C \dashrightarrow \mathbb{A}^1_k$ be the rational map $t = v/u$, so that $\varphi \circ \nu$ is the identity wherever it is defined.

The map φ is not defined at the cusp. Otherwise $t \in \mathcal{O}_{C,(0,0)}$, so $t = a/b$ with $a, b \in k[t^2, t^3]$ and $b(0) \neq 0$. Then $tb = a$ in $k[t]$, and the coefficient of t in tb is $b(0) \neq 0$, while a has no linear term. Thus the locus of indeterminacy is the single closed point $(0,0)$, of codimension 1 in the curve C.

Composing with $\mathbb{A}^1_k \subseteq \mathbb{P}^1_k$ replaces the target by a proper one without changing the conclusion: if the resulting map extended to a morphism $\psi : C \to \mathbb{P}^1_k$, then $\psi \circ \nu$ and the identity would agree on a dense open subset of the reduced scheme $\mathbb{A}^1_k$, hence everywhere, giving $\psi((0,0)) = 0$. Then ψ would carry a neighborhood of the cusp into $\mathbb{A}^1_k$ and so exhibit t as an element of $\mathcal{O}_{C,(0,0)}$, which was just excluded. The hypothesis of theorem 4.7.4 that fails is normality of C, and the cusp is precisely where normality fails.

Passing from codimension 2 to codimension infinity is where the group law enters, and it does so through a statement about rational functions. The following lemma is the divisor-theoretic input, proved here from unique factorization rather than from the theory of divisors developed in chapter 5, which is not yet available.

Lemma 4.7.6: Poles Are Detected by the Denominator

Let R be a unique factorization domain with fraction field K, and let $f \in K$ be written as $f = a/b$ with $a, b \in R$, $b \neq 0$, and a and b having no common irreducible factor. Then for every prime ideal $\mathfrak{p} \subseteq R$,

$$f \in R_{\mathfrak{p}} \qquad \text{if and only if} \qquad b \notin \mathfrak{p}$$

***Proof*:**

If $b \notin \mathfrak{p}$, then $f = a/b$ lies in $R_{\mathfrak{p}}$ by the definition of the localization. Conversely, suppose $b \in \mathfrak{p}$ and $f \in R_{\mathfrak{p}}$, say $f = c/s$ with $c \in R$ and $s \notin \mathfrak{p}$. Then $as = cb$ in R. The element b is a nonzero nonunit, since $b \in \mathfrak{p}$, so it has an irreducible factorization, and $\mathfrak{p}$ being prime forces some irreducible factor π of b to lie in $\mathfrak{p}$. Now π divides $cb = as$, and π does not divide a because a and b have no common irreducible factor, so π divides s. Then $s \in \mathfrak{p}$, contrary to the choice of s, completing the proof.

Lemma 4.7.7: The Polar Locus is Pure of Codimension One

Let X be a regular integral Noetherian scheme and let $f \in K(X)$ be a nonzero rational function. Then the *polar locus*

$$P(f) = \{x \in X \mid f \notin \mathcal{O}_{X,x}\}$$

is closed, and every irreducible component of $P(f)$ has codimension 1 in X.

***Proof*:**
Step 1: the polar locus is closed. If $f \in \mathcal{O}_{X,x}$, then f is represented by a section of $\mathcal{O}_X$ over some open neighborhood of x, and hence lies in $\mathcal{O}_{X,x'}$ for every x' in that neighborhood. The complement of $P(f)$ is therefore open.

Step 2: a local factorization. Let W be an irreducible component of $P(f)$ with generic point w, and put $R = \mathcal{O}_{X,w}$. This is a regular local ring, hence a unique factorization domain by proposition 2.7.27, and its fraction field is $K(X)$. Write $f = a/b$ with $a, b \in R$, $b \neq 0$, and a and b having no common irreducible factor. Since $w \in P(f)$ the element f does not lie in R, so b lies in the maximal ideal of R by lemma 4.7.6; in particular b is a nonzero nonunit.

Step 3: Krull's theorem produces a codimension-one pole. Being a nonzero nonunit of a domain, b is neither a unit nor a zero-divisor, so every minimal prime over (b) has height exactly 1 by theorem 2.6.13. Fix such a prime $\mathfrak{q}$ and let $w' \in X$ be the corresponding point, so that $\mathcal{O}_{X,w'} = R_{\mathfrak{q}}$ and $\operatorname{codim}\overline{\{w'\}} = \operatorname{height}(\mathfrak{q}) = 1$. Since $b \in \mathfrak{q}$, lemma 4.7.6 gives $f \notin \mathcal{O}_{X,w'}$, so $w' \in P(f)$ and therefore $\overline{\{w'\}} \subseteq P(f)$.

Step 4: conclusion. The primes of R correspond to the generizations of w in X, so $w \in \overline{\{w'\}}$. Thus $\overline{\{w'\}}$ is an irreducible closed subset of $P(f)$ containing $W = \overline{\{w\}}$, and maximality of the component W forces $W = \overline{\{w'\}}$. Hence $\operatorname{codim} W = 1$, as we sought to show.

Everything is now in place for the central lemma. The idea is to trade the rational map φ for the rational map $(x, y) \mapsto \varphi(x)\varphi(y)^{-1}$ on $X \times_k X$, which takes the constant value e along the diagonal. A pole of a function pulled back from a neighborhood of e therefore misses the diagonal generically, and lemma 4.7.7 converts that into a codimension-one statement back on X.

Lemma 4.7.8: Indeterminacy of a Rational Map to a Group Scheme

Let k be an algebraically closed field, let X be a smooth integral k-scheme of finite type, and let G be a group scheme over k that is separated and of finite type. Then the locus of indeterminacy of a rational map $\varphi\colon X \dashrightarrow G$ is either empty or of pure codimension 1 in X.

***Proof*:**
Write $U = \operatorname{dom}(\varphi)$ and $N = X \setminus U$, and assume $N \neq \emptyset$. Since k is algebraically closed, X is geometrically integral, so $X \times_k X$ is integral; it is smooth over k and therefore regular, and it is Noetherian.

Step 1: the comparison map. Let $\Phi\colon X \times_k X \dashrightarrow G$ be the rational map represented on $U \times_k U$ by the composite

$$U \times_k U \xrightarrow{\varphi\times\varphi} G \times_k G \xrightarrow{\mathrm{id}\times\iota} G \times_k G \xrightarrow{\mu} G$$

informally $(x, y) \mapsto \varphi(x)\varphi(y)^{-1}$. If φ is defined at a point x, then $(x, x) \in U \times_k U$ and Φ is defined at (x, x).

Step 2: Φ is trivial along the diagonal. Let $\Delta\colon X \to X \times_k X$ be the diagonal, a closed immersion since X is separated. The composite $\Phi \circ \Delta|_U$ is $\mu \circ (\mathrm{id}, \iota) \circ \varphi|_U$, which is the constant morphism e by the inverse axiom of definition 4.1.1. In particular $\Delta(U) \subseteq \mathrm{dom}(\Phi)$ and Φ takes the value e there.

Step 3: indeterminacy of φ at a closed point forces indeterminacy of Φ. Let $x \in N$ be a closed point and suppose Φ were defined at (x, x). Because k is algebraically closed and x is closed, $\kappa(x) = k$ and the fibre of the first projection over x is a copy of X. Its intersection with $\mathrm{dom}(\Phi)$ is open and nonempty, as it contains (x, x), so its intersection with the dense open U is a nonempty open subset of X and hence contains a closed point b, which is k-rational. Then $(x, b) \in \mathrm{dom}(\Phi)$ with $b \in U(k)$. Let $\sigma_b\colon X \to X \times_k X$ be the section $y \mapsto (y, b)$ of the first projection. On U the composite $\Phi \circ \sigma_b$ is $y \mapsto \varphi(y)\varphi(b)^{-1}$, so

$$\varphi = \mu \circ \big(\Phi \circ \sigma_b,\ \varphi(b)\big)$$

as rational maps, the second entry being a constant k-point of G. The right-hand side is a morphism on $\sigma_b^{-1}(\mathrm{dom}(\Phi))$, which is an open set containing x, so φ is defined at x, contradicting $x \in N$. Hence Φ is not defined at (x, x).

Step 4: a pole near the identity. Keep $x \in N$ closed and put $p = (x, x)$. Pullback along Φ makes sense on functions regular near e: by Step 2 the map Φ carries the nonempty open set $\Delta(U)$ to e, so for $f \in \mathcal{O}_{G,e}$ the composite $f \circ \Phi$ is regular on a nonempty, hence dense, open subset of the integral scheme $X \times_k X$, and therefore defines an element $\Phi^* f \in K(X \times_k X)$.

Suppose every $f \in \mathcal{O}_{G,e}$ satisfied $\Phi^* f \in \mathcal{O}_{X\times_k X,p}$. Choosing an affine open $\operatorname{Spec} B \subseteq G$ containing e, the composite $B \to \mathcal{O}_{G,e} \to \mathcal{O}_{X\times_k X,p}$ would be a morphism $\operatorname{Spec} \mathcal{O}_{X\times_k X,p} \to \operatorname{Spec} B \subseteq G$, which lemma 4.7.3 descends to a morphism on a neighborhood of p. That morphism agrees with Φ at the generic point by construction, hence agrees with it wherever both are defined because G is separated, so it represents Φ near p and makes Φ defined at p, contradicting Step 3. Hence some $f \in \mathcal{O}_{G,e}$ has $g = \Phi^* f \notin \mathcal{O}_{X\times_k X,p}$; in particular $g \neq 0$.

Step 5: restricting the pole to the diagonal. Put $R = \mathcal{O}_{X\times_k X,p}$, a regular local ring, and write $g = a/b$ with $a, b \in R$ having no common irreducible factor, as in lemma 4.7.7; by Step 4 the element b is a nonzero nonunit. Let $\delta = \Delta(\eta)$, where η is the generic point of X. Then δ is the generic point of $\Delta(X)$ and a generization of p, so it corresponds to a prime $\mathfrak{p}_\delta \subseteq R$, and $\mathfrak{p}_\delta$ is the ideal cutting out the integral closed subscheme $\Delta(X)$ near p; thus $R/\mathfrak{p}_\delta = \mathcal{O}_{X,x}$. Since $\eta \in U$, Step 2 puts δ in $\mathrm{dom}(\Phi)$ with $\Phi(\delta) = e$, so g is regular at δ, that is $g \in R_{\mathfrak{p}_\delta}$. By lemma 4.7.6, $b \notin \mathfrak{p}_\delta$. Writing $S = R/\mathfrak{p}_\delta = \mathcal{O}_{X,x}$, the image $\overline{b} \in S$ is therefore nonzero, and it is a nonunit because b lies in the maximal ideal of R. The ring S is a regular local Noetherian domain, so

theorem 2.6.13 gives a minimal prime over $(\bar{b})$ of height 1, say corresponding to a point $x' \in X$ with $\operatorname{codim}\overline{\{x'\}} = 1$.

Step 6: conclusion. The minimal prime produced in Step 5 is the image of a prime $\tilde{\mathfrak{q}} \subseteq R$ that contains both $\mathfrak{p}_\delta$ and b, and $\tilde{\mathfrak{q}}$ is the prime of R at the point $\Delta(x') = (x', x')$. Since $b \in \tilde{\mathfrak{q}}$, lemma 4.7.6 gives $g \notin \mathcal{O}_{X \times_k X,(x',x')}$, and hence Φ is not defined at (x', x'). By the contrapositive of the last sentence of Step 1, φ is not defined at x', so $\overline{\{x'\}} \subseteq N$ is an irreducible closed subset of codimension 1 containing x.

Now let W be an irreducible component of N. The complement in W of the remaining components is a nonempty open subset of a scheme of finite type over k, so it contains a point x that is closed in X and lies on no other component. The previous paragraph produces an irreducible closed $\overline{\{x'\}} \subseteq N$ of codimension 1 with $x \in \overline{\{x'\}}$; being irreducible and contained in N, it lies in a single component, which must be W. Hence $\operatorname{codim} W \leq \operatorname{codim}\overline{\{x'\}} = 1$, and $\operatorname{codim} W \geq 1$ because U is dense, completing the proof.

Theorem 4.7.9: Weil's Extension Theorem

Let k be an algebraically closed field, let X be a smooth integral k-scheme of finite type, and let G be a group scheme over k that is separated and of finite type. If a rational map $\varphi\colon X \dashrightarrow G$ is defined at every point of X of codimension 1, then φ is a morphism $X \to G$.

***Proof*:**
By lemma 4.7.8 the locus of indeterminacy is empty or of pure codimension 1. In the second case it has an irreducible component of codimension 1, whose generic point is a point of X of codimension 1 at which φ is not defined, contrary to hypothesis. So the locus is empty and proposition 4.7.2 delivers the morphism, as we sought to show.

Corollary 4.7.10: Rational Maps to a Proper Group Scheme

Let k be an algebraically closed field, let X be a smooth integral k-scheme of finite type, and let G be a group scheme over k that is proper over k. Then every rational map $\varphi\colon X \dashrightarrow G$ is a morphism.

***Proof*:**
A regular local ring is a unique factorization domain by proposition 2.7.27, and a unique factorization domain is integrally closed [10, A Unique Factorization Domain Is Integrally Closed], so the smooth scheme X is normal. A proper k-scheme is separated and of finite type, so theorem 4.7.4 applies and the locus of indeterminacy has codimension at least 2. In particular φ is defined at every point of codimension 1, and theorem 4.7.9 makes it a morphism, completing the proof.

Corollary 4.7.10 is the form in which the theorem is used downstream. A proper group scheme over a field that is in addition smooth and geometrically connected is an abelian variety, and the corollary then says that every rational map to one from a smooth source is a morphism. That is what makes the Néron mapping property reasonable to ask for, and it is why a rational map from a smooth variety to an abelian variety may be manipulated as though it were a morphism from the start.

4.7.11 Note: The Hypothesis on the Ground Field Algebraic closure is used at exactly one place in the proof of lemma 4.7.8: Step 3 produces a k-rational point b in a nonempty open set and uses the section $y \mapsto (y, b)$ of the first projection to convert a statement about Φ into one about φ. Over an arbitrary field such a point need not exist, and the section is replaced by faithfully flat descent along the projection. The theorem holds in that generality, and indeed over an arbitrary base scheme for a smooth source and a smooth separated group scheme; see Bosch, Lütkebohmert and Raynaud [3], 4.4/1.

Example 4.7.12: The Blow-Up of the Plane at a Point
Let k be algebraically closed, let $P = (1 : 0 : 0) \in \mathbb{P}^2_k$, and let

$$Y = \left\{ \big((x_0 : x_1 : x_2),\, (y_1 : y_2)\big) \in \mathbb{P}^2_k \times \mathbb{P}^1_k \;\middle|\; x_1 y_2 = x_2 y_1 \right\}$$

be the blow-up of $\mathbb{P}^2_k$ at P, with projection $\pi\colon Y \to \mathbb{P}^2_k$. Then π restricts to an isomorphism over $\mathbb{P}^2_k \setminus \{P\}$, while $\pi^{-1}(P)$ is a copy of $\mathbb{P}^1_k$. Both Y and $\mathbb{P}^2_k$ are smooth, integral and proper.

The inverse of that isomorphism is a rational map $\rho\colon \mathbb{P}^2_k \dashrightarrow Y$, and it is not a morphism. If it were, then $\pi \circ \rho$ and the identity would agree on a dense open subset of the reduced scheme $\mathbb{P}^2_k$, hence everywhere, so ρ would be a section of π and therefore a closed immersion. Its image would be a closed subscheme of the irreducible surface Y isomorphic to $\mathbb{P}^2_k$, hence of dimension 2, hence all of Y; so ρ would be an isomorphism and π its inverse, contradicting the fact that π contracts a curve to a point.

The locus of indeterminacy of ρ is thus the single point P, of codimension 2. This shows that the bound in theorem 4.7.4 is attained and cannot be improved, and that the group hypothesis in lemma 4.7.8 is not decorative: source and target here are smooth and proper, and the indeterminacy locus is nonempty of codimension 2, so it is not of pure codimension 1. What Y lacks is a group law.

Theorem 4.7.9 is a statement about the domain of definition of a rational map into a group scheme. The group law on the *target* converts a criterion in codimension 1 into an extension, and nothing is asked about a group law on the source, which need not have one. A second mechanism becomes available when the source is itself a group and the rational map respects that group law generically. Multiplicativity on a dense open set forces the extension on its own, with no criterion in codimension 1, no smoothness of the source, and no hypothesis on the target beyond separatedness. The two statements are not comparable and neither implies the other, so a reader who runs them together will look for the wrong hypothesis: Weil's theorem sees only the group law on the target, the theorem below sees the group laws on both.

What replaces the divisor theory is homogeneity. A group is swept out by the translates of any nonempty open subset, provided the translating points are drawn from a dense enough set, and that is what turns a formula valid near one point into a morphism defined everywhere.

Lemma 4.7.13: Translates of an Open Set Along a Dense Set of Points Cover the Group

Let k be an algebraically closed field and let G be a group scheme over k whose underlying scheme is integral and of finite type over k. For $a \in G(k)$ write $\lambda_a \colon G \to G$ for left translation by a (example 4.3.2), and write $aB = \lambda_a(B)$ for a subset $B \subseteq G$. Call a subset $S \subseteq G(k)$ dense when the set of points of G it determines is dense. Let $B \subseteq G$ be a nonempty open subset.

1. If $S \subseteq G(k)$ is dense, then every $g \in G(k)$ factors as $g = ab$ with $a \in S$ and $b \in B(k)$, and the open sets aB for $a \in S$ cover G.
2. If $A \subseteq G$ is a nonempty open subset, then $A(k)$ is dense. Consequently $A(k)B(k) = G(k)$, and the open sets aB for $a \in A(k)$ cover G.

***Proof*:**
Two facts are used throughout. Since G is irreducible, a nonempty open subset of G is dense and any two nonempty open subsets meet. And a nonempty open subset U of G has $U(k) \neq \emptyset$: being a nonempty open subset of a k-scheme of finite type it contains a closed point, and such a point is k-rational because k is algebraically closed. A k-point of G whose image lies in an open subscheme U is a k-point of U, and this identification is used without further comment.

Left translation is recalled next. For $a \in G(k)$ the morphism λ_a sends $x \mapsto ax$ on T-valued points, for every k-scheme T, with a read in $G(T)$ through the group homomorphism $G(k) \to G(T)$. So $\lambda_a \circ \lambda_{a^{-1}}$ and $\lambda_{a^{-1}} \circ \lambda_a$ induce the identity on T-valued points for every T and are therefore both id_G, and the same computation gives $\iota \circ \iota = \mathrm{id}_G$. Thus λ_a, with inverse $\lambda_{a^{-1}}$, and the inversion ι are automorphisms of the scheme G; each carries an open subscheme isomorphically onto its image and so carries k-points bijectively onto k-points.

1. Fix $g \in G(k)$ and put $C = (\lambda_g \circ \iota)(B)$, a nonempty open subset of G whose k-points are exactly the gb^{-1} for $b \in B(k)$. The set S is dense and C is nonempty open, so some $a \in S$ has its image in C; as a k-point of C it is $a = gb^{-1}$ for a unique $b \in B(k)$, whence $b = a^{-1}g$ and $g = ab$.

   Put $V = \bigcup_{a \in S} aB$, an open subset of G. The factorization just produced gives $g = \lambda_a(b) \in aB$, so V contains the image of every k-point of G. Were the closed complement $Z = G \setminus V$ nonempty, then Z with its reduced induced structure (definition 3.1.26) would be a nonempty k-scheme of finite type and would contain a closed point, again k-rational; that point would be a k-point of G outside V. Hence $Z = \emptyset$ and $V = G$.
2. Let $D \subseteq G$ be a nonempty open subset. Then $A \cap D$ is a nonempty open subset of G, so $(A \cap D)(k) \neq \emptyset$, and $A(k)$ meets D. Thus $A(k)$ is dense, and part (1) with $S = A(k)$ covers G by the sets aB and gives $G(k) \subseteq A(k)B(k)$; the reverse inclusion is immediate, completing the proof.

The lemma is why no hypothesis in codimension 1 is needed below. Weil's theorem has to work inward from the divisors of X because a general X offers no way of moving one point to another;

a group offers exactly that, and the price is that the rational map has to be compatible with the motion.

Stating that compatibility takes one piece of bookkeeping. A rational map φ out of a group G has a domain of definition Ω, and the equation $\varphi(xy) = \varphi(x)\varphi(y)$ is not even well formed at a pair (x, y) unless all three of x, y and xy lie in Ω. The pairs for which they do form an open subset of $G \times_k G$, and the hypothesis is that the two sides agree on a dense open subset of it.

Theorem 4.7.14: A Rational Homomorphism Is a Homomorphism

Let k be an algebraically closed field, let G be a group scheme over k whose underlying scheme is integral and of finite type over k, and let H be a separated group scheme over k. Let $\varphi\colon G \dashrightarrow H$ be a rational map, write $\Omega = \mathrm{dom}(\varphi)$, and write φ also for the morphism $\Omega \to H$ representing it (definition 4.7.1 and proposition 4.7.2). Let $p_1, p_2\colon G \times_k G \to G$ be the two projections and let

$$\Omega_2 = p_1^{-1}(\Omega) \cap p_2^{-1}(\Omega) \cap \mu_G^{-1}(\Omega) \subseteq G \times_k G$$

be the open subset over which x, y and xy all lie in Ω. On Ω_2 put

$$\alpha = \varphi \circ \mu_G|_{\Omega_2}, \qquad \beta = \mu_H \circ \left(\varphi \circ p_1|_{\Omega_2},\ \varphi \circ p_2|_{\Omega_2}\right)$$

informally $\alpha(x, y) = \varphi(xy)$ and $\beta(x, y) = \varphi(x)\varphi(y)$. Suppose there is a dense open subset W of $G \times_k G$ with $W \subseteq \Omega_2$ and $\alpha|_W = \beta|_W$. Then φ is represented by a unique morphism $\widetilde{\varphi}\colon G \to H$, and $\widetilde{\varphi}$ is a homomorphism of group schemes.

***Proof*:**
Since k is algebraically closed, G is geometrically integral (definition 3.2.8), so $G \times_k G$ is integral; it is of finite type over k because G is. Every closed point of G or of $G \times_k G$ is therefore k-rational, and a k-point of $G \times_k G$ is a pair of k-points of G. Since H is separated, proposition 4.7.2 applies and Ω is a dense open subset of G carrying the morphism φ.

One principle is used four times below, and it is worth naming once: if Y is an integral scheme and $f, g\colon Y \to H$ are morphisms agreeing on a dense open subscheme of Y, then $f = g$. This is proposition 3.5.10, applicable because an integral scheme is reduced and H is separated by hypothesis. Call it the *agreement principle* below.

Step 1: a dense set of translating points. For $a \in G(k)$ let $\sigma_a = (a \circ \pi_G,\ \mathrm{id}_G)\colon G \to G \times_k G$ be the section $y \mapsto (a, y)$ of p_2, where $\pi_G\colon G \to \operatorname{Spec} k$ is the structure morphism, and put $W_a = \sigma_a^{-1}(W)$, an open subset of G. Let

$$P = \{a \in G(k) \mid a \in \Omega \text{ and } W_a \neq \emptyset\}$$

Then P is dense. Let $D \subseteq G$ be a nonempty open subset. Since G is irreducible, $D \cap \Omega$ is nonempty, so $p_1^{-1}(D \cap \Omega)$ is a nonempty open subset of $G \times_k G$; it meets the nonempty open set W because $G \times_k G$ is irreducible, and their intersection, a nonempty open subset of a k-scheme of finite type, contains a closed point, which is k-rational. Write that point as (a, b) with $a, b \in G(k)$. Then $a \in (D \cap \Omega)(k)$, and $\sigma_a \circ b = (a, b)$ has its image in W, so $b \in W_a$. Hence $a \in P \cap D$.

Step 2: the translated maps agree with φ. Fix $a \in P$. The k-point a factors through Ω, so composing it with φ gives a k-point $\varphi(a) \in H(k)$ and hence an automorphism $\lambda_{\varphi(a)}$ of H; λ_a is an automorphism of G. Put $a\Omega = \lambda_a(\Omega)$, a dense open subset of G, and

$$\rho_a = \lambda_{\varphi(a)} \circ \varphi \circ \lambda_{a^{-1}}|_{a\Omega} \colon a\Omega \longrightarrow H$$

informally $\rho_a(x) = \varphi(a)\,\varphi(a^{-1}x)$; this is a morphism because $\lambda_{a^{-1}}$ carries $a\Omega$ isomorphically onto Ω.

Three inclusions are needed before the two maps can be compared. By construction $\sigma_a(W_a) \subseteq W \subseteq \Omega_2$. Since $p_2 \circ \sigma_a = \mathrm{id}_G$ and $\Omega_2 \subseteq p_2^{-1}(\Omega)$, this gives $W_a = p_2(\sigma_a(W_a)) \subseteq \Omega$. Since $\mu_G \circ \sigma_a = \lambda_a$, which on T-valued points is the identity $\mu_G(a, y) = ay$, and $\mu_G(\Omega_2) \subseteq \Omega$, it also gives $aW_a = \mu_G(\sigma_a(W_a)) \subseteq \Omega$. So both W_a and aW_a lie in Ω, and consequently $aW_a \subseteq a\Omega \cap \Omega$.

Now restrict α and β along $\sigma_a|_{W_a} \colon W_a \to W$. Using $p_1 \circ \sigma_a = a \circ \pi_G$ and $p_2 \circ \sigma_a = \mathrm{id}_G$, together with the inclusions just established,

$$\alpha \circ \sigma_a|_{W_a} = \varphi \circ \mu_G \circ \sigma_a|_{W_a} = \varphi \circ \lambda_a|_{W_a}, \qquad \beta \circ \sigma_a|_{W_a} = \mu_H \circ \big(\varphi(a) \circ \pi_G|_{W_a},\ \varphi|_{W_a}\big) = \lambda_{\varphi(a)} \circ \varphi|_{W_a}$$

the second equality on the right holding because μ_H with a constant first entry is left translation by that constant. The hypothesis $\alpha|_W = \beta|_W$ therefore reads

$$\varphi \circ \lambda_a|_{W_a} = \lambda_{\varphi(a)} \circ \varphi|_{W_a}$$

and precomposing with the isomorphism $\lambda_{a^{-1}} \colon aW_a \to W_a$ turns the left side into $\varphi|_{aW_a}$ and the right side into $\rho_a|_{aW_a}$. So φ and ρ_a agree on aW_a, which is nonempty because $a \in P$ and λ_a is a homeomorphism.

Both φ and ρ_a are morphisms on $a\Omega \cap \Omega$, a nonempty open subscheme of the integral scheme G and hence integral, and aW_a is a nonempty open subset of it, hence dense in it. The agreement principle gives

$$\varphi = \rho_a \quad \text{on } a\Omega \cap \Omega$$

Step 3: the translated maps agree with one another. Let $a, b \in P$. The sets $a\Omega$, $b\Omega$ and Ω are dense open subsets of the irreducible G, so $a\Omega \cap b\Omega \cap \Omega$ is a nonempty open subset of the integral scheme $a\Omega \cap b\Omega$ and hence dense in it. On it Step 2 gives $\rho_a = \varphi = \rho_b$, and the agreement principle upgrades this to $\rho_a = \rho_b$ on $a\Omega \cap b\Omega$. This compatibility is what permits the next step.

Step 4: gluing. By Step 1 the set P is a dense set of k-points of G, so lemma 4.7.13(1), applied with $S = P$ and $B = \Omega$, says the open sets $a\Omega$ for $a \in P$ cover G. By Step 3 the ρ_a agree on overlaps, so they glue to a morphism $\widetilde{\varphi} \colon G \to H$. Step 2 gives $\widetilde{\varphi} = \varphi$ on $a\Omega \cap \Omega$ for every $a \in P$, and those sets cover Ω, so $\widetilde{\varphi}|_\Omega = \varphi$ and $\widetilde{\varphi}$ represents the rational map.

Step 5: uniqueness. If $\psi \colon G \to H$ is a morphism representing φ, then ψ and $\widetilde{\varphi}$ agree on a dense open subset of G by definition 3.6.1, so $\psi = \widetilde{\varphi}$ by the agreement principle.

Step 6: multiplicativity. Consider the morphisms $\widetilde{\varphi} \circ \mu_G$ and $\mu_H \circ (\widetilde{\varphi} \times \widetilde{\varphi})$ from $G \times_k G$ to H.

On W one has $\mu_G(W) \subseteq \Omega$ and $p_i(W) \subseteq \Omega$ for $i = 1, 2$, and $\widetilde{\varphi}|_\Omega = \varphi$, so the two restrict on W to $\alpha|_W$ and $\beta|_W$, which agree by hypothesis. As W is a dense open subset of the integral scheme $G \times_k G$, the agreement principle gives

$$\widetilde{\varphi} \circ \mu_G = \mu_H \circ (\widetilde{\varphi} \times \widetilde{\varphi})$$

Step 7: the remaining axioms. For a k-scheme T the identification $(G \times_k G)(T) = G(T) \times G(T)$ turns μ_G and μ_H into the group laws on $G(T)$ and $H(T)$, so Step 6 says exactly that $\widetilde{\varphi}(T)\colon G(T) \to H(T)$ is multiplicative, that is, a homomorphism of abstract groups. A homomorphism of groups carries the identity to the identity and inverses to inverses, so $\widetilde{\varphi} \circ \epsilon_G$ and ϵ_H induce the same map on T-valued points for every T, as do $\widetilde{\varphi} \circ \iota_G$ and $\iota_H \circ \widetilde{\varphi}$. By the Yoneda lemma the corresponding identities of morphisms hold, so $\widetilde{\varphi}$ satisfies the three conditions of definition 4.2.1, completing the proof.

Since $\widetilde{\varphi}$ is defined on all of G and represents φ, the conclusion contains the assertion that $\mathrm{dom}(\varphi) = G$. The confinement of W to Ω_2 is not a technicality that could be dropped: outside Ω_2 one of $\varphi(x)$, $\varphi(y)$, $\varphi(xy)$ is undefined and there is no equation to impose. Inside Ω_2, on the other hand, the choice of W carries no information. Both α and β are morphisms on all of Ω_2, which is a nonempty open subscheme of the integral $G \times_k G$ and hence integral, so agreement on a dense open subset propagates to Ω_2 because H is separated. The theorem could have been stated with $W = \Omega_2$; it is stated with a general W because that is the shape in which the hypothesis is usually met, as an identity of rational functions valid away from some divisor (example 4.7.15).

Both hypotheses on the two group schemes are weaker than they look. Integrality of G holds for every connected group variety, since a smooth connected group scheme of finite type over a field is geometrically integral (corollary 4.1.3); smoothness is used for nothing else, and the density arguments see only integrality. Separatedness of H is automatic as soon as H is of finite type over k (proposition 4.1.2), and is stated in place of finite type because nothing in the proof bounds H.

Example 4.7.15: Additive Rational Functions in One Variable

Let k be algebraically closed and let $f \in k(x)$ be a rational function satisfying

$$f(x + y) = f(x) + f(y)$$

as an identity in $k(x, y)$. Write $f = g/h$ with $g, h \in k[x]$ having no common irreducible factor. The additive group $\mathbb{G}_a = \mathbb{A}^1_k$ (example 4.1.4) is a group scheme over k whose underlying scheme is integral and of finite type, and $k[x]$ is a unique factorization domain, so lemma 4.7.6 identifies the domain of definition of the rational map $\varphi\colon \mathbb{G}_a \dashrightarrow \mathbb{G}_a$ determined by f as $\Omega = \mathbb{A}^1_k \setminus V(h)$.

Multiplication on $\mathbb{G}_a$ is $(x, y) \mapsto x + y$, so the set Ω_2 of theorem 4.7.14 is the complement in $\mathbb{A}^2_k$ of the three closed sets cut out by $h(x)$, $h(y)$ and $h(x + y)$, each a proper closed subset; hence Ω_2 is a dense open subset of $\mathbb{A}^2_k$. On it both $(x, y) \mapsto f(x + y)$ and $(x, y) \mapsto f(x) + f(y)$ are regular, and a morphism to $\mathbb{A}^1_k$ is the same thing as a regular function; since $\mathbb{A}^2_k$ is integral, the regular functions on Ω_2 inject into $k(x, y)$, so the displayed identity says precisely that the two morphisms agree on Ω_2. Taking $W = \Omega_2$, theorem 4.7.14 makes φ a morphism $\mathbb{A}^1_k \to \mathbb{A}^1_k$, that is, a global section of $\mathcal{O}_{\mathbb{A}^1_k}$. Thus $f \in k[x]$: an additive rational function has no poles at all.

Which polynomials survive is a coefficient comparison. Write $f = \sum_{n \geq 0} c_n x^n$. Substituting

$x = y = 0$ gives $c_0 = 0$, and for $n \geq 2$ the coefficient of xy^{n-1} in $f(x+y) - f(x) - f(y)$ is nc_n, since $f(x)$ and $f(y)$ contribute no mixed monomials. In characteristic 0 this forces $c_n = 0$ for all $n \geq 2$, so the additive rational functions are exactly the $c_1 x$ with $c_1 \in k$. In characteristic p the relation $nc_n = 0$ is vacuous whenever $p \mid n$, and there are more of them: $(x+y)^p = x^p + y^p$, so $x \mapsto x^p$ is additive, as is every k-linear combination of the x^{p^e}.

4.7.16 Remark: Why PGL_n Needs This Theorem Corollary 4.7.10 requires a proper target and so says nothing about $\mathrm{PGL}_{n,k}$, which is a closed subgroup scheme of $\mathrm{GL}_{n^2,k}$ (definition 4.1.7) and therefore affine and of finite type over k. For $n \geq 2$ it is not proper over k. An affine morphism is proper exactly when it is finite (proposition 3.5.25), and a finite k-scheme is $\operatorname{Spec} A$ for a finite-dimensional k-algebra A; distinct maximal ideals of A are comaximal, so A surjects onto the product of the corresponding residue fields and there are at most $\dim_k A$ of them. A k-point of $\operatorname{Spec} A$ is a k-algebra map $A \to k$ and is determined by its kernel, a maximal ideal, so a finite k-scheme has finitely many k-points. But $\mathrm{PGL}_n(k) = \mathrm{Aut}_{k\text{-alg}}(M_n(k))$ is infinite for $n \geq 2$: conjugation by $\operatorname{diag}(t, 1, \ldots, 1)$ fixes e_{11} and sends e_{12} to $t\,e_{12}$, so these automorphisms are pairwise distinct as t runs over $k^\times$, which is infinite because an algebraically closed field is (the case $n = 2$ is the computation in example 4.1.8).

Theorem 4.7.14 is what is left, and it suffices. Being a group scheme of finite type over a field, $\mathrm{PGL}_{n,k}$ is separated (proposition 4.1.2), so a rational map into $\mathrm{PGL}_{n,k}$ from a group scheme whose underlying scheme is integral and of finite type over k, multiplicative on a dense open set, is a morphism and a homomorphism of group schemes.

Exercise 4.7.1

1. Let k be a field and let $\varphi\colon \mathbb{P}^2_k \dashrightarrow \mathbb{P}^1_k$ be the rational map $(x_0 : x_1 : x_2) \mapsto (x_0 : x_1)$. Determine the locus of indeterminacy of φ and verify that it is consistent with theorem 4.7.4.

2. Let k be a field, let X be a normal integral k-scheme of finite type of dimension 1, and let Y be a proper k-scheme. Show that every rational map $X \dashrightarrow Y$ is a morphism.

3. Let X be a regular integral Noetherian scheme and let $f \in K(X)$ be nonzero with $P(f) = \emptyset$. Show that $f \in \mathcal{O}_X(X)$. (*Hint:* for an integral scheme, $\mathcal{O}_X(X)$ is the intersection of the $\mathcal{O}_{X,x}$ inside $K(X)$.)

4. The target $\mathbb{A}^1_k = \mathbb{G}_a$ of example 4.7.5 is a group scheme, and the locus of indeterminacy there is a single closed point. Explain why this is not a counterexample to lemma 4.7.8, and identify the hypothesis of that lemma which the cuspidal cubic fails.

5. Deduce from example 4.7.12 that the surface Y appearing there carries no group scheme structure over k, and identify the step in the proof of lemma 4.7.8 at which the group law is used.

5

Schemes III: Sheaves of Modules, Divisors, and Differentials

So far, we have worked exclusively with the structure sheaf $\mathcal{O}_X$ of a scheme X. But just as the geometry of a smooth manifold M is enriched by studying vector bundles, principal bundles[1], and their sections, the geometry of a scheme is enriched by studying additional sheaves that carry an $\mathcal{O}_X$-module structure. Every vector bundle on M gives rise to a sheaf of sections that is a module over the ring of smooth functions $C^\infty(M)$; in the same way, every "bundle-like" object on a scheme gives rise to a sheaf of $\mathcal{O}_X$-modules.

The most important such sheaves are those that are *locally* isomorphic to the sheaf associated to a module: on each affine open $U_i = \operatorname{Spec} R_i$, the sheaf restricts to $\widetilde{M_i}$ for some R_i-module M_i. These are the *quasi-coherent sheaves*, and when the modules M_i satisfy a finite presentation condition they are called *coherent sheaves*[2].

Quasi-coherent sheaves can be understood as a natural extension of vector bundles. *Locally free sheaves* (the algebraic incarnation of vector bundles) are not closed under kernels and cokernels — the category of vector bundles is not abelian. The category $\mathbf{QCoh}(X)$ is obtained by adding precisely those kernels and cokernels; it is the smallest abelian subcategory of $\mathcal{O}_X$-modules containing the locally free sheaves, just as the category of all R-modules is the abelian completion of the category of free R-modules. This perspective is developed in section 5.1.3.

By working with quasi-coherent sheaves, we gain access to a wealth of new geometric objects: ideal sheaves (encoding closed subschemes with scheme structure), twisted sheaves on projective space (from which one can recover the graded ring $S_\bullet$ of $\operatorname{Proj} S_\bullet$), skyscraper-like sheaves remembering

[1] more generally, gerbes and n-gerbes, for which see *Moduli, Deformation Theory, and Stacks* [20]

[2] The term "coherent" originates with Serre [65], who required sheaves to "cohere" (stick together) in a controlled way. The prefix "quasi-" signals that finite generation is dropped, which loses some finiteness results — for instance, quasi-coherent sheaves are not in general preserved under pushforward.

multiplicities (generalizing example 1.1.12), and sheaves of differentials (from which we recover differential forms, tangent bundles, and smoothness criteria). The study of these objects occupies the present chapter and sets the stage for sheaf cohomology in chapter 6.

There shall be sheaves on schemes that are *not* bundle-like (not quasi-coherent), such as the sheaf given by taking all units at each local section. Therefore, we start with actions of abelian sheave on ringed spaces.

5.1 Definition and Basic Properties

We begin by sheafifying the notion of a module over a ring. If $(X, \mathcal{O}_X)$ is a ringed space, an $\mathcal{O}_X$-module is a sheaf of abelian groups on which $\mathcal{O}_X$ acts section-wise, compatibly with restriction.

Definition 5.1.1: $\mathcal{O}_X$-Module

Let $(X, \mathcal{O}_X)$ be a ringed space. An *$\mathcal{O}_X$-module* is a sheaf $\mathscr{F} \in \mathbf{Sh}_{\mathbf{Ab}}(X)$ equipped with the following data:

1. For each open $U \subseteq X$, the abelian group $\mathscr{F}(U)$ carries the structure of an $\mathcal{O}_X(U)$-module.
2. For each inclusion $U \subseteq V$ of open sets, the restriction map $\mathrm{res}_{V,U} : \mathscr{F}(V) \to \mathscr{F}(U)$ is compatible with the module structure:

$$\begin{array}{ccc} \mathcal{O}_X(V) \times \mathscr{F}(V) & \xrightarrow{\curvearrowright} & \mathscr{F}(V) \\ \downarrow{\scriptstyle \mathrm{res}_{V,U} \times \mathrm{res}_{V,U}} & & \downarrow{\scriptstyle \mathrm{res}_{V,U}} \\ \mathcal{O}_X(U) \times \mathscr{F}(U) & \xrightarrow{\curvearrowright} & \mathscr{F}(U) \end{array}$$

A *morphism of $\mathcal{O}_X$-modules* $\varphi : \mathscr{F} \to \mathscr{G}$ is a morphism of sheaves of abelian groups such that for each open $U \subseteq X$, the map $\varphi(U) : \mathscr{F}(U) \to \mathscr{G}(U)$ is an $\mathcal{O}_X(U)$-module homomorphism. The collection of all such morphisms is denoted $\mathrm{Hom}_{\mathcal{O}_X}(\mathscr{F}, \mathscr{G})$, or simply $\mathrm{Hom}_X(\mathscr{F}, \mathscr{G})$ when the structure sheaf is understood.

The reader should think of $\mathcal{O}_X$-modules as the scheme-theoretic analogue of (sections of) fibre bundles: the structure sheaf plays the role of smooth functions, and the module structure encodes how sections transform under multiplication by functions.

Given a morphism of ringed spaces $(f, f^\#) : (X, \mathcal{O}_X) \to (Y, \mathcal{O}_Y)$, there are natural operations that transport $\mathcal{O}_X$-modules to $\mathcal{O}_Y$-modules and vice versa:

Definition 5.1.2: $\mathcal{O}_X$-module Pushforward and Pullback

Let $(f, f^\#) : (X, \mathcal{O}_X) \to (Y, \mathcal{O}_Y)$ be a morphism of ringed spaces.

1. *(Direct image.)* If $\mathscr{F}$ is an $\mathcal{O}_X$-module, then the pushforward sheaf $f_*\mathscr{F}$ is naturally an $f_*\mathcal{O}_X$-module on Y. Via the sheaf map $f^\# : \mathcal{O}_Y \to f_*\mathcal{O}_X$, this acquires the structure of an $\mathcal{O}_Y$-module: for $V \subseteq Y$ open and $g \in \mathcal{O}_Y(V)$, $s \in f_*\mathscr{F}(V) = \mathscr{F}(f^{-1}(V))$, the action is $g \cdot s := f^\#(V)(g) \cdot s$.

2. *(Inverse image.)* If $\mathscr{G}$ is a $\mathcal{O}_Y$-module, then the sheaf $f^{-1}\mathscr{G}$ is an $f^{-1}\mathcal{O}_Y$-module on X. The *pullback* (or *inverse image $\mathcal{O}_X$-module*) is

$$f^*\mathscr{G} \;:=\; f^{-1}\mathscr{G} \otimes_{f^{-1}\mathcal{O}_Y} \mathcal{O}_X.$$

   The tensor product re-bases the module structure from $\mathcal{O}_Y$ to $\mathcal{O}_X$.

These operations form an adjoint pair: there is a natural isomorphism $\mathrm{Hom}_{\mathcal{O}_X}(f^*\mathscr{G}, \mathscr{F}) \cong \mathrm{Hom}_{\mathcal{O}_Y}(\mathscr{G}, f_*\mathscr{F})$.

Example 5.1.3: General $\mathcal{O}_X$-Module Constructions

Fix a ringed space $(X, \mathcal{O}_X)$.

1. *The structure sheaf.* The sheaf $\mathcal{O}_X$ is an $\mathcal{O}_X$-module with the action given by ring multiplication. Since each restriction map is a ring homomorphism, compatibility with the module structure is automatic.

2. *Abelian groups as modules.* Just as every abelian group is a $\mathbb{Z}$-module, every sheaf of abelian groups $\mathscr{G}$ on X is a module over the constant sheaf $\underline{\mathbb{Z}}$. The category of $\mathcal{O}_X$-modules therefore generalizes $\mathbf{Sh}_{\mathbf{Ab}}(X)$.

3. *Kernels, images, cokernels.* If $\varphi : \mathscr{F} \to \mathscr{G}$ is a morphism of $\mathcal{O}_X$-modules, then $\ker\varphi$, $\mathrm{im}\varphi$, and $\mathrm{coker}\varphi$ are all $\mathcal{O}_X$-modules (by the same arguments as for sheaves of abelian groups, since the module structure is inherited section-wise). The category of $\mathcal{O}_X$-modules is abelian, with products and coproducts computed section-wise.

4. *Sheaf Hom.* For $\mathcal{O}_X$-modules $\mathscr{F}, \mathscr{G}$, the assignment

$$U \;\longmapsto\; \mathrm{Hom}_{\mathcal{O}_X|_U}(\mathscr{F}|_U, \mathscr{G}|_U)$$

   defines a sheaf of $\mathcal{O}_X$-modules, the *sheaf Hom*, denoted $\mathit{Hom}_{\mathcal{O}_X}(\mathscr{F}, \mathscr{G})$.

5. *Tensor product.* The *tensor product* $\mathscr{F} \otimes_{\mathcal{O}_X} \mathscr{G}$ is the sheafification of the presheaf $U \mapsto \mathscr{F}(U) \otimes_{\mathcal{O}_X(U)} \mathscr{G}(U)$. Sheafification is essential here: the naive presheaf need not satisfy the gluing axiom, and the resulting sections can differ dramatically from the section-wise tensor product. See example 5.3.19 for a instance of this phenomenon.

The following result is a useful way

Proposition 5.1.4: Evaluation via Skyscraper Sheaves

Let X be a scheme and $x \in X$ a point. Let $\mathcal{O}_x$ denote the skyscraper sheaf at x with stalk $\kappa(x)$ (formally, the pushforward of $\mathcal{O}_{\operatorname{Spec}(\kappa(x))}$ along the inclusion $i : \operatorname{Spec}(\kappa(x)) \hookrightarrow X$). Then there is a natural isomorphism:

$$\operatorname{Hom}_{\mathcal{O}_X}(\mathcal{O}_X, \mathcal{O}_x) \;\cong\; \kappa(x).$$

***Proof*:**
For any sheaf of $\mathcal{O}_X$-modules $\mathcal{F}$, morphisms from the structure sheaf $\mathcal{O}_X$ to $\mathcal{F}$ are naturally identified with the global sections of $\mathcal{F}$. Explicitly, a morphism $\varphi : \mathcal{O}_X \to \mathcal{F}$ is uniquely determined by the global section $\varphi(X)(1) \in \Gamma(X, \mathcal{F})$, giving an isomorphism:

$$\operatorname{Hom}_{\mathcal{O}_X}(\mathcal{O}_X, \mathcal{F}) \;\cong\; \Gamma(X, \mathcal{F})$$

Applying this general fact to the skyscraper sheaf $\mathcal{F} = \mathcal{O}_x$, we obtain:

$$\operatorname{Hom}_{\mathcal{O}_X}(\mathcal{O}_X, \mathcal{O}_x) \;\cong\; \Gamma(X, \mathcal{O}_x)$$

By the definition of the skyscraper sheaf, its global sections evaluate exactly to the stalk at the point x, which is the residue field $\kappa(x)$. Therefore:

$$\operatorname{Hom}_{\mathcal{O}_X}(\mathcal{O}_X, \mathcal{O}_x) \;\cong\; \kappa(x)$$

This proposition demonstrates that taking the global section of a function $f \in \Gamma(X, \mathcal{O}_X) \cong \operatorname{Hom}_{\mathcal{O}_X}(\mathcal{O}_X, \mathcal{O}_X)$ and composing it with the natural projection $\mathcal{O}_X \to \mathcal{O}_x$ precisely yields the evaluation $f(x) \in \kappa(x)$.

5.1.1 Quasi-Coherent and Coherent Sheaves

A natural hope is that, for an affine scheme $X = \operatorname{Spec} R$, the category of $\mathcal{O}_X$-modules is equivalent to R-Mod. This turns out to be false: the category of $\mathcal{O}_X$-modules is too large.

Example 5.1.5: Too Many $\mathcal{O}_X$-modules Over Affine Schemes

For any such equivalence to work, the sheaf $\widetilde{M}$ associated to an R-module M would need to satisfy $\widetilde{M}_{\mathfrak{p}} \cong M_{\mathfrak{p}}$ at each prime $\mathfrak{p}$. We show that not every $\mathcal{O}_X$-module has this form.

Let $X = \operatorname{Spec} \mathbb{Z}_{(2)}$, which has two points: the closed point (2) and the generic point $\eta = (0)$. The nontrivial open sets are $\{(2), \eta\} = X$ and $\{\eta\}$. The structure sheaf satisfies $\mathcal{O}_X(X) = \mathbb{Z}_{(2)}$ and $\mathcal{O}_X(\{\eta\}) = \mathbb{Q}$.

To specify an $\mathcal{O}_X$-module $\mathscr{F}$, we must give a $\mathbb{Z}_{(2)}$-module $\mathscr{F}(X)$, a $\mathbb{Q}$-module $\mathscr{F}(\{\eta\})$, and a restriction (module) homomorphism $\mathscr{F}(X) \to \mathscr{F}(\{\eta\})$. Consider the choice

$$\mathscr{F}(X) = \mathbb{Z}_{(2)}, \qquad \mathscr{F}(\{\eta\}) = 0, \qquad \mathbb{Z}_{(2)} \twoheadrightarrow 0.$$

This defines a valid $\mathcal{O}_X$-module. However, if $\mathscr{F} = \widetilde{M}$ for some $\mathbb{Z}_{(2)}$-module M, then $M = \Gamma(X, \widetilde{M}) = \mathbb{Z}_{(2)}$ and $\widetilde{M}(\{\eta\}) = M_{(0)} = M[1/2] = \mathbb{Q} \neq 0$. This contradicts $\mathscr{F}(\{\eta\}) = 0$, so $\mathscr{F}$ does not arise from any module.

The problem is that a general $\mathcal{O}_X$-module need not preserve the localization structure in compatible with the affine structure. The remedy is to restrict attention to $\mathcal{O}_X$-modules that are locally of the form $\widetilde{M}$. We first define the "affine building blocks."

Definition 5.1.6: Sheaf Associated to M

Let R be a ring and M an R-module. The *sheaf associated to* M on $\operatorname{Spec} R$ is the $\mathcal{O}_{\operatorname{Spec} R}$-module $\widetilde{M}$ defined on the basis of distinguished open sets by

$$\widetilde{M}(D(f)) \;=\; M_f \qquad \text{(the localization of } M \text{ at } f\text{)},$$

with the natural restriction maps induced by further localization. One checks (by the same argument as for the structure sheaf) that this defines a sheaf on $\operatorname{Spec} R$.

Since we now have multiple sheaves on a scheme, we adopt the standard notation $\Gamma(X, \mathscr{F})$ for the global sections of a sheaf $\mathscr{F}$ on X. In particular, $\Gamma(\operatorname{Spec} R, \widetilde{M}) \cong M$ and $\Gamma(X, \mathcal{O}_X) = \mathcal{O}_X(X)$.

Definition 5.1.7: Quasi-coherent Sheaf

Let $(X, \mathcal{O}_X)$ be a scheme. An $\mathcal{O}_X$-module $\mathscr{F}$ is *quasi-coherent* if X can be covered by affine open subsets $U_i = \operatorname{Spec} R_i$ such that for each i there exists an R_i-module M_i with

$$\mathscr{F}\big|_{U_i} \;\cong\; \widetilde{M_i}.$$

The full subcategory of quasi-coherent sheaves on X is denoted $\mathbf{QCoh}(X)$.

For the finitely generated analogue, a subtlety arises: the kernel of a map between finitely generated modules need not be finitely generated. For instance, the surjection $k[x_1, x_2, \dots] \to k[x_1, x_2, \dots]$ sending each x_i to 0 has kernel $(x_1, x_2, \dots)$, which is not finitely generated. (Images and cokernels of maps between finitely generated modules *are* finitely generated, so this is specifically a kernel problem.) The correct notion must therefore be stronger than finite generation:

Definition 5.1.8: Coherent Module

Let R be a ring and M an R-module. Then M is *coherent* if it is finitely generated and for every $n \geq 1$, every R-module homomorphism $R^n \to M$ has finitely generated kernel.
A ring R is called *coherent* if it is coherent as a module over itself.

Coherence is strictly stronger than finite presentation: the latter only requires *some* surjection $R^n \twoheadrightarrow M$ to have finitely generated kernel, whereas coherence requires this for *all* maps $R^n \to M$. Over a Noetherian ring, coherence, finite presentation, and finite generation are all equivalent.

Definition 5.1.9: Coherent Sheaf

Let $(X, \mathcal{O}_X)$ be a scheme. An $\mathcal{O}_X$-module $\mathscr{F}$ is *coherent* if X can be covered by affine open subsets $U_i = \operatorname{Spec} R_i$ such that for each i there exists a coherent R_i-module M_i with $\mathscr{F}|_{U_i} \cong \widetilde{M_i}$.

5.1.10 Distinction from Hartshorne Hartshorne defines "coherent sheaf" to mean "sheaf associated to a finitely generated module," but then immediately restricts to Noetherian schemes where the distinction vanishes. Our definition agrees with the one used by EGA and most modern references: coherence is an intrinsic condition on the sheaf (finitely generated with finitely generated kernels of all maps from free sheaves), which coincides with Hartshorne's definition precisely when X is locally Noetherian.

A key feature of quasi-coherence is that it is *affine-local*: checking it on one affine cover suffices.

Proposition 5.1.11: Quasi-coherence is Affine-local

Let $(X, \mathcal{O}_X)$ be a scheme and $\mathscr{F}$ an $\mathcal{O}_X$-module. Then $\mathscr{F}$ is quasi-coherent if and only if for *every* affine open subset $U = \operatorname{Spec} R \subseteq X$, there is an isomorphism $\mathscr{F}|_U \cong \widetilde{M}$ for some R-module M.
If X is locally Noetherian, then $\mathscr{F}$ is coherent if and only if this holds with each M finitely generated (equivalently, coherent).

***Proof*:**
The "if" direction holds because definition 5.1.7 requires the isomorphism on only one affine cover. For the "only if" direction, since the property is local and every affine scheme is covered by distinguished opens of any given affine cover, it suffices to prove the claim when $X = \operatorname{Spec} R$ is itself affine.

Let $M = \Gamma(\operatorname{Spec} R, \mathscr{F})$. We must show $\mathscr{F} \cong \widetilde{M}$. By hypothesis, there is an affine cover $\{D(g_i)\}$ of X with $\mathscr{F}|_{D(g_i)} \cong \widetilde{M_i}$ for some R_{g_i}-module M_i. Taking global sections over $D(g_i)$ gives $\mathscr{F}(D(g_i)) \cong M_i$. But $\mathscr{F}(D(g_i))$ is also obtained from M by localization at g_i: the restriction map $M = \mathscr{F}(X) \to \mathscr{F}(D(g_i))$ factors through M_{g_i}, and the quasi-coherence assumption on overlaps $D(g_i g_j)$ ensures this identification is consistent. Hence $M_i \cong M_{g_i}$, and by corollary 1.3.4 the natural map $\widetilde{M} \to \mathscr{F}$ is an isomorphism on a basis, hence an isomorphism of sheaves.

For the coherent case, if X is locally Noetherian then each R_{g_i} is Noetherian, so M_{g_i} is finitely generated if and only if it is coherent. The result follows by the same argument with the additional finite generation condition.

The affine-local property immediately yields the scheme-theoretic analogue of corollary 3.1.5:

Corollary 5.1.12: Equivalence of Categories: Module Version

If $X = \operatorname{Spec} R$, the functor $M \mapsto \widetilde{M}$ is an equivalence of categories:

$$\mathbf{QCoh}(\operatorname{Spec} R) \underset{\widetilde{(-)}}{\overset{\Gamma}{\rightleftarrows}} R\text{-Mod}$$

with inverse $\mathscr{F} \mapsto \Gamma(\operatorname{Spec} R, \mathscr{F})$. If R is Noetherian, this restricts to an equivalence between $\mathbf{Coh}(\operatorname{Spec} R)$ and finitely generated R-modules.
More generally, for any $\mathcal{O}_X$-module $\mathscr{F}$ on X and any R-module M, there is a natural adjunction isomorphism:

$$\operatorname{Hom}_{\mathcal{O}_X}(\widetilde{M}, \mathscr{F}) \cong \operatorname{Hom}_R(M, \Gamma(X, \mathscr{F})).$$

***Proof*:**

By proposition 5.1.11, every quasi-coherent sheaf on $\operatorname{Spec} R$ is of the form $\widetilde{M}$. Since $\Gamma(\operatorname{Spec} R, \widetilde{M}) \cong M$ and the tilde construction is functorial and exact, the two functors are inverse equivalences. The adjunction follows from the universal property of localization: a map $\widetilde{M} \to \mathscr{F}$ is determined by its value on global sections $M \to \Gamma(X, \mathscr{F})$, and conversely any such module map extends uniquely to a sheaf map by localization.

Example 5.1.13: Quasi-Coherent Sheaves

1. *The structure sheaf.* The structure sheaf $\mathcal{O}_X$ is quasi-coherent (even coherent): on each affine open $\operatorname{Spec} R$, it is $\widetilde{R}$. At each stalk, the $\mathcal{O}_{X,\mathfrak{p}}$-module $\mathcal{O}_{X,\mathfrak{p}} \cong R_{\mathfrak{p}}$ is free of rank one.

2. *Sheaves supported on a subvariety.* Let $X = \operatorname{Spec} k[x,y]$ and let $f \in k[x,y]$ be irreducible. The sheaf $\widetilde{k[x,y]/(f)}$ has stalks

$$\widetilde{k[x,y]/(f)}_{\mathfrak{p}} = \begin{cases} (k[x,y]/(f))_{\mathfrak{p}} = k[x,y]_{\mathfrak{p}}/(f)_{\mathfrak{p}} & \text{if } f \in \mathfrak{p}, \\ 0 & \text{if } f \notin \mathfrak{p}. \end{cases}$$

   Geometrically, this is a "line bundle on $V(f)$, extended by zero to all of X." More generally, if $Y = V(\mathfrak{a}) \subseteq X$ is a closed subscheme defined by an ideal $\mathfrak{a} \subseteq R$, then the pushforward $\iota_* \mathcal{O}_Y$ is a quasi-coherent $\mathcal{O}_X$-module isomorphic to $\widetilde{R/\mathfrak{a}}$.

3. *Torsion modules over* $\operatorname{Spec} \mathbb{Z}$. Let $X = \operatorname{Spec} \mathbb{Z}$ and $M = \mathbb{Z}/12\mathbb{Z}$. The stalks of $\widetilde{M}$ are:

$$\widetilde{M}_{(2)} \cong \mathbb{Z}/4\mathbb{Z}, \qquad \widetilde{M}_{(3)} \cong \mathbb{Z}/3\mathbb{Z}, \qquad \widetilde{M}_{(p)} = 0 \text{ for } p \neq 2,3, \qquad \widetilde{M}_{(0)} = 0.$$

   The sheaf is supported on the two closed points (2) and (3), with "fibre dimensions" reflecting the prime factorization $12 = 2^2 \cdot 3$. More generally, if M is any torsion R-module, then $\widetilde{M}_{(0)} = 0$: torsion modules are not "spread" across the generic point.

4. *Eigenvalues and Jordan blocks.* Let $\alpha \in \operatorname{End}_{\mathbb{C}}(V)$ be an endomorphism of an n-dimensional $\mathbb{C}$-vector space, and view V as a $\mathbb{C}[x]$-module via $p(x) \cdot v = p(\alpha)(v)$. The associated sheaf $\widetilde{V}$ on $\mathbb{A}^1_{\mathbb{C}} = \operatorname{Spec} \mathbb{C}[x]$ is a quasi-coherent sheaf supported on the eigenvalues of α.

At a closed point λ, the stalk $\widetilde{V}_{(\lambda)} = V_{(x-\lambda)}$ is nonzero precisely when λ is an eigenvalue: if $\alpha - \lambda I$ is invertible, every element of V can be divided by $(x-\lambda)$, so the localization vanishes. If λ is an eigenvalue, the generalized eigenspace decomposition gives

$$\widetilde{V}_{(\lambda)} \cong \bigoplus_{j=1}^{r} \frac{\mathbb{C}[x]}{(x-\lambda)^{m_j}},$$

where $m_1, \dots, m_r$ are the sizes of the Jordan blocks for λ. The sheaf $\widetilde{V}$ thus encodes the full Jordan normal form of α: the support is the spectrum of eigenvalues, and the stalk structure records the Jordan block sizes. One can make this even cleaner by descending to $\operatorname{Spec}(\mathbb{C}[x]/m_\alpha(x))$, where m_α is the minimal polynomial.

5. *Extension by zero is not quasi-coherent.* Let $U \subseteq X$ be an open subscheme of an integral scheme with inclusion $\iota : U \hookrightarrow X$. The sheaf $\iota_!\mathcal{O}_U$ (extension by zero) is an $\mathcal{O}_X$-module but is typically *not* quasi-coherent. Indeed, if $V = \operatorname{Spec} R$ is an affine open not contained in U but with $U \cap V \neq \emptyset$, then $\iota_!\mathcal{O}_U|_V$ has no nonzero global sections (since elements must vanish outside U) but is not the zero sheaf on V, and hence cannot equal $\widetilde{M}$ for any R-module M.

6. *Restriction vs. pushforward.* If Y is a closed subscheme of X, the restriction $\mathcal{O}_X|_Y$ is typically *not* an $\mathcal{O}_Y$-module. For example, let $X = \operatorname{Spec} k[x,y]$ and $Y = \operatorname{Spec} k[x,y]/(y^2)$. In $\mathcal{O}_Y$, the function y is nilpotent ($y^2 = 0$), but in $\mathcal{O}_X|_Y$ there is no such nilpotence — restriction treats Y as a topological subspace, not as a scheme. The correct construction is the pushforward $\iota_*\mathcal{O}_Y$, which *is* a quasi-coherent $\mathcal{O}_X$-module (isomorphic to $\widetilde{R/\mathfrak{a}}$ where $\mathfrak{a}$ is the ideal of Y).

7. *The Hedgehog Theorem, algebraically.* Take $R = \mathbb{R}[x,y,z]/(x^2+y^2+z^2-1)$ and let M be the submodule of R^3 consisting of triples (X,Y,Z) with $xX + yY + zZ = 0$: the module of "tangent vectors to the sphere." On each affine open, $\widetilde{M}$ is locally free of rank 2 (isomorphic to $\mathcal{O}_X^2$), but it is not globally free — its complexification and analytification would contradict the Hairy Ball Theorem.

8. *Free and locally free sheaves.* The simplest quasi-coherent sheaves are the *free* sheaves $\mathcal{O}_X^{\oplus I}$ and the *locally free* sheaves, which restrict to free sheaves on an open cover. Locally free sheaves correspond to vector bundles (theorem 5.1.27). The category of locally free sheaves is not abelian: the multiplication-by-t map $\mathcal{O}_{\mathbb{A}^1_{\mathbb{C}}} \xrightarrow{\cdot t} \mathcal{O}_{\mathbb{A}^1_{\mathbb{C}}}$ has cokernel the skyscraper sheaf at the origin, which is not locally free. Quasi-coherent sheaves arise as the "abelian completion" — obtained by adjoining all kernels and cokernels of maps between locally free sheaves.

9. *Quillen–Suslin.* On affine space $\mathbb{A}^n_k = \operatorname{Spec} k[x_1, \dots, x_n]$, every locally free sheaf is in fact free. This is the geometric content of the Quillen–Suslin theorem (formerly Serre's conjecture): every projective module over a polynomial ring is free. Nontrivial quasi-coherent sheaves on affine space therefore arise only as kernels and cokernels of maps between free sheaves.

10. *Projective and flat modules.* Let $X = \operatorname{Spec} R$ and M a projective R-module. Since projective modules over local rings are free, each stalk $\widetilde{M}_{\mathfrak{p}} \cong M_{\mathfrak{p}}$ is a free $R_{\mathfrak{p}}$-module: projective modules give the closest algebraic analogue of nontrivial vector bundles. If instead M is flat, the localization $M_{\mathfrak{p}}$ remains flat at each point. Geometrically, flatness is the condition that fibres vary "continuously" (cf. section 6.8.1).

11. *Number-theoretic line bundles.* Let $R = \mathbb{Z}[\sqrt{-5}]$, which is not a UFD: $2 \cdot 3 = (1+\sqrt{-5})(1-\sqrt{-5})$. The ideal $J = (2, 1+\sqrt{-5})$ is not principal, so the sheaf $\widetilde{J}$ is a line bundle on $\operatorname{Spec} R$

that is not globally free. On the distinguished opens $D(2)$ and $D(3)$, the ideal J becomes principal:

$$J \cdot R[1/2] = R[1/2] \qquad \text{(since 2 is a unit)},$$
$$J \cdot R[1/3] = (1+\sqrt{-5}) \cdot R[1/3] \qquad \text{(since } 2 = (1+\sqrt{-5})(1-\sqrt{-5})/3\text{)}.$$

Hence $\widetilde{J}$ is locally free of rank 1 but not globally free — the failure of unique factorization is detected by the nontriviality of this line bundle. More generally, the Picard group $\operatorname{Pic}(\operatorname{Spec}\mathcal{O}_K)$ of a number ring is the ideal class group of K (see section 5.4).

12. *Line bundles on $\mathbb{P}^n_k$.* There is a natural line bundle on $\mathbb{P}^n_k$ whose fibre over a point p is the one-dimensional subspace of $\mathbb{A}^{n+1}_k$ that p represents. We shall see in section 5.3 that every line bundle on $\mathbb{P}^n_k$ is isomorphic to a tensor power of this bundle or its dual.

13. *Skyscraper at the generic point.* Let X be an integral Noetherian scheme with function field $K(X)$. The constant sheaf $\underline{K(X)}$ is a quasi-coherent $\mathcal{O}_X$-module (on each affine open $\operatorname{Spec} R$, it corresponds to the R-module $\operatorname{Frac}(R)$), but it is not coherent unless X is a point.

5.1.2 Properties of Quasi-Coherent Sheaves

The following collects the basic algebraic properties of the tilde construction.

Proposition 5.1.14: Properties of Sheaf Module

Let R be a ring, $X = \operatorname{Spec} R$, M an R-module, and $\varphi : R \to S$ a ring homomorphism with $f = \varphi^a : \operatorname{Spec} S \to \operatorname{Spec} R$. Then:

1. $\widetilde{M}_{\mathfrak{p}} \cong M_{\mathfrak{p}}$ for every $\mathfrak{p} \in \operatorname{Spec} R$.
2. $\Gamma(\operatorname{Spec} R, \widetilde{M}) \cong M$.
3. The functor $M \mapsto \widetilde{M}$ is exact and fully faithful from R-Mod to $\mathcal{O}_{\operatorname{Spec} R}$-Mod.
4. $\widetilde{M \otimes_R N} \cong \widetilde{M} \otimes_{\mathcal{O}_{\operatorname{Spec} R}} \widetilde{N}$ for any R-module N.
5. $\widetilde{\bigoplus_{i\in I} M_i} \cong \bigoplus_{i\in I} \widetilde{M_i}$ for any family $\{M_i\}_{i\in I}$ of R-modules.
6. If N is an S-module, then $f_*(\widetilde{N}) \cong \widetilde{({}_R N)}$, where ${}_R N$ denotes N viewed as an R-module via φ.
7. $f^*(\widetilde{M}) \cong \widetilde{M \otimes_R S}$.

***Proof*:**
Parts (1)–(2) follow from the definition: stalks are computed as colimits of localizations M_f over distinguished opens containing $\mathfrak{p}$, which yields $M_{\mathfrak{p}}$; global sections are $M_1 = M$. Part (3): full faithfulness follows from (2) and corollary 5.1.12; exactness follows from the exactness of localization. Part (4): the isomorphism $(M \otimes_R N)_f \cong M_f \otimes_{R_f} N_f$ on each distinguished open

gives the result on a basis, hence globally. Part (5) is similar using $(\bigoplus M_i)_f \cong \bigoplus (M_i)_f$. Parts (6)–(7) follow from the canonical identifications of pushforward and pullback with restriction and extension of scalars.

It is important for exactness in (6) that the source is an affine scheme. For non-affine schemes, f_* fails to be exact — this failure is precisely what sheaf cohomology measures (chapter 6).

The next result shows that sections of quasi-coherent sheaves on affine schemes behave like elements of modules: they can be "cleared of denominators" and extended from distinguished opens.

Lemma 5.1.15: Quasi-coherence and Extending From $D(f)$

Let $X = \operatorname{Spec} R$, $f \in R$, and $\mathscr{F}$ a quasi-coherent sheaf on X. Then:

1. If $s \in \Gamma(X, \mathscr{F})$ satisfies $s|_{D(f)} = 0$, then $f^n s = 0$ for some $n > 0$.
2. If $t \in \mathscr{F}(D(f))$, then $f^n t$ extends to a global section in $\Gamma(X, \mathscr{F})$ for some $n > 0$.

Part (2) should be thought of as "clearing denominators": the section $t \in \mathscr{F}(D(f))$ may involve f in the denominator, and multiplying by a sufficiently high power of f eliminates the denominator to produce a global section. Part (1) is the quasi-coherent analogue of analytic continuation, tempered by the possibility of nilpotent elements: a section vanishing on a distinguished open must be "almost zero" globally.

***Proof*:**

By proposition 5.1.11, $\mathscr{F} \cong \widetilde{M}$ for $M = \Gamma(X, \mathscr{F})$.

(1). The condition $s|_{D(f)} = 0$ means $s/1 = 0$ in M_f. By the definition of localization, $f^n s = 0$ in M for some $n > 0$.

(2). The section $t \in \mathscr{F}(D(f)) = M_f$ can be written as $t = m/f^n$ for some $m \in M$ and $n > 0$. Then $f^n t = m/1 \in M = \Gamma(X, \mathscr{F})$, which is the desired global section.

The global section functor on an affine scheme is exact when restricted to quasi-coherent sheaves — a crucial property that distinguishes affine schemes:

Proposition 5.1.16: Exactness of Global Sections for Quasi-Coherent Sheaves

Let $X = \operatorname{Spec} R$ be an affine scheme and $0 \to \mathscr{F}' \to \mathscr{F} \to \mathscr{F}'' \to 0$ an exact sequence of $\mathcal{O}_X$-modules with $\mathscr{F}'$ quasi-coherent. Then the sequence of global sections

$$0 \longrightarrow \Gamma(X, \mathscr{F}') \longrightarrow \Gamma(X, \mathscr{F}) \longrightarrow \Gamma(X, \mathscr{F}'') \longrightarrow 0$$

is exact.

***Proof*:**

Left-exactness of Γ holds for all sheaves (proposition 1.2.16), so it suffices to prove surjectivity on the right.

Let $s \in \Gamma(X, \mathscr{F}'')$. Since $\mathscr{F} \to \mathscr{F}''$ is surjective as a sheaf map, for each $x \in X$ there exists a distinguished open $D(g_x) \ni x$ such that $s|_{D(g_x)}$ lifts to some $\hat{s}_x \in \mathscr{F}(D(g_x))$. By

quasicompactness, finitely many such opens $D(g_1), \dots, D(g_r)$ cover X. Choose lifts $t_i \in \mathscr{F}(D(g_i))$ of $s|_{D(g_i)}$.

On overlaps $D(g_i g_j)$, the sections t_i and t_j both lift s, so their difference $t_i - t_j$ lies in $\mathscr{F}'(D(g_i g_j))$. Since $\mathscr{F}'$ is quasi-coherent, lemma 5.1.15(1) gives an integer n_{ij} with $g_i^{n_{ij}}(t_i - t_j) = 0$ on $D(g_j)$ (after appropriate further localization). Taking n larger than all n_{ij} and applying lemma 5.1.15(2) to replace each t_i by $g_i^n t_i$ (which now lifts $g_i^n s$), one obtains local sections that agree on overlaps and hence glue to a global section $\bar{t} \in \Gamma(X, \mathscr{F})$ lifting $g_i^n s$ for each i.

Finally, since $D(g_1), \dots, D(g_r)$ cover X, we have $(g_1^n, \dots, g_r^n) = R$ (by corollary 2.1.18). Write $1 = \sum_i a_i g_i^n$ and let $\bar{t}_i$ be the global lift of $g_i^n s$ constructed above. Then $t := \sum_i a_i \bar{t}_i \in \Gamma(X, \mathscr{F})$ maps to $\sum_i a_i g_i^n s = s$, proving surjectivity.

This exactness characterizes affine schemes: in chapter 6, we shall see that for Noetherian X, $H^i(X, \mathscr{F}) = 0$ for all $i > 0$ and all quasi-coherent $\mathscr{F}$ if and only if X is affine (theorem 6.2.11).

The following collects closure properties of quasi-coherent and coherent sheaves:

Proposition 5.1.17: Closure Properties of Quasi-Coherent Sheaves

Let X be a scheme.

1. The kernel, cokernel, and image of any morphism of quasi-coherent sheaves are quasi-coherent. Any extension of quasi-coherent sheaves by quasi-coherent sheaves is quasi-coherent. In particular, $\mathbf{QCoh}(X)$ is an abelian category.
2. If X is locally Noetherian, the same statements hold with "quasi-coherent" replaced by "coherent."

Let $f : X \to Y$ be a morphism of schemes. Then:

3. If $\mathscr{G}$ is a quasi-coherent $\mathcal{O}_Y$-module, then $f^*\mathscr{G}$ is a quasi-coherent $\mathcal{O}_X$-module. If additionally X and Y are locally Noetherian and $\mathscr{G}$ is coherent, then $f^*\mathscr{G}$ is coherent.
4. If $\mathscr{F}$ is a quasi-coherent $\mathcal{O}_X$-module and either X is Noetherian or f is quasi-compact and quasi-separated, then $f_*\mathscr{F}$ is a quasi-coherent $\mathcal{O}_Y$-module.

***Proof*:**
All claims are local on X, so we may assume $X = \operatorname{Spec} R$ is affine.

(1) Kernels, cokernels, images. Under the equivalence $\mathbf{QCoh}(\operatorname{Spec} R) \simeq R\text{-Mod}$, kernels, cokernels, and images of $\widetilde{M} \to \widetilde{N}$ correspond to $\widetilde{\ker}$, $\widetilde{\operatorname{coker}}$, and $\widetilde{\operatorname{im}}$ respectively, by exactness of the tilde functor.

Extensions. Let $0 \to \mathscr{F}' \to \mathscr{F} \to \mathscr{F}'' \to 0$ be an exact sequence with $\mathscr{F}'$ and $\mathscr{F}''$ quasi-coherent. Set $M' = \Gamma(X, \mathscr{F}')$, $M = \Gamma(X, \mathscr{F})$, and $M'' = \Gamma(X, \mathscr{F}'')$. By proposition 5.1.16 (applied with $\mathscr{F}'$ quasi-coherent), the sequence $0 \to M' \to M \to M'' \to 0$ is exact. Applying the exact functor

$\widetilde{(-)}$ gives a commutative diagram

$$\begin{array}{ccccccccc} 0 & \longrightarrow & \widetilde{M'} & \longrightarrow & \widetilde{M} & \longrightarrow & \widetilde{M''} & \longrightarrow & 0 \\ & & \downarrow\cong & & \downarrow & & \downarrow\cong & & \\ 0 & \longrightarrow & \mathscr{F}' & \longrightarrow & \mathscr{F} & \longrightarrow & \mathscr{F}'' & \longrightarrow & 0 \end{array}$$

where the outer vertical maps are isomorphisms (since $\mathscr{F}'$ and $\mathscr{F}''$ are quasi-coherent). By the five lemma, the middle map $\widetilde{M} \to \mathscr{F}$ is an isomorphism, so $\mathscr{F}$ is quasi-coherent.

(2) follows by adding finite generation hypotheses throughout.

(3) On affine opens, pullback corresponds to extension of scalars: $f^*(\widetilde{M}) \cong \widetilde{M \otimes_R S}$ by proposition 5.1.14(7).

(4) The key observation is that quasi-compactness and quasi-separatedness ensure $f^{-1}(V)$ is covered by finitely many affine opens whose pairwise intersections are again finitely covered by affine opens. This yields an exact sequence

$$0 \longrightarrow f_*\mathscr{F} \longrightarrow \bigoplus_i f_*(\mathscr{F}|_{U_i}) \longrightarrow \bigoplus_{i,j,k} f_*(\mathscr{F}|_{U_{ijk}})$$

where the second and third terms are quasi-coherent (since each U_i and U_{ijk} is affine), and $f_*\mathscr{F}$ is their kernel, hence quasi-coherent by (1).

Without the quasi-compactness or quasi-separatedness hypotheses, the pushforward can fail to preserve quasi-coherence:

Example 5.1.18: Pushforward Not Quasi-Coherent
Let $X = \coprod_{i\in\mathbb{N}} \operatorname{Spec}\mathbb{Z}$, $Y = \operatorname{Spec}\mathbb{Z}$, and $f : X \to Y$ the identity on each component. Take $\mathscr{F} = \mathcal{O}_X$. If $f_*\mathcal{O}_X$ were quasi-coherent, then the natural map

$$\Gamma(Y, f_*\mathcal{O}_X) \otimes_{\mathbb{Z}} \mathbb{Z}[\tfrac{1}{2}] \longrightarrow (f_*\mathcal{O}_X)(D(2))$$

would be an isomorphism. The left side is $(\prod_{i\in\mathbb{N}} \mathbb{Z}) \otimes_{\mathbb{Z}} \mathbb{Z}[\frac{1}{2}]$: every element has the form $(a_i/2^n)_{i\in\mathbb{N}}$ for a *fixed* exponent n. The right side is $\prod_{i\in\mathbb{N}} \mathbb{Z}[\frac{1}{2}]$, which contains elements $(b_i/2^{n_i})$ where n_i grows without bound. Such elements have no preimage, so the map is not surjective.

Even between Noetherian schemes, the pushforward of a coherent sheaf need not be coherent:

Example 5.1.19: Pushforward of Noetherian Not Coherent
The structure morphism $f : \operatorname{Spec} k[x] \to \operatorname{Spec} k$ gives $f_*\mathcal{O}_{\operatorname{Spec} k[x]} = k[x]$, which is not a finitely generated k-module. The issue is that f is not a finite morphism. Sufficient conditions for f_* to preserve coherence include f being finite, projective, or proper (see theorem 6.6.6 for the projective case and theorem 6.6.13 for the proper case).

For later use, we record the pushforward result from the proof of proposition 5.1.17(4) as a separate statement:

Proposition 5.1.20: Pushforward is Quasi-Coherent for Qcqs Morphisms

Let $f : X \to Y$ be a quasi-compact quasi-separated morphism of schemes. If $\mathscr{F}$ is a quasi-coherent $\mathcal{O}_X$-module, then $f_*\mathscr{F}$ is a quasi-coherent $\mathcal{O}_Y$-module.

***Proof*:**
It suffices to show that for each affine open $\operatorname{Spec} A \subseteq Y$ and each $g \in A$, the natural map

$$\Gamma(f^{-1}(\operatorname{Spec} A), \mathscr{F}) \otimes_A A_g \xrightarrow{\cong} \Gamma(f^{-1}(D(g)), \mathscr{F})$$

is an isomorphism. This can be verified using the exact sequence from the proof of proposition 5.1.17(4): the quasi-compact quasi-separated hypothesis ensures that $f^{-1}(\operatorname{Spec} A)$ admits a finite affine cover with affine pairwise intersections, reducing the claim to the affine case where it follows from the exactness of localization.

5.1.3 Locally Free Sheaves and Vector Bundles

An important class of quasi-coherent sheaves — historically the first studied — arises from vector bundles. In this subsection, we establish the equivalence between vector bundles and locally free sheaves, and explain how $\mathbf{QCoh}(X)$ can be viewed as the abelian completion of the category of vector bundles on X.

Recall from section 0.5 that a vector bundle is locally trivial: over each point x, there is an open neighborhood U and an isomorphism $\pi^{-1}(U) \cong U \times \mathbb{A}^r_k$ respecting projection and linear on fibres. The algebraic translation of "locally trivial" is "locally free."

Definition 5.1.21: Vector Bundle

Let X be a scheme over a field k. A *vector bundle of rank r* on X is a scheme E together with a morphism $\pi : E \to X$ and an open covering $\{U_i\}$ of X with isomorphisms

$$\varphi_i : \pi^{-1}(U_i) \xrightarrow{\sim} U_i \times \mathbb{A}^r_k$$

such that for every pair i, j, the composite $\varphi_j \circ \varphi_i^{-1}|_{U_i \cap U_j}$ is of the form $(\mathrm{id}, \varphi_{ij})$ where $\varphi_{ij} \in \mathrm{GL}_r(\mathcal{O}_X(U_i \cap U_j))$. A *section* of E over $U \subseteq X$ is a morphism $s : U \to E$ with $\pi \circ s = \mathrm{id}_U$.

The transition matrices φ_{ij} satisfy the cocycle conditions $\varphi_{ii} = I$ and $\varphi_{jk} \circ \varphi_{ij} = \varphi_{ik}$ on triple overlaps. These are precisely the data needed to glue a sheaf from local pieces.

Definition 5.1.22: Free and Locally Free $\mathcal{O}_X$-module

Let $(X, \mathcal{O}_X)$ be a ringed space and $\mathscr{F}$ an $\mathcal{O}_X$-module. Then $\mathscr{F}$ is *free of rank r* if $\mathscr{F} \cong \mathcal{O}_X^r$. It is *locally free of rank r* if there exists an open cover $\{U_i\}$ of X with $\mathscr{F}|_{U_i} \cong \mathcal{O}_{U_i}^r$ for each i. If X is connected, the rank is globally constant.

To identify which quasi-coherent sheaves are locally free, we need to know that "freeness of the stalk" propagates to a neighborhood. In differential geometry, this propagation is immediate: a smooth

map of constant rank at a point has the same rank nearby, because the rank of a matrix is detected by the nonvanishing of minors, which is an open condition. The algebraic analogue is provided by Nakayama's lemma. We now formulate a sheaf-theoretic version that will serve as our main tool.

First, some notation. For a quasi-coherent sheaf $\mathscr{F}$ on a scheme X and a point $x \in X$ with residue field $\kappa(x) = \mathcal{O}_{X,x}/\mathfrak{m}_x$, the *fibre* of $\mathscr{F}$ at x is the $\kappa(x)$-vector space

$$\mathscr{F}(x) \;:=\; \mathscr{F}_x \otimes_{\mathcal{O}_{X,x}} \kappa(x) \;=\; \mathscr{F}_x \,/\, \mathfrak{m}_x \mathscr{F}_x.$$

This is the "value" of the sheaf at the point: a vector bundle of rank r has fibres that are r-dimensional $\kappa(x)$-vector spaces, while a general coherent sheaf may have fibres of varying dimension.

Lemma 5.1.23: Geometric Nakayama Lemma

Let X be a scheme, $\mathscr{F}$ a quasi-coherent $\mathcal{O}_X$-module of finite type *(*i.e., locally finitely generated*)*, and $x \in X$.

1. *(Vanishing.)* If $\mathscr{F}(x) = 0$, then there exists an open neighborhood $U \ni x$ with $\mathscr{F}|_U = 0$.
2. *(Generators lift.)* If sections $s_1, \dots, s_n \in \mathscr{F}(U)$ are defined on a neighborhood $U \ni x$ and their images $\overline{s}_1, \dots, \overline{s}_n$ generate the fibre $\mathscr{F}(x)$, then after shrinking U the induced map $\mathcal{O}_U^n \to \mathscr{F}|_U$ is surjective.
3. *(Semicontinuity of fibre dimension.)* The function
$$x \;\longmapsto\; \dim_{\kappa(x)} \mathscr{F}(x)$$
is upper semicontinuous: for each $n \geq 0$, the set $\{x \in X : \dim \mathscr{F}(x) \leq n\}$ is open.

***Proof*:**

All three statements are local, so we may assume $X = \operatorname{Spec} R$ is affine and $\mathscr{F} = \widetilde{M}$ for a finitely generated R-module $M = Rm_1 + \cdots + Rm_k$. Let $\mathfrak{p} \in \operatorname{Spec} R$ correspond to x.

(1) Vanishing. The hypothesis $\mathscr{F}(x) = 0$ translates to $M_\mathfrak{p}/\mathfrak{p}M_\mathfrak{p} = 0$, i.e., $M_\mathfrak{p} = \mathfrak{p}M_\mathfrak{p}$. By the classical Nakayama lemma (applied to the finitely generated $R_\mathfrak{p}$-module $M_\mathfrak{p}$), we conclude $M_\mathfrak{p} = 0$.

Since M is finitely generated and each generator m_i satisfies $m_i/1 = 0$ in $M_\mathfrak{p}$, there exists $f_i \notin \mathfrak{p}$ with $f_i m_i = 0$ in M. Setting $f = f_1 \cdots f_k$ gives $f \notin \mathfrak{p}$ and $M_f = 0$, so $\mathscr{F}|_{D(f)} = \widetilde{M_f} = 0$.

(2) Generators lift. After shrinking U to a distinguished open and clearing denominators (using lemma 5.1.15), we may assume the s_i arise from elements $m_1, \dots, m_n \in M$. Define $\varphi : R^n \to M$ by sending the ith basis vector to m_i, and let $C = \operatorname{coker}(\varphi) = M/(Rm_1 + \cdots + Rm_n)$.

The hypothesis that $\overline{s}_1, \dots, \overline{s}_n$ generate $\mathscr{F}(x)$ translates to $C_\mathfrak{p}/\mathfrak{p}C_\mathfrak{p} = 0$, i.e., $C(x) = 0$. Since C is a quotient of the finitely generated module M, it is itself finitely generated. Applying part (1) to $\widetilde{C}$ yields $g \notin \mathfrak{p}$ with $C_g = 0$, meaning φ is surjective after localizing at g. Hence the map $\mathcal{O}_{D(g)}^n \to \mathscr{F}|_{D(g)}$ is surjective.

(3) Semicontinuity. Suppose $\dim_{\kappa(x)} \mathscr{F}(x) \leq n$. Choose n elements $s_1, \dots, s_n$ in $\mathscr{F}(U)$ (for some neighborhood $U \ni x$) whose images generate $\mathscr{F}(x)$. By (2), after shrinking U, the map $\mathcal{O}_U^n \to \mathscr{F}|_U$ is surjective. For any $y \in U$, applying $- \otimes_{\mathcal{O}_{X,y}} \kappa(y)$ to this surjection gives a

surjection $\kappa(y)^n \twoheadrightarrow \mathscr{F}(y)$, so $\dim_{\kappa(y)} \mathscr{F}(y) \leq n$. Hence $\{x \in X : \dim \mathscr{F}(x) \leq n\}$ is open.

The classical Nakayama lemma is a statement about a single local ring; the geometric upgrade converts it into a *neighborhood* statement by exploiting finite generation. The mechanism is simple but powerful: finitely many generators can be simultaneously controlled by inverting finitely many elements, i.e., by passing to a single distinguished open.

Part (3) should be compared with the situation for smooth vector bundles, which have *constant* fibre dimension. For coherent sheaves on schemes, constancy of fibre dimension is a strong condition — it is closely related to local freeness, as the next proposition makes precise.

Proposition 5.1.24: Locally Free if and only if Stalks are Free

Let X be a scheme and $\mathscr{F}$ a quasi-coherent sheaf of finite presentation *(*i.e., locally the cokernel of a map between finitely generated free modules*)*. Then $\mathscr{F}$ is locally free of rank r if and only if $\mathscr{F}_x$ is a free $\mathcal{O}_{X,x}$-module of rank r for every $x \in X$.
In particular, this applies to all coherent sheaves on locally Noetherian schemes, since over a Noetherian ring every finitely generated module is finitely presented.

The finite presentation hypothesis enters in a precise way: the geometric Nakayama lemma (lemma 5.1.23) requires finite type (= finitely generated) to show that generators of the fibre extend to generators of the sheaf nearby. This gives the surjectivity step. For the injectivity step, we must also apply the vanishing part of the lemma to the *kernel*, which requires the kernel to be finitely generated — and this is precisely what finite presentation guarantees.

***Proof*:**
($\Rightarrow$). If $\mathscr{F}|_U \cong \mathcal{O}_U^r$ for an open $U \ni x$, then $\mathscr{F}_x \cong \mathcal{O}_{X,x}^r$, since localization preserves free modules.

($\Leftarrow$). Suppose $\mathscr{F}_x \cong \mathcal{O}_{X,x}^r$ for all $x \in X$. Fix a point x and choose an affine neighborhood $\operatorname{Spec} R \ni x$ with $\mathscr{F}|_{\operatorname{Spec} R} \cong \widetilde{M}$, where M is a finitely presented R-module. Let $\mathfrak{p} \in \operatorname{Spec} R$ correspond to x, so that $M_{\mathfrak{p}} \cong R_{\mathfrak{p}}^r$ by hypothesis.

Step 1: Construct generators. Choose a basis $e_1, \ldots, e_r$ of the free $R_{\mathfrak{p}}$-module $M_{\mathfrak{p}}$. Write $e_i = m_i/g_i$ with $m_i \in M$ and $g_i \notin \mathfrak{p}$. Replacing $\operatorname{Spec} R$ by the distinguished open $D(g_1 \cdots g_r)$ (still an affine neighborhood of x on which $\mathscr{F}$ corresponds to a finitely presented module), we may assume $e_i = m_i \in M$.

The images $\overline{m}_1, \ldots, \overline{m}_r$ in the fibre $\mathscr{F}(x) = M_{\mathfrak{p}}/\mathfrak{p}M_{\mathfrak{p}} \cong \kappa(\mathfrak{p})^r$ form a basis.

Step 2: Surjectivity via geometric Nakayama. Define the map

$$\varphi : R^r \longrightarrow M, \qquad (a_1, \ldots, a_r) \longmapsto a_1 m_1 + \cdots + a_r m_r.$$

Since $\overline{m}_1, \ldots, \overline{m}_r$ generate $\mathscr{F}(x)$, lemma 5.1.23(2) gives a distinguished open $D(f) \ni x$ on which $\widetilde{\varphi} : \mathcal{O}_{D(f)}^r \to \mathscr{F}|_{D(f)}$ is surjective. Replacing $\operatorname{Spec} R$ by $D(f)$, we may assume φ is surjective.

Step 3: Injectivity via geometric Nakayama. Let $K = \ker(\varphi)$, so that $\widetilde{K}$ is a quasi-coherent sheaf on $\operatorname{Spec} R$ and we have the short exact sequence

$$0 \longrightarrow \widetilde{K} \longrightarrow \mathcal{O}_{\operatorname{Spec} R}^r \xrightarrow{\widetilde{\varphi}} \widetilde{M} \longrightarrow 0.$$

Since M is finitely presented, K is a finitely generated R-module[a]. At the stalk over x,

$$K_{\mathfrak{p}} \;=\; \ker(\varphi_{\mathfrak{p}} : R_{\mathfrak{p}}^{r} \to M_{\mathfrak{p}}).$$

But $\varphi_{\mathfrak{p}}$ sends $e_i \mapsto m_i$, which is a basis of $M_{\mathfrak{p}}$, so $\varphi_{\mathfrak{p}}$ is an isomorphism and $K_{\mathfrak{p}} = 0$. In particular, $\widetilde{K}(x) = K_{\mathfrak{p}}/\mathfrak{p}K_{\mathfrak{p}} = 0$. Since K is finitely generated, lemma 5.1.23(1) provides a distinguished open $D(g) \ni x$ with $K_g = 0$.

Conclusion. On the distinguished open $D(fg)$, the map φ is both surjective (by Step 2) and injective (by Step 3), hence an isomorphism $\mathcal{O}_{D(fg)}^{r} \xrightarrow{\sim} \mathscr{F}|_{D(fg)}$. Since x was arbitrary, $\mathscr{F}$ is locally free of rank r.

[a] If $R^{s} \xrightarrow{\alpha} R^{t} \to M \to 0$ is a finite presentation and $\varphi : R^{r} \twoheadrightarrow M$ is any surjection from a finitely generated free module, then Schanuel's lemma gives $K \oplus R^{t} \cong \ker(\alpha) \oplus R^{r}$; since $\ker(\alpha)$ is finitely generated by hypothesis, so is $K \oplus R^{t}$, hence K. Equivalently, being finitely presented is equivalent to *every* surjection from a finitely generated free module having finitely generated kernel.

The correspondence between vector bundles and locally free sheaves generalizes the classical fact that a vector bundle is determined by its transition functions. The key input from the scheme-theoretic side is the *relative spectrum* construction, which we develop in full generality in section 5.1.5 below. For the present theorem, we use it as follows: given a locally free sheaf $\mathcal{E}$ of rank r, the $\mathcal{O}_X$-algebra $\mathrm{Sym}(\mathcal{E}^{\vee})$ is locally isomorphic to a polynomial ring in r variables, and $\underline{\mathrm{Spec}}_X(\mathrm{Sym}(\mathcal{E}^{\vee}))$ is the corresponding "total space" scheme over X.

Lemma 5.1.25: Coherent Sheaf Locally Free Criterion

Let $\mathcal{E}$ be a coherent sheaf on a locally Noetherian scheme X. Then:

1. A subsheaf of a locally free sheaf is locally free.
2. If $\mathcal{E}$ and $\mathcal{E}'$ are locally free of rank 1, then every nonzero morphism $\mathcal{E} \to \mathcal{E}'$ is injective.

***Proof*:**
Both statements are local and reduce to the corresponding facts about modules over local rings: a torsion-free module over a DVR is free, and a nonzero map between rank-1 free modules over a domain is injective.

The category of locally free sheaves is not abelian: quotients may fail to be locally free due to torsion. The following lemma shows how to "correct" this:

Lemma 5.1.26: Natural Quotient of Locally Free Sheaf

Let X be a Noetherian integral scheme, $\mathcal{E}$ a locally free sheaf of rank r, and $\mathcal{E}' \subseteq \mathcal{E}$ a locally free subsheaf of rank $r' \leq r$. Then there exists a locally free subsheaf $\overline{\mathcal{E}}' \subseteq \mathcal{E}$ of rank r' containing $\mathcal{E}'$ such that $\mathcal{E}/\overline{\mathcal{E}}'$ is locally free of rank $r - r'$.

***Proof*:**

Let $M = \mathcal{E}/\mathcal{E}'$ and $T \subseteq M$ its torsion subsheaf. Define $\overline{\mathcal{E}}'$ to be the kernel of the composition $\mathcal{E} \twoheadrightarrow M \twoheadrightarrow M/T$. The quotient $\mathcal{E}/\overline{\mathcal{E}}' \cong M/T$ is torsion-free and coherent, hence locally free by lemma 5.1.25(1). By the snake lemma applied to the diagram

$$\begin{array}{ccccccccc} 0 & \longrightarrow & \mathcal{E}' & \longrightarrow & \mathcal{E} & \longrightarrow & M & \longrightarrow & 0 \\ & & \downarrow & & \| & & \downarrow & & \\ 0 & \longrightarrow & \overline{\mathcal{E}}' & \longrightarrow & \mathcal{E} & \longrightarrow & M/T & \longrightarrow & 0 \end{array}$$

we get $\overline{\mathcal{E}}'/\mathcal{E}' \cong T$, which is torsion, so $\overline{\mathcal{E}}'$ and $\mathcal{E}'$ have the same rank.

Theorem 5.1.27: Vector Bundle / Locally Free Sheaf Correspondence

Let X be a scheme over a field k. There is an equivalence of categories

$$\left\{ \begin{array}{c} \text{vector bundles} \\ \text{on } X \end{array} \right\} \quad \begin{array}{c} E \mapsto \mathcal{E}_E \\ \rightleftarrows \\ \underline{\mathrm{Spec}}_X \, \mathrm{Sym}(\mathscr{F}^\vee) \hookleftarrow \mathscr{F} \end{array} \quad \left\{ \begin{array}{c} \text{locally free} \\ \text{sheaves on } X \end{array} \right\}$$

where morphisms on the left are bundle maps (linear on fibres) and on the right are $\mathcal{O}_X$-module homomorphisms.

***Proof*:**

From bundles to sheaves. Let $\pi : E \to X$ be a vector bundle of rank r. Define the *sheaf of sections* $\mathcal{E}_E$ by $\mathcal{E}_E(U) = \{s : U \to E \big| \pi \circ s = \mathrm{id}_U\}$, with $\mathcal{O}_X$-action $(f \cdot s)_p = f(p) \cdot s_p$ and addition defined fibre-wise. On a trivializing open U_i with $\pi^{-1}(U_i) \cong U_i \times \mathbb{A}^r_k$, every section decomposes uniquely as $s = f_1 e_1 + \cdots + f_r e_r$ where e_j are the coordinate sections and $f_j \in \mathcal{O}_X(U_i)$. The map $s \mapsto (f_1, \ldots, f_r)$ gives an $\mathcal{O}_X(U_i)$-module isomorphism $\mathcal{E}_E(U_i) \cong \mathcal{O}_X(U_i)^r$, so $\mathcal{E}_E$ is locally free of rank r.

From sheaves to bundles. Let $\mathcal{E}$ be a locally free sheaf of rank r. The total space is constructed via the *relative spectrum* (section 5.1.5):

$$E \;:=\; \underline{\mathrm{Spec}}_X(\mathrm{Sym}(\mathcal{E}^\vee)),$$

where $\mathcal{E}^\vee = \mathit{Hom}_{\mathcal{O}_X}(\mathcal{E}, \mathcal{O}_X)$ is the dual sheaf and $\mathrm{Sym}(\mathcal{E}^\vee)$ is the symmetric algebra. On a trivializing open U_i where $\mathcal{E}|_{U_i} \cong \mathcal{O}^r_{U_i}$, we have $\mathrm{Sym}(\mathcal{E}^\vee)|_{U_i} \cong \mathcal{O}_X(U_i)[t_1, \ldots, t_r]$, and hence

$$\pi^{-1}(U_i) \;=\; \mathrm{Spec}\,(\mathcal{O}_X(U_i)[t_1, \ldots, t_r]) \;\cong\; U_i \times \mathbb{A}^r_k.$$

The transition functions of $\mathcal{E}$ give GL_r-valued cocycles on overlaps, furnishing E with the structure of a vector bundle. As a set,

$$E \;=\; \left\{(p, v) \;\middle|\; p \in X,\; v \in \mathcal{E}_p \otimes_{\mathcal{O}_{X,p}} \kappa(p)\right\},$$

which the reader should compare with the classical construction for curves where $E = \{(p, t) : p \in X,\; t \in \mathcal{E}_p/\mathfrak{m}_p\mathcal{E}_p\}$.

Functoriality. A bundle map $\varphi : E \to E'$ induces a sheaf morphism on sections by composition: $s \mapsto \varphi \circ s$. Conversely, a sheaf morphism $\mathcal{E} \to \mathcal{E}'$ induces a morphism of symmetric algebras

(contravariantly on the duals), hence a bundle map via relative Spec. These operations are inverse to each other.

Definition 5.1.28: Line Bundle

A locally free $\mathcal{O}_X$-module of rank 1 is called a *line bundle* or *invertible sheaf*.

The term "invertible" is justified by the existence of a tensor-product inverse: if $\mathcal{L}$ is a line bundle, then $\mathcal{L}^\vee = \mathit{Hom}_{\mathcal{O}_X}(\mathcal{L}, \mathcal{O}_X)$ satisfies $\mathcal{L} \otimes_{\mathcal{O}_X} \mathcal{L}^\vee \cong \mathcal{O}_X$. The group of isomorphism classes of line bundles under tensor product is the *Picard group* $\mathrm{Pic}\,(X)$, studied in depth in section 5.4.

5.1.29 K-Theory The monoid of isomorphism classes of locally free sheaves under direct sum can be group-completed to form the *Grothendieck group* $K_0(X)$ (see [6, K_0 of a Scheme, Ring and Module Structures]). The tensor product makes $K_0(X)$ into a ring. Line bundles correspond to the units of this ring under the tensor product, and the inclusion $\mathrm{Pic}\,(X) \hookrightarrow K_0(X)^\times$ connects divisor theory to K-theory. For affine space, the Quillen–Suslin theorem ($K_0(\mathbb{A}^n_k) \cong \mathbb{Z}$) reflects the triviality of all vector bundles.

5.1.4 Ideal Sheaves and Closed Subschemes

Having developed the general theory of quasi-coherent sheaves, we return to closed subschemes. Recall from definition 3.1.22 that a closed immersion $\iota : Y \hookrightarrow X$ is a morphism for which $\iota^\# : \mathcal{O}_X \to \iota_*\mathcal{O}_Y$ is surjective. The kernel of this surjection is an $\mathcal{O}_X$-submodule of $\mathcal{O}_X$ — an *ideal sheaf* — and it encodes the scheme structure of Y inside X.

Definition 5.1.30: Ideal Sheaf

Let $(X, \mathcal{O}_X)$ be a scheme. An *ideal sheaf* on X is an $\mathcal{O}_X$-submodule $\mathcal{I} \subseteq \mathcal{O}_X$, i.e., a sheaf of ideals: for each open $U \subseteq X$, $\mathcal{I}(U)$ is an ideal of $\mathcal{O}_X(U)$, compatibly with restriction.

Lemma 5.1.31: Kernel and Ideal Sheaves

Let $\iota : Y \hookrightarrow X$ be a closed immersion. Then the kernel

$$\mathcal{I}_{Y/X} := \ker\left(\iota^\# : \mathcal{O}_X \to \iota_*\mathcal{O}_Y\right)$$

is an ideal sheaf on X, and there is a short exact sequence of $\mathcal{O}_X$-modules:

$$0 \longrightarrow \mathcal{I}_{Y/X} \longrightarrow \mathcal{O}_X \longrightarrow \iota_*\mathcal{O}_Y \longrightarrow 0.$$

***Proof*:**
As $\iota^\#$ is a surjective morphism of $\mathcal{O}_X$-modules, its kernel is an $\mathcal{O}_X$-submodule of $\mathcal{O}_X$, hence an ideal sheaf. The short exact sequence follows immediately.

Not every ideal sheaf arises from a closed immersion. The obstruction is quasi-coherence:

Example 5.1.32: Not All Ideal Sheaves Correspond To Closed Immersions
Let $X = \operatorname{Spec} k[x]_{(x)}$, which has two points: the closed point $\mathfrak{m} = (x)$ and the generic point $\eta = (0)$. Define an ideal sheaf $\mathcal{I}$ by

$$\mathcal{I}(X) = 0 \subseteq \mathcal{O}_X(X) = k[x]_{(x)}, \qquad \mathcal{I}(\{\eta\}) = k(x) = \mathcal{O}_X(\{\eta\}).$$

This is a valid $\mathcal{O}_X$-submodule, but it is not quasi-coherent: if $\mathcal{I} = \widetilde{I}$ for some ideal $I \subseteq k[x]_{(x)}$, then $\mathcal{I}(X) = I = 0$ forces $\mathcal{I}(\{\eta\}) = I_{(0)} = 0$, contradicting $\mathcal{I}(\{\eta\}) = k(x)$. By proposition 5.1.33 below, $\mathcal{I}$ cannot define a closed subscheme.

Proposition 5.1.33: Ideal Sheaves and Closed Immersions

Let X be a scheme and $\mathcal{I} \subseteq \mathcal{O}_X$ an ideal sheaf. Then $\mathcal{I}$ defines a closed immersion $Y \hookrightarrow X$ (with $\mathcal{O}_Y = \mathcal{O}_X/\mathcal{I}$) if and only if for every affine open $\operatorname{Spec} S \subseteq X$ and every $f \in S$, the restriction map induces an isomorphism $\mathcal{I}(\operatorname{Spec} S)_f \cong \mathcal{I}(D(f))$. This condition holds precisely when $\mathcal{I}$ is quasi-coherent.

***Proof Key ideas*:**
($\Rightarrow$) A closed immersion $\iota\colon Y \to X$ is quasicompact, the preimage of an affine open being homeomorphic to a closed subset of it, and separated (lemma 3.5.11(2)). So $\iota_*\mathcal{O}_Y$ is quasi-coherent (proposition 5.1.17(4)), and so is the kernel $\mathcal{I}_{Y/X}$ of $\mathcal{O}_X \to \iota_*\mathcal{O}_Y$ (proposition 5.1.17(1)). On an affine open $\operatorname{Spec} S$ it is then $\widetilde{I}$ with $I = \mathcal{I}(\operatorname{Spec} S)$ (corollary 5.1.12), and $\widetilde{I}(D(f)) = I_f$ is the localization condition. Conversely, the localization condition identifies $\mathcal{I}$ on each affine open $\operatorname{Spec} S$ with $\widetilde{\mathcal{I}(\operatorname{Spec} S)}$, which is quasi-coherence.

($\Leftarrow$) For quasi-coherent $\mathcal{I}$, the support Y of $\mathcal{O}_X/\mathcal{I}$ is closed. On an affine open $\operatorname{Spec} S$, where $\mathcal{I} = \widetilde{I}$, exactness of the tilde functor (proposition 5.1.14(3)) gives $\mathcal{O}_X/\mathcal{I} = \widetilde{S/I}$, and $(Y \cap \operatorname{Spec} S, (\mathcal{O}_X/\mathcal{I})|_{Y\cap\operatorname{Spec} S})$ is identified with the closed subscheme $\operatorname{Spec}(S/I)$. These identifications make $(Y, (\mathcal{O}_X/\mathcal{I})|_Y)$ a closed subscheme with ideal sheaf $\mathcal{I}$.

Reason for deferral. The identification of $(Y\cap\operatorname{Spec} S, (\mathcal{O}_X/\mathcal{I})|_{Y\cap\operatorname{Spec} S})$ with $\operatorname{Spec}(S/I)$ compares the restriction to Y of a sheaf on X supported on Y with a pushforward from $\operatorname{Spec}(S/I)$, which this book does not develop. The complete argument is [36, Proposition II.5.9], whose proof of ($\Rightarrow$) uses [36, Proposition II.5.8]. The equivalence of the localization condition with quasi-coherence follows from [36, Proposition II.5.4].

We can now state the precise correspondence:

Proposition 5.1.34: Quasi-coherent Ideal Sheaves

Let X be a scheme. The assignment $Y \mapsto \mathcal{I}_{Y/X}$ defines a bijection

$$\{\text{closed subschemes of } X\} \xleftrightarrow{1:1} \{\text{quasi-coherent ideal sheaves on } X\}.$$

If X is locally Noetherian, then $\mathcal{I}_{Y/X}$ is coherent.

***Proof Key ideas*:**
By proposition 5.1.33, the ideal sheaf $\mathcal{I}_{Y/X}$ of a closed subscheme is quasi-coherent, and every quasi-coherent ideal sheaf $\mathcal{I}$ is the ideal sheaf of the closed subscheme $(Y, (\mathcal{O}_X/\mathcal{I})|_Y)$, Y being the support of $\mathcal{O}_X/\mathcal{I}$. A closed subscheme is recovered from its ideal sheaf in this way, because $\iota^\#$ induces an isomorphism $\mathcal{O}_X/\mathcal{I}_{Y/X} \cong \iota_*\mathcal{O}_Y$ (lemma 5.1.31). If X is locally Noetherian, then on an affine open $\operatorname{Spec} S$ with S Noetherian the ideal sheaf is $\widetilde{I}$ with I finitely generated, so it is coherent.

Reason for deferral. Both directions rest on proposition 5.1.33, and recovering Y from $\iota_*\mathcal{O}_Y$ needs the same comparison of a sheaf supported on Y with its restriction to Y. This is [36, Proposition II.5.9].

Corollary 5.1.35: Ideal Sheaves On Affine Schemes

For an affine scheme $X = \operatorname{Spec} R$, there is a bijection between ideals $\mathfrak{a} \subseteq R$ and closed subschemes of X, given by $\mathfrak{a} \mapsto \operatorname{Spec}(R/\mathfrak{a})$.

***Proof Key ideas*:**
A quasi-coherent ideal sheaf on $\operatorname{Spec} R$ is $\widetilde{\mathfrak{a}}$ for the ideal $\mathfrak{a}$ of its global sections (corollary 5.1.12). The closed immersion $\operatorname{Spec}(R/\mathfrak{a}) \to \operatorname{Spec}(R)$ induced by $R \to R/\mathfrak{a}$ (proposition 3.1.24(2)) has ideal sheaf $\widetilde{\mathfrak{a}}$, by exactness of the tilde functor (proposition 5.1.14(3), (6)). So the bijection of proposition 5.1.34 becomes $\mathfrak{a} \mapsto \operatorname{Spec}(R/\mathfrak{a})$.

Reason for deferral. The bijection itself is proposition 5.1.34, whose proof is deferred. The statement is [36, Corollary II.5.10].

With this correspondence in hand, we can define scheme-theoretic analogues of vanishing sets:

Definition 5.1.36: Vanishing Scheme

Let X be a scheme and $s \in \mathcal{O}_X(X)$ a global section. The *vanishing scheme* $V(s)$ is the closed subscheme corresponding to the quasi-coherent ideal sheaf generated by s: on each affine open $\operatorname{Spec} R \subseteq X$, it is defined by $\operatorname{Spec}(R/(s|_{\operatorname{Spec} R}))$. More generally, for a collection $S \subseteq \mathcal{O}_X(X)$, the vanishing scheme $V(S)$ corresponds to the ideal sheaf generated by S.

The advantage of vanishing *schemes* over vanishing *sets* is that they remember scheme-theoretic information such as intersection multiplicities.

Proposition 5.1.37: Properties of Vanishing Schemes

Let X be a scheme and $S_1, \dots, S_n, \{S_i\}_{i \in \mathcal{I}}$ collections of global sections. Then:

1. $V(S_1) \cup \cdots \cup V(S_n) = V(S_1 \cap \cdots \cap S_n)$ (finite unions correspond to ideal intersections).
2. $\bigcap_{i \in \mathcal{I}} V(S_i) = V(\langle S_i | i \in \mathcal{I} \rangle)$ (intersections correspond to ideal sums).

***Proof*:**
The underlying topological statement follows from the corresponding properties of $V(-)$ and $I(-)$ (proposition 2.1.6). The scheme-theoretic content is that finite unions use ideal *intersections* (not products): we need $V(S) \cap V(S) = V(S)$, which holds for $S \cap S = S$ but would fail for $S \cdot S = S^2$. The key algebraic input is the inclusion $IJ \subseteq I \cap J \subseteq \sqrt{IJ}$.

Example 5.1.38: Vanishing Scheme Intersection

1. Consider $V(y - x^2) \cap V(y) \subseteq \mathbb{A}^2_k$. As a set, this is the single point $(0, 0)$. As a scheme:

$$V(y - x^2) \cap V(y) \;=\; V(y - x^2,\, y) \;=\; \operatorname{Spec} \frac{k[x, y]}{(y - x^2,\, y)} \;\cong\; \operatorname{Spec} \frac{k[x]}{(x^2)}.$$

   The intersection point has *multiplicity* 2, reflecting the tangency between the parabola and the x-axis. This information is invisible to the underlying set but recorded by the scheme structure.

2. The intersection $V(y^2 - x^2) \cap V(y)$ gives an example where distributivity of unions and intersections fails at the scheme level. Since $y^2 - x^2 = (y - x)(y + x)$, we have $V(y^2 - x^2) = V(y - x) \cup V(y + x)$ (two lines through the origin). The scheme-theoretic computation shows

$$(V(y - x) \cup V(y + x)) \cap V(y) \;\neq\; (V(y - x) \cap V(y)) \cup (V(y + x) \cap V(y)),$$

   because the left side is $\operatorname{Spec} k[x]/(x^2)$ (a double point) while the right side is $\operatorname{Spec} k[x]/(x) \sqcup \operatorname{Spec} k[x]/(x) = \operatorname{Spec} k$ (a reduced point).

5.1.5 Relative Spectrum

In the proof of theorem 5.1.27, we constructed the total space of a vector bundle as $\underline{\operatorname{Spec}}_X(\operatorname{Sym}(\mathcal{E}^\vee))$. This is an instance of a general construction that associates a scheme over X to any quasi-coherent $\mathcal{O}_X$-algebra. Relative Spec is the scheme-theoretic analogue of the total space of a fibre bundle: it globalizes Spec from a single ring to a sheaf of rings over a base.

Definition 5.1.39: $\mathcal{O}_X$-algebra

Let $(X, \mathcal{O}_X)$ be a scheme. A *quasi-coherent $\mathcal{O}_X$-algebra* is a sheaf of commutative rings $\mathcal{A}$ on X that is simultaneously a quasi-coherent $\mathcal{O}_X$-module, with the ring multiplication $\mathcal{A}(U) \otimes_{\mathcal{O}_X(U)} \mathcal{A}(U) \to \mathcal{A}(U)$ being $\mathcal{O}_X(U)$-bilinear for each open $U \subseteq X$.
Equivalently, $\mathcal{A}$ is a ring object in the category $\mathbf{QCoh}(X)$. On an affine open $\operatorname{Spec} R \subseteq X$, a quasi-coherent $\mathcal{O}_X$-algebra restricts to $\widetilde{A}$ for some R-algebra A.

Example 5.1.40: Quasi-Coherent $\mathcal{O}_X$-Algebras

1. The structure sheaf $\mathcal{O}_X$ is a quasi-coherent $\mathcal{O}_X$-algebra.

2. If $\mathcal{E}$ is a locally free sheaf of rank r, then $\operatorname{Sym}(\mathcal{E}) = \bigoplus_{n \geq 0} \operatorname{Sym}^n(\mathcal{E})$ is a quasi-coherent $\mathcal{O}_X$-algebra. On a trivializing open where $\mathcal{E} \cong \mathcal{O}_X^r$, we have $\operatorname{Sym}(\mathcal{E}) \cong \mathcal{O}_X[t_1, \ldots, t_r]$.

3. If $f : Y \to X$ is an affine morphism, then $f_*\mathcal{O}_Y$ is a quasi-coherent $\mathcal{O}_X$-algebra.

The construction of relative Spec proceeds by gluing:

Theorem 5.1.41: Relative Spectrum

Let X be a scheme and $\mathcal{A}$ a quasi-coherent $\mathcal{O}_X$-algebra. There exists a scheme $\underline{\operatorname{Spec}}_X(\mathcal{A})$ together with an affine morphism $\pi : \underline{\operatorname{Spec}}_X(\mathcal{A}) \to X$ and an isomorphism of $\mathcal{O}_X$-algebras $\pi_*\mathcal{O}_{\underline{\operatorname{Spec}}_X(\mathcal{A})} \cong \mathcal{A}$, characterized by the following properties:

1. For each affine open $U = \operatorname{Spec} R \subseteq X$ with $\mathcal{A}|_U \cong \widetilde{A}$, there is a canonical isomorphism $\pi^{-1}(U) \cong \operatorname{Spec} A$.
2. As a set, $\underline{\operatorname{Spec}}_X(\mathcal{A}) = \{(x, \mathfrak{q}) : x \in X,\ \mathfrak{q} \in \operatorname{Spec}(\mathcal{A}_x)\}$.
3. The fibre over a point $x \in X$ with residue field $\kappa(x)$ is
$$\pi^{-1}(x) \ \cong\ \operatorname{Spec}(\mathcal{A}_x \otimes_{\mathcal{O}_{X,x}} \kappa(x)).$$

Furthermore, the construction gives an equivalence of categories:
$$\{\text{quasi-coherent } \mathcal{O}_X\text{-algebras}\} \ \simeq\ \{\text{affine morphisms } Y \to X\}^{\mathrm{op}}$$
with inverse $(\pi : Y \to X) \mapsto \pi_*\mathcal{O}_Y$.

***Proof*:**

Construction. Cover X by affine opens $U_i = \operatorname{Spec} R_i$ and write $\mathcal{A}|_{U_i} \cong \widetilde{A_i}$ for R_i-algebras A_i. On overlaps $U_i \cap U_j$, the quasi-coherence of $\mathcal{A}$ gives canonical isomorphisms between the restrictions of $\widetilde{A_i}$ and $\widetilde{A_j}$, hence isomorphisms between the corresponding spectra on the overlap. These isomorphisms satisfy the cocycle condition (being induced by identity maps of the sheaf $\mathcal{A}$), so by theorem 2.2.56 the schemes $\operatorname{Spec} A_i$ glue to a scheme $\underline{\operatorname{Spec}}_X(\mathcal{A})$ with a natural projection π to X.

Affine morphism. By construction, $\pi^{-1}(U_i) = \operatorname{Spec} A_i$ is affine for each i, so π is an affine morphism.

Equivalence. Given an affine morphism $\pi : Y \to X$, the pushforward $\pi_*\mathcal{O}_Y$ is a quasi-coherent $\mathcal{O}_X$-algebra (by proposition 5.1.20, since affine morphisms are quasi-compact and separated). On each affine open $\operatorname{Spec} R \subseteq X$, we have $\pi^{-1}(\operatorname{Spec} R) = \operatorname{Spec} A$ and $\pi_*\mathcal{O}_Y|_{\operatorname{Spec} R} \cong \widetilde{A}$. The two operations are inverse by the equivalence $\mathbf{QCoh}(\operatorname{Spec} R) \simeq R\text{-Mod}$ applied to algebras.

Description of points and fibres. The set-theoretic description follows from the gluing construction: a point of $\underline{\operatorname{Spec}}_X(\mathcal{A})$ over $x \in U_i$ is a prime ideal of $(A_i)_{\mathfrak{p}}$ where $\mathfrak{p}$ corresponds to x. The fibre computation follows from $\operatorname{Spec}(A_i \otimes_{R_i} \kappa(\mathfrak{p})) \cong \operatorname{Spec}(A_i/\mathfrak{p}A_i)_{\mathrm{red}}$ after base change to the residue field.

The relative spectrum provides a unified perspective on several constructions we have already seen:

Example 5.1.42: Relative Spectrum

1. *Closed subschemes.* If $\mathcal{I} \subseteq \mathcal{O}_X$ is a quasi-coherent ideal sheaf, then $\underline{\mathrm{Spec}}_X(\mathcal{O}_X/\mathcal{I})$ is the closed subscheme defined by $\mathcal{I}$.
2. *Vector bundles.* If $\mathcal{E}$ is locally free of rank r, then $\underline{\mathrm{Spec}}_X(\mathrm{Sym}(\mathcal{E}^\vee))$ is the total space of the corresponding vector bundle, with fibres $\mathbb{A}^r$.
3. *Affine morphisms.* Every affine morphism $Y \to X$ arises as $\underline{\mathrm{Spec}}_X(\pi_*\mathcal{O}_Y) \to X$ for a unique quasi-coherent $\mathcal{O}_X$-algebra.
4. *Base change.* For any morphism $f : X' \to X$,

$$\underline{\mathrm{Spec}}_{X'}(f^*\mathcal{A}) \;\cong\; \underline{\mathrm{Spec}}_X(\mathcal{A}) \times_X X'.$$

The relative spectrum thus plays the same role for quasi-coherent algebras that Spec plays for rings: it is the universal "total space" construction, and it provides the bridge between the algebraic data (a sheaf of algebras) and the geometric data (a scheme over X).

5.1.6 Regular Embeddings and Local Complete Intersections

Relative Spec attaches a scheme over X to a sheaf of algebras, and the second item of example 5.1.42 identifies the total space of a vector bundle as the case of a symmetric algebra. One closed subscheme in particular deserves that treatment: a closed subscheme cut out, locally, by exactly as many equations as its codimension. Such subschemes behave like linear subspaces, and the sheaf of equations modulo its square turns out to be locally free, so relative Spec converts it into a vector bundle along the subscheme. This is the class of embeddings for which the geometry near Z is controlled by a bundle rather than by a general cone, and the adjunction and duality computations later in this book already assume it.

Definition 5.1.43: Regular Embedding

Let $i\colon Z \hookrightarrow X$ be a closed immersion of Noetherian schemes with quasi-coherent ideal sheaf $\mathcal{I} \subseteq \mathcal{O}_X$ (definition 5.1.30, proposition 5.1.34). The immersion is a *regular embedding of codimension* d if every point of Z has an affine open neighbourhood $U = \operatorname{Spec} A$ in X for which the ideal $I = \mathcal{I}(U)$ is generated by a regular sequence $f_1, \dots, f_d$ of length d in A: that is, f_j is a nonzerodivisor on $A/(f_1, \dots, f_{j-1})$ for each j, and $A/I \neq 0$ ([18, Regular Sequence]).

The condition is local on Z and the codimension is required to be the same d at every point; a closed immersion that is regular of different codimensions on different components of Z is treated by working one component at a time.

Proposition 5.1.44: The Conormal Sheaf of a Regular Embedding

Let $i\colon Z \hookrightarrow X$ be a regular embedding of codimension d with ideal sheaf $\mathcal{I}$. Then $\mathcal{I}/\mathcal{I}^2$, regarded as a sheaf of $\mathcal{O}_Z$-modules, is locally free of rank d. If $f_1, \dots, f_d$ generate $\mathcal{I}$ over an affine open U, their classes form a basis of $(\mathcal{I}/\mathcal{I}^2)|_{Z\cap U}$.

***Proof*:**
The statement is local, so take $U = \operatorname{Spec} A$ affine with $I = (f_1, \dots, f_d)$ generated by a regular sequence. The quotient I/I^2 is the degree-one piece of the associated graded ring $\bigoplus_{n\geq 0} I^n/I^{n+1}$, and for an ideal generated by a regular sequence that graded ring is the polynomial ring: [10, Associated Graded Ring of a Regular Sequence] gives an isomorphism of graded A/I-algebras

$$(A/I)[T_1, \dots, T_d] \xrightarrow{\sim} \bigoplus_{n\geq 0} I^n/I^{n+1}, \qquad T_j \longmapsto \overline{f_j}$$

Comparing degree-one pieces, I/I^2 is free over A/I on the classes of $f_1, \dots, f_d$. Since $\mathcal{I}/\mathcal{I}^2$ is annihilated by $\mathcal{I}$, it is a sheaf of $\mathcal{O}_Z$-modules, and these local descriptions exhibit it as locally free of rank d, as we sought to show.

Local freeness is exactly what relative Spec needs, so the equations of a regular embedding assemble into a vector bundle over Z.

Definition 5.1.45: Conormal Sheaf and Normal Bundle

Let $i\colon Z \hookrightarrow X$ be a regular embedding of codimension d. The *conormal sheaf* is the locally free rank-d sheaf $\mathcal{I}/\mathcal{I}^2$ of proposition 5.1.44, and the *normal sheaf* is its dual $\mathcal{N}_{Z/X} = (\mathcal{I}/\mathcal{I}^2)^\vee$. The *normal bundle* of Z in X is the total space

$$N_Z X \;=\; \underline{\operatorname{Spec}}_Z\big(\operatorname{Sym}(\mathcal{I}/\mathcal{I}^2)\big)$$

a vector bundle of rank d over Z whose sheaf of sections is $\mathcal{N}_{Z/X}$, by example 5.1.42(2) applied to the locally free sheaf $\mathcal{N}_{Z/X}$.

Example 5.1.46: Regular and Irregular Embeddings

1. *Effective Cartier divisors.* A closed subscheme whose ideal sheaf is locally generated by a single nonzerodivisor is a regular embedding of codimension 1, a regular sequence of length one being exactly a nonzerodivisor. Its normal bundle is a line bundle.
2. *Linear subspaces.* For $\mathbb{P}^m \subseteq \mathbb{P}^n$ cut out by $x_{m+1}, \dots, x_n$, those $n-m$ coordinates form a regular sequence in each standard affine chart meeting $\mathbb{P}^m$, so the inclusion is a regular embedding of codimension $n-m$.
3. *A point on a smooth curve.* If X is a smooth curve over a field (definition 2.7.13) and $P \in X$ a closed point, then $\mathcal{O}_{X,P}$ is a regular local ring of dimension one by proposition 2.7.14, so its maximal ideal is generated by one element, and $\{P\} \hookrightarrow X$ is a regular embedding of codimension 1.
4. *A failure.* Let $X = \operatorname{Spec} \overline{k}[x,y]/(xy)$ be the union of the two coordinate axes and $Z = \{P\}$ the origin, so $\dim X = 1$ and Z has codimension 1. The embedding is *not* regular. A regular embedding of codimension 1 would force the ideal $\mathfrak{m} = (x,y)$ to be locally generated by a single element, but $\mathfrak{m}^2 = (x^2, xy, y^2) = (x^2, y^2)$ because $xy = 0$, so $\mathfrak{m}/\mathfrak{m}^2$ has $\overline{k}$-dimension 2 and $\mathfrak{m}$ cannot be generated by fewer than two elements. Equally, no generator can be a nonzerodivisor: x kills y and y kills x. Singular points obstruct regularity of the embedding, which is why the constructions resting on a normal *bundle* require this hypothesis and the general case needs a cone instead.

Regular embeddings and smooth morphisms are the two classes of maps along which geometry is controlled by a bundle, and the morphisms built from them are the ones for which Riemann-Roch and the intersection-theoretic Gysin maps are available.

Two properties of regular embeddings are used by the intersection theory downstream and are proved here, where they belong, rather than assumed there.

Proposition 5.1.47: The Conormal Sequence of a Chain of Regular Embeddings

Let $i\colon Z \hookrightarrow Y$ and $j\colon Y \hookrightarrow P$ be regular embeddings of codimensions d and e. Then $j \circ i$ is a regular embedding of codimension $d+e$, and there is a short exact sequence of locally free sheaves on Z,

$$0 \longrightarrow N_Z Y \longrightarrow N_Z P \longrightarrow \big(N_Y P\big)\big|_Z \longrightarrow 0$$

which is locally split. Dually, $0 \to (\mathcal{I}_Y/\mathcal{I}_Y^2)|_Z \to \mathcal{I}_Z/\mathcal{I}_Z^2 \to \mathcal{I}_{Z/Y}/\mathcal{I}_{Z/Y}^2 \to 0$ is exact, where $\mathcal{I}_Y \subseteq \mathcal{I}_Z$ are the ideal sheaves in $\mathcal{O}_P$ and $\mathcal{I}_{Z/Y} = \mathcal{I}_Z/\mathcal{I}_Y$ is the ideal sheaf of Z in Y.

***Proof*:**
Work locally on P, on an affine open $\operatorname{Spec} A$ over which $\mathcal{I}_Y$ is generated by a regular sequence $g_1, \dots, g_e$ and $\mathcal{I}_{Z/Y}$ by a regular sequence of length d in $A/(g)$; lift the latter to $h_1, \dots, h_d \in A$. Then $\mathcal{I}_Z = (g_1, \dots, g_e, h_1, \dots, h_d)$: a section of $\mathcal{I}_Z$ reduces modulo $\mathcal{I}_Y$ to a combination of the $\bar{h}$'s, so differs from a combination of the h's by a section of $\mathcal{I}_Y$. The concatenation is a regular sequence, since the g's are one in A and the h's are one on $A/(g)$ by hypothesis. Hence $j \circ i$ is a regular embedding of codimension $d+e$.

For the sequence, write $J = \mathcal{I}_Y$ and $I = \mathcal{I}_Z$, so $J \subseteq I$. The ideal of Z in Y is $\bar{I} = I/J$, and

$$\bar{I}/\bar{I}^2 \;=\; (I/J)\big/\big((I^2+J)/J\big) \;=\; I/(I^2+J),$$

so $I/I^2 \to \bar{I}/\bar{I}^2$ is surjective with kernel $(I^2+J)/I^2 = J/(J \cap I^2)$. Moreover $J/J^2 \otimes_{\mathcal{O}_Y} \mathcal{O}_Z = J/(J^2+IJ) = J/IJ$, because $J \subseteq I$ forces $J^2 \subseteq IJ$, and $IJ \subseteq J \cap I^2$; so J/IJ surjects onto $J/(J \cap I^2)$. The composite

$$\big(J/J^2\big)\big|_Z \longrightarrow I/I^2 \longrightarrow \bar{I}/\bar{I}^2 \longrightarrow 0$$

is therefore exact in the middle and on the right, with no hypothesis beyond $J \subseteq I$.

Regularity gives injectivity on the left and the splitting. In the local presentation above, $(J/J^2)|_Z$ is free on the classes of $g_1, \dots, g_e$ and I/I^2 is free on the classes of $g_1, \dots, g_e, h_1, \dots, h_d$, both by proposition 5.1.44, and the left-hand map sends the class of g_m to the class of g_m. It is therefore the inclusion of a free direct summand, split by the projection killing the h's. Dualizing a locally split short exact sequence of locally free sheaves gives the sequence of normal bundles in the statement, with the arrows reversed: the surjection onto $\bar{I}/\bar{I}^2$ dualizes to the inclusion $N_Z Y \subseteq N_Z P$, and the inclusion of $(J/J^2)|_Z$ dualizes to the surjection onto $(J/J^2)^\vee|_Z = (N_Y P)|_Z$ (definition 5.1.45).

Proposition 5.1.48: A Section of a Smooth Morphism Is a Regular Embedding

Let $p\colon P \to S$ be smooth of relative dimension e and let $s\colon S \to P$ be a section, so $p \circ s = \mathrm{id}_S$. Then s is a regular embedding of codimension e.

***Proof*:**
A section of a separated morphism is a closed immersion: s is the base change of the diagonal $P \to P \times_S P$ along $(\mathrm{id}_P, s \circ p)$, and the diagonal of a separated morphism is a closed immersion. So $s(S)$ is a closed subscheme; write $\mathcal{I}$ for its ideal sheaf.

The question is local, so take a point of $s(S)$ and a factorization as in definition 2.7.10: an open $U \subseteq P$ around it, an affine open $V = \operatorname{Spec} A \subseteq S$, and a closed immersion of U into $\operatorname{Spec} A[x_1, \dots, x_{e+r}]/(f_1, \dots, f_r)$ over V whose Jacobian has rank r. Shrinking further, the section is cut out inside U by the e functions $x_j - s^{\#}(x_j)$ for the e coordinates surviving after the r relations, since a point of U lies on $s(S)$ exactly when its coordinates agree with those of the section over the same point of V. These e functions generate $\mathcal{I}(U)$, and they form a regular sequence: modulo the first $j-1$ of them the ring is again of the same shape with one fewer free coordinate, and the next function is a coordinate minus a constant of the base, hence a nonzerodivisor there. So $\mathcal{I}$ is locally generated by a regular sequence of length e, as we sought to show.

Proposition 5.1.49: Regular Embeddings Pull Back Along Flat Morphisms

Let $i\colon Z \hookrightarrow X$ be a regular embedding of codimension d and let $g\colon X' \to X$ be flat. Then the base change $i'\colon Z' = g^{-1}(Z) \hookrightarrow X'$ is a regular embedding of codimension d, and its conormal sheaf is the pullback: the canonical surjection $h^*(\mathcal{I}/\mathcal{I}^2) \to \mathcal{I}_{Z'}/\mathcal{I}_{Z'}^2$, for $h\colon Z' \to Z$ the induced morphism, is an isomorphism.

***Proof*:**
Base change of a closed immersion is a closed immersion, and the ideal sheaf of Z' is $\mathcal{I}\mathcal{O}_{X'}$. Work on an affine open $\operatorname{Spec} A \subseteq X$ over which $\mathcal{I}$ is generated by a regular sequence $f_1, \dots, f_d$, and let $\operatorname{Spec} A' \subseteq X'$ be an affine open mapping into it, so A' is flat over A. The images of $f_1, \dots, f_d$ generate $\mathcal{I}\mathcal{O}_{X'}$ on $\operatorname{Spec} A'$, and they remain a regular sequence: multiplication by f_j on $A/(f_1, \dots, f_{j-1})$ is injective by hypothesis, and tensoring that injection with the flat A-module A' keeps it injective, which says exactly that f_j is a nonzerodivisor on $A'/(f_1, \dots, f_{j-1})$. Flatness also gives $(\mathcal{I}/\mathcal{I}^2) \otimes_A A' \cong \mathcal{I}\mathcal{O}_{X'}/(\mathcal{I}\mathcal{O}_{X'})^2$, and this isomorphism is the canonical map, being induced by $f \mapsto f \otimes 1$ on generators; that is the asserted identification of conormal sheaves, completing the proof.

Definition 5.1.50: Local Complete Intersection Morphism

A morphism $f\colon X \to Y$ of Noetherian schemes is a *local complete intersection morphism*, or *l.c.i. morphism*, if every point of X has an open neighbourhood U for which $f|_U$ admits a factorization

$$U \xhookrightarrow{i} P \xrightarrow{p} Y$$

with i a regular embedding (definition 5.1.43) and p smooth (definition 2.7.10).

Proposition 5.1.51: Regular Embeddings and Smooth Morphisms Are l.c.i.

Every regular embedding is an l.c.i. morphism, and every smooth morphism is an l.c.i. morphism.

***Proof*:**
For a regular embedding $i\colon Z \hookrightarrow X$, take $U = Z$, $P = X$, and $p = \mathrm{id}_X$, which is smooth of relative dimension 0: the identity is covered by the factorization of definition 2.7.10 with $n = r = 0$, the empty Jacobian having rank 0.

For a smooth morphism $p\colon X \to Y$, take $U = X$, $P = X$, and $i = \mathrm{id}_X$. The identity is a closed immersion with zero ideal sheaf, which is generated by the empty regular sequence, so it is a regular embedding of codimension 0, completing the proof.

A single morphism can admit many such factorizations. Nothing in this book depends on comparing them: the definition above is the only thing used here, and it is independent of any choice because it only asserts that a factorization exists. The constructions that do depend on the choice, the Gysin maps and the Riemann-Roch correction terms, are cycle-level rather than scheme-level, and a reader who wants them will find the independence proved alongside them in *Intersection Theory* [19]. That pointer is a signpost, not a dependency.

5.2 Images of Scheme Morphisms

In the theory of sets, the image of a function is unambiguous: if $f : X \to Y$, then $f(X) = \{f(x) \big| x \in X\} \subseteq Y$, and it is automatically a subset. In algebraic geometry, things are not so clean. The image of a morphism of schemes need not be open, need not be closed, and need not even be locally closed—so it cannot, in general, carry a scheme structure coming from Y. This section develops two complementary responses to this problem:

1. *Chevalley's theorem*: the image is always *constructible*—a finite union of locally closed sets. This is the best purely topological statement one can make, and it has surprisingly powerful consequences for the geometry of fibers.
2. *The scheme-theoretic image*: when we insist on a scheme structure, we replace the set-theoretic image with the smallest closed subscheme through which the morphism factors. This is the "correct" notion of image in the category of schemes.

Both ideas require some commutative algebra—in particular, Grothendieck's Generic Freeness lemma, which says that finitely generated modules over a domain become free after inverting a single element. This is the key algebraic engine behind Chevalley's theorem.

5.2.1 Chevalley's Theorem

The reader who has worked primarily with affine morphisms may expect the image of a scheme morphism to be something reasonable—an open set, a closed set, or at worst a locally closed set. The following example shows that none of these hold in general.

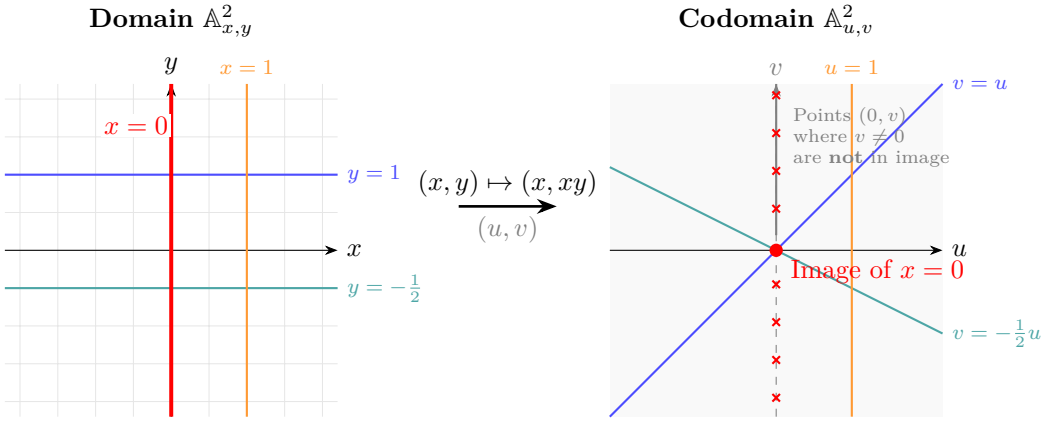
Example 5.2.1: Image That is Not Locally Closed

Let $f : \mathbb{A}^2_k \to \mathbb{A}^2_k$ be the morphism given by $(x, y) \mapsto (x, xy)$, corresponding to the ring map $k[u, v] \to k[x, y]$ with $u \mapsto x$, $v \mapsto xy$. Its image is:

$$\operatorname{im}(f) = \left(\mathbb{A}^2_k \setminus \{(u, v) \mid u = 0\}\right) \cup \{(0, 0)\}$$

In words: the entire plane with the v-axis removed, plus the origin put back in.

To see this, note that if $u = x \neq 0$, then $v = xy$ can be any element of k (by choosing $y = v/x$), so every point (u, v) with $u \neq 0$ is in the image. On the other hand, if $u = x = 0$, then $v = xy = 0$ regardless of y, so the only point on the v-axis that lies in the image is the origin.



This set is *not* locally closed. Recall that a locally closed subset is the intersection of an open set and a closed set. If $\operatorname{im}(f)$ were locally closed, its closure would be $\mathbb{A}^2_k$ (since $\operatorname{im}(f)$ is dense), and so $\operatorname{im}(f)$ would have to be open. But the complement $\mathbb{A}^2_k \setminus \operatorname{im}(f) = \{(0, v) \big| v \neq 0\}$ is not closed (its closure is the entire v-axis), so $\operatorname{im}(f)$ is not open either.

Because $\operatorname{im}(f)$ is not locally closed, it does not arise as the underlying set of any subscheme of $\mathbb{A}^2_k$.

However, the image in this example is not completely wild: it is the union of the open set $D(u)$ and the closed point $\{(0, 0)\}$—a union of two locally closed sets. Chevalley's theorem says this is always the case.

Constructible Sets

To state Chevalley's theorem, we need the notion of a constructible set. The idea is simple: constructible sets are exactly the subsets you can build from open sets using complements and finite intersections. Equivalently (and more geometrically), they are finite unions of locally closed sets.

Definition 5.2.2: Constructible Set

Let X be a Noetherian topological space. A subset $S \subseteq X$ is *constructible* if it belongs to the smallest family of subsets containing all open sets and closed under:

1. complements
2. finite intersections

The first condition alone would give all open and all closed sets. Adding finite intersections then adds all locally closed sets (an open intersected with a closed set), and then all finite unions of locally closed sets (by De Morgan: a union of two constructible sets $S_1 \cup S_2 = (S_1^c \cap S_2^c)^c$ is constructible). In fact, these are *all* the constructible sets:

Lemma 5.2.3: Constructible Sets as Locally Closed Sets

Let X be a Noetherian topological space. Then a subset $S \subseteq X$ is constructible if and only if it is a finite union of locally closed subsets. Consequently:

1. Finite unions and finite intersections of constructible sets are constructible.
2. If $f : X \to Y$ is continuous, then the preimage of a constructible set is constructible.

***Proof*:**
Every locally closed set is constructible (it is the intersection of an open and a closed set, both of which are in the family). Conversely, we must show that the collection of finite unions of locally closed sets is closed under complements and finite intersections.

Finite intersections. The intersection of two locally closed sets $(U_1 \cap Z_1) \cap (U_2 \cap Z_2) = (U_1 \cap U_2) \cap (Z_1 \cap Z_2)$ is locally closed. By distributing, the intersection of two finite unions of locally closed sets is again a finite union of locally closed sets.

Complements. It suffices to show the complement of a single locally closed set $U \cap Z$ is a finite union of locally closed sets. We have:

$$X \setminus (U \cap Z) = (X \setminus U) \cup (X \setminus Z)$$

The first is closed (hence locally closed) and the second is open (hence locally closed), so their union is a finite union of locally closed sets.

The two consequences are immediate: (1) follows from the definition, and (2) follows because continuous preimages preserve open sets, closed sets, intersections, and complements.

If the reader finds this reminiscent of the definition of measurable sets in measure theory, the analogy is apt: constructible sets form the "Boolean algebra" generated by open sets, just as measurable

sets form the σ-algebra generated by open sets. The key difference is the finiteness constraint—no countable unions allowed.

For completeness, we record the definition given in EGA for the non-Noetherian case. A *retrocompact* open set $U \subseteq X$ is one where the inclusion $U \hookrightarrow X$ is quasi-compact. In the general definition, one replaces "open" by "retrocompact open" in definition 5.2.2. This ensures that constructible sets are described by "finite amounts of data" even when X is not Noetherian. For schemes of finite type over a Noetherian base—which includes essentially all schemes we care about—retrocompact open sets are the same as quasi-compact open sets, and the two definitions agree. We shall not need this generality.

5.2.4 Assumption For This Section We assume all topological spaces and schemes are Noetherian for the rest of this section, unless otherwise stated.

Generic Freeness

The algebraic engine behind Chevalley's theorem is Grothendieck's Generic Freeness lemma, which says that finitely generated modules over a domain become free after inverting a single element. Geometrically, this means that a coherent sheaf on an integral scheme is "generically" free—it is a vector bundle over a dense open subset.

Theorem 5.2.5: Generic Freeness (Grothendieck)

Let A be a Noetherian domain, S a finitely generated A-algebra, and M a finitely generated S-module. Then there exists a nonzero element $f \in A$ such that M_f is a free A_f-module.

Before giving the proof, let us unpack the geometric content. Taking $S = B$ and $M = B$ for a finitely generated A-algebra B: the theorem says that after shrinking $\operatorname{Spec} A$ to a distinguished open subset $D(f)$, the A_f-module B_f becomes free. In particular, $A_f \to B_f$ is faithfully flat (a nonzero free module is faithfully flat), and so the induced map $\operatorname{Spec} B_f \to \operatorname{Spec} A_f$ is *surjective*. This is the key input for Chevalley's theorem: a dominant finite type morphism from an integral scheme is surjective over a dense open.

***Proof*:**
We induct on the number of A-algebra generators of S. Write $S = A[t_1, \ldots, t_n]$ (up to taking a quotient, which only makes M a module over a ring with fewer generators; explicitly, if $S = A[t_1, \ldots, t_n]/I$ then we view M as an $A[t_1, \ldots, t_n]$-module annihilated by I, and the result for $A[t_1, \ldots, t_n]$ implies the result for S).

Base case: $n = 0$, so $S = A$. Then M is a finitely generated module over the Noetherian domain A. Let $K = \operatorname{Frac}(A)$, and let $r = \dim_K(M \otimes_A K)$. Choose elements $m_1, \ldots, m_r \in M$ whose images form a K-basis of $M \otimes_A K$, and consider the A-linear map:

$$\varphi : A^r \longrightarrow M, \qquad e_i \longmapsto m_i.$$

After tensoring with K, this becomes an isomorphism. Both $\ker(\varphi)$ and $\operatorname{coker}(\varphi)$ are finitely generated A-modules that vanish after tensoring with K. For each generator of $\ker(\varphi)$ and $\operatorname{coker}(\varphi)$, there is a nonzero element of A annihilating it (since the generator maps to zero in the localization at (0)). Let f be the product of all such elements. Then $\ker(\varphi)_f = 0$ and

$\operatorname{coker}(\varphi)_f = 0$, so $\varphi_f : A_f^r \xrightarrow{\sim} M_f$ is an isomorphism. Hence M_f is free of rank r.

Inductive step: $S = S_0[t]$ where $S_0 = A[t_1, \ldots, t_{n-1}]$. Let M be a finitely generated $S_0[t]$-module. We split M into its t-torsion and t-torsion-free parts.

Step 1: The t-torsion part. Let $T = \{m \in M : t^k m = 0 \text{ for some } k \geq 1\}$. Since M is Noetherian, T is finitely generated, and there exists $N \geq 1$ with $t^N T = 0$. Hence T is annihilated by t^N, so T is a module over $S_0[t]/(t^N)$. Now, $S_0[t]/(t^N)$ is a finitely generated S_0-module (with S_0-basis $1, t, t^2, \ldots, t^{N-1}$). Since T is finitely generated over $S_0[t]/(t^N)$, it is finitely generated over S_0. By the inductive hypothesis (applied with S_0 in place of S), there exists $f_1 \in A \setminus \{0\}$ such that T_{f_1} is a free A_{f_1}-module.

Step 2: The torsion-free part. The quotient $\overline{M} = M/T$ is a finitely generated $S_0[t]$-module on which t acts injectively. Let $K = \operatorname{Frac}(A)$. Then $\overline{M}_K = \overline{M} \otimes_A K$ is a finitely generated module over $K[t_1, \ldots, t_{n-1}, t]$, and since t acts injectively, $\overline{M}_K$ is a finitely generated *torsion-free* module over $K[t]$ (torsion-free with respect to t, viewing $\overline{M}_K$ as a $K[t]$-module via the other generators). Since $K[t]$ is a PID, torsion-free and finitely generated implies free:

$$\overline{M}_K \cong K[t]^s$$

for some $s \geq 0$[a]. Choose $\overline{m}_1, \ldots, \overline{m}_s \in \overline{M}$ whose images form a $K[t]$-basis of $\overline{M}_K$. The map $S_0[t]^s \to \overline{M}$ sending $e_i \mapsto \overline{m}_i$ becomes an isomorphism after tensoring with K. As in the base case, the kernel and cokernel are finitely generated and K-torsion, so there exists $f_2 \in A \setminus \{0\}$ with:

$$\overline{M}_{f_2} \cong A_{f_2}[t_1, \ldots, t_{n-1}, t]^s$$

The right side is a free A_{f_2}-module (of countably infinite rank when $s > 0$, with A_{f_2}-basis given by all monomials times the standard basis vectors).

Step 3: Reassembly. Taking $f = f_1 f_2$, we have that T_f is free over A_f (from Step 1) and $\overline{M}_f$ is free over A_f (from Step 2). The short exact sequence $0 \to T_f \to M_f \to \overline{M}_f \to 0$ splits, since $\overline{M}_f$ is free (hence projective). Therefore $M_f \cong T_f \oplus \overline{M}_f$ is free over A_f.

[a] Strictly speaking, $\overline{M}_K$ is a module over $K[t_1, \ldots, t_{n-1}][t]$, and we are using that it is torsion-free over $K[t_1, \ldots, t_{n-1}][t]$, which is a PID when we view it as a localization of the polynomial ring. The key point is that after localizing A to K, the t-torsion-free part becomes free over the resulting ring, which is a PID in the variable t.

Corollary 5.2.6: Dominant Morphisms Hit a Dense Open

Let $f : X \to Y$ be a dominant morphism of finite type between Noetherian integral schemes. Then $f(X)$ contains a dense open subset of Y.

***Proof*:**
We may assume $X = \operatorname{Spec} B$ and $Y = \operatorname{Spec} A$ are affine. Since f is dominant, the ring map $A \to B$ is injective (proposition 3.1.34), and B is a finitely generated A-algebra. By theorem 5.2.5 (with $S = B$ and $M = B$), there exists $a \in A \setminus \{0\}$ with B_a a free A_a-module. Since $B \neq 0$, we have $B_a \neq 0$ (localizing a nonzero module at a nonzero element of a domain gives a nonzero module). A nonzero free module is faithfully flat, so $A_a \to B_a$ is faithfully flat, and the induced map $\operatorname{Spec} B_a \to \operatorname{Spec} A_a$ is surjective. Thus $f(X) \supseteq D(a)$, a dense open subset of Y.

Chevalley's Theorem

We can now state and prove the main result. The proof is an application of Noetherian induction (theorem 0.3.3): the corollary above gives us control over a dense open subset, and the inductive hypothesis handles the closed complement.

Theorem 5.2.7: Chevalley's Theorem

Let $f : X \to Y$ be a morphism of finite type of Noetherian schemes. Then the image of every constructible subset of X is a constructible subset of Y. In particular, $f(X)$ is constructible.

***Proof*:**
Since a constructible set is a finite union of locally closed sets, and a finite union of constructible sets is constructible, it suffices to show that the image of a locally closed set is constructible. A locally closed subset of a Noetherian scheme is again a Noetherian scheme (with its reduced induced structure, definition 3.1.26), and the restriction of a finite type morphism to a locally closed subscheme is still of finite type. Thus it suffices to show that $f(X)$ is constructible.

Step 1: Reduction to the affine case. The question is local on Y: if $Y = \bigcup_j V_j$ is an open cover, then $f(X)$ is constructible in Y if and only if $f(X) \cap V_j$ is constructible in V_j for each j. So we may assume $Y = \operatorname{Spec} A$ is affine. Since f is of finite type, X is quasi-compact, so we can cover X by finitely many affine opens $X = U_1 \cup \cdots \cup U_m$. Then $f(X) = f(U_1) \cup \cdots \cup f(U_m)$, and a finite union of constructible sets is constructible. So we may assume $X = \operatorname{Spec} B$ is also affine, with B a finitely generated A-algebra.

Step 2: Noetherian induction on Y. We prove by Noetherian induction on Y: the statement holds for Y, assuming it holds for every proper closed subset $Z \subsetneq Y$ (equipped with any scheme structure making it a Noetherian scheme). This is valid by corollary 0.3.5, applied to the property of closed subsets $Z \subseteq Y$ that the theorem holds over Z, because Y is a Noetherian topological space (lemma 2.4.2).

Step 3: Decompose the base. Decompose $Y = \operatorname{Spec} A$ into its (finitely many) irreducible components $Y_1, \ldots, Y_r$, where $Y_i = V(\mathfrak{p}_i)$ for the minimal primes $\mathfrak{p}_1, \ldots, \mathfrak{p}_r$ of A (proposition 2.1.25). The image $f(X)$ is the union of the images $f(f^{-1}(Y_i)) \to Y_i$ for $i = 1, \ldots, r$. Each $f^{-1}(Y_i) \to Y_i$ is a morphism of finite type (base change preserves finite type). If $r > 1$, each Y_i is a proper closed subset of Y, and by the inductive hypothesis the image of $f^{-1}(Y_i) \to Y_i$ is constructible in Y_i. A constructible subset of a closed subspace is constructible in the ambient space (it is still a finite union of locally closed sets), so $f(X)$ is constructible. If $r = 1$, then Y is irreducible.

Step 4: The irreducible case. We may now assume $Y = \operatorname{Spec} A$ is irreducible. Replacing A by $A/\operatorname{Nil}(A)$ does not change the underlying topological space (proposition 2.1.10), so we may further assume A is a domain (Y is integral). Similarly, replacing B by $B/\ker(A \to B) \cdot B$ does not change the image of the map on spectra. After this, the morphism $\operatorname{Spec} B \to \operatorname{Spec} A$ factors as $\operatorname{Spec} B \to \operatorname{Spec}(A/J) \hookrightarrow \operatorname{Spec} A$ for some ideal J. If $J \neq 0$, then $V(J) \subsetneq Y$ and we are done by the inductive hypothesis. If $J = 0$, then $A \hookrightarrow B$ and the morphism is dominant.

Step 5: The dominant case. We have A a Noetherian domain, B a finitely generated A-algebra with $A \hookrightarrow B$ injective, and $f : \operatorname{Spec} B \to \operatorname{Spec} A$ dominant. By corollary 5.2.6, there exists $a \in A \setminus \{0\}$ with $D(a) \subseteq f(X)$.

Consider the closed complement $Z = V(a) \subsetneq \operatorname{Spec} A$. Since $a \neq 0$ and A is a domain, Z is a proper closed subset. The preimage $f^{-1}(Z) = \operatorname{Spec}(B/aB)$ maps to $Z = \operatorname{Spec}(A/aA)$ by a morphism of finite type. By the Noetherian inductive hypothesis, the image of $f^{-1}(Z) \to Z$ is constructible in Z, hence constructible in Y. Therefore:

$$f(X) = D(a) \cup \big(\text{constructible subset of } V(a)\big)$$

is a union of two constructible sets, hence constructible.

Constructibility controls the image of a single morphism. Families ask for something transverse to that: a property visible on the generic fiber of $f\colon X \to S$ ought to persist on the fiber over every point of some nonempty open subset of the base. The pattern has appeared once already, in proposition 3.2.12, where an empty generic fiber was shown to force empty fibers over a whole neighborhood of η. Emptiness is the crudest property one can spread out this way. The next one up, and the one that carries most of the weight in practice, is density: given an open subscheme $U \subseteq X$, does density of U_η inside X_η propagate to the nearby fibers?

The question is not idle. Constructions on a family are routinely carried out on an open subset U, such as the locus where a chosen section does not vanish or the complement of a bad divisor, and what is then needed is that U still meets every part of X_s for most s. Density is exactly the condition that U_s meet every irreducible component of X_s, so the question is whether the components of a fiber can conspire, as s moves, to hide inside the closed complement $X \setminus U$. They can, but only over a proper closed subset of the base.

The engine is generic freeness again, applied in a shape slightly different from the one used for corollary 5.2.6: rather than tracking a whole image, it tracks a single equation, and it reports that the equation stays a nonzerodivisor on all nearby fibers.

Lemma 5.2.8: Nonzerodivisors Persist on Nearby Fibers

Let A be a Noetherian domain, let C be a finitely generated A-algebra, and let $c \in C$ be a nonzerodivisor. Write $g\colon \operatorname{Spec} C \to \operatorname{Spec} A$ for the induced morphism. Then there is a nonzero $a \in A$ such that for every $s \in D(a)$:

1. the image of c in $C \otimes_A \kappa(s)$ is a nonzerodivisor;
2. $D(c) \cap g^{-1}(s)$ is dense in $g^{-1}(s)$.

***Proof*:**

Step 1: A nonzerodivisor cuts out a dense distinguished open.
Let R be any ring and $r \in R$ a nonzerodivisor; we claim $D(r)$ is dense in $\operatorname{Spec} R$. The distinguished opens form a basis (corollary 2.1.13), so it suffices to check that $D(r) \cap D(h) = D(rh)$ is nonempty whenever $D(h)$ is. Recall that $D(x) = \emptyset$ says precisely that x lies in every prime of R, that is, that x is nilpotent ([22, Radical Through Prime Ideals]). So suppose $D(rh) = \emptyset$, say $(rh)^n = r^n h^n = 0$. A power of a nonzerodivisor is a nonzerodivisor, so $h^n = 0$ and $D(h) = \emptyset$. This proves the claim, and with it part (2) of the statement, granted part (1).

Step 2: Choosing a.
The quotient C/cC is a finitely generated module over the finitely generated A-algebra C. By theorem 5.2.5 there is a nonzero $a \in A$ such that $(C/cC)_a$ is a free A_a-module. Fix this a for

the rest of the proof.

Step 3: The ideal cC_a is an A_a-module direct summand.
Localization at a is exact, so

$$0 \longrightarrow cC_a \longrightarrow C_a \longrightarrow (C/cC)_a \longrightarrow 0$$

is an exact sequence of A_a-modules. Its right-hand term is free, hence projective, so the sequence splits and cC_a is a direct summand of C_a as an A_a-module.

Moreover c remains a nonzerodivisor on C_a: if $c \cdot (x/a^m) = 0$ then $a^k cx = 0$ in C for some $k \geq 0$, so $c \cdot (a^k x) = 0$, so $a^k x = 0$, so $x/a^m = 0$. Multiplication by c is therefore an isomorphism of A_a-modules

$$\mu \colon C_a \xrightarrow{\sim} cC_a$$

Step 4: Base change to a fiber.
Fix $s \in D(a)$. Since a is invertible in $\kappa(s)$, the residue field $\kappa(s)$ is an A_a-algebra and $C \otimes_A \kappa(s) = C_a \otimes_{A_a} \kappa(s)$. Apply $- \otimes_{A_a} \kappa(s)$ to Step 3. The isomorphism μ becomes an isomorphism, and the inclusion $cC_a \hookrightarrow C_a$ is a split injection of A_a-modules, so it stays injective. Their composite is multiplication by the image of c, whence multiplication by c is injective on $C \otimes_A \kappa(s)$. That is part (1).

Step 5: Reading it on the fiber.
By proposition 3.2.10 the fiber $g^{-1}(s)$ is homeomorphic to $\operatorname{Spec}(C \otimes_A \kappa(s))$, and under this homeomorphism $D(c) \cap g^{-1}(s)$ corresponds to the distinguished open cut out by the image of c. Step 1 applied to $R = C \otimes_A \kappa(s)$ and r the image of c gives part (2), completing the proof.

Proposition 5.2.9: Fiberwise Density Spreads Out

Let S be a Noetherian integral scheme with generic point η, let $f \colon X \to S$ be a morphism of finite type, and let $U \subseteq X$ be an open subscheme. If U_η is dense in the generic fiber X_η, then there is a nonempty open $V \subseteq S$ such that U_s is dense in X_s for every $s \in V$.

***Proof*:**
Throughout, proposition 3.2.10 lets us identify the underlying space of the fiber X_s (definition 3.2.9) with the subspace $f^{-1}(s) \subseteq X$, and U_s with $U \cap f^{-1}(s)$; only the open subset underlying U matters. We record one fact used repeatedly: since $\overline{\{\eta\}} = S$ (definition 2.1.9), every nonempty open of S contains η, for otherwise it would be disjoint from a closed set containing η and hence from S. Consequently a finite intersection of nonempty opens of S is nonempty.

Step 1: Reduction to an affine base.
As S is Noetherian it has a cover by affine opens $\operatorname{Spec} A_i$ with A_i Noetherian (definition 2.4.1); choose one, $S_0 = \operatorname{Spec} A$. Then $A = \mathcal{O}_S(S_0)$ is a domain by definition 2.4.19, and $\eta \in S_0$ with $\overline{\{\eta\}} \cap S_0 = S_0$, so η is the generic point of S_0, that is, the zero ideal of A. Replace S by S_0, X by $f^{-1}(S_0)$ and U by $U \cap f^{-1}(S_0)$. This alters neither the generic fiber nor any fiber over a point of S_0, the restricted morphism is still of finite type, and a nonempty open of S_0 is a nonempty open of S. So assume $S = \operatorname{Spec} A$ with A a Noetherian domain and $\eta = (0)$.

Step 2: Reduction to an affine total space.
Two observations about an open cover $T = W_1 \cup \cdots \cup W_m$ of a topological space and an open

subset $O \subseteq T$. First, if O is dense in T then $O \cap W_\alpha$ is dense in W_α, since a nonempty open of W_α is a nonempty open of T. Second, if $O \cap W_\alpha$ is dense in W_α for every α then O is dense in T, since a nonempty open of T meets some W_α in a nonempty open of W_α.

Now f is quasicompact (definition 3.4.29) and S is affine, so X is quasicompact and admits a finite cover by affine opens $W_1, \dots, W_m$ with $\mathcal{O}_X(W_\alpha)$ a finitely generated A-algebra. The first observation, applied to the cover of X_η by the sets $X_\eta \cap W_\alpha$, says the hypothesis is inherited by each W_α. The second, applied to the cover of $f^{-1}(s)$ by the sets $f^{-1}(s) \cap W_\alpha$, says that if the conclusion holds for each W_α over a nonempty open $V_\alpha \subseteq S$, then it holds for X over $V = V_1 \cap \cdots \cap V_m$, which is nonempty by the opening remark. So assume $X = \operatorname{Spec} B$ with B a finitely generated A-algebra, and write $Z = X \setminus U = V(\mathfrak{b})$ for an ideal $\mathfrak{b} \subseteq B$.

Step 3: Components of X, and which of them dominate.
If $B = 0$ take $V = S$ and stop. Otherwise B is Noetherian by the Hilbert basis theorem, so X is a Noetherian topological space and proposition 2.1.25 writes $X = X_1 \cup \cdots \cup X_n$ with each X_i closed and irreducible and $X_i \not\subseteq X_j$ for $i \neq j$. Set $\mathfrak{p}_i = \bigcap_{\mathfrak{q} \in X_i} \mathfrak{q}$. Writing $X_i = V(\mathfrak{a}_i)$, we have $\mathfrak{a}_i \subseteq \mathfrak{p}_i$ and every $\mathfrak{q} \in X_i$ contains $\mathfrak{p}_i$, so $X_i = V(\mathfrak{p}_i)$. Each $\mathfrak{p}_i$ is prime: it is proper because $X_i \neq \emptyset$, and if $gh \in \mathfrak{p}_i$ then X_i is the union of the closed sets $X_i \cap V(g)$ and $X_i \cap V(h)$, so irreducibility puts X_i inside one of them, that is, $g \in \mathfrak{p}_i$ or $h \in \mathfrak{p}_i$. Finally $X_i \not\subseteq X_j$ reads $\mathfrak{p}_j \not\subseteq \mathfrak{p}_i$.

The morphism f sends $\mathfrak{q}$ to $\mathfrak{q} \cap A$. Call X_i *dominating* when $\mathfrak{p}_i \cap A = (0)$, and let I collect the dominating indices. For $i \notin I$ pick a nonzero $a_i \in \mathfrak{p}_i \cap A$; every $\mathfrak{q} \in V(\mathfrak{p}_i)$ contains a_i, so $f(\mathfrak{q}) \in V(a_i)$ and therefore $V(\mathfrak{p}_i) \cap f^{-1}(s) = \emptyset$ for $s \in D(a_i)$. Put $a' = \prod_{i \notin I} a_i$, an empty product being 1; it is nonzero because A is a domain. Then

$$f^{-1}(s) = \bigcup_{i \in I} \big(V(\mathfrak{p}_i) \cap f^{-1}(s)\big) \qquad \text{for all } s \in D(a')$$

Step 4: What the hypothesis says at η.
We claim $\mathfrak{b} \not\subseteq \mathfrak{p}_i$ for every $i \in I$. Suppose instead $\mathfrak{b} \subseteq \mathfrak{p}_i$ with $\mathfrak{p}_i \cap A = (0)$. Then $f(\mathfrak{p}_i) = (0) = \eta$, so $\mathfrak{p}_i \in f^{-1}(\eta)$. Consider

$$O = f^{-1}(\eta) \setminus \bigcup_{j \neq i} V(\mathfrak{p}_j)$$

an open subset of X_η. It contains $\mathfrak{p}_i$, since $\mathfrak{p}_i \in V(\mathfrak{p}_j)$ would say $\mathfrak{p}_j \subseteq \mathfrak{p}_i$, excluded in Step 3. Every prime of B lies in some X_j, so any point of O lies in $V(\mathfrak{p}_i)$; and $\mathfrak{b} \subseteq \mathfrak{p}_i$ gives $V(\mathfrak{p}_i) \subseteq V(\mathfrak{b}) = Z$. Thus O is a nonempty open subset of X_η disjoint from U, contradicting the density of U_η. For each $i \in I$ fix $b_i \in \mathfrak{b} \setminus \mathfrak{p}_i$.

Step 5: Spreading out one dominating component at a time.
Fix $i \in I$ and set $B_i = B/\mathfrak{p}_i$, a domain and a finitely generated A-algebra. The canonical homeomorphism $\operatorname{Spec} B_i \to V(\mathfrak{p}_i)$ carries the morphism $g_i \colon \operatorname{Spec} B_i \to \operatorname{Spec} A$ induced by $A \to B_i$ onto the restriction of f, and $A \to B_i$ is injective because $\mathfrak{p}_i \cap A = (0)$. The image β_i of b_i in B_i is nonzero, hence a nonzerodivisor. By lemma 5.2.8 there is a nonzero $a_i' \in A$ such that $D(\beta_i) \cap g_i^{-1}(s)$ is dense in $g_i^{-1}(s)$ for every $s \in D(a_i')$. Under the homeomorphism, $g_i^{-1}(s)$ is $V(\mathfrak{p}_i) \cap f^{-1}(s)$ and $D(\beta_i)$ is the set of primes $\mathfrak{q} \supseteq \mathfrak{p}_i$ with $b_i \notin \mathfrak{q}$. Since $b_i \in \mathfrak{b}$, such a $\mathfrak{q}$ does not contain $\mathfrak{b}$ and so lies in U. A superset of a dense subset is dense, so

$$U \cap V(\mathfrak{p}_i) \cap f^{-1}(s) \quad \text{is dense in} \quad V(\mathfrak{p}_i) \cap f^{-1}(s) \qquad \text{for all } s \in D(a_i')$$

Step 6: Assembly.
Put $a = a' \prod_{i \in I} a_i'$, a nonzero element of the domain A, and let $V = D(a)$, a nonempty open of

S. Fix $s \in V$ and abbreviate $C_i = V(\mathfrak{p}_i) \cap f^{-1}(s)$ for $i \in I$; these are closed in $f^{-1}(s)$ and, by Step 3, they cover it. Taking closures inside $f^{-1}(s)$, and using that the closure of a subset of the closed set C_i is the same computed in C_i or in $f^{-1}(s)$, Step 5 gives

$$\overline{U_s} \supseteq \bigcup_{i \in I} \overline{U_s \cap C_i} = \bigcup_{i \in I} C_i = f^{-1}(s)$$

So U_s is dense in X_s for every $s \in V$, as we sought to show.

The proof produces the open set explicitly. It is $D(a)$ for a a product of two kinds of factors: elements of A that kill the components of X lying over proper closed subsets of S, and, for each component that dominates, the element given by generic freeness for the quotient $B_i/\beta_i B_i$. Both kinds are computable in examples. Take $S = \operatorname{Spec} k[t]$, $X = \operatorname{Spec} k[t, x, y]/(xy - t)$ and $U = D(x)$. Here $B/xB = k[y]$ carried by the base as a t-torsion module, which becomes free, indeed zero, only after inverting t; and the answer $V = D(t)$ is correct, because the fiber over $t = 0$ is the union of the two axes $xy = 0$ while U misses the axis $x = 0$ entirely. Every other fiber is a single hyperbola on which x is invertible.

Integrality of S is used twice. Irreducibility gives the membership relation exploited in Step 2, so that finitely many shrinkings of the base can be intersected without emptying it; and A being a domain is both what generic freeness requires and what makes $a \neq 0$ equivalent to $D(a) \neq \emptyset$. Over a base that is only Noetherian the statement must be applied to each irreducible component of S_{red} in turn.

The hypothesis cannot be relaxed from density of U_η to nonemptiness. With $S = \operatorname{Spec} k[t]$, $X = \operatorname{Spec} k[t, x, y]/(xy)$ and $U = D(x)$, the generic fiber has two components and U_η meets only one of them; every closed fiber looks the same, so no shrinking of the base helps. The content of the proposition is that this failure, once it is absent at η, is absent over a whole neighborhood.

One word on what this replaces. EGA IV_3, section 9, builds a general machine for questions of this shape: a scheme of finite presentation is written as a pullback from a Noetherian base of finite type over $\mathbb{Z}$, and a long list of properties is shown to ascend and descend along such limits, so that almost any reasonable condition holding at η holds over a neighborhood. That machine gives uniformity across many properties at once, but needs the limit formalism, which this book does not develop. The substitute used here is narrower and adequate for the families met later: generic freeness converts a statement about one module over the generic point into a statement over a distinguished open, and the constructible sets of definition 5.2.2 ensure that a condition assembled from finitely many such data is either empty or contains a nonempty open of an irreducible base. The price is that each property must be spread out by hand, as was done above for density.

The proof reveals more than the statement: it shows that the image of a dominant finite type morphism *contains* a dense open subset of the target. Let us record this and a few other consequences.

Corollary 5.2.10: Image Contains a Dense Open of its Closure

Let $f : X \to Y$ be a morphism of finite type of Noetherian schemes. Then $f(X)$ contains a dense open subset of $\overline{f(X)}$.

***Proof*:**
By theorem 5.2.7, $f(X)$ is constructible, hence a finite union of locally closed sets $f(X) = \bigsqcup_{i=1}^{n}(U_i \cap Z_i)$, where U_i is open and Z_i is closed. We may assume this decomposition is irredundant. Let $\overline{f(X)}$ be the closure of $f(X)$. Since $f(X)$ is dense in $\overline{f(X)}$, at least one of the locally closed pieces, say $U_1 \cap Z_1$, must be dense in some irreducible component of $\overline{f(X)}$. Then $U_1 \cap Z_1$ contains a dense open subset of that component. Applying this argument to each irreducible component gives a dense open subset of $\overline{f(X)}$ contained in $f(X)$.

Alternatively, one can argue directly: replace X by the reduced induced structure on X and Y by $\overline{f(X)}$ with its reduced induced structure (definition 3.1.26), through which the morphism factors (proposition 3.1.25(3)). Then $f : X \to \overline{f(X)}$ is dominant, and the result follows from Step 5 of the proof of theorem 5.2.7 (applied to each irreducible component).

Corollary 5.2.11: Open and Closed Morphisms via Constructibility

Let $f : X \to Y$ be a morphism of finite type of Noetherian schemes.

1. If f is an open map, then $f(X)$ is open. If f is a closed map (e.g., if f is proper by proposition 3.5.23), then $f(X)$ is closed.
2. More generally, $f(X)$ is open (resp. closed) if and only if it is stable under generization (resp. specialization).

***Proof*:**
Part (1) applies the definition of an open (resp. closed) map to X itself, which is both open and closed in X. For (2), a constructible set that is stable under generization is open, proved in full in lemma 6.8.9. The closed case is dual.

5.2.2 Application: Fiber Dimension

One of the most important applications of Chevalley's theorem is to the study of how fibers vary in a family. Given a morphism $f : X \to Y$, the *fiber* over a point $y \in Y$ is the scheme $X_y = X \times_Y \operatorname{Spec} \kappa(y)$ (definition 3.2.9). The dimension of this fiber can jump as y varies, but Chevalley's theorem constrains *how* it can jump.

Definition 5.2.12: Fiber Dimension

Let $f : X \to Y$ be a morphism of schemes and $x \in X$ a point with $y = f(x)$. The *fiber dimension of f at x* is:

$$\dim_x(f) = \dim_x(X_y)$$

where $X_y = f^{-1}(y)$ is the scheme-theoretic fiber over y, and $\dim_x$ denotes the local dimension at x (the maximum of the dimensions of the irreducible components of X_y containing x).

Theorem 5.2.13: Upper Semicontinuity of Fiber Dimension

Let $f : X \to Y$ be a morphism of finite type of Noetherian schemes. Then for each integer $n \geq 0$, the set:

$$\{x \in X : \dim_x(f) \geq n\}$$

is closed in X. In other words, the function $x \mapsto \dim_x(f)$ is upper semicontinuous.

***Proof Key ideas*:**
The question is local on X and on Y, so one may take $Y = \operatorname{Spec} A$ and $X = \operatorname{Spec} B$ with B a finitely generated A-algebra. It suffices to show that $\{x \in X \mid \dim_x(f) \leq d\}$ is open for every $d \geq 0$, its complement being $\{x \in X \mid \dim_x(f) \geq d+1\}$. Let $x \in X$ have prime $\mathfrak{q} \subseteq B$ and $\dim_x(f) = d$. Noether normalization of the fiber ring $B \otimes_A \kappa(f(x))$, near x, followed by a lift of the normalizing elements to B, produces $g \in B \setminus \mathfrak{q}$ and an A-algebra map $A[t_1, \ldots, t_d] \to B_g$ that is quasi-finite at $\mathfrak{q}$. It is then quasi-finite on a neighbourhood of x, because the locus where a morphism of finite type is quasi-finite is open. On that neighbourhood each fiber of f is quasi-finite over an affine space of dimension d over a field, hence of dimension at most d, which gives the required open neighbourhood of x.

Reason for deferral. The openness of the quasi-finite locus is the algebraic form of Zariski's Main Theorem, which this book does not prove. Theorem 3.4.38 factors a morphism that is already known to be separated and quasi-finite, and it does not give this openness. The argument above is [70, Tag 00QH], the statement for schemes is [70, Tag 02FZ], and the original reference is [32, Théorème 13.1.3].

The upshot is that fibers can "jump up" in dimension on closed subsets, but not on open ones. Intuitively, the dimension of fibers can only increase as you specialize.

One consequence deserves its own statement, because downstream books reach for it by name and the series has not carried it: a dominant morphism between varieties of the same dimension is, over a dense open subset of the target, as simple as it could be.

Proposition 5.2.14: Generic Finiteness and Étaleness

Let $f\colon V \to V'$ be a proper dominant morphism of varieties over an algebraically closed field of characteristic zero, with $\dim V = \dim V'$. Then $d = [k(V) : k(V')]$ is finite, and there is a dense open $U' \subseteq V'$ such that

$$f^{-1}(U') \longrightarrow U'$$

is finite and étale of degree d; in particular every fibre over U' consists of exactly d reduced points.

***Proof*:**
Step 1: the extension is finite. Both function fields have transcendence degree $\dim V = \dim V'$ over the ground field, and $k(V)$ is finitely generated over $k(V')$; a finitely generated extension of the same transcendence degree is algebraic, hence finite. Write d for its degree.

Step 2: quasi-finite over a dense open. The generic fibre has dimension $\dim V - \dim V' = 0$ by

lemma 3.8.2. By theorem 5.2.13 the locus in V where the fibre dimension is positive is closed, and it is not all of V; its image is closed in V' because f is proper, and proper because it misses the generic point. Deleting that image leaves a dense open U_1' over which every fibre is finite, so $f^{-1}(U_1') \to U_1'$ is quasi-finite.

Step 3: finite. That restriction is proper, being a base change of f, and separated and of finite type. Theorem 3.4.38 factors it as an open immersion j followed by a finite morphism g. Properness makes the image of j closed as well as open, hence a union of connected components of the target of j, so j is finite and therefore so is the composite. Thus $f^{-1}(U_1') \to U_1'$ is finite.

Step 4: étale after one more shrink. Write $U_1' = \operatorname{Spec} B$ locally, with B a domain, and $f^{-1}(U_1') = \operatorname{Spec} A$ with A finite over B; A is torsion-free over B, so after inverting one nonzero element of B it is free, of rank d by Step 1. Characteristic zero makes $k(V)/k(V')$ separable, so the trace form of $A \otimes_B k(V')$ over $k(V')$ is nondegenerate and the discriminant of the free algebra A over B is a nonzero element of B. Inverting it gives a dense open $U' \subseteq U_1'$ over which A is finite, free and of nonvanishing discriminant, which is precisely the condition that f be étale there (definition 2.7.10, relative dimension zero). Each fibre is then Spec of a rank-d étale algebra over an algebraically closed field, that is d reduced points.

Example 5.2.15: Fiber Dimension in Practice

1. *Blowing up.* Let $f : \operatorname{Bl}_0 \mathbb{A}_k^2 \to \mathbb{A}_k^2$ be the blow-up of the plane at the origin (see definition 8.4.1). The generic fiber (over any point $y \neq 0$) is a single point, so $\dim(f^{-1}(y)) = 0$. But the exceptional fiber $f^{-1}(0) \cong \mathbb{P}_k^1$ has dimension 1. The fiber dimension jumps from 0 to 1 at the origin—consistent with semicontinuity, since $\{0\}$ is closed.

2. *Projection of a surface.* Let $X = V(xz-y) \subseteq \mathbb{A}_k^3$ and $f : X \to \mathbb{A}_k^1$ the projection $(x, y, z) \mapsto x$. For $x = a \neq 0$, the fiber is $V(az - y) \cong \mathbb{A}_k^1$ (a line). For $x = 0$, the fiber is $V(y) \subseteq \mathbb{A}_k^2$, which is a copy of $\mathbb{A}_k^1$ parametrized by z. So the fiber dimension is 1 everywhere—no jumping. This is consistent with f being flat (the ring map $k[x] \to k[x, y, z]/(xz - y) \cong k[x, z]$ makes $k[x, z]$ a free $k[x]$-module).

3. *The $(x, y) \mapsto (x, xy)$ example revisited.* For the morphism $f : \mathbb{A}_k^2 \to \mathbb{A}_k^2$ from example 5.2.1, the fiber over a point (a, b) with $a \neq 0$ is a single point $(a, b/a)$, so $\dim = 0$. The fiber over $(0, 0)$ is $V(x, xy) = V(x) = \mathbb{A}_k^1$ (the entire y-axis), so $\dim = 1$. The fiber over $(0, b)$ with $b \neq 0$ is $V(x, xy - b) = V(x, b) = \emptyset$. The locus where $\dim_x(f) \geq 1$ is $\{x = 0\} \subseteq \mathbb{A}_k^2$, which is indeed closed.

The examples above suggest a link between flatness and the constancy of fiber dimension. We know that flat morphisms between nice schemes are equidimensional (the fiber dimension does not jump). under the right algebraic hypotheses, the converse is true. This is the celebrated "Miracle Flatness" theorem.

Theorem 5.2.16: Miracle Flatness Theorem

Let $f : X \to Y$ be a morphism of finite type between locally Noetherian schemes. Suppose that:

1. Y is a regular scheme (e.g., a smooth variety),
2. X is Cohen-Macaulay.

Then f is flat if and only if all non-empty fibers X_y have the exact same "expected" dimension, namely $\dim X - \dim Y$. (In other words, f is flat $\iff$ f is equidimensional).

***Proof*:**
The question is entirely local, so we may assume $X = \operatorname{Spec} B$ and $Y = \operatorname{Spec} A$ are affine, and we can localize at a point $x \in X$ and its image $y = f(x) \in Y$. This reduces the problem to a local homomorphism of local Noetherian rings $\varphi : (A, \mathfrak{m}) \to (B, \mathfrak{n})$.

By hypothesis, A is a regular local ring, B is a Cohen-Macaulay local ring, and the fiber dimension condition translates to the dimension formula:

$$\dim B = \dim A + \dim(B/\mathfrak{m}B).$$

We want to show that B is a flat A-module.

Since A is regular, its maximal ideal $\mathfrak{m}$ is generated by a regular sequence of parameters $a_1, \dots, a_d$, where $d = \dim A$.

Because B is Cohen-Macaulay, it is universally catenary and the dimension formula shows that quotienting B by the ideal $\mathfrak{m}B = (a_1, \dots, a_d)B$ drops the dimension of B by exactly d. In a Cohen-Macaulay ring, any sequence of elements that drops the dimension by its length must be a regular sequence. Therefore, the images $\varphi(a_1), \dots, \varphi(a_d)$ form a regular sequence in B.

We now invoke the *Regular Sequence Criterion for Flatness* (see [18, Regular Sequence Criterion for Flatness]): if $(A, \mathfrak{m}) \to (B, \mathfrak{n})$ is a local homomorphism, and an ideal $I \subseteq A$ is generated by a regular sequence that maps to a regular sequence in B, and B/IB is flat over A/I, then B is flat over A. Applying this to $I = \mathfrak{m}$, the quotient $A/\mathfrak{m}$ is a field, so the flatness of $B/\mathfrak{m}B$ over it is automatic, completing the proof.

5.2.17 Remark: Why is it a "Miracle"? Checking flatness algebraically can be incredibly difficult, as it requires computing Tor groups or checking exactness for all modules. Checking that fibers have constant dimension is a purely topological/geometric task. The "miracle" is that geometry implies algebra here.
The hypotheses are absolutely necessary to prevent algebraic pathologies from hiding behind the geometry:

- If X is not Cohen-Macaulay, it might have embedded points (fuzz) that do not affect the topological dimension of the fibers but destroy algebraic flatness.
- If Y is not regular, the base might have singularities that force any flat family over it to also be highly singular, breaking the dimension count.

The interplay between fiber dimension and cohomology runs deep. In section 6.8.3 we shall prove a more refined semicontinuity result: for a projective morphism $f : X \to Y$ and a coherent sheaf $\mathscr{F}$ flat over Y, the function $y \mapsto h^i(X_y, \mathscr{F}_y)$ is upper semicontinuous. The Euler characteristic $y \mapsto \chi(X_y, \mathscr{F}_y)$ is locally constant. This is a far-reaching generalization whose proof uses a completely different technique (reducing to linear algebra on a complex of free modules), but the philosophical point is the same: cohomological invariants of fibers can only jump up on closed subsets.

5.2.3 The Scheme-Theoretic Image

Chevalley's theorem shows that the set-theoretic image is constructible, but it says nothing about scheme structure. For many purposes—particularly when studying families or intersections—we need an image that lives in the category of schemes, not just the category of topological spaces. The set-theoretic image rarely has a natural scheme structure (as example 5.2.1 illustrates), so we do the next best thing: we take the smallest *closed* subscheme containing the image.

The idea is natural from the sheaf-theoretic perspective developed in this chapter. A regular function g on Y should "vanish on the image" if and only if g pulls back to zero on X, i.e., $\pi^\#(g) = 0$ in $\mathcal{O}_X$. The collection of all such g forms an ideal sheaf, and the corresponding closed subscheme is the scheme-theoretic image.

Definition 5.2.18: Scheme-Theoretic Image

Let $\pi : X \to Y$ be a morphism of schemes. We say the *image of π lies in a closed subscheme* $\iota : Z \hookrightarrow Y$ if the composition:

$$\mathcal{I}_{Z/Y} \longrightarrow \mathcal{O}_Y \xrightarrow{\pi^\#} \pi_*\mathcal{O}_X$$

is zero, where $\mathcal{I}_{Z/Y} = \ker(\mathcal{O}_Y \to \iota_*\mathcal{O}_Z)$ is the ideal sheaf of Z (definition 5.1.30). Equivalently, π factors through Z: every function on Y that vanishes on Z pulls back to zero on X.
The *scheme-theoretic image* of π is the closed subscheme of Y defined by the ideal sheaf:

$$\mathcal{I} = \ker\bigl(\mathcal{O}_Y \xrightarrow{\pi^\#} \pi_*\mathcal{O}_X\bigr)$$

That is, it is the smallest closed subscheme of Y through which π factors.

The scheme-theoretic image is well-defined: if the image lies in closed subschemes Z_1 and Z_2, it lies in their scheme-theoretic intersection $Z_1 \cap Z_2$ (the closed subscheme defined by $\mathcal{I}_{Z_1} + \mathcal{I}_{Z_2}$, see proposition 5.1.37). Hence the intersection of all closed subschemes containing the image exists and is the scheme-theoretic image.

Note that the scheme-theoretic image depends on the *scheme structures* of both X and Y, not just on the set-theoretic image. In particular, a non-reduced source can produce a non-reduced image, even when the set-theoretic image is a single point.

There is an important subtlety: the ideal sheaf $\mathcal{I} = \ker(\mathcal{O}_Y \to \pi_*\mathcal{O}_X)$ need not be quasi-coherent in general (the pushforward $\pi_*\mathcal{O}_X$ is not always quasi-coherent). However, by proposition 5.1.20, if π is quasi-compact and quasi-separated, then $\pi_*\mathcal{O}_X$ is quasi-coherent, and hence so is $\mathcal{I}$. In this case, the scheme-theoretic image is a closed subscheme (proposition 5.1.34).

The relationship between the scheme-theoretic image and the set-theoretic image is controlled by the

quasi-compactness of π:

Proposition 5.2.19: Set-Theoretic Image and Scheme-Theoretic Image

Let $\pi : X \to Y$ be a quasi-compact morphism of schemes, and let Z be the scheme-theoretic image. Then the underlying set of Z is the closure of the set-theoretic image: $|Z| = \overline{\pi(X)}$.

***Proof*:**
Since π is quasi-compact, $\pi_*\mathcal{O}_X$ is quasi-coherent (proposition 5.1.20), and the question is local on Y. So assume $Y = \operatorname{Spec} A$ and $\pi_*\mathcal{O}_X(Y) = B$ (some A-algebra). The ideal sheaf $\mathcal{I}$ corresponds to $I = \ker(A \to B)$, and the scheme-theoretic image is $\operatorname{Spec}(A/I)$.

We must show $V(I) = \overline{\pi(X)}$. A prime $\mathfrak{p} \in \operatorname{Spec} A$ is in $V(I)$ if and only if $I \subseteq \mathfrak{p}$, i.e., every element of A that maps to zero in B lies in $\mathfrak{p}$. On the other hand, $\mathfrak{p} \in \overline{\pi(X)}$ if and only if $\mathfrak{p}$ is in the closure of the set-theoretic image.

If $\mathfrak{p} \in \pi(X)$, then $\mathfrak{p} = \pi(\mathfrak{q})$ for some $\mathfrak{q} \in X$, and the composition $A \to B \to \kappa(\mathfrak{q})$ sends I to zero, so $I \subseteq \mathfrak{p}$. Hence $\pi(X) \subseteq V(I)$. Since $V(I)$ is closed, $\overline{\pi(X)} \subseteq V(I)$.

For the reverse inclusion, suppose $\mathfrak{p} \supseteq I$. We must show $\mathfrak{p} \in \overline{\pi(X)}$, i.e., every open set containing $\mathfrak{p}$ meets $\pi(X)$. Let $D(a)$ be a basic open neighborhood of $\mathfrak{p}$, so $a \notin \mathfrak{p}$ and in particular $a \notin I$. Then the image of a in B is nonzero, which means $\pi^{-1}(D(a)) \neq \emptyset$ (the function a does not vanish on X). Hence $D(a)$ meets $\pi(X)$.

Chevalley's theorem (theorem 5.2.7) says that the image of a morphism of finite type is constructible, and example 5.2.1 shows that constructible is the most one can ask for in general: images of morphisms of schemes are rarely closed, and rarely even locally closed. A group law changes the situation entirely. If the morphism is a homomorphism of group schemes (definition 4.2.1), then its image absorbs its own translates, and a constructible set that is stable under translation by its own points has no room left to be anything but closed. The classical form of this statement is proved for affine algebraic groups over an algebraically closed field of characteristic zero, where smoothness and generic smoothness are on hand; none of those three hypotheses is needed.

Proposition 5.2.20: The Image of a Homomorphism of Group Schemes Is Closed

Let k be a field and let $f\colon G \to H$ be a homomorphism of group schemes of finite type over k. Then:

1. the set-theoretic image $f(G)$ is a closed subset of H;
2. the scheme-theoretic image of f (definition 5.2.18) is a closed subgroup scheme of H whose underlying set is $f(G)$;
3. if k is perfect, the set $f(G)$ with its reduced induced structure (definition 3.1.26) is a closed subgroup scheme of H as well.

***Proof*:**
Both G and H are Noetherian, and f is a morphism of finite type.[a]

Step 1: Reduction to an algebraically closed field. Write $\pi_G\colon G_{\overline{k}} \to G$ and $\pi_H\colon H_{\overline{k}} \to H$ for the

projections and $f' = f_{\overline{k}}$, a homomorphism of group schemes of finite type over $\overline{k}$. Every element of $\overline{k}$ is algebraic over k, so $\operatorname{Spec}\overline{k} \to \operatorname{Spec} k$ is an integral morphism; integrality is stable under base change (proposition 3.4.25(3)), so π_H is integral and therefore a closed map by proposition 3.4.23. Both projections are surjective, the fiber over x being $\operatorname{Spec}(\kappa(x) \otimes_k \overline{k})$, the spectrum of a nonzero ring. From $\pi_H \circ f' = f \circ \pi_G$ and surjectivity of π_G we get $\pi_H(f'(G_{\overline{k}})) = f(\pi_G(G_{\overline{k}})) = f(G)$. So if $f'(G_{\overline{k}})$ is closed then so is its image $f(G)$ under the closed map π_H, and we may assume $k = \overline{k}$.

Step 2: Closed points. Set $E = f(G) \subseteq H$ and let $\Gamma \subseteq H(k)$ be the image of $G(k) \to H(k)$, a subgroup. Since H is of finite type over $k = \overline{k}$, the residue field at a closed point is a finite extension of k, hence k itself, so the closed points of H are exactly the elements of $H(k)$; moreover H is a Jacobson scheme (definition 3.7.4, by lemma 2.4.5), and the same holds for G. Two consequences will be used repeatedly. First, every nonempty constructible subset of H contains a closed point of H, because it contains a nonempty locally closed set $U \cap C$ and the closed points of C are dense in C; hence a constructible set all of whose closed points lie in a closed set C is contained in C. Second, $E \cap H(k) = \Gamma$: if $h \in H(k)$ lies in E, its fiber $G \times_{H,h} \operatorname{Spec} k$ is a nonempty k-scheme of finite type, so it has a closed point, and that point is a k-point mapping to h.

Step 3: Translations preserve the image. Let $\gamma = f(g)$ with $g \in G(k)$. On T-valued points the left translation λ_γ of example 4.3.2 satisfies $\lambda_\gamma(f(x)) = f(g)f(x) = f(gx)$, so $\lambda_\gamma \circ f = f \circ \lambda_g$ as morphisms of k-schemes. As λ_g is an automorphism of G, this gives $\lambda_\gamma(E) = f(\lambda_g(G)) = E$, and the same computation with right translations and with $\iota_H \circ f = f \circ \iota_G$ gives $\rho_\gamma(E) = E$ and $\iota_H(E) = E$.

Step 4: The closure. Let $Z = \overline{E}$, nonempty since the identity lies in E. Each of λ_γ, ρ_γ and ι_H is a homeomorphism of H carrying E onto E, hence carrying Z onto Z. By theorem 5.2.7 the set E is constructible, and its closed points are exactly the elements of Γ by Step 2, so $E \subseteq \overline{\Gamma}$ by the first consequence recorded there; therefore $Z = \overline{\Gamma}$.

Step 5: The closed points of Z act on Z. Let $z \in Z \cap H(k)$. Since $\lambda_\gamma(Z) = Z$ for $\gamma \in \Gamma$, the point γz lies in Z, so $\rho_z(\Gamma) \subseteq Z$; continuity and $Z = \overline{\Gamma}$ then give $\rho_z(Z) \subseteq Z$, and symmetrically $\lambda_z(Z) \subseteq Z$. Applying $\iota_H(Z) = Z$ shows $z^{-1} \in Z \cap H(k)$, whence $Z = \lambda_z(\lambda_{z^{-1}}(Z)) \subseteq \lambda_z(Z)$ and $\lambda_z(Z) = Z$, and likewise $\rho_z(Z) = Z$. In particular $Z \cap H(k)$ is a subgroup of $H(k)$.

Step 6: The covering argument. By corollary 5.2.10 there is a dense open $V \subseteq Z$ with $V \subseteq E$. Fix a closed point $z \in Z \cap H(k)$ and put $\theta = \lambda_z \circ \iota_H$, an automorphism of H over k carrying Z onto Z by Steps 4 and 5. Then $\theta(V)$ is a dense open subset of Z. In any topological space the intersection of two dense open subsets is dense open, so $V \cap \theta(V)$ is a nonempty open subset of Z and therefore contains a closed point v of Z, which is a closed point of H since Z is closed. Let $w = \theta^{-1}(v) \in V$; as θ is an automorphism over k, the point w is again k-rational. In the group $H(k)$ the relation $v = \theta(w) = zw^{-1}$ reads $z = vw$, and $v, w \in V \cap H(k) \subseteq E \cap H(k) = \Gamma$ by Step 2, so $z \in \Gamma \subseteq E$. Thus every closed point of Z lies in E. The difference $Z \setminus E$ is constructible and contains no closed point of H, so it is empty. Hence $E = Z$ is closed, which with Step 1 proves (1).

Step 7: The scheme-theoretic image. Return to an arbitrary field k and let $I \subseteq H$ be the scheme-theoretic image of f. Since f is quasi-compact, $|I| = \overline{f(G)} = f(G)$ by proposition 5.2.19 and part (1). The identity section $\epsilon_H = f \circ \epsilon_G$ factors through I by the defining property of I, and ι_H, being an automorphism of H with $\iota_H \circ f = f \circ \iota_G$ and ι_G an automorphism of G, carries I onto the scheme-theoretic image of $\iota_H \circ f$, which is I again. For multiplication, note

that $\mathcal{O}_I \to f_*\mathcal{O}_G$ is injective, and that formation of the scheme-theoretic image of a quasi-compact separated morphism commutes with flat base change: on an affine open $\operatorname{Spec} A \subseteq H$ the image ideal is $\ker(A \to B)$ with $B = \Gamma(f^{-1}(\operatorname{Spec} A), \mathcal{O}_G)$ computed as the equalizer of a finite Čech complex, and tensoring by a flat A-algebra is exact. Base changing f along the flat projection $H \times_k G \to H$ shows that $I \times_k G$ is the scheme-theoretic image of $f \times \mathrm{id}_G$, and base changing along $I \times_k H \to H$ shows that $I \times_k I$ is the scheme-theoretic image of $\mathrm{id}_I \times f$. Pushforward is left exact, so the composite $G \times_k G \to I \times_k I$ again has scheme-theoretic image $I \times_k I$. Now $\mu_H \circ (f \times f) = f \circ \mu_G$ factors through I, so $G \times_k G$ factors through the closed subscheme $C = (I \times_k I) \times_{\mu_H, H} I$ of $I \times_k I$; the ideal sheaf of C is contained in the kernel of $\mathcal{O}_{I \times_k I} \to (f \times f)_*\mathcal{O}_{G \times_k G}$, which vanishes, so $C = I \times_k I$ and μ_H restricts to $I \times_k I \to I$. This proves (2).

Step 8: The reduced structure over a perfect field. Let Y be $f(G)$ with its reduced induced structure (definition 3.1.26). Over a perfect field a reduced k-scheme is geometrically reduced, so $Y \times_k Y$ is reduced. Base change along $\operatorname{Spec} \overline{k} \to \operatorname{Spec} k$ carries $Y \times_k Y$ to $Y_{\overline{k}} \times_{\overline{k}} Y_{\overline{k}}$, whose underlying space is $Z \times Z$ on closed points; Step 5 sends such a pair to a closed point of Z, and the image of μ_H there is constructible, so it lies in Z by Step 2. Pushing forward along the surjection π_H as in Step 1 gives $\mu_H(Y \times_k Y) \subseteq f(G)$. A morphism from a reduced scheme whose set-theoretic image lies in a closed subset factors through the reduced induced structure on that subset (proposition 3.1.25(3)), so μ_H restricts to $Y \times_k Y \to Y$, and the same applies to ι_H and ϵ_H, completing the proof.

[a] A group scheme of finite type over a field is separated: the identity is a k-rational point, hence a closed point with residue field k, so ϵ_H is a closed immersion, and the diagonal of H is its pullback along $(x, y) \mapsto xy^{-1}$. Since G is quasi-compact and H is quasi-separated, f is quasi-compact, and it is locally of finite type because G is of finite type over k.

The group law entered only through translations, and the field only through the passage to $\overline{k}$, where closed points become $\overline{k}$-points and the Jacobson property lets a constructible set be tested on them. Nothing in the argument sees affineness, characteristic, reducedness or smoothness: G and H may be abelian varieties, k may have characteristic p, and either group may be infinitesimal, as μ_p and α_p are. What replaces the generic smoothness of the classical proof is corollary 5.2.10, which is available over any field and for any finite type morphism.

The gap between (2) and (3) is real. Over an imperfect field a product of reduced k-schemes can fail to be reduced, and a group scheme of finite type over such a field need not have its underlying reduced subscheme be a subgroup scheme; taking $f = \mathrm{id}_G$ for such a G turns it into a counterexample to (3). The scheme-theoretic image has no such defect, which is one more reason to treat it, rather than the reduced image, as the image of a morphism of schemes.

With (1) in hand the corestriction $G \to f(G)$ is a homomorphism of group schemes of finite type onto a closed subgroup scheme, and the kernel of definition 4.2.2 sits on the left of the resulting sequence. The caution about exactness stands: surjectivity on the right is faithful flatness of $G \to f(G)$, not surjectivity of $G(T) \to f(G)(T)$, and the two differ already for the n-th power map on $\mathbb{G}_m$ over $\mathbb{Q}$.

Without the quasi-compactness hypothesis, the scheme-theoretic image can be strictly larger than the closure of the set-theoretic image, as the pathological example in the following illustrates:

Example 5.2.21: Scheme-Theoretic Image

1. *Non-reduced source.* The morphism $\pi : \operatorname{Spec}(k[\epsilon]/(\epsilon^2)) \to \mathbb{A}^1_k = \operatorname{Spec} k[x]$ given by $x \mapsto \epsilon$ has scheme-theoretic image $V(x^2) = \operatorname{Spec}(k[x]/(x^2))$: a polynomial $g(x)$ pulls back to zero

precisely when $g(\epsilon) = 0$ in $k[\epsilon]/(\epsilon^2)$, which forces $g \in (x^2)$. The set-theoretic image is just the origin, but the scheme-theoretic image is a "fat point" of length 2, remembering the tangent direction.

Compare this with the discussion of nilpotent thickening in example 2.2.8: the scheme-theoretic image captures the first-order infinitesimal data of the source.

2. *Reduced image from non-reduced source.* The morphism $\operatorname{Spec}(k[\epsilon]/(\epsilon^2)) \to \mathbb{A}^1_k$ given by $x \mapsto 0$ has scheme-theoretic image $V(x) = \operatorname{Spec}(k)$: now $g(x)$ pulls back to $g(0)$, which vanishes precisely when $g \in (x)$. Although the source is non-reduced, the image is reduced. The difference from (1) is that the tangent direction is "crushed" by the map—the non-reduced structure of the source is not reflected in the image.

3. *Dense image = scheme-theoretic closure.* The open immersion $j : \mathbb{A}^1_k \setminus \{0\} = \operatorname{Spec} k[t, t^{-1}] \hookrightarrow \mathbb{A}^1_k = \operatorname{Spec} k[u]$ given by $u \mapsto t$ has scheme-theoretic image all of $\mathbb{A}^1_k$: a polynomial $g(u)$ that becomes zero in $k[t, t^{-1}]$ must be the zero polynomial. This is consistent with proposition 5.2.19: the set-theoretic image is $\mathbb{A}^1_k \setminus \{0\}$, whose closure is $\mathbb{A}^1_k$. In this sense, the scheme-theoretic image is the *scheme-theoretic closure* of the image.

4. *Pathological example (non-quasi-compact source).* Consider the morphism:

$$\pi : \coprod_{n \geq 1} \operatorname{Spec}\left(\frac{k[\epsilon_n]}{(\epsilon_n^n)}\right) \longrightarrow \mathbb{A}^1_k = \operatorname{Spec} k[x]$$

given by $x \mapsto \epsilon_n$ on the nth component. A polynomial $g(x)$ pulls back to zero on the nth component precisely when g vanishes to order $\geq n$ at the origin. Since this must hold for *all* n, the only such g is 0. The scheme-theoretic image is therefore all of $\mathbb{A}^1_k$, even though the set-theoretic image is just the origin.

This pathology arises because the source is not quasi-compact. For quasi-compact morphisms, the scheme-theoretic image has underlying set equal to the closure of the set-theoretic image (proposition 5.2.19).

The scheme-theoretic image has a clean universal property: it is the smallest closed subscheme $Z \hookrightarrow Y$ such that π factors through Z. This makes it functorial and well-behaved in diagram chases. For instance, if $\pi = \iota \circ g$ where $\iota : Z \hookrightarrow Y$ is a closed immersion, then the scheme-theoretic image of π is contained in Z (as a closed subscheme). This is the algebraic incarnation of the geometric intuition that "the image of a composition lies in the image of the outer map."

We close with a remark connecting the two threads of this section:

5.2.22 Set-Theoretic vs. Scheme-Theoretic: When Do They Agree? For a morphism $\pi : X \to Y$ of finite type between Noetherian schemes, the set-theoretic image $\pi(X)$ is constructible (Chevalley), and the scheme-theoretic image Z has underlying set $\overline{\pi(X)}$ (the closure). These agree—i.e., $|\text{scheme-theoretic image}| = \pi(X)$—precisely when the image is already closed. This happens, for instance, when π is proper (definition 3.5.21), or more generally when π is universally closed.
Even when the underlying sets agree, the scheme-theoretic image may carry non-reduced structure (as in example (1) above). The scheme-theoretic image is the "right" notion of image whenever one cares about scheme structure—for instance, when computing intersections, studying flatness, or constructing moduli spaces.

5.3 Q.C. Sheaves on Projective Schemes

On an affine scheme $\operatorname{Spec} R$, the tilde construction $M \mapsto \widetilde{M}$ establishes an equivalence between R-modules and quasi-coherent sheaves (corollary 5.1.12). For a projective scheme $X = \operatorname{Proj} S_\bullet$, the analogous construction starts with *graded* $S_\bullet$-modules and produces quasi-coherent sheaves on X. The main result of this section is that every quasi-coherent sheaf on $\operatorname{Proj} S_\bullet$ arises in this way (proposition 5.3.14), and that — when $S_\bullet$ is a polynomial ring — the graded ring itself is recovered as a direct sum of global sections of certain natural line bundles, the *twisting sheaves* (proposition 5.3.10).

5.3.1 The Tilde Construction on Proj

Recall from theorem 2.3.3 that the structure sheaf of $X = \operatorname{Proj} S_\bullet$ is defined on distinguished opens by $\mathcal{O}_X(D_+(f)) = (S_f)_0$: we localize at a homogeneous element f and then take the degree-zero part. The same recipe generalizes from the ring $S_\bullet$ to any graded module over it.

Definition 5.3.1: Sheaf Associated to M [on Projective Scheme]

Let $S_\bullet$ be a graded ring and $M_\bullet$ a graded $S_\bullet$-module. The *sheaf associated to* $M_\bullet$ on $X = \operatorname{Proj} S_\bullet$, denoted $\widetilde{M}$, is the $\mathcal{O}_X$-module defined on the basis of projective distinguished opens by

$$\widetilde{M}\big(D_+(f)\big) \;=\; \big(M[f^{-1}]\big)_0,$$

where $f \in S_\bullet$ is a nonzero homogeneous element and $(-)_0$ denotes the degree-zero component. The restriction maps are induced by further localization.

Let us unpack this. The localization $M[f^{-1}]$ inverts f in the graded sense: a typical element is m/f^k where $m \in M$ is homogeneous. This element has degree $\deg m - k \deg f$. Taking the degree-zero part selects those fractions m/f^k where $\deg m = k \deg f$. When $M = S_\bullet$ this recovers the structure sheaf $(S_f)_0 = \mathcal{O}_X(D_+(f))$; for a general module, the degree-zero localization $(M_f)_0$ is a module over the ring $(S_f)_0$.

Concretely, if $S_\bullet = R[x_0, \ldots, x_r]$ is the standard polynomial ring graded by degree, then $D_+(x_i) \cong \operatorname{Spec}(S_{x_i})_0$ and

$$\widetilde{M}(D_+(x_i)) \;=\; (M_{x_i})_0,$$

which is a module over the affine coordinate ring $(S_{x_i})_0 \cong R[x_0/x_i, \ldots, \widehat{x_i/x_i}, \ldots, x_r/x_i]$.

Example 5.3.2: The Tilde Construction on $\mathbb{P}^1$

Let $S_\bullet = k[x,y]$ with the standard grading, so that $X = \mathbb{P}^1_k$. Take $M = S_\bullet/(x^2 - y^2)$, graded by the quotient grading, so that M has one generator in degree 0 and the relation $x^2 = y^2$ in degree 2.

On the chart $D_+(y) \cong \operatorname{Spec} k[t]$ (where $t = x/y$), the module of sections is

$$(M_y)_0 \;=\; \frac{k[x/y]}{((x/y)^2 - 1)} \;=\; \frac{k[t]}{(t^2-1)},$$

which is a $k[t]$-module. Similarly, on $D_+(x) \cong \operatorname{Spec} k[s]$ (where $s = y/x$):

$$(M_x)_0 \;=\; \frac{k[s]}{(1-s^2)}.$$

Thus $\widetilde{M}$ restricts on each affine chart to the sheaf associated to a familiar quotient ring — the coordinate ring of two points $\{t = \pm 1\}$ on $\mathbb{A}^1$. The gluing of these local pieces recovers the sheaf on all of $\mathbb{P}^1$.

Proposition 5.3.3: Sheaf on Projective Scheme is Quasi-Coherent

Let $S_\bullet$ be a graded ring, $M_\bullet$ a graded $S_\bullet$-module, and $X = \operatorname{Proj} S_\bullet$. Then:

1. The stalk of $\widetilde{M}$ at a homogeneous prime $\mathfrak{p} \in X$ is $\widetilde{M}_{\mathfrak{p}} \cong M_{(\mathfrak{p})}$, the homogeneous localization of M at $\mathfrak{p}$.
2. $\widetilde{M}$ is a quasi-coherent $\mathcal{O}_X$-module. If $S_\bullet$ is Noetherian and $M_\bullet$ is finitely generated, then $\widetilde{M}$ is coherent.

***Proof*:**

On each basic open $D_+(f) \cong \operatorname{Spec}(S_f)_0$, the sheaf $\widetilde{M}$ restricts to the sheaf associated to the $(S_f)_0$-module $(M_f)_0$, which is quasi-coherent by the affine theory. The stalk computation follows from the same colimit argument as for the structure sheaf (theorem 2.3.3). For coherence, if $M_\bullet$ is finitely generated over a Noetherian ring $S_\bullet$, then each $(M_f)_0$ is finitely generated over the Noetherian ring $(S_f)_0$.

The structure sheaf $\mathcal{O}_X = \widetilde{S_\bullet}$ captures the degree-zero part of $S_\bullet$. To access sections of other degrees, we shift the grading:

Definition 5.3.4: Twisting Sheaf

Let $S_\bullet = \bigoplus_n S_n$ be a graded ring and $X = \operatorname{Proj} S_\bullet$. For each $n \in \mathbb{Z}$, define the *nth twisting sheaf* to be

$$\mathcal{O}_X(n) \;:=\; \widetilde{S(n)},$$

where $S(n)$ is the graded module with $S(n)_d = S_{n+d}$. The sheaf $\mathcal{O}_X(1)$ is called *the twisting sheaf* (or *Serre's twisting sheaf*) of X.

Note that $S(0) = S_\bullet$, so $\mathcal{O}_X(0) = \mathcal{O}_X$. Let us work out what the sections of $\mathcal{O}_X(n)$ look like concretely. On a distinguished open $D_+(f)$ for a homogeneous element f of degree e, the definition gives

$$\mathcal{O}_X(n)(D_+(f)) \;=\; \big(S(n)[f^{-1}]\big)_0 \;=\; \big(S[f^{-1}]\big)_n \;=\; (S_f)_n,$$

the elements of degree n in the localized ring S_f. For $n \neq 0$, this is no longer a ring, but it is a module over $\mathcal{O}_X(D_+(f)) = (S_f)_0$.

Let us see what this means explicitly on the standard projective space. Take $S_\bullet = k[x_0, \dots, x_r]$ with the usual grading by total degree. On the chart $D_+(x_i) \cong \operatorname{Spec} k[x_0/x_i, \dots, x_r/x_i]$, the ring $S_{x_i} = k[x_0, \dots, x_r, x_i^{-1}]$ contains elements of every integer degree. An element of $(S_{x_i})_n$ is a sum of monomials in $x_0, \dots, x_r, x_i^{-1}$ of total degree n. Since x_i is invertible, every such monomial can be written as x_i^n times a degree-zero monomial. More precisely, the map

$$(S_{x_i})_0 \xrightarrow[\cdot\, x_i^n]{\sim} (S_{x_i})_n, \qquad g \longmapsto x_i^n \cdot g, \tag{5.1}$$

is an isomorphism of $(S_{x_i})_0$-modules: multiplication by the unit x_i^n shifts the degree from 0 to n. This proves the following:

Proposition 5.3.5: $\mathcal{O}_X(n)$ is an Invertible Sheaf

Let $S_\bullet$ be a graded ring generated by S_1 as an S_0-algebra, and $X = \operatorname{Proj} S_\bullet$. Then $\mathcal{O}_X(n)$ is an invertible sheaf *(*line bundle*)* for each $n \in \mathbb{Z}$.

***Proof*:**

For each $f \in S_1$, multiplication by f^n defines an isomorphism $(S_f)_0 \xrightarrow{\sim} (S_f)_n$ of $(S_f)_0$-modules (as in (5.1)), giving $\mathcal{O}_X(n)|_{D_+(f)} \cong \mathcal{O}_{D_+(f)}$. Since $S_\bullet$ is generated by S_1, the opens $\{D_+(f) : f \in S_1\}$ cover X, so $\mathcal{O}_X(n)$ is locally free of rank 1.

Although $\mathcal{O}_X(n)$ is locally isomorphic to $\mathcal{O}_X$ on each chart, the sheaves $\mathcal{O}_X(n)$ for different n are *not* globally isomorphic — the local trivializations glue together differently, producing genuinely distinct line bundles. We shall now prove this by computing global sections. Before treating the general case, let us work through $\mathbb{P}^1$ in full detail.

Let $S_\bullet = k[x, y]$ and $X = \mathbb{P}^1_k = \operatorname{Proj} S_\bullet$, covered by the two charts

$$D_+(y) \cong \operatorname{Spec} k[t], \quad t = x/y,$$
$$D_+(x) \cong \operatorname{Spec} k[s], \quad s = y/x.$$

On the overlap $D_+(x) \cap D_+(y) \cong \operatorname{Spec} k[t, t^{-1}]$, we have $s = 1/t$.

A section of $\mathcal{O}(n)$ over $D_+(y)$ is an element of $(S_y)_n$: a degree-n element of $k[x, y, y^{-1}]$. Any such element can be written uniquely as $y^n \cdot p(x/y) = y^n \cdot p(t)$ for a polynomial $p \in k[t]$. Similarly, a section over $D_+(x)$ is $x^n \cdot q(y/x) = x^n \cdot q(s)$ for a polynomial $q \in k[s]$. A global section must give a pair $(p(t), q(s))$ that *agree on the overlap*. On $D_+(x) \cap D_+(y)$, both expressions represent the same element of $(S_{xy})_n$:

$$y^n \cdot p(t) \;=\; x^n \cdot q(s) \;=\; (ty)^n \cdot q(1/t) \;=\; y^n \cdot t^n\, q(1/t).$$

Cancelling y^n (which is a unit on the overlap), the gluing condition is

$$p(t) \;=\; t^n \cdot q(1/t). \tag{5.2}$$

Let us now examine this condition for several values of n.

Case $n = 0$ (the structure sheaf). The condition becomes $p(t) = q(1/t)$. Writing $p(t) = \sum_{i \geq 0} a_i t^i$ and $q(s) = \sum_{j \geq 0} b_j s^j$, the right side is $\sum_j b_j t^{-j}$. For this to equal a polynomial in t (no negative powers), we need $b_j = 0$ for $j \geq 1$, so $q(s) = b_0$ is constant. Then $p(t) = b_0$ is also constant. Hence $\Gamma(\mathbb{P}^1, \mathcal{O}) = k$, recovering the familiar fact that the only global regular functions on $\mathbb{P}^1$ are constants (proposition 2.3.6).

Case $n = 1$ (the twisting sheaf). Now $p(t) = t \cdot q(1/t)$. Write $q(s) = b_0 + b_1 s + b_2 s^2 + \cdots$. Then

$$t \cdot q(1/t) \;=\; t\left(b_0 + \frac{b_1}{t} + \frac{b_2}{t^2} + \cdots\right) \;=\; b_0 t + b_1 + \frac{b_2}{t} + \cdots$$

For $p(t)$ to be a polynomial (no negative powers of t), we need $b_2 = b_3 = \cdots = 0$, giving $q(s) = b_0 + b_1 s$. Then $p(t) = b_1 + b_0 t$. The global section is thus parametrized by the pair $(b_0, b_1) \in k^2$, or equivalently by the degree-1 polynomial $b_0 x + b_1 y \in S_1$. Hence $\Gamma(\mathbb{P}^1, \mathcal{O}(1)) \cong S_1 \cong k^2$.

Case $n = 2$. By the same analysis with $p(t) = t^2 \cdot q(1/t)$, one finds $q(s) = b_0 + b_1 s + b_2 s^2$ and $p(t) = b_2 + b_1 t + b_0 t^2$. The global sections are parametrized by $(b_0, b_1, b_2) \in k^3$, corresponding to degree-2 polynomials $b_0 x^2 + b_1 xy + b_2 y^2 \in S_2$. So $\Gamma(\mathbb{P}^1, \mathcal{O}(2)) \cong S_2 \cong k^3$.

Case $n = -1$ (negative twist). Now $p(t) = t^{-1} \cdot q(1/t)$. Write $q(s) = b_0 + b_1 s + \cdots$. Then

$$t^{-1} \cdot q(1/t) \;=\; \frac{1}{t}\left(b_0 + \frac{b_1}{t} + \cdots\right) \;=\; \frac{b_0}{t} + \frac{b_1}{t^2} + \cdots$$

For this to be a polynomial in t, *every* coefficient must vanish: $b_0 = b_1 = \cdots = 0$. Hence $q = 0$ and $p = 0$: the only global section of $\mathcal{O}(-1)$ is zero. Geometrically, the transition function t^{-1} introduces a pole that no polynomial can cancel, so no nonzero section can be defined on both charts simultaneously.

General $n < 0$. The same argument applies: $p(t) = t^n q(1/t)$ for $n < 0$ forces all coefficients of q to vanish, so $\Gamma(\mathbb{P}^1, \mathcal{O}(n)) = 0$ for every $n < 0$. Note that these sheaves often be distinguished by having different local sections, where poles of different degrees can occur.

General $n \geq 0$. The gluing condition $p(t) = t^n q(1/t)$ is satisfied by polynomials q of degree $\leq n$, giving an $(n+1)$-dimensional space. Hence $\Gamma(\mathbb{P}^1, \mathcal{O}(n)) \cong S_n \cong k^{n+1}$ for $n \geq 0$.

The pattern generalizes to $\mathbb{P}^r_k$ with $S_\bullet = k[x_0, \ldots, x_r]$. The same argument — covering by the $r+1$ standard charts and imposing the gluing conditions — yields $\Gamma(\mathbb{P}^r_k, \mathcal{O}(n)) \cong S_n$ for $n \geq 0$, the vector space of homogeneous polynomials of degree n in $r+1$ variables, which has dimension $\binom{n+r}{r}$. For $n < 0$ and $r \geq 1$, the only global section is zero (the transition functions introduce poles that cannot be compensated on every chart).

The exceptional case is $\mathbb{P}^0_k = \operatorname{Proj} k[x_0] \cong \operatorname{Spec} k$: since there is only one chart, there are no nontrivial gluing conditions, and $\Gamma(\mathbb{P}^0_k, \mathcal{O}(n)) \cong k$ for all n.

Collecting these results:

$\dim_k \Gamma(\mathbb{P}^r_k,\, \mathcal{O}(n))$	$\mathbb{P}^0$	$\mathbb{P}^1$	$\mathbb{P}^2$	$\mathbb{P}^3$	$\mathbb{P}^4$
$n = -2$	1	0	0	0	0
$n = -1$	1	0	0	0	0
$n = 0$	1	1	1	1	1
$n = 1$	1	2	3	4	5
$n = 2$	1	3	6	10	15
general $n \geq 0$, $r \geq 1$	—	$\binom{n+1}{1}$	$\binom{n+2}{2}$	$\binom{n+3}{3}$	$\binom{n+4}{4}$

Since the spaces of global sections have different dimensions for different n (when $r \geq 1$), the line bundles $\mathcal{O}_X(n)$ are pairwise non-isomorphic. These dimensions will be computed systematically using sheaf cohomology in theorem 6.3.1, where we shall also determine $H^i(\mathbb{P}^r, \mathcal{O}(n))$ for $i > 0$: the vanishing of H^0 for negative twists is only the beginning of the story.

5.3.6 Why "Twisting"? The name comes from the transition functions. On the overlap $D_+(x) \cap D_+(y)$ of $\mathbb{P}^1_k$, the local trivializations (5.1) identify sections of $\mathcal{O}(n)$ with elements of the structure sheaf via multiplication by y^n (on the y-chart) and x^n (on the x-chart). The transition function relating these two trivializations is

$$\varphi_{xy} \;=\; \frac{x^n}{y^n} \;=\; t^n \;=\; s^{-n},$$

which is precisely the factor appearing in the gluing condition (5.2).
For $n = 0$, the transition function is 1: the bundle is trivial. For $n = 1$, the transition function is t: traversing the base $\mathbb{P}^1$ and returning to the starting chart, the fibre is multiplied by $t \cdot (1/t) = 1$ only after passing through *both* charts. Over $\mathbb{R}$, the total space of $\mathcal{O}(1)$ on $\mathbb{P}^1_{\mathbb{R}}$ is a Möbius-like band: the fibre must be "twisted" once to close up. For general n, the transition involves $|n|$ such twists (with opposite orientation for $n < 0$).
More precisely, the transition functions define an element of the Čech cohomology group $\check{H}^1(\mathbb{P}^1, \mathcal{O}_X^\times) \cong \operatorname{Pic}(\mathbb{P}^1) \cong \mathbb{Z}$, and the integer n is exactly the degree of the line bundle. This connection between transition functions, line bundles, and Čech cohomology will be developed fully in section 5.4 and section 6.1.1.

Definition 5.3.7: n-Twisted Sheaf of $\mathscr{F}$

For any $\mathcal{O}_X$-module $\mathscr{F}$ on $X = \operatorname{Proj} S_\bullet$, the *$n$-twisted sheaf* is

$$\mathscr{F}(n) \;:=\; \mathscr{F} \otimes_{\mathcal{O}_X} \mathcal{O}_X(n).$$

Twisting by $\mathcal{O}_X(n)$ is a fundamental operation: it shifts the "degree of homogeneity" of sections. Since $\mathcal{O}_X(n)$ is locally free of rank 1, tensoring with it preserves quasi-coherence, coherence, and local freeness.

Proposition 5.3.8: Properties of Twisted Sheaves

Let $S_\bullet$ be a graded ring generated by S_1 as an S_0-algebra, and $X = \operatorname{Proj} S_\bullet$. Then:

1. $\mathcal{O}_X(n) \otimes_{\mathcal{O}_X} \mathcal{O}_X(m) \cong \mathcal{O}_X(n+m)$ for all $n, m \in \mathbb{Z}$. In particular, $\mathcal{O}_X(n)$ and $\mathcal{O}_X(-n)$ are tensor-product inverses, confirming that each $\mathcal{O}_X(n)$ is invertible.
2. For any graded $S_\bullet$-module $M_\bullet$, $\widetilde{M}(n) \cong \widetilde{M(n)}$.
3. Let $T_\bullet$ be another graded ring generated by T_1 over T_0, let $\varphi : S_\bullet \to T_\bullet$ be a graded homomorphism, and let $U \subseteq \operatorname{Proj} T_\bullet$ be the open locus where $f : U \to X$ is induced by φ. Then

$$f^*(\mathcal{O}_X(n)) \;\cong\; \mathcal{O}_{\operatorname{Proj} T}(n)\big|_U.$$

***Proof*:**
(1). On each $D_+(f)$ for $f \in S_1$, both sides restrict to $(S_f)_{n+m}$: the left side via the isomorphism $(S_f)_n \otimes_{(S_f)_0} (S_f)_m \xrightarrow{\sim} (S_f)_{n+m}$ given by multiplication in S_f, and the right side by definition. These local isomorphisms are compatible with restriction (further localization commutes with multiplication), hence glue to a global isomorphism by corollary 1.3.4.

(2). On $D_+(f)$, the left side gives $(M_f)_0 \otimes_{(S_f)_0} (S_f)_n \cong (M_f)_n$, and the right gives $(M(n)_f)_0 = (M_f)_n$.

(3). Reduce to the affine case: on $D_+(g) \subseteq \operatorname{Proj} T_\bullet$ mapping into $D_+(\varphi(g)) \subseteq X$, pullback corresponds to extension of scalars, and the grading shift commutes with this operation.

Having seen that $\Gamma(\mathbb{P}^r_k, \mathcal{O}(n)) \cong S_n$, it is natural to reassemble all degrees into a single graded object and ask whether we recover the entire graded ring $S_\bullet$.

Definition 5.3.9: Graded Module Associated to $\mathscr{F}$

Let $S_\bullet$ be a graded ring, $X = \operatorname{Proj} S_\bullet$, and $\mathscr{F}$ an $\mathcal{O}_X$-module. The *graded $S_\bullet$-module associated to $\mathscr{F}$* is

$$\Gamma_*(\mathscr{F}) := \bigoplus_{n\in\mathbb{Z}} \Gamma(X, \mathscr{F}(n)),$$

with the $S_\bullet$-module structure induced by the multiplication maps $S_d \otimes \Gamma(X, \mathscr{F}(n)) \to \Gamma(X, \mathscr{F}(n+d))$ coming from proposition 5.3.8(1)–(2).

For polynomial rings, this recovers $S_\bullet$ exactly:

Proposition 5.3.10: Recovering the Graded Ring

Let R be a ring, $r \geq 1$, $S_\bullet = R[x_0, \ldots, x_r]$, and $X = \operatorname{Proj} S_\bullet$. Then $\Gamma_*(\mathcal{O}_X) \cong S_\bullet$.

***Proof*:**

A global section $t \in \Gamma(X, \mathcal{O}_X(n))$ is a family of sections $t_i \in \mathcal{O}_X(n)(D_+(x_i)) = (S_{x_i})_n$ that agree on overlaps. The variables x_i are not zero divisors in $S_\bullet$, so the localization maps $S_\bullet \hookrightarrow S_{x_i} \hookrightarrow S_{x_0\cdots x_r}$ are all injective, and these rings are subrings of the common localization $S' = S_{x_0\cdots x_r}$. The agreement condition on overlaps means t lies in $\bigcap_{i=0}^r (S_{x_i})_n$ inside S'.

It remains to show $\bigcap_i S_{x_i} = S_\bullet$ inside S'. Every homogeneous element of S' can be written uniquely as $x_0^{a_0}\cdots x_r^{a_r}\cdot g(x_0,\ldots,x_r)$ with $a_j \in \mathbb{Z}$ and g not divisible by any x_i. Such an element lies in S_{x_i} if and only if $a_j \geq 0$ for all $j \neq i$. Belonging to *every* S_{x_i} requires $a_j \geq 0$ for all j (for each j, choose any $i \neq j$, which exists as $r \geq 1$), which means the element lies in $S_\bullet$. Summing over degrees gives $\Gamma_*(\mathcal{O}_X) \cong S_\bullet$.

The hypothesis $r \geq 1$ is needed. For $r = 0$, the scheme $X = D_+(x_0)$ has $\Gamma(X, \mathcal{O}_X(n)) = (S_{x_0})_n = Rx_0^n$ for every $n \in \mathbb{Z}$, so $\Gamma_*(\mathcal{O}_X) = R[x_0, x_0^{-1}]$, which is larger than $S_\bullet$.

The reader should note two important features of this proof: the injectivity of the localization maps (which uses the fact that x_i are not zero divisors) and the combinatorial intersection argument. For a general graded ring $S_\bullet$ where the generators may be zero divisors, the conclusion can fail.

More fundamentally, the Γ_*–tilde correspondence is *not* an equivalence in either direction:

- The map $M_\bullet \mapsto \Gamma_*(\widetilde{M})$ need not recover M. For instance, if $M_0 = k$ and $M_n = 0$ for $n \neq 0$, then $\widetilde{M} = 0$ (there are no nonzero elements to localize in positive degrees), so $\Gamma_*(\widetilde{M}) = 0 \neq M$. The graded module M is concentrated in a single degree that is "invisible" to Proj.
- The map $M_\bullet \mapsto \widetilde{M}$ is not injective: the above example shows two different modules (M and 0) giving the same sheaf.

However, the other direction works perfectly: starting from a quasi-coherent sheaf and applying Γ_*, one can always recover the sheaf by applying the tilde construction. This is the key structural result of this section.

The proof requires a generalization of lemma 5.1.15 from distinguished opens of affine schemes to the projective setting. The role of the element $f \in R$ (which defines the distinguished open $D(f)$

in the affine case) is played by a global section $f \in \Gamma(X, \mathcal{L})$ of an invertible sheaf $\mathcal{L}$. The open set $X_f := \{x \in X : f_x \notin \mathfrak{m}_x \mathcal{L}_x\}$ is the locus where f "does not vanish" (cf. the evaluation of sections in definition 2.2.25); when $\mathcal{L} = \mathcal{O}_X$ this recovers the ordinary distinguished open, and when $\mathcal{L} = \mathcal{O}_X(1)$ on $\operatorname{Proj} S_\bullet$ this recovers $D_+(f)$.

Lemma 5.3.11: Extending Sections for Line Bundles

Let X be a scheme, $\mathcal{L}$ an invertible sheaf on X, $f \in \Gamma(X, \mathcal{L})$, and X_f the open set of points $x \in X$ where $f_x \notin \mathfrak{m}_x \mathcal{L}_x$. Let $\mathscr{F}$ be a quasi-coherent sheaf on X.

1. Suppose X is quasi-compact. If $s \in \Gamma(X, \mathscr{F})$ satisfies $s|_{X_f} = 0$, then $f^n s = 0$ in $\Gamma(X, \mathscr{F} \otimes \mathcal{L}^{\otimes n})$ for some $n > 0$.
2. Suppose additionally that X has a finite cover by affine opens $\{U_i\}$ with $\mathcal{L}|_{U_i}$ free and $U_i \cap U_j$ quasi-compact for all i, j. Then for any $t \in \Gamma(X_f, \mathscr{F})$, there exists $n > 0$ such that $f^n t \in \Gamma(X_f, \mathscr{F} \otimes \mathcal{L}^{\otimes n})$ extends to a global section of $\mathscr{F} \otimes \mathcal{L}^{\otimes n}$.

The reader should think of this lemma as the projective analogue of lemma 5.1.15: part (1) says "sections that vanish on X_f are killed by a power of f," and part (2) says "sections defined only on X_f can be extended globally after multiplying by a power of f." The new feature is that f is a section of a line bundle rather than a global function, so "multiplying by f^n" lands in the twisted sheaf $\mathscr{F} \otimes \mathcal{L}^{\otimes n}$ rather than in $\mathscr{F}$ itself. In the special case $\mathcal{L} = \mathcal{O}_X$, the section f is an ordinary global function, the open X_f is the distinguished open $D(f)$, and we recover lemma 5.1.15 exactly.

Before giving the proof, let us verify the hypotheses in the case we care about most.

Example 5.3.12: Hypotheses for Proj

Let $S_\bullet$ be a graded ring finitely generated by S_1 over S_0, $X = \operatorname{Proj} S_\bullet$, $\mathcal{L} = \mathcal{O}_X(1)$, and $f \in S_1$ a degree-1 element viewed as a section $f \in \Gamma(X, \mathcal{O}_X(1))$.

The open set X_f. A point $x \in X$ (corresponding to a homogeneous prime $\mathfrak{p}$) lies in X_f precisely when f_x generates the free $\mathcal{O}_{X,x}$-module $\mathcal{L}_x = \mathcal{O}_X(1)_x$, i.e., when f_x is a unit times the local generator. On the chart $D_+(g)$ for $g \in S_1$, the generator of $\mathcal{O}_X(1)|_{D_+(g)}$ is g (via (5.1)), and f_x is a unit times g precisely when $f/g \notin \mathfrak{p}$, i.e., when $\mathfrak{p} \in D_+(f)$. Hence $X_f = D_+(f)$.

The finite cover. Since $S_\bullet$ is generated by S_1, choose generators $f_0, \ldots, f_r \in S_1$; then $U_i = D_+(f_i)$ is affine (theorem 2.3.3), $\mathcal{L}|_{U_i}$ is free (by proposition 5.3.5), and $U_i \cap U_j = D_+(f_i f_j)$ is affine (hence quasi-compact).

Thus the hypotheses of both parts of lemma 5.3.11 are satisfied for any $f \in S_1$ on $\operatorname{Proj} S_\bullet$.

Proof of lemma 5.3.11:
The strategy for both parts is to reduce to the affine case by trivializing $\mathcal{L}$ on a finite cover, applying lemma 5.1.15, and then patching the results.

(1) Annihilation. Cover X by finitely many affine opens $U_\alpha = \operatorname{Spec} A_\alpha$ (possible since X is quasi-compact) on which $\mathcal{L}$ is free; choose trivializations $\psi_\alpha : \mathcal{L}|_{U_\alpha} \xrightarrow{\sim} \mathcal{O}_{U_\alpha}$. Set $g_\alpha = \psi_\alpha(f|_{U_\alpha}) \in A_\alpha$, so that $X_f \cap U_\alpha = D(g_\alpha)$:

$$\mathcal{L}(U_\alpha) \xrightarrow[\sim]{\psi_\alpha} \mathcal{O}_X(U_\alpha) = A_\alpha$$

$$f|_{U_\alpha} \longmapsto g_\alpha$$

Write $\mathscr{F}|_{U_\alpha} \cong \widetilde{M_\alpha}$ for an A_α-module M_α. The section $s|_{U_\alpha}$ corresponds to an element $s_\alpha \in M_\alpha$, and the hypothesis $s|_{X_f} = 0$ gives $s_\alpha|_{D(g_\alpha)} = 0$.

By the affine annihilation lemma (lemma 5.1.15(1)), $g_\alpha^{n_\alpha} s_\alpha = 0$ in M_α for some $n_\alpha > 0$. Using the trivialization $\psi_\alpha^{\otimes n} : \mathcal{L}^{\otimes n}|_{U_\alpha} \xrightarrow{\sim} \mathcal{O}_{U_\alpha}$, the element $g_\alpha^{n_\alpha} s_\alpha$ corresponds to $(f^{n_\alpha} s)|_{U_\alpha}$ intrinsically (independent of the choice of ψ_α). Taking $n = \max_\alpha n_\alpha$ gives $f^n s|_{U_\alpha} = 0$ for every α. Since the U_α cover X, the locality axiom gives $f^n s = 0$ globally.

(2) Extension. This is more involved: we must first extend the section locally, then correct the local extensions so that they glue.

Step 1: Local extensions. On each U_i of the given finite cover, the trivialization ψ_i converts $f|_{U_i}$ to $g_i \in A_i$ and gives $X_f \cap U_i = D(g_i)$. The section $t|_{D(g_i)} \in \mathscr{F}(D(g_i)) = (M_i)_{g_i}$. By the affine extension lemma (lemma 5.1.15(2)), there exists $n_i > 0$ such that $g_i^{n_i} \cdot t|_{D(g_i)}$ extends to an element $\hat{t}_i \in M_i = \mathscr{F}(U_i)$. Intrinsically, this means $f^{n_i} t|_{X_f \cap U_i}$ extends to a section of $\mathscr{F} \otimes \mathcal{L}^{\otimes n_i}$ over U_i. Taking $N = \max_i n_i$ (and multiplying by further powers of f where needed), we may assume each $\hat{t}_i$ is a section of $\mathscr{F} \otimes \mathcal{L}^{\otimes N}$ over U_i that restricts to $f^N t$ on $X_f \cap U_i$.

Step 2: Correcting overlaps. On $U_i \cap U_j$, both $\hat{t}_i$ and $\hat{t}_j$ restrict to $f^N t$ on $X_f \cap U_i \cap U_j$, so their difference $\delta_{ij} = \hat{t}_i - \hat{t}_j$ vanishes on $X_f \cap U_i \cap U_j$. Since $U_i \cap U_j$ is quasi-compact by hypothesis, part (1) applied to δ_{ij} on $U_i \cap U_j$ gives $m_{ij} > 0$ with $f^{m_{ij}} \delta_{ij} = 0$ on $U_i \cap U_j$. Set $m = \max_{i,j} m_{ij}$ (a finite maximum since the cover is finite).

Then the sections $f^m \hat{t}_i \in \Gamma(U_i, \mathscr{F} \otimes \mathcal{L}^{\otimes(N+m)})$ satisfy $f^m \hat{t}_i = f^m \hat{t}_j$ on every overlap $U_i \cap U_j$. By the gluability axiom, they glue to a global section $\bar{t} \in \Gamma(X, \mathscr{F} \otimes \mathcal{L}^{\otimes(N+m)})$. On X_f, this restricts to $f^{N+m} t$, so $\bar{t}$ is the desired extension with $n = N + m$.

5.3.13 The Mechanism of Extension The reader may find it helpful to compare the affine and projective extension results side by side:

Affine (lemma 5.1.15)	**Projective** (lemma 5.3.11)
$f \in R$ (ring element)	$f \in \Gamma(X, \mathcal{L})$ (section of line bundle)
Open: $D(f) \subseteq \operatorname{Spec} R$	Open: $X_f \subseteq X$
$t \in M_f$ (localization)	$t \in \mathscr{F}(X_f)$ (sections on X_f)
$f^n t \in M$ (cleared denominator)	$f^n t \in \Gamma(X, \mathscr{F} \otimes \mathcal{L}^{\otimes n})$ (twisted extension)

In the affine case, multiplying by f^n "clears the denominator" and lands back in M. In the projective case, the same multiplication lands in a *different* module — the global sections of the nth twist — because f is a section of $\mathcal{L}$, not of $\mathcal{O}_X$. The price we pay for working on a non-affine scheme is that extension requires passing to a twist; the benefit is that this works for *any* quasi-compact scheme with a suitable line bundle.

We now prove the key structural result: the tilde functor applied to $\Gamma_*(\mathscr{F})$ recovers $\mathscr{F}$.

Proposition 5.3.14: Partial Equivalence for Graded Rings

Let $S_\bullet$ be a graded ring, finitely generated by S_1 as an S_0-algebra. Let $X = \operatorname{Proj} S_\bullet$ and $\mathscr{F}$ a quasi-coherent sheaf on X. Then there is a natural isomorphism

$$\beta : \widetilde{\Gamma_*(\mathscr{F})} \xrightarrow{\sim} \mathscr{F}.$$

In particular, every quasi-coherent sheaf on $\operatorname{Proj} S_\bullet$ arises from a graded $S_\bullet$-module.

The hypothesis that $S_\bullet$ is generated by S_1 is essential: it ensures that X is covered by the affine opens $D_+(f)$ for $f \in S_1$, on which the twisting sheaf $\mathcal{O}_X(1)$ becomes free.

***Proof*:**

Step 1: Constructing β. We define β on the basis of distinguished opens $D_+(f)$ for $f \in S_+$ homogeneous. A section of $\widetilde{\Gamma_*(\mathscr{F})}$ over $D_+(f)$ is an element of the homogeneous localization $\Gamma_*(\mathscr{F})_{(f)}$: a fraction m/f^d where $m \in \Gamma(X, \mathscr{F}(d))$ is a global section of the dth twist and $\deg f \cdot d$ matches $\deg m$ in the graded module.

Now f^{-d} can be viewed as a section of $\mathcal{O}_X(-d)$ over $D_+(f)$: since $\mathcal{O}_X(d)|_{D_+(f)}$ is trivial, the element f^d is a generator, and f^{-d} is its inverse in the localized sections. Tensoring, the element $m \otimes f^{-d}$ is a section of $\mathscr{F}(d) \otimes \mathcal{O}_X(-d) \cong \mathscr{F}$ over $D_+(f)$. Define

$$\beta_{D_+(f)}\left(\frac{m}{f^d}\right) := m \otimes f^{-d} \in \mathscr{F}(D_+(f)).$$

Let us verify this is well-defined. If $m/f^d = m'/f^{d'}$ in $\Gamma_*(\mathscr{F})_{(f)}$, then $f^N(f^{d'}m - f^d m') = 0$ for some N, i.e., the sections $f^{d'}m$ and $f^d m'$ agree in $\mathscr{F}(d + d' + N\deg f)$ after multiplying by f^N. Restricting to $D_+(f)$ and dividing by $f^{d+d'+N}$ (which is a unit there), we get $m \otimes f^{-d} = m' \otimes f^{-d'}$ in $\mathscr{F}(D_+(f))$. Compatibility with restriction holds because $m/f^d = mg^d/(fg)^d$ and $mg^d \otimes (fg)^{-d} = m \otimes f^{-d}$ in $\mathscr{F}(D_+(fg))$, so β is a well-defined morphism of sheaves.

Step 2: β is an isomorphism. It suffices to show that $\beta_{D_+(f)}$ is bijective for each $f \in S_1$. By example 5.3.12, the hypotheses of lemma 5.3.11 are satisfied with $\mathcal{L} = \mathcal{O}_X(1)$ and $X_f = D_+(f)$.

Surjectivity. Let $t \in \mathscr{F}(D_+(f))$. By lemma 5.3.11(2), there exists $n > 0$ such that $f^n t$ extends to a global section $s \in \Gamma(X, \mathscr{F} \otimes \mathcal{O}_X(n)) = \Gamma(X, \mathscr{F}(n))$. Then $s/f^n \in \Gamma_*(\mathscr{F})_{(f)}$ maps under β to $s \otimes f^{-n} = t$ on $D_+(f)$.

Injectivity. Suppose $m/f^d \in \Gamma_*(\mathscr{F})_{(f)}$ maps to zero, i.e., $m|_{D_+(f)} = 0$ in $\mathscr{F}(d)(D_+(f))$. By lemma 5.3.11(1) applied to the global section m of $\mathscr{F}(d)$ (which vanishes on $X_f = D_+(f)$), there exists $n > 0$ with $f^n m = 0$ in $\Gamma(X, \mathscr{F}(d+n))$. Hence $m/f^d = 0$ in $\Gamma_*(\mathscr{F})_{(f)}$.

Let us pause to summarize the relationship between graded modules and quasi-coherent sheaves on Proj. The two functors

$$\{\text{graded } S_\bullet\text{-modules}\} \underset{\Gamma_*(-)}{\overset{\widetilde{(-)}}{\rightleftarrows}} \mathbf{QCoh}(\operatorname{Proj} S_\bullet)$$

satisfy $\widetilde{\Gamma_*(\mathscr{F})} \cong \mathscr{F}$ for every quasi-coherent $\mathscr{F}$ (proposition 5.3.14), but $\Gamma_*(\widetilde{M})$ need not be isomorphic to M. In the language of category theory, Γ_* is a *right adjoint* that is faithful but not full, and the

tilde functor is a *left adjoint* that is essentially surjective. The failure of the unit $M \to \Gamma_*(\widetilde{M})$ to be an isomorphism is measured by the *saturation* of M: two graded modules give the same sheaf if and only if they agree in sufficiently large degrees. This is analogous to the fact that Proj itself depends only on the "tail" of the graded ring (definition 2.3.1).

5.3.2 Characterizing Projective Subschemes

Combining the partial equivalence with the theory of ideal sheaves from section 5.1.4, we obtain concrete descriptions of closed subschemes of projective space and of projective schemes themselves.

Corollary 5.3.15: Characterizing Projective Subschemes

Let R be a ring and $S_\bullet = R[x_0, \dots, x_r]$. Then:

1. A surjection $\varphi\colon S_\bullet \to S'_\bullet$ of graded R-algebras preserving degrees induces a closed immersion $\operatorname{Proj} S'_\bullet \to \mathbb{P}^r_R$ over $\operatorname{Spec} R$, which over each $D_+(x_i)$ is given by the surjection $S_{(x_i)} \to S'_{(\varphi(x_i))}$, $a/x_i^k \mapsto \varphi(a)/\varphi(x_i)^k$.
2. Every closed subscheme $Y \subseteq \mathbb{P}^r_R$ is determined by a homogeneous ideal $I \subseteq S_\bullet$: namely $Y \cong \operatorname{Proj}(S_\bullet/I)$ over $\mathbb{P}^r_R$, for the closed immersion of (1).
3. A scheme Y over $\operatorname{Spec} R$ admits a closed immersion into some $\mathbb{P}^n_R$ over $\operatorname{Spec} R$ if and only if $Y \cong \operatorname{Proj} S'_\bullet$ over $\operatorname{Spec} R$ for some graded ring $S'_\bullet$ with $S'_0 = R$ that is finitely generated by S'_1 as an S'_0-algebra.

***Proof*:**

1. Put $f_i = \varphi(x_i)$. Every $S'_d = \varphi(S_d)$ is spanned by monomials in the f_i, so a homogeneous prime of $S'_\bullet$ not containing S'_+ misses some f_i, and the sets $D_+(f_i)$ cover $\operatorname{Proj} S'_\bullet$. The map $\varphi_i\colon S_{(x_i)} \to S'_{(f_i)}$, $a/x_i^k \mapsto \varphi(a)/f_i^k$, is the degree-0 part of the ring map $S_{x_i} \to S'_{f_i}$ induced by φ, and it is surjective, since every element of $S'_{(f_i)}$ is b/f_i^k with $b \in S'_k = \varphi(S_k)$. Through the charts of theorem 2.3.3 it induces a closed immersion $g_i\colon D_+(f_i) \to D_+(x_i)$ (proposition 3.1.24(2), theorem 3.1.4). Under the charts, $D_+(x_ix_j)$ corresponds to $D(x_j/x_i) \subseteq \operatorname{Spec} S_{(x_i)}$ (lemma 2.3.2), whose preimage under the map induced by φ_i is $D(f_j/f_i)$ (lemma 2.1.28), which corresponds to $D_+(f_if_j)$. So $g_i^{-1}(D_+(x_ix_j)) = D_+(f_if_j)$, likewise $g_j^{-1}(D_+(x_ix_j)) = D_+(f_if_j)$, and on this set g_i and g_j are both morphisms into $D_+(x_ix_j) \cong \operatorname{Spec} S_{(x_ix_j)}$ induced by the map $S_{(x_ix_j)} \to S'_{(f_if_j)}$ given by φ, because the restriction maps of theorem 2.3.3 are localization maps, which commute with the maps induced by φ. So g_i and g_j agree on $D_+(f_if_j)$ (theorem 3.1.4), and lemma 3.1.9 glues the g_i to $g\colon \operatorname{Proj} S'_\bullet \to \mathbb{P}^r_R$. A point of $D_+(f_j)$ mapping into $D_+(x_i)$ lies in $g_j^{-1}(D_+(x_ix_j)) = D_+(f_if_j) \subseteq D_+(f_i)$, so $g^{-1}(D_+(x_i)) = D_+(f_i)$, over which g is the closed immersion g_i, and g is a closed immersion by proposition 3.1.24(4). On each chart the structure maps from R are $R \to S_{(x_i)} \to S'_{(f_i)}$, as φ is R-linear, so g is a morphism over $\operatorname{Spec} R$.
2. Let $X = \mathbb{P}^r_R$. If $r = 0$, then $X = D_+(x_0)$, and $S_{(x_0)} = R$, a degree-0 fraction a/x_0^k with $a \in S_k = Rx_0^k$ being a scalar. So $X \cong \operatorname{Spec} R$, and $Y \cong \operatorname{Spec}(R/\mathfrak{a})$ over X for an ideal $\mathfrak{a} \subseteq R$ (corollary 5.1.35). With $I = \mathfrak{a}S_\bullet$ we have $S_\bullet/I = (R/\mathfrak{a})[x_0]$, whose Proj is $D_+(x_0)$

with ring $(S_\bullet/I)_{(x_0)} = R/\mathfrak{a}$, and by (1) its closed immersion into X is given by $R \to R/\mathfrak{a}$. So $Y \cong \operatorname{Proj}(S_\bullet/I)$ over X.

Let $r \geq 1$ and $\mathcal{I}_Y \subseteq \mathcal{O}_X$ be the ideal sheaf of Y in X, which is quasi-coherent (proposition 5.1.34). Since the twisting functor is exact (tensor with a locally free sheaf preserves exact sequences) and Γ is left exact, the inclusion $\mathcal{I}_Y \hookrightarrow \mathcal{O}_X$ gives an inclusion of graded modules

$$\Gamma_*(\mathcal{I}_Y) \hookrightarrow \Gamma_*(\mathcal{O}_X) \cong S_\bullet,$$

where the isomorphism is proposition 5.3.10, which needs $r \geq 1$. Set $I = \Gamma_*(\mathcal{I}_Y)$, a homogeneous ideal of $S_\bullet$. By proposition 5.3.14, $\mathcal{I}_Y \cong \widetilde{I}$, so Y is the closed subscheme of X defined by I, which is the closed immersion of (1) for $S_\bullet \to S_\bullet/I$, i.e., $Y \cong \operatorname{Proj}(S_\bullet/I)$. Moreover, I is the *largest* (or *saturated*) homogeneous ideal defining Y: by construction, it contains every homogeneous polynomial that vanishes on Y in every degree.

3. Let Y admit a closed immersion into $\mathbb{P}^n_R$ over $\operatorname{Spec} R$, and let $S_\bullet = R[x_0, \ldots, x_n]$. By (2), $Y \cong \operatorname{Proj}(S_\bullet/I)$ over $\mathbb{P}^n_R$, hence over $\operatorname{Spec} R$, for a homogeneous ideal I. Put $I_+ = \bigoplus_{d \geq 1} I_d$, a homogeneous ideal. For $a \in I_0$ and f homogeneous of positive degree, $af \in I_+$, so $a/1 = af/f$ vanishes in $(S_\bullet/I_+)_f$. The kernel of the surjection $S_\bullet/I_+ \to S_\bullet/I$ is I/I_+, which is I_0 in degree 0, so localization at f gives isomorphisms $(S_\bullet/I_+)_f \cong (S_\bullet/I)_f$ ([10, Localization Is Exact]) and $(S_\bullet/I_+)_{(f)} \cong (S_\bullet/I)_{(f)}$. By (1), the closed immersions of $\operatorname{Proj}(S_\bullet/I_+)$ and $\operatorname{Proj}(S_\bullet/I)$ into $\mathbb{P}^n_R$ are given over each $D_+(x_i)$ by surjections from $S_{(x_i)}$ that differ by these isomorphisms, which commute with the restriction maps of theorem 2.3.3. So the chartwise isomorphisms glue (lemma 3.1.9) to an isomorphism $\operatorname{Proj}(S_\bullet/I_+) \cong \operatorname{Proj}(S_\bullet/I)$ over $\mathbb{P}^n_R$, hence over $\operatorname{Spec} R$. Setting $S'_\bullet = S_\bullet/I_+$, we have $S'_0 = R$, and $S'_\bullet$ is generated as an R-algebra by the images of $x_0, \ldots, x_n$ in S'_1.

   Conversely, if $S'_0 = R$ and $S'_\bullet$ is generated as an R-algebra by $f_0, \ldots, f_n \in S'_1$, the map $R[x_0, \ldots, x_n] \to S'_\bullet$, $x_i \mapsto f_i$, is a surjection of graded R-algebras preserving degrees, and (1) gives a closed immersion $\operatorname{Proj} S'_\bullet \to \mathbb{P}^n_R$ over $\operatorname{Spec} R$, completing the proof.

Lemma 5.3.16: Reduced Affine Schemes of Finite Type Are Quasiprojective

Let k be a field and let X be a reduced affine scheme of finite type over k. Then X is quasiprojective over k (definition 2.3.5).

***Proof*:**
Write $X = \operatorname{Spec} A$ with $A = k[T_1, \ldots, T_m]/I$ for an ideal I. This identifies X with a closed subscheme of $\mathbb{A}^m_k$ (proposition 3.1.24(2)), and $\mathbb{A}^m_k$ is the open subscheme $D_+(x_0)$ of $\mathbb{P}^m_k$, whose ring of sections $k[x_1/x_0, \ldots, x_m/x_0]$ is a polynomial ring in m variables (theorem 2.3.3). Let Z be the closure of X in $\mathbb{P}^m_k$ with its reduced induced structure (definition 3.1.26). As a closed subscheme of $\mathbb{P}^m_k$ it is projective over k (corollary 5.3.15(3)). Its open subscheme $Z \cap D_+(x_0)$ is reduced, and its inclusion into $D_+(x_0)$ is a closed immersion by proposition 3.1.24(1), every affine open subset of $D_+(x_0)$ being an affine open subset of $\mathbb{P}^m_k$. The underlying set of $Z \cap D_+(x_0)$ is the closure of X in $D_+(x_0)$, which is X because X is closed in $D_+(x_0)$. The scheme X is reduced by hypothesis. So X and $Z \cap D_+(x_0)$ are reduced closed subschemes of $D_+(x_0)$ with the same underlying set, and both are isomorphic over $D_+(x_0)$ to the reduced induced structure

on that set (proposition 3.1.25). Hence X is isomorphic to an open subscheme of the projective k-scheme Z, that is, it is quasiprojective, as claimed.

Corollary 5.3.17: Characterizing Hypersurfaces in Projective Space

Let R be a UFD. Then every irreducible hypersurface $X \subseteq \mathbb{P}^n_R$ corresponds to a principal homogeneous ideal $(f) \subseteq R[x_0, \ldots, x_n]$.

***Proof*:**
By corollary 5.3.15(2), the hypersurface X corresponds to a homogeneous prime ideal $\mathfrak{p} \subseteq S_\bullet = R[x_0, \ldots, x_n]$ of height 1 (since X has codimension 1 in $\mathbb{P}^n_R$). Since R is a UFD, so is $S_\bullet$ by [12, Polynomial Rings in Several Variables over a UFD are UFDs], and in a UFD every height-1 prime ideal is principal ([10, Factoriality by Height-One Primes]). Hence $\mathfrak{p} = (f)$ for some irreducible homogeneous polynomial f.

Definition 5.3.18: Degree of Projective Hypersurface

Let R be a UFD and $X \subseteq \mathbb{P}^n_R$ an irreducible hypersurface corresponding to $(f) \subseteq R[x_0, \ldots, x_n]$ via corollary 5.3.17. The *degree* of X is $\deg X := \deg f$.

This is well-defined because f is determined up to a unit in R, which does not affect the degree. When $R = k$ is an algebraically closed field, the degree of a hypersurface $V(f) \subseteq \mathbb{P}^n_k$ counts (with multiplicity) the number of intersection points with a generic line, in accordance with Bézout's theorem ([22, Degree Counts Points of a Linear Section]).

We conclude this section with the promised example showing that the tensor product of $\mathcal{O}_X$-modules cannot, in general, be computed section-wise.

Example 5.3.19: Tensor Products Require Sheafification

Let $X = \mathbb{P}^1_k$ for a field k. Consider $\mathscr{F} = \mathcal{O}_X(-1)$ and $\mathscr{G} = \mathcal{O}_X(1)$. By proposition 5.3.8(1),

$$\mathscr{F} \otimes_{\mathcal{O}_X} \mathscr{G} \;=\; \mathcal{O}_X(-1) \otimes \mathcal{O}_X(1) \;\cong\; \mathcal{O}_X(0) \;=\; \mathcal{O}_X,$$

so the global sections of the tensor product sheaf satisfy

$$(\mathscr{F} \otimes \mathscr{G})(\mathbb{P}^1_k) \;=\; \Gamma(\mathbb{P}^1_k, \mathcal{O}_X) \;\cong\; k.$$

On the other hand, the section-wise tensor product on global sections gives

$$\mathscr{F}(\mathbb{P}^1_k) \otimes_k \mathscr{G}(\mathbb{P}^1_k) \;=\; \underbrace{\Gamma(\mathbb{P}^1_k, \mathcal{O}(-1))}_{=0} \otimes_k \underbrace{\Gamma(\mathbb{P}^1_k, \mathcal{O}(1))}_{\cong k^2} \;=\; 0.$$

Hence $(\mathscr{F} \otimes \mathscr{G})(X) \cong k \;\ncong\; 0 = \mathscr{F}(X) \otimes \mathscr{G}(X)$.

The mechanism behind this is illuminating. Consider the presheaf $\mathbb{C}\mathrm{P} : U \mapsto \mathscr{F}(U) \otimes_{\mathcal{O}_X(U)} \mathscr{G}(U)$. On each chart $U_i = D_+(x_i)$, we have $\mathscr{F}(U_i) \cong \mathscr{G}(U_i) \cong \mathcal{O}_X(U_i)$ (both are free of rank 1), so $\mathbb{C}\mathrm{P}(U_i) \cong \mathcal{O}_X(U_i)$. But on global sections, $\mathbb{C}\mathrm{P}(X) = 0 \otimes k^2 = 0$. The presheaf $\mathbb{C}\mathrm{P}$ has "plenty of local sections" but "no global sections" — it violates the gluing axiom. Sheafification repairs this: the sheaf $\mathscr{F} \otimes \mathscr{G}$ is constructed by gluing the local sections of $\mathbb{C}\mathrm{P}$, and the global sections of the sheafification can be strictly larger than those of the presheaf.

This phenomenon is purely projective: on affine schemes, the presheaf tensor product is already a sheaf, so $\widetilde{M} \otimes \widetilde{N} \cong \widetilde{M \otimes N}$ and global sections commute with tensor products (proposition 5.1.14(4)).

Exercise 5.3.1

1. *(Krull's Principal Ideal Theorem.)* Let R be a Noetherian ring and $(x) \subsetneq R$ a proper principal ideal. Show that every minimal prime $\mathfrak{p}$ over (x) has height at most 1. *(Hint: localize at $\mathfrak{p}$ to reduce to a local ring with exactly one minimal prime over its principal ideal. Use the Artin–Rees lemma or Nakayama's lemma to show the chain of primes below $\mathfrak{p}$ has length at most 1.)*

Another example I came across is very interesting:

Example 5.3.20: Free Sheaves Need Not Be Projective Objects
Really cool! here

5.3.3 Maps to Projective Space

In example 3.1.14, we observed a fundamental asymmetry: a morphism $X \to \mathbb{A}^n_k$ is determined by n global functions $f_1, \ldots, f_n \in \mathcal{O}_X(X)$, but a morphism $X \to \mathbb{P}^n_k$ cannot be described by global functions alone — the homogeneous coordinates x_i on $\mathbb{P}^n$ are sections of the twisting sheaf $\mathcal{O}(1)$, not of the structure sheaf. We commented that defining a map to projective space requires a *line bundle* $\mathcal{L}$ on X together with global sections $s_0, \ldots, s_n \in \Gamma(X, \mathcal{L})$ that do not vanish simultaneously.

This subsection makes that observation precise. The main result (theorem 5.3.43) establishes a bijection between morphisms $X \to \mathbb{P}^n$ and equivalence classes of line bundles with generating global sections. Along the way, we develop two important tools: the notion of a *very ample sheaf* (which characterizes projective schemes), and a fundamental finite generation theorem for coherent sheaves on projective schemes (theorem 5.3.26).

Maps *from* projective space are better understood using cohomological methods (see chapter 6); theorem 3.5.34 shows that all proper schemes admit a surjective birational morphism from a projective scheme, making projective schemes the natural "compact" objects.

Twisting Sheaves on Relative Projective Space

The twisting sheaves of definition 5.3.4 were defined for $\operatorname{Proj} S_\bullet$ over the base $\operatorname{Spec} S_0$. For a general base scheme X, the projective space $\mathbb{P}^n_X$ is defined by base change, and the twisting sheaf is obtained by pulling back from the universal case:

Definition 5.3.21: Twisting Sheaf on $\mathbb{P}^n_X$

Let X be a scheme. The *twisting sheaf* $\mathcal{O}(1)$ on $\mathbb{P}^n_X = X \times_{\mathbb{Z}} \mathbb{P}^n_{\mathbb{Z}}$ is the pullback

$$\mathcal{O}_{\mathbb{P}^n_X}(1) := g^*\big(\mathcal{O}_{\mathbb{P}^n_{\mathbb{Z}}}(1)\big),$$

where $g : \mathbb{P}^n_X \to \mathbb{P}^n_{\mathbb{Z}}$ is the projection. More generally, $\mathcal{O}_{\mathbb{P}^n_X}(d) := \mathcal{O}_{\mathbb{P}^n_X}(1)^{\otimes d}$ for $d \in \mathbb{Z}$.

By proposition 5.3.8(3), if $X = \operatorname{Spec} R$ and $\mathbb{P}^n_R = \operatorname{Proj} R[x_0, \dots, x_n]$, this recovers the twisting sheaf $\widetilde{S(1)}$ defined earlier. The base-change definition ensures that all properties established over $\mathbb{Z}$ (invertibility, non-isomorphism for different twists, etc.) propagate to $\mathbb{P}^n_X$ for any base X.

Definition 5.3.22: Very Ample Sheaf

Let X be a scheme over a base scheme Y, and $\mathcal{L}$ a line bundle on X. Then $\mathcal{L}$ is *very ample relative to* Y if there exists a closed immersion $\iota : X \hookrightarrow \mathbb{P}^n_Y$ (for some n) such that

$$\iota^*\big(\mathcal{O}_{\mathbb{P}^n_Y}(1)\big) \;\cong\; \mathcal{L}.$$

The condition says that $\mathcal{L}$ is the restriction of $\mathcal{O}(1)$ under an embedding into projective space. Very ample sheaves characterize projectivity:

Lemma 5.3.23: Characterizing Projective Schemes via Very Ample Sheaves

Let Y be a Noetherian scheme. A Y-scheme X is projective over Y if and only if X is proper over Y and possesses a very ample invertible sheaf relative to Y.

***Proof*:**
($\Rightarrow$). If X is projective over Y, then by definition there exists a closed immersion $\iota : X \hookrightarrow \mathbb{P}^n_Y$. The sheaf $\mathcal{L} = \iota^*(\mathcal{O}(1))$ is very ample, and X is proper by theorem 3.5.32.

($\Leftarrow$). Suppose X is proper over Y and $\mathcal{L}$ is very ample, so $\mathcal{L} \cong \iota^*(\mathcal{O}(1))$ for some locally closed immersion $\iota : X \to \mathbb{P}^n_Y$. Since X is proper over Y and $\mathbb{P}^n_Y$ is separated over Y, the morphism ι is proper (proposition 3.5.23). A proper locally closed immersion has closed image (definition 3.5.21), so ι is in fact a closed immersion. Hence X is projective.

Global Generation

We need a notion expressing when a sheaf has "enough global sections" to control its local behaviour.

Definition 5.3.24: Generated by Global Sections

Let X be a scheme and $\mathscr{F}$ an $\mathcal{O}_X$-module. Then $\mathscr{F}$ is *generated by global sections* if there exists a family of global sections $\{s_i\}_{i \in I} \subseteq \Gamma(X, \mathscr{F})$ whose images in the stalk $\mathscr{F}_x$ generate $\mathscr{F}_x$ as an $\mathcal{O}_{X,x}$-module for every $x \in X$.
Equivalently, $\mathscr{F}$ is generated by global sections if and only if it is a quotient of a free sheaf: the surjection $\bigoplus_{i \in I} \mathcal{O}_X \twoheadrightarrow \mathscr{F}$ sending the ith generator to s_i is surjective on stalks, hence surjective as a sheaf morphism.

Example 5.3.25: Generated by Global Sections

1. *Affine schemes.* Let $X = \operatorname{Spec} R$ and $\mathscr{F} = \widetilde{M}$. Choose any set of generators $\{m_i\}$ for M as an R-module. The corresponding global sections generate $\mathscr{F}$, since generation is checked on stalks (localizations), and generators of M remain generators after localization. Thus *every* quasi-coherent sheaf on an affine scheme is generated by global sections.

2. *The twisting sheaf.* Let $X = \operatorname{Proj} S_\bullet$ where $S_\bullet$ is generated by S_1 over S_0. The degree-1 elements $f_0, \dots, f_r \in S_1$ define global sections of $\mathcal{O}_X(1)$ (they lie in $\Gamma(X, \mathcal{O}_X(1)) \supseteq S_1$). On $D_+(f_i)$, the section f_i generates the free module $\mathcal{O}_X(1)|_{D_+(f_i)} \cong \mathcal{O}_{D_+(f_i)}$. Since the $D_+(f_i)$ cover X, the sections $f_0, \dots, f_r$ generate $\mathcal{O}_X(1)$.

3. *Negative twists.* The sheaf $\mathcal{O}_{\mathbb{P}^r_k}(-1)$ is *not* generated by global sections for $r \geq 1$: it has no nonzero global sections at all, as we computed in definition 5.3.4.

The next theorem is one of the most important results about coherent sheaves on projective schemes: after twisting by a sufficiently large power of a very ample sheaf, any coherent sheaf becomes generated by *finitely many* global sections. This should be compared with the affine case, where global generation is automatic but says nothing about projective schemes.

Theorem 5.3.26: Global Section Generation of Twisted Sheaves

Let X be a projective scheme over a Noetherian ring R, let $\mathcal{O}_X(1)$ be a very ample invertible sheaf on X, and let $\mathscr{F}$ be a coherent $\mathcal{O}_X$-module. Then there exists an integer n_0 such that for all $n \geq n_0$, the twisted sheaf $\mathscr{F}(n)$ is generated by a finite number of global sections.

***Proof*:**

Reduction to $X = \mathbb{P}^r_R$. Let $\iota : X \hookrightarrow \mathbb{P}^r_R$ be a closed immersion with $\iota^*(\mathcal{O}(1)) = \mathcal{O}_X(1)$. The pushforward $\iota_*\mathscr{F}$ is coherent on $\mathbb{P}^r_R$ (closed immersions are finite, hence preserve coherence), and $\iota_*(\mathscr{F}(n)) \cong (\iota_*\mathscr{F})(n)$ by the projection formula, $\mathcal{O}(n)$ being invertible and $\mathscr{F}(n) = \mathscr{F} \otimes \iota^*\mathcal{O}(n)$. Moreover, $\mathscr{F}(n)$ is generated by global sections if and only if $\iota_*(\mathscr{F}(n))$ is (their global sections agree, and generation is checked on the support). We may therefore assume $X = \mathbb{P}^r_R = \operatorname{Proj} R[x_0, \dots, x_r]$.

Local generators. Cover X by the standard affine charts $U_i = D_+(x_i)$ for $i = 0, \dots, r$. Since $\mathscr{F}$ is coherent, on each U_i there is a finitely generated $(S_{x_i})_0$-module M_i with $\mathscr{F}|_{U_i} \cong \widetilde{M_i}$. Choose finite sets of generators $\{s_{i,1}, \dots, s_{i,k_i}\} \subseteq M_i$.

Extending to global sections. Each $s_{i,j}$ is a section of $\mathscr{F}$ over the affine open $U_i = D_+(x_i)$. By lemma 5.3.11(2) (with $\mathcal{L} = \mathcal{O}_X(1)$ and $f = x_i$), there exists $n_{i,j} > 0$ such that $x_i^{n_{i,j}} s_{i,j}$ extends to a global section $t_{i,j} \in \Gamma(X, \mathscr{F}(n_{i,j}))$. Taking $n_0 = \max_{i,j} n_{i,j}$ (and multiplying by further powers of x_i where needed), we may assume all extensions live in $\Gamma(X, \mathscr{F}(n_0))$.

Generation. The twisted sheaf $\mathscr{F}(n_0)$ restricts on U_i to a module M_i', and the trivialization $\mathcal{O}_X(n_0)|_{U_i} \cong \mathcal{O}_{U_i}$ (via multiplication by $x_i^{n_0}$) gives an isomorphism $M_i \xrightarrow{\sim} M_i'$ sending $s_{i,j}$ to $x_i^{n_0} s_{i,j}$. Since $\{s_{i,j}\}_j$ generates M_i, the sections $\{x_i^{n_0} s_{i,j}\}_j$ generate M_i'. Hence the global sections $\{t_{i,j}\}_{i,j}$ generate $\mathscr{F}(n_0)$ on every chart U_i, and therefore everywhere. There are finitely many such sections (at most $\sum_{i=0}^r k_i$).

For $n > n_0$, the sheaf $\mathscr{F}(n) = \mathscr{F}(n_0)(n - n_0)$ is generated by products of the $t_{i,j}$ with degree-$(n - n_0)$ monomials in the x_i, which are again finitely many.

This result has two immediate and important consequences.

Corollary 5.3.27: Coherent Sheaves as Quotients of Twisted Free Sheaves

Let X be a projective scheme over a Noetherian ring R and $\mathscr{F}$ a coherent sheaf on X. Then $\mathscr{F}$ can be written as a quotient of a sheaf of the form $\bigoplus_{i=1}^{m} \mathcal{O}_X(n_i)$ for finitely many integers n_i.

***Proof*:**
By theorem 5.3.26, there exist $n \geq 0$ and a surjection $\mathcal{O}_X^m \twoheadrightarrow \mathscr{F}(n)$. Tensoring with $\mathcal{O}_X(-n)$ gives a surjection $\mathcal{O}_X(-n)^m \twoheadrightarrow \mathscr{F}$.

This should be viewed as a projective analogue of the fact that every finitely generated module over a ring is a quotient of a free module. On projective space, free modules are replaced by direct sums of line bundles. Iterating, one obtains a *free resolution* of $\mathscr{F}$ by direct sums of twisted structure sheaves — the projective analogue of a free resolution of a module. When $R = k$ is a field and $\mathscr{F} \cong \widetilde{M}$ for a finitely generated graded $k[x_0, \ldots, x_r]$-module M, choosing homogeneous generators at each step gives a resolution $F_r \to \cdots \to F_0 \to M \to 0$ by finitely generated graded free modules. The ring $k[x_0, \ldots, x_r]$ has global dimension $r+1$ ([18, Hilbert's Syzygy Theorem]), so the kernel K of $F_r \to F_{r-1}$ is projective. It is also finitely generated and graded, and a finitely generated graded projective module P over $k[x_0, \ldots, x_r]$ is graded free. Indeed, with $\mathfrak{m} = (x_0, \ldots, x_r)$, lifting a homogeneous k-basis of $P/\mathfrak{m}P$ to homogeneous elements gives, by the graded Nakayama lemma, a surjection $F \to P$ from a graded free module that is an isomorphism modulo $\mathfrak{m}$. Its kernel L is a direct summand of F with $L/\mathfrak{m}L = 0$, so $L = 0$ by graded Nakayama again. Hence $0 \to K \to F_r \to \cdots \to F_0 \to M \to 0$ is a graded free resolution of length at most $r+1$, and applying $\widetilde{(\cdot)}$ to it gives a resolution of $\mathscr{F}$ of the same length on $\mathbb{P}_k^r$.

Finite Generation of Global Sections

On affine schemes, coherence of $\widetilde{M}$ corresponds to finite generation of M as an R-module. In particular, $\Gamma(\operatorname{Spec} R, \widetilde{M})$ is finitely generated. On projective schemes, global sections are *not* the same as the underlying module, so finite generation requires a separate argument. The following theorem provides it:

Theorem 5.3.28: Finite Generation of Coherent Sheaves over Projective Schemes

Let k be a field, R a finitely generated k-algebra, X a projective R-scheme, and $\mathscr{F}$ a coherent $\mathcal{O}_X$-module. Then $\Gamma(X, \mathscr{F})$ is a finitely generated R-module.
In particular, if $R = k$, then $\Gamma(X, \mathscr{F})$ is a finite-dimensional k-vector space.

The contrast with the affine case is stark: $\Gamma(\mathbb{A}_k^n, \mathcal{O}) = k[x_1, \ldots, x_n]$ is infinite-dimensional over k, reflecting the fact that $\mathbb{A}_k^n$ is not projective. The finiteness of global sections is a manifestation of the "compactness" of projective schemes.

***Proof*:**
By definition 2.3.4, we may write $X = \operatorname{Proj} S_\bullet$, over $\operatorname{Spec} R$, where $S_0 = R$ and $S_\bullet$ is finitely generated by S_1 as an S_0-algebra. Let $M_\bullet = \Gamma_*(\mathscr{F})$. By proposition 5.3.14, $\widetilde{M} \cong \mathscr{F}$. The proof proceeds in three steps: we first replace $M_\bullet$ by a finitely generated submodule, then reduce to a special case via filtration, and finally prove that special case using integral dependence.

Step 1: Replacing $M_\bullet$ by a finitely generated submodule. By theorem 5.3.26, there exists n_0 such that $\mathscr{F}(n)$ is generated by finitely many global sections for all $n \geq n_0$. Let $M'_\bullet \subseteq M_\bullet$ be the graded submodule generated by these finitely many sections (for a single n_0). Then $M'_\bullet$ is a finitely generated $S_\bullet$-module, the inclusion $M' \hookrightarrow M$ induces $\widetilde{M'} \hookrightarrow \widetilde{M} = \mathscr{F}$, and twisting by n_0 gives $\widetilde{M'}(n_0) \hookrightarrow \mathscr{F}(n_0)$. Since the generators of M' in degree n_0 are exactly the global sections generating $\mathscr{F}(n_0)$, this inclusion is an isomorphism in degree n_0, hence an isomorphism of sheaves (a map of quasi-coherent sheaves that is an isomorphism after twisting is an isomorphism). Twisting back gives $\widetilde{M'} \cong \mathscr{F}$.

We have shown: every coherent sheaf on $\operatorname{Proj} S_\bullet$ is the tilde of a finitely generated graded $S_\bullet$-module. It remains to show that if $M_\bullet$ is finitely generated, then $\Gamma(X, \widetilde{M}) = (M_\bullet)_0$ ... but this is *not* true in general; $\Gamma(X, \widetilde{M})$ can differ from M_0. Instead, we must show $\Gamma(X, \widetilde{M})$ is finitely generated over R directly.

Step 2: Reduction via filtration. Since $S_\bullet$ is a Noetherian ring and $M_\bullet$ is a finitely generated graded $S_\bullet$-module, there exists a finite filtration of graded submodules (see [10])

$$0 = M^{(0)} \subseteq M^{(1)} \subseteq \cdots \subseteq M^{(r)} = M_\bullet$$

such that each successive quotient $M^{(i)}/M^{(i-1)}$ is isomorphic to $(S_\bullet/\mathfrak{p}_i)(n_i)$ for some homogeneous prime $\mathfrak{p}_i$ and integer n_i. Applying the tilde construction and taking global sections, the short exact sequences $0 \to \widetilde{M^{(i-1)}} \to \widetilde{M^{(i)}} \to \widetilde{M^{(i)}/M^{(i-1)}} \to 0$ give left-exact sequences

$$0 \longrightarrow \Gamma(X, \widetilde{M^{(i-1)}}) \longrightarrow \Gamma(X, \widetilde{M^{(i)}}) \longrightarrow \Gamma(X, \widetilde{M^{(i)}/M^{(i-1)}}).$$

By induction, it suffices to show that each $\Gamma(X, \widetilde{S_\bullet/\mathfrak{p}}(n))$ is finitely generated over R.

Step 3: The integral domain case. We are thus reduced to proving the following: let $S_\bullet$ be a graded integral domain, finitely generated by $S_1 = Rx_0 + \cdots + Rx_r$ over $S_0 = R$ (a finitely generated k-algebra), and $X = \operatorname{Proj} S_\bullet$. Then $\Gamma(X, \mathcal{O}_X(n))$ is a finitely generated R-module for each $n \in \mathbb{Z}$.

Since $S_\bullet$ is a domain, multiplication by any x_i gives an injection $S(n) \hookrightarrow S(n+1)$, hence an injection $\Gamma(X, \mathcal{O}(n)) \hookrightarrow \Gamma(X, \mathcal{O}(n+1))$. It therefore suffices to prove finite generation for all $n \geq 0$.

Set $S'_\bullet = \bigoplus_{n \geq 0} \Gamma(X, \mathcal{O}_X(n)) = \Gamma_*(\mathcal{O}_X)$. This is a graded ring containing $S_\bullet$ (via the natural map $S_n \to \Gamma(X, \mathcal{O}(n))$) and contained in $\bigcap_i S_{x_i}$ (each global section restricts to a section on each $D_+(x_i)$).

$S'_\bullet$ is integral over $S_\bullet$. Let $s' \in S'_d$ for some $d \geq 0$. Since $s' \in S_{x_i}$ for each i, there exists N (independent of i, by taking a maximum) such that $x_i^N s' \in S_\bullet$. Since the monomials in $x_0, \ldots, x_r$ of degree N generate S_N, we may enlarge N so that $y \cdot s' \in S_\bullet$ for all $y \in S_{\geq N} := \bigoplus_{e \geq N} S_e$.

By induction, $y \cdot (s')^q \in S_{\geq N}$ for all $q \geq 1$ and all $y \in S_{\geq N}$. Taking $y = x_0^N$ gives $(s')^q \in x_0^{-N} \cdot S_\bullet$ for all $q \geq 1$. The module $x_0^{-N} \cdot S_\bullet$ is a finitely generated $S_\bullet$-submodule of the quotient field of $S_\bullet$. By the standard criterion for integral dependence (see [10, Integral Element Equivalences]), s' is integral over $S_\bullet$.

Conclusion. Since $S_\bullet$ is a finitely generated k-algebra (hence a Nagata ring), its integral closure in the quotient field is a finitely generated $S_\bullet$-module ([10, Finiteness of Integral Closure]). Since

$S'_\bullet$ is contained in this integral closure, it is a finitely generated $S_\bullet$-module as well[a]. In particular, each graded piece $S'_n = \Gamma(X, \mathcal{O}_X(n))$ is a finitely generated S_0-module, i.e., a finitely generated R-module.

[a]More precisely, $S'_\bullet$ is a graded submodule of a finitely generated module over the Noetherian ring $S_\bullet$, hence itself finitely generated.

Corollary 5.3.29: Pushforward of Coherent Sheaf by Projective Morphism

Let $f : X \to Y$ be a projective morphism of schemes of finite type over a field k. Let $\mathscr{F}$ be a coherent sheaf on X. Then $f_*\mathscr{F}$ is coherent on Y.

Recall that this fails for non-projective morphisms: example 5.1.19 showed that the pushforward of $\mathcal{O}_{\mathbb{A}^1}$ along $\mathbb{A}^1 \to \operatorname{Spec} k$ is $k[x]$, which is not finitely generated over k.

***Proof*:**
The question is local on Y, so we may assume $Y = \operatorname{Spec} R$ where R is a finitely generated k-algebra. Since f is projective, X is a projective R-scheme. By proposition 5.1.20, $f_*\mathscr{F}$ is quasi-coherent, so $f_*\mathscr{F} \cong \widetilde{N}$ for $N = \Gamma(Y, f_*\mathscr{F}) = \Gamma(X, \mathscr{F})$. By theorem 5.3.28, N is a finitely generated R-module, so $f_*\mathscr{F}$ is coherent.

5.3.4 Ample Sheaves

Very ampleness of definition 5.3.22 is tied to a chosen projective embedding, and theorem 5.3.26 extracts from it the property that gets used: twisting far enough makes a coherent sheaf globally generated. That property is worth isolating, because it survives on schemes that are not projective (where no very ample sheaf need exist) and because it is exactly what is needed to resolve coherent sheaves by locally free ones.

Definition 5.3.30: Ample Invertible Sheaf

Let X be a Noetherian scheme and $\mathcal{L}$ an invertible sheaf on X. Then $\mathcal{L}$ is *ample* if for every coherent sheaf $\mathscr{F}$ on X there is an integer n_0 such that $\mathscr{F} \otimes \mathcal{L}^{\otimes n}$ is generated by its global sections for every $n \geq n_0$.

Ampleness asks nothing about an embedding, and nothing about properness. On a projective scheme it is implied by very ampleness, which is the content of the theorem already proved.

Proposition 5.3.31: Very Ample Implies Ample

Let X be a projective scheme over a Noetherian ring R and let $\mathcal{O}_X(1)$ be very ample relative to $\operatorname{Spec} R$. Then $\mathcal{O}_X(1)$ is ample.

***Proof*:**
Very ampleness gives a closed immersion $\iota\colon X \to \mathbb{P}^N_R$ with $\mathcal{O}_X(1) = \iota^*\mathcal{O}(1)$ (definition 5.3.22). Then X is Noetherian: $\mathbb{P}^N_R$ is covered by finitely many copies of $\operatorname{Spec} R[x_1, \ldots, x_N]$, Noetherian by the Hilbert basis theorem, and a closed subscheme of a Noetherian scheme is Noetherian.

Pullback is monoidal, so $\mathcal{O}_X(n) = \iota^*\big(\mathcal{O}(1)^{\otimes n}\big) \cong \mathcal{O}_X(1)^{\otimes n}$, without needing X presented as a Proj. With that identification theorem 5.3.26 is definition 5.3.30 verbatim.

The first payoff is a version of corollary 5.3.27 that does not mention twists of a projective embedding.

Proposition 5.3.32: Coherent Sheaves as Quotients of Locally Free Sheaves

Let X be a Noetherian scheme carrying an ample invertible sheaf $\mathcal{L}$, and let $\mathscr{F}$ be coherent on X. Then there are an integer n and a surjection

$$\big(\mathcal{L}^{\otimes -n}\big)^{\oplus m} \twoheadrightarrow \mathscr{F}$$

for some m. In particular every coherent sheaf on X is a quotient of a locally free sheaf of finite rank.

***Proof*:**
Choose n with $\mathscr{G} = \mathscr{F} \otimes \mathcal{L}^{\otimes n}$ generated by its global sections, and let $U_1, \dots, U_r$ be a finite affine cover, available because a Noetherian scheme is quasi-compact. Write $U_i = \operatorname{Spec} A_i$ and $\mathscr{G}|_{U_i} = \widetilde{M_i}$ with M_i finitely generated. The A_i-submodule $M_i' \subseteq M_i$ generated by the restrictions of all global sections has $\widetilde{M_i'} \to \widetilde{M_i}$ surjective on stalks, by global generation, so $M_i' = M_i$; as M_i is finitely generated, finitely many global sections already generate $\mathscr{G}$ over U_i. Collecting these finite sets over the r charts gives global sections $s_1, \dots, s_m$ generating $\mathscr{G}$ everywhere, hence a surjection $\mathcal{O}_X^{\oplus m} \twoheadrightarrow \mathscr{F} \otimes \mathcal{L}^{\otimes n}$, which twists to the one asserted.

Ampleness is also insensitive to passing to a power, which is what lets a definition phrased through "some multiple" agree with definition 5.3.30.

Lemma 5.3.33: Ampleness Descends from a Power

Let X be a Noetherian scheme and $\mathcal{L}$ an invertible sheaf. If $\mathcal{L}^{\otimes n}$ is ample for some $n \geq 1$, then $\mathcal{L}$ is ample.

***Proof*:**
Let $\mathscr{F}$ be coherent. Apply ampleness of $\mathcal{L}^{\otimes n}$ to each of the n coherent sheaves $\mathscr{F}, \mathscr{F} \otimes \mathcal{L}, \dots, \mathscr{F} \otimes \mathcal{L}^{\otimes (n-1)}$, obtaining a single m_0 such that $\mathscr{F} \otimes \mathcal{L}^{\otimes r} \otimes \mathcal{L}^{\otimes nm}$ is globally generated for every $0 \leq r < n$ and every $m \geq m_0$. Every integer $s \geq nm_0 + n$ is $nm + r$ with $m \geq m_0$ and $0 \leq r < n$, so $\mathscr{F} \otimes \mathcal{L}^{\otimes s}$ is globally generated for all such s.

Whether a quasi-projective scheme carries an ample sheaf is a genuine question: the obvious candidate, the restriction of $\mathcal{O}(1)$ along an immersion into projective space, cannot be handled by theorem 5.3.26 directly, because that theorem needs the scheme to be projective. The missing step is that a coherent sheaf on an open subscheme is the restriction of a coherent sheaf on the whole. That is lemma 5.3.35 below, reached through the next lemma, and it is the reason Noetherian hypotheses are being carried.

Lemma 5.3.34: Quasi-Coherent Sheaves are Unions of Coherent Subsheaves

Let X be a Noetherian scheme and $\mathscr{G}$ a quasi-coherent sheaf on X. Then the coherent subsheaves of $\mathscr{G}$ form a directed family whose union is $\mathscr{G}$.

***Proof*:**
The sum of two coherent subsheaves is the image of a map out of a coherent sheaf, hence coherent, so the family is directed. A subsheaf whose stalks all agree with those of $\mathscr{G}$ is $\mathscr{G}$, so it suffices to produce, for each point x and each germ at x, a coherent subsheaf containing it. Choosing an affine open $U \ni x$ and a section of $\mathscr{G}$ over U with that germ, and letting $\mathscr{H}$ be the coherent subsheaf of $\mathscr{G}|_U$ it generates, the claim reduces to the following extension statement.

Claim. For every open $U \subseteq X$ and every coherent $\mathscr{H} \subseteq \mathscr{G}|_U$ there is a coherent $\mathscr{H}' \subseteq \mathscr{G}$ with $\mathscr{H}'|_U = \mathscr{H}$.

Step 1: enlarging from an open into an affine. Let $V = \operatorname{Spec} B$ be affine, $\mathscr{G}|_V = \widetilde{N}$, let $W \subseteq V$ be open, and let $\mathscr{K} \subseteq \mathscr{G}|_W$ be coherent. We find a coherent $\mathscr{K}' \subseteq \mathscr{G}|_V$ with $\mathscr{K}'|_W \supseteq \mathscr{K}$. Cover W by finitely many basic opens $D(f_1), \dots, D(f_s)$, possible because W is quasi-compact. Each $D(f_i)$ is affine, so the coherent sheaf $\mathscr{K}|_{D(f_i)}$ is generated by finitely many sections $s_{i1}, \dots, s_{i,m_i} \in \mathscr{K}(D(f_i))$, and these live in $\Gamma(D(f_i), \widetilde{N}) = N_{f_i}$; write $s_{ij} = n_{ij}/f_i^{k_{ij}}$ with $n_{ij} \in N$. Let $\mathscr{K}'$ be the coherent subsheaf of $\widetilde{N}$ generated by the finitely many n_{ij}. On $D(f_i)$ the element f_i is invertible, so $\mathscr{K}'$ contains every s_{ij} there and hence $\mathscr{K}'|_{D(f_i)} \supseteq \mathscr{K}|_{D(f_i)}$. The $D(f_i)$ cover W, so $\mathscr{K}'|_W \supseteq \mathscr{K}$.

Step 2: trimming back down. Keep the notation of Step 1. The subsheaf of $\mathscr{K}'$ of sections restricting into $\mathscr{K}$ over W is the kernel of $\mathscr{K}' \to \iota_*\big((\mathscr{G}|_W)/\mathscr{K}\big)$, where $\iota\colon W \to V$ is the inclusion; that pushforward is quasi-coherent by proposition 5.1.20, ι being quasi-compact (W is Noetherian) and separated (an open immersion), so the kernel $\mathscr{K}''$ is quasi-coherent. Restriction to an open is exact and $(\iota_*\mathcal{A})|_W = \mathcal{A}$, so $\mathscr{K}''|_W = \mathscr{K}'|_W \cap \mathscr{K} = \mathscr{K}$ by Step 1. Applying Step 1 once more, with $\mathscr{K}''$ in place of $\mathscr{G}$ (legitimate, since Step 1 uses only the restriction of its input to V) produces a coherent $\mathrm{SL} \subseteq \mathscr{K}''$ with $\mathrm{SL}|_W \supseteq \mathscr{K}$; since also $\mathrm{SL}|_W \subseteq \mathscr{K}''|_W = \mathscr{K}$, we get $\mathrm{SL}|_W = \mathscr{K}$ exactly.

Step 3: induction over an affine cover. Fix a finite affine cover $X = V_1 \cup \cdots \cup V_r$ and set $X_j = U \cup V_1 \cup \cdots \cup V_j$, so $X_0 = U$ and $X_r = X$. Suppose inductively that $\mathscr{H}_j \subseteq \mathscr{G}|_{X_j}$ is coherent with $\mathscr{H}_j|_U = \mathscr{H}$, the case $j = 0$ being $\mathscr{H}$ itself. Apply Step 2 with $V = V_{j+1}$, $W = X_j \cap V_{j+1}$ and $\mathscr{K} = \mathscr{H}_j|_W$: it yields a coherent $\mathrm{SL} \subseteq \mathscr{G}|_{V_{j+1}}$ with $\mathrm{SL}|_W = \mathscr{H}_j|_W$. Since W is the entire overlap of X_j and V_{j+1}, and the two agree there as subsheaves of $\mathscr{G}$, they glue to a coherent $\mathscr{H}_{j+1} \subseteq \mathscr{G}|_{X_{j+1}}$ with $\mathscr{H}_{j+1}|_{X_j} = \mathscr{H}_j$, hence with $\mathscr{H}_{j+1}|_U = \mathscr{H}$ because $U \subseteq X_j$. After r steps we reach X, which proves the claim and with it the lemma.

Lemma 5.3.35: Extending a Coherent Sheaf Across an Open Immersion

Let X be a Noetherian scheme, $j\colon U \hookrightarrow X$ an open immersion, and $\mathscr{F}$ a coherent sheaf on U. Then there is a coherent sheaf $\overline{\mathscr{F}}$ on X with $\overline{\mathscr{F}}\big|_U \cong \mathscr{F}$.

***Proof*:**
The immersion j is quasi-compact and separated, X being Noetherian, so $j_*\mathscr{F}$ is quasi-coherent by proposition 5.1.20, and $j^*j_*\mathscr{F} = \mathscr{F}$. By lemma 5.3.34 the sheaf $j_*\mathscr{F}$ is the directed union of its coherent subsheaves $\mathscr{H}_\lambda$; restricting, the $\mathscr{H}_\lambda|_U$ are a directed family of subsheaves of $\mathscr{F}$ whose stalks exhaust those of $\mathscr{F}$.

Cover U by finitely many affine opens $U_1, \dots, U_t$ on each of which $\mathscr{F}$ is generated by finitely many sections. Fix i and one such section s. Each germ of s lies in some $\mathscr{H}_\lambda$, so U_i is covered by opens on each of which s is a section of some $\mathscr{H}_\lambda|_U$; as U_i is quasi-compact, finitely many suffice, and directedness gives a single λ working on all of them, whence $s \in \mathscr{H}_\lambda(U_i)$ by the sheaf axiom for the subsheaf $\mathscr{H}_\lambda$. Running over the finitely many sections and the finitely many U_i, and using directedness once more, gives a single λ with $\mathscr{H}_\lambda|_U = \mathscr{F}$. Then $\overline{\mathscr{F}} = \mathscr{H}_\lambda$ serves.

Proposition 5.3.36: Quasi-Projective Schemes Carry an Ample Sheaf

Let X be a scheme quasi-projective over a Noetherian ring R, that is, admitting an immersion $\iota\colon X \to \mathbb{P}^N_R$. Then $\mathcal{O}_X(1) = \iota^*\mathcal{O}(1)$ is ample.

***Proof*:**
The scheme $\mathbb{P}^N_R$ is Noetherian, being covered by finitely many copies of $\operatorname{Spec} R[x_1, \dots, x_N]$, so its locally closed subscheme X is Noetherian too and definition 5.3.30 applies to X. Every morphism out of a Noetherian scheme is quasi-compact and quasi-separated, so ι is both. Factor ι as a closed immersion $X \to V$ followed by an open immersion $V \subseteq \mathbb{P}^N_R$ (definition 3.1.27), and let $\overline{X}$ be the scheme-theoretic image of ι (definition 5.2.18). Because ι is quasi-compact and quasi-separated, $\iota_*\mathcal{O}_X$ is quasi-coherent (proposition 5.1.20), hence so is the kernel of $\mathcal{O}_{\mathbb{P}^N_R} \to \iota_*\mathcal{O}_X$, kernels of maps of quasi-coherent sheaves being quasi-coherent; a quasi-coherent ideal sheaf cuts out a closed subscheme (proposition 5.1.34), so $\overline{X}$ is a genuine closed subscheme of $\mathbb{P}^N_R$, and being closed in $\mathbb{P}^N_R$ it is projective over R. Write $\bar{\iota}\colon \overline{X} \to \mathbb{P}^N_R$ for that closed immersion and $\mathcal{O}_{\overline{X}}(1) = \bar{\iota}^*\mathcal{O}(1)$.

Forming the scheme-theoretic image commutes with restricting to an open of the target: restriction is exact and $(\iota_*\mathcal{O}_X)|_V = \iota'_*\mathcal{O}_X$ for the induced $\iota'\colon X \to V$, so the ideal sheaf of $\overline{X}$ restricts to that of the scheme-theoretic image of ι'. As ι' is a closed immersion it is its own scheme-theoretic image, so $\overline{X} \cap V = X$. Hence X is an open subscheme of $\overline{X}$, and $\mathcal{O}_{\overline{X}}(1)$ restricts to $\mathcal{O}_X(1)$.

Let $\mathscr{F}$ be coherent on X. By lemma 5.3.35 it extends to a coherent $\overline{\mathscr{F}}$ on $\overline{X}$. By theorem 5.3.26 there is n_0 with $\overline{\mathscr{F}} \otimes \mathcal{O}_{\overline{X}}(1)^{\otimes n}$ globally generated for all $n \geq n_0$, and global generation is inherited by the restriction to the open subscheme X. Hence $\mathscr{F} \otimes \mathcal{O}_X(1)^{\otimes n}$ is globally generated for $n \geq n_0$, which is definition 5.3.30.

The second payoff is the one that matters for K-theory and for Riemann–Roch: on a regular scheme, iterating proposition 5.3.32 terminates.

Proposition 5.3.37: Finite Locally Free Resolutions on a Regular Scheme

Let X be a regular Noetherian scheme of finite Krull dimension d carrying an ample invertible sheaf. Then every coherent sheaf $\mathscr{F}$ on X admits a resolution

$$0 \longrightarrow \mathcal{E}_d \longrightarrow \mathcal{E}_{d-1} \longrightarrow \cdots \longrightarrow \mathcal{E}_0 \longrightarrow \mathscr{F} \longrightarrow 0$$

by locally free sheaves of finite rank.

***Proof*:**
If $d = 0$ every local ring is a field, so $\mathscr{F}$ has free stalks and Step 3 below already makes it locally free; assume $d \geq 1$.

Step 1: build d steps. By proposition 5.3.32 choose a surjection $\mathcal{E}_0 \twoheadrightarrow \mathscr{F}$ with $\mathcal{E}_0$ locally free of finite rank. Its kernel is coherent, X being Noetherian, so the construction iterates; after d steps we have an exact sequence

$$0 \to \mathscr{K} \to \mathcal{E}_{d-1} \to \cdots \to \mathcal{E}_0 \to \mathscr{F} \to 0$$

with all $\mathcal{E}_i$ locally free of finite rank and $\mathscr{K}$ coherent.

Step 2: the last syzygy is locally free. Fix $x \in X$. The local ring $\mathcal{O}_{X,x}$ is regular of dimension at most d, so $\operatorname{pd}_{\mathcal{O}_{X,x}} \mathscr{F}_x \leq d$ by [18, Global Dimension of a Regular Local Ring]; in particular $\operatorname{Ext}^{d+1}_{\mathcal{O}_{X,x}}(\mathscr{F}_x, -) = 0$. Localizing the sequence at x makes each $\mathcal{E}_{i,x}$ free, so dimension shifting along it gives $\operatorname{Ext}^1(\mathscr{K}_x, -) \cong \operatorname{Ext}^{d+1}(\mathscr{F}_x, -) = 0$, and $\mathscr{K}_x$ is a projective $\mathcal{O}_{X,x}$-module, hence free by [10, Projective Module over Local Ring], being finitely generated.

Step 3: free stalks give local freeness. Lift a basis of $\mathscr{K}_x$ to sections of $\mathscr{K}$ on a neighbourhood of x, giving $\varphi\colon \mathcal{O}^{\oplus r} \to \mathscr{K}$ with φ_x an isomorphism. Both $\ker\varphi$ and $\operatorname{coker}\varphi$ are coherent with vanishing stalk at x, hence vanish on a neighbourhood of x. So $\mathscr{K}$ is locally free of finite rank, and setting $\mathcal{E}_d = \mathscr{K}$ completes the resolution.

Proposition 5.3.31 passes from very ample to ample. On a proper scheme the converse holds up to a power: the definition of ampleness mentions no embedding, yet some tensor power of an ample sheaf is the restriction of $\mathcal{O}(1)$ along a closed immersion into a projective space. Together with lemma 5.3.33, which descends ampleness from a power, it follows that on a scheme proper over a Noetherian ring an invertible sheaf is ample exactly when one of its powers is very ample. The power is what turns an ample sheaf on a complete variety into a projective embedding, the step on which the projectivity of abelian varieties rests (*Abelian Varieties* [5]).

Lemma 5.3.38: Ample Sheaves Give Affine Non-Vanishing Loci

Let X be a Noetherian scheme and $\mathcal{L}$ an ample invertible sheaf on X. Then for every point $x \in X$ there are an integer $n \geq 1$ and a section $s \in \Gamma(X, \mathcal{L}^{\otimes n})$ such that x lies in the non-vanishing locus X_s of lemma 5.3.11 and X_s is affine.

***Proof*:**
Step 1: an affine trivializing neighbourhood. Since $\mathcal{L}$ is invertible there is an open $V \ni x$ with $\mathcal{L}|_V \cong \mathcal{O}_V$. Choose an affine open $W \ni x$. The distinguished opens of W form a basis of its topology, so there is an affine open U with $x \in U \subseteq V \cap W$, and $\mathcal{L}|_U$ is free. Write $U = \operatorname{Spec} A$.

Step 2: the ideal sheaf of the complement. The complement $Y = X \setminus U$ is closed. Let $\mathcal{I}_Y \subseteq \mathcal{O}_X$ be the ideal sheaf cutting out the reduced closed subscheme structure on Y (exercise set 3.1.1): its sections over an open V' are the functions whose image in $\kappa(y)$ vanishes for every $y \in Y \cap V'$, and on an affine open it is the sheaf associated to the radical ideal of functions vanishing on the corresponding closed set, hence quasi-coherent. It is coherent because X is locally Noetherian (proposition 5.1.34). The constant 1 is a section of $\mathcal{I}_Y$ over $U = X \setminus Y$, so $\mathcal{I}_{Y,x} = \mathcal{O}_{X,x}$, while for $y \in Y$ every germ of $\mathcal{I}_Y$ at y lies in $\mathfrak{m}_y$.

Step 3: a section through x vanishing on Y. By definition 5.3.30 there is $n \geq 1$ such that $\mathcal{I}_Y \otimes \mathcal{L}^{\otimes n}$ is generated by its global sections. Its stalk at x is $\mathcal{L}_x^{\otimes n}$, free of rank one. If every global section had germ in $\mathfrak{m}_x \mathcal{L}_x^{\otimes n}$, global generation would give $\mathcal{L}_x^{\otimes n} = \mathfrak{m}_x \mathcal{L}_x^{\otimes n}$, forcing $\mathcal{L}_x^{\otimes n} = 0$ by [10, Nakayama's Lemma I], which is impossible for an invertible sheaf. So some $t \in \Gamma(X, \mathcal{I}_Y \otimes \mathcal{L}^{\otimes n})$ has germ at x outside $\mathfrak{m}_x \mathcal{L}_x^{\otimes n}$. Let $s \in \Gamma(X, \mathcal{L}^{\otimes n})$ be the image of t under the map $\mathcal{I}_Y \otimes \mathcal{L}^{\otimes n} \to \mathcal{L}^{\otimes n}$ obtained by tensoring the inclusion $\mathcal{I}_Y \subseteq \mathcal{O}_X$ with $\mathcal{L}^{\otimes n}$. Then $s_x = t_x \notin \mathfrak{m}_x \mathcal{L}_x^{\otimes n}$, so $x \in X_s$, while for every $y \in Y$ we have $s_y \in \mathcal{I}_{Y,y} \mathcal{L}_y^{\otimes n} \subseteq \mathfrak{m}_y \mathcal{L}_y^{\otimes n}$, so $y \notin X_s$. Hence $X_s \subseteq U$.

Step 4: X_s is a principal open of U. A trivialization $\mathcal{L}|_U \cong \mathcal{O}_U$ induces $\mathcal{L}^{\otimes n}|_U \cong \mathcal{O}_U$, under which $s|_U$ corresponds to some $g \in A$. For $u \in U$ the germ s_u generates $\mathcal{L}_u^{\otimes n}$ exactly when g_u is a unit of $\mathcal{O}_{X,u}$, that is exactly when $u \in D(g)$. Since $X_s \subseteq U$, this gives $X_s = D(g)$, a distinguished open of the affine scheme U, so X_s is affine.

Lemma 5.3.39: Closed Immersions Are Local on the Target

Let $f\colon Z \to T$ be a morphism of schemes and $\{T_\alpha\}$ an open cover of T such that each restriction $f^{-1}(T_\alpha) \to T_\alpha$ is a closed immersion. Then f is a closed immersion.

***Proof*:**
Both conditions of definition 3.1.22 are checked on the cover. The image $f(Z)$ meets each T_α in $f(f^{-1}(T_\alpha))$, which is closed in T_α, and a subset of T whose intersection with every member of an open cover is closed in that member is closed. The map f is injective because two points with the same image lie in a common $f^{-1}(T_\alpha)$, where the restriction is injective, and the inverse map $f(Z) \to Z$ is continuous because its restriction to each $f(Z) \cap T_\alpha$ inverts a homeomorphism. So f is a homeomorphism onto the closed subset $f(Z)$. Surjectivity of $f^{\#}\colon \mathcal{O}_T \to f_* \mathcal{O}_Z$ is a stalkwise condition, and over T_α the sheaf $f_* \mathcal{O}_Z$ restricts to the pushforward of $\mathcal{O}_{f^{-1}(T_\alpha)}$, whose stalks receive surjections by hypothesis.

Lemma 5.3.40: Charts of a Morphism to Projective Space

Let R be a ring, X an R-scheme, $\mathcal{L}$ an invertible sheaf on X, and $s_0, \dots, s_n \in \Gamma(X, \mathcal{L})$ global sections generating $\mathcal{L}$. Let $\varphi\colon X \to \mathbb{P}^n_R$ be the R-morphism they determine, with $\mathcal{L} \cong \varphi^*(\mathcal{O}(1))$ and $s_i = \varphi^*(x_i)$ (theorem 5.3.43). Then for each i:

1. $\varphi^{-1}(D_+(x_i)) = X_{s_i}$, the non-vanishing locus of s_i.
2. On X_{s_i} every s_j factors uniquely as $s_j = (s_j/s_i) \cdot s_i$ with $s_j/s_i \in \mathcal{O}_X(X_{s_i})$, and the restriction $X_{s_i} \to D_+(x_i) \cong \operatorname{Spec} R[x_0/x_i, \dots, x_n/x_i]$ of φ corresponds under proposition 3.1.12 to the R-algebra homomorphism sending x_j/x_i to s_j/s_i.

***Proof*:**
(1). Fix $x \in X$ and an index j with $\varphi(x) \in D_+(x_j)$. On $D_+(x_j)$ the sheaf $\mathcal{O}(1)$ is free with generator x_j (proposition 5.3.5), so $x_i = (x_i/x_j) \cdot x_j$ there for a function $x_i/x_j \in \mathcal{O}_{\mathbb{P}^n_R}(D_+(x_j))$, and for $p \in D_+(x_j)$ the germ of x_i generates $\mathcal{O}(1)_p$ exactly when $(x_i/x_j)_p$ is a unit, that is exactly when $p \in D_+(x_i)$. The trivialization of $\mathcal{O}(1)$ by x_j pulls back to a trivialization of $\varphi^*\mathcal{O}(1)$ by $\varphi^*(x_j) = s_j$ on $\varphi^{-1}(D_+(x_j))$, so s_j generates $\mathcal{L}$ there and $s_i = \varphi^{\#}(x_i/x_j) \cdot s_j$ on $\varphi^{-1}(D_+(x_j))$. The stalk homomorphism $\mathcal{O}_{\mathbb{P}^n_R,\varphi(x)} \to \mathcal{O}_{X,x}$ is local, so it carries units to units and $\mathfrak{m}_{\varphi(x)}$ into $\mathfrak{m}_x$. Hence $s_{i,x}$ generates $\mathcal{L}_x$ exactly when $(x_i/x_j)_{\varphi(x)}$ is a unit, that is exactly when $\varphi(x) \in D_+(x_i)$. As the charts $D_+(x_j)$ cover $\mathbb{P}^n_R$, this proves (1).

(2). On X_{s_i} the germ of s_i generates $\mathcal{L}$ at every point, so multiplication by s_i is an isomorphism $\mathcal{O}_{X_{s_i}} \to \mathcal{L}|_{X_{s_i}}$ and the ratios s_j/s_i exist and are unique. By (1) the morphism φ carries X_{s_i} into $D_+(x_i) \cong \operatorname{Spec}(S_{x_i})_0 = \operatorname{Spec} R[x_0/x_i, \dots, x_n/x_i]$ (lemma 2.3.2), and a morphism into an affine scheme corresponds to the ring homomorphism it induces on global sections (proposition 3.1.12). On $D_+(x_i)$ there is the factorization $x_j = (x_j/x_i) \cdot x_i$, which pulls back to $s_j = \varphi^{\#}(x_j/x_i) \cdot s_i$ on X_{s_i}, so $\varphi^{\#}(x_j/x_i) = s_j/s_i$ by uniqueness of the ratio.

Theorem 5.3.41: Ample Sheaves Have Very Ample Powers

Let A be a Noetherian ring and let X be a scheme proper over $\operatorname{Spec} A$ carrying an ample invertible sheaf $\mathcal{L}$. Then there is an integer $m \geq 1$ such that $\mathcal{L}^{\otimes m}$ is very ample relative to $\operatorname{Spec} A$.

***Proof*:**
If $X = \emptyset$ then $X \to \mathbb{P}^0_A$ is a closed immersion, its image the empty closed set and its sheaf map the surjection onto the zero pushforward, and every sheaf on $\emptyset$ is zero, so the pullback condition of definition 5.3.22 holds and every power of $\mathcal{L}$ is very ample. Assume $X \neq \emptyset$.

Step 1: X is Noetherian. Properness includes finite type (definition 3.5.21), so lemma 3.5.27(1) applied to $X \to \operatorname{Spec} A$, the scheme $\operatorname{Spec} A$ being Noetherian (definition 2.4.1), makes X a Noetherian scheme in which every open subset is quasi-compact, and definition 5.3.30 applies to $\mathcal{L}$.

Step 2: a finite affine cover by non-vanishing loci in a single power. By lemma 5.3.38 and quasi-compactness there are finitely many sections $s'_i \in \Gamma(X, \mathcal{L}^{\otimes n_i})$, $i = 1, \dots, k$, whose affine non-vanishing loci cover X. Set $n = n_1 \cdots n_k$ and $s_i = (s'_i)^{n/n_i} \in \Gamma(X, \mathcal{L}^{\otimes n})$. A germ of s'_i

generates $\mathcal{L}^{\otimes n_i}$ at a point exactly when the corresponding power generates $\mathcal{L}^{\otimes n}$ there, since a generator raises to a generator of the tensor power and a germ in $\mathfrak{m}_x\mathcal{L}_x^{\otimes n_i}$ raises into $\mathfrak{m}_x\mathcal{L}_x^{\otimes n}$. Hence $X_{s_i} = X_{s_i'}$, so $X = X_{s_1} \cup \cdots \cup X_{s_k}$ with every $U_i := X_{s_i} = \operatorname{Spec} B_i$ affine.

Step 3: extending algebra generators. Each $B_i = \mathcal{O}_X(U_i)$ is a finitely generated A-algebra (definition 3.4.28). Fix generators $b_{i1}, \ldots, b_{ig_i}$ of B_i over A with $g_i \geq 1$, enlarging a generating set if necessary. The hypotheses of lemma 5.3.11(2) hold for the invertible sheaf $\mathcal{L}^{\otimes n}$, the section s_i and the quasi-coherent sheaf $\mathcal{O}_X$: the scheme X is quasi-compact, every point has an affine open neighbourhood on which $\mathcal{L}$, and hence $\mathcal{L}^{\otimes n}$, is free (Step 1 of the proof of lemma 5.3.38), finitely many such neighbourhoods cover X, and their pairwise intersections are quasi-compact, every open subset of X being quasi-compact by Step 1. For each pair (i, j) there is therefore $r_{ij} > 0$ such that $s_i^{r_{ij}} b_{ij} \in \Gamma(U_i, \mathcal{L}^{\otimes n r_{ij}})$ extends to a global section of $\mathcal{L}^{\otimes n r_{ij}}$. With $r = \max_{i,j} r_{ij}$, multiplying each extension by $s_i^{r - r_{ij}}$ produces sections $c_{ij} \in \Gamma(X, \mathcal{L}^{\otimes nr})$ with $c_{ij}|_{U_i} = s_i^r\, b_{ij}$.

Step 4: the morphism to projective space. The sections $\sigma_i := s_i^r$ of $\mathcal{L}^{\otimes nr}$ generate it: each x lies in some U_i, where the germ of s_i generates $\mathcal{L}^{\otimes n}$, so $\sigma_{i,x}$ generates $\mathcal{L}_x^{\otimes nr}$ as in Step 2. The list $(\sigma_1, \ldots, \sigma_k, c_{11}, \ldots, c_{kg_k})$ then also generates $\mathcal{L}^{\otimes nr}$, and theorem 5.3.43(2) with $R = A$ gives an A-morphism $\varphi\colon X \to \mathbb{P}_A^q$, where $q + 1 = k + \sum_i g_i$, with $\varphi^*\mathcal{O}(1) \cong \mathcal{L}^{\otimes nr}$, $\varphi^*(z_i) = \sigma_i$ and $\varphi^*(w_{ij}) = c_{ij}$, writing $z_1, \ldots, z_k, w_{11}, \ldots, w_{kg_k}$ for the homogeneous coordinates. Lemma 5.3.40(1) gives $\varphi^{-1}(D_+(z_i)) = X_{\sigma_i} = X_{s_i} = U_i$, the middle equality again by the power argument of Step 2.

Step 5: each chart is a closed immersion. By lemma 5.3.40(2) the restriction $U_i \to D_+(z_i) \cong \operatorname{Spec} A\big[\{z_l/z_i\}_{l \neq i}, \{w_{jm}/z_i\}\big]$ corresponds to the A-algebra homomorphism sending every coordinate ratio to the matching ratio of sections. In particular $w_{ij}/z_i \mapsto c_{ij}/\sigma_i = b_{ij}$, because $c_{ij} = b_{ij}\,\sigma_i$ on U_i and ratios are unique. The image of this ring homomorphism is an A-subalgebra of B_i containing every b_{ij}, hence is all of B_i, so the ring homomorphism is surjective and $U_i \to D_+(z_i)$ is a closed immersion (proposition 3.1.24(2)).

Step 6: the image is closed and φ is a closed immersion. The scheme $\mathbb{P}_A^q$ is Noetherian, its structure morphism being of finite type (example 3.4.31(2) and lemma 3.5.27(1)), and it is separated over $\operatorname{Spec} A$ (corollary 3.5.5). The composite of φ with the projection $\mathbb{P}_A^q \to \operatorname{Spec} A$ is the proper structure morphism of X, so φ is proper by proposition 3.5.23(5), the Noetherian standing hypothesis holding by Step 1 and the present step. In particular φ is a closed map and $\varphi(X)$ is closed in $\mathbb{P}_A^q$. Since the U_i cover X, the image lies in $D_+(z_1) \cup \cdots \cup D_+(z_k)$. The opens $D_+(z_1), \ldots, D_+(z_k)$ together with $\mathbb{P}_A^q \setminus \varphi(X)$ cover $\mathbb{P}_A^q$, the preimage of the last is empty, and the empty scheme maps to it by a closed immersion, being homeomorphic to the closed subset $\emptyset$ with sheaf map surjecting onto the zero pushforward. By Step 5 and lemma 5.3.39, φ is a closed immersion. With $\varphi^*\mathcal{O}(1) \cong \mathcal{L}^{\otimes nr}$, definition 5.3.22 makes $\mathcal{L}^{\otimes nr}$ very ample relative to $\operatorname{Spec} A$, so $m = nr$ serves.

Corollary 5.3.42: Proper Schemes with an Ample Sheaf Are Projective

Let A be a Noetherian ring and let X be a scheme proper over $\operatorname{Spec} A$ carrying an ample invertible sheaf. Then X is projective over $\operatorname{Spec} A$.

***Proof*:**
Theorem 5.3.41 produces an invertible sheaf on X very ample relative to $\operatorname{Spec} A$, and lemma 5.3.23 with $Y = \operatorname{Spec} A$ upgrades properness together with a very ample sheaf to projectivity.

Conversely, a scheme projective over $\operatorname{Spec} A$ is Noetherian, being a closed subscheme of the Noetherian $\mathbb{P}^n_A$ (example 3.4.31(2) and lemma 3.5.27(1)), hence proper over $\operatorname{Spec} A$ (theorem 3.5.32) and carries an ample invertible sheaf, the restriction of $\mathcal{O}(1)$ along the closed immersion (proposition 5.3.36). Over a Noetherian ring, projectivity is therefore the conjunction of properness and the existence of an ample invertible sheaf.

Properness cannot be dropped from theorem 5.3.41. On $X = \mathbb{A}^1_k$ the structure sheaf is ample, every quasi-coherent sheaf on an affine scheme being generated by its global sections (example 5.3.25), but no power of $\mathcal{O}_X$ is very ample in the sense of definition 5.3.22: a closed immersion $\mathbb{A}^1_k \to \mathbb{P}^n_k$ would make $\mathbb{A}^1_k$ projective, hence proper, over k (theorem 3.5.32), and it is not proper (example 3.5.26). Hartshorne defines very ample through an immersion that need not be closed, and with that definition every ample sheaf on a scheme of finite type over a Noetherian ring has a very ample power [36, p. II.7.6].

5.3.5 Characterizing Morphisms to Projective Space

With the notion of global generation in hand (definition 5.3.24), we can now make precise the correspondence between morphisms to projective space and line bundles with generating sections that was previewed in the introduction to this subsection.

Recall that for a line bundle $\mathcal{L}$ (locally free of rank 1), being generated by sections $s_0, \dots, s_n$ has a particularly clean geometric interpretation: at each point $x \in X$, the fibre $\mathcal{L}(x)$ is a one-dimensional $\kappa(x)$-vector space, and the sections span this fibre if and only if at least one s_i is nonzero at x. Equivalently, the *base locus* $\{x \in X : s_0(x) = \cdots = s_n(x) = 0\}$ is empty. This is exactly the condition ensuring that the "ratio map" $x \mapsto [s_0(x) : \cdots : s_n(x)]$ is well-defined at every point.

Before stating the theorem, let us trace through the construction on a concrete case. Consider the Veronese map $\mathbb{P}^1_k \to \mathbb{P}^2_k$, classically given by $[s : t] \mapsto [s^2 : st : t^2]$. In our language:

- The line bundle is $\mathcal{L} = \mathcal{O}_{\mathbb{P}^1}(2)$ on $\mathbb{P}^1_k$.
- The global sections are the three degree-2 monomials: $s_0 = x_0^2$, $s_1 = x_0x_1$, $s_2 = x_1^2$ in $\Gamma(\mathbb{P}^1_k, \mathcal{O}(2)) \cong k[x_0, x_1]_2$.
- At each point of $\mathbb{P}^1_k$, at least one of x_0^2, x_0x_1, x_1^2 is nonzero, so these sections generate $\mathcal{O}(2)$.
- On the open set $X_0 = D_+(x_0)$ where $s_0 = x_0^2$ does not vanish, the ratios are $s_1/s_0 = x_1/x_0$ and $s_2/s_0 = (x_1/x_0)^2$ — these are the "affine coordinates" of the map, sending $D_+(x_0)$ into the affine chart $U_0 \subseteq \mathbb{P}^2_k$ by $t \mapsto (t, t^2)$.

The image is the conic $V(y_0y_2 - y_1^2) \subseteq \mathbb{P}^2_k$, and the map is a closed immersion. The general theorem below formalizes this pattern.

Theorem 5.3.43: Determining Maps To Projective Space

Let R be a ring and X an R-scheme. Then:

1. *(From morphisms to line bundles.)* If $\varphi : X \to \mathbb{P}^n_R$ is an R-morphism, then $\mathcal{L} := \varphi^*(\mathcal{O}(1))$ is an invertible sheaf on X, and the pullback sections $s_i := \varphi^*(x_i) \in \Gamma(X, \mathcal{L})$ for $i = 0, \ldots, n$ generate $\mathcal{L}$.
2. *(From line bundles to morphisms.)* If $\mathcal{L}$ is an invertible sheaf on X and $s_0, \ldots, s_n \in \Gamma(X, \mathcal{L})$ are global sections that generate $\mathcal{L}$, then there exists a unique R-morphism $\varphi : X \to \mathbb{P}^n_R$ such that $\mathcal{L} \cong \varphi^*(\mathcal{O}(1))$ and $s_i = \varphi^*(x_i)$ under this isomorphism.

In particular, there is a natural bijection:

$$\mathrm{Hom}_R(X, \mathbb{P}^n_R) \overset{1:1}{\longleftrightarrow} \left\{ (\mathcal{L}, s_0, \ldots, s_n) \;\middle|\; \begin{array}{c} \mathcal{L} \text{ invertible on } X, \\ s_i \in \Gamma(X, \mathcal{L}) \text{ generating } \mathcal{L} \end{array} \right\} \Big/ \cong$$

where $(\mathcal{L}, s_0, \ldots, s_n) \cong (\mathcal{L}', s'_0, \ldots, s'_n)$ if there is an isomorphism $\alpha : \mathcal{L} \xrightarrow{\sim} \mathcal{L}'$ with $\alpha(s_i) = s'_i$ for all i.

***Proof*:**

Part (1): morphisms yield generating sections. Since pullback preserves locally free sheaves and their rank (proposition 5.3.8(3)), $\mathcal{L} = \varphi^*(\mathcal{O}(1))$ is an invertible sheaf on X. The homogeneous coordinates $x_0, \ldots, x_n$ generate $\mathcal{O}(1)$ on $\mathbb{P}^n_R$, and pullback preserves surjections, so the sections $s_i = \varphi^*(x_i)$ generate $\mathcal{L}$.

Part (2): generating sections yield a morphism. This is the substantial direction. We construct φ by defining it on an open cover and gluing.

Step 1: The open cover. For each $i = 0, \ldots, n$, define

$$X_i \;:=\; \{x \in X \mid (s_i)_x \notin \mathfrak{m}_x \mathcal{L}_x\},$$

the locus where s_i does not vanish. Since $\mathcal{L}$ is locally free of rank 1, the section s_i generates $\mathcal{L}_x$ as an $\mathcal{O}_{X,x}$-module at precisely the points where it maps to a nonzero element of the fibre. The generating hypothesis means that for each $x \in X$, some s_i does not vanish at x, so $X = X_0 \cup \cdots \cup X_n$.

Each X_i is open: on any affine open $\operatorname{Spec} A \subseteq X$ where $\mathcal{L}|_{\operatorname{Spec} A} \cong \mathcal{O}_{\operatorname{Spec} A}$, the section s_i corresponds to an element $a_i \in A$, and $X_i \cap \operatorname{Spec} A = D(a_i)$.

Step 2: Trivializing $\mathcal{L}$ on each X_i. On X_i, the section s_i generates $\mathcal{L}|_{X_i}$: for each $x \in X_i$, the element $(s_i)_x$ is a generator of the free rank-1 module $\mathcal{L}_x$. Therefore, the map

$$\psi_i : \mathcal{O}_{X_i} \xrightarrow{\sim} \mathcal{L}|_{X_i}, \qquad 1 \longmapsto s_i|_{X_i},$$

is an isomorphism of $\mathcal{O}_{X_i}$-modules. Intuitively, s_i provides a "local coordinate" for the line bundle on X_i.

Step 3: Constructing regular functions s_j/s_i. For each j, the section $s_j|_{X_i}$ is an element of

$\mathcal{L}(X_i)$. Via the trivialization ψ_i, it corresponds to a unique regular function:

$$\frac{s_j}{s_i} \;:=\; \psi_i^{-1}(s_j|_{X_i}) \;\in\; \mathcal{O}_X(X_i).$$

At each stalk, this is the unique element $f \in \mathcal{O}_{X,x}$ satisfying $(s_j)_x = f \cdot (s_i)_x$ in $\mathcal{L}_x$. Such f exists and is unique because $\mathcal{L}_x$ is free of rank 1 with generator $(s_i)_x$.

In particular, $s_i/s_i = 1$, and on $X_i \cap X_j$ the ratios satisfy the multiplicative relation

$$\frac{s_k}{s_i} \;=\; \frac{s_k}{s_j} \cdot \frac{s_j}{s_i} \qquad \text{on } X_i \cap X_j, \tag{5.3}$$

since $(s_k)_x = (s_k/s_j)_x \cdot (s_j)_x = (s_k/s_j)_x \cdot (s_j/s_i)_x \cdot (s_i)_x$ and generators are unique.

Step 4: Local morphisms. The ith standard affine chart of $\mathbb{P}^n_R$ is $U_i = D_+(x_i) \cong \operatorname{Spec} R[y_0, \ldots, \hat{y}_i, \ldots, y_n]$, where $y_j = x_j/x_i$. Define a ring homomorphism

$$\varphi_i^{\#} :\; R[y_0, \ldots, \hat{y}_i, \ldots, y_n] \;\longrightarrow\; \mathcal{O}_X(X_i), \qquad y_j \;\longmapsto\; \frac{s_j}{s_i}.$$

This is well-defined: each s_j/s_i is a regular function on X_i by Step 3, and the map extends R-linearly (since X is an R-scheme). By proposition 3.1.12, this determines a morphism of schemes $\varphi_i : X_i \to U_i$.

Step 5: Gluing. We must show that $\varphi_i|_{X_i \cap X_j} = \varphi_j|_{X_i \cap X_j}$ as morphisms into $\mathbb{P}^n_R$. The transition functions on $\mathbb{P}^n_R$ between charts U_i and U_j are given by $y_k^{(j)} = y_k^{(i)}/y_j^{(i)}$, where $y_k^{(i)} = x_k/x_i$. Under φ_i^*, we have $y_k^{(i)} \mapsto s_k/s_i$. Hence

$$\varphi_i^*\left(\frac{y_k^{(i)}}{y_j^{(i)}}\right) \;=\; \frac{s_k/s_i}{s_j/s_i} \;=\; \frac{s_k}{s_j} \;=\; \varphi_j^*(y_k^{(j)}),$$

using (5.3). Since the local morphisms agree on overlaps, they glue by lemma 3.1.9 to a unique morphism $\varphi : X \to \mathbb{P}^n_R$.

Step 6: Identifying $\mathcal{L} \cong \varphi^*(\mathcal{O}(1))$. On U_i, the sheaf $\mathcal{O}(1)$ is trivial with generator x_i (recall from proposition 5.3.5 that multiplication by x_i trivializes $\mathcal{O}(1)$ on $D_+(x_i)$). The pullback $\varphi^*(\mathcal{O}(1))|_{X_i}$ is therefore trivial with generator $\varphi^*(x_i) = s_i$. But $\mathcal{L}|_{X_i}$ is also trivial with generator s_i (by ψ_i). These identifications are compatible on overlaps: the transition function for $\varphi^*(\mathcal{O}(1))$ between X_i and X_j is $\varphi^*(x_j/x_i) = s_j/s_i$, which is the same as the transition function for $\mathcal{L}$ (since $\psi_j^{-1} \circ \psi_i$ sends $1 \mapsto s_i/s_j$, i.e., the transition function of $\mathcal{L}$ between the ψ_i and ψ_j trivializations is s_i/s_j). Hence the local isomorphisms glue to a global isomorphism $\mathcal{L} \cong \varphi^*(\mathcal{O}(1))$ sending s_i to $\varphi^*(x_i)$.

Step 7: Uniqueness. Any R-morphism $\varphi' : X \to \mathbb{P}^n_R$ with $(\varphi')^*(\mathcal{O}(1)) \cong \mathcal{L}$ and $(\varphi')^*(x_i) = s_i$ must restrict on X_i to the morphism $X_i \to U_i$ induced by the ring map $y_j \mapsto s_j/s_i$ (since $(\varphi')^*(x_j/x_i) = s_j/s_i$ is forced). By lemma 3.1.9, φ' is uniquely determined by its restrictions to the cover $\{X_i\}$, so $\varphi' = \varphi$.

5.3.44 Comparison with the Affine Case The theorem establishes a clean parallel:

Target $\mathbb{A}^n_R$	**Target $\mathbb{P}^n_R$**
$\mathrm{Hom}_R(X, \mathbb{A}^n_R) \cong \mathcal{O}_X(X)^n$	$\mathrm{Hom}_R(X, \mathbb{P}^n_R) \cong \{(\mathcal{L}, s_0, \ldots, s_n)\}/\cong$
n global functions	line bundle + $(n+1)$ generating sections
always well-defined	need base locus to be empty

In the affine case, the structure sheaf $\mathcal{O}_X$ plays the role of the line bundle, and it is always generated by a single section (1), so no additional choice is needed. In the projective case, the choice of line bundle $\mathcal{L}$ is genuine extra data — different line bundles yield morphisms into *different* projective spaces (with different embeddings). The study of which line bundles on X are generated by global sections, and how the resulting morphisms to projective space behave, is a central theme of algebraic geometry; it is developed further in section 5.4.

Example 5.3.45: Maps Into Projective Space

1. *Linear automorphisms of $\mathbb{P}^n_k$.* Let $X = \mathbb{P}^n_k$, $\mathcal{L} = \mathcal{O}(1)$, and choose generating sections $s_i = \sum_{j=0}^n a_{ij}\, x_j$ for an invertible matrix $(a_{ij}) \in \mathrm{GL}_{n+1}(k)$. The resulting morphism $\varphi : \mathbb{P}^n_k \to \mathbb{P}^n_k$ sends $[x_0 : \cdots : x_n] \mapsto [s_0 : \cdots : s_n]$, which is a linear change of coordinates — an automorphism of $\mathbb{P}^n_k$. Conversely, any automorphism of $\mathbb{P}^n_k$ arising from $\mathcal{O}(1)$ with linear sections must be of this form. Rescaling all sections by a common scalar $\lambda \in k^*$ does not change the morphism (since $[\lambda s_0 : \cdots : \lambda s_n] = [s_0 : \cdots : s_n]$), so we obtain $\mathrm{PGL}_{n+1}(k) \subseteq \mathrm{Aut}(\mathbb{P}^n_k)$. In fact, equality holds: every automorphism of $\mathbb{P}^n_k$ is linear (this follows from $\mathrm{Pic}\,(\mathbb{P}^n_k) \cong \mathbb{Z}$, proved in corollary 5.4.46).

2. *The Veronese embedding.* Let $X = \mathbb{P}^n_k$, $\mathcal{L} = \mathcal{O}(d)$ for $d \geq 1$, and take the sections to be all $N+1 = \binom{n+d}{d}$ monomials of degree d. These generate $\mathcal{O}(d)$, so the theorem gives a morphism $\nu_d : \mathbb{P}^n_k \to \mathbb{P}^N_k$. For $n = 1$, $d = 2$:

$$\nu_2 : \mathbb{P}^1_k \longrightarrow \mathbb{P}^2_k, \qquad [s : t] \longmapsto [s^2 : st : t^2].$$

   The image is the conic $V(y_0 y_2 - y_1^2)$, and ν_2 is a closed immersion. More generally, ν_d is always a closed immersion (see definition 3.1.38); this is the *d-uple Veronese embedding.* A key consequence: every degree-d hypersurface in $\mathbb{P}^n$ becomes a *hyperplane section* of the Veronese image, reducing questions about higher-degree geometry to linear geometry in a larger projective space.

3. *Maps from affine schemes.* Let $X = \mathrm{Spec}\, A$ be affine. Every line bundle on an affine scheme is trivial: $\mathcal{L} \cong \mathcal{O}_X$. A trivialization $\psi : \mathcal{O}_X \xrightarrow{\sim} \mathcal{L}$ sends the generating sections s_i to elements $a_i = \psi^{-1}(s_i) \in A$. The generating condition $X = X_0 \cup \cdots \cup X_n$ translates to $D(a_0) \cup \cdots \cup D(a_n) = \mathrm{Spec}\, A$, i.e., $(a_0, \ldots, a_n) = A$. The morphism $\varphi : \mathrm{Spec}\, A \to \mathbb{P}^n_R$ is then given on $D(a_i)$ by the ring map $x_j/x_i \mapsto a_j/a_i \in A_{a_i}$.

   For a concrete case take $A = k[s,t]$ with $a_0 = s$, $a_1 = t$, $a_2 = 1$. Since $(s, t, 1) = A$, this defines a morphism $\mathbb{A}^2_k \to \mathbb{P}^2_k$. On $D(s)$: $y_1 \mapsto t/s$, $y_2 \mapsto 1/s$; on $D(1) = \mathrm{Spec}\, A$: $y_0 \mapsto s$, $y_1 \mapsto t$. This is the standard open embedding $\mathbb{A}^2_k \hookrightarrow \mathbb{P}^2_k$ sending $(s,t) \mapsto [s : t : 1]$.

4. *Projections.* Let $X = \mathbb{P}^n_k$ and choose a linear subspace $\Lambda \cong \mathbb{P}^{n-m-1}_k$ (the "centre of projection"). After a change of coordinates, $\Lambda = V(x_0, \ldots, x_m)$. The sections $x_0, \ldots, x_m \in$

$\Gamma(\mathbb{P}^n_k, \mathcal{O}(1))$ generate $\mathcal{O}(1)$ at every point *outside* Λ, but they all vanish on Λ. Thus, the theorem applies to $U = \mathbb{P}^n_k \setminus \Lambda$, giving a morphism $\pi : U \to \mathbb{P}^m_k$. This is the classical *projection from* Λ: the fibre $\pi^{-1}([a])$ is the $(n-m)$-plane spanned by Λ and the point $[a_0 : \cdots : a_m : 0 : \cdots : a_m : 0 : \cdots : 0]$ and Λ; geometrically, π sends a point $P \notin \Lambda$ to the unique point where the linear span $\langle P, \Lambda \rangle$ meets a complementary $\mathbb{P}^m$.

Note that π is *not* defined on Λ itself: the base locus of the sections $x_0, \ldots, x_m$ is exactly Λ. This illustrates the general principle that a collection of sections that fails to generate $\mathcal{L}$ everywhere still defines a morphism on the complement of the base locus — a *rational map* $X \dashrightarrow \mathbb{P}^m$ in the sense of section 3.6.

5. *The Segre embedding.* Let $X = \mathbb{P}^m_k \times_k \mathbb{P}^n_k$ with the line bundle $\mathcal{L} = \mathcal{O}(1,1) := \mathrm{pr}_1^*\mathcal{O}(1) \otimes \mathrm{pr}_2^*\mathcal{O}(1)$. The global sections $\{x_i \otimes y_j\}_{0 \le i \le m,\, 0 \le j \le n}$ generate $\mathcal{L}$ (at any point (P, Q), some x_i is nonzero at P and some y_j is nonzero at Q, so $x_i \otimes y_j$ is nonzero). The resulting morphism

$$\sigma : \mathbb{P}^m_k \times \mathbb{P}^n_k \longrightarrow \mathbb{P}^{(m+1)(n+1)-1}_k$$

is the *Segre embedding*. . It is a closed immersion, and its image is cut out by the 2×2 minors of the matrix (z_{ij}). The Segre embedding shows, in particular, that the product of projective schemes is again projective.

5.3.46 Base-Point-Free Linear Systems The data $(\mathcal{L}, s_0, \ldots, s_n)$ appearing in theorem 5.3.43 is closely related to the classical notion of a *linear system* on X. The vector subspace $V = \mathrm{span}_k\{s_0, \ldots, s_n\} \subseteq \Gamma(X, \mathcal{L})$ determines a linear system $|V|$, and the generating condition (empty base locus) says precisely that this linear system is *base-point free*. The morphism φ depends only on $\mathcal{L}$ and the subspace V (up to the choice of basis, which amounts to a linear automorphism of the target $\mathbb{P}^n$). Choosing a different basis of V gives a morphism that differs from φ by an element of PGL_{n+1}.
When $V = \Gamma(X, \mathcal{L})$ is the *complete* linear system and $\mathcal{L}$ is very ample, the resulting morphism is a closed immersion — this is essentially the definition of very ampleness (definition 5.3.22). Choosing a proper subspace $V \subsetneq \Gamma(X, \mathcal{L})$ gives a morphism to a *smaller* projective space, obtained by composing the complete embedding with a projection (as in example 5.3.45(4)).

5.3.6 Relative Proj and Projective Bundles

Relative Spec (theorem 5.1.41) globalized Spec : it turned a sheaf of algebras over a base into a scheme over that base, and recovered the total space of a vector bundle as $\underline{\mathrm{Spec}}_X(\mathrm{Sym}(\mathcal{E}^\vee))$. The same move applies to Proj , and it is the one this book has been using informally. A ruled surface is a family of projective lines over a curve; the blow-up of an ideal sheaf is Proj of the Rees algebra $\bigoplus_{n \ge 0} \mathcal{I}^n$; and the fibrewise projective space of a vector bundle is what turns a bundle into a variety one can do geometry on. All three are instances of a single construction, and this subsection gives it.

The input is a sheaf of algebras carrying a grading, since Proj sees only the graded structure.

Definition 5.3.47: Graded Quasi-Coherent $\mathcal{O}_X$-Algebra

Let X be a scheme. A *graded quasi-coherent $\mathcal{O}_X$-algebra* is a quasi-coherent $\mathcal{O}_X$-algebra $\mathcal{S}$ (definition 5.1.39) equipped with a decomposition

$$\mathcal{S} = \bigoplus_{n\geq 0} \mathcal{S}_n$$

into quasi-coherent $\mathcal{O}_X$-submodules, such that the multiplication carries $\mathcal{S}_m \otimes_{\mathcal{O}_X} \mathcal{S}_n$ into $\mathcal{S}_{m+n}$ and such that $\mathcal{S}_0 = \mathcal{O}_X$.
The algebra is *generated in degree one* if $\mathcal{S}_1$ is of finite type and the multiplication maps induce a surjection $\mathrm{Sym}^\bullet(\mathcal{S}_1) \twoheadrightarrow \mathcal{S}$.

The generation hypothesis is not decoration. On an affine open $U = \operatorname{Spec} A$ a graded quasi-coherent algebra restricts to $\widetilde{S}$ for a graded A-algebra S with $S_0 = A$, and it is exactly when S is generated by S_1 as an A-algebra, with S_1 finitely generated, that $\operatorname{Proj} S$ carries an invertible $\mathcal{O}(1)$ and is projective over $\operatorname{Spec} A$ (example 5.3.12, proposition 5.3.5). Everything below assumes it.

Example 5.3.48: Graded Algebras from Sheaves and Ideals

1. *Symmetric algebras.* For a quasi-coherent $\mathcal{E}$, the symmetric algebra $\mathrm{Sym}^\bullet \mathcal{E} = \bigoplus_{n\geq 0} \mathrm{Sym}^n \mathcal{E}$ of example 5.1.40 is graded, with $\mathrm{Sym}^0 \mathcal{E} = \mathcal{O}_X$ and $\mathrm{Sym}^1 \mathcal{E} = \mathcal{E}$. It is generated in degree one by construction, and $\mathcal{E}$ of finite type makes $\mathcal{S}_1$ so. On an open where $\mathcal{E} \cong \mathcal{O}_U^{r+1}$ it is the polynomial algebra $\mathcal{O}_U[x_0, \ldots, x_r]$ with the usual grading.

2. *Rees algebras.* For a quasi-coherent ideal sheaf $\mathcal{I} \subseteq \mathcal{O}_X$, the algebra $\bigoplus_{n\geq 0} \mathcal{I}^n$ is graded, with $\mathcal{I}^0 = \mathcal{O}_X$. It is generated in degree one because $\mathcal{I}^n$ is the n-fold product of $\mathcal{I}^1 = \mathcal{I}$. This is the algebra whose Proj is the blow-up (definition 8.4.1).

3. *A generator in the wrong degree.* Let $\mathcal{S} = \mathcal{O}_X[t]$ with t placed in degree two, so $\mathcal{S}_1 = 0$ and $\mathrm{Sym}^\bullet(\mathcal{S}_1) = \mathcal{O}_X \neq \mathcal{S}$: this is not generated in degree one. Its relative Proj is X itself, since on an affine $U = \operatorname{Spec} A$ the only standard open is $D_+(t) = \operatorname{Spec}(A[t]_t)_0 = \operatorname{Spec} A$. But every element of $A[t, t^{-1}]$ has even degree, so $(A[t]_t)_1 = 0$ and $\mathcal{O}(1)$ is the zero sheaf, not a line bundle. The hypothesis is exactly what makes the twisting sheaf invertible.

The construction now proceeds exactly as relative Spec did: build the answer over each affine open, and check that the pieces glue.

Theorem 5.3.49: Relative Proj

Let X be a scheme and $\mathcal{S}$ a graded quasi-coherent $\mathcal{O}_X$-algebra generated in degree one. There exists a scheme $\underline{\operatorname{Proj}}_X(\mathcal{S})$ with a morphism $\pi\colon \underline{\operatorname{Proj}}_X(\mathcal{S}) \to X$ and an invertible sheaf $\mathcal{O}_{\underline{\operatorname{Proj}}_X(\mathcal{S})}(1)$, characterized up to canonical isomorphism by:

1. for every affine open $U = \operatorname{Spec} A \subseteq X$ with $\mathcal{S}|_U \cong \widetilde{S}$, an isomorphism $\pi^{-1}(U) \cong \operatorname{Proj} S$ over U, carrying $\mathcal{O}_{\underline{\operatorname{Proj}}_X(\mathcal{S})}(1)|_{\pi^{-1}(U)}$ to $\mathcal{O}_{\operatorname{Proj} S}(1)$ (definition 5.3.4);
2. compatibility of these isomorphisms on overlaps.

The morphism π is projective over every affine open of X; if X is Noetherian, π is proper.

***Proof*:**

Step 1: the affine pieces and their overlaps. Cover X by affine opens $U_i = \operatorname{Spec} A_i$ with $\mathcal{S}|_{U_i} \cong \widetilde{S^{(i)}}$ for graded A_i-algebras $S^{(i)}$ generated in degree one. Set $P_i = \operatorname{Proj} S^{(i)}$ with its projection $\pi_i\colon P_i \to U_i$.

The gluing data comes from a single compatibility, which is where the generation hypothesis and quasi-coherence do their work. Let $f \in A_i$ and let $D(f) \subseteq U_i$ be the corresponding basic open. Quasi-coherence gives $\mathcal{S}|_{D(f)} \cong \widetilde{S^{(i)}_f}$, where $S^{(i)}_f = S^{(i)} \otimes_{A_i} (A_i)_f$ is graded by the grading of $S^{(i)}$, since f has degree zero. Localizing at a degree-zero element commutes with the Proj construction: a homogeneous prime of $S^{(i)}_f$ not containing the irrelevant ideal is the expansion of a unique such prime of $S^{(i)}$ meeting $\{1, f, f^2, \dots\}$ trivially, and the standard opens $D_+(g)$ for g homogeneous of positive degree correspond likewise. Hence

$$\pi_i^{-1}(D(f)) \;\cong\; \operatorname{Proj}\left(S^{(i)}_f\right)$$

as schemes over $D(f)$, compatibly with the degree-one parts and therefore with $\mathcal{O}(1)$.

Step 2: gluing. On an overlap $U_i \cap U_j$, cover by basic opens that are simultaneously basic in U_i and in U_j. Over such an open, Step 1 identifies both π_i^{-1} and π_j^{-1} with the Proj of the same localization of $\mathcal{S}$, because $\mathcal{S}$ is one sheaf and the two descriptions restrict to it. The resulting isomorphisms $\varphi_{ji}\colon \pi_i^{-1}(U_i \cap U_j) \to \pi_j^{-1}(U_i \cap U_j)$ are induced by the identity of $\mathcal{S}$, so φ_{ii} is the identity and $\varphi_{kj} \circ \varphi_{ji} = \varphi_{ki}$ on triple overlaps. The cocycle condition holds, and theorem 2.2.56 produces a scheme $\underline{\operatorname{Proj}}_X(\mathcal{S})$ with a morphism π to X restricting to each π_i. The sheaves $\mathcal{O}_{P_i}(1)$ glue by the same cocycle, since the identifications of Step 1 respect $\mathcal{O}(1)$; each is invertible by proposition 5.3.5, using generation in degree one, so the glued sheaf is invertible.

Step 3: properness. Over $U_i = \operatorname{Spec} A_i$, generation in degree one gives a map $A_i[x_0, \dots, x_r] \to S^{(i)}$ of graded A_i-algebras, sending the variables to a finite generating set of the A_i-module $S^{(i)}_1$. It is surjective because $S^{(i)}$ is generated by $S^{(i)}_1$ as an A_i-algebra. So π_i is projective by corollary 5.3.15(1), and hence proper by theorem 3.5.32. Separatedness is part of properness (definition 3.5.21). Properness, unlike projectivity, is local on the target (proposition 3.5.23(6)), so for Noetherian X (the standing hypothesis of proposition 3.5.23) properness of every π_i gives properness of π, completing the proof.

Uniqueness. A scheme over X satisfying (1) and (2) is determined on each $\pi^{-1}(U_i)$ up to canonical isomorphism, and the compatibility in (2) makes these agree on overlaps, so any two such glue to canonically isomorphic schemes.

Relative Proj is the projective counterpart of relative Spec, and the contrast between them is the usual one. Relative Spec produces affine morphisms and loses nothing; relative Proj produces proper ones, and pays for properness by discarding the irrelevant ideal. What makes the projective version the more useful of the two here is precisely that properness: proper morphisms have pushforwards that preserve coherence (theorem 6.6.13), which is what makes the fibres of a family of projective spaces computable.

Two further properties are what make relative Proj usable downstream, and both are invisible in the construction as stated: it is compatible with arbitrary base change, and its twisting sheaf carries a universal quotient. The first says that a family of projective spaces restricts to a family over any base that maps in, which is the property every fibrewise argument silently uses.

Proposition 5.3.50: Relative Proj Commutes with Base Change

Let $\mathcal{S}$ be a graded quasi-coherent $\mathcal{O}_X$-algebra generated in degree one and let $h\colon Y \to X$ be *any* morphism of schemes. Then $h^*\mathcal{S}$ is a graded quasi-coherent $\mathcal{O}_Y$-algebra generated in degree one, and there is a canonical isomorphism

$$\underline{\operatorname{Proj}}_Y(h^*\mathcal{S}) \;\cong\; \underline{\operatorname{Proj}}_X(\mathcal{S}) \times_X Y$$

over Y, carrying $\mathcal{O}_{\underline{\operatorname{Proj}}_Y(h^*\mathcal{S})}(1)$ to the pullback of $\mathcal{O}_{\underline{\operatorname{Proj}}_X(\mathcal{S})}(1)$ along the projection. In particular, for $\mathcal{E}$ locally free of rank $r+1$ there is a canonical isomorphism $\mathbb{P}(h^*\mathcal{E}) \cong \mathbb{P}(\mathcal{E}) \times_X Y$ over Y, compatible with the twisting sheaves.

***Proof*:**
Both sides are local on Y and on X, so it suffices to treat $X = \operatorname{Spec} A$, $Y = \operatorname{Spec} A'$ with $\mathcal{S}|_X \cong \widetilde{S}$ for a graded A-algebra S generated by S_1. Then $h^*\mathcal{S} \cong \widetilde{S'}$ with $S' = S \otimes_A A'$, graded by the grading of S (the factor A' sits in degree zero) and generated by $S'_1 = S_1 \otimes_A A'$, so the hypotheses transfer.

Since S is generated in degree one, the opens $D_+(f)$ for $f \in S_1$ cover $\operatorname{Proj} S$, and likewise the $D_+(f \otimes 1)$ cover $\operatorname{Proj} S'$. Localization commutes with the base change and preserves degrees, so

$$(S'_{f\otimes 1})_0 \;\cong\; (S_f)_0 \otimes_A A'$$

which says exactly that $D_+(f \otimes 1) \cong D_+(f) \times_{\operatorname{Spec} A} \operatorname{Spec} A'$. These isomorphisms are induced by the identity of S on each piece, so they agree on overlaps and glue to the asserted isomorphism. In degree one the same computation gives $(S'_{f\otimes 1})_1 \cong (S_f)_1 \otimes_A A'$, which is the compatibility of the twisting sheaves, completing the proof.

The case the rest of the book needs is the one where the graded algebra is a symmetric algebra.

Definition 5.3.51: Projective Bundle

Let X be a scheme and $\mathcal{E}$ a locally free $\mathcal{O}_X$-module of rank $r+1$. The *projective bundle* associated to $\mathcal{E}$ is

$$\mathbb{P}(\mathcal{E}) \;=\; \underline{\mathrm{Proj}}_X\big(\mathrm{Sym}^\bullet \mathcal{E}\big)$$

with its projection $\pi\colon \mathbb{P}(\mathcal{E}) \to X$ and the invertible sheaf $\mathcal{O}_{\mathbb{P}(\mathcal{E})}(1)$ given by theorem 5.3.49. The latter is the *tautological*, or *Serre*, twisting sheaf of the bundle.

A $\kappa(x)$-rational point of the fibre $\pi^{-1}(x)$ corresponds to a one-dimensional *quotient* of the fibre $\mathcal{E}(x) = \mathcal{E}_x \otimes_{\mathcal{O}_{X,x}} \kappa(x)$: this is Grothendieck's convention, used for ruled surfaces in section 8.5. The opposite convention, in which $\mathbb{P}(\mathcal{E})$ is the bundle of one-dimensional subspaces, is equally standard and is the one used in *Intersection Theory* [19]. The two differ by a dual. See box 8.5.2 for the consequences of confusing them.

Proposition 5.3.52: Local Triviality of a Projective Bundle

Let $\mathcal{E}$ be locally free of rank $r+1$ on X and let $U \subseteq X$ be an open over which $\mathcal{E}|_U \cong \mathcal{O}_U^{r+1}$. Then there is an isomorphism

$$\pi^{-1}(U) \;\cong\; \mathbb{P}^r_U$$

over U, carrying $\mathcal{O}_{\mathbb{P}(\mathcal{E})}(1)$ to $\mathcal{O}_{\mathbb{P}^r_U}(1)$ (definition 5.3.21). In particular every fibre of π over a point $x \in X$ is $\mathbb{P}^r_{\kappa(x)}$.

***Proof*:**

A trivialization $\mathcal{E}|_U \cong \mathcal{O}_U^{r+1}$ induces an isomorphism of graded $\mathcal{O}_U$-algebras $\mathrm{Sym}^\bullet(\mathcal{E}|_U) \cong \mathcal{O}_U[x_0,\dots,x_r]$, since the symmetric algebra of a free module of rank $r+1$ is the polynomial algebra on a basis and the isomorphism respects degrees. Relative Proj depends only on the graded algebra, and its formation commutes with restriction to an open by characterization (1) of theorem 5.3.49, so $\pi^{-1}(U) \cong \underline{\mathrm{Proj}}_U(\mathcal{O}_U[x_0,\dots,x_r]) = \mathbb{P}^r_U$. The isomorphism of algebras is the identity in degree one, so it carries $\mathcal{O}(1)$ to $\mathcal{O}(1)$.

For the fibre statement, take U a trivializing neighbourhood of x and base change along $\operatorname{Spec}\kappa(x) \to U$; the fibre of $\mathbb{P}^r_U \to U$ over x is $\mathbb{P}^r_{\kappa(x)}$, as we sought to show.

Local triviality is what earns the name *bundle*: a projective bundle is not a proper morphism whose fibres happen to be projective spaces, but one that is Zariski-locally on the base a product. That is a genuinely stronger statement, and it is what makes the next proposition a one-line reduction.

The twisting sheaf is not a line bundle that happens to exist: it is a quotient of the pulled-back bundle, and that is what pins down its geometric meaning and, with it, the convention above.

Proposition 5.3.53: The Tautological Quotient

Let $\mathcal{E}$ be locally free of rank $r+1$ on X, with $\pi\colon \mathbb{P}(\mathcal{E}) \to X$. There is a canonical surjection of $\mathcal{O}_{\mathbb{P}(\mathcal{E})}$-modules

$$\pi^*\mathcal{E} \twoheadrightarrow \mathcal{O}_{\mathbb{P}(\mathcal{E})}(1)$$

whose kernel is locally free of rank r. Over a point $x \in X$ it specializes to the one-dimensional quotient of $\mathcal{E}(x)$ named by the corresponding point of the fibre. Dually, $\mathcal{O}_{\mathbb{P}(\mathcal{E})}(-1)$ is canonically a subsheaf of $\pi^*\mathcal{E}^\vee$, locally free of rank one.

***Proof*:**
The degree-one piece of $\operatorname{Sym}^\bullet\mathcal{E}$ is $\mathcal{E}$ itself, and for a graded ring S generated by S_1 the sheaf $\mathcal{O}_{\operatorname{Proj} S}(1)$ is generated by the images of S_1: on $D_+(f)$ with $f \in S_1$ the module $(S_f)_1$ is free of rank one on f (this is the computation behind proposition 5.3.5). The multiplication $S_1 \otimes (S_f)_0 \to (S_f)_1$ therefore gives a surjection, and these glue over the affine opens of X by theorem 5.3.49(2) to the asserted map $\pi^*\mathcal{E} \to \mathcal{O}_{\mathbb{P}(\mathcal{E})}(1)$, surjective because surjectivity is checked on stalks and holds on each $D_+(f)$.

A surjection of locally free sheaves of ranks $r+1$ and 1 has locally free kernel of rank r: the sequence splits locally, since a surjection onto a locally free sheaf admits local sections. Restricting the surjection to the fibre over x gives a one-dimensional quotient of $\mathcal{E}(x)$, which is the quotient the point names by definition 5.3.51. Dualizing the surjection gives the injection $\mathcal{O}_{\mathbb{P}(\mathcal{E})}(-1) \hookrightarrow \pi^*\mathcal{E}^\vee$ with locally free cokernel, as we sought to show.

Proposition 5.3.54: A Projective Bundle is Proper and Flat

Let $\mathcal{E}$ be locally free of rank $r+1$ on a scheme X. Then $\pi\colon \mathbb{P}(\mathcal{E}) \to X$ is flat with all fibres of dimension r, and it is proper when X is Noetherian.

***Proof*:**
Properness is theorem 5.3.49. For flatness, both flatness (definition 6.8.3) and the dimension of the fibres are local on the base, so by proposition 5.3.52 it suffices to treat $\mathbb{P}^r_A \to \operatorname{Spec} A$ for a ring A. That morphism is covered by the standard opens $D_+(x_i) = \operatorname{Spec} A[x_0/x_i, \dots, x_r/x_i]$, each a polynomial ring in r variables over A, hence free as an A-module and so flat over A. The local ring of $\mathbb{P}^r_A$ at any point is a localization of one of these, and a localization of a flat A-algebra is again flat over A, so every local ring of $\mathbb{P}^r_A$ is flat over A, which is flatness of the morphism. The fibre over a point of $\operatorname{Spec} A$ with residue field κ is $\mathbb{P}^r_\kappa$, of dimension r, completing the proof.

Example 5.3.55: Projective Bundles

1. *The trivial bundle.* For $\mathcal{E} = \mathcal{O}_X^{r+1}$, the definition gives $\mathbb{P}(\mathcal{E}) = \mathbb{P}^r_X$, recovering relative projective space and its twisting sheaf (definition 5.3.21). Relative Proj is thus a genuine generalization of the construction already in use.
2. *Ruled surfaces.* For $\mathcal{E}$ locally free of rank 2 on a smooth projective curve C over an algebraically closed field, the surface $\mathbb{P}(\mathcal{E}) \to C$ is a ruled surface in the sense of definition 8.5.1, and every ruled surface is of this form [36, Proposition V.2.2]. The construction itself is available for

bundles of every rank over every base.

3. *Twisting the bundle changes nothing.* For an invertible $\mathcal{L}$ there is a canonical isomorphism $\mathbb{P}(\mathcal{E} \otimes \mathcal{L}) \cong \mathbb{P}(\mathcal{E})$ over X, because tensoring by $\mathcal{L}$ induces a degree-preserving isomorphism $\mathrm{Sym}^n(\mathcal{E} \otimes \mathcal{L}) \cong (\mathrm{Sym}^n \mathcal{E}) \otimes \mathcal{L}^{\otimes n}$ for every n. Relative Proj is unchanged by twisting the degree-n piece by $\mathcal{L}^{\otimes n}$: over an open trivializing $\mathcal{L}$ the two graded algebras are identified compatibly with the grading, and those identifications glue. The twisting sheaves differ: the isomorphism carries $\mathcal{O}_{\mathbb{P}(\mathcal{E}\otimes\mathcal{L})}(1)$ to $\mathcal{O}_{\mathbb{P}(\mathcal{E})}(1) \otimes \pi^*\mathcal{L}$. A projective bundle therefore remembers less than the vector bundle it came from, which is exactly why ruled surfaces are classified by $\mathcal{E}$ only up to twist.

4. *The blow-up.* Taking $\mathcal{S} = \bigoplus_{n \geq 0} \mathcal{I}^n$ for $\mathcal{I}$ the ideal sheaf of a closed subscheme $Z \subseteq X$, rather than a symmetric algebra, gives $\underline{\mathrm{Proj}}_X(\mathcal{S}) = \mathrm{Bl}_Z X$ (definition 8.4.1); the exceptional divisor is the fibre of this Proj over Z. Blow-ups and projective bundles are the same construction applied to different graded algebras.

Relative Proj was built to package families of projective spaces, and it also settles a question left open much earlier. The normalization of a variety was produced as a *finite* morphism (theorem 3.4.41), and finiteness gives properness (proposition 3.5.24) but says nothing about projectivity. Every finiteness theorem for coherent sheaves proved so far asks for a projective scheme rather than a proper one (theorem 5.3.28, corollary 5.3.29), so a normalization sits outside their reach until the distance from finite to projective is measured. The construction of this subsection measures it: a finite morphism is a relative Proj .

The graded algebra to feed to theorem 5.3.49 is the one with $\mathcal{A} = f_*\mathcal{O}_Y$ placed in every positive degree. What matters is the choice in degree zero. Putting $\mathcal{A}$ there, as in the algebra $\mathcal{A}[t]$, violates the requirement $\mathcal{S}_0 = \mathcal{O}_X$ of definition 5.3.47 and would produce a Proj over Y rather than over X; keeping $\mathcal{O}_X$ in degree zero and $\mathcal{A}$ above it gives a graded algebra over the intended base, whose Proj has no points at infinity because the unit of $\mathcal{A}$ is a nowhere-vanishing section of degree one.

5.3.7 The Grassmannian and Its Tautological Sequence

Projective space parametrizes lines through the origin; the Grassmannian does the same for subspaces of any fixed dimension. [22, The Grassmannian Is a Projective Variety] constructs it as a projective variety through the Plücker embedding, and that construction is not repeated here. What is added is the bundle-theoretic content the variety carries and which the classical account, having no vector bundles, cannot state: a Grassmannian comes with a tautological subbundle, and every natural class on it is built from that bundle's Chern classes.

Theorem 5.3.56: The Tautological Sequence on a Grassmannian

Let V be an n-dimensional vector space over $\overline{k}$ and let $G = \mathrm{Gr}(r, V)$ be the Grassmannian of r-dimensional subspaces of V, a projective variety by [22, The Grassmannian Is a Projective Variety]. There is a short exact sequence of vector bundles on G,

$$0 \longrightarrow S \longrightarrow V \otimes_{\overline{k}} \mathcal{O}_G \longrightarrow Q \longrightarrow 0$$

in which S is locally free of rank r with fibre $S([W]) = W$ at the point $[W]$, and Q is locally free of rank $n - r$ with fibre V/W. Moreover $\det(S^\vee) \cong \mathcal{O}_G(1)$, the restriction of $\mathcal{O}(1)$ under the Plücker embedding.

***Proof*:**

Fix a basis of V and, for an r-element subset $I \subseteq \{1, \dots, n\}$, let $U_I \subseteq G$ be the locus of subspaces W projecting isomorphically onto the coordinate subspace V_I spanned by the basis vectors indexed by I. These are the standard affine charts of the Plücker construction: on U_I a subspace W has a unique basis whose I-block is the identity matrix, and the remaining $r(n-r)$ entries are affine coordinates, so $U_I \cong \mathbb{A}^{r(n-r)}$ and the U_I cover G.

Define S by declaring $S|_{U_I}$ to be the free $\mathcal{O}_{U_I}$-module on the r rows of that normalized basis, viewed inside $V \otimes \mathcal{O}_{U_I}$. The inclusion into $V \otimes \mathcal{O}_G$ is visibly $\mathcal{O}$-linear and, fibrewise at $[W]$, has image exactly W. On an overlap $U_I \cap U_J$ the two normalized bases differ by the invertible $r \times r$ matrix expressing one I-block in terms of the other, whose entries are regular on the overlap; so the local pieces glue and S is locally free of rank r with the stated fibres.

The quotient $Q = (V \otimes \mathcal{O}_G)/S$ is locally free of rank $n - r$: on U_I the inclusion of S is split by projection onto the complementary coordinates, so the quotient is free there. Fibrewise $Q([W]) = V/W$ because taking fibres is right exact and the rank count leaves no room for a kernel.

For the last claim, the Plücker embedding sends $[W]$ to the line $\Lambda^r W \subseteq \Lambda^r V$, that is to the fibre of $\det S = \Lambda^r S$. So $\det S$ is the restriction of the tautological subbundle $\mathcal{O}(-1)$ of $\mathbb{P}(\Lambda^r V)$, whence $\det(S^\vee) \cong \mathcal{O}_G(1)$.

The rank of S and the rank of Q add to $\dim V$, and the sequence is the only structure the Grassmannian carries that is natural in V; every construction on G in [19] is assembled from $c(S)$ and $c(Q)$, which the Whitney formula ties together as $c(S)c(Q) = 1$.

Proposition 5.3.57: Finite Morphisms Are Projective

Let X be a Noetherian scheme, let $f\colon Y \to X$ be a finite morphism, and set $\mathcal{A} = f_*\mathcal{O}_Y$. Let $\mathcal{S} = \mathcal{O}_X + t\,\mathcal{A}[t] \subseteq \mathcal{A}[t]$ be the graded $\mathcal{O}_X$-algebra with $\mathcal{S}_0 = \mathcal{O}_X$ and $\mathcal{S}_d = \mathcal{A}t^d$ for $d \geq 1$.

1. $\mathcal{A}$ is a coherent $\mathcal{O}_X$-algebra, $\mathcal{S}$ is generated in degree one, and there is an isomorphism
$$\underline{\operatorname{Proj}}_X(\mathcal{S}) \;\cong\; \underline{\operatorname{Spec}}_X(\mathcal{A}) \;\cong\; Y$$
over X. So f is a relative Proj, and it is projective over every affine open of X.
2. Suppose in addition that X is projective over a Noetherian ring R, with $\mathcal{O}_X(1)$ very ample relative to $\operatorname{Spec} R$. Then there are integers $m \geq 1$ and $n \geq 0$ and a closed immersion over X
$$\iota\colon Y \hookrightarrow \mathbb{P}\big(\mathcal{O}_X(-n)^{\oplus m}\big) \;\cong\; \mathbb{P}^{m-1}_X$$
so f is projective.

***Proof*:**
Step 1: the algebra $\mathcal{A}$. A finite morphism is affine: by definition 3.4.15 the preimage of an affine open $U = \operatorname{Spec} B \subseteq X$ is $\operatorname{Spec} A$ with A finite as a B-module. So $\mathcal{A} = f_*\mathcal{O}_Y$ is a quasi-coherent $\mathcal{O}_X$-algebra (example 5.1.40(3)) with $\mathcal{A}|_U \cong \widetilde{A}$, and $Y \cong \underline{\operatorname{Spec}}_X(\mathcal{A})$ over X by example 5.1.42(3). Each A is a finite module over the Noetherian ring B, so $\mathcal{A}$ is coherent by proposition 5.1.11 and definition 5.1.9.

Step 2: $\mathcal{S}$ is graded and generated in degree one. Over $U = \operatorname{Spec} B$ one has $\mathcal{S}|_U \cong \widetilde{S}$ for the graded ring $S = B \oplus \bigoplus_{d\geq 1} At^d \subseteq A[t]$ with $S_0 = B$, and these identifications commute with localization at elements of B because $\mathcal{A}$ is quasi-coherent and t is a formal variable. Hence $\mathcal{S}$ is a graded quasi-coherent $\mathcal{O}_X$-algebra in the sense of definition 5.3.47, its degree-zero part being $\mathcal{O}_X$. Its degree-one part $\mathcal{S}_1 = \mathcal{A}$ is coherent, hence of finite type, and the multiplication $\operatorname{Sym}^d(\mathcal{S}_1) \to \mathcal{S}_d$ is the d-fold product $\mathcal{A}^{\otimes d} \to \mathcal{A}$, which is surjective because $a \otimes 1 \otimes \cdots \otimes 1 \mapsto a$. So $\operatorname{Sym}^\bullet(\mathcal{S}_1) \twoheadrightarrow \mathcal{S}$ is surjective.

Step 3: the Proj *is the* Spec. Work over $U = \operatorname{Spec} B$ as above and put $t = 1 \cdot t \in S_1$. If $\mathfrak{p}$ is a homogeneous prime containing t and $x = at^d \in S_d$ with $d \geq 1$, then $x^2 = (a^2t^{2d-1}) \cdot t \in \mathfrak{p}$, since $2d - 1 \geq 1$ makes a^2t^{2d-1} an element of S; primeness gives $x \in \mathfrak{p}$, so $\mathfrak{p}$ contains S_+ and is irrelevant. Hence $\operatorname{Proj} S = D_+(t)$. The map $A \to (S_t)_0$ sending $a \mapsto at/t$ is a homomorphism of B-algebras; it is surjective because a degree-zero element of S_t is s/t^d with $s = at^d \in S_d$, and injective because $at^{N+1} = 0$ in $A[t]$ forces $a = 0$. Therefore
$$\operatorname{Proj} S \;=\; D_+(t) \;=\; \operatorname{Spec}\,(S_t)_0 \;=\; \operatorname{Spec} A$$
over $\operatorname{Spec} B$. This isomorphism is induced by the unit of $\mathcal{A}$ and commutes with restriction to basic opens of $\operatorname{Spec} B$, so the local isomorphisms agree on overlaps and glue, by the characterizations in theorem 5.3.49(1),(2) and theorem 5.1.41(1), to an isomorphism $\underline{\operatorname{Proj}}_X(\mathcal{S}) \cong \underline{\operatorname{Spec}}_X(\mathcal{A}) \cong Y$ over X. Projectivity over each affine open is part of theorem 5.3.49, proving (1).

Step 4: a locally free sheaf mapping onto $\mathcal{A}$. Assume now the hypotheses of (2). Since $\mathcal{A}$ is

coherent, theorem 5.3.26 gives $n \geq 0$ and $m \geq 1$ with $\mathcal{A}(n)$ generated by m global sections, that is (definition 5.3.24) a surjection $\mathcal{O}_X^{\oplus m} \twoheadrightarrow \mathcal{A}(n)$. Tensoring with the invertible sheaf $\mathcal{O}_X(-n)$ preserves surjectivity and undoes the twist (definition 5.3.7, proposition 5.3.8(1)), so there is a surjection $q\colon \mathcal{E} \twoheadrightarrow \mathcal{A} = \mathcal{S}_1$ with $\mathcal{E} = \mathcal{O}_X(-n)^{\oplus m}$ locally free of rank m. Composing $\mathrm{Sym}^\bullet(q)$ with the surjection of Step 2 gives a surjection of graded $\mathcal{O}_X$-algebras $\mathrm{Sym}^\bullet \mathcal{E} \twoheadrightarrow \mathcal{S}$.

Step 5: the closed immersion. Cover X by affine opens $U = \operatorname{Spec} B$ over which $\mathcal{O}_X(-n)$ is trivial. Over such a U one has $\mathrm{Sym}^\bullet \mathcal{E}|_U \cong B[x_1, \dots, x_m]$ with its usual grading (example 5.3.48(1)) and $\pi^{-1}(U) \cong \mathbb{P}^{m-1}_U$ for the projection $\pi\colon \mathbb{P}(\mathcal{E}) \to X$ of definition 5.3.51, by proposition 5.3.52. Step 4 restricts to a graded surjection $B[x_1, \dots, x_m] \twoheadrightarrow S$, which induces a closed immersion $\operatorname{Proj} S \hookrightarrow \mathbb{P}^{m-1}_B$ by corollary 5.3.15(1). All of these are induced by the single map of graded algebras, so they agree on overlaps and glue to a morphism $\iota\colon Y \to \mathbb{P}(\mathcal{E})$ over X; being a closed immersion can be checked on the members of an open cover of the target (proposition 3.1.24(1)), so ι is one. Finally $\mathcal{E} = \mathcal{O}_X^{\oplus m} \otimes \mathcal{O}_X(-n)$, so $\mathbb{P}(\mathcal{E}) \cong \mathbb{P}(\mathcal{O}_X^{\oplus m}) = \mathbb{P}^{m-1}_X$ over X by example 5.3.55(3),(1), as we sought to show.

Part (1) holds over any Noetherian base, the algebra $\mathcal{S}$ being manufactured from $\mathcal{A}$ alone. Projectivity of the base is spent in part (2) on one thing, namely producing a single locally free sheaf mapping onto $\mathcal{A}$ globally; over an arbitrary Noetherian X the local embeddings given by (1) have no reason to assemble into one closed immersion into a fixed $\mathbb{P}^{m-1}_X$, which is why the two parts are separated. The projective bundle carries the assembly: the target $\mathbb{P}(\mathcal{O}_X(-n)^{\oplus m})$ absorbs the twist that makes $\mathcal{A}$ a quotient and then loses it again, since twisting a bundle by a line bundle does not change its projectivization.

Corollary 5.3.58: Normalization of a Projective Scheme Is Projective

Let k be a field, let X be an integral projective k-scheme, and let $\nu\colon \widetilde{X} \to X$ be its normalization (theorem 3.4.41). Then $\widetilde{X}$ is projective over k.

***Proof*:**

Step 1: $\widetilde{X}$ is finite over $\mathbb{P}^r_k$. Fix a closed immersion $j\colon X \hookrightarrow \mathbb{P}^r_k$ over $\operatorname{Spec} k$, which exists by definition 2.3.4 and corollary 5.3.15(3). Each standard chart $\operatorname{Spec} S$ of $\mathbb{P}^r_k$ is the spectrum of a finitely generated k-algebra and $j^{-1}(\operatorname{Spec} S) = \operatorname{Spec}(S/I)$ for an ideal I (proposition 3.1.24(1)), so X is of finite type over k; being also integral, it is a legitimate input for theorem 3.4.41, which makes ν finite. The quotient S/I is generated by 1 as an S-module, so j is itself finite by definition 3.4.15. For an affine open $\operatorname{Spec} S \subseteq \mathbb{P}^r_k$ one gets $(j \circ \nu)^{-1}(\operatorname{Spec} S) = \nu^{-1}(\operatorname{Spec} S/I) = \operatorname{Spec} C$ with C finite over S/I and S/I finite over S, hence C finite over S. So $g = j \circ \nu$ is finite.

Step 2: apply the proposition over $\mathbb{P}^r_k$. The base $\mathbb{P}^r_k$ is Noetherian (lemma 2.4.8), is projective over the Noetherian ring k, and carries the very ample sheaf $\mathcal{O}(1)$ (definition 5.3.22, taking the identity as the closed immersion). By proposition 5.3.57(2) applied to g there is a closed immersion $\widetilde{X} \hookrightarrow \mathbb{P}^{m-1}_{\mathbb{P}^r_k}$ over $\mathbb{P}^r_k$ for some $m \geq 1$.

Step 3: down to k. Writing $\mathbb{P}^{m-1}_{\mathbb{P}^r_k} = \mathbb{P}(\mathcal{O}^{\oplus m})$ (example 5.3.55(1)) and applying proposition 5.3.50

to the structure morphism $h\colon \mathbb{P}^r_k \to \operatorname{Spec} k$ and the free sheaf $\mathcal{O}^{\oplus m}$ on $\operatorname{Spec} k$ gives

$$\mathbb{P}^{m-1}_{\mathbb{P}^r_k} \;\cong\; \mathbb{P}^{m-1}_k \times_k \mathbb{P}^r_k$$

The Segre embedding (example 5.3.45(5)) is a closed immersion of the right-hand side into $\mathbb{P}^M_k$ with $M = m(r+1)-1$. A composite of closed immersions is a closed immersion (proposition 3.1.24(3)), so $\widetilde{X}$ admits a closed immersion into $\mathbb{P}^M_k$ and is projective over k, as we sought to show.

The route through a projective bundle is not decoration. The obvious candidate for an embedding, the pullback $\nu^*\mathcal{O}_X(1)$, does not serve: a very ample sheaf pulled back along a finite morphism need not be very ample[3], so lemma 5.3.23 cannot be invoked with that sheaf in hand. What the proposition produces instead is an embedding into a projective bundle over the base, and the Segre embedding flattens the result.

With corollary 5.3.58 the finiteness statements available for projective schemes, among them finite generation of global sections of a coherent sheaf (theorem 5.3.28) and coherence of pushforwards (corollary 5.3.29), apply verbatim to the normalization of a projective variety, which is what makes normalizing a harmless operation for every finiteness argument that follows.

5.4 Divisors: Geometric Representation of Line Bundles

Having established the correspondence between line bundles and maps to projective space (section 5.3.3), we now turn to the *geometry* of line bundles themselves. The central idea is simple: just as a regular function $f \in \mathcal{O}_X(X)$ determines a vanishing scheme $V(f)$ of codimension 1, a section of a line bundle determines a "generalized hypersurface" called a *divisor*. Since regular functions are sections of the trivial line bundle $\mathcal{O}_X$, divisors are a genuine generalization of zero sets.

Divisors measure how far the geometry of a scheme is from being controlled by global functions alone. There are two natural approaches to formalizing this idea:

1. *Weil divisors* are formal integer combinations of codimension-1 subvarieties — the "geometric" approach. They record *where* a section vanishes or has poles, and with what multiplicity.
2. *Cartier divisors* encode the same data "algebraically," as collections of local rational functions with prescribed vanishing behaviour, glued together by invertible transition functions.

On sufficiently nice schemes (integral, Noetherian, locally factorial), these two notions coincide. Both give rise to an abelian group — the *divisor class group* — that classifies line bundles up to isomorphism, and this group is isomorphic to the Picard group $\operatorname{Pic}(X)$ under mild hypotheses. If the reader has seen the divisor theory for Dedekind domains in [10], the present section generalizes that theory to arbitrary schemes and connects it to the geometry of line bundles[4].

The intuition for divisors is best developed by starting with examples from complex analysis (see [58]), where the two notions of divisor coincide and the key ideas can be seen concretely.

[3] The degree-2 morphism from a curve of genus 2 to $\mathbb{P}^1_k$ pulls $\mathcal{O}(1)$ back to a line bundle of degree 2; were that bundle very ample it would embed the curve as a curve of degree 2 in a projective space, hence as a plane conic, which has genus 0.

[4] This connection is foundational for the *Riemann–Roch theorem* (theorem 7.2.10), one of the most important tools for understanding the geometry of curves and, more broadly, for classifying projective varieties.

Divisors on $\mathbb{C}$. Let $X = \mathbb{C}$ and consider the free abelian group generated by the points of X: each element is a finite formal sum $D = \sum_i n_i[p_i]$ with $n_i \in \mathbb{Z}$ and $p_i \in \mathbb{C}$. Every such divisor determines a rational (meromorphic) function

$$f_D(z) \;=\; \prod_i (z - p_i)^{n_i},$$

which has a zero of order n_i at p_i when $n_i > 0$ and a pole of order $|n_i|$ when $n_i < 0$. Conversely, every nonzero rational function on $\mathbb{C}$ (i.e., every nonzero element of $\mathbb{C}(z)$) has finitely many zeros and poles, and hence determines a divisor. Such divisors are called *principal*. Since every divisor on $\mathbb{C}$ is principal, the quotient group is trivial:

$$\frac{\{\text{divisors on } \mathbb{C}\}}{\{\text{principal divisors}\}} \;=\; 0.$$

Divisors on $\mathbb{P}^1$. Now let $X = \mathbb{P}^1_{\mathbb{C}}$, the Riemann sphere. The divisor group is again the free abelian group on the points of X, but the principal divisors are more constrained: a meromorphic function on a compact Riemann surface must satisfy $\sum_{p \in X} \operatorname{ord}_p(f) = 0$ (the number of zeros equals the number of poles, counted with multiplicity)[5]. The *degree* map $D = \sum n_i[p_i] \mapsto \sum n_i$ therefore vanishes on principal divisors, and one can show it induces an isomorphism

$$\frac{\{\text{divisors on } \mathbb{P}^1\}}{\{\text{principal divisors}\}} \;\cong\; \mathbb{Z}.$$

The generator is the class of any single point $[p]$: two divisors are linearly equivalent if and only if they have the same total degree. Under the correspondence between divisors and line bundles, the divisor class of degree n corresponds to the twisting sheaf $\mathcal{O}_{\mathbb{P}^1}(n)$.

Divisors on an elliptic curve. Let $X = \mathbb{C}/\Lambda$ be an elliptic curve, where $\Lambda \subseteq \mathbb{C}$ is a lattice. As before, a meromorphic function on X has equal numbers of zeros and poles (counted with multiplicity), so principal divisors have degree zero. Hence the degree map again gives a surjection onto $\mathbb{Z}$, and the divisor class group fits into a short exact sequence

$$0 \longrightarrow \operatorname{Pic}^0(X) \longrightarrow \operatorname{Pic}(X) \xrightarrow{\deg} \mathbb{Z} \longrightarrow 0.$$

The degree-zero part $\operatorname{Pic}^0(X)$ is *nontrivial*: by Abel's theorem, a degree-zero divisor $D = \sum n_i[p_i]$ is principal if and only if $\sum n_i p_i = 0$ in the group law of X. This gives an isomorphism $\operatorname{Pic}^0(X) \cong X$ (as a group), recovering the classical fact that an elliptic curve is its own Jacobian. In particular, $\operatorname{Pic}(X)$ is *not* simply $\mathbb{Z}$ — the divisor class group detects genuinely new geometric information beyond the degree.

The progression from $\mathbb{C}$ (trivial divisor class group) to $\mathbb{P}^1$ (divisor class group $\cong \mathbb{Z}$) to elliptic curves (divisor class group $\cong \mathbb{Z} \oplus X$) illustrates a general principle: the more geometrically interesting a scheme is, the richer its divisor theory becomes. The remainder of this section develops the algebraic theory in full generality.

5.4.1 Weil Divisors

Weil divisors generalize the divisors on curves of [22] to higher-dimensional schemes; the classical theory, including linear equivalence and the Picard group of a curve, is developed there and is

[5] For instance, z^2 viewed on $\mathbb{P}^1$ has a double zero at 0 and a double pole at ∞, summing to zero.

the case to keep in mind throughout. The key requirement is a well-defined notion of "order of vanishing" at each codimension-1 point. On a nonsingular curve, every local ring is a DVR, so the valuation provides this order directly. In higher dimensions, the codimension-1 points play the role of "prime divisors," and we need the local rings at these points to be DVRs — equivalently, regular of dimension 1.

Definition 5.4.1: Regular in Codimension 1

A scheme X is *regular in codimension* 1 if every local ring $\mathcal{O}_{X,\mathfrak{p}}$ of dimension 1 (i.e., at each codimension-1 point $\mathfrak{p}$) is a regular local ring.

The condition just defined is checked one point at a time, which makes it easy to state and useless as a hypothesis: no one verifies a condition at every codimension-1 point of a scheme by hand. What makes it usable is that a single global hypothesis implies it. Normality (definition 2.4.28) is that hypothesis, and it does more than give the codimension-1 points: it pushes the whole non-regular locus down two steps in codimension, so that the theory of divisors developed below never meets a singular point along a prime divisor.

Proposition 5.4.2: Normal Schemes Are Regular Outside Codimension Two

Let X be a Noetherian normal integral scheme. Then:

1. for every point $\mathfrak{p} \in X$ with $\operatorname{kdim}\mathcal{O}_{X,\mathfrak{p}} = 1$, the local ring $\mathcal{O}_{X,\mathfrak{p}}$ is a discrete valuation ring; in particular X is regular in codimension 1 (definition 5.4.1);
2. every point at which X fails to be regular has codimension at least 2 (definition 2.6.5);
3. if in addition X is locally of finite type over a field, then $\operatorname{Sing}(X)$ (definition 2.7.22) is closed and each of its irreducible components has codimension at least 2 in X.

***Proof*:**
(1). Let $\mathfrak{p} \in X$ have $\operatorname{kdim}\mathcal{O}_{X,\mathfrak{p}} = 1$. Since X is integral, the local ring $\mathcal{O}_{X,\mathfrak{p}}$ is a domain, and since X is normal it is integrally closed in its fraction field (definition 2.4.28); it is Noetherian because X is. A Noetherian integrally closed local domain of dimension 1 is a discrete valuation ring [10, Characterizations of a DVR]. Such a ring has principal maximal ideal $\mathfrak{m} = (\pi)$, so $\mathfrak{m}/\mathfrak{m}^2$ is generated by the class of π over the residue field and has dimension at most 1; the general inequality $\operatorname{kdim} R \leq \dim_{R/\mathfrak{m}}(\mathfrak{m}/\mathfrak{m}^2)$ recorded in definition 2.7.3 forces equality, so $\mathcal{O}_{X,\mathfrak{p}}$ is regular.
(2). By definition 2.6.5 the codimension of a point $\mathfrak{p}$ is $\operatorname{kdim}\mathcal{O}_{X,\mathfrak{p}}$, so the claim is that $\mathcal{O}_{X,\mathfrak{p}}$ is regular whenever its dimension is 0 or 1. Dimension 1 is part (1). For dimension 0: an integral scheme has a unique generic point η (lemma 2.4.21), the only point whose local ring has dimension 0, and $\mathcal{O}_{X,\eta}$ is the function field $K(X)$ (corollary 2.4.25). A field is a regular local ring, its maximal ideal being zero.
(3). Under the additional hypothesis, $\operatorname{Sing}(X)$ is closed by proposition 2.7.23. Let Z be an irreducible component of $\operatorname{Sing}(X)$. Being an irreducible closed subset of the integral scheme X, it has a unique generic point η_Z with $\overline{\{\eta_Z\}} = Z$ (lemma 2.4.21), and η_Z lies in the closed set $\operatorname{Sing}(X)$, so $\mathcal{O}_{X,\eta_Z}$ is not regular. Part (2) gives $\operatorname{codim}_X Z = \operatorname{kdim}\mathcal{O}_{X,\eta_Z} \geq 2$, completing the proof.

Part (1) is the statement the theory of Weil divisors runs on: it says that the order function of the next subsection is defined at every codimension-1 point of a normal Noetherian scheme, with no further hypothesis on the singularities. Parts (2) and (3) say where the singularities went. On a normal surface they can only be isolated points; on a normal threefold they occupy a curve or less. This is why normality, rather than regularity, is the standing hypothesis in divisor theory: it is weak enough to hold for the schemes that arise in practice, and strong enough that the codimension-1 geometry is untouched.

The hypothesis carried by part (3) is not decoration. Parts (1) and (2) are statements about individual local rings, and for a general Noetherian scheme the non-regular locus need not be a closed subset at all; a bound on the codimension of each of its points is then no bound on the codimension of its closure. Closedness is what converts the pointwise statement into a statement about a closed subset, and the finite-type hypothesis is one standard way to obtain it. Over a general Noetherian base one asks instead that the scheme be excellent, a condition designed to make loci of this kind closed; see [70].

The converse of part (1) fails. Regularity in codimension 1 is one half of Serre's criterion for normality, and by itself it permits schemes such as the union of two planes in $\mathbb{A}^4_k$ meeting at a point, whose local rings are regular at all codimension-1 points and whose local ring at the origin is not a domain. Normality is the conjunction of that condition with a depth condition in higher codimension, and it is the conjunction that yields the extension property for rational functions recorded earlier.

Example 5.4.3: Regular in Codimension 1

1. Every regular scheme is regular in codimension 1, since all local rings are regular. In particular, every nonsingular curve satisfies this condition.
2. Every normal Noetherian scheme is regular in codimension 1. Indeed, if $\mathcal{O}_{X,\mathfrak{p}}$ is a one-dimensional integrally closed local domain, then it is a DVR (see [10]). This is the most common hypothesis encountered in practice.
3. The scheme $X = \operatorname{Spec} k[x,y]/(xy)$ (the union of two coordinate axes) is *not* regular in codimension 1: the local ring at the origin is not a domain, hence not regular. However, each irreducible component separately is regular.

When $\mathcal{O}_{X,\mathfrak{p}}$ is a DVR with valuation $\nu_{\mathfrak{p}}$, the integer $\nu_{\mathfrak{p}}(f)$ for a nonzero rational function f measures the order of vanishing (if positive) or the order of the pole (if negative) of f along the codimension-1 subscheme $\overline{\{\mathfrak{p}\}}$. This is the only datum we need to define Weil divisors.

5.4.4 Remark: Beyond the Regular Case If $\mathcal{O}_{X,\mathfrak{p}}$ is not regular, one can still define an "order of vanishing" using the *length* $\ell_{\mathcal{O}_{X,\mathfrak{p}}}(\mathcal{O}_{X,\mathfrak{p}}/(f))$ in place of the valuation. The key property $\operatorname{ord}(fg) = \operatorname{ord}(f) + \operatorname{ord}(g)$ is preserved, as one sees from the short exact sequence

$$0 \longrightarrow \mathcal{O}_{X,\mathfrak{p}}/(g) \xrightarrow{\cdot f} \mathcal{O}_{X,\mathfrak{p}}/(fg) \longrightarrow \mathcal{O}_{X,\mathfrak{p}}/(f) \longrightarrow 0,$$

where the left map is multiplication by f (which is injective when f is a non-zero-divisor) and additivity of length gives the result. This approach works on any Noetherian scheme, but the valuation-based definition is cleaner when available, and suffices for the applications in this text.

For narrative clarity, we impose the following standing hypotheses on the ambient scheme. The

conditions are chosen to ensure that: the scheme has a function field to draw rational functions from (integral), restriction maps are injective so that rational functions are unambiguous (separated), the topology is well-behaved (Noetherian), and the order of vanishing is well-defined (regular in codimension 1).

Definition 5.4.5: Weil Divisor

Let X be a Noetherian, integral, separated scheme that is regular in codimension 1. A *prime divisor* on X is a closed integral subscheme of codimension 1. A *Weil divisor* is an element of the free abelian group generated by the prime divisors:

$$\operatorname{WDiv} X := \bigoplus_{\substack{Y \subseteq X \\ \operatorname{codim}(Y)=1}} \mathbb{Z} \cdot [Y].$$

A Weil divisor $D = \sum_i n_i [Y_i]$ is called *effective* if $n_i \geq 0$ for all i, written $D \geq 0$.

Let $Y \subseteq X$ be a prime divisor with generic point η_Y. Since Y has codimension 1 and X is regular in codimension 1, the local ring $\mathcal{O}_{X,\eta_Y}$ is a one-dimensional regular local ring, hence a DVR. Denote its valuation by ν_Y. Since X is separated, the prime divisor Y is uniquely determined by its generic point (cf. exercise 3.5.1), and hence by its valuation ring $\mathcal{O}_{X,\eta_Y} \subseteq K(X)$.

Now let $K(X)$ denote the function field of X and let $f \in K(X)^*$ be a nonzero rational function. The integer $\nu_Y(f)$ records the behaviour of f along Y:

- $\nu_Y(f) > 0$: f vanishes along Y to order $\nu_Y(f)$ (a *zero* of order $\nu_Y(f)$).
- $\nu_Y(f) < 0$: f has a *pole* of order $|\nu_Y(f)|$ along Y.
- $\nu_Y(f) = 0$: f is regular and nonvanishing at the generic point of Y.

For the formal sum $\sum_Y \nu_Y(f)[Y]$ to be a well-defined Weil divisor, we need it to be a *finite* sum. This is guaranteed by the following lemma:

Lemma 5.4.6: Almost All Valuations Vanish

Let X be a Noetherian, integral, separated scheme that is regular in codimension 1, and let $f \in K(X)^*$. Then $\nu_Y(f) = 0$ for all but finitely many prime divisors Y.

***Proof*:**
Since f is a rational function on an integral scheme, it is regular on some nonempty open subset; choose an affine open $U = \operatorname{Spec} R \subseteq X$ on which f is regular. The complement $C = X \setminus U$ is a proper closed subset. Since X is Noetherian, C is a finite union of irreducible closed subsets, each of codimension ≥ 1. In particular, C contains at most finitely many prime divisors of X.

It remains to show that only finitely many prime divisors of U have $\nu_Y(f) \neq 0$. On U, the function f is regular (i.e., $f \in R \subseteq K(X)$), so $\nu_Y(f) \geq 0$ for every prime divisor Y meeting U. Moreover, $\nu_Y(f) > 0$ if and only if f vanishes along $Y \cap U$, meaning $Y \cap U$ is contained in the closed subset $V(f) \subseteq U$. Since $f \neq 0$, this closed subset is proper, and a Noetherian space has only finitely many irreducible components of any given codimension. Hence only finitely many prime divisors Y of U satisfy $\nu_Y(f) > 0$.

Combining, $\nu_Y(f) \neq 0$ for at most finitely many prime divisors (those contained in C together with those in $V(f) \cap U$).

This ensures that the following formal sum is a well-defined Weil divisor:

Definition 5.4.7: Principal Divisor

Let X be as above and $f \in K(X)^*$ a nonzero rational function. The *principal divisor* (or *divisor of* f) is

$$\operatorname{div}(f) \;:=\; \sum_{Y} \nu_Y(f)\,[Y] \;\in\; \operatorname{WDiv} X,$$

where the sum runs over all prime divisors Y of X. The subgroup of principal divisors is

$$\operatorname{PDiv}(X) \;:=\; \{\operatorname{div}(f) \mid f \in K(X)^*\} \;\leq\; \operatorname{WDiv} X.$$

The assignment $f \mapsto \operatorname{div}(f)$ is a group homomorphism $K(X)^* \to \operatorname{WDiv} X$, since $\nu_Y(fg) = \nu_Y(f) + \nu_Y(g)$ for every prime divisor Y.

Note that $\operatorname{Div}(f/g) = \operatorname{Div} f - \operatorname{Div} g$ by the properties of discrete valuations, and hence this is even a group homomorphism from the multiplicative group $(\kappa(X))^*$ to the additive group $\operatorname{Div} X$!

Two Weil divisors D, D' are called *linearly equivalent*, written $D \sim D'$, if their difference is principal: $D - D' = \operatorname{div}(f)$ for some $f \in K(X)^*$. This is an equivalence relation (since principal divisors form a subgroup), and the quotient measures the "non-trivial" divisors on X.

Definition 5.4.8: Divisor Class Group

Let X be a Noetherian, integral, separated scheme that is regular in codimension 1. The *divisor class group* (or *Weil divisor class group*) of X is

$$\operatorname{Cl} X \;:=\; \frac{\operatorname{WDiv} X}{\operatorname{PDiv}(X)}.$$

Two divisors representing the same class in $\operatorname{Cl} X$ are said to be *linearly equivalent*.

A divisor is more than a bookkeeping device: it cuts out a space of rational functions, namely those whose poles it dominates. On a smooth projective curve that space is the one Riemann-Roch computes. It can be defined for a Weil divisor on any scheme regular in codimension 1, with no Cartier hypothesis, and the point of what follows is that on a projective scheme it is finite-dimensional even then.

Definition 5.4.9: The Sheaf of a Weil Divisor

Let X be a Noetherian, integral, separated scheme, regular in codimension 1, and let $D \in \operatorname{WDiv} X$. The *sheaf associated to* D is the subsheaf $\mathcal{O}_X(D)$ of the constant sheaf $K(X)$ given on an open $V \subseteq X$ by

$$\mathcal{O}_X(D)(V) \;=\; \{f \in K(X)^* \mid (\operatorname{div} f + D)|_V \geq 0\} \cup \{0\}$$

Its global sections are written $L(D) := \Gamma(X, \mathcal{O}_X(D))$, the *Riemann-Roch space* of D.

The condition is imposed only at codimension-1 points, since that is where $\operatorname{div} f$ lives, so $\mathcal{O}_X(D)$ sees nothing of what happens in codimension ≥ 2. That insensitivity is recorded in corollary 5.4.12 below and is what makes the sheaf useful on schemes that are singular in high codimension.

Proposition 5.4.10: The Sheaf of a Weil Divisor Is Coherent

In the situation of definition 5.4.9, assume in addition that X is *normal*. Then the sheaf $\mathcal{O}_X(D)$ is a coherent $\mathcal{O}_X$-module.

Normality is doing real work and cannot be relaxed to regularity in codimension 1. The proof clears denominators and then appeals to Algebraic Hartogs' Lemma (theorem 2.6.18) to conclude that a rational function with no poles in codimension 1 is regular, and that identity, $A = \bigcap_{\operatorname{height}(\mathfrak{p})=1} A_{\mathfrak{p}}$, is a property of normal domains. Regularity in codimension 1 alone gives the R_1 half of Serre's criterion and not the S_2 half, which is exactly what the intersection formula needs.

***Proof*:**
The defining condition is local and closed under addition and under multiplication by sections of $\mathcal{O}_X$, so $\mathcal{O}_X(D)$ is a sheaf of $\mathcal{O}_X$-modules; quasi-coherence is likewise local. Since X is Noetherian it suffices to show that $M := \mathcal{O}_X(D)(\operatorname{Spec} A)$ is a finite A-module for each affine open $\operatorname{Spec} A \subseteq X$ meeting the support of D, with A a Noetherian domain.

Write $D|_{\operatorname{Spec} A} = D_+ - D_-$ with $D_\pm$ effective. Choose a nonzero $a \in A$ vanishing on the support of D_+ to order at least that of D_+ at each of its finitely many codimension-1 points; such an a exists because D_+ has finitely many components in $\operatorname{Spec} A$ and A is a domain, so a product of suitable elements of the corresponding height-1 primes will do. Then for $f \in M$ the element af has $\operatorname{div}(af) \geq \operatorname{div}(a) - D_+ \geq 0$ at every codimension-1 point of $\operatorname{Spec} A$, so af lies in A by Algebraic Hartogs' Lemma (theorem 2.6.18), A being a normal domain and therefore the intersection of its height-one localizations. Hence $M \subseteq a^{-1}A$, a free A-module of rank one, and M is finite because A is Noetherian, as we sought to show.

Corollary 5.4.11: Riemann-Roch Spaces Are Finite-Dimensional

Let X be a normal integral projective scheme over a field k, and let $D \in \operatorname{WDiv} X$. Then $L(D)$ is a finite-dimensional k-vector space.

***Proof*:**
The sheaf $\mathcal{O}_X(D)$ is coherent by proposition 5.4.10, and $L(D) = H^0(X, \mathcal{O}_X(D))$, which is finite-dimensional over k by corollary 6.6.7, completing the proof.

The last statement is the one that makes divisor theory usable on a scheme whose singular locus is small but nonempty: passing to the regular locus loses nothing.

Corollary 5.4.12: Sections over a Big Open Subset

Let X be as in definition 5.4.9 and let $U \subseteq X$ be an open subset whose complement has codimension ≥ 2. Then restriction is an isomorphism

$$\Gamma\big(X, \mathcal{O}_X(D)\big) \xrightarrow{\sim} \Gamma\big(U, \mathcal{O}_X(D)|_U\big)$$

for every $D \in \operatorname{WDiv} X$. Consequently, if X is in addition normal and projective over a field k, then $\Gamma(U, \mathcal{O}_X(D)|_U)$ is a finite-dimensional k-vector space.

***Proof*:**
Both sides are subsets of $K(X)$, since U is a nonempty open of an integral scheme, and the map is restriction, hence injective. It is surjective because the defining condition of definition 5.4.9 is imposed only at codimension-1 points of X, and every such point lies in U: a codimension-1 point outside U would lie in $X \setminus U$, which has codimension ≥ 2. So a section over U already satisfies the condition defining a section over X. The final statement follows from corollary 5.4.11, completing the proof.

Before computing examples, we establish a geometric characterization of unique factorization domains in terms of divisor class groups as these characterize trivial divisor class groups of affine schemes. The underlying algebraic input is that a Noetherian domain is a UFD if and only if every height-1 prime ideal is principal ([10, Factoriality by Height-One Primes]). The following lemma provides the geometric bridge.

Lemma 5.4.13: Normal Domain: Intersection of Height-One Localizations

Let R be a Noetherian normal domain (i.e., an integrally closed Noetherian domain). Then

$$R = \bigcap_{\substack{\mathfrak{p} \in \operatorname{Spec} R \\ \operatorname{ht}(\mathfrak{p}) = 1}} R_{\mathfrak{p}},$$

where the intersection is taken inside $\operatorname{Frac}(R)$.

***Proof*:**
This is [10, A Normal Domain Is the Intersection of Its Height-One Localizations]. The mechanism is that every associated prime of a principal ideal in a Noetherian normal domain has height one, which is where normality enters: if $x = a/b$ lies in every height-one localization but $a \notin (b)$, then $(b : a)$ consists of zero-divisors on $R/(b)$ and so lies in a single associated prime $\mathfrak{q}$ of (b), necessarily of height one, while $x \in R_{\mathfrak{q}}$ produces an element of $(b : a)$ outside $\mathfrak{q}$.

This lemma says that on a normal scheme, a rational function that is regular at every codimension-1 point is in fact a regular function. Put geometrically: regular functions on a normal scheme are determined by their behaviour along prime divisors. This is the algebraic analogue of the Hartogs extension theorem in complex analysis (compare theorem 2.6.18), and it is the key input for the following characterization.

Proposition 5.4.14: Characterizing UFDs Geometrically

Let R be a Noetherian domain. Then R is a UFD if and only if $\operatorname{Spec} R$ is normal and $\operatorname{Cl} \operatorname{Spec} R = 0$.

***Proof*:**
($\Rightarrow$) A UFD is integrally closed (see [10, A Unique Factorization Domain Is Integrally Closed]), so $\operatorname{Spec} R$ is normal. To show $\operatorname{Cl} \operatorname{Spec} R = 0$, it suffices to show that every prime divisor is principal. Let Y be a prime divisor with generic point η_Y corresponding to a height-1 prime $\mathfrak{p} \subseteq R$. Since R is a UFD, $\mathfrak{p}$ is principal: $\mathfrak{p} = (p)$ for some irreducible element $p \in R$. Then $\operatorname{div}(p) = [Y]$ (since $\nu_Y(p) = 1$ and $\nu_{Y'}(p) = 0$ for every other prime divisor Y', as p lies in no other height-1 prime of a UFD). Hence $[Y] = 0$ in $\operatorname{Cl} \operatorname{Spec} R$.

($\Leftarrow$) It suffices to show that every height-1 prime $\mathfrak{p}$ of R is principal (this characterizes Noetherian UFDs; cf. corollary 5.3.17). Let Y be the prime divisor corresponding to $\mathfrak{p}$. Since $\operatorname{Cl} \operatorname{Spec} R = 0$, the divisor $[Y]$ is principal: there exists $f \in K(X)^*$ with $\operatorname{div}(f) = [Y]$. This means:

- $\nu_Y(f) = 1$, so $f \in R_{\mathfrak{p}}$ and f generates $\mathfrak{p}R_{\mathfrak{p}}$;
- $\nu_{Y'}(f) = 0$ for every other prime divisor Y' (corresponding to every other height-1 prime $\mathfrak{q} \neq \mathfrak{p}$), so $f \in R_{\mathfrak{q}}^*$.

In particular, $\nu_{Y'}(f) \geq 0$ for all height-1 primes, so $f \in \bigcap_{\operatorname{ht}(\mathfrak{q})=1} R_{\mathfrak{q}} = R$ by lemma 5.4.13. Similarly, $\nu_{Y'}(f^{-1}) \geq 0$ for all $\mathfrak{q} \neq \mathfrak{p}$, so there is no obstruction from other primes. We claim $\mathfrak{p} = (f)$. Take any $g \in \mathfrak{p}$; then $\nu_Y(g) \geq 1$ and $\nu_{Y'}(g) \geq 0$ for all other prime divisors Y'. Hence

$$\nu_{Y'}(g/f) \;\geq\; 0$$

for *every* height-1 prime, and so $g/f \in R$ by lemma 5.4.13. This gives $g \in fR = (f)$, so $\mathfrak{p} \subseteq (f)$. Since $f \in \mathfrak{p}$ (as $\nu_Y(f) = 1 > 0$) and $\mathfrak{p}$ is prime, we conclude $\mathfrak{p} = (f)$.

Example 5.4.15: Divisor Class Groups

1. *Affine space.* The class group $\operatorname{Cl} \mathbb{A}^n_k = 0$, since $k[x_1, \ldots, x_n]$ is a UFD and proposition 5.4.14 applies. More generally, $\operatorname{Cl} \operatorname{Spec} R = 0$ for any UFD R.
2. *Dedekind domains.* Let R be a Dedekind domain (e.g., the ring of integers $\mathcal{O}_K$ of a number field K). The scheme $X = \operatorname{Spec} R$ is a normal Noetherian integral scheme of dimension 1, so it satisfies our standing hypotheses. The prime divisors are exactly the closed points of X, corresponding to the nonzero prime ideals of R. Every Weil divisor is thus a formal sum of nonzero primes — in other words, a fractional ideal written additively. Under this identification, the principal divisors correspond to principal fractional ideals, and the divisor

class group $\operatorname{Cl} X$ is precisely the *ideal class group* $\operatorname{Cl}(R)$ from algebraic number theory [7, Ideal Class Group]. In particular:

- $\operatorname{Cl}\operatorname{Spec}\mathbb{Z} = 0$, since $\mathbb{Z}$ is a PID (equivalently, a UFD).
- $\operatorname{Cl}\operatorname{Spec}\mathbb{Z}[\sqrt{-5}] \cong \mathbb{Z}/2\mathbb{Z}$, the classical example of a non-trivial ideal class group.

This recovers the fundamental theorem of algebraic number theory: R is a UFD if and only if $\operatorname{Cl}(R) = 0$.

We now compute the divisor class group of projective space — the fundamental example for the rest of the text. Recall that every prime divisor of $\mathbb{P}^n_k$ is a hypersurface $V_+(f)$ for some irreducible homogeneous polynomial $f \in k[x_0, \dots, x_n]$ (unique up to scalar), and its *degree* is $\deg f$.

Proposition 5.4.16: Characterizing Divisors of Projective Space

Let k be a field, $X = \mathbb{P}^n_k$, and $H = V_+(x_0)$ the standard hyperplane. For a Weil divisor $D = \sum_i n_i[Y_i]$, define $\deg D = \sum_i n_i \deg Y_i$. Then:

1. For every $f \in K(X)^*$, one has $\deg(\operatorname{div}(f)) = 0$.
2. If D is a divisor of degree d, then $[D] = d[H]$ in $\operatorname{Cl}\mathbb{P}^n_k$.
3. The degree map induces an isomorphism $\deg\colon \operatorname{Cl}\mathbb{P}^n_k \xrightarrow{\sim} \mathbb{Z}$.

***Proof*:**

Part (1). A rational function on $\mathbb{P}^n_k$ is a ratio f/g of homogeneous polynomials of the same degree d. Factor $f = \prod_i f_i^{a_i}$ and $g = \prod_j g_j^{b_j}$ into irreducibles in the UFD $k[x_0, \dots, x_n]$. Then

$$\operatorname{div}(f/g) \;=\; \sum_i a_i\,[V_+(f_i)] \;-\; \sum_j b_j\,[V_+(g_j)],$$

so $\deg(\operatorname{div}(f/g)) = \sum_i a_i \deg f_i - \sum_j b_j \deg g_j = \deg f - \deg g = 0$.

Part (2). Let $D = \sum_i n_i[Y_i]$ with $\deg D = d$. Each prime divisor Y_i is cut out by an irreducible homogeneous polynomial f_i of degree $\deg Y_i$. The expression

$$r \;=\; \frac{\prod_i f_i^{n_i}}{x_0^d}$$

is a well-defined element of $K(X)^*$: after collecting positive and negative exponents, the numerator and denominator are homogeneous of the same total degree (since $\sum_i n_i \deg f_i = d$). Computing its divisor:

$$\operatorname{div}(r) \;=\; \sum_i n_i\,[Y_i] \;-\; d\,[H] \;=\; D - d[H].$$

Hence $[D] = d[H]$ in $\operatorname{Cl}\mathbb{P}^n_k$.

Part (3). By (1), the degree map descends to a well-defined homomorphism $\deg\colon \operatorname{Cl}\mathbb{P}^n_k \to \mathbb{Z}$. It is surjective (since $\deg[H] = 1$) and injective (since any D of degree 0 satisfies $[D] = 0 \cdot [H] = 0$ by (2)).

Under the correspondence between line bundles and Cartier divisors (proposition 5.4.43), the class $n[H] \in \operatorname{Cl} \mathbb{P}^n_k$ corresponds to the twisting sheaf $\mathcal{O}_{\mathbb{P}^n_k}(n)$. This recovers the classification of line bundles on projective space from corollary 5.4.46.

The next result is the primary computational tool for divisor class groups. It describes how $\operatorname{Cl} X$ changes when we pass to an open subset.

Before comparing class groups, it is worth isolating the bookkeeping fact on which the comparison rests. Passing from X to an open subset U can only destroy the prime divisors lying entirely in the complement; those that survive are recovered from their traces on U, and nothing appears on U that was not already present on X. The statement is used often enough, and in enough places where the ambient scheme is chosen for convenience rather than fixed in advance, that it deserves a name of its own.

Lemma 5.4.17: Extension of Weil Divisors by Closure

Let X be a Noetherian, integral, separated scheme that is regular in codimension 1, and let $U \subseteq X$ be a nonempty open subset.

1. Let $P(U)$ be the set of prime divisors of U, and let $P_U(X) := \{Y \mid Y \text{ a prime divisor of } X,\ Y \cap U \neq \emptyset\}$. Then $Y' \mapsto \overline{Y'}$, the closure taken in X, and $Y \mapsto Y \cap U$ are mutually inverse bijections between $P(U)$ and $P_U(X)$.
2. Extension by zero, defined on generators by $[Y'] \mapsto [\overline{Y'}]$, is a group homomorphism $\operatorname{WDiv} U \to \operatorname{WDiv} X$, and it is a section of the restriction homomorphism $\operatorname{WDiv} X \to \operatorname{WDiv} U$ that sends $[Y]$ to $[Y \cap U]$ when Y meets U and to 0 otherwise.

In particular the restriction homomorphism $\operatorname{WDiv} X \to \operatorname{WDiv} U$ is surjective.

***Proof*:**
First, U inherits the standing hypotheses of definition 5.4.5: open subschemes of Noetherian schemes are Noetherian, a nonempty open subscheme of an integral scheme is integral, separatedness is preserved by open immersions, and the local rings of U are local rings of X. So $\operatorname{WDiv} U$ is defined, and U is dense in the irreducible space X. Throughout, closed integral subschemes carry the reduced induced structure (definition 3.1.26), and codimension is measured at the generic point.

Let Y' be a prime divisor of U, with generic point η, and put $Y := \overline{Y'}$. The closure of an irreducible subset is irreducible (lemma 2.1.24(1)), so Y is a closed integral subscheme of X with generic point η, and $\mathcal{O}_{U,\eta} = \mathcal{O}_{X,\eta}$ because U is open, whence

$$\operatorname{codim}(Y, X) = \dim \mathcal{O}_{X,\eta} = \dim \mathcal{O}_{U,\eta} = \operatorname{codim}(Y', U) = 1$$

So Y is a prime divisor of X meeting U, as $\eta \in U$. Conversely, if Y is a prime divisor of X meeting U, then $Y \cap U$ is a nonempty open subset of the irreducible space Y, hence irreducible and dense in Y, and it is closed in U; it is therefore a closed integral subscheme of U with the same generic point as Y, and the displayed computation run backwards gives $\operatorname{codim}(Y \cap U, U) = 1$. The assignments are mutually inverse: $Y \cap U$ is dense in Y, so $\overline{Y \cap U} = Y$, and Y' is closed in U, so $\overline{Y'} \cap U = Y'$. This proves (1).

For (2), the prime divisors of U form a basis of the free abelian group $\operatorname{WDiv} U$, so $[Y'] \mapsto [\overline{Y'}]$ extends uniquely to a homomorphism. Restricting it sends $[\overline{Y'}]$ to $[\overline{Y'} \cap U] = [Y']$ by (1), so the composite is the identity on a basis and hence on all of $\operatorname{WDiv} U$. A homomorphism admitting a section is surjective, as we sought to show.

The lemma is what lets divisor computations be transported between a scheme and a large open subset of it. A divisor written down on U may be read as a divisor on X with no further choices, and the two descriptions of one prime divisor differ only by the passage between Y and $Y \cap U$. The case of interest is that in which $X \setminus U$ has codimension at least 2: then no prime divisor of X is disjoint from U, restriction and extension are mutually inverse isomorphisms $\operatorname{WDiv} X \cong \operatorname{WDiv} U$, and X may be replaced by its regular locus, or by any convenient open piece with small complement, without loss of divisor-theoretic information. This is the mechanism behind proposition 5.4.18(2) and corollary 5.4.12.

Proposition 5.4.18: Divisor Class Group and Closed Subsets

Let X be a Noetherian, integral, separated scheme that is regular in codimension 1, and let $Z \subsetneq X$ be a proper closed subset with complement $U = X \setminus Z$.

1. There is a surjective group homomorphism $\operatorname{Cl} X \twoheadrightarrow \operatorname{Cl} U$ induced by restriction: $\sum n_i[Y_i] \mapsto \sum n_i[Y_i \cap U]$ (discarding components that do not meet U).
2. If $\operatorname{codim}(Z, X) \geq 2$, then this map is an isomorphism.
3. If Z is irreducible of codimension 1 (hence a prime divisor), there is an exact sequence

$$\mathbb{Z} \xrightarrow{1 \mapsto [Z]} \operatorname{Cl} X \longrightarrow \operatorname{Cl} U \longrightarrow 0.$$

Point (2) explains why the class group is insensitive to removing closed subsets of "small" codimension. It is the divisor-theoretic form of the algebraic Hartogs lemma (theorem 2.6.18): regular functions extend across codimension-2 subsets, and correspondingly, the divisor theory "does not see" such subsets.

***Proof*:**

Part (1). Since U is a dense open subset of the integral scheme X, we have $K(U) = K(X)$. Every prime divisor of U extends uniquely to a prime divisor of X (take the closure in X), so the restriction map $\operatorname{WDiv} X \to \operatorname{WDiv} U$ is surjective on generators and hence surjective. Moreover, a principal divisor $\operatorname{div}(f)$ on X restricts to $\operatorname{div}_U(f)$ on U (the valuations at prime divisors meeting U are unchanged), so the map descends to a surjection $\operatorname{Cl} X \twoheadrightarrow \operatorname{Cl} U$.

Part (2). If $\operatorname{codim}(Z, X) \geq 2$, then Z contains no codimension-1 points of X. Hence every prime divisor of X meets U, and the restriction $\operatorname{WDiv} X \to \operatorname{WDiv} U$ is an isomorphism (not a surjection). For injectivity on class groups: if $D \in \operatorname{WDiv} X$ becomes principal on U, say $D|_U = \operatorname{div}_U(f)$, then $D - \operatorname{div}_X(f)$ is a Weil divisor on X supported on Z. Since Z contains no prime divisors, this forces $D - \operatorname{div}_X(f) = 0$, so D is principal. This is the divisor-theoretic counterpart of the algebraic Hartogs lemma (theorem 2.6.18): regular functions extend across codimension-2 subsets, and correspondingly the divisor theory "does not see" such subsets.

Part (3). The kernel of $\operatorname{Cl} X \to \operatorname{Cl} U$ consists of classes representable by divisors supported entirely on Z. Since Z is irreducible of codimension 1, every such divisor is an integer multiple of $[Z]$, so the kernel is exactly the image of the map $\mathbb{Z} \to \operatorname{Cl} X$, $1 \mapsto [Z]$.

By strategically removing closed subsets and applying the exact sequence in part (3), one can often reduce the computation of $\operatorname{Cl} X$ to an affine (and hence often factorial) situation. Part (2) shows that removing "small" closed subsets does not change the class group, which frequently allows us to simplify X without losing divisor-theoretic information.

Example 5.4.19: More Divisor Class Groups

1. *Punctured affine space.* Let $X = \mathbb{A}^n_k \setminus \{0\}$ for $n \geq 2$. Since $\operatorname{Cl} \mathbb{A}^n_k = 0$ and $\{0\}$ has codimension $n \geq 2$, the surjection $\operatorname{Cl} \mathbb{A}^n_k \twoheadrightarrow \operatorname{Cl} X$ is an isomorphism (no prime divisors are lost). Hence $\operatorname{Cl} X = 0$.

2. *Complements of hypersurfaces in $\mathbb{P}^2$.* Let $Y \subseteq \mathbb{P}^2_k$ be an irreducible curve of degree d. Then Y is a prime divisor, and proposition 5.4.18(3) gives

$$\mathbb{Z} \xrightarrow{1 \mapsto [Y]} \operatorname{Cl} \mathbb{P}^2_k \longrightarrow \operatorname{Cl}(\mathbb{P}^2_k \setminus Y) \longrightarrow 0.$$

   Since $[Y] = d[H]$ in $\operatorname{Cl} \mathbb{P}^2_k \cong \mathbb{Z}$ by proposition 5.4.16, the image of $\mathbb{Z}$ is $d\mathbb{Z} \subseteq \mathbb{Z}$, and hence

$$\operatorname{Cl}(\mathbb{P}^2_k \setminus Y) \;\cong\; \mathbb{Z}/d\mathbb{Z}.$$

   For instance, the complement of a conic ($d = 2$) has class group $\mathbb{Z}/2\mathbb{Z}$, while the complement of a line ($d = 1$) has trivial class group — geometrically, every codimension-1 subvariety of $\mathbb{P}^2 \setminus L$ is already a principal divisor. The d-torsion can be understood via Bézout's theorem ([22, Bézout's Theorem]): any line L meets Y in exactly d points (counted with multiplicity), so $d[L \cap (\mathbb{P}^2 \setminus Y)]$ is principal on the complement, but $[L \cap (\mathbb{P}^2 \setminus Y)]$ itself need not be.

3. *The quadric cone.* Let $\operatorname{char} k \neq 2$ and consider the quadric cone $X = \operatorname{Spec} R$ where $R = k[x, y, z]/(xy - z^2)$. The ring R is a normal domain[a], so X is normal, Noetherian, integral, and regular in codimension 1. However, R is *not* a UFD: the equation $xy = z^2$ exhibits a failure of unique factorization. By proposition 5.4.14, we expect $\operatorname{Cl} X \neq 0$.

   Let $Y = V(y, z) \subseteq X$ be the "ruling" of the cone (see fig. 5.1). Since $\mathfrak{p} = (y, z)$ is a height-1 prime of R, the subscheme Y is a prime divisor. We claim that $\operatorname{div}(y) = 2[Y]$. Indeed, in the local ring $\mathcal{O}_{X,\eta_Y} = R_{\mathfrak{p}}$, the relation $xy = z^2$ and the fact that $x \notin \mathfrak{p}$ (so x is a unit in $R_{\mathfrak{p}}$) give $y = z^2 x^{-1} \in (z)^2$. Since z generates the maximal ideal of the DVR $R_{\mathfrak{p}}$, the uniformizer is z and $\nu_Y(y) = 2$.

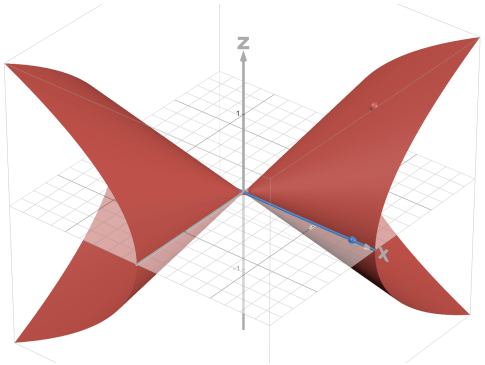


Figure 5.1: A ruling on the quadratic cone

Now apply the exact sequence to $Z = Y$:

$$\mathbb{Z} \xrightarrow{1\mapsto[Y]} \operatorname{Cl} X \longrightarrow \operatorname{Cl}(X \setminus Y) \longrightarrow 0.$$

The complement is $X \setminus Y = \operatorname{Spec} R_y$ (inverting y). In R_y, the relation $xy = z^2$ allows us to eliminate $x = z^2 y^{-1}$, yielding $R_y \cong k[y, y^{-1}, z]$. This is a localization of the polynomial ring $k[y, z]$ (hence a UFD), so $\operatorname{Cl}(X \setminus Y) = 0$ by proposition 5.4.14. The exact sequence collapses to

$$\mathbb{Z} \longrightarrow \operatorname{Cl} X \longrightarrow 0,$$

so $\operatorname{Cl} X$ is cyclic, generated by $[Y]$. Since $\operatorname{div}(y) = 2[Y]$, we have $2[Y] = 0$ in $\operatorname{Cl} X$, giving $\operatorname{Cl} X \cong \mathbb{Z}/2\mathbb{Z}$ or $\operatorname{Cl} X = 0$.

To rule out the latter, we show that $\mathfrak{p} = (y, z)$ is not principal. Let $\mathfrak{m} = (x, y, z)$ be the maximal ideal at the vertex of the cone. The images $\overline{y}$ and $\overline{z}$ are linearly independent in the cotangent space $\mathfrak{m}/\mathfrak{m}^2 \cong k^3$ (the relation $xy - z^2 \in \mathfrak{m}^2$ imposes no constraint at the level of $\mathfrak{m}/\mathfrak{m}^2$). Hence $\mathfrak{p}$ requires at least two generators and cannot be principal. We conclude

$$\operatorname{Cl} X \;\cong\; \mathbb{Z}/2\mathbb{Z}.$$

Geometrically, a single ruling cannot be cut out by one equation, but the divisor $2[Y] = \operatorname{div}(y)$ *can*: the function y vanishes along Y with multiplicity 2.

4. *The smooth quadric surface.* Let $Q = V_+(xw - yz) \subseteq \mathbb{P}^3_k$. Under the Segre embedding (example 0.2.28), $Q \cong \mathbb{P}^1_k \times \mathbb{P}^1_k$, and Q carries two families of lines (rulings): "horizontal" lines $L_p = \{p\} \times \mathbb{P}^1$ and "vertical" lines $M_q = \mathbb{P}^1 \times \{q\}$ for $p, q \in \mathbb{P}^1$. Fix one line L_0 from the first family. Then $Q \setminus L_0 \cong \mathbb{A}^1 \times \mathbb{P}^1$ (removing one point from the first factor).

   Claim: $\operatorname{Cl}(\mathbb{A}^1 \times \mathbb{P}^1) \cong \mathbb{Z}$, generated by a vertical line. Removing a vertical section $M_0' = \mathbb{A}^1 \times \{q_0\}$ gives $\mathbb{A}^1 \times (\mathbb{P}^1 \setminus \{q_0\}) \cong \mathbb{A}^2$, which has $\operatorname{Cl} = 0$. The exact sequence

$$\mathbb{Z} \xrightarrow{1\mapsto[M_0']} \operatorname{Cl}(\mathbb{A}^1 \times \mathbb{P}^1) \longrightarrow 0$$

   shows $\operatorname{Cl}(\mathbb{A}^1 \times \mathbb{P}^1)$ is cyclic, generated by $[M_0']$. To see that $[M_0']$ has infinite order, restrict to a fiber: the inclusion $\iota\colon \{s_0\} \times \mathbb{P}^1 \hookrightarrow \mathbb{A}^1 \times \mathbb{P}^1$ induces $\iota^*\colon \operatorname{Cl}(\mathbb{A}^1 \times \mathbb{P}^1) \to \operatorname{Cl}(\mathbb{P}^1) \cong \mathbb{Z}$ sending $[M_0'] \mapsto [\{q_0\}]$, which is a generator. Since ι^* is surjective, $[M_0']$ cannot be torsion.

   Returning to Q, the exact sequence for removing L_0 is

$$\mathbb{Z} \xrightarrow{1\mapsto[L_0]} \operatorname{Cl} Q \longrightarrow \operatorname{Cl}(\mathbb{A}^1 \times \mathbb{P}^1) \cong \mathbb{Z} \longrightarrow 0.$$

   A vertical line M_0 on Q restricts to a generator of $\operatorname{Cl}(\mathbb{A}^1 \times \mathbb{P}^1)$, so the map $\mathbb{Z} \to \operatorname{Cl} Q$, $1 \mapsto [M_0]$ provides a splitting. Hence $\operatorname{Cl} Q \cong \mathbb{Z}[L_0] \oplus \mathbb{Z}[M_0]$, provided $[L_0]$ has infinite order. This follows by symmetry: swapping the two $\mathbb{P}^1$ factors exchanges horizontal and vertical rulings, so the same argument with M_0 removed shows $[L_0]$ has infinite order. Therefore

$$\operatorname{Cl} Q \;\cong\; \mathbb{Z} \oplus \mathbb{Z},$$

   generated by the two families of rulings. A hyperplane section of Q has class $[L_0] + [M_0]$[b]. In contrast to $\operatorname{Cl} \mathbb{P}^2_k \cong \mathbb{Z}$, the "degree" of a curve on Q is a *pair* of integers (a, b) recording how it meets each family of rulings — a first glimpse of how the Picard group can carry genuinely higher-rank information.

[a] A standard exercise: if $\operatorname{char} k \neq 2$ and $f \in k[x_1, \ldots, x_n]$ is square-free, then $k[x_1, \ldots, x_n, z]/(z^2 - f)$ is integrally closed.

[b] This can be checked directly: the hyperplane $V_+(x)$ meets Q in $V_+(x, yz) = V_+(x, y) \cup V_+(x, z)$, which is a horizontal ruling plus a vertical ruling.

The exact sequence in proposition 5.4.18 relates the class groups of a scheme and its open subsets. The following result provides a complementary tool, showing that the class group is insensitive to "thickening" in the affine-line direction.

Proposition 5.4.20: Class Group Stably Isomorphic

Let X be a Noetherian, integral, separated scheme that is regular in codimension 1. Then $X \times \mathbb{A}^1$ also satisfies all of these conditions, and the projection $\pi\colon X \times \mathbb{A}^1 \to X$ induces an isomorphism

$$\pi^*\colon \operatorname{Cl} X \xrightarrow{\sim} \operatorname{Cl}(X \times \mathbb{A}^1).$$

***Proof*:**
The scheme $X \times \mathbb{A}^1$ is Noetherian (locally $\operatorname{Spec} R[t]$, which is Noetherian by Hilbert's basis theorem), integral (with function field $K(X)(t)$), and separated (since separatedness is preserved under base change). It remains to verify regularity in codimension 1. The codimension-1 points of $X \times \mathbb{A}^1$ fall into two types:

- *Type 1:* the generic point of $Y \times \mathbb{A}^1$, where Y is a prime divisor of X. The local ring is $\mathcal{O}_{X,\eta_Y}[t]$ localized at $\mathfrak{m}_{\eta_Y} \cdot \mathcal{O}_{X,\eta_Y}[t]$. Since $\mathcal{O}_{X,\eta_Y}$ is a DVR (say with uniformizer u), this localization is a DVR with the same uniformizer u.
- *Type 2:* a point ξ whose image under π is the generic point of X. The local ring is $K(X)[t]$ localized at an irreducible polynomial, hence a DVR (since $K(X)[t]$ is a PID).

Thus $X \times \mathbb{A}^1$ is regular in codimension 1.

Definition of π^*. For a Weil divisor $D = \sum n_i[Y_i]$ on X, set $\pi^*(D) = \sum n_i[Y_i \times \mathbb{A}^1]$. Each $Y_i \times \mathbb{A}^1$ is a type 1 prime divisor of $X \times \mathbb{A}^1$. If $f \in K(X)^*$, then $\pi^*(\operatorname{div}_X(f)) = \operatorname{div}_{X\times\mathbb{A}^1}(f)$ (viewing f as an element of $K(X) \subseteq K(X)(t) = K(X \times \mathbb{A}^1)$), so π^* descends to a well-defined homomorphism $\pi^*\colon \operatorname{Cl} X \to \operatorname{Cl}(X \times \mathbb{A}^1)$.

Injectivity. Suppose $D \in \operatorname{WDiv} X$ and $\pi^*(D) = \operatorname{div}(h)$ for some $h \in K(X)(t)^*$. Since $\pi^*(D)$ involves only type 1 prime divisors, h must have $\nu_Z(h) = 0$ for every type 2 prime divisor Z. Write $h = g_1/g_2$ with $g_1, g_2 \in K(X)[t]$ coprime. Any irreducible factor of g_i of positive degree in t would contribute a nonzero valuation at a type 2 prime divisor, contradicting $\nu_Z(h) = 0$. Hence $g_1, g_2 \in K(X)^*$, so $h = g_1/g_2 \in K(X)^*$. It follows that $D = \operatorname{div}_X(h)$ is principal, and π^* is injective.

Surjectivity. It suffices to show that every type 2 prime divisor on $X \times \mathbb{A}^1$ is linearly equivalent to a $\mathbb{Z}$-linear combination of type 1 prime divisors (which lie in the image of π^*). Let $Z \subseteq X \times \mathbb{A}^1$ be a type 2 prime divisor. Localizing at the generic point of X gives a prime divisor of $\operatorname{Spec} K(X)[t]$. Since $K(X)[t]$ is a PID, this corresponds to a principal ideal (f) for some irreducible $f \in K(X)[t]$. Viewed as an element of $K(X)(t)^*$, the divisor $\operatorname{div}_{X\times\mathbb{A}^1}(f)$ involves Z with coefficient 1, plus possibly type 1 prime divisors arising from the denominators of the coefficients of f. Crucially, f cannot vanish along any *other* type 2 prime divisor $Z' \neq Z$: such a Z' would correspond to a different irreducible in $K(X)[t]$, and f is irreducible. Hence

$$\operatorname{div}(f) \;=\; [Z] + (\text{type 1 divisors}),$$

showing that $[Z]$ is linearly equivalent to a combination of type 1 divisors. Since type 1 divisors are in the image of π^*, the map π^* is surjective.

This result is the scheme-theoretic upgrade of the intuition that "adding a free variable does not change divisor theory." As immediate applications: $\operatorname{Cl}(\mathbb{A}^n_k) = 0$ for all n (by induction from $\operatorname{Cl}(\operatorname{Spec} k) = 0$), and $\operatorname{Cl}(X \times \mathbb{A}^n) \cong \operatorname{Cl} X$ for any X satisfying our hypotheses.

5.4.21 Foreshadowing: $\mathbb{A}^1$-Invariance By proposition 5.4.20, the functor $X \mapsto \operatorname{Cl} X$ is $\mathbb{A}^1$-*invariant*: pullback along $X \times \mathbb{A}^1 \to X$ is an isomorphism. This is the defining property of the invariants studied in motivic ($\mathbb{A}^1$-)homotopy theory (Morel-Voevodsky), in which $\mathbb{A}^1$ takes the place of the unit interval $[0,1]$. The map $H\colon \mathbb{A}^n \times \mathbb{A}^1 \to \mathbb{A}^n$, $(x,t) \mapsto tx$, is the identity at $t = 1$ and the constant map to 0 at $t = 0$, so every $\mathbb{A}^1$-invariant functor takes the same value on $\mathbb{A}^n_k$ as on $\operatorname{Spec} k$. The computation $\operatorname{Cl}(\mathbb{A}^n_k) = 0$ above is the first instance.
The same question can be asked of three invariants of increasing difficulty, with X regular:

1. *Line bundles.* For X locally factorial, $\operatorname{Cl} X \cong \operatorname{Pic} X$ (corollary 5.4.45), so Pic is $\mathbb{A}^1$-invariant. When X is a variety over a field k, $\operatorname{Cl} X = \mathrm{CH}^1(X)$, and this is the codimension-1 case of the $\mathbb{A}^1$-invariance of Chow groups: for a vector bundle $E \to X$ of rank r, flat pullback $\mathrm{CH}_j(X) \to \mathrm{CH}_{j+r}(E)$ is an isomorphism for every j, over any field k [29, Thm. 3.3(a)]. *Intersection Theory* proves it over an algebraically closed field [19, Affine-Bundle Invariance].

2. *Stable classes.* $K_0(X) \cong K_0(X \times \mathbb{A}^1)$ (Grothendieck). For $X = \operatorname{Spec} k$ this is Serre's theorem that every finitely generated projective $k[x_1, \dots, x_n]$-module is stably free [6].

3. *Vector bundles of fixed rank.* $\operatorname{Vect}_r(X) \cong \operatorname{Vect}_r(X \times \mathbb{A}^1)$ for X smooth and affine over a field (Lindel). For $X = \operatorname{Spec} k$ this is the Quillen-Suslin theorem (example 5.1.13).

The hypotheses are needed. Without regularity, invariance fails already in rank 1: for the cusp $R = k[x,y]/(y^2 - x^3)$, the map $\operatorname{Pic} R \to \operatorname{Pic} R[t]$ is not surjective (Traverso), which is one reason motivic homotopy theory is built on smooth schemes. Without affineness, rank 1 survives (proposition 5.4.20 holds for $X = \mathbb{P}^1$) but rank 2 fails: the extension of $\mathcal{O}(1)$ by $\mathcal{O}(-1)$ on $\mathbb{P}^1 \times \mathbb{A}^1$ with class $t \in H^1(\mathbb{P}^1 \times \mathbb{A}^1, \mathcal{O}(-2)) \cong k[t]$ is a rank-2 bundle whose fiber over $t \neq 0$ is $\mathcal{O} \oplus \mathcal{O}$ and over $t = 0$ is $\mathcal{O}(1) \oplus \mathcal{O}(-1)$, so it is not pulled back from $\mathbb{P}^1$.

We now record several further applications that illustrate the interplay between the class group, unique factorization, and geometric structure.

Example 5.4.22: Applications of Stable Isomorphism

1. *Punctured affine space.* Let $X = \mathbb{A}^n_k \setminus \{0\}$ for $n \geq 2$. The complement $\{0\}$ has codimension $n \geq 2$ in $\mathbb{A}^n_k$, so proposition 5.4.18(2) gives $\operatorname{Cl} X \cong \operatorname{Cl} \mathbb{A}^n_k = 0$.

2. *Products with affine space.* The class group of the smooth quadric surface $Q \cong \mathbb{P}^1 \times \mathbb{P}^1$ is $\mathbb{Z} \oplus \mathbb{Z}$ (example 5.4.19). By proposition 5.4.20, $\operatorname{Cl}(\mathbb{P}^1 \times \mathbb{P}^1 \times \mathbb{A}^n) \cong \mathbb{Z} \oplus \mathbb{Z}$ for all $n \geq 0$.

3. *Normality is necessary.* Let $C = \operatorname{Spec} k[x,y]/(y^2 - x^3)$ be the cuspidal cubic. The local ring at the origin is not integrally closed (y/x satisfies $(y/x)^2 = x$ but does not lie in the local ring), so C is not normal. The class group formalism as developed here does not directly apply; one must pass to the normalization $\widetilde{C} \cong \mathbb{A}^1_k$ (via $t \mapsto (t^2, t^3)$), which has $\operatorname{Cl} = 0$. The failure of normality at a single point introduces pathologies that the Cartier divisor formalism (definition 5.4.24) handles more gracefully.

5.4.2 Cartier Divisors

Weil divisors provide a geometric language for codimension-1 phenomena, but their definition requires strong hypotheses on the ambient scheme: integrality (for the function field), regularity in codimension 1 (for the valuations), and the Noetherian condition (for finiteness). On a general scheme — possibly reducible, non-reduced, or with non-regular local rings — none of these tools are available.

Cartier divisors provide an alternative that works on any scheme. The idea is to abandon the "global" perspective of Weil divisors (formal sums of codimension-1 subvarieties) in favour of a "local" one: a Cartier divisor is the data of rational functions on an open cover that are "locally principal," glued together by units on overlaps. Since principal divisors carry no interesting divisor-class information, the gluing data is precisely what the class group detects.

On sufficiently nice schemes (integral, Noetherian, locally factorial), the two theories agree and the distinction is a matter of perspective. The advantage of Cartier divisors, beyond their greater generality, is their direct connection to line bundles: we will see in proposition 5.4.43 that Cartier divisor classes are in natural bijection with isomorphism classes of invertible sheaves.

To define Cartier divisors on schemes that may not be integral, we need a replacement for the function field. The correct substitute is the *sheaf of total quotient rings*, which inverts all non-zero-divisors rather than all nonzero elements.

Definition 5.4.23: Sheaf of Total Quotient Rings

Let X be a scheme. For each affine open $U = \operatorname{Spec} R \subseteq X$, let $S_U \subseteq \Gamma(U, \mathcal{O}_X)$ denote the set of non-zero-divisors, and form the total quotient ring $S_U^{-1}\Gamma(U, \mathcal{O}_X)$. This assignment defines a presheaf on the affine opens of X; its sheafification is the *sheaf of total quotient rings*, denoted $\mathscr{K}_X$.
Write $\mathscr{K}_X^*$ for the subsheaf of abelian groups whose sections over an open U are the invertible elements of $\mathscr{K}_X(U)$, and $\mathcal{O}_X^*$ for the subsheaf of units of $\mathcal{O}_X$. The inclusion $\mathcal{O}_X^* \hookrightarrow \mathscr{K}_X^*$ makes $\mathcal{O}_X^*$ a subsheaf of abelian groups, and we may form the quotient sheaf $\mathscr{K}_X^*/\mathcal{O}_X^*$.

When X is integral, every nonzero element of an integral domain is a non-zero-divisor, so the total quotient ring of $\Gamma(U, \mathcal{O}_X)$ is simply its fraction field — which is $K(X)$ for every nonempty open U. Thus $\mathscr{K}_X$ is the constant sheaf associated to $K(X)$, and we recover the usual notion of rational function. In the non-integral case, $\mathscr{K}_X$ is genuinely more subtle: its sections are "rational functions" that may have poles but are not allowed to divide by zero-divisors.

Definition 5.4.24: Cartier Divisor

Let X be a scheme. A *Cartier divisor* on X is a global section of the quotient sheaf $\mathscr{K}_X^*/\mathcal{O}_X^*$:

$$\operatorname{CDiv} X \;:=\; \Gamma(X,\ \mathscr{K}_X^*/\mathcal{O}_X^*).$$

This is an abelian group under the multiplication induced from $\mathscr{K}_X^*$. A Cartier divisor is *principal* if it lies in the image of the natural map

$$\Gamma(X, \mathscr{K}_X^*) \longrightarrow \Gamma(X, \mathscr{K}_X^*/\mathcal{O}_X^*) = \operatorname{CDiv} X.$$

Two Cartier divisors are *linearly equivalent* if they differ by a principal Cartier divisor.

Unwinding the definition of global sections of a quotient sheaf, a Cartier divisor can be represented by the following data: an open cover $\{U_i\}$ of X together with sections $f_i \in \Gamma(U_i, \mathscr{K}_X^*)$ such that on every overlap

$$\frac{f_i}{f_j} \;\in\; \Gamma(U_i \cap U_j,\ \mathcal{O}_X^*).$$

Two such collections $\{(U_i, f_i)\}$ and $\{(V_j, g_j)\}$ represent the same Cartier divisor if and only if, after passing to a common refinement, the ratios f_i/g_j are units wherever defined. The condition $f_i/f_j \in \mathcal{O}_X^*$ says that f_i and f_j agree up to an invertible regular function on the overlap — and since units contribute neither zeros nor poles, the "divisorial data" (zeros and poles of the f_i) glue to a well-defined global object.

A Cartier divisor is principal precisely when the cover can be taken to be $\{X\}$ itself: it is represented by a single global section $f \in \Gamma(X, \mathscr{K}_X^*)$.

5.4.25 Warning The group $\operatorname{CDiv} X = \Gamma(X, \mathscr{K}_X^*/\mathcal{O}_X^*)$ is *not* the same as the quotient $\Gamma(X, \mathscr{K}_X^*)/\Gamma(X, \mathcal{O}_X^*)$ in general. The former involves sections of the *quotient sheaf*, which may be represented by locally defined data that cannot be globalized. The discrepancy is measured by $H^1(X, \mathcal{O}_X^*)$, which we will identify with the Picard group in theorem 5.4.44.

In practice, the most frequently encountered Cartier divisors are *effective* — those with no poles:

Definition 5.4.26: Effective Cartier Divisor

A Cartier divisor $D \in \operatorname{CDiv} X$ is *effective* if it can be represented by data $\{(U_i, f_i)\}$ with each $f_i \in \Gamma(U_i, \mathcal{O}_X)$ (not in $\mathscr{K}_X^*$). Since $f_i \in \mathscr{K}_X^*$, each f_i is a non-zero-divisor in $\mathcal{O}_X(U_i)$.
An effective Cartier divisor determines a closed subscheme $D \hookrightarrow X$ whose ideal sheaf $\mathcal{I}_D \subseteq \mathcal{O}_X$ is locally generated by f_i on each U_i. This is a closed subscheme that is *locally principal*: on each U_i, it is cut out by a single equation.

An effective Cartier divisor is a closed subscheme cut out, locally, by one equation that is not a zero divisor. Both halves of that description are conditions on a single scheme, and neither is preserved in the passage to a family on its own. If X lies over a base S and $D \subseteq X$ is locally principal, nothing prevents the local equation from degenerating over a special point of S: a non-zero-divisor on the total space may restrict to a zero divisor, or to 0, on one fiber. The subscheme D is then a divisor on X while D_s is not a divisor on X_s, and every construction that reads numerical data off the fibers

collapses at that point of the base. The repair is a single condition on the morphism $D \to S$ rather than a list of conditions on the individual fibers.

Definition 5.4.27: Relative Effective Cartier Divisor

Let $f\colon X \to S$ be a morphism of schemes. A closed subscheme $D \subseteq X$ is an *effective Cartier divisor on X relative to S* (a *relative effective Cartier divisor on X/S*) if

1. the ideal sheaf $\mathcal{I}_D \subseteq \mathcal{O}_X$ is invertible, i.e. locally free of rank 1 (definition 5.1.28), so that D is an effective Cartier divisor on X; and
2. the composite $D \hookrightarrow X \xrightarrow{f} S$ is flat: for every $d \in D$, the local ring $\mathcal{O}_{D,d}$ is flat over $\mathcal{O}_{S,f(d)}$.

Both conditions are affine-local, and it pays to record what they say in coordinates. Let $\operatorname{Spec} A \subseteq S$ and $\operatorname{Spec} B \subseteq X$ be affine opens with $f(\operatorname{Spec} B) \subseteq \operatorname{Spec} A$, corresponding to a ring map $A \to B$, and shrink $\operatorname{Spec} B$ so that $\mathcal{I}_D$ is free there (definition 5.1.22). By the correspondence between closed subschemes and quasi-coherent ideal sheaves (proposition 5.1.34) and the module-sheaf dictionary (corollary 5.1.12), the restriction $\mathcal{I}_D|_{\operatorname{Spec} B}$ is $\widetilde{gB}$ for a single $g \in B$, and freeness of rank 1 says exactly that $B \to gB$, $1 \mapsto g$, is an isomorphism, i.e. $\operatorname{Ann}_B(g) = 0$. So

$$D \cap \operatorname{Spec} B \;=\; \operatorname{Spec}\big(B/gB\big)$$

with g a non-zero-divisor, and condition (2) says that B/gB is flat as an A-module.[6] Proposition 5.4.29 converts this into the condition one checks in practice: for every $s \in S$, the image $\overline{g}$ of g in the fiber ring $B \otimes_A \kappa(s)$ is again a non-zero-divisor. The converse implication is not formal; recovering flatness of B/gB from the fiberwise condition requires $X \to S$ itself to be flat, together with finiteness hypotheses and the local criterion of flatness (see [18]).

Both results below rest on one piece of commutative algebra: a flat quotient forces the defining equation to stay injective after every change of coefficients.

Lemma 5.4.28: Non-Zero-Divisors and Flat Quotients

Let $A \to B$ be a ring homomorphism and let $g \in B$ satisfy $\operatorname{Ann}_B(g) = 0$. Assume $C := B/gB$ is flat as an A-module. Then for every A-module M the sequence of B-modules

$$0 \longrightarrow B \otimes_A M \xrightarrow{g\otimes \mathrm{id}} B \otimes_A M \longrightarrow C \otimes_A M \longrightarrow 0$$

is exact. In particular multiplication by g is injective on $B \otimes_A M$, and if M is an A-algebra, the image of g in the ring $B \otimes_A M$ is a non-zero-divisor.

***Proof*:**

Since $\operatorname{Ann}_B(g) = 0$, multiplication by g is injective on B, so $0 \to B \xrightarrow{g} B \to C \to 0$ is a short exact sequence of A-modules. Applying $- \otimes_A M$ and invoking the long exact sequence of

[6] Flatness of a module may be tested at primes, so the stalkwise condition of (2) is equivalent to flatness of B/gB over A for every such pair of affine opens; see [10].

$\operatorname{Tor}^A_\bullet(-, M)$ (see [18, Long Exact Sequence From Derived Functor]) yields exactness of

$$\operatorname{Tor}^A_1(C, M) \longrightarrow B \otimes_A M \xrightarrow{g\otimes \mathrm{id}} B \otimes_A M \longrightarrow C \otimes_A M \longrightarrow 0$$

Flatness of C over A says precisely that $\operatorname{Tor}^A_1(C, -)$ vanishes, so the leftmost term is zero and multiplication by g on $B \otimes_A M$ is injective. When M carries an A-algebra structure, $B \otimes_A M$ is a ring and that injectivity is the assertion that the image of g is a non-zero-divisor, as we sought to show.

The first payoff is the statement that motivated the definition: the members of the family really are divisors.

Proposition 5.4.29: Relative Cartier Divisors and Fibers

Let $f\colon X \to S$ be a morphism of schemes and let $D \subseteq X$ be an effective Cartier divisor on X relative to S. For every $s \in S$, the fiber $D_s = D \times_S \operatorname{Spec} \kappa(s)$ is an effective Cartier divisor on the fiber $X_s = X \times_S \operatorname{Spec} \kappa(s)$ (definition 3.2.9).

***Proof*:**
Step 1: D_s is a closed subscheme of X_s. The morphism $D \to S$ factors through X, so associativity of fiber products (theorem 3.2.1) gives $D_s = D \times_S \operatorname{Spec} \kappa(s) = D \times_X X_s$. Thus $D_s \to X_s$ is the base change of the closed immersion $D \hookrightarrow X$ (definition 3.1.22) along $X_s \to X$, hence a closed immersion by proposition 3.2.5(2). It remains to show that $\mathcal{I}_{D_s}$ is invertible, which may be checked on an affine open cover of X_s.
Step 2: the local model. Let $x \in X_s$, and choose affine opens $\operatorname{Spec} A \subseteq S$ containing s and $\operatorname{Spec} B \subseteq X$ containing the image of x, with $f(\operatorname{Spec} B) \subseteq \operatorname{Spec} A$ and $\mathcal{I}_D$ free on $\operatorname{Spec} B$. As in the local description above, $D \cap \operatorname{Spec} B = \operatorname{Spec}(B/gB)$ with $\operatorname{Ann}_B(g) = 0$ and $C = B/gB$ flat over A. Put $\overline{B} = B \otimes_A \kappa(s)$ and let $\overline{g}$ be the image of g in $\overline{B}$. The open $\operatorname{Spec} \overline{B} = \operatorname{Spec} B \times_{\operatorname{Spec} A} \operatorname{Spec} \kappa(s)$ of X_s contains x, and such opens cover X_s as $\operatorname{Spec} B$ ranges over the chosen cover of X. Right exactness of the tensor product gives

$$C \otimes_A \kappa(s) \;=\; (B/gB) \otimes_A \kappa(s) \;=\; \overline{B}/\overline{g}\,\overline{B}$$

so $D_s \cap \operatorname{Spec} \overline{B} = \operatorname{Spec}(\overline{B}/\overline{g}\,\overline{B})$, and the ideal of D_s on this open is $\overline{g}\,\overline{B}$.
Step 3: invertibility. Apply lemma 5.4.28 with the A-algebra $M = \kappa(s)$: the element $\overline{g}$ is a non-zero-divisor in $\overline{B}$. Multiplication by $\overline{g}$ is then an isomorphism $\overline{B} \to \overline{g}\,\overline{B}$ of $\overline{B}$-modules, so $\mathcal{I}_{D_s}$ is free of rank 1 on $\operatorname{Spec} \overline{B}$ by corollary 5.1.12. These opens cover X_s, so $\mathcal{I}_{D_s}$ is invertible and D_s is an effective Cartier divisor on X_s, as we sought to show.

Proposition 5.4.29 runs the cheap implication: flatness of $\mathcal{O}_D$ over the base is a hypothesis about every A-module at once, and specializing it to $M = \kappa(s)$ needs nothing. The converse runs against the grain, since it must produce flatness, a statement quantified over all A-modules, out of information carried by one fiber at a time. Two extra inputs close the gap. The first is a source of flatness on the total space: the morphism f is assumed flat, so the ambient local ring contributes no Tor of its own. The second is a theorem reducing flatness to the vanishing of a single Tor against the residue field of the base.

The form of that theorem used below is the local criterion for flatness proved in *Homological Algebra* [18]: for a local homomorphism $(A, \mathfrak{m}) \to (B, \mathfrak{n})$ of Noetherian local rings and a finitely

generated B-module M, the module M is flat over A if and only if $\operatorname{Tor}_1^A(M, A/\mathfrak{m}) = 0$. The same criterion, in the same local-homomorphism form, is recorded in *Commutative Algebra* [10], where its proof is referred back to the homological account. The Noetherian and finiteness hypotheses in the proposition are present for one reason: they are what the criterion asks for.

Proposition 5.4.30: Fiberwise Criterion for a Relative Cartier Divisor

Let S be a Noetherian scheme and let $f\colon X \to S$ be a morphism locally of finite presentation, which over a Noetherian base is the same as locally of finite type (definition 3.4.28). Assume X is flat over S, and let $D \subseteq X$ be a closed subscheme whose ideal sheaf $\mathcal{I}_D \subseteq \mathcal{O}_X$ is invertible. If for every $s \in S$ the fiber D_s is an effective Cartier divisor on the fiber X_s (definition 3.2.9), then D is an effective Cartier divisor on X relative to S (definition 5.4.27); that is, $D \to S$ is flat.

***Proof*:**
Step 1: reduction to stalks. Invertibility of $\mathcal{I}_D$ is a hypothesis, so only the flatness of $D \to S$ is at issue, and that condition is imposed one point at a time. Fix $d \in D$, put $s = f(d)$, and write

$$A = \mathcal{O}_{S,s}, \qquad B = \mathcal{O}_{X,d}, \qquad C = \mathcal{O}_{D,d}$$

Step 2: the hypotheses of the criterion. Since S is Noetherian, A is a Noetherian local ring. Choose affine opens $\operatorname{Spec} A_0 \subseteq S$ containing s and $\operatorname{Spec} B_0 \subseteq f^{-1}(\operatorname{Spec} A_0)$ containing d; then B_0 is a finitely generated algebra over the Noetherian ring A_0, hence Noetherian by the Hilbert basis theorem [10, Finitely Generated Algebras Are Noetherian], and B is a localization of B_0, hence Noetherian local. The homomorphism $A \to B$ is local because $f(d) = s$, and it is flat because X is flat over S.
Step 3: the local equation. The sheaf $\mathcal{I}_D$ is invertible, so its stalk at d is free of rank 1 over B. By the local description following definition 5.4.27, $\mathcal{I}_{D,d} = gB$ for an element $g \in B$ with $\operatorname{Ann}_B(g) = 0$, and $C = B/gB$. In particular C is a finitely generated B-module and

$$0 \longrightarrow B \xrightarrow{\,g\,} B \longrightarrow C \longrightarrow 0$$

is a short exact sequence of A-modules.
Step 4: what the fiber hypothesis says. Write $\kappa = \kappa(s)$ and $\overline{B} = B \otimes_A \kappa$, and let $\overline{g}$ be the image of g. Localization commutes with the tensor product, so $\overline{B}$ is the local ring of X_s at d, and right exactness gives $C \otimes_A \kappa = \overline{B}/\overline{g}\,\overline{B}$; hence $\overline{g}\,\overline{B}$ is the stalk of $\mathcal{I}_{D_s}$ at d. By hypothesis D_s is an effective Cartier divisor on X_s, so this ideal is free of rank 1 over $\overline{B}$. It is generated by $\overline{g}$, so composing the surjection $\overline{B} \to \overline{g}\,\overline{B}$, $1 \mapsto \overline{g}$, with an isomorphism $\overline{g}\,\overline{B} \cong \overline{B}$ exhibits a surjective $\overline{B}$-linear endomorphism of $\overline{B}$, which is multiplication by a unit and therefore injective. Thus $\operatorname{Ann}_{\overline{B}}(\overline{g}) = 0$.
Step 5: the Tor *computation.* Apply $\operatorname{Tor}_\bullet^A(-, \kappa)$ to the sequence of Step 3 and use $B \otimes_A \kappa = \overline{B}$:

$$\operatorname{Tor}_1^A(B, \kappa) \longrightarrow \operatorname{Tor}_1^A(C, \kappa) \longrightarrow \overline{B} \xrightarrow{\,\overline{g}\,} \overline{B}$$

The leftmost term vanishes because B is flat over A, and the rightmost map is injective by Step 4. Exactness therefore forces $\operatorname{Tor}_1^A(C, \kappa) = 0$. By Step 2 the local criterion for flatness of *Homological Algebra* [18, Local Criterion for Flatness] applies to the local homomorphism $A \to B$ and the finitely generated B-module C, and gives that $C = \mathcal{O}_{D,d}$ is flat over $A = \mathcal{O}_{S,s}$. As

$d \in D$ was arbitrary, $D \to S$ is flat, which together with the invertibility of $\mathcal{I}_D$ is the definition, as we sought to show.

Each hypothesis is doing one job. Flatness of X over S kills $\operatorname{Tor}_1^A(B, \kappa)$, and without it the fiberwise condition says nothing about the total space; the Noetherian base and the finite presentation of f put the stalks inside the range of the local criterion; and the fiber hypothesis is used only through the injectivity of $\overline{g}$ on $\overline{B}$.

Flatness of X over S cannot be dropped. Take $S = \operatorname{Spec} k[t]$, $X = \operatorname{Spec} B$ with $B = k[t, x]/(t^2)$, and $D = V(x) \subseteq X$. The element $t \in B$ is nonzero and killed by t, so B is not flat over $k[t]$. The element $g = x$ satisfies $\operatorname{Ann}_B(x) = 0$, so $\mathcal{I}_D = xB$ is invertible. The fiber over a point $s \neq [(t)]$ is empty, and over $s = [(t)]$ one has $\overline{B} = k[x]$ with $\overline{g} = x$, so D_s is a reduced point on $\mathbb{A}^1_k$ and is an effective Cartier divisor there. Every fiber condition holds, yet $\mathcal{O}_D = k[t]/(t^2)$ has t-torsion and is not flat over $k[t]$, so D is not a relative effective Cartier divisor.

Nothing in that argument sees the geometry of X_s. Flatness of $\mathcal{O}_D$ over the base is converted, by one application of the vanishing of Tor_1, into the statement that the local equation stays a non-zero-divisor modulo $\mathfrak{m}_s$. The same mechanism gives stability under arbitrary base change.

Proposition 5.4.31: Base Change of a Relative Cartier Divisor

Let $f\colon X \to S$ be a morphism of schemes, let $D \subseteq X$ be an effective Cartier divisor on X relative to S, and let $S' \to S$ be an arbitrary morphism. Write $X' = X \times_S S'$ and $D' = D \times_S S'$, and let $p\colon X' \to X$ be the projection (definition 3.2.3). Then D' is an effective Cartier divisor on X' relative to S', and the canonical map $p^*\mathcal{I}_D \to \mathcal{O}_{X'}$ is injective with image $\mathcal{I}_{D'}$; in particular

$$\mathcal{I}_{D'} \;\cong\; p^*\mathcal{I}_D$$

***Proof*:**

Step 1: D' is a closed subscheme of X'. As in the previous proof, $D' = D \times_S S' = D \times_X X'$ by theorem 3.2.1, so $D' \to X'$ is the base change of $D \hookrightarrow X$ along p and is a closed immersion by proposition 3.2.5(2).

Step 2: the local model. Fix affine opens $\operatorname{Spec} A \subseteq S$, then $\operatorname{Spec} A' \subseteq S'$ mapping into $\operatorname{Spec} A$, and $\operatorname{Spec} B \subseteq X$ mapping into $\operatorname{Spec} A$ with $\mathcal{I}_D$ free on $\operatorname{Spec} B$, so that $D \cap \operatorname{Spec} B = \operatorname{Spec}(B/gB)$ with $\operatorname{Ann}_B(g) = 0$ and $C = B/gB$ flat over A. Set $B' = B \otimes_A A'$ and let g' be the image of g in B'. Then $\operatorname{Spec} B'$ is an open of X', such opens cover X', and $D' \cap \operatorname{Spec} B' = \operatorname{Spec}(C \otimes_A A')$.

Step 3: flatness over S'. Let $M' \hookrightarrow N'$ be an injection of A'-modules. The identification $(C \otimes_A A') \otimes_{A'} M' \cong C \otimes_A M'$ is functorial in M', and $M' \to N'$ is in particular an injection of A-modules, so flatness of C over A makes $C \otimes_A M' \to C \otimes_A N'$ injective. Hence $C \otimes_A A'$ is flat over A', and $D' \to S'$ is flat.

Step 4: the ideal sheaf on X'. Apply lemma 5.4.28 with $M = A'$ viewed as an A-module. Since $B \otimes_A A' = B'$ and $g \otimes \mathrm{id}$ is multiplication by g', the sequence

$$0 \longrightarrow B' \xrightarrow{g'} B' \longrightarrow C \otimes_A A' \longrightarrow 0$$

is exact. Exactness on the left says $\operatorname{Ann}_{B'}(g') = 0$; exactness on the right identifies $C \otimes_A A'$ with $B'/g'B'$. So the ideal of D' on $\operatorname{Spec} B'$ is $g'B'$, free of rank 1 over B'. As these opens cover X', the sheaf $\mathcal{I}_{D'}$ is invertible, and with Step 3 this proves D' is an effective Cartier divisor on X' relative to S'.

Step 5: comparison with the pullback. Pull the ideal sequence

$$0 \longrightarrow \mathcal{I}_D \longrightarrow \mathcal{O}_X \longrightarrow \iota_*\mathcal{O}_D \longrightarrow 0$$

of lemma 5.1.31, in which $\iota\colon D \hookrightarrow X$ is the closed immersion, back along p. The functor p^* is right exact, and on $\operatorname{Spec} B'$ it is $- \otimes_B B'$ by proposition 5.1.14(7). The isomorphism $B \to gB$, $1 \mapsto g$, gives $(gB) \otimes_B B' \cong B'$, under which the canonical map $p^*\mathcal{I}_D \to p^*\mathcal{O}_X = \mathcal{O}_{X'}$ becomes multiplication by g' on B'. Step 4 shows this map is injective with image $g'B' = \mathcal{I}_{D'}$, which is the assertion that the pulled-back ideal sequence stays exact on the left. Injectivity and the identification of the image are local statements about a globally defined map of sheaves, so they hold on all of X', completing the proof.

Proposition 5.4.29 is the case $S' = \operatorname{Spec} \kappa(s)$ of proposition 5.4.31, and the two proofs are one computation run with different coefficients. Note which half of the definition does the work. Invertibility transports for free, since the pullback of an invertible sheaf along any morphism is invertible. What is not free is that $p^*\mathcal{I}_D$ still *embeds* in $\mathcal{O}_{X'}$, so that it is the ideal sheaf of a closed subscheme rather than an abstract line bundle equipped with a map to $\mathcal{O}_{X'}$ that kills something. That injectivity is the vanishing of $\operatorname{Tor}_1^A(C, A')$, which is flatness of $\mathcal{O}_D$ over the base.

Without the flatness hypothesis the notion is unstable in families, in a way that is easy to exhibit. Take $S = \operatorname{Spec} k[t]$, $X = \operatorname{Spec} k[t, x, y]$, and $D = V(t) \subseteq X$. The element t is a non-zero-divisor in $k[t, x, y]$, so D is an effective Cartier divisor on X in the absolute sense. But $D \to S$ is the inclusion of a point, $\operatorname{Spec} k[x, y] \to \operatorname{Spec} k[t]$, which is not flat, and the fibers of D jump from empty over $t \neq 0$ to all of $X_0 = \mathbb{A}^2_k$ over $t = 0$. Over the origin the ideal sheaf of D_0 is zero, so D_0 is not a divisor on X_0 at all. Flatness of $D \to S$ rules this out by forcing the local equation to remain a non-zero-divisor after restriction to every fiber, and that is the one thing needed for the divisorial character of the family to be constant across the base.

The two propositions together say that the assignment

$$S' \longmapsto \{D' \subseteq X \times_S S' \mid D' \text{ relative effective Cartier on } (X \times_S S')/S'\}$$

is a contravariant functor on S-schemes. Representability of that functor produces the scheme of relative divisors and, through it, the relative Picard functor (over a field this is [20, Relative Divisors Form an Open Subscheme of the Hilbert Scheme]); the same stability is what lets divisor classes be intersected in families (see [19]).

The effective Cartier divisors are precisely the closed subschemes whose ideal sheaf is an invertible $\mathcal{O}_X$-module (i.e., locally free of rank 1). This characterization — which requires no reference to $\mathscr{K}_X$ at all — is the starting point adopted by the Stacks project[7], and it has the advantage of working without any integrality or Noetherian hypotheses. We will return to this perspective in proposition 5.4.47.

When X satisfies the standing hypotheses of the previous subsection (Noetherian, integral, separated, regular in codimension 1), Cartier divisors can be compared directly with Weil divisors.

[7] https://stacks.math.columbia.edu, Tag 01WR.

Proposition 5.4.32: Cartier Divisors to Weil Divisors

Let X be a Noetherian, integral, separated scheme that is regular in codimension 1. There is a group homomorphism

$$\operatorname{CDiv} X \longrightarrow \operatorname{WDiv} X$$

that sends principal Cartier divisors to principal Weil divisors.

***Proof*:**

Since X is integral, $\mathscr{K}_X$ is the constant sheaf $K(X)$, so a Cartier divisor is represented by data $\{(U_i, f_i)\}$ with $f_i \in K(X)^*$ and $f_i/f_j \in \mathcal{O}_X^*(U_i \cap U_j)$. For each prime divisor Y of X, choose an index i with $\eta_Y \in U_i$ (such an i exists since the U_i cover X) and define the coefficient of $[Y]$ to be $\nu_Y(f_i)$. This is independent of the choice: if $\eta_Y \in U_j$ as well, then f_i/f_j is a unit near η_Y, so $\nu_Y(f_i/f_j) = 0$ and $\nu_Y(f_i) = \nu_Y(f_j)$.

The resulting sum $\sum_Y \nu_Y(f_i)[Y]$ is finite: since X is quasi-compact (it is Noetherian), we may take the cover $\{U_i\}$ to be finite, and on each U_i the rational function f_i has nonzero valuation at only finitely many prime divisors (by the same argument as lemma 5.4.6).

If $\{(U_i, f_i)\}$ and $\{(U_i, f_i')\}$ represent Cartier divisors D and D' (after passing to a common refinement), then $D + D'$ is represented by $\{(U_i, f_i f_i')\}$, and $\nu_Y(f_i f_i') = \nu_Y(f_i) + \nu_Y(f_i')$, so the map is a homomorphism. A principal Cartier divisor is represented by a single $f \in K(X)^*$, which maps to $\operatorname{div}(f) \in \operatorname{WDiv} X$.

The key question is: when is this map an isomorphism? It is surjective precisely when every Weil divisor is *locally principal* — that is, when each prime divisor is locally cut out by a single equation. The natural hypothesis guaranteeing this is that every local ring is a UFD (so every height-1 prime is principal).

Proposition 5.4.33: Weil and Cartier Divisors Under Locally Factorial Hypotheses

Let X be an integral, separated, Noetherian scheme that is locally factorial (i.e., $\mathcal{O}_{X,x}$ is a UFD for all $x \in X$; cf. definition 2.4.30). Then the homomorphism of proposition 5.4.32 is an isomorphism

$$\operatorname{CDiv} X \xrightarrow{\sim} \operatorname{WDiv} X,$$

and principal Cartier divisors correspond to principal Weil divisors. In particular, there is an induced isomorphism of class groups.

***Proof*:**

Since every UFD is integrally closed, X is normal, and hence regular in codimension 1 (example 5.4.3). Because X is integral, $\mathscr{K}_X$ is the constant sheaf $K(X)$.

Injectivity. Suppose a Cartier divisor D, represented by $\{(U_i, f_i)\}$, maps to the zero Weil divisor. Then $\nu_Y(f_i) = 0$ for every prime divisor Y meeting U_i. For each affine open $V = \operatorname{Spec} R \subseteq U_i$, the element $f_i \in K(X)^*$ satisfies $\nu_{\mathfrak{p}}(f_i) = 0$ for every height-1 prime $\mathfrak{p}$ of R. Since R is a UFD (hence normal), lemma 5.4.13 gives $f_i \in R^*$, so f_i is a unit on V. As this holds for every affine open in every U_i, we have $f_i \in \mathcal{O}_X^*(U_i)$ for all i, meaning $D = 0$ in $\operatorname{CDiv} X$.

Surjectivity. Let $D = \sum n_j[Y_j]$ be a Weil divisor. For each point $x \in X$, the prime divisors

through x correspond to height-1 primes of $\mathcal{O}_{X,x}$. Since $\mathcal{O}_{X,x}$ is a UFD, each such prime is principal, so $\mathrm{Cl}\,(\operatorname{Spec}\mathcal{O}_{X,x}) = 0$ by proposition 5.4.14. Hence the restriction of D to $\operatorname{Spec}\mathcal{O}_{X,x}$ is principal: there exists $f_x \in K(X)^*$ with

$$\nu_Y(f_x) \;=\; \begin{cases} n_j & \text{if } Y = Y_j \text{ passes through } x, \\ 0 & \text{for all other prime divisors through } x. \end{cases}$$

Now, D and $\operatorname{div}(f_x)$ can differ only at prime divisors not passing through x. Both D and $\operatorname{div}(f_x)$ involve finitely many prime divisors (the latter by lemma 5.4.6), and those not passing through x form a closed subset $Z_x \subseteq X$ with $x \notin Z_x$. Set $U_x = X \setminus Z_x$; then $D|_{U_x} = \operatorname{div}(f_x)|_{U_x}$.

It remains to verify that the collection $\{(U_x, f_x)\}_{x\in X}$ defines a Cartier divisor, i.e., that $f_x/f_{x'} \in \mathcal{O}_X^*(U_x \cap U_{x'})$ for all x, x'. On $U_x \cap U_{x'}$, both f_x and $f_{x'}$ represent D, so $\operatorname{div}(f_x/f_{x'}) = 0$ on $U_x \cap U_{x'}$. For any affine open $V = \operatorname{Spec} R \subseteq U_x \cap U_{x'}$, the ring R is a normal domain (each stalk is a UFD, hence integrally closed). By lemma 5.4.13, $f_x/f_{x'} \in \bigcap_{\mathrm{ht}(\mathfrak{p})=1} R_{\mathfrak{p}} = R$, and the same argument applied to $f_{x'}/f_x$ gives the inverse. Hence $f_x/f_{x'} \in R^* = \mathcal{O}_X^*(V)$, and since this holds on every affine open in $U_x \cap U_{x'}$, we conclude $f_x/f_{x'} \in \mathcal{O}_X^*(U_x \cap U_{x'})$.

These two constructions are mutually inverse (one verifies directly that composing in either order yields the identity), and both send principal divisors to principal divisors.

The locally factorial hypothesis is satisfied by all regular schemes (since regular local rings are UFDs by the Auslander–Buchsbaum theorem; cf. proposition 2.7.27) and by all smooth varieties over a field. In particular, the Weil and Cartier theories agree on any nonsingular variety — the setting where divisors are most commonly used. On a normal but non-factorial scheme (such as the quadric cone of example 5.4.19), the map $\mathrm{CDiv}\,X \to \mathrm{WDiv}\,X$ is a proper inclusion: the ruling $[Y]$ is a Weil divisor that is *not* locally principal at the vertex, hence has no Cartier counterpart.

When X is not locally factorial, the map $\mathrm{CDiv}\,X \to \mathrm{WDiv}\,X$ identifies Cartier divisors with a distinguished subgroup of Weil divisors:

Definition 5.4.34: Locally Principal Weil Divisor

Let X be a Noetherian, integral, separated scheme that is regular in codimension 1. A Weil divisor $D \in \mathrm{WDiv}\,X$ is *locally principal* if for every $x \in X$, there exists an open neighbourhood U of x and $f \in K(X)^*$ such that $D|_U = \operatorname{div}(f)|_U$.

By construction, the image of $\mathrm{CDiv}\,X \to \mathrm{WDiv}\,X$ consists precisely of the locally principal Weil divisors. We therefore have a chain of inclusions

$$\mathrm{PDiv}(X) \;\subseteq\; \{\text{locally principal Weil divisors}\} \;\subseteq\; \mathrm{WDiv}\,X,$$

and the locally factorial hypothesis of proposition 5.4.33 collapses the second inclusion to an equality.

Definition 5.4.35: Cartier Divisor Class Group

Let X be a scheme. The *Cartier divisor class group* of X is

$$\operatorname{CaCl} X := \frac{\operatorname{CDiv} X}{\{\text{principal Cartier divisors}\}}.$$

When X is integral, separated, Noetherian, and locally factorial, proposition 5.4.33 gives an isomorphism $\operatorname{CaCl} X \cong \operatorname{Cl} X$.

Example 5.4.36: Cartier Divisors

1. *Projective space.* On $\mathbb{P}^n_k$, every local ring is a localization of a polynomial ring, hence a UFD. By proposition 5.4.33, every Weil divisor is Cartier, and $\operatorname{CaCl} \mathbb{P}^n_k \cong \operatorname{Cl} \mathbb{P}^n_k \cong \mathbb{Z}$. A hyperplane $H = V_+(x_0)$ is represented by the Cartier data $\{(U_i, x_0/x_i)\}_{i=0}^n$: on the standard affine chart $U_i = D_+(x_i) \cong \mathbb{A}^n$, the hyperplane is cut out by the single function $x_0/x_i \in \mathcal{O}_X(U_i)$.

2. *Smooth curves.* Every nonsingular curve is locally factorial (one-dimensional regular local rings are DVRs, hence UFDs). Thus Weil and Cartier divisors coincide, and a divisor is simply a formal sum of closed points — the classical notion of a divisor on a curve.

3. *Normal, non-factorial schemes.* On the quadric cone $X = \operatorname{Spec} k[x,y,z]/(xy - z^2)$ (example 5.4.19), the ruling $Y = V(y,z)$ is a Weil divisor that is *not* locally principal at the vertex (x,y,z): the prime ideal (y,z) is not principal in the local ring at that point. However, $2[Y] = \operatorname{div}(y)$ *is* Cartier. In this case, $\operatorname{CDiv} X \subsetneq \operatorname{WDiv} X$, and $\operatorname{CaCl} X = 0 \neq \mathbb{Z}/2\mathbb{Z} = \operatorname{Cl} X$.[a]

4. *Non-reduced schemes.* Let $X = \operatorname{Spec} k[x]/(x^2)$, the "double point." Since x is a zero-divisor, the element $x \in \mathcal{O}_X(X)$ does not define an effective Cartier divisor. The total quotient ring is $k[x]/(x^2)$ itself (no non-zero-divisors to invert beyond units), so $\mathscr{K}_X = \mathcal{O}_X$ and $\operatorname{CDiv} X = 0$. This illustrates the limitations of the Cartier formalism on highly non-reduced schemes.

[a] The Cartier class group is trivial because the only Cartier divisors on X are principal: any locally principal divisor is an even multiple of $[Y]$, hence already principal.

Example 5.4.37: Weil ≠ Cartier on Singular Curves

Example. Consider the nodal cubic $X = V(y^2 - x^3 - x^2) \subseteq \mathbb{P}^2_k$ (with $\operatorname{char} k \neq 2$), which has a node at the origin $P = (0,0)$. The local ring $\mathcal{O}_{X,P} \cong k[x,y]_{(x,y)}/(y^2 - x^2 - x^3)$ is *not* a UFD: in the completion $\widehat{\mathcal{O}}_{X,P} \cong k[\![u,v]\!]/(uv)$, the element $y^2 - x^2 = x^3$ factors into two branches that cannot be separated in $\mathcal{O}_{X,P}$ itself[a]. The Weil divisor $1 \cdot P$ (a single closed point, considered as a codimension-1 subvariety) is *not* Cartier at P: its ideal sheaf $\mathcal{I}_P$ is not locally principal near P, because the maximal ideal $\mathfrak{m}_P \subseteq \mathcal{O}_{X,P}$ is not principal (it requires two generators). Hence $\mathcal{I}_P$ is a coherent ideal sheaf, but *not* an invertible sheaf.

More concretely, on the normalization $\nu\colon \widetilde{X} \cong \mathbb{P}^1 \to X$, the preimage $\nu^{-1}(P)$ consists of two points $\{P_1, P_2\}$ (the two branches), and the Weil divisor $P_1 + P_2$ on $\mathbb{P}^1$ is Cartier—but it is the pullback of P only as a *cycle*, not as a Cartier divisor on X.

The upshot:

[a] See example 3.1.32 for the formal branch structure of a node.

5.4.3 The Picard Group

The discussion of divisors on projective schemes has made it clear that there is a strong connection to the twisting sheaves $\mathcal{O}(d)$. On an integral scheme, Cartier divisors up to linear equivalence and line bundles up to isomorphism determine each other (theorem 5.4.44).

In a Dedekind domain R, fractional ideals are nonzero finitely generated R-submodules of $\mathrm{Frac}(R)$, and they form a group under multiplication. The ideal class group $\mathrm{Cl}\,(R)$, the group of fractional ideals modulo principal ones, measures the failure of unique factorization (example 5.4.15). The Picard group is precisely the scheme-theoretic generalization of this construction: invertible sheaves ("line bundles") play the role of fractional ideals, the tensor product replaces ideal multiplication, and the Picard group classifies them up to isomorphism.

Proposition 5.4.38: Invertible Sheaves Form a Group

Let X be a ringed space and let $\mathcal{L}, \mathcal{L}'$ be invertible sheaves on X. Then:

1. The tensor product $\mathcal{L} \otimes_{\mathcal{O}_X} \mathcal{L}'$ is invertible.
2. The *dual* $\mathcal{L}^\vee := \mathit{Hom}_{\mathcal{O}_X}(\mathcal{L}, \mathcal{O}_X)$ is invertible, and $\mathcal{L} \otimes_{\mathcal{O}_X} \mathcal{L}^\vee \cong \mathcal{O}_X$.
3. The tensor product is associative and commutative (up to natural isomorphism) with unit $\mathcal{O}_X$.

Hence the set of isomorphism classes of invertible sheaves on X forms an abelian group under $\otimes$.

***Proof*:**
For (1), choose trivialisations $\mathcal{L}|_U \cong \mathcal{O}_U$ and $\mathcal{L}'|_V \cong \mathcal{O}_V$. On $U \cap V$, $(\mathcal{L} \otimes \mathcal{L}')|_{U\cap V} \cong \mathcal{O}_{U\cap V} \otimes \mathcal{O}_{U\cap V} \cong \mathcal{O}_{U\cap V}$, so $\mathcal{L} \otimes \mathcal{L}'$ is locally free of rank 1.

For (2), on a trivialising open where $\mathcal{L}|_U \cong \mathcal{O}_U$, the dual restricts as $\mathcal{L}^\vee|_U \cong \mathit{Hom}_{\mathcal{O}_U}(\mathcal{O}_U, \mathcal{O}_U) \cong \mathcal{O}_U$, so $\mathcal{L}^\vee$ is invertible. The evaluation map

$$\mathcal{L} \otimes_{\mathcal{O}_X} \mathcal{L}^\vee \longrightarrow \mathcal{O}_X, \qquad s \otimes \varphi \longmapsto \varphi(s)$$

is an isomorphism on each trivialising open (where it is the multiplication $R \otimes_R R \to R$), hence a global isomorphism.

Part (3) follows from the corresponding properties of the tensor product of modules.

Definition 5.4.39: Picard Group

The *Picard group* of a ringed space X is the abelian group

$$\mathrm{Pic}\, X \;:=\; \frac{\{\text{invertible sheaves on } X\}}{\text{isomorphism}}$$

with group operation $[\mathcal{L}] \cdot [\mathcal{L}'] = [\mathcal{L} \otimes_{\mathcal{O}_X} \mathcal{L}']$, identity element $[\mathcal{O}_X]$, and inverse $[\mathcal{L}]^{-1} = [\mathcal{L}^\vee]$.

We shall see in theorem 5.4.44 that for integral schemes, $\mathrm{Pic}\, X \cong H^1(X, \mathcal{O}_X^*)$, giving a cohomological

interpretation.

Proposition 5.4.40: Functoriality of the Picard Group

Let $f\colon X \to Y$ be a morphism of ringed spaces. The pullback of $\mathcal{O}_Y$-modules induces a well-defined group homomorphism

$$f^*\colon \operatorname{Pic} Y \longrightarrow \operatorname{Pic} X, \qquad [\mathcal{L}] \longmapsto [f^*\mathcal{L}].$$

Consequently, Pic defines a contravariant functor from the category of ringed spaces to the category of abelian groups.

***Proof*:**

Recall that the pullback of an $\mathcal{O}_Y$-module $\mathcal{L}$ is given by $f^*\mathcal{L} = f^{-1}\mathcal{L} \otimes_{f^{-1}\mathcal{O}_Y} \mathcal{O}_X$. We must first verify that $f^*\mathcal{L}$ is an invertible sheaf on X whenever $\mathcal{L}$ is an invertible sheaf on Y.

Choose an open cover $\{V_i\}$ of Y that trivialises $\mathcal{L}$, meaning $\mathcal{L}|_{V_i} \cong \mathcal{O}_{V_i}$. For each i, let $U_i = f^{-1}(V_i)$, which forms an open cover of X. Because the pullback operation commutes with restriction to open subsets, we have

$$(f^*\mathcal{L})|_{U_i} \;\cong\; (f|_{U_i})^*(\mathcal{L}|_{V_i}) \;\cong\; (f|_{U_i})^*\mathcal{O}_{V_i} \;\cong\; \mathcal{O}_{U_i}.$$

Thus, $f^*\mathcal{L}$ is locally free of rank 1, making it an invertible sheaf on X. This ensures the assignment $[\mathcal{L}] \mapsto [f^*\mathcal{L}]$ is well-defined.

To see that f^* is a group homomorphism, we rely on the fact that the pullback functor for modules commutes with the tensor product. For any two invertible sheaves $\mathcal{L}, \mathcal{L}'$ on Y, there is a canonical isomorphism

$$f^*(\mathcal{L} \otimes_{\mathcal{O}_Y} \mathcal{L}') \;\cong\; f^*\mathcal{L} \otimes_{\mathcal{O}_X} f^*\mathcal{L}'.$$

Therefore, $f^*([\mathcal{L}] \cdot [\mathcal{L}']) = f^*([\mathcal{L}]) \cdot f^*([\mathcal{L}'])$.

Finally, the canonical isomorphisms $(g \circ f)^* \cong f^* \circ g^*$ and $(\mathrm{id}_Y)^* \cong \mathrm{id}_{\operatorname{Pic} Y}$ follow directly from the general categorical properties of module pullbacks, confirming that Pic is a contravariant functor.

Definition 5.4.41: Line Bundle Associated to a Cartier Divisor

Let X be a scheme and $D \in \operatorname{CDiv} X$ a Cartier divisor represented by data $\{(U_i, f_i)\}$ with $f_i \in \Gamma(U_i, \mathscr{K}_X^*)$ and $f_i/f_j \in \mathcal{O}_X^*(U_i \cap U_j)$. The *invertible sheaf associated to* D is the sub-$\mathcal{O}_X$-module $\mathcal{O}_X(D) \subseteq \mathscr{K}_X$ defined by

$$\mathcal{O}_X(D)|_{U_i} \;:=\; f_i^{-1} \cdot \mathcal{O}_X|_{U_i} \;\subseteq\; \mathscr{K}_X|_{U_i}.$$

On the overlap $U_i \cap U_j$, the generators satisfy $f_i^{-1} = (f_j/f_i) \cdot f_j^{-1}$ with $f_j/f_i \in \mathcal{O}_X^*$, so the local descriptions glue. The trivialisation $\mathcal{O}_X(D)|_{U_i} \xrightarrow{\sim} \mathcal{O}_X|_{U_i}$ is given by $g \mapsto g f_i$.[a]

[a] The construction does not need X to be integral: the sheaf $\mathscr{K}_X$ of total quotient rings replaces the function field.

When X is integral, one can describe the sections of $\mathcal{O}_X(D)$ more concretely. Since $\mathscr{K}_X$ is the

constant sheaf $K(X)$, the sections over an open U are

$$\mathcal{O}_X(D)(U) \;=\; \{g \in K(X)^* \mid \operatorname{div}(g) + D \geq 0 \text{ on } U\} \;\cup\; \{0\},$$

where $\operatorname{div}(g) + D \geq 0$ means the Weil divisor is effective on U. Intuitively, $\mathcal{O}_X(D)(U)$ consists of all rational functions whose poles are "bounded by D": the divisor D prescribes how many poles are allowed along each prime divisor, and g must compensate. Divisors first arose on Riemann surfaces as records of the poles and zeros of meromorphic functions, and the space of global sections $\mathcal{O}_X(D)(X)$ is the classical *Riemann-Roch space* $L(D)$.

Example 5.4.42: Number-Theoretic Line Bundle

Let $R = \mathbb{Z}[\sqrt{-5}]$ and $X = \operatorname{Spec} R$. Since R is a Dedekind domain, X is a normal Noetherian integral scheme of dimension 1, and every fractional ideal of R defines an invertible sheaf via the tilde construction (definition 5.1.6).

Consider the prime ideal $\mathfrak{p} = (2, 1+\sqrt{-5})$, which lies over 2 and satisfies $\mathfrak{p}^2 = (2)$ (example 5.4.15). As a Weil divisor, $[V(\mathfrak{p})]$ corresponds to the single closed point $\mathfrak{p} \in X$. This Weil divisor is Cartier (since every local ring of a Dedekind domain is a DVR, hence a UFD): at the point $\mathfrak{p}$ itself, the uniformizer $1+\sqrt{-5}$ generates the ideal locally, while at every other closed point $\mathfrak{q} \neq \mathfrak{p}$, the ideal $\mathfrak{p}$ is the full local ring and is generated by 1.

The invertible sheaf $\widetilde{\mathfrak{p}}$ associated to this fractional ideal is *not* isomorphic to $\mathcal{O}_X$, since $\mathfrak{p}$ is not principal. Its class $[\widetilde{\mathfrak{p}}] \in \operatorname{Pic} X \cong \operatorname{Cl}(R) \cong \mathbb{Z}/2\mathbb{Z}$ is the nontrivial element, and $\widetilde{\mathfrak{p}}^{\otimes 2} \cong \widetilde{\mathfrak{p}^2} = \widetilde{(2)} \cong \mathcal{O}_X$ (the last isomorphism since (2) is principal as a fractional ideal). This recovers the ideal class group of $\mathbb{Z}[\sqrt{-5}]$ as the Picard group.

Proposition 5.4.43: Correspondence Between Line Bundles and Cartier Divisors

Let X be a scheme. The assignment $D \mapsto \mathcal{O}_X(D)$ defines a group homomorphism $\operatorname{CDiv} X \to \operatorname{Pic} X$ satisfying:

1. $\mathcal{O}_X(D + D') \cong \mathcal{O}_X(D) \otimes_{\mathcal{O}_X} \mathcal{O}_X(D')$ for all $D, D' \in \operatorname{CDiv} X$.
2. $\mathcal{O}_X(-D) \cong \mathcal{O}_X(D)^\vee$ for all $D \in \operatorname{CDiv} X$.
3. $\mathcal{O}_X(D) \cong \mathcal{O}_X$ if and only if D is a principal Cartier divisor.

In particular, the homomorphism descends to an injection $\operatorname{CaCl} X \hookrightarrow \operatorname{Pic} X$.

Part (1) pairs well with the stalkwise intuition for the tensor product: if D prescribes poles and zeros along certain prime divisors and D' prescribes its own, then $\mathcal{O}_X(D) \otimes \mathcal{O}_X(D')$ "combines" the two prescriptions, gaining the zeros and poles of both.

***Proof*:**

Part (1). If $D = \{(U_i, f_i)\}$ and $D' = \{(U_i, g_i)\}$ (after passing to a common refinement), then $D + D' = \{(U_i, f_i g_i)\}$. On U_i, the multiplication map in $\mathscr{K}_X$ gives

$$\mathcal{O}_X(D)|_{U_i} \otimes \mathcal{O}_X(D')|_{U_i} \;=\; f_i^{-1}\mathcal{O}_{U_i} \otimes g_i^{-1}\mathcal{O}_{U_i} \;\xrightarrow{\sim}\; (f_i g_i)^{-1}\mathcal{O}_{U_i} \;=\; \mathcal{O}_X(D + D')|_{U_i}.$$

These local isomorphisms are compatible on overlaps (the transition functions multiply), yielding a global isomorphism.

Part (2). The dual $\mathcal{O}_X(D)^\vee = \mathit{Hom}_{\mathcal{O}_X}(\mathcal{O}_X(D), \mathcal{O}_X)$ is locally generated by f_i (the functional

sending $f_i^{-1} \mapsto 1$), which is $\mathcal{O}_X(-D)|_{U_i}$.

Part (3). If $D = \operatorname{div}(f)$ is principal (represented by a single global $f \in \Gamma(X, \mathscr{K}_X^*)$), then $\mathcal{O}_X(D) = f^{-1}\mathcal{O}_X \cong \mathcal{O}_X$. Conversely, if $\varphi\colon \mathcal{O}_X(D) \xrightarrow{\sim} \mathcal{O}_X$, then φ sends the local generator f_i^{-1} to some $h_i \in \mathcal{O}_X^*(U_i)$. The element $f = f_i h_i^{-1} \in \mathscr{K}_X^*$ is independent of i (on $U_i \cap U_j$, the transition functions force $f_i h_i^{-1} = f_j h_j^{-1}$), and $D = \operatorname{div}(f)$.

For a general scheme X, the injection $\operatorname{CaCl} X \hookrightarrow \operatorname{Pic} X$ may be strict: there can exist invertible sheaves that do not embed as sub-$\mathcal{O}_X$-modules of $\mathscr{K}_X$. The map is surjective, hence an isomorphism, whenever X is integral:

Theorem 5.4.44: Isomorphism of Cartier Class Group and Picard Group

Let X be an integral scheme. The map $D \mapsto [\mathcal{O}_X(D)]$ induces an isomorphism of abelian groups

$$\operatorname{CaCl} X \xrightarrow{\sim} \operatorname{Pic} X.$$

***Proof*:**

Injectivity follows from proposition 5.4.43(3). For surjectivity, let $\mathcal{L}$ be an invertible sheaf. Since X is integral, the constant sheaf $\mathscr{K}_X = K(X)$ satisfies $\mathcal{L} \otimes_{\mathcal{O}_X} \mathscr{K}_X \cong \mathscr{K}_X$ (both sides are locally free $\mathscr{K}_X$-modules of rank 1, and $\mathscr{K}_X$ has a unique such module up to isomorphism). The natural map $\mathcal{L} \hookrightarrow \mathcal{L} \otimes \mathscr{K}_X$ (sending $s \mapsto s \otimes 1$) is injective since X is integral. Composing with any isomorphism $\mathcal{L} \otimes \mathscr{K}_X \xrightarrow{\sim} \mathscr{K}_X$ embeds $\mathcal{L}$ as a sub-$\mathcal{O}_X$-module of $\mathscr{K}_X$. On a trivialising open U_i, this embedding sends the local generator of $\mathcal{L}|_{U_i}$ to some $f_i^{-1} \in K(X)^*$, and the collection $\{(U_i, f_i)\}$ defines a Cartier divisor D with $\mathcal{L} \cong \mathcal{O}_X(D)$. A different choice of isomorphism $\mathcal{L} \otimes \mathscr{K}_X \cong \mathscr{K}_X$ changes each f_i by a common factor $h \in K(X)^*$, replacing D by $D + \operatorname{div}(h)$.

Cohomological perspective. Consider the short exact sequence of abelian sheaves

$$1 \longrightarrow \mathcal{O}_X^* \longrightarrow \mathscr{K}_X^* \longrightarrow \mathscr{K}_X^*/\mathcal{O}_X^* \longrightarrow 1.$$

Since X is integral, $\mathscr{K}_X^* = K(X)^*$ is a constant sheaf on the irreducible space X, hence flasque. The long exact sequence in cohomology gives

$$\Gamma(X, \mathscr{K}_X^*) \longrightarrow \underbrace{\Gamma(X, \mathscr{K}_X^*/\mathcal{O}_X^*)}_{\operatorname{CDiv} X} \xrightarrow{\delta} \underbrace{H^1(X, \mathcal{O}_X^*)}_{\operatorname{Pic} X} \longrightarrow \underbrace{H^1(X, \mathscr{K}_X^*)}_{=0}$$

where the vanishing follows from flasqueness (corollary 6.1.7). The connecting homomorphism δ sends a Cartier divisor $\{(U_i, f_i)\}$ to the Čech 1-cocycle $\{f_i/f_j\} \in \check{H}^1(\{U_i\}, \mathcal{O}_X^*)$, which classifies the line bundle $\mathcal{O}_X(D)$. Its cokernel is zero and its kernel is the image of $\Gamma(X, \mathscr{K}_X^*)$, the group of principal divisors. Hence $\operatorname{CaCl} X \cong \operatorname{Pic} X$.

The cohomological proof explains the warning following definition 5.4.24: the discrepancy between global sections of the quotient sheaf $\mathscr{K}_X^*/\mathcal{O}_X^*$ and the literal quotient of global sections is precisely the connecting homomorphism δ, and the group $H^1(X, \mathcal{O}_X^*)$ classifies line bundles.

Corollary 5.4.45: Divisor Class Group and Picard Group

Let X be an integral, separated, Noetherian, locally factorial scheme. Then there is a chain of isomorphisms

$$\operatorname{Cl} X \xrightarrow[\text{proposition } 5.4.33]{\sim} \operatorname{CaCl} X \xrightarrow[\text{theorem } 5.4.44]{\sim} \operatorname{Pic} X.$$

Corollary 5.4.46: Classifying Line Bundles on Projective Space

Let k be a field. Every invertible sheaf on $\mathbb{P}^n_k$ is isomorphic to $\mathcal{O}_{\mathbb{P}^n_k}(d)$ for a unique $d \in \mathbb{Z}$. Hence

$$\operatorname{Pic} \mathbb{P}^n_k \cong \mathbb{Z},$$

with generator $[\mathcal{O}_{\mathbb{P}^n_k}(1)]$ corresponding to the hyperplane class.

***Proof*:**
The scheme $\mathbb{P}^n_k$ is integral, separated, Noetherian, and regular (hence locally factorial by proposition 2.7.27). By corollary 5.4.45 and proposition 5.4.16, $\operatorname{Pic} \mathbb{P}^n_k \cong \operatorname{Cl} \mathbb{P}^n_k \cong \mathbb{Z}$, generated by the hyperplane class. Since the hyperplane divisor $[H]$ corresponds to $\mathcal{O}_{\mathbb{P}^n_k}(1)$ under the construction of definition 5.4.41, the class $d[H]$ corresponds to $\mathcal{O}_{\mathbb{P}^n_k}(d)$, confirming the earlier computation of proposition 5.3.5.

On an integral scheme, effective Cartier divisors correspond to line bundles with a nonzero global section. This is the algebraic counterpart of the fact that the zero locus of a nonzero holomorphic section of a holomorphic line bundle on a connected complex manifold is an effective divisor.

Proposition 5.4.47: Effective Cartier Divisors and Line Bundles

Let X be an integral scheme and D an effective Cartier divisor with associated closed subscheme $Y \hookrightarrow X$ (definition 5.4.26).

1. The ideal sheaf of Y satisfies $\mathcal{I}_Y \cong \mathcal{O}_X(-D)$.
2. There is a natural bijection

$$\{ \text{ effective Cartier divisors on } X \ \} \longleftrightarrow \left\{ \begin{array}{c} \text{pairs } (\mathcal{L},\, s) \text{ with } \mathcal{L} \text{ invertible and} \\ 0 \neq s \in \Gamma(X, \mathcal{L}), \\ \text{up to isomorphism} \end{array} \right\}$$

Under this bijection, D corresponds to $(\mathcal{O}_X(D),\, s_D)$, where s_D, the *canonical section*, is the element $1 \in K(X)$ viewed as a global section of $\mathcal{O}_X(D)$.

***Proof*:**
Part (1). Let D be represented by $\{(U_i, f_i)\}$ with $f_i \in \mathcal{O}_X(U_i)$ non-zero-divisors. Then $\mathcal{O}_X(-D)|_{U_i} = f_i \cdot \mathcal{O}_{U_i}$, which is precisely the ideal sheaf of Y on U_i (by definition 5.4.26, $\mathcal{I}_Y$ is locally generated by f_i). Since both sides glue by the same transition functions, we get $\mathcal{I}_Y \cong \mathcal{O}_X(-D)$ as subsheaves of $\mathcal{O}_X$.

Part (2). Given an effective $D = \{(U_i, f_i)\}$, the element $1 \in K(X)$ lies in $\mathcal{O}_X(D)(U_i) = f_i^{-1}\mathcal{O}_{U_i}$ (since $1 \cdot f_i = f_i \in \mathcal{O}_X(U_i)$), so $s_D := 1$ is a well-defined nonzero global section. Under the trivialisation $\mathcal{O}_X(D)|_{U_i} \cong \mathcal{O}_{U_i}$, the section s_D corresponds to f_i, so s_D vanishes precisely along Y.

Conversely, given $(\mathcal{L}, s)$ with $s \neq 0$, choose trivialisations $\psi_i \colon \mathcal{L}|_{U_i} \xrightarrow{\sim} \mathcal{O}_{U_i}$ and set $f_i = \psi_i(s|_{U_i})$. Since $s \neq 0$ and X is integral, each f_i is a non-zero-divisor. On overlaps, $f_i/f_j = \psi_i \circ \psi_j^{-1}(1) \in \mathcal{O}_X^*(U_i \cap U_j)$, so $\{(U_i, f_i)\}$ is an effective Cartier divisor. The two constructions are mutually inverse.

Part (1) provides a useful dictionary: the ideal sheaf of an effective Cartier divisor is the "negative" line bundle $\mathcal{O}_X(-D)$, and the short exact sequence of $\mathcal{O}_X$-modules

$$0 \longrightarrow \mathcal{O}_X(-D) \longrightarrow \mathcal{O}_X \longrightarrow \mathcal{O}_Y \longrightarrow 0$$

will be a fundamental tool in cohomological computations (section 6.3). By part (2), line bundles with a nonzero section and effective divisors are equivalent data, and linear systems (box 5.3.46) and the Riemann-Roch theorem (theorem 7.2.10) use both descriptions.

Exercise 5.4.1

1. *(The twisting line bundle.)* Let R be a ring and φ an R-algebra automorphism of $M_n(R)$, with matrix units e_{ij}. Let $L_\varphi := \varphi(e_{11})R^n$.
   a) Show that the $\varphi(e_{ii})$ are orthogonal idempotents summing to 1 and that $\varphi(e_{ji})$ restricts to an isomorphism $\varphi(e_{ii})R^n \to \varphi(e_{jj})R^n$. Deduce $R^n \cong L_\varphi^{\oplus n}$, so that L_φ is finitely generated projective, and by localizing show that $\widetilde{L_\varphi}$ is invertible (proposition 5.1.24).
   b) Show that $\varphi = c_g$ for some $g \in \mathrm{GL}_n(R)$ if and only if L_φ is free. *(Hint: given a generator v, send $e_j \mapsto \varphi(e_{j1})v$.)* Conclude that every automorphism of $M_n(R)$ is inner when R is local, recovering the field case proved after definition 4.1.7.
2. *(The Rosenberg-Zelinsky sequence [63].)* Keep the notation of exercise 5.4.1 (1), and let $T_\varphi := \{f \in M_n(R) \mid fa = \varphi(a)f \text{ for all } a\}$.
   a) Show that $f \mapsto f(e_1)$ is an isomorphism $T_\varphi \xrightarrow{\sim} L_\varphi$.
   b) Show that composition $f \otimes f' \mapsto f \circ f'$ defines a map $T_\varphi \otimes_R T_\psi \to T_{\varphi\circ\psi}$, and prove that it is an isomorphism by localizing and using exercise 5.4.1 (1)(2).
   c) Deduce the exact sequence of groups

$$1 \to R^\times \to \mathrm{GL}_n(R) \xrightarrow{g \mapsto c_g} \mathrm{Aut}_R\big(M_n(R)\big) \xrightarrow{\varphi \mapsto [\widetilde{L_\varphi}]} \mathrm{Pic}\,(\operatorname{Spec} R).$$

   When $R = k$ is a field, $\operatorname{Spec} k$ is a point, so $\mathrm{Pic}\,(\operatorname{Spec} k)$ is trivial and every k-algebra automorphism of $M_n(k)$ is inner. This is the case $A = B = M_n(k)$ of the Skolem-Noether Theorem [13, Skolem-Noether Theorem].
3. Show that the image of the last map in exercise 5.4.1 (2) is $\left\{[\widetilde{L}] \;\middle|\; L^{\oplus n} \cong R^n\right\}$, and by taking $\wedge^n$ show that it lies in the n-torsion subgroup $\mathrm{Pic}\,(\operatorname{Spec} R)[n]$. *(For the inclusion $\supseteq$: an isomorphism $\theta \colon L^{\oplus n} \to R^n$ gives $\varphi(a) := \theta a \theta^{-1}$.)*
4. Let $R = \mathbb{Z}[\sqrt{-5}]$ and $L = (2, 1 + \sqrt{-5})$. Show that L is not principal but $L^2 = (2)$. Choose a surjection $R^2 \to L$ with kernel K, and use $\wedge^2$ to show $K \cong L^{-1} \cong L$, hence $L \oplus L \cong R^2$.

Conclude that $M_2(R)$ has an R-algebra automorphism that is not conjugation by any element of $\mathrm{GL}_2(R)$.

5. *(Globalization.)* Let X be a scheme and φ an $\mathcal{O}_X$-algebra automorphism of $M_n(\mathcal{O}_X)$.
   a) Using exercise 5.4.1 (1)(2) on stalks, show that there are an open cover $\{U_i\}$ and $g_i \in \mathrm{GL}_n(\mathcal{O}_X(U_i))$ with $\varphi|_{U_i} = c_{g_i}$.
   b) Show that $\lambda_{ij} := g_j^{-1} g_i$ lies in $\mathcal{O}_X^*(U_i \cap U_j)$ and satisfies $\lambda_{ij}\lambda_{jk} = \lambda_{ik}$, so it is the transition data of an invertible sheaf $\mathcal{L}_\varphi$, whose class does not depend on the choices made.
   c) Deduce an exact sequence

$$1 \to \mathcal{O}_X(X)^* \to \mathrm{GL}_n(\mathcal{O}_X(X)) \to \mathrm{Aut}\big(M_n(\mathcal{O}_X)\big) \to \mathrm{Pic}\, X$$

   and show that for $X = \operatorname{Spec} R$ it agrees with exercise 5.4.1 (2).

5.4.4⋆ Families of Weil Divisors and Birational Transforms

⋆ *Optional. This subsection is an aside; nothing later in the book depends on it.*

Two operations carry a Weil divisor from one normal variety to another, and they are easily confused. The first is specialization inside a family. A divisor $\mathbf{D}$ on a product $S \times X$ has a fiber $\mathbf{D}_s$ over each parameter $s \in S$, got by restricting $\mathbf{D}$ to the regular locus of the fiber, where it is Cartier and restriction is defined, and taking the closure. The second is transport along a birational map $g\colon X \dashrightarrow Y$. Every prime divisor of X carries a discrete valuation of the function field $K(X)$; the birational transform g_* sends it to the prime divisor of Y carrying the same valuation when there is one, and to zero when there is none.

These two operations disagree, and they disagree by exceptional terms. A birational map contracts some prime divisors and extracts others; a fiber of a family records the contracted ones, and a birational transform discards them. Example 5.4.59 computes the smallest instance, on the blow-up of $\mathbb{P}^2$ at a point: the fiber of a family of lines whose special member passes through the blown-up point acquires the exceptional curve as a component, while the birational transform of that same line does not. The general accounting of the discrepancy, a finite list of linear subspaces of the parameter space outside of which the two agree, is lemma 5.4.64; that is a forward reference, and nothing below depends on it.

The distinction has consequences: Chevalley's structure theorem, that a connected algebraic group over a perfect field is an extension of an abelian variety by an affine group, is proved in *Abelian Varieties* [5] by an argument that manufactures a projective model X on which a group G acts only rationally and then compares the divisors $\mathbf{D}_g$ as g varies. The 2002 proof of Conrad [24] treats the translate of a divisor by a group element as if the two operations above agreed, and a translation of a rationally acting group can contract a boundary divisor; the repair, following Kollár [45], needs the two notions kept apart from the start. That is what this subsection does.

Definition 5.4.48: Family of Weil Divisors

Let k be an algebraically closed field, let X be a normal variety over k (definition 2.4.28), and let S be a smooth variety over k (definition 2.7.13). Write $\pi\colon S \times_k X \to S$ for the first projection and $X^{\mathrm{reg}} \subseteq X$ for the complement of $\mathrm{Sing}(X)$ (definition 2.7.22).
A *family of Weil divisors on X parametrized by S* is a Weil divisor $\mathbf{D} \in \mathrm{WDiv}\,(S \times_k X)$ (definition 5.4.5) such that $\operatorname{Supp}\mathbf{D}$ contains no fiber $\pi^{-1}(s)$, $s \in S$. The family is *effective* if $\mathbf{D} \geq 0$.
For a closed point $s \in S$, identify the fiber $\pi^{-1}(s)$ with X. The restriction of $\mathbf{D}$ to $S \times X^{\mathrm{reg}}$ is a Cartier divisor (proposition 5.4.49(1)), so it restricts to a Cartier divisor on $\{s\} \times X^{\mathrm{reg}} = X^{\mathrm{reg}}$. The *fiber of* $\mathbf{D}$ *over* s is the Weil divisor

$$\mathbf{D}_s \;\in\; \mathrm{WDiv}\,X$$

obtained from that Cartier divisor by proposition 5.4.32 and then extension by closure from X^{reg} to X (lemma 5.4.17).

Three points about the definition deserve to be separated. The hypothesis on the support is what makes the restriction to a fiber legitimate: a local equation for $\mathbf{D}$ that vanished identically on $\pi^{-1}(s)$ would restrict to zero, not to a divisor. The passage through X^{reg} is forced, since a Weil divisor on a singular X need not be Cartier anywhere near a singular point and there is then nothing to restrict; on X^{reg} the local rings are regular, hence factorial (proposition 2.7.27), and proposition 5.4.33 identifies the two theories. Finally, nothing is lost by the detour: X is normal, so $\mathrm{Sing}(X)$ has codimension at least 2 (proposition 5.4.2(3)), and restriction $\mathrm{WDiv}\,X \to \mathrm{WDiv}\,X^{\mathrm{reg}}$ is then an isomorphism with inverse extension by closure (lemma 5.4.17 and proposition 5.4.18(2)).

Kollár flags a delicacy at exactly this point, warning that the identification of Weil divisors on X with Cartier divisors on X^{reg} "can be misleading for families of divisors" and referring to the theory of K-flatness for the general case. The warning is real, and it is worth saying precisely when it applies. It applies when the parameter space is only normal rather than smooth, and when the ambient family is not a product, so that the fibers X_s themselves vary: in either case Sing of the total space need not be π^{-1}(something), the regular locus of the total space need not meet a given fiber in the regular locus of that fiber, and the fiber-by-fiber recipe above can fail to be flat or even to be additive in the family. None of that happens here. The parameter space is smooth, the ambient family is the constant product $S \times X$, and the next proposition shows that under those two hypotheses the naive definition produces an effective Cartier divisor relative to S in the sense already established for this book, definition 5.4.27, so that its fibers are governed by proposition 5.4.29 and no further foundational theory is needed.

Proposition 5.4.49: A Family of Weil Divisors Is a Relative Cartier Divisor

Let k, X, S and π be as in definition 5.4.48, and put

$$Y \;:=\; S \times_k X^{\mathrm{reg}}, \qquad \pi_Y \;:=\; \pi|_Y \colon Y \longrightarrow S$$

1. $S \times_k X$ is a Noetherian, integral, separated scheme that is regular in codimension 1, so that $\mathrm{WDiv}\,(S \times_k X)$ is defined, and its non-regular locus lies in $S \times_k \mathrm{Sing}(X)$, of codimension at least 2. The open subscheme Y is regular, integral, separated and Noetherian; every Weil divisor on $S \times_k X$ restricts on Y to a Cartier divisor, and the restriction of an effective Weil divisor is an effective Cartier divisor.
2. If $\mathbf{D}$ is an effective family of Weil divisors on X parametrized by S, then the closed subscheme $D \subseteq Y$ cut out by the effective Cartier divisor $\mathbf{D}|_Y$ is an effective Cartier divisor on Y relative to S (definition 5.4.27).
3. For every closed point $s \in S$, the fiber D_s given by proposition 5.4.29 is an effective Cartier divisor on X^{reg}, and its extension by closure to X is $\mathbf{D}_s$.

***Proof*:**

Step 1: Y is regular. Since X is normal, X^{reg} is a regular scheme, locally of finite type over k. An algebraically closed field is perfect, so theorem 2.7.15 makes X^{reg} smooth over k. The scheme S is smooth over k by hypothesis, so $Y = S \times_k X^{\mathrm{reg}}$ is smooth over k by proposition 2.7.11(3), and hence regular by proposition 2.7.14. It is Noetherian and separated because S and X^{reg} are and both properties pass to a fiber product of finite type k-schemes. It is integral: k is algebraically closed, so the integral k-schemes S and X^{reg} are geometrically integral (definition 3.2.8), and a product of geometrically integral k-schemes is integral.

The same three arguments, with X in place of X^{reg}, make $S \times_k X$ Noetherian, integral and separated. Its non-regular locus is contained in $S \times_k \mathrm{Sing}(X)$, because Y is regular. Let T be an irreducible component of $\mathrm{Sing}(X)$, so $\dim T \leq \dim X - 2$ by proposition 5.4.2(3). Both $S \times_k T$ and $S \times_k X$ are integral by the argument just given, and transcendence degree is additive along a tensor product of finitely generated k-domains with domain tensor product, so their function fields have transcendence degrees $\dim S + \dim T$ and $\dim S + \dim X$ over k. Now theorem 2.6.11 computes both dimensions and theorem 2.6.12 gives

$$\mathrm{codim}\big(S \times_k T,\; S \times_k X\big) \;=\; \dim X - \dim T \;\geq\; 2$$

Every codimension-1 point of $S \times_k X$ therefore lies in Y and has a regular local ring, which is regularity in codimension 1.

Step 2: Weil divisors on Y are Cartier. Each local ring of Y is regular, hence a unique factorization domain by proposition 2.7.27, so Y is locally factorial (definition 2.4.30). By proposition 5.4.33 the homomorphism $\mathrm{CDiv}\, Y \to \mathrm{WDiv}\, Y$ of proposition 5.4.32 is an isomorphism. The restriction map $\mathrm{WDiv}\,(S \times_k X) \to \mathrm{WDiv}\, Y$ is defined because Y is a nonempty open subscheme of $S \times_k X$ (lemma 5.4.17); composing with the inverse isomorphism gives the asserted Cartier divisor.

For effectivity, let $\{(U_i, f_i)\}$ be Cartier data on Y whose associated Weil divisor is effective, so $\nu_\Gamma(f_i) \geq 0$ for every prime divisor Γ of Y meeting U_i. Fix an affine open $\mathrm{Spec}\, R \subseteq U_i$. A regular local ring is normal (proposition 2.7.26), so R is a Noetherian normal domain, and every height-1 prime of R is the generic point of a prime divisor of Y meeting U_i. Thus f_i lies in $R_{\mathfrak{p}}$

for every height-1 prime $\mathfrak{p}$, and lemma 5.4.13 puts $f_i \in R$. As such opens cover U_i, each f_i is regular, which is definition 5.4.26.
Step 3: the hypotheses of the fiberwise criterion. The projection $\pi_Y : Y = S \times_k X^{\mathrm{reg}} \to S$ is the base change of the smooth morphism $X^{\mathrm{reg}} \to \operatorname{Spec} k$ along $S \to \operatorname{Spec} k$, hence smooth by proposition 2.7.11(1) and flat by proposition 2.7.12. It is of finite type, and S is Noetherian, so π_Y is locally of finite presentation. The ideal sheaf $\mathcal{I}_D$ is invertible by Step 2 and definition 5.4.26. So proposition 5.4.30 applies to $D \subseteq Y$ once its fiberwise hypothesis is verified.
Step 4: the fiberwise hypothesis. Let $s \in S$ be any point, not necessarily closed, and write $\kappa = \kappa(s)$. The fiber $Y_s = X^{\mathrm{reg}} \times_k \kappa$ is smooth over κ by proposition 2.7.11(1) and therefore regular (proposition 2.7.14); it is integral because X^{reg} is geometrically integral over the algebraically closed field k, so that $X^{\mathrm{reg}} \times_k \kappa$ is integral. Take an affine open $V \subseteq Y$ on which $\mathcal{I}_D$ is free, say $\mathcal{I}_D|_V = gB$ with $B = \mathcal{O}_Y(V)$, and suppose $V_s \neq \emptyset$. If the image $\overline{g}$ of g in $\mathcal{O}_{Y_s}(V_s)$ were zero, then V_s would be contained in $\operatorname{Supp} \mathbf{D}$; since V_s is a nonempty open of the irreducible space Y_s it is dense there, and $\operatorname{Supp} \mathbf{D}$ is closed, so $Y_s \subseteq \operatorname{Supp} \mathbf{D}$. The fiber $\pi^{-1}(s)$ of $S \times_k X \to S$ is integral with Y_s as a nonempty open subscheme, so Y_s is dense in $\pi^{-1}(s)$ and $\pi^{-1}(s) \subseteq \operatorname{Supp} \mathbf{D}$, contradicting definition 5.4.48. Hence $\overline{g} \neq 0$, and $\overline{g}$ is a non-zero-divisor because Y_s is integral. So D_s is an effective Cartier divisor on Y_s for every $s \in S$, and proposition 5.4.30 gives (2).
Step 5: identification of the fiber. Let $s \in S$ be a closed point. Since k is algebraically closed, $\kappa(s) = k$ and $Y_s = X^{\mathrm{reg}}$. Unwinding Step 4 in this case, the local ideal of D_s on V_s is generated by the image of a local equation for $\mathbf{D}|_Y$, which is exactly the restriction of the Cartier divisor $\mathbf{D}|_Y$ to $\{s\} \times X^{\mathrm{reg}}$ used in definition 5.4.48. Its extension by closure is therefore $\mathbf{D}_s$, as we sought to show.

The second operation moves a divisor along a birational map rather than along a family. It is defined valuation by valuation, which is what makes it available with no hypothesis whatever on the map.

Definition 5.4.50: Birational Transform of a Weil Divisor

Let X and Y be Noetherian, integral, separated schemes that are regular in codimension 1 (definition 5.4.5), and let $g\colon X \dashrightarrow Y$ be a birational map (definition 3.6.2). Write $g^\sharp\colon K(Y) \xrightarrow{\sim} K(X)$ for the induced isomorphism of function fields (corollary 3.6.6).
Let $D \subseteq X$ be a prime divisor with generic point η_D, so that $\mathcal{O}_{X,\eta_D} \subseteq K(X)$ is a discrete valuation ring with fraction field $K(X)$. Set

$$g_*[D] := \begin{cases} [D'] & \text{if a prime divisor } D' \subseteq Y \text{ satisfies } g^\sharp\big(\mathcal{O}_{Y,\eta_{D'}}\big) = \mathcal{O}_{X,\eta_D}, \\ 0 & \text{if no prime divisor of } Y \text{ satisfies this,} \end{cases}$$

and extend $\mathbb{Z}$-linearly to a group homomorphism

$$g_*\colon \operatorname{WDiv} X \longrightarrow \operatorname{WDiv} Y$$

called the *birational transform* (also the *proper* or *strict transform*). A prime divisor D with $g_*[D] = 0$ is *contracted* by g.

The definition asks nothing of g beyond being birational. It does not ask Y to be proper, it does not ask g to be a morphism, and it does not ask g to be an automorphism or even to have a codimension-2 indeterminacy locus. This is what makes it the right instrument for a group G acting on X only rationally: each $g \in G$ gives a birational self-map of X, and g_* is defined for every one of them, whether or not the action extends to a morphism anywhere near the boundary. The usual

description, as the closure of the image of D under an isomorphism between dense opens, is recovered in lemma 5.4.51(1), which also settles that the recipe above is unambiguous.

Lemma 5.4.51: Functoriality of the Birational Transform

Let X, Y and W be Noetherian, integral, separated schemes that are regular in codimension 1, and let $g\colon X \dashrightarrow Y$ and $h\colon Y \dashrightarrow W$ be birational maps.

1. For each prime divisor $D \subseteq X$ there is at most one prime divisor $D' \subseteq Y$ with $g^\sharp(\mathcal{O}_{Y,\eta_{D'}}) = \mathcal{O}_{X,\eta_D}$, so g_* is well defined. If moreover g is represented by a morphism on an open neighbourhood of η_D, then
$$g_*[D] \;=\; \begin{cases} \big[\,\overline{\{g(\eta_D)\}}\,\big] & \text{if } \operatorname{codim}\big(\overline{\{g(\eta_D)\}}, Y\big) = 1, \\ 0 & \text{otherwise.} \end{cases}$$
2. The composite $h \circ g$ is a birational map and $(h \circ g)^\sharp = g^\sharp \circ h^\sharp$.
3. If the prime divisor $D \subseteq X$ is not contracted by g, then
$$(h \circ g)_*[D] \;=\; h_*\big(g_*[D]\big)$$
Hence $h_* \circ g_*$ and $(h \circ g)_*$ agree on the subgroup of $\operatorname{WDiv} X$ generated by the prime divisors that g does not contract.
4. $h_* \circ g_* = (h \circ g)_*$ on all of $\operatorname{WDiv} X$ if and only if every prime divisor contracted by g is contracted by $h \circ g$. Without that condition the two differ (example 5.4.52).

***Proof*:**

Throughout, η_X and η_Y denote generic points, and K denotes $K(X)$, with $K(Y)$ and $K(W)$ identified with K by the relevant function-field isomorphisms whenever that is unambiguous.

(1) Uniqueness. Let $D'_1, D'_2 \subseteq Y$ be prime divisors with $g^\sharp(\mathcal{O}_{Y,\eta_{D'_i}}) = \mathcal{O}_{X,\eta_D} =: R$. Then R is a discrete valuation ring with fraction field K, and for each i the canonical morphism $\operatorname{Spec}\mathcal{O}_{Y,\eta_{D'_i}} \to Y$ becomes, after the identification $g^\sharp$, a morphism $\varphi_i\colon \operatorname{Spec} R \to Y$ carrying the closed point to $\eta_{D'_i}$ and the generic point to η_Y. The two agree on $\operatorname{Spec} K$, both being the canonical morphism $\operatorname{Spec} K(Y) \to Y$, and Y is separated over $\operatorname{Spec}\mathbb{Z}$ with Y Noetherian, so theorem 3.5.15 allows at most one such extension over $\operatorname{Spec} R$. Hence $\varphi_1 = \varphi_2$ and $\eta_{D'_1} = \eta_{D'_2}$; a prime divisor is the closure of its generic point, so $D'_1 = D'_2$.

(1) The description at a point of definition. Suppose g is represented by a morphism $g_0\colon U \to Y$ with $\eta_D \in U$, and put $y := g_0(\eta_D)$. Since g is dominant, $g_0(\eta_X) = \eta_Y$ (lemma 3.6.4) and the local homomorphism $\mathcal{O}_{Y,y} \to \mathcal{O}_{X,\eta_D} = R$ induced by g_0 agrees with $g^\sharp$ on fraction fields; write $V := g^\sharp(\mathcal{O}_{Y,y}) \subseteq R$, a local subring of K with $\mathfrak{m}_V \subseteq \mathfrak{m}_R$.

Assume first $\operatorname{codim}(\overline{\{y\}}, Y) = 1$. Then $\mathcal{O}_{Y,y}$, hence V, is a discrete valuation ring with fraction field K, and $V \subseteq R$. Let $x \in R$, the case $x = 0$ being trivial, and suppose $x \notin V$. A valuation ring of K contains x or $1/x$, so $x \notin V$ forces $1/x \in \mathfrak{m}_V \subseteq \mathfrak{m}_R$, making $1/x$ a nonunit of R; but $x \in R$ and $1/x \in R$ make $1/x$ a unit of R, a contradiction. Therefore $R \subseteq V$, so $V = R$, and $\overline{\{y\}}$ is a prime divisor of Y realizing the defining condition: $g_*[D] = [\overline{\{y\}}]$.

Assume instead $\operatorname{codim}(\overline{\{y\}}, Y) \neq 1$, and suppose some prime divisor $D' \subseteq Y$ had $g^\sharp(\mathcal{O}_{Y,\eta_{D'}}) = R$. The composite $\operatorname{Spec} R = \operatorname{Spec} \mathcal{O}_{X,\eta_D} \to U \xrightarrow{g_0} Y$ and the morphism $\operatorname{Spec} R \to Y$ attached to D' as in the uniqueness argument are two extensions over $\operatorname{Spec} R$ of the canonical $\operatorname{Spec} K(Y) \to Y$, so theorem 3.5.15 identifies them. Comparing images of the closed point gives $y = \eta_{D'}$, whose closure has codimension 1, a contradiction. So $g_*[D] = 0$.
(2). Represent g by a dominant morphism $g_0 \colon U \to Y$ and h by a morphism $h_0 \colon V \to W$ with $V \subseteq Y$ dense open. By lemma 3.6.4, $g_0(\eta_X) = \eta_Y$, and $\eta_Y \in V$ because V is a nonempty open of the irreducible space Y. Hence $g_0^{-1}(V)$ is a nonempty open of U and $h_0 \circ g_0|_{g_0^{-1}(V)}$ represents a rational map $h \circ g$; it is dominant, since it carries η_X to $h_0(\eta_Y) = \eta_W$. The same construction applied to g^{-1} and h^{-1} produces a dominant rational inverse, so $h \circ g$ is birational. On function fields, the isomorphism induced by a dominant rational map is the map on local rings at the generic points, so the isomorphism induced by $h_0 \circ g_0$ is $g^\sharp \circ h^\sharp$.
(3). Let $g_*[D] = [D'] \neq 0$, so $g^\sharp(\mathcal{O}_{Y,\eta_{D'}}) = \mathcal{O}_{X,\eta_D}$. Two cases.
If $h_*[D'] = [D''] \neq 0$, then $h^\sharp(\mathcal{O}_{W,\eta_{D''}}) = \mathcal{O}_{Y,\eta_{D'}}$, and applying $g^\sharp$ and using (2),

$$(h \circ g)^\sharp\big(\mathcal{O}_{W,\eta_{D''}}\big) \;=\; g^\sharp\Big(h^\sharp\big(\mathcal{O}_{W,\eta_{D''}}\big)\Big) \;=\; g^\sharp\big(\mathcal{O}_{Y,\eta_{D'}}\big) \;=\; \mathcal{O}_{X,\eta_D}$$

so $(h \circ g)_*[D] = [D''] = h_*(g_*[D])$.
If $h_*[D'] = 0$, then $h_*(g_*[D]) = 0$, and it must be shown that $(h \circ g)_*[D] = 0$ as well. Were it $[D''] \neq 0$, the displayed computation read backwards would give $h^\sharp(\mathcal{O}_{W,\eta_{D''}}) = \mathcal{O}_{Y,\eta_{D'}}$, since $g^\sharp$ is injective, contradicting $h_*[D'] = 0$.
(4). Both $h_* \circ g_*$ and $(h \circ g)_*$ are homomorphisms $\operatorname{WDiv} X \to \operatorname{WDiv} W$, and $\operatorname{WDiv} X$ is free on the prime divisors, so they are equal precisely when they agree on every prime divisor D. On a D not contracted by g they agree by (3). On a D contracted by g the left side is $h_*(0) = 0$, so agreement there says exactly $(h \circ g)_*[D] = 0$, which is the stated condition, completing the proof.

Part (3) is the positive statement in the form it is used: a prime divisor that persists under the first map is transported by the composite exactly as the two maps transport it in turn, and this needs nothing of the second map. What fails is the other half. A birational map can destroy a prime divisor, and a later birational map can create it again, and the birational transform has no memory of what was destroyed. The plane provides the smallest example, and it is the mechanism the Chevalley argument has to control.

Example 5.4.52: The Quadratic Transformation of the Plane
Let k be a field and let

$$\sigma \colon \mathbb{P}^2_k \dashrightarrow \mathbb{P}^2_k, \qquad \sigma(x_0 : x_1 : x_2) \;=\; (x_1x_2 : x_0x_2 : x_0x_1)$$

The three forms x_1x_2, x_0x_2, x_0x_1 vanish simultaneously exactly when at least two coordinates vanish, so σ is a morphism on the complement of the three coordinate points $p_0 = (1 : 0 : 0)$, $p_1 = (0 : 1 : 0)$, $p_2 = (0 : 0 : 1)$. On the open set where all $x_i \neq 0$,

$$\sigma\big(\sigma(x_0 : x_1 : x_2)\big) \;=\; \big(x_0^2x_1x_2 \;:\; x_0x_1^2x_2 \;:\; x_0x_1x_2^2\big) \;=\; (x_0 : x_1 : x_2)$$

so $\sigma \circ \sigma$ and the identity agree on a dense open subset and are therefore the same rational map: σ is a birational involution of the plane.

Let $L_0 = V_+(x_0)$, a prime divisor of $\mathbb{P}^2_k$ with generic point η_{L_0}. A point of L_0 other than p_1 and p_2 has $x_1x_2 \neq 0$, and σ sends it to $(x_1x_2 : 0 : 0) = p_0$. Thus σ is a morphism near η_{L_0} and carries it to the closed point p_0, whose closure has codimension 2 in $\mathbb{P}^2_k$. By lemma 5.4.51(1),

$$\sigma_*[L_0] \;=\; 0$$

The identity map has $(\mathrm{id})_*[L_0] = [L_0]$ directly from definition 5.4.50. Consequently

$$\sigma_*\big(\sigma_*[L_0]\big) \;=\; \sigma_*(0) \;=\; 0 \qquad \text{while} \qquad (\sigma\circ\sigma)_*[L_0] \;=\; [L_0] \;\neq\; 0$$

so the birational transform is not functorial.

The geometry behind the computation is that σ contracts L_0 to the point p_0 and, run a second time, extracts L_0 back out of p_0: it is a blow-down followed by a blow-up, packaged as one self-inverse map of the plane. The example is stated with $\mathbb{P}^2$ and homogeneous coordinates only, so that nothing about blow-ups is needed to verify it.

Families of Weil divisors arise most often not one at a time but as the total space of a linear system, and the parameter space is then a projective space rather than an abstract smooth variety. Setting that up requires care about one thing only, which is the dimension convention.

Definition 5.4.53: Linear System of Weil Divisors

Let X be a normal, projective, geometrically integral scheme over a field k (definition 3.2.8), and let $D_0 \in \mathrm{WDiv}\, X$. Write $L(D_0) = \Gamma(X, \mathcal{O}_X(D_0))$ for the Riemann-Roch space (definition 5.4.9), a finite-dimensional k-vector space (corollary 5.4.11).
Let $W \subseteq L(D_0)$ be a k-linear subspace with $\dim_k W = n \geq 1$, and let $\mathbb{P}(W)$ denote the projective space of *lines* in W, so $\dim \mathbb{P}(W) = n - 1$. For $0 \neq f \in W$ the Weil divisor $\mathrm{div}(f) + D_0$ is effective (definition 5.4.9) and linearly equivalent to D_0 (definition 5.4.8), and it depends only on the line $[f] \in \mathbb{P}(W)$, since $\mathrm{div}(cf) = \mathrm{div}(f)$ for $c \in k^*$. The *linear system spanned by* W is the resulting set of effective Weil divisors

$$|W| \;:=\; \{\mathrm{div}(f) + D_0 \mid 0 \neq f \in W\}$$

together with the bijection $\mathbb{P}(W) \to |W|$, $[f] \mapsto \mathrm{div}(f) + D_0$. Its members are written D_λ for $\lambda \in \mathbb{P}(W)$, and $[D_\lambda]$ denotes the corresponding point of $\mathbb{P}(W)$. The system is *complete* if $W = L(D_0)$, and a *pencil* if $n = 2$.
A prime divisor $F \subseteq X$ is a *fixed component* of $|W|$ if $F \subseteq \mathrm{Supp}\, D_\lambda$ for every λ; the system is *without fixed components* if it has none. The *base locus* is $\mathrm{Bs}\,|W| = \bigcap_\lambda \mathrm{Supp}\, D_\lambda$.

The map $\mathbb{P}(W) \to |W|$ is injective, and this is where geometric integrality earns its place: $\mathrm{div}(f)+D_0 = \mathrm{div}(f') + D_0$ says $\mathrm{div}(f/f') = 0$, so f/f' and its inverse are regular by lemma 5.4.13 applied on each affine open of the normal scheme X, whence $f/f' \in \Gamma(X, \mathcal{O}_X)^*$; and $\Gamma(X, \mathcal{O}_X) = k$ because X is projective, hence proper (theorem 3.5.32), and geometrically integral (theorem 3.5.33). Over a field that is not algebraically closed and an X that is integral but not geometrically so, $\Gamma(X, \mathcal{O}_X)$ can be a proper finite extension of k and the map collapses.

5.4.54 Convention: Vector Dimension and Projective Dimension Two dimensions are attached to a linear system and they differ by one. The space W has a *vector* dimension $\dim_k W = n$; the system $|W|$, as a parameter space, has *projective* dimension $\dim \mathbb{P}(W) = n - 1$. The convention here is:

- $\dim |W| := \dim \mathbb{P}(W) = \dim_k W - 1$, so that a pencil has $\dim |W| = 1$;
- a Grassmannian of subsystems is *always* written with the *vector space* as its second argument. Thus $\mathrm{Grass}(r, W)$, the Grassmannian of *Classical Algebraic Geometry* [22, Grassmannian], is the variety of r-dimensional linear *subspaces of the vector space* W, whose points correspond to subsystems $|W'| \subseteq |W|$ of projective dimension $r - 1$.

The notation $\mathrm{Grass}(r, |W|)$ is never used, because it is ambiguous and the ambiguity is not harmless. In the literature the same symbol is read both ways, and reading it projectively when it was meant vectorially shifts every subsystem by one dimension. Kollár's Corollary 24 in [45] writes $\mathrm{Grass}(n-1, |H|)$ where the argument computes with subspaces of vector dimension $n - 1$; under his own Definition 15 the symbol reads projectively, which would make the intersection of the corresponding members zero-dimensional rather than a curve. Anything imported from that source must be reread with the vector convention.

A linear system carries a divisor on $X \times \mathbb{P}(W)$ whose fiber over λ is D_λ. It is the object that turns statements about "a general member" into statements about a dense open subset of a parameter space, and it is a family of Weil divisors in the sense of definition 5.4.48.

Definition 5.4.55: Incidence Correspondence of a Linear System

Let X, D_0 and W be as in definition 5.4.53, and fix a basis $f_1, \dots, f_n$ of W, identifying $\mathbb{P}(W)$ with $\mathbb{P}^{n-1}_k$ carrying homogeneous coordinates $\lambda_1, \dots, \lambda_n$ so that λ corresponds to $f_\lambda := \sum_i \lambda_i f_i$. Choose an open cover $\{U_\alpha\}$ of X^{reg} and $h_\alpha \in K(X)^*$ with $D_0|_{U_\alpha} = \mathrm{div}(h_\alpha)|_{U_\alpha}$, which is possible by proposition 5.4.33 applied on X^{reg}.
The *incidence correspondence* of $|W|$ is the effective Cartier divisor $\mathbf{D}_{|W|}$ on $X^{\mathrm{reg}} \times_k \mathbb{P}(W)$ represented by the data

$$\left\{ \left(U_\alpha \times \{\lambda_j \neq 0\},\ \ G_{\alpha j} := h_\alpha \sum_{i=1}^{n} \frac{\lambda_i}{\lambda_j} f_i \right) \right\}_{\alpha,\, j}$$

together with the Weil divisor on $X \times_k \mathbb{P}(W)$ obtained from it by proposition 5.4.32 and extension by closure (lemma 5.4.17), for which the same symbol is used. The classical order of the factors is used here; interchanging them puts this in the situation of definition 5.4.48 with parameter space $S = \mathbb{P}(W)$.

Proposition 5.4.56: The Incidence Correspondence Is a Family with the Expected Fibers

In the situation of definition 5.4.55:

1. the data $\{(U_\alpha \times \{\lambda_j \neq 0\}, G_{\alpha j})\}$ define an effective Cartier divisor, independent of the choice of the (U_α, h_α);
2. $\operatorname{Supp} \mathbf{D}_{|W|}$ contains no fiber of the projection $q\colon X \times_k \mathbb{P}(W) \to \mathbb{P}(W)$;
3. if k is algebraically closed, $\mathbf{D}_{|W|}$ is an effective family of Weil divisors on X parametrized by $S = \mathbb{P}(W)$ in the sense of definition 5.4.48, and its fiber over $\lambda \in \mathbb{P}(W)$ is D_λ.

***Proof*:**

(1). Each f_i lies in $L(D_0)$, so $\operatorname{div}(f_i) + D_0 \geq 0$, and on U_α this reads $\operatorname{div}(h_\alpha f_i)|_{U_\alpha} \geq 0$. The scheme X^{reg} is regular, hence normal (proposition 2.7.26, proposition 2.4.29), so lemma 5.4.13 applied on each affine open of U_α gives $h_\alpha f_i \in \mathcal{O}_X(U_\alpha)$. The functions λ_i/λ_j are regular on $\{\lambda_j \neq 0\}$, so $G_{\alpha j}$ is a regular function on $U_\alpha \times \{\lambda_j \neq 0\}$.

It is nowhere the zero function. The elements $h_\alpha f_1, \ldots, h_\alpha f_n$ of $\mathcal{O}_X(U_\alpha)$ are linearly independent over k, because the f_i are and $h_\alpha \neq 0$; they therefore span a k-subspace E of dimension n. Restricting $G_{\alpha j}$ to the fiber over a point $\mu \in \{\lambda_j \neq 0\}$ of the second factor gives $\sum_i c_i \cdot (h_\alpha f_i \otimes 1)$ in $\mathcal{O}_X(U_\alpha) \otimes_k \kappa(\mu)$, with $c_i \in \kappa(\mu)$ the values of λ_i/λ_j at μ and $c_j = 1$. A field extension is a free, hence flat, module, so $E \otimes_k \kappa(\mu)$ injects into $\mathcal{O}_X(U_\alpha) \otimes_k \kappa(\mu)$ and the $h_\alpha f_i \otimes 1$ stay linearly independent over $\kappa(\mu)$. Hence the restriction is nonzero, so $G_{\alpha j} \neq 0$.

It is moreover a non-zero-divisor. Shrink U_α to an affine open $\operatorname{Spec} R \subseteq X^{\mathrm{reg}}$, with R a domain since X is integral. Then $\operatorname{Spec} R \times \{\lambda_j \neq 0\}$ is the spectrum of the polynomial ring $R[\lambda_1/\lambda_j, \ldots, \lambda_n/\lambda_j]$ in the $n-1$ variables other than $\lambda_j/\lambda_j = 1$, which is again a domain. A nonzero element of a domain is a non-zero-divisor, and such opens cover $U_\alpha \times \{\lambda_j \neq 0\}$.

On overlaps,

$$\frac{G_{\alpha j}}{G_{\beta j'}} \;=\; \frac{h_\alpha}{h_\beta} \cdot \frac{\lambda_{j'}}{\lambda_j}$$

The second factor is a unit on $\{\lambda_j \lambda_{j'} \neq 0\}$. For the first, $\operatorname{div}(h_\alpha/h_\beta) = 0$ on $U_\alpha \cap U_\beta$, so h_α/h_β and its inverse are regular there by lemma 5.4.13, making it a unit. This is definition 5.4.24 together with definition 5.4.26. A second choice (U'_γ, h'_γ) produces data whose ratios with the first are again of the form h_α/h'_γ, units by the same argument, so the Cartier divisor is unchanged.

(2). Fix $\mu \in \mathbb{P}(W)$ and choose j with $\lambda_j(\mu) \neq 0$. The computation in (1) shows that $G_{\alpha j}$ restricts to a nonzero function on $(U_\alpha \times \{\lambda_j \neq 0\}) \cap q^{-1}(\mu)$, a nonempty open subset of $q^{-1}(\mu)$. So that open subset is not contained in $\operatorname{Supp} \mathbf{D}_{|W|}$, and neither is $q^{-1}(\mu)$.

(3). The projective space $\mathbb{P}(W) = \mathbb{P}^{n-1}_k$ is covered by copies of $\mathbb{A}^{n-1}_k$, each smooth over k by definition 2.7.10 with $r = 0$, so $\mathbb{P}(W)$ is a smooth variety. With (2), the two conditions of definition 5.4.48 hold. For the fiber, let $\lambda \in \mathbb{P}(W)$ be a closed point and pick j with $\lambda_j \neq 0$. Restricting $G_{\alpha j}$ to $U_\alpha \times \{\lambda\}$ replaces each λ_i/λ_j by a scalar in k and gives $h_\alpha f_\lambda/\lambda_j$, whose divisor on U_α is

$$\operatorname{div}(h_\alpha)|_{U_\alpha} + \operatorname{div}(f_\lambda)|_{U_\alpha} \;=\; \big(D_0 + \operatorname{div}(f_\lambda)\big)\big|_{U_\alpha} \;=\; D_\lambda|_{U_\alpha}$$

because a nonzero scalar has zero divisor. These opens cover X^{reg}, so the Cartier fiber is $D_\lambda|_{X^{\mathrm{reg}}}$, whose extension by closure is D_λ, as we sought to show.

Two features of the incidence correspondence are worth separating from the proof. Its fiber over λ recovers the member D_λ exactly, with multiplicities, which is what justifies the interchange of "a property of a general member of $|W|$" with "a property of the fiber of $\mathbf{D}_{|W|}$ over a general point of $\mathbb{P}(W)$". And by proposition 5.4.56(3) together with proposition 5.4.49, it is an effective Cartier divisor on $X^{\mathrm{reg}} \times \mathbb{P}(W)$ relative to $\mathbb{P}(W)$, so its formation commutes with base change on the parameter space (proposition 5.4.31). Kollár's Definition 15 asserts in addition that $\mathbf{D}_{|W|}$ is irreducible when $|W|$ has no fixed components; that assertion is not proved there, is not used below, and is not claimed here.

Finally, the recurring technical need: a construction is performed on a convenient dense open subset of the total space, and the conclusion has to be transported back to the fibers. The following lemma is what does the transporting, and it is Kollár's Claim 13.1 with the hypothesis that his statement omits.

Lemma 5.4.57: Fibers of a Family Are Recovered from a Dense Open

Let k, X, S and π be as in definition 5.4.48, let $\mathbf{D}$ be an effective family of Weil divisors on X parametrized by S, and let $U \subseteq S \times_k X$ be an open subset that meets every irreducible component of $\operatorname{Supp} \mathbf{D}$. Then there is a dense open subset $S^\circ \subseteq S$ such that for every closed point $s \in S^\circ$, writing $U_s = U \cap \pi^{-1}(s) \subseteq X$:

1. no irreducible component of $\operatorname{Supp} \mathbf{D}_s$ is contained in $X \setminus U_s$, so $\operatorname{Supp} \mathbf{D}_s = \overline{\operatorname{Supp} \mathbf{D}_s \cap U_s}$;
2. $\mathbf{D}_s$ is the extension by closure (lemma 5.4.17) of its restriction to U_s.

The hypothesis on U cannot be dropped, even when $\operatorname{Supp} \mathbf{D}$ is irreducible (example 5.4.58).

***Proof*:**

Let $Z := \operatorname{Supp} \mathbf{D}$ carry its reduced induced structure (definition 3.1.26), so $Z \to S$ is of finite type, and let η be the generic point of S. Since $S \times_k X$ is Noetherian, proposition 2.1.25 writes $Z = Z_1 \cup \cdots \cup Z_r$ with each Z_i closed and irreducible, say with generic point ζ_i.

Step 1: $U \cap Z_\eta$ is dense in the generic fiber Z_η. Fix i. If some point of Z_i maps to η, then $\overline{\{\eta\}} = S$ forces $\overline{\pi(Z_i)} = S$; and conversely if $\overline{\pi(Z_i)} = S$ then $\overline{\{\pi(\zeta_i)\}} = \overline{\pi(\overline{\{\zeta_i\}})} = S$, so $\pi(\zeta_i) = \eta$. Thus $(Z_i)_\eta$ is empty unless Z_i dominates S, in which case it contains ζ_i.

Let Z_i dominate S. By hypothesis $U \cap Z_i$ is a nonempty open subset of the irreducible space Z_i, so it contains ζ_i. Any nonempty open subset of $(Z_i)_\eta$ has the form $O \cap (Z_i)_\eta$ with $O \subseteq Z_i$ open and nonempty, hence contains ζ_i; so U meets it. Therefore $U \cap (Z_i)_\eta$ is dense in $(Z_i)_\eta$. Since $Z_\eta = \bigcup_i (Z_i)_\eta$ is a finite union of closed subsets and a nonempty open of Z_η meets one of them in a nonempty open of it, $U \cap Z_\eta$ is dense in Z_η.

Step 2: spreading out. The scheme S is Noetherian and integral and $Z \to S$ is of finite type, so proposition 5.2.9 applied to the open subscheme $U \cap Z \subseteq Z$ produces a nonempty, hence dense, open $S^\circ \subseteq S$ such that $U \cap Z_s$ is dense in Z_s for every $s \in S^\circ$.

Step 3: the components of $\operatorname{Supp} \mathbf{D}_s$ are components of Z_s. Let $s \in S^\circ$ be a closed point and identify $\pi^{-1}(s)$ with X. Because $\mathbf{D}$ is effective, a local equation for $\mathbf{D}$ on $S \times X^{\mathrm{reg}}$ cuts out Z there, so its restriction to X^{reg} cuts out $Z_s \cap X^{\mathrm{reg}}$ and

$$\operatorname{Supp} \mathbf{D}_s \;=\; \overline{Z_s \cap X^{\mathrm{reg}}} \;\subseteq\; Z_s$$

Let T be an irreducible component of $\operatorname{Supp} \mathbf{D}_s$. By lemma 2.6.1, $T \cap X^{\mathrm{reg}}$ is then an irreducible

component of $Z_s \cap X^{\mathrm{reg}}$, which on X^{reg} is cut out by a single regular non-zero-divisor; so $T \cap X^{\mathrm{reg}}$ has codimension 1 in X^{reg} by corollary 2.6.14. Codimension is the dimension of the local ring at the generic point, and T and $T \cap X^{\mathrm{reg}}$ have the same generic point with the same local ring, so $\operatorname{codim}(T, X) = 1$ and $\dim T = \dim X - 1$ by theorem 2.6.12.

On the other hand $Z_s \neq X$ by definition 5.4.48 and X is irreducible, so every irreducible closed subset of Z_s has codimension at least 1 in X and hence dimension at most $\dim X - 1$, again by theorem 2.6.12. If $T \subseteq T'$ with T' irreducible and closed in Z_s and $T \neq T'$, the same theorem applied inside T' would give $\dim T \leq \dim T' - 1 \leq \dim X - 2$, which is false. So T is maximal among the irreducible closed subsets of Z_s, that is, an irreducible component of Z_s.
Step 4: conclusion. Let T be an irreducible component of $\operatorname{Supp} \mathbf{D}_s$, hence of Z_s by Step 3. The set $T \setminus \bigcup_{T' \neq T} T'$, the union running over the other components of Z_s, is a nonempty open subset of Z_s, so Step 2 makes U_s meet it. Thus $U_s \cap T$ is a nonempty open subset of the irreducible space T and is dense in it, which is (1).

For (2), restriction $\operatorname{WDiv} X \to \operatorname{WDiv} U_s$ followed by extension by closure sends $[T]$ to $[\overline{T \cap U_s}]$ when T meets U_s and to 0 otherwise (lemma 5.4.17). By (1) every component of $\operatorname{Supp} \mathbf{D}_s$ meets U_s and satisfies $\overline{T \cap U_s} = T$, so the composite fixes $\mathbf{D}_s$, completing the proof.

Example 5.4.58: A Dense Open Missing Every Fiber

Take k algebraically closed, $X = \mathbb{P}^1_k$, $S = \mathbb{A}^1_k$, and let $q \in \mathbb{P}^1_k$ be a closed point. Put

$$\mathbf{D} \;:=\; \big[\, S \times \{q\} \,\big] \;\in\; \operatorname{WDiv}\left(S \times_k \mathbb{P}^1_k\right)$$

which is a prime divisor, so $\mathbf{D}$ is effective with irreducible support. Its intersection with a fiber $\pi^{-1}(s)$ is the single point (s, q), so no fiber lies in its support and $\mathbf{D}$ is a family of Weil divisors. Since $\mathbb{P}^1_k$ is regular, $X^{\mathrm{reg}} = X$, and restricting a local equation for $\mathbf{D}$ to a fiber gives

$$\mathbf{D}_s \;=\; [q] \qquad \text{for every closed point } s \in S$$

Now let $U := (S \times_k \mathbb{P}^1_k) \setminus \operatorname{Supp} \mathbf{D}$, the complement of a proper closed subset of an irreducible scheme, hence a dense open subset. Then $U_s = \mathbb{P}^1_k \setminus \{q\}$, so $\operatorname{Supp} \mathbf{D}_s \cap U_s = \emptyset$ and the closure of $\mathbf{D}_s \cap U_s$ is 0, not $\mathbf{D}_s$. This happens for *every* $s \in S$, so no shrinking of the parameter space repairs it.

Density of U is therefore not enough, and irreducibility of $\mathbf{D}$ is not what is missing: what lemma 5.4.57 needs, and what fails here, is that U meet $\operatorname{Supp} \mathbf{D}$ at all.

Everything above can now be put on one surface. The example below exhibits the two operations of this subsection on the same divisor and measures their difference: it is exactly one exceptional curve. It also shows why the difference is not a bookkeeping artefact, since the two answers are not even linearly equivalent.

Example 5.4.59: Flat Limit and Birational Transform on a Blown-Up Plane

Let k be algebraically closed, let $p = (1 : 0 : 0) \in \mathbb{P}^2_k$, and let

$$Y \;:=\; \left\{ \big((x_0 : x_1 : x_2),\, (u_1 : u_2)\big) \in \mathbb{P}^2_k \times \mathbb{P}^1_k \;\middle|\; x_1 u_2 = x_2 u_1 \right\}$$

with $\pi \colon Y \to \mathbb{P}^2_k$ the first projection. This is the blow-up of the plane at p, presented so that only homogeneous coordinates are needed.

The surface. If $(x_1, x_2) \neq (0, 0)$ the equation forces $(u_1 : u_2) = (x_1 : x_2)$, so π restricts to

an isomorphism over $\mathbb{P}^2_k \setminus \{p\}$, with inverse $(x_0 : x_1 : x_2) \mapsto ((x_0 : x_1 : x_2), (x_1 : x_2))$, and $E := \pi^{-1}(p) = \{p\} \times \mathbb{P}^1_k$. Over the chart $\{x_0 \neq 0\}$, put $z_1 = x_1/x_0$ and $z_2 = x_2/x_0$; the equation reads $z_1 u_2 = z_2 u_1$, so Y is covered there by the two charts

$$(z_1, w),\ w = u_2/u_1, \quad \pi(z_1, w) = (z_1, z_1 w), \quad E = \{z_1 = 0\}$$

$$(z_2, w'),\ w' = u_1/u_2, \quad \pi(z_2, w') = (z_2 w', z_2), \quad E = \{z_2 = 0\}$$

each isomorphic to $\mathbb{A}^2_k$. Together with $\pi^{-1}(\mathbb{P}^2_k \setminus \{p\})$ these cover Y, so Y is smooth over k, hence regular (proposition 2.7.14) and normal (proposition 2.7.26, proposition 2.4.29). It is irreducible: in the chart (z_1, w) the complement of E is $\{z_1 \neq 0\}$, dense in $\mathbb{A}^2_k$, and similarly in the other chart, so $Y = \overline{Y \setminus E}$ is the closure of an irreducible set (lemma 2.1.24(1)).

The family. Let $S = \mathbb{A}^1_k$ with coordinate t and let $\mathbf{L}$ be the Cartier divisor on $S \times_k Y$ given by the data

$$\left\{ \left(S \times \pi^{-1}(\{x_i \neq 0\}),\ \ \pi^\sharp\left(\frac{x_2 - t x_0}{x_i}\right) \right) \right\}_{i=0,1,2}$$

Each entry is a regular function, and on an overlap the ratio of two of them is $\pi^\sharp(x_{i'}/x_i)$, a unit; so this is an effective Cartier divisor (definition 5.4.26). Its restriction to the fiber over any point of S is the pullback of a nonzero linear form, hence nonzero, so $\operatorname{Supp} \mathbf{L}$ contains no fiber and $\mathbf{L}$ is an effective family of Weil divisors on Y parametrized by S (definition 5.4.48; here $Y^{\mathrm{reg}} = Y$). Its members are the total transforms of the lines

$$L_t \ = \ V_+(x_2 - t x_0) \ \subseteq \ \mathbb{P}^2_k$$

and $p \in L_t$ if and only if $t = 0$.

The general fiber. Let $t \neq 0$. In the chart (z_1, w) the local equation is $z_1 w - t$, which is a unit at every point of $E = \{z_1 = 0\}$, and in the chart (z_2, w') it is $z_2 - t$, again a unit along $E = \{z_2 = 0\}$. So $\operatorname{Supp} \mathbf{L}_t$ misses E entirely and lies in $\pi^{-1}(\mathbb{P}^2_k \setminus \{p\})$, where π is an isomorphism. Hence

$$\mathbf{L}_t \ = \ \left(\pi^{-1}\right)_* [L_t] \qquad (t \neq 0)$$

The special fiber. Let $t = 0$, so the local equations become $\pi^\sharp(x_2/x_i)$. In the chart (z_1, w) this is $z_1 w$, whose divisor is $\{z_1 = 0\} + \{w = 0\}$, that is E plus the curve $\widetilde{L_0}$ on which $\pi(z_1, 0) = (z_1, 0)$ runs through $L_0 \cap \{x_0 \neq 0\}$. In the chart (z_2, w') it is z_2, whose divisor is E with multiplicity 1; and over $\mathbb{P}^2_k \setminus \{p\}$ it cuts out $L_0 \setminus \{p\}$. Assembling,

$$\mathbf{L}_0 \ = \ \widetilde{L_0} \ + \ E$$

where $\widetilde{L_0}$ is the closure in Y of $\pi^{-1}(L_0 \setminus \{p\})$.

The birational transform. The map $\pi^{-1} \colon \mathbb{P}^2_k \dashrightarrow Y$ is birational and is a morphism on $\mathbb{P}^2_k \setminus \{p\}$, which contains the generic point of L_0; that generic point goes to the generic point of $\widetilde{L_0}$, of codimension 1. By lemma 5.4.51(1),

$$\left(\pi^{-1}\right)_* [L_0] \ = \ \widetilde{L_0} \ = \ \mathbf{L}_0 - E$$

The fiber of the family and the birational transform of the same line differ by the exceptional curve.

The difference is not principal. Comparing the Cartier data for two values of t, the ratio of the local equations is the single global rational function $\pi^\sharp\big((x_2 - tx_0)/x_2\big)$, so

$$\mathbf{L}_t - \mathbf{L}_0 \;=\; \operatorname{div}_Y\Big(\pi^\sharp\Big(\frac{x_2 - tx_0}{x_2}\Big)\Big)$$

and all fibers of the family are linearly equivalent. On the other hand $[E] \neq 0$ in $\operatorname{Cl} Y$. Indeed, suppose $E = \operatorname{div}_Y(f)$ with $f \in K(Y)^* = K(\mathbb{P}^2_k)^*$. Every prime divisor of Y other than E meets $Y \setminus E$, and $Y \setminus E \cong \mathbb{P}^2_k \setminus \{p\}$, whose prime divisors correspond to those of $\mathbb{P}^2_k$ with the same local rings (lemma 5.4.17(1) applied twice). So $\operatorname{div}_{\mathbb{P}^2_k}(f) = 0$; on each standard affine chart of $\mathbb{P}^2_k$, which is the spectrum of a polynomial ring and so a normal domain, lemma 5.4.13 makes f and f^{-1} regular, whence $f \in \Gamma(\mathbb{P}^2_k, \mathcal{O})^* = k^*$ by proposition 2.3.6 and $\operatorname{div}_Y(f) = 0 \neq E$, a contradiction.

Consequently, although $L_0 \sim L_t$ on $\mathbb{P}^2_k$, the birational transforms satisfy

$$\big(\pi^{-1}\big)_*[L_0] \;=\; \mathbf{L}_0 - E \;\not\sim\; \mathbf{L}_t \;=\; \big(\pi^{-1}\big)_*[L_t]$$

so the birational transform does not preserve linear equivalence, while the fibers of the family do. No intersection numbers are used anywhere above: every claim is an identity between divisors or a principal-divisor computation.

The mechanism is now visible in its simplest form. Passing to the fiber of a family keeps whatever the local equation picks up along the exceptional locus, and passing to the birational transform throws it away. In the Chevalley argument the role of π is played by a translation L_g of a group acting only rationally on a projective model, the role of E by a component of the boundary that L_g contracts, and the role of t by the group element; the conclusion is that the two families of divisors that an argument might reasonably call "the translate of D by g" are different divisors, and differ by boundary terms that must be accounted for rather than assumed away. That accounting is lemma 5.4.64.

5.4.5⋆ Birational Transforms of Linear Systems

⋆ *Optional. This subsection is an aside; nothing later in the book depends on it.*

A birational map moves a single Weil divisor by definition 5.4.50, and the accounting promised at the end of the previous subsection is about what happens when it moves a whole linear system at once. The obstacle is that the birational transform does not preserve linear equivalence (example 5.4.59), so the images $g_*(D_\lambda)$ of the members of a linear system $|W|$ on X need not be the members of any linear system on Y. There is nonetheless a linear system on Y that deserves to be called the transform of $|W|$, and the two agree for all λ outside a finite union of proper linear subspaces of the parameter space. That is lemma 5.4.64, and its corollary is the statement the Chevalley argument uses: an irreducible family of members transforms as expected.

Throughout, for a Weil divisor $D = \sum_\Gamma n_\Gamma[\Gamma]$ on a scheme as in definition 5.4.5, the integer n_Γ is written $\operatorname{mult}_\Gamma D$. Thus $D \geq 0$ says $\operatorname{mult}_\Gamma D \geq 0$ for every prime divisor Γ, and for $D \geq 0$ the support of D is the union of those Γ with $\operatorname{mult}_\Gamma D > 0$.

The construction rests on a normalization that is worth isolating first. A linear system without fixed components carries no information beyond the k-subspace of $K(X)$ that defines it: the divisor D_0 of definition 5.4.53 is forced.

Lemma 5.4.60: The Normalized Linear System of a Function Space

Let X be a normal, projective, geometrically integral scheme over a field k, and let $W \subseteq K(X)$ be a k-linear subspace with $\dim_k W = n \geq 1$. For a prime divisor $\Gamma \subseteq X$ set

$$m_\Gamma(W) \;:=\; \min\{\nu_\Gamma(f) \mid 0 \neq f \in W\}$$

1. The minimum exists, and $m_\Gamma(W) = 0$ for all but finitely many prime divisors Γ, so that

$$\Delta_X(W) \;:=\; \sum_\Gamma m_\Gamma(W)\,[\Gamma] \;\in\; \mathrm{WDiv}\, X$$

is a well-defined Weil divisor.

2. Put $D_0 := -\Delta_X(W)$. Then $W \subseteq L(D_0)$ (definition 5.4.9), and the linear system $|W|$ that W and D_0 define by definition 5.4.53 has no fixed components. It is called the *normalized system* of W.

3. Conversely, if $D_0' \in \mathrm{WDiv}\, X$ satisfies $W \subseteq L(D_0')$ and the linear system defined by W and D_0' has no fixed components, then $D_0' = -\Delta_X(W)$. Every linear system without fixed components is therefore the normalized system of its own space of functions.

4. For every prime divisor $\Gamma \subseteq X$, the set

$$W_\Gamma \;:=\; \{f \in W \mid \nu_\Gamma(f) > m_\Gamma(W)\} \cup \{0\}$$

is a k-linear subspace of W with $W_\Gamma \neq W$, and for $\lambda \in \mathbb{P}(W)$ one has $\Gamma \subseteq \mathrm{Supp}\, D_\lambda$ if and only if $\lambda \in \mathbb{P}(W_\Gamma)$.

5. For $h \in K(X)^*$ put $hW := \{hf \mid f \in W\}$. Then $\Delta_X(hW) = \Delta_X(W) + \mathrm{div}(h)$, and the normalized systems of W and of hW have the same members, matched by the isomorphism $\mathbb{P}(W) \to \mathbb{P}(hW)$, $[f] \mapsto [hf]$. A linear system without fixed components therefore determines its space of functions only up to such a scaling.

***Proof*:**
(1). Fix a k-basis $f_1, \ldots, f_n$ of W and let $0 \neq f = \sum_i c_i f_i$ with $c_i \in k$. For $c_i \neq 0$ the scalar c_i is a unit of the discrete valuation ring $\mathcal{O}_{X,\eta_\Gamma}$, so $\nu_\Gamma(c_i f_i) = \nu_\Gamma(f_i)$, and the ultrametric inequality gives

$$\nu_\Gamma(f) \;\geq\; \min\{\nu_\Gamma(f_i) \mid c_i \neq 0\} \;\geq\; \min_{1 \leq i \leq n} \nu_\Gamma(f_i)$$

So the set of values $\nu_\Gamma(f)$ is a nonempty set of integers bounded below and the minimum exists. The same two bounds give

$$\min_i \nu_\Gamma(f_i) \;\leq\; m_\Gamma(W) \;\leq\; \nu_\Gamma(f_1)$$

and by lemma 5.4.6 both ends vanish for every Γ outside the finite set of prime divisors occurring in one of the $\mathrm{div}(f_i)$.
(2). For $0 \neq f \in W$ the coefficient of Γ in $\mathrm{div}(f) + D_0$ is $\nu_\Gamma(f) - m_\Gamma(W) \geq 0$, so $\mathrm{div}(f) + D_0 \geq 0$ and $f \in L(D_0)$ by definition 5.4.9. Given a prime divisor Γ, choose $f \in W$ attaining $m_\Gamma(W)$; then the coefficient of Γ in $D_{[f]}$ is zero, so Γ is not contained in $\mathrm{Supp}\, D_{[f]}$ and Γ is not a fixed

component (definition 5.4.53).
(3). With D_0' in place of D_0 the coefficient of Γ in D_λ is $\nu_\Gamma(f_\lambda) + \mathrm{mult}_\Gamma(D_0')$, and all members are effective, so

$$\min_{\lambda \in \mathbb{P}(W)} \mathrm{mult}_\Gamma(D_\lambda) \;=\; m_\Gamma(W) + \mathrm{mult}_\Gamma(D_0') \;\geq\; 0$$

Absence of fixed components says exactly that this minimum is zero for every Γ, which is $\mathrm{mult}_\Gamma(D_0') = -m_\Gamma(W)$.
(4). For $c \in k^*$ one has $\nu_\Gamma(cf) = \nu_\Gamma(f)$ and for $f, f' \in W_\Gamma$ the ultrametric inequality gives $\nu_\Gamma(f+f') \geq \min\{\nu_\Gamma(f), \nu_\Gamma(f')\} > m_\Gamma(W)$, so W_Γ is a subspace; it is proper because the minimum defining $m_\Gamma(W)$ is attained. By (2) the coefficient of Γ in D_λ is $\nu_\Gamma(f_\lambda) - m_\Gamma(W)$, and D_λ is effective, so that coefficient is positive precisely when $f_\lambda \in W_\Gamma$.
(5). From $\nu_\Gamma(hf) = \nu_\Gamma(h) + \nu_\Gamma(f)$ the minimum defining $m_\Gamma(hW)$ is $\nu_\Gamma(h) + m_\Gamma(W)$, so $\Delta_X(hW) = \Delta_X(W) + \mathrm{div}(h)$, and then

$$\mathrm{div}(hf) - \Delta_X(hW) \;=\; \mathrm{div}(h) + \mathrm{div}(f) - \Delta_X(W) - \mathrm{div}(h) \;=\; \mathrm{div}(f) - \Delta_X(W)$$

which is the asserted matching of members, completing the proof.

The normalization removes an ambiguity that would otherwise obstruct the whole construction. Transporting a linear system along a birational map transports the space of functions, which is unproblematic because $g^\sharp$ is an isomorphism of function fields; it is the divisor D_0 that has no canonical image, since $g_*(D_0)$ and the fibers of the associated family differ by exceptional terms. Lemma 5.4.60(3) says the choice may be dispensed with entirely.

Definition 5.4.61: Birational Transform of a Linear System

Let X and Y be normal, projective, geometrically integral schemes over a field k, let $g\colon X \dashrightarrow Y$ be a birational map of k-schemes (definition 3.6.2), and write $g^\sharp\colon K(Y) \xrightarrow{\sim} K(X)$ for the induced k-algebra isomorphism of function fields (corollary 3.6.6). Let $|W|$ be a linear system of Weil divisors on X without fixed components (definition 5.4.53), so that $W \subseteq K(X)$ is a k-linear subspace of dimension $n \geq 1$ and, by lemma 5.4.60(3), $|W|$ is the normalized system of W.
The *birational transform of* $|W|$ *along* g is the normalized system on Y of the k-linear subspace

$$W^Y \;:=\; \left(g^\sharp\right)^{-1}(W) \;\subseteq\; K(Y)$$

written $g_*|W| := |W^Y|$. The isomorphism $\left(g^\sharp\right)^{-1}\big|_W\colon W \to W^Y$ is k-linear, so it induces an isomorphism of parameter spaces

$$\Phi_g\colon \mathbb{P}(W) \xrightarrow{\sim} \mathbb{P}(W^Y)$$

For $\lambda \in \mathbb{P}(W)$ the corresponding members are written D_λ^X, the member of $|W|$ indexed by λ, and D_λ^Y, the member of $g_*|W|$ indexed by $\Phi_g(\lambda)$.

By lemma 5.4.60(2) the transform is again a linear system without fixed components, of the same dimension as $|W|$, and it depends on nothing but g and $|W|$. What it is *not* is the set of birational transforms of the members of $|W|$: the map $\lambda \mapsto D_\lambda^Y$ moves the parameter space by a linear isomorphism, while $\lambda \mapsto g_*(D_\lambda^X)$ need not be linear at all, and the whole point of lemma 5.4.64 is to locate where the two disagree.

5.4.62 Note: Comparison with Kollár's Construction Kollár [45] defines the birational transform of $|W|$ differently: he takes the incidence correspondence $\mathbf{D}_{|W|}$ on $X \times \mathbb{P}(W)$ (definition 5.4.55), applies the birational transform along $g \times \mathrm{id}_{\mathbb{P}(W)}$ to obtain a divisor on $Y \times \mathbb{P}(W)$, and declares the transform to be the linear system to which that divisor corresponds. The last step is the difficulty. Recovering a linear system from a divisor on $Y \times \mathbb{P}(W)$ is the converse half of the assertion in his Definition 15 that a divisor on a product is the incidence correspondence of a unique linear system without fixed components, and that assertion is stated there without proof; the previous subsection records that the forward half, irreducibility of $\mathbf{D}_{|W|}$, is likewise unproved and is not used in this book. The description through W^Y above avoids both: it needs only that $g^\sharp$ is an isomorphism of function fields, and lemma 5.4.60 gives the divisor D_0^Y that Kollár's route would have to extract from the geometry.

The proof of the lemma uses one fact about birational maps that the book has so far recorded only in the weaker form of proposition 3.6.9, which produces isomorphic dense open subsets of two birationally equivalent schemes but does not say that the isomorphism represents a prescribed birational map. That refinement is what is needed, both here and for the base change in corollary 5.4.67.

Lemma 5.4.63: A Birational Map Is an Isomorphism on Dense Open Subsets

Let X and Y be integral separated schemes and let $g\colon X \dashrightarrow Y$ be a birational map. Then there are dense open subschemes $U \subseteq X$ and $V \subseteq Y$ and an isomorphism $\gamma\colon U \xrightarrow{\sim} V$ such that γ (followed by the inclusion of V) represents g and γ^{-1} represents g^{-1}.

***Proof*:**

Let $\varphi\colon U_1 \to Y$ represent g and $\psi\colon V_1 \to X$ represent g^{-1}, with $U_1 \subseteq X$ and $V_1 \subseteq Y$ nonempty open; a nonempty open subset of an irreducible space is dense, so U_1 and V_1 are dense. By lemma 3.6.4 the morphism φ carries the generic point η_X to η_Y and ψ carries η_Y to η_X.

Step 1: the two composites are inclusions. Put $U_2 := \varphi^{-1}(V_1)$ and $V_2 := \psi^{-1}(U_1)$. Both are open and nonempty, since $\eta_X \in U_2$ and $\eta_Y \in V_2$, hence both are dense. The composite $\psi \circ \varphi|_{U_2}\colon U_2 \to X$ represents the rational map $g^{-1} \circ g = \mathrm{id}_X$, so by definition 3.6.1 it agrees with the inclusion $U_2 \hookrightarrow X$ on a dense open subset of U_2. Now U_2 is reduced, being an open subscheme of the integral scheme X, and X is separated, so proposition 3.5.10 gives

$$\psi \circ \varphi|_{U_2} \;=\; \iota_{U_2}$$

where ι_{U_2} is the inclusion. Symmetrically $\varphi \circ \psi|_{V_2} = \iota_{V_2}$.
Step 2: cutting down. Put

$$U \;:=\; U_2 \cap \varphi^{-1}(V_2), \qquad V \;:=\; V_2 \cap \psi^{-1}(U_2)$$

Again $\eta_X \in U$ and $\eta_Y \in V$, so both are dense open. If $u \in U$ then $\varphi(u) \in V_2$ and, by Step 1, $\psi(\varphi(u)) = u \in U_2$, so $\varphi(u) \in \psi^{-1}(U_2)$ and hence $\varphi(u) \in V$. Thus $\varphi(U) \subseteq V$, and since V is open the restriction factors as a morphism $\gamma\colon U \to V$. Symmetrically $\psi(V) \subseteq U$, giving $\delta\colon V \to U$.
Step 3: conclusion. The identities of Step 1 restrict to $\delta \circ \gamma = \mathrm{id}_U$ and $\gamma \circ \delta = \mathrm{id}_V$, so γ is an isomorphism with inverse δ. It represents g because it is a restriction of φ to a dense open subset, and $\delta = \gamma^{-1}$ represents g^{-1} for the same reason, as we sought to show.

Everything is now in place. The lemma says that a birational transform changes the members of a linear system only along the divisors of Y that no divisor of X maps to, and that for each such divisor the members it swallows form a proper linear subspace of the parameter space.

Lemma 5.4.64: Exceptional Terms in the Transform of a Linear System

Let X, Y, g and $|W|$ be as in definition 5.4.61, and let $g_*|W|$ be the birational transform of $|W|$ along g. Let

$$\mathcal{E}_g \;:=\; \{E \subseteq Y \text{ a prime divisor} \mid [E] \neq g_*[\Gamma] \text{ for every prime divisor } \Gamma \subseteq X\}$$

the set of prime divisors of Y that are the transform of no prime divisor of X, equivalently those contracted by g^{-1}. Then:

1. $\mathcal{E}_g$ is finite; write $\mathcal{E}_g = \{E_1, \dots, E_r\}$. More precisely, if $\gamma\colon U \to V$ is an isomorphism of dense open subschemes representing g as in lemma 5.4.63, then every element of $\mathcal{E}_g$ is an irreducible component of $Y \setminus V$.
2. For every $\lambda \in \mathbb{P}(W)$,
$$D_\lambda^Y \;=\; g_*\big(D_\lambda^X\big) \;+\; \sum_{i=1}^{r} m_{i,\lambda}\, E_i, \qquad m_{i,\lambda} := \operatorname{mult}_{E_i} D_\lambda^Y \;\in\; \mathbb{Z}_{\geq 0}$$
3. For each i there is a proper k-linear subspace $W_i \subsetneq W$ such that, writing $L_i := \mathbb{P}(W_i) \subsetneq \mathbb{P}(W)$,
$$m_{i,\lambda} > 0 \iff E_i \subseteq \operatorname{Supp} D_\lambda^Y \iff \lambda \in L_i$$
Consequently
$$g_*\big(D_\lambda^X\big) \;=\; D_\lambda^Y \;\in\; g_*|W| \qquad \text{for every } \lambda \in \mathbb{P}(W) \setminus \bigcup_{i=1}^{r} L_i$$

***Proof*:**
Write $\sigma := g^\sharp\colon K(Y) \to K(X)$ and, for a prime divisor $\Gamma \subseteq X$ or $E \subseteq Y$, abbreviate $\mathcal{O}_\Gamma := \mathcal{O}_{X,\eta_\Gamma} \subseteq K(X)$ and $\mathcal{O}_E := \mathcal{O}_{Y,\eta_E} \subseteq K(Y)$.

Step 1: the dictionary between uncontracted prime divisors. Applying lemma 5.4.51(2) to g and $h = g^{-1}$ and using $g^{-1} \circ g = \mathrm{id}_X$ gives $\mathrm{id} = g^\sharp \circ (g^{-1})^\sharp$, so

$$\big(g^{-1}\big)^\sharp \;=\; \sigma^{-1}$$

By definition 5.4.50, $g_*[\Gamma] = [E]$ holds precisely when $\sigma(\mathcal{O}_E) = \mathcal{O}_\Gamma$, and $\big(g^{-1}\big)_*[E] = [\Gamma]$ holds precisely when $\sigma^{-1}(\mathcal{O}_\Gamma) = \mathcal{O}_E$. These two conditions are the same, and lemma 5.4.51(1) makes the partner unique on each side. So g_* and $(g^{-1})_*$ are mutually inverse bijections

$$\{\Gamma \subseteq X \text{ prime divisor} \mid g_*[\Gamma] \neq 0\} \;\longleftrightarrow\; \big\{E \subseteq Y \text{ prime divisor} \;\big|\; \big(g^{-1}\big)_*[E] \neq 0\big\}$$

and the complement of the right-hand set is $\mathcal{E}_g$, which justifies the two descriptions of $\mathcal{E}_g$ in the statement. Moreover, if $g_*[\Gamma] = [E]$ then σ restricts to an isomorphism of the discrete valuation

ring $\mathcal{O}_E$ onto $\mathcal{O}_\Gamma$, hence carries $\mathfrak{m}_E$ onto $\mathfrak{m}_\Gamma$, and therefore

$$\nu_E(h) \;=\; \nu_\Gamma(\sigma h) \qquad \text{for all } h \in K(Y)^* \tag{5.4}$$

Step 2: $\mathcal{E}_g$ is finite. By lemma 5.4.63 there are dense opens $U \subseteq X$ and $V \subseteq Y$ and an isomorphism $\gamma\colon U \to V$ representing g; the isomorphism of function fields induced by γ is σ, since γ represents g.

Let E be a prime divisor of Y with $\eta_E \in V$ and put $x := \gamma^{-1}(\eta_E) \in U$. An isomorphism induces isomorphisms of local rings compatible with the identification of function fields, so $\sigma(\mathcal{O}_{Y,\eta_E}) = \mathcal{O}_{X,x}$. In particular $\mathcal{O}_{X,x}$ is a one-dimensional regular local ring, so x has codimension 1 in X and $\Gamma := \overline{\{x\}}$ is a prime divisor with $\sigma(\mathcal{O}_E) = \mathcal{O}_\Gamma$. By Step 1, $E = g_*[\Gamma] \notin \mathcal{E}_g$.

Every $E \in \mathcal{E}_g$ therefore satisfies $\eta_E \notin V$, that is, $E \subseteq Y \setminus V$. The scheme Y is Noetherian, so proposition 2.1.25 writes $Y \setminus V$ as a finite union of irreducible closed subsets, and E, being irreducible, lies in one of them, say C. Since C is a proper closed irreducible subset of the irreducible finite type k-scheme Y, theorem 2.6.12 gives $\dim C \le \dim Y - 1$, while the same theorem applied to E gives $\dim E = \dim Y - 1$. From $E \subseteq C$ it follows that $E = C$. So $\mathcal{E}_g$ is contained in the finite set of irreducible components of $Y \setminus V$, which is (1).

Step 3: the members agree away from $\mathcal{E}_g$. Write $\Delta_X := \Delta_X(W)$ and $\Delta_Y := \Delta_Y(W^Y)$, so that by lemma 5.4.60(2)

$$D^X_\lambda \;=\; \operatorname{div}_X(f_\lambda) - \Delta_X, \qquad D^Y_\lambda \;=\; \operatorname{div}_Y\bigl(\sigma^{-1} f_\lambda\bigr) - \Delta_Y$$

for $\lambda = [f_\lambda] \in \mathbb{P}(W)$.

Let $E \notin \mathcal{E}_g$ and let Γ be the prime divisor of X with $g_*[\Gamma] = [E]$. Since $\sigma(W^Y) = W$, (5.4) turns the minimum defining $m_E(W^Y)$ into the minimum defining $m_\Gamma(W)$:

$$m_E\bigl(W^Y\bigr) \;=\; \min\bigl\{\nu_E(h) \;\big|\; 0 \neq h \in W^Y\bigr\} \;=\; \min\bigl\{\nu_\Gamma(\sigma h) \;\big|\; 0 \neq h \in W^Y\bigr\} \;=\; m_\Gamma(W)$$

Applying (5.4) once more,

$$\operatorname{mult}_E D^Y_\lambda \;=\; \nu_E\bigl(\sigma^{-1} f_\lambda\bigr) - m_E\bigl(W^Y\bigr) \;=\; \nu_\Gamma(f_\lambda) - m_\Gamma(W) \;=\; \operatorname{mult}_\Gamma D^X_\lambda$$

On the other hand g_* is a homomorphism $\operatorname{WDiv} X \to \operatorname{WDiv} Y$ killing the contracted prime divisors and, by Step 1, carrying the remaining ones bijectively to the prime divisors outside $\mathcal{E}_g$. Hence

$$\operatorname{mult}_E g_*\bigl(D^X_\lambda\bigr) \;=\; \begin{cases} \operatorname{mult}_\Gamma D^X_\lambda & \text{if } E = g_*[\Gamma] \notin \mathcal{E}_g, \\ 0 & \text{if } E \in \mathcal{E}_g \end{cases}$$

Comparing the two displays, the divisor $D^Y_\lambda - g_*(D^X_\lambda)$ has multiplicity zero at every prime divisor outside $\mathcal{E}_g$, so it is supported on $E_1, \dots, E_r$ with multiplicities $m_{i,\lambda} = \operatorname{mult}_{E_i} D^Y_\lambda$; and these are nonnegative because D^Y_λ is effective (lemma 5.4.60(2)). This is (2).

Step 4: the bad loci are proper linear subspaces. Apply lemma 5.4.60(4) on Y to the space W^Y and the prime divisor E_i: the subspace $\bigl(W^Y\bigr)_{E_i} \subsetneq W^Y$ is proper and, for $\mu \in \mathbb{P}(W^Y)$, one has $E_i \subseteq \operatorname{Supp} D_\mu$ exactly when $\mu \in \mathbb{P}\bigl((W^Y)_{E_i}\bigr)$. Put $W_i := \sigma\bigl((W^Y)_{E_i}\bigr) \subsetneq W$, a proper k-subspace since $\sigma|_{W^Y}\colon W^Y \to W$ is an isomorphism, and $L_i := \mathbb{P}(W_i) = \Phi_g^{-1}\bigl(\mathbb{P}((W^Y)_{E_i})\bigr)$. Since D^Y_λ is the member of $g_*|W|$ indexed by $\Phi_g(\lambda)$, the three conditions in (3) are equivalent, the first two by effectivity of D^Y_λ.

If λ lies in none of the L_i then every $m_{i,\lambda}$ vanishes, so (2) reads $g_*(D^X_\lambda) = D^Y_\lambda$, and the right-hand side is a member of $g_*|W|$, completing the proof.

Part (2) is an inequality in disguise: since the $m_{i,\lambda}$ are nonnegative, the transformed member D_λ^Y always *dominates* the birational transform $g_*(D_\lambda^X)$, and the excess is carried entirely by the finitely many prime divisors of Y that g extracts out of codimension ≥ 2. Example 5.4.59 is the case $r = 1$. There $g = \pi^{-1}$ is the inverse of the blow-up of $\mathbb{P}^2_k$ at a point, and every prime divisor of Y other than the exceptional curve E has its generic point in $Y \setminus E \cong \mathbb{P}^2_k \setminus \{p\}$, so the argument of Step 2 above makes it a transform and $\mathcal{E}_{\pi^{-1}} = \{E\}$. The identity $\mathbf{L}_0 = \widetilde{L_0} + E$ computed there has exactly the shape of (2): the birational transform $\widetilde{L_0} = (\pi^{-1})_*[L_0]$, plus a positive multiple of the one available exceptional divisor.

This is the general form of the phenomenon that makes the Chevalley argument delicate. A group G acting on a projective model X only rationally acts by birational self-maps, and translation by a group element can contract a component of the boundary $X \setminus U$ and extract another; the extracted components are exactly the elements of $\mathcal{E}_g$ for that translation. A construction that defines the translate of a divisor by a local equation, be it the fiber of a family (definition 5.4.48) or the normalization D_0^Y of lemma 5.4.60, can pick up a contribution along those components, and the birational transform discards it; that is the discrepancy computed on one surface in example 5.4.59 and measured in general by (2). Conrad's 2002 proof [24] treats the two notions as interchangeable. Part (3) says how much of that is true: they do agree, but only outside a finite union of proper linear subspaces of the parameter space, and example 5.4.65 exhibits a case where those subspaces are not empty.

Example 5.4.65: The Standard Quadratic Transformation on the System of Lines

Let k be a field and let $\sigma\colon \mathbb{P}^2_k \dashrightarrow \mathbb{P}^2_k$ be the birational involution of example 5.4.52. Write x_i for the homogeneous coordinates on the source and y_i for those on the target, so that $\sigma(x_0 : x_1 : x_2) = (x_1x_2 : x_0x_2 : x_0x_1)$, and put

$$H_i \;:=\; V_+(x_i) \subseteq X = \mathbb{P}^2_k, \qquad \ell_i \;:=\; V_+(y_i) \subseteq Y = \mathbb{P}^2_k$$

(the divisor called L_0 in example 5.4.52 is H_0 here; the letter L is reserved for the subspaces of lemma 5.4.64).

The system of lines. Set $u_i := x_i/x_0 \in K(X)$ and $W := \langle 1, u_1, u_2\rangle_k \subseteq K(X)$, of dimension 3. For $\lambda = (a : b : c)$,

$$a + bu_1 + cu_2 \;=\; \frac{ax_0 + bx_1 + cx_2}{x_0}$$

so $\operatorname{div}_X(a + bu_1 + cu_2) = [V_+(ax_0 + bx_1 + cx_2)] - H_0$. Hence ν_{H_0} takes the value -1 on every element of W except the multiples of 1, where it is 0, and every other prime divisor Γ has $\nu_\Gamma(1) = 0$ while $\nu_\Gamma \geq 0$ on W. So $\Delta_X(W) = -H_0$, the normalized system is

$$D^X_{(a:b:c)} \;=\; V_+(ax_0 + bx_1 + cx_2)$$

the complete system of lines, and it has no fixed components.

The transform. With $v_i := y_i/y_0$ one has $\sigma^\sharp(v_1) = x_0x_2/(x_1x_2) = 1/u_1$ and likewise $\sigma^\sharp(v_2) = 1/u_2$, so $\sigma^\sharp(1/v_i) = u_i$ and

$$W^Y \;=\; \left(\sigma^\sharp\right)^{-1}(W) \;=\; \langle 1,\, 1/v_1,\, 1/v_2\rangle_k$$

Writing $Q_{abc} := ay_1y_2 + by_0y_2 + cy_0y_1$, a quadratic form,

$$a + \frac{b}{v_1} + \frac{c}{v_2} \;=\; a + \frac{by_0}{y_1} + \frac{cy_0}{y_2} \;=\; \frac{Q_{abc}}{y_1y_2}$$

so for every prime divisor Γ of Y,

$$\nu_\Gamma\Big(a + \frac{b}{v_1} + \frac{c}{v_2}\Big) \;=\; \operatorname{mult}_\Gamma\big[V_+(Q_{abc})\big] \;-\; \operatorname{mult}_\Gamma(\ell_1 + \ell_2)$$

The first term is nonnegative, and it vanishes for a suitable choice of $(a : b : c)$ in each case: $Q_{010} = y_0y_2$ is not divisible by y_1, $Q_{001} = y_0y_1$ is not divisible by y_2, and $Q_{100} = y_1y_2$ is divisible by no prime divisor other than ℓ_1 and ℓ_2. Hence $m_{\ell_1}(W^Y) = m_{\ell_2}(W^Y) = -1$ and $m_\Gamma(W^Y) = 0$ for every other prime divisor Γ, so $\Delta_Y(W^Y) = -\ell_1 - \ell_2$ and

$$D^Y_{(a:b:c)} \;=\; \big[V_+(Q_{abc})\big]$$

so $\sigma_*|W|$ is the net of conics through the three coordinate points $e_0 = (1 : 0 : 0)$, $e_1 = (0 : 1 : 0)$, $e_2 = (0 : 0 : 1)$ of Y.

The exceptional divisors. On $\Omega := \{x_0x_1x_2 \neq 0\}$ every coordinate of $\sigma(x)$ is nonzero, and $\sigma\circ\sigma = \mathrm{id}$ there (example 5.4.52), so σ restricts to an isomorphism of Ω onto $\Omega' := \{y_0y_1y_2 \neq 0\}$, with inverse $\sigma|_{\Omega'}$. Hence lemma 5.4.64(1) puts $\mathcal{E}_\sigma$ among the irreducible components of $Y \setminus \Omega'$, namely ℓ_0, ℓ_1 and ℓ_2. All three occur: a point of ℓ_1 with $y_0y_2 \neq 0$ is carried by σ to $(y_0y_2 : 0 : 0) = e_0$, so σ is a morphism at η_{ℓ_1} and carries it to a closed point, whence $\big(\sigma^{-1}\big)_*[\ell_1] = \sigma_*[\ell_1] = 0$ by lemma 5.4.51(1); the same computation applies to ℓ_0 and ℓ_2. So $\mathcal{E}_\sigma = \{\ell_0, \ell_1, \ell_2\}$.

The bad loci. Setting $y_0 = 0$ in Q_{abc} leaves ay_1y_2, so $\ell_0 \subseteq \operatorname{Supp} D^Y_{(a:b:c)}$ if and only if $a = 0$; similarly for ℓ_1 and b, and for ℓ_2 and c. Thus

$$L_0 = \{a = 0\}, \qquad L_1 = \{b = 0\}, \qquad L_2 = \{c = 0\}$$

three lines in $\mathbb{P}(W) = \mathbb{P}^2$. Under the identification above, L_i is the pencil of lines of X through the i-th coordinate point.

A member that leaves the system. Let $\lambda = (0 : b : c)$ with $bc \neq 0$, so that $D^X_\lambda = V_+(bx_1 + cx_2)$, a line through $(1 : 0 : 0)$, and $Q_{0bc} = y_0(by_2 + cy_1)$. Then

$$D^Y_\lambda \;=\; \ell_0 + \big[V_+(cy_1 + by_2)\big], \qquad \sigma_*\big(D^X_\lambda\big) \;=\; \big[V_+(cy_1 + by_2)\big]$$

the second by lemma 5.4.64(2), since $m_{0,\lambda} = 1$ and $m_{1,\lambda} = m_{2,\lambda} = 0$. The transform of the member is a line, and it is not a member of $\sigma_*|W|$ at all: two members of a linear system differ by a principal divisor, so $\sigma_*(D^X_\lambda) \in \sigma_*|W|$ would make $\ell_0 = D^Y_\lambda - \sigma_*(D^X_\lambda)$ principal, contradicting $\deg \ell_0 = 1$ and proposition 5.4.16(1).

The exceptional locus $\bigcup_i L_i$ is a finite union of proper linear subspaces, so it is small in the parameter space; a general member transforms as expected. Turning that into a usable statement means asking when a *prescribed* family of members avoids the locus, and the answer is that it suffices for the family to be parametrized irreducibly and to span. Both words have to be read geometrically.

Definition 5.4.66: Linear Span and Irreducibly Parametrized Systems

Let X be as in definition 5.4.53 and let $|W|$ be a linear system of Weil divisors on X with parameter space $\mathbb{P}(W)$.
A *linear subspace* of $\mathbb{P}(W)$ is a closed subscheme of the form $\mathbb{P}(W')$ for a k-linear subspace $W' \subseteq W$, and it is *proper* if $W' \neq W$. Since $\mathbb{P}(W_1') \cap \mathbb{P}(W_2') = \mathbb{P}(W_1' \cap W_2')$, every subset $T \subseteq \mathbb{P}(W)$ is contained in a smallest linear subspace, its *linear span* $\langle T \rangle$; and T *spans* $|W|$ if $\langle T \rangle = \mathbb{P}(W)$.
Let $\overline{k}$ be an algebraic closure of k and identify $\mathbb{P}(W)_{\overline{k}}$ with $\mathbb{P}(W \otimes_k \overline{k})$. The system $|W|$ is *irreducibly parametrized* by a closed subvariety $T \subseteq \mathbb{P}(W)$ if $T_{\overline{k}}$ is irreducible and spans $\mathbb{P}(W \otimes_k \overline{k})$; that is, if T is geometrically irreducible and geometrically spanning.

If $T_{\overline{k}}$ spans $\mathbb{P}(W \otimes_k \overline{k})$ then T spans $|W|$, since a proper linear subspace $\mathbb{P}(W') \subseteq \mathbb{P}(W)$ containing T would give the proper linear subspace $\mathbb{P}(W' \otimes_k \overline{k})$ containing $T_{\overline{k}}$. Over an algebraically closed field the definition reads simply that T is irreducible and spans. Over a general field both readings must be geometric, and example 5.4.69 shows that k-irreducibility is not enough.

One further hypothesis appears below. Applying lemma 5.4.64 over $\overline{k}$ requires $X_{\overline{k}}$ and $Y_{\overline{k}}$ to be normal, so that is assumed; the assumption is vacuous when k is algebraically closed, which is the case the Chevalley argument uses.

Corollary 5.4.67: Transform of an Irreducibly Parametrized System

Let k be a field with algebraic closure $\overline{k}$, let X and Y be normal, projective, geometrically integral k-schemes such that $X_{\overline{k}}$ and $Y_{\overline{k}}$ are normal, let $g\colon X \dashrightarrow Y$ be a birational map of k-schemes, and let $|W|$ be a linear system of Weil divisors on X without fixed components. Suppose $|W|$ is irreducibly parametrized by $T \subseteq \mathbb{P}(W)$ (definition 5.4.66). Then:

1. $W_{\overline{k}} := W \otimes_k \overline{k}$ is an $\overline{k}$-linear subspace of $K(X_{\overline{k}})$ of dimension $\dim_k W$, its normalized system $|W_{\overline{k}}|$ is a linear system of Weil divisors on $X_{\overline{k}}$ without fixed components, and g base changes to a birational map $g_{\overline{k}}\colon X_{\overline{k}} \dashrightarrow Y_{\overline{k}}$. In particular lemma 5.4.64 applies to $g_{\overline{k}}$ and $|W_{\overline{k}}|$; let $L_1, \dots, L_r \subsetneq \mathbb{P}(W_{\overline{k}})$ be the proper linear subspaces it produces.
2. The set $T^\circ := T_{\overline{k}} \setminus \bigcup_{i=1}^{r} L_i$ is a dense open subvariety of $T_{\overline{k}}$.
3. For every closed point $\lambda \in T^\circ$, which is a line in $W_{\overline{k}}$ because $\overline{k}$ is algebraically closed and hence indexes a member of $|W_{\overline{k}}|$,
$$\big(g_{\overline{k}}\big)_*\big(D_\lambda^{X_{\overline{k}}}\big) \;=\; D_\lambda^{Y_{\overline{k}}} \;\in\; \big(g_{\overline{k}}\big)_*|W_{\overline{k}}|$$
4. $\Phi_{g_{\overline{k}}}(T^\circ)$ spans the parameter space of $\big(g_{\overline{k}}\big)_*|W_{\overline{k}}|$. Together with (3): the birational transforms of the members of $|W_{\overline{k}}|$ indexed by T° are members of $(g_{\overline{k}})_*|W_{\overline{k}}|$, and they are indexed by a spanning subset of its parameter space.

***Proof*:**
Step 1: base change of the birational map. By lemma 5.4.63 there are dense open subschemes $U \subseteq X$, $V \subseteq Y$ and an isomorphism $\gamma\colon U \to V$ representing g, with γ^{-1} representing g^{-1}.

Since X and Y are geometrically integral (definition 3.2.8), the schemes $X_{\overline{k}}$ and $Y_{\overline{k}}$ are integral; they are projective over $\overline{k}$, and normal by hypothesis. As $\overline{k}$ is algebraically closed, an integral $\overline{k}$-scheme is geometrically integral, so $X_{\overline{k}}$ and $Y_{\overline{k}}$ satisfy the hypotheses of definition 5.4.61.

Base change carries the isomorphism γ to an isomorphism $\gamma_{\overline{k}}\colon U_{\overline{k}} \to V_{\overline{k}}$ of nonempty, hence dense, open subschemes of $X_{\overline{k}}$ and $Y_{\overline{k}}$. Let $g_{\overline{k}}$ be the rational map it represents; its image contains the dense subset $V_{\overline{k}}$, so it is dominant, and $\gamma_{\overline{k}}^{-1}$ represents a dominant rational inverse, so $g_{\overline{k}}$ is birational (definition 3.6.2). It does not depend on the choice of (U, V, γ): two such isomorphisms represent the same rational map, so by definition 3.6.1 they agree on a dense open subset of the intersection of their domains and hence, by proposition 3.5.10, on the whole intersection, and the same is then true after base change.

Step 2: base change of the linear system. Choose a k-basis $f_1, \dots, f_n$ of W and a nonempty affine open $\operatorname{Spec} R \subseteq X$ on which every f_i is regular, so that $W \subseteq R$. Then $(\operatorname{Spec} R)_{\overline{k}} = \operatorname{Spec}(R \otimes_k \overline{k})$ is a nonempty open subscheme of the integral scheme $X_{\overline{k}}$, so $R \otimes_k \overline{k}$ is a domain with fraction field $K(X_{\overline{k}})$. The field $\overline{k}$ is free, hence flat, over k, so the inclusion $W \subseteq R$ gives an inclusion $W_{\overline{k}} \subseteq R \otimes_k \overline{k} \subseteq K(X_{\overline{k}})$, and $W_{\overline{k}}$ is an $\overline{k}$-subspace of dimension n. Now lemma 5.4.60 applied over $\overline{k}$ produces the normalized system $|W_{\overline{k}}|$ on $X_{\overline{k}}$ and shows it has no fixed components. This is (1).

Step 3: the general member of T. By hypothesis $T_{\overline{k}}$ is irreducible. Each L_i is a proper linear subspace of $\mathbb{P}(W_{\overline{k}})$ by lemma 5.4.64(3), so $T_{\overline{k}} \not\subseteq L_i$: were $T_{\overline{k}} \subseteq L_i$, the span $\langle T_{\overline{k}} \rangle$ would be contained in L_i, contradicting the assumption that $T_{\overline{k}}$ spans $\mathbb{P}(W_{\overline{k}})$. A linear subspace is closed, so $T_{\overline{k}} \cap L_i$ is a proper closed subset of $T_{\overline{k}}$ for each i, and

$$T^\circ \;=\; T_{\overline{k}} \setminus \bigcup_{i=1}^{r} \left(T_{\overline{k}} \cap L_i\right)$$

is a nonempty open subset of the irreducible space $T_{\overline{k}}$, hence dense. This is (2), and (3) is then lemma 5.4.64(3) applied to the closed points of T°, which lie outside every L_i.

Step 4: spanning. A dense subset and its closure have the same linear span, because a linear subspace is closed: any linear subspace containing T° contains $\overline{T^\circ} = T_{\overline{k}}$, hence contains $\langle T_{\overline{k}} \rangle$. Therefore $\langle T^\circ \rangle = \langle T_{\overline{k}} \rangle = \mathbb{P}(W_{\overline{k}})$. Finally $\Phi_{g_{\overline{k}}}$ is induced by an isomorphism of $\overline{k}$-vector spaces, so it carries linear subspaces to linear subspaces bijectively and hence satisfies $\langle \Phi_{g_{\overline{k}}}(S) \rangle = \Phi_{g_{\overline{k}}}(\langle S \rangle)$ for every subset $S \subseteq \mathbb{P}(W_{\overline{k}})$. Applying this to $S = T^\circ$ gives that $\Phi_{g_{\overline{k}}}(T^\circ)$ spans the parameter space of $\left(g_{\overline{k}}\right)_* |W_{\overline{k}}|$, which is (4), as we sought to show.

Neither hypothesis can be dropped, and the standard quadratic transformation gives a counterexample to each. The first shows that a connected spanning set is not enough, the second that irreducibility over k is not enough. Both are Kollár's Example 17 in [45].

Example 5.4.68: A Connected Reducible Spanning Set in the System of Lines
Keep the notation of example 5.4.65 and let

$$T \;:=\; L_0 \cup L_1 \cup L_2 \;=\; V_+(abc) \;\subseteq\; \mathbb{P}(W) = \mathbb{P}^2_k$$

the set of lines of X passing through at least one of the three coordinate points.

T is connected and spans. Write $L_i = \mathbb{P}(W_i)$ with $W_i \subseteq W$ of dimension 2; the three W_i are pairwise distinct, so $W_0 + W_1 = W$ and any linear subspace containing T contains $L_0 \cup L_1$, hence equals $\mathbb{P}(W)$. Also $\dim_k(W_i \cap W_j) = 1$ for $i \neq j$, so the three lines meet pairwise and T is

connected. It is not irreducible: abc is a product of three pairwise non-proportional linear forms.

The conclusion fails. By construction $T \setminus \bigcup_i L_i = \emptyset$, so there is no dense open subset of T on which corollary 5.4.67(3) could hold, and indeed nothing survives: for $\lambda \in T$ at least one $m_{i,\lambda}$ is positive, so

$$D^Y_\lambda - \sigma_*\big(D^X_\lambda\big) \;=\; \sum_i m_{i,\lambda}\,\ell_i$$

is effective and nonzero, of degree $\sum_i m_{i,\lambda} > 0$. Two members of a linear system differ by a principal divisor, and a principal divisor on $\mathbb{P}^2_k$ has degree 0 (proposition 5.4.16(1)), so $\sigma_*(D^X_\lambda)$ is not a member of $\sigma_*|W|$ for any $\lambda \in T$. The degrees show this: every member of $\sigma_*|W|$ is a conic, of degree 2, while $\deg \sigma_*(D^X_\lambda) = 2 - \sum_i m_{i,\lambda} \le 1$.

The field plays no role here: the example works over every field, including an algebraically closed one, so what fails is irreducibility and nothing else.

The second failure needs a field over which a k-irreducible variety can break into conjugate pieces. The mechanism is that the three exceptional divisors of the quadratic transformation, and with them the three subspaces L_i, become a single Galois orbit, so their union is k-irreducible while $\bigcup_i L_i$ is not defined over k component by component. To keep the construction explicit, the quadratic transformation is presented as the inversion map of a cubic étale algebra.

Example 5.4.69: Conjugate Base Points and a k-Irreducible Spanning Set

Let k be a field admitting a separable cubic field extension A/k; for instance $k = \mathbb{Q}$ and $A = \mathbb{Q}[t]/(t^3 - 2)$. Write $\mathrm{tr}_{A/k}$ and $N_{A/k}$ for the trace and norm, and let $s_2 \colon A \to k$ be the coefficient defined by the characteristic polynomial of multiplication by a,

$$\chi_a(t) \;=\; t^3 - \mathrm{tr}_{A/k}(a)\,t^2 + s_2(a)\,t - N_{A/k}(a)$$

Let $X = Y := \mathbb{P}(A)$, the projective plane of lines in the three-dimensional k-vector space A; a k-basis of A identifies it with $\mathbb{P}^2_k$.

The map. Put $\mathrm{adj}(a) := a^2 - \mathrm{tr}_{A/k}(a)\,a + s_2(a)\cdot 1 \in A$. The assignment $a \mapsto a^2$ is quadratic, and $s_2(a)$ is the sum of the 2×2 principal minors of the matrix of multiplication by a, hence also quadratic in the coordinates of a; so each component of adj is a quadratic form over k and

$$\iota \colon \mathbb{P}(A) \dashrightarrow \mathbb{P}(A), \qquad \iota([a]) \;=\; \big[\mathrm{adj}(a)\big]$$

is a rational map defined over k.

Over $\overline{k}$. Since A/k is separable of degree 3, the algebra $A \otimes_k \overline{k}$ is isomorphic to $\overline{k}\times\overline{k}\times\overline{k}$; let e_0, e_1, e_2 be its primitive idempotents and write $a = (x_0, x_1, x_2)$ in the corresponding coordinates. Multiplication by a is then diagonal, so $\chi_a(t) = \prod_i (t - x_i)$ and

$$\mathrm{tr}(a) = x_0 + x_1 + x_2, \qquad s_2(a) = x_0x_1 + x_0x_2 + x_1x_2, \qquad N(a) = x_0x_1x_2$$

whence $\mathrm{adj}(a) = (x_1x_2,\, x_0x_2,\, x_0x_1)$. So the idempotent basis identifies $\mathbb{P}(A)_{\overline{k}}$ with $\mathbb{P}^2_{\overline{k}}$ in such a way that $\iota_{\overline{k}}$ is the quadratic transformation σ of example 5.4.65, with base points the three idempotents.

Birationality over k. In those coordinates $\mathrm{adj}(\mathrm{adj}(a))$ has first component $(x_0x_2)(x_0x_1) = N(a)\,x_0$, and likewise in the other two, while $N(\mathrm{adj}(a)) = (x_1x_2)(x_0x_2)(x_0x_1) = N(a)^2$. Both sides of

$$\mathrm{adj}\big(\mathrm{adj}(a)\big) = N_{A/k}(a)\,a, \qquad N_{A/k}\big(\mathrm{adj}(a)\big) = N_{A/k}(a)^2$$

are polynomial maps with coefficients in k, and A injects into $A \otimes_k \overline{k}$, so these identities hold over k. Consequently ι carries the open subscheme $D_+(N_{A/k})$, nonempty because $N_{A/k}(1) = 1$, into itself, and does so by an isomorphism equal to its own inverse. Hence ι is a birational involution of $\mathbb{P}(A)$.

The system and its bad loci. Let $|W|$ be the complete system of hyperplanes of $\mathbb{P}(A)$: fix $0 \neq \varphi_0 \in A^\vee$ and put $W := \{\varphi/\varphi_0 \mid \varphi \in A^\vee\} \subseteq K(\mathbb{P}(A))$, so that $D^X_{[\varphi/\varphi_0]} = V_+(\varphi)$ and $\mathbb{P}(W) \cong \mathbb{P}(A^\vee)$. Changing φ_0 replaces W by a scalar multiple hW and W^Y by $\left(\iota^\sharp\right)^{-1}(h)\, W^Y$, so lemma 5.4.60(5), applied on X and on Y, leaves the members of $|W|$ and of $\iota_*|W|$ and the subspaces L_i unchanged. Over $\overline{k}$, in the idempotent coordinates, $W_{\overline{k}}$ is a scaling of the space $\langle 1, x_1/x_0, x_2/x_0\rangle$ of example 5.4.65, so that example computes everything: $\mathcal{E}_{\iota_{\overline{k}}}$ consists of the three hyperplanes $V_+(x_i)$, and the associated subspaces are

$$L_i \;=\; \{[\varphi] \mid \varphi(e_i) = 0\} \qquad (i = 0, 1, 2)$$

the pencil of hyperplanes through the i-th idempotent.

The spanning set. The trace pairing $\langle a, b\rangle := \mathrm{tr}_{A/k}(ab)$ is nondegenerate because A/k is separable, so $c \mapsto \varphi_c := \mathrm{tr}_{A/k}(c\,\cdot)$ is a k-isomorphism $A \to A^\vee$; in idempotent coordinates $\varphi_c(e_i) = \mathrm{tr}(ce_i) = c_i$. Let

$$T \;:=\; \left\{[\varphi_c] \in \mathbb{P}(A^\vee) \cong \mathbb{P}(W) \;\middle|\; N_{A/k}(c) = 0\right\}$$

the norm cubic. In idempotent coordinates its equation is $c_0c_1c_2 = 0$, so

$$T_{\overline{k}} \;=\; L_0 \cup L_1 \cup L_2$$

exactly the set of example 5.4.68.

T is irreducible over k but not geometrically. Suppose $N_{A/k} = \ell \cdot q$ with ℓ a nonzero linear form on A over k. The kernel of ℓ is a 2-dimensional k-subspace of A, and every c in it has $N_{A/k}(c) = 0$, hence $c = 0$ because A is a field. This is absurd, so $N_{A/k}$ has no linear factor over k; a reducible homogeneous cubic has one, so $N_{A/k}$ is irreducible, the ideal $(N_{A/k})$ is prime, and T is an integral closed subvariety of $\mathbb{P}(W)$. Over $\overline{k}$ it acquires three components. It spans: if T lay in a proper linear subspace, some nonzero k-linear form ℓ on A would vanish on $T = V_+(N_{A/k})$, and since $(N_{A/k})$ is a homogeneous prime not containing the irrelevant ideal it is itself a point of $V_+(N_{A/k})$, forcing $\ell \in (N_{A/k})$, which is impossible for degree reasons. And $T_{\overline{k}}$ spans by example 5.4.68.

The conclusion fails. Since $T_{\overline{k}} \setminus \bigcup_i L_i = \emptyset$, there is again no dense open subset of $T_{\overline{k}}$ on which corollary 5.4.67(3) holds, and for every closed point λ of $T_{\overline{k}}$ the divisor $\left(\iota_{\overline{k}}\right)_*\left(D^{X_{\overline{k}}}_\lambda\right)$ is not a member of $\left(\iota_{\overline{k}}\right)_*|W_{\overline{k}}|$ (example 5.4.68). The only hypothesis of corollary 5.4.67 that fails is geometric irreducibility.

The failure is invisible over k. A k-point of T would be $[\varphi_c]$ with $c \neq 0$ and $N_{A/k}(c) = 0$, impossible in a field, so $T(k) = \emptyset$: no member of $|W|$ itself is indexed by T, and only the geometric points detect the obstruction.

5.5 Differentials

The reader arrives at this section already fluent in its geometry. On a smooth manifold M, the cotangent space T^*_pM (definition 0.5.11) and the differential forms assembled from it (definition 0.5.14)

carry the infinitesimal geometry of M, and the differential of a function $df_p \in T_p^*M$ (definition 0.5.12) records its first-order behavior at p. The point of contact with algebra is a description of these objects that uses no analysis at all. Writing $\mathfrak{m}_p$ for the ideal of functions vanishing at p,

$$T_pM \;\cong\; (\mathfrak{m}_p/\mathfrak{m}_p^2)^\vee, \qquad T_p^*M \;\cong\; \mathfrak{m}_p/\mathfrak{m}_p^2$$

([59]), and under the second identification the covector df_p is the class of $f - f(p)$ modulo $\mathfrak{m}_p^2$. No limit and no smooth structure enter the right-hand sides: only an ideal, its square, and a quotient. This is verbatim the Zariski cotangent space introduced earlier for varieties (definition 0.1.25), and that coincidence is why differentials belong to algebraic geometry at all.

The same description settles which of the two objects is primitive. The assignment $f \mapsto df = [\, f - f(p) \,]$ produces a covector from a function by a single quotient; the tangent vector, an element of the dual, is what one obtains *afterward*, by dualizing. In the smooth category the distinction is immaterial, since both spaces are finite-dimensional and either determines the other. But the construction that survives with no appeal to limits, and so transports to an arbitrary ring, is the cotangent one. Functoriality points the same way: functions pull back along a morphism $f\colon X \to Y$, so the cotangent construction, covariant in the ring of functions, transforms along with them, while tangent vectors push forward, in the opposite direction. Algebraic geometry therefore takes the cotangent side as basic. The *sheaf of differentials* $\Omega_{X/Y}$ (definition 5.5.14) is the object it builds first; the universal derivation $d\colon b \mapsto db$ is the Leibniz and chain rules promoted to a definition; and the tangent sheaf (definition 5.5.28) appears only afterward, as its dual.

The payoff is an exact algebraic stand-in for the cotangent bundle. Once $\Omega_{X/Y}$ is in hand, its fiber at a point is canonically $\mathfrak{m}_p/\mathfrak{m}_p^2 = T_p^*$ (proposition 5.5.10): the sheaf of differentials is the cotangent bundle glued together globally, with no coordinate charts and no partitions of unity. Being the cotangent object and not its dual is what lets it see singularities. The dimension of $\mathfrak{m}_p/\mathfrak{m}_p^2$ jumps above the dimension of X exactly at a singular point (definition 0.1.27), the sheaf $\Omega_{X/Y}$ fails to be locally free there (theorem 5.5.12), and this failure is the algebraic test for smoothness (proposition 5.5.25); the tangent sheaf, a dual that is well behaved only in the smooth case, is the wrong place to look for it. Further downstream, the top exterior power of $\Omega_{X/Y}$ is the canonical sheaf standing behind Serre duality (theorem 6.5.9) and Riemann–Roch (theorem 7.2.10), and $\Omega_{X/Y}$ itself governs deformation theory. These are the section's eventual destinations; the road to them begins in commutative algebra.

We begin with that algebra: the module of Kähler differentials $\Omega_{B/A}$ attached to a ring map $A \to B$, before globalizing to schemes. The reader who knows forms on manifolds will recognize the parallel at once, since the Leibniz rule and the chain rule are built directly into the definition.

5.5.1 Kähler Differentials

The module of relative differentials is pure commutative algebra and is developed in [12]: derivations, $\Omega_{B/A}$ and its universal property, the diagonal construction, base change and localization, both fundamental exact sequences, and finite generation. The statements are recalled here because the sheaf-theoretic constructions below rest on them, but the proofs are not repeated; what this book proves is the passage from the module to the sheaf $\Omega_{X/S}$ and everything after it.

Definition 5.5.1: A-Derivation

Let A be a commutative ring, B an A-algebra, and M a B-module. An *A-derivation* of B into M is a map $d\colon B \to M$ satisfying:

1. *(Additivity)* $d(b+b') = d(b) + d(b')$,
2. *(Leibniz rule)* $d(bb') = b\,d(b') + b'\,d(b)$,
3. *(Vanishing on A)* $d(a) = 0$ for all $a \in A$ (treating A as mapped into B)

The set of all A-derivations $B \to M$ is denoted $\mathrm{Der}_A(B, M)$ and carries a natural B-module structure via $(b \cdot d)(b') = b\,d(b')$.

Condition (3) is the algebraic analogue of "the derivative of a constant is zero": the elements of A, viewed as "constants" in the A-algebra B, have trivial differential. Note that (3) is automatic when A is a prime ring or prime field (generated by a single element) since then $d(1) = d(1 \cdot 1) = 2\,d(1)$ forces $d(1) = 0$. Condition (3) forces the derivation to be A-linear since for $a \in A$:

$$d(ab) = d(a)b + a d(b) = 0 + a d(b) = a d(b)$$

Definition 5.5.2: Module of Relative Differential Forms

Let A be a commutative ring and B an A-algebra. A *module of (relative) Kähler differentials* for B over A is a B-module $\Omega_{B/A}$ equipped with an A-derivation $d\colon B \to \Omega_{B/A}$ satisfying the following universal property: for every B-module M and every A-derivation $d'\colon B \to M$, there exists a unique B-module homomorphism $\varphi\colon \Omega_{B/A} \to M$ such that $d' = \varphi \circ d$:

$$\begin{array}{ccc} B & \xrightarrow{\;d\;} & \Omega_{B/A} \\ & \searrow^{d'} & \downarrow{\scriptstyle \exists!\,\varphi} \\ & & M \end{array}$$

Equivalently, d induces a B-module isomorphism

$$\mathrm{Hom}_B(\Omega_{B/A},\, M) \xrightarrow{\;\sim\;} \mathrm{Der}_A(B,\, M), \qquad \varphi \longmapsto \varphi \circ d.$$

The universal property determines $(\Omega_{B/A}, d)$ up to unique isomorphism, as with any object defined by a universal property. There are two general constructions to $\Omega_{B/A}$: an explicit generators-and-relations presentation, and a more geometric one via the diagonal ideal.

Construction 1: generators and relations. Let F be the free B-module on symbols $\{db : b \in B\}$, and let R be the submodule generated by the relations

$$d(b+b') - db - db', \qquad d(bb') - b\,db' - b'\,db, \qquad da \quad (a \in A).$$

Set $\Omega_{B/A} = F/R$ and define $d\colon B \to \Omega_{B/A}$ by $b \mapsto db$. By construction, d is an A-derivation, and the universal property is immediate: any A-derivation $d'\colon B \to M$ extends uniquely to a B-module map $F \to M$ (sending $db \mapsto d'(b)$), which kills R and hence factors through $\Omega_{B/A}$.

Construction 2: the diagonal ideal. The following construction has a geometric interpretation that globalizes directly to schemes.

Proposition 5.5.3: Differentials via the Diagonal

Let B be an A-algebra. Let $\mu\colon B \otimes_A B \to B$ be the multiplication map $b \otimes b' \mapsto bb'$, and let $I = \ker \mu$. View $B \otimes_A B$ as a B-module via left multiplication $(b \cdot (b_1 \otimes b_2) = bb_1 \otimes b_2)$. Then:

1. I/I^2 inherits the structure of a B-module.
2. The map $d\colon B \to I/I^2$ defined by $d(b) = 1 \otimes b - b \otimes 1 \pmod{I^2}$ is an A-derivation.
3. The pair $(I/I^2, d)$ satisfies the universal property of $\Omega_{B/A}$.

***Proof*:**
This is proved in [12, Natural Differential Structure], where the diagonal construction is verified against the universal property. The verification is pure commutative algebra and is not repeated here.

The geometric content of Construction 2 is the following: if $X = \operatorname{Spec} B$ and $Y = \operatorname{Spec} A$, then $B \otimes_A B$ is the coordinate ring of the fiber product $X \times_Y X$, and the multiplication map $\mu\colon B \otimes_A B \to B$ corresponds to the diagonal morphism $\Delta\colon X \to X \times_Y X$. The ideal $I = \ker \mu$ is the ideal sheaf of the diagonal, and I/I^2 is the *conormal sheaf* of the diagonal embedding. Thus the module of differentials is the algebraic incarnation of "functions vanishing on the diagonal, modulo those vanishing to second order", precisely the data captured by first-order Taylor expansions. This interpretation globalizes immediately to the sheaf of differentials on a morphism of schemes (cf. definition 5.5.14).

Note that we can define the higher order equivalently (these are known as a jet bundle):

$$\mathcal{P}^n_{B/A} = \frac{B \otimes_A B}{I^{n+1}}$$

but we do not focus on these in this section.

Example 5.5.4: Relative Differentials

1. *Polynomial rings.* Let $B = A[x_1, \dots, x_n]$. The Leibniz rule and the chain rule force $\Omega_{B/A}$ to be generated by $dx_1, \dots, dx_n$. Conversely, the partial derivatives $\partial/\partial x_i\colon B \to B$ are A-derivations, and the universal property provides B-linear maps $\Omega_{B/A} \to B$ sending $dx_i \mapsto 1$ and $dx_j \mapsto 0$ for $j \neq i$. Hence $dx_1, \dots, dx_n$ are linearly independent, and
$$\Omega_{B/A} \cong \bigoplus_{i=1}^{n} B\, dx_i,$$
a free B-module of rank n. The universal derivation is $d(f) = \sum_{i=1}^n \frac{\partial f}{\partial x_i}\, dx_i$, exactly as in multivariable calculus.
2. *Quotient rings.* Let $B = A[x_1, \dots, x_n]/J$ where $J = (f_1, \dots, f_r)$. Using the two exact sequences below (propositions 5.5.5 and 5.5.6), one obtains
$$\Omega_{B/A} \cong \frac{\bigoplus_{i=1}^n B\, dx_i}{\sum_{j=1}^r B\, df_j},$$
where $df_j = \sum_i \frac{\partial f_j}{\partial x_i}\, dx_i$. This is the algebraic analogue of computing the cotangent space of a submanifold by imposing the constraint equations.

For instance, if $B = k[x,y]/(y^2 - x^3)$ (the cuspidal cubic), then $\Omega_{B/k} = B\,dx \oplus B\,dy \,/\, (2y\,dy - 3x^2\,dx)$. At the origin $(x,y) = (0,0)$, the relation $2y\,dy - 3x^2\,dx$ vanishes identically, so $\Omega_{B/k} \otimes_B k \cong k^2$, two-dimensional rather than the expected one. This detects the singularity (cf. example 2.7.6).

3. *Separable field extensions.* Let K/k be a finite field extension. Then $\Omega_{K/k} = 0$ if and only if K/k is separable. To see this, first consider a simple extension $K = k(\alpha) \cong k[x]/(f)$, where f is the minimal polynomial of α. By the quotient ring computation above, $\Omega_{K/k} \cong K\,dx/(f'(\alpha)\,dx)$. If K/k is separable, f has no multiple roots, meaning $f'(\alpha) \neq 0$. Since K is a field, $f'(\alpha)$ is a unit, forcing $dx = 0$ and hence $\Omega_{K/k} = 0$. Conversely, if K/k is inseparable (which only occurs if $\operatorname{char} k = p > 0$), there exists some $\alpha \in K$ whose minimal polynomial takes the form $f(x) = g(x^p)$. Its formal derivative vanishes identically ($f'(x) = 0$), so the relation $f'(\alpha)\,dx = 0$ imposes no constraint on dx. Thus $\Omega_{k(\alpha)/k} \neq 0$, and hence $\Omega_{K/k} \neq 0$.

   Geometrically, a separable field extension represents a discrete covering of an isolated points, meaning there is no continuous space to move and therefore a trivial cotangent space ($\Omega_{K/k} = 0$). In contrast, an inseparable extension contains "infinitesimal fuzz" where points have collided in positive characteristic, spawning a hidden, non-trivial differential direction even in a zero-dimensional space.

4. *Localization.* If $S \subseteq B$ is a multiplicative set, then $\Omega_{S^{-1}B/A} \cong S^{-1}\Omega_{B/A}$. This follows from the universal property: an A-derivation $S^{-1}B \to M$ is determined by its restriction to B, using the quotient rule $d(b/s) = (s\,db - b\,ds)/s^2$.

The following two exact sequences are used to compute differentials. They are the algebraic counterparts of the cotangent exact sequence and the conormal exact sequence in differential geometry:

Proposition 5.5.5: First Exact Sequence (Base Change, or Fibrations)

Let $A \to B \to C$ be ring homomorphisms. Then there is a natural exact sequence of C-modules

$$\Omega_{B/A} \otimes_B C \longrightarrow \Omega_{C/A} \longrightarrow \Omega_{C/B} \longrightarrow 0.$$

The first map sends $db \otimes 1 \mapsto d(\varphi(b))$ (where $\varphi\colon B \to C$ is the structure map), and the second map sends $dc \mapsto dc$.

***Proof*:**
Proved in [12, Differentials and Exact Sequence I] by applying $\operatorname{Hom}_C(-, M)$ and checking exactness of the resulting sequence of derivation modules.

Note that $A \to B \to C$ is has no exactness conditions set on it: geometrically this has many interpretations (this can be a fibration, it can be closed immersions, etc). This generality is partly to build good tools to construct differentials, and partly because singularities often break exactness (as demonstrated in examples in chapter 3). Geometrically, if you take the total differential of a space ($\Omega_{C/A}$) and quotient out the differentials coming from the base space ($\Omega_{B/A}$), you are left exactly with the differentials of the fibers ($\Omega_{C/B}$). In terms of tangent vectors (the dual picture), the proposition says the tangent vectors of a fiber are exactly the tangent vectors of the total space that project to zero on the base.

Proposition 5.5.6: Second Exact Sequence (Conormal)

Let B be an A-algebra, $J \subseteq B$ an ideal, and $C = B/J$. There is a natural exact sequence of C-modules

$$J/J^2 \xrightarrow{\delta} \Omega_{B/A} \otimes_B C \longrightarrow \Omega_{C/A} \longrightarrow 0,$$

where $\delta(\overline{f}) = df \otimes 1$ for $f \in J$.

***Proof*:**
Proved in [12, Differentials and Exact Sequence II], by the same $\mathrm{Hom}_C(-, M)$ argument as the first sequence, together with the observation that a derivation into a C-module descends to C exactly when it kills I.

Geometrically, this tells you how to link the intrinsic cotangent space of the subspace $\Omega_{C/A}$ to the ambient cotangent space $\Omega_{B/A} \otimes_B C$. In other word, to find the differentials of a subspace ($\Omega_{C/A}$), start with the differentials of the ambient space ($\Omega_{B/A}$) and kill the gradients of the constraint equations ($\delta(J/J^2)$).

Corollary 5.5.7: Finite Generation of Differentials

If B is a finitely generated A-algebra, then $\Omega_{B/A}$ is a finitely generated B-module.

***Proof*:**
Immediate from the construction of $\Omega_{B/A}$ by generators and relations; see [12, Finite Generation of Differential Forms].

The next two results relate differentials to base change and to field extensions, the latter being crucial for detecting smoothness.

Proposition 5.5.8: Base Change for Differentials

Let $A \to B$ and $A \to A'$ be ring homomorphisms, and set $B' = B \otimes_A A'$. Then

$$\Omega_{B'/A'} \;\cong\; \Omega_{B/A} \otimes_B B'.$$

***Proof*:**
Proved in [12, Base Change For Differentials].

Proposition 5.5.9: Differentials and Field Extensions

Let k be a field and K/k a finitely generated[a] field extension. Then $\Omega_{K/k}$ is a K-vector space and:

$$\dim_K \Omega_{K/k} \geq \operatorname{trdeg}(K/k)$$

with equality if and only if K is separably generated over k (i.e., K is a separable algebraic extension of a purely transcendental extension of k).

[a] be sure not to confuse with finite! example 5.5.4 was a special case with finite extensions

5.5.2 Differentials and Local Rings

The module of differentials detects regularity of local rings, providing the algebraic foundation for the Jacobian criterion.

Proposition 5.5.10: Differentials and Local Rings

Let $(B, \mathfrak{m})$ be the localization at a *maximal* ideal of a finitely generated algebra over a field k, with residue field $\kappa = B/\mathfrak{m}$, and assume κ is separable *algebraic* over k (automatic when k is perfect, and trivially so when k is algebraically closed, where $\kappa = k$). Then there is a natural isomorphism of κ-vector spaces

$$\Omega_{B/k} \otimes_B \kappa \;\cong\; \mathfrak{m}/\mathfrak{m}^2$$

***Proof*:**
Apply the second exact sequence (proposition 5.5.6) to the surjection $B \twoheadrightarrow \kappa$ with kernel $\mathfrak{m}$:

$$\mathfrak{m}/\mathfrak{m}^2 \xrightarrow{\delta} \Omega_{B/k} \otimes_B \kappa \longrightarrow \Omega_{\kappa/k} \longrightarrow 0$$

Since $\mathfrak{m}$ is maximal in a finitely generated k-algebra, κ/k is finite by the Nullstellensatz, in particular finitely generated; being also separable algebraic it has $\operatorname{trdeg}(\kappa/k) = 0$ and is separably generated, so $\Omega_{\kappa/k} = 0$ by proposition 5.5.9 and δ is surjective. Injectivity is the substantial half, and it is exactly where separability is spent: it is the statement that the second fundamental sequence stays exact on the left for $B \twoheadrightarrow \kappa$, which holds because a separable field extension is formally smooth ([49, Thm. 62 with Thm. 58(iii)]: separable makes κ smooth over k, which trivializes the extension and so gives a left inverse for δ). Concluding equality of dimensions from surjectivity alone would be circular, since that equality is the statement being proved.

What is not removable is that κ/k be *algebraic*. Over $k = \mathbb{Q}$ take $B = k[x]_{(0)} = k(x)$. Then $\mathfrak{m} = 0$, so $\mathfrak{m}/\mathfrak{m}^2 = 0$, while $\Omega_{B/k} \otimes_B \kappa = \Omega_{k(x)/k}$ is one-dimensional: $\kappa = k(x)$ is separable over k but transcendental, and $\Omega_{\kappa/k}$ does not vanish. Given the algebraicity, maximality of $\mathfrak{m}$ comes for free: if κ is algebraic over k then the domain $A/\mathfrak{p}$, sitting inside κ, is algebraic over k and hence a field, so $\mathfrak{p}$ is maximal. Compare box 5.5.13.

Lemma 5.5.11: Characterizing Freeness via Residue and Quotient Fields

Let $(R, \mathfrak{m})$ be a Noetherian local domain with fraction field K, and let M be a finitely generated R-module. Then M is free of rank r if and only if

$$\dim_{R/\mathfrak{m}}(M \otimes_R R/\mathfrak{m}) \;=\; r \;=\; \dim_K(M \otimes_R K).$$

***Proof*:**
By Nakayama's lemma, M is generated by r elements if and only if $\dim_{R/\mathfrak{m}}(M/\mathfrak{m}M) = r$. If additionally $\dim_K(M \otimes_R K) = r$, then a minimal generating set of M is K-linearly independent, hence R-linearly independent (since R is a domain), so $M \cong R^r$.

Theorem 5.5.12: Characterizing Regular Local Rings via Differentials

Let B be a localization of a finitely generated k-algebra at a maximal ideal, where k is a perfect field. Set $n = \dim B$. Then B is a regular local ring if and only if $\Omega_{B/k}$ is a free B-module of rank n.

***Proof*:**
By the Nullstellensatz, the residue field κ is a finite extension of k. Since k is perfect, κ/k is separable, so $\Omega_{\kappa/k} = 0$. Thus, by proposition 5.5.10, $\dim_\kappa(\Omega_{B/k} \otimes_B \kappa) = \dim_\kappa(\mathfrak{m}/\mathfrak{m}^2)$.

Let $K = \mathrm{Frac}(B)$; then $\Omega_{B/k} \otimes_B K \cong \Omega_{K/k}$. Because B is localized at a maximal ideal, we have $\mathrm{trdeg}(K/k) = \dim B = n$, so $\Omega_{K/k}$ has dimension n by proposition 5.5.9 (using that k is perfect).

If B is regular, then $\dim_\kappa(\mathfrak{m}/\mathfrak{m}^2) = n = \dim B$, so both the fibre and the generic fibre of $\Omega_{B/k}$ have the same dimension n. By lemma 5.5.11, $\Omega_{B/k}$ is free of rank n.

Conversely, if $\Omega_{B/k}$ is free of rank n, then $\dim_\kappa(\mathfrak{m}/\mathfrak{m}^2) = n = \dim B$, which is the definition of regularity.

We shall see by box 5.5.24 that the result for maximal points is sufficient.

5.5.13 Non-Closed Points and Arbitrary Primes If B is localized at a non-maximal prime P, it represents the local ring of a positive-dimensional subvariety (i.e., its generic point). The residue field $\kappa(P)$ is then a transcendental function field over k, meaning its differentials no longer vanish ($\Omega_{\kappa(P)/k} \neq 0$). Consequently the sequence of proposition 5.5.6 does not yield the isomorphism of proposition 5.5.10, whose hypothesis of a maximal ideal fails here.
If B is a regular local ring at such a prime, $\Omega_{B/k}$ is indeed still a free module, but its rank is $\dim B + \mathrm{trdeg}_k\kappa(P)$. Geometrically, $\dim B$ captures the codimension (the normal directions transverse to the subvariety), while the transcendence degree captures the tangent directions along the subvariety itself. Restricting to maximal ideals sets the transcendence degree to zero, isolating the local geometry of a single point.

5.5.3 The Sheaf of Differentials

The algebraic constructions of the previous subsection globalize to schemes via the diagonal ideal, following the geometric interpretation of proposition 5.5.3. Let $f\colon X \to Y$ be a morphism of schemes and $\Delta\colon X \to X \times_Y X$ the diagonal morphism. By proposition 3.5.1, Δ is a locally closed immersion: the image $\Delta(X)$ is isomorphic to a closed subscheme of an open subset $W \subseteq X \times_Y X$. (When f is separated, Δ is a closed immersion into all of $X \times_Y X$, and we may take $W = X \times_Y X$.)

Definition 5.5.14: Sheaf of Differentials

Let $f\colon X \to Y$ be a morphism of schemes and let $\mathscr{I}$ be the ideal sheaf of $\Delta(X)$ in the open subset $W \subseteq X \times_Y X$ described above. The *sheaf of relative differentials* (or *cotangent sheaf*) of X over Y is the quasi-coherent $\mathcal{O}_X$-module

$$\Omega_{X/Y} \;:=\; \Delta^*(\mathscr{I}/\mathscr{I}^2),$$

The *universal derivation* $d\colon \mathcal{O}_X \to \Omega_{X/Y}$ is defined locally by $d(b) = 1 \otimes b - b \otimes 1 \pmod{\mathscr{I}^2}$, exactly as in proposition 5.5.3.

The sheaf $\mathscr{I}/\mathscr{I}^2$ is naturally an $\mathcal{O}_{\Delta(X)}$-module, and the isomorphism $X \cong \Delta(X)$ gives it the structure of an $\mathcal{O}_X$-module. Since $\mathscr{I}$ is a quasi-coherent ideal sheaf (proposition 5.1.34), the quotient $\mathscr{I}/\mathscr{I}^2$ and its pullback are quasi-coherent (proposition 5.1.17). When Y is Noetherian and f is of finite type, $\Omega_{X/Y}$ is coherent.

Proposition 5.5.15: Differentials on Affine Opens

Let $f\colon X \to Y$ be a morphism of schemes, and let $V \subseteq Y$ and $U \subseteq f^{-1}(V)$ be affine opens with $V \cong \operatorname{Spec} A$ and $U \cong \operatorname{Spec} B$. Then

$$\Omega_{X/Y}|_U \;\cong\; \widetilde{\Omega_{B/A}}$$

compatibly with the derivations d, and in particular $\Omega_{X/Y}(U) \cong \Omega_{B/A}$.

***Proof*:**
By proposition 3.2.5(1), applied twice, and the universal property of theorem 3.2.1, $U \times_V U$ is an open subscheme of $X \times_Y X$, isomorphic to $\operatorname{Spec}(B \otimes_A B)$ by the affine case of theorem 3.2.1. As in the proof of proposition 3.5.1, $\Delta^{-1}(U \times_V U) = U$, and $\Delta|_U\colon U \to U \times_V U$ is the closed immersion given by the surjection $\mu\colon B \otimes_A B \to B$, $b \otimes b' \mapsto bb'$. The open set $W \cap (U \times_V U)$ contains $\Delta(U)$, and on it $\mathscr{I}$ restricts to the ideal sheaf of the closed subscheme $\Delta(U)$, the kernel of $\widetilde{B \otimes_A B} \to (\Delta|_U)_*\mathcal{O}_U = \widetilde{B}$ (proposition 5.1.14(6)). The tilde functor is exact (proposition 5.1.14(3)), so this kernel is $\widetilde{I}$ for $I = \ker\mu$. Then $\mathscr{I}^2$ restricts to $\widetilde{I^2}$ and, again by exactness, $\mathscr{I}/\mathscr{I}^2$ restricts to $\widetilde{I/I^2}$. The inverse image along $\Delta|_U$ depends only on the restriction to an open set containing $\Delta(U)$, so proposition 5.1.14(7) gives

$$\Omega_{X/Y}|_U = (\Delta|_U)^*\widetilde{I/I^2} \cong \widetilde{(I/I^2) \otimes_{B\otimes_A B} B} = \widetilde{I/I^2}$$

the last equality because I/I^2 is annihilated by $I = \ker\mu$. By proposition 5.5.3, I/I^2 with $d(b) = 1 \otimes b - b \otimes 1$ is $\Omega_{B/A}$ with its universal derivation. This is the computation of [36,

Remark II.8.9.2], completing the proof.

Hence the sheaf of differentials can equivalently be constructed by gluing the modules $\widetilde{\Omega_{B/A}}$ over an affine cover. Similarly, the derivations $d\colon B \to \Omega_{B/A}$ glue to the global derivation $d\colon \mathcal{O}_X \to \Omega_{X/Y}$.

The three fundamental properties of Kähler differentials (base change, the cotangent sequence, and the conormal sequence) globalize immediately.

Proposition 5.5.16: Sheaf of Differentials: Base Change

Let $f\colon X \to Y$ be a morphism, $g\colon Y' \to Y$ another morphism, and $f'\colon X' = X \times_Y Y' \to Y'$ the base change of f. Let $g'\colon X' \to X$ denote the projection. Then

$$\Omega_{X'/Y'} \cong (g')^* \Omega_{X/Y}.$$

***Proof*:**
On affine patches, this is proposition 5.5.8. The local isomorphisms are compatible on overlaps (by the uniqueness in the universal property of differentials), so they glue to a global isomorphism.

Proposition 5.5.17: Cotangent Exact Sequence

Let $X \xrightarrow{f} Y \xrightarrow{g} Z$ be morphisms of schemes. There is a natural exact sequence of $\mathcal{O}_X$-modules

$$f^*\Omega_{Y/Z} \longrightarrow \Omega_{X/Z} \longrightarrow \Omega_{X/Y} \longrightarrow 0.$$

***Proof*:**
Follows from proposition 5.5.5 on affine patches and gluing.

Proposition 5.5.18: Conormal Exact Sequence

Let $f\colon X \to Y$ be a morphism and $\iota\colon Z \hookrightarrow X$ a closed immersion with ideal sheaf $\mathscr{I}_Z \subseteq \mathcal{O}_X$. There is a natural exact sequence of $\mathcal{O}_Z$-modules

$$\mathscr{I}_Z/\mathscr{I}_Z^2 \xrightarrow{\delta} \iota^*\Omega_{X/Y} \longrightarrow \Omega_{Z/Y} \longrightarrow 0,$$

where $\delta(\overline{f}) = df \otimes 1$ for local sections f of $\mathscr{I}_Z$. The sheaf $\mathscr{I}_Z/\mathscr{I}_Z^2$ is the *conormal sheaf* of Z in X.

***Proof*:**
Follows from proposition 5.5.6 on affine patches and gluing.

The next result is specific to projective space and will be indispensable for cohomological computations throughout the rest of the book.

Theorem 5.5.19: The Euler Sequence

Let A be a ring, $Y = \operatorname{Spec} A$, and $X = \mathbb{P}^n_A$. There is an exact sequence of sheaves on X:

$$0 \longrightarrow \Omega_{X/Y} \longrightarrow \mathcal{O}_X(-1)^{\oplus(n+1)} \longrightarrow \mathcal{O}_X \longrightarrow 0.$$

In particular, $\Omega_{X/Y}$ is locally free of rank n.

Recall that $\mathcal{O}_X(-1) \cong \mathcal{O}_X(1)^\vee = \mathit{Hom}_{\mathcal{O}_X}(\mathcal{O}_X(1), \mathcal{O}_X)$ (proposition 5.3.5), so sections of $\mathcal{O}_X(-1)$ are naturally linear functionals on $\mathcal{O}_X(1)$. The map $\mathcal{O}_X(-1)^{n+1} \to \mathcal{O}_X$ is an evaluation map: in homogeneous coordinates, it sends an $(n+1)$-tuple of linear functionals to their pairing with the tautological coordinates $(x_0, \dots, x_n)$. The kernel of this evaluation (the "relations among the coordinates") is precisely the sheaf of differentials.

Dualizing presents the tangent sheaf as the quotient of $\mathcal{O}_X(1)^{\oplus(n+1)}$ by a copy of $\mathcal{O}_X$, the sheaf-theoretic form of the classical description $T_\ell\mathbb{P}(V) = \operatorname{Hom}(\ell, V/\ell)$ of the tangent space at a line $\ell \subseteq V$ to the space of lines in V, and this dual sequence is corollary 5.5.21.

***Proof*:**

Let $S = A[x_0, \dots, x_n]$ be the homogeneous coordinate ring of X, and let $E = S(-1)^{n+1}$ be the free graded S-module with basis $e_0, \dots, e_n$ in degree 1. Define a degree-0 homomorphism of graded S-modules $\varphi\colon E \to S$ by $e_i \mapsto x_i$, and let $M = \ker\varphi$. The exact sequence of graded modules

$$0 \longrightarrow M \longrightarrow E \xrightarrow{\varphi} S$$

gives rise to an exact sequence of sheaves on X:

$$0 \longrightarrow \widetilde{M} \longrightarrow \mathcal{O}_X(-1)^{n+1} \longrightarrow \mathcal{O}_X \longrightarrow 0.$$

Note that φ is not surjective as a map of graded modules (it misses degree 0), but it is surjective in all degrees ≥ 1 and hence surjective on all stalks of the associated sheaves.

It remains to show that $\widetilde{M} \cong \Omega_{X/Y}$. We construct compatible isomorphisms on each standard affine chart and verify that they glue.

Step 1: local structure of M. On the chart $U_i = D_+(x_i) \cong \operatorname{Spec} A[x_0/x_i, \dots, x_n/x_i]$, localizing at x_i makes $\varphi_{x_i}\colon E_{x_i} \to S_{x_i}$ a surjection of free S_{x_i}-modules. The kernel M_{x_i} is free of rank n, generated by $\{x_i e_j - x_j e_i \mid j \neq i\}$. Passing to the associated sheaf (and extracting degree-0 elements), $\widetilde{M}|_{U_i}$ is free over $\mathcal{O}_{U_i}$ with basis

$$\left\{ \frac{1}{x_i}\, e_j - \frac{x_j}{x_i^2}\, e_i \right\}_{j \neq i}$$

(the factor $1/x_i$ ensures these have degree 0 in M_{x_i}).

Step 2: defining the isomorphism φ_i. On U_i, the sheaf $\Omega_{X/Y}|_{U_i}$ is the free $\mathcal{O}_{U_i}$-module with basis $\{d(x_j/x_i)\}_{j \neq i}$ (example 5.5.4). Define $\varphi_i\colon \Omega_{X/Y}|_{U_i} \to \widetilde{M}|_{U_i}$ by

$$\varphi_i\left(d\left(\frac{x_j}{x_i}\right)\right) \;=\; \frac{1}{x_i^2}(x_i\, e_j - x_j\, e_i).$$

Since both sides are free of rank n and φ_i sends a basis to a basis, this is an isomorphism.

Step 3: gluing. On the overlap $U_i \cap U_j$, the chain rule gives

$$d\left(\frac{x_k}{x_i}\right) \;=\; \frac{x_k}{x_j}\,d\left(\frac{x_j}{x_i}\right) \;+\; \frac{x_j}{x_i}\,d\left(\frac{x_k}{x_j}\right).$$

Applying φ_i to the left-hand side and φ_j to the right-hand side, both yield $\frac{1}{x_i x_j}(x_j\,e_k - x_k\,e_j)$. Hence $\varphi_i|_{U_i\cap U_j} = \varphi_j|_{U_i\cap U_j}$, and the local isomorphisms glue to a global isomorphism $\Omega_{X/Y} \cong \widetilde{M}$.

The tangent form of the Euler sequence is obtained by dualizing, and dualizing is exact on sequences of locally free sheaves because such sequences split locally.

Lemma 5.5.20: Duals of Locally Split Exact Sequences

Let $(X, \mathcal{O}_X)$ be a ringed space and let

$$0 \longrightarrow \mathscr{F}' \xrightarrow{\alpha} \mathscr{F} \xrightarrow{\beta} \mathscr{F}'' \longrightarrow 0$$

be an exact sequence of $\mathcal{O}_X$-modules in which $\mathscr{F}''$ is locally free of finite rank (definition 5.1.22). Then the sequence is *locally split*: every point has an open neighbourhood U over which β admits a section $\sigma\colon \mathscr{F}''|_U \to \mathscr{F}|_U$ with $\beta\circ\sigma = \mathrm{id}$, and then $\alpha + \sigma\colon \mathscr{F}'|_U \oplus \mathscr{F}''|_U \to \mathscr{F}|_U$ is an isomorphism. Consequently, for every $\mathcal{O}_X$-module $\mathscr{G}$, the sequence

$$0 \longrightarrow \mathit{Hom}_{\mathcal{O}_X}(\mathscr{F}'', \mathscr{G}) \xrightarrow{\beta^*} \mathit{Hom}_{\mathcal{O}_X}(\mathscr{F}, \mathscr{G}) \xrightarrow{\alpha^*} \mathit{Hom}_{\mathcal{O}_X}(\mathscr{F}', \mathscr{G}) \longrightarrow 0$$

is exact. In particular, writing $\mathscr{E}^\vee := \mathit{Hom}_{\mathcal{O}_X}(\mathscr{E}, \mathcal{O}_X)$ for the dual of an $\mathcal{O}_X$-module $\mathscr{E}$ (the sheaf hom of example 5.1.3 into $\mathcal{O}_X$), the dual sequence $0 \to (\mathscr{F}'')^\vee \to \mathscr{F}^\vee \to (\mathscr{F}')^\vee \to 0$ is exact.

***Proof*:**

Let $x \in X$ and choose an open $U \ni x$ with $\mathscr{F}''|_U \cong \mathcal{O}_U^{\oplus r}$, with basis $e_1, \dots, e_r \in \mathscr{F}''(U)$. Since β_x is surjective, each $(e_k)_x$ is the germ of a section f_k of $\mathscr{F}$ near x with $\beta(f_k) = e_k$ near x, and as there are finitely many k we may shrink U so that all $f_k \in \mathscr{F}(U)$ satisfy $\beta(f_k) = e_k$. The $\mathcal{O}_U$-linear map $\sigma\colon \mathscr{F}''|_U \to \mathscr{F}|_U$ with $\sigma(e_k) = f_k$ satisfies $\beta\circ\sigma = \mathrm{id}$ on the basis, hence everywhere. On each stalk, $\alpha_x + \sigma_x$ is injective, because $\alpha_x(f') + \sigma_x(f'') = 0$ gives $f'' = \beta_x\alpha_x(f') + \beta_x\sigma_x(f'') = 0$ and then $\alpha_x(f') = 0$ forces $f' = 0$, and surjective, because $f - \sigma_x\beta_x(f) \in \ker\beta_x = \mathrm{im}\alpha_x$ for every $f \in \mathscr{F}_x$. So $\alpha + \sigma$ is an isomorphism over U.

Exactness of the *Hom* sequence may be checked over the opens U just found, and $\mathit{Hom}_{\mathcal{O}_X}(\mathscr{E}, \mathscr{G})|_U = \mathit{Hom}_{\mathcal{O}_U}(\mathscr{E}|_U, \mathscr{G}|_U)$ (example 5.1.3), so we may assume $\mathscr{F} = \mathscr{F}' \oplus \mathscr{F}''$ with α the inclusion of the first summand and β the projection onto the second. Then $\lambda \mapsto (\lambda\circ\alpha, \lambda\circ\sigma)$ identifies $\mathit{Hom}_{\mathcal{O}_X}(\mathscr{F}' \oplus \mathscr{F}'', \mathscr{G})$ with $\mathit{Hom}_{\mathcal{O}_X}(\mathscr{F}', \mathscr{G}) \oplus \mathit{Hom}_{\mathcal{O}_X}(\mathscr{F}'', \mathscr{G})$, under which β^* becomes the inclusion of the second summand and α^* the projection onto the first, and the sequence is split exact. The dual sequence is the case $\mathscr{G} = \mathcal{O}_X$.

Corollary 5.5.21: The Euler Sequence for the Tangent Sheaf

Let A be a ring, $Y = \operatorname{Spec} A$, and $X = \mathbb{P}^n_A$, and let $\mathcal{T}_{X/Y} := \Omega^\vee_{X/Y} = \mathit{Hom}_{\mathcal{O}_X}(\Omega_{X/Y}, \mathcal{O}_X)$ be the *relative tangent sheaf* of X over Y. When $A = k$ is an algebraically closed field, $\mathbb{P}^n_k$ is a nonsingular variety of dimension n (theorem 5.5.19 and proposition 5.5.25), and $\mathcal{T}_{X/Y}$ is then the tangent sheaf $\mathcal{T}_X$ of definition 5.5.28. There is an exact sequence of sheaves on X

$$0 \longrightarrow \mathcal{O}_X \longrightarrow \mathcal{O}_X(1)^{\oplus(n+1)} \longrightarrow \mathcal{T}_{X/Y} \longrightarrow 0$$

whose first map sends 1 to $(x_0, \ldots, x_n)$, and $\mathcal{T}_{X/Y}$ is locally free of rank n.

***Proof*:**
All three terms of the sequence of theorem 5.5.19 are locally free of finite rank, so lemma 5.5.20 gives an exact sequence

$$0 \longrightarrow \mathcal{O}_X^\vee \longrightarrow \left(\mathcal{O}_X(-1)^{\oplus(n+1)}\right)^\vee \longrightarrow \Omega^\vee_{X/Y} \longrightarrow 0$$

and $\Omega^\vee_{X/Y}$ is locally free of rank n because $\mathit{Hom}_{\mathcal{O}_U}(\mathcal{O}_U^{\oplus n}, \mathcal{O}_U) \cong \mathcal{O}_U^{\oplus n}$. Evaluation at 1 identifies $\mathcal{O}_X^\vee$ with $\mathcal{O}_X$, and *Hom* out of a finite direct sum is the direct sum of the *Hom*'s, so the middle term is $(\mathcal{O}_X(-1)^\vee)^{\oplus(n+1)}$. The multiplication map $\mathcal{O}_X(1) \to \mathcal{O}_X(-1)^\vee$, $g \mapsto (f \mapsto fg)$, is an isomorphism: on $U_i = D_+(x_i)$ the sheaf $\mathcal{O}_X(1)$ is free on x_i and $\mathcal{O}_X(-1)$ is free on x_i^{-1} (proposition 5.3.5), and multiplication by x_i carries x_i^{-1} to 1, so it is the dual basis element. Finally, the first map is the dual of the evaluation map $(f_0, \ldots, f_n) \mapsto \sum_i f_i x_i$ of theorem 5.5.19, so it sends $\lambda \in \mathcal{O}_X^\vee$ to the tuple of the maps $f_i \mapsto \lambda(f_i x_i)$, and for $\lambda = \mathrm{id}$, the element $1 \in \mathcal{O}_X$, the ith entry is multiplication by x_i, which is $x_i \in \mathcal{O}_X(1)$ under the identification just made.

The canonical sheaf of $\mathbb{P}^n_A$ is computed from the differentials form of the sequence in box 5.5.22.

5.5.22 Relation To Canonical Sheaf Taking the top exterior power of the Euler sequence (the determinant bundle) and applying the formula:

$$\bigwedge^{\text{top}} B \;\cong\; \left(\bigwedge^{\text{top}} A\right) \otimes \left(\bigwedge^{\text{top}} C\right)$$

yields the important formula:

$$\omega_{\mathbb{P}^n_A/A} \;:=\; \bigwedge^n \Omega_{\mathbb{P}^n_A/A} \;\cong\; \mathcal{O}_{\mathbb{P}^n_A}(-n-1),$$

which will be central to Serre duality (section 6.5).

5.5.4 Smoothness and the Cotangent Sheaf

We now focus on varieties over an algebraically closed field, where the cotangent sheaf provides a clean characterization of nonsingularity.

Definition 5.5.23: Abstract Nonsingular Variety

Let X be an abstract variety over an algebraically closed field k. Then X is *nonsingular* (or *smooth over* k) if every local ring $\mathcal{O}_{X,x}$ is a regular local ring.

5.5.24 Non-Closed Points? The definition requires regularity at *all* points, not just closed ones. However, since the local ring at a non-closed point $\mathfrak{p}$ is a localization of a local ring at a closed point (namely, $\mathcal{O}_{X,\mathfrak{p}} \cong (\mathcal{O}_{X,x})_{\mathfrak{p}}$ for a suitable closed point $x \in \overline{\{\mathfrak{p}\}}$), and localization preserves regularity (see [18]), regularity at all closed points already implies regularity everywhere. Thus the two conditions are equivalent.

Proposition 5.5.25: Nonsingular Varieties and the Cotangent Sheaf

Let X be an irreducible separated scheme of finite type over an algebraically closed field k, and let $n = \dim X$. Then $\Omega_{X/k}$ is locally free of rank n if and only if X is a nonsingular variety.

***Proof*:**
The question is local, so we may work at a closed point $x \in X$ with local ring $B = \mathcal{O}_{X,x}$ and residue field $\kappa(x) = k$ (since k is algebraically closed). The stalk $(\Omega_{X/k})_x = \Omega_{B/k}$, and by proposition 5.1.24, $\Omega_{X/k}$ is locally free of rank n at x if and only if $\Omega_{B/k}$ is a free B-module of rank n. By theorem 5.5.12, this holds if and only if B is a regular local ring of dimension n. Combining over all closed points and invoking box 5.5.24 for the non-closed points gives the result.

This characterization yields a short proof that the smooth locus is open and dense:

Corollary 5.5.26: Nonsingular Points are Open and Dense

Let X be a variety over an algebraically closed field k. Then the set of nonsingular points of X is a nonempty open subset.

***Proof*:**
Let $n = \dim X$ and let $K = K(X)$ be the function field. Since k is algebraically closed, the extension K/k is separably generated, so $\dim_K \Omega_{K/k} = \operatorname{trdeg}(K/k) = n$ by proposition 5.5.9. Now, $\Omega_{K/k}$ is the stalk of $\Omega_{X/k}$ at the generic point η of X.

Now $\Omega_{X/k}$ is locally free of rank n on a nonempty open subset U containing η. This does not follow from proposition 5.1.24, which compares local freeness with freeness of *every* stalk; the passage from the generic stalk to a neighbourhood is the following argument. Choose n elements of $(\Omega_{X/k})_\eta$ forming a K-basis and spread them out to sections over some affine open $V \ni \eta$, giving $\varphi\colon \mathcal{O}_V^n \to \Omega_{X/k}|_V$ that is an isomorphism on the stalk at η. Its kernel and cokernel are coherent, X being Noetherian, and both have zero stalk at η; a coherent sheaf has closed support, and a support not containing the generic point of the irreducible X is a proper closed subset. Removing the union of the two supports from V leaves a nonempty open $U \ni \eta$ on which φ is an

isomorphism, so $\Omega_{X/k}|_U$ is free of rank n. By proposition 5.5.25, the points of U are nonsingular.

The conormal exact sequence imposes strong constraints on nonsingular closed subvarieties. The following theorem refines proposition 5.5.18 when both ambient and subvariety are nonsingular.

Theorem 5.5.27: Nonsingular Closed Subschemes and Differentials

Let X be a nonsingular variety over an algebraically closed field k, and let $Y \subseteq X$ be an irreducible closed subscheme defined by the ideal sheaf $\mathscr{I}_Y$. Then Y is nonsingular if and only if the following two conditions hold:

1. $\Omega_{Y/k}$ is locally free.
2. The conormal sequence of proposition 5.5.18 is exact on the left:

$$0 \longrightarrow \mathscr{I}_Y/\mathscr{I}_Y^2 \longrightarrow \Omega_{X/k} \otimes \mathcal{O}_Y \longrightarrow \Omega_{Y/k} \longrightarrow 0.$$

When Y is nonsingular, $\mathscr{I}_Y$ is locally generated by $r = \operatorname{codim}_X(Y)$ elements, and the conormal sheaf $\mathscr{I}_Y/\mathscr{I}_Y^2$ is locally free of rank r on Y.

***Proof*:**
($\Leftarrow$) Assume (1) and (2) hold, and let $q = \operatorname{rank}\Omega_{Y/k}$. Since $\Omega_{X/k}$ is locally free of rank $n = \dim X$ (proposition 5.5.25), the short exact sequence in (2) forces $\mathscr{I}_Y/\mathscr{I}_Y^2$ to be locally free of rank $n-q$. By Nakayama's lemma, $\mathscr{I}_Y$ can be locally generated by $n-q$ elements, so $\dim Y \geq n-(n-q) = q$ (see [10]). On the other hand, at any closed point $y \in Y$, proposition 5.5.10 gives $q = \dim_k(\mathfrak{m}_y/\mathfrak{m}_y^2) \geq \dim \mathcal{O}_{Y,y} = \dim Y$. Hence $q = \dim Y$, so $\dim_k(\mathfrak{m}_y/\mathfrak{m}_y^2) = \dim Y$ and Y is nonsingular. Moreover, $n - q = \operatorname{codim}_X(Y)$.

($\Rightarrow$) Assume Y is nonsingular of dimension $q = \dim Y$. Then (1) is immediate from proposition 5.5.25. It remains to show that the conormal map $\delta\colon \mathscr{I}_Y/\mathscr{I}_Y^2 \to \Omega_{X/k} \otimes \mathcal{O}_Y$ is injective. Let $r = n - q = \operatorname{codim}_X(Y)$.

At a closed point $y \in Y$, the cokernel of δ is $\Omega_{Y/k}$, which is locally free of rank q. Hence $\ker(\Omega_{X/k} \otimes \mathcal{O}_Y \to \Omega_{Y/k})$ is locally free of rank r near y. Choose local sections $f_1, \ldots, f_r \in \mathscr{I}_Y$ in a neighbourhood of y such that $df_1, \ldots, df_r$ generate this kernel, and let $\mathscr{I}'$ be the ideal sheaf generated by $f_1, \ldots, f_r$ with associated closed subscheme Y'.

By construction, the images $\delta(\overline{f_1}), \ldots, \delta(\overline{f_r})$ span a free subsheaf of rank r in $\Omega_{X/k} \otimes \mathcal{O}_{Y'}$ near y. Applying the conormal sequence (proposition 5.5.18) to Y':

$$\mathscr{I}'/(\mathscr{I}')^2 \xrightarrow{\delta'} \Omega_{X/k} \otimes \mathcal{O}_{Y'} \longrightarrow \Omega_{Y'/k} \longrightarrow 0,$$

the map δ' is injective (its image is free of rank r), so by the argument of the forward direction, Y' is nonsingular of dimension q near y. Since $Y \subseteq Y'$ and both are integral of the same dimension, we must have $Y = Y'$ and $\mathscr{I}_Y = \mathscr{I}'$ locally near y. Hence δ is injective near every closed point, giving the global left-exactness.

5.5.5 Tangent Sheaf, Canonical Sheaf, and the Adjunction Formula

Definition 5.5.28: Tangent Sheaf and Canonical Sheaf

Let X be a nonsingular variety of dimension n over a field k. The *tangent sheaf* is

$$\mathcal{T}_X := \mathit{Hom}_{\mathcal{O}_X}(\Omega_{X/k}, \mathcal{O}_X),$$

a locally free sheaf of rank n. The *canonical sheaf* (or *dualizing sheaf*) is the invertible sheaf

$$\omega_X := \bigwedge^n \Omega_{X/k}.$$

For projective space, the Euler sequence (theorem 5.5.19) gives $\omega_{\mathbb{P}^n_k} \cong \mathcal{O}_{\mathbb{P}^n_k}(-n-1)$. Since this sheaf has no nonzero global sections (the degree is negative), the geometric genus of projective space is zero, a first indication that $\mathbb{P}^n$ carries no "interesting" top-degree differential forms.

In [57], we established a connection between the topology of a space and its analytical structure: the dimension of the space of globally defined holomorphic 1-forms on a compact Riemann surface of genus g is exactly g. This result is historically significant since it provides one of the first explicit algebraic characterizations of a geometric and topological invariant. Rather than simply counting the "number of holes" in a surface, this equivalence demonstrates that the macroscopic shape of the manifold dictates and constrains the algebraic structure of the differential forms that can exist upon it. In modern algebraic geometry, this perspective becomes the starting point: the geometric genus of a curve is formally defined exactly by this property, as the dimension of the space of global regular 1-forms. The abstraction allows for defining the genus of varieties over arbitrary fields: by replacing complex-analytic holomorphic forms with the purely algebraic machinery of Kähler differentials, we can formally measure the "topology" of spaces:

Definition 5.5.29: Geometric Genus

Let X be a nonsingular projective variety of dimension n over a field k. The *geometric genus* of X is

$$p_g(X) := \dim_k \Gamma(X, \omega_X) = \dim_k H^0\big(X, \bigwedge^n \Omega_{X/k}\big).$$

5.5.30 Foreshadowing: Arithmetic and Geometric Genus In [22, Hilbert Function of a Projective Variety, Degree and Arithmetic Genus], we introduced the Hilbert function and the *arithmetic genus* p_a. Serre duality (section 6.5) will show that $p_g = p_a$ for nonsingular projective *curves*, but these two invariants can differ in higher dimensions. The arithmetic genus is defined in terms of the Hilbert polynomial (definition 6.7.6), while the geometric genus measures global sections of the canonical sheaf, a more intrinsic quantity.

The following fundamental result shows that the geometric genus is a birational invariant: it cannot change under birational maps. This makes it a useful tool for showing that varieties are *not* rational.

Theorem 5.5.31: Birational Invariance of Geometric Genus

Let X and X' be nonsingular projective varieties over a field k. If X and X' are birational, then $p_g(X) = p_g(X')$.

***Proof*:**
Let $F\colon X \dashrightarrow X'$ be a birational map, and let $V \subseteq X$ be the largest open subset on which F is represented by a morphism $f\colon V \to X'$.

Step 1: the complement $X \setminus V$ has codimension ≥ 2. Let $P \in X$ be a point of codimension 1. Since X is nonsingular, $\mathcal{O}_{X,P}$ is a DVR. The birational map gives a morphism from the generic point η of X to X', i.e., a map $\operatorname{Spec} K(X) \to X'$. Since X' is projective (hence proper over k), the valuative criterion of properness (theorem 3.5.22) provides a unique extension $\operatorname{Spec} \mathcal{O}_{X,P} \to X'$, which extends to a morphism on some open neighbourhood of P. Hence $P \in V$, and every codimension-1 point lies in V.

Step 2: pullback of canonical forms. The cotangent sequence (proposition 5.5.17) gives $f^*\Omega_{X'/k} \to \Omega_{V/k}$. Since X and X' are birational, both cotangent sheaves are locally free of the same rank $n = \dim X$, and f is an isomorphism on a dense open $U \subseteq V$. Taking n-th exterior powers gives $f^*\omega_{X'} \to \omega_V$, which restricts to an isomorphism on U. This induces a map on global sections:

$$f^*\colon \Gamma(X', \omega_{X'}) \longrightarrow \Gamma(V, \omega_V).$$

A nonzero global section of an invertible sheaf on an integral scheme cannot vanish on a dense open subset (U is dense in V), so f^* is injective.

Step 3: extending sections across codimension ≥ 2. We claim that the restriction map $\Gamma(X, \omega_X) \to \Gamma(V, \omega_V)$ is an isomorphism. It suffices to check this on affine opens: for any affine open $W = \operatorname{Spec} R \subseteq X$ with $\omega_X|_W \cong \mathcal{O}_W$, the map $\Gamma(W, \mathcal{O}_W) \to \Gamma(W \cap V, \mathcal{O}_{W\cap V})$ is bijective. Since X is nonsingular, R is a normal domain, and $W \setminus (W \cap V) = W \cap (X \setminus V)$ has codimension ≥ 2 in W by Step 1. The bijectivity now follows from lemma 5.4.13: a regular function on $W \cap V$ that is bounded at all codimension-1 points extends uniquely to W.

Conclusion. Combining Steps 2 and 3, we have an injection $\Gamma(X', \omega_{X'}) \hookrightarrow \Gamma(X, \omega_X)$, so $p_g(X') \leq p_g(X)$. By symmetry (applying the same argument to the inverse birational map F^{-1}), we obtain $p_g(X) \leq p_g(X')$. Hence $p_g(X) = p_g(X')$.

The geometric genus is therefore a birational invariant: any variety with $p_g > 0$ cannot be rational (i.e., birational to $\mathbb{P}^n$), since $p_g(\mathbb{P}^n) = 0$. For example, a smooth hypersurface of degree $d \geq n+2$ in $\mathbb{P}^n$ has $p_g > 0$ and is therefore not rational. The converse fails in general: the geometric genus alone does not determine the birational equivalence class.

We close this section with the normal and conormal sheaves and the adjunction formula, which computes the canonical sheaf of a subvariety from that of the ambient variety.

Definition 5.5.32: Normal and Conormal Sheaf

Let $Y \subseteq X$ be a nonsingular closed subvariety of a nonsingular variety X over k, with ideal sheaf $\mathscr{I}_Y$. The *conormal sheaf* of Y in X is $\mathscr{I}_Y/\mathscr{I}_Y^2$, a locally free $\mathcal{O}_Y$-module of rank $r = \operatorname{codim}_X(Y)$ (theorem 5.5.27). The *normal sheaf* is its dual:

$$\mathcal{N}_{Y/X} := \mathit{Hom}_{\mathcal{O}_Y}(\mathscr{I}_Y/\mathscr{I}_Y^2,\ \mathcal{O}_Y),$$

also locally free of rank r.

Dualizing the exact sequence of theorem 5.5.27 gives the familiar normal bundle sequence from differential geometry (definition 0.5.7):

$$0 \longrightarrow \mathcal{T}_Y \longrightarrow \mathcal{T}_X \otimes \mathcal{O}_Y \longrightarrow \mathcal{N}_{Y/X} \longrightarrow 0.$$

This is the algebraic incarnation of the splitting of the tangent bundle of X along Y into tangential and normal directions.

Proposition 5.5.33: Adjunction Formula

Let $Y \subseteq X$ be a nonsingular closed subvariety of codimension r in a nonsingular variety X over k. Then

$$\omega_Y \cong \omega_X \otimes \bigwedge^r \mathcal{N}_{Y/X} \otimes \mathcal{O}_Y.$$

In the important special case $r = 1$ (i.e., Y is a divisor), let $\mathcal{L} = \mathcal{O}_X(Y)$ be the invertible sheaf associated to Y (definition 5.4.41). Then

$$\omega_Y \cong (\omega_X \otimes \mathcal{L})\big|_Y.$$

***Proof*:**
The short exact sequence from theorem 5.5.27,

$$0 \longrightarrow \mathscr{I}_Y/\mathscr{I}_Y^2 \longrightarrow \Omega_{X/k} \otimes \mathcal{O}_Y \longrightarrow \Omega_{Y/k} \longrightarrow 0,$$

involves locally free sheaves of ranks r, $n = \dim X$, and $q = \dim Y$ respectively, with $n = q + r$. Taking top exterior powers of a short exact sequence of locally free sheaves gives

$$\bigwedge^n(\Omega_{X/k} \otimes \mathcal{O}_Y) \cong \bigwedge^r(\mathscr{I}_Y/\mathscr{I}_Y^2) \otimes \bigwedge^q \Omega_{Y/k}.$$

The left-hand side is $\omega_X|_Y$, and the right-hand side is $\bigwedge^r(\mathscr{I}_Y/\mathscr{I}_Y^2) \otimes \omega_Y$. Tensoring both sides with $\bigwedge^r \mathcal{N}_{Y/X} = (\bigwedge^r(\mathscr{I}_Y/\mathscr{I}_Y^2))^\vee$ yields

$$\omega_Y \cong \omega_X|_Y \otimes \bigwedge^r \mathcal{N}_{Y/X}.$$

When $r = 1$, the conormal sheaf is $\mathscr{I}_Y/\mathscr{I}_Y^2 \cong \mathcal{L}^{-1}|_Y$ by proposition 5.4.47 (since $\mathscr{I}_Y \cong \mathcal{O}_X(-Y) = \mathcal{L}^{-1}$). Its dual is $\mathcal{N}_{Y/X} \cong \mathcal{L}|_Y$, so

$$\omega_Y \cong \omega_X|_Y \otimes \mathcal{L}|_Y = (\omega_X \otimes \mathcal{L})\big|_Y.$$

Example 5.5.34: Canonical Sheaves via the Adjunction Formula

1. *Hypersurfaces in $\mathbb{P}^n$.* Let $Y = V_+(f) \subseteq \mathbb{P}^n_k$ be a nonsingular hypersurface of degree d. Then $\mathcal{L} = \mathcal{O}_{\mathbb{P}^n}(d)$ and $\omega_{\mathbb{P}^n} = \mathcal{O}_{\mathbb{P}^n}(-n-1)$, so the adjunction formula gives
$$\omega_Y \;\cong\; \mathcal{O}_{\mathbb{P}^n}(-n-1+d)\big|_Y \;=\; \mathcal{O}_Y(d-n-1).$$
For a nonsingular plane curve of degree d in $\mathbb{P}^2$, $\omega_Y \cong \mathcal{O}_Y(d-3)$, and the geometric genus is $p_g = \dim_k H^0(Y, \mathcal{O}_Y(d-3)) = \binom{d-1}{2}$. This recovers the classical genus formula $g = (d-1)(d-2)/2$ for plane curves.

2. *Complete intersections.* More generally, if $Y = V_+(f_1) \cap \cdots \cap V_+(f_r)$ is a nonsingular complete intersection of hypersurfaces of degrees $d_1, \ldots, d_r$ in $\mathbb{P}^n_k$, iterating the adjunction formula gives
$$\omega_Y \;\cong\; \mathcal{O}_Y(d_1 + \cdots + d_r - n - 1).$$
In particular, Y is a *Fano variety* (i.e., $\omega_Y^\vee$ is ample) precisely when $\sum d_i < n+1$.

3. *Non-rational varieties in every dimension.* A nonsingular hypersurface Y of degree $d \geq n+2$ in $\mathbb{P}^n_k$ satisfies $\omega_Y \cong \mathcal{O}_Y(d-n-1)$ with $d-n-1 \geq 1$, so $p_g(Y) \geq 1$. By theorem 5.5.31, Y is not rational. Since such hypersurfaces exist in every dimension $n-1 \geq 1$, this produces non-rational varieties in all dimensions, a consequence of the interplay between differentials, the adjunction formula, and birational invariance.

5.5.6 *Invariant Differentials on Group Schemes

We saw in example 6.5.8 that the canonical sheaf (and more generally, the cotangent sheaf $\Omega_{X/k}$) carries geometric information about a variety X. However, when the scheme in question possesses a group structure, the situation becomes especially rigid. Algebraic group schemes are "everywhere locally the same," and this global symmetry forces their sheaves of differentials to be uniform across the entire space.

Let G be a group scheme over a field k (for instance, an affine algebraic group or an abelian variety). For any T-valued point $g \in G(T)$, the group operation defines a *left translation* morphism:
$$\ell_g \colon G_T \longrightarrow G_T, \qquad x \longmapsto g \cdot x.$$
Because G is a group, ℓ_g is an isomorphism of schemes (with inverse $\ell_{g^{-1}}$). This allows us to "transport" local geometric data from the identity element $e \in G(k)$ to any other point.

Definition 5.5.35: Left-Invariant Differentials

Let G be a group scheme over k, and let $\mu \colon G \times_k G \to G$ be the multiplication morphism. A global differential form $\omega \in \Gamma(G, \Omega_{G/k})$ is *left-invariant* if
$$\mu^*\omega \;=\; 1 \otimes \omega \quad \text{in} \quad \Gamma(G \times_k G,\, \Omega_{(G\times_k G)/k}).$$
(In the functor of points perspective, this means $\ell_g^*\omega = \omega$ for all points g.)

Let $\mathfrak{g} = T_eG$ be the Lie algebra of G, which is the Zariski tangent space at the identity. Its dual vector space, $\mathfrak{g}^\vee = (\mathfrak{m}_e/\mathfrak{m}_e^2)$, is the cotangent space at e. Any left-invariant differential form is

completely determined by its value at the identity, giving a natural k-linear map from the space of left-invariant differentials to $\mathfrak{g}^\vee$. In fact, this map is an isomorphism: every cotangent vector at the identity extends uniquely to a global left-invariant differential form via translation.

This process of spreading out the tangent space at the identity to the entire group scheme yields a profound structural result.

Theorem 5.5.36: Group Schemes are Parallelizable

Let G be a group scheme of finite type over a field k. Then the cotangent sheaf $\Omega_{G/k}$ is globally free. Specifically, evaluation at the identity induces an isomorphism of $\mathcal{O}_G$-modules:

$$\Omega_{G/k} \;\cong\; \mathcal{O}_G \otimes_k \mathfrak{g}^\vee.$$

***Proof*:**
The formal proof relies on the multiplication map $\mu\colon G \times_k G \to G$ and the first projection $p_1\colon G \times_k G \to G$. Consider the map $\Phi\colon G \times_k G \to G \times_k G$ defined by $(g, h) \mapsto (g, g\cdot h)$. This is an isomorphism of schemes over G (relative to the first projection), with inverse $(g, x) \mapsto (g, g^{-1}\cdot x)$.

The isomorphism Φ transforms the second projection $p_2\colon G \times_k G \to G$ into the multiplication map μ. Therefore, it must induce an isomorphism on the pullbacks of the cotangent sheaf:

$$\Phi^* p_2^* \Omega_{G/k} \;\cong\; \mu^* \Omega_{G/k}.$$

Restricting this isomorphism to the slice $G \times \{e\}$ (i.e., pulling back along $\mathrm{id}_G \times e\colon G \to G \times_k G$) trivializes the differential components from the base, leaving an isomorphism between $\Omega_{G/k}$ and the pullback of the fiber at e, which is precisely $\mathcal{O}_G \otimes_k \mathfrak{g}^\vee$.

Corollary 5.5.37: Triviality of the Canonical Sheaf for Groups

If G is a nonsingular group scheme over k of dimension n, then its tangent sheaf $\mathcal{T}_G$ and its canonical sheaf ω_G are both trivial:

$$\mathcal{T}_G \;\cong\; \mathcal{O}_G^{\oplus n}, \qquad \text{and} \qquad \omega_G \;\cong\; \mathcal{O}_G.$$

This corollary provides a massive constraint on which varieties can be endowed with a group structure. For example, among all projective curves, only those of geometric genus $p_g = 1$ (elliptic curves) have a trivial canonical sheaf. Thus, it is geometrically impossible to put a group scheme structure on the projective line $\mathbb{P}^1_k$ (where $\omega \cong \mathcal{O}(-2)$) or on a nonsingular curve of genus $g \geq 2$. Similarly, an abelian variety of dimension g must have $\omega_A \cong \mathcal{O}_A$, which immediately distinguishes it from most other projective varieties of the same dimension and restricts its intersection theory.

5.6⋆ *Formal Schemes and Local Geometry

⋆ *Optional. This section is an aside; nothing later in the book depends on it.*

One feature distinguishing the theory of schemes from the older classical theory of varieties is the possibility of having nilpotent elements in the structure sheaf. In particular, if $Y \subseteq X$ is a closed

subvariety defined by a sheaf of ideals $\mathcal{I}$, then for any $n \geq 1$ we can consider the closed subscheme Y_n defined by the n-th power $\mathcal{I}^n$. For $n \geq 2$, this is a scheme with nilpotent elements. It carries information about Y together with the infinitesimal properties of the embedding of Y in X.

We wish to construct an object which carries information about all the infinitesimal neighborhoods Y_n of Y simultaneously. Such an object would be "thicker" than any individual Y_n, but contained inside any actual open neighborhood of Y in X. We might naturally call this the formal neighborhood of Y in X.

To see why a new geometric construction is necessary to capture this, recall that the algebraic construction of the formal power series ring $k[[x]]$ is given by the limit of the truncated polynomial rings $k[x]/(x^n)$. We shall now explore the geometric consequences of this algebraic operation.

Topologically, the scheme $\operatorname{Spec}(k[x]/(x^n))$ consists of exactly one point, as the ring is Artinian and possesses a unique prime ideal (x). Geometrically, one views this as a "fat point" – a point equipped with infinitesimal nilpotents that remember $n-1$ orders of derivatives. Since $k[[x]]$ is the algebraic limit of these rings as $n \to \infty$, one might intuitively expect the geometric limit of a sequence of one-point spaces to remain a one-point space. However, passing to the infinite limit crosses a structural threshold: the limit of infinite nilpotents coalesces into a genuine coordinate variable, making $k[[x]]$ a domain of Krull dimension 1. Consequently, the affine scheme $\operatorname{Spec}(k[[x]])$ possesses *two* points. One should visualize $\operatorname{Spec}(k[[x]])$ as the "germ" of a curve, an unimaginably small neighborhood that has lost all global properties but is just large enough to have a center and an infinitesimal edge.

Formal Completions

The fact that the Spec functor transforms a limit of one-point spaces into a two-point space demonstrates that Spec does not commute nicely with infinite limits. To rectify this topological mismatch, Grothendieck introduced formal schemes. We shall follow Hartshorne and define this globally as a completion along a closed subscheme.

Definition 5.6.1: Formal Completion

Let X be a Noetherian scheme, and let Y be a closed subscheme defined by a sheaf of ideals $\mathcal{I}$. Then we define the *formal completion* of X along Y, denoted $(\hat{X}, \mathcal{O}_{\hat{X}})$, to be the following ringed space: we take the topological space Y, and on it the sheaf of rings $\mathcal{O}_{\hat{X}} = \varprojlim \mathcal{O}_X/\mathcal{I}^n$. Here we consider each $\mathcal{O}_X/\mathcal{I}^n$ as a sheaf of rings on Y, and make them into an inverse system in the natural way.

The reader should notice that the structure sheaf $\mathcal{O}_{\hat{X}}$ depends only on the closed subset Y, and not on the particular scheme structure on Y. For if $\mathcal{J}$ is another sheaf of ideals defining a closed subscheme structure on Y, then since X is a Noetherian scheme, there are integers m, n such that $\mathcal{I} \supseteq \mathcal{J}^m$ and $\mathcal{J} \supseteq \mathcal{I}^n$. Thus the inverse systems $(\mathcal{O}_X/\mathcal{I}^n)$ and $(\mathcal{O}_X/\mathcal{J}^m)$ are cofinal with each other, and hence have the same inverse limit.

Furthermore, one sees easily that the stalks of the sheaf $\mathcal{O}_{\hat{X}}$ are local rings, so in fact $(\hat{X}, \mathcal{O}_{\hat{X}})$ is a locally ringed space. If $U = \operatorname{Spec} A$ is an open affine subset of X, and if $I \subseteq A$ is the ideal $\Gamma(U, \mathcal{I})$, then evaluating the sections reveals that $\Gamma(\hat{X} \cap U, \mathcal{O}_{\hat{X}}) = \hat{A}$, the I-adic completion of A. Thus the process of completing X along Y is strictly analogous to the I-adic completion of a ring. However, one should note that the local rings of $\hat{X}$ are in general *not* complete, and their dimension (which is

equal to $\dim X$) is *not* equal to the dimension of the underlying topological space Y.

We shall call a formal scheme obtained by completing a single Noetherian scheme X along a closed subscheme Y *algebraizable*.[8]

Example 5.6.2: Formal Completions

1. Take X to be a Noetherian scheme, and let $Y = X$. Then $\mathcal{I} = 0$, so $\mathcal{O}_X/\mathcal{I}^n = \mathcal{O}_X$ for every n and $\hat{X} = X$. Thus the category of Noetherian formal schemes naturally includes all Noetherian schemes.
2. Take X to be a Noetherian scheme, and let $Y = \{P\}$ be a closed point. Then $\hat{X}$ is a one point space $\{P\}$ with the completion $\hat{\mathcal{O}}_P$ of the local ring at P as its structure sheaf. An $\hat{\mathcal{O}}_P$-module M, considered as a sheaf on $\hat{X}$, is coherent if and only if M is a finitely generated module. This directly resolves our earlier algebraic paradox: $\hat{X}$ is topologically a single point, capturing the geometric intuition of a true limit of fat points without accidentally spawning a generic point.

5.6.3 Comparing Local Geometries To summarize the distinctions between these deeply related objects for a point P on a curve:

1. $\operatorname{Spec}(k[x]/(x^n))$ is a single point representing a "fat point" of finite thickness.
2. $\hat{X}$ (the formal completion at P) is a single point representing the true geometric limit of fat points (infinite formal thickness).
3. $\operatorname{Spec}(k[[x]])$ is a two-point space representing a "formal disk" or germ (a center point plus a surrounding generic neighborhood).

Universal Smoothness and Singularities

The utility of formal schemes shines when studying the local properties of varieties. In classical algebraic geometry, curves can exhibit wildly different global shapes, such as a line $y = x$ or a parabola $y = x^2$. However, if one selects a smooth point on any such curve over a field k and takes the formal completion of the local ring at that point, the resulting ring is always isomorphic to $k[[t]]$, where t is a local parameter.

Because their local algebra is identical, their formal geometry is identical. Thus, $\operatorname{Spec}(k[[x]])$ does not represent *a* curve; it represents the fundamental atomic structure of *any* smooth curve at a smooth point. Under an infinite-power formal microscope, every smooth curve is indistinguishable from a straight line.

This universal behavior fails precisely when the geometry ceases to be smooth.

[8] It is apparently quite difficult to find non-algebraizable formal schemes, maybe add an exercise based on Hartshorne p.194 later?

Example 5.6.4: Singularities and Formal Completions

If a curve possesses a singularity, its formal completion at that point acts as a perfect algebraic memory of the singularity, remaining distinct from $k[[x]]$.

1. *The Cusp:* Consider the curve cut out by $y^2 = x^3$, which has a sharp cusp at the origin. The formal completion of the local ring at the origin is

$$\frac{k[[x, y]]}{(y^2 - x^3)}$$

   This ring is not isomorphic to $k[[t]]$. Even infinitesimally, the formal scheme remembers the sharp turn.

2. *The Node:* Consider a curve with a self-intersection at the origin, locally modeled by $xy = 0$. Its formal completion at the crossing is

$$\frac{k[[x, y]]}{(xy)}$$

   This represents two distinct microscopic branches crossing each other. Topologically, the formal completion is still a single point, but the underlying sheaf of topological rings completely captures the crossing branches.

6

Cohomology of Schemes

Let X be a scheme with a sheaf $\mathscr{F}$. A natural question is: can we quantify the obstruction to gluing local sections into global sections? On an affine scheme, there is no obstruction: the global section functor is exact for quasi-coherent sheaves, and all local data glues. But on a projective variety of positive dimension, or on a scheme built by gluing affine pieces in a non-trivial way, the functor $\Gamma(X, -)$ fails to be exact, and its failure is measured by the *sheaf cohomology groups* $H^i(X, \mathscr{F}) = R^i\Gamma(X, \mathscr{F})$, the right derived functors of the global section functor. These groups are the central objects of study in this chapter.

The sheaves to which this machinery applies include the structure sheaf $\mathcal{O}_X$, the twisted sheaves $\mathcal{O}_X(n)$ when X is projective, the sheaves of differentials $\Omega_{X/S}$ of a morphism $X \to S$ and their exterior powers, and the pushforwards of sheaves along closed immersions. Each choice of sheaf extracts different geometric information: the cohomology of $\mathcal{O}_X$ governs the arithmetic genus, the cohomology of line bundles governs linear systems and the Riemann–Roch theorem, and the cohomology of the canonical sheaf ω_X encodes duality phenomena that constrain the topology of the variety.

Though the derived-functor definition is theoretically clean, it is computationally impractical: it requires an injective resolution, and injective sheaves are typically enormous and non-explicit. For practical computations, we introduce *Čech cohomology*, which replaces injective resolutions with the combinatorial data of an open cover. For quasi-coherent sheaves on separated Noetherian schemes with an affine cover, Čech cohomology and derived-functor cohomology coincide, and this is our main computational tool. Using it, we compute the cohomology of projective space $\mathbb{P}^n_R$ with coefficients in the twisted sheaves $\mathcal{O}(d)$ — the single most important explicit computation in the subject.

With these foundations, the chapter develops in four main directions.

Vanishing and finiteness. Two structural results govern the cohomology of schemes: the higher cohomology of quasi-coherent sheaves vanishes on affine Noetherian schemes (section 6.2), and the cohomology of coherent sheaves on projective schemes over a field is finite-dimensional (section 6.6). The first confirms the philosophy that cohomology measures gluing; the second ensures that the

invariants we extract are "finitely complicated." Together, they justify defining the Euler characteristic $\chi(X, \mathscr{F})$, the Hilbert polynomial, and the arithmetic genus (section 6.7).

Duality. Serre duality (section 6.5) is the algebraic analogue of Poincaré duality: for a smooth projective variety X of dimension n over a field k, the cohomology groups satisfy $H^i(X, \mathscr{F}) \cong H^{n-i}(X, \mathscr{F}^\vee \otimes \omega_X)^*$ for locally free $\mathscr{F}$ of finite rank. The proof passes through the Ext groups and the local-to-global spectral sequence (section 6.4), which provide the natural framework for the duality pairing. As a consequence, we obtain the Riemann–Roch theorem for curves and the classical genus formulas.

Families and variation. The higher direct image sheaves $R^i f_* \mathscr{F}$ (section 6.6) relativize cohomology over a base scheme. For a proper morphism $f\colon X \to Y$ of Noetherian schemes and a coherent sheaf $\mathscr{F}$ on X flat over Y, the semicontinuity theorem (section 6.8) shows that the fiber cohomology $h^i(X_y, \mathscr{F}_y)$ is upper semicontinuous in y, and Grauert's theorem guarantees that, when Y is reduced, constancy of h^i implies local freeness of $R^i f_* \mathscr{F}$ and compatibility with base change. These results are the technical heart of moduli theory.

GAGA. For projective varieties over $\mathbb{C}$, Serre's GAGA theorems (section 6.9) identify the algebraic cohomology developed in this chapter with the analytic cohomology of the complex geometry: the analytification functor induces an equivalence $\mathbf{Coh}(X) \simeq \mathbf{Coh}(X^{\mathrm{an}})$ and isomorphisms $H^i(X, \mathscr{F}) \cong H^i(X^{\mathrm{an}}, \mathscr{F}^{\mathrm{an}})$. Through GAGA, the Hodge decomposition and the de Rham theorem pass from the analytic theory to algebraic cohomology, and Chow's theorem follows, closing the loop between the algebraic and analytic perspectives on cohomology.

Prerequisites. The reader should be comfortable with the theory of sheaves and schemes (chapters 1–3), quasi-coherent and coherent sheaves (chapter 5), and the basic properties of projective schemes and differentials (sections 2.3 and 5.5). For the GAGA section, familiarity with the complex geometry review (section 0.6) is assumed. The reader may also want to review [18] for the derived-functor approach to (co)homology theory.

6.1 Definition and Basic Properties

Recall that a category $\mathbf{C}$ is *pre-additive* (or enriched over $\mathbf{Ab}$) if for every $A, B \in \mathrm{Obj}(\mathbf{C})$, the set $\mathrm{Hom}_{\mathbf{C}}(A, B)$ carries the structure of an abelian group, and composition of morphisms is bilinear:

$$f \circ (g + h) = (f \circ g) + (f \circ h) \qquad \text{and} \qquad (f + g) \circ h = (f \circ h) + (g \circ h).$$

An *additive category* is a pre-additive category that admits all finite products and coproducts, and in which these coincide (this common construction is sometimes called the *biproduct*).

An abelian category is an additive category with the "right" conditions to do cohomology[1]. Concretely (cf. definition 0.4.29), a category $\mathbf{A}$ is *abelian* if:

1. for every $A, B \in \mathrm{Obj}(\mathbf{A})$, the set $\mathrm{Hom}_{\mathbf{A}}(A, B)$ is an abelian group and composition is bilinear;
2. finite products and coproducts exist and coincide;
3. every morphism $f \in \mathrm{Hom}_{\mathbf{A}}(A, B)$ has a kernel and a cokernel;

[1] Technically, one can do cohomology on exact categories that are not abelian, such as the category of vector bundles, but it is more natural to see this as a special generalization of the abelian setting.

4. every monomorphism is the kernel of its cokernel, and every epimorphism is the cokernel of its kernel;

5. defining $\mathrm{im}(f) = \ker(\mathrm{coker}(f))$, the first isomorphism theorem gives

$$\mathrm{im}(f) = \ker(\mathrm{coker}(f)) = \mathrm{coker}(\ker(f)),$$

so every morphism factors as an epimorphism followed by a monomorphism.

The prototypical abelian category is R-**Mod**. By the Freyd-Mitchell embedding theorem, any small abelian category $\mathbf{A}$ fully-faithfully embeds into R-**Mod** via an exact functor for an appropriate (not necessarily commutative) ring R. This is useful theoretically, but for our purposes it is more practical to think of the abelian category of sheaves $\mathbf{Sh}_{\mathbf{Ab}}(X)$ as its own creature. The "sheafification" of R-**Mod** gives $\mathcal{O}_X$-**Mod**. The important full subcategories $\mathbf{QCoh}(X)$ and, for X locally Noetherian, $\mathbf{Coh}(X)$ are both abelian (proposition 5.1.17(1) and (2)). Though not critical at the moment, it should be noted that $\mathbf{QCoh}(X)$ is not closed under the internal Hom sheaf *Hom*, while $\mathbf{Coh}(X)$ is.

A functor between abelian categories $F : \mathbf{A} \to \mathbf{B}$ is *additive* if for every $X, Y \in \mathrm{Obj}(\mathbf{A})$, the induced map

$$\mathrm{Hom}_{\mathbf{A}}(X, Y) \longrightarrow \mathrm{Hom}_{\mathbf{B}}(FX, FY)$$

is a group homomorphism. An additive functor F is *left exact* if it preserves kernels, equivalently if a short exact sequence $0 \to X \to Y \to Z \to 0$ yields an exact sequence

$$0 \longrightarrow FX \longrightarrow FY \longrightarrow FZ,$$

and *right exact* if it preserves cokernels, equivalently if the same short exact sequence yields

$$FX \longrightarrow FY \longrightarrow FZ \longrightarrow 0.$$

If F is both left and right exact, it is called *exact*. The most common (and arguably the most important) left-exact functors are the covariant $\mathrm{Hom}(X, -)$ and the contravariant $\mathrm{Hom}(-, Y)$. The most important right-exact functor is $- \otimes Y$[2].

Given a left-exact functor $F : \mathbf{A} \to \mathbf{B}$, we can form its *right derived functors* $R^iF : \mathbf{A} \to \mathbf{B}$ for $i \geq 0$. Abstractly, one passes from $\mathbf{A}$ to its bounded-below derived category $D^+(\mathbf{A})$ — constructed by taking the category of bounded-below cochain complexes $\mathrm{Kom}^+(\mathbf{A})$, passing to the homotopy category $K^+(\mathbf{A})$, and then localizing at quasi-isomorphisms — and the total derived functor $\mathbf{R}F : D^+(\mathbf{A}) \to D^+(\mathbf{B})$ captures the right information to "fix" exactness. The derived category does not lend itself well to direct computation, but if $\mathbf{A}$ has enough injectives, then injective resolutions are universal up to homotopy, and the right derived functors R^iF can be computed as follows: given an object $A \in \mathbf{A}$, choose an injective resolution $0 \to A \to I^0 \to I^1 \to \cdots$, apply F, and take cohomology:

$$R^iF(A) = h^i(F(I^\bullet)).$$

This is independent of the choice of injective resolution (up to canonical isomorphism). The details of these constructions can be found in [18, Total Derived Functor], and a "classical" approach using δ-functors in [18, Universal δ-Functor].

Recall that if R is a ring, then every R-module embeds into an injective module (see [18]). From this, we shall be able to conclude that the module categories over ringed spaces have enough injectives:

[2]The functor $X \otimes -$ is also right exact and is left adjoint to $\mathrm{Hom}(X, -)$; this adjunction is the tensor–Hom adjunction.

Proposition 6.1.1: Ringed Space Has Enough Injectives

Let $(X, \mathcal{O}_X)$ be a ringed space. Then the category $\mathcal{O}_X$-**Mod** of sheaves of $\mathcal{O}_X$-modules has enough injectives.

***Proof*:**
Let $\mathscr{F}$ be a sheaf of $\mathcal{O}_X$-modules. For each $x \in X$, the stalk $\mathscr{F}_x$ is an $\mathcal{O}_{X,x}$-module, so there exists an injective $\mathcal{O}_{X,x}$-module I_x together with an embedding $\mathscr{F}_x \hookrightarrow I_x$. For each point $x \in X$, let $j_x : \{x\} \hookrightarrow X$ denote the inclusion of the one-point space $\{x\}$, and equip $\{x\}$ with the $\mathcal{O}_{X,x}$-module I_x (viewed as a sheaf on a point). Define:

$$\mathscr{I} = \prod_{x \in X} (j_x)_*(I_x).$$

We claim that $\mathscr{I}$ is an injective $\mathcal{O}_X$-module and that $\mathscr{F}$ embeds into it.

The embedding. For each $x \in X$, the stalk of $(j_x)_*(I_x)$ at x is I_x, and the embedding $\mathscr{F}_x \hookrightarrow I_x$ extends (by the adjunction below) to a sheaf map $\mathscr{F} \to (j_x)_*(I_x)$. Taking the product over all x yields a morphism $\mathscr{F} \to \mathscr{I}$ that is injective on every stalk, hence injective as a morphism of sheaves (by theorem 1.2.27).

Injectivity of $\mathscr{I}$. For any $\mathcal{O}_X$-module $\mathscr{G}$, we have:

$$\operatorname{Hom}_{\mathcal{O}_X}(\mathscr{G}, \mathscr{I}) \cong \prod_{x \in X} \operatorname{Hom}_{\mathcal{O}_X}\big(\mathscr{G}, (j_x)_*(I_x)\big) \cong \prod_{x \in X} \operatorname{Hom}_{\mathcal{O}_{X,x}}(\mathscr{G}_x, I_x).$$

The first isomorphism is the universal property of products. The second uses the fact that for the inclusion $j_x : \{x\} \hookrightarrow X$, the stalk functor $\mathscr{G} \mapsto \mathscr{G}_x$ is left adjoint to $(j_x)_*$[a].

Now, the functor $\operatorname{Hom}_{\mathcal{O}_X}(-, \mathscr{I})$ is the product of functors $\operatorname{Hom}_{\mathcal{O}_{X,x}}((-)_x, I_x)$. Each factor is the composition of the stalk functor $\mathscr{G} \mapsto \mathscr{G}_x$, which is exact (by proposition 1.2.22), with the functor $\operatorname{Hom}_{\mathcal{O}_{X,x}}(-, I_x)$, which is exact because I_x is injective. A product of exact functors is exact, so $\operatorname{Hom}_{\mathcal{O}_X}(-, \mathscr{I})$ is exact, making $\mathscr{I}$ injective.

[a] More precisely, a morphism $\mathscr{G} \to (j_x)_*(I_x)$ is determined by its value on stalks at x, since $(j_x)_*(I_x)$ is supported at $\overline{\{x\}}$ and its stalk at x is I_x. The formal adjunction is $\operatorname{Hom}_{\mathcal{O}_X}(\mathscr{G}, (j_x)_*M) \cong \operatorname{Hom}_{\mathcal{O}_{X,x}}(\mathscr{G}_x, M)$ for any $\mathcal{O}_{X,x}$-module M.

Corollary 6.1.2: Sheaves of Abelian Groups Has Enough Injectives

Let X be a topological space. Then the category of sheaves of abelian groups on X has enough injectives.

***Proof*:**
View X as the ringed space $(X, \underline{\mathbb{Z}})$, where $\underline{\mathbb{Z}}$ is the constant sheaf. Sheaves of abelian groups on X are precisely $\underline{\mathbb{Z}}$-modules, so the result follows from proposition 6.1.1.

Definition 6.1.3: Global Section Functor and Sheaf Cohomology

Let $(X, \mathcal{O}_X)$ be a ringed space. The *global section functor*

$$\Gamma(X, -) : \mathcal{O}_X\textbf{-Mod} \longrightarrow \textbf{Ab}$$

is left exact (by proposition 1.2.16) but not right exact in general (see Example 1.2.7). Since $\mathcal{O}_X$-**Mod** has enough injectives, we may form the right derived functors. The i-th right derived functor

$$H^i(X, -) = R^i\Gamma(X, -)$$

is called the i-th *sheaf cohomology functor*. For a sheaf $\mathscr{F}$, the abelian group $H^i(X, \mathscr{F})$ is the i-th *cohomology group* of $\mathscr{F}$. By general properties of derived functors, $H^0(X, \mathscr{F}) = \Gamma(X, \mathscr{F})$ and a short exact sequence $0 \to \mathscr{F}' \to \mathscr{F} \to \mathscr{F}'' \to 0$ of sheaves induces a long exact sequence:

$$0 \to H^0(X, \mathscr{F}') \to H^0(X, \mathscr{F}) \to H^0(X, \mathscr{F}'') \xrightarrow{\delta} H^1(X, \mathscr{F}') \to H^1(X, \mathscr{F}) \to \cdots$$

To compute these cohomology groups, we need resolutions. Recall that it suffices to resolve by *acyclic* objects — those $\mathscr{G}$ with $H^i(X, \mathscr{G}) = 0$ for $i > 0$ — rather than by injective objects (see [18]). We shall show that the following class of sheaves is acyclic:

Definition 6.1.4: Flasque Sheaf

Let X be a topological space. A sheaf $\mathscr{F}$ on X is called *flasque* (or *flabby*) if for every inclusion of open sets $U \subseteq V$, the restriction map $\mathrm{res}_{V,U} : \mathscr{F}(V) \to \mathscr{F}(U)$ is surjective.

Proposition 6.1.5: Injective Sheaf Modules Are Flasque

Let $(X, \mathcal{O}_X)$ be a ringed space. Then any injective $\mathcal{O}_X$-module is flasque.

***Proof*:**
Let $\mathscr{I}$ be an injective $\mathcal{O}_X$-module, and let $U \subseteq V$ be open subsets of X. For any open set $W \subseteq X$, let $j_W : W \hookrightarrow X$ denote the inclusion and define $\mathcal{O}_W = (j_W)_!(\mathcal{O}_X|_W)$, the extension by zero of $\mathcal{O}_X|_W$ to all of X[a]. The inclusion $U \subseteq V$ induces a natural monomorphism of $\mathcal{O}_X$-modules $0 \to \mathcal{O}_U \to \mathcal{O}_V$. Since $\mathscr{I}$ is injective, the functor $\mathrm{Hom}_{\mathcal{O}_X}(-, \mathscr{I})$ is exact, giving a surjection:

$$\mathrm{Hom}_{\mathcal{O}_X}(\mathcal{O}_V, \mathscr{I}) \longrightarrow \mathrm{Hom}_{\mathcal{O}_X}(\mathcal{O}_U, \mathscr{I}) \longrightarrow 0.$$

But $\mathrm{Hom}_{\mathcal{O}_X}(\mathcal{O}_W, \mathscr{I}) \cong \mathscr{I}(W)$ for any open W, since a morphism $\mathcal{O}_W \to \mathscr{I}$ is determined by the image of the identity section $1 \in \mathcal{O}_X(W)$. Hence the restriction $\mathscr{I}(V) \to \mathscr{I}(U)$ is surjective, so $\mathscr{I}$ is flasque.

[a] Recall from definition 1.3.11 that $(j_W)_!$ extends a sheaf by zero outside W. In particular, $(j_W)_!(\mathcal{O}_X|_W)(V') = \{s \in \mathcal{O}_X(V' \cap W) \,|\, \mathrm{supp}(s) \text{ is closed in } V'\}$ for open $V' \subseteq X$.

Before proving that flasque sheaves are acyclic, we need one key property: the global section functor is *exact* on short exact sequences of flasque sheaves. Let us record this precisely.

Lemma 6.1.6: Exactness of Global Sections on Flasque Sequences

Let $0 \to \mathscr{F}' \to \mathscr{F} \to \mathscr{F}'' \to 0$ be a short exact sequence of sheaves on a topological space X. If $\mathscr{F}'$ is flasque, then the sequence

$$0 \longrightarrow \Gamma(X, \mathscr{F}') \longrightarrow \Gamma(X, \mathscr{F}) \longrightarrow \Gamma(X, \mathscr{F}'') \longrightarrow 0$$

is exact. Furthermore, if both $\mathscr{F}'$ and $\mathscr{F}$ are flasque, then $\mathscr{F}''$ is also flasque.

***Proof*:**
Left exactness of $\Gamma(X, -)$ gives exactness at $\Gamma(X, \mathscr{F}')$ and $\Gamma(X, \mathscr{F})$. We must show surjectivity on the right. Let $s'' \in \Gamma(X, \mathscr{F}'')$. Consider the collection of pairs (U, s) where $U \subseteq X$ is open and $s \in \mathscr{F}(U)$ maps to $s''|_U$ under the surjection $\mathscr{F} \to \mathscr{F}''$. This collection is non-empty (by surjectivity on stalks, we can find such pairs locally) and is partially ordered by extension. By Zorn's lemma, there is a maximal such pair (U_0, s_0).

We claim $U_0 = X$. Suppose not; pick $x \in X \setminus U_0$. By surjectivity on stalks, there is an open $V \ni x$ and $t \in \mathscr{F}(V)$ mapping to $s''|_V$. On $U_0 \cap V$, the difference $s_0|_{U_0 \cap V} - t|_{U_0 \cap V}$ maps to zero in $\mathscr{F}''(U_0 \cap V)$, so it lies in $\mathscr{F}'(U_0 \cap V)$. Since $\mathscr{F}'$ is flasque, this section extends to some $\sigma \in \mathscr{F}'(V)$. Replacing t by $t + \sigma$, we may assume s_0 and t agree on $U_0 \cap V$, and they glue to a section on $U_0 \cup V$, contradicting maximality. Hence $U_0 = X$.

For the second claim, suppose $\mathscr{F}'$ and $\mathscr{F}$ are both flasque. Let $U \subseteq V$ be open sets and $s'' \in \mathscr{F}''(V)$. By the surjectivity just proved (applied to V), there exists $s \in \mathscr{F}(V)$ mapping to s''. Since $\mathscr{F}$ is flasque, the restriction $\mathscr{F}(V) \to \mathscr{F}(U)$ is surjective, so the image of $s|_U$ in $\mathscr{F}''(U)$ equals $s''|_U$. But we already have the section s defined on all of V, and any element of $\mathscr{F}''(U)$ can be lifted to $\mathscr{F}(V)$ and then restricted. More precisely, given any $t'' \in \mathscr{F}''(U)$, apply the surjectivity result to U to get a lift $t \in \mathscr{F}(U)$, then use the flasqueness of $\mathscr{F}$ to extend t to all of V, giving a preimage in $\mathscr{F}''(V)$. Hence $\mathscr{F}''$ is flasque.

Corollary 6.1.7: Trivial Cohomology For Flasque Sheaves

Let $\mathscr{F}$ be a flasque sheaf on a topological space X. Then

$$H^i(X, \mathscr{F}) = 0 \qquad \text{for all } i > 0.$$

***Proof*:**
Since $\mathcal{O}_X$-**Mod** has enough injectives, we can embed $\mathscr{F}$ into an injective sheaf $\mathscr{I}$ and form the short exact sequence:

$$0 \longrightarrow \mathscr{F} \longrightarrow \mathscr{I} \longrightarrow \mathscr{G} \longrightarrow 0$$

where $\mathscr{G} = \mathscr{I}/\mathscr{F}$. By proposition 6.1.5, the sheaf $\mathscr{I}$ is flasque. Since both $\mathscr{F}$ and $\mathscr{I}$ are flasque, the quotient $\mathscr{G}$ is also flasque by the lemma above.

Since $\mathscr{F}$ is flasque, the global section functor preserves the exactness of this sequence (again by the lemma above):

$$0 \longrightarrow \Gamma(X, \mathscr{F}) \longrightarrow \Gamma(X, \mathscr{I}) \longrightarrow \Gamma(X, \mathscr{G}) \longrightarrow 0.$$

Comparing with the long exact sequence in cohomology and using $H^0 = \Gamma$, we see that the connecting homomorphism $\delta : H^0(X, \mathscr{G}) \to H^1(X, \mathscr{F})$ is zero (since the map $\Gamma(X, \mathscr{I}) \to \Gamma(X, \mathscr{G})$

is already surjective). Hence $H^1(X, \mathscr{F}) = 0$.

For $i \geq 2$, the long exact sequence gives isomorphisms $H^i(X, \mathscr{F}) \cong H^{i-1}(X, \mathscr{G})$ because $\mathscr{I}$ is injective and therefore $H^j(X, \mathscr{I}) = 0$ for all $j > 0$. Since $\mathscr{G}$ is flasque, we may repeat the argument with $\mathscr{G}$ in place of $\mathscr{F}$, obtaining $H^{i-1}(X, \mathscr{G}) = 0$ by induction on i. Hence $H^i(X, \mathscr{F}) = 0$ for all $i > 0$.

Flasque sheaves are therefore acyclic for $\Gamma(X, -)$, and we may compute cohomology using flasque resolutions rather than injective ones. Let us see some examples:

Example 6.1.8: Flasque Sheaves

1. It was just shown that all injective $\mathcal{O}_X$-modules are flasque. The converse fails: flasque sheaves need not be injective.

2. *Constant sheaves on irreducible spaces.* Let X be an irreducible topological space and A an abelian group. Then the constant sheaf $\underline{A}$ is flasque. Indeed, for any non-empty open $U \subseteq V$, both U and V are connected (every open subset of an irreducible space is connected), so $\underline{A}(V) = A = \underline{A}(U)$ and the restriction map is the identity. This need not hold on reducible spaces: on $X = \mathbb{R}$, take $U = (0, 1) \cup (2, 3)$. The section $(1, 0) \in \underline{\mathbb{Z}}(U) = \mathbb{Z} \times \mathbb{Z}$ is not the restriction of any element of $\underline{\mathbb{Z}}(\mathbb{R}) = \mathbb{Z}$, so $\underline{\mathbb{Z}}$ is not flasque on $\mathbb{R}$. The key point is that irreducibility forces all opens to overlap, making extension trivial.

3. *The Godement resolution.* Let $\mathscr{F}$ be any sheaf of abelian groups on X. We can always construct a canonical flasque resolution called the *Godement resolution*. For each $x \in X$, let $\iota_x : \{x\} \hookrightarrow X$ denote the inclusion. Define:

$$C^0(\mathscr{F}) = \prod_{x \in X} (\iota_x)_*(\mathscr{F}_x),$$

   so that $\Gamma(U, C^0(\mathscr{F})) = \prod_{x \in U} \mathscr{F}_x$ for every open $U \subseteq X$. This sheaf is flasque because a section over U is simply a choice of element in each stalk $\mathscr{F}_x$ for $x \in U$, with no compatibility condition — so restriction just forgets the stalks outside U, which is always surjective.

   Note, however, that $C^0(\mathscr{F})$ need *not* be injective. For example, if $X \neq \emptyset$ and $\mathscr{F} = \underline{\mathbb{Z}}$, then $\Gamma(X, C^0(\mathscr{F})) = \prod_{x \in X} \mathbb{Z}$ is not a divisible group, so $C^0(\mathscr{F})$ is not injective in the category of sheaves of abelian groups (where every group of sections of an injective sheaf is divisible).

   The natural map $\epsilon : \mathscr{F} \to C^0(\mathscr{F})$, sending a section $s \in \mathscr{F}(U)$ to the tuple of its germs $(s_x)_{x \in U}$, is injective by the locality axiom for sheaves. Let $\mathscr{Q}^0 = C^0(\mathscr{F})/\mathrm{im}(\epsilon)$. Applying the same construction to $\mathscr{Q}^0$ yields $C^1(\mathscr{F}) = C^0(\mathscr{Q}^0)$, and iterating:

$$0 \longrightarrow \mathscr{F} \xrightarrow{\epsilon} C^0(\mathscr{F}) \longrightarrow C^1(\mathscr{F}) \longrightarrow C^2(\mathscr{F}) \longrightarrow \cdots$$

   This is a flasque resolution of $\mathscr{F}$, and one can compute $H^i(X, \mathscr{F})$ as the cohomology of the complex $\Gamma(X, C^\bullet(\mathscr{F}))$. The Godement resolution is completely canonical (no choices involved), but the resulting sheaves are typically enormous and *not* quasi-coherent.

4. *The divisor sheaf.* Let X be an integral, separated, Noetherian, locally factorial scheme with function field K. The sheaf of Weil divisors is:

$$\mathscr{D}iv \cong \bigoplus_{x \in X^{(1)}} (\iota_x)_*(\underline{\mathbb{Z}})$$

where $X^{(1)}$ denotes the set of codimension-1 points of X. The exact sequence of sheaves:

$$1 \longrightarrow \mathcal{O}_X^* \longrightarrow \mathcal{K}^* \longrightarrow \mathscr{D}iv \longrightarrow 0$$

relates the invertible regular functions, the non-zero rational functions (here $\mathcal{K}^*$ is the constant sheaf associated to K^*, hence flasque by item (2) above), and the divisor sheaf. Since $\mathscr{D}iv$ is a direct sum of pushforwards of flasque sheaves, it is flasque. The long exact sequence in cohomology, together with the acyclicity of $\mathcal{K}^*$ and $\mathscr{D}iv$, gives:

$$\Gamma(X, \mathcal{K}^*) \longrightarrow \Gamma(X, \mathscr{D}iv) \longrightarrow H^1(X, \mathcal{O}_X^*) \longrightarrow 0$$

and hence an isomorphism:

$$H^1(X, \mathcal{O}_X^*) \;\cong\; \frac{\mathrm{WDiv}\,(X)}{\mathrm{PDiv}(X)} \;=\; \mathrm{Cl}\,X,$$

recovering the divisor class group from section 5.4 as cohomological data. This illustrates a recurring theme: the H^1 of a sheaf of units is a "class group" measuring the failure of local-to-global extension for invertible objects.

Acyclic does not imply flasque. Though not central to our development, it is worth recording the main class of acyclic-but-not-flasque sheaves, as it explains why sheaf cohomology in the smooth category is far simpler than in the algebraic category:

Example 6.1.9: Acyclic, But Not Injective Or Flasque

Let M be a smooth manifold of positive dimension and consider the sheaf C^∞_M of smooth real-valued functions. This sheaf is *not* flasque: for instance, the function $1/x$ is a smooth section of $C^\infty_{\mathbb{R}}$ over the open set $(0,1)$, but it cannot be extended to any smooth function on $(-1,1)$. Nor is C^∞_M injective in general.

However, C^∞_M is acyclic. The key reason is the existence of *partitions of unity* subordinate to any open cover (see theorem 0.5.47): given a C^∞_M-linear surjection $\mathscr{F} \to \mathscr{G} \to 0$ of sheaves of C^∞_M-modules and a global section $s \in \Gamma(M, \mathscr{G})$, we can lift s locally and then glue the local lifts using a partition of unity. More precisely, any sheaf of C^∞_M-modules admits partitions of unity, and such sheaves (called *fine sheaves*) are acyclic for Γ.

This is exactly the mechanism that makes de Rham cohomology computable on manifolds. The sheaves Ω^p_M of smooth differential p-forms are fine (being C^∞_M-modules), hence acyclic, and the de Rham complex $0 \to \underline{\mathbb{R}} \to \Omega^0 \to \Omega^1 \to \cdots$ is an acyclic resolution of the constant sheaf $\underline{\mathbb{R}}$. Since acyclic resolutions compute derived functors (see [18, Hypercohomology Recovers the Ordinary Derived Functors]), this gives the isomorphism $H^i_{\mathrm{dR}}(M) \cong H^i(M, \underline{\mathbb{R}})$, and theorem 0.5.23 identifies both with singular cohomology $H^i(M; \mathbb{R})$.

In algebraic geometry, no such luxury exists: the structure sheaf $\mathcal{O}_X$ of an irreducible scheme admits no partition of unity subordinate to a cover by proper open subsets (the Zariski topology is too coarse, and algebraic functions are too rigid). This is why we must work harder — using flasque resolutions, Čech cohomology, or injective resolutions — to compute sheaf cohomology in the algebraic setting.

Let us link this to the module structure on cohomology. If $(X, \mathcal{O}_X)$ is a ringed space, then $\Gamma(X, \mathcal{O}_X)$ is a ring, and for any $\mathcal{O}_X$-module $\mathscr{F}$, the global sections $\Gamma(X, \mathscr{F})$ carry a natural $\Gamma(X, \mathcal{O}_X)$-module

structure. This structure propagates to cohomology:

Corollary 6.1.10: Section and Hom Functor

Let $(X, \mathcal{O}_X)$ be a ringed space and let $R = \Gamma(X, \mathcal{O}_X)$. Then for any $\mathcal{O}_X$-module $\mathscr{F}$, the cohomology groups $H^i(X, \mathscr{F})$ carry a natural R-module structure. In particular:

1. If X is a scheme over a ring A (i.e., equipped with a structure morphism $X \to \operatorname{Spec} A$), then each $H^i(X, \mathscr{F})$ is naturally an A-module.
2. If X is a scheme over a field k, then each $H^i(X, \mathscr{F})$ is a k-vector space.

Moreover, for any $\mathcal{O}_X$-module $\mathscr{F}$, the group $H^i(X, \mathscr{F})$ is naturally isomorphic to the i-th right derived functor of $\Gamma(X, -)\colon \mathbf{Sh}_{\mathbf{Ab}}(X) \to \mathbf{Ab}$ evaluated on the underlying sheaf of abelian groups of $\mathscr{F}$, and the connecting homomorphisms in the long exact sequence are R-module maps.

***Proof*:**
The global section functor $\Gamma(X, -)$ lands in R-Mod (not just **Ab**), and the category $\mathcal{O}_X$-**Mod** has enough injectives. The right derived functors of $\Gamma(X, -)$ computed in $\mathbf{Sh}_{\mathbf{Ab}}(X)$ and in $\mathcal{O}_X$-**Mod** coincide, since an injective $\mathcal{O}_X$-module is flasque (proposition 6.1.5), hence $\Gamma(X, -)$-acyclic (corollary 6.1.7). Hence the cohomology groups inherit the R-module structure. For a scheme over A, the structure morphism $X \to \operatorname{Spec} A$ gives a ring homomorphism $A \to \Gamma(X, \mathcal{O}_X) = R$, making every R-module an A-module by restriction of scalars.

6.1.1 Computation: Čech Cohomology

The derived-functor definition of sheaf cohomology is clean and canonical, but it is difficult to use in practice because injective (or even flasque) resolutions are typically non-explicit. Čech cohomology provides a more "hands-on" approach: it computes the obstruction to extending sections from an open cover $\mathcal{U}$ of a topological space X to larger open sets, using only the sections of the sheaf over finite intersections of the cover. When X is a Noetherian separated scheme, $\mathscr{F}$ is quasi-coherent, and the cover is affine, the Čech groups coincide with the derived-functor groups. This will be our main computational tool.

The following construction will be familiar to those who have seen group cohomology or simplicial cohomology in algebraic topology (see [57, chapter 4]).

Let X be a topological space, $\mathcal{U} = (U_i)_{i \in I}$ an open covering of X, and fix a well-ordering on the index set I[3]. For any finite set of indices $i_0 < \cdots < i_p$ in I, write:

$$U_{i_0,\dots,i_p} = U_{i_0} \cap \cdots \cap U_{i_p}.$$

Let $\mathscr{F}$ be a sheaf of abelian groups on X. Define the *Čech cochain groups*:

$$C^p(\mathcal{U}, \mathscr{F}) = \prod_{i_0 < \cdots < i_p} \mathscr{F}(U_{i_0,\dots,i_p}).$$

[3]Alternatively, one can work with the "alternating" convention: allow all tuples of indices, require that the cochain α is alternating (i.e. $\alpha_{\sigma(i_0),\dots,\sigma(i_p)} = \operatorname{sgn}(\sigma) \cdot \alpha_{i_0,\dots,i_p}$) and vanishes when any two indices coincide. The two approaches yield isomorphic cohomology groups.

An element $\alpha \in C^p(\mathcal{U}, \mathscr{F})$ is a collection of sections $\alpha_{i_0,\dots,i_p} \in \mathscr{F}(U_{i_0,\dots,i_p})$, one for each ordered $(p+1)$-tuple. For an arbitrary tuple of indices, set $\alpha_{i_0,\dots,i_p} = 0$ if two indices coincide, and otherwise $\alpha_{i_0,\dots,i_p} = \operatorname{sgn}(\sigma)\,\alpha_{i_{\sigma(0)},\dots,i_{\sigma(p)}}$, where σ is the permutation with $i_{\sigma(0)} < \cdots < i_{\sigma(p)}$. The *coboundary map* $d : C^p(\mathcal{U}, \mathscr{F}) \to C^{p+1}(\mathcal{U}, \mathscr{F})$ is defined by:

$$(d\alpha)_{i_0,\dots,i_{p+1}} = \sum_{k=0}^{p+1} (-1)^k \, \alpha_{i_0,\dots,\widehat{i_k},\dots,i_{p+1}}\big|_{U_{i_0,\dots,i_{p+1}}}$$

where $\widehat{i_k}$ means the index i_k is omitted. A standard computation shows $d^2 = 0$, so $(C^\bullet(\mathcal{U}, \mathscr{F}), d)$ is a cochain complex.

Definition 6.1.11: Čech Cohomology

Let X be a topological space, $\mathcal{U}$ an open covering of X, and $\mathscr{F}$ a sheaf of abelian groups on X. The *Čech cohomology groups* of $\mathscr{F}$ with respect to $\mathcal{U}$ are:

$$\check{H}^p(\mathcal{U}, \mathscr{F}) = h^p\big(C^\bullet(\mathcal{U}, \mathscr{F})\big) = \frac{\ker\big(d : C^p \to C^{p+1}\big)}{\operatorname{im}\big(d : C^{p-1} \to C^p\big)}.$$

An important warning: $\check{H}^p(\mathcal{U}, -)$ is *not* in general a δ-functor. That is, a short exact sequence of sheaves need not induce a long exact sequence in Čech cohomology:

Example 6.1.12: Čech Cohomology is Not a δ-Functor

Suppose $\mathcal{U}$ consists of the single open set X itself. Then every intersection $U_{i_0,\dots,i_p}$ equals X, and there is only one index, so $C^p(\mathcal{U}, \mathscr{F}) = 0$ for $p > 0$ while $C^0(\mathcal{U}, \mathscr{F}) = \mathscr{F}(X) = \Gamma(X, \mathscr{F})$. Hence $\check{H}^0(\mathcal{U}, \mathscr{F}) = \Gamma(X, \mathscr{F})$ and $\check{H}^p(\mathcal{U}, \mathscr{F}) = 0$ for all $p > 0$.

Now let $0 \to \mathscr{F}' \to \mathscr{F} \to \mathscr{F}'' \to 0$ be a short exact sequence of sheaves. If $\check{H}^*(\mathcal{U}, -)$ were a δ-functor, it would produce a long exact sequence, which in this case would read:

$$0 \longrightarrow \Gamma(X, \mathscr{F}') \longrightarrow \Gamma(X, \mathscr{F}) \longrightarrow \Gamma(X, \mathscr{F}'') \longrightarrow 0.$$

But $\Gamma(X, -)$ is not right exact in general — this is the whole point of sheaf cohomology! — so this need not be exact on the right. Hence, whenever some surjection of sheaves on X is not surjective on global sections, $\check{H}^*$ cannot be a δ-functor for this covering.

This failure is not a defect of the theory but a feature of the choice of cover. As we shall see, choosing an appropriate cover (an affine cover for quasi-coherent sheaves on a Noetherian separated scheme) restores the connection to derived-functor cohomology. Before proving this, let us see Čech cohomology in action:

Example 6.1.13: Čech Cohomology

1. *Differentials on $\mathbb{P}^1_k$.* Let $X = \mathbb{P}^1_k$, let $\mathscr{F} = \Omega_{X/k}$ be the sheaf of differentials, and let $\mathcal{U} = \{U, V\}$ be the standard affine cover with $U = D_+(x_0) \cong \operatorname{Spec} k[t]$ and $V = D_+(x_1) \cong \operatorname{Spec} k[u]$, where $t = x_1/x_0$ and $u = x_0/x_1 = 1/t$. The Čech complex has two terms:

$$C^0 = \Gamma(U, \Omega) \times \Gamma(V, \Omega), \qquad C^1 = \Gamma(U \cap V, \Omega).$$

The sections of the sheaf of differentials are:

$$\Gamma(U,\Omega) = k[t]\,dt, \qquad \Gamma(V,\Omega) = k[u]\,du, \qquad \Gamma(U\cap V,\Omega) = k[t,t^{-1}]\,dt.$$

On the overlap, the coordinate change gives $u = 1/t$, so $du = -t^{-2}\,dt$. The coboundary map $d : C^0 \to C^1$ sends $(f(t)\,dt,\ g(u)\,du)$ to:

$$g(u)\,du\big|_{U\cap V} - f(t)\,dt\big|_{U\cap V} \;=\; \big(-t^{-2}g(1/t) - f(t)\big)\,dt.$$

Computing $\check{H}^0$: A cocycle $(f(t)\,dt,\, g(u)\,du) \in \ker d$ satisfies $f(t) = -t^{-2}g(1/t)$. The left side is a polynomial in t, while the right side is a polynomial in t^{-1} with no constant term (the t^{-2} factor shifts degrees). These can only agree if both are zero. Hence $\check{H}^0(\mathcal{U},\Omega) = 0$, confirming that $\mathbb{P}^1_k$ has no nonzero global differential forms.

Computing $\check{H}^1$: The image of d consists of all Laurent polynomials of the form $\big(-t^{-2}g(1/t) - f(t)\big)\,dt$ where $f \in k[t]$ and $g \in k[u]$. Writing a general element of $k[t,t^{-1}]\,dt$ as $\sum_{n\in\mathbb{Z}} a_n t^n\,dt$, the image of d is spanned by all $t^n\,dt$ with $n \neq -1$: the terms $t^n\,dt$ for $n \geq 0$ come from $f(t)$, and the terms $t^n\,dt$ for $n \leq -2$ come from $g(u)$. The single "missing" basis element is $t^{-1}\,dt$. Therefore:

$$\check{H}^1(\mathcal{U},\Omega) \cong k,$$

generated by the class of $t^{-1}\,dt$. It is the vanishing of $\check{H}^0$ that reflects the genus of $\mathbb{P}^1$ being zero (for a smooth projective curve of genus g, one has $\dim_k H^0(X,\Omega) = g$). The one-dimensional $H^1(X,\Omega)$ is the domain of the trace map $H^1(X,\Omega) \to k$, and we will revisit it when we discuss Serre duality.

2. *The circle.* Let $X = S^1$ with the Euclidean topology, and consider the constant sheaf $\underline{\mathbb{Z}}$. Take the cover $\mathcal{U} = \{U, V\}$ where U and V are two overlapping open arcs whose intersection $U \cap V = W_1 \coprod W_2$ consists of two disjoint small open intervals. Since U, V, W_1, and W_2 are all connected, the constant sheaf gives:

$$C^0 = \underline{\mathbb{Z}}(U) \times \underline{\mathbb{Z}}(V) = \mathbb{Z}\times\mathbb{Z}, \qquad C^1 = \underline{\mathbb{Z}}(U\cap V) = \underline{\mathbb{Z}}(W_1)\times\underline{\mathbb{Z}}(W_2) = \mathbb{Z}\times\mathbb{Z}.$$

The coboundary $d : C^0 \to C^1$ sends (a,b) to $(b-a,\ b-a)$, since on each connected component of $U\cap V$ the sections restrict to constants. Thus:

$$\ker d = \{(a,b) : b-a = 0\} \cong \mathbb{Z} \qquad \text{and} \qquad \mathrm{im} d = \{(c,c) : c \in \mathbb{Z}\} \cong \mathbb{Z}.$$

The cokernel $(\mathbb{Z}\times\mathbb{Z})/\{(c,c)\} \cong \mathbb{Z}$ (via $(m,n) \mapsto m-n$). Hence:

$$\check{H}^0(\mathcal{U},\underline{\mathbb{Z}}) = \mathbb{Z}, \qquad \check{H}^1(\mathcal{U},\underline{\mathbb{Z}}) = \mathbb{Z},$$

which agrees with the singular cohomology $H^*(S^1;\mathbb{Z})$. The class in $\check{H}^1$ is generated by the cocycle $(1,0) \in \mathbb{Z}\times\mathbb{Z}$: the two components of $U\cap V$ see different "transition data," reflecting the non-trivial loop in S^1.

Let us now build towards the theorem that Čech cohomology computes derived-functor cohomology under appropriate hypotheses. The strategy is to show that the Čech complex arises from a resolution of the sheaf, and that this resolution can be compared to any injective resolution. The key condition is that each open set in the cover, and each finite intersection of them, must be "cohomologically trivial" for the sheaf in question, so that the main obstruction comes from their combinations.

Lemma 6.1.14: 0th Čech Cohomology

Let X be a topological space, $\mathcal{U}$ an open covering, and $\mathscr{F}$ a sheaf of abelian groups on X. Then:

$$\check{H}^0(\mathcal{U}, \mathscr{F}) = \Gamma(X, \mathscr{F}).$$

***Proof*:**

By definition, $\check{H}^0(\mathcal{U}, \mathscr{F}) = \ker(d : C^0(\mathcal{U}, \mathscr{F}) \to C^1(\mathcal{U}, \mathscr{F}))$. An element $\alpha \in C^0$ is a collection $\{\alpha_i \in \mathscr{F}(U_i)\}_{i \in I}$. The coboundary condition $d\alpha = 0$ says that for each $i < j$,

$$(d\alpha)_{ij} = \alpha_j\big|_{U_i \cap U_j} - \alpha_i\big|_{U_i \cap U_j} = 0,$$

i.e. the sections α_i and α_j agree on the overlap $U_i \cap U_j$. Since $\mathscr{F}$ is a sheaf, the gluability axiom gives a unique global section $s \in \Gamma(X, \mathscr{F})$ with $s|_{U_i} = \alpha_i$ for all i. Hence $\ker d = \Gamma(X, \mathscr{F})$.

We next address the lack of a δ-functor by "sheafifying" the Čech construction. Let $X, \mathcal{U}$, and $\mathscr{F}$ be as above. For each $p \geq 0$, define the *sheaf of Čech p-cochains* by:

$$\mathscr{C}^p(\mathcal{U}, \mathscr{F}) = \prod_{i_0 < \cdots < i_p} (f_{i_0,\ldots,i_p})_*\big(\mathscr{F}\big|_{U_{i_0,\ldots,i_p}}\big)$$

where $f_{i_0,\ldots,i_p} : U_{i_0,\ldots,i_p} \hookrightarrow X$ is the inclusion and $(f_{i_0,\ldots,i_p})_*$ is the pushforward. The coboundary maps $d : \mathscr{C}^p \to \mathscr{C}^{p+1}$ are defined by the same alternating-sum formula as before. The key property of this construction is:

$$\Gamma\big(X,\, \mathscr{C}^p(\mathcal{U}, \mathscr{F})\big) = C^p(\mathcal{U}, \mathscr{F}),$$

so the Čech cochain groups are the global sections of the Čech sheaf complex.

Lemma 6.1.15: Čech Cohomology Resolution

Let X be a topological space, $\mathcal{U}$ an open covering, and $\mathscr{F}$ a sheaf of abelian groups. Then the augmented complex

$$0 \longrightarrow \mathscr{F} \xrightarrow{\epsilon} \mathscr{C}^0(\mathcal{U}, \mathscr{F}) \xrightarrow{d} \mathscr{C}^1(\mathcal{U}, \mathscr{F}) \xrightarrow{d} \mathscr{C}^2(\mathcal{U}, \mathscr{F}) \longrightarrow \cdots$$

is exact. That is, $\mathscr{C}^\bullet(\mathcal{U}, \mathscr{F})$ is a resolution of $\mathscr{F}$.

***Proof*:**

The augmentation $\epsilon : \mathscr{F} \to \mathscr{C}^0(\mathcal{U}, \mathscr{F})$ is defined by taking the product of the natural restriction maps $\mathscr{F} \to (f_i)_*(\mathscr{F}|_{U_i})$ for each $i \in I$. Exactness at $\mathscr{C}^0$ (i.e. $\ker d^0 = \mathrm{im}\epsilon$) follows from the sheaf axioms for $\mathscr{F}$, exactly as in lemma 6.1.14.

For the exactness of the complex at $\mathscr{C}^p$ for $p \geq 1$, it suffices to check on stalks. Let $x \in X$ and choose an index $j \in I$ such that $x \in U_j$. For each $p \geq 0$, we define a *contracting homotopy* $k : \mathscr{C}^p(\mathcal{U}, \mathscr{F})_x \to \mathscr{C}^{p-1}(\mathcal{U}, \mathscr{F})_x$ as follows. A germ $\alpha_x \in \mathscr{C}^p(\mathcal{U}, \mathscr{F})_x$ is represented by a section $\alpha \in \Gamma(W, \mathscr{C}^p(\mathcal{U}, \mathscr{F}))$ over some open neighborhood W of x that we may shrink until $W \subseteq U_j$.

For any p-tuple $i_0 < \cdots < i_{p-1}$, set:

$$(k\alpha)_{i_0,\ldots,i_{p-1}} = \alpha_{j,i_0,\ldots,i_{p-1}}\big|_W.$$

This is well-defined because $W \subseteq U_j$ implies $W \cap U_{i_0,\ldots,i_{p-1}} = W \cap U_{j,i_0,\ldots,i_{p-1}}$, so the section $\alpha_{j,i_0,\ldots,i_{p-1}}$ is defined on W. Passing to germs at x gives the map k on stalks. A direct computation shows:

$$(dk + kd)(\alpha) = \alpha \qquad \text{for all } \alpha \in \mathscr{C}^p(\mathcal{U}, \mathscr{F})_x,\ p \geq 1.$$

Hence k is a chain homotopy between the identity and the zero map on the stalk complex $\mathscr{C}^\bullet(\mathcal{U}, \mathscr{F})_x$ in degrees $p \geq 1$. It follows that the cohomology of this stalk complex vanishes in positive degrees, which is precisely the exactness of the resolution.

Lemma 6.1.16: Čech Cohomology and Flasque Sheaves

Let X be a topological space, $\mathcal{U}$ an open covering, and $\mathscr{F}$ a flasque sheaf of abelian groups on X. Then $\check{H}^p(\mathcal{U}, \mathscr{F}) = 0$ for all $p > 0$.

***Proof*:**
By lemma 6.1.15, the complex $0 \to \mathscr{F} \to \mathscr{C}^\bullet(\mathcal{U}, \mathscr{F})$ is a resolution. Since $\mathscr{F}$ is flasque, the sheaves $\mathscr{C}^p(\mathcal{U}, \mathscr{F})$ are flasque for each $p \geq 0$: indeed, $\mathscr{F}|_{U_{i_0,\ldots,i_p}}$ is flasque (restriction of a flasque sheaf is flasque), the pushforward $(f_{i_0,\ldots,i_p})_*$ preserves flasque sheaves, and a product of flasque sheaves is flasque. Thus $\mathscr{C}^\bullet(\mathcal{U}, \mathscr{F})$ is a flasque resolution of $\mathscr{F}$, and we may use it to compute derived-functor cohomology:

$$H^p(X, \mathscr{F}) = h^p\big(\Gamma(X, \mathscr{C}^\bullet(\mathcal{U}, \mathscr{F}))\big) = h^p\big(C^\bullet(\mathcal{U}, \mathscr{F})\big) = \check{H}^p(\mathcal{U}, \mathscr{F}).$$

But $\mathscr{F}$ is flasque, so $H^p(X, \mathscr{F}) = 0$ for $p > 0$ by corollary 6.1.7. Hence $\check{H}^p(\mathcal{U}, \mathscr{F}) = 0$ for $p > 0$.

Lemma 6.1.17: Čech Cohomology Comparison Map

Let X be a topological space and $\mathcal{U}$ an open covering. Then for each $p \geq 0$, there is a natural map, functorial in $\mathscr{F}$:

$$\check{H}^p(\mathcal{U}, \mathscr{F}) \longrightarrow H^p(X, \mathscr{F}).$$

***Proof*:**
Choose an injective resolution $0 \to \mathscr{F} \to \mathscr{I}^\bullet$. We also have the Čech resolution $0 \to \mathscr{F} \to \mathscr{C}^\bullet(\mathcal{U}, \mathscr{F})$ from lemma 6.1.15. By the comparison lemma for resolutions, [18, Projective Resolution As Initial Objects], applied in the opposite category of sheaves of abelian groups on X, where $\mathscr{I}^\bullet$ becomes a complex of projectives, there exists a morphism of complexes $\mathscr{C}^\bullet(\mathcal{U}, \mathscr{F}) \to \mathscr{I}^\bullet$ extending the identity on $\mathscr{F}$, unique up to homotopy. Applying $\Gamma(X, -)$ and taking cohomology gives the desired maps $\check{H}^p(\mathcal{U}, \mathscr{F}) \to H^p(X, \mathscr{F})$, and functoriality follows from the naturality of the comparison.

The hypothesis that makes the argument work is not affineness or quasi-coherence but a vanishing condition, and the general form is what later applications need: a covering computes derived-functor

cohomology as soon as the sheaf has no higher cohomology on the members of the covering and their intersections. Such a covering is called *acyclic* for the sheaf, and the result is Leray's theorem.

Theorem 6.1.18: Leray's Acyclic Covering Theorem

Let X be a topological space, $\mathcal{U} = \{U_i\}_{i\in I}$ an open covering, and $\mathscr{F}$ a sheaf of abelian groups on X. Suppose the covering is *acyclic* for $\mathscr{F}$: for every $q > 0$ and every finite tuple $i_0, \dots, i_p$,

$$H^q\big(U_{i_0,\dots,i_p}, \mathscr{F}\big) = 0$$

Then the comparison maps of lemma 6.1.17 are isomorphisms

$$\check{H}^p(\mathcal{U}, \mathscr{F}) \xrightarrow{\sim} H^p(X, \mathscr{F}) \qquad \text{for all } p \geq 0$$

***Proof*:**
For $p = 0$ both sides are $\Gamma(X, \mathscr{F})$ by lemma 6.1.14. The rest is induction on p, by dimension shifting.

Setup. The category of abelian sheaves on X has enough injectives, so embed $\mathscr{F}$ into an injective sheaf $\mathscr{G}$, which is flasque by proposition 6.1.5, and let $\mathscr{Q}$ be the quotient:

$$0 \longrightarrow \mathscr{F} \longrightarrow \mathscr{G} \longrightarrow \mathscr{Q} \longrightarrow 0$$

The hypothesis passes to $\mathscr{Q}$. Fix a tuple and write $U = U_{i_0,\dots,i_p}$. Since $\mathscr{G}$ is flasque, $H^q(U, \mathscr{G}) = 0$ for $q > 0$ by corollary 6.1.7, so the long exact sequence on U gives $H^q(U, \mathscr{Q}) \cong H^{q+1}(U, \mathscr{F})$ for $q \geq 1$, which vanishes by hypothesis. So $\mathcal{U}$ is acyclic for $\mathscr{Q}$ as well, and the induction has somewhere to go.

Exactness of the Čech complex. The same long exact sequence in degree zero, together with $H^1(U, \mathscr{F}) = 0$, gives exactness of

$$0 \longrightarrow \mathscr{F}(U) \longrightarrow \mathscr{G}(U) \longrightarrow \mathscr{Q}(U) \longrightarrow 0$$

for every intersection U. This is exactly where the hypothesis is spent: without it the right-hand map need not be surjective. Taking products over all tuples, the Čech cochain complexes form a short exact sequence

$$0 \longrightarrow C^\bullet(\mathcal{U}, \mathscr{F}) \longrightarrow C^\bullet(\mathcal{U}, \mathscr{G}) \longrightarrow C^\bullet(\mathcal{U}, \mathscr{Q}) \longrightarrow 0$$

An adapted resolution. Write $q\colon \mathscr{G} \to \mathscr{Q}$ for the quotient map and choose an injective resolution $\mathscr{Q} \to \mathscr{J}^\bullet$ with augmentation $\varepsilon\colon \mathscr{Q} \to \mathscr{J}^0$. Then

$$\mathscr{I}^\bullet : \quad \mathscr{G} \xrightarrow{\varepsilon q} \mathscr{J}^0 \longrightarrow \mathscr{J}^1 \longrightarrow \cdots,$$

with $\mathscr{G}$ in degree 0 and $\mathscr{I}^{n+1} = \mathscr{J}^n$, is an injective resolution of $\mathscr{F}$. For $p \geq 1$ the groups $H^{p+1}(X, \mathscr{F})$ and $H^p(X, \mathscr{Q})$ are therefore both the degree-p cohomology of $\Gamma(X, \mathscr{J}^\bullet)$. Since $\Gamma(X, -)$ is left exact, ε identifies $\Gamma(X, \mathscr{Q})$ with the 0-cocycles of $\Gamma(X, \mathscr{J}^\bullet)$, so

$$H^1(X, \mathscr{F}) = \Gamma(X, \mathscr{Q})/q\big(\Gamma(X, \mathscr{G})\big).$$

The chain maps of lemma 6.1.17 are unique up to homotopy, so the comparison maps for $\mathscr{F}$ and $\mathscr{Q}$ may be computed with $\mathscr{I}^\bullet$ and $\mathscr{J}^\bullet$.

Adapted chain maps. Let $\varphi_{\mathscr{Q}} \colon \mathscr{C}^\bullet(\mathcal{U}, \mathscr{Q}) \to \mathscr{J}^\bullet$ be a chain map extending the identity on $\mathscr{Q}$, and let i and π be the maps of Čech resolutions induced by $\mathscr{F} \to \mathscr{G}$ and by q. As $\mathscr{G}$ is injective, the augmentation $\mathscr{G} \to \mathscr{C}^0(\mathcal{U}, \mathscr{G})$ has a retraction ρ. The chain maps $\varphi_{\mathscr{Q}}\pi$ and $\varepsilon q \rho$ from $\mathscr{C}^\bullet(\mathcal{U}, \mathscr{G})$ to $\mathscr{J}^\bullet$, the second concentrated in degree 0, both extend q, so the comparison lemma used in lemma 6.1.17 gives maps $h^n \colon \mathscr{C}^n(\mathcal{U}, \mathscr{G}) \to \mathscr{J}^{n-1}$, with $h^0 = 0$, such that

$$\varphi_{\mathscr{Q}}\pi - \varepsilon q \rho = dh + hd.$$

Since $\pi i = 0$, this identity makes $\varphi^0_{\mathscr{F}} = \rho\, i$ and $\varphi^{n+1}_{\mathscr{F}} = (-1)^{n+1} h\, i$ a chain map $\varphi_{\mathscr{F}} \colon \mathscr{C}^\bullet(\mathcal{U}, \mathscr{F}) \to \mathscr{I}^\bullet$ extending the identity on $\mathscr{F}$.

Compatibility with the connecting maps. Write λ for the comparison maps and δ for the connecting maps of the Čech long exact sequence. Let $c \in C^p(\mathcal{U}, \mathscr{Q})$ be a cocycle, lift it to $b \in C^p(\mathcal{U}, \mathscr{G})$, and write $db = i(a)$, so that $\delta[c] = [a]$. Evaluating the identity above on b gives

$$\varphi_{\mathscr{F}}(a) = (-1)^{p+1} h(db) = (-1)^{p+1}\big(\varphi_{\mathscr{Q}}(c) - \varepsilon q \rho(b) - dh(b)\big).$$

For $p \geq 1$ the term $\varepsilon q \rho(b)$ vanishes, and for $p = 0$ it lies in the image of $\Gamma(X, \mathscr{G})$ while $h(b) = 0$. Hence $\lambda_{\mathscr{F}} \delta = (-1)^{p+1} \lambda_{\mathscr{Q}}$ on $\check{H}^p(\mathcal{U}, \mathscr{Q})$, with $H^p(X, \mathscr{Q})$ read as $H^{p+1}(X, \mathscr{F})$ for $p \geq 1$ and mapped onto $H^1(X, \mathscr{F})$ for $p = 0$.

Induction. Since $\mathscr{G}$ is flasque, $\check{H}^p(\mathcal{U}, \mathscr{G}) = 0$ for $p > 0$ by lemma 6.1.16. The Čech long exact sequence therefore makes δ a surjection $\Gamma(X, \mathscr{Q}) \to \check{H}^1(\mathcal{U}, \mathscr{F})$ with kernel $q(\Gamma(X, \mathscr{G}))$, and an isomorphism $\check{H}^p(\mathcal{U}, \mathscr{Q}) \to \check{H}^{p+1}(\mathcal{U}, \mathscr{F})$ for $p \geq 1$. In degree one, $\lambda_{\mathscr{F}}$ is thus, up to sign, the identity of $\Gamma(X, \mathscr{Q})/q(\Gamma(X, \mathscr{G}))$. For $p \geq 1$, the inductive hypothesis applied to $\mathscr{Q}$ (legitimate, since $\mathcal{U}$ is acyclic for $\mathscr{Q}$) makes $\lambda_{\mathscr{Q}}$ an isomorphism in degree p, and then $\lambda_{\mathscr{F}}$ is one in degree $p+1$. This completes the induction, as we sought to show.

The scheme-theoretic case now follows by producing the acyclicity: on a separated scheme the intersections of affine opens are again affine, and a quasi-coherent sheaf has no higher cohomology on an affine. Separatedness is precisely what makes the intersections affine, which is where that hypothesis enters.

Theorem 6.1.19: Čech Cohomology Equivalence

Let X be a Noetherian separated scheme, $\mathcal{U}$ an open affine covering of X, and $\mathscr{F}$ a quasi-coherent sheaf on X. Then for all $p \geq 0$, the natural maps

$$\check{H}^p(\mathcal{U}, \mathscr{F}) \longrightarrow H^p(X, \mathscr{F})$$

are isomorphisms.

***Proof*:**
By theorem 6.1.18 it suffices to check that $\mathcal{U}$ is acyclic for $\mathscr{F}$. Fix a tuple $i_0, \ldots, i_p$. The intersection $U_{i_0,\ldots,i_p}$ is a finite intersection of affine opens in a separated scheme, hence affine by proposition 3.5.8. A quasi-coherent sheaf on a Noetherian affine scheme has vanishing higher cohomology by theorem 6.2.9, so $H^q(U_{i_0,\ldots,i_p}, \mathscr{F}) = 0$ for every $q > 0$. That is the hypothesis of Leray's theorem, and the conclusion is the claim, as we sought to show.

6.1.2 Refinement and the Degree-One Comparison

Everything so far has been relative to a fixed covering, and theorem 6.1.18 gives agreement with derived-functor cohomology only when a covering acyclic for the sheaf can be found. For many sheaves no such covering exists. What survives without any acyclicity hypothesis is a comparison in degree one, and reaching it means first letting the covering vary.

Definition 6.1.20: Refinement of a Covering

Let $\mathcal{U} = (U_i)_{i\in I}$ and $\mathcal{V} = (V_j)_{j\in J}$ be open coverings of X. Then $\mathcal{V}$ *refines* $\mathcal{U}$ if there is a map $\varepsilon : J \to I$ with $V_j \subseteq U_{\varepsilon(j)}$ for every j; such an ε is a *refinement function*. Write $\mathcal{V} \preceq \mathcal{U}$.

A refinement function transports cochains, and the point of the next lemma is that the transported map on cohomology does not depend on which refinement function was chosen. Without that, the coverings could not be assembled into a single invariant.

Lemma 6.1.21: Refinement Induces a Well-Defined Map

Let $\mathscr{F}$ be a sheaf of abelian groups on X and let $\mathcal{V} \preceq \mathcal{U}$.

1. A refinement function ε induces a morphism of complexes $\varepsilon^\bullet : C^\bullet(\mathcal{U}, \mathscr{F}) \to C^\bullet(\mathcal{V}, \mathscr{F})$, by
$$(\varepsilon^p \alpha)_{j_0,\dots,j_p} = \alpha_{\varepsilon(j_0),\dots,\varepsilon(j_p)}\big|_{V_{j_0,\dots,j_p}}$$
2. Any two refinement functions ε, η induce chain homotopic morphisms. Hence the induced map $\check{H}^p(\mathcal{U}, \mathscr{F}) \to \check{H}^p(\mathcal{V}, \mathscr{F})$ is independent of the choice, and these maps are compatible with composition of refinements.

***Proof*:**
For (1), $\varepsilon^\bullet$ commutes with d because the coboundary formula, which under the sign convention preceding definition 6.1.11 holds at every index tuple, is a signed sum over omitted indices and ε acts on indices, so omitting the k-th index before or after applying ε gives the same term; the restrictions agree since $V_{j_0,\dots,j_p} \subseteq U_{\varepsilon(j_0),\dots,\varepsilon(j_p)}$.
(2). Define $h^p : C^p(\mathcal{U}, \mathscr{F}) \to C^{p-1}(\mathcal{V}, \mathscr{F})$ by

$$(h^p \alpha)_{j_0,\dots,j_{p-1}} = \sum_{k=0}^{p-1} (-1)^k \, \alpha_{\varepsilon(j_0),\dots,\varepsilon(j_k),\eta(j_k),\dots,\eta(j_{p-1})}\big|_{V_{j_0,\dots,j_{p-1}}}$$

which makes sense because $V_{j_0,\dots,j_{p-1}}$ is contained in every open set appearing on the right. The standard simplicial computation, identical to the one that shows two simplicial approximations are homotopic, gives $dh + hd = \eta^\bullet - \varepsilon^\bullet$. Hence the two morphisms agree on cohomology. Compatibility with composition is immediate, since a composite of refinement functions is one, completing the proof.

Definition 6.1.22: Čech Cohomology of a Space

Let $\mathscr{F}$ be a sheaf of abelian groups on X. The open coverings of X, ordered by refinement, form a filtered preorder: two coverings $\mathcal{U}$ and $\mathcal{U}'$ admit the common refinement $(U_i \cap U'_{i'})_{i,i'}$. The *Čech cohomology of* X is the filtered colimit

$$\check{H}^p(X, \mathscr{F}) = \varinjlim_{\mathcal{U}} \check{H}^p(\mathcal{U}, \mathscr{F})$$

over that preorder, the transition maps being those of lemma 6.1.21.

The comparison with derived-functor cohomology now holds in degrees zero and one for an arbitrary sheaf on an arbitrary space, with no acyclicity hypothesis anywhere. This is the form to use when Leray is unavailable.

Theorem 6.1.23: The Degree-One Comparison

Let X be a topological space and $\mathscr{F}$ a sheaf of abelian groups on X. Then

$$\check{H}^0(X, \mathscr{F}) \cong H^0(X, \mathscr{F}), \qquad \check{H}^1(X, \mathscr{F}) \cong H^1(X, \mathscr{F})$$

naturally in $\mathscr{F}$.

***Proof*:**
Degree zero is lemma 6.1.14 for each covering, and the transition maps are the identity on $\Gamma(X, \mathscr{F})$, so the colimit is $\Gamma(X, \mathscr{F}) = H^0(X, \mathscr{F})$.

Setup for degree one. The category of abelian sheaves on X has enough injectives; embed $\mathscr{F}$ into an injective $\mathscr{I}$, which is flasque by proposition 6.1.5, and let $\mathscr{Q} = \mathscr{I}/\mathscr{F}$. Since $H^1(X, \mathscr{I}) = 0$ by corollary 6.1.7, the long exact sequence gives

$$H^1(X, \mathscr{F}) \cong \operatorname{coker}\big(\Gamma(X, \mathscr{I}) \to \Gamma(X, \mathscr{Q})\big)$$

A class gives a cocycle. Let $\xi \in H^1(X, \mathscr{F})$ be represented by $t \in \Gamma(X, \mathscr{Q})$. Because $\mathscr{I} \to \mathscr{Q}$ is an epimorphism of sheaves, it is surjective on stalks, so there is an open covering $\mathcal{U} = (U_i)$ and sections $s_i \in \mathscr{I}(U_i)$ mapping to $t|_{U_i}$. On U_{ij} the difference $s_j - s_i$ maps to 0 in $\mathscr{Q}$, hence lies in $\mathscr{F}(U_{ij})$; set $c_{ij} = s_j - s_i$. Then c is a Čech 1-cocycle, since $c_{jk} - c_{ik} + c_{ij} = 0$ by inspection, and its class in $\check{H}^1(X, \mathscr{F})$ is independent of the choices: changing s_i by $a_i \in \mathscr{F}(U_i)$ changes c by the coboundary da, and passing to a refinement changes nothing by lemma 6.1.21.

The two constructions are mutually inverse. In the other direction, a cocycle $c \in \check{C}^1(\mathcal{U}, \mathscr{F})$ may be written $c_{ij} = s_j - s_i$ for some $s_i \in \mathscr{I}(U_i)$, because $\mathscr{I}$ is flasque and hence $\check{H}^1(\mathcal{U}, \mathscr{I}) = 0$ for every covering by theorem 6.1.18; the images of the s_i in $\mathscr{Q}(U_i)$ then agree on overlaps and glue to a global section t, whose class in the cokernel is the corresponding element of $H^1(X, \mathscr{F})$. Chasing the definitions shows the two assignments invert one another, and both are natural in $\mathscr{F}$ because every step is, completing the proof.

6.1.3 Soft Sheaves and the Paracompact Comparison

With no hypothesis on the space, the comparison map from Čech to derived-functor cohomology need not be an isomorphism in degree two, so theorem 6.1.23 does not extend to higher degrees. On a paracompact Hausdorff space Čech cohomology agrees with derived-functor cohomology in every degree (theorem 6.1.29). Such spaces also carry a class of sheaves that is to the topological setting what flasque sheaves are to the general one, and which was promised earlier when partitions of unity were contrasted with the rigidity of schemes.

Throughout this subsection X is a paracompact Hausdorff space. Recall that this means every open cover admits a locally finite open refinement; the consequence used below is that such a cover admits a shrinking, an open cover $\{V_i\}$ with $\overline{V_i} \subseteq U_i$.

Definition 6.1.24: Soft Sheaf

A sheaf $\mathscr{F}$ of abelian groups on a paracompact Hausdorff space X is *soft* if for every closed $K \subseteq X$ the restriction $\mathscr{F}(X) \to \mathscr{F}(K) := \varinjlim_{U \supseteq K} \mathscr{F}(U)$ is surjective: every section defined on a neighbourhood of a closed set extends to all of X.

A *fine* sheaf, in the sense of example 6.1.9, is one admitting partitions of unity subordinate to any locally finite cover. Fineness is the property one can check; softness is the property that makes cohomology vanish, and the first implies the second.

Proposition 6.1.25: Flasque and Fine Sheaves Are Soft

On a paracompact Hausdorff space, every flasque sheaf is soft, and every fine sheaf of modules over a sheaf of rings is soft.

***Proof*:**
A flasque sheaf restricts surjectively from X to any open set, hence a fortiori to the colimit over neighbourhoods of a closed set.

Let $\mathscr{F}$ be fine and let s be a section over an open $U \supseteq K$ with K closed. The two opens U and $X \setminus K$ cover X; take a locally finite refinement and a subordinate partition of unity, and let η be the sum of the members supported in U, so that η is an endomorphism of $\mathscr{F}$ with $\eta = \mathrm{id}$ on a neighbourhood of K and $\eta = 0$ off U. Then $\eta(s)$, extended by zero, is a global section restricting to s near K.

Lemma 6.1.26: Sections Near a Closed Set

Let $\mathscr{F}$ be a sheaf of abelian groups on X and $K \subseteq X$ closed. Suppose given germs $\sigma_z \in \mathscr{F}_z$ for $z \in K$ such that every $z \in K$ has an open neighbourhood O and a section $u \in \mathscr{F}(O)$ with $u_y = \sigma_y$ for all $y \in O \cap K$. Then some section of $\mathscr{F}$ over an open neighbourhood of K has germ σ_z at every $z \in K$, and its class in $\mathscr{F}(K)$ is unique.

***Proof*:**
Two sections over open neighbourhoods of K with the same germ at every point of K agree on the open set of points of their common domain where their germs agree, which contains K. This is the uniqueness.

For existence, choose for each $z \in K$ an open O_z and $u_z \in \mathscr{F}(O_z)$ as in the hypothesis. The O_z and $X \setminus K$ cover X. Take a locally finite open refinement $\{U_i\}_{i \in I}$ and a shrinking $\{V_i\}$ with $\overline{V_i} \subseteq U_i$. The family $\{\overline{V_i}\}$ is again locally finite. Let I' consist of the i for which U_i lies in some O_z, and let u_i be the restriction of that u_z, so that $(u_i)_y = \sigma_y$ for $y \in U_i \cap K$. For $i \notin I'$ we have $U_i \subseteq X \setminus K$. Let W be the set of $x \in X$ that lie in no $\overline{V_i}$ with $i \notin I'$ and satisfy $(u_i)_x = (u_j)_x$ whenever $i, j \in I'$ and $x \in \overline{V_i} \cap \overline{V_j}$.

Every $z \in K$ lies in W, because $\overline{V_i} \subseteq X \setminus K$ for $i \notin I'$, and for $i, j \in I'$ both germs at z equal σ_z. The set W is open. Given $x \in W$, local finiteness gives an open neighbourhood N of x that meets no $\overline{V_i}$ with $x \notin \overline{V_i}$. For each of the finitely many pairs $i, j \in I'$ with $x \in \overline{V_i} \cap \overline{V_j}$ the sections u_i and u_j agree on a neighbourhood of x in $U_i \cap U_j$, and we shrink N into each of these. A point $y \in N$ then lies only in sets $\overline{V_i}$ that contain x, so $y \in W$.

The sets $W \cap V_i$ with $i \in I'$ cover W, since each point of W lies in some V_i and not in any $\overline{V_i}$ with $i \notin I'$. On $W \cap V_i \cap V_j$ the sections u_i and u_j have the same germ at every point, so they agree. They glue to a section over W whose germ at each $z \in K$ is σ_z, as claimed.

Lemma 6.1.27: Sections and Quotients of Soft Sheaves

Let $0 \to \mathscr{F}' \to \mathscr{F} \to \mathscr{F}'' \to 0$ be exact on X with $\mathscr{F}'$ soft. Then $\mathscr{F}(X) \to \mathscr{F}''(X)$ is surjective. If $\mathscr{F}$ is soft as well, then $\mathscr{F}''$ is soft.

***Proof*:**
Let $K \subseteq X$ be closed and t a section of $\mathscr{F}''$ over an open $U \supseteq K$. We first lift t over a neighbourhood of K. A shrinking of the cover $\{U, X \setminus K\}$ gives an open W with $K \subseteq W \subseteq \overline{W} \subseteq U$. The open subsets of U over which t lifts, together with $X \setminus \overline{W}$, cover X. Choose a locally finite open refinement $\{U_i\}_{i \in I}$ and a shrinking $\{V_i\}$ with $\overline{V_i} \subseteq U_i$. Let $I' \subseteq I$ consist of the i for which U_i lies in an open subset of U over which t lifts, and choose lifts $s_i \in \mathscr{F}(U_i)$ for $i \in I'$. Every V_i with $i \notin I'$ misses $\overline{W}$, so $\overline{W} \subseteq \bigcup_{i \in I'} V_i$. For $J \subseteq I'$ the set $\overline{V_J} := \bigcup_{i \in J} \overline{V_i}$ is closed, because the family $\{\overline{V_i}\}$ is locally finite. Consider the set of pairs (J, s) where $J \subseteq I'$ and $s \in \mathscr{F}(\overline{V_J})$ maps to the image of t in $\mathscr{F}''(\overline{V_J})$, with $(J, s) \le (J', s')$ when $J \subseteq J'$ and s' restricts to s in $\mathscr{F}(\overline{V_J})$.

A chain $\{(J_\alpha, s_\alpha)\}$ has an upper bound. Put $J = \bigcup_\alpha J_\alpha$. Each $z \in \overline{V_J}$ has an open neighbourhood O_z meeting only finitely many $\overline{V_i}$, so $O_z \cap \overline{V_J} \subseteq \overline{V_{J_\alpha}}$ for some α. For $y \in \overline{V_J}$ let σ_y be the germ of s_α at y for any α with $y \in \overline{V_{J_\alpha}}$, which does not depend on α because the pairs form a chain. Near each z these germs are those of a single s_α, so lemma 6.1.26 gives $s \in \mathscr{F}(\overline{V_J})$ with germ σ_y at every $y \in \overline{V_J}$. By its uniqueness clause s restricts to each s_α and maps to the image of t, so (J, s) bounds the chain, and Zorn gives a maximal (J, s).

Suppose some $k \in I' \setminus J$. On an open neighbourhood of $\overline{V_J} \cap \overline{V_k}$ both s and s_k lift t, so $s - s_k$ is a section of $\mathscr{F}'$ there. As $\mathscr{F}'$ is soft, it extends to some $u \in \mathscr{F}'(X)$, and s and $s_k + u$ agree on an open neighbourhood M of $\overline{V_J} \cap \overline{V_k}$. Let σ_z be the germ of s at $z \in \overline{V_J}$ and of $s_k + u$ at $z \in \overline{V_k}$, which is consistent on the overlap. Near a point of $\overline{V_J} \cap \overline{V_k}$ these germs are those of s on M, and near any other point they are those of s off $\overline{V_k}$ or of $s_k + u$ off $\overline{V_J}$. By lemma 6.1.26 they come from an element of $\mathscr{F}(\overline{V_{J \cup \{k\}}})$, which by the uniqueness clause restricts to s and maps to the

image of t, contradicting maximality. So $J = I'$, and s lifts t over a neighbourhood of $\overline{V_{I'}} \supseteq K$.

For $K = U = X$ this is the first claim. For the second, suppose $\mathscr{F}$ is soft. The lift s then extends to a global section of $\mathscr{F}$, whose image in $\mathscr{F}''(X)$ agrees with t near K.

Theorem 6.1.28: Soft Sheaves Are Acyclic

Let $\mathscr{F}$ be a soft sheaf on a paracompact Hausdorff space X. Then $H^i(X, \mathscr{F}) = 0$ for all $i > 0$.

***Proof*:**
Embed $\mathscr{F}$ in a flasque sheaf, $0 \to \mathscr{F} \to \mathscr{G} \to \mathscr{Q} \to 0$, which is possible because there are enough injectives (corollary 6.1.2) and injectives are flasque (proposition 6.1.5). By proposition 6.1.25 the sheaf $\mathscr{G}$ is soft, so $\mathscr{Q}$ is soft by lemma 6.1.27.

The long exact sequence in cohomology, together with $H^i(X, \mathscr{G}) = 0$ for $i > 0$ (corollary 6.1.7), gives

$$\mathscr{G}(X) \to \mathscr{Q}(X) \to H^1(X, \mathscr{F}) \to 0, \qquad H^i(X, \mathscr{F}) \cong H^{i-1}(X, \mathscr{Q}) \quad \text{for } i \geq 2$$

The first map is surjective by lemma 6.1.27, so $H^1(X, \mathscr{F}) = 0$, and this holds for every soft sheaf. Applying it to the soft sheaf $\mathscr{Q}$ and inducting on i kills the higher groups as well.

Theorem 6.1.29: Čech Cohomology on a Paracompact Space

Let X be a paracompact Hausdorff space and $\mathscr{F}$ a sheaf of abelian groups on X. Then for every i

$$\check{H}^i(X, \mathscr{F}) \cong H^i(X, \mathscr{F})$$

naturally in $\mathscr{F}$.

***Proof*:**
Step 1: Čech cohomology vanishes on flasque sheaves. If $\mathscr{G}$ is flasque, then $\check{H}^q(\mathcal{U}, \mathscr{G}) = 0$ for every open cover $\mathcal{U}$ and every $q \geq 1$ by lemma 6.1.16, so passing to the colimit over covers, $\check{H}^q(X, \mathscr{G}) = 0$.

Step 2: presheaves that vanish locally. This is where paracompactness enters. The complexes $C^\bullet(\mathcal{U}, \mathscr{P})$ and the groups $\check{H}^\bullet(X, \mathscr{P})$ are defined for a presheaf $\mathscr{P}$ by the same formulas. Suppose the sheafification of $\mathscr{P}$ is zero, so that every section of $\mathscr{P}$ vanishes locally. We show $\check{H}^q(X, \mathscr{P}) = 0$ for every q. Let $c \in C^q(\mathcal{U}, \mathscr{P})$ be a cocycle. After refining, $\mathcal{U} = \{U_i\}_{i \in I}$ is locally finite. Choose a shrinking $\{W_i\}$ with $\overline{W_i} \subseteq U_i$. The family $\{\overline{W_i}\}$ is again locally finite. For each $x \in X$ choose $\tau(x)$ with $x \in W_{\tau(x)}$ and an open neighbourhood V_x of x such that

(a) $V_x \subseteq W_{\tau(x)}$, and $V_x \subseteq U_i$ for each of the finitely many i with $x \in U_i$;

(b) V_x meets no $\overline{W_i}$ with $x \notin \overline{W_i}$;

(c) $c_{i_0 \cdots i_q}|_{V_x} = 0$ whenever $x \in U_{i_0} \cap \cdots \cap U_{i_q}$.

Condition (b) is possible because the union of the $\overline{W_i}$ not containing x is closed, and (c) consists of finitely many conditions, each satisfiable because sections of $\mathscr{P}$ vanish locally. The

cover $\mathcal{V} = \{V_x\}_{x \in X}$ refines $\mathcal{U}$ via τ. If $V_{x_0} \cap \cdots \cap V_{x_q} \neq \emptyset$, then V_{x_0} meets each $W_{\tau(x_k)}$, so $x_0 \in \overline{W_{\tau(x_k)}} \subseteq U_{\tau(x_k)}$ by (b), and (c) at x_0 gives $(\tau^q c)_{x_0 \cdots x_q} = c_{\tau(x_0)\cdots\tau(x_q)}|_{V_{x_0\cdots x_q}} = 0$. So the class of c dies in the colimit.

Step 3: Čech cohomology has a long exact sequence. Let $0 \to \mathscr{F}' \to \mathscr{F} \to \mathscr{F}'' \to 0$ be exact and let $\mathscr{P}$ be the presheaf $U \mapsto \mathscr{F}(U)/\mathscr{F}'(U)$, a subpresheaf of $\mathscr{F}''$. For every cover the sequence $0 \to C^\bullet(\mathcal{U}, \mathscr{F}') \to C^\bullet(\mathcal{U}, \mathscr{F}) \to C^\bullet(\mathcal{U}, \mathscr{P}) \to 0$ is exact, and filtered colimits are exact, so $\check{H}^\bullet(X, -)$ has a long exact sequence for $0 \to \mathscr{F}' \to \mathscr{F} \to \mathscr{P} \to 0$, and likewise for $0 \to \mathscr{P} \to \mathscr{F}'' \to \mathscr{F}''/\mathscr{P} \to 0$. The presheaf $\mathscr{F}''/\mathscr{P}$ has zero sheafification because $\mathscr{F} \to \mathscr{F}''$ is surjective on stalks, so Step 2 gives $\check{H}^\bullet(X, \mathscr{P}) \cong \check{H}^\bullet(X, \mathscr{F}'')$. Substituting yields a long exact sequence for $0 \to \mathscr{F}' \to \mathscr{F} \to \mathscr{F}'' \to 0$, natural in the sequence.

Step 4: comparison. Both $\check{H}^\bullet(X, -)$ and $H^\bullet(X, -)$ are now δ-functors on sheaves of abelian groups over X; they agree in degree 0, both being $\Gamma(X, -)$; and both vanish in positive degrees on flasque sheaves, by Step 1 and corollary 6.1.7. Since every sheaf embeds in a flasque one, both are effaceable, hence universal, so the identification in degree 0 extends to an isomorphism of δ-functors (*Homological Algebra* [18, An Effaceable δ-Functor Is Universal, A Universal δ-Functor Is Determined by Its Degree-Zero Part]).

This is the form that applies outside algebraic geometry. A complex manifold is paracompact, its sheaves of smooth forms are fine, and theorem 6.1.28 together with proposition 6.1.25 is what makes a resolution by smooth forms compute sheaf cohomology.

6.2 Vanishing Theorems

The comparison of the previous section rested on the vanishing in positive degrees of the cohomology of quasi-coherent sheaves on affine Noetherian schemes. This section proves that result, and before it a second: cohomology vanishes above the dimension of the space. These are the foundations on which virtually all further cohomological arguments rest.

The first result — the *dimension theorem* — says that if X is a Noetherian topological space of dimension n, then $H^i(X, \mathscr{F}) = 0$ for all $i > n$ and all sheaves of abelian groups $\mathscr{F}$. This is the scheme-theoretic analogue of the general principle that cohomology should not exceed the dimension of the underlying space[4].

The second result — the *affine vanishing theorem* — says that on an affine Noetherian scheme $X = \operatorname{Spec} R$, all quasi-coherent sheaves have trivial higher cohomology: $H^i(X, \mathscr{F}) = 0$ for $i > 0$. This confirms the philosophy that cohomology measures the obstruction to gluing: on an affine scheme, there is no gluing to be done. Conversely, we shall show that this vanishing property *characterizes* affine schemes among Noetherian schemes.

6.2.1 Dimension Theorem for Schemes

We begin with two lemmas about the behaviour of cohomology on Noetherian spaces. The Noetherian hypothesis enters through a single but crucial property: on Noetherian topological spaces, filtered colimits of sheaves commute with the global section functor. This fails in general.

[4]Recall from [57] that the corresponding statement in singular cohomology is that $H^i(M; \mathbb{Z}) = 0$ for $i > \dim M$ when M is a manifold; in de Rham cohomology, it is the vanishing of the de Rham groups above the manifold dimension.

Lemma 6.2.1: Noetherian Space and Flasque Sheaves

Let X be a Noetherian topological space. Then a filtered colimit of flasque sheaves on X is flasque.

***Proof*:**
Let $(\mathscr{F}_\alpha)$ be a filtered (directed) system of flasque sheaves on X, and let $U \subseteq V$ be open subsets. For each α, the restriction map $\mathscr{F}_\alpha(V) \to \mathscr{F}_\alpha(U)$ is surjective. Since the system is filtered, the colimit functor on abelian groups is exact, so:

$$\varinjlim_\alpha \mathscr{F}_\alpha(V) \longrightarrow \varinjlim_\alpha \mathscr{F}_\alpha(U)$$

is surjective. Now, the key point is that on a Noetherian topological space, every open subset is quasi-compact, so filtered colimits of sheaves commute with the section functor:

$$\big(\varinjlim_\alpha \mathscr{F}_\alpha\big)(U) \;\cong\; \varinjlim_\alpha \mathscr{F}_\alpha(U)$$

for every open U. This is *not* true in general: for a presheaf the colimit is computed section-wise, but for a sheaf one may need to sheafify. On a Noetherian space, however, every open cover has a finite subcover, so any section of the sheafification already comes from some $\mathscr{F}_\alpha$, and sheafification is unnecessary. Combining the two displayed equations, the restriction map

$$\big(\varinjlim_\alpha \mathscr{F}_\alpha\big)(V) \longrightarrow \big(\varinjlim_\alpha \mathscr{F}_\alpha\big)(U)$$

is surjective, so $\varinjlim_\alpha \mathscr{F}_\alpha$ is flasque.

Lemma 6.2.2: Colimit and Cohomology Commuting

Let X be a Noetherian topological space and $(\mathscr{F}_\alpha)$ a filtered direct system of sheaves of abelian groups. Then for each $i \geq 0$ there is a natural isomorphism:

$$\varinjlim_\alpha H^i(X, \mathscr{F}_\alpha) \;\cong\; H^i(X, \varinjlim_\alpha \mathscr{F}_\alpha).$$

***Proof*:**
For each $\mathscr{F}_\alpha$, choose a flasque resolution $0 \to \mathscr{F}_\alpha \to \mathscr{I}_\alpha^\bullet$ functorially (for instance, using the Godement resolution from Example 6.1.8(3), which is canonical). The cohomology groups are then:

$$H^i(X, \mathscr{F}_\alpha) = h^i\big(\Gamma(X, \mathscr{I}_\alpha^\bullet)\big).$$

Since filtered colimits are exact in the category of abelian sheaves, taking the colimit of these resolutions yields an exact sequence:

$$0 \longrightarrow \varinjlim_\alpha \mathscr{F}_\alpha \longrightarrow \varinjlim_\alpha \mathscr{I}_\alpha^\bullet.$$

By lemma 6.2.1, since X is Noetherian, each term $\varinjlim_\alpha \mathscr{I}_\alpha^p$ is flasque. Hence $\varinjlim_\alpha \mathscr{I}_\alpha^\bullet$ is a flasque resolution of $\varinjlim_\alpha \mathscr{F}_\alpha$, and we may use it to compute cohomology:

$$H^i(X, \varinjlim_\alpha \mathscr{F}_\alpha) = h^i\big(\Gamma(X, \varinjlim_\alpha \mathscr{I}_\alpha^\bullet)\big).$$

On the other hand, filtered colimits of abelian groups commute with the cohomology functor h^i (since filtered colimits are exact):

$$\varinjlim_\alpha H^i(X, \mathscr{F}_\alpha) = \varinjlim_\alpha h^i\big(\Gamma(X, \mathscr{I}_\alpha^\bullet)\big) \cong h^i\big(\varinjlim_\alpha \Gamma(X, \mathscr{I}_\alpha^\bullet)\big).$$

It remains to show $\varinjlim_\alpha \Gamma(X, \mathscr{I}_\alpha^p) \cong \Gamma(X, \varinjlim_\alpha \mathscr{I}_\alpha^p)$ for each p. This holds because global sections commute with filtered colimits of sheaves on Noetherian spaces (the same property used in the proof of lemma 6.2.1). Combining these identifications completes the proof.

Corollary 6.2.3: Direct Sum and Cohomology Commuting

Let X be a Noetherian topological space and $(\mathscr{F}_\alpha)_{\alpha \in A}$ a family of sheaves of abelian groups. Then for each $i \geq 0$:

$$\bigoplus_{\alpha \in A} H^i(X, \mathscr{F}_\alpha) \;\cong\; H^i\Big(X, \bigoplus_{\alpha \in A} \mathscr{F}_\alpha\Big).$$

***Proof*:**
A direct sum is a filtered colimit of finite partial sums, and cohomology commutes with finite direct sums (since it is an additive functor). Hence the result follows from lemma 6.2.2.

The following result shows that cohomology computations on a closed subset can be performed on the ambient space by pushing forward:

Proposition 6.2.4: Cohomology of Closed Subsets

Let X be a topological space, $Y \subseteq X$ a closed subset, $j : Y \to X$ the inclusion, and $\mathscr{F}$ a sheaf of abelian groups on Y. Then for all $i \geq 0$:

$$H^i(Y, \mathscr{F}) \cong H^i(X, j_*\mathscr{F}).$$

***Proof*:**
Let $0 \to \mathscr{F} \to \mathscr{I}^\bullet$ be a flasque resolution of $\mathscr{F}$ on Y. Each $j_*\mathscr{I}^p$ is flasque, since for $U \subseteq V$ open in X the surjection $\mathscr{I}^p(V \cap Y) \twoheadrightarrow \mathscr{I}^p(U \cap Y)$ is precisely the restriction map for $j_*\mathscr{I}^p$. Because Y is closed, $(j_*\mathscr{G})_x = \mathscr{G}_x$ for $x \in Y$ and $(j_*\mathscr{G})_x = 0$ for $x \notin Y$, for any sheaf $\mathscr{G}$ on Y, so j_* is exact. Hence the sequence $0 \to j_*\mathscr{F} \to j_*\mathscr{I}^\bullet$ is a flasque resolution of $j_*\mathscr{F}$ on X. Now:

$$H^i(X, j_*\mathscr{F}) = h^i\big(\Gamma(X, j_*\mathscr{I}^\bullet)\big) = h^i\big(\Gamma(Y, \mathscr{I}^\bullet)\big) = H^i(Y, \mathscr{F}),$$

where the middle equality uses $\Gamma(X, j_*\mathscr{G}) = \mathscr{G}(j^{-1}(X)) = \mathscr{G}(Y) = \Gamma(Y, \mathscr{G})$ for any sheaf $\mathscr{G}$ on Y.

By virtue of this result, we shall often write $\mathscr{F}$ in place of $j_*\mathscr{F}$ when Y is a closed subset of X, without risk of ambiguity: the cohomology groups are the same. With these tools, we can prove our first main vanishing theorem:

Theorem 6.2.5: Noetherian Space and Dimension

Let X be a Noetherian topological space of dimension n. Then for all sheaves of abelian groups $\mathscr{F}$ on X:

$$H^i(X, \mathscr{F}) = 0 \qquad \text{for all } i > n.$$

***Proof*:**
We shall need the following notation. For a closed subset $Y \subseteq X$ with inclusion $j : Y \hookrightarrow X$, write $\mathscr{F}_Y = j_*(j^{-1}\mathscr{F})$ for the "restriction of $\mathscr{F}$ to Y, pushed back to X." For an open subset $U \subseteq X$ with inclusion $i : U \hookrightarrow X$, write $\mathscr{F}_U = i_!(i^{-1}\mathscr{F})$, the extension by zero of $\mathscr{F}|_U$. If $U = X \setminus Y$, checking on stalks gives a short exact sequence:

$$0 \longrightarrow \mathscr{F}_U \longrightarrow \mathscr{F} \longrightarrow \mathscr{F}_Y \longrightarrow 0.$$

We prove the theorem by a double induction: on the dimension of X and on the number of irreducible components.

Step 1: Reduction to X irreducible. Suppose X is reducible. Let Y be an irreducible component of X and $U = X \setminus Y$. The short exact sequence above gives a long exact sequence:

$$\cdots \to H^i(X, \mathscr{F}_U) \to H^i(X, \mathscr{F}) \to H^i(X, \mathscr{F}_Y) \to \cdots$$

It suffices to show vanishing for $\mathscr{F}_Y$ and $\mathscr{F}_U$ separately. For $\mathscr{F}_Y$: since Y is closed and irreducible with $\dim Y \leq n$, proposition 6.2.4 gives $H^i(X, \mathscr{F}_Y) = H^i(Y, j^{-1}\mathscr{F})$, which vanishes for $i > n$ by the inductive hypothesis applied to the irreducible space Y. For $\mathscr{F}_U$: the support of $\mathscr{F}_U$ is contained in $\overline{U}$, which is a closed subset with strictly fewer irreducible components than X (we have removed the component Y). By proposition 6.2.4 applied to $\overline{U}$ and induction on the number of irreducible components, we get the required vanishing. Hence we may assume X is irreducible.

Step 2: Base case $\dim X = 0$. If X is irreducible of dimension 0, its only open subsets are $\emptyset$ and X itself (a proper closed irreducible subset would give $\dim X \geq 1$). Hence the global section functor $\Gamma(X, -) : \mathbf{Ab}(X) \to \mathbf{Ab}$ is an equivalence of categories (a sheaf on X is determined by a single abelian group). In particular, $\Gamma(X, -)$ is exact, so all its derived functors vanish: $H^i(X, \mathscr{F}) = 0$ for $i > 0$.

Step 3: Reduction to $\underline{\mathbb{Z}}_U$ and its subsheaves. Now let X be irreducible of dimension $n \geq 1$ and $\mathscr{F}$ arbitrary. For a finite set α of sections of $\mathscr{F}$, each over some open subset, let $\mathscr{F}_\alpha \subseteq \mathscr{F}$ be the subsheaf they generate. These subsheaves form a filtered system under inclusion with colimit $\mathscr{F}$, so by lemma 6.2.2 it suffices to prove the vanishing for each $\mathscr{F}_\alpha$. If $\alpha = \alpha' \cup \{s\}$, then $\mathscr{F}_\alpha/\mathscr{F}_{\alpha'}$ is generated by the image of the single section s, and the long exact sequence of

$$0 \longrightarrow \mathscr{F}_{\alpha'} \longrightarrow \mathscr{F}_\alpha \longrightarrow \mathscr{F}_\alpha/\mathscr{F}_{\alpha'} \longrightarrow 0$$

reduces the vanishing for $\mathscr{F}_\alpha$, by induction on the number of elements of α, to the case of a sheaf $\mathscr{G}$ generated by a single section $s \in \mathscr{G}(U)$. Such a sheaf is a quotient of $\underline{\mathbb{Z}}_U$, by the map

sending 1 to s:

$$0 \longrightarrow \mathcal{K} \longrightarrow \underline{\mathbb{Z}}_U \longrightarrow \mathscr{G} \longrightarrow 0,$$

with $\mathcal{K}$ a subsheaf of $\underline{\mathbb{Z}}_U$. The long exact sequence $H^i(X, \underline{\mathbb{Z}}_U) \to H^i(X, \mathscr{G}) \to H^{i+1}(X, \mathcal{K})$ shows that it suffices to prove, for every open $U \subseteq X$ and every $i > n$, that $H^i(X, \mathcal{K}) = 0$ for each subsheaf $\mathcal{K} \subseteq \underline{\mathbb{Z}}_U$ and that $H^i(X, \underline{\mathbb{Z}}_U) = 0$.

Step 4: Subsheaves of $\underline{\mathbb{Z}}_U$. Let $U \subseteq X$ be open and $\mathcal{K} \subseteq \underline{\mathbb{Z}}_U$ a subsheaf. If $\mathcal{K} = 0$, there is nothing to prove. Otherwise, for each $x \in U$, the stalk $\mathcal{K}_x$ is a subgroup of $\mathbb{Z}$, hence either 0 or $d_x\mathbb{Z}$ for some positive integer d_x. Let d be the smallest positive integer occurring among the stalks. There is a non-empty open subset $V \subseteq U$ on which $\mathcal{K}|_V \cong d \cdot \underline{\mathbb{Z}}|_V \cong \underline{\mathbb{Z}}_V$ (as a subsheaf of $\underline{\mathbb{Z}}|_V$). Thus $\mathcal{K}_V \cong \underline{\mathbb{Z}}_V$, and we have:

$$0 \longrightarrow \underline{\mathbb{Z}}_V \longrightarrow \mathcal{K} \longrightarrow \mathcal{K}/\underline{\mathbb{Z}}_V \longrightarrow 0.$$

The quotient $\mathcal{K}/\underline{\mathbb{Z}}_V$ is supported on the closure $\overline{U \setminus V}$, which is a proper closed subset of X. Since X is irreducible, $\dim \overline{U \setminus V} < n$. By proposition 6.2.4 and the inductive hypothesis on dimension, we get $H^i(X, \mathcal{K}/\underline{\mathbb{Z}}_V) = 0$ for $i \geq n$. From the long exact sequence, it suffices to prove vanishing for $\underline{\mathbb{Z}}_V$.

Step 5: Completing the proof for $\underline{\mathbb{Z}}_U$. It remains to show that for any open subset $U \subseteq X$, we have $H^i(X, \underline{\mathbb{Z}}_U) = 0$ for $i > n$. Let $Y = X \setminus U$ and consider the exact sequence:

$$0 \longrightarrow \underline{\mathbb{Z}}_U \longrightarrow \underline{\mathbb{Z}} \longrightarrow \underline{\mathbb{Z}}_Y \longrightarrow 0.$$

Since X is irreducible, the constant sheaf $\underline{\mathbb{Z}}$ on X is flasque (every non-empty open subset of an irreducible space is connected, so the constant sheaf has the same sections on every non-empty open; see Example 6.1.8(2)). Hence $H^i(X, \underline{\mathbb{Z}}) = 0$ for all $i > 0$. Furthermore, Y is a proper closed subset of the irreducible space X, so $\dim Y < n$. By proposition 6.2.4 and the inductive hypothesis on dimension, $H^i(X, \underline{\mathbb{Z}}_Y) = 0$ for $i \geq n$. The long exact sequence now gives $H^i(X, \underline{\mathbb{Z}}_U) = 0$ for $i > n$, completing the proof.

6.2.2 Vanishing Theorem for Affine Schemes

We now turn to the second fundamental vanishing result: on an affine Noetherian scheme $X = \operatorname{Spec} R$, all quasi-coherent sheaves have trivial higher cohomology. The key insight is that if I is an injective R-module, then the associated sheaf $\widetilde{I}$ on $\operatorname{Spec} R$ is flasque. This is not obvious: the sections of $\widetilde{I}$ on a general open set involve a limit over distinguished opens, and there is no a priori reason for restriction maps to be surjective. The proof requires the Noetherian hypothesis in an essential way, via the Artin–Rees lemma and Noetherian induction.

Let R be a Noetherian ring, M an R-module, and $\mathfrak{a} \subseteq R$ an ideal. The $\mathfrak{a}$-*torsion submodule* of M is:

$$\Gamma_{\mathfrak{a}}(M) = \left\{ m \in M \;\middle|\; \mathfrak{a}^k m = 0 \text{ for some } k > 0 \right\}.$$

Lemma 6.2.6: $\mathfrak{a}$-Torsion of an Injective Module is Injective

Let R be a Noetherian ring, $\mathfrak{a} \subseteq R$ an ideal, and I an injective R-module. Then the submodule $J = \Gamma_{\mathfrak{a}}(I)$ is injective.

***Proof*:**
By Baer's criterion, it suffices to show that for any ideal $\mathfrak{b} \subseteq R$ and any homomorphism $\varphi : \mathfrak{b} \to J$, there exists a homomorphism $\psi : R \to J$ extending φ.

Since R is Noetherian, $\mathfrak{b}$ is finitely generated, and every element of J is annihilated by some power of $\mathfrak{a}$. Hence there exists $n > 0$ such that $\mathfrak{a}^n \cdot \varphi(\mathfrak{b}) = 0$, i.e. $\varphi(\mathfrak{a}^n \mathfrak{b}) = 0$. By the Artin–Rees lemma (see [10, Artin-Rees Lemma]) applied to the inclusion $\mathfrak{b} \subseteq R$ and the $\mathfrak{a}$-adic filtration, there exists $n' \geq n$ such that $\mathfrak{a}^n \mathfrak{b} \supseteq \mathfrak{b} \cap \mathfrak{a}^{n'}$. Hence $\varphi(\mathfrak{b} \cap \mathfrak{a}^{n'}) = 0$, and φ factors through $\mathfrak{b}/(\mathfrak{b} \cap \mathfrak{a}^{n'})$.

Consider the commutative diagram:

$$\begin{array}{ccccccc} R & \twoheadrightarrow & R/\mathfrak{a}^{n'} & & & & \\ \uparrow & & \uparrow & \overset{\psi'}{\searrow} & & & \\ \mathfrak{b} & \twoheadrightarrow & \mathfrak{b}/(\mathfrak{b} \cap \mathfrak{a}^{n'}) & \longrightarrow & J & \hookrightarrow & I \end{array}$$

Since I is injective, the composed map $\mathfrak{b}/(\mathfrak{b} \cap \mathfrak{a}^{n'}) \to I$ extends to a map $\psi' : R/\mathfrak{a}^{n'} \to I$. The image of ψ' is annihilated by $\mathfrak{a}^{n'}$, so it is contained in $J = \Gamma_{\mathfrak{a}}(I)$. Composing with the natural projection $R \twoheadrightarrow R/\mathfrak{a}^{n'}$, we obtain the required extension $\psi : R \to J$.

Lemma 6.2.7: Localization of Injective Modules is Surjective

Let R be a Noetherian ring, I an injective R-module, and $f \in R$. Then the natural localization map $I \to I_f$ is surjective.

***Proof*:**
Let $\varphi : I \to I_f$ be the natural map and let $x \in I_f$. By definition of localization, there exist $y \in I$ and $n \geq 0$ such that $x = \varphi(y)/f^n$.

For each $i > 0$, let $\mathfrak{b}_i = \operatorname{Ann}_R(f^i)$. Since R is Noetherian, the ascending chain $\mathfrak{b}_1 \subseteq \mathfrak{b}_2 \subseteq \cdots$ stabilizes: there exists m such that $\mathfrak{b}_{m+j} = \mathfrak{b}_m$ for all $j \geq 0$. Define a homomorphism $\psi : (f^{n+m}) \to I$ by $f^{n+m} \mapsto f^m y$. This is well-defined: if $r f^{n+m} = 0$, then $r \in \mathfrak{b}_{n+m} = \mathfrak{b}_m$, so $r f^m = 0$, hence $r f^m y = 0$.

Since I is injective, Baer's criterion extends ψ to a map $\Psi : R \to I$. Let $z = \Psi(1)$. Then:

$$f^{n+m} z = f^{n+m} \Psi(1) = \Psi(f^{n+m}) = \psi(f^{n+m}) = f^m y,$$

so in I_f we have $\varphi(z) = z/1 = f^m y / f^{n+m} = y/f^n = x$. Hence φ is surjective.

Lemma 6.2.8: Associated Sheaf of Injective Module is Flasque

Let R be a Noetherian ring and I an injective R-module. Then the associated sheaf $\widetilde{I}$ on $X = \operatorname{Spec} R$ is flasque.

***Proof*:**

The space X is Noetherian (lemma 2.4.2). For a closed subset $Y \subseteq X$ let $P(Y)$ say that $\widetilde{I}$ is flasque for every injective R-module I whose support is contained in Y. By corollary 0.3.5 it suffices to prove $P(Y)$ assuming $P(Y')$ for every closed $Y' \subsetneq Y$, and taking $Y = X$ then gives the lemma. So let I be injective with support in Y.

It suffices to show that for any open set $U \subseteq X$, the restriction $\Gamma(X, \widetilde{I}) \to \Gamma(U, \widetilde{I})$ is surjective. If $Y \cap U = \emptyset$, then $\Gamma(U, \widetilde{I}) = 0$ and there is nothing to prove. Otherwise, choose $f \in R$ such that the distinguished open $D(f)$ satisfies $D(f) \subseteq U$ and $D(f) \cap Y \neq \emptyset$. Let $Z = X \setminus D(f)$, and for any open W let $\Gamma_Z(W, \widetilde{I})$ denote the sections of $\widetilde{I}$ over W with support in Z.

Given a section $s \in \Gamma(U, \widetilde{I})$, its restriction to $D(f)$ yields an element $s' \in \Gamma(D(f), \widetilde{I})$. By the identification $\Gamma(D(f), \widetilde{I}) \cong I_f$ (from definition 5.1.6) and the surjectivity of $I \to I_f$ (lemma 6.2.7), there exists $t \in I = \Gamma(X, \widetilde{I})$ (proposition 5.1.14(2)) with $t|_{D(f)} = s'$. Let t' be the restriction of t to U. Then $s - t'$ restricts to zero on $D(f)$, so it has support in Z: $s - t' \in \Gamma_Z(U, \widetilde{I})$.

It remains to show that $\Gamma_Z(X, \widetilde{I}) \to \Gamma_Z(U, \widetilde{I})$ is surjective. Let $J = \Gamma_{\mathfrak{a}}(I)$ where $\mathfrak{a} = (f)$. Then $\Gamma_Z(X, \widetilde{I}) = J$ (the sections of $\widetilde{I}$ over all of X supported on $Z = V(f)$ are precisely the f-torsion elements of I). By lemma 6.2.6, J is an injective R-module. The support of $\widetilde{J}$ is contained in $Y \cap Z$, a closed set strictly contained in Y (since $D(f) \cap Y \neq \emptyset$, we have removed a non-empty open piece of Y). By the hypothesis $P(Y \cap Z)$, $\widetilde{J}$ is flasque. Since $\Gamma(U, \widetilde{J}) = \Gamma_Z(U, \widetilde{I})$, the flasqueness of $\widetilde{J}$ gives the required surjectivity.

Theorem 6.2.9: Affine Noetherian Scheme: Trivial Cohomology

Let $X = \operatorname{Spec} R$ be the spectrum of a Noetherian ring R, and let $\mathscr{F}$ be a quasi-coherent sheaf on X. Then:

$$H^i(X, \mathscr{F}) = 0 \qquad \text{for all } i > 0.$$

***Proof*:**

Since $\mathscr{F}$ is quasi-coherent on the affine scheme $X = \operatorname{Spec} R$, there exists an R-module M with $\mathscr{F} \cong \widetilde{M}$ (by corollary 5.1.12). Choose an injective resolution $0 \to M \to I^\bullet$ in the category of R-modules. The associated-sheaf functor $\widetilde{(-)}$ is exact (by proposition 5.1.14), so:

$$0 \longrightarrow \widetilde{M} \longrightarrow \widetilde{I}^\bullet$$

is an exact sequence of $\mathcal{O}_X$-modules. By lemma 6.2.8, each $\widetilde{I}^p$ is flasque. Hence $\widetilde{I}^\bullet$ is a flasque resolution of $\mathscr{F} = \widetilde{M}$, and we may use it to compute cohomology:

$$H^i(X, \mathscr{F}) = h^i\big(\Gamma(X, \widetilde{I}^\bullet)\big) = h^i(I^\bullet).$$

But $I^\bullet$ is an injective resolution of M, so its cohomology is $h^0(I^\bullet) = M$ and $h^i(I^\bullet) = 0$ for $i > 0$. Hence:

$$H^0(X, \mathscr{F}) = M = \Gamma(X, \mathscr{F}) \qquad \text{and} \qquad H^i(X, \mathscr{F}) = 0 \text{ for } i > 0.$$

As a useful corollary, we can embed any quasi-coherent sheaf on a Noetherian scheme into a quasi-coherent flasque sheaf.

Corollary 6.2.10: Injective Embedding of Sheaves On Noetherian Schemes

Let X be a Noetherian scheme and $\mathscr{F}$ a quasi-coherent sheaf on X. Then $\mathscr{F}$ can be embedded in a quasi-coherent flasque sheaf.

***Proof*:**
Since X is Noetherian, it is quasi-compact, so it has a finite affine open cover $X = \bigcup_{j=1}^{r} U_j$ with $U_j = \operatorname{Spec} R_j$. For each j, let $\mathscr{F}|_{U_j} \cong \widetilde{M_j}$ for the R_j-module $M_j = \Gamma(U_j, \mathscr{F})$. Embed each M_j into an injective R_j-module I_j, and define:

$$\mathscr{G} = \bigoplus_{j=1}^{r} (\iota_j)_*(\widetilde{I_j})$$

where $\iota_j : U_j \hookrightarrow X$ is the inclusion. For each j, the embedding $\mathscr{F}|_{U_j} \hookrightarrow \widetilde{I_j}$ induces $\mathscr{F} \to (\iota_j)_*(\widetilde{I_j})$ (by adjunction of restriction and pushforward). Taking the direct sum gives an injective morphism $\mathscr{F} \hookrightarrow \mathscr{G}$.

Each $\widetilde{I_j}$ is flasque by lemma 6.2.8 and quasi-coherent on U_j. The pushforward $(\iota_j)_*(\widetilde{I_j})$ is flasque (the pushforward of a flasque sheaf along any continuous map is flasque) and quasi-coherent by proposition 5.1.20[a]. A finite direct sum of flasque quasi-coherent sheaves is flasque and quasi-coherent, so $\mathscr{G}$ has the required properties.

[a] This is where the Noetherian hypothesis enters: the pushforward of a quasi-coherent sheaf along an open immersion is quasi-coherent when the target is Noetherian (or more generally, when the morphism is quasi-compact and quasi-separated), but can fail for non-Noetherian schemes.

The vanishing theorem shows that the higher cohomology of quasi-coherent sheaves detects non-affineness. The converse is also true:

Theorem 6.2.11: Characterizing Affine Schemes via Cohomology

Let X be a Noetherian scheme. Then the following are equivalent:

1. X is affine.
2. $H^i(X, \mathscr{F}) = 0$ for all $i > 0$ and all quasi-coherent sheaves $\mathscr{F}$ on X.
3. $H^1(X, \mathscr{I}) = 0$ for all coherent sheaves of ideals $\mathscr{I} \subseteq \mathcal{O}_X$.

***Proof*:**
(1) ⇒ (2): This is theorem 6.2.9.

(2) ⇒ (3): Every coherent sheaf of ideals is quasi-coherent, so this is immediate.

(3) ⇒ (1): We verify the hypotheses of corollary 3.1.15: we shall find elements $f_1, \dots, f_r \in R = \Gamma(X, \mathcal{O}_X)$ such that each X_{f_j} is affine and $(f_1, \dots, f_r) = R$.

Step A: Finding the f_j. Let $\mathfrak{p}$ be a closed point of X and let U be an affine open neighborhood of $\mathfrak{p}$. Let $Y = X \setminus U$. Then Y and $Y \cup \{\mathfrak{p}\}$ are closed subsets, and their ideal sheaves $\mathscr{I}_Y \supseteq \mathscr{I}_{Y \cup \{\mathfrak{p}\}}$

fit in a short exact sequence:

$$0 \longrightarrow \mathscr{I}_{Y\cup\{\mathfrak{p}\}} \longrightarrow \mathscr{I}_Y \longrightarrow \kappa(\mathfrak{p}) \longrightarrow 0,$$

where $\kappa(\mathfrak{p}) = \mathcal{O}_{X,\mathfrak{p}}/\mathfrak{m}_\mathfrak{p}$ is the skyscraper sheaf at $\mathfrak{p}$. The long exact sequence gives:

$$\Gamma(X, \mathscr{I}_Y) \longrightarrow \Gamma(X, \kappa(\mathfrak{p})) \longrightarrow H^1(X, \mathscr{I}_{Y\cup\{\mathfrak{p}\}}) = 0,$$

where the last group vanishes by hypothesis (3). Hence the map $\Gamma(X, \mathscr{I}_Y) \to \kappa(\mathfrak{p})$ is surjective, so there exists $f \in \Gamma(X, \mathscr{I}_Y) \subseteq R$ with $f_\mathfrak{p} \equiv 1 \pmod{\mathfrak{m}_\mathfrak{p}}$.

By construction, f vanishes on Y, so $\mathfrak{p} \in X_f \subseteq U$. Furthermore, $X_f = U_{\bar{f}}$ where $\bar{f}$ is the image of f in $\Gamma(U, \mathcal{O}_U)$, so X_f is a distinguished open of the affine scheme U, hence affine.

Every closed point of X has an affine neighborhood of the form X_f with $f \in R$. The complement of the union of these X_f is closed and contains no closed point of X, so it is empty, since the closure of any point of the quasi-compact X contains a closed point (proposition 2.4.4). By quasi-compactness we can cover X by finitely many such opens: $X = X_{f_1} \cup \cdots \cup X_{f_r}$.

Step B: Showing $(f_1, \ldots, f_r) = R$. The elements $f_1, \ldots, f_r \in R$ define a morphism of sheaves $\alpha : \mathcal{O}_X^r \to \mathcal{O}_X$ by $(a_1, \ldots, a_r) \mapsto \sum_j f_j a_j$. Since the X_{f_j} cover X, the map α is surjective on stalks, hence surjective. Let $\mathscr{F} = \ker \alpha$:

$$0 \longrightarrow \mathscr{F} \longrightarrow \mathcal{O}_X^r \xrightarrow{\alpha} \mathcal{O}_X \longrightarrow 0.$$

We need to show $H^1(X, \mathscr{F}) = 0$, for then $\Gamma(X, \mathcal{O}_X^r) \to \Gamma(X, \mathcal{O}_X)$ is surjective, giving $1 \in \operatorname{im}(\alpha)$, i.e. $\sum a_j f_j = 1$ for some $a_j \in R$, proving $(f_1, \ldots, f_r) = R$.

Filter $\mathscr{F}$ by:

$$0 = \mathscr{F} \cap \mathcal{O}_X^0 \subseteq \mathscr{F} \cap \mathcal{O}_X^1 \subseteq \cdots \subseteq \mathscr{F} \cap \mathcal{O}_X^r = \mathscr{F}$$

where $\mathcal{O}_X^j$ denotes the first j summands of $\mathcal{O}_X^r$. Each successive quotient $(\mathscr{F} \cap \mathcal{O}_X^{j+1})/(\mathscr{F} \cap \mathcal{O}_X^j)$ embeds into $\mathcal{O}_X^{j+1}/\mathcal{O}_X^j \cong \mathcal{O}_X$, so it is a coherent sheaf of ideals. By hypothesis (3), its H^1 vanishes. Climbing up the filtration via the long exact sequence, we deduce $H^1(X, \mathscr{F}) = 0$.

Hence $(f_1, \ldots, f_r) = R$, and by corollary 3.1.15, X is affine.

6.3 Application: Sheaf Cohomology of Projective Spaces

Let us now compute the cohomology of twisted sheaves on projective space — the single most important explicit computation in the cohomology of schemes.

6.3.1 Cohomology of Twisted Sheaves

Let R be a Noetherian ring, $S_\bullet = R[x_0, \ldots, x_n]$ with the standard grading, and $X = \operatorname{Proj} S_\bullet = \mathbb{P}^n_R$. Recall from definition 5.3.4 that the Serre twisting sheaf $\mathcal{O}_X(1)$ is the invertible sheaf whose sections on the basic open $D_+(x_i)$ are the degree-1 elements $(S_{x_i})_1$ of S_{x_i}. For any sheaf $\mathscr{F}$ of $\mathcal{O}_X$-modules, we write:

$$\Gamma_*(\mathscr{F}) = \bigoplus_{d\in\mathbb{Z}} \Gamma(X, \mathscr{F}(d)),$$

which carries a natural structure of graded $S_\bullet$-module (multiplication by x_i corresponds to the natural map $\mathscr{F}(d) \to \mathscr{F}(d+1)$).

Theorem 6.3.1: Twisted Sheaf Cohomology

Let R be a Noetherian ring and let $X = \mathbb{P}^n_R$, with $n \geq 1$. Then:

1. $H^0(X, \mathcal{O}_X(d)) \cong (S_\bullet)_d$ for all $d \geq 0$, and $H^0(X, \mathcal{O}_X(d)) = 0$ for $d < 0$.
2. $H^n(X, \mathcal{O}_X(-n-1)) \cong R$.
3. The natural multiplication map
$$H^0(X, \mathcal{O}_X(d)) \times H^n(X, \mathcal{O}_X(-d-n-1)) \longrightarrow H^n(X, \mathcal{O}_X(-n-1)) \cong R$$
is a perfect pairing of finitely generated free R-modules, for each $d \geq 0$.
4. $H^i(X, \mathcal{O}_X(d)) = 0$ for $0 < i < n$ and all $d \in \mathbb{Z}$.

In summary:
$$H^i\big(\mathbb{P}^n_R,\, \mathcal{O}_{\mathbb{P}^n_R}(d)\big) = \begin{cases} (R[x_0, \dots, x_n])_d & i = 0,\ d \geq 0, \\ \left(\dfrac{1}{x_0 x_1 \cdots x_n} R[x_0^{-1}, \dots, x_n^{-1}]\right)_d & i = n,\ d \leq -(n+1), \\ 0 & \text{otherwise.} \end{cases}$$

6.3.2 Remark: Sheaf vs. Singular Cohomology The reader should note that this result is fundamentally different from the singular cohomology of complex projective space. When $R = \mathbb{C}$ and $d = 0$, the theorem says $H^i(\mathbb{P}^n_{\mathbb{C}}, \mathcal{O}_{\mathbb{P}^n_{\mathbb{C}}}) = 0$ for all $i > 0$, whereas the singular cohomology $H^i_{\text{sing}}(\mathbb{CP}^n; \mathbb{Z})$ is nontrivial: it equals $\mathbb{Z}$ in each even degree $0 \leq i \leq 2n$ and vanishes in odd degrees (see [57, The Cohomology Ring of Complex Projective Space]). The discrepancy arises because sheaf cohomology with coefficients in $\mathcal{O}_X$ measures obstructions to extending *holomorphic* (or algebraic) sections, not topological ones. Singular cohomology is instead the sheaf cohomology of the constant sheaf $\underline{\mathbb{Z}}$ on the analytic space X^{an} with its classical topology, and section 6.9 relates it, with $\mathbb{C}$ coefficients, to algebraic de Rham cohomology.

***Proof*:**

We use Čech cohomology with respect to the standard affine cover. Let $\mathscr{F} = \bigoplus_{d \in \mathbb{Z}} \mathcal{O}_X(d)$. By corollary 6.2.3, cohomology commutes with direct sums on Noetherian spaces, so $H^i(X, \mathscr{F}) = \bigoplus_{d \in \mathbb{Z}} H^i(X, \mathcal{O}_X(d))$. Hence it suffices to compute the cohomology of $\mathscr{F}$ as a graded module and then read off each graded piece.

Let $\mathcal{U} = \{U_0, \dots, U_n\}$ where $U_j = D_+(x_j)$ is the standard affine cover of $\operatorname{Proj} S_\bullet$. By theorem 6.1.19, the Čech cohomology $\check{H}^p(\mathcal{U}, \mathscr{F})$ computes the derived-functor cohomology $H^p(X, \mathscr{F})$. The sections of $\mathscr{F}$ on each intersection are:
$$\mathscr{F}(U_{i_0, \dots, i_p}) = S_{x_{i_0} \cdots x_{i_p}}$$

(the localization of $S_\bullet$ at the product $x_{i_0}\cdots x_{i_p}$, retaining only degree-0 elements for each summand $\mathcal{O}_X(d)$, but over all d we obtain the full localization). The Čech complex is therefore:

$$C^\bullet(\mathcal{U},\mathscr{F}): \quad \prod_i S_{x_i} \xrightarrow{d^0} \prod_{i<j} S_{x_ix_j} \xrightarrow{d^1} \cdots \xrightarrow{d^{n-1}} S_{x_0x_1\cdots x_n}.$$

Proof of (1): Computing H^0. By lemma 6.1.14, $H^0(X,\mathscr{F}) = \Gamma(X,\mathscr{F})$. The global sections of $\mathscr{F}$ are the elements lying in all localizations simultaneously: $\Gamma(X,\mathscr{F}) = \bigcap_{i=0}^n S_{x_i} \cap S = S$ by proposition 5.3.10. The graded piece of degree d is $(S_\bullet)_d$, which is a free R-module with basis $\{x_0^{m_0}\cdots x_n^{m_n} \big| m_i \geq 0, \sum m_i = d\}$ when $d \geq 0$, and is zero when $d < 0$ (there are no monomials of negative degree in a polynomial ring). Hence $H^0(X,\mathcal{O}_X(d)) \cong (S_\bullet)_d$ for $d \geq 0$ and $H^0(X,\mathcal{O}_X(d)) = 0$ for $d < 0$.

Proof of (2): Computing H^n. The group $H^n(X,\mathscr{F})$ is the cokernel of the last differential:

$$d^{n-1} : \prod_{j=0}^n S_{x_0\cdots\widehat{x_j}\cdots x_n} \longrightarrow S_{x_0x_1\cdots x_n}.$$

The target $S_{x_0\cdots x_n}$ is a free R-module with basis consisting of all Laurent monomials $\{x_0^{l_0}\cdots x_n^{l_n} \big| l_i \in \mathbb{Z}\}$. The image of d^{n-1} is the R-submodule generated by those Laurent monomials $x_0^{l_0}\cdots x_n^{l_n}$ for which at least one $l_j \geq 0$: indeed, a monomial with $l_j \geq 0$ lies in $S_{x_0\cdots\widehat{x_j}\cdots x_n}$, so it is in the image of the j-th summand. Conversely, every element of each summand has at least one non-negative exponent.

Therefore, $H^n(X,\mathscr{F})$ is the free R-module with basis

$$\{x_0^{l_0}\cdots x_n^{l_n} \big| l_i < 0 \text{ for all } i\},$$

graded by total degree $\sum l_i$. In degree $-(n+1)$, there is exactly one such monomial, namely $x_0^{-1}x_1^{-1}\cdots x_n^{-1}$. Hence:

$$H^n(X,\mathcal{O}_X(-n-1)) \cong R.$$

More generally, for $d \leq -(n+1)$, the group $H^n(X,\mathcal{O}_X(d))$ is a free R-module of rank $\binom{-d-1}{n}$, and $H^n(X,\mathcal{O}_X(d)) = 0$ for $d > -(n+1)$ (since there are no "all-negative" monomials of degree greater than $-(n+1)$).

Proof of (3): The perfect pairing. For $d \geq 0$, the module $H^0(X,\mathcal{O}_X(d))$ is free over R with basis $\{x_0^{m_0}\cdots x_n^{m_n} \big| m_i \geq 0, \sum m_i = d\}$, and $H^n(X,\mathcal{O}_X(-d-n-1))$ is free with basis $\{x_0^{l_0}\cdots x_n^{l_n} \big| l_i < 0, \sum l_i = -d-n-1\}$. The multiplication map

$$H^0(X,\mathcal{O}_X(d)) \times H^n(X,\mathcal{O}_X(-d-n-1)) \longrightarrow H^n(X,\mathcal{O}_X(-n-1)) \cong R$$

sends $(x_0^{m_0}\cdots x_n^{m_n})\cdot(x_0^{l_0}\cdots x_n^{l_n}) = x_0^{m_0+l_0}\cdots x_n^{m_n+l_n}$, and this product is nonzero in H^n if and only if all exponents remain strictly negative, i.e. $m_j + l_j < 0$ for all j. In particular, the element $x_0^{-m_0-1}\cdots x_n^{-m_n-1}$ (which has total degree $-d-n-1$) pairs with $x_0^{m_0}\cdots x_n^{m_n}$ to give $x_0^{-1}\cdots x_n^{-1}$, the generator of $H^n(X,\mathcal{O}_X(-n-1)) \cong R$. Any other basis monomial in $H^n(X,\mathcal{O}_X(-d-n-1))$ pairs with $x_0^{m_0}\cdots x_n^{m_n}$ to produce a monomial with at least one non-negative exponent, hence zero in H^n.

This shows that the monomial bases are dual to each other under the pairing, so the pairing is perfect.

Proof of (4): Vanishing of intermediate cohomology. We proceed by induction on n. For $n = 1$, the range $0 < i < 1$ is empty, so there is nothing to prove.

Let $n > 1$. We first show that every element of $H^i(X, \mathscr{F})$, for $0 < i < n$, is annihilated by some power of x_n. Localizing the Čech complex $C^\bullet(\mathcal{U}, \mathscr{F})$ at x_n yields the Čech complex of $\mathscr{F}|_{U_n}$ on the affine scheme $U_n = D_+(x_n)$ with respect to the open cover $\{U_j \cap U_n \big| 0 \le j \le n\}$. Each set in this cover is a distinguished affine open of U_n, and U_n itself is one of them. Since U_n is affine and $\mathscr{F}|_{U_n}$ is quasi-coherent, $H^i(U_n, \mathscr{F}|_{U_n}) = 0$ for $i > 0$ by theorem 6.2.9. Since localization is exact, it follows that $H^i(X, \mathscr{F})_{x_n} = 0$ for $i > 0$. In other words, every element of $H^i(X, \mathscr{F})$ is annihilated by some power of x_n.

Next, we show that for $0 < i < n$, multiplication by x_n induces a bijection on $H^i(X, \mathscr{F})$, forcing this module to be zero. Consider the exact sequence of graded $S_\bullet$-modules:

$$0 \longrightarrow S_\bullet(-1) \xrightarrow{\cdot x_n} S_\bullet \longrightarrow S_\bullet/(x_n) \longrightarrow 0,$$

which gives the short exact sequence of sheaves on X:

$$0 \longrightarrow \mathcal{O}_X(-1) \xrightarrow{\cdot x_n} \mathcal{O}_X \longrightarrow \mathcal{O}_H \longrightarrow 0,$$

where $H = V_+(x_n) \cong \mathbb{P}^{n-1}_R$ is the hyperplane $x_n = 0$. Twisting by all $d \in \mathbb{Z}$ and taking the direct sum:

$$0 \longrightarrow \mathscr{F}(-1) \xrightarrow{\cdot x_n} \mathscr{F} \longrightarrow \mathscr{F}_H \longrightarrow 0,$$

where $\mathscr{F}_H = \bigoplus_{d \in \mathbb{Z}} \mathcal{O}_H(d)$. The long exact sequence in cohomology gives:

$$\cdots \longrightarrow H^i(X, \mathscr{F}(-1)) \xrightarrow{\cdot x_n} H^i(X, \mathscr{F}) \longrightarrow H^i(X, \mathscr{F}_H) \longrightarrow \cdots$$

As graded $S_\bullet$-modules, $H^i(X, \mathscr{F}(-1))$ is simply $H^i(X, \mathscr{F})$ shifted by one degree, and the map $H^i(X, \mathscr{F}(-1)) \to H^i(X, \mathscr{F})$ induced by $\cdot x_n$ is multiplication by x_n.

Now $H \cong \mathbb{P}^{n-1}_R$, and $H^i(X, \mathscr{F}_H) = H^i(H, \bigoplus_d \mathcal{O}_H(d))$ by proposition 6.2.4. Applying the inductive hypothesis to $\mathbb{P}^{n-1}_R$, we get $H^i(X, \mathscr{F}_H) = 0$ for $0 < i < n - 1$.

For $0 < i < n - 1$: Since $H^i(X, \mathscr{F}_H) = 0$, multiplication by x_n maps $H^i(X, \mathscr{F}(-1))$ onto $H^i(X, \mathscr{F})$. For injectivity, the preceding term $H^{i-1}(X, \mathscr{F}_H)$ vanishes by the inductive hypothesis when $i \ge 2$. When $i = 1$ it is $H^0(X, \mathscr{F}_H)$, and the map $H^0(X, \mathscr{F}) \to H^0(X, \mathscr{F}_H)$ before it is the surjection $S_\bullet \to S_\bullet/(x_n)$ by part (1), so the connecting map $H^0(X, \mathscr{F}_H) \to H^1(X, \mathscr{F}(-1))$ is zero. So multiplication by x_n is bijective. Since every element is also killed by a power of x_n, we conclude $H^i(X, \mathscr{F}) = 0$.

For $i = n-1$ (when $n \ge 2$): We must show multiplication by x_n is still bijective on $H^{n-1}(X, \mathscr{F})$. The relevant portion of the long exact sequence is:

$$H^{n-2}(X, \mathscr{F}_H) \longrightarrow H^{n-1}(X, \mathscr{F}(-1)) \xrightarrow{\cdot x_n} H^{n-1}(X, \mathscr{F}) \longrightarrow H^{n-1}(X, \mathscr{F}_H) \xrightarrow{\partial} H^n(X, \mathscr{F}(-1)).$$

First, we establish injectivity on the left. If $n \ge 3$, then $H^{n-2}(X, \mathscr{F}_H) = 0$ by the inductive hypothesis (since $0 < n-2 < \dim H = n-1$). If $n = 2$, the term on the left is $H^0(X, \mathscr{F}_H)$; in this case, the preceding map in the sequence is $H^0(X, \mathscr{F}) \to H^0(X, \mathscr{F}_H)$, which is the natural surjection $S_\bullet \to S_\bullet/(x_n)$ by part (1). Exactness then forces the map $H^0(X, \mathscr{F}_H) \to H^1(X, \mathscr{F}(-1))$ to be the zero map. In either case, multiplication by x_n is injective on $H^{n-1}(X, \mathscr{F}(-1))$.

Next, we establish surjectivity on the right. The term $H^{n-1}(X, \mathscr{F}_H)$ is the top cohomology of $H \cong \mathbb{P}^{n-1}_R$, which by part (2) is a free R-module generated by all-negative Laurent monomials in

the variables $x_0, \dots, x_{n-1}$. The connecting homomorphism ∂ maps a basis element $x_0^{l_0} \cdots x_{n-1}^{l_{n-1}}$ to the element $x_0^{l_0} \cdots x_{n-1}^{l_{n-1}} x_n^{-1}$ in $H^n(X, \mathscr{F}(-1))$. This map is visibly injective. Because ∂ is injective, exactness forces the preceding map $H^{n-1}(X, \mathscr{F}) \to H^{n-1}(X, \mathscr{F}_H)$ to be the zero map.

Therefore, multiplication by x_n is surjective on $H^{n-1}(X, \mathscr{F})$. Being both injective and surjective, multiplication by x_n is bijective on $H^{n-1}(X, \mathscr{F})$, and the annihilation argument completes the proof.

6.3.2 Cohomology of Twisted Sheaves of Differentials

The cohomology of the line bundles computes that of every twist $\Omega^p_X(m)$ on projective space. The mechanism is the Euler sequence (theorem 5.5.19): its exterior powers express each Ω^p in terms of sums of line bundles, and theorem 6.3.1 finishes each step. The vanishing statement below is the one used in practice, and it is sharp: the exceptions listed are genuinely nonzero.

Theorem 6.3.3: Bott Vanishing

Let k be a field, $X = \mathbb{P}^n_k$ with $n \geq 1$, and $0 \leq p \leq n$. Then

$$H^q\big(X,\ \Omega^p_X(m)\big) = 0$$

for every q and m except in the following three cases:

1. $q = 0$ and $m > p$;
2. $q = n$ and $m < p - n$;
3. $q = p$ and $m = 0$, where the group is one-dimensional over k.

In particular $H^q(X, \Omega^p_X) = 0$ for $q \neq p$ and $H^p(X, \Omega^p_X) \cong k$, and $H^q(X, \Omega^p_X(m)) = 0$ for all $0 < q < n$ whenever $m \neq 0$.

***Proof*:**

Write $N = \binom{n+1}{p}$. Taking p-th exterior powers in the Euler sequence

$$0 \longrightarrow \Omega_X \longrightarrow \mathcal{O}_X(-1)^{\oplus(n+1)} \longrightarrow \mathcal{O}_X \longrightarrow 0$$

of theorem 5.5.19 gives, for each $p \geq 1$, a short exact sequence

$$0 \longrightarrow \Omega^p_X(m) \longrightarrow \mathcal{O}_X(m-p)^{\oplus N} \longrightarrow \Omega^{p-1}_X(m) \longrightarrow 0 \tag{6.1}$$

The exterior power of a short exact sequence of locally free sheaves with locally free quotient is exact in this form: locally the sequence splits, and for a split sequence $0 \to A \to A \oplus B \to B \to 0$ the filtration of $\Lambda^p(A \oplus B)$ by the number of factors from A has $\Lambda^p A$ as its bottom piece and $\Lambda^{p-1} A \otimes B$ as the next; here $B = \mathcal{O}_X$ has rank one, so the filtration has exactly two steps and (6.1) is what remains after twisting by $\mathcal{O}_X(m)$ and using $\Lambda^p(\mathcal{O}_X(-1)^{\oplus(n+1)}) \cong \mathcal{O}_X(-p)^{\oplus N}$. The identification is independent of the local splitting because the two extreme pieces of the filtration are canonical.

Now induct on p. For $p = 0$ the claim is theorem 6.3.1: the cohomology of $\mathcal{O}_X(m)$ vanishes except in degree 0 for $m \geq 0$ and degree n for $m \leq -n-1$, which is cases (1) and (2) with $p = 0$,

and case (3) reads $H^0(X, \mathcal{O}_X) = k$.

Let $p \geq 1$ and assume the statement for $p-1$. The long exact sequence of (6.1) reads

$$\cdots \to H^{q-1}\big(\Omega_X^{p-1}(m)\big) \to H^q\big(\Omega_X^p(m)\big) \to H^q\big(\mathcal{O}_X(m-p)\big)^{\oplus N} \to H^q\big(\Omega_X^{p-1}(m)\big) \to \cdots$$

so $H^q(\Omega_X^p(m))$ is squeezed between $H^{q-1}(\Omega_X^{p-1}(m))$ and $H^q(\mathcal{O}_X(m-p))^{\oplus N}$, and vanishes as soon as both do.

Global sections. For $m \geq 1$ the map $H^0(\mathcal{O}_X(m-p))^{\oplus N} \to H^0(\Omega_X^{p-1}(m))$ of (6.1) is surjective. Composed with the inclusion of $\Omega_X^{p-1}(m)$ into $\mathcal{O}_X(m-p+1)^{\oplus\binom{n+1}{p-1}}$ given by the same sequence for $p-1$, it is the contraction of Λ^p against the Euler map $(x_0, \ldots, x_n)$. With $S = k[x_0, \ldots, x_n]$ and $H^0(\mathcal{O}_X(d)) = S_d$ (theorem 6.3.1(1)), left exactness of H^0 identifies the global sections of the spliced sequences with the degree-m part of the Koszul complex of $x_0, \ldots, x_n$ over S. The variables form a regular sequence, so for $p \geq 2$ the kernel of the Koszul differential on Λ^{p-1}, which is $H^0(\Omega_X^{p-1}(m))$, is the image of Λ^p [12, Exactness of the Koszul Complex]. For $p = 1$ the map is $(f_j) \mapsto \sum_j x_j f_j$ from $S_{m-1}^{\oplus(n+1)}$ to S_m, onto for $m \geq 1$.

Middle degrees. Let $0 < q < n$. Then $H^q(\mathcal{O}_X(m-p)) = 0$ by theorem 6.3.1(4). By induction $H^{q-1}(\Omega_X^{p-1}(m))$ vanishes unless $q = p$ and $m = 0$, or $q = 1$ and $m \geq p$. In the second case the map $H^0(\mathcal{O}_X(m-p))^{\oplus N} \to H^0(\Omega_X^{p-1}(m))$ before it is surjective, as $m \geq p \geq 1$, so $H^0(\Omega_X^{p-1}(m)) \to H^1(\Omega_X^p(m))$ is zero and $H^1(\Omega_X^p(m))$ vanishes all the same. So for $m \neq 0$ all middle-degree groups vanish, and for $m = 0$ the only possibly nonzero one is at $q = p$, where the sequence gives $H^p(\Omega_X^p) \cong H^{p-1}(\Omega_X^{p-1}) \cong k$ by induction, the neighbouring $\mathcal{O}_X(-p)$ terms vanishing since $0 < p < n$ forces H^{p-1} and H^p of $\mathcal{O}_X(-p)$ to be zero. (For $p = n$ the group $H^n(\mathcal{O}_X(-n))$ also vanishes, as $-n > -n-1$.)

Degree zero. $H^0(\Omega_X^p(m))$ injects into $H^0(\mathcal{O}_X(m-p))^{\oplus N}$, which is zero for $m < p$; so H^0 vanishes unless $m \geq p$. At $m = p$ the sequence identifies $H^0(\Omega_X^p(p))$ with the kernel of $k^{\oplus N} \to H^0(\Omega_X^{p-1}(p))$, and that map is injective, as one checks on the standard basis of Λ^p of the coordinate forms. So $H^0(\Omega_X^p(m)) = 0$ for $m \leq p$. For $m > p$ the group is nonzero, which is (1). The identification of the global-sections step, applied to the sequences for p and $p-1$, makes $H^0(\Omega_X^p(m))$ the kernel of the Koszul differential $\Lambda^p \otimes S_{m-p} \to \Lambda^{p-1} \otimes S_{m-p+1}$, and exactness of the Koszul complex makes this kernel the image of $\Lambda^{p+1} \otimes S_{m-p-1}$. With $e_0, \ldots, e_n$ the basis on which the Koszul complex is built, the image of $e_0 \wedge \cdots \wedge e_p \otimes x_0^{m-p-1}$ has component x_0^{m-p} along $e_1 \wedge \cdots \wedge e_p$, so it is nonzero.

Degree n. The group $H^n(\Omega_X^p(m))$ maps to $H^n(\mathcal{O}_X(m-p))^{\oplus N}$ with kernel the image of $H^{n-1}(\Omega_X^{p-1}(m))$, and $H^n(\mathcal{O}_X(m-p)) = 0$ unless $m - p \leq -n-1$, that is unless $m \leq p-n-1$, which is (2). Combined with the induction hypothesis for $H^{n-1}(\Omega_X^{p-1}(m))$, which for $n \geq 2$ vanishes unless $n-1 = p-1$ and $m = 0$, and which for $n = 1$ and $m \geq 1$ maps to zero in $H^1(\Omega_X^1(m))$ by the surjectivity on global sections, this leaves only the stated cases. For $m < p-n$ the group is nonzero. By corollary 6.5.10, with $\omega_X = \Omega_X^n$ and $(\Omega_X^p)^\vee \otimes \Omega_X^n \cong \Omega_X^{n-p}$, it is dual to $H^0(\Omega_X^{n-p}(-m))$, which is nonzero by case (1) for $n-p$, as $-m > n-p$.

The diagonal case $H^p(X, \Omega_X^p) \cong k$ is the algebraic form of a topological fact: it is generated by the p-th power of the hyperplane class, and with the off-diagonal vanishing it is what makes the Hodge diamond of $\mathbb{P}^n$ a single line of 1s down the diagonal. The vanishing in the middle degrees is what allows the cohomology of a hypersurface in $\mathbb{P}^{n+1}$ to be read off a graded ring, which is how Hodge theory computes the Hodge numbers of hypersurfaces.

6.3.3⋆ Separating Points and Jets

⋆ *Optional. This subsection is an aside; nothing later in the book depends on it.*

A linear system prescribes local data at a point through its jets. The jet of a section is its class in the quotient of its stalk by a power of the maximal ideal, which records the value of the section and its derivatives up to a fixed order. The construction of Lefschetz pencils in *Hodge Theory* [17] needs the sections of a large twist to prescribe every 2-jet at every point of a smooth projective variety, and every pair of 1-jets at every pair of points, with a single twist serving all points at once. Explicit monomials on projective space give a threshold depending only on the order of the jets, uniform in the point, and restriction carries the statement to every closed subscheme.

Definition 6.3.4: Jets of Sections

Let X be a scheme over a field k, let $x \in X$ be a closed point with maximal ideal $\mathfrak{m}_x \subseteq \mathcal{O}_{X,x}$, let $\mathcal{L}$ be an invertible sheaf on X, and let $r \geq 0$ be an integer. The *space of r-jets* of $\mathcal{L}$ at x is the quotient

$$J^r_x(\mathcal{L}) := \mathcal{L}_x \,/\, \mathfrak{m}_x^{r+1}\mathcal{L}_x$$

regarded as a module over $\mathcal{O}_{X,x}/\mathfrak{m}_x^{r+1}$. A k-subspace $V \subseteq \Gamma(X,\mathcal{L})$ *realizes every r-jet at x* if the k-linear map $V \to J^r_x(\mathcal{L})$ sending a section to the class of its germ is surjective, and *realizes every pair of r-jets* at distinct closed points $x \neq y$ if the map $V \to J^r_x(\mathcal{L}) \oplus J^r_y(\mathcal{L})$ is surjective.

When X is locally of finite type over an algebraically closed field k, each $J^r_x(\mathcal{L})$ is a finite-dimensional k-vector space. Indeed x lies in an affine open $\operatorname{Spec} B$ with B a finitely generated k-algebra and stays closed there, so its maximal ideal pulls back along a presentation $k[t_1,\dots,t_l] \twoheadrightarrow B$ to a maximal ideal of the polynomial ring, and $\kappa(x) = k$ by theorem 0.1.12(3). The ring $\mathcal{O}_{X,x}$ is Noetherian: B is Noetherian by [10, Finitely Generated Algebras Are Noetherian], and every ideal of a localization is the extension of its contraction, so ascending chains stabilize. Hence each $\mathfrak{m}_x^i/\mathfrak{m}_x^{i+1}$ with $i \leq r$ is a finitely generated module over $\kappa(x) = k$, and a trivialization of $\mathcal{L}$ at x identifies $J^r_x(\mathcal{L})$ with $\mathcal{O}_{X,x}/\mathfrak{m}_x^{r+1}$, an iterated extension of these finitely many quotients. Jets at distinct closed points are independent conditions in the following sense: if $x \neq y$ lie in a common affine open $\operatorname{Spec} B$ with maximal ideals $\mathfrak{p} \neq \mathfrak{q}$, then $\mathfrak{p}^{r+1}$ and $\mathfrak{q}^{r+1}$ are comaximal, since a maximal ideal containing $\mathfrak{p}^{r+1}+\mathfrak{q}^{r+1}$ would contain $\mathfrak{p}$ and $\mathfrak{q}$ by primality and hence equal each of them, so $B/(\mathfrak{p}^{r+1} \cap \mathfrak{q}^{r+1}) \cong B/\mathfrak{p}^{r+1} \times B/\mathfrak{q}^{r+1}$ by [12, Chinese Remainder Theorem].

Lemma 6.3.5: Forms of Degree d Realize Jets on Projective Space

Let k be an algebraically closed field, let $N \geq 1$ and $r \geq 0$ be integers, and write $S = k[x_0,\dots,x_N]$, so that $\Gamma(\mathbb{P}^N_k, \mathcal{O}(d)) \cong S_d$ for $d \geq 0$ (theorem 6.3.1(1)).

1. For every $d \geq r$ and every closed point $x \in \mathbb{P}^N_k$, the space $\Gamma(\mathbb{P}^N_k, \mathcal{O}(d))$ realizes every r-jet of $\mathcal{O}(d)$ at x.
2. For every $d \geq 2r+1$ and all distinct closed points $x \neq y$ of $\mathbb{P}^N_k$, it realizes every pair of r-jets of $\mathcal{O}(d)$ at x and y.

***Proof*:**

Step 1: closed points are k-points. A closed point x lies in a chart $D_+(x_i) \cong \operatorname{Spec} k[t_j : j \neq i]$ with $t_j = x_j/x_i$ (lemma 2.3.2) and is closed there, so theorem 0.1.12(3) gives $\kappa(x) = k$ and identifies x with a point of k^N, that is with homogeneous coordinates $[v_0 : \cdots : v_N]$ for a vector $v \in k^{N+1} \setminus \{0\}$ with $v_i = 1$. Distinct closed points have non-proportional coordinate vectors: proportional vectors normalize to the same tuple in a common chart, hence give the same maximal ideal and the same point.

Step 2: monomials at a k-point. Let x have homogeneous coordinates $[v]$ with $v_i = 1$, and work on $U_i = D_+(x_i) = \operatorname{Spec} k[t_j : j \neq i]$ with $t_j = x_j/x_i$ (lemma 2.3.2), where $\mathcal{O}(d)$ is free with generator x_i^d (proposition 5.3.5) and division by x_i^d carries $F \in S_d$ to its dehomogenization, a polynomial of total degree at most d. The point x is the k-point $a = (a_j)_{j \neq i}$ with $a_j = v_j$, and maximal ideal $\mathfrak{n}_a = (t_j - a_j : j \neq i)$. The quotient $k[t]/\mathfrak{n}_a^{r+1}$ is local, so localizing at $\mathfrak{n}_a$ leaves it unchanged, and x_i^d identifies $J_x^r(\mathcal{O}(d))$ with $k[t]/\mathfrak{n}_a^{r+1}$, whose k-basis is the set of classes of $(t-a)^\beta := \prod_{j \neq i}(t_j - a_j)^{\beta_j}$ with $|\beta| \leq r$. Under these identifications the jet map sends F to the class of its dehomogenization. For $|\beta| \leq r \leq d$ the polynomial $(t-a)^\beta$ is the dehomogenization of $x_i^{d-|\beta|} \prod_{j \neq i}(x_j - a_j x_i)^{\beta_j} \in S_d$, so the jet map is surjective, proving (1).

Step 3: pairs. Let $x = [v]$ and $y = [w]$ be distinct closed points, let $d \geq 2r+1$, and set $d' = d - r - 1 \geq r$. The linear forms vanishing at w form the kernel of the evaluation $S_1 \to k$ at w, and their common zeros in k^{N+1} are exactly the scalar multiples of w. Since v is not one, there is a linear form h with $h(w) = 0$ and $h(v) \neq 0$. On a chart containing y the function h/x_i vanishes at y, so the germ of h lies in $\mathfrak{m}_y \mathcal{O}(1)_y$, and for every $F \in S_{d'}$ the product Fh^{r+1}, formed with the multiplication of proposition 5.3.8(1), has germ in $\mathfrak{m}_y^{r+1}\mathcal{O}(d)_y$ and zero image in $J_y^r(\mathcal{O}(d))$. On a chart containing x the function h/x_i has nonzero value at x, hence is a unit in $\mathcal{O}_{\mathbb{P}^N_k,x}$, so the germ of h generates $\mathcal{O}(1)_x$, multiplication by the germ of h^{r+1} is an isomorphism $\mathcal{O}(d')_x \to \mathcal{O}(d)_x$, and it induces an isomorphism $J_x^r(\mathcal{O}(d')) \to J_x^r(\mathcal{O}(d))$. By (1) applied to $d' \geq r$, the sections Fh^{r+1} with $F \in S_{d'}$ realize every element of $J_x^r(\mathcal{O}(d)) \oplus 0$. A linear form h' with $h'(v) = 0$ and $h'(w) \neq 0$ symmetrically yields sections realizing every element of $0 \oplus J_y^r(\mathcal{O}(d))$, and sums realize every pair.

For a closed subscheme $\iota\colon X \hookrightarrow \mathbb{P}^N_k$ write $\mathcal{O}_X(d) := \iota^*\mathcal{O}(d)$, extending the notation $\mathcal{O}_X(1) = \iota^*\mathcal{O}(1)$ of proposition 5.3.36. Every $\sigma \in \Gamma(\mathbb{P}^N_k, \mathcal{O}(d))$ has a restriction $\sigma|_X := \iota^*(\sigma) \in \Gamma(X, \mathcal{O}_X(d))$, as in theorem 5.3.43(1): on an open V on which $\mathcal{O}(d)$ is free with generator e, the sheaf $\mathcal{O}_X(d)$ is free on $X \cap V$ with generator ι^*e, and $\sigma = fe$ restricts to $\iota^\#(f) \cdot \iota^*e$. In particular a germ σ_x generating $\mathcal{O}(d)_x$ restricts to a generator of $\mathcal{O}_X(d)_x$, and a germ in $\mathfrak{m}^{r+1}_{\mathbb{P}^N,x}\mathcal{O}(d)_x$ restricts into $\mathfrak{m}_x^{r+1}\mathcal{O}_X(d)_x$, the stalk map $\iota^\#$ being a local homomorphism.

Theorem 6.3.6: Large Twists Separate Points and Jets

Let k be an algebraically closed field, let $\iota\colon X \hookrightarrow \mathbb{P}^N_k$ be a closed subscheme with $N \geq 1$, and let $r \geq 0$ be an integer.

1. For every $d \geq r$, the space $\Gamma(X, \mathcal{O}_X(d))$ realizes every r-jet of $\mathcal{O}_X(d)$ at every closed point of X.
2. For every $d \geq 2r+1$, it realizes every pair of r-jets of $\mathcal{O}_X(d)$ at every two distinct closed points of X.
3. There is an integer d_1, depending on X, such that the restriction map $\Gamma(\mathbb{P}^N_k, \mathcal{O}(d)) \to \Gamma(X, \mathcal{O}_X(d))$, $\sigma \mapsto \sigma|_X$, is surjective for every $d \geq d_1$.

***Proof*:**

Step 1: jets restrict surjectively. Let $x \in X$ be a closed point. Then $\{x\}$ is closed in $\mathbb{P}^N_k$, the subset X being closed. The stalk homomorphism $\iota^{\#}\colon \mathcal{O}_{\mathbb{P}^N_k,x} \to \mathcal{O}_{X,x}$ is surjective (definition 3.1.22) and local, so it carries $\mathfrak{m}_{\mathbb{P}^N,x}$ onto $\mathfrak{m}_x$ and induces a surjection $\mathcal{O}_{\mathbb{P}^N_k,x}/\mathfrak{m}^{r+1}_{\mathbb{P}^N,x} \to \mathcal{O}_{X,x}/\mathfrak{m}^{r+1}_x$. A trivialization of $\mathcal{O}(d)$ near x with generator e, together with the induced trivialization of $\mathcal{O}_X(d)$ by ι^*e, identifies the jet spaces with these two quotients and yields a surjection $\pi_x\colon J^r_x(\mathcal{O}(d)) \to J^r_x(\mathcal{O}_X(d))$ carrying the class of σ_x to the class of $(\sigma|_X)_x$ for every global section σ.

Step 2: parts (1) and (2). Let $d \geq r$ and let $x \in X$ be closed. The composite of the jet map $\Gamma(\mathbb{P}^N_k, \mathcal{O}(d)) \to J^r_x(\mathcal{O}(d))$ with π_x is surjective by lemma 6.3.5(1) and Step 1, and by the compatibility of Step 1 it equals the jet map $\Gamma(X, \mathcal{O}_X(d)) \to J^r_x(\mathcal{O}_X(d))$ preceded by restriction. A surjection factoring through $\Gamma(X, \mathcal{O}_X(d))$ forces the second factor to be surjective, which is (1). For (2), let $d \geq 2r+1$ and let $x \neq y$ be closed points of X. They are distinct closed points of $\mathbb{P}^N_k$, so lemma 6.3.5(2) makes $\Gamma(\mathbb{P}^N_k, \mathcal{O}(d)) \to J^r_x(\mathcal{O}(d)) \oplus J^r_y(\mathcal{O}(d))$ surjective, and composing with the surjection $\pi_x \oplus \pi_y$ and factoring as before proves (2).

Step 3: part (3). The ideal sheaf sequence of the closed subscheme X is

$$0 \longrightarrow \mathcal{I}_X \longrightarrow \mathcal{O}_{\mathbb{P}^N_k} \longrightarrow \iota_*\mathcal{O}_X \longrightarrow 0$$

with $\mathcal{I}_X$ quasi-coherent (proposition 5.1.33) and coherent (proposition 5.1.34), the scheme $\mathbb{P}^N_k$ being Noetherian (example 3.4.31(2) and lemma 3.5.27(1)). Tensoring with $\mathcal{O}(d)$ preserves exactness, the stalks of $\mathcal{O}(d)$ being free of rank one, and gives the exact sequence $0 \to \mathcal{I}_X(d) \to \mathcal{O}(d) \to (\iota_*\mathcal{O}_X)(d) \to 0$ in the notation of definition 5.3.7. The sheaf $(\iota_*\mathcal{O}_X)(d)$ is $\iota_*(\mathcal{O}_X(d))$: on an open V trivializing $\mathcal{O}(d)$ with generator e, a section of the source is uniquely $t \otimes e$ with $t \in \mathcal{O}_X(X \cap V)$, the assignment $t \otimes e \mapsto t \cdot \iota^*e$ is an isomorphism onto the sections of the target, and replacing e by ue for a unit u changes both descriptions by the same factor, so the local isomorphisms agree on overlaps and glue. Under this identification the quotient map $\mathcal{O}(d) \to \iota_*(\mathcal{O}_X(d))$ sends fe to $\iota^{\#}(f) \cdot \iota^*e$, so on global sections it is $\sigma \mapsto \sigma|_X$, using $\Gamma(\mathbb{P}^N_k, \iota_*(\mathcal{O}_X(d))) = \Gamma(X, \mathcal{O}_X(d))$. By lemma 6.8.23 applied to the coherent sheaf $\mathcal{I}_X$ over the Noetherian ring k there is d_1 with $H^1(\mathbb{P}^N_k, \mathcal{I}_X(d)) = 0$ for all $d \geq d_1$. The long exact sequence in cohomology ([18, Long Exact Sequence From a Right-Derived Functor]) applied to the twisted ideal sheaf sequence gives exactness of

$$\Gamma(\mathbb{P}^N_k, \mathcal{O}(d)) \longrightarrow \Gamma(X, \mathcal{O}_X(d)) \longrightarrow H^1(\mathbb{P}^N_k, \mathcal{I}_X(d)) = 0$$

for $d \geq d_1$, which is (3).

Corollary 6.3.7: Lefschetz-Type Jet Conditions Hold after Re-embedding

Let k be an algebraically closed field and let $\iota\colon X \hookrightarrow \mathbb{P}^N_k$ be a closed subscheme that is irreducible of dimension $n \geq 1$ and whose local rings are all regular. Let d_1 be as in theorem 6.3.6(3) and let $d \geq \max(d_1, 3)$. List the monomials of degree d in $x_0, \ldots, x_N$ as $M_0, \ldots, M_{N'}$. Then:

1. The restrictions $M_\alpha|_X$ span $\Gamma(X, \mathcal{O}_X(d))$ and generate $\mathcal{O}_X(d)$, and the morphism $\varphi_d\colon X \to \mathbb{P}^{N'}_k$ they determine (theorem 5.3.43) is a closed immersion with $\varphi_d^*\mathcal{O}(1) \cong \mathcal{O}_X(d)$, the embedding by the complete linear system of $\mathcal{O}_X(d)$.
2. Under φ_d the restrictions to X of the linear forms on $\mathbb{P}^{N'}_k$ are exactly the sections in $\Gamma(X, \mathcal{O}_X(d))$, and they realize every 2-jet at every closed point of X and every pair of 1-jets at every two distinct closed points of X.
3. At every closed point $x \in X$, $\dim_k J^1_x(\mathcal{O}_X(d)) = 1 + n$ and $\dim_k J^2_x(\mathcal{O}_X(d)) = 1 + n + \binom{n+1}{2}$.

***Proof*:**

(1). The monomials form a basis of $S_d \cong \Gamma(\mathbb{P}^N_k, \mathcal{O}(d))$ (theorem 6.3.1(1)) and generate $\mathcal{O}(d)$, each point lying in some $D_+(x_i)$ where the monomial x_i^d generates (proposition 5.3.5). Their restrictions generate $\mathcal{O}_X(d)$, generators restricting to generators, and they span $\Gamma(X, \mathcal{O}_X(d))$ because $d \geq d_1$ makes the restriction map surjective (theorem 6.3.6(3)). Theorem 5.3.43(2) therefore gives the morphism φ_d with $\varphi_d^*\mathcal{O}(1) \cong \mathcal{O}_X(d)$ and $\varphi_d^*(y_\alpha) = M_\alpha|_X$, writing $y_0, \ldots, y_{N'}$ for the homogeneous coordinates of $\mathbb{P}^{N'}_k$.

The immersion ι is itself the morphism determined by $\mathcal{O}_X(1)$ and the sections $x_0|_X, \ldots, x_N|_X$, which generate $\mathcal{O}_X(1)$ as restrictions of the generating x_i, by the uniqueness clause of theorem 5.3.43(2), so lemma 5.3.40(1) applied to ι gives $X_{x_i|_X} = \iota^{-1}(D_+(x_i)) = X \cap D_+(x_i) =: X_i$. A germ of $x_i|_X$ generates $\mathcal{O}_X(1)$ exactly when its d-th power generates $\mathcal{O}_X(d)$, a generator raising to a generator and a germ in $\mathfrak{m}_x\mathcal{O}_X(1)_x$ raising into $\mathfrak{m}_x\mathcal{O}_X(d)_x$, so also $X_{x_i^d|_X} = X_i$. Each X_i is affine and the ring homomorphism $k[t_j : j \neq i] \twoheadrightarrow B_i := \mathcal{O}_X(X_i)$, $t_j \mapsto \bar{t}_j$, induced by ι over the chart $D_+(x_i) \cong \operatorname{Spec} k[t_j : j \neq i]$, with $t_j = x_j/x_i$ (lemma 2.3.2), is surjective (proposition 3.1.24(1)), so the classes $\bar{t}_j$ generate B_i as a k-algebra. Writing α_i for the exponent vector of x_i^d, lemma 5.3.40 applied to φ_d gives $\varphi_d^{-1}(D_+(y_{\alpha_i})) = X_{x_i^d|_X} = X_i$, and the restriction $X_i \to D_+(y_{\alpha_i})$ corresponds to the k-algebra map sending each y_α/y_{α_i} to the ratio $(M_\alpha|_X)/(x_i^d|_X)$. For $M_\alpha = x_j x_i^{d-1}$ this ratio is $\bar{t}_j$, since $x_j x_i^{d-1} = (x_j/x_i)\cdot x_i^d$ on $D_+(x_i)$ restricts to $\bar{t}_j \cdot (x_i^d|_X)$ on X_i. The image of the chart ring map therefore contains every $\bar{t}_j$, so the map is surjective and $X_i \to D_+(y_{\alpha_i})$ is a closed immersion (proposition 3.1.24(2)).

Both X and $\mathbb{P}^{N'}_k$ are Noetherian, the first as a closed subscheme of the Noetherian $\mathbb{P}^N_k$ and the second by example 3.4.31(2) and lemma 3.5.27(1). The scheme X is projective, hence proper, over $\operatorname{Spec} k$ (theorem 3.5.32), and $\mathbb{P}^{N'}_k$ is separated over $\operatorname{Spec} k$ (corollary 3.5.5), so φ_d is proper (proposition 3.5.23(5)) and $\varphi_d(X)$ is closed. The union of the opens $D_+(y_{\alpha_0}), \ldots, D_+(y_{\alpha_N})$ contains $\varphi_d(X)$, the X_i covering X, so together with $\mathbb{P}^{N'}_k \setminus \varphi_d(X)$ they cover $\mathbb{P}^{N'}_k$. The preimage of the last member is empty, and the empty scheme maps to it by a closed immersion. By

lemma 5.3.39, φ_d is a closed immersion.

(2). The pullback of the linear form $\sum_\alpha c_\alpha y_\alpha$ along φ_d is $\sum_\alpha c_\alpha M_\alpha|_X$, so by the spanning in (1) the restrictions of the linear forms are exactly the sections in $\Gamma(X, \mathcal{O}_X(d))$. Theorem 6.3.6(1) with $r = 2$ and $d \geq 3 > 2$ realizes every 2-jet at every closed point, and theorem 6.3.6(2) with $r = 1$ and $d \geq 3$ realizes every pair of 1-jets at every two distinct closed points.

(3). Let $x \in X$ be a closed point. As in the discussion after definition 6.3.4, $\kappa(x) = k$ and $\mathcal{O}_{X,x}$ is Noetherian. The subset $\{x\}$ is irreducible and closed with generic point x, and X is locally of finite type over k, being a closed subscheme of $\mathbb{P}^N_k$, so theorem 2.6.12 gives $\dim \mathcal{O}_{X,x} = \dim X = n$. The local ring is regular by hypothesis, so [10, Characterizations of a Regular Local Ring] identifies its associated graded ring with the polynomial ring $k[u_1, \ldots, u_n]$, whence $\dim_k \mathfrak{m}_x^i/\mathfrak{m}_x^{i+1} = \binom{n+i-1}{n-1}$, the number of monomials of degree i in n variables. A trivialization of $\mathcal{O}_X(d)$ at x identifies $J^r_x(\mathcal{O}_X(d))$ with $\mathcal{O}_{X,x}/\mathfrak{m}_x^{r+1}$, of k-dimension $\sum_{i=0}^r \binom{n+i-1}{n-1}$. For $r = 1$ this is $1 + n$ and for $r = 2$ it is $1 + n + \binom{n+1}{2}$.

Over a field that is not algebraically closed the same arguments apply unchanged at k-rational points. At closed points with residue field larger than k no claim is made.

6.4 Ext Groups and Sheaf Ext

The cohomology groups $H^i(X, \mathscr{F})$ are the derived functors of $\Gamma(X, -) = \mathrm{Hom}_{\mathcal{O}_X}(\mathcal{O}_X, -)$. From this perspective, cohomology measures the failure of the global-section functor to be exact. But $\mathrm{Hom}_{\mathcal{O}_X}(\mathcal{O}_X, -)$ is a special case of $\mathrm{Hom}_{\mathcal{O}_X}(\mathscr{F}, -)$ for a general $\mathcal{O}_X$-module $\mathscr{F}$; the derived functors of this more general functor are the *Ext groups*. These carry strictly more information than cohomology alone: while $H^i(X, \mathscr{G})$ measures the obstruction to extending global sections of $\mathscr{G}$, the group $\mathrm{Ext}^i_{\mathcal{O}_X}(\mathscr{F}, \mathscr{G})$ measures the obstruction to extending morphisms from $\mathscr{F}$ to $\mathscr{G}$. In Serre duality, Ext groups will serve as the natural "dual" objects to cohomology groups, and computing them on projective space will be the key step.

There are two kinds of Ext: the *global Ext* groups $\mathrm{Ext}^i_{\mathcal{O}_X}(\mathscr{F}, \mathscr{G})$, which are abelian groups (or modules over the global sections ring), and the *sheaf Ext* $\mathcal{E}xt^i_{\mathcal{O}_X}(\mathscr{F}, \mathscr{G})$, which is itself a sheaf on X. These are related by a spectral sequence, and in the most important case — when $\mathscr{F}$ is locally free of finite rank — this spectral sequence degenerates, reducing Ext computations to ordinary cohomology. This reduction is what makes the theory practical.

6.4.1 Global and Local Ext

Recall from example 5.1.3 that for $\mathcal{O}_X$-modules $\mathscr{F}$ and $\mathscr{G}$ on a ringed space $(X, \mathcal{O}_X)$, the *sheaf Hom* is the sheaf:

$$\mathit{Hom}_{\mathcal{O}_X}(\mathscr{F}, \mathscr{G})(U) = \mathrm{Hom}_{\mathcal{O}_U}(\mathscr{F}|_U, \mathscr{G}|_U),$$

and the *global Hom* is:

$$\mathrm{Hom}_{\mathcal{O}_X}(\mathscr{F}, \mathscr{G}) = \Gamma(X, \mathit{Hom}_{\mathcal{O}_X}(\mathscr{F}, \mathscr{G})) = \mathit{Hom}_{\mathcal{O}_X}(\mathscr{F}, \mathscr{G})(X).$$

Both $\mathrm{Hom}_{\mathcal{O}_X}(\mathscr{F}, -)$ and $\mathit{Hom}_{\mathcal{O}_X}(\mathscr{F}, -)$ are left-exact functors. Since the category $\mathcal{O}_X$-**Mod** has enough injectives (by proposition 6.1.1), we may form their right derived functors:

Definition 6.4.1: Ext Groups for Sheaves

Let $(X, \mathcal{O}_X)$ be a ringed space and let $\mathscr{F}, \mathscr{G}$ be $\mathcal{O}_X$-modules.

1. The *global Ext groups* are the right derived functors of $\mathrm{Hom}_{\mathcal{O}_X}(\mathscr{F}, -)$:

$$\mathrm{Ext}^i_{\mathcal{O}_X}(\mathscr{F}, \mathscr{G}) = R^i \mathrm{Hom}_{\mathcal{O}_X}(\mathscr{F}, -)(\mathscr{G}).$$

   Concretely, choose an injective resolution $0 \to \mathscr{G} \to \mathscr{I}^\bullet$ and compute:

$$\mathrm{Ext}^i_{\mathcal{O}_X}(\mathscr{F}, \mathscr{G}) = h^i\big(\mathrm{Hom}_{\mathcal{O}_X}(\mathscr{F}, \mathscr{I}^\bullet)\big).$$

2. The *sheaf Ext* (or *local Ext*) functors are the right derived functors of $\mathit{Hom}_{\mathcal{O}_X}(\mathscr{F}, -)$:

$$\mathcal{E}xt^i_{\mathcal{O}_X}(\mathscr{F}, \mathscr{G}) = R^i \mathit{Hom}_{\mathcal{O}_X}(\mathscr{F}, -)(\mathscr{G}).$$

   Concretely, choose an injective resolution $0 \to \mathscr{G} \to \mathscr{I}^\bullet$ and compute:

$$\mathcal{E}xt^i_{\mathcal{O}_X}(\mathscr{F}, \mathscr{G}) = h^i\big(\mathit{Hom}_{\mathcal{O}_X}(\mathscr{F}, \mathscr{I}^\bullet)\big).$$

   This is a sheaf of $\mathcal{O}_X$-modules on X.

By convention, $\mathrm{Ext}^0_{\mathcal{O}_X}(\mathscr{F}, \mathscr{G}) = \mathrm{Hom}_{\mathcal{O}_X}(\mathscr{F}, \mathscr{G})$ and $\mathcal{E}xt^0_{\mathcal{O}_X}(\mathscr{F}, \mathscr{G}) = \mathit{Hom}_{\mathcal{O}_X}(\mathscr{F}, \mathscr{G})$.

Note the asymmetry: we derive in the *second* variable. Resolving $\mathscr{F}$ by locally free sheaves computes the sheaf Ext, but not the global Ext[5]. Both Ext groups come equipped with the standard long exact sequences of derived functors:

Proposition 6.4.2: Long Exact Sequences for Ext

Let $(X, \mathcal{O}_X)$ be a ringed space and let $\mathscr{F}$ be an $\mathcal{O}_X$-module.

1. *(Covariant variable.)* A short exact sequence $0 \to \mathscr{G}' \to \mathscr{G} \to \mathscr{G}'' \to 0$ of $\mathcal{O}_X$-modules induces long exact sequences:

$$0 \to \mathrm{Hom}(\mathscr{F}, \mathscr{G}') \to \mathrm{Hom}(\mathscr{F}, \mathscr{G}) \to \mathrm{Hom}(\mathscr{F}, \mathscr{G}'') \xrightarrow{\delta} \mathrm{Ext}^1(\mathscr{F}, \mathscr{G}') \to \cdots$$

   and similarly for $\mathcal{E}xt$.

2. *(Contravariant variable.)* A short exact sequence $0 \to \mathscr{F}' \to \mathscr{F} \to \mathscr{F}'' \to 0$ of $\mathcal{O}_X$-modules induces a long exact sequence:

$$0 \to \mathrm{Hom}(\mathscr{F}'', \mathscr{G}) \to \mathrm{Hom}(\mathscr{F}, \mathscr{G}) \to \mathrm{Hom}(\mathscr{F}', \mathscr{G}) \xrightarrow{\delta} \mathrm{Ext}^1(\mathscr{F}'', \mathscr{G}) \to \cdots$$

   and similarly for $\mathcal{E}xt$.

[5] If $\mathscr{F}$ admits a resolution $\mathcal{E}^\bullet \to \mathscr{F} \to 0$ by locally free sheaves of finite rank (e.g., if $\mathscr{F}$ is coherent on projective space or on a smooth projective variety), then $\mathcal{E}xt^i_{\mathcal{O}_X}(\mathscr{F}, \mathscr{G}) \cong h^i(\mathit{Hom}_{\mathcal{O}_X}(\mathcal{E}^\bullet, \mathscr{G}))$ (proposition 6.4.3). The global $\mathrm{Ext}^i_{\mathcal{O}_X}(\mathscr{F}, \mathscr{G})$ is assembled from these sheaves by theorem 6.4.8.

***Proof*:**
Part (1) is the standard long exact sequence of a right derived functor applied to a short exact sequence. For part (2), choose an injective resolution $0 \to \mathscr{G} \to \mathscr{I}^\bullet$. Each $\mathrm{Hom}(-, \mathscr{I}^p)$ is exact because $\mathscr{I}^p$ is injective, and each $\mathit{Hom}(-, \mathscr{I}^p)$ is exact because every $\mathscr{I}^p|_U$ is an injective $\mathcal{O}_U$-module (lemma 6.4.5). So $0 \to \mathrm{Hom}(\mathscr{F}'', \mathscr{I}^\bullet) \to \mathrm{Hom}(\mathscr{F}, \mathscr{I}^\bullet) \to \mathrm{Hom}(\mathscr{F}', \mathscr{I}^\bullet) \to 0$ is a short exact sequence of complexes, and so is its *Hom* analogue. Their long exact cohomology sequences are the ones claimed.

The next result shows that the sheaf Ext can be computed from a locally free resolution of the first argument. The global Ext cannot: on $X = \mathbb{P}^1_k$ the resolution $\mathcal{E}^0 = \mathcal{O}_X$ of $\mathscr{F} = \mathcal{O}_X$ gives $h^1(\mathrm{Hom}_{\mathcal{O}_X}(\mathcal{E}^\bullet, \mathcal{O}_X(-2))) = 0$, while $\mathrm{Ext}^1_{\mathcal{O}_X}(\mathcal{O}_X, \mathcal{O}_X(-2)) \cong H^1(X, \mathcal{O}_X(-2)) \cong k$ by corollary 6.4.9 and theorem 6.3.1.

Proposition 6.4.3: Computing Sheaf Ext via Locally Free Resolutions

Let $(X, \mathcal{O}_X)$ be a ringed space and let $\mathscr{F}, \mathscr{G}$ be $\mathcal{O}_X$-modules. Suppose $\mathscr{F}$ admits a resolution by locally free sheaves of finite rank:

$$\cdots \to \mathcal{E}^{-1} \to \mathcal{E}^0 \to \mathscr{F} \to 0.$$

Then:

$$\mathcal{E}xt^i_{\mathcal{O}_X}(\mathscr{F}, \mathscr{G}) \cong h^i\big(\mathit{Hom}_{\mathcal{O}_X}(\mathcal{E}^\bullet, \mathscr{G})\big).$$

In particular, if $\mathscr{F}$ is locally free of finite rank, then $\mathcal{E}xt^i_{\mathcal{O}_X}(\mathscr{F}, \mathscr{G}) = 0$ for all $i > 0$.

***Proof*:**
Both sides are δ-functors in $\mathscr{G}$: the right side because each $\mathit{Hom}_{\mathcal{O}_X}(\mathcal{E}^p, -)$ is exact, $\mathcal{E}^p$ being locally free of finite rank, so a short exact sequence in $\mathscr{G}$ gives a short exact sequence of complexes. They agree for $i = 0$, since $\mathit{Hom}_{\mathcal{O}_X}(-, \mathscr{G})$ is contravariant and left exact. Both vanish for $i > 0$ when $\mathscr{G} = \mathscr{I}$ is injective: the right side because $\mathit{Hom}_{\mathcal{O}_X}(-, \mathscr{I})$ is exact, each $\mathscr{I}|_U$ being an injective $\mathcal{O}_U$-module (lemma 6.4.5). So both are effaceable, hence universal, and they agree in every degree (*Homological Algebra* [18, An Effaceable δ-Functor Is Universal, A Universal δ-Functor Is Determined by Its Degree-Zero Part]). For $\mathscr{F}$ locally free of finite rank, take $\mathcal{E}^\bullet = [\mathscr{F}]$ concentrated in degree 0.

The sheaf Ext is a "local" invariant, while the global Ext is obtained by "assembling" the local data via global sections. The following result makes this precise at the stalk level:

Proposition 6.4.4: Stalks of Sheaf Ext

Let X be a Noetherian scheme, $\mathscr{F}$ a coherent $\mathcal{O}_X$-module, $\mathscr{G}$ any $\mathcal{O}_X$-module, and $x \in X$. Then there is a natural isomorphism:

$$\mathcal{E}xt^i_{\mathcal{O}_X}(\mathscr{F}, \mathscr{G})_x \;\cong\; \mathrm{Ext}^i_{\mathcal{O}_{X,x}}(\mathscr{F}_x, \mathscr{G}_x).$$

In particular, if $\mathscr{F}$ is locally free, then $\mathcal{E}xt^i_{\mathcal{O}_X}(\mathscr{F}, \mathscr{G}) = 0$ for all $i > 0$ (since the stalks $\mathscr{F}_x$ are free $\mathcal{O}_{X,x}$-modules, and Ext over a local ring vanishes for free first arguments).

***Proof*:**
The question is local by proposition 6.4.6, so we may assume $X = \operatorname{Spec} A$ with A Noetherian. Then $\mathscr{F} = \widetilde{M}$ with M finitely generated, and a resolution of M by finitely generated free A-modules gives a resolution $\mathcal{E}^\bullet \to \mathscr{F} \to 0$ by free sheaves of finite rank, whose stalks at x form a free resolution of $\mathscr{F}_x$. By proposition 6.4.3 and the exactness of the stalk functor (proposition 1.2.22),

$$\mathcal{E}xt^i_{\mathcal{O}_X}(\mathscr{F},\mathscr{G})_x = h^i\big(Hom_{\mathcal{O}_X}(\mathcal{E}^\bullet,\mathscr{G})\big)_x = h^i\big(Hom_{\mathcal{O}_X}(\mathcal{E}^\bullet,\mathscr{G})_x\big).$$

For $\mathcal{E}$ free of finite rank r, both $Hom_{\mathcal{O}_X}(\mathcal{E},\mathscr{G})_x$ and $\operatorname{Hom}_{\mathcal{O}_{X,x}}(\mathcal{E}_x,\mathscr{G}_x)$ are $\mathscr{G}_x^{\oplus r}$, so the right side is $h^i(\operatorname{Hom}_{\mathcal{O}_{X,x}}(\mathcal{E}^\bullet_x,\mathscr{G}_x)) = \operatorname{Ext}^i_{\mathcal{O}_{X,x}}(\mathscr{F}_x,\mathscr{G}_x)$.

Lemma 6.4.5: Injective Sheaves and Restriction

Let $(X,\mathcal{O}_X)$ be a ringed space and let $\mathscr{I}$ be an injective $\mathcal{O}_X$-module. For any open set $U \subseteq X$, the restriction $\mathscr{I}|_U$ is an injective $\mathcal{O}_U$-module.

***Proof*:**
Let $j : U \hookrightarrow X$ denote the inclusion. The functor $j_! : \mathcal{O}_U\textbf{-Mod} \to \mathcal{O}_X\textbf{-Mod}$ (extension by zero) is exact and left adjoint to the restriction functor j^* (by proposition 1.3.12). Hence j^* preserves injectives: for any monomorphism $\alpha : \mathscr{F} \hookrightarrow \mathscr{G}$ of $\mathcal{O}_U$-modules, the map

$$\operatorname{Hom}_{\mathcal{O}_U}(\mathscr{G},\mathscr{I}|_U) = \operatorname{Hom}_{\mathcal{O}_X}(j_!\mathscr{G},\mathscr{I}) \longrightarrow \operatorname{Hom}_{\mathcal{O}_X}(j_!\mathscr{F},\mathscr{I}) = \operatorname{Hom}_{\mathcal{O}_U}(\mathscr{F},\mathscr{I}|_U)$$

is surjective because $j_!\alpha$ is a monomorphism (exactness of $j_!$) and $\mathscr{I}$ is injective.

As a consequence, sheaf Ext behaves well with respect to restricting to open subsets:

Proposition 6.4.6: Open Subsets and Ext Groups

Let $(X,\mathcal{O}_X)$ be a ringed space, $\mathscr{F}$ and $\mathscr{G}$ be $\mathcal{O}_X$-modules, and $U \subseteq X$ an open subset. Then there is a natural isomorphism of sheaves on U:

$$\mathcal{E}xt^i_{\mathcal{O}_X}(\mathscr{F},\mathscr{G})\big|_U \;\cong\; \mathcal{E}xt^i_{\mathcal{O}_U}(\mathscr{F}|_U,\mathscr{G}|_U).$$

***Proof*:**
Let $0 \to \mathscr{G} \to \mathscr{I}^\bullet$ be an injective resolution on X. By lemma 6.4.5, the restricted sequence $0 \to \mathscr{G}|_U \to \mathscr{I}^\bullet|_U$ is an injective resolution on U. Then:

$$\mathcal{E}xt^i_{\mathcal{O}_U}(\mathscr{F}|_U,\mathscr{G}|_U) = h^i\big(Hom_{\mathcal{O}_U}(\mathscr{F}|_U,\mathscr{I}^\bullet|_U)\big) = h^i\big(Hom_{\mathcal{O}_X}(\mathscr{F},\mathscr{I}^\bullet)|_U\big) = \mathcal{E}xt^i_{\mathcal{O}_X}(\mathscr{F},\mathscr{G})\big|_U,$$

where the middle equality uses the fact that $Hom_{\mathcal{O}_X}(\mathscr{F},\mathscr{I}^p)|_U = Hom_{\mathcal{O}_U}(\mathscr{F}|_U,\mathscr{I}^p|_U)$ (the sheaf Hom is computed open-set by open-set), and the last equality uses the exactness of restriction (which commutes with taking cohomology of a complex of sheaves).

6.4.2 The Local-to-Global Spectral Sequence

The global Ext and the sheaf Ext are related by the factorization:

$$\mathrm{Hom}_{\mathcal{O}_X}(\mathscr{F}, -) = \Gamma(X, -) \circ \mathit{Hom}_{\mathcal{O}_X}(\mathscr{F}, -).$$

This is simply the statement that a global morphism $\mathscr{F} \to \mathscr{G}$ is a global section of the sheaf Hom. Both functors on the right are left exact, and this composition of functors is the standard setting for the Grothendieck spectral sequence[6]. For this spectral sequence to exist, we need $\mathit{Hom}_{\mathcal{O}_X}(\mathscr{F}, -)$ to send injective objects to $\Gamma(X, -)$-acyclic objects. The following key lemma establishes this:

Lemma 6.4.7: Sheaf Hom of Injectives is Flasque

Let $(X, \mathcal{O}_X)$ be a ringed space, $\mathscr{F}$ an $\mathcal{O}_X$-module, and $\mathscr{I}$ an injective $\mathcal{O}_X$-module. Then $\mathit{Hom}_{\mathcal{O}_X}(\mathscr{F}, \mathscr{I})$ is a flasque sheaf.

***Proof*:**
Let $U \subseteq V$ be open subsets of X. We must show that the restriction map

$$\mathit{Hom}_{\mathcal{O}_X}(\mathscr{F}, \mathscr{I})(V) = \mathrm{Hom}_{\mathcal{O}_V}(\mathscr{F}|_V, \mathscr{I}|_V) \longrightarrow \mathrm{Hom}_{\mathcal{O}_U}(\mathscr{F}|_U, \mathscr{I}|_U) = \mathit{Hom}_{\mathcal{O}_X}(\mathscr{F}, \mathscr{I})(U)$$

is surjective. Let $\varphi : \mathscr{F}|_U \to \mathscr{I}|_U$ be a morphism of $\mathcal{O}_U$-modules. Let $j : U \hookrightarrow V$ be the inclusion. By the adjunction $j_! \dashv j^*$, the morphism φ corresponds to a morphism $j_!(\mathscr{F}|_U) \to \mathscr{I}|_V$. The natural map $j_!(\mathscr{F}|_U) \to \mathscr{F}|_V$ gives a monomorphism in the category of $\mathcal{O}_V$-modules. Since $\mathscr{I}|_V$ is injective (by lemma 6.4.5), the morphism $j_!(\mathscr{F}|_U) \to \mathscr{I}|_V$ extends to a morphism $\psi : \mathscr{F}|_V \to \mathscr{I}|_V$. By construction, $\psi|_U = \varphi$, so the restriction map is surjective.

Since flasque sheaves are $\Gamma(X, -)$-acyclic (by corollary 6.1.7), the hypotheses of the Grothendieck spectral sequence are satisfied. We obtain:

Theorem 6.4.8: Local-to-Global Ext Spectral Sequence

Let $(X, \mathcal{O}_X)$ be a ringed space and let $\mathscr{F}, \mathscr{G}$ be $\mathcal{O}_X$-modules. Then there is a convergent spectral sequence:

$$E_2^{p,q} = H^p\big(X, \mathcal{E}xt^q_{\mathcal{O}_X}(\mathscr{F}, \mathscr{G})\big) \implies \mathrm{Ext}^{p+q}_{\mathcal{O}_X}(\mathscr{F}, \mathscr{G}).$$

This is natural in both $\mathscr{F}$ and $\mathscr{G}$.

***Proof*:**
This is the Grothendieck spectral sequence for the composite functor $\mathrm{Hom}_{\mathcal{O}_X}(\mathscr{F}, -) = \Gamma(X, -) \circ \mathit{Hom}_{\mathcal{O}_X}(\mathscr{F}, -)$. By lemma 6.4.7, the functor $\mathit{Hom}_{\mathcal{O}_X}(\mathscr{F}, -)$ sends injective $\mathcal{O}_X$-modules to flasque sheaves, which are $\Gamma(X, -)$-acyclic. The E_2-page of the Grothendieck spectral sequence is $R^p\Gamma \circ R^q \mathit{Hom}(\mathscr{F}, -) = H^p(X, \mathcal{E}xt^q(\mathscr{F}, -))$, and the abutment is $R^{p+q}(\Gamma \circ \mathit{Hom}(\mathscr{F}, -)) = \mathrm{Ext}^{p+q}(\mathscr{F}, -)$.

[6] See *Homological Algebra* [18] for the construction of the Grothendieck spectral sequence and the conditions under which it is defined.

The spectral sequence simplifies dramatically in the most important case:

Corollary 6.4.9: Degeneration for Locally Free Sheaves

Let $(X, \mathcal{O}_X)$ be a ringed space and let $\mathcal{L}$ be a locally free $\mathcal{O}_X$-module of finite rank, with dual $\mathcal{L}^\vee = \mathit{Hom}_{\mathcal{O}_X}(\mathcal{L}, \mathcal{O}_X)$. Then for all $\mathcal{O}_X$-modules $\mathscr{G}$ and all $i \geq 0$:

$$\mathrm{Ext}^i_{\mathcal{O}_X}(\mathcal{L}, \mathscr{G}) \;\cong\; H^i(X,\, \mathcal{L}^\vee \otimes_{\mathcal{O}_X} \mathscr{G}).$$

In particular, setting $\mathcal{L} = \mathcal{O}_X$ recovers sheaf cohomology: $\mathrm{Ext}^i_{\mathcal{O}_X}(\mathcal{O}_X, \mathscr{G}) \cong H^i(X, \mathscr{G})$.

***Proof*:**
The vanishing $\mathcal{E}xt^q_{\mathcal{O}_X}(\mathcal{L}, \mathscr{G}) = 0$ for $q > 0$ is local by proposition 6.4.6, so we may assume $\mathcal{L} = \mathcal{O}_X^{\oplus r}$. Then $\mathit{Hom}_{\mathcal{O}_X}(\mathcal{L}, -) = (-)^{\oplus r}$ is exact, so its right derived functors vanish in positive degrees.

The spectral sequence of theorem 6.4.8 collapses at the E_2-page: only the $q = 0$ row survives, giving:

$$\mathrm{Ext}^i_{\mathcal{O}_X}(\mathcal{L}, \mathscr{G}) \;\cong\; H^i\big(X,\, \mathcal{E}xt^0_{\mathcal{O}_X}(\mathcal{L}, \mathscr{G})\big) = H^i\big(X,\, \mathit{Hom}_{\mathcal{O}_X}(\mathcal{L}, \mathscr{G})\big).$$

Finally, for a locally free sheaf of finite rank, the natural map $\mathcal{L}^\vee \otimes_{\mathcal{O}_X} \mathscr{G} \to \mathit{Hom}_{\mathcal{O}_X}(\mathcal{L}, \mathscr{G})$ is an isomorphism (this can be checked on stalks, where it reduces to the standard isomorphism $M^\vee \otimes N \xrightarrow{\sim} \mathrm{Hom}_R(M, N)$ for a free module M of finite rank over a ring R). Hence $\mathrm{Ext}^i(\mathcal{L}, \mathscr{G}) \cong H^i(X, \mathcal{L}^\vee \otimes \mathscr{G})$.

6.4.10 Remark: Why Locally Free is the Key Case The reader should pause to appreciate how much corollary 6.4.9 simplifies matters. For a general coherent sheaf $\mathscr{F}$, computing $\mathrm{Ext}^i(\mathscr{F}, \mathscr{G})$ requires either an injective resolution of $\mathscr{G}$ (typically enormous and non-explicit) or the full spectral sequence. But for a locally free $\mathscr{F}$ of finite rank, the Ext groups reduce to ordinary sheaf cohomology, which we can compute using the Čech machinery of section 6.1.1. This is not a marginal simplification: on a smooth projective variety, *every* coherent sheaf has a finite locally free resolution (proposition 5.3.37). So the sheaf Ext between any two coherent sheaves can be computed by resolving the first argument by locally free sheaves (proposition 6.4.3), and the global Ext is assembled from it by theorem 6.4.8. On projective space, the situation is even better: every coherent sheaf has a resolution by direct sums of twisted sheaves $\mathcal{O}(d)$ (by corollary 5.3.27), and the cohomology of these is known by theorem 6.3.1.

One can also give a direct proof of corollary 6.4.9 without invoking spectral sequences. Since this argument is short and illuminating, we record it:

***Proof*:**
(Alternative proof of corollary 6.4.9)
Since $\mathcal{L}$ is locally free of finite rank, the functor $\mathit{Hom}_{\mathcal{O}_X}(\mathcal{L}, -)$ is exact (checking on stalks: if $\mathcal{L}_x \cong \mathcal{O}^r_{X,x}$, then $\mathrm{Hom}_{\mathcal{O}_{X,x}}(\mathcal{O}^r_{X,x}, -) = (-)^r$ is exact). Hence $\mathit{Hom}_{\mathcal{O}_X}(\mathcal{L}, -)$ preserves injective resolutions up to exactness. Let $0 \to \mathscr{G} \to \mathscr{I}^\bullet$ be an injective resolution. Then:

$$0 \to \mathit{Hom}(\mathcal{L}, \mathscr{G}) \to \mathit{Hom}(\mathcal{L}, \mathscr{I}^\bullet)$$

is exact, and each $\mathit{Hom}(\mathcal{L}, \mathscr{I}^p)$ is flasque by lemma 6.4.7. Thus this is a flasque resolution of

$\mathit{Hom}(\mathcal{L}, \mathscr{G}) \cong \mathcal{L}^\vee \otimes \mathscr{G}$, and:

$$\mathrm{Ext}^i(\mathcal{L}, \mathscr{G}) = h^i(\mathrm{Hom}(\mathcal{L}, \mathscr{I}^\bullet)) = h^i(\Gamma(X, \mathit{Hom}(\mathcal{L}, \mathscr{I}^\bullet))) = H^i(X, \mathcal{L}^\vee \otimes \mathscr{G}).$$

6.4.3 Computations on Projective Space

We now combine the degeneration result with the computation of twisted sheaf cohomology (theorem 6.3.1) to obtain explicit Ext groups on projective space. These calculations are the foundation for Serre duality.

Proposition 6.4.11: Ext of Twisted Sheaves on Projective Space

Let R be a Noetherian ring, $X = \mathbb{P}^n_R$ with $n \geq 1$, and $a, b \in \mathbb{Z}$. Then:

$$\mathrm{Ext}^i_{\mathcal{O}_X}\big(\mathcal{O}_X(a),\, \mathcal{O}_X(b)\big) \;\cong\; H^i\big(X,\, \mathcal{O}_X(b-a)\big),$$

and hence by theorem 6.3.1:

$$\mathrm{Ext}^i_{\mathcal{O}_X}\big(\mathcal{O}_X(a),\, \mathcal{O}_X(b)\big) \;\cong\; \begin{cases} \big(R[x_0, \ldots, x_n]\big)_{b-a} & i = 0,\ b-a \geq 0, \\ R^{\binom{a-b-1}{n}} & i = n,\ b-a \leq -(n+1), \\ 0 & \text{otherwise.} \end{cases}$$

In particular:

1. $\mathrm{Ext}^i(\mathcal{O}_X, \mathcal{O}_X(d)) = H^i(X, \mathcal{O}_X(d))$ for all i, d.
2. $\mathrm{Hom}(\mathcal{O}_X(a), \mathcal{O}_X(b)) = 0$ if $b < a$, and is a free R-module of rank $\binom{b-a+n}{n}$ if $b \geq a$.
3. $\mathrm{Ext}^n(\mathcal{O}_X(a), \mathcal{O}_X(b)) \neq 0$ only when $b - a \leq -(n+1)$, i.e. $a \geq b + n + 1$.

***Proof*:**
The line bundle $\mathcal{O}_X(a)$ is locally free of rank 1, with dual $\mathcal{O}_X(a)^\vee \cong \mathcal{O}_X(-a)$ (by proposition 5.3.8). Hence by corollary 6.4.9:

$$\mathrm{Ext}^i(\mathcal{O}_X(a), \mathcal{O}_X(b)) \cong H^i(X, \mathcal{O}_X(-a) \otimes \mathcal{O}_X(b)) = H^i(X, \mathcal{O}_X(b-a)),$$

where we used $\mathcal{O}_X(c) \otimes \mathcal{O}_X(d) \cong \mathcal{O}_X(c+d)$. The explicit values follow from theorem 6.3.1. The rank in the $i = n$ case is the number of "all-negative" monomials $x_0^{l_0} \cdots x_n^{l_n}$ of total degree $b - a$ with each $l_j < 0$, which by the substitution $l_j = -m_j - 1$ (so $m_j \geq 0$ and $\sum m_j = a - b - n - 1$) equals $\binom{a-b-1}{n}$.

Now let us see how to handle sheaves that are not themselves line bundles but can be resolved by them:

Example 6.4.12: Ext Computations on Projective Space

1. *Self-Ext of the structure sheaf.* Taking $a = b = 0$ in proposition 6.4.11:

$$\mathrm{Ext}^i(\mathcal{O}_X, \mathcal{O}_X) \cong H^i(X, \mathcal{O}_X) = \begin{cases} R & i = 0, \\ 0 & i > 0. \end{cases}$$

   This says that the structure sheaf has no non-trivial self-extensions, which is consistent with $\mathcal{O}_X$ being "rigid" as a line bundle on projective space.

2. *The dualizing line bundle.* Let k be a field and $X = \mathbb{P}^n_k$. We shall see in the next section that the *dualizing sheaf* of X is $\omega_X = \mathcal{O}_X(-n-1)$ (proposition 6.5.3). Note the "dual" behaviour: by proposition 6.4.11,

$$\mathrm{Ext}^n(\mathcal{O}_X(d), \omega_X) = H^n(X, \mathcal{O}_X(-n-1-d)) \cong \begin{cases} k^{\binom{d+n}{n}} & d \geq 0, \\ 0 & d < 0, \end{cases}$$

   and for $d \geq 0$ this is the only nonzero Ext from $\mathcal{O}_X(d)$ to ω_X. Meanwhile $H^0(X, \mathcal{O}_X(d)) \cong k^{\binom{d+n}{n}}$ for $d \geq 0$, so the dimensions agree, and the *pairing* between $H^0(X, \mathcal{O}_X(d))$ and $\mathrm{Ext}^n(\mathcal{O}_X(d), \omega_X)$ is perfect by theorem 6.3.1(3). For $d = 0$:

$$\mathrm{Ext}^n(\mathcal{O}_X, \omega_X) = H^n(X, \omega_X) \cong k,$$

   which will serve as the "trace" isomorphism $H^n(X, \omega_X) \xrightarrow{\sim} k$ in the duality theorem (theorem 6.5.5).

3. *Ext from the cotangent sheaf (using the Euler sequence).* Let k be a field and $X = \mathbb{P}^n_k$. The Euler sequence (by theorem 5.5.19) is:

$$0 \longrightarrow \Omega_{X/k} \longrightarrow \mathcal{O}_X(-1)^{\oplus(n+1)} \longrightarrow \mathcal{O}_X \longrightarrow 0.$$

   Although $\Omega_{X/k}$ is locally free (of rank n), let us use this short exact sequence to compute $\mathrm{Ext}^i(\Omega_{X/k}, \mathcal{O}_X)$ via the long exact sequence in the *contravariant* variable (proposition 6.4.2(2)):

$$\cdots \to \mathrm{Ext}^i(\mathcal{O}_X, \mathcal{O}_X) \to \mathrm{Ext}^i(\mathcal{O}_X(-1)^{n+1}, \mathcal{O}_X) \to \mathrm{Ext}^i(\Omega_{X/k}, \mathcal{O}_X) \to \mathrm{Ext}^{i+1}(\mathcal{O}_X, \mathcal{O}_X) \to \cdots$$

   By proposition 6.4.11:

$$\mathrm{Ext}^i(\mathcal{O}_X(-1)^{n+1}, \mathcal{O}_X) \cong H^i(X, \mathcal{O}_X(1))^{n+1},$$

   which is $(k^{n+1})^{n+1}$ for $i = 0$ and zero for $0 < i < n$. Also, $\mathrm{Ext}^i(\mathcal{O}_X, \mathcal{O}_X) = k$ for $i = 0$ and zero for $i > 0$.

   For $i = 0$, the long exact sequence gives:

$$0 \to \mathrm{Hom}(\mathcal{O}_X, \mathcal{O}_X) \to \mathrm{Hom}(\mathcal{O}_X(-1)^{n+1}, \mathcal{O}_X) \to \mathrm{Hom}(\Omega, \mathcal{O}_X) \to 0$$

   so $\mathrm{Hom}(\Omega_{X/k}, \mathcal{O}_X) \cong k^{(n+1)^2}/k \cong k^{n^2+2n}$. On the other hand, by corollary 6.4.9, $\mathrm{Hom}(\Omega, \mathcal{O}_X) = H^0(X, \Omega^\vee) = H^0(X, \mathcal{T}_X)$, where $\mathcal{T}_X$ is the tangent sheaf. This is the space of global vector fields on $\mathbb{P}^n_k$, which has dimension $n^2 + 2n = \dim_k \mathfrak{gl}_{n+1}(k) - 1$: the global vector fields on projective space are precisely those induced by the action of GL_{n+1}, modulo scalars.

4. *Ext between non-locally-free sheaves (sketch).* Let $X = \mathbb{P}^2_k$ and let $C \hookrightarrow X$ be a smooth plane curve of degree d, with ideal sheaf $\mathscr{I}_C$. By corollaries 5.3.15 and 5.3.17, $\mathscr{I}_C$ is the sheaf associated to a principal homogeneous ideal $(f) \subseteq S = k[x_0, x_1, x_2]$ with $\deg f = d$. Multiplication by f is an isomorphism of graded modules $S(-d) \to (f)$, so $\mathscr{I}_C \cong \widetilde{S(-d)} \cong \mathcal{O}_X(-d)$ by proposition 5.3.8(2). The short exact sequence $0 \to \mathscr{I}_C \to \mathcal{O}_X \to \mathcal{O}_C \to 0$ therefore becomes
$$0 \to \mathcal{O}_X(-d) \to \mathcal{O}_X \to \mathcal{O}_C \to 0.$$
This is a locally free resolution of $\mathcal{O}_C$ as an $\mathcal{O}_X$-module (of length 1). The long exact sequence in the contravariant variable (proposition 6.4.2(2)), with the Ext groups out of $\mathcal{O}_X$ and $\mathcal{O}_X(-d)$ computed by corollary 6.4.9, reads
$$0 \to \operatorname{Hom}(\mathcal{O}_C, \mathscr{G}) \to H^0(X, \mathscr{G}) \xrightarrow{\cdot f} H^0(X, \mathscr{G}(d)) \to \operatorname{Ext}^1(\mathcal{O}_C, \mathscr{G}) \to H^1(X, \mathscr{G}) \xrightarrow{\cdot f} H^1(X, \mathscr{G}(d)) \to \cdots$$
for any $\mathcal{O}_X$-module $\mathscr{G}$. Taking $\mathscr{G} = \omega_X = \mathcal{O}_X(-3)$, the group $H^1(X, \mathcal{O}_X(-3))$ vanishes by theorem 6.3.1(4), so $\operatorname{Ext}^1(\mathcal{O}_C, \omega_X) = \operatorname{coker}\big(H^0(X, \mathcal{O}(-3)) \to H^0(X, \mathcal{O}(d-3))\big)$. For $d \geq 3$, this is a k-vector space of dimension $\binom{d-1}{2}$, which (as we will verify via Serre duality) equals $\dim_k H^0(C, \omega_C) = g$, the genus of the curve. This is not a coincidence: Serre duality will identify $\operatorname{Ext}^1(\mathcal{O}_C, \omega_X)$ with $H^0(C, \omega_C)$ via the adjunction between the closed embedding and the ambient duality.

Proposition 6.4.13: Finiteness of Ext on Projective Schemes

Let X be a projective scheme over a Noetherian ring R, and let $\mathscr{F}$ and $\mathscr{G}$ be coherent sheaves on X. Then:

1. Each $\mathcal{E}xt^q_{\mathcal{O}_X}(\mathscr{F}, \mathscr{G})$ is a coherent sheaf on X.
2. Each $\operatorname{Ext}^i_{\mathcal{O}_X}(\mathscr{F}, \mathscr{G})$ is a finitely generated R-module.

***Proof*:**
(1) This is a local statement: on any affine open $U = \operatorname{Spec} A$ of X, the sheaves $\mathscr{F}|_U$ and $\mathscr{G}|_U$ correspond to finitely generated A-modules M and N (by corollary 5.1.12), and the sheaf Ext on U is:
$$\mathcal{E}xt^q_{\mathcal{O}_U}(\widetilde{M}, \widetilde{N}) \cong \widetilde{\operatorname{Ext}^q_A(M, N)}$$
(this follows from proposition 6.4.6 and the standard computation of Ext over a Noetherian ring via a free resolution of the finitely generated module M; each $\operatorname{Ext}^q_A(M, N)$ is finitely generated). Hence $\mathcal{E}xt^q(\mathscr{F}, \mathscr{G})$ is coherent.

(2) The local-to-global spectral sequence (theorem 6.4.8) gives $H^p(X, \mathcal{E}xt^q(\mathscr{F}, \mathscr{G})) \Rightarrow \operatorname{Ext}^{p+q}(\mathscr{F}, \mathscr{G})$. Since $\mathcal{E}xt^q(\mathscr{F}, \mathscr{G})$ is coherent and X is projective over a Noetherian ring, each cohomology group $H^p(X, \mathcal{E}xt^q(\mathscr{F}, \mathscr{G}))$ is a finitely generated R-module[a]. Since the E_2-page consists of finitely generated R-modules and the spectral sequence converges, the abutment $\operatorname{Ext}^i(\mathscr{F}, \mathscr{G})$ is finitely generated.

[a] This uses the finiteness theorem for cohomology of coherent sheaves on projective schemes, which we will prove in section 6.6 (see theorem 6.6.6(1)). No circularity arises: the finiteness theorem depends on the twisted sheaf computation (theorem 6.3.1) and the "coherent sheaves are quotients of twisted free sheaves" result (corollary 5.3.27), both of which are already established.

6.4.14 Remark: What the Spectral Sequence Encodes The local-to-global spectral sequence should be thought of as expressing the passage from local to global information for morphisms between sheaves. On the E_2-page:

- The *bottom row* $E_2^{p,0} = H^p(X, Hom(\mathscr{F}, \mathscr{G}))$ captures the "global extension classes that are locally trivial." When $\mathscr{F}$ is locally free of finite rank, this is the only nonzero row, and the spectral sequence degenerates.
- The *left column* $E_2^{0,q} = \Gamma(X, \mathcal{E}xt^q(\mathscr{F}, \mathscr{G}))$ captures the "local extension classes that are globally defined." These are the obstructions arising from the local algebra (e.g., the singularities of $\mathscr{F}$ or the non-regularity of the local rings).

The differentials in the spectral sequence measure the interaction between these two types of obstruction. For smooth projective varieties with locally free sheaves of finite rank, all the interesting information is in the bottom row — this is why Serre duality can be stated purely in terms of cohomology groups, without mentioning Ext.

6.5 Serre Duality Theorem

In classical topology, Poincaré duality asserts that for a compact connected oriented manifold M of dimension n, the singular cohomology groups satisfy $H^i(M;\mathbb{R}) \cong H^{n-i}(M;\mathbb{R})^*$. The duality is mediated by integration: the cup product $H^i \times H^{n-i} \to H^n$, followed by integration over the fundamental class $H^n(M;\mathbb{R}) \xrightarrow{\sim} \mathbb{R}$, gives a perfect pairing. In Kähler geometry it takes the refined form of Hodge duality: for a compact Kähler manifold X of complex dimension n, $H^{p,q}(X) \cong H^{n-p,\,n-q}(X)^*$, with the pairing mediated by the wedge product of (p,q)-forms followed by integration of the resulting top form.

Serre duality is the algebraic analogue. For a smooth projective variety X of dimension n over an algebraically closed field k, the role of the top differential form is played by the *canonical sheaf* $\omega_X = \det \Omega_{X/k}$, and the role of integration is played by a "trace" isomorphism $H^n(X, \omega_X) \xrightarrow{\sim} k$. The resulting duality

$$H^i(X, \mathscr{F}) \;\cong\; H^{n-i}(X,\, \mathscr{F}^\vee \otimes \omega_X)^*$$

for locally free $\mathscr{F}$ of finite rank is the central structural result in the cohomology of algebraic varieties. It constrains the Hilbert polynomial, underpins the Riemann–Roch theorem, and governs the geometry of linear systems.

We first define the dualizing sheaf of a proper scheme and prove the duality theorem for projective space, where the dualizing sheaf is simply $\mathcal{O}(-n-1)$ and the computations are completely explicit. We then extend the theorem to projective schemes, smooth projective varieties among them.

6.5.1 Duality on Projective Space

The twisted sheaf computation (theorem 6.3.1) already contains the seed of duality: the perfect pairing of part (3) shows that $H^0(X, \mathcal{O}(d))$ and $H^n(X, \mathcal{O}(-d-n-1))$ are dual to each other. Our task is to recognize this as the $i = 0$ case of a more general duality, then extend to all i and to all coherent sheaves.

The first step is to abstract the key property of $\mathcal{O}(-n-1)$ that makes the pairing work:

Definition 6.5.1: Dualizing Sheaf

Let X be a proper scheme of dimension n over a field k. A *dualizing sheaf* for X is a coherent sheaf ω_X on X together with a k-linear map $t : H^n(X, \omega_X) \to k$, called the *trace map*, such that for every coherent sheaf $\mathscr{F}$ on X, the natural pairing:

$$\mathrm{Hom}_{\mathcal{O}_X}(\mathscr{F}, \omega_X) \times H^n(X, \mathscr{F}) \xrightarrow{\text{apply } H^n} H^n(X, \omega_X) \xrightarrow{t} k$$

is a perfect pairing of finite-dimensional k-vector spaces. In other words, the induced map

$$\mathrm{Hom}_{\mathcal{O}_X}(\mathscr{F}, \omega_X) \xrightarrow{\sim} H^n(X, \mathscr{F})^*$$

is an isomorphism, where $(-)^*$ denotes the k-linear dual.
Equivalently, ω_X *represents* the functor $\mathscr{F} \mapsto H^n(X, \mathscr{F})^*$ on the category of coherent sheaves on X.

Proposition 6.5.2: Uniqueness of the Dualizing Sheaf

If a dualizing sheaf exists, it is unique up to unique isomorphism. That is, if (ω_X, t) and (ω'_X, t') are two dualizing sheaves for X, then there is a unique isomorphism $\varphi : \omega_X \xrightarrow{\sim} \omega'_X$ such that $t = t' \circ H^n(X, \varphi)$.

***Proof*:**
Both ω_X and ω'_X represent the same functor $\mathscr{F} \mapsto H^n(X, \mathscr{F})^*$. By the Yoneda lemma (theorem 0.4.22), any two representing objects are uniquely isomorphic via an isomorphism that intertwines the representing natural transformations.

Proposition 6.5.3: Dualizing Sheaf of Projective Space

Let k be a field and $X = \mathbb{P}^n_k$ with $n \geq 1$. The twisted sheaf $\omega_X = \mathcal{O}_X(-n-1)$ is a dualizing sheaf for X, with trace map $t : H^n(X, \mathcal{O}(-n-1)) \xrightarrow{\sim} k$ being the isomorphism that sends the class of $x_0^{-1}x_1^{-1}\cdots x_n^{-1}$ to $1 \in k$.

***Proof*:**
This is precisely the content of theorem 6.3.1(3) (with $R = k$). Indeed, that theorem establishes that for each $e \geq 0$, the multiplication pairing:

$$H^0(X, \mathcal{O}(e)) \times H^n(X, \mathcal{O}(-e-n-1)) \longrightarrow H^n(X, \mathcal{O}(-n-1)) \cong k$$

is a perfect pairing of finite-dimensional k-vector spaces. Since $\mathrm{Hom}(\mathcal{O}(d), \mathcal{O}(-n-1)) \cong H^0(X, \mathcal{O}(-d-n-1))$ (by corollary 6.4.9), taking $e = -d-n-1$ gives $\mathrm{Hom}(\mathcal{O}(d), \omega_X) \cong H^n(X, \mathcal{O}(d))^*$ for every $d \leq -n-1$. When $d > -n-1$, both sides are zero (the left because $H^0(X, \mathcal{O}(-d-n-1)) = 0$, the right by the summary table of theorem 6.3.1).

For a general coherent $\mathscr{F}$, one reduces to line bundles by choosing a presentation of $\mathscr{F}$ by direct sums of $\mathcal{O}(d)$'s (using corollary 5.3.27) and comparing the two sides via the five lemma. The

argument is identical to Step 2 of the proof of theorem 6.5.5 below, so we defer the details.

The dualizing sheaf definition captures the "$i = 0$" case of duality: it says $\mathrm{Hom}(\mathscr{F}, \omega) = \mathrm{Ext}^0(\mathscr{F}, \omega) \cong H^n(X, \mathscr{F})^*$. On $\mathbb{P}^n_k$, and more generally on a Cohen-Macaulay equidimensional projective X, Serre duality extends this to all Ext groups. To state it, we need the *Yoneda pairing*:

6.5.4 The Yoneda Pairing For $\mathcal{O}_X$-modules $\mathscr{F}, \mathscr{G}, \mathscr{H}$, the *Yoneda product* ([70, Tag 06XP]) is a bilinear map:

$$\mathrm{Ext}^i_{\mathcal{O}_X}(\mathscr{F}, \mathscr{G}) \times \mathrm{Ext}^j_{\mathcal{O}_X}(\mathscr{H}, \mathscr{F}) \longrightarrow \mathrm{Ext}^{i+j}_{\mathcal{O}_X}(\mathscr{H}, \mathscr{G}).$$

This is defined by splicing extensions: an element of $\mathrm{Ext}^i(\mathscr{F}, \mathscr{G})$ is represented by an equivalence class of i-fold extensions $0 \to \mathscr{G} \to \mathcal{E}^{i-1} \to \cdots \to \mathcal{E}^0 \to \mathscr{F} \to 0$, and the Yoneda product splices two such sequences together.
Now, since $H^j(X, \mathscr{F}) = \mathrm{Ext}^j_{\mathcal{O}_X}(\mathcal{O}_X, \mathscr{F})$, the Yoneda product specializes to give a pairing:

$$\mathrm{Ext}^i_{\mathcal{O}_X}(\mathscr{F}, \omega_X) \times H^{n-i}(X, \mathscr{F}) \longrightarrow H^n(X, \omega_X) \xrightarrow{t} k.$$

This defines for each i a natural map:

$$\eta^i_{\mathscr{F}} : \mathrm{Ext}^i_{\mathcal{O}_X}(\mathscr{F}, \omega_X) \longrightarrow H^{n-i}(X, \mathscr{F})^*.$$

When $\mathscr{F}$ is locally free of finite rank, $\mathrm{Ext}^i(\mathscr{F}, \omega_X) \cong H^i(X, \mathscr{F}^\vee \otimes \omega_X)$ by corollary 6.4.9, and the Yoneda pairing reduces to the cup product:

$$H^i(X, \mathscr{F}^\vee \otimes \omega_X) \times H^{n-i}(X, \mathscr{F}) \xrightarrow{\cup} H^n(X, \mathscr{F}^\vee \otimes \omega_X \otimes \mathscr{F}) \xrightarrow{\mathrm{ev}\otimes\mathrm{id}} H^n(X, \omega_X) \xrightarrow{t} k,$$

where $\mathrm{ev} : \mathscr{F}^\vee \otimes \mathscr{F} \to \mathcal{O}_X$ is the evaluation map. In the Čech framework, the cup product of cochains $\alpha \in C^p(\mathcal{U}, \mathscr{F}^\vee \otimes \omega_X)$ and $\beta \in C^q(\mathcal{U}, \mathscr{F})$ is defined by:

$$(\alpha \cup \beta)_{i_0,\ldots,i_{p+q}} = \alpha_{i_0,\ldots,i_p} \otimes \beta_{i_p,\ldots,i_{p+q}}\big|_{U_{i_0,\ldots,i_{p+q}}},$$

followed by the evaluation and trace maps.

Theorem 6.5.5: Duality for Projective Space

Let k be a field, $X = \mathbb{P}^n_k$ with $n \geq 1$, and $\omega_X = \mathcal{O}_X(-n-1)$. Then for every coherent sheaf $\mathscr{F}$ on X, the Yoneda pairing induces isomorphisms:

$$\eta^i_{\mathscr{F}} : \mathrm{Ext}^i_{\mathcal{O}_X}(\mathscr{F}, \omega_X) \xrightarrow{\sim} H^{n-i}(X, \mathscr{F})^* \qquad \text{for all } 0 \leq i \leq n.$$

Both sides vanish for $i > n$.

***Proof*:**
We verify the theorem in two stages: first for line bundles, then for all coherent sheaves.

Step 1: Verification for $\mathscr{F} = \mathcal{O}_X(d)$. By corollary 6.4.9, $\mathrm{Ext}^i(\mathcal{O}(d), \omega_X) \cong H^i(X, \mathcal{O}(-d-n-1))$. The Yoneda pairing (which reduces to the cup product for locally free sheaves of finite rank)

gives a bilinear map:

$$H^i(X, \mathcal{O}(-d-n-1)) \times H^{n-i}(X, \mathcal{O}(d)) \longrightarrow H^n(X, \mathcal{O}(-n-1)) \xrightarrow{t} k.$$

We must show this pairing is perfect for each i. By theorem 6.3.1(4), the intermediate cohomology vanishes: $H^j(X, \mathcal{O}(m)) = 0$ for $0 < j < n$ and all $m \in \mathbb{Z}$. So for $0 < i < n$, both $H^i(X, \mathcal{O}(-d-n-1))$ and $H^{n-i}(X, \mathcal{O}(d))$ are zero and the pairing is trivially perfect (both sides vanish). For $i > n$, both $H^i(X, \mathcal{O}(-d-n-1))$ (by theorem 6.2.5) and $H^{n-i}(X, \mathcal{O}(d))$ are zero.

For $i = 0$: the pairing is $H^0(X, \mathcal{O}(-d-n-1)) \times H^n(X, \mathcal{O}(d)) \to k$. If $-d-n-1 < 0$ (i.e. $d > -(n+1)$), then $H^0 = 0$ and $H^n = 0$, and the pairing is perfect. If $-d-n-1 \geq 0$ (i.e. $d \leq -(n+1)$), this is the pairing of theorem 6.3.1(3) (with the degree parameter set to $-d-n-1 \geq 0$), which is perfect.

For $i = n$: by symmetry, the pairing is $H^n(X, \mathcal{O}(-d-n-1)) \times H^0(X, \mathcal{O}(d)) \to k$. If $d < 0$, both sides vanish. If $d \geq 0$, this is again the perfect pairing of theorem 6.3.1(3) with the two factors interchanged (the pairing is symmetric up to the sign conventions in the cup product, which can be checked to be trivial in this case since all sheaves involved are line bundles).

Hence $\eta^i_{\mathcal{O}(d)}$ is an isomorphism for all i and all $d \in \mathbb{Z}$. By the additivity of Ext and cohomology in direct sums, the same holds for $\mathscr{F} = \bigoplus_j \mathcal{O}(d_j)$ (any finite direct sum of line bundles).

Step 2: Extension to all coherent sheaves. Let $\mathscr{F}$ be an arbitrary coherent sheaf on $X = \mathbb{P}^n_k$. By theorem 5.3.26, $\mathscr{F}(q)$ is generated by finitely many global sections for some $q \geq 1$, which gives a surjection $\mathcal{E}_0 = \mathcal{O}(-q)^{\oplus r} \twoheadrightarrow \mathscr{F}$ with coherent kernel $\mathcal{K}$. For $j > 0$, theorem 6.3.1 gives $\operatorname{Ext}^j(\mathcal{E}_0, \omega_X) \cong H^j(X, \mathcal{O}(q-n-1))^{\oplus r} = 0$ and $H^{n-j}(X, \mathcal{E}_0) = 0$, since $q \geq 1$. We prove that $\eta^i_{\mathscr{F}}$ is an isomorphism for every coherent $\mathscr{F}$ by induction on i.

Base case ($i = 0$): Applying the same construction to $\mathcal{K}$ gives a presentation $\mathcal{E}_1 \to \mathcal{E}_0 \to \mathscr{F} \to 0$ by finite direct sums of line bundles. The functor $\operatorname{Hom}(-, \omega_X)$ is left exact, and $H^n(X, -)$ is right exact because H^{n+1} vanishes on X (theorem 6.2.5), so the rows $0 \to \operatorname{Hom}(\mathscr{F}, \omega_X) \to \operatorname{Hom}(\mathcal{E}_0, \omega_X) \to \operatorname{Hom}(\mathcal{E}_1, \omega_X)$ and $0 \to H^n(X, \mathscr{F})^* \to H^n(X, \mathcal{E}_0)^* \to H^n(X, \mathcal{E}_1)^*$ are exact. The maps η^0 form a morphism between them, which is an isomorphism on the $\mathcal{E}_j$ by Step 1, hence also on $\mathscr{F}$.

Inductive step: Suppose $i \geq 1$ and that η^{i-1} is an isomorphism for every coherent sheaf. The short exact sequence $0 \to \mathcal{K} \to \mathcal{E}_0 \to \mathscr{F} \to 0$ induces two long exact sequences: one from $\operatorname{Ext}^*(-, \omega_X)$ (contravariant in the first argument) and one from $H^{n-*}(X, -)^*$ (the dual of the covariant cohomology sequence, which is exact since dualization is exact over a field). The naturality of the Yoneda pairing ensures that the maps η^i assemble into a morphism between these long exact sequences:

$$\begin{array}{ccccccc}
\operatorname{Ext}^{i-1}(\mathcal{E}_0, \omega) & \longrightarrow & \operatorname{Ext}^{i-1}(\mathcal{K}, \omega) & \longrightarrow & \operatorname{Ext}^i(\mathscr{F}, \omega) & \longrightarrow & \operatorname{Ext}^i(\mathcal{E}_0, \omega) = 0 \\
\sim\downarrow{\scriptstyle \eta^{i-1}_{\mathcal{E}_0}} & & \sim\downarrow{\scriptstyle \eta^{i-1}_{\mathcal{K}}} & & \downarrow{\scriptstyle \eta^i_{\mathscr{F}}} & & \\
H^{n-i+1}(\mathcal{E}_0)^* & \longrightarrow & H^{n-i+1}(\mathcal{K})^* & \longrightarrow & H^{n-i}(\mathscr{F})^* & \longrightarrow & H^{n-i}(\mathcal{E}_0)^* = 0
\end{array}$$

The two left vertical maps are isomorphisms by the inductive hypothesis, and the right-hand terms vanish by the choice of q. So $\eta^i_{\mathscr{F}}$ is the map induced on the cokernels of the first horizontal maps, and it is an isomorphism, completing the induction.

Vanishing for $i > n$: $H^{n-i}(X, \mathscr{F}) = 0$ for $n - i < 0$, so $\operatorname{Ext}^i(\mathscr{F}, \omega_X) = 0$ for $i > n$ because $\eta^i_{\mathscr{F}}$ is an isomorphism.

Example 6.5.6: Serre Duality on Projective Space

1. *Structure sheaf.* For $\mathscr{F} = \mathcal{O}_X$ on $X = \mathbb{P}^n_k$:

$$\operatorname{Ext}^i(\mathcal{O}_X, \mathcal{O}(-n-1)) \cong H^i(X, \mathcal{O}(-n-1)) \cong H^{n-i}(X, \mathcal{O}_X)^*.$$

   For $i = n$, this gives $H^n(X, \mathcal{O}(-n-1)) \cong H^0(X, \mathcal{O}_X)^* = k$. For $0 < i < n$, both sides are zero.

2. *Cotangent sheaf on $\mathbb{P}^2$.* Let $X = \mathbb{P}^2_k$ and $\mathscr{F} = \Omega_{X/k}$. Since Ω is locally free of rank 2, duality gives:

$$H^i(X, \Omega^\vee \otimes \mathcal{O}(-3)) \cong H^{2-i}(X, \Omega)^*.$$

   Now $\Omega^\vee = \mathcal{T}_X$, the tangent sheaf, and $\mathcal{T}_X \cong \mathcal{T}_{\mathbb{P}^2}$. From the dual of the Euler sequence (theorem 5.5.19):

$$0 \to \mathcal{O}_X \to \mathcal{O}_X(1)^3 \to \mathcal{T}_X \to 0,$$

   we get $\mathcal{T}_X \otimes \mathcal{O}(-3) \cong \mathcal{T}_X(-3)$, whose cohomology can be computed from the twisted Euler sequence. In particular:

$$H^0(X, \Omega) \cong H^2(X, \mathcal{T}_X(-3))^* = 0$$

   (no nonzero global 1-forms on $\mathbb{P}^2$), the analogue for $\mathbb{P}^2$ of the computation for $\mathbb{P}^1$ in Example 6.1.13(1).

3. *Skyscraper sheaves and residues.* Let $p \in \mathbb{P}^1_k$ be a k-rational point and let $\mathscr{F} = \kappa(p)$ be the skyscraper sheaf at p. Then $H^0(X, \kappa(p)) = k$ and $H^i(X, \kappa(p)) = 0$ for $i > 0$ (a skyscraper sheaf has no higher cohomology). Serre duality gives:

$$\operatorname{Ext}^1(\kappa(p), \mathcal{O}(-2)) \cong H^0(X, \kappa(p))^* = k, \qquad \operatorname{Hom}(\kappa(p), \mathcal{O}(-2)) = 0.$$

   The non-trivial Ext group $\operatorname{Ext}^1(\kappa(p), \omega_X) \cong k$ has a concrete interpretation: the short exact sequence $0 \to \mathcal{O}(-2) \to \mathcal{O}(-1) \to \kappa(p) \to 0$ (where the first map is multiplication by the linear form vanishing at p, suitably twisted) represents a generator. This is the algebraic avatar of the *residue map* at p: the element of $\operatorname{Ext}^1(\kappa(p), \omega)$ corresponds to extracting the residue of a meromorphic differential form at p.

6.5.2 Duality for Projective Schemes

We now extend Serre duality from projective space to projective schemes. The key new ingredient is identifying the correct dualizing sheaf. For smooth varieties, this is the canonical sheaf. For a singular projective scheme the dualizing sheaf still exists but need not be invertible, and duality in positive degrees needs X to be Cohen-Macaulay and equidimensional.

Recall from definition 5.5.28 that for a smooth variety X of dimension n over a field k, the *canonical sheaf* (or *canonical bundle*) is:

$$\omega_X = \det \Omega_{X/k} = \bigwedge^n \Omega_{X/k},$$

the top exterior power of the sheaf of differentials. This is an invertible sheaf. For $X = \mathbb{P}^n_k$, the computation of theorem 5.5.19 shows that $\omega_{\mathbb{P}^n} \cong \mathcal{O}_{\mathbb{P}^n}(-n-1)$, consistent with our earlier identification.

For a smooth projective variety X of dimension n embedded in $\mathbb{P}^N_k$ as a closed subscheme, the canonical sheaf is related to $\omega_{\mathbb{P}^N}$ by the *adjunction formula*. Recall from proposition 5.5.18 that if $X \hookrightarrow \mathbb{P}^N$ is a regular embedding of codimension $c = N - n$, there is an exact sequence:

$$\mathscr{I}_X/\mathscr{I}_X^2 \longrightarrow \Omega_{\mathbb{P}^N}\big|_X \longrightarrow \Omega_X \longrightarrow 0,$$

where $\mathscr{I}_X$ is the ideal sheaf of X. Taking top exterior powers (and using that the conormal sheaf $\mathcal{N}^\vee_{X/\mathbb{P}^N} = \mathscr{I}_X/\mathscr{I}_X^2$ is locally free of rank c for a regular embedding), one obtains:

Proposition 6.5.7: Adjunction Formula in Projective Space

Let X be a smooth closed subvariety of dimension n in $\mathbb{P}^N_k$, with codimension $c = N - n$. Then:

$$\omega_X \;\cong\; \omega_{\mathbb{P}^N}\big|_X \otimes \det\big(\mathcal{N}^\vee_{X/\mathbb{P}^N}\big)^{-1} \;\cong\; \mathcal{O}_{\mathbb{P}^N}(-N-1)\big|_X \otimes \det(\mathcal{N}_{X/\mathbb{P}^N}).$$

If X is a complete intersection, cut out by hypersurfaces of degrees $d_1, \ldots, d_c$, then $\mathcal{N}_{X/\mathbb{P}^N} \cong \bigoplus_{j=1}^c \mathcal{O}_X(d_j)$ and hence:

$$\omega_X \cong \mathcal{O}_X\left(\sum_{j=1}^c d_j - N - 1\right).$$

***Proof*:**

The conormal–cotangent exact sequence for the regular embedding $i : X \hookrightarrow \mathbb{P}^N$ is:

$$0 \longrightarrow \mathcal{N}^\vee_{X/\mathbb{P}^N} \longrightarrow \Omega_{\mathbb{P}^N}\big|_X \longrightarrow \Omega_X \longrightarrow 0.$$

This is exact on the left because X is smooth over k [70, Tag 06AA], and the embedding is a regular immersion [70, Tag 067U]. Taking the top exterior power of the middle term (which has rank N) and using the multiplicativity of determinants on short exact sequences of locally free sheaves:

$$\det(\Omega_{\mathbb{P}^N}|_X) \cong \det(\mathcal{N}^\vee_{X/\mathbb{P}^N}) \otimes \det(\Omega_X),$$

so $\omega_X = \det(\Omega_X) \cong \det(\Omega_{\mathbb{P}^N}|_X) \otimes \det(\mathcal{N}^\vee_{X/\mathbb{P}^N})^{-1} = \omega_{\mathbb{P}^N}|_X \otimes \det(\mathcal{N}_{X/\mathbb{P}^N})$.

For a complete intersection, the ideal sheaf $\mathscr{I}_X$ is generated by global sections $f_1, \ldots, f_c$ of degrees $d_1, \ldots, d_c$. The conormal sheaf is $\mathscr{I}_X/\mathscr{I}_X^2 \cong \bigoplus_{j=1}^c \mathcal{O}_X(-d_j)$ (the surjection $\bigoplus \mathcal{O}(-d_j) \twoheadrightarrow \mathscr{I}_X$ induces an isomorphism on the conormal level), so $\mathcal{N}_{X/\mathbb{P}^N} = (\mathscr{I}_X/\mathscr{I}_X^2)^\vee \cong \bigoplus \mathcal{O}_X(d_j)$. Hence $\det(\mathcal{N}_{X/\mathbb{P}^N}) = \mathcal{O}_X(\sum d_j)$ and $\omega_X \cong \mathcal{O}_X(-N - 1 + \sum d_j)$.

Example 6.5.8: Canonical Sheaves via Adjunction

1. *Smooth plane curve of degree d.* Let $C \subset \mathbb{P}^2_k$ be a smooth curve cut out by a polynomial of degree d. Then $\omega_C \cong \mathcal{O}_C(d-3)$. For $d = 1$ (a line, genus 0): $\omega_C \cong \mathcal{O}_C(-2)$, so $H^0(C, \omega_C) = 0$, confirming $g = 0$. For $d = 3$ (a smooth cubic, genus 1): $\omega_C \cong \mathcal{O}_C$, so $H^0(C, \omega_C) = k$, confirming $g = 1$. For general $d \geq 1$, the genus is $g = \dim_k H^0(C, \omega_C) = \binom{d-1}{2}$ (example 5.5.34(1)).

2. *Smooth surface of degree d in $\mathbb{P}^3$.* Let $S \subset \mathbb{P}^3_k$ be a smooth surface of degree d. Then $\omega_S \cong \mathcal{O}_S(d-4)$. For $d = 4$ (a K3 surface): $\omega_S \cong \mathcal{O}_S$, so S has trivial canonical bundle. For

$d < 4$, ω_S has negative degree (in an appropriate sense), and for $d > 4$, the canonical bundle is ample.

3. *Hypersurface of degree d in $\mathbb{P}^n$.* More generally, a smooth hypersurface $X \subset \mathbb{P}^n_k$ of degree d has $\omega_X \cong \mathcal{O}_X(d-n-1)$. For $n \geq 2$, the canonical class is trivial (i.e. X is Calabi–Yau in dimension $n-1$) precisely when $d = n+1$.

We can now state the full Serre duality theorem. For the proof, we shall use the dualizing sheaf of projective space together with a closed embedding to transfer duality from $\mathbb{P}^N$ to X:

Theorem 6.5.9: Serre Duality for a Projective Scheme

Let X be a projective scheme of dimension n over a field k, and suppose X admits a dualizing sheaf ω_X (with trace map $t\colon H^n(X,\omega_X) \to k$). For every coherent sheaf $\mathscr{F}$ on X, consider the Yoneda pairing

$$\operatorname{Ext}^i_{\mathcal{O}_X}(\mathscr{F},\,\omega_X) \times H^{n-i}(X,\mathscr{F}) \longrightarrow H^n(X,\omega_X) \xrightarrow{t} k$$

For $i = 0$ it induces an isomorphism $\operatorname{Hom}_{\mathcal{O}_X}(\mathscr{F},\omega_X) \cong H^n(X,\mathscr{F})^*$. If moreover X is Cohen-Macaulay and equidimensional, it induces isomorphisms

$$\operatorname{Ext}^i_{\mathcal{O}_X}(\mathscr{F},\omega_X) \;\cong\; H^{n-i}(X,\mathscr{F})^* \qquad \text{for all } 0 \leq i \leq n$$

Furthermore, a dualizing sheaf exists in each of the following cases:

1. X is a closed subscheme of $\mathbb{P}^N_k$ that is equidimensional and Cohen-Macaulay. In this case:
$$\omega_X \cong \mathcal{E}xt^c_{\mathcal{O}_{\mathbb{P}^N}}(\mathcal{O}_X,\,\omega_{\mathbb{P}^N}),$$
where $c = N - n$ is the codimension. In particular, if X is a smooth variety, then $\omega_X \cong \det \Omega_{X/k}$.
2. $X = \mathbb{P}^n_k$, with $\omega_X = \mathcal{O}(-n-1)$.

***Proof Key ideas*:**
For $X = \mathbb{P}^n_k$ this is theorem 6.5.5.

Existence of the dualizing sheaf. For a closed subscheme $i\colon X \hookrightarrow \mathbb{P}^N_k$ of codimension $c = N - n$, put

$$\omega_X = \mathcal{E}xt^c_{\mathcal{O}_{\mathbb{P}^N}}(i_*\mathcal{O}_X,\,\omega_{\mathbb{P}^N})$$

a coherent sheaf on $\mathbb{P}^N$ annihilated by the ideal of X, hence a coherent sheaf on X. The sheaves $\mathcal{E}xt^j_{\mathcal{O}_{\mathbb{P}^N}}(i_*\mathcal{O}_X,\omega_{\mathbb{P}^N})$ vanish for $j < c$ with no hypothesis on X [36, Lemma III.7.3], and from this one obtains an isomorphism $\operatorname{Hom}_{\mathcal{O}_X}(\mathscr{F},\omega_X) \cong \operatorname{Ext}^c_{\mathcal{O}_{\mathbb{P}^N}}(i_*\mathscr{F},\omega_{\mathbb{P}^N})$, functorial in the $\mathcal{O}_X$-module $\mathscr{F}$ [36, Lemma III.7.4]. For coherent $\mathscr{F}$, theorem 6.5.5 identifies the right side with $H^{N-c}(\mathbb{P}^N, i_*\mathscr{F})^* = H^n(X,\mathscr{F})^*$ (proposition 6.2.4). The trace t is the image of id_{ω_X} under this isomorphism for $\mathscr{F} = \omega_X$, and (ω_X, t) is a dualizing sheaf [36, Proposition III.7.5]. When X is Cohen-Macaulay and equidimensional, the sheaves $\mathcal{E}xt^j_{\mathcal{O}_{\mathbb{P}^N}}(i_*\mathcal{O}_X,\omega_{\mathbb{P}^N})$ also vanish for $j > c$: at a closed point x the local ring $\mathcal{O}_{X,x}$ has depth n, hence projective dimension $N - n = c$ over the regular local ring $\mathcal{O}_{\mathbb{P}^N,x}$.

When X is smooth, the identification $\omega_X \cong \det \Omega_{X/k}$ follows from proposition 6.5.7 and the fact that $\mathcal{E}xt^c(\mathcal{O}_X, \omega_{\mathbb{P}^N}) \cong \det(\mathcal{N}_{X/\mathbb{P}^N}) \otimes \omega_{\mathbb{P}^N}|_X$ for a regular embedding of codimension c [36, Theorem III.7.11].

Duality for the general case. The case $i = 0$ is the defining property of a dualizing sheaf. For $i > 0$ the argument is [36, Theorem III.7.6], whose proof uses the algebraic closure of k only to see that the local rings of $\mathbb{P}^N_k$ at closed points are regular of dimension N, which holds over any field. Both $\operatorname{Ext}^i_{\mathcal{O}_X}(-, \omega_X)$ and $H^{n-i}(X, -)^*$ are contravariant δ-functors in $\mathscr{F}$ that agree for $i = 0$. The first is universal, because every coherent sheaf is a quotient of a sheaf $\bigoplus \mathcal{O}_X(-q)$ with $q \gg 0$ (theorem 5.3.26), and $\operatorname{Ext}^i(\mathcal{O}_X(-q), \omega_X) = H^i(X, \omega_X(q))$ vanishes for $i > 0$ and $q \gg 0$ by Serre vanishing (lemma 6.8.23, through the closed embedding). The second is universal if and only if $H^{n-i}(X, \mathcal{O}_X(-q)) = 0$ for $i > 0$ and $q \gg 0$, and through duality on $\mathbb{P}^N_k$ (theorem 6.5.5) this vanishing is equivalent to X being Cohen-Macaulay and equidimensional. When it holds, the two universal δ-functors agree in every degree.

For smooth varieties, the Ext groups simplify and Serre duality takes its most elegant form:

Corollary 6.5.10: Serre Duality: Nonsingular Projective Variety

Let X be a projective variety of dimension n over a field k that is smooth over k (definition 2.7.13), with canonical sheaf $\omega_X = \det \Omega_{X/k}$. Then for every locally free coherent sheaf $\mathscr{F}$ on X:

$$H^i(X, \mathscr{F}) \;\cong\; H^{n-i}(X,\, \mathscr{F}^\vee \otimes \omega_X)^* \qquad \text{for all } 0 \le i \le n.$$

In particular:

1. $H^n(X, \omega_X) \cong H^0(X, \mathcal{O}_X)^*$, which is k when $H^0(X, \mathcal{O}_X) = k$, for instance when k is algebraically closed (theorem 3.5.33).
2. $H^i(X, \mathcal{O}_X) \cong H^{n-i}(X, \omega_X)^*$ for all i.
3. If $H^0(X, \mathcal{O}_X) = k$, the *arithmetic genus* $p_a(X) = (-1)^n(\chi(\mathcal{O}_X) - 1)$ is computed by the cohomology of the canonical sheaf:

$$p_a(X) = \sum_{j=0}^{n-1} (-1)^j \dim_k H^j(X, \omega_X).$$

***Proof*:**

Since $\mathscr{F}$ is locally free, $\operatorname{Ext}^i(\mathscr{F}, \omega_X) \cong H^i(X, \mathscr{F}^\vee \otimes \omega_X)$ by corollary 6.4.9. The variety X is integral, hence equidimensional, and its local rings are regular (proposition 2.7.14), hence Cohen-Macaulay [36, Theorem II.8.21A(a)], so theorem 6.5.9 applies in every degree, with $\omega_X \cong \det \Omega_{X/k}$ by its part (1). Substituting into it:

$$H^i(X,\, \mathscr{F}^\vee \otimes \omega_X) \cong H^{n-i}(X, \mathscr{F})^*.$$

Replacing $\mathscr{F}$ by $\mathscr{F}^\vee \otimes \omega_X$ (which is also locally free) and using $(\mathscr{F}^\vee \otimes \omega_X)^\vee \otimes \omega_X \cong \mathscr{F}$:

$$H^i(X, \mathscr{F}) \cong H^{n-i}(X,\, \mathscr{F}^\vee \otimes \omega_X)^*.$$

Part (1) follows by setting $\mathscr{F} = \mathcal{O}_X$ and $i = 0$: $H^n(X, \omega_X) \cong H^0(X, \mathcal{O}_X)^*$. When k is algebraically closed, X is reduced and connected, so $H^0(X, \mathcal{O}_X) = k$ by theorem 3.5.33. Part (2)

is the special case $\mathscr{F} = \mathcal{O}_X$. For part (3), write $h^j = \dim_k H^j$. Part (2) gives $\chi(\mathcal{O}_X) = \sum_{i=0}^{n}(-1)^i h^{n-i}(\omega_X) = (-1)^n \sum_{j=0}^{n}(-1)^j h^j(\omega_X)$, so

$$p_a(X) = \sum_{j=0}^{n-1}(-1)^j h^j(\omega_X) + (-1)^n\big(h^n(\omega_X) - 1\big),$$

and $h^n(\omega_X) = h^0(\mathcal{O}_X) = 1$ by part (1), as claimed.

Example 6.5.11: Applications of Serre Duality

1. *Curves: Riemann–Roch.* Let C be a smooth projective curve of genus $g = \dim_k H^1(C, \mathcal{O}_C)$ over k with $H^0(C, \mathcal{O}_C) = k$. Serre duality gives $H^1(C, \mathcal{L}) \cong H^0(C, \mathcal{L}^\vee \otimes \omega_C)^*$ for any line bundle $\mathcal{L}$. If $\mathcal{L} = \mathcal{O}_C(D)$ for a divisor D, this becomes:

   $$h^1(C, \mathcal{O}(D)) = h^0(C, \omega_C(-D)) = h^0(C, \mathcal{O}(K - D)),$$

   where K is a canonical divisor (with $\mathcal{O}(K) \cong \omega_C$) and $h^i = \dim_k H^i$. Substituting into $\chi(\mathcal{O}(D)) = h^0(\mathcal{O}(D)) - h^1(\mathcal{O}(D))$, and using $\chi(\mathcal{O}(D)) = \deg D + 1 - g$ (lemma 6.7.11 below), gives the *Riemann-Roch theorem* for curves (theorem 6.7.12 below):

   $$h^0(C, \mathcal{O}(D)) - h^0(C, \mathcal{O}(K - D)) = \deg D + 1 - g$$

   In particular, setting $D = K$: $h^0(\omega_C) - h^0(\mathcal{O}_C) = \deg K + 1 - g$, giving $\deg K = 2g - 2$ (since $h^0(\omega_C) = h^1(\mathcal{O}_C) = g$ by Serre duality, corollary 6.5.10(2), and $h^0(\mathcal{O}_C) = 1$).

2. *The Hodge numbers of a smooth surface.* Let S be a smooth projective surface over $\mathbb{C}$. Serre duality gives:

   $$H^0(S, \mathcal{O}_S) \cong H^2(S, \omega_S)^*,$$
   $$H^1(S, \mathcal{O}_S) \cong H^1(S, \omega_S)^*,$$
   $$H^2(S, \mathcal{O}_S) \cong H^0(S, \omega_S)^*.$$

   Since $h^{p,q} = \dim H^q(S, \Omega^p)$ by the Dolbeault theorem (theorem 0.6.17) and GAGA, these translate to symmetries of the Hodge diamond: $h^{0,0} = h^{2,2}$, $h^{0,1} = h^{2,1}$, and $h^{0,2} = h^{2,0}$. These are the instances $h^{0,q} = h^{2,2-q}$ of the duality in definition 0.6.29(2).

3. *Kodaira vanishing (preview).* Let X be a smooth projective variety of dimension n over $\mathbb{C}$ and let $\mathcal{L}$ be an ample line bundle. The Kodaira vanishing theorem (which requires analytic or characteristic-0 methods beyond our present scope) asserts:

   $$H^i(X, \omega_X \otimes \mathcal{L}) = 0 \qquad \text{for } i > 0.$$

   By Serre duality, this is equivalent to:

   $$H^{n-i}(X, \mathcal{L}^{-1}) = 0 \qquad \text{for } i > 0,$$

   i.e. $H^j(X, \mathcal{L}^{-1}) = 0$ for $j < n$. Thus Kodaira vanishing can be stated either as acyclicity of $\omega_X \otimes \mathcal{L}$ or as "low-degree vanishing" of $\mathcal{L}^{-1}$. Serre duality reduces one formulation to the other.

4. *Self-duality of abelian varieties.* Let A be an abelian variety of dimension g over k. Then $\Omega_{A/k}$ is trivial: $\Omega_{A/k} \cong \mathcal{O}_A^g$ (the tangent bundle of a group scheme is trivial, by translation). Hence $\omega_A = \det(\mathcal{O}_A^g) \cong \mathcal{O}_A$, and Serre duality gives:

$$H^i(A, \mathcal{O}_A) \cong H^{g-i}(A, \mathcal{O}_A)^*.$$

This symmetry gives only $h^i = h^{g-i}$. It is consistent with $\dim_k H^i(A, \mathcal{O}_A) = \binom{g}{i}$, which needs further input: $H^*(A, \mathcal{O}_A)$ is the exterior algebra on the g-dimensional space $H^1(A, \mathcal{O}_A)$.

6.5.12 Remark: Beyond Smooth Varieties For singular projective schemes, Serre duality still holds when X is Cohen-Macaulay and equidimensional (theorem 6.5.9), but the dualizing sheaf ω_X is no longer necessarily invertible. It is a coherent sheaf, and its local structure reflects the singularities of X. The theory of dualizing complexes (due to Grothendieck and Hartshorne) extends duality to arbitrary proper schemes, replacing the dualizing sheaf with a complex of sheaves $\omega_X^\bullet \in D^b(\mathbf{Coh}(X))$. In this generality, Serre duality becomes Grothendieck duality:

$$\mathbf{R}\mathrm{Hom}_{\mathcal{O}_X}(\mathscr{F}, \omega_X^\bullet) \cong \mathbf{R}\Gamma(X, \mathscr{F})^*$$

in the derived category. For Cohen-Macaulay equidimensional X, the dualizing complex is concentrated in a single degree and is the dualizing sheaf we have defined. For Gorenstein schemes (where ω_X is invertible), the theory is especially clean. We shall not pursue derived duality in these notes, but the reader should be aware that theorem 6.5.9 is a special case of this more general structure.

6.6 Higher Direct Images

The sheaf cohomology groups $H^i(X, \mathscr{F})$ encode global information about a single scheme X. But in practice, schemes arise in *families*: a morphism $f : X \to Y$ presents X as a family of fibers $X_y = f^{-1}(y)$ parameterized by points of Y, and we want to understand how the cohomology $H^i(X_y, \mathscr{F}|_{X_y})$ varies as y moves. The *higher direct image sheaves* $R^i f_* \mathscr{F}$ are sheaves on the base Y that give the "relative" version of sheaf cohomology, and they reduce to ordinary cohomology when Y is a point.

The central result of this section is the *finiteness theorem*: if $f : X \to Y$ is a projective morphism of Noetherian schemes and $\mathscr{F}$ is a coherent sheaf on X, then each $R^i f_* \mathscr{F}$ is a coherent sheaf on Y. When $Y = \operatorname{Spec} k$ is a point, this says that the cohomology groups of coherent sheaves on projective varieties are finite-dimensional — a fact we have already used (via forward reference) in proposition 6.4.13 and implicitly throughout our Serre duality arguments.

6.6.1 Definition and First Properties

Recall from definition 1.3.1 that the *direct image* (or pushforward) of a sheaf $\mathscr{F}$ on X along a continuous map $f : X \to Y$ is the sheaf $f_*\mathscr{F}$ on Y defined by $f_*\mathscr{F}(V) = \mathscr{F}(f^{-1}(V))$ for open $V \subseteq Y$. When f is a morphism of ringed spaces and $\mathscr{F}$ is an $\mathcal{O}_X$-module, the pushforward $f_*\mathscr{F}$ is an $\mathcal{O}_Y$-module (via the sheaf map $f^\# : \mathcal{O}_Y \to f_*\mathcal{O}_X$). The functor $f_* : \mathcal{O}_X\text{-}\mathbf{Mod} \to \mathcal{O}_Y\text{-}\mathbf{Mod}$ is left exact but not right exact in general.

Definition 6.6.1: Higher Direct Image Sheaves

Let $f : X \to Y$ be a morphism of ringed spaces. The *higher direct image sheaves* are the right derived functors of f_*:

$$R^i f_* : \mathcal{O}_X\text{-}\mathbf{Mod} \longrightarrow \mathcal{O}_Y\text{-}\mathbf{Mod}, \qquad R^i f_*(\mathscr{F}) = h^i(f_* \mathscr{I}^\bullet),$$

where $0 \to \mathscr{F} \to \mathscr{I}^\bullet$ is an injective resolution of $\mathscr{F}$ in the category of $\mathcal{O}_X$-modules. By construction, $R^0 f_* = f_*$, and a short exact sequence $0 \to \mathscr{F}' \to \mathscr{F} \to \mathscr{F}'' \to 0$ of $\mathcal{O}_X$-modules induces a long exact sequence of $\mathcal{O}_Y$-modules:

$$0 \to f_*\mathscr{F}' \to f_*\mathscr{F} \to f_*\mathscr{F}'' \xrightarrow{\delta} R^1 f_* \mathscr{F}' \to R^1 f_* \mathscr{F} \to \cdots$$

When $Y = \{pt\}$ is a point, the morphism $f : X \to \{pt\}$ is the unique map to the terminal space, and $f_* = \Gamma(X, -)$. Hence $R^i f_* \mathscr{F} = H^i(X, \mathscr{F})$, recovering ordinary sheaf cohomology. This is the sense in which higher direct images "relativize" cohomology.

The following result is fundamental: it says that higher direct images can be computed "stalkwise" by computing cohomology on the preimages of open sets.

Proposition 6.6.2: Sheafification of Presheaf Direct Images

Let $f : X \to Y$ be a continuous map of topological spaces and let $\mathscr{F}$ be a sheaf of abelian groups on X. Then $R^i f_* \mathscr{F}$ is the sheafification of the presheaf on Y defined by:

$$V \longmapsto H^i(f^{-1}(V), \mathscr{F}|_{f^{-1}(V)})$$

for open $V \subseteq Y$. If this presheaf is already a sheaf (e.g., when Y has a basis of open sets V for which $f^{-1}(V)$ is "cohomologically well-behaved" in an appropriate sense), then no sheafification is needed.

***Proof*:**
Let $0 \to \mathscr{F} \to \mathscr{I}^\bullet$ be an injective resolution. For each open $V \subseteq Y$:

$$(f_* \mathscr{I}^p)(V) = \mathscr{I}^p(f^{-1}(V)).$$

The presheaf $V \mapsto h^i(\mathscr{I}^\bullet(f^{-1}(V)))$ computes $H^i(f^{-1}(V), \mathscr{F}|_{f^{-1}(V)})$ provided the restriction $\mathscr{I}^\bullet|_{f^{-1}(V)}$ is an injective (or at least acyclic) resolution of $\mathscr{F}|_{f^{-1}(V)}$. By lemma 6.4.5, the restriction of an injective sheaf to an open subset is injective, so this is indeed the case.

Now, the derived functors $R^i f_* \mathscr{F}$ are defined as the cohomology sheaves $h^i(f_* \mathscr{I}^\bullet)$. The presheaf $V \mapsto h^i((f_* \mathscr{I}^\bullet)(V))$ need not be a sheaf (taking cohomology does not commute with the sheaf condition in general), so $R^i f_* \mathscr{F}$ is the *sheafification* of this presheaf. Since $(f_* \mathscr{I}^\bullet)(V) = \mathscr{I}^\bullet(f^{-1}(V))$, the presheaf in question is $V \mapsto H^i(f^{-1}(V), \mathscr{F})$, as claimed.

Proposition 6.6.3: Stalks of Higher Direct Images

Let $f : X \to Y$ be a continuous map of topological spaces, $\mathscr{F}$ a sheaf of abelian groups on X, and $y \in Y$. Then there is a natural map:

$$(R^i f_* \mathscr{F})_y \longrightarrow H^i(X_y, \mathscr{F}|_{X_y}),$$

where $X_y = f^{-1}(y)$ is the (set-theoretic) fiber. This map is an isomorphism if y has a basis of open neighborhoods $\{V_\alpha\}$ such that the natural maps $\varinjlim_\alpha H^i(f^{-1}(V_\alpha), \mathscr{F}) \to H^i(X_y, \mathscr{F}|_{X_y})$ are isomorphisms.

***Proof*:**
By proposition 6.6.2, the stalk of $R^i f_* \mathscr{F}$ at y is:

$$(R^i f_* \mathscr{F})_y = \varinjlim_{V \ni y} H^i(f^{-1}(V), \mathscr{F}|_{f^{-1}(V)}).$$

The inclusion $X_y = f^{-1}(y) \hookrightarrow f^{-1}(V)$ for each open $V \ni y$ induces a restriction map $H^i(f^{-1}(V), \mathscr{F}) \to H^i(X_y, \mathscr{F}|_{X_y})$, and these are compatible as V varies, yielding the natural map on the colimit. The condition for this to be an isomorphism is precisely that cohomology "commutes with the limit" of shrinking V to $\{y\}$.

6.6.4 Remark: When Do Stalks Compute Fiber Cohomology? The natural map $(R^i f_* \mathscr{F})_y \to H^i(X_y, \mathscr{F}|_{X_y})$ is *not* an isomorphism in general, and the discrepancy is important. The stalk sees the cohomology of all "thickenings" of the fiber (i.e., the preimages of open neighborhoods of y), while the fiber cohomology sees only the fiber itself. These can differ when nearby fibers contribute non-trivially.
In the Zariski topology, neither properness nor a local base makes the map an isomorphism. If $Y = \operatorname{Spec} R$ for a local ring $(R, \mathfrak{m})$ and $y = [\mathfrak{m}]$ is the closed point, then the only open set containing y is Y itself, so the stalk is $H^i(X, \mathscr{F})$, the cohomology of all of X. Take $R = \mathbb{Z}_{(5)}$, $S = \mathbb{Z}_{(5)}[t]/(t^2 + 1)$ and the finite, hence proper (proposition 3.5.24), morphism $f\colon \operatorname{Spec} S \to \operatorname{Spec} R$ with $\mathscr{F} = \mathcal{O}$. Since $5 = (2 + t)(2 - t)$ in S, the fiber over the closed point is two closed points $\mathfrak{m}_1, \mathfrak{m}_2$, and for $i = 0$ the map is the diagonal $S \to S_{\mathfrak{m}_1} \times S_{\mathfrak{m}_2}$, which is not surjective.
In the scheme-theoretic setting, the correct fiber to consider is the *scheme-theoretic fiber* $X_y = X \times_Y \operatorname{Spec} \kappa(y)$ (see definition 3.2.9). The comparison between $R^i f_* \mathscr{F}$ and the cohomology of X_y is the subject of the *formal functions theorem* (Grothendieck's theorem on formal functions), which we shall not develop in full here, and, for $\mathscr{F}$ flat over Y, of theorem 6.8.19.

We now establish the key properties that make higher direct images useful in the scheme-theoretic setting:

Proposition 6.6.5: Quasi-Coherence of Higher Direct Images

Let $f : X \to Y$ be a quasi-compact and quasi-separated morphism of schemes, and let $\mathscr{F}$ be a quasi-coherent sheaf on X. Then each $R^i f_* \mathscr{F}$ is quasi-coherent on Y.

***Proof Key ideas*:**
The question is local on Y, so we may assume $Y = \operatorname{Spec} R$ is affine. Since f is quasi-compact and quasi-separated, the preimage $X = f^{-1}(Y)$ is a quasi-compact quasi-separated scheme. The intersections of a finite affine open cover of X need not be affine, so its Čech complex need not compute the cohomology of X. The proof instead builds X up one affine open at a time. For $X = U \cup V$ with V affine, the relative Mayer-Vietoris sequence places $R^i f_* \mathscr{F}$ between a cokernel and a kernel of maps among the higher direct images from U, V and $U \cap V$, and the case $i = 0$ is proposition 5.1.20. The induction that organizes this is in [70, Tag 01XJ], and we do not reproduce it.

6.6.2 The Finiteness Theorem

The most important structural result about higher direct images is that they preserve coherence for projective morphisms. This is the "relative" version of the statement that cohomology groups of coherent sheaves on projective varieties are finite-dimensional, and it is the technical engine behind semicontinuity theorems, base change, and the construction of Hilbert schemes.

The strategy is to reduce to the computation of cohomology of twisted sheaves on projective space (theorem 6.3.1), which we already know completely. The key tools are:

1. Every coherent sheaf on a projective scheme is a quotient of a direct sum of twisted sheaves (corollary 5.3.27).
2. Cohomology of twisted sheaves on $\mathbb{P}^n_R$ gives finitely generated R-modules (theorem 6.3.1).
3. Descending induction on i, starting from the vanishing of H^i of a quasi-coherent sheaf on $\mathbb{P}^N_R$ for $i > N$, since $\mathbb{P}^N_R$ is covered by $N + 1$ affines (theorem 6.1.19).

Theorem 6.6.6: Finiteness of Higher Direct Images

Let $f : X \to Y$ be a projective morphism of Noetherian schemes, and let $\mathscr{F}$ be a coherent sheaf on X. Then:

1. Each $R^i f_* \mathscr{F}$ is a coherent sheaf on Y.
2. $R^i f_* \mathscr{F} = 0$ for $i > \dim X$.

***Proof*:**
Part (2) follows immediately from theorem 6.2.5: the stalks $(R^i f_* \mathscr{F})_y$ are filtered colimits of $H^i(f^{-1}(V), \mathscr{F})$ over open neighborhoods V of y (proposition 6.6.2), and $\dim f^{-1}(V) \leq \dim X$.

For part (1), the question is local on Y, so we may assume $Y = \operatorname{Spec} R$ is affine. Since f is projective, there is a closed immersion $i : X \hookrightarrow \mathbb{P}^N_R$ for some $N \geq 1$, and f factors as $f = \pi \circ i$ where $\pi : \mathbb{P}^N_R \to \operatorname{Spec} R$ is the structure morphism. By corollary 6.6.12, $R^i f_* \mathscr{F} \cong R^i \pi_* (i_* \mathscr{F})$. Since i is a closed immersion, $i_* \mathscr{F}$ is coherent on $\mathbb{P}^N_R$: over an affine open $\operatorname{Spec} S \subseteq \mathbb{P}^N_R$ we have $i^{-1}(\operatorname{Spec} S) = \operatorname{Spec} T$ with $S \to T$ surjective (proposition 3.1.24(1)), so there $i_* \mathscr{F}$ is the sheaf associated to the finitely generated T-module $\mathscr{F}(\operatorname{Spec} T)$ viewed as an S-module (proposition 5.1.14(6)), which is finitely generated over S because $S \to T$ is surjective. Hence it suffices to prove the theorem for $\mathbb{P}^N_R$ over $\operatorname{Spec} R$. Since π and $\operatorname{Spec} R$ are separated,

proposition 6.6.14 identifies $R^i\pi_*\mathscr{G}$, for a coherent sheaf $\mathscr{G}$ on $\mathbb{P}^N_R$, with the quasi-coherent sheaf associated to the R-module $H^i(\mathbb{P}^N_R, \mathscr{G})$, which is coherent exactly when that module is finitely generated. So it suffices to show:

Claim: For any coherent sheaf $\mathscr{G}$ on $\mathbb{P}^N_R$, the R-module $H^i(\mathbb{P}^N_R, \mathscr{G})$ is finitely generated.

We prove the claim by descending induction on i. For $i > N$, we have $H^i = 0$, and there is nothing to prove: $\mathbb{P}^N_R$ is covered by the $N+1$ standard affine opens, whose Čech complex vanishes in degrees above N, and theorem 6.1.19 identifies its cohomology with $H^i(\mathbb{P}^N_R, \mathscr{G})$.

For the inductive step, assume that $H^j(\mathbb{P}^N_R, \mathscr{H})$ is finitely generated for all $j > i$ and all coherent $\mathscr{H}$. By corollary 5.3.27, $\mathscr{G}$ admits a surjection from a direct sum of twisted sheaves:

$$\mathcal{E} = \bigoplus_{j=1}^{s} \mathcal{O}_{\mathbb{P}^N}(d_j) \twoheadrightarrow \mathscr{G} \to 0$$

for some integers $d_1, \ldots, d_s$. Let $\mathcal{K} = \ker(\mathcal{E} \twoheadrightarrow \mathscr{G})$, which is coherent (since $\mathbf{Coh}(X)$ is an abelian category). The short exact sequence $0 \to \mathcal{K} \to \mathcal{E} \to \mathscr{G} \to 0$ induces a long exact sequence:

$$H^i(\mathbb{P}^N, \mathcal{E}) \longrightarrow H^i(\mathbb{P}^N, \mathscr{G}) \longrightarrow H^{i+1}(\mathbb{P}^N, \mathcal{K}) \longrightarrow H^{i+1}(\mathbb{P}^N, \mathcal{E}).$$

By theorem 6.3.1, $H^i(\mathbb{P}^N, \mathcal{E}) = \bigoplus_j H^i(\mathbb{P}^N, \mathcal{O}(d_j))$ is a finitely generated R-module (it is even free). By the inductive hypothesis, $H^{i+1}(\mathbb{P}^N, \mathcal{K})$ is finitely generated. Hence $H^i(\mathbb{P}^N, \mathscr{G})$, which sits between two finitely generated R-modules in the exact sequence, is itself finitely generated (using the standard fact that a subquotient of a finitely generated module over a Noetherian ring is finitely generated).

The most commonly used special case is when the base is a field:

Corollary 6.6.7: Finite-Dimensionality of Cohomology

Let X be a projective scheme over a field k, and let $\mathscr{F}$ be a coherent sheaf on X. Then each cohomology group $H^i(X, \mathscr{F})$ is a finite-dimensional k-vector space.

***Proof*:**
Apply theorem 6.6.6 with $Y = \operatorname{Spec} k$. The higher direct images $R^i f_*\mathscr{F}$ are coherent sheaves on $\operatorname{Spec} k$, i.e., finite-dimensional k-vector spaces. Since $R^i f_*\mathscr{F} = H^i(X, \mathscr{F})$ when Y is a point, the result follows.

This result justifies the use of the notation $h^i(X, \mathscr{F}) = \dim_k H^i(X, \mathscr{F})$ for the *Betti numbers* of the sheaf $\mathscr{F}$, and underpins the entire theory of Hilbert polynomials and Euler characteristics. Let us also record the general version for the base that we referenced earlier:

Corollary 6.6.8: (Reminder) Pushforward of Coherent Sheaves by Projective Morphisms

Let $f : X \to Y$ be a projective morphism of Noetherian schemes. Then f_* sends coherent sheaves on X to coherent sheaves on Y, and more generally, each $R^i f_*$ sends $\mathbf{Coh}(X)$ to $\mathbf{Coh}(Y)$.

***Proof*:**
This is a restatement of theorem 6.6.6(1).

6.6.9 Remark: Properness vs. Projectivity The finiteness theorem as stated requires f to be *projective*. The natural level of generality is *proper*: Grothendieck proved that theorem 6.6.6 holds whenever f is a proper morphism of Noetherian schemes. Since every projective morphism is proper (theorem 3.5.32), the statement above covers the cases most commonly encountered in practice. The proper case is proved in section 6.6.3. Later in this book it enters the semicontinuity and base change theorems of section 6.8, which are stated for proper morphisms. *Algebraic K-Theory* [6] needs it in order to define the pushforward along a proper morphism on the Grothendieck group $K_0(\mathbf{Coh}(X))$ of coherent sheaves.

6.6.3 The Proper Case

The projective hypothesis entered the proof of theorem 6.6.6 through the closed immersion into $\mathbb{P}^N_Y$, which is what made the twisting sheaves available. A proper morphism need not admit such an embedding. Chow's lemma (theorem 3.5.34) is the substitute: it produces a projective morphism that agrees with f over a dense open set, and the discrepancy is then pushed into a smaller closed subset where an inductive hypothesis applies.

Two preliminaries are needed. The first records that a coherent sheaf can be broken into pieces supported on integral closed subschemes, which is what makes the induction bite.

Lemma 6.6.10: Filtration of a Coherent Sheaf by Integral Subschemes

Let X be a Noetherian scheme and $\mathscr{F}$ a coherent $\mathcal{O}_X$-module. Then $\mathscr{F}$ admits a finite filtration by coherent subsheaves

$$0 = \mathscr{F}_0 \subseteq \mathscr{F}_1 \subseteq \cdots \subseteq \mathscr{F}_n = \mathscr{F}$$

such that each quotient $\mathscr{F}_j/\mathscr{F}_{j-1}$ is annihilated by the ideal sheaf of an integral closed subscheme $Z_j \subseteq \operatorname{Supp}\mathscr{F}$, so that each quotient is the pushforward of a coherent sheaf on such a Z_j.

Two remarks on what the lemma does *not* say. It does not claim the quotients are copies of $\mathcal{O}_{Z_j}$: on a smooth projective curve a line bundle L of negative degree admits no such filtration, since the bottom step would embed $\mathcal{O}_{Z_1}$ in L, impossible for $Z_1 = X$ because $H^0(L) = 0$ and impossible for $Z_1 \subsetneq X$ because $\mathcal{O}_{Z_1}$ is torsion while L is not. And it is not proved by taking the affine statement, that a finitely generated module over a Noetherian ring is filtered with quotients $A/\mathfrak{p}$, and gluing: subsheaves chosen independently over overlapping affines need not agree, so filtrations do not glue.

***Proof*:**
The filtration is built in two stages. The first makes $\mathscr{F}$ a module over the reduced support, the second breaks that support into irreducible pieces.

Stage 1: reduce to a module on the reduced support. Let $Z = \operatorname{Supp}\mathscr{F}$ with its reduced induced structure (definition 3.1.26) and let $\mathcal{I} \subseteq \mathcal{O}_X$ be its ideal sheaf. On an affine open $\operatorname{Spec} A$ with $\mathscr{F}$

given by a finitely generated module M, the support is $V(\mathrm{Ann}M)$ and $\mathcal{I}$ is $\sqrt{\mathrm{Ann}M}$. Since A is Noetherian that radical is finitely generated, so $\mathcal{I}^m \subseteq \mathrm{Ann}M$ for some m; taking the maximum over a finite affine cover gives an n with $\mathcal{I}^n\mathscr{F} = 0$ globally. The chain

$$\mathscr{F} \supseteq \mathcal{I}\mathscr{F} \supseteq \mathcal{I}^2\mathscr{F} \supseteq \cdots \supseteq \mathcal{I}^n\mathscr{F} = 0$$

is therefore finite, its terms are coherent, and each quotient $\mathcal{I}^j\mathscr{F}/\mathcal{I}^{j+1}\mathscr{F}$ is annihilated by $\mathcal{I}$, hence is a coherent module on Z. So it suffices to filter a coherent module on a *reduced* closed subscheme.

Stage 2: Noetherian induction on the reduced support. Argue by corollary 0.3.5 over the closed subsets of X, which satisfy the descending chain condition (lemma 2.4.2). Let $\mathscr{G}$ be coherent on a reduced closed Z, and assume the lemma for every coherent sheaf whose support is properly contained in Z. If Z is irreducible then, being reduced, it is integral, and the one-step filtration $0 \subseteq \mathscr{G}$ already has the required form. Otherwise write $Z = Z_1 \cup \cdots \cup Z_m$ for the irreducible components, $m \geq 2$, and let $\mathcal{J}$ be the ideal sheaf of Z_1 inside Z. Then:

1. $\mathscr{G}/\mathcal{J}\mathscr{G}$ is annihilated by $\mathcal{J}$, so it is a coherent module on Z_1, which is integral.
2. $\mathrm{Supp}(\mathcal{J}\mathscr{G}) \subseteq Z_2 \cup \cdots \cup Z_m$, a proper closed subset of Z. Indeed, let $z \in Z_1$ lie on no other component. Some open neighbourhood of z in Z meets only Z_1, so on it Z and Z_1 agree as *reduced* schemes and $\mathcal{J}$ vanishes there; hence $(\mathcal{J}\mathscr{G})_z = 0$. This is where Stage 1 is used: for non-reduced Z the ideal $\mathcal{J}$ need not vanish at such a z, which is why the naive one-stage argument fails.

The inductive hypothesis applied to $\mathcal{J}\mathscr{G}$ gives a filtration of it of the required shape, and appending $\mathscr{G}$ itself adds the quotient $\mathscr{G}/\mathcal{J}\mathscr{G}$ from (1). Splicing this into the Stage 1 chain gives a finite filtration of the original $\mathscr{F}$ whose quotients are all coherent modules on integral closed subschemes of $\mathrm{Supp}\mathscr{F}$, as we sought to show.

The second preliminary is the relative form of the Leray spectral sequence. The version recorded in theorem 6.6.16 computes cohomology on the total space; what is needed here is the version comparing higher direct images along a composite, which is the Grothendieck spectral sequence of [18, The Grothendieck Spectral Sequence] applied to the pair of pushforward functors.

Lemma 6.6.11: Higher Direct Images Along a Composite

Let $\pi\colon X' \to X$ and $f\colon X \to Y$ be morphisms of Noetherian schemes and $\mathscr{G}$ a quasi-coherent sheaf on X'. Then there is a convergent first-quadrant spectral sequence, natural in $\mathscr{G}$,

$$E_2^{p,q} = R^p f_*\big(R^q\pi_*\mathscr{G}\big) \implies R^{p+q}(f\circ\pi)_*\mathscr{G}$$

***Proof*:**
Apply [18, The Grothendieck Spectral Sequence] with $F = \pi_*$ and $G = f_*$, both left exact additive functors between categories of $\mathcal{O}$-modules, which have enough injectives. The hypothesis to discharge is that π_* carries injective $\mathcal{O}_{X'}$-modules to f_*-acyclic $\mathcal{O}_X$-modules. An injective $\mathcal{O}_{X'}$-module is flasque, the pushforward of a flasque sheaf is flasque, and flasque sheaves are acyclic for every pushforward functor, so the hypothesis holds, completing the proof.

Corollary 6.6.12: Higher Direct Images Along a Closed Immersion

Let $i\colon Z \hookrightarrow X$ be a closed immersion and $f\colon X \to Y$ a morphism of Noetherian schemes, and let $\mathcal{H}$ be a quasi-coherent sheaf on Z. Then i_* is exact, and $R^p f_*(i_*\mathcal{H}) \cong R^p(f\circ i)_*\mathcal{H}$ for every $p \geq 0$.

***Proof*:**
Since Z is closed in X, the stalk $(i_*\mathscr{G})_x$ of a sheaf $\mathscr{G}$ on Z is $\mathscr{G}_x$ for $x \in Z$ and 0 for $x \notin Z$, so i_* is exact. For an injective resolution $0 \to \mathcal{H} \to \mathscr{I}^\bullet$ this gives $R^q i_*\mathcal{H} = h^q(i_*\mathscr{I}^\bullet) = i_* h^q(\mathscr{I}^\bullet) = 0$ for $q > 0$. The spectral sequence of lemma 6.6.11 for i and f is therefore concentrated in the row $q = 0$, and its edge maps give $R^p f_*(i_*\mathcal{H}) \cong R^p(f\circ i)_*\mathcal{H}$.

Theorem 6.6.13: Finiteness of Higher Direct Images, Proper Case

Let $f\colon X \to Y$ be a *proper* morphism of Noetherian schemes and let $\mathscr{F}$ be a coherent sheaf on X. Then:

1. $R^i f_*\mathscr{F}$ is coherent on Y for every $i \geq 0$;
2. $R^i f_*\mathscr{F} = 0$ for $i > \dim X$.

Part (2) is the easy half and is proved first, since it is independent of everything below. By proposition 6.6.2 the sheaf $R^i f_*\mathscr{F}$ is the sheafification of $V \mapsto H^i(f^{-1}V, \mathscr{F}|_{f^{-1}V})$. Each $f^{-1}V$ is a Noetherian topological space of dimension at most $\dim X$, so theorem 6.2.5 makes those groups vanish for $i > \dim X$, and a presheaf that is zero sheafifies to zero. Note that this half needs nothing about coherence or properness beyond X being Noetherian; it is (1) that carries the content.

***Proof*:**
Only (1) remains. Coherence is local on Y, so assume $Y = \operatorname{Spec} A$ with A Noetherian. For a closed subset $Z \subseteq X$ let $P(Z)$ be the statement that $R^i f_*\mathscr{G}$ is coherent for every i and every coherent $\mathscr{G}$ on X with $\operatorname{Supp}\mathscr{G} \subseteq Z$. The closed subsets of the Noetherian space X satisfy the descending chain condition (lemma 2.4.2), so by corollary 0.3.5 it suffices to prove $P(Z)$ assuming $P(Z')$ for every closed $Z' \subsetneq Z$. Taking $Z = X$ then gives the theorem.

Step 1: reduction to a coherent sheaf on an integral closed subscheme. By lemma 6.6.10 the sheaf $\mathscr{G}$ has a finite filtration whose quotients are pushforwards of coherent sheaves on integral closed subschemes of Z. Higher direct images take a short exact sequence to a long exact sequence, and all the sheaves in sight are quasi-coherent, f being proper and hence quasi-compact and quasi-separated. Inside quasi-coherent sheaves on a Noetherian scheme the coherent ones are closed under subobjects, quotients and extensions, so in a long exact sequence a term flanked by coherent neighbours is itself coherent. Induction along the filtration therefore reduces the claim to the case where $\mathscr{G}$ is a coherent sheaf on an integral closed subscheme of Z. If that subscheme is proper inside Z the inductive hypothesis applies, so we may assume it is Z itself and that Z is integral.

Step 2: Chow's lemma. The composite $Z \hookrightarrow X \to Y$ is proper, a closed immersion being proper and properness being stable under composition (proposition 3.5.23). Apply theorem 3.5.34 to it: there is a surjective projective morphism $\pi\colon Z' \to Z$ with $g := f\circ\pi\colon Z' \to Y$ projective, and a

dense open $U \subseteq Z$ over which π is an isomorphism.

Step 3: comparing $\mathscr{G}$ with its pullback. Let $\mathscr{G}' = \pi^*\mathscr{G}$, a coherent sheaf on Z', and consider the unit map $u\colon \mathscr{G} \to \pi_*\mathscr{G}'$. Over U the morphism π is an isomorphism, so $u|_U$ is an isomorphism. Now $\pi_*\mathscr{G}'$ is coherent, by theorem 6.6.6(1) in degree 0 applied to the projective morphism π, so $\ker u$ and $\operatorname{coker} u$ are coherent; and both are supported in the closed set $Z \setminus U$, which is a proper closed subset of Z since U is dense. The inductive hypothesis therefore gives coherence of $R^i f_*(\ker u)$ and $R^i f_*(\operatorname{coker} u)$ for all i. By the long exact sequences attached to

$$0 \to \ker u \to \mathscr{G} \to \operatorname{im} u \to 0, \qquad 0 \to \operatorname{im} u \to \pi_*\mathscr{G}' \to \operatorname{coker} u \to 0$$

it now suffices to prove that $R^i f_*(\pi_*\mathscr{G}')$ is coherent for every i.

Step 4: the spectral sequence. Apply lemma 6.6.11 to π and f with the sheaf $\mathscr{G}'$:

$$E_2^{p,q} = R^p f_*\big(R^q \pi_* \mathscr{G}'\big) \implies R^{p+q} g_* \mathscr{G}'$$

The abutment is coherent for every degree, because g is projective and theorem 6.6.6 applies to it. For $q > 0$ the sheaf $R^q\pi_*\mathscr{G}'$ is coherent, again by theorem 6.6.6 applied to the projective morphism π, and it is supported in $Z \setminus U$, since π is an isomorphism over U and higher direct images along an isomorphism vanish. So the inductive hypothesis makes every $E_2^{p,q}$ with $q > 0$ coherent. Throughout, sheaves living on the closed subscheme Z are fed to the hypothesis $P(-)$, which is stated for sheaves on X, via the identification $R^i f_*(i_*\mathcal{H}) \cong R^i(f|_Z)_*\mathcal{H}$ for the closed immersion $i\colon Z \hookrightarrow X$ (corollary 6.6.12).

It remains to treat the bottom row $E_2^{p,0} = R^p f_*(\pi_*\mathscr{G}')$, which is the object we want. For $p = 0$ the edge map $E_2^{0,0} \to R^0 g_*\mathscr{G}'$ is an isomorphism, so $f_*(\pi_*\mathscr{G}')$ is coherent. Fix $p \geq 1$. The differential *into* $E_r^{p,0}$ is $d_r\colon E_r^{p-r,\,r-1} \to E_r^{p,0}$, whose source sits in row $r - 1 \geq 1$, hence is coherent by the previous paragraph. The differential *out* of it lands in $E_r^{p+r,\,1-r}$, which vanishes because the sequence is first-quadrant and $1 - r < 0$. So each page is obtained from the previous one by a *quotient*, $E_{r+1}^{p,0} = E_r^{p,0}/\operatorname{im} d_r$, with $\operatorname{im} d_r$ a quotient of a coherent sheaf. Closure of the coherent sheaves under subobjects, quotients and extensions is what lets coherence travel back down the pages. Since $E_\infty^{p,0}$ is a graded piece of the coherent sheaf $R^p g_*\mathscr{G}'$ it is coherent, and the tower stabilizes at $E_{p+1}^{p,0} = E_\infty^{p,0}$. Descending from $r = p + 1$ to $r = 2$, each stage sits in a short exact sequence

$$0 \to \operatorname{im} d_r \to E_r^{p,0} \to E_{r+1}^{p,0} \to 0$$

whose outer terms are coherent, the left one a quotient of the coherent $E_r^{p-r,\,r-1}$ and the right one coherent by descending induction. Closure under extensions then gives $E_2^{p,0}$ coherent.

Therefore $R^i f_*(\pi_*\mathscr{G}')$ is coherent for every i, which by Step 3 gives $P(Z)$ and completes the Noetherian induction, as we sought to show.

6.6.4 Computation of Higher Direct Images

Higher direct images can be computed using the Čech machinery, provided the morphism is sufficiently well-behaved:

Proposition 6.6.14: Čech Computation of Higher Direct Images

Let $f : X \to Y$ be a separated morphism of Noetherian schemes with Y separated, $\mathscr{F}$ a quasi-coherent sheaf on X, and $\mathcal{U} = \{U_1, \ldots, U_r\}$ a finite open affine cover of X. Then for each open affine $V = \operatorname{Spec} R \subseteq Y$, the higher direct images are computed by:

$$R^i f_*(\mathscr{F})(V) \;\cong\; \check{H}^i(\mathcal{U}|_{f^{-1}(V)},\, \mathscr{F}|_{f^{-1}(V)}),$$

where $\mathcal{U}|_{f^{-1}(V)} = \{U_j \cap f^{-1}(V)\}$ is the restricted cover.

***Proof*:**
For affine $V \subseteq Y$, the preimage $f^{-1}(V)$ is a Noetherian scheme, separated over V (separatedness is preserved under base change). Each $U_j \cap f^{-1}(V) = U_j \times_Y V$ is affine because Y is separated, and the intersection of two affine opens in $f^{-1}(V)$ is affine by proposition 3.5.8. By theorem 6.1.19, the Čech cohomology with respect to this affine cover computes the derived-functor cohomology:

$$\check{H}^i(\mathcal{U}|_{f^{-1}(V)}, \mathscr{F}) \cong H^i(f^{-1}(V), \mathscr{F}).$$

For $g \in R$, the restricted cover of $f^{-1}(D(g))$ has Čech complex $C^\bullet(\mathcal{U}|_{f^{-1}(V)}, \mathscr{F})_g$, since $\mathscr{F}$ is quasi-coherent, and localization is exact, so $H^i(f^{-1}(D(g)), \mathscr{F}) \cong H^i(f^{-1}(V), \mathscr{F})_g$. On the distinguished opens of V the presheaf of proposition 6.6.2 therefore agrees with the quasi-coherent sheaf associated to $H^i(f^{-1}(V), \mathscr{F})$, so its sheafification $R^i f_* \mathscr{F}|_V$ is that sheaf, with sections $H^i(f^{-1}(V), \mathscr{F})$ over V.

Example 6.6.15: Higher Direct Images

1. *Structure morphism of projective space.* Let R be a Noetherian ring, let $n \geq 1$, and let $\pi : \mathbb{P}^n_R \to \operatorname{Spec} R$ be the structure morphism. By propositions 6.6.5 and 6.6.14:

   $$R^i \pi_*\big(\mathcal{O}_{\mathbb{P}^n}(d)\big) \;\cong\; \widetilde{H^i(\mathbb{P}^n_R, \mathcal{O}(d))},$$

   the quasi-coherent sheaf associated to the R-module $H^i(\mathbb{P}^n_R, \mathcal{O}(d))$, and theorem 6.3.1 computes these modules. In particular, $\pi_*(\mathcal{O}_{\mathbb{P}^n}) \cong \widetilde{R} = \mathcal{O}_{\operatorname{Spec} R}$ and $R^i \pi_*(\mathcal{O}_{\mathbb{P}^n}) = 0$ for $i > 0$. More generally:

   $$\pi_*(\mathcal{O}(d)) \cong \begin{cases} \widetilde{(R[x_0, \ldots, x_n])_d} & d \geq 0, \\ 0 & d < 0, \end{cases}$$

   and all higher direct images vanish for $0 < i < n$.

2. *Blowing up a point.* Let $Y = \mathbb{A}^2_k = \operatorname{Spec} k[x, y]$ and let $f : X = \operatorname{Bl}_0 Y \to Y$ be the blowup of the origin. This is a projective morphism with f an isomorphism over $Y \setminus \{0\}$ and fiber $f^{-1}(0) \cong \mathbb{P}^1_k$ (the exceptional divisor E). Then:

   $$f_* \mathcal{O}_X \cong \mathcal{O}_Y \qquad \text{and} \qquad R^i f_* \mathcal{O}_X = 0 \text{ for } i > 0.$$

   The first statement holds because f is an isomorphism over $Y \setminus \{0\}$. A section of $f_* \mathcal{O}_X$ over an open $V \subseteq Y$ is therefore a rational function regular on $V \setminus \{0\}$, hence at every

codimension-1 point of V, the origin having codimension 2. Since Y is normal, the algebraic Hartogs lemma (theorem 2.6.18) makes it regular on V.

For the vanishing, $R^i f_* \mathcal{O}_X$ is quasi-coherent (proposition 6.6.5) on the affine Y, so it is enough that its global sections vanish, and proposition 6.6.14 with $V = Y$ computes them by the Čech complex of the two charts $U_1 = \operatorname{Spec} k[x,t]$ with $t = y/x$ and $U_2 = \operatorname{Spec} k[y,s]$ with $s = x/y$, which meet in $U_{12} = \operatorname{Spec} k[x,t,t^{-1}]$. With two charts the complex stops in degree 1. The monomials of $\mathcal{O}(U_{12})$ are the $x^a t^b$ with $a \geq 0$ and $b \in \mathbb{Z}$. Those from U_1 are the ones with $b \geq 0$, and since $y^c s^e = x^c t^{c-e}$, those from U_2 are the ones with $b \leq a$. Every monomial is of one of the two kinds, so the Čech differential is surjective and $R^1 f_* \mathcal{O}_X = 0$.

The ideal sheaf $\mathcal{O}_X(-E) = \mathfrak{m}_0 \mathcal{O}_X$ of the exceptional divisor is $xk[x,t]$ on U_1 and $yk[y,s]$ on U_2, and the same computation runs with $a \geq 0$ replaced by $a \geq 1$. Again $R^1 f_*(\mathcal{O}_X(-E)) = 0$, and the global sections are spanned by the $x^a t^b = x^{a-b} y^b$ with $a \geq 1$ and $0 \leq b \leq a$, so $f_*(\mathcal{O}_X(-E)) \cong \mathfrak{m}_0$, the maximal ideal of the origin in $\mathcal{O}_Y$.

3. *Projection of a product.* Let C be a smooth projective curve of genus g over k with $H^0(C, \mathcal{O}_C) = k$, for instance with k algebraically closed (theorem 3.5.33), and consider the projection $f = \operatorname{pr}_2 : C \times_k \mathbb{P}^1_k \to \mathbb{P}^1_k$. Then for the structure sheaf:

$$f_* \mathcal{O}_{C \times \mathbb{P}^1} \cong \mathcal{O}_{\mathbb{P}^1}, \qquad R^1 f_* \mathcal{O}_{C \times \mathbb{P}^1} \cong \mathcal{O}_{\mathbb{P}^1}^g.$$

Cover C by finitely many affines U_j. For $V \subseteq \mathbb{P}^1$ affine, proposition 6.6.14 and theorem 6.1.19 give $R^i f_* \mathcal{O}(V) \cong H^i(C \times V, \mathcal{O})$, computed by the affine cover $\{U_j \times V\}$, whose Čech complex is $C^\bullet(\{U_j\}, \mathcal{O}_C) \otimes_k \mathcal{O}(V)$. Tensoring over k is exact, so $R^i f_* \mathcal{O}(V) \cong H^i(C, \mathcal{O}_C) \otimes_k \mathcal{O}(V)$, and both isomorphisms follow from $H^0(C, \mathcal{O}_C) = k$ and $H^1(C, \mathcal{O}_C) \cong k^g$.

Here every fiber has $h^1 = g$ and $R^1 f_* \mathcal{O}$ is locally free of rank g. For a proper morphism of Noetherian schemes and a coherent $\mathscr{F}$ flat over the base, section 6.8 makes precise how $R^i f_* \mathscr{F}$ relates to the fiber cohomology $H^i(X_y, \mathscr{F}|_{X_y})$.

4. *Failure for non-proper morphisms.* Let $j : \mathbb{A}^1_k \setminus \{0\} \hookrightarrow \mathbb{A}^1_k$ be the open immersion. Then $j_* \mathcal{O} = \mathcal{O}_{\mathbb{A}^1}[1/x]$ (rational functions regular away from the origin), which is *not* coherent as an $\mathcal{O}_{\mathbb{A}^1}$-module: it is quasi-coherent (corresponding to the $k[x]$-module $k[x, x^{-1}]$) but not finitely generated. Also, $R^1 j_* \mathcal{O} = 0$ (since $\mathbb{A}^1 \setminus \{0\}$ is affine, hence acyclic), but the pushforward already fails coherence. This shows that the projectivity hypothesis in theorem 6.6.6 cannot be dropped.

To close this section, we record the connection between higher direct images and the Leray spectral sequence, which is the fundamental tool for computing cohomology of a space in terms of a morphism to a simpler space:

Theorem 6.6.16: Leray Spectral Sequence

Let $f : X \to Y$ be a morphism of ringed spaces and let $\mathscr{F}$ be a sheaf of $\mathcal{O}_X$-modules. Then there is a convergent spectral sequence:

$$E_2^{p,q} = H^p(Y,\, R^q f_* \mathscr{F}) \implies H^{p+q}(X, \mathscr{F}).$$

This is natural in $\mathscr{F}$.

***Proof*:**
This is the Grothendieck spectral sequence for the composite functor $\Gamma(X,-) = \Gamma(Y,-)\circ f_*$. We need f_* to send injective $\mathcal{O}_X$-modules to $\Gamma(Y,-)$-acyclic $\mathcal{O}_Y$-modules. Indeed, if $\mathscr{I}$ is an injective $\mathcal{O}_X$-module, then $f_*\mathscr{I}$ is flasque: for $U \subseteq V$ open in Y, the restriction $f_*\mathscr{I}(V) = \mathscr{I}(f^{-1}(V)) \to \mathscr{I}(f^{-1}(U)) = f_*\mathscr{I}(U)$ is surjective because $\mathscr{I}$ is flasque (by proposition 6.1.5) and $f^{-1}(U) \subseteq f^{-1}(V)$. Hence $f_*\mathscr{I}$ is acyclic for $\Gamma(Y,-)$, and the Grothendieck spectral sequence applies.

Example 6.6.17: Applications of the Leray Spectral Sequence

1. *Affine morphisms.* If $f : X \to Y$ is an affine morphism of Noetherian schemes and $\mathscr{F}$ is quasi-coherent, then $R^q f_*\mathscr{F} = 0$ for all $q > 0$: $f^{-1}(V)$ is affine for every affine open $V \subseteq Y$, so $H^q(f^{-1}(V), \mathscr{F}) = 0$ by theorem 6.2.9, and the sheafification $R^q f_*\mathscr{F}$ of the presheaf $V \mapsto H^q(f^{-1}(V), \mathscr{F})$ (proposition 6.6.2) vanishes. The Leray spectral sequence degenerates at E_2, giving $H^p(X, \mathscr{F}) \cong H^p(Y, f_*\mathscr{F})$. This reduces the cohomology of X to the cohomology of Y with coefficients in the pushforward.

2. *Projective bundles.* Let $\mathcal{E}$ be a locally free sheaf of rank $r+1$ on a Noetherian scheme Y and let $f : X = \mathbb{P}(\mathcal{E}) \to Y$ be the associated projective bundle. The higher direct images of the twisted sheaves $\mathcal{O}_X(d)$ are computed by the relative version of theorem 6.3.1: $f_*\mathcal{O}_X(d) \cong \mathrm{Sym}^d \mathcal{E}$ for $d \geq 0$, and $R^i f_*\mathcal{O}_X(d) = 0$ for $0 < i < r$. The Leray spectral sequence then computes $H^*(X, \mathcal{O}_X(d))$ in terms of $H^*(Y, \mathrm{Sym}^d \mathcal{E})$.

3. *Recovering the projection formula.* For a proper morphism $f : X \to Y$, an $\mathcal{O}_X$-module $\mathscr{F}$, and a locally free sheaf $\mathcal{E}$ on Y, the *projection formula* gives:

$$R^i f_*(\mathscr{F} \otimes f^*\mathcal{E}) \;\cong\; (R^i f_*\mathscr{F}) \otimes \mathcal{E}.$$

   This follows from the fact that tensoring with a locally free sheaf is exact and commutes with the formation of higher direct images (since $f^*\mathcal{E}$ is locally free on X and the tensor product with a flat module commutes with cohomology). Combined with the Leray spectral sequence, the projection formula reduces many cohomological computations to known cases.

6.6.18 Remark: Finiteness Results in Context The finiteness theorem (theorem 6.6.6) is one of several finiteness results that govern the cohomology of schemes. Let us place it in context:

1. *Affine vanishing* (theorem 6.2.9): $H^i = 0$ for $i > 0$ on Noetherian affine schemes with quasi-coherent coefficients. This is the strongest vanishing result, but it applies only to affine schemes.

2. *Dimension bound* (theorem 6.2.5): $H^i = 0$ for $i > \dim X$ on Noetherian spaces. This holds for *all* sheaves of abelian groups, not just quasi-coherent ones.

3. *Finiteness* (theorem 6.6.6): H^i is finitely generated for coherent sheaves on projective schemes over Noetherian rings. This does *not* say the groups vanish, only that they are "finitely complicated."

4. *Serre vanishing* (lemma 6.8.23, through a closed embedding $X \hookrightarrow \mathbb{P}^n_A$): for a coherent sheaf $\mathscr{F}$ on a projective scheme X over a Noetherian ring A, there exists d_0 such that $H^i(X, \mathscr{F}(d)) = 0$ for all $i > 0$ and all $d \geq d_0$. This says that twisting "enough" kills all higher cohomology.

Together, these results are what cohomological arguments rest on in algebraic geometry. The finiteness theorem sits at the center: it is what allows us to define numerical invariants (Euler characteristics, Hilbert polynomials, Hodge numbers) that can then be studied using the vanishing results.

6.7 Euler Characteristics and Hilbert Polynomials

With the finiteness theorem (theorem 6.6.6) in hand, we know that the cohomology groups $H^i(X, \mathscr{F})$ of a coherent sheaf on a projective scheme over a field are finite-dimensional vector spaces. This means we can *count*: we can form the alternating sum of their dimensions, obtaining the *Euler characteristic* $\chi(X, \mathscr{F})$. The Euler characteristic is a cruder invariant than the individual cohomology groups, but it has a decisive advantage: it is *additive* on short exact sequences, while the individual h^i are not. This additivity makes the Euler characteristic computable by purely algebraic means, without explicitly resolving the sheaf.

When X is a projective scheme and $\mathscr{F}$ is twisted by a line bundle $\mathcal{O}_X(d)$, the Euler characteristic $\chi(X, \mathscr{F}(d))$ is a polynomial in d: the *Hilbert polynomial*. For $\mathscr{F} = \mathcal{O}_X$ this polynomial encodes the dimension, degree, and arithmetic genus of X, and it is the basis for the theory of Hilbert schemes and moduli spaces (cf. the discussion in section 3.3.6).

For curves, the interplay between the Euler characteristic and Serre duality yields the *Riemann–Roch theorem*, the single most important computational tool in the theory of algebraic curves.

6.7.1 The Euler Characteristic

Definition 6.7.1: Euler Characteristic

Let X be a projective scheme over a field k and let $\mathscr{F}$ be a coherent sheaf on X. The *Euler characteristic* of $\mathscr{F}$ is:

$$\chi(X,\mathscr{F}) = \sum_{i=0}^{\dim X} (-1)^i \dim_k H^i(X,\mathscr{F}) = \sum_{i\geq 0} (-1)^i h^i(X,\mathscr{F}),$$

where $h^i(X,\mathscr{F}) = \dim_k H^i(X,\mathscr{F})$. The sum is finite by theorem 6.2.5 and each term is finite by corollary 6.6.7.

The key property that makes the Euler characteristic tractable is the following:

Proposition 6.7.2: Additivity of the Euler Characteristic

Let X be a projective scheme over a field k. If $0 \to \mathscr{F}' \to \mathscr{F} \to \mathscr{F}'' \to 0$ is a short exact sequence of coherent sheaves on X, then:

$$\chi(X,\mathscr{F}) = \chi(X,\mathscr{F}') + \chi(X,\mathscr{F}'').$$

***Proof*:**
The short exact sequence induces a long exact sequence in cohomology:

$$0 \to H^0(X,\mathscr{F}') \to H^0(X,\mathscr{F}) \to H^0(X,\mathscr{F}'') \to H^1(X,\mathscr{F}') \to \cdots \to H^n(X,\mathscr{F}'') \to 0,$$

where $n = \dim X$. This is a finite exact sequence of finite-dimensional k-vector spaces. The alternating sum of dimensions in any finite exact sequence $0 \to V_0 \xrightarrow{d_0} V_1 \xrightarrow{d_1} \cdots \xrightarrow{d_{m-1}} V_m \to 0$ is zero. For such a sequence put $r_j = \dim \operatorname{im}(d_j)$ for $0 \leq j \leq m-1$ and $r_{-1} = r_m = 0$. Exactness says $\ker(d_j) = \operatorname{im}(d_{j-1})$, with $d_m = 0$ and $d_{-1} = 0$, so rank and nullity give $\dim V_j = r_{j-1} + r_j$ for $0 \leq j \leq m$, and the alternating sum $\sum_j (-1)^j (r_{j-1} + r_j)$ telescopes to $r_{-1} + (-1)^m r_m = 0$. Grouping the terms by sheaf:

$$\sum_{i=0}^{n} \left[(-1)^i h^i(\mathscr{F}') - (-1)^i h^i(\mathscr{F}) + (-1)^i h^i(\mathscr{F}'')\right] = 0,$$

which rearranges to $\chi(\mathscr{F}) = \chi(\mathscr{F}') + \chi(\mathscr{F}'')$.

6.7.3 Remark: Euler Characteristic as a Group Homomorphism The additivity property can be rephrased as follows. The Grothendieck group $K_0(X)$ of coherent sheaves on X is the free abelian group generated by isomorphism classes $[\mathscr{F}]$ of coherent sheaves, modulo the relation $[\mathscr{F}] = [\mathscr{F}'] + [\mathscr{F}'']$ for every short exact sequence $0 \to \mathscr{F}' \to \mathscr{F} \to \mathscr{F}'' \to 0$. The Euler characteristic defines a group homomorphism:

$$\chi : K_0(X) \longrightarrow \mathbb{Z}, \qquad [\mathscr{F}] \longmapsto \chi(X, \mathscr{F}).$$

This is the starting point for K-theory in algebraic geometry. The Hilbert polynomial also factors through $K_0(X)$, and for smooth X so does the Chern character.

Example 6.7.4: Euler Characteristics

1. *Twisted sheaves on $\mathbb{P}^n_k$.* By theorem 6.3.1, the only nonzero cohomology groups of $\mathcal{O}_{\mathbb{P}^n}(d)$ are H^0 (when $d \geq 0$) and H^n (when $d \leq -(n+1)$). Hence:

$$\chi(\mathbb{P}^n_k, \mathcal{O}(d)) = \begin{cases} \binom{d+n}{n} & d \geq 0, \\ 0 & -(n+1) < d < 0, \\ (-1)^n \binom{-d-1}{n} & d \leq -(n+1). \end{cases}$$

   A fact (which we will prove in the next subsection) is that all three cases are given by a single polynomial in d:

$$\chi(\mathbb{P}^n_k, \mathcal{O}(d)) = \binom{d+n}{n} = \frac{(d+1)(d+2)\cdots(d+n)}{n!}$$

   for *all* $d \in \mathbb{Z}$. For instance, when $n = 1$: $\chi(\mathbb{P}^1, \mathcal{O}(d)) = d+1$. When $d = -1$: $\chi(\mathbb{P}^1, \mathcal{O}(-1)) = 0$, which is correct since both H^0 and H^1 vanish. When $d = -2$: $\chi(\mathbb{P}^1, \mathcal{O}(-2)) = -1$, reflecting $h^0 = 0$ and $h^1 = 1$.

2. *Structure sheaf of a curve.* Let C be a smooth projective curve of genus g over k with $H^0(C, \mathcal{O}_C) = k$, for instance with k algebraically closed (theorem 3.5.33). Then $h^0(C, \mathcal{O}_C) = 1$ and $h^1(C, \mathcal{O}_C) = g$ (this is one definition of the genus; see definition 7.2.4). Hence:

$$\chi(C, \mathcal{O}_C) = 1 - g.$$

   By Serre duality (corollary 6.5.10), $h^1(C, \mathcal{O}_C) = h^0(C, \omega_C) = p_g$, the geometric genus (definition 5.5.29). So $\chi(C, \mathcal{O}_C) = 1 - p_g$, confirming the equivalence of the arithmetic and geometric genus for smooth curves (proposition 7.2.5).

3. *Structure sheaf of a surface.* Let S be a smooth projective surface over k with $H^0(S, \mathcal{O}_S) = k$, for instance with k algebraically closed (theorem 3.5.33). Then:

$$\chi(S, \mathcal{O}_S) = h^0(\mathcal{O}_S) - h^1(\mathcal{O}_S) + h^2(\mathcal{O}_S) = 1 - q + p_g,$$

   where $q = h^1(\mathcal{O}_S)$ is the *irregularity* and $p_g = h^2(\mathcal{O}_S) = h^0(\omega_S)$ is the geometric genus (the last equality by Serre duality). The integer $\chi(\mathcal{O}_S) = 1 - q + p_g$ is a fundamental invariant in the classification of surfaces.

4. *Additivity in action.* Let $C \subset \mathbb{P}^2_k$ be a smooth plane curve of degree d. The short exact sequence $0 \to \mathcal{O}_{\mathbb{P}^2}(-d) \to \mathcal{O}_{\mathbb{P}^2} \to \mathcal{O}_C \to 0$ gives:

$$\chi(C, \mathcal{O}_C) = \chi(\mathbb{P}^2, \mathcal{O}_{\mathbb{P}^2}) - \chi(\mathbb{P}^2, \mathcal{O}(-d)) = 1 - \binom{-d+2}{2} = 1 - \frac{(d-1)(d-2)}{2}.$$

Hence $g = \frac{(d-1)(d-2)}{2}$, the classical genus formula for plane curves. For $d = 1$: $g = 0$ (a line). For $d = 3$: $g = 1$ (a smooth cubic). For $d = 4$: $g = 3$ (a quartic).

6.7.2 Hilbert Polynomials

The Euler characteristic of a twisted sheaf $\mathscr{F}(d)$ is a polynomial in d. This is the most important non-obvious property of the Euler characteristic. Polynomiality, the degree and the sign of the leading term all come from the additivity of χ, by induction on hyperplane sections.

Theorem 6.7.5: Hilbert Polynomial

Let X be a projective scheme over a field k, equipped with a very ample sheaf $\mathcal{O}_X(1)$, and let $\mathscr{F}$ be a coherent sheaf on X. Then there exists a polynomial $P_{\mathscr{F}} \in \mathbb{Q}[t]$, called the *Hilbert polynomial* of $\mathscr{F}$, such that:

$$P_{\mathscr{F}}(d) = \chi(X, \mathscr{F}(d)) \qquad \text{for all } d \in \mathbb{Z}.$$

If $\mathscr{F} \neq 0$, the degree of $P_{\mathscr{F}}$ equals the dimension of the support of $\mathscr{F}$, and the leading coefficient is positive.

***Proof*:**

For every field extension K/k, example 6.8.14(3) gives $\chi(X, \mathscr{F}(d)) = \chi(X_K, \mathscr{F}_K(d))$. Take $K = \overline{k}$, which is infinite. The support of $\mathscr{F}_K$ is the preimage of $\operatorname{supp}\mathscr{F}$. Over an affine open $\operatorname{Spec} A$ of X on which $\operatorname{supp}\mathscr{F}$ is $V(\mathfrak{a})$, that preimage is the underlying set of $\operatorname{Spec}(A/\mathfrak{a} \otimes_k K)$, an integral extension of $A/\mathfrak{a}$, so $\dim(\operatorname{supp}\mathscr{F}_K) = \dim(\operatorname{supp}\mathscr{F})$ by lemma 2.6.3. We may therefore replace k by K, and we assume k infinite. If $\mathscr{F} = 0$ then $P_{\mathscr{F}} = 0$, so let $\mathscr{F} \neq 0$. We prove the whole statement by induction on $r = \dim(\operatorname{supp}\mathscr{F})$.

Base case ($r = 0$): If $\operatorname{supp}\mathscr{F}$ is zero-dimensional, it consists of finitely many closed points (since X is Noetherian). On a zero-dimensional space, H^0 is the only nonzero cohomology group, and twisting by $\mathcal{O}_X(d)$ has no effect on a sheaf supported at finitely many points (since the stalk of $\mathcal{O}_X(d)$ at a closed point is isomorphic to $\mathcal{O}_{X,x}$, with the twist absorbed into the local isomorphism). Hence $\chi(X, \mathscr{F}(d)) = h^0(X, \mathscr{F}) = \dim_k \Gamma(X, \mathscr{F})$, a constant polynomial, positive because $\Gamma(X, \mathscr{F})$ is the sum of the nonzero stalks of $\mathscr{F}$ at the points of its support.

Inductive step: Let $r \geq 1$. Choose a hyperplane section $H \in |\mathcal{O}_X(1)|$ that does not contain any associated point of $\mathscr{F}$ (this is possible because $\mathscr{F}$ has finitely many associated points, $\mathcal{O}_X(1)$ is generated by the pullbacks of the coordinates $x_0, \ldots, x_N$ along the embedding (theorem 5.3.43(1)), and k is infinite). Then multiplication by the section $s \in H^0(X, \mathcal{O}_X(1))$ defining H gives an injective map $\mathscr{F}(-1) \xrightarrow{\cdot s} \mathscr{F}$. Let $\mathscr{G}$ be the cokernel:

$$0 \longrightarrow \mathscr{F}(-1) \xrightarrow{\cdot s} \mathscr{F} \longrightarrow \mathscr{G} \longrightarrow 0.$$

Since s vanishes on H, the sheaf $\mathscr{G}$ is supported on $\operatorname{supp}(\mathscr{F}) \cap H$, and at each point x of that set $\mathscr{G}_x = \mathscr{F}_x/s\mathscr{F}_x \neq 0$ by Nakayama's lemma, so $\operatorname{supp}\mathscr{G} = \operatorname{supp}(\mathscr{F}) \cap H$. This set has dimension exactly $r-1$. The section s is the restriction of a linear form ℓ in $x_0, \ldots, x_N$, so $H = X \cap V_+(\ell)$. Let Z be an irreducible component of $\operatorname{supp}\mathscr{F}$ with $\dim Z \geq 1$, a closed subvariety of $\mathbb{P}^N_k$ with its reduced structure. Its generic point is an associated point of $\mathscr{F}$, so $Z \not\subseteq V_+(\ell)$, and every irreducible component of $Z \cap V_+(\ell)$ has dimension $\dim Z - 1$ by lemma 3.7.11. The zero-dimensional components of $\operatorname{supp}\mathscr{F}$ are associated points and miss H. This gives the bound $\leq r-1$. When $\dim Z = r$, the set $Z \cap V_+(\ell)$ is nonempty: otherwise Z is a closed subscheme of the affine scheme $D_+(\ell)$, hence affine, and proper over k by theorem 3.5.32, hence finite over k by proposition 3.5.25 and of dimension 0. Twisting the whole sequence by $\mathcal{O}_X(d)$:

$$0 \longrightarrow \mathscr{F}(d-1) \longrightarrow \mathscr{F}(d) \longrightarrow \mathscr{G}(d) \longrightarrow 0.$$

By the additivity of χ:

$$\chi(X, \mathscr{F}(d)) - \chi(X, \mathscr{F}(d-1)) = \chi(X, \mathscr{G}(d)).$$

By the inductive hypothesis, $\chi(X, \mathscr{G}(d))$ is a polynomial in d of degree $r-1$ with positive leading coefficient c. The function $\Delta P(d) = P(d) - P(d-1)$ is the *difference operator*; if ΔP is a polynomial of degree $r-1$ with leading coefficient c, then P is a polynomial of degree r with leading coefficient c/r[a]. Since $\chi(X, \mathscr{F}(d))$ is an integer for all d and agrees with a polynomial in d after taking differences, it is itself a polynomial in d (with rational coefficients), of degree r with leading coefficient $c/r > 0$.

[a] This is the discrete analogue of the fact that if $f'(x)$ is a polynomial of degree $r-1$, then $f(x)$ is a polynomial of degree r. Concretely: the binomial coefficients $\binom{d}{j}$ form a basis for $\mathbb{Q}[d]$, and $\Delta\binom{d}{j} = \binom{d}{j} - \binom{d-1}{j} = \binom{d-1}{j-1}$, so the difference operator lowers the degree by exactly one.

Definition 6.7.6: Hilbert Polynomial and Arithmetic Genus

Let X be a projective scheme of dimension n over a field k. The *Hilbert polynomial* of X (with respect to $\mathcal{O}_X(1)$) is $P_X = P_{\mathcal{O}_X}$. The *arithmetic genus* of X is:

$$p_a(X) = (-1)^n\big(\chi(X, \mathcal{O}_X) - 1\big) = (-1)^n(P_X(0) - 1).$$

When $H^0(X, \mathcal{O}_X) = k$: for a curve ($n = 1$), $p_a = 1 - \chi(\mathcal{O}_C) = h^1(\mathcal{O}_C)$, which is the genus g when C is smooth, and for a surface ($n = 2$), $p_a = \chi(\mathcal{O}_S) - 1 = h^2(\mathcal{O}_S) - h^1(\mathcal{O}_S)$, which is $p_g - q$ when S is smooth.

Example 6.7.7: Hilbert Polynomials

1. *Projective space.* By Example 6.7.4(1), $P_{\mathbb{P}^n}(d) = \binom{d+n}{n}$. This is a polynomial of degree n in d with leading coefficient $1/n!$. The arithmetic genus is $p_a(\mathbb{P}^n) = (-1)^n\big(\binom{n}{n} - 1\big) = 0$.

2. *Plane curve of degree d.* Let $C \subset \mathbb{P}^2_k$ be a curve of degree d. The short exact sequence $0 \to \mathcal{O}_{\mathbb{P}^2}(-d) \to \mathcal{O}_{\mathbb{P}^2} \to \mathcal{O}_C \to 0$ gives, after twisting by $\mathcal{O}(m)$ and taking χ:

$$P_C(m) = \binom{m+2}{2} - \binom{m-d+2}{2} = dm - \frac{d(d-3)}{2}.$$

This is a polynomial of degree 1 (as expected, since $\dim C = 1$) with leading coefficient d, which is the *degree* of C. The arithmetic genus is $p_a = 1 - P_C(0) = \frac{(d-1)(d-2)}{2}$.

3. *Hilbert polynomial of a line bundle on a curve.* Let C be a smooth projective curve of genus g with $H^0(C, \mathcal{O}_C) = k$ and let $\mathcal{L}$ be a line bundle of degree ℓ on C. By lemma 6.7.11 below:

$$P_{\mathcal{L}}(d) = \chi(C, \mathcal{L}(d)) = \ell + d \cdot \deg(\mathcal{O}_C(1)) + 1 - g.$$

In particular, $\chi(C, \mathcal{L}) = \deg \mathcal{L} + 1 - g$, confirming theorem 6.7.5.

4. *Twisted ideal sheaf.* Let $Z \subset \mathbb{P}^n_k$ be a closed subscheme with ideal sheaf $\mathscr{I}_Z$. Then the short exact sequence $0 \to \mathscr{I}_Z \to \mathcal{O}_{\mathbb{P}^n} \to \mathcal{O}_Z \to 0$ gives $P_{\mathscr{I}_Z}(d) = P_{\mathbb{P}^n}(d) - P_Z(d)$. The Hilbert polynomial of $\mathscr{I}_Z$ encodes the "difference" between the ambient space and the subscheme, and this is the key to defining Hilbert schemes: the closed subschemes of $\mathbb{P}^n_k$ with a given Hilbert polynomial P all lie in the one Hilbert scheme $\mathrm{Hilb}^P_{\mathbb{P}^n_k}$ (theorem 3.3.11).

Proposition 6.7.8: Hilbert Polynomial and Hilbert Function

Let $X \subseteq \mathbb{P}^N_k$ be a projective scheme and $\mathscr{F}$ a coherent sheaf on X. Then for $d \gg 0$:

$$P_{\mathscr{F}}(d) = h^0(X, \mathscr{F}(d)).$$

That is, the Hilbert polynomial agrees with the function $d \mapsto h^0(X, \mathscr{F}(d))$ for all sufficiently large d. If X is cut out by a homogeneous ideal $I \subseteq S = k[x_0, \ldots, x_N]$, then also $P_{\mathcal{O}_X}(d) = \dim_k(S/I)_d$ for $d \gg 0$, so P_X is the Hilbert polynomial of the graded ring S/I, and for a projective variety it is the classical Hilbert polynomial of [22, Hilbert Polynomial].

***Proof*:**
By Serre vanishing (lemma 6.8.23, through the closed embedding $X \subseteq \mathbb{P}^N_k$), there exists d_0 such that $H^i(X, \mathscr{F}(d)) = 0$ for all $i > 0$ and all $d \geq d_0$. For such d:

$$P_{\mathscr{F}}(d) = \chi(X, \mathscr{F}(d)) = h^0(X, \mathscr{F}(d)) - 0 + 0 - \cdots = h^0(X, \mathscr{F}(d)).$$

For the second statement, $\mathcal{O}_X$ is the sheaf associated to the graded S-module S/I, and the natural map $(S/I)_d \to H^0(X, \mathcal{O}_X(d))$ is an isomorphism for $d \gg 0$ [36, Exercise II.5.9(b)], as claimed.

The degree and leading coefficient of the Hilbert polynomial have geometric meaning:

Proposition 6.7.9: Degree and Leading Coefficient

Let $X \subseteq \mathbb{P}^N_k$ be a projective scheme of dimension n and let $\mathscr{F} \neq 0$ be a coherent sheaf with $\dim(\operatorname{supp}\mathscr{F}) = r$. Write:

$$P_{\mathscr{F}}(d) = \frac{a_r}{r!}\, d^r + \text{lower order terms.}$$

Then a_r is a positive integer. When $\mathscr{F} = \mathcal{O}_X$, the integer a_n is the *degree* of X in $\mathbb{P}^N$. When k is algebraically closed and X is a variety, it is the number of points in the intersection of X with a general linear subspace of complementary dimension.

***Proof*:**
The positivity of a_r is part of theorem 6.7.5. The integrality of a_r follows from the inductive proof of theorem 6.7.5, run over an infinite extension of k, which does not change $P_{\mathscr{F}}$: with $\mathscr{G}$ the cokernel of that proof, the difference $\Delta P_{\mathscr{F}} = P_{\mathscr{G}}$ has leading coefficient $r \cdot a_r/r! = a_r/(r-1)!$, so a_r equals the coefficient a_{r-1} of $P_{\mathscr{G}}$ in the same normalization, an integer by induction on r.

When k is algebraically closed and X is a variety, P_X is the classical Hilbert polynomial of X (proposition 6.7.8), so a_n is the classical degree, and [22, Degree Counts Points of a Linear Section] identifies it with the number of points of a general linear section of complementary dimension.

6.7.10 Remark: Invariance of the Hilbert Polynomial The Hilbert polynomial depends on the choice of embedding (or more precisely, on the choice of very ample sheaf $\mathcal{O}_X(1)$). However, the *constant term* $P_X(0) = \chi(X, \mathcal{O}_X)$ is independent of the embedding: it depends only on the abstract scheme X, since it is defined purely in terms of the cohomology of the structure sheaf. Similarly, the arithmetic genus $p_a(X) = (-1)^n(\chi(\mathcal{O}_X) - 1)$ is an intrinsic invariant. In fact, by theorem 5.5.31, the geometric genus $p_g = h^0(X, \omega_X)$ is a birational invariant for smooth projective varieties, and Serre duality relates it to the "top" cohomology of $\mathcal{O}_X$.
The higher coefficients of P_X do depend on the embedding, but they carry geometric information about the embedding itself (degree, sectional genus, etc.).

6.7.3 Riemann–Roch for Curves

Serre duality and the additivity of the Euler characteristic combine into the Riemann-Roch theorem for a smooth projective curve C over any field k with $H^0(C, \mathcal{O}_C) = k$. Chapter 7 develops its consequences over an algebraically closed field.

Here a *smooth projective curve* over a field k is a smooth, projective, integral scheme of dimension 1 over k (the general notion of a curve is definition 7.1.1). Its *genus* is $g = \dim_k H^1(C, \mathcal{O}_C)$.

For a smooth projective curve C over k, the cohomology groups have the following structure: H^0 and H^1 are the only potentially nonzero groups (since $\dim C = 1$), the canonical sheaf $\omega_C = \Omega_{C/k}$ is an invertible sheaf, and Serre duality (corollary 6.5.10) gives $H^1(C, \mathcal{L}) \cong H^0(C, \mathcal{L}^\vee \otimes \omega_C)^*$ for any line bundle $\mathcal{L}$.

Lemma 6.7.11: Degree and Euler Characteristic

Let C be a smooth projective curve over a field k and let $\mathcal{L}$ be a line bundle on C. Then

$$\chi(C, \mathcal{L}) = \deg \mathcal{L} + \chi(C, \mathcal{O}_C)$$

If $H^0(C, \mathcal{O}_C) = k$, for instance if k is algebraically closed (theorem 3.5.33), then $\chi(C, \mathcal{L}) = \deg \mathcal{L} + 1 - g$. Here $\deg \mathcal{L} = \deg D$ for a Weil divisor $D = \sum_p n_p\, p$ with $\mathcal{L} \cong \mathcal{O}_C(D)$ (corollary 5.4.45), where $\deg D = \sum_p n_p\, [\kappa(p) : k]$, and the proof shows that this does not depend on D.

***Proof*:**
The curve C is integral, separated, Noetherian and regular, hence locally factorial (proposition 2.7.27), so every line bundle on C is $\mathcal{O}_C(D)$ for a Weil divisor D (corollary 5.4.45). Let D be a divisor and p a closed point, with $\delta_p = [\kappa(p) : k]$. If t is a uniformizer of the discrete valuation ring $\mathcal{O}_{C,p}$ and n is the coefficient of p in D, then near p the sheaves $\mathcal{O}_C(D)$ and $\mathcal{O}_C(D + p)$ are generated by t^{-n} and t^{-n-1}, and away from p they agree. So the inclusion $\mathcal{O}_C(D) \subseteq \mathcal{O}_C(D + p)$ has cokernel the skyscraper sheaf $\mathcal{O}_{C,p}/(t) = \kappa(p)$ at p, and the sequence

$$0 \to \mathcal{O}_C(D) \to \mathcal{O}_C(D + p) \to \kappa(p) \to 0$$

is exact. The skyscraper sheaf has $H^0 = \kappa(p)$, of dimension δ_p over k, and no higher cohomology (proposition 6.2.4), so additivity of the Euler characteristic (proposition 6.7.2) gives $\chi(\mathcal{O}_C(D + p)) = \chi(\mathcal{O}_C(D)) + \delta_p$. As $\deg(D+p) = \deg D + \delta_p$, the formula $\chi(\mathcal{O}_C(D)) = \deg D + \chi(\mathcal{O}_C)$ holds for D if and only if it holds for $D + p$. It holds for $D = 0$, and every divisor is reached from 0 by finitely many additions and subtractions of closed points, so it holds for every D. Since $\chi(\mathcal{O}_C(D))$ depends only on the isomorphism class of $\mathcal{O}_C(D)$, so does $\deg D$, and $\deg \mathcal{O}_C(D) = \deg D$ is well defined. When $H^0(C, \mathcal{O}_C) = k$, the Euler characteristic $\chi(\mathcal{O}_C)$ is $1 - g$, which gives the last formula, completing the proof.

Theorem 6.7.12: Riemann–Roch Theorem

Let C be a smooth projective curve of genus g over a field k with $H^0(C, \mathcal{O}_C) = k$, and let D be a divisor on C with associated line bundle $\mathcal{L} = \mathcal{O}_C(D)$. Let K be a canonical divisor (so $\mathcal{O}_C(K) \cong \omega_C$). Then:

$$h^0(C, \mathcal{O}_C(D)) - h^0(C, \mathcal{O}_C(K - D)) = \deg D + 1 - g.$$

Equivalently, using the notation $\ell(D) = h^0(C, \mathcal{O}_C(D))$:

$$\ell(D) - \ell(K - D) = \deg D + 1 - g.$$

***Proof*:**
By Serre duality (corollary 6.5.10):

$$h^1(C, \mathcal{O}_C(D)) = \dim_k H^1(C, \mathcal{O}_C(D)) = \dim_k H^0(C, \mathcal{O}_C(D)^\vee \otimes \omega_C)^* = h^0(C, \mathcal{O}_C(K - D)),$$

where we used $\mathcal{O}_C(D)^\vee \otimes \omega_C \cong \mathcal{O}_C(-D) \otimes \mathcal{O}_C(K) \cong \mathcal{O}_C(K-D)$. Now apply lemma 6.7.11:

$$h^0(C, \mathcal{O}_C(D)) - h^1(C, \mathcal{O}_C(D)) = \chi(C, \mathcal{O}_C(D)) = \deg D + 1 - g.$$

Substituting $h^1(C, \mathcal{O}_C(D)) = h^0(C, \mathcal{O}_C(K-D))$ gives the formula, as claimed.

Example 6.7.13: Riemann–Roch in Practice
Let C be as in theorem 6.7.12.

1. *Degree of the canonical divisor.* Setting $D = K$ in Riemann–Roch: $\ell(K) - \ell(0) = \deg K + 1 - g$. Since $\ell(K) = h^0(\omega_C) = g$ and $\ell(0) = h^0(\mathcal{O}_C) = 1$:

   $$\deg K = 2g - 2.$$

   For a smooth plane curve of degree d: $\omega_C \cong \mathcal{O}_C(d-3)$ (by proposition 6.5.7), so $\deg K = d(d-3)$. Setting $g = (d-1)(d-2)/2$, we verify: $2g-2 = (d-1)(d-2)-2 = d^2-3d = d(d-3)$.

2. *Very ample line bundles on curves.* When k is algebraically closed, a line bundle $\mathcal{L}$ on C is very ample (hence gives an embedding into projective space) if $\deg \mathcal{L} \geq 2g+1$. To see this: by Riemann–Roch, $\ell(D) = \deg D + 1 - g$ whenever $\deg D > 2g - 2 = \deg K$ (since then $\deg(K-D) < 0$, forcing $\ell(K-D) = 0$). If $\deg D \geq 2g+1$ and p, q are closed points, equal or not, then $D - p - q$ has degree $\geq 2g-1$, so $\ell(D-p-q) = \ell(D) - 2$. By the criterion of [36, Proposition IV.3.1], the linear system $|D|$ therefore separates points and tangent vectors, and $\mathcal{O}_C(D)$ is very ample (corollary 7.3.4).

3. *Elliptic curves.* Let C be a smooth projective curve of genus 1 with a rational point $O \in C(k)$. By Riemann–Roch with $D = 3O$:

   $$\ell(3O) = 3 + 1 - 1 = 3$$

   (since $\deg(K - 3O) = 2 \cdot 1 - 2 - 3 = -3 < 0$, so $\ell(K - 3O) = 0$). When k is algebraically closed, the three-dimensional space $H^0(C, \mathcal{O}(3O))$ gives a closed embedding $C \hookrightarrow \mathbb{P}^2$, and the image is a smooth plane cubic (proposition 7.6.2).

4. *Rational curves.* Let C have genus 0. Then $\chi(\mathcal{O}_C) = 1$ and for any line bundle $\mathcal{L}$ of degree $d \geq 0$:

   $$h^0(C, \mathcal{L}) = d + 1$$

   (since $h^1 = h^0(\omega_C \otimes \mathcal{L}^{-1})$ and $\deg(\omega_C \otimes \mathcal{L}^{-1}) = -2 - d < 0$). In particular, if C has a k-rational point p, then $h^0(\mathcal{O}_C(p)) = 2$ gives a morphism $C \to \mathbb{P}^1_k$ of degree 1, so $C \cong \mathbb{P}^1_k$.

6.7.14 Remark: Beyond Curves The Riemann–Roch theorem for curves is the "$n = 1$" case of a family of results that express the Euler characteristic in terms of topological and geometric invariants. In higher dimensions, the *Hirzebruch–Riemann–Roch theorem* asserts that for a smooth projective variety X of dimension n and a locally free sheaf $\mathcal{E}$:

$$\chi(X, \mathcal{E}) = \int_X \mathrm{ch}(\mathcal{E}) \cdot \mathrm{td}(T_X),$$

where $\mathrm{ch}(\mathcal{E})$ is the Chern character of $\mathcal{E}$, $\mathrm{td}(T_X)$ is the Todd class of the tangent bundle, and the integral is the degree of the component in codimension n (i.e., the intersection with the fundamental class). For curves, the Chern character of a line bundle is $1 + c_1(\mathcal{L})$ and the Todd class of T_C is $1 + \frac{1}{2}c_1(T_C) = 1 + \frac{1}{2}(2 - 2g)$, so the formula reduces to $\chi(\mathcal{L}) = \deg \mathcal{L} + 1 - g$. For surfaces, it gives the *Noether formula*:

$$\chi(\mathcal{O}_S) = \frac{1}{12}(K_S^2 + e(S)),$$

where K_S^2 is the self-intersection of the canonical class and $e(S)$ is the topological Euler characteristic. These generalizations belong to the intersection-theoretic framework developed in *Intersection Theory* [19].

6.8 Flat Morphisms and Semicontinuity

In the previous sections, we studied the cohomology of a *single* scheme. In practice, schemes often arise in *families*: a morphism $f : X \to Y$ presents X as a family of fibers X_y parameterized by points $y \in Y$, and for a coherent sheaf $\mathscr{F}$ on X with restrictions $\mathscr{F}_y = \mathscr{F}|_{X_y}$ to the fibers, the higher direct images $R^i f_* \mathscr{F}$ relativize cohomology over the base. A fundamental question is: how does the function $y \mapsto h^i(X_y, \mathscr{F}_y)$ behave as y varies?

6.8.1 Relation to Algebraic Topology This idea is already familiar to those who seen Serre spectral sequences where the cohomology of a space E is calculated given a fibration $0 \to F \to E \to B \to 0$.

The answer involves two intertwined ideas. The first is *flatness*: the condition that the fibers of f vary "continuously," without sudden jumps in dimension. Flat families are the algebraic analogue of fiber bundles, and they are the correct notion of "continuously varying family" in algebraic geometry. The second idea is *semicontinuity*: for f proper, Y Noetherian and $\mathscr{F}$ coherent and flat over Y, the dimensions $h^i(X_y, \mathscr{F}_y)$ can jump *up* at special points but not down, and when they are constant on a reduced base, the higher direct images are locally free and commute with base change.

These results are essential for moduli theory. The semicontinuity theorem ensures that the loci where cohomological invariants jump form closed subsets, and Grauert's theorem guarantees that the "generic" behavior of cohomology is well-controlled.

6.8.1 Flat Families

Definition 6.8.2: Family of Schemes

Let X, B be schemes. Then a *family of schemes* is given by a morphism $\pi : X \to B$. The scheme B is sometimes called the *parameter space*.

The notion of a family of schemes as it stands is much too general: consider any family $\pi : X \to B$, choose a closed point $b \in B$, and create a new family by taking the disjoint union of $X \setminus \pi^{-1}(b)$ and Y for some arbitrary $\kappa(b)$-scheme Y, mapped to B through $\operatorname{Spec} \kappa(b) \to B$. So a bare family has no properties worth deducing, and further conditions are needed.

Recall from commutative algebra (see [10]) that an R-module M is *flat* if the functor $M \otimes_R -$ is exact. Geometrically, flatness of a morphism means that the fibers do not "degenerate" in an uncontrolled way:

Definition 6.8.3: Flat Morphism

Let $f : X \to Y$ be a morphism of schemes.

1. An $\mathcal{O}_X$-module $\mathscr{F}$ is *flat over* Y (or f*-flat*) if for every point $x \in X$, the stalk $\mathscr{F}_x$ is a flat $\mathcal{O}_{Y,f(x)}$-module (via the local homomorphism $f_x^{\#} : \mathcal{O}_{Y,f(x)} \to \mathcal{O}_{X,x}$).
2. The morphism f itself is *flat* if $\mathcal{O}_X$ is flat over Y, i.e., if $\mathcal{O}_{X,x}$ is a flat $\mathcal{O}_{Y,f(x)}$-module for every $x \in X$.

Example 6.8.4: Flat and Non-Flat Morphisms

1. *Open immersions.* Any open immersion $j : U \hookrightarrow X$ is flat: the stalk map $\mathcal{O}_{X,x} \to \mathcal{O}_{U,x}$ is the identity for $x \in U$.
2. *Localization.* The morphism $\operatorname{Spec} R_f \to \operatorname{Spec} R$ induced by the localization $R \to R_f$ is flat, since localization is an exact functor.
3. *Finite free morphisms.* If $f : \operatorname{Spec} S \to \operatorname{Spec} R$ makes S a free R-module, then f is flat. More generally, if S is a finitely generated projective R-module, then f is flat.
4. *The family of conics.* Consider the morphism $f : V(xy - t) \to \mathbb{A}^1_k = \operatorname{Spec} k[t]$, where the fiber over $t = a \neq 0$ is the smooth hyperbola $xy = a$, and the fiber over $t = 0$ is the union of coordinate axes $xy = 0$. The coordinate ring is $k[x, y, t]/(xy - t) \cong k[x, y]$, and the map to $k[t]$ sends $t \mapsto xy$. This is flat: $k[x, y]$ is free over $k[xy] = k[t]$ (with basis given by the monomials $x^i y^j$ with either $i = 0$ or $j = 0$). The fiber over $t = 0$ is reducible, but the family is still flat.
5. *A non-flat family.* The blowup morphism $f : \operatorname{Bl}_0 \mathbb{A}^2 \to \mathbb{A}^2$ is *not* flat: the fiber over the origin is $\mathbb{P}^1$ (one-dimensional), while the fiber over every other point is a single point (zero-dimensional). For a flat morphism of schemes of finite type over k with irreducible source X and target Y, every irreducible component of every nonempty fiber has dimension $\dim X - \dim Y$ [36, Corollary III.9.6], so this jump is an obstruction. More precisely, the structure sheaf $\mathcal{O}_{\operatorname{Bl}_0 \mathbb{A}^2}$ is not flat over $\mathcal{O}_{\mathbb{A}^2,0}$.

6. *Families of subschemes.* Let $X = V(x^2 - t) \subset \mathbb{A}^1_x \times \mathbb{A}^1_t$ and let $f : X \to \mathbb{A}^1_t$ be the projection. For k algebraically closed of characteristic $\neq 2$ and a closed point $t = a \neq 0$, the fiber is two reduced points $x = \pm\sqrt{a}$; for $t = 0$, the fiber is the non-reduced scheme $\operatorname{Spec} k[x]/(x^2)$. This family *is* flat: $k[x,t]/(x^2 - t) \cong k[x]$ is free of rank 2 over $k[t]$ (with basis $\{1, x\}$). The non-reduced fiber at $t = 0$ is the "correct" flat limit of the two-point fibers: it remembers the multiplicity. This is exactly the phenomenon we encountered with intersection multiplicities in section 2.2.2.

Flatness as defined in definition 6.8.3 is a condition on stalks. Four facts make it usable in families: flatness is affine-local, it is stable under composition and base change, flat morphisms lift generizations, and a flat morphism locally of finite type over a locally Noetherian base is an open map. The last of these, with finite type sharpened to finite presentation, is the openness statement the theory of étale morphisms rests on ([16]).

Lemma 6.8.5: Transitivity and Base Change for Flat Modules

Let A be a ring.

1. If B is an A-algebra that is flat as an A-module, and M is a flat B-module, then M is flat as an A-module.
2. If M is a flat A-module and $A \to A'$ is any ring homomorphism, then $M \otimes_A A'$ is a flat A'-module.
3. For any multiplicatively closed subset $D \subseteq A$, the A-module $D^{-1}A$ is flat.

***Proof*:**
(1) Let $N \hookrightarrow N'$ be an injection of A-modules. There are natural isomorphisms $N \otimes_A M \cong N \otimes_A (B \otimes_B M) \cong (N \otimes_A B) \otimes_B M$, the second by [12, Associativity of the Tensor Product]. The map $N \otimes_A B \to N' \otimes_A B$ is injective because B is flat over A, and applying $- \otimes_B M$ preserves the injection because M is flat over B.

(2) Let $N \hookrightarrow N'$ be an injection of A'-modules. Using [12, Commutativity of the Tensor Product] to write $M \otimes_A A' \cong A' \otimes_A M$, associativity and $N \otimes_{A'} A' \cong N$ give natural isomorphisms

$$N \otimes_{A'} (A' \otimes_A M) \cong (N \otimes_{A'} A') \otimes_A M \cong N \otimes_A M$$

The map $N \to N'$ stays injective after restricting scalars along $A \to A'$, and M is flat over A, so $N \otimes_A M \to N' \otimes_A M$ is injective.

(3) For any A-module N there is a natural isomorphism $D^{-1}A \otimes_A N \cong D^{-1}N$ ([10, Characterization of Localization]), and localization preserves injections ([10, Localization Is Exact]).

Lemma 6.8.6: Flatness Is Affine-Local

Let $\pi : X \to Y$ be a scheme morphism. The following are equivalent:

1. π is flat (definition 6.8.3).
2. For every pair of affine opens $\operatorname{Spec} B \subseteq X$ and $\operatorname{Spec} A \subseteq Y$ with $\pi(\operatorname{Spec} B) \subseteq \operatorname{Spec} A$, the induced ring map makes B a flat A-module.
3. There are an affine open cover $\{\operatorname{Spec} A_j\}$ of Y and, for each j, an affine open cover $\{\operatorname{Spec} B_{ij}\}$ of $\pi^{-1}(\operatorname{Spec} A_j)$ such that each B_{ij} is a flat A_j-module.

***Proof*:**

Step 1: (2) implies (3). Every $x \in X$ has an affine open neighborhood mapping into an affine open of Y: choose an affine open $\operatorname{Spec} A \ni \pi(x)$, and then an affine open neighborhood of x inside the open set $\pi^{-1}(\operatorname{Spec} A)$, affine opens forming a basis of the topology. Such neighborhoods assemble into covers as in (3), and (2) gives the flatness.

Step 2: (3) implies (1). Let $x \in X$, and choose $\operatorname{Spec} B := \operatorname{Spec} B_{ij} \ni x$ over $\operatorname{Spec} A := \operatorname{Spec} A_j$. The restriction of π is the morphism of affine schemes induced by the ring map $A \to B$ (proposition 3.1.12). Writing $\mathfrak{q} \subseteq B$ for the prime of x and $\mathfrak{p} \subseteq A$ for its contraction, the prime of $y = \pi(x)$, the stalk map is $A_\mathfrak{p} \to B_\mathfrak{q}$. Set $B_\mathfrak{p} := (A \setminus \mathfrak{p})^{-1}B$. Since B is flat over A, the module $B_\mathfrak{p}$ is flat over $A_\mathfrak{p}$ by [10, Flatness Is A Local Property]. The ring $B_\mathfrak{q}$ is a further localization of $B_\mathfrak{p}$, hence flat over $B_\mathfrak{p}$ by lemma 6.8.5(3), and therefore flat over $A_\mathfrak{p}$ by lemma 6.8.5(1). So every stalk map of π is flat.

Step 3: (1) implies (2). Let $\operatorname{Spec} B$ over $\operatorname{Spec} A$ be as in (2), let $N \hookrightarrow N'$ be an injection of A-modules, and let K be the kernel of $N \otimes_A B \to N' \otimes_A B$, a B-module. Fix a prime $\mathfrak{q} \subseteq B$ with contraction $\mathfrak{p} \subseteq A$. Localization is exact ([10, Localization Is Exact]), so $K_\mathfrak{q}$ is the kernel of $(N \otimes_A B)_\mathfrak{q} \to (N' \otimes_A B)_\mathfrak{q}$. The image of $A \setminus \mathfrak{p}$ is invertible in $B_\mathfrak{q}$, making $B_\mathfrak{q}$ an $A_\mathfrak{p}$-module, and [10, Characterization of Localization] together with [12, Associativity of the Tensor Product] identifies

$$(N \otimes_A B)_\mathfrak{q} \cong N \otimes_A B_\mathfrak{q} \cong (N \otimes_A A_\mathfrak{p}) \otimes_{A_\mathfrak{p}} B_\mathfrak{q} \cong N_\mathfrak{p} \otimes_{A_\mathfrak{p}} B_\mathfrak{q}$$

naturally in N. The map $N_\mathfrak{p} \to N'_\mathfrak{p}$ is injective, and $B_\mathfrak{q}$ is a flat $A_\mathfrak{p}$-module, being a stalk map of the flat π. So $K_\mathfrak{q} = 0$ for every prime $\mathfrak{q}$, and $K = 0$ by [10, Zero Module Is A Local Property].

Proposition 6.8.7: Flat Morphisms Are Stable under Composition and Base Change

1. If $\pi : X \to Y$ and $\rho : Y \to Z$ are flat, then $\rho \circ \pi$ is flat.
2. If $\pi : X \to Y$ is flat and $g : Y' \to Y$ is any morphism, then the base change $X \times_Y Y' \to Y'$ is flat.

***Proof*:**
(1) For $x \in X$ the stalk maps form the tower $\mathcal{O}_{Z,\rho(\pi(x))} \to \mathcal{O}_{Y,\pi(x)} \to \mathcal{O}_{X,x}$, in which $\mathcal{O}_{X,x}$ is flat over $\mathcal{O}_{Y,\pi(x)}$ and $\mathcal{O}_{Y,\pi(x)}$ is flat over $\mathcal{O}_{Z,\rho(\pi(x))}$. Lemma 6.8.5(1) gives flatness of $\mathcal{O}_{X,x}$ over $\mathcal{O}_{Z,\rho(\pi(x))}$.

(2) Cover Y' by affine opens $\operatorname{Spec} A'$, each mapping into an affine open $\operatorname{Spec} A \subseteq Y$, as in Step 1 of the proof of lemma 6.8.6. As in the proof of proposition 3.4.25, proposition 3.2.5(1) and the affine construction of theorem 3.2.1 identify the preimage of $\operatorname{Spec} A'$ in $X \times_Y Y'$ with $\pi^{-1}(\operatorname{Spec} A) \times_{\operatorname{Spec} A} \operatorname{Spec} A'$, and covering $\pi^{-1}(\operatorname{Spec} A)$ by affine opens $\operatorname{Spec} B$ covers that preimage by the affine opens $\operatorname{Spec}(B \otimes_A A')$. Each B is flat over A by lemma 6.8.6, so each $B \otimes_A A'$ is flat over A' by lemma 6.8.5(2), and lemma 6.8.6(3) makes the base change flat.

Lemma 6.8.8: Flat Morphisms Lift Generizations

Let $\pi : X \to Y$ be a flat morphism of schemes, let $x \in X$ with $y = \pi(x)$, and let $y' \in Y$ be a generization of y (definition 2.1.8). Then there is a generization x' of x with $\pi(x') = y'$.

***Proof*:**
Step 1: reduction to rings. Choose affine opens $\operatorname{Spec} B \subseteq X$ containing x and $\operatorname{Spec} A \subseteq Y$ containing y with $\pi(\operatorname{Spec} B) \subseteq \operatorname{Spec} A$. An open set contains every generization of each of its points: if $y' \notin \operatorname{Spec} A$, then y' lies in the closed complement, hence so does every point of $\overline{\{y'\}}$, including y, a contradiction. So $y' \in \operatorname{Spec} A$. Let $\mathfrak{q} \subseteq B$ and $\mathfrak{p}, \mathfrak{p}' \subseteq A$ be the primes of x, y, y'. That y' generizes y means $\mathfrak{p}' \subseteq \mathfrak{p}$ (definition 2.1.8), and a generization of x inside $\operatorname{Spec} B$ is a generization in X, so it suffices to find a prime $\mathfrak{q}' \subseteq \mathfrak{q}$ of B contracting to $\mathfrak{p}'$.

Step 2: the local stalk map is faithfully flat. The stalk map $A_{\mathfrak{p}} \to B_{\mathfrak{q}}$ is flat, and it carries $\mathfrak{p}A_{\mathfrak{p}}$ into $\mathfrak{q}B_{\mathfrak{q}}$, since $\mathfrak{p}$ is the contraction of $\mathfrak{q}$. Hence $\mathfrak{p}A_{\mathfrak{p}} \cdot B_{\mathfrak{q}} \subseteq \mathfrak{q}B_{\mathfrak{q}} \neq B_{\mathfrak{q}}$, which is condition (3) of [10, Characterizations of Faithful Flatness] for the unique maximal ideal of the local ring $A_{\mathfrak{p}}$. That result then makes $B_{\mathfrak{q}}$ faithfully flat over $A_{\mathfrak{p}}$ and, for the flat ring map $A_{\mathfrak{p}} \to B_{\mathfrak{q}}$, makes $\operatorname{Spec} B_{\mathfrak{q}} \to \operatorname{Spec} A_{\mathfrak{p}}$ surjective.

Step 3: contract. The primes of $A_{\mathfrak{p}}$ are the primes of A contained in $\mathfrak{p}$, so $\mathfrak{p}'A_{\mathfrak{p}}$ is one of them, and surjectivity provides a prime of $B_{\mathfrak{q}}$ lying over it. Its contraction $\mathfrak{q}' \subseteq B$ satisfies $\mathfrak{q}' \subseteq \mathfrak{q}$ and $\mathfrak{q}' \cap A = \mathfrak{p}'$. The point $x' = [\mathfrak{q}']$ is a generization of x with $\pi(x') = y'$, as we sought to show.

For ring maps this is the going-down property of flatness. *Commutative Algebra* proves going-down for integral extensions of a normal domain by a domain ([10, Integral Extension Prime Correspondence (Cohen-Seidenberg Theorem)]). Flatness is the other standard hypothesis that yields it.

Lemma 6.8.9: Constructible Sets Stable under Generization Are Open

Let Y be a Noetherian scheme and $E \subseteq Y$ a constructible subset (definition 5.2.2).

1. If E is stable under specialization, then E is closed.
2. If E is stable under generization, then E is open.

***Proof*:**
The underlying space of Y is a Noetherian topological space: Y is a finite union of spectra of Noetherian rings, descending chains of closed subsets of a spectrum correspond to ascending chains of radical ideals, and a finite union of Noetherian subspaces is Noetherian.

Step 1: the closed statement. By lemma 5.2.3, $E = \bigcup_{i=1}^{n} U_i \cap Z_i$ with U_i open and Z_i closed, and, discarding the indices with $U_i \cap Z_i = \emptyset$ and replacing each remaining Z_i by the closure of $U_i \cap Z_i$, we may assume $Z_i = \overline{U_i \cap Z_i} \neq \emptyset$. Since $\overline{E} \subseteq Z_1 \cup \cdots \cup Z_n$, it suffices to show $Z_i \subseteq E$ for each i. Fix i and write $T := Z_i$, so that $U_i \cap T = U_i \cap Z_i$ is a dense open subset of T. Decompose $T = T_1 \cup \cdots \cup T_m$ into irreducible closed subsets, none contained in another (proposition 2.1.25). For each j the set $V_j := T \setminus \bigcup_{l \neq j} T_l$ is open in T with $V_j \subseteq T_j$, and nonempty: otherwise the irreducible T_j, covered by the finitely many closed sets $T_j \cap T_l$ with $l \neq j$, would lie in a single T_l, against proposition 2.1.25. The dense open $U_i \cap T$ meets V_j, so $W := U_i \cap T \cap V_j$ is a nonempty subset of T_j, open in T and hence open in T_j. With its reduced induced structure (definition 3.1.26), T_j is an integral scheme (proposition 2.4.26), so it has a generic point η_j with $\overline{\{\eta_j\}} = T_j$ (lemma 2.4.21). The point η_j lies in every nonempty open subset of T_j: otherwise $\overline{\{\eta_j\}}$ would be contained in a proper closed subset. So $\eta_j \in W \subseteq U_i \cap Z_i \subseteq E$. Every point of T_j is a specialization of η_j, so stability under specialization gives $T_j \subseteq E$. Hence $T \subseteq E$, and $\overline{E} = E$.

Step 2: the open statement. The complement of a constructible set is constructible (definition 5.2.2), and E is stable under generization exactly when $Y \setminus E$ is stable under specialization. Step 1 makes $Y \setminus E$ closed, so E is open.

The openness criterion proves in full the fact underlying corollary 5.2.11(2).

Theorem 6.8.10: Flat Morphisms of Finite Type Are Open

Let Y be a locally Noetherian scheme (definition 2.4.1) and let $\pi : X \to Y$ be flat and locally of finite type. Then π is an open map. Moreover, for every morphism $Y' \to Y$ with Y' locally Noetherian, the base change $X \times_Y Y' \to Y'$ is an open map.

***Proof*:**
Step 1: a basis of good charts. Let $U \subseteq X$ be open and $x \in U$. Choose an affine open $\operatorname{Spec} A \subseteq Y$ with A Noetherian containing $\pi(x)$ (definition 2.4.1), then an affine open $\operatorname{Spec} B \subseteq \pi^{-1}(\operatorname{Spec} A)$ containing x with B a finitely generated A-algebra (definition 3.4.28), then $g \in B$ with $x \in \operatorname{Spec} B_g \subseteq U \cap \operatorname{Spec} B$, distinguished opens forming a basis. The localization $B_g \cong B[t]/(tg - 1)$ is again a finitely generated A-algebra. As x ranges over U, the sets $\operatorname{Spec} B_g$ cover U, so $\pi(U)$ is the union of the images $\pi(\operatorname{Spec} B_g)$, and it suffices to show each such image is open in $\operatorname{Spec} A$, hence in Y.

Step 2: the image of a chart is constructible. The morphism $\operatorname{Spec} B_g \to \operatorname{Spec} A$ induced by $A \to B_g$ is of finite type: the one-chart cover exhibits it as locally of finite type (definition 3.4.28 via proposition 3.4.30), and it is quasicompact because B_g is Noetherian, as a quotient of a polynomial ring over A ([10, HBT on multi-variable Polynomials]), so that $\operatorname{Spec} B_g$ is a Noetherian topological space, every subset of which is quasicompact. Both schemes are Noetherian, so

theorem 5.2.7 makes the image $\pi(\operatorname{Spec} B_g)$ constructible in $\operatorname{Spec} A$.

Step 3: the image is stable under generization. The morphism $\operatorname{Spec} B_g \to \operatorname{Spec} A$ is flat: its stalk maps are stalk maps of π, both schemes being open subschemes. Given $y \in \pi(\operatorname{Spec} B_g)$, a preimage $u \in \operatorname{Spec} B_g$, and a generization y' of y in $\operatorname{Spec} A$, lemma 6.8.8 produces $u' \in \operatorname{Spec} B_g$ with image y'. So $\pi(\operatorname{Spec} B_g)$ is stable under generization, and lemma 6.8.9 makes it open in the Noetherian scheme $\operatorname{Spec} A$. This proves that π is open.

Step 4: base change. Let $Y' \to Y$ have Y' locally Noetherian. The base change $X \times_Y Y' \to Y'$ is flat (proposition 6.8.7) and locally of finite type (proposition 3.4.35), so the first part of the proof applies to it, as we sought to show.

Corollary 6.8.11: Flat Morphisms of Finite Presentation Are Open over Noetherian Bases

Let Y be a locally Noetherian scheme and let $\pi : X \to Y$ be flat and locally of finite presentation (definition 3.4.33). Then π is an open map, and so is every base change of π along a morphism from a locally Noetherian scheme.

***Proof*:**
Locally of finite presentation implies locally of finite type (proposition 3.4.34), so theorem 6.8.10 applies.

The locally Noetherian hypothesis enters through Chevalley's theorem and through the openness criterion of lemma 6.8.9. The statement is true over an arbitrary base: a flat morphism locally of finite presentation is universally open ([70, Tag 01UA]). That proof requires machinery not developed in this book and is not reproduced.

The key property of flat families for cohomology is that they interact well with base change. Recall from definition 3.2.3 that for a morphism $f : X \to Y$ and a base change $g : Y' \to Y$, we form the fiber product:

$$\begin{array}{ccc} X' = X \times_Y Y' & \xrightarrow{g'} & X \\ {\scriptstyle f'}\downarrow & & \downarrow{\scriptstyle f} \\ Y' & \xrightarrow[g]{} & Y \end{array}$$

For any $\mathcal{O}_X$-module $\mathscr{F}$, the *pullback* $g'^*\mathscr{F}$ is a sheaf on X', and we can form both $R^i f'_*(g'^*\mathscr{F})$ (the higher direct image along f') and $g^*(R^i f_*\mathscr{F})$ (pull back the higher direct image). These are both sheaves on Y', and there is a natural *base change map*:

$$\beta^i : \; g^*(R^i f_*\mathscr{F}) \longrightarrow R^i f'_*(g'^*\mathscr{F}).$$

This map is not an isomorphism in general. When it is, we say that *cohomology commutes with base change in degree i*.

6.8.2 The Flat Base Change Theorem

The following theorem says that, for a separated morphism of finite type between Noetherian schemes and a quasi-coherent sheaf flat over the base, cohomology commutes with flat base change to a Noetherian scheme:

Theorem 6.8.12: Flat Base Change

Let $f : X \to Y$ be a separated morphism of finite type between Noetherian schemes, and let $\mathscr{F}$ be a quasi-coherent sheaf on X that is flat over Y. Let $g : Y' \to Y$ be a flat morphism with Y' Noetherian, and form the Cartesian square:

$$\begin{array}{ccc} X' & \xrightarrow{g'} & X \\ {\scriptstyle f'}\downarrow & & \downarrow{\scriptstyle f} \\ Y' & \xrightarrow[g]{} & Y \end{array}$$

Then for all $i \geq 0$, the base change map is an isomorphism:

$$g^*(R^i f_* \mathscr{F}) \xrightarrow{\sim} R^i f'_*(g'^* \mathscr{F}).$$

***Proof*:**

The question is local on Y and Y', so we may assume both are affine: $Y = \operatorname{Spec} R$ and $Y' = \operatorname{Spec} R'$, with the flat map g corresponding to a flat ring homomorphism $\varphi : R \to R'$.

Step 1: Reduction to Čech cohomology. Since f is separated and of finite type, X admits a finite open affine cover $\mathcal{U} = \{U_1, \ldots, U_r\}$. By proposition 6.6.14, the higher direct images are computed by the Čech complex: $R^i f_* \mathscr{F}$ is the quasi-coherent sheaf on Y associated to the R-module $h^i(C^\bullet(\mathcal{U}, \mathscr{F}))$, where:

$$C^p(\mathcal{U}, \mathscr{F}) = \prod_{i_0 < \cdots < i_p} \mathscr{F}(U_{i_0} \cap \cdots \cap U_{i_p}).$$

Each intersection $U_{i_0,\ldots,i_p}$ is affine (by separatedness), and $\mathscr{F}$ is quasi-coherent, so $\mathscr{F}(U_{i_0,\ldots,i_p})$ is an R-module.

Step 2: Base change of the Čech complex. The base change $X' = X \times_Y Y'$ has the affine cover $\mathcal{U}' = \{U'_1, \ldots, U'_r\}$ where $U'_j = U_j \times_Y Y' = \operatorname{Spec}(A_j \otimes_R R')$ for $U_j = \operatorname{Spec} A_j$. The Čech complex for $g'^* \mathscr{F}$ on X' is:

$$C^p(\mathcal{U}', g'^* \mathscr{F}) = \prod_{i_0 < \cdots < i_p} (g'^* \mathscr{F})(U'_{i_0,\ldots,i_p}) = \prod_{i_0 < \cdots < i_p} \mathscr{F}(U_{i_0,\ldots,i_p}) \otimes_R R'.$$

The last equality uses the fact that for an affine base change $\operatorname{Spec}(A \otimes_R R') \to \operatorname{Spec} A$, the pullback of a quasi-coherent sheaf $\widetilde{M}$ is $\widetilde{M \otimes_R R'}$. Hence:

$$C^\bullet(\mathcal{U}', g'^* \mathscr{F}) \cong C^\bullet(\mathcal{U}, \mathscr{F}) \otimes_R R'.$$

Step 3: Flatness and cohomology. Since R' is a flat R-module (by hypothesis), the functor $- \otimes_R R'$ is exact. Exact functors commute with the formation of cohomology (i.e., with taking

kernels modulo images):

$$h^i(C^\bullet(\mathcal{U},\mathscr{F})\otimes_R R') \cong h^i(C^\bullet(\mathcal{U},\mathscr{F}))\otimes_R R'.$$

The left side computes $R^i f'_*(g'^*\mathscr{F})(Y')$, while the right side computes $g^*(R^i f_*\mathscr{F})(Y') = (R^i f_*\mathscr{F})(Y)\otimes_R R'$. Hence the base change map is an isomorphism on global sections over affine opens, which suffices.

One special case is base change to a generic point:

Corollary 6.8.13: Cohomology of Fibers via Base Change

Under the hypotheses of theorem 6.8.12, let $y \in Y$ be a point whose local ring $\mathcal{O}_{Y,y}$ is a field (for instance the generic point of an integral Y), with residue field $\kappa(y)$, and let $X_y = f^{-1}(y)$ be the scheme-theoretic fiber (definition 3.2.9). Then:

$$(R^i f_*\mathscr{F})\otimes_{\mathcal{O}_{Y,y}} \kappa(y) \;\cong\; H^i(X_y,\, \mathscr{F}_y),$$

where $\mathscr{F}_y = \mathscr{F}|_{X_y}$ is the restriction to the fiber, provided that $\mathscr{F}$ is flat over Y.

***Proof*:**
Apply theorem 6.8.12 to the base change $g : \operatorname{Spec}\kappa(y) = \operatorname{Spec}\mathcal{O}_{Y,y} \to Y$, which is flat. Then $X' = X_y$, and the left side is $g^*(R^i f_*\mathscr{F}) = (R^i f_*\mathscr{F})\otimes\kappa(y)$, while the right side is $H^i(X_y,\mathscr{F}_y)$.

Example 6.8.14: Flat Base Change in Practice

1. *Constant families.* Let $f = \mathrm{pr}_2 : X \times_k Y \to Y$ be the projection of a product (with X projective over k and Y Noetherian). Then for any coherent sheaf $\mathscr{F}$ on X, pulled back to $X\times Y$ via the first projection, the sheaf $\mathscr{F}$ is flat over Y (trivially, since it does not depend on Y at all). The base change theorem gives:

   $$R^i f_*(\mathrm{pr}_1^*\mathscr{F}) \cong H^i(X,\mathscr{F})\otimes_k \mathcal{O}_Y,$$

   i.e., the higher direct images are the constant (free) sheaves with fiber $H^i(X,\mathscr{F})$. This is the trivial case, but it illustrates the principle: for a constant family, the cohomology does not vary.

2. *Base change to the generic point.* Let $f : X \to Y$ be a projective morphism of Noetherian schemes with Y integral, let $\mathscr{F}$ be a coherent sheaf on X flat over Y, and let $\eta \in Y$ be the generic point. The base change $\operatorname{Spec}K(Y) \to Y$ (where $K(Y)$ is the function field) is flat, so:

   $$(R^i f_*\mathscr{F})\otimes\kappa(\eta) \cong H^i(X_\eta,\mathscr{F}_\eta).$$

   This says the *generic rank* of $R^i f_*\mathscr{F}$ equals the cohomology of the generic fiber. Combined with semicontinuity, this shows that the generic fiber has the "smallest" cohomology, and special fibers can only have more.

3. *Flat base change for field extensions.* Let X be a projective scheme over a field k, let $\mathscr{F}$ be a coherent sheaf on X, and let K/k be a field extension. The base change $X_K =$

$X \times_k \operatorname{Spec} K \to X$ is flat, so:

$$H^i(X_K, \mathscr{F}_K) \cong H^i(X, \mathscr{F}) \otimes_k K.$$

In particular, $h^i(X_K, \mathscr{F}_K) = h^i(X, \mathscr{F})$: the Betti numbers do not change under field extension. This is essential for reducing problems over arbitrary fields to the algebraically closed case.

Proposition 6.8.15: Associated Points and Morphisms

Let $f : X \to Y$ be a flat morphism of locally Noetherian schemes. Then:

$$\operatorname{Ass}(X) = \bigcup_{y \in \operatorname{Ass}(Y)} \operatorname{Ass}(X_y)$$

where $X_y = f^{-1}(y)$ is the fiber over y. In particular, associated points of X map to associated points of Y (among the points in the image).

***Proof Key ideas*:**
Let $x \in X$ with $y = f(x)$, and set $A = \mathcal{O}_{Y,y}$ and $B = \mathcal{O}_{X,x}$, so that $A \to B$ is a flat local homomorphism of Noetherian local rings and $\mathcal{O}_{X_y,x} = B/\mathfrak{m}_A B$. The statement at x is that $\mathfrak{m}_B \in \operatorname{Ass}_B(B)$ if and only if $\mathfrak{m}_A \in \operatorname{Ass}_A(A)$ and $\mathfrak{m}_B/\mathfrak{m}_A B \in \operatorname{Ass}(B/\mathfrak{m}_A B)$. This equivalence is a theorem of commutative algebra on associated primes under flat ring maps [70, Tag 05DB], which we do not prove here.

6.8.3 Semicontinuity

The flat base change theorem shows that, under its finiteness hypotheses, cohomology commutes with flat base change. But what happens for non-flat base changes, such as the inclusion of a point $y \hookrightarrow Y$? In general, the base change map $(R^i f_* \mathscr{F}) \otimes \kappa(y) \to H^i(X_y, \mathscr{F}_y)$ need not be an isomorphism, and the fiber cohomology can be *larger* than what the stalk of the higher direct image predicts. The semicontinuity theorem makes this precise: the function $y \mapsto h^i(X_y, \mathscr{F}_y)$ is *upper semicontinuous*, meaning it can jump up on closed subsets but not on open ones.

To state the result precisely, we need a piece of commutative algebra. Let R be a Noetherian ring and $C^\bullet$ a bounded complex of finitely generated free R-modules. For any R-module M, write $h^i(C^\bullet \otimes_R M) = h^i(C^\bullet; M)$. When R is a local ring and $M = R/\mathfrak{m}$ is the residue field, the function $i \mapsto h^i(C^\bullet; R/\mathfrak{m})$ measures the fiber cohomology. The following result governs how this varies:

Lemma 6.8.16: Semicontinuity on Complexes

Let R be a Noetherian ring, and let $C^\bullet$ be a bounded complex of finitely generated free R-modules. Then:

1. The function $\mathfrak{p} \mapsto \dim_{\kappa(\mathfrak{p})} h^i(C^\bullet \otimes_R \kappa(\mathfrak{p}))$ is upper semicontinuous on $\operatorname{Spec} R$: for each n, the set $\{\mathfrak{p} \in \operatorname{Spec} R \mid \dim h^i(C^\bullet \otimes \kappa(\mathfrak{p})) \geq n\}$ is closed.
2. The Euler characteristic $\mathfrak{p} \mapsto \chi(C^\bullet \otimes \kappa(\mathfrak{p})) = \sum(-1)^i \dim h^i(C^\bullet \otimes \kappa(\mathfrak{p}))$ is *locally constant* (it equals $\sum(-1)^i \mathrm{rk} C^i$, which is independent of $\mathfrak{p}$).
3. If $h^i(C^\bullet \otimes \kappa(\mathfrak{p}))$ is constant as a function of $\mathfrak{p}$ (say, equal to r), and R is local and reduced, then $h^i(C^\bullet)$ is a free R-module of rank r.

***Proof*:**

(1) The complex $C^\bullet$ is a sequence of maps between free modules: $\cdots \to R^{a_{i-1}} \xrightarrow{d^{i-1}} R^{a_i} \xrightarrow{d^i} R^{a_{i+1}} \to \cdots$. After tensoring with $\kappa(\mathfrak{p})$, these become linear maps between vector spaces: $\kappa(\mathfrak{p})^{a_{i-1}} \to \kappa(\mathfrak{p})^{a_i} \to \kappa(\mathfrak{p})^{a_{i+1}}$. Now, $\dim h^i(C^\bullet \otimes \kappa(\mathfrak{p})) = a_i - \mathrm{rk}(d^{i-1} \otimes \kappa(\mathfrak{p})) - \mathrm{rk}(d^i \otimes \kappa(\mathfrak{p}))$. The rank of a matrix can only *drop* under specialization (the vanishing of minors is a closed condition), so $\mathrm{rk}(d^j \otimes \kappa(\mathfrak{p}))$ is lower semicontinuous in $\mathfrak{p}$. Hence $\dim h^i$ is upper semicontinuous.

(2) The Euler characteristic is $\sum(-1)^i \dim h^i(C^\bullet \otimes \kappa(\mathfrak{p})) = \sum(-1)^i a_i$ (the alternating sum of the ranks of the free modules), which is independent of $\mathfrak{p}$. This uses the rank-nullity theorem: the terms involving ranks of differentials cancel in the alternating sum.

(3) For each j let $W^j = \operatorname{coker}(d^{j-1})$. The differential d^i induces $W^i \to C^{i+1}$ with kernel $h^i(C^\bullet)$ and cokernel W^{i+1}. Tensor product is right exact, so tensoring with $\kappa(\mathfrak{p})$ gives an exact sequence

$$0 \to h^i(C^\bullet \otimes \kappa(\mathfrak{p})) \to W^i \otimes \kappa(\mathfrak{p}) \to C^{i+1} \otimes \kappa(\mathfrak{p}) \to W^{i+1} \otimes \kappa(\mathfrak{p}) \to 0,$$

and $\dim h^i(C^\bullet \otimes \kappa(\mathfrak{p})) = \dim W^i \otimes \kappa(\mathfrak{p}) + \dim W^{i+1} \otimes \kappa(\mathfrak{p}) - \mathrm{rk} C^{i+1}$. The two fiber dimensions on the right are upper semicontinuous in $\mathfrak{p}$ (lemma 5.1.23(3)) with constant sum, so each is locally constant. As R is reduced, W^i and W^{i+1} are then locally free [36, Exercise II.5.8(c)], hence free over the local ring R. So $\mathrm{im} d^i = \ker(C^{i+1} \to W^{i+1})$ is a direct summand of C^{i+1}, hence free, and the surjection $W^i \to \mathrm{im} d^i$ splits: $W^i \cong h^i(C^\bullet) \oplus \mathrm{im} d^i$. Thus $h^i(C^\bullet)$ is a direct summand of a free module over a local ring, hence free, and tensoring the splitting with $\kappa(\mathfrak{p})$ identifies $h^i(C^\bullet) \otimes \kappa(\mathfrak{p})$ with $h^i(C^\bullet \otimes \kappa(\mathfrak{p}))$, which has dimension r.

The geometric content of this algebraic lemma is the semicontinuity theorem:

Theorem 6.8.17: Semicontinuity of Cohomology

Let $f : X \to Y$ be a proper morphism of Noetherian schemes, and let $\mathscr{F}$ be a coherent sheaf on X that is flat over Y. Then:

1. *(Upper semicontinuity.)* For each $i \geq 0$, the function
$$y \longmapsto h^i(X_y, \mathscr{F}_y)$$
is upper semicontinuous on Y: for every $n \geq 0$, the set $\{y \in Y \,|\, h^i(X_y, \mathscr{F}_y) \geq n\}$ is a closed subset of Y.

2. *(Euler characteristic is locally constant.)* The function
$$y \longmapsto \chi(X_y, \mathscr{F}_y) = \sum_{i \geq 0} (-1)^i h^i(X_y, \mathscr{F}_y)$$
is locally constant on Y.

***Proof*:**
The question is local on Y, so assume $Y = \operatorname{Spec} R$ is affine.

Constructing the representing complex. The key technical input is the existence of a *ech complex computing fiber cohomology uniformly.* Choose a finite affine open cover $\mathcal{U}$ of X. By proposition 6.6.14, the cohomology $H^i(X, \mathscr{F})$ is computed by the Čech complex $C^\bullet(\mathcal{U}, \mathscr{F})$. Since $\mathscr{F}$ is flat over $Y = \operatorname{Spec} R$, and each intersection $U_{i_0,\dots,i_p}$ is affine, the R-modules $\mathscr{F}(U_{i_0,\dots,i_p})$ are flat over R. The cohomology modules $H^i(X, \mathscr{F})$ of this complex are finitely generated by theorem 6.6.13 (by theorem 6.6.6 when f is projective).

One can replace the Čech complex by a quasi-isomorphic bounded complex $K^\bullet$ of finitely generated *projective* R-modules[a]. Finitely generated projective modules over the Noetherian ring R are locally free, so after shrinking Y around a given point the K^i are free. This complex has the property that:
$$h^i(K^\bullet \otimes_R \kappa(y)) \cong H^i(X_y, \mathscr{F}_y)$$
for every $y \in Y$. Indeed, tensoring with $\kappa(y)$ corresponds to restricting to the fiber, and the flatness of $\mathscr{F}$ ensures that the Čech complex commutes with base change (as in the proof of theorem 6.8.12).

Both statements now follow from lemma 6.8.16: part (1) is the upper semicontinuity of $\dim h^i(K^\bullet \otimes \kappa(y))$, and part (2) is the constancy of the Euler characteristic.

[a] Hartshorne's construction [36, Lemma III.12.3] gives a bounded above complex $L^\bullet$ of finitely generated free R-modules with $h^i(L^\bullet \otimes_R M) \cong h^i(C^\bullet \otimes_R M)$ for every R-module M. These groups vanish for $i < 0$, so $Q = L^0/d(L^{-1})$ satisfies $\operatorname{Tor}_1^R(Q, M) = h^{-1}(L^\bullet \otimes_R M) = 0$. Thus Q is flat and finitely generated, hence projective, and $0 \to Q \to L^1 \to L^2 \to \cdots$ is the required bounded complex.

Example 6.8.18: Semicontinuity in Families

1. *Jumping h^1 for line bundles on curves.* Let $f : \mathcal{C} \to B$ be a flat projective family of smooth curves of genus g over a Noetherian scheme B, with $H^0(C_b, \mathcal{O}_{C_b}) = \kappa(b)$ for every $b \in B$ (for

instance with geometrically connected fibers, theorem 3.5.33), and let $\mathcal{L}$ be a line bundle on $\mathcal{C}$ of relative degree d (i.e., $\deg \mathcal{L}|_{C_b} = d$ for each fiber C_b). By Riemann–Roch (theorem 6.7.12):

$$h^0(C_b, \mathcal{L}_b) - h^1(C_b, \mathcal{L}_b) = d + 1 - g$$

for every $b \in B$. Since the Euler characteristic is constant, any jump in h^0 is exactly compensated by a jump in h^1. By semicontinuity, h^0 is upper semicontinuous, so the locus where h^0 is "large" (and correspondingly h^1 is large) is closed.

Concretely, take a pencil of plane cubics whose general member is smooth, restricted to the open subset of $\mathbb{P}^1$ over which the members are smooth (genus 1), and a line bundle $\mathcal{L}$ of degree 0 on every fiber. At a fiber where $\mathcal{L}_b$ is nontrivial, $h^0 = 0$, as a nontrivial line bundle of degree 0 has no nonzero sections, and Riemann–Roch then gives $h^1 = 0$. At fibers where $\mathcal{L}_b$ is trivial, $h^0 = 1$ and $h^1 = 1$. The locus of such fibers is a closed subset of the base.

2. *Degenerating an extension.* Let X be a projective scheme over k, and let $0 \to \mathscr{F}' \to \mathscr{F} \to \mathscr{F}'' \to 0$ be a short exact sequence of coherent sheaves on X with extension class $\xi \in \operatorname{Ext}^1(\mathscr{F}'', \mathscr{F}')$. On $X \times \mathbb{A}^1_k$, with coordinate t on $\mathbb{A}^1_k$, the extension of the pullback of $\mathscr{F}''$ by the pullback of $\mathscr{F}'$ with class $t\xi$ is flat over $\mathbb{A}^1_k$. Its fiber over $a \neq 0$ is isomorphic to $\mathscr{F}$, and its fiber over 0 is $\mathscr{F}' \oplus \mathscr{F}''$. Semicontinuity (theorem 6.8.17) then gives $h^0(\mathscr{F}) \leq h^0(\mathscr{F}') + h^0(\mathscr{F}'')$, and equality need not hold. The "excess" is controlled by $H^1(\mathscr{F}')$, which is the obstruction to lifting global sections.

3. *Jumping h^0 for the structure sheaf of a degeneration.* Let k be algebraically closed. Projecting a twisted cubic in $\mathbb{P}^3_k$ from a point onto a plane gives a flat projective family $f : X \to \mathbb{A}^1_k$ whose fiber over each $a \neq 0$ is a twisted cubic, with $h^0(\mathcal{O}) = 1$, and whose fiber over 0 is a plane cubic with an embedded point at its singular point [36, Examples III.9.8.3 and III.9.8.4]. The special fiber is connected, and the embedded point contributes a nilpotent global section, so $h^0(\mathcal{O}) = 2$ there. Thus h^0 has jumped up at 0, and the locus $\{h^0 \geq 2\}$ is the closed point $\{0\}$, consistent with semicontinuity.

6.8.4 Grauert's Theorem and Cohomology and Base Change

The semicontinuity theorem says h^i can jump up. The natural follow-up question is: what happens when it does *not* jump? The answer is Grauert's theorem: over a reduced base, constancy of h^i implies that the higher direct image is locally free and commutes with *all* base changes from locally Noetherian schemes, not just flat ones.

Theorem 6.8.19: Grauert's Theorem (Cohomology and Base Change)

Let $f : X \to Y$ be a proper morphism of Noetherian schemes, and let $\mathscr{F}$ be a coherent sheaf on X that is flat over Y. Let $y \in Y$ be a point, and suppose the natural base change map:

$$\beta^i(y) : \ (R^i f_* \mathscr{F}) \otimes \kappa(y) \longrightarrow H^i(X_y, \mathscr{F}_y)$$

is surjective. Then:

1. $\beta^i(y)$ is an isomorphism. That is, surjectivity implies bijectivity.
2. There is an open neighborhood U of y over which the formation of $R^i f_* \mathscr{F}$ commutes with every base change $g : Y' \to U$ with Y' locally Noetherian. In particular $\beta^i(y')$ is an isomorphism for every $y' \in U$.
3. $\beta^{i-1}(y)$ is surjective if and only if $R^i f_* \mathscr{F}$ is locally free in a neighborhood of y.

Moreover, if $h^i(X_y, \mathscr{F}_y)$ is constant on Y and Y is reduced, then $R^i f_* \mathscr{F}$ is a locally free sheaf of rank $h^i(X_y, \mathscr{F}_y)$, and the formation of $R^i f_* \mathscr{F}$ commutes with every base change $g : Y' \to Y$ with Y' locally Noetherian.

***Proof Key ideas*:**
We do not give a complete proof. Parts (1) and (3), and the last sentence of (2), are [36, Theorem III.12.11], stated there for projective f. Its proof uses projectivity only through the complex $K^\bullet$ below and the theorem on formal functions, which holds for proper f [70, Tag 02OC]. The final assertion strengthens [36, Corollary III.12.9], stated there for projective f and integral Y. The question is local on Y, so take $Y = \operatorname{Spec} A$ affine. As in the proof of theorem 6.8.17, there is a bounded complex $K^\bullet$ of finitely generated projective A-modules with $T^i(M) := h^i(K^\bullet \otimes_A M) \cong H^i(X, \mathscr{F} \otimes_A M)$ for every A-module M. So $R^i f_* \mathscr{F}$ corresponds to $T^i(A)$, and $\beta^i(y)$ is the natural map $T^i(A) \otimes \kappa(y) \to T^i(\kappa(y))$.

The functors T^i form a δ-functor, so T^i is right exact exactly when T^{i+1} is left exact, and T^i is left exact exactly when $W^i = \operatorname{coker}(K^{i-1} \to K^i)$ is projective. If T^i is right exact, then $T^i(A) \otimes M \to T^i(M)$ is an isomorphism for every M, and T^i is exact exactly when $T^i(A)$ is in addition projective. The substantive step is that surjectivity of $\beta^i(y)$ makes T^i right exact on a neighborhood of y. It is proved for modules of finite length by induction on length, and passes to finitely generated modules through the theorem on formal functions, which this book does not develop. Right exactness near y gives (1) and (2). For (3), surjectivity of $\beta^{i-1}(y)$ makes T^{i-1} right exact near y, that is T^i left exact, so T^i is exact and $T^i(A)$ is projective. Conversely, if $T^i(A)$ is projective near y, then the right exact T^i is exact, so T^{i-1} is right exact and $\beta^{i-1}(y)$ is surjective.

For the final assertion, the dimension count in the proof of lemma 6.8.16(3) writes $h^i(X_y, \mathscr{F}_y)$ as $\dim W^i \otimes \kappa(y) + \dim W^{i+1} \otimes \kappa(y) - \operatorname{rk} K^{i+1}$. When the left side is constant, both fiber dimensions are locally constant, and on a reduced base W^i and W^{i+1} are then locally free [36, Exercise II.5.8(c)]. So T^i and T^{i+1} are left exact, T^i is exact, $T^i(A)$ is projective, and (1) and (2) apply at every point.

Corollary 6.8.20: Cohomology and Base Change: Clean Form

Let $f : X \to Y$ be a projective morphism of Noetherian schemes with Y integral, and let $\mathscr{F}$ be coherent and f-flat. Suppose $h^i(X_y, \mathscr{F}_y)$ is constant for all $y \in Y$. Then:

1. $R^i f_* \mathscr{F}$ is locally free of rank $h^i(X_y, \mathscr{F}_y)$.
2. For any morphism $g : Y' \to Y$ with Y' locally Noetherian, the base change map $g^*(R^i f_* \mathscr{F}) \xrightarrow{\sim} R^i f'_*(g'^* \mathscr{F})$ is an isomorphism.
3. If additionally $h^{i-1}(X_y, \mathscr{F}_y)$ is also constant, then $R^{i-1} f_* \mathscr{F}$ is locally free and commutes with base change as well.

***Proof*:**
Part (1) follows from theorem 6.8.19 together with the observation that on an integral (hence reduced) scheme, a coherent sheaf of constant fiber dimension is locally free. Part (2) is the final assertion of theorem 6.8.19. Part (3) is the final assertion of theorem 6.8.19 with $i-1$ in place of i.

Example 6.8.21: Grauert's Theorem in Practice

1. *Line bundles on a constant family of curves.* Let C be a smooth projective curve of genus g over an algebraically closed field k. Grant the degree-d component $\mathrm{Pic}^d(C)$ of the Picard scheme, a smooth proper k-scheme of pure dimension g whose k-points are the degree-d line bundles on C, and the universal (Poincaré) line bundle $\mathcal{L}$ on $C \times \mathrm{Pic}^d(C)$. Both exist because C has a k-point ([20, Grothendieck's Theorem: the Picard Scheme of a Projective Variety], [20, The Degree Decomposition], [20, Properness of the Degree-Zero Picard Scheme of a Curve]). Consider the family $f : C \times \mathrm{Pic}^d(C) \to \mathrm{Pic}^d(C)$. If $d > 2g-2$, then $h^1(C, \mathcal{L}_b) = 0$ for all $b \in \mathrm{Pic}^d(C)$ (by Serre duality and degree considerations). By Grauert's theorem, $R^1 f_* \mathcal{L} = 0$ and $f_* \mathcal{L}$ is a locally free sheaf of rank $h^0 = d + 1 - g$ on $\mathrm{Pic}^d(C)$. This locally free sheaf is the *Picard bundle*, a fundamental object in the study of Jacobians.
2. *The Euler characteristic without flatness.* Consider the blowup $f : X = \mathrm{Bl}_0 \mathbb{A}^2 \to \mathbb{A}^2$ and the line bundle $\mathcal{O}_X(-E)$ from example 6.6.15(2). Like $\mathcal{O}_X$, it is *not* flat over $\mathbb{A}^2$ (the fiber dimension jumps). Over a point $a \neq 0$ the fiber is a reduced point, and $\chi = 1$. The fiber over the origin is $E \cong \mathbb{P}^1$, on which $\mathcal{O}_X(-E)$ restricts to $\mathcal{O}_{\mathbb{P}^1}(1)$, so $\chi = 2$. The Euler characteristic of the fibers is not locally constant, and theorem 6.8.17(2) fails without its flatness hypothesis.
3. *When constancy fails: the Brill–Noether locus.* Let C be a general smooth curve of genus g over an algebraically closed field k, and consider the family of degree-d line bundles of item (1). The *Brill–Noether locus* $W^r_d(C) = \{L \in \mathrm{Pic}^d(C) \,|\, h^0(C, L) \geq r+1\}$ is a closed subvariety of $\mathrm{Pic}^d(C)$ by the semicontinuity theorem applied to the universal line bundle. With the "expected dimension" $\rho = g - (r+1)(g-d+r)$, the Brill–Noether theorem asserts that for a general curve of genus g, $W^r_d(C)$ has dimension $\min(\rho, g)$ when $\rho \geq 0$, and is empty when $\rho < 0$. This is a result in the geometry of algebraic curves, and semicontinuity is the foundational tool that ensures W^r_d is an algebraic variety.

6.8.22 Remark: The Coherent Base Change Complex The key technical tool behind both the semicontinuity and Grauert theorems is the existence, over an affine base $Y = \operatorname{Spec} R$, of a bounded complex $K^\bullet$ of finitely generated projective R-modules representing the derived pushforward $\mathbf{R}f_*\mathscr{F}$ in the derived category of R-modules. Its terms become free after shrinking Y around a given point, and are free when R is local. Such a complex exists whenever f is proper and $\mathscr{F}$ is coherent and f-flat: one takes the Čech complex associated to a finite affine cover, resolves it by a bounded above complex of finitely generated free modules, and truncates (as in the proof of theorem 6.8.17).
In the language of derived categories (which we have not formally developed), the statement is: $\mathbf{R}f_*\mathscr{F} \in D^b(\mathrm{Mod}_R)$ is a *perfect complex* (quasi-isomorphic to a bounded complex of finitely generated projective modules). The fiber cohomology is then $H^i(X_y, \mathscr{F}_y) = h^i(K^\bullet \otimes^{\mathbf{L}} \kappa(y))$, where $\otimes^{\mathbf{L}}$ is the derived tensor product (which equals the ordinary tensor product since $K^\bullet$ consists of projective modules).
This perspective explains why the semicontinuity theorem is fundamentally a result in commutative algebra (about cohomology of complexes of finitely generated projective modules under base change) rather than algebraic geometry. The geometry enters only in constructing the complex $K^\bullet$, and this construction uses the properness of f and the coherence of $\mathscr{F}$ in an essential way.

6.8.5 Recognizing Flatness Cohomologically

The base-change theorems of this section all take flatness as a hypothesis and study the cohomology that results. The reverse question is equally useful. Given a coherent sheaf on a projective family, how can flatness over the base be recognized from cohomology alone? The answer is a criterion on the pushforwards of high twists. Nothing else in this chapter uses it, but the construction of the Quot and Hilbert schemes, through the flattening stratification, does, so it is proved here rather than left to that construction.

The criterion runs on Serre vanishing over a general Noetherian base, which is recorded first in the form the proof uses.

Lemma 6.8.23: Serre Vanishing on Projective Space

Let A be a Noetherian ring and $\mathscr{F}$ a coherent sheaf on $\mathbb{P}^n_A$. Then there is an integer m_0 such that

$$H^i(\mathbb{P}^n_A, \mathscr{F}(m)) = 0 \qquad \text{for all } i \geq 1 \text{ and all } m \geq m_0$$

***Proof*:**
We argue by descending induction on i. For $i > n$ every such group vanishes, uniformly in m: $\mathbb{P}^n_A$ is covered by the $n+1$ standard affine opens, whose Čech complex vanishes in degrees above n, and theorem 6.1.19 identifies its cohomology with $H^i(\mathbb{P}^n_A, \mathscr{F}(m))$. Assume the vanishing holds in all degrees above i, with a threshold depending on the coherent sheaf. By corollary 5.3.27 choose a surjection $\mathcal{E} = \bigoplus_{j=1}^{s} \mathcal{O}_{\mathbb{P}^n_A}(d_j) \twoheadrightarrow \mathscr{F}$ with kernel $\mathcal{K}$, a coherent sheaf. The twisted short exact sequence $0 \to \mathcal{K}(m) \to \mathcal{E}(m) \to \mathscr{F}(m) \to 0$ gives

$$H^i(\mathbb{P}^n_A, \mathcal{E}(m)) \longrightarrow H^i(\mathbb{P}^n_A, \mathscr{F}(m)) \longrightarrow H^{i+1}(\mathbb{P}^n_A, \mathcal{K}(m))$$

By theorem 6.3.1 the left group is $\bigoplus_j H^i(\mathbb{P}^n_A, \mathcal{O}(m+d_j))$, and the only nonzero higher cohomology

of a line bundle on $\mathbb{P}^n_A$ sits in degree n at twists at most $-n-1$, so this vanishes for $i \geq 1$ once m is large. By the inductive hypothesis applied to $\mathcal{K}$ the right group vanishes for m large. Hence $H^i(\mathbb{P}^n_A, \mathscr{F}(m)) = 0$ for m large, and taking m_0 to dominate the finitely many thresholds met for $i = 1, \dots, n$ completes the induction.

Theorem 6.8.24: Flatness Criterion via Twisted Global Sections

Let A be a Noetherian ring, $\pi\colon \mathbb{P}^n_A \to \operatorname{Spec} A$ the projection, and $\mathscr{F}$ a coherent sheaf on $\mathbb{P}^n_A$. Then $\mathscr{F}$ is flat over A if and only if there is an integer m_0 such that the A-module

$$M_m := H^0(\mathbb{P}^n_A, \mathscr{F}(m)) = \Gamma\big(\operatorname{Spec} A,\, \pi_*\mathscr{F}(m)\big)$$

is locally free for every $m \geq m_0$. When $\mathscr{F}$ is flat, one may take m_0 to be any Serre vanishing threshold from lemma 6.8.23, and then for every homomorphism $A \to A'$ the base change map

$$M_m \otimes_A A' \longrightarrow H^0(\mathbb{P}^n_{A'}, \mathscr{F}'(m))$$

is an isomorphism, where $\mathscr{F}'$ is the pullback of $\mathscr{F}$ to $\mathbb{P}^n_{A'}$.

***Proof*:**

Flat implies locally free and compatible with base change. Suppose $\mathscr{F}$ is flat over A, and take m_0 as in lemma 6.8.23. Fix $m \geq m_0$ and let $C^\bullet = C^\bullet(\mathcal{U}, \mathscr{F}(m))$ be the Čech complex of $\mathscr{F}(m)$ for the standard affine cover $\mathcal{U}$ of $\mathbb{P}^n_A$, which computes $H^\bullet(\mathbb{P}^n_A, \mathscr{F}(m))$ by theorem 6.1.19. Each term C^p is a finite product of modules of sections of $\mathscr{F}(m)$ over affine opens, and these are flat over A because $\mathscr{F}$ is A-flat. By the choice of m_0 the cohomology of $C^\bullet$ is concentrated in degree 0, where it equals M_m, so prepending M_m yields an exact sequence

$$0 \longrightarrow M_m \longrightarrow C^0 \longrightarrow C^1 \longrightarrow \cdots \longrightarrow C^n \longrightarrow 0$$

Its terms $C^0, \dots, C^n$ are flat, and a kernel that sits in an exact sequence whose other terms are flat is itself flat, by dimension-shifting on Tor ([18]). Thus M_m is flat, and being finitely generated over the Noetherian ring A (theorem 6.6.6) it is locally free.

For the base change assertion, all terms of the displayed sequence are now flat over A, so it stays exact after applying $- \otimes_A A'$. The Čech complex base changes term by term, $C^p \otimes_A A' = C^p(\mathcal{U}', \mathscr{F}'(m))$, since forming sections of a quasi-coherent sheaf over an affine open commutes with the base change $A \to A'$. Exactness at the first two terms then identifies $M_m \otimes_A A'$ with the kernel of $C^0(\mathcal{U}', \mathscr{F}'(m)) \to C^1(\mathcal{U}', \mathscr{F}'(m))$, which is $\Gamma(\mathbb{P}^n_{A'}, \mathscr{F}'(m))$ by lemma 6.1.14. Hence $M_m \otimes_A A' = H^0(\mathbb{P}^n_{A'}, \mathscr{F}'(m))$.

Locally free twists imply flat. Conversely, suppose M_m is locally free over A for all $m \geq m_0$. Set $M = \bigoplus_{m \geq m_0} M_m$, a graded module over the homogeneous coordinate ring $A[x_0, \dots, x_n]$ whose associated sheaf is $\mathscr{F}$, since twisting identifies M with a truncation of $\bigoplus_m \Gamma(\mathbb{P}^n_A, \mathscr{F}(m))$ and the associated sheaf is unchanged by truncation in high degrees. Flatness of $\mathscr{F}$ over A is local on $\mathbb{P}^n_A$, so it suffices to check it on each standard chart $U_i = D_+(x_i)$. There $\mathscr{F}|_{U_i}$ is the sheaf associated to the degree-0 part $M_{(x_i)}$ of the localization $M[x_i^{-1}]$, and as an A-module

$$M_{(x_i)} = \varinjlim\big(M_{m_0} \xrightarrow{\cdot x_i} M_{m_0+1} \xrightarrow{\cdot x_i} M_{m_0+2} \xrightarrow{\cdot x_i} \cdots\big)$$

is a directed colimit of the locally free, hence flat, A-modules M_m. A directed colimit of flat modules is flat [70, Tag 05UU], so $M_{(x_i)}$ is A-flat, and the stalks of $\mathscr{F}$ on U_i, being localizations

of $M_{(x_i)}$, are flat over A and hence over the relevant localization of A. Therefore $\mathscr{F}$ is flat over A.

The criterion is the cohomological recognition of flatness that the construction of the Quot and Hilbert schemes turns on. There the base $\operatorname{Spec} A$ is stratified by the loci over which a coherent sheaf on a projective family becomes flat with a fixed Hilbert polynomial, and theorem 6.8.24 is what converts flatness on a stratum into the local freeness of the modules M_m, a condition cut out by Fitting ideals.

That construction needs one further input that is not established here. It uses the compatibility of M_m with base change *before* flatness is known, for a sheaf not assumed flat over the base. The flat case proved above gives that compatibility only under the flatness it is meant to detect, since the argument runs on the flatness of the Čech complex. The flatness-free statement instead rests on a uniform bound for the vanishing of higher cohomology on the fibres, that is, on Castelnuovo-Mumford regularity, which the present chapter does not treat. Its proof is the missing companion to the criterion above.

6.9 *GAGA

There are two parallel cohomological machines: the algebraic cohomology of coherent sheaves on schemes (the subject of this chapter), and the analytic cohomology of coherent analytic sheaves on complex manifolds (via the Dolbeault complex, Hodge theory, and the analytic structure sheaf). Serre's GAGA theorems — *Géométrie Algébrique et Géométrie Analytique* [66] — form the bridge between these two worlds. They assert that for projective varieties over $\mathbb{C}$, the algebraic and analytic theories carry *exactly the same information*: the category of coherent algebraic sheaves is equivalent to the category of coherent analytic sheaves, and this equivalence preserves cohomology.

This is a surprising result. The algebraic world uses the Zariski topology (which is coarse: open sets are large and plentiful) and polynomial or rational functions, while the analytic world uses the classical (Euclidean) topology (which is fine: open sets can be arbitrarily small) and holomorphic functions. There are far more analytic open sets, far more analytic functions, and far more analytic sheaves than algebraic ones. Yet on projective varieties, these extra objects contribute nothing new to the cohomological theory. The rigidity of projective varieties forces the algebraic and analytic categories to coincide.

We state the GAGA theorems precisely, outline the key ideas of the proof, and develop the main consequences. This section draws heavily on the complex geometry review of section 0.6[7]; we now sketch why the algebraic cohomology developed in this chapter matches the analytic cohomology reviewed there.

6.9.1 Analytification

We begin by recalling the analytification functor, which passes from algebraic geometry over $\mathbb{C}$ to complex-analytic geometry. The reader who has studied section 0.6 in detail may skim this subsection; it is included for self-containedness.

[7]Specifically: analytification (definition 0.6.46), analytic sheaves (definition 0.6.47), and the GAGA statement (theorem 0.6.48) were introduced there.

Recall the analytification X^{an} of a scheme X of finite type over $\mathbb{C}$ (definition 0.6.46): the same underlying set as $X(\mathbb{C})$, the classical topology, and the analytic structure sheaf described below, so that X^{an} is a complex manifold when X is smooth.

Let X be a scheme of finite type over $\mathbb{C}$. The set of $\mathbb{C}$-valued points $X(\mathbb{C}) = \mathrm{Hom}_{\mathbf{Sch}_{\mathbb{C}}}(\operatorname{Spec}\mathbb{C}, X)$ carries a natural structure of complex-analytic space: locally, if $X = \operatorname{Spec}\mathbb{C}[x_1,\dots,x_n]/(f_1,\dots,f_r)$, then $X(\mathbb{C}) = V(f_1,\dots,f_r) \subseteq \mathbb{C}^n$ with the subspace topology from $\mathbb{C}^n$ and the structure sheaf $\mathcal{O}_{\mathbb{C}^n}/(f_1,\dots,f_r)\mathcal{O}_{\mathbb{C}^n}$ restricted to $V(f_1,\dots,f_r)$, where $\mathcal{O}_{\mathbb{C}^n}$ is the sheaf of holomorphic functions on $\mathbb{C}^n$. When X is smooth, $X(\mathbb{C})$ is a complex manifold (definition 0.6.1).

Definition 6.9.1: The Analytification Functor

Let X be a scheme of finite type over $\mathbb{C}$.

1. The *analytification* of X, denoted X^{an}, is the complex-analytic space $X(\mathbb{C})$ equipped with its natural analytic structure sheaf $\mathcal{O}_{X^{\mathrm{an}}}$ (cf. definition 0.6.46).

2. There is a natural continuous map of topological spaces $\varphi : X^{\mathrm{an}} \to X$ (the "identity on points," where X carries the Zariski topology and X^{an} carries the classical topology), and a morphism of sheaves of rings $\varphi^{\#} : \mathcal{O}_X \to \varphi_*\mathcal{O}_{X^{\mathrm{an}}}$ (sending algebraic regular functions to holomorphic functions).

3. For a coherent algebraic sheaf $\mathscr{F}$ on X, the *analytification* is the coherent analytic sheaf:

$$\mathscr{F}^{\mathrm{an}} = \varphi^*\mathscr{F} = \mathscr{F} \otimes_{\mathcal{O}_X} \mathcal{O}_{X^{\mathrm{an}}}$$

on X^{an}. This defines an exact functor:

$$(-)^{\mathrm{an}} : \mathbf{Coh}(X) \longrightarrow \mathbf{Coh}(X^{\mathrm{an}}).$$

The analytification functor has good formal properties: it is exact (since $\mathcal{O}_{X^{\mathrm{an}}}$ is flat over $\mathcal{O}_X$ — this is a non-trivial fact, reflecting the faithfully flat nature of the completion of a local ring), it preserves tensor products, duals, and exterior powers of locally free sheaves, and it sends line bundles to line bundles. Furthermore, for each $i \geq 0$, there is a natural *comparison map*:

$$\gamma^i : \ H^i(X, \mathscr{F}) \longrightarrow H^i(X^{\mathrm{an}}, \mathscr{F}^{\mathrm{an}}),$$

induced by the morphism of ringed spaces $\varphi : (X^{\mathrm{an}}, \mathcal{O}_{X^{\mathrm{an}}}) \to (X, \mathcal{O}_X)$. The GAGA theorems assert that, for projective X, this map is an isomorphism and the functor $(-)^{\mathrm{an}}$ is an equivalence.

6.9.2 Statement of the GAGA Theorems

Theorem 6.9.2: Serre's GAGA Theorems

Let X be a projective scheme over $\mathbb{C}$. Then:

1. *(Equivalence of categories.)* The analytification functor
$$(-)^{\mathrm{an}} : \mathbf{Coh}(X) \longrightarrow \mathbf{Coh}(X^{\mathrm{an}})$$
is an equivalence of categories. In particular:
   (a) Every coherent analytic sheaf on X^{an} is the analytification of a unique (up to isomorphism) coherent algebraic sheaf on X.
   (b) Every morphism of coherent analytic sheaves on X^{an} is the analytification of a unique morphism of coherent algebraic sheaves on X.
2. *(Comparison of cohomology.)* For every coherent algebraic sheaf $\mathscr{F}$ on X and every $i \geq 0$, the comparison map
$$\gamma^i : H^i(X, \mathscr{F}) \xrightarrow{\sim} H^i(X^{\mathrm{an}}, \mathscr{F}^{\mathrm{an}})$$
is an isomorphism of finite-dimensional $\mathbb{C}$-vector spaces.
3. *(Equivalence for subschemes.)* The analytification functor induces a bijection between closed algebraic subschemes of X and closed analytic subspaces of X^{an}.

Part (2) is typically proved first and is the most important for applications. Full faithfulness in part (1) follows from it formally, while essential surjectivity in part (1) needs an analytic input beyond part (2), and part (3) is deduced from part (1). Let us outline the proof of part (2):

***Proof*:**
(Proof sketch of theorem 6.9.2(2))
The proof proceeds in three stages.

Stage 1: The case $X = \mathbb{P}^n_{\mathbb{C}}$ and $\mathscr{F} = \mathcal{O}(d)$. The algebraic cohomology $H^i(\mathbb{P}^n, \mathcal{O}(d))$ was computed in theorem 6.3.1. The analytic cohomology $H^i((\mathbb{P}^n)^{\mathrm{an}}, \mathcal{O}^{\mathrm{an}}(d))$ can be computed using the Dolbeault resolution (theorem 0.6.44): since $\mathbb{CP}^n$ is a compact Kähler manifold (example 0.6.22), the Hodge decomposition (theorem 0.6.28) gives $H^i(\mathbb{CP}^n, \mathcal{O}^{\mathrm{an}}) \cong H^{0,i}_{\bar\partial}(\mathbb{CP}^n)$, and the Hodge diamond of $\mathbb{CP}^n$ (example 0.6.30) shows this vanishes for $i > 0$, while H^0 is $\mathbb{C}$ on both sides and γ^0 sends the constant function 1 to the constant function 1, so γ^i is an isomorphism for $\mathcal{O}$ in every degree. For the twists, induct on n, the case $n = 0$ being trivial. A hyperplane $H \cong \mathbb{P}^{n-1}$ gives the exact sequence $0 \to \mathcal{O}(d-1) \to \mathcal{O}(d) \to \mathcal{O}_H(d) \to 0$ and its analytification, with compatible long exact sequences. By induction γ^i is an isomorphism for $\mathcal{O}_H(d)$, so the five lemma makes γ^i an isomorphism for $\mathcal{O}(d)$ if and only if it is one for $\mathcal{O}(d-1)$. Starting from $d = 0$, this covers every d.

Stage 2: Extension to all coherent sheaves on $\mathbb{P}^n_{\mathbb{C}}$. Given a coherent sheaf $\mathscr{F}$ on $\mathbb{P}^n$, write it as a quotient of a finite direct sum $\mathcal{E} = \bigoplus \mathcal{O}(d_l)$ of twisted sheaves (corollary 5.3.27), with coherent kernel $\mathcal{K}$. Since the analytification functor is exact, $0 \to \mathcal{K} \to \mathcal{E} \to \mathscr{F} \to 0$ gives compatible long

exact sequences on both sides. Both sides vanish for $i > 2n$, the analytic side because $(\mathbb{P}^n)^{\mathrm{an}}$ is a manifold of real dimension $2n$. By descending induction on i, assume γ^{i+1} is an isomorphism for every coherent sheaf. The four lemma applied to $H^i(\mathcal{E}) \to H^i(\mathscr{F}) \to H^{i+1}(\mathcal{K}) \to H^{i+1}(\mathcal{E})$, with $\gamma^i_{\mathcal{E}}$ an isomorphism by Stage 1, shows that γ^i is surjective for every coherent sheaf, in particular for $\mathcal{K}$, and the five lemma then makes $\gamma^i_{\mathscr{F}}$ an isomorphism.

Stage 3: Extension to projective schemes. For a general projective scheme $i : X \hookrightarrow \mathbb{P}^N_{\mathbb{C}}$, the pushforward $i_*\mathscr{F}$ is a coherent algebraic sheaf on $\mathbb{P}^N$, and $(i_*\mathscr{F})^{\mathrm{an}} = i_*^{\mathrm{an}}(\mathscr{F}^{\mathrm{an}})$ (the analytification of a pushforward along a closed embedding is the analytic pushforward of the analytification). By proposition 6.2.4, $H^i(X, \mathscr{F}) = H^i(\mathbb{P}^N, i_*\mathscr{F})$ and similarly for the analytic side. Stage 2 gives the isomorphism for $i_*\mathscr{F}$ on $\mathbb{P}^N$, hence for $\mathscr{F}$ on X.

***Proof*:**
(Proof sketch of theorem 6.9.2(1))

Full faithfulness. For coherent sheaves $\mathscr{F}, \mathscr{G}$ on X, the natural map $\mathrm{Hom}_{\mathcal{O}_X}(\mathscr{F}, \mathscr{G}) \to \mathrm{Hom}_{\mathcal{O}_{X^{\mathrm{an}}}}(\mathscr{F}^{\mathrm{an}}, \mathscr{G}^{\mathrm{an}})$ is an isomorphism. Indeed, $\mathrm{Hom}_{\mathcal{O}_X}(\mathscr{F}, \mathscr{G}) = H^0(X, \mathit{Hom}(\mathscr{F}, \mathscr{G}))$ and $\mathrm{Hom}_{\mathcal{O}_{X^{\mathrm{an}}}}(\mathscr{F}^{\mathrm{an}}, \mathscr{G}^{\mathrm{an}}) = H^0(X^{\mathrm{an}}, \mathit{Hom}(\mathscr{F}, \mathscr{G})^{\mathrm{an}})$, and part (2) with $i = 0$ gives the isomorphism.

Essential surjectivity. Let $\mathscr{H}$ be a coherent analytic sheaf on X^{an}. One must produce a coherent algebraic sheaf $\mathscr{G}$ with $\mathscr{G}^{\mathrm{an}} \cong \mathscr{H}$. For $d \gg 0$, Serre's analytic counterpart of theorem 5.3.26 on $(\mathbb{P}^N)^{\mathrm{an}}$, applied to the pushforward along $X^{\mathrm{an}} \hookrightarrow (\mathbb{P}^N)^{\mathrm{an}}$ [66], gives that $\mathscr{H}(d) = \mathscr{H} \otimes \mathcal{O}^{\mathrm{an}}(d)$ is generated by its global sections. Finitely many of them suffice, X^{an} being compact, so there is a surjection $\mathcal{O}_{X^{\mathrm{an}}}(-d)^r \to \mathscr{H}$. Its kernel is again coherent, and the same step applied to it gives a presentation

$$\mathcal{O}_{X^{\mathrm{an}}}(-d')^s \xrightarrow{\varphi} \mathcal{O}_{X^{\mathrm{an}}}(-d)^r \longrightarrow \mathscr{H} \longrightarrow 0.$$

By full faithfulness, $\varphi = \psi^{\mathrm{an}}$ for a morphism $\psi\colon \mathcal{O}_X(-d')^s \to \mathcal{O}_X(-d)^r$, and since analytification is exact, $\mathscr{G} = \mathrm{coker}\psi$ satisfies $\mathscr{G}^{\mathrm{an}} \cong \mathrm{coker}\varphi \cong \mathscr{H}$. See [35] for a full account.

6.9.3 Remark: The Properness Hypothesis is Essential The GAGA theorems fail dramatically for non-proper varieties. The simplest example is the affine line: on $\mathbb{A}^1_{\mathbb{C}}$, the algebraic structure sheaf has $\Gamma(\mathbb{A}^1, \mathcal{O}) = \mathbb{C}[x]$ (polynomial functions), while the analytic structure sheaf has $\Gamma(\mathbb{C}, \mathcal{O}^{\mathrm{an}}) = \mathcal{O}(\mathbb{C})$ (entire holomorphic functions), which is an enormously larger ring (containing e^x, $\sin x$, etc.). Hence the global section comparison already fails.
More strikingly, there exist coherent analytic sheaves on $\mathbb{C}$ that have no algebraic counterpart: for the ideal sheaf $\mathcal{I}_{\mathbb{Z}}$ of the analytic subset $\mathbb{Z} \subseteq \mathbb{C}$ (the zero set of $\sin \pi z$), the quotient $\mathcal{O}^{\mathrm{an}}/\mathcal{I}_{\mathbb{Z}}$ is coherent analytic but not of the form $\mathscr{F}^{\mathrm{an}}$ with $\mathscr{F}$ coherent on $\mathbb{A}^1_{\mathbb{C}}$, since its support $\mathbb{Z}$ is infinite and discrete, while the support of $\mathscr{F}^{\mathrm{an}}$ is the support of $\mathscr{F}$, which is finite or all of $\mathbb{C}$.
For *proper* but non-projective varieties, a version of GAGA still holds (due to Grothendieck): the analytification functor is an equivalence on coherent sheaves, and cohomology comparison holds. The key requirement is properness — which ensures compactness of the analytic space — not projectivity. However, our sketch relied on the embedding into $\mathbb{P}^N$ and the explicit computation of twisted sheaf cohomology, so we have stated the theorem in the projective case.

6.9.3 Consequences

The GAGA theorems have far-reaching consequences that connect algebraic geometry, complex analysis, and topology. We develop the most important ones.

Corollary 6.9.4: Chow's Theorem

Every closed analytic subspace of $\mathbb{CP}^n$ is algebraic. That is, if $Z \subseteq \mathbb{CP}^n$ is a closed complex-analytic subset, then Z is the zero set of finitely many homogeneous polynomials.

***Proof*:**
A closed analytic subspace $Z \subseteq (\mathbb{P}^n_{\mathbb{C}})^{\mathrm{an}}$ is determined by its ideal sheaf $\mathscr{I}_Z^{\mathrm{an}}$, which is a coherent analytic sheaf on $(\mathbb{P}^n)^{\mathrm{an}}$. By theorem 6.9.2(1), there exists a coherent algebraic sheaf $\mathscr{I}$ on $\mathbb{P}^n_{\mathbb{C}}$ with $\mathscr{I}^{\mathrm{an}} \cong \mathscr{I}_Z^{\mathrm{an}}$. By full faithfulness in theorem 6.9.2(1), the composite $\mathscr{I}^{\mathrm{an}} \cong \mathscr{I}_Z^{\mathrm{an}} \hookrightarrow \mathcal{O}_{\mathbb{P}^n}^{\mathrm{an}}$ is the analytification of a morphism $u\colon \mathscr{I} \to \mathcal{O}_{\mathbb{P}^n}$. Since $(-)^{\mathrm{an}}$ is exact and an equivalence, $(\ker u)^{\mathrm{an}} = \ker(u^{\mathrm{an}}) = 0$ forces $\ker u = 0$. So $\mathscr{I}$ is an ideal sheaf on $\mathbb{P}^n_{\mathbb{C}}$, defining a closed algebraic subscheme Z_{alg} with $Z_{\mathrm{alg}}^{\mathrm{an}} = Z$. By theorem 6.9.2(3), this correspondence is a bijection.

Both comparisons below rest on one analytic fact, which is used three times in this section. It is the holomorphic counterpart of the Poincaré lemma: on a complex manifold a closed holomorphic form is locally exact, so the holomorphic de Rham complex resolves the constant sheaf. The local statement is several-variable complex analysis; the sheaf statement is one line from it.

Lemma 6.9.5: The Holomorphic Poincaré Lemma

Let $P \subseteq \mathbb{C}^n$ be an open polydisc and let ω be a holomorphic p-form on P with $d\omega = 0$. If $p \geq 1$ then $\omega = d\eta$ for some holomorphic $(p-1)$-form η on P; if $p = 0$ then ω is constant.

***Proof*:**
Translating coordinates, we may assume P is centred at 0. Write $\omega = \sum_{|I|=p} f_I \, dz_I$ with each f_I holomorphic on P, and expand each f_I in a power series converging normally on compact subsets of P ([58, Higher-dimensional Power Series Expansion]). Since d acts coefficientwise on such a series and the homotopy operator below is continuous for normal convergence, it suffices to treat a single monomial term, and then to observe that the classical Poincaré homotopy

$$(h\omega)(z) = \sum_{|I|=p} \sum_{k=1}^{p} (-1)^{k-1} \Big(\int_0^1 t^{p-1} f_I(tz)\, dt \Big) z_{i_k} \, dz_{i_1} \wedge \cdots \widehat{dz_{i_k}} \cdots \wedge dz_{i_p}$$

produces a form that is again *holomorphic*, because $f_I(tz)$ is holomorphic in z for each fixed t and the integral in t is over a compact interval, so differentiation under the integral sign preserves $\partial/\partial \bar{z}_j = 0$. The identity $d(h\omega) + h(d\omega) = \omega$ is the same computation as in the smooth case, carried out on a star-shaped domain, and a polydisc is star-shaped about its centre 0. With $d\omega = 0$ this gives $\omega = d(h\omega)$. For $p = 0$ a closed holomorphic function has vanishing differential, hence is constant on the connected P.

Corollary 6.9.6: The Holomorphic de Rham Resolution

Let X be a complex manifold. Then

$$0 \longrightarrow \underline{\mathbb{C}}_X \longrightarrow \mathcal{O}_X^{\mathrm{an}} \longrightarrow \Omega_X^{1,\mathrm{an}} \longrightarrow \Omega_X^{2,\mathrm{an}} \longrightarrow \cdots$$

is a resolution of the constant sheaf $\underline{\mathbb{C}}_X$ by sheaves of holomorphic forms. Consequently $\mathbb{H}^i(X, \Omega_X^{\bullet,\mathrm{an}}) \cong H^i(X, \underline{\mathbb{C}}_X)$.

***Proof*:**
Exactness is a statement about stalks, and every point of a complex manifold has a polydisc neighbourhood, so lemma 6.9.5 gives exactness in positive degrees and identifies the kernel in degree zero with the locally constant functions. The displayed isomorphism is then the hypercohomology of a resolution.

Corollary 6.9.7: Algebraic de Rham Theorem

Let X be a smooth projective variety over $\mathbb{C}$. Then the algebraic de Rham cohomology

$$H^i_{\mathrm{dR}}(X/\mathbb{C}) = \mathbb{H}^i(X,\, \Omega^{\bullet}_{X/\mathbb{C}})$$

(the hypercohomology of the algebraic de Rham complex $0 \to \mathcal{O}_X \to \Omega^1_{X/\mathbb{C}} \to \Omega^2_{X/\mathbb{C}} \to \cdots$) is naturally isomorphic to the singular cohomology:

$$H^i_{\mathrm{dR}}(X/\mathbb{C}) \;\cong\; H^i_{\mathrm{sing}}(X(\mathbb{C});\, \mathbb{C}).$$

***Proof*:**
The morphism φ sends the algebraic de Rham complex to the holomorphic de Rham complex on X^{an}, compatibly with the filtrations by degree, so it induces a map of hypercohomology spectral sequences whose E_1 terms are

$$E_1^{p,q} = H^q(X,\, \Omega^p_{X/\mathbb{C}}) \longrightarrow H^q(X^{\mathrm{an}},\, \Omega_X^{p,\mathrm{an}}).$$

Each $\Omega^p_{X/\mathbb{C}}$ is coherent with analytification $\Omega_X^{p,\mathrm{an}}$, so theorem 6.9.2(2) makes every E_1 map an isomorphism, and the comparison theorem ([18, Comparison Theorem]) gives

$$\mathbb{H}^i(X,\, \Omega^{\bullet}_{X/\mathbb{C}}) \;\cong\; \mathbb{H}^i(X^{\mathrm{an}},\, \Omega_X^{\bullet,\mathrm{an}}).$$

Since X^{an} is a complex manifold, the holomorphic Poincaré lemma (lemma 6.9.5) gives an exact sequence $0 \to \underline{\mathbb{C}} \to \Omega^{0,\mathrm{an}} \to \Omega^{1,\mathrm{an}} \to \cdots$, i.e., the holomorphic de Rham complex is a resolution of the constant sheaf $\underline{\mathbb{C}}$. Hence:

$$\mathbb{H}^i(X^{\mathrm{an}}, \Omega^{\bullet,\mathrm{an}}) \cong H^i(X^{\mathrm{an}}, \underline{\mathbb{C}}).$$

Finally, the complex-valued smooth forms on the manifold X^{an} are fine, hence acyclic (proposition 6.1.25 and theorem 6.1.28), and they resolve $\underline{\mathbb{C}}$, so $H^i(X^{\mathrm{an}}, \underline{\mathbb{C}})$ is the smooth de Rham cohomology with complex coefficients. The de Rham theorem (theorem 0.5.23), tensored with $\mathbb{C}$, identifies this with singular cohomology: $H^i(X^{\mathrm{an}}, \underline{\mathbb{C}}) \cong H^i_{\mathrm{sing}}(X(\mathbb{C}); \mathbb{C})$.

Corollary 6.9.7 used properness only through GAGA. The identification of analytic hypercohomology with singular cohomology holds on every complex manifold, by corollary 6.9.6 and the last step of that proof. GAGA does not survive on an affine variety, which is never proper of positive dimension. Yet the affine case holds, for a reason that has nothing to do with GAGA: on an affine scheme the algebraic side degenerates completely, so the whole content sits in the comparison with the analytic topology. This is Grothendieck's theorem, and the affine case is the one that arises in practice, because the complement of a hypersurface in $\mathbb{P}^{n+1}$ is affine.

Theorem 6.9.8: Grothendieck's Comparison Theorem

Let U be a smooth affine variety over $\mathbb{C}$. Then the complex of *global algebraic* differential forms computes the complex cohomology of the associated analytic space:

$$H^i\big(\Gamma(U, \Omega^\bullet_{U/\mathbb{C}})\big) \;\cong\; H^i\big(U^{\mathrm{an}}; \mathbb{C}\big)$$

***Proof Key ideas*:**

The algebraic side degenerates, and this half is complete. Each $\Omega^p_{U/\mathbb{C}}$ is quasi-coherent on the affine U, so $H^q(U, \Omega^p_{U/\mathbb{C}}) = 0$ for every $q > 0$ by theorem 6.2.9. In the hypercohomology spectral sequence

$$E_1^{p,q} = H^q\big(U, \Omega^p_{U/\mathbb{C}}\big) \Longrightarrow \mathbb{H}^{p+q}\big(U, \Omega^\bullet_{U/\mathbb{C}}\big)$$

the entire $q > 0$ region vanishes, so the sequence degenerates and

$$\mathbb{H}^i\big(U, \Omega^\bullet_{U/\mathbb{C}}\big) \;\cong\; H^i\big(\Gamma(U, \Omega^\bullet_{U/\mathbb{C}})\big)$$

The left side of the theorem is therefore the algebraic de Rham hypercohomology, exactly the object of corollary 6.9.7. No properness is used, and nothing about U beyond affineness.

The comparison is the content, and is not proved here. What remains is $\mathbb{H}^i(U, \Omega^\bullet_{U/\mathbb{C}}) \cong H^i(U^{\mathrm{an}}; \mathbb{C})$. On the analytic side the holomorphic Poincaré lemma (corollary 6.9.6) still makes $\Omega^{\bullet,\mathrm{an}}$ a resolution of $\underline{\mathbb{C}}$, so the right side is analytic hypercohomology; but algebraic and analytic hypercohomology cannot be compared through GAGA, which needs properness. The analytic side also degenerates, since U^{an} is a Stein manifold and Cartan's Theorem B makes each $\Omega^{p,\mathrm{an}}$ acyclic on it, so the content is that the inclusion of the complex of global algebraic forms into the far larger complex of global holomorphic forms on U^{an} is a quasi-isomorphism. The standard route chooses a smooth compactification $U \subseteq \overline{U}$ whose boundary is a normal-crossings divisor D, replaces $\Omega^\bullet$ by the complex of meromorphic forms with poles of arbitrary order along D, and compares on $\overline{U}$, where properness is restored. That needs resolution of singularities and the meromorphic de Rham complex, neither developed in this book. The argument is [33, the comparison theorem for smooth affine varieties].

Corollary 6.9.9: Comparison of Euler Characteristics

Let X be a smooth projective variety of dimension n over $\mathbb{C}$ and let $\mathscr{F}$ be a coherent algebraic sheaf on X. Then:

$$\chi_{\text{alg}}(X, \mathscr{F}) = \sum_{i=0}^{n} (-1)^i \dim_{\mathbb{C}} H^i(X, \mathscr{F}) = \sum_{i=0}^{n} (-1)^i \dim_{\mathbb{C}} H^i(X^{\text{an}}, \mathscr{F}^{\text{an}}) = \chi_{\text{an}}(X^{\text{an}}, \mathscr{F}^{\text{an}}).$$

In particular, the algebraic and analytic Hilbert polynomials coincide: the algebraic Hilbert polynomial $P_{\mathscr{F}}$ of theorem 6.7.5 equals the analytic one.

***Proof*:**
Immediate from theorem 6.9.2(2): the comparison isomorphism preserves dimensions.

Corollary 6.9.10: Hodge Decomposition via GAGA

Let X be a smooth projective variety of dimension n over $\mathbb{C}$. Then the singular cohomology of $X(\mathbb{C})$ admits a Hodge decomposition:

$$H^k_{\text{sing}}(X(\mathbb{C}); \mathbb{C}) \cong \bigoplus_{p+q=k} H^q(X, \Omega^p_{X/\mathbb{C}}),$$

where $H^q(X, \Omega^p_{X/\mathbb{C}})$ is the algebraic sheaf cohomology of the p-th sheaf of algebraic differentials. The *Hodge numbers* are:

$$h^{p,q}(X) = \dim_{\mathbb{C}} H^q(X, \Omega^p_{X/\mathbb{C}}),$$

and they satisfy the symmetries $h^{p,q} = h^{q,p}$ (Hodge symmetry) and $h^{p,q} = h^{n-p,\,n-q}$ (Serre duality).

***Proof*:**
By GAGA, $H^q(X, \Omega^p_{X/\mathbb{C}}) \cong H^q(X^{\text{an}}, (\Omega^p)^{\text{an}})$. By the Dolbeault theorem (theorem 0.6.17), $H^q(X^{\text{an}}, (\Omega^p)^{\text{an}}) \cong H^{p,q}_{\bar{\partial}}(X^{\text{an}})$, the Dolbeault cohomology. Since X^{an} is a compact Kähler manifold, the Hodge decomposition theorem (theorem 0.6.28) gives:

$$H^k_{\text{dR}}(X^{\text{an}}) \cong \bigoplus_{p+q=k} H^{p,q}_{\bar{\partial}}(X^{\text{an}}),$$

and the de Rham theorem identifies the left side with $H^k_{\text{sing}}(X(\mathbb{C}); \mathbb{C})$. Combining these isomorphisms gives the decomposition.

The symmetry $h^{p,q} = h^{q,p}$ is the Hodge symmetry from complex conjugation on a Kähler manifold (theorem 0.6.28). The symmetry $h^{p,q} = h^{n-p,\,n-q}$ follows from Serre duality (corollary 6.5.10): since $\omega_X = \Omega^n_{X/\mathbb{C}}$, we have $H^q(X, \Omega^p) \cong H^{n-q}(X, (\Omega^p)^\vee \otimes \Omega^n)^* \cong H^{n-q}(X, \Omega^{n-p})^*$, giving $h^{p,q} = h^{n-p,\,n-q}$.

Example 6.9.11: GAGA in Action

1. *Line bundles and the exponential sequence.* On a compact complex manifold M, the *exponential sequence* of analytic sheaves:

$$0 \longrightarrow \underline{\mathbb{Z}} \xrightarrow{2\pi i} \mathcal{O}_M^{\mathrm{an}} \xrightarrow{\exp} (\mathcal{O}_M^{\mathrm{an}})^* \longrightarrow 0$$

induces a long exact sequence in cohomology:

$$\cdots \to H^1(M, \mathcal{O}^{\mathrm{an}}) \to H^1(M, (\mathcal{O}^{\mathrm{an}})^*) \xrightarrow{c_1} H^2(M, \mathbb{Z}) \to H^2(M, \mathcal{O}^{\mathrm{an}}) \to \cdots$$

The group $H^1(M, (\mathcal{O}^{\mathrm{an}})^*)$ classifies analytic line bundles (it is the analytic Picard group), and $c_1 : \mathrm{Pic}^{\mathrm{an}}(M) \to H^2(M, \mathbb{Z})$ is the first Chern class map (definition 0.6.41).

For a smooth projective variety X over $\mathbb{C}$, GAGA identifies the analytic Picard group with the algebraic one: $\mathrm{Pic}(X) \cong \mathrm{Pic}^{\mathrm{an}}(X^{\mathrm{an}})$. The exponential sequence is *not* algebraic (the exponential map exp is transcendental), but GAGA carries its consequences over to algebraic line bundles: the Chern class map becomes a homomorphism $c_1 \colon \mathrm{Pic}(X) \to H^2(X(\mathbb{C}); \mathbb{Z})$, and by the Lefschetz theorem on $(1,1)$-classes every integral class of type $(1,1)$ is the first Chern class of an algebraic line bundle.

2. *Holomorphic vs. algebraic vector bundles.* Let X be a smooth projective variety over $\mathbb{C}$. By GAGA, the categories of algebraic and holomorphic vector bundles on X are equivalent (vector bundles correspond to locally free sheaves algebraically by theorem 5.1.27 and holomorphically by proposition 0.6.36). Hence every holomorphic vector bundle on X^{an} is algebraic. This fails for non-proper varieties. On $U = \mathbb{A}^2_{\mathbb{C}} \setminus \{0\}$ every algebraic line bundle is trivial, since $\mathrm{Pic}(U) = \mathrm{Pic}(\mathbb{A}^2_{\mathbb{C}}) = 0$. But $U^{\mathrm{an}} \simeq S^3$ has $H^1(U^{\mathrm{an}}; \mathbb{Z}) = H^2(U^{\mathrm{an}}; \mathbb{Z}) = 0$, so the exponential sequence gives $\mathrm{Pic}^{\mathrm{an}}(U^{\mathrm{an}}) \cong H^1(U^{\mathrm{an}}, \mathcal{O}^{\mathrm{an}})$, which is infinite-dimensional. Hence U^{an} carries holomorphic line bundles that are not algebraic.

3. *Sheaf cohomology and singular cohomology of $\mathbb{P}^n$.* Let $X = \mathbb{P}^n_{\mathbb{C}}$. The algebraic sheaf cohomology groups are: $H^i(\mathbb{P}^n, \mathcal{O}) = \mathbb{C}$ for $i = 0$ and 0 for $i > 0$ (by theorem 6.3.1). By GAGA, the same holds analytically. The Hodge decomposition gives $H^k_{\mathrm{sing}}(\mathbb{CP}^n; \mathbb{C}) = \bigoplus_{p+q=k} H^q(\mathbb{P}^n, \Omega^p)$. By theorem 6.3.3, $h^{p,p} = 1$ for $0 \le p \le n$ and all other $h^{p,q} = 0$, so:

$$H^k_{\mathrm{sing}}(\mathbb{CP}^n; \mathbb{C}) = \begin{cases} \mathbb{C} & k \text{ even},\ 0 \le k \le 2n, \\ 0 & k \text{ odd}. \end{cases}$$

This agrees with the classical computation from algebraic topology (see [57, The Cohomology Ring of Complex Projective Space]). The agreement is only a consistency check, since GAGA on $\mathbb{P}^n$ rests on the Hodge diamond of $\mathbb{CP}^n$ (example 0.6.30), which is read off from that same computation.

4. *The Lefschetz theorem on hyperplane sections.* Let $X \subset \mathbb{P}^N_{\mathbb{C}}$ be a smooth projective variety of dimension n and let $H \subset \mathbb{P}^N$ be a hyperplane with $Y = X \cap H$ smooth. The Lefschetz hyperplane theorem asserts that the restriction map:

$$H^i_{\mathrm{sing}}(X(\mathbb{C}); \mathbb{Z}) \longrightarrow H^i_{\mathrm{sing}}(Y(\mathbb{C}); \mathbb{Z})$$

is an isomorphism for $i < n-1$ and injective for $i = n-1$. The proof uses Morse theory and is fundamentally analytic. Restriction is compatible with the Hodge decompositions

of corollary 6.9.10, so the restriction maps $H^q(X, \Omega^p_X) \to H^q(Y, \Omega^p_Y)$ are isomorphisms for $p + q < n - 1$. This has purely algebraic applications — for instance, it implies that the Picard group of a smooth hypersurface $X \subset \mathbb{P}^n$ with $n \geq 4$ is isomorphic to $\mathbb{Z}$, generated by $\mathcal{O}_X(1)$.

6.9.12 Remark: GAGA as a Principle Beyond the precise theorems stated above, GAGA functions as a *guiding principle* in algebraic geometry over $\mathbb{C}$: any "reasonable" statement about coherent sheaves and their cohomology on projective varieties that can be proved analytically also holds algebraically, and vice versa. This principle is invoked (often implicitly) throughout the subject:

- Kodaira vanishing ($H^i(X, \omega_X \otimes \mathcal{L}) = 0$ for $i > 0$ and $\mathcal{L}$ ample) is proved analytically using L^2-estimates for the $\bar{\partial}$-operator, but applies to algebraic cohomology via GAGA.
- The degeneration of the Hodge-to-de Rham spectral sequence for smooth projective varieties is proved using Kähler geometry (or, for a variety over a perfect field F of characteristic exceeding its dimension that lifts to $W_2(F)$, by the method of Deligne and Illusie), but its algebraic consequences are what matter for classification: $\sum_{p+q=k} h^{p,q}(X) = \dim_{\mathbb{C}} H^k_{\mathrm{dR}}(X/\mathbb{C})$, and, with semicontinuity, the Hodge numbers are locally constant in smooth projective families.
- The construction of moduli spaces often proceeds by first constructing an analytic moduli space (using complex-analytic deformation theory) and then invoking GAGA to conclude that it is algebraic.

In this way, GAGA allows free passage between the algebraic and analytic worlds, using whichever tools are most natural for the problem at hand.

6.9.4 Summary

We close with a summary of the cohomology that this chapter has developed, viewed through GAGA. Let X be a smooth projective variety of dimension n over $\mathbb{C}$, and let $\mathscr{F}$ be a coherent sheaf on X. The following table summarizes the relationships between the various cohomology theories:

Algebraic	**Analytic**	**Via**
$H^i(X, \mathscr{F})$	$H^i(X^{\mathrm{an}}, \mathscr{F}^{\mathrm{an}})$	GAGA
$H^q(X, \Omega^p_{X/\mathbb{C}})$	$H^{p,q}_{\bar{\partial}}(X^{\mathrm{an}})$	GAGA + Dolbeault
$\mathbb{H}^k(X, \Omega^\bullet)$	$H^k_{\mathrm{sing}}(X(\mathbb{C}); \mathbb{C})$	Algebraic de Rham
$\chi(X, \mathscr{F})$	$\chi(X^{\mathrm{an}}, \mathscr{F}^{\mathrm{an}})$	GAGA
$\mathrm{Pic}\,(X)$	$\mathrm{Pic}^{\,\mathrm{an}}(X^{\mathrm{an}})$	GAGA

The algebraic side (left column) uses only the Zariski topology, coherent sheaves, and the machinery of this chapter. The analytic side (middle column) uses the classical topology, Dolbeault cohomology, and Hodge theory from section 0.6. The right column indicates which bridge theorem connects them.

Serre duality, $\mathrm{Ext}^i(\mathscr{F}, \omega_X) \cong H^{n-i}(X, \mathscr{F})^*$, and the symmetry $h^{p,q} = h^{n-p,\,n-q}$ that it gives stay inside the algebraic column.

For non-projective or non-smooth varieties, the dictionary partially breaks down: GAGA requires properness, the Hodge decomposition requires smoothness and properness, and Serre duality requires properness, with X Cohen-Macaulay and equidimensional for duality in every degree. The interplay between these hypotheses — determining which parts of the dictionary survive under which conditions — is a central theme of modern algebraic geometry.

7

Application: Algebraic Curves

This chapter applies the cohomology of chapter 6 to complete nonsingular curves over an algebraically closed field. For such a curve X, the dimension of $H^1(X, \mathcal{O}_X)$ is the genus g, and the additivity of the Euler characteristic together with Serre duality gives the Riemann-Roch theorem $\ell(D) - \ell(K - D) = \deg D + 1 - g$. Section 7.1 identifies a curve with its function field and defines the degree of a morphism of curves, and section 7.2 proves Riemann-Roch. Section 7.3 derives from it the degree $2g - 2$ of the canonical divisor, criteria for a divisor to be base-point-free or very ample, and the canonical map, and section 7.4 proves Hurwitz's formula for a finite separable morphism of curves. Section 7.5 treats the curves of genus $g \geq 2$ whose canonical map is not an embedding, and section 7.6 obtains the group law of a curve of genus 1 from Riemann-Roch. Section 7.7 gives Castelnuovo's bound on the genus of a curve in $\mathbb{P}^r$, the semicontinuity of cohomology in families of curves, and the changes over a field that is not algebraically closed.

Families of curves and the moduli space of curves of genus g are the subject of *Moduli, Deformation Theory, and Stacks* [20]. The Jacobian of a curve of genus g, an abelian variety of dimension g whose k-points are the classes of divisors of degree 0, is constructed in *Abelian Varieties* [5], and the arithmetic of elliptic curves is developed in *Elliptic Curves* [15]. Section 8.7, at the end of chapter 8, bounds the number of points of a curve C over a finite field by applying the Hodge index theorem to the surface $C \times C$.

7.0.1 Conventions Unless otherwise stated, k denotes an algebraically closed field and all schemes are k-schemes. A *curve* is a complete nonsingular curve over k (definition 7.1.1 gives the scheme-theoretic definition). A *point* is a closed point: the only other point of an integral curve is its generic point η, which is named explicitly when it occurs. Statements that relax these hypotheses, in particular the algebraic closure of k, say so.

7.1 Curves and Their Function Fields

A complete nonsingular curve over k admits three descriptions, the second only over $\mathbb{C}$:

1. *The geometric perspective.* A curve is a one-dimensional, complete, nonsingular scheme over k (definition 7.1.1), and the theorems of this chapter are proved in this language. It continues the classical theory of curves as varieties, with divisors, the genus and Bézout's theorem, developed in *Classical Algebraic Geometry* [22].

2. *The analytic perspective.* Over $\mathbb{C}$, analytification (definition 0.6.46) is an equivalence between the category of smooth projective curves and the category of compact Riemann surfaces with holomorphic maps (box 7.1.5). Forgetting the complex structure of a compact Riemann surface leaves a compact, connected, oriented smooth surface, and such a surface is determined up to diffeomorphism by its genus, the number of handles.[1]

3. *The algebraic perspective.* Two complete nonsingular curves are isomorphic exactly when their function fields are isomorphic over k, and every finitely generated field extension K/k of transcendence degree 1 is isomorphic over k to the function field of such a curve. On morphisms the correspondence is contravariant: $X \mapsto K(X)$ is an anti-equivalence between the category of complete nonsingular curves with nonconstant morphisms and the category of these field extensions with k-algebra homomorphisms (theorem 7.1.4). Studying curves is therefore equivalent to studying function fields. This is the perspective closest to number theory: the function field of a curve over $\mathbb{F}_q$ shares many properties with a number field, and several results of this chapter have arithmetic counterparts (box 7.1.11). For curves as varieties, *Classical Algebraic Geometry* [22, Function Field and Morphism Correspondence, Characterizing Nonsingular Projective Curves] states the bijection on morphisms and the isomorphism criterion.

Recall that a scheme is *integral* if it is both reduced and irreducible (proposition 2.4.20), and *regular* if all of its local rings are regular local rings (definition 2.7.5).

[1] Every topological surface admits a smooth structure, unique up to diffeomorphism, so the topological and smooth classifications of surfaces agree.

Definition 7.1.1: Curve

Let k be a field. A *curve over* k is a Noetherian, integral, separated scheme X of dimension 1 and of finite type over k.

1. The curve X is *complete* (or *proper*) if the structure morphism $X \to \operatorname{Spec} k$ is proper.
2. The curve X is *nonsingular* (or *regular*) if all local rings $\mathcal{O}_{X,x}$ are regular. When k is perfect, in particular when $k = \overline{k}$, this holds if and only if the structure morphism is smooth (theorem 2.7.15), and a nonsingular curve is then also called *smooth*.

Let $\mathbf{Crv}_k$ be the category of all curves over k, $\mathbf{Crv}_k^{\mathrm{prop}}$ the subcategory of proper curves, and $\mathbf{SmProjCrv}_k$ the subcategory of curves that are smooth and projective over k. When k is perfect, the curves in $\mathbf{SmProjCrv}_k$ are exactly the complete nonsingular curves (theorem 2.7.15, and proposition 7.1.7 below), and $\mathbf{SmProjCrv}_k$ is the default category of this chapter.

The definition admits curves that are not complete, such as the affine line $\mathbb{A}^1_k$, and curves that are singular, such as the affine nodal cubic $V(y^2 - x^3 - x^2) \subseteq \mathbb{A}^2_k$. Under the conventions of this chapter an unqualified curve is complete and nonsingular.

The integrality assumption gives a curve a unique generic point η with $\overline{\{\eta\}} = X$, and the function field $\kappa(\eta) = K(X)$ is a finitely generated field extension of k of transcendence degree 1 (theorem 2.6.11). When k is algebraically closed, every such extension is isomorphic over k to the function field of a smooth projective curve (theorem 7.1.4(3) below).

At a closed point x of a nonsingular curve, the local ring $\mathcal{O}_{X,x}$ is a regular local ring of dimension 1, that is, a discrete valuation ring (DVR) [10, Characterizations of a DVR]. When X is moreover complete, every DVR of $K(X)/k$ dominates, and hence equals, the local ring of exactly one closed point of X (corollary 7.3.5 below). The closed points of a complete nonsingular curve are therefore in bijection with the DVRs of $K(X)/k$, the classical starting point for the theory of abstract curves. Every complete nonsingular curve is projective (proposition 7.1.7 below).

Singular curves arise as fibers of families, as images of morphisms, and as boundary points of moduli spaces. At a singular point x of an integral curve the local ring $\mathcal{O}_{X,x}$ is not a DVR, and the bijection between closed points and DVRs of $K(X)/k$ fails. A singular curve is related to the nonsingular theory through its *normalization*. A Noetherian local domain of dimension 1 is a DVR if and only if it is integrally closed [10, Characterizations of a DVR], so the normalization of an integral curve (theorem 3.4.41) is a nonsingular curve with the same function field, together with a finite birational morphism onto the original curve.

Proposition 7.1.2: Affine Curves and Unique Proper Curve

Let X be an affine nonsingular curve over k. Then there is a complete nonsingular curve Y birational to X (definition 3.6.2), so that $K(Y) \cong K(X)$ (corollary 3.6.6), and Y is unique up to isomorphism.

Existence is the case $V = X$ of theorem 7.1.3, whose proof carries out the steps of the proof below in full, and uniqueness follows from part (2) of that theorem applied to a dense open subscheme common to two candidates.

***Proof Key ideas*:**
As X is affine and of finite type, $X \cong \operatorname{Spec} k[x_1, \dots, x_n]/I$, so X is a closed subscheme of $\mathbb{A}^n_k$ and hence a locally closed subscheme of $\mathbb{P}^n_k$ (lemma 5.3.16). Its closure $\overline{X} \subseteq \mathbb{P}^n_k$, with the reduced induced structure, is an integral projective curve containing X as a dense open subscheme. It may be singular, so take its normalization

$$\nu\colon \widetilde{X} \longrightarrow \overline{X}$$

(theorem 3.4.41), obtained by replacing the coordinate ring of each affine open of $\overline{X}$ by its integral closure in $K(X)$ (definition 2.4.28). The scheme $\widetilde{X}$ is normal of dimension 1, hence nonsingular, and it is complete because ν is finite, hence proper (proposition 3.5.24), and $\overline{X}$ is proper over k (proposition 3.5.23(2)). Over the normal open subscheme X the morphism ν is an isomorphism, so $Y = \widetilde{X}$ contains X as a dense open subscheme. For uniqueness, two such curves Y and Y' contain dense open subschemes isomorphic to one and the same dense open subscheme $W \subseteq X$ (proposition 3.6.9). Both are projective (proposition 7.1.7 below), so both are completions of W in the sense of theorem 7.1.3(1), and part (2) of that theorem gives $Y \cong Y'$, as claimed.

The normalization separates the branches of a curve at a node and replaces a cusp by a nonsingular point:

1. The normalization of the *nodal cubic* $V(y^2z - x^3 - x^2z) \subseteq \mathbb{P}^2_k$ with $\operatorname{char} k \neq 2$, the projective closure of $y^2 = x^3 + x^2$, is $\mathbb{P}^1_k$, with two points over the node, one for each branch through the origin.

2. The normalization of the *cuspidal cubic* $V(y^2z - x^3) \subseteq \mathbb{P}^2_k$ is again $\mathbb{P}^1_k$, with a single point over the cusp.

At a node two branches with distinct tangent lines cross, and at a cusp a single branch has one tangent line. More complicated singularities occur (higher-order cusps, tacnodes, points where several branches meet), and the normalization separates their branches in the same way.

Both cubics above have arithmetic genus 1 and normalization $\mathbb{P}^1_k$, so both are rational. The difference between the arithmetic genus of X and the genus of its normalization $\widetilde{X}$ is the sum of the δ-invariants of the singular points (box 7.2.9). When X is complete, so is $\widetilde{X}$, being finite over X, and $\widetilde{X}$ is then the complete nonsingular model of the function field $K(X)$ (corollary 7.3.5 below), unique up to isomorphism by theorem 7.1.4(3) below.

Proposition 7.1.2 applies only to an affine curve, and its statement relates Y to X only through a birational map, with no open immersion $X \hookrightarrow Y$. A curve arising as a fiber of a family, as an orbit, or as a one-dimensional slice of a larger variety is quasiprojective and rarely affine, and the arguments that use a completion need the original curve to be an open subscheme of the completed one. Theorem 7.1.3 removes both restrictions. Over an arbitrary field it produces a regular curve, and regularity coincides with smoothness when the field is perfect (theorem 2.7.15).

Theorem 7.1.3: Projective Completion of a Smooth Curve

Let k be a field and let V be a quasiprojective curve over k (definition 7.1.1, definition 2.3.5) which is smooth over k. Then:

1. There exist a regular projective curve $\overline{V}$ over k, in particular integral and hence connected, and an open immersion $j\colon V \hookrightarrow \overline{V}$ with dense image, and the map j identifies $K(\overline{V})$ with $K(V)$.
2. The pair $(\overline{V}, j)$ is unique up to unique isomorphism under V: for any second such pair $(\overline{V}', j')$ there is exactly one isomorphism $f\colon \overline{V} \to \overline{V}'$ with $f \circ j = j'$.
3. If k is perfect, then $\overline{V}$ is smooth over k.

***Proof*:**

Step 1: an integral projective curve containing V as a dense open. By definition 2.3.5 there are a projective k-scheme Y and an open immersion $V \subseteq Y$. Let $\overline{X} \subseteq Y$ be the closure of V, taken with its reduced induced structure (definition 3.1.26). It is irreducible, being the closure of the irreducible V (lemma 2.1.24(1)), and reduced, hence integral (proposition 2.4.20). A composite of closed immersions is a closed immersion (proposition 3.1.24(3)), so the closed subscheme $\overline{X}$ of Y admits a closed immersion into some $\mathbb{P}^n_k$ and is projective (corollary 5.3.15(3)), hence of finite type and proper over k (theorem 3.5.32). The subset V is open in $\overline{X}$, and the induced structure agrees with the given one, both being reduced with the same underlying space (proposition 3.1.25). So V is a dense open subscheme of $\overline{X}$ and $K(\overline{X}) = K(V)$. That field has transcendence degree 1 over k, so $\operatorname{kdim}\overline{X} = 1$ (theorem 2.6.11).

Step 2: normalize. Since $\overline{X}$ is integral of finite type over k, theorem 3.4.41 gives a normalization $\nu\colon \widetilde{X} \to \overline{X}$: the scheme $\widetilde{X}$ is integral, normal and of finite type over k, the morphism ν is finite, surjective and birational, and ν is an isomorphism over the normal locus of $\overline{X}$. As $\overline{X}$ is projective, corollary 5.3.58 makes $\widetilde{X}$ projective over k, hence separated, Noetherian and proper. Birationality gives $K(\widetilde{X}) = K(V)$ and therefore $\operatorname{kdim}\widetilde{X} = 1$ (theorem 2.6.11).

Step 3: V is unchanged by the normalization. Smoothness implies regularity (proposition 2.7.14), and a regular local ring is integrally closed (proposition 2.7.26), so every local ring of V is normal. Those local rings are the local rings of $\overline{X}$ at the points of V, so V lies in the normal locus of $\overline{X}$ and ν restricts to an isomorphism $\nu^{-1}(V) \to V$. Set $\overline{V} = \widetilde{X}$ and let $j\colon V \to \overline{V}$ be the inverse of that isomorphism followed by the open immersion $\nu^{-1}(V) \subseteq \overline{V}$. Its image is a nonempty open subset of the irreducible scheme $\overline{V}$, hence dense, and j identifies $K(V)$ with $K(\overline{V})$.

Step 4: $\overline{V}$ is regular, and smooth over a perfect field. The scheme $\overline{V}$ is Noetherian, integral and normal, so by proposition 5.4.2(2) every point at which it fails to be regular has codimension at least 2. No point has codimension 2: for $x \in \overline{V}$ the closure $\overline{\{x\}}$ is an irreducible closed subset, so $\operatorname{codim}_{\overline{V}}\overline{\{x\}} \leq \operatorname{kdim}\overline{V} = 1$ by lemma 2.6.7, and this codimension is $\operatorname{kdim}\mathcal{O}_{\overline{V},x}$ by definition 2.6.5. Hence $\overline{V}$ is regular (definition 2.7.5) and is a projective curve over k in the sense of definition 7.1.1. If k is perfect, regularity and smoothness coincide for k-schemes locally of finite type (theorem 2.7.15), so $\overline{V}$ is then smooth over k. This proves (1) and (3).

Step 5: a morphism between two completions. Let $(\overline{V}', j')$ be a second pair as in (1). The composite $j' \circ j^{-1}$, defined on the dense open $j(V) \subseteq \overline{V}$, is a rational map $\varphi\colon \overline{V} \dashrightarrow \overline{V}'$. Its source is integral, normal (proposition 2.7.26) and of finite type over k, and its target is proper over k (theorem 3.5.32), so by theorem 4.7.4 the indeterminacy locus of φ has codimension at least 2 in $\overline{V}$. By the computation of Step 4 no point of $\overline{V}$ has codimension 2, so that locus is empty and φ is a morphism $f\colon \overline{V} \to \overline{V}'$ with $f \circ j = j'$.

Step 6: the morphism is the unique isomorphism under V. The source $\overline{V}$ is proper over k and the target $\overline{V}'$ is separated over k, so f is proper (proposition 3.5.23(5)) and therefore separated, of finite type and closed (definition 3.5.21). Its image is closed and contains the dense subset $j'(V)$, so f is surjective, and $f \circ j = j'$ forces f to induce the identity on $K(V)$, so f is birational. Each fiber is finite. The fiber over the generic point of $\overline{V}'$ is the generic point of $\overline{V}$, because f is closed and so carries closed points to closed points. The fiber over a closed point is a closed subset of $\overline{V}$ not containing the generic point, so each of its irreducible components Z satisfies $\operatorname{codim}_{\overline{V}} Z \geq 1$ and hence $\operatorname{kdim} Z = 0$ by lemma 2.6.7, making Z a closed point, and the components are finite in number because $\overline{V}$ is Noetherian. So f is quasifinite. Both schemes are integral and Noetherian and $\overline{V}'$ is normal, so corollary 3.4.39 makes f an open immersion, and a surjective open immersion is an isomorphism. Finally, two isomorphisms $\overline{V} \to \overline{V}'$ under V agree on the dense open $j(V)$ of the reduced scheme $\overline{V}$, and $\overline{V}'$ is projective over k and so separated, hence the two agree (proposition 3.5.10). This proves (2), completing the proof.

The theorem contains proposition 7.1.2: an affine nonsingular curve is quasiprojective (lemma 5.3.16), so it has a completion, which contains it as a dense open subscheme, and part (2), applied to a common dense open subscheme, gives the uniqueness.

Part (3) needs k to be perfect. Over an imperfect field a smooth curve can fail to have any smooth projective completion, so the general conclusion is regularity, and example 2.7.16 shows the phenomenon at a single point. For a curve, let p be odd, let $k = \mathbb{F}_p(t)$ and let $C = \operatorname{Spec} k[x,y]/(y^2 - x^p + t)$, an integral affine curve. The Jacobian of $y^2 - x^p + t$ is $(0, 2y)$, so C fails to be smooth over k exactly along $y = 0$, which on C is the single closed point P cut out by y, with residue field $k(t^{1/p})$. Thus $V = C \setminus \{P\}$ is a smooth quasiprojective curve. But C is regular at P as well: the relation $y^2 = x^p - t$ makes the maximal ideal of $\mathcal{O}_{C,P}$ equal to (y), a principal ideal in a one-dimensional Noetherian local domain. The argument of Step 5, applied to the regular, hence normal, curve C in place of $\overline{V}$, extends $j\colon V \hookrightarrow \overline{V}$ to a birational morphism $C \to \overline{V}$. This morphism is quasifinite, since its fiber over a closed point is a proper closed subset of C and its fiber over the generic point lies in $\{\eta_C, P\}$, so it is an open immersion by corollary 3.4.39. Hence $\overline{V}$ contains an open subscheme isomorphic to C and is not smooth over k, and by part (2) no smooth projective curve contains V as a dense open subscheme.

A nonsingular curve is normal, and each of its points other than the generic point has codimension 1, so theorem 4.7.4 leaves a rational map from a nonsingular curve to a complete curve no indeterminacy. Every dominant rational map between complete nonsingular curves is therefore a morphism, and lemma 3.6.8 identifies these maps with the homomorphisms between the function fields.

Theorem 7.1.4: Curves and Their Function Fields

Let k be algebraically closed.

1. A k-morphism $f\colon X \to Y$ of curves over k in the sense of definition 7.1.1, not necessarily complete or nonsingular, is dominant if and only if it is *nonconstant*, that is, $f(X)$ is not a single point.

2. Let $\mathcal{C}_k$ be the category whose objects are the complete nonsingular curves over k, which are exactly the smooth projective curves over k (definition 7.1.1 and proposition 7.1.7 below), and whose morphisms are the nonconstant, equivalently by (1) the dominant, k-morphisms. Let $\mathcal{F}_k$ be the category whose objects are the finitely generated field extensions of k of transcendence degree 1 and whose morphisms are the k-algebra homomorphisms. Then the functor sending X to $K(X)$ and a morphism $f\colon X \to Y$ to the induced homomorphism $f^\sharp\colon K(Y) \to K(X)$ is an anti-equivalence, that is, an equivalence of categories $\mathcal{C}_k \to \mathcal{F}_k^{\mathrm{op}}$.

3. Two complete nonsingular curves over k are isomorphic if and only if their function fields are isomorphic over k, and every finitely generated field extension of k of transcendence degree 1 is isomorphic over k to the function field of a complete nonsingular curve over k.

***Proof*:**

Step 1: the irreducible closed subsets of a curve. Let Y be a curve over k and let $Z \subsetneq Y$ be an irreducible closed subset. The chain $Z \subsetneq Y$ gives $\operatorname{codim}_Y Z \geq 1$, so $\operatorname{kdim} Z = 0$ by lemma 2.6.7. For $z \in Z$ the closure of $\{z\}$ is an irreducible closed subset of Z (lemma 2.1.24(1)), and it is all of Z, since otherwise the two would form a chain of length 1 in Z. So every point of Z is a generic point of Z, and the generic point is unique (lemma 2.4.21). Hence Z is a single closed point. In particular every point $y \neq \eta_Y$ of Y is closed: $\overline{\{y\}}$ is an irreducible closed subset of Y, different from $Y = \overline{\{\eta_Y\}}$ by the uniqueness of the generic point, so it is the single point y.

Step 2: part (1). If f is not dominant, the closure of $f(X)$ is an irreducible closed subset of Y (lemma 2.1.24(1),(2)) other than Y, so $f(X)$ is a single point by Step 1. Suppose that f is dominant, so that $f(\eta_X) = \eta_Y$ (lemma 3.6.4). As $\operatorname{kdim} X = 1$, the curve X contains an irreducible closed subset other than X, and a point x of that subset differs from η_X, so x is closed by Step 1. Choose an affine open $\operatorname{Spec} A \ni x$ with $A = k[T_1, \ldots, T_n]/I$ a finitely generated k-algebra (definition 2.2.18). The closed point x is a maximal ideal of A, the image of a maximal ideal of $k[T_1, \ldots, T_n]$ containing I, which has the form $(T_1 - a_1, \ldots, T_n - a_n)$ by theorem 0.1.12(3), so $\kappa(x) = k$. If $f(x)$ were η_Y, the local homomorphism $\mathcal{O}_{Y,\eta_Y} = K(Y) \to \mathcal{O}_{X,x}$ would induce a k-algebra homomorphism of residue fields $K(Y) \to \kappa(x) = k$. It would be injective, $K(Y)$ being a field, and so $\dim_k K(Y) \leq 1$. This is impossible, because $K(Y)$ has transcendence degree 1 over k (theorem 2.6.11) and the powers of an element transcendental over k are linearly independent over k. Hence $f(x) \neq f(\eta_X)$ and f is nonconstant. By part (1) and lemma 3.6.4, a composite of nonconstant morphisms of curves carries the generic point to the generic point and is again nonconstant, and identities are nonconstant, so $\mathcal{C}_k$ is a category.

Step 3: the functor. Let X be a curve over k and $\operatorname{Spec} A \subseteq X$ a nonempty affine open with A a

finitely generated k-algebra. Then $K(X) = \mathrm{Frac}A$ (corollary 2.4.25) is generated as a field over k by the finitely many generators of A, and it has transcendence degree 1 over k by theorem 2.6.11, a curve being a k-variety (definition 3.7.2) of dimension 1. A morphism $f\colon X \to Y$ of $\mathcal{C}_k$ is dominant by part (1), so it represents a dominant rational map, and proposition 3.6.5 gives its stalk map at η_X as a k-algebra homomorphism $f^\sharp\colon K(Y) \to K(X)$, with $(g \circ f)^\sharp = f^\sharp \circ g^\sharp$ and $\mathrm{id}_X^\sharp = \mathrm{id}_{K(X)}$. So $X \mapsto K(X)$ and $f \mapsto f^\sharp$ form a functor $\mathcal{C}_k \to \mathcal{F}_k^{\mathrm{op}}$.

Step 4: the functor is faithful. Let $f, g\colon X \to Y$ be morphisms of $\mathcal{C}_k$ with $f^\sharp = g^\sharp$. The curve Y is of finite type over k, so by lemma 3.6.8 the dominant rational maps represented by f and g are equal, that is, f and g agree on a dense open subset of X (definition 3.6.1). The scheme X is reduced (proposition 2.4.20) and Y is separated over k, so $f = g$ by proposition 3.5.10.

Step 5: the functor is full. Let X and Y be objects of $\mathcal{C}_k$ and let $\sigma\colon K(Y) \to K(X)$ be a k-algebra homomorphism. By lemma 3.6.8 there is a dominant rational map $\pi\colon X \dashrightarrow Y$ of k-schemes with $\pi^\sharp = \sigma$. The curve X is integral and of finite type over k, and it is normal because its local rings are regular, hence integrally closed (proposition 2.7.26). The curve Y is proper over k. By theorem 4.7.4 the locus of indeterminacy of π has codimension at least 2 in X. No nonempty irreducible closed subset of X has codimension at least 2, since $\mathrm{codim} + \dim \leq \mathrm{kdim} X = 1$ (lemma 2.6.7), so that locus is empty. As Y is separated, proposition 4.7.2 shows that the representatives of π glue to a morphism $f\colon X \to Y$ representing π, and f is a k-morphism because its restrictions to the domains of the representatives are. It carries η_X to η_Y (lemma 3.6.4), so it is dominant, hence nonconstant by part (1). Finally $f^\sharp = \pi^\sharp = \sigma$, because the homomorphism of proposition 3.6.5 does not depend on the representative.

Step 6: isomorphisms. An isomorphism $f\colon X \to Y$ of curves over k is surjective, hence nonconstant because Y is not a single point, and $f^\sharp$ is an isomorphism with inverse $(f^{-1})^\sharp$ by Step 3. Conversely, let $\sigma\colon K(Y) \to K(X)$ be an isomorphism of k-algebras between the function fields of two objects of $\mathcal{C}_k$. Step 5 gives morphisms $f\colon X \to Y$ and $g\colon Y \to X$ of $\mathcal{C}_k$ with $f^\sharp = \sigma$ and $g^\sharp = \sigma^{-1}$, and

$$(g \circ f)^\sharp = f^\sharp \circ g^\sharp = \sigma \circ \sigma^{-1} = \mathrm{id}_{K(X)} = \mathrm{id}_X^\sharp$$

so $g \circ f = \mathrm{id}_X$ by Step 4, and likewise $f \circ g = \mathrm{id}_Y$. This proves the first assertion of part (3).

Step 7: essential surjectivity. Let K be a finitely generated field extension of k of transcendence degree 1, generated by $t_1, \ldots, t_n$, and let $A = k[t_1, \ldots, t_n] \subseteq K$. Then A is a finitely generated k-algebra and a domain, and $\mathrm{Frac}A = K$ because K is generated as a field by $t_1, \ldots, t_n$. So $\operatorname{Spec} A$ is an integral scheme of finite type over k. Let $\nu\colon \widetilde{X} \to \operatorname{Spec} A$ be its normalization (theorem 3.4.41): the scheme $\widetilde{X}$ is integral, normal and of finite type over k, the morphism ν is finite, and it induces an isomorphism $K \to K(\widetilde{X})$ of k-algebras. Since ν is finite and $\operatorname{Spec} A$ is affine, $\widetilde{X} = \operatorname{Spec} \widetilde{A}$ for a finite A-algebra $\widetilde{A}$ (definition 3.4.15). So $\widetilde{A}$ is a finitely generated k-algebra, Noetherian by the Hilbert basis theorem, and, $\widetilde{X}$ being integral, a domain with fraction field $K(\widetilde{X}) \cong K$ (corollary 2.4.25). Hence $\mathrm{kdim}\widetilde{X} = 1$ by theorem 2.6.11. Being affine, $\widetilde{X}$ is separated over k (example 3.5.3(2)), so it is a curve over k.

The curve $\widetilde{X}$ is regular. Its local ring at the generic point is the field $K(\widetilde{X})$, a regular local ring of dimension 0. Every other point x is closed by Step 1, so the chain $\{x\} \subsetneq \widetilde{X}$ and lemma 2.6.7

give $\operatorname{codim}_{\widetilde{X}}\{x\} = 1$, and this is $\operatorname{kdim}\mathcal{O}_{\widetilde{X},x}$ (definition 2.6.5). Thus $\mathcal{O}_{\widetilde{X},x}$, a localization of the Noetherian domain $\widetilde{A}$, is a Noetherian local domain of dimension 1, integrally closed because $\widetilde{X}$ is normal (definition 2.4.28), hence regular [10, Characterizations of a DVR]. As k is perfect, $\widetilde{X}$ is smooth over k (theorem 2.7.15).

The curve $\widetilde{X}$ is integral, hence reduced, and affine of finite type over k, so it is quasiprojective (lemma 5.3.16).

By theorem 7.1.3(1) and (3) there are a regular projective curve C over k, smooth over k, and an open immersion $\widetilde{X} \hookrightarrow C$ over k identifying $K(C)$ with $K(\widetilde{X}) \cong K$. The curve C is proper over k (theorem 3.5.32), so it is an object of $\mathcal{C}_k$, and $K(C) \cong K$ over k. This proves the second assertion of part (3).

Step 8: conclusion. By Steps 4 and 5 the functor of Step 3 is fully faithful, and by Step 7 every object of $\mathcal{F}_k$ is isomorphic to the function field of an object of $\mathcal{C}_k$. The functor is therefore an equivalence $\mathcal{C}_k \to \mathcal{F}_k^{\mathrm{op}}$ (definition 0.4.6), which is part (2), completing the proof.

Under the conventions of this chapter, where $k = \overline{k}$, theorem 7.1.4(3) gives a bijection between isomorphism classes

$$\left\{\begin{array}{c}\text{Smooth Projective Curves}\\ \text{over } k\end{array}\right\} \underset{K \mapsto C_K}{\overset{X \mapsto K(X)}{\rightleftarrows}} \left\{\begin{array}{c}\text{Finitely Generated Field Extensions}\\ \text{of } k\\ \text{of Transcendence Degree 1}\end{array}\right\} \tag{7.1}$$

where C_K is a complete nonsingular curve whose function field is isomorphic to K over k, unique up to isomorphism.

7.1.5 The Analytic Perspective Over $\mathbb{C}$, the analytification X^{an} of a smooth projective curve X (definition 0.6.46) is a compact Riemann surface, and $X \mapsto X^{\mathrm{an}}$ is an equivalence between the category of smooth projective curves over $\mathbb{C}$ and the category of compact Riemann surfaces with holomorphic maps. Full faithfulness comes from GAGA (theorem 6.9.2): by part (3) of that theorem, the graph of a holomorphic map $X^{\mathrm{an}} \to Y^{\mathrm{an}}$, a closed analytic subspace of $(X \times_{\mathbb{C}} Y)^{\mathrm{an}}$, is the analytification of a closed subscheme Γ of $X \times_{\mathbb{C}} Y$. The projection $\Gamma \to X$ is an isomorphism because its analytification is [34, Exposé XII, Proposition 3.1(ix)], so Γ is the graph of a morphism $X \to Y$. Essential surjectivity is not part of GAGA. It is Riemann's theorem that every compact Riemann surface is projective algebraic [36, Appendix B, Theorem 3.1], whose analytic input is the existence of a nonconstant meromorphic function on every compact Riemann surface [28, §14].

A compact Riemann surface is in particular a compact, connected, oriented surface, determined up to diffeomorphism by its topological genus g_{top}, the number of handles. A compact Riemann surface is a compact Kähler manifold, so by the Hodge decomposition (theorem 0.6.28) and GAGA its first Betti number is $2g_{\mathrm{top}} = h^{1,0} + h^{0,1} = 2\dim_{\mathbb{C}} H^1(X, \mathcal{O}_X)$ (example 0.6.31). The topological genus therefore equals $\dim_{\mathbb{C}} H^1(X, \mathcal{O}_X)$, which proposition 7.2.5 identifies with the arithmetic and the geometric genus.

Proposition 7.1.6: Image of Complete Nonsingular Curve is Closed

Let k be a field, let X be a complete nonsingular curve over k, and let Y be any curve over k. Suppose $f\colon X \to Y$ is a morphism. Then either:

1. $f(X)$ is a single closed point of Y, or
2. f is surjective, Y is necessarily complete, and f is a finite morphism.

***Proof*:**
Since X is proper over k and Y is separated over k, the morphism f is proper (proposition 3.5.23(5)), so the image $f(X) \subseteq Y$ is closed. As X is irreducible, so is $f(X)$ (lemma 2.1.24(2)). A closed irreducible subset of the curve Y other than Y has codimension at least 1, hence dimension 0 by lemma 2.6.7, and is a single closed point. So either $f(X)$ is a closed point or f is surjective. Assume from now on that f is surjective.

Step 1: Y is complete. The scheme Y is separated and of finite type over k, so it remains to show that $Y \to \operatorname{Spec} k$ is universally closed. Let T be a k-scheme and let $W \subseteq Y \times_k T$ be a closed subset. The base change $f_T\colon X \times_k T \to Y \times_k T$ is surjective, since surjectivity is stable under base change [70, Tag 01S1]. The image of W in T is therefore the image of the closed subset $f_T^{-1}(W)$ of $X \times_k T$, and this image is closed because X is proper over k.

Step 2: f is finite. The fiber of f over the generic point of Y is the generic point of X, because f is closed and so carries closed points to closed points. The fiber over a closed point is a closed subset of X not containing the generic point, so each of its irreducible components Z satisfies $\operatorname{codim}_X Z \geq 1$, hence $\operatorname{kdim} Z = 0$ by lemma 2.6.7, and Z is a closed point. There are finitely many components because X is Noetherian, and f is of finite type, being proper, so f is quasifinite (definition 3.4.36). It is separated because X is separated over k. By theorem 3.4.38, $f = g \circ j$ with $j\colon X \to M$ an open immersion and $g\colon M \to Y$ finite. The morphism g is proper (proposition 3.5.24), hence separated, and $g \circ j = f$ is proper, so j is proper (proposition 3.5.23(5)) and $j(X)$ is closed as well as open in M. An open immersion onto a closed and open subset is a closed immersion, so j is finite, and $f = g \circ j$ is finite (proposition 3.4.25(1)).

Step 3: the affine pieces. For an affine open $V = \operatorname{Spec} S \subseteq Y$, finiteness gives $f^{-1}(V) = \operatorname{Spec} R$ with R a finite S-module, and R is the integral closure of S in $K(X)$. Indeed, every element of R is integral over S, and R is integrally closed in its fraction field $K(X)$ because its localizations at prime ideals are local rings of X, which are integrally closed (proposition 2.7.26, proposition 2.4.29). Since f is dominant, it induces an inclusion of function fields $K(Y) \hookrightarrow K(X)$, and both have transcendence degree 1 over k, so $K(X)/K(Y)$ is a finite extension, completing the proof.

Properness of a nonsingular curve already forces projectivity, through a finite morphism to $\mathbb{P}^1_k$.

Proposition 7.1.7: Complete Nonsingular Curves Are Projective

Let k be a field and let X be a complete nonsingular curve over k (definition 7.1.1). Then X is projective over k: there is a closed immersion $X \hookrightarrow \mathbb{P}^n_k$ for some n.

***Proof*:**
Step 1: a nonconstant function. The scheme X is integral of finite type over k with $\dim X = 1$, so its function field $K(X)$ has transcendence degree 1 over k (theorem 2.6.11). Choose $f \in K(X)$ transcendental over k. The function field is the local ring at the generic point, a filtered colimit of the rings of sections (corollary 2.4.25), so f is a regular function on some nonempty affine open $V \subseteq X$, and f defines a morphism $V \to \mathbb{A}^1_k \subseteq \mathbb{P}^1_k$, hence a rational map $\varphi\colon X \dashrightarrow \mathbb{P}^1_k$.
Step 2: the rational map is a morphism. Every local ring of X is regular, hence integrally closed (proposition 2.7.26), so X is a normal integral k-scheme of finite type, and $\mathbb{P}^1_k$ is proper over k. By theorem 4.7.4 the locus of indeterminacy of φ has codimension at least 2 in X. No nonempty irreducible closed subset of X has codimension at least 2, since $\operatorname{codim} + \dim \leq \dim X = 1$ (lemma 2.6.7), so that locus is empty and φ is a morphism $\varphi\colon X \to \mathbb{P}^1_k$.
Step 3: the morphism is finite. The restriction $\varphi|_V\colon V \to \mathbb{A}^1_k = \operatorname{Spec} k[t]$ corresponds to the k-algebra map $k[t] \to \mathcal{O}_X(V)$ sending t to f, which is injective because f is transcendental over k. So $\varphi|_V$ is dominant (proposition 3.1.34), and in particular $\varphi(X)$ is not a single closed point. The target $\mathbb{P}^1_k$ is a curve over k in the sense of definition 7.1.1, so proposition 7.1.6 applies and gives that φ is surjective and finite.
Step 4: descending projectivity along a finite morphism. The scheme $\mathbb{P}^1_k$ is Noetherian and projective over k, with $\mathcal{O}(1)$ very ample, so proposition 5.3.57(2) applied to φ gives a closed immersion $X \hookrightarrow \mathbb{P}^{m-1}_{\mathbb{P}^1_k}$ over $\mathbb{P}^1_k$ for some $m \geq 1$. As in the proof of corollary 5.3.58, $\mathbb{P}^{m-1}_{\mathbb{P}^1_k} \cong \mathbb{P}^{m-1}_k \times_k \mathbb{P}^1_k$ (example 5.3.55(1), proposition 5.3.50), the Segre embedding (example 5.3.45(5)) is a closed immersion of this product into $\mathbb{P}^{2m-1}_k$, and a composite of closed immersions is a closed immersion (proposition 3.1.24(3)). So X is projective over k, as we sought to show.

The proof does not use the Riemann-Roch theorem (theorem 7.2.10, proved in section 7.2 below). The very ample criterion corollary 7.3.4, also below, would give projectivity at once through lemma 5.3.23, but its proof rests on theorem 7.2.10, which is derived from Serre duality for projective schemes (theorem 6.5.9), so that route assumes projectivity. The descent of projectivity along a finite morphism is proposition 5.3.57 from chapter 5, and every complete nonsingular curve admits a finite morphism to $\mathbb{P}^1_k$, because its function field contains an element transcendental over k (Steps 1 to 3). Since complete nonsingular curves are projective, Serre duality, and with it the Riemann-Roch theorem of section 7.2, applies to every complete nonsingular curve.

Nonsingularity is not needed for the conclusion: every proper k-scheme of dimension at most 1 is projective over k [70, Tag 0A26]. In dimension at least 2 the conclusion fails. Over an algebraically closed field, complete nonsingular surfaces are projective, but there are complete singular surfaces and complete nonsingular threefolds that are not projective [36, Remark II.4.10.2]. Hironaka's example (proposition 8.4.12, in section 8.4) is such a threefold: an effective curve on it is algebraically equivalent to the negative of another, so no divisor on it is ample.

By proposition 7.1.6, a morphism between complete nonsingular curves is either constant or finite and surjective, and a finite one has a degree.

Definition 7.1.8: Degree of Morphism of Curves

Let $f\colon X \to Y$ be a finite morphism of complete nonsingular curves. The *degree* of f is the degree of the finite field extension $K(X)/K(Y)$:

$$\deg f \;=\; [K(X) : K(Y)]$$

Over a closed point $y \in Y$ the fiber $f^{-1}(y)$ consists of $\deg f$ points counted with multiplicity: $\sum_{x\in f^{-1}(y)} e_x = \deg f$ (proposition 7.1.9), where e_x is the ramification index defined below. In the analytic picture over $\mathbb{C}$, a degree-n map between compact Riemann surfaces is an n-sheeted branched cover.

A finite morphism of complete nonsingular curves pulls back divisors. On a nonsingular curve X Weil and Cartier divisors coincide (proposition 5.4.33, the local rings being DVRs and hence UFDs), so divisors are elements of $\operatorname{WDiv} X = \bigoplus_{x\in X_{(1)}} \mathbb{Z}\cdot x$, and their classes form $\operatorname{Cl} X$. Since $\dim X = 1$, every divisor is a finite formal sum of closed points.

Let $f\colon X \to Y$ be a finite morphism between complete nonsingular curves. For a closed point $y \in Y$, choose a uniformizer $t \in \mathcal{O}_{Y,y}$ (a generator of the maximal ideal $\mathfrak{m}_y$, so that $\nu_y(t) = 1$). For each point $x \in f^{-1}(y)$, the map $f^{\#}\colon \mathcal{O}_{Y,y} \to \mathcal{O}_{X,x}$ sends t to an element of the maximal ideal $\mathfrak{m}_x$. Define the map $f^*\colon \operatorname{WDiv}(Y) \to \operatorname{WDiv}(X)$ on closed points as

$$f^*(y) \;=\; \sum_{x\in f^{-1}(y)} \nu_x(f^{\#}(t))\cdot x \;=\; \sum_{f(x)=y} e_x \cdot x$$

where $e_x = \nu_x(f^{\#}(t))$ is the *ramification index* at x (studied in section 7.4). This is a finite sum since f is a finite morphism, and it is independent of the choice of uniformizer t: any other uniformizer has the form $t' = ut$ for a unit $u \in \mathcal{O}_{Y,y}^*$, and $\nu_x(f^{\#}(u)) = 0$ since $f^{\#}(u)$ is a unit in $\mathcal{O}_{X,x}$.

Extending by $\mathbb{Z}$-linearity gives a group homomorphism $f^*\colon \operatorname{WDiv} Y \to \operatorname{WDiv} X$. It sends principal divisors to principal divisors: if $g \in K(Y)^*$ and $g = ut^m$ with u a unit at y, then $\nu_x(f^{\#}g) = me_x$ for every x over y, so $f^*(\operatorname{div}_Y g) = \operatorname{div}_X(f^{\#}g)$. Hence f^* descends to a homomorphism

$$f^*\colon \operatorname{Cl} Y \to \operatorname{Cl} X$$

Proposition 7.1.9: Finite Map and Degrees

Let $f\colon X \to Y$ be a finite morphism of complete nonsingular curves. For any divisor D on Y:

$$\deg f^*(D) \;=\; (\deg f)\cdot(\deg D)$$

***Proof*:**

By $\mathbb{Z}$-linearity, it suffices to prove $\deg f^*(y) = (\deg f)(\deg y)$ for a single closed point $y \in Y$, where the degree of a closed point is $[\kappa(y) : k]$. Let $V = \operatorname{Spec} S \subseteq Y$ be an affine open neighbourhood of y. Since Y is a nonsingular curve, S is a Dedekind domain. Let R be the integral closure of S in $K(X)$. Then $f^{-1}(V) = \operatorname{Spec} R$ and R is a finitely generated S-module, by Step 3 of the proof of proposition 7.1.6.

Localize at the maximal ideal $\mathfrak{m}_y \subseteq S$. Set $B = S_{\mathfrak{m}_y} = \mathcal{O}_{Y,y}$, which is a DVR with uniformizer t, and let $R' = R \otimes_S B$, the semi-local ring (R' has finitely many maximal ideals) obtained

by localizing R at $\mathfrak{m}_y$. The S-module R is finitely generated and torsion-free, and its rank is $n = [K(X) : K(Y)] = \deg f$, because $R \otimes_S K(Y)$ is a domain finite over the field $K(Y)$, hence a field lying between R and $\operatorname{Frac} R = K(X)$, hence equal to $K(X)$. The localization R' is therefore a free B-module of rank n (torsion-free and finitely generated over a PID).

The maximal ideals of R' correspond to the points $x_1, \dots, x_r \in X$ lying over y, with $(R')_{\mathfrak{m}_i} = \mathcal{O}_{X,x_i}$ for each i, and each of them contains t. The ring R'/tR' is a finite-dimensional algebra over the residue field $\kappa(y) = B/tB$, hence Artinian, and an Artinian ring is the product of its localizations at its maximal ideals. This gives an isomorphism of k-algebras

$$R'/tR' \cong \bigoplus_{i=1}^{r} R'_{\mathfrak{m}_i}/tR'_{\mathfrak{m}_i} = \bigoplus_{i=1}^{r} \mathcal{O}_{X,x_i}/t\,\mathcal{O}_{X,x_i}$$

where t stands for its image $f^{\#}(t)$. Since $R'/tR' \cong \kappa(y)^n$, taking dimensions over k of both sides gives

$$n[\kappa(y) : k] = \dim_k R'/tR' = \sum_{i=1}^{r} \dim_k(\mathcal{O}_{X,x_i}/t\,\mathcal{O}_{X,x_i}) = \sum_{i=1}^{r} \nu_{x_i}(f^{\#}(t))[\kappa(x_i) : k] = \sum_{i=1}^{r} e_{x_i} \deg(x_i)$$

The left-hand side is $(\deg f)(\deg y)$, and the right-hand side is $\deg f^*(y)$ by definition, as claimed.

Over $\mathbb{C}$, this is the statement that a degree-n branched cover of compact Riemann surfaces has n points over each point, counted with multiplicity.

Corollary 7.1.10: Degree Map and Integers

Let X be a complete nonsingular curve. Then principal divisors have degree 0, and the degree map induces a surjective homomorphism

$$\deg\colon \operatorname{Cl} X \twoheadrightarrow \mathbb{Z}$$

***Proof*:**
Let $g \in K(X)^*$ with $g \notin k^*$ (if $g \in k^*$, then $\operatorname{div} g = 0$ and the result is trivial). Since k is algebraically closed, g is transcendental over k, and it defines a finite morphism $\varphi\colon X \to \mathbb{P}^1_k$.[a] With t the coordinate on $\mathbb{P}^1_k$, the function t is a uniformizer at 0 and $1/t$ is a uniformizer at ∞, and $\varphi^{\#}(t) = g$. So the coefficient of a point x in $\varphi^*([0])$ is $\nu_x(g)$ when $\varphi(x) = 0$, its coefficient in $\varphi^*([\infty])$ is $-\nu_x(g)$ when $\varphi(x) = \infty$, and $\nu_x(g) = 0$ at the remaining points. Hence $\operatorname{div} g = \varphi^*([0]) - \varphi^*([\infty])$, where $[0]$ and $[\infty]$ are the divisors of the points 0 and ∞ on $\mathbb{P}^1_k$. By proposition 7.1.9:

$$\deg(\operatorname{div} g) = \deg \varphi^*([0]) - \deg \varphi^*([\infty]) = \deg \varphi - \deg \varphi = 0$$

Thus the degree map $\deg\colon \operatorname{WDiv} X \to \mathbb{Z}$ kills principal divisors and descends to $\operatorname{Cl} X$. It is surjective because a closed point $x \in X$ is a divisor of degree 1, k being algebraically closed, completing the proof.

[a] The function g is regular on a nonempty open $V \subseteq X$, so it defines a morphism $V \to \mathbb{A}^1_k \subseteq \mathbb{P}^1_k$ and hence a rational map $X \dashrightarrow \mathbb{P}^1_k$. Its locus of indeterminacy has codimension at least 2 by theorem 4.7.4, since X is normal and $\mathbb{P}^1_k$ is proper, and X has no points of codimension 2, so the rational map is a morphism φ. It is dominant because g is transcendental over k, and finite by proposition 7.1.6.

The kernel $\mathrm{Cl}^0 X = \ker(\deg\colon \mathrm{Cl}\, X \to \mathbb{Z})$ is the group of divisor classes of degree 0. For an elliptic curve E it is isomorphic to the group $E(k)$ of points (example 7.2.12, theorem 7.6.3). In general, $\mathrm{Cl}^0 X$ is the group of k-points of an abelian variety of dimension equal to the genus of X (definition 7.2.4), the *Jacobian* $\mathrm{Jac}(X)$, constructed in *Abelian Varieties* [5, The Jacobian Is an Abelian Variety].[2]

7.1.11 Arithmetic Motif: Covers of $\operatorname{Spec}\mathbb{Z}$ The inclusion $\mathbb{Z} \hookrightarrow \mathbb{Z}[i]$ of the integers into the Gaussian integers induces a finite morphism of schemes

$$\pi\colon \operatorname{Spec}\mathbb{Z}[i] \longrightarrow \operatorname{Spec}\mathbb{Z}$$

of degree $[\mathbb{Q}(i):\mathbb{Q}] = 2$. Its fibers over the closed points (p) of $\operatorname{Spec}\mathbb{Z}$ show the three behaviours of a prime in a quadratic extension familiar from algebraic number theory:

1. A prime $p \equiv 1 \pmod 4$ *splits*. For $p = 5$, $5 = (2+i)(2-i)$, and the fiber $\pi^{-1}((5))$ consists of the two points $(2+i)$ and $(2-i)$, each with ramification index 1 and residue field $\mathbb{F}_5$.
2. A prime $p \equiv 3 \pmod 4$ is *inert*. For $p = 3$, the ideal (3) remains prime in $\mathbb{Z}[i]$, and the fiber is a single point with ramification index 1 and residue field $\mathbb{F}_9$, of degree 2 over $\mathbb{F}_3$.
3. The prime 2 *ramifies*: $(1+i)^2 = 2i$, so $2 = -i(1+i)^2$, and the fiber is the single point $(1+i)$, with ramification index 2 and residue field $\mathbb{F}_2$.

In each case the sum over the points of the fiber of the ramification index times the residue field degree equals $\deg \pi = 2$, the arithmetic counterpart of proposition 7.1.9. Over an algebraically closed field the residue field degrees are all 1, and only splitting and ramification occur. The ramified primes are detected by the discriminant: a prime ramifies in $\mathbb{Z}[i]$ if and only if it divides the discriminant -4 of $\mathbb{Z}[i]$ [7, Discriminant And Ramified Primes]. See [62] for the arithmetic side of the analogy between number fields and function fields of curves.

7.2 The Riemann-Roch Theorem

For a divisor D on a complete nonsingular curve X of genus g, the Riemann-Roch theorem computes the Euler characteristic $\chi(\mathcal{L}(D)) = \dim_k H^0(X,\mathcal{L}(D)) - \dim_k H^1(X,\mathcal{L}(D))$ as $\deg D + 1 - g$. Serre duality identifies $H^1(X,\mathcal{L}(D))$ with the dual of $H^0(X,\omega_X \otimes \mathcal{L}(D)^\vee)$, so the theorem relates the sections of $\mathcal{L}(D)$ to the sections of $\omega_X \otimes \mathcal{L}(D)^\vee$. When $\deg D > 2g-2$ the second space vanishes and $\dim_k H^0(X,\mathcal{L}(D)) = \deg D + 1 - g$ (proposition 7.3.2). The theorem gives the degree of the canonical divisor, which enters Hurwitz's formula (section 7.4), and the very ample criterion and the embeddings of section 7.3.

7.2.1 Divisors on Curves: Degree, Linear Systems, and $\ell(D)$

Let X be a complete nonsingular curve over $k = \overline{k}$. Since X is a regular scheme of dimension 1, every local ring $\mathcal{O}_{X,x}$ at a closed point is a DVR, hence a UFD. In particular, X is locally factorial, and the

[2] The Jacobian is the identity component of the Picard scheme $\mathrm{Pic}_{X/k}$, treated in *Moduli, Deformation Theory, and Stacks* ([20, Grothendieck's Theorem: the Picard Scheme of a Projective Variety]). When $k = \mathbb{C}$, it is a complex torus $\mathbb{C}^g/\Lambda$ for a lattice Λ of rank $2g$, the period lattice of the curve. For $g = 1$, a choice of base point identifies the Jacobian with the curve itself, and this identification gives the group law on an elliptic curve.

natural map from Cartier divisors to Weil divisors of section 5.4 is an isomorphism (proposition 5.4.33 and corollary 5.4.45):

$$\operatorname{CaCl} X \;\cong\; \operatorname{Cl} X \;\cong\; \operatorname{Pic} X$$

On a singular curve the local ring at a singular point is not a DVR, so the valuation at that point used to define Weil divisors in section 5.4 is not available, and a point need not be a Cartier divisor. At the cusp P of $y^2 = x^3$ the maximal ideal (x, y) of $\mathcal{O}_{X,P}$ is not principal: since $y^2 - x^3$ lies in $(x, y)^2 \subseteq k[x, y]$, the space $\mathfrak{m}_P/\mathfrak{m}_P^2$ has dimension 2, and a principal maximal ideal would give dimension at most 1 (Nakayama). Example 5.4.37 treats the node. On a singular curve, Cartier divisors (equivalently, line bundles, elements of $\operatorname{Pic} X$) remain the objects for which Riemann-Roch is stated (section 7.2.3) and from which morphisms to projective space are built.

Divisors are written additively as $D = \sum_{x \in X_{\mathrm{cl}}} n_x \cdot x$, where all but finitely many n_x are zero. The *degree* of D is $\deg D = \sum n_x$. Each divisor D has an associated invertible sheaf $\mathcal{L}(D)$, and for divisors D, D' (proposition 5.4.43)

$$\mathcal{L}(D + D') \cong \mathcal{L}(D) \otimes \mathcal{L}(D'), \qquad \mathcal{L}(-D) \cong \mathcal{L}(D)^\vee,$$
$$\mathcal{L}(D) \cong \mathcal{O}_X \iff D \sim 0 \quad \text{(linearly equivalent to zero)}$$

Recall that D is *effective*, written $D \geq 0$, if all $n_x \geq 0$. A global section $s \in H^0(X, \mathcal{L}(D))$ corresponds to a rational function $f \in K(X)^*$ with $\operatorname{div}(f) + D \geq 0$, together with the zero section.

Definition 7.2.1: Complete Linear System

The *complete linear system* of D is the set of effective divisors linearly equivalent to D:

$$|D| \;=\; \{D' \geq 0 : D' \sim D\} \;=\; \{\operatorname{div}(f) + D : f \in K(X)^*,\ \operatorname{div}(f) + D \geq 0\}$$

There is a natural bijection $|D| \cong \mathbb{P}\big(H^0(X, \mathcal{L}(D))\big)$: the effective divisors in the class of D are parametrized by the projectivization of the space of global sections of $\mathcal{L}(D)$, since two sections $s, s' \in H^0(X, \mathcal{L}(D)) \setminus \{0\}$ define the same effective divisor if and only if $s' = \lambda s$ for some $\lambda \in k^*$. (If s and s' correspond to f and f' with $\operatorname{div}(f) = \operatorname{div}(f')$, then f'/f has no zeros or poles, so it lies in $H^0(X, \mathcal{O}_X)^* = k^*$ by theorem 3.5.33.)

The classical notation is

$$\ell(D) \;:=\; \dim_k H^0(X, \mathcal{L}(D))$$

so $|D|$ is a projective space of dimension $\ell(D) - 1$ (and $|D| = \emptyset$ precisely when $\ell(D) = 0$). The number $\ell(D)$ is finite by theorem 5.3.28, X being projective (proposition 7.1.7). A *linear system* is any linear subspace of $|D|$, and a linear system of dimension 1 is called a *pencil*.

Lemma 7.2.2: Divisors and Degree

Let D be a divisor on a complete nonsingular curve X.

1. If $\ell(D) > 0$, then $\deg D \geq 0$.
2. If $\ell(D) > 0$ and $\deg D = 0$, then $D \sim 0$ (i.e. $\mathcal{L}(D) \cong \mathcal{O}_X$).

***Proof*:**
If $\ell(D) > 0$, then $|D| \neq \emptyset$, so $D \sim D'$ for some effective $D' \geq 0$. Linearly equivalent divisors have the same degree (corollary 7.1.10), so $\deg D = \deg D' = \sum n_x \geq 0$ (with $n_x \geq 0$ for all x), which is (1). For (2), if additionally $\deg D = 0$, then $D' = \sum n_x \cdot x$ with all $n_x \geq 0$ and $\sum n_x = 0$, forcing $n_x = 0$ for all x, i.e. $D' = 0$, so $D \sim 0$, as claimed.

7.2.2 The Canonical Divisor and Genus

Recall from section 5.5 that the sheaf of relative differentials $\Omega_{X/k}$ on a nonsingular variety X of dimension n over $k = \overline{k}$ is locally free of rank n (proposition 5.5.25). When X is a nonsingular curve, $n = 1$, so $\Omega_{X/k}$ is an invertible sheaf.

By the definition of the canonical sheaf (definition 5.5.28), $\omega_X = \bigwedge^{\dim X} \Omega_{X/k}$, which for a curve is $\omega_X = \Omega_{X/k}$. For a nonsingular projective variety the dualizing sheaf of Serre duality (definition 6.5.1) is $\det \Omega_{X/k}$ (theorem 6.5.9(1)), so on a nonsingular curve the canonical sheaf and the dualizing sheaf are the same invertible sheaf $\Omega_{X/k}$, and Serre duality applies to it directly.

Definition 7.2.3: Canonical Divisor

Let X be a complete nonsingular curve. A *canonical divisor* K (or K_X) is any divisor in the linear equivalence class corresponding to $\omega_X = \Omega_{X/k}$, i.e. $\mathcal{L}(K) \cong \omega_X$.

The canonical divisor is defined only up to linear equivalence, but its degree and the dimension $\ell(K)$ are well-defined invariants of X.

Definition 7.2.4: Genus of a Curve

Let X be a complete nonsingular curve over k, which is projective by proposition 7.1.7. The following three numbers are defined:

1. The *arithmetic genus*: $p_a(X) = 1 - P_X(0)$, where P_X is the Hilbert polynomial of X with respect to any very ample sheaf (definition 6.7.6). The constant term $P_X(0) = \chi(\mathcal{O}_X)$ does not depend on that choice.
2. The *geometric genus*: $p_g(X) = \dim_k H^0(X, \omega_X)$ (definition 5.5.29).
3. The *cohomological genus*: $\dim_k H^1(X, \mathcal{O}_X)$.

By proposition 7.2.5 these three numbers coincide, and their common value is called the *genus* of X, denoted g (or $g(X)$).

Proposition 7.2.5: Equivalence of Genus

Let X be a complete nonsingular curve over $k = \overline{k}$. Then:

$$p_a(X) \;=\; p_g(X) \;=\; \dim_k H^1(X, \mathcal{O}_X)$$

***Proof*:**
Step 1: $p_a = \dim_k H^1(X, \mathcal{O}_X)$. The curve X is projective (proposition 7.1.7). By definition, $p_a(X) = 1 - P_X(0)$ where $P_X(n) = \chi(\mathcal{O}_X(n))$ is the Hilbert polynomial (theorem 6.7.5 and definition 6.7.6). Since $\dim X = 1$, the groups $H^i(X, \mathcal{O}_X)$ vanish for $i \geq 2$ (theorem 6.2.5), so setting $n = 0$ gives

$$P_X(0) = \chi(\mathcal{O}_X) = \dim_k H^0(X, \mathcal{O}_X) - \dim_k H^1(X, \mathcal{O}_X)$$

Since X is a projective integral scheme over $k = \overline{k}$, the global sections are $H^0(X, \mathcal{O}_X) = k$ (theorem 3.5.33), so $P_X(0) = 1 - \dim_k H^1(X, \mathcal{O}_X)$, and $p_a = 1 - P_X(0) = \dim_k H^1(X, \mathcal{O}_X)$.

Step 2: $\dim_k H^1(X, \mathcal{O}_X) = p_g$. Serre duality for a nonsingular projective variety X of dimension n over k gives, for every locally free coherent sheaf $\mathscr{F}$, (corollary 6.5.10):

$$H^i(X, \mathscr{F}) \;\cong\; H^{n-i}(X, \mathscr{F}^\vee \otimes \omega_X)^\vee$$

Taking $n = 1$, $i = 1$, $\mathscr{F} = \mathcal{O}_X$:

$$H^1(X, \mathcal{O}_X) \;\cong\; H^0(X, \omega_X)^\vee$$

Therefore $\dim_k H^1(X, \mathcal{O}_X) = \dim_k H^0(X, \omega_X) = p_g(X)$, as we sought to show.

Write $g = g(X)$ for the genus. Being the dimension $\dim_k H^1(X, \mathcal{O}_X)$, it is a nonnegative integer.

The genus of a smooth plane curve depends only on its degree. In *Classical Algebraic Geometry* the formula below first defines the genus of a plane curve and is then shown to agree with the genus of Riemann's theorem [22, Genus of a Nonsingular Plane Curve]. Here it is a theorem about $\dim_k H^1(C, \mathcal{O}_C)$.

Proposition 7.2.6: Degree Formula for Plane Curves

Let $C = V(F) \subseteq \mathbb{P}^2_k$ be a smooth projective curve cut out by a homogeneous polynomial F of degree d. Then the genus of C is

$$g(C) \;=\; \frac{(d-1)(d-2)}{2}$$

***Proof*:**
Since C is the closed subscheme of $\mathbb{P}^2$ defined by the single homogeneous polynomial F of degree d, its ideal sheaf is $\mathcal{I}_C \cong \mathcal{O}_{\mathbb{P}^2}(-d)$. This gives the short exact sequence of sheaves on $\mathbb{P}^2$

$$0 \longrightarrow \mathcal{O}_{\mathbb{P}^2}(-d) \longrightarrow \mathcal{O}_{\mathbb{P}^2} \longrightarrow \mathcal{O}_C \longrightarrow 0$$

where $\mathcal{O}_C$ stands for its pushforward to $\mathbb{P}^2$, which has the same cohomology (proposition 6.2.4). The Euler characteristic $\chi(\mathscr{F}) = \sum_i (-1)^i \dim_k H^i(\mathbb{P}^2, \mathscr{F})$ is additive on short exact sequences (proposition 6.7.2), so

$$\chi(\mathcal{O}_C) \;=\; \chi(\mathcal{O}_{\mathbb{P}^2}) - \chi(\mathcal{O}_{\mathbb{P}^2}(-d))$$

From the cohomology of projective space (theorem 6.3.1), the Hilbert polynomial of $\mathbb{P}^2$ is

$\chi(\mathcal{O}_{\mathbb{P}^2}(n)) = \frac{(n+1)(n+2)}{2}$ for all $n \in \mathbb{Z}$. Evaluating this at $n = 0$ and $n = -d$ yields

$$\chi(\mathcal{O}_{\mathbb{P}^2}) = 1,$$
$$\chi(\mathcal{O}_{\mathbb{P}^2}(-d)) = \frac{(-d+1)(-d+2)}{2} = \frac{(d-1)(d-2)}{2}$$

Therefore, $\chi(\mathcal{O}_C) = 1 - \frac{(d-1)(d-2)}{2}$.

On the other hand, since C is a complete nonsingular curve, proposition 7.2.5 establishes that the genus g equals the arithmetic genus $p_a(C) = 1 - \chi(\mathcal{O}_C)$. Thus

$$1 - g = 1 - \frac{(d-1)(d-2)}{2} \quad \Longrightarrow \quad g = \frac{(d-1)(d-2)}{2}$$

as claimed.

The computation of $\chi(\mathcal{O}_C)$ uses only that C is cut out by a form of degree d, so every plane curve of degree d, singular or not, has arithmetic genus $(d-1)(d-2)/2$. The hypothesis $C \subseteq \mathbb{P}^2_k$ is needed for the genus: the twisted cubic in $\mathbb{P}^3_k$ has degree 3 and is isomorphic to $\mathbb{P}^1_k$, so it has genus 0, while a smooth plane cubic has genus 1. For a curve of degree d in $\mathbb{P}^r_k$ not contained in a hyperplane, Castelnuovo's bound (theorem 7.7.1) is an upper bound for the genus, and for $r = 2$ it is the formula of proposition 7.2.6.

Example 7.2.7: Genus Computations

1. *The projective line.* $g(\mathbb{P}^1_k) = 0$. Indeed, $H^1(\mathbb{P}^1_k, \mathcal{O}_{\mathbb{P}^1}) = 0$ by the computation of cohomology of projective space (theorem 6.3.1).

2. *Smooth plane curves.* Let $C \subseteq \mathbb{P}^2_k$ be a smooth curve of degree d. By the degree formula (proposition 7.2.6):
$$g(C) = \frac{(d-1)(d-2)}{2}$$
This gives $g = 0$ for $d = 1, 2$ (lines and conics, both isomorphic to $\mathbb{P}^1_k$), $g = 1$ for $d = 3$ (smooth plane cubics), $g = 3$ for $d = 4$, and so on. The genus of a smooth plane curve of degree d grows quadratically in d.

   The degree of the canonical divisor of a plane curve can also be computed from the adjunction formula (proposition 6.5.7), which gives $\omega_C \cong (\omega_{\mathbb{P}^2} \otimes \mathcal{O}_{\mathbb{P}^2}(C))|_C \cong \mathcal{O}_C(d-3)$. Twisting the exact sequence in the proof of proposition 7.2.6 by $\mathcal{O}_{\mathbb{P}^2}(n)$ gives $\chi(\mathcal{O}_C(n)) = \frac{(n+1)(n+2)}{2} - \frac{(n-d+1)(n-d+2)}{2} = dn + 1 - g$, and comparing with the Riemann-Roch theorem proved below (theorem 7.2.10) shows that $\mathcal{O}_C(n)$ has degree dn. So ω_C has degree $d(d-3)$, which agrees with $2g - 2 = 2\left(\frac{d^2-3d+2}{2}\right) - 2 = d(d-3)$, the case of plane curves of the general formula $\deg K_X = 2g - 2$ of item (3).

3. *The degree of the canonical divisor.* Every complete nonsingular curve of genus g satisfies (proposition 7.3.1, deduced from theorem 7.2.10 below)
$$\deg K_X = 2g - 2$$

For $\mathbb{P}^1$ this gives $\deg K = -2$, consistent with $\omega_{\mathbb{P}^1} \cong \mathcal{O}_{\mathbb{P}^1}(-2)$ (from the Euler sequence, theorem 5.5.19). For a smooth plane cubic, $\deg K = 0$, consistent with $\omega_C \cong \mathcal{O}_C(3-3) = \mathcal{O}_C$ from item (2).

7.2.8 Forward Reference: The Moduli Space $\mathcal{M}_g$ The genus is a discrete invariant, and for $g \geq 1$ it does not determine a curve up to isomorphism: the smooth projective curves of a fixed genus $g \geq 1$ form a continuous family. The coarse moduli space $\mathcal{M}_g$ parametrizing them has dimension

$$\dim \mathcal{M}_g = \begin{cases} 0 & \text{if } g = 0 \quad (\text{only } \mathbb{P}^1), \\ 1 & \text{if } g = 1 \quad (\text{the } j\text{-line}), \\ 3g-3 & \text{if } g \geq 2 \end{cases}$$

and the deformation-theoretic computation behind these numbers is sketched in box 7.7.4. Some curves of every genus have nontrivial automorphisms ($\mathbb{P}^1$, the involution $[-1]$ of an elliptic curve, the hyperelliptic involution), so no scheme represents the moduli functor of smooth curves of genus g. For $g \geq 2$ the moduli problem is a Deligne-Mumford stack, and its construction is carried out in *Moduli, Deformation Theory, and Stacks* [20].

7.2.9 Warning: The Genera Diverge for Singular Curves The equality $p_a = p_g = \dim_k H^1(X, \mathcal{O}_X)$ of proposition 7.2.5 uses nonsingularity. For an integral projective curve X with singular points, the arithmetic genus and the genus of the normalization differ.

The *geometric genus* of X is the genus $g(\widetilde{X})$ of its normalization $\nu\colon \widetilde{X} \to X$, and it depends only on the function field $K(X)$. The *arithmetic genus* $p_a(X) = 1 - \chi(\mathcal{O}_X)$ depends on the singularities, and it is constant in a flat projective family over a connected base, because the Euler characteristic of the fibers is locally constant (theorem 6.8.17(2)). They are related by the *δ-invariant*: for each singular point $P \in X$, set

$$\delta_P \;=\; \dim_k\Big(\widetilde{\mathcal{O}}_{X,P} \,/\, \mathcal{O}_{X,P}\Big)$$

where $\widetilde{\mathcal{O}}_{X,P}$ denotes the integral closure of $\mathcal{O}_{X,P}$ in $K(X)$ (equivalently, the stalk $(\nu_*\mathcal{O}_{\widetilde{X}})_P$ of the pushforward of the structure sheaf along the normalization map ν). Then [36, Exercise IV.1.8]

$$p_a(X) \;=\; g(\widetilde{X}) \;+\; \sum_{P \in X_{\mathrm{sing}}} \delta_P$$

A node (two branches crossing transversally) has $\delta = 1$, and an ordinary cusp has $\delta = 1$ as well.[a]

For example, the nodal cubic $y^2z = x^3 + x^2z$ (with $\mathrm{char}k \neq 2$) and the cuspidal cubic $y^2z = x^3$ are plane cubics, so $p_a = \frac{(3-1)(3-2)}{2} = 1$ by the computation following proposition 7.2.6. Both have normalization $\mathbb{P}^1$, giving $g(\widetilde{X}) = 0$. The difference $p_a - g(\widetilde{X}) = 1$ is the δ-invariant of the single singular point.

The difference appears in families: when a smooth plane cubic ($g = 1$) degenerates in a flat family to a nodal cubic, the arithmetic genus stays equal to 1 by theorem 6.8.17(2), while the geometric genus drops to 0. For this reason the compactification $\overline{\mathcal{M}}_g$ of the moduli space of curves, for $g \geq 2$, parametrizes *stable* curves, the nodal curves of arithmetic genus g with finitely many automorphisms (*Moduli, Deformation Theory, and Stacks* [20, Stable Curve]).

[a] Larger values occur: the curve $y^2 = x^{2m+1}$ has $\delta = m$ at the origin, since its coordinate ring is $k[t^2, t^{2m+1}]$, which misses exactly $t, t^3, \ldots, t^{2m-1}$ from its normalization $k[t]$. Over $\mathbb{C}$, the space X near P is obtained from $\widetilde{X}$ by identifying the r_P points over P, one for each branch through P, and $\delta_P \geq r_P - 1$, with strict inequality at a cusp, where $r_P = 1$ and $\delta_P = 1$.

7.2.3 Statement and Proof of Riemann-Roch

Recall that Serre duality (corollary 6.5.10) gives, for any invertible sheaf $\mathcal{L}$ on a complete nonsingular curve X of genus g, which is projective by proposition 7.1.7,

$$H^1(X, \mathcal{L}) \;\cong\; H^0(X, \omega_X \otimes \mathcal{L}^\vee)^\vee$$

In divisor language, if $\mathcal{L} = \mathcal{L}(D)$, then $\omega_X \otimes \mathcal{L}(D)^\vee \cong \mathcal{L}(K - D)$, and hence

$$\dim_k H^1(X, \mathcal{L}(D)) \;=\; \ell(K - D)$$

So the Euler characteristic $\chi(\mathcal{L}(D)) = \ell(D) - \ell(K - D)$ is a difference of dimensions of spaces of global sections.

Theorem 7.2.10: Riemann-Roch Theorem over an Algebraically Closed Field

Let X be a complete nonsingular curve of genus g over $k = \overline{k}$, and let K be a canonical divisor. For any divisor D on X:

$$\ell(D) - \ell(K - D) \;=\; \deg D + 1 - g$$

Equivalently, in sheaf-cohomological language:

$$\chi(\mathcal{L}(D)) \;=\; \deg D + 1 - g$$

The right-hand side depends only on the degree of D and the genus of X, and over $\mathbb{C}$ the genus is a topological invariant, the number of handles of the Riemann surface (box 7.1.5). The two terms on the left are not determined by $\deg D$ and g: $\ell(D)$ and $\ell(K - D)$ are dimensions of spaces of rational functions with poles bounded by D and by $K - D$, and they depend on D and on the curve. Only their difference is fixed by the theorem.

The term $\ell(K - D)$ is called the *index of speciality* of D and is denoted $i(D) := \ell(K - D)$. A divisor with $i(D) = 0$ is called *non-special*, and for such divisors Riemann-Roch reads $\ell(D) = \deg D + 1 - g$. Every divisor of degree greater than $2g - 2$ is non-special (proposition 7.3.2).

***Proof*:**

We prove the cohomological form $\chi(\mathcal{L}(D)) = \deg D + 1 - g$, from which the classical form follows by Serre duality as explained above.

Base case: $D = 0$. The groups $H^i(X, \mathcal{O}_X)$ vanish for $i \geq 2$ (theorem 6.2.5), so $\chi(\mathcal{O}_X) = \dim_k H^0(X, \mathcal{O}_X) - \dim_k H^1(X, \mathcal{O}_X) = 1 - g$, since $H^0(X, \mathcal{O}_X) = k$ (theorem 3.5.33) and $g = \dim_k H^1(X, \mathcal{O}_X)$ (proposition 7.2.5). This equals $\deg 0 + 1 - g = 1 - g$.

Induction step. We show: the formula holds for D if and only if it holds for $D + P$, where P is any closed point of X.

Consider P as a closed subscheme of X, defined by the ideal sheaf $\mathcal{L}(-P)$ (a single effective Cartier divisor). The structure sheaf $\mathcal{O}_P$ is a skyscraper sheaf supported at P with stalk $\kappa(P) = k$, which we denote k_P. This gives the short exact sequence

$$0 \to \mathcal{L}(-P) \to \mathcal{O}_X \to k_P \to 0$$

Tensoring with the invertible sheaf $\mathcal{L}(D + P)$, which is locally free of rank 1, hence flat, preserves exactness:

$$0 \to \mathcal{L}(-P) \otimes \mathcal{L}(D + P) \to \mathcal{L}(D + P) \to k_P \otimes \mathcal{L}(D + P) \to 0$$

Now $\mathcal{L}(-P) \otimes \mathcal{L}(D + P) \cong \mathcal{L}(D)$, and $k_P \otimes \mathcal{L}(D + P) \cong k_P$, because the invertible sheaf $\mathcal{L}(D + P)$ is isomorphic to $\mathcal{O}_X$ on a neighbourhood of P. So we have

$$0 \to \mathcal{L}(D) \to \mathcal{L}(D + P) \to k_P \to 0 \tag{7.2}$$

Taking Euler characteristics and using additivity (proposition 6.7.2):

$$\chi(\mathcal{L}(D + P)) = \chi(\mathcal{L}(D)) + \chi(k_P)$$

The sheaf k_P is the pushforward of the sheaf k on the one-point space P, so $H^0(X, k_P) = k$ and $H^i(X, k_P) = H^i(P, k) = 0$ for $i > 0$ (proposition 6.2.4 and theorem 6.2.5), and $\chi(k_P) = 1$.

Therefore

$$\chi(\mathcal{L}(D+P)) \;=\; \chi(\mathcal{L}(D)) + 1$$

On the other hand, $\deg(D+P) = \deg D + 1$. So the right-hand side of the Riemann-Roch formula also increases by 1 when D is replaced by $D+P$. This proves the claim: Riemann-Roch holds for D if and only if it holds for $D+P$.

Every divisor D can be reached from the zero divisor by finitely many steps of adding or subtracting a point[a]. By the induction step, the formula holds for D if and only if it holds for 0. Since we verified it for $D = 0$, it holds for all D, as we sought to show.

[a] Write $D = \sum n_i P_i = \sum_{n_i>0} n_i P_i - \sum_{n_j<0} |n_j| P_j$. Starting from 0, add the positive points and subtract the negative points one at a time.

The proof of the cohomological form uses only the additivity of the Euler characteristic (proposition 6.7.2) and the value $\chi(\mathcal{O}_X) = 1 - g$. It shows that $D \mapsto \chi(\mathcal{L}(D)) - \chi(\mathcal{O}_X)$ is additive in D and takes the value 1 on every point, hence equals $\deg D$. Serre duality enters in two places: in identifying the genus $\dim_k H^1(X, \mathcal{O}_X)$ with $p_g = \dim_k H^0(X, \omega_X)$ (proposition 7.2.5), and in rewriting $\chi(\mathcal{L}(D))$ as $\ell(D) - \ell(K - D)$.

7.2.11 Riemann-Roch as an Index Theorem Over $\mathbb{C}$, Riemann-Roch is an index theorem: it equates the index of a differential operator, the dimension of its kernel minus the dimension of its cokernel, with a topological quantity. By the Dolbeault isomorphism and GAGA, $\chi(\mathcal{L}) = h^0 - h^1$ is the index of the $\overline{\partial}$-operator (the Cauchy-Riemann operator) on the line bundle $\mathcal{L}$, and Riemann-Roch computes it from the degree of $\mathcal{L}$ and the genus of the curve. Two generalizations are proved in *Intersection Theory* [19, Hirzebruch-Riemann-Roch, Grothendieck-Riemann-Roch]:

1. The *Hirzebruch-Riemann-Roch theorem* computes $\chi(\mathscr{E})$ for any vector bundle $\mathscr{E}$ on a smooth projective variety X in terms of the Chern character $\mathrm{ch}(\mathscr{E})$ and the Todd class $\mathrm{td}(T_X)$: $\chi(\mathscr{E}) = \int_X \mathrm{ch}(\mathscr{E}) \cdot \mathrm{td}(T_X)$.
2. The *Grothendieck-Riemann-Roch theorem* is the relative version: for a proper morphism $f\colon X \to Y$ of smooth quasi-projective varieties, it computes the Chern character of the alternating sum $\sum_i (-1)^i [R^i f_* \mathscr{E}]$ of the higher direct images in terms of $\mathrm{ch}(\mathscr{E})$, $\mathrm{td}(T_X)$ and $\mathrm{td}(T_Y)$.

For a curve, $\mathrm{ch}(\mathcal{L}(D)) = 1 + c_1(\mathcal{L}(D))$ and $\mathrm{td}(T_X) = 1 + \frac{1}{2}c_1(T_X)$, whose degree-one parts have degrees $\deg D$ and $\frac{1}{2}(2-2g) = 1 - g$ (proposition 7.3.1), so Hirzebruch-Riemann-Roch reduces to $\chi(\mathcal{L}(D)) = \deg D + 1 - g$.

Example 7.2.12: Points of a Plane Cubic as Divisor Classes

Let $E \subseteq \mathbb{P}^2_k$ be the nonsingular cubic $y^2 z = x^3 - xz^2$ with $\mathrm{char}(k) \neq 2$. This curve is not rational: it has genus 1 by proposition 7.2.6, while $\mathbb{P}^1_k$ has genus 0, and two birational complete nonsingular curves are isomorphic (corollary 3.6.6, theorem 7.1.4(3)).[a] Let $P_0 = [0:1:0]$, the unique point at infinity, which is an inflection point: the tangent line $\{z = 0\}$ meets E with multiplicity 3, giving the divisor relation $3P_0 \sim$ (hyperplane section). Since $\omega_E \cong \mathcal{O}_E$ (example 7.2.7), a canonical divisor is $K = 0$, so every divisor of degree 1 is non-special: for $\deg D = 1$, $\deg(K - D) = -1$ and $\ell(K - D) = 0$ by lemma 7.2.2, and Riemann-Roch gives $\ell(D) = 1 + 1 - 1 = 1$.

Define a map φ from the set of k-rational points $E(k)$ to $\mathrm{Cl}^0 E = \ker(\deg\colon \mathrm{Cl}\, E \to \mathbb{Z})$ by $\varphi(P) = [P - P_0]$. This map is *injective*: if $[P - P_0] = [Q - P_0]$, then $Q \sim P$, so $Q \in |P|$. Since $\ell(P) = 1$, the complete linear system $|P|$ has dimension $\ell(P) - 1 = 0$, meaning it consists of exactly one effective divisor, namely P itself. Thus $P = Q$.

To see *surjectivity*, take any $D \in \mathrm{Cl}^0 E$, so $\deg D = 0$. Then $\deg(D + P_0) = 1$, so $\ell(D + P_0) = 1$, and there exists a unique effective divisor of degree 1, which must be a single k-rational point P, such that $D + P_0 \sim P$, meaning $D \sim P - P_0$.

The chord-tangent construction realizes the group law on points. If a line through P and Q meets E in a third point R, then $P + Q + R \sim 3P_0$, so $[P - P_0] + [Q - P_0] = [T - P_0]$ where T is the third intersection of the line P_0R with E. Applied repeatedly, this reduces any sum $\sum n_i(P_i - P_0)$ to a single class $[P - P_0]$ with $P \in E(k)$, together with $-[P - P_0] = [P' - P_0]$, where P' is the third intersection of the line P_0P with E.

Thus φ is a bijection, and the group structure on $\mathrm{Cl}^0 E$ transfers to a group structure on $E(k)$, with identity P_0. The addition and inversion maps $+\colon E \times E \to E$ and $[-1]\colon E \to E$ are morphisms of schemes (theorem 7.6.3), making E a *group scheme* over k (example 4.1.12). Section 7.6 develops this group law.

For any complete nonsingular curve X, the group $\mathrm{Cl}^0 X$ is the group of k-points of the Jacobian variety $\mathrm{Jac}(X)$, and $\mathrm{Cl}\, X$ sits in a short exact sequence

$$0 \to \mathrm{Jac}(X)(k) \to \mathrm{Cl}\, X \xrightarrow{\deg} \mathbb{Z} \to 0$$

For surfaces and higher-dimensional varieties, the quotient of the divisor class group by the classes algebraically equivalent to zero is the *Néron-Severi group* $\mathrm{NS}(X)$, a finitely generated abelian group that can have rank greater than 1. It is defined in definition 8.1.4 and used throughout chapter 8.

[a] The classical proof is [22, Nonsingular Plane Cubics Are Not Rational].

Riemann-Roch extends to singular curves once $\Omega_{X/k}$ is replaced by the dualizing sheaf. On a nonsingular curve, $\Omega_{X/k}$ is an invertible sheaf (proposition 5.5.25) and is the dualizing sheaf for Serre duality (theorem 6.5.9(1)). On an integral curve X over $k = \overline{k}$, the sheaf $\Omega_{X/k}$ is not locally free of rank 1 at a singular point (proposition 5.5.25), and it can have torsion there.[3] The dualizing sheaf of definition 6.5.1 takes over the role of $\Omega_{X/k}$.

Let X be an integral projective curve over $k = \overline{k}$. Its local rings are one-dimensional domains, hence Cohen-Macaulay, so theorem 6.5.9(1) provides a dualizing sheaf ω_X°, and

$$H^1(X, \mathscr{F}) \;\cong\; \mathrm{Hom}_{\mathcal{O}_X}(\mathscr{F}, \omega_X^\circ)^\vee$$

for every coherent sheaf $\mathscr{F}$ on X (definition 6.5.1). This sheaf has an explicit description due to

[3] On the cuspidal cubic $X = \operatorname{Spec} A$, $A = k[x,y]/(y^2 - x^3)$, the module of differentials is $\Omega_{A/k} = (A\,dx \oplus A\,dy)/A\,(3x^2\,dx - 2y\,dy)$. Consider the differential $\tau = 2x\,dy - 3y\,dx$. It is nonzero: writing $x = t^2$, $y = t^3$ identifies A with $k[t^2, t^3]$, and $\tau = a\,(3x^2\,dx - 2y\,dy)$ in $A\,dx \oplus A\,dy$ would force $a = -t^{-1} \notin A$. But $y \cdot \tau = 2xy\,dy - 3y^2\,dx = x(3x^2\,dx) - 3x^3\,dx = 0$ in $\Omega_{A/k}$, and $x^2 \cdot \tau = -y\,(3x^2\,dx - 2y\,dy) = 0$ as well. Thus τ generates a torsion submodule supported at the cusp $(0,0)$, so $\Omega_{A/k}$ is not locally free there.

Rosenlicht. If $\nu\colon \widetilde{X} \to X$ is the normalization, a rational differential $\eta \in \Omega_{K(X)/k}$ is a section of ω_X° over an open $U \subseteq X$ exactly when $\sum_{Q\in\nu^{-1}(P)} \operatorname{Res}_Q(f\eta) = 0$ for every closed point $P \in U$ and every $f \in \mathcal{O}_{X,P}$ [64, Chapter IV]. At a nonsingular point P the condition says that η is regular at P, so $\omega_X^\circ = \Omega_{X/k}$ on a nonsingular curve. On a singular curve, ω_X° is a torsion-free sheaf of rank 1, a subsheaf of the constant sheaf $\Omega_{K(X)/k}$, and it need not be invertible.

Riemann-Roch for an integral projective curve X, possibly singular, reads

$$\chi(\mathcal{L}) \;=\; \deg \mathcal{L} + 1 - p_a(X)$$

for every invertible sheaf $\mathcal{L}$ on X, where $p_a(X) = 1 - \chi(\mathcal{O}_X)$ is the arithmetic genus. Here $\deg \mathcal{L}$ is the degree of a divisor D supported on the nonsingular points of X with $\mathcal{L} \cong \mathcal{L}(D)$. Every invertible sheaf has this form, and the proof of theorem 7.2.10 applies verbatim to such D, since each nonsingular point is an effective Cartier divisor [36, Exercise IV.1.9]. Serre duality then gives $\dim_k H^0(X,\mathcal{L}) - \dim_k \operatorname{Hom}_{\mathcal{O}_X}(\mathcal{L},\omega_X^\circ) = \deg \mathcal{L} + 1 - p_a(X)$ [47, Theorem 7.3.26]. The genus in the formula is the arithmetic genus p_a, not the geometric genus $g(\widetilde{X})$. The duality term is computed with ω_X°, and it equals $\ell(K-D)$ for a divisor K only when $\omega_X^\circ \cong \mathcal{L}(K)$ is invertible.

7.3 Applications of Riemann-Roch

The criteria of this section compare $\deg D$ with the degree $2g-2$ of the canonical divisor (proposition 7.3.1). When $\deg D > 2g-2$, Riemann-Roch computes $\ell(D)$, and it shows that $|D|$ is base-point-free once $\deg D \geq 2g$ and very ample once $\deg D \geq 2g+1$. When $\deg D \leq 2g-2$, it bounds $\ell(D)$ only from below, and Clifford's theorem (theorem 7.3.10) bounds $\ell(D)$ from above for effective special divisors.

Proposition 7.3.1: Degree of the Canonical Divisor

Let X be a complete nonsingular curve of genus g. Then

$$\deg K_X \;=\; 2g-2, \qquad \ell(K_X) = g$$

***Proof*:**
Apply Riemann-Roch (theorem 7.2.10) with $D = K$:

$$\ell(K) - \ell(K-K) = \deg K + 1 - g$$

Now $\ell(K-K) = \ell(0) = \dim_k H^0(X,\mathcal{O}_X) = 1$ (theorem 3.5.33), and $\ell(K) = p_g = g$ by proposition 7.2.5. So $g - 1 = \deg K + 1 - g$, giving $\deg K = 2g-2$, as claimed.

Over $\mathbb{C}$, the compact Riemann surface $X(\mathbb{C})$ has topological Euler characteristic $\chi_{\text{top}} = 2-2g$, so $\deg K_X = -\chi_{\text{top}}$. Equivalently, the holomorphic tangent bundle $\omega_X^\vee$ has degree χ_{top}, which is the Gauss-Bonnet theorem for $X(\mathbb{C})$ stated through the first Chern class of its tangent bundle.

Proposition 7.3.2: Non-Speciality for Large Degree

Let D be a divisor on a curve X of genus g. If $\deg D > 2g-2$, then $\ell(K-D)=0$, and Riemann-Roch gives

$$\ell(D) \;=\; \deg D + 1 - g$$

For every divisor D, $\ell(D) \geq \deg D + 1 - g$, so $\ell(D) > 0$ whenever $\deg D \geq g$.

***Proof*:**
If $\ell(K-D) > 0$, then $\deg(K-D) \geq 0$ by lemma 7.2.2, that is, $\deg D \leq \deg K = 2g-2$ (proposition 7.3.1). Hence $\deg D > 2g-2$ forces $\ell(K-D)=0$, and theorem 7.2.10 reads $\ell(D) = \deg D + 1 - g$. For an arbitrary divisor D, theorem 7.2.10 and $\ell(K-D) \geq 0$ give $\ell(D) = \deg D + 1 - g + \ell(K-D) \geq \deg D + 1 - g$, which is at least 1 when $\deg D \geq g$, completing the proof.

Riemann-Roch decides when a complete linear system embeds a curve in projective space. Recall (definitions 5.3.22 and 5.3.24) that an invertible sheaf $\mathcal{L}$ on X is:

1. *generated by global sections* (or *base-point-free*) if the natural map $H^0(X,\mathcal{L}) \otimes_k \mathcal{O}_X \to \mathcal{L}$ is surjective,
2. *very ample* if $\mathcal{L} \cong \iota^*\mathcal{O}(1)$ for some closed immersion $\iota\colon X \hookrightarrow \mathbb{P}^N_k$.

A base-point-free $\mathcal{L}$ defines a morphism $\varphi_{\mathcal{L}}\colon X \to \mathbb{P}^N_k$ with $N+1 = \dim_k H^0(X,\mathcal{L})$ (theorem 5.3.43), and for $\mathcal{L} = \mathcal{L}(D)$ this is the morphism $\varphi_{|D|}$ of the complete linear system $|D|$. On a complete nonsingular curve both conditions are numerical [36, Proposition IV.3.1]. The sheaf $\mathcal{L}(D)$ is base-point-free if and only if $\ell(D-P) = \ell(D) - 1$ for every closed point P, since $H^0(X,\mathcal{L}(D-P))$ is the subspace of sections of $\mathcal{L}(D)$ vanishing at P. It is very ample if and only if $\ell(D-P-Q) = \ell(D) - 2$ for every pair of closed points $P, Q \in X$, equal or not. For $P \neq Q$ this condition says that the sections of $\mathcal{L}(D)$ separate P and Q, and for $P = Q$ that they separate tangent directions at P, which are the two conditions under which $\varphi_{|D|}$ is a closed immersion [36, Proposition II.7.3].

Proposition 7.3.3: Base-Point-Free Criterion for Curves

Let D be a divisor on a complete nonsingular curve X of genus g. If $\deg D \geq 2g$, then $\mathcal{L}(D)$ is base-point-free.

***Proof*:**
Let $P \in X$ be any closed point. We need $\ell(D) > \ell(D-P)$, i.e. a section of $\mathcal{L}(D)$ not vanishing at P. By Riemann-Roch,

$$\ell(D) = \deg D + 1 - g + \ell(K-D),$$
$$\ell(D-P) = \deg D - 1 + 1 - g + \ell(K-D+P)$$

so $\ell(D) - \ell(D-P) = 1 + \ell(K-D) - \ell(K-D+P)$. Since $\deg D \geq 2g$, we have $\deg(K-D) = 2g-2-\deg D \leq -2$ and $\deg(K-D+P) = 2g-1-\deg D \leq -1$, so $\ell(K-D) = \ell(K-D+P) = 0$ by lemma 7.2.2. Therefore $\ell(D) - \ell(D-P) = 1$ for every closed point P, and $\mathcal{L}(D)$ is base-point-free, as claimed.

Corollary 7.3.4: Very Ample Criterion for Curves

If $\deg D \geq 2g+1$, then $\mathcal{L}(D)$ is very ample, and $\varphi_{|D|}$ embeds X as a closed subvariety of $\mathbb{P}_k^{\ell(D)-1}$.

***Proof*:**
Let $P, Q \in X$ be closed points, possibly equal. Riemann-Roch applied to D and to $D-P-Q$ gives $\ell(D)-\ell(D-P-Q) = 2+\ell(K-D)-\ell(K-D+P+Q)$. Since $\deg D \geq 2g+1$, we have $\deg(K-D+P+Q) \leq (2g-2)-(2g+1)+2 = -1$ and $\deg(K-D) \leq -3$, so both terms vanish by lemma 7.2.2. Hence $\ell(D-P-Q) = \ell(D)-2$ for all P and Q, and $\mathcal{L}(D)$ is very ample by the criterion above, as claimed.

Both bounds are sharp once $g \geq 1$. For a closed point P, Riemann-Roch gives $\ell(K+P) = g = \ell(K)$, as $\ell(-P) = 0$, so the divisor $K+P$ of degree $2g-1$ has the base point P. For closed points P and Q, the divisor $K+P+Q$ has degree $2g$ and $\ell(K+P+Q) = g+1$, so it is base-point-free by proposition 7.3.3, but $\ell((K+P+Q)-P-Q) = \ell(K) = \ell(K+P+Q)-1$, so it is not very ample. On a hyperelliptic curve of genus g the divisor $g(P+Q)$, for $P+Q$ a member of the degree-2 pencil, is linearly equivalent to $K+P+Q$ (theorem 7.3.8(3) below). Hyperelliptic curves are the subject of section 7.5.

Three descriptions of a nonsingular curve agree: as a projective scheme, as a proper scheme, and through the discrete valuation rings of its function field (corollary 7.3.5). Properness is the valuative condition of theorem 3.5.22 and says nothing about equations, while projectivity is the existence of a closed immersion into a projective space, that is, of equations. The passage from properness to projectivity is proposition 7.1.7.

Corollary 7.3.5: Equivalence of Curve Models

Let X be a nonsingular curve over $k = \overline{k}$, with function field $K = K(X)$. Then the following are equivalent:

1. X is projective.
2. X is proper (complete).
3. For every DVR R of K/k, the valuation ring R dominates $\mathcal{O}_{X,x}$ for some (necessarily unique) closed point $x \in X$.

***Proof*:**
(1) $\Rightarrow$ (2) holds because projective morphisms are proper (theorem 3.5.32). The converse (2) $\Rightarrow$ (1) is proposition 7.1.7.

(2) $\Rightarrow$ (3): Let R be a DVR of K/k. The generic point gives $\operatorname{Spec} K \to X$, and the valuative criterion of properness (theorem 3.5.22) for $X \to \operatorname{Spec} k$ gives a unique lift $\operatorname{Spec} R \to X$. It sends the closed point of $\operatorname{Spec} R$ to a point x and induces a local homomorphism $\mathcal{O}_{X,x} \to R$ compatible with the inclusions into K. The point x is not the generic point, since $\mathcal{O}_{X,\eta} = K$ would then map locally to R and force R to be a field. So x is a closed point and R dominates $\mathcal{O}_{X,x}$. A second closed point whose local ring is dominated by R would give a second lift, so x is

unique.

(3) ⇒ (2): The morphism $X \to \operatorname{Spec} k$ is separated and of finite type, so by theorem 3.5.31 it suffices to lift every diagram formed by a DVR R with fraction field L, a morphism $\operatorname{Spec} L \to X$ and a morphism $\operatorname{Spec} R \to \operatorname{Spec} k$. The points of X are its closed points and its generic point. If $\operatorname{Spec} L$ maps to a closed point ξ, then $\kappa(\xi) = k$ and $\operatorname{Spec} R \to \operatorname{Spec} k = \operatorname{Spec} \kappa(\xi) \to X$ is a lift. Otherwise $\operatorname{Spec} L$ maps to the generic point and $K \subseteq L$. Then $R' = R \cap K$ is a valuation ring of K containing k, whose value group is a subgroup of the value group $\mathbb{Z}$ of R. If $R' = K$, then $\operatorname{Spec} R \to \operatorname{Spec} K \to X$ is a lift. Otherwise R' is a DVR of K/k, by (3) it dominates $\mathcal{O}_{X,x}$ for a closed point x, and $\operatorname{Spec} R \to \operatorname{Spec} R' \to \operatorname{Spec} \mathcal{O}_{X,x} \to X$ is a lift. Hence X is proper, completing the proof.

In the language of classical algebraic geometry, such an X is the *complete nonsingular model* of the function field K, unique up to isomorphism (theorem 7.1.4(3)).

Proposition 7.3.6: Genus Zero Means Rational

Let X be a complete nonsingular curve of genus $g = 0$ over $k = \overline{k}$. Then $X \cong \mathbb{P}^1_k$.

***Proof*:**
Let P be any closed point of X. By proposition 7.3.1 $\deg K = -2$, so $\deg(K - P) = -3 < 0$, forcing $\ell(K - P) = 0$ by lemma 7.2.2. Riemann-Roch then gives

$$\ell(P) = \deg P + 1 - g + \ell(K - P) = 2$$

Choose $f \in H^0(X, \mathcal{L}(P)) \setminus k$, viewed as a rational function with $\operatorname{div}(f) + P \geq 0$. It has a pole, since otherwise $f \in \Gamma(X, \mathcal{O}_X) = k$ (theorem 3.5.33), so its divisor of poles is exactly P. As in the proof of corollary 7.1.10, f defines a finite morphism $\varphi\colon X \to \mathbb{P}^1_k$ with $\operatorname{div}(f) = \varphi^*([0]) - \varphi^*([\infty])$, so $\varphi^*([\infty]) = P$ and $\deg \varphi = \deg \varphi^*([\infty]) = 1$ by proposition 7.1.9. Thus φ is birational. It is finite, hence separated and quasifinite, and $\mathbb{P}^1_k$ is normal, so φ is an open immersion by corollary 3.4.39. Being surjective (proposition 7.1.6), it is an isomorphism, as claimed.

Thus $\mathbb{P}^1_k$ is the *unique* smooth projective curve of genus 0 over $\overline{k}$. Over a field k that is not algebraically closed, a geometrically integral projective curve of genus 0 is a nonsingular conic in $\mathbb{P}^2_k$, and it is isomorphic to $\mathbb{P}^1_k$ if and only if it has a k-rational point, that is, a point with residue field k [47, Proposition 7.4.1]. Over $\mathbb{Q}$, for instance, the conic $x^2 + y^2 + z^2 = 0$ has genus 0 but is not isomorphic to $\mathbb{P}^1_{\mathbb{Q}}$, since it has no rational points.

Example 7.3.7: Embedding Low-Genus Curves

1. *Genus* 0*:* By proposition 7.3.6, $X \cong \mathbb{P}^1$. Taking $D = n \cdot P$ for $n \geq 1$ gives $\ell(D) = n + 1$, and $\varphi_{|D|}\colon \mathbb{P}^1 \to \mathbb{P}^n$ is the Veronese embedding of degree n (definition 3.1.38), whose image is the rational normal curve of degree n. For $n = 2$ this is a conic in $\mathbb{P}^2$.

2. *Genus* 1*:* Let E be a complete nonsingular curve of genus one and $O \in E$ a closed point. Then $\deg K = 0$ and $\ell(K) = 1$ (proposition 7.3.1), so $K \sim 0$ by lemma 7.2.2 and $\omega_E \cong \mathcal{O}_E$. The divisor $3O$ is non-special, as $\deg(3O) = 3 > 0 = 2g - 2$, so $\ell(3O) = 3 + 1 - 1 = 3$. Since $\deg(3O) = 2g + 1$, corollary 7.3.4 shows that $|3O|$ embeds E in $\mathbb{P}^2$, and the image, a nonsingular plane curve of genus 1, is a cubic by proposition 7.2.6. This is the Weierstrass

embedding of section 7.6.

3. *Genus* 2*:* $\deg K = 2$ and $\ell(K) = g = 2$, so $|K|$ is a pencil (a linear system of dimension 1), base-point-free: for a closed point P, Riemann-Roch gives $\ell(K - P) = \ell(P) = 1 = \ell(K) - 1$, since $\ell(P) \geq 2$ would give $X \cong \mathbb{P}^1$ by the argument of proposition 7.3.6. The canonical map $\varphi_{|K|} \colon X \to \mathbb{P}^1$ pulls a point back to a divisor of $|K|$, so it has degree 2 (proposition 7.1.9), and every curve of genus 2 is hyperelliptic. By corollary 7.3.4 every divisor of degree $5 = 2g + 1$ is very ample, and for $D = 5P$, $\ell(D) = 4$, so $X \hookrightarrow \mathbb{P}^3$ as a curve of degree 5. No divisor D of smaller degree is very ample. For $\deg D \in \{3, 4\}$, $\ell(D) = \deg D - 1$ by proposition 7.3.2, and for $\deg D \leq 2$, $\ell(D) \leq \deg D + 1$, since each closed point imposes at most one condition. So a very ample divisor of degree at most 4 would embed X in $\mathbb{P}^m$ with $m \leq 2$, while X is neither $\mathbb{P}^1$ nor a nonsingular plane curve, because 2 is not of the form $(d-1)(d-2)/2$ (proposition 7.2.6).

4. *Genus* 3*:* $\deg K = 4$ and $\ell(K) = 3$, so $|K|$ maps $X \to \mathbb{P}^2$. If X is non-hyperelliptic, this is an embedding (theorem 7.3.8(2)), and X is realized as a nonsingular quartic in $\mathbb{P}^2$, consistent with $g = \frac{(4-1)(4-2)}{2} = 3$ (proposition 7.2.6). If X is hyperelliptic, $|K|$ factors through the degree-2 map to $\mathbb{P}^1$ and maps X onto a conic in $\mathbb{P}^2$ (theorem 7.3.8(3)).

The canonical linear system $|K|$ is attached to X without any choice: a canonical divisor is determined by X up to linear equivalence, so the canonical map is determined up to a linear automorphism of the target.

Theorem 7.3.8: The Canonical Map and Embedding

Let X be a complete nonsingular curve of genus $g \geq 2$. Then:

1. *Existence:* The canonical linear system $|K|$ is base-point-free. Consequently, it defines a morphism
$$\varphi_K \colon X \longrightarrow \mathbb{P}^{g-1}$$

2. *The non-hyperelliptic case:* If $\ell(P + Q) = 1$ for all closed points $P, Q \in X$, equal or not, then φ_K is a closed immersion, and X is embedded as a curve of degree $2g - 2$ in $\mathbb{P}^{g-1}$. This is the *canonical embedding* of X.

3. *The hyperelliptic case:* If $\ell(P + Q) \geq 2$ for some closed points $P, Q \in X$, then $\varphi_{|P+Q|} \colon X \to \mathbb{P}^1$ is a morphism of degree 2, $K \sim (g-1)(P + Q)$, and φ_K factors as
$$X \xrightarrow{\varphi_{|P+Q|}} \mathbb{P}^1 \xrightarrow{\nu_{g-1}} \mathbb{P}^{g-1}$$
where ν_{g-1} is the Veronese embedding of degree $g-1$. Thus φ_K is a morphism of degree 2 onto a rational normal curve of degree $g - 1$, and it is not a closed immersion.

The curves in case (3) are the *hyperelliptic* curves (definition 7.5.1), since a degree-2 morphism $\varphi \colon X \to \mathbb{P}^1$ pulls a point y back to a divisor $P + Q$ with $\ell(P + Q) \geq 2$ (the pullbacks of 1 and of a coordinate with a pole at y are independent sections), and $\varphi_{|P+Q|}$ is the *hyperelliptic double cover*.

***Proof*:**
(1) For $|K|$ to be base-point-free means that for every closed point $P \in X$ there is a global differential form that does not vanish at P, that is, $\ell(K) - \ell(K - P) = 1$. By Riemann-Roch,

$$\ell(K - P) = \deg(K - P) + 1 - g + \ell(P) = (2g - 3) + 1 - g + \ell(P) = g - 2 + \ell(P)$$

Since $g \geq 1$, $\ell(P) = 1$ for every closed point P: if $\ell(P) \geq 2$, the argument in the proof of proposition 7.3.6 would give an isomorphism $X \cong \mathbb{P}^1$, implying $g = 0$. Substituting $\ell(P) = 1$ yields $\ell(K - P) = g - 1 = \ell(K) - 1$, so $|K|$ has no base points, and theorem 5.3.43 gives the morphism φ_K.

(2) By the criterion stated before proposition 7.3.3, φ_K is a closed immersion if and only if $\ell(K) - \ell(K - P - Q) = 2$ for all closed points P, Q, with $P = Q$ corresponding to separating tangent vectors. Riemann-Roch applied to $K - P - Q$ gives

$$\ell(K-P-Q) = \deg(K-P-Q)+1-g+\ell(P+Q) = (2g-4)+1-g+\ell(P+Q) = g-3+\ell(P+Q)$$

and subtracting this from $\ell(K) = g$,

$$\ell(K) - \ell(K - P - Q) = 3 - \ell(P + Q)$$

Under the hypothesis of (2) this drop is 2 for every P and Q, so φ_K is a closed immersion. Its hyperplane sections are the canonical divisors, of degree $2g - 2$ (proposition 7.3.1), so the image has degree $2g - 2$.

(3) Suppose $\ell(P + Q) \geq 2$ and choose $h \in H^0(X, \mathcal{L}(P + Q)) \setminus k$. Its divisor of poles is at most $P + Q$ and is not a single point, since $\ell(P') = 1$ for every closed point P' by (1), so it equals $P + Q$. As in the proof of proposition 7.3.6, h defines a finite morphism $X \to \mathbb{P}^1$ pulling ∞ back to $P + Q$, of degree 2 by proposition 7.1.9. Each closed point imposes at most one condition on sections, so $\ell(P + Q) \leq \ell(P) + 1 = 2$, and $1, h$ is a basis of $H^0(X, \mathcal{L}(P + Q))$. The pencil $|P + Q|$ has no base point, since a base point R would give $\ell(P + Q - R) = 2$ for a divisor of degree 1, which is then linearly equivalent to a closed point P' with $\ell(P') = 2$, contradicting (1). So this morphism is $\varphi_{|P+Q|} = [1 : h]$.

The elements $1, h, \ldots, h^{g-1}$ lie in $H^0(X, \mathcal{L}((g - 1)(P + Q)))$, since $\operatorname{div}(h^i) + (g - 1)(P + Q) = i(\operatorname{div}(h) + P + Q) + (g - 1 - i)(P + Q) \geq 0$, and they are linearly independent over k because h is transcendental over the algebraically closed field k. So $\ell((g - 1)(P + Q)) \geq g$. Riemann-Roch for $(g - 1)(P + Q)$ reads

$$\ell((g-1)(P+Q)) - \ell(K-(g-1)(P+Q)) = \deg((g-1)(P+Q)) + 1 - g = (2g-2)+1-g = g-1$$

so $\ell(K - (g - 1)(P + Q)) \geq 1$. The divisor $K - (g - 1)(P + Q)$ has degree $(2g - 2) - 2(g - 1) = 0$, so $K \sim (g-1)(P+Q)$ by lemma 7.2.2(2). Then $\ell((g-1)(P+Q)) = \ell(K) = g$, and $1, h, \ldots, h^{g-1}$ is a basis of the sections of $\mathcal{L}((g - 1)(P + Q)) \cong \omega_X$. With this basis, $\varphi_K = [1 : h : \cdots : h^{g-1}]$, which is the composite of $[1 : h] = \varphi_{|P+Q|}$ with the Veronese map ν_{g-1} given by all monomials of degree $g - 1$ (example 5.3.45(2)). The image of φ_K is the rational normal curve $\nu_{g-1}(\mathbb{P}^1) \cong \mathbb{P}^1$, and φ_K maps onto it with degree $\deg \varphi_{|P+Q|} = 2$. A closed immersion is an isomorphism onto its image, so φ_K is not one, completing the proof.

An isomorphism $X \cong X'$ of non-hyperelliptic curves pulls $\omega_{X'}$ back to ω_X, so it carries the canonical image of X to that of X' up to a linear automorphism of $\mathbb{P}^{g-1}$. The target $\mathbb{P}^{g-1}$ grows with the

genus, while every curve already embeds in $\mathbb{P}^3$.

Proposition 7.3.9: Every Curve Embeds in Projective 3-Space

Let X be a complete nonsingular curve over an algebraically closed field k. Then X is isomorphic to a closed subvariety of $\mathbb{P}^3_k$.

***Proof*:**

By corollary 7.3.4, a divisor D of degree $\deg D \geq 2g+1$ gives a closed immersion $\varphi_{|D|}\colon X \hookrightarrow \mathbb{P}^N_k$, where $N = \ell(D) - 1$. If $N \leq 3$, compose with a linear inclusion $\mathbb{P}^N_k \hookrightarrow \mathbb{P}^3_k$.

Suppose $N \geq 4$. For a closed point $O \in \mathbb{P}^N_k \setminus X$, the linear projection from O (example 5.3.45(4)) is a morphism

$$\pi_O\colon \mathbb{P}^N_k \setminus \{O\} \longrightarrow \mathbb{P}^{N-1}_k$$

and $\pi_O|_X$ is a closed immersion if and only if O lies on no secant line of X (a line joining two distinct points of X) and on no tangent line of X [36, Proposition IV.3.4]. The first condition says that $\pi_O|_X$ separates points, and the second that it separates tangent vectors. The union $\mathrm{Sec}(X)$ of the secant lines is locally the image of a morphism from $(X \times X \setminus \Delta) \times \mathbb{P}^1$, so it has dimension at most 3, and the union of the tangent lines is locally the image of $X \times \mathbb{P}^1$, so it has dimension at most 2 [36, Proposition IV.3.5]. As $N \geq 4$, the closures of these two sets do not cover $\mathbb{P}^N_k$, and since k is algebraically closed the nonempty open complement contains a closed point O. It lies off X, which is contained in the closure of $\mathrm{Sec}(X)$, and projecting from it gives a closed immersion $X \hookrightarrow \mathbb{P}^{N-1}_k$.

Repeating this step $N-3$ times gives a closed immersion $X \hookrightarrow \mathbb{P}^3_k$, as we sought to show.

Not every curve embeds in $\mathbb{P}^2_k$: a nonsingular plane curve of degree d has genus $\frac{(d-1)(d-2)}{2}$ (proposition 7.2.6), so a curve of genus 2, for instance, is not a nonsingular plane curve. Every curve is birational to a plane curve whose only singularities are nodes [36, Corollary IV.3.11]. The dimension count in the proof is the one behind the weak Whitney embedding theorem: projection from points off the secant variety embeds a compact smooth n-manifold in $\mathbb{R}^{2n+1}$, and for the curve X, of dimension $n = 1$, the same count gives $\mathbb{P}^{2n+1} = \mathbb{P}^3$.

Clifford's theorem bounds $\ell(D)$ for *special* divisors D, those with $\ell(K - D) > 0$, for which theorem 7.2.10 does not determine $\ell(D)$ from $\deg D$. Special divisors have $\deg D \leq 2g - 2$ (proposition 7.3.2).

Theorem 7.3.10: Clifford's Theorem

Let D be an effective divisor on a complete nonsingular curve X of genus $g \geq 1$ with $0 \leq \deg D \leq 2g - 2$ and $\ell(K - D) > 0$ (i.e. D is *special*). Then

$$\ell(D) - 1 \;\leq\; \frac{\deg D}{2}$$

Equality holds if $D = 0$, if $D \sim K$, or if there are closed points $P, Q \in X$ with $\ell(P + Q) \geq 2$ and $D \sim r(P + Q)$ for an integer $r \geq 0$ (so that $|P + Q|$ is a g^1_2 and D is a multiple of it).

***Proof*:**
Throughout, k is algebraically closed, as in theorem 7.2.10, and $H^0(X, \mathcal{L}(E))$ is identified with $L(E) = \{f \in K(X)^* \mid \operatorname{div}(f) + E \geq 0\} \cup \{0\}$ (definition 5.4.9).

Step 1: a dimension count in $K(X)$. Fix a closed point P. The local ring $\mathcal{O}_{X,P}$ is a discrete valuation ring with valuation ν_P on $K(X)$. Its residue field is k: the point P lies in an affine open $\operatorname{Spec} B$ with B a finitely generated k-algebra, its maximal ideal $\mathfrak{m} \subseteq B$ pulls back to a maximal ideal of a polynomial ring $k[x_1, \ldots, x_N]$ surjecting onto B, so $B/\mathfrak{m} \cong k$ by theorem 0.1.12(3), and the residue field of $\mathcal{O}_{X,P} \cong B_{\mathfrak{m}}$ (proposition 2.2.22) is $B/\mathfrak{m}$ by [10, Localization and Quotient Commute]. For a finite-dimensional k-subspace $V \subseteq K(X)$ put $\nu_P(V) = \{\nu_P(v) \mid v \in V \setminus \{0\}\}$. We claim $|\nu_P(V)| = \dim_k V$. Elements $v_1, \ldots, v_s \in V$ with distinct valuations are linearly independent, since a combination $\sum_i c_i v_i$ with some $c_i \neq 0$ has valuation $\min\{\nu_P(v_i) \mid c_i \neq 0\}$ and so is nonzero. In particular $\nu_P(V)$ has at most $\dim_k V$ elements. For each value in $\nu_P(V)$ choose an element of V with that valuation, giving independent elements $v_1, \ldots, v_s$. If some element of V lies outside their span, choose such a v with $\nu_P(v)$ largest, which is possible because $\nu_P(V)$ is finite. Then $\nu_P(v) = \nu_P(v_i)$ for some i, the quotient v/v_i is a unit of $\mathcal{O}_{X,P}$, and its residue class is that of some $c \in k^\times$. So $v - cv_i = v_i(v/v_i - c)$ lies outside the span and has larger valuation, a contradiction. Hence the v_i form a basis of V.

Now let $V, W \subseteq K(X)$ be nonzero finite-dimensional subspaces and $V \cdot W$ the span of the products vw. Since $\nu_P(vw) = \nu_P(v) + \nu_P(w)$, the set $\nu_P(V \cdot W)$ contains every sum of an element of $\nu_P(V) = \{a_1 < \cdots < a_s\}$ and an element of $\nu_P(W) = \{b_1 < \cdots < b_t\}$, and the sums $a_1 + b_1 < a_1 + b_2 < \cdots < a_1 + b_t < a_2 + b_t < \cdots < a_s + b_t$ are $s + t - 1$ distinct values. Hence

$$\dim_k(V \cdot W) \geq \dim_k V + \dim_k W - 1$$

Step 2: the inequality. For $f \in L(D)$ and $h \in L(K - D)$ we have $\operatorname{div}(fh) + K = (\operatorname{div}(f) + D) + (\operatorname{div}(h) + K - D) \geq 0$, so the span $L(D) \cdot L(K - D)$, which is the image of the multiplication map $H^0(X, \mathcal{L}(D)) \otimes H^0(X, \mathcal{L}(K - D)) \to H^0(X, \mathcal{L}(K))$, lies in $L(K)$. Here $L(D) \neq 0$ because $1 \in L(D)$ for effective D, and $L(K - D) \neq 0$ by hypothesis. Step 1 and $\ell(K) = g$ (proposition 7.3.1) give

$$\ell(D) + \ell(K - D) - 1 \leq g$$

By theorem 7.2.10, $\ell(K - D) = \ell(D) - \deg D - 1 + g$, and substituting yields $2\ell(D) \leq \deg D + 2$, that is, $\ell(D) - 1 \leq \deg D/2$.

Step 3: the cases of equality. The number $\ell(E)$ depends only on the linear equivalence class of E: if $E' = E + \operatorname{div}(\varphi)$, then $f \mapsto f\varphi^{-1}$ is a k-linear bijection $L(E) \to L(E')$. If $D = 0$, then $\ell(D) = \dim_k \Gamma(X, \mathcal{O}_X) = 1$ by theorem 3.5.33, and both sides of the inequality vanish. If $D \sim K$, then $\ell(D) = \ell(K) = g$ and $\deg D = 2g - 2$ (proposition 7.3.1 and corollary 7.1.10), so both sides equal $g - 1$. Finally let $\ell(P + Q) \geq 2$ and $D \sim r(P + Q)$, and choose a nonconstant $h \in L(P + Q)$. The elements $1, h, \ldots, h^r$ lie in $L(r(P + Q))$, since $\operatorname{div}(h^i) + r(P + Q) = i(\operatorname{div}(h) + P + Q) + (r - i)(P + Q) \geq 0$. They are linearly independent over k, because h is transcendental over k: an element of $K(X)$ algebraic over the algebraically closed field k lies in k. Hence $\ell(D) = \ell(r(P + Q)) \geq r + 1$, while $\deg D = 2r$ (corollary 7.1.10). With Step 2 this gives $\ell(D) - 1 = r = \deg D/2$, completing the proof.

The converse also holds: if equality holds, then $D = 0$, $D \sim K$, or X is hyperelliptic and D is a multiple of its g^1_2 [36, Theorem IV.5.4].

7.3.11 Petri's Theorem and Syzygies The canonical embedding $X \hookrightarrow \mathbb{P}^{g-1}$ of a non-hyperelliptic curve raises the question of which equations define the image. A *quadric* is a hypersurface defined by a homogeneous polynomial of degree 2. Petri's theorem says that the homogeneous ideal of the canonical curve is generated by quadrics unless X is trigonal (admits a degree-3 morphism to $\mathbb{P}^1$) or is a nonsingular plane quintic ($g = 6$).
The finer structure of the ideal is recorded by its *syzygies*, the terms of a minimal graded free resolution of the homogeneous coordinate ring of the canonical curve over the polynomial ring in g variables, which is finite by Hilbert's syzygy theorem (*Homological Algebra* [18, Hilbert's Syzygy Theorem]) together with the graded argument following corollary 5.3.27. Green's conjecture predicts that the graded Betti numbers of this resolution determine the *Clifford index* of X. Voisin proved it for a general curve of each genus.

7.4 Ramification and Hurwitz's Formula

Riemann-Roch relates the invariants of a single curve: its genus, its divisors and their spaces of sections. Hurwitz's theorem relates the genera of two curves X and Y through a finite morphism $f\colon X \to Y$, its degree and its ramification.

Over $\mathbb{C}$, a finite morphism of smooth projective curves is a branched covering of compact Riemann surfaces. Away from finitely many *branch points*, the map is a covering space with $n = \deg f$ sheets, and over a branch point several sheets meet. The Euler characteristic (equivalently, the genus) of X is then determined by the Euler characteristic of Y, the degree of f and the ramification. Hurwitz's formula is the algebraic form of this count over an arbitrary algebraically closed field.

7.4.1 Ramification

Throughout this section, $f\colon X \to Y$ is a finite morphism of complete nonsingular curves over $k = \overline{k}$, with $n = \deg f = [K(X) : K(Y)]$.

Every local ring $\mathcal{O}_{X,x}$ (resp. $\mathcal{O}_{Y,y}$) at a closed point of a smooth curve is a DVR with discrete valuation ν_x (resp. ν_y). The ramification index $e_x = \nu_x(f^{\#}(t))$, for t a uniformizer at $y = f(x)$, already entered the pullback of divisors preceding proposition 7.1.9.

Definition 7.4.1: Ramification Index

Let $f\colon X \to Y$ be a finite morphism of complete nonsingular curves, $x \in X$ a closed point, and $y = f(x)$. The *ramification index* of f at x is

$$e_x \;=\; \nu_x(f^{\#}(t))$$

where t is any uniformizer of $\mathcal{O}_{Y,y}$. This is independent of the choice of t.

1. f is *unramified* at x if $e_x = 1$.
2. f is *ramified* at x if $e_x > 1$. The point $y = f(x)$ is then called a *branch point*.
3. If $\mathrm{char} k = 0$, all ramification is *tame*. If $\mathrm{char} k = p > 0$, the ramification at x is *tame* if $p \nmid e_x$, and *wild* if $p \mid e_x$.

The index does not depend on t: if $t' = ut$ for a unit $u \in \mathcal{O}_{Y,y}^*$, then $f^{\#}(u)$ is a unit in $\mathcal{O}_{X,x}$ (local maps send units to units), so $\nu_x(f^{\#}(t')) = \nu_x(f^{\#}(u)) + \nu_x(f^{\#}(t)) = 0 + e_x$. The index e_x is the multiplicity of x in the divisor $f^*(y)$. If s is a uniformizer of $\mathcal{O}_{X,x}$, then $f^{\#}(t) = v \cdot s^{e_x}$ for a unit v, so in these local parameters f is $s \mapsto v s^{e_x}$.

Example 7.4.2: Ramification in Practice

1. *The squaring map.* Let $\mathrm{char} k \neq 2$ and consider $f\colon \mathbb{P}^1 \to \mathbb{P}^1$, $[x : y] \mapsto [x^2 : y^2]$, which in the affine charts $y \neq 0$ is $t \mapsto u = t^2$. This has degree 2. At $t = 0$ the uniformizer of the target is u, and $f^{\#}(u) = t^2$, so $e_0 = 2$ (ramified). Similarly at $t = \infty$ (using the chart $x \neq 0$, the map is $s \mapsto s^2$ in the uniformizer $s = 1/t$), so $e_\infty = 2$. At every other point $t = a \neq 0, \infty$, the uniformizer of the target at a^2 is $u - a^2$, and $f^{\#}(u - a^2) = t^2 - a^2 = (t - a)(t + a)$. Since $t + a$ is a unit at $t = a$ (as $2a \neq 0$), $\nu_a((t - a)(t + a)) = 1$ and $e_a = 1$ (unramified). The fiber over a^2 consists of the two points a and $-a$, each with $e = 1$, confirming $\sum e_x = 2 = \deg f$. In characteristic 2 the same formula defines a purely inseparable morphism with $e_x = 2$ at every point (box 7.4.4).

2. *Wild ramification.* Let k have characteristic $p > 0$ and let $f\colon \mathbb{P}^1 \to \mathbb{P}^1$ be given in the affine coordinate by $t \mapsto u = t^p - t$ (the Artin-Schreier map). It has degree p, the degree of $t^p - t$ as a rational function of t. At a point $a \in \mathbb{A}^1$, the uniformizer of the target at $f(a) = a^p - a$ is $u - (a^p - a)$, and $f^{\#}(u - (a^p - a)) = (t - a)^p - (t - a)$, whose lowest-order term in $t - a$ is $-(t - a)$. So $e_a = 1$ at every point of $\mathbb{A}^1$, and the fiber over each point c of $\mathbb{A}^1$ consists of the p distinct roots of the separable polynomial $T^p - T - c$. At ∞, with uniformizers $s = 1/t$ and $1/u$,

$$f^{\#}(1/u) = \frac{1}{t^p - t} = \frac{s^p}{1 - s^{p-1}}$$

has valuation p, so $e_\infty = p$: the ramification at ∞ is wild, and ∞ is the only point over ∞. The restriction $\mathbb{A}^1 \to \mathbb{A}^1$ is therefore a connected cover of degree p, unramified at every point. Over $\mathbb{C}$ the affine line has no such cover, since $\mathbb{A}^1(\mathbb{C}) = \mathbb{C}$ is simply connected.

Differentials measure ramification. The first map of the cotangent sequence (proposition 5.5.17) for

$X \xrightarrow{f} Y \to \operatorname{Spec} k$ is a map of sheaves on X

$$f^*\Omega_{Y/k} \longrightarrow \Omega_{X/k} \tag{7.3}$$

between invertible sheaves. Call f *separable* if $K(X)/K(Y)$ is a separable field extension, which is automatic in characteristic 0. If f is separable, (7.3) is injective. Indeed, at the generic point it is a $K(X)$-linear map $K(X) \otimes_{K(Y)} \Omega_{K(Y)/k} \to \Omega_{K(X)/k}$ between one-dimensional vector spaces, whose cokernel $\Omega_{K(X)/K(Y)}$ vanishes because $K(X)/K(Y)$ is separable algebraic (proposition 5.5.9). So it is an isomorphism there, and a map of invertible sheaves on an integral scheme that is nonzero at the generic point is injective. Its cokernel $\Omega_{X/Y}$ is then a coherent sheaf vanishing at the generic point, hence supported at finitely many closed points. If f is inseparable, (7.3) is zero (box 7.4.4).

In local coordinates, let $x \in X$ be a closed point with $y = f(x)$, let s be a uniformizer of $\mathcal{O}_{X,x}$ and t a uniformizer of $\mathcal{O}_{Y,y}$. By proposition 5.5.10 the class of ds spans $\Omega_{X/k,x} \otimes k \cong \mathfrak{m}_x/\mathfrak{m}_x^2$, so ds generates the free rank-one module $\Omega_{X/k,x}$ (proposition 5.5.25) by Nakayama's lemma, and likewise dt generates $\Omega_{Y/k,y}$. Write $f^{\#}(t) = v \cdot s^{e_x}$ for a unit $v \in \mathcal{O}_{X,x}^*$ and $dv = v'\,ds$. Then

$$f^*(dt) \;=\; d(f^{\#}(t)) \;=\; d(v \cdot s^{e_x}) \;=\; (v' s^{e_x} + e_x v s^{e_x - 1})\,ds \;=\; s^{e_x - 1}(v's + e_x v)\,ds$$

The factor s^{e_x-1} vanishes to order $e_x - 1$ at x. The factor $(v's + e_x v)$ is a unit at x if and only if $e_x \neq 0$ in k, that is, if and only if the ramification at x is tame. In the case of tame ramification, the map (7.3) vanishes to order exactly $e_x - 1$ at x.

Definition 7.4.3: Ramification and Branch Divisors

Let f be separable. For a closed point $x \in X$, with uniformizers s at x and t at $f(x)$, write $f^*(dt) = h_x\,ds$ with $h_x \in \mathcal{O}_{X,x}$, and set $r_x = \nu_x(h_x)$, the length of $\Omega_{X/Y,x} \cong \mathcal{O}_{X,x}/(h_x)$. Replacing s or t by a unit multiple multiplies h_x by a unit, so r_x does not depend on these choices, and $r_x = 0$ outside the finite support of $\Omega_{X/Y}$. The *ramification divisor* of f is

$$R \;=\; \sum_{x \in X_{\mathrm{cl}}} r_x \cdot x$$

so that the image of (7.3) is $\Omega_{X/k} \otimes \mathcal{L}(-R)$ and $f^*\Omega_{Y/k} \cong \Omega_{X/k} \otimes \mathcal{L}(-R)$. The *branch divisor* is $B = f_*(R) = \sum_x r_x \cdot f(x) \in \operatorname{WDiv} Y$.
When the ramification at x is tame (in particular, always in characteristic 0),

$$r_x = e_x - 1$$

When it is wild ($\operatorname{char} k = p > 0$ and $p \mid e_x$), $r_x \geq e_x$, and the precise value depends on higher-order ramification data.

The relation $f^*\Omega_{Y/k} \cong \Omega_{X/k}(-R)$ can be rewritten in terms of canonical divisors. Here $\Omega_{Y/k} \cong \mathcal{L}(K_Y)$ and $\Omega_{X/k} \cong \mathcal{L}(K_X)$, and $f^*\mathcal{L}(D) \cong \mathcal{L}(f^*D)$ for every divisor D on Y: near $f(x)$, the sheaf $\mathcal{L}(D)$ is generated by t^{-m}, where m is the coefficient of $f(x)$ in D, and $f^*\mathcal{L}(D)$ by $f^{\#}(t)^{-m} = v^{-m} s^{-e_x m}$, a generator of $\mathcal{L}(f^*D)$. This gives the *divisor relation*

$$K_X \;\sim\; f^*K_Y + R \tag{7.4}$$

and Hurwitz's formula follows from it by taking degrees.

7.4.4 Remark: Inseparable Maps If the field extension $K(X)/K(Y)$ is inseparable (which can only happen in characteristic $p > 0$), the map (7.3) is the *zero map*. At the generic point its cokernel $\Omega_{K(X)/K(Y)}$ is nonzero (proposition 5.5.9), and a linear map into a one-dimensional space that is not surjective is zero, so the map of sheaves, whose target $\Omega_{X/k}$ is invertible and hence torsion-free, is zero. The simplest example is $F\colon \mathbb{P}^1 \to \mathbb{P}^1$, $[x : y] \mapsto [x^p : y^p]$, which is purely inseparable of degree p, as $k(t)$ has degree p over $k(t^p)$. In the affine coordinates, t on the source and u on the target, $F^{\#}(u) = t^p$, so $F^*(du) = d(t^p) = pt^{p-1}\,dt = 0$, and $e_x = p$ at every point.
Hurwitz's formula is stated for separable morphisms. A general finite morphism of curves factors, through the separable closure of $K(Y)$ in $K(X)$, as a purely inseparable morphism followed by a separable one, and a purely inseparable finite morphism of curves is a composite of k-linear Frobenius morphisms, so its source and target have the same genus [36, Proposition IV.2.5 and Example IV.2.5.4].

7.4.2 Hurwitz's Formula

Theorem 7.4.5: Hurwitz's Formula

Let $f\colon X \to Y$ be a separable finite morphism of complete nonsingular curves of genera g_X and g_Y respectively, with $n = \deg f$. Then

$$2g_X - 2 \;=\; n(2g_Y - 2) \;+\; \deg R$$

where $R = \sum r_x \cdot x$ is the ramification divisor. In the tamely ramified case (in particular, whenever $\mathrm{char} k = 0$),

$$2g_X - 2 \;=\; n(2g_Y - 2) \;+\; \sum_{x \in X_{\mathrm{cl}}} (e_x - 1)$$

***Proof*:**
Taking degrees of both sides of (7.4) gives

$$\deg K_X = \deg(f^* K_Y) + \deg R$$

By proposition 7.3.1, $\deg K_X = 2g_X - 2$ and $\deg K_Y = 2g_Y - 2$. By the degree formula for pullbacks (proposition 7.1.9), $\deg(f^* K_Y) = n \cdot \deg K_Y = n(2g_Y - 2)$. Therefore

$$2g_X - 2 = n(2g_Y - 2) + \deg R$$

The second formula follows since $r_x = e_x - 1$ in the tame case, completing the proof.

The substance of the proof is the relation (7.4) of section 7.4.1, which rests on the Kähler differentials of section 5.5. The same exact sequence gives the formula through Euler characteristics, and this route identifies $\deg R$ with $\dim_k H^0(X, \Omega_{X/Y})$.

***Proof Alternative proof via Euler characteristics*:**

Since f is separable, the cotangent sequence (proposition 5.5.17) for $X \xrightarrow{f} Y \to \operatorname{Spec} k$ and the injectivity of (7.3) give a short exact sequence on X

$$0 \to f^*\Omega_{Y/k} \to \Omega_{X/k} \to \Omega_{X/Y} \to 0$$

Taking Euler characteristics (proposition 6.7.2),

$$\chi(\Omega_{X/k}) = \chi(f^*\Omega_{Y/k}) + \chi(\Omega_{X/Y})$$

Now $\Omega_{X/k} = \omega_X \cong \mathcal{L}(K_X)$, and by Riemann-Roch (theorem 7.2.10)

$$\chi(\omega_X) = \deg K_X + 1 - g_X = (2g_X - 2) + 1 - g_X = g_X - 1$$

For the pullback, $f^*\Omega_{Y/k} \cong f^*\omega_Y \cong \mathcal{L}(f^*K_Y)$, and

$$\chi(f^*\omega_Y) = \deg(f^*K_Y) + 1 - g_X = n(2g_Y - 2) + 1 - g_X$$

The sheaf $\Omega_{X/Y}$ is supported at finitely many closed points, with stalks $\Omega_{X/Y,x} \cong \mathcal{O}_{X,x}/(h_x)$ of dimension r_x over k (definition 7.4.3). A coherent sheaf supported at finitely many closed points is a finite direct sum of skyscraper sheaves, which have no higher cohomology, so $\chi(\Omega_{X/Y}) = \dim_k H^0(X, \Omega_{X/Y}) = \sum_x r_x = \deg R$. Putting it all together,

$$g_X - 1 = n(2g_Y - 2) + 1 - g_X + \deg R$$

which rearranges to $2g_X - 2 = n(2g_Y - 2) + \deg R$, as claimed.

Example 7.4.6: Hurwitz in Action

1. *Double covers of* $\mathbb{P}^1$. Let $f\colon X \to \mathbb{P}^1$ be a degree-2 morphism in characteristic $\neq 2$, which is separable. Then $n = 2$, $g_Y = 0$, and all ramification is tame with $e_x \in \{1, 2\}$. Let b be the number of branch points in $\mathbb{P}^1$ (since $n = 2$, each corresponds to exactly one ramification point in X, where $e_x = 2$). Hurwitz gives

   $$2g_X - 2 = 2(0 - 2) + b = -4 + b, \qquad \text{so } g_X = \frac{b-2}{2}$$

   Since g_X is a non-negative integer, b is even and $b \geq 2$. For $b = 2$: $g_X = 0$ (the map $[x : y] \mapsto [x^2 : y^2]$). For $b = 4$: $g_X = 1$. For $b = 6$: $g_X = 2$ (genus-2 curves, which are all hyperelliptic). More generally, a *hyperelliptic curve* of genus g is a double cover of $\mathbb{P}^1$ with $2g + 2$ branch points.

2. *Cyclic covers.* Let $\operatorname{char} k \nmid n$ and let $f\colon X \to \mathbb{P}^1$ be a cyclic cover of degree n (that is, $K(X)/K(\mathbb{P}^1)$ is Galois with cyclic group of order n), totally ramified ($e_x = n$) over exactly b points and unramified elsewhere. Then

   $$2g_X - 2 = n(-2) + b(n-1), \quad \text{so} \quad g_X = \frac{(n-1)(b-2)}{2}$$

   For $n = 3$, $b = 4$: $g_X = 2$. For $n = 3$, $b = 5$: $g_X = 3$ (a trigonal curve of genus 3). For $n = 3$, $b = 3$: $g_X = 1$, and a generator of the Galois group is an automorphism of order 3 of this curve of genus 1.

3. *Genus bound.* If f is separable and $g_Y \geq 1$, then $\deg R \geq 0$ gives $2g_X - 2 \geq n(2g_Y - 2)$, so $g_X \geq n(g_Y - 1) + 1 \geq g_Y$. A purely inseparable morphism preserves the genus (box 7.4.4), so $g_X \geq g_Y$ for every finite morphism $X \to Y$ (for $g_Y = 0$ there is nothing to prove). In particular, there are no nonconstant morphisms from a curve of genus g to a curve of genus $g' > g$.

7.4.3 Applications of Hurwitz's Formula

Hurwitz's formula bounds the order of a finite group of automorphisms of a curve of genus $g \geq 2$ by a linear function of g, provided the characteristic of k does not divide the order of the group.

Theorem 7.4.7: Hurwitz's Automorphism Bound

Let X be a complete nonsingular curve of genus $g \geq 2$, and let G be a finite group of automorphisms of X whose order is not divisible by $\mathrm{char}k$ (no condition if $\mathrm{char}k = 0$). Then

$$|G| \;\leq\; 84(g-1)$$

The bound is attained: over $\mathbb{C}$, the Klein quartic $x^3y + y^3z + z^3x = 0$ in $\mathbb{P}^2$ has genus 3 and an automorphism group of order $168 = 84 \cdot 2$. In characteristic 0 the whole group $\mathrm{Aut}(X)$ is finite for $g \geq 2$ [36, Exercise IV.5.2], so $|\mathrm{Aut}(X)| \leq 84(g-1)$. In characteristic $p \geq 5$ the condition $p \nmid |G|$ cannot be dropped: the curve $y^2 = x^p - x$ has genus $(p-1)/2$ and $2p(p^2-1)$ automorphisms, more than $84(g-1)$, by a theorem of Roquette recorded in the note to [36, Exercise IV.2.5].

***Proof*:**

Write $n = |G|$. The fixed field $K(X)^G$ is the function field of a complete nonsingular curve Y, the extension $K(X)/K(X)^G$ is Galois with group G, and the inclusion $K(X)^G \subseteq K(X)$ corresponds to a finite separable morphism $\pi\colon X \to Y$ of degree n. If $x \in X$ has ramification index e, then the fiber $\pi^{-1}(\pi(x))$ consists of exactly n/e points, each with ramification index e. These facts are the setup and part (a) of [36, Exercise IV.2.5], posed there in characteristic 0, and are not proved here. Their proofs do not use the characteristic: the key input, that G permutes the points of each fiber transitively, is [62, Chapter I, Proposition (9.1)] applied to the Dedekind domains of affine opens of Y. Each ramification index divides n, and $\mathrm{char}k \nmid n$, so all ramification of π is tame.

Let $y_1, \ldots, y_s \in Y$ be the branch points of π and $e_1, \ldots, e_s \geq 2$ the ramification indices over them. The fiber over y_i contributes $(n/e_i)(e_i - 1)$ to the sum in theorem 7.4.5, which therefore gives

$$2g - 2 = n(2g_Y - 2) + \sum_{i=1}^{s} n\left(1 - \frac{1}{e_i}\right)$$

that is,

$$\frac{2g-2}{n} = 2g_Y - 2 + \sum_{i=1}^{s}\left(1 - \frac{1}{e_i}\right) \tag{7.5}$$

The left side is positive since $g \geq 2$. Call the right side μ. It suffices to show $\mu \geq 1/42$. Each term $1 - 1/e_i$ lies in $[1/2, 1)$.

1. If $g_Y \geq 2$, then $\mu \geq 2$.

2. If $g_Y = 1$, then $\mu > 0$ forces $s \geq 1$, so $\mu \geq 1/2$.
3. If $g_Y = 0$ and $s \leq 2$, then $\mu < 0$, since each of the at most two terms $1 - 1/e_i$ is less than 1, which is excluded.
4. If $g_Y = 0$ and $s \geq 5$, then $\mu \geq -2 + 5/2 = 1/2$.
5. If $g_Y = 0$ and $s = 4$, then $\mu = 2 - \sum_i 1/e_i$, which is positive only if some $e_i \geq 3$, and then $\mu \geq 2 - 3/2 - 1/3 = 1/6$.
6. If $g_Y = 0$ and $s = 3$, order the indices as $e_1 \leq e_2 \leq e_3$, so that $\mu = 1 - 1/e_1 - 1/e_2 - 1/e_3$, which increases with each e_i. If $e_1 = e_2 = 2$, then $\mu < 0$. If $e_1 = 2$ and $e_2 = 3$, then $\mu = 1/6 - 1/e_3 > 0$ forces $e_3 \geq 7$, so $\mu \geq 1/6 - 1/7 = 1/42$. If $e_1 = 2$ and $e_2 = 4$, then $\mu > 0$ forces $e_3 \geq 5$, so $\mu \geq 1/4 - 1/5 = 1/20$. If $e_1 = 2$ and $e_2 \geq 5$, then $\mu \geq 1/2 - 2/5 = 1/10$. If $e_1 \geq 3$, then $\mu > 0$ excludes $(3, 3, 3)$, so $e_3 \geq 4$ and $\mu \geq 1 - 2/3 - 1/4 = 1/12$.

So $\mu \geq 1/42$, with equality only for $g_Y = 0$, $s = 3$ and $(e_1, e_2, e_3) = (2, 3, 7)$. Therefore $n = (2g - 2)/\mu \leq 42(2g - 2) = 84(g - 1)$, as we sought to show.

Over $\mathbb{C}$, the number μ in (7.5) is the hyperbolic area of the quotient orbifold divided by 2π. The triple $(2, 3, 7)$ is the signature of the triangle group of area $2\pi/42 = \pi/21$, which by a theorem of Siegel is the lattice in $\mathrm{PSL}_2(\mathbb{R})$ of smallest co-area.

Example 7.4.8: Automorphism Groups in Low Genus

1. *Genus* 0*:* $\mathrm{Aut}(\mathbb{P}^1_k) = \mathrm{PGL}_2(k)$, which is infinite (and 3-dimensional as an algebraic group). Hurwitz's bound does not apply since $g < 2$.
2. *Genus* 1*:* If E is an elliptic curve with origin O, then the group $\mathrm{Aut}(E, O)$ of automorphisms fixing O is finite and contains the inversion $P \mapsto -P$. Over $\mathbb{C}$, $\mathrm{Aut}(E, O) \cong \mathbb{Z}/2\mathbb{Z}$ unless $j = 0$ or $j = 1728$, where the automorphism group is $\mathbb{Z}/6\mathbb{Z}$ (resp. $\mathbb{Z}/4\mathbb{Z}$), corresponding to the extra symmetries of the hexagonal (resp. square) lattice.

   As a curve, E also has the translations $\tau_P \colon Q \mapsto Q + P$ for every $P \in E(k)$, so the full automorphism group $\mathrm{Aut}(E)$ is an extension of $\mathrm{Aut}(E, O)$ by $E(k)$, which is infinite.
3. *Genus* 2*:* Every genus-2 curve is hyperelliptic, so it comes with an involution ι (the hyperelliptic involution, definition 7.5.3), giving $|\mathrm{Aut}(X)| \geq 2$. In characteristic 0 the group $\mathrm{Aut}(X)$ is finite and Hurwitz's bound gives $|\mathrm{Aut}(X)| \leq 84$. Over $\mathbb{C}$ the maximum for $g = 2$ is 48, attained by the Bolza curve $y^2 = x^5 - x$, so the bound 84 is not attained in genus 2. In characteristic 5 the same equation defines a curve of genus 2 with 240 automorphisms, the case $p = 5$ of the remark after theorem 7.4.7.

A finite morphism $f \colon X \to Y$ of smooth curves is *unramified* if $\Omega_{X/Y} = 0$ (definition 2.7.19). For separable f this holds if and only if $R = 0$, since $\Omega_{X/Y,x} \cong \mathcal{O}_{X,x}/(h_x)$, and if and only if $e_x = 1$ for all $x \in X$, since $r_x = e_x - 1$ in the tame case and $r_x \geq e_x \geq 2$ in the wild case. A nonconstant finite morphism between smooth curves over a field is flat[4], so an unramified finite morphism is *étale*. For separable f with $R = 0$, Hurwitz's formula reads $2g_X - 2 = n(2g_Y - 2)$.

[4] A finite torsion-free module over a Dedekind domain is flat. Flatness also follows from the miracle flatness theorem in dimension 1.

Corollary 7.4.9: Étale Covers and Genus

Let $f\colon X \to Y$ be a finite étale morphism of complete nonsingular curves of degree n.

1. If $g_Y = 0$, then $2g_X - 2 = -2n < 0$, which is impossible for $n \geq 2$ (since $g_X \geq 0$). So $\mathbb{P}^1$ has no nontrivial finite étale covers: $\mathbb{P}^1$ is *simply connected* in the algebraic sense.
2. If $g_Y = 1$, then $g_X = 1$ as well. Étale covers of elliptic curves have genus 1.
3. If $g_Y \geq 2$, then $g_X = n(g_Y - 1) + 1$, and $g_X > g_Y$ when $n \geq 2$. A nontrivial cover has strictly larger genus.

***Proof*:**
By definition 2.7.19, $\Omega_{X/Y} = 0$.

Step 1: f is separable. Let $\operatorname{Spec} A \subseteq Y$ be a nonempty affine open. As f is finite, $f^{-1}(\operatorname{Spec} A) = \operatorname{Spec} B$ with $A \subseteq B$ domains whose fraction fields are $K(Y)$ and $K(X)$. By proposition 5.5.15, $\Omega_{B/A} \cong \Omega_{X/Y}(\operatorname{Spec} B) = 0$. With $S = B \setminus \{0\}$, [12, Base Change For Differentials] gives $\Omega_{K(X)/A} \cong S^{-1}\Omega_{B/A} = 0$. The universal derivation $d\colon K(X) \to \Omega_{K(X)/A}$ vanishes on $K(Y)$, since for $a, b \in A$ with $b \neq 0$ we have $b\,d(a/b) = d(a) - (a/b)\,d(b) = 0$. So d is a $K(Y)$-derivation, the universal properties identify $\Omega_{K(X)/K(Y)}$ with $\Omega_{K(X)/A}$, and $\Omega_{K(X)/K(Y)} = 0$. The extension $K(X)/K(Y)$ is finite of degree n, so it has transcendence degree 0, and proposition 5.5.9 shows that it is separably generated, that is, separable algebraic.

Step 2: $R = 0$. Let $x \in X$ be a closed point and $y = f(x)$, which is closed, f being finite and hence closed (corollary 3.4.24), and let s and t be uniformizers of $\mathcal{O}_{X,x}$ and $\mathcal{O}_{Y,y}$. These local rings are localizations of finitely generated k-algebras at maximal ideals, with residue field k since k is algebraically closed, so proposition 5.5.10 applies to them. By proposition 5.5.25, $\Omega_{X/k,x}$ is free of rank one over $\mathcal{O}_{X,x}$. By proposition 5.5.10, the class of ds generates $\Omega_{X/k,x} \otimes_{\mathcal{O}_{X,x}} k \cong \mathfrak{m}_x/\mathfrak{m}_x^2$, so ds generates $\Omega_{X/k,x}$ by Nakayama's lemma, and in the same way dt generates $\Omega_{Y/k,y}$. The cotangent sequence (proposition 5.5.17) for $X \to Y \to \operatorname{Spec} k$ is exact, $f^*\Omega_{Y/k} \to \Omega_{X/k} \to \Omega_{X/Y} = 0$. At x its first map sends the generator $dt \otimes 1$ of $(f^*\Omega_{Y/k})_x = \Omega_{Y/k,y} \otimes_{\mathcal{O}_{Y,y}} \mathcal{O}_{X,x}$ to $f^*(dt) = d(f^\#(t))$, which therefore generates $\Omega_{X/k,x} = \mathcal{O}_{X,x}\,ds$. So $f^*(dt) = v\,ds$ with $v \in \mathcal{O}_{X,x}^\times$, and $r_x = \nu_x(v) = 0$ (definition 7.4.3). Hence $R = 0$.

Step 3: the genus relation. By Steps 1 and 2, theorem 7.4.5 applies with $\deg R = 0$ and gives $2g_X - 2 = n(2g_Y - 2)$.

1. If $g_Y = 0$, then $2g_X - 2 = -2n$, and $g_X \geq 0$ forces $n \leq 1$, so a finite étale cover of $\mathbb{P}^1$ has degree one.
2. If $g_Y = 1$, then $2g_X - 2 = 0$ and $g_X = 1$.
3. If $g_Y \geq 2$, then $g_X = n(g_Y - 1) + 1$, which for $n \geq 2$ is at least $2g_Y - 1 > g_Y$, as claimed.

Corollary 7.4.9(1) holds in every characteristic and is the algebraic counterpart of the topological fact $\pi_1(\mathbb{P}^1(\mathbb{C})) = 0$ (the Riemann sphere is simply connected). Corollary 7.4.9(2) matches $\pi_1(E(\mathbb{C})) \cong \mathbb{Z}^2$,

which has a finite quotient of every order. The multiplication-by-n maps of an elliptic curve with $\mathrm{char} k \nmid n$, treated in section 7.6, are examples.

7.4.10 Forward Reference: The Étale Fundamental Group The étale fundamental group $\pi_1^{\text{ét}}(X, \overline{x})$ is the profinite group classifying finite étale covers of X (see box 3.1.19 for an earlier preview). For X connected and of finite type over $\mathbb{C}$, it is the *profinite completion* of the topological fundamental group,

$$\pi_1^{\text{ét}}(X_{\mathbb{C}}) \;\cong\; \widehat{\pi_1^{\text{top}}(X(\mathbb{C}))}$$

by [16, Comparison with the topological fundamental group]. For projective X the comparison rests on GAGA (theorem 6.9.2), applied to the finite covering spaces of $X(\mathbb{C})$. For a curve of genus g, π_1^{top} is the surface group with $2g$ generators $a_1, b_1, \ldots, a_g, b_g$ and the single relation $\prod_{i=1}^{g} a_i b_i a_i^{-1} b_i^{-1} = 1$. Over fields that are not algebraically closed and over arithmetic bases, the étale fundamental group also records Galois-theoretic information: for the spectrum of a field it is the absolute Galois group of that field. The theory is developed in *Étale Cohomology* [16].

Hurwitz's formula has an arithmetic counterpart. Let L/K be a finite extension of number fields with $[L : K] = n$, and consider the associated map of spectra of rings of integers

$$\pi \colon \operatorname{Spec} \mathcal{O}_L \;\longrightarrow\; \operatorname{Spec} \mathcal{O}_K$$

The scheme $\operatorname{Spec} \mathcal{O}_K$ is not a curve in the sense of definition 7.1.1: it is of finite type over $\operatorname{Spec} \mathbb{Z}$, not over a field, and no genus is attached to it in the same way. The local structure of π follows the geometric pattern. For each nonzero prime $\mathfrak{p}$ of $\mathcal{O}_K$, the fiber $\pi^{-1}(\mathfrak{p})$ consists of primes $\mathfrak{q}_1, \ldots, \mathfrak{q}_r$ of $\mathcal{O}_L$ with ramification indices e_i and residue field degrees f_i satisfying the fundamental identity

$$\sum_{i=1}^{r} e_i f_i \;=\; n$$

the analogue of proposition 7.1.9. The *different* $\mathfrak{d}_{L/K}$ is the analogue of the ramification divisor R. It is the annihilator of the $\mathcal{O}_L$-module $\Omega_{\mathcal{O}_L/\mathcal{O}_K}$, as R is cut out by the annihilator of $\Omega_{X/Y}$, and its norm is the *discriminant* $\Delta_{L/K}$, an ideal of $\mathcal{O}_K$ playing the role of the branch divisor $B = f_*(R)$ [62, Chapter III, Proposition (2.7) and Theorem (2.9)]. If $\mathfrak{q}_i^{s_i}$ is the exact power of $\mathfrak{q}_i$ dividing $\mathfrak{d}_{L/K}$, then

$$v_{\mathfrak{p}}(\Delta_{L/K}) \;=\; \sum_{i=1}^{r} f_i s_i$$

with $s_i = e_i - 1$ when $\mathfrak{q}_i$ is tamely ramified and $s_i \geq e_i$ when it is wildly ramified [62, Chapter III, Theorem (2.6)]. In the tame case the exponent of $\mathfrak{p}$ in $\Delta_{L/K}$ is therefore $\sum_i f_i(e_i - 1)$, which matches the coefficient $\sum_{f(x)=y}(e_x - 1)$ of y in the branch divisor of a tamely ramified morphism of curves, where every residue degree is 1. In the wild case the exponent is larger, as $r_x \geq e_x$ is in the geometric setting.

Under this correspondence, number fields play the role of function fields of curves over finite fields, primes the role of closed points, and the discriminant the role of the branch divisor.

7.5 Hyperelliptic Curves

By theorem 7.3.8, the canonical map $\varphi_K\colon X \to \mathbb{P}^{g-1}$ of a curve of genus $g \geq 2$ is either a closed immersion or a 2-to-1 map onto a rational normal curve, and the curves of the second kind are the hyperelliptic curves (proposition 7.5.2). When $\mathrm{char} k \neq 2$, every hyperelliptic curve has an affine model $y^2 = f(x)$ (proposition 7.5.4), and much of the classical theory of curves was developed on such models.

Definition 7.5.1: Hyperelliptic Curve

A complete nonsingular curve X of genus $g \geq 2$ over $k = \overline{k}$ is *hyperelliptic* if there exists a finite morphism $\varphi\colon X \to \mathbb{P}^1$ of degree 2.

The condition $g \geq 2$ excludes $\mathbb{P}^1$, which maps to itself with degree 2 by $[s:t] \mapsto [s^2:t^2]$, and the curves of genus 1, which map to $\mathbb{P}^1$ with degree 2 by the complete linear system $|2O|$ of any closed point O (a base-point-free pencil by proposition 7.3.3) and are treated in section 7.6.

Proposition 7.5.2: Characterizations of Hyperelliptic Curves

Let X be a complete nonsingular curve of genus $g \geq 2$ over $k = \overline{k}$. The following are equivalent:

1. X is hyperelliptic (there exists a degree-2 morphism $X \to \mathbb{P}^1$).
2. There exist two closed points $P, Q \in X$ (possibly equal) with $\ell(P+Q) \geq 2$, that is, X admits a g^1_2 (a linear system of degree 2 and dimension 1).
3. The canonical linear system $|K_X|$ is *not* very ample.
4. The canonical map $\varphi_K\colon X \to \mathbb{P}^{g-1}$ is not a closed immersion.

Moreover, when these conditions hold, the g^1_2 is unique: any two degree-2 maps $X \to \mathbb{P}^1$ differ by an automorphism of $\mathbb{P}^1$.

***Proof*:**

(1) $\Rightarrow$ (2): A degree-2 morphism $\varphi\colon X \to \mathbb{P}^1$ corresponds to a base-point-free linear system of degree 2 and dimension 1. Taking $P + Q = \varphi^*(y)$ for some $y \in \mathbb{P}^1$, we get $\ell(P+Q) \geq 2$.

(2) $\Rightarrow$ (1): This is theorem 7.3.8(3).

(1) $\Leftrightarrow$ (3): The proof of theorem 7.3.8(2) shows that $\ell(K) - \ell(K - P - Q) = 3 - \ell(P+Q)$ for all closed points P, Q, so $|K|$ fails to be very ample precisely when there exist P, Q (possibly $P = Q$) with $\ell(P+Q) \geq 2$. That is condition (2), which is equivalent to (1).

(3) $\Leftrightarrow$ (4): A line bundle is very ample if and only if the associated morphism is a closed immersion.

Uniqueness of the g^1_2: Let $f, f'\colon X \to \mathbb{P}^1$ be two degree-2 morphisms, and let s, t be the pullbacks under f of the homogeneous coordinates of $\mathbb{P}^1$, so that $f = [s:t]$. The proof of theorem 7.3.8(3), applied to the g^1_2 defined by f, shows that the g monomials $s^{g-1}, s^{g-2}t, \ldots, t^{g-1}$ form a basis of $H^0(X, \omega_X)$. In this basis $\varphi_K = \nu_{g-1} \circ f$, where $\nu_{g-1}\colon \mathbb{P}^1 \to \mathbb{P}^{g-1}$ is the Veronese embedding of

degree $g-1$ (definition 3.1.38). The same holds for f' in a second basis of $H^0(X,\omega_X)$. Two bases differ by an invertible matrix, so $\nu_{g-1}\circ f' = A\circ\nu_{g-1}\circ f$ for a projective linear automorphism A of $\mathbb{P}^{g-1}$. Both f and f' are surjective, so A maps the rational normal curve $R=\nu_{g-1}(\mathbb{P}^1)$ onto itself, and $\alpha = \nu_{g-1}^{-1}\circ A|_R\circ\nu_{g-1}$ is an automorphism of $\mathbb{P}^1$ with $f'=\alpha\circ f$, completing the proof.

For the rest of section 7.5, assume $\mathrm{char}k\neq 2$. Let $\varphi\colon X\to\mathbb{P}^1$ have degree 2. The extension $K(X)/K(\mathbb{P}^1)=K(X)/k(t)$ is then separable, since a generator has minimal polynomial $T^2+c_1T+c_0$, whose derivative $2T+c_1$ is nonzero. A separable quadratic extension is Galois with group $\mathbb{Z}/2\mathbb{Z}$, and by the anti-equivalence between complete nonsingular curves and their function fields (theorem 7.1.4) its nontrivial automorphism is induced by an automorphism of X over $\mathbb{P}^1$.

Definition 7.5.3: Hyperelliptic Involution

Let X be a hyperelliptic curve with g^1_2 given by $\varphi\colon X\to\mathbb{P}^1$. The *hyperelliptic involution* is the unique nontrivial automorphism $\iota\colon X\to X$ such that $\varphi\circ\iota=\varphi$. Equivalently, ι is the nontrivial element of $\mathrm{Gal}(K(X)/k(t))\cong\mathbb{Z}/2\mathbb{Z}$. The ramification points of φ are the *Weierstrass points* of X.

For each closed point $y\in\mathbb{P}^1$, the fiber $\varphi^{-1}(y)$ is either two distinct points $\{x,\iota(x)\}$ (when $e_x=1$) or a single point fixed by ι (when $e_x=2$, a ramification point), since the Galois group acts transitively on the points over y. So the fixed points of ι are exactly the Weierstrass points. Since $\mathrm{char}k\neq 2$, Hurwitz's formula (theorem 7.4.5) holds in its tame form:

$$2g-2 = 2(0-2)+\sum(e_x-1) = -4+b$$

where b is the number of ramification points. Therefore $b=2g+2$, and a hyperelliptic curve of genus g has exactly $2g+2$ Weierstrass points.

The hyperelliptic structure gives an explicit affine model. Let y_0 generate $K(X)$ over $k(t)$, with $y_0^2+c_1y_0+c_0=0$ for some $c_0,c_1\in k(t)$. Completing the square gives $(y_0+c_1/2)^2=c_1^2/4-c_0$, and writing $c_1^2/4-c_0=fh^2$ with $f\in k[t]$ squarefree and $h\in k(t)^\times$, the element $y=(y_0+c_1/2)/h$ generates $K(X)$ over $k(t)$ and satisfies $y^2=f(t)$. The polynomial f is not constant, since k is algebraically closed and $y\notin k(t)$. The degree of f is determined by the genus:

Proposition 7.5.4: Weierstrass Models

Let X be a hyperelliptic curve of genus g over k with $\mathrm{char}k\neq 2$. Then X is birational to the affine curve

$$y^2=f(x),\qquad f\in k[x],\quad \deg f\in\{2g+1,2g+2\},\quad f\text{ squarefree}$$

Conversely, for every $g\geq 2$ and every squarefree $f\in k[x]$ of degree $2g+1$ or $2g+2$, the complete nonsingular curve with function field $k(x)[y]/(y^2-f(x))$ has genus g and is hyperelliptic. The two cases are related by a change of variable at infinity:

1. $\deg f=2g+1$: there is a single point at infinity, which is a Weierstrass point (a ramification point of φ).
2. $\deg f=2g+2$: there are two points at infinity, neither of which is a Weierstrass point (the map φ is unramified at infinity).

***Proof*:**
Let $f \in k[x]$ be squarefree and nonconstant of degree d, let $m = \lceil d/2 \rceil$, and let X be the complete nonsingular curve with function field $K = k(x)[y]/(y^2 - f(x))$, which is a field because f is not a square in $k(x)$. The construction above produces such an f for every hyperelliptic X, so it suffices to show that X has genus $m-1$ and to locate the ramification of $\varphi = x \colon X \to \mathbb{P}^1$.

Step 1: two affine charts. The affine curve $U = \operatorname{Spec} k[x,y]/(y^2 - f(x))$ is integral with function field K. By proposition 2.7.7, U is singular at a closed point (a,b) only if $2b = 0$ and $f'(a) = 0$. Then $f(a) = b^2 = 0$, so a would be a repeated root of f, and hence U is nonsingular. Put $u = 1/x$ and $v = yu^m$. Then $v^2 = F(u)$ with $F(u) = u^{2m} f(1/u) \in k[u]$, whose roots are the inverses of the nonzero roots of f together with $u = 0$ when $d = 2m-1$, all simple. So F is squarefree and $V = \operatorname{Spec} k[u,v]/(v^2 - F(u))$ is nonsingular by the same argument. Each of U and V is an open subscheme of its projective completion (theorem 7.1.3), which is a complete nonsingular curve with function field K and hence isomorphic to X (theorem 7.1.4(3)). So U and V are open in X, in particular X is birational to U, and on these charts φ is $(x,y) \mapsto [x:1]$ and $(u,v) \mapsto [1:u]$ respectively.

Step 2: ramification. For every closed point $z \in \mathbb{P}^1$, the ramification indices of the points over z sum to $\deg \varphi^*(z) = 2$ (proposition 7.1.9). Over $x = a$ with $f(a) \neq 0$, the chart U contains the two points $(a, \pm\sqrt{f(a)})$, distinct because $\operatorname{char} k \neq 2$, so both are unramified. Over a root a of f, write $f = (x-a)q(x)$ with $q(a) \neq 0$. In the local ring of U at $(a,0)$ we have $x - a = y^2/q(x)$ with $q(x)$ a unit, so the maximal ideal $(x-a, y)$ is generated by y, and $x - a$ has valuation 2. So $e_{(a,0)} = 2$, and $(a,0)$ is the only point over a. The same two computations in the chart V over $u = 0$ show that φ has two unramified points over ∞ when $F(0) \neq 0$, that is, when $d = 2m$, and a single ramification point over ∞ when $d = 2m-1$.

Step 3: the genus. The ramification points are the d roots of f, together with the point at infinity when d is odd, so there are $b = 2m$ of them. As $\operatorname{char} k \neq 2$, the tame form of theorem 7.4.5 gives $2g(X) - 2 = 2(0-2) + 2m$, so $g(X) = m - 1$. Hence $g(X) = g$ exactly when $d \in \{2g+1, 2g+2\}$, and for $g \geq 2$ the map φ makes X hyperelliptic. Step 2 gives the description of the points at infinity, as we sought to show.

Example 7.5.5: Hyperelliptic Curves in Practice

1. Every curve of genus 2 is hyperelliptic (example 7.3.7). By proposition 7.5.4, X has a model $y^2 = f(x)$ with f squarefree of degree 5 or 6, for instance $y^2 = x^5 - x$.
2. A hyperelliptic curve of genus 3 has a model $y^2 = f(x)$ with f squarefree of degree 7 or 8. Its canonical map is $X \xrightarrow{2:1} \mathbb{P}^1 \xrightarrow{\nu_2} \mathbb{P}^2$, with ν_2 the Veronese embedding, so $\varphi_K(X)$ is a conic in $\mathbb{P}^2$. A curve of genus 3 that is not hyperelliptic is a smooth quartic in $\mathbb{P}^2$ (example 7.3.7).
3. Over a finite field $\mathbb{F}_q$ with q odd, an equation $y^2 = f(x)$ with $f \in \mathbb{F}_q[x]$ squarefree defines a curve over $\mathbb{F}_q$, and its number of $\mathbb{F}_q$-points obeys Weil's bound (theorem 8.7.12, proved in chapter 8). The groups of $\mathbb{F}_q$-points of the Jacobians of such curves are used in discrete-logarithm cryptography.

Let D_0 be a divisor in the unique g^1_2, for instance $D_0 = \varphi^*(y)$ for a closed point $y \in \mathbb{P}^1$, so that $\deg D_0 = 2$ and $\ell(D_0) = 2$.

Proposition 7.5.6: Canonical Class of a Hyperelliptic Curve

$K_X \sim (g-1)D_0$. In particular, the canonical linear system is $|K_X| = |(g-1)D_0|$.

***Proof*:**
By theorem 7.3.8(3) applied to $D_0 = P + Q$, $K_X \sim (g-1)D_0$, and linearly equivalent divisors have the same complete linear system, as claimed.

The canonical map therefore factors as

$$X \xrightarrow[2:1]{\varphi} \mathbb{P}^1 \xhookrightarrow{\nu_{g-1}} \mathbb{P}^{g-1}, \qquad \varphi_K = \nu_{g-1} \circ \varphi$$

where $\nu_{g-1}\colon \mathbb{P}^1 \hookrightarrow \mathbb{P}^{g-1}$ is the Veronese embedding of degree $g-1$ (definition 3.1.38), mapping $[s:t] \mapsto [s^{g-1} : s^{g-2}t : \cdots : t^{g-1}]$. The image $\nu_{g-1}(\mathbb{P}^1) \subseteq \mathbb{P}^{g-1}$ is the rational normal curve of degree $g-1$, and $\varphi_K(X)$ is this rational normal curve, with φ_K being 2-to-1 onto it.

7.6 Elliptic Curves

A curve of genus 1 with a chosen k-point carries a group law, and this section derives it from Riemann-Roch. Section 5.4 identified Pic^0 of a complex elliptic curve $\mathbb{C}/\Lambda$ with the curve itself through Abel's theorem, and example 4.1.12 recorded that elliptic curves are group schemes. Here the identification $E(k) \cong \mathrm{Pic}^0(E)$ is proved over an arbitrary algebraically closed field.

Definition 7.6.1: Elliptic Curve

An *elliptic curve* over k is a pair (E, O) where E is a complete nonsingular curve of genus 1 over k, and $O \in E(k)$ is a chosen k-rational point, called the *origin* or *identity*.

Over $k = \overline{k}$ every closed point is a k-point, so every curve of genus 1 becomes an elliptic curve once a point is chosen. Over a field that is not algebraically closed, a curve of genus 1 can have no k-point at all[5], and it is then not an elliptic curve over k.

Proposition 7.6.2: Elliptic Curves are Plane Cubics

Let (E, O) be an elliptic curve over k. Then the complete linear system $|3O|$ is very ample and embeds E as a smooth cubic curve in $\mathbb{P}^2_k$.

***Proof*:**
Since $g = 1$, proposition 7.3.1 gives $\deg K_E = 2g - 2 = 0$ and $\ell(K_E) = g = 1$, so $K_E \sim 0$ by lemma 7.2.2(2), that is, $\omega_E \cong \mathcal{O}_E$. Hence $\ell(K_E - D) = \ell(-D) = 0$ whenever $\deg D > 0$ (lemma 7.2.2(1)), and Riemann-Roch (theorem 7.2.10) gives $\ell(nO) = n + 1 - 1 = n$ for every

[5] For example, the Selmer curve $3x^3 + 4y^3 + 5z^3 = 0$ in $\mathbb{P}^2_{\mathbb{Q}}$ has genus 1 and no $\mathbb{Q}$-rational point, a theorem of Selmer. It is a *torsor* (principal homogeneous space) for its Jacobian, an elliptic curve over $\mathbb{Q}$ with its natural origin (section 7.7.2).

$n \geq 1$. In particular $\ell(O) = 1$, $\ell(2O) = 2$ and $\ell(3O) = 3$. As $\deg(3O) = 3 = 2g+1$, corollary 7.3.4 shows that $|3O|$ is very ample, so it defines a closed immersion $E \hookrightarrow \mathbb{P}_k^{\ell(3O)-1} = \mathbb{P}_k^2$. Its image is isomorphic to E, hence nonsingular, and has degree $\deg(3O) = 3$, so it is a smooth plane cubic, as claimed.

In coordinates, the embedding comes from functions with poles only at O:

- $H^0(E, \mathcal{L}(O))$ is spanned by the constant function 1, since $\ell(O) = 1$.
- Since $\ell(2O) = 2 > \ell(O)$, there is a function $x \in H^0(E, \mathcal{L}(2O))$ with a pole of order exactly 2 at O and no other poles, and $1, x$ is a basis.
- Since $\ell(3O) = 3 > \ell(2O)$, there is a function $y \in H^0(E, \mathcal{L}(3O))$ with a pole of order exactly 3 at O, and $1, x, y$ is a basis.

The space $H^0(E, \mathcal{L}(6O))$ has dimension $\ell(6O) = 6$ and contains the seven functions $1, x, y, x^2, xy, x^3, y^2$, whose pole orders at O are $0, 2, 3, 4, 5, 6, 6$. So they satisfy a nontrivial linear relation. Functions with distinct pole orders are linearly independent, so both x^3 and y^2 occur in the relation with nonzero coefficients. Replacing x and y by suitable nonzero scalar multiples and dividing by a constant puts the relation in the form

$$y^2 + a_1xy + a_3y \;=\; x^3 + a_2x^2 + a_4x + a_6 \tag{7.6}$$

This is the *generalized Weierstrass equation*. In the basis $1, x, y$ of $H^0(E, \mathcal{L}(3O))$, the embedding of proposition 7.6.2 is $P \mapsto [x(P) : y(P) : 1]$, its image is the plane cubic obtained by homogenizing (7.6) with a variable z, and O maps to $[0 : 1 : 0]$.

When $\mathrm{char} k \neq 2$, completing the square in y removes the xy and y terms. Grouping the y terms on the left gives

$$y^2 + (a_1x + a_3)y = x^3 + a_2x^2 + a_4x + a_6$$

and adding $\left(\frac{a_1x+a_3}{2}\right)^2$ to both sides gives

$$\left(y + \frac{a_1x + a_3}{2}\right)^2 = x^3 + a_2x^2 + a_4x + a_6 + \frac{(a_1x + a_3)^2}{4}$$

With $y_1 = y + \frac{a_1x+a_3}{2}$, expanding the right side and grouping by powers of x yields

$$y_1^2 = x^3 + \left(a_2 + \frac{a_1^2}{4}\right)x^2 + \left(a_4 + \frac{a_1a_3}{2}\right)x + \left(a_6 + \frac{a_3^2}{4}\right)$$

Call the new coefficients a_2', a_4' and a_6', so that the equation becomes

$$y_1^2 = x^3 + a_2'x^2 + a_4'x + a_6'$$

If also $\mathrm{char} k \neq 3$, the substitution $x = x_1 - \frac{a_2'}{3}$ completes the cube in x:

$$y_1^2 = \left(x_1 - \frac{a_2'}{3}\right)^3 + a_2'\left(x_1 - \frac{a_2'}{3}\right)^2 + a_4'\left(x_1 - \frac{a_2'}{3}\right) + a_6'$$

Expanding the cubic and quadratic terms gives

$$y_1^2 = \left(x_1^3 - a_2'x_1^2 + \frac{a_2'^2}{3}x_1 - \frac{a_2'^3}{27}\right) + \left(a_2'x_1^2 - \frac{2a_2'^2}{3}x_1 + \frac{a_2'^3}{9}\right) + a_4'\left(x_1 - \frac{a_2'}{3}\right) + a_6'$$

The x_1^2 terms cancel, and grouping the remaining coefficients of x_1 and the constants leaves an equation of the form

$$y_1^2 = x_1^3 + Ax_1 + B$$

Dropping the subscripts, this transforms (7.6) into the *short Weierstrass form*

$$y^2 = x^3 + Ax + B, \qquad \Delta = -16(4A^3 + 27B^2) \neq 0$$

The condition $\Delta \neq 0$ is equivalent to the cubic having no repeated roots, which is equivalent to the curve being nonsingular. In characteristic 2 completing the square divides by 2, and in characteristic 3 completing the cube divides by 3, so there the short form is not available in general and one works with (7.6).

The group structure on $E(k)$ comes from the theory of divisors, by identifying $E(k)$ with $\mathrm{Pic}^0(E)$ through $P \mapsto [P - O]$.

Theorem 7.6.3: The Group Law on an Elliptic Curve

Let (E, O) be an elliptic curve over $k = \overline{k}$. The map

$$\sigma\colon E(k) \to \mathrm{Pic}^0(E), \qquad P \mapsto \text{class of } [P] - [O]$$

is a bijection. Transporting the group structure from $\mathrm{Pic}^0(E)$ to $E(k)$, the curve E becomes an abelian group with identity O, where:

1. The addition law $P + Q = R$ is characterized by $[P] + [Q] - [R] \sim [O]$. In the embedding of proposition 7.6.2, let R' be the third point of intersection of E with the line through P and Q (the tangent line if $P = Q$), so that $P + Q + R' \sim 3O$. Then R is the third point of intersection of E with the line through R' and O (the tangent line at O if $R' = O$).
2. The inverse $-P$ is the third point of intersection of E with the line through P and O (the tangent line at O if $P = O$).
3. The maps $+\colon E \times E \to E$ and $[-1]\colon E \to E$ are morphisms of k-schemes.

In particular, (E, O) is a group scheme over k (definition 4.1.1).

***Proof*:**

As in the proof of proposition 7.6.2, $K_E \sim 0$, so Riemann-Roch reads $\ell(D) = \deg D + \ell(-D)$, and $\ell(-D) = 0$ when $\deg D > 0$.

Bijectivity of σ: Injectivity: if $[P - O] = [Q - O]$ in $\mathrm{Pic}^0(E)$, then $P \sim Q$, so P is an effective divisor in the complete linear system $|Q|$. But $\ell(Q) = 1 + \ell(-Q) = 1$, so $|Q|$ has dimension $\ell(Q) - 1 = 0$ and consists of exactly one effective divisor, which is Q itself. Thus $P = Q$.

Surjectivity: let $D \in \mathrm{Pic}^0(E)$, that is, D is a divisor of degree 0 modulo linear equivalence. Then $D + O$ has degree 1, so $\ell(D + O) = 1 + \ell(-D - O) = 1$, since $\deg(-D - O) = -1 < 0$. So $|D + O|$ contains a unique effective divisor, which is a single point P (an effective divisor of degree 1 on a curve is a single point). Thus $D + [O] \sim [P]$, that is, $D \sim [P] - [O] = \sigma(P)$.

The geometric addition law: The chord-tangent description in (1) follows from the fact that three collinear points P, Q, R' on the plane cubic satisfy $P + Q + R' \sim 3O$: they are the

intersection of E with a line L, and L is linearly equivalent to the line $z = 0$, which meets E in $3O$ because z corresponds to the section 1 of $\mathcal{L}(3O)$. From $P + Q + R' \sim 3O$ we get $\sigma(P) + \sigma(Q) = [P] + [Q] - 2[O] = [3O - R'] - 2[O] = [O] - [R']$, and we need $\sigma(R) = [O] - [R']$, that is, $[R] - [O] = [O] - [R']$, giving $R + R' \sim 2O$. The line through O and R' meets E in $O + R' + R''$ with $O + R' + R'' \sim 3O$, so $R'' \sim 2O - R' \sim R$ and $R'' = R$ by injectivity of σ. For (2), the line through P and O meets E in $P + O + R''$ with $P + O + R'' \sim 3O$, so $\sigma(R'') = [O] - [P] = -\sigma(P)$ and $R'' = -P$.

Morphism property: Part (3) is not proved here. On the model (7.6), the line through $P = (x_0, y_0)$ and O is $x = x_0$, so by (2) the map $[-1]$ is the restriction of the linear automorphism $[x : y : z] \mapsto [x : -y - a_1x - a_3z : z]$ of $\mathbb{P}^2$, which preserves E, so it is a morphism. The chord-tangent construction expresses the coordinates of $P + Q$ as rational functions of those of P and Q, and showing that these define a morphism on all of the surface $E \times E$ is a separate argument, given in [69, Theorem III.3.6]. A second route goes through the functor of points (section 3.3): the divisor-theoretic construction, carried out for T-valued points (definition 3.3.1), needs the relative Picard functor of [20, Grothendieck's Theorem: the Picard Scheme of a Projective Variety], and the Yoneda lemma (theorem 0.4.22) then produces the morphism $E \times E \to E$. Granting (3), (E, O) is a group scheme, completing the proof.

The group law on an elliptic curve is *commutative*, because $\mathrm{Pic}^0(E)$ is an abelian group. An elliptic curve is therefore a smooth, connected, projective group scheme of dimension 1 over k, that is, an abelian variety of dimension 1 (*Abelian Varieties* [5]).

The chord-tangent construction also applies to the smooth locus of an irreducible singular plane cubic, and there it produces a linear algebraic group. Elliptic curves are smooth by definition, so the singular objects here are called singular cubic curves.

Let $C \subseteq \mathbb{P}^2_k$ be an irreducible cubic curve with a singular point P_{sing}, and assume $\mathrm{char} k \neq 2, 3$. A line not contained in C meets C in three points counted with multiplicity, and with multiplicity at least 2 at each singular point. So P_{sing} is the only singular point, and a line through two smooth points of C (or tangent at one) meets C in a third point that is again smooth. The chord-tangent recipe of theorem 7.6.3 therefore makes sense on the smooth locus $C_{\mathrm{reg}} = C \setminus \{P_{\mathrm{sing}}\}$, with a chosen smooth point O as identity. What changes is compactness: C_{reg} is not proper. The identification with the Picard group survives once divisors are taken to be Cartier divisors: $P \mapsto [P - O]$ is a bijection from C_{reg} onto the group $\mathrm{Pic}^0(C)$ of degree-zero line bundles on C, and the group law it transports is the chord-tangent law ([36, Example II.6.11.4] for the cusp, [36, Exercise II.6.7] for the node). The resulting group depends on the singularity type:

1. Node ($C\colon y^2 = x^3 + x^2$, say): the normalization of C is $\mathbb{P}^1$, and C is obtained from it by identifying the two preimages of the node. Removing the singular point leaves $C_{\mathrm{reg}} \cong \mathbb{P}^1 \setminus \{0, \infty\} \cong \mathbb{G}_m = \mathrm{Spec}\, k[t, t^{-1}]$, the *multiplicative group* (example 4.1.5), and the group law on the smooth locus is isomorphic to $(k^\times, \cdot)$.
2. Cusp ($C\colon y^2 = x^3$): the normalization is $\mathbb{P}^1$ with a single preimage of the cusp, and removing the singular point gives $C_{\mathrm{reg}} \cong \mathbb{G}_a = \mathrm{Spec}\, k[t]$, the *additive group* (example 4.1.4). The group law on the smooth locus is isomorphic to $(k, +)$.

Along a degeneration of an elliptic curve to a nodal cubic, and of a nodal cubic to a cuspidal one, the group of smooth points changes as

$$\underbrace{E}_{\text{abelian variety}} \rightsquigarrow \underbrace{\mathbb{G}_m}_{\text{node}} \rightsquigarrow \underbrace{\mathbb{G}_a}_{\text{cusp}}$$

The same trichotomy classifies reduction. Let E be an elliptic curve over a local field K, say $K = \mathbb{Q}_p$, given by a minimal Weierstrass equation. Reducing its coefficients modulo p gives a cubic curve $\overline{E}$ over the residue field $\mathbb{F}_p$, and the reduction type of E is determined by $\overline{E}$ and its smooth locus $\overline{E}_{\mathrm{reg}}$ ([15, Reduction Types]):

1. *Good reduction:* $\overline{E}$ is smooth, and $\overline{E}_{\mathrm{reg}} = \overline{E}$ is an elliptic curve over $\mathbb{F}_p$.
2. *Multiplicative (semistable) reduction:* $\overline{E}$ has a node, and $\overline{E}_{\mathrm{reg}}$ becomes isomorphic to $\mathbb{G}_m$ over $\mathbb{F}_{p^2}$. The reduction is *split* when $\overline{E}_{\mathrm{reg}} \cong \mathbb{G}_m$ over $\mathbb{F}_p$ and *non-split* otherwise, in which case $\overline{E}_{\mathrm{reg}}$ is a twist of $\mathbb{G}_m$.
3. *Additive (unstable) reduction:* $\overline{E}$ has a cusp, and $\overline{E}_{\mathrm{reg}} \cong \mathbb{G}_a$.

The *Néron model* of E over $\mathbb{Z}_p$ is a smooth group scheme over $\mathbb{Z}_p$ with generic fiber E, and the identity component of its special fiber is $\overline{E}_{\mathrm{reg}}$ ([68, Corollary IV.9.1]). The *conductor* of E records the reduction type: its exponent at p is 0, 1, or at least 2 according as the reduction is good, multiplicative or additive [15, The Conductor via the Galois Representation].

Different Weierstrass equations can define isomorphic elliptic curves, and over $\overline{k}$ a single invariant determines the isomorphism class.

Definition 7.6.4: The j-Invariant

Let E be an elliptic curve over k with $\mathrm{char} k \neq 2, 3$, given in short Weierstrass form $y^2 = x^3 + Ax + B$. The *j-invariant* of E is

$$j(E) \;=\; -1728 \cdot \frac{(4A)^3}{\Delta} \;=\; -1728 \cdot \frac{64A^3}{-16(4A^3 + 27B^2)} \;=\; 1728 \cdot \frac{4A^3}{4A^3 + 27B^2}$$

With the factor $1728 = 12^3$, the j-invariant equals c_4^3/Δ, where c_4 and Δ are polynomials with integer coefficients in the a_i of (7.6) (for the short form, $c_4 = -48A$). The formula $j = c_4^3/\Delta$ defines j in every characteristic, including 2 and 3 [15, The c-Invariants, The j-Invariant].

Proposition 7.6.5: j-Invariant Classifies Elliptic Curves

Let $k = \overline{k}$ with $\mathrm{char} k \neq 2, 3$.

1. Two elliptic curves E_1, E_2 over k are isomorphic (as curves with marked point) if and only if $j(E_1) = j(E_2)$.
2. For every $j_0 \in k$, there exists an elliptic curve E over k with $j(E) = j_0$.

In other words, the j-invariant defines a bijection between isomorphism classes of elliptic curves over k and the elements of k.

***Proof*:**
By the reduction above, every elliptic curve over k has a short Weierstrass model.

(1) Let $E\colon y^2 = x^3 + Ax + B$ and $E'\colon y'^2 = x'^3 + A'x' + B'$, and let $\psi\colon E \to E'$ be an isomorphism with $\psi(O) = O'$. Pulling back along ψ maps $H^0(E', \mathcal{L}(nO'))$ isomorphically onto $H^0(E, \mathcal{L}(nO))$ for every n. So $\psi^* x' = \alpha x + r$ and $\psi^* y' = \beta y + sx + t$ with $\alpha, \beta \in k^\times$ and $r, s, t \in k$. Substituting

into $y'^2 = x'^3 + A'x' + B'$ and using $y^2 = x^3 + Ax + B$ gives a relation among $1, x, x^2, x^3, y, xy$, which have the distinct pole orders $0, 2, 4, 6, 3, 5$ at O and are therefore linearly independent. Comparing coefficients of xy, y, x^3 and x^2 gives $2\beta s = 0$, $2\beta t = 0$, $\beta^2 = \alpha^3$ and $s^2 = 3\alpha^2 r$. As $\mathrm{char} k \neq 2, 3$, this forces $s = t = 0$ and then $r = 0$, and with $u = \beta/\alpha$ we get $\alpha = u^2$ and $\beta = u^3$. The coefficients of x and 1 then give $A' = u^4 A$ and $B' = u^6 B$, so

$$j(E') = 1728 \cdot \frac{4(u^4A)^3}{4(u^4A)^3 + 27(u^6B)^2} = 1728 \cdot \frac{4A^3u^{12}}{(4A^3 + 27B^2)u^{12}} = j(E)$$

Conversely, suppose $j(E) = j(E')$. Then $A^3B'^2 = A'^3B^2$, and $j = 0$ exactly when $A = 0$, while $j = 1728$ exactly when $B = 0$. If $A = A' = 0$, then $BB' \neq 0$, and we choose u with $u^6 = B'/B$. If $B = B' = 0$, then $AA' \neq 0$, and we choose u with $u^4 = A'/A$. If $ABA'B' \neq 0$, we choose u with $u^4 = A'/A$. Then $u^{12} = (A'/A)^3 = (B'/B)^2$, so $u^6 = \pm B'/B$, and replacing u by iu with $i^2 = -1$ if necessary gives $u^6 = B'/B$ without changing u^4. In each case $A' = u^4A$ and $B' = u^6B$, and $[x : y : z] \mapsto [u^2x : u^3y : z]$ is an isomorphism $E \to E'$ sending O to O'.

(2) For $j_0 \neq 0, 1728$, take $A = 3j_0(1728 - j_0)$ and $B = 2j_0(1728 - j_0)^2$. Then $4A^3 = 108j_0^3(1728 - j_0)^3$ and $27B^2 = 108j_0^2(1728 - j_0)^4$, so $4A^3 + 27B^2 = 108 \cdot 1728\, j_0^2(1728 - j_0)^3 \neq 0$ and $j = j_0$. For $j_0 = 0$, take $y^2 = x^3 + 1$, and for $j_0 = 1728$, take $y^2 = x^3 + x$. Both are nonsingular because $\mathrm{char} k \neq 2, 3$, completing the proof.

Over a perfect field k of characteristic different from 2 and 3 that is not algebraically closed, equality of j-invariants is necessary but need not be sufficient for isomorphism over k. Two elliptic curves over k with the same j-invariant become isomorphic over $\overline{k}$, and the curves over k that become isomorphic to E over $\overline{k}$, the *twists* of E, are classified by $H^1(\mathrm{Gal}(\overline{k}/k), \mathrm{Aut}_{\overline{k}}(E))$ [15, Twists are Classified by H^1].

7.6.6 The j-Line as $\mathcal{M}_{1,1}$ Since elliptic curves over $\overline{k}$ are classified by a single invariant $j \in k$, the coarse moduli space of elliptic curves is the affine line: $\mathcal{M}_{1,1}^{\mathrm{coarse}} \cong \mathbb{A}^1_k$. Elliptic curves have nontrivial automorphisms (at least the inversion $[-1]$, and more at $j = 0$ and $j = 1728$ when $\mathrm{char} k \neq 2, 3$), so the fine moduli space does not exist as a scheme. It exists as a Deligne-Mumford stack $\mathcal{M}_{1,1}$, whose coarse space is $\mathbb{A}^1$, a basic example of a moduli stack in *Moduli, Deformation Theory, and Stacks* [20].

Theorem 7.6.3 identifies $E(k)$ with $\mathrm{Pic}^0(E)$. For a smooth projective curve X of genus g, the group $\mathrm{Pic}^0(X)$ is the group of k-points of an abelian variety $\mathrm{Jac}(X)$ of dimension g, called the *Jacobian* of X:

$$\mathrm{Jac}(X)(k) \;\cong\; \mathrm{Pic}^0(X)$$

For $g = 1$ with a marked point O, the Jacobian is isomorphic to (X, O) itself, and on k-points this is theorem 7.6.3. For $g \geq 2$, the *Abel-Jacobi map*

$$\alpha \colon X \hookrightarrow \mathrm{Jac}(X), \qquad P \mapsto [P - O] \quad \text{(for a choice of base point } O\text{)}$$

is a closed immersion, and every morphism from X to an abelian variety A sending O to 0 factors uniquely as $h \circ \alpha$ with $h \colon \mathrm{Jac}(X) \to A$ a homomorphism (the *Albanese property* of the Jacobian, proved over $\mathbb{C}$ in [2, Theorem 11.4.1]). The other facts are proved in [5, The Jacobian Is an Abelian Variety, The Jacobian of an Elliptic Curve, The Curve Embeds in Its Jacobian].

7.6.7 Forward Reference: The Picard Scheme and Moduli The group $\mathrm{Pic}^0(X)$ is the group of k-points of a scheme. For a projective, geometrically integral k-scheme X with a k-point, the relative Picard functor $T \mapsto \mathrm{Pic}(X_T)/\pi_T^*\mathrm{Pic}(T)$ is represented by a group scheme $\mathrm{Pic}_{X/k}$, the *Picard scheme* ([20, Grothendieck's Theorem: the Picard Scheme of a Projective Variety]), and for a smooth projective curve its identity component $\mathrm{Pic}^0_{X/k}$ is an abelian variety, the Jacobian.
Through the Jacobian, curves give principally polarized abelian varieties: the Torelli map $X \mapsto (\mathrm{Jac}(X), \Theta)$, where Θ is the theta divisor, is injective on isomorphism classes of curves of genus $g \geq 1$ [5, The Torelli Theorem].
The Néron-Severi group $\mathrm{NS}(X) = \mathrm{Pic}(X)/\mathrm{Pic}^0(X)$ of a curve is isomorphic to $\mathbb{Z}$ through the degree. For surfaces and higher-dimensional varieties it is a finitely generated abelian group that can have large rank, and for surfaces it is studied in section 8.1.

The nonconstant morphisms between elliptic curves that respect the origins are the isogenies.

Definition 7.6.8: Isogeny

An *isogeny* $\varphi\colon E_1 \to E_2$ between elliptic curves is a morphism of schemes that sends O_1 to O_2 and is nonconstant (hence surjective and finite, by proposition 7.1.6).

Every isogeny is a homomorphism of group schemes, $\varphi(P+Q) = \varphi(P) + \varphi(Q)$ [15, Isogenies are Group Homomorphisms][6].

Proposition 7.6.9: Multiplication by n

For any integer $n \neq 0$, the map $[n]\colon E \to E$ defined by $P \mapsto \underbrace{P + P + \cdots + P}_{n}$ (and $[-n] = [n] \circ [-1]$) is a finite morphism of degree n^2.

The map $[n]$ is a morphism by theorem 7.6.3(3). That its degree is n^2 in every characteristic, including $\mathrm{char} k \mid n$, is proved in [15, Degree of the Multiplication Map]. In particular $[n]$ is nonconstant, hence finite and surjective by proposition 7.1.6.

The *n-torsion subgroup* $E[n]$ is the kernel of $[n]$. As a group scheme it is finite of order (rank) $\deg[n] = n^2$, and its structure depends on the characteristic:

[6] The argument is rigidity. The map $h(P,Q) = \varphi(P+Q) - \varphi(P) - \varphi(Q)$ is a morphism $E_1 \times E_1 \to E_2$ by theorem 7.6.3(3), and $h(P, O_1) = O_2$ for every P. By the rigidity lemma [5, Rigidity Lemma], $h(P,Q) = h(O_1, Q) = O_2$ for all P, Q.

Proposition 7.6.10: Structure of the n-Torsion

Let E be an elliptic curve over $k = \overline{k}$.

1. If $\mathrm{char} k \nmid n$ (including $\mathrm{char} k = 0$): $E[n](\overline{k}) \cong (\mathbb{Z}/n\mathbb{Z})^2$ as abstract groups (and $E[n]$ is an étale group scheme over k).
2. If $\mathrm{char} k = p$ and $n = p^m$: the group scheme $E[p^m]$ has rank p^{2m}, but its group of $\overline{k}$-points is either $E[p^m](\overline{k}) \cong \mathbb{Z}/p^m\mathbb{Z}$ (the *ordinary* case) or $E[p^m](\overline{k}) \cong \{0\}$ (the *supersingular* case), depending on the Hasse invariant of E. In the supersingular case, the group scheme is connected (supported entirely at the origin).

The statements about $\overline{k}$-points are [15, Structure of the n-Torsion, p-Torsion Structure]. For $n = p^m n'$ with $p \nmid n'$ they combine to give $E[n](\overline{k}) \cong E[p^m](\overline{k}) \times E[n'](\overline{k})$.

The *endomorphism ring* $\mathrm{End}(E) = \mathrm{Hom}(E, E)$, consisting of the isogenies $E \to E$ and the zero map, is a ring under addition and composition, and it has one of three forms:

Proposition 7.6.11: Endomorphism Ring of an Elliptic Curve

Let E be an elliptic curve over $k = \overline{k}$. Then $\mathrm{End}(E)$ is isomorphic to one of the following:

1. $\mathbb{Z}$ (the generic case).
2. An order in an imaginary quadratic field $\mathbb{Q}(\sqrt{-d})$ (the *complex multiplication* or *CM* case). Over $\mathbb{C}$, with $E \cong \mathbb{C}/\Lambda$, this occurs exactly when the ratio τ of a basis of Λ is an imaginary quadratic number.
3. An order in a definite quaternion algebra over $\mathbb{Q}$. This occurs only in characteristic $p > 0$, exactly for the *supersingular* elliptic curves, and for $k = \overline{\mathbb{F}}_p$ the order is maximal in the quaternion algebra ramified exactly at p and ∞.

These are [15, Classification of Endomorphism Rings (Algebraic), Endomorphism Ring of Supersingular Curves, Classification of Endomorphism Rings (Analytic)].

7.6.12 Forward Reference: The Tate Module and ℓ-adic Representations Let E be an elliptic curve over a perfect field k, not necessarily algebraically closed, and fix a prime $\ell \neq \mathrm{char} k$. The *$\ell$-adic Tate module* of E is the limit

$$T_\ell(E) \;=\; \varprojlim_m E[\ell^m](\overline{k}) \;\cong\; \mathbb{Z}_\ell^2$$

where the isomorphism comes from proposition 7.6.10(1). It is a free $\mathbb{Z}_\ell$-module of rank 2 carrying a continuous action of $\mathrm{Gal}(\overline{k}/k)$, and the resulting representation $\rho_\ell\colon \mathrm{Gal}(\overline{k}/k) \to \mathrm{GL}_2(\mathbb{Z}_\ell)$ is a basic example of an *ℓ-adic Galois representation*.
When $k = \mathbb{F}_q$ is a finite field, the Frobenius element $\mathrm{Frob}_q \in \mathrm{Gal}(\overline{\mathbb{F}}_q/\mathbb{F}_q)$ acts on $T_\ell(E)$ as the q-power Frobenius endomorphism of E does, with characteristic polynomial $t^2 - a_q t + q$, where $a_q = q + 1 - |E(\mathbb{F}_q)|$ [15, Characteristic Polynomial of Frobenius, Determinant of an Endomorphism]. The *Riemann hypothesis for elliptic curves over finite fields*, proved by Hasse, states that $|a_q| \leq 2\sqrt{q}$ [15, Hasse's Theorem], or equivalently that both roots of $t^2 - a_q t + q$ have absolute value $\sqrt{q}$. It is the genus-1 case of the Weil conjectures, which are developed, with Deligne's proof, in *Étale Cohomology* [16].
For a curve X of higher genus g, the Tate module is replaced by the ℓ-adic cohomology group $H^1_{\text{ét}}(X_{\overline{k}}, \mathbb{Z}_\ell)$, a free $\mathbb{Z}_\ell$-module of rank $2g$. The Riemann hypothesis part of the Weil conjectures, proved by Deligne for smooth projective varieties over $\mathbb{F}_q$ of every dimension, states that the eigenvalues of Frobenius on $H^i_{\text{ét}}(X_{\overline{\mathbb{F}}_q}, \mathbb{Q}_\ell)$ have absolute value $q^{i/2}$. For $i = 1$ and $X = E$ this is Hasse's bound.

Example 7.6.13: Elliptic Curves over Number Fields
If E is an elliptic curve over a number field K (for example $K = \mathbb{Q}$), the group $E(K)$ of K-rational points is a finitely generated abelian group by the *Mordell-Weil theorem* [15, The Mordell-Weil Theorem]:

$$E(K) \;\cong\; \mathbb{Z}^r \oplus E(K)_{\text{tors}}$$

where $r \geq 0$ is the *Mordell-Weil rank* and $E(K)_{\text{tors}}$ is finite. No algorithm is known that provably computes r for every E. The *Birch and Swinnerton-Dyer conjecture* (one of the Millennium Prize Problems) predicts that r equals the order of vanishing of the L-function $L(E, s)$ at $s = 1$.

7.7 Further Topics on Curves

Four topics complete the account of curves: Castelnuovo's bound on the genus of a curve in $\mathbb{P}^r$, the behavior of cohomology in families of curves, the stable curves that compactify the moduli of curves, and curves over fields that are not algebraically closed.

The embedding theorem corollary 7.3.4 embeds every curve in some projective space, but it does not say which pairs (d, g) of degree and genus occur for curves in a given $\mathbb{P}^r$. Castelnuovo's bound gives the largest genus of a nondegenerate curve of degree d in $\mathbb{P}^r$.

Theorem 7.7.1: Castelnuovo's Bound

Let $X \subseteq \mathbb{P}^r$, $r \geq 2$, be a nondegenerate curve of degree d, that is, a curve not contained in any hyperplane. Write $d-1 = m(r-1)+\epsilon$ with $0 \leq \epsilon \leq r-2$ (Euclidean division). Then:

$$g(X) \;\leq\; \pi(d,r) \;:=\; \binom{m}{2}(r-1) + m\epsilon$$

The bound $\pi(d,r)$ is called the *Castelnuovo bound*, and curves attaining it are called *Castelnuovo curves* or *curves of maximal genus*. The bound is attained for every $d \geq r$, and for $d \geq 2r+1$ every Castelnuovo curve lies on a surface of degree $r-1$ in $\mathbb{P}^r$, the least degree of an irreducible surface not contained in a hyperplane.

Example 7.7.2: Castelnuovo's Bound in Low Dimension

1. For plane curves ($r=2$), $d-1 = m\cdot 1 + 0$, so $\pi(d,2) = \binom{d-1}{2} = \frac{(d-1)(d-2)}{2}$, the genus of a smooth plane curve of degree d (proposition 7.2.6). So smooth plane curves are Castelnuovo curves.
2. For space curves ($r=3$), $d-1 = 2m+\epsilon$ with $\epsilon \in \{0,1\}$. For a curve of degree 6 in $\mathbb{P}^3$, $d-1 = 5 = 2\cdot 2+1$ gives $\pi(6,3) = \binom{2}{2}\cdot 2 + 2\cdot 1 = 4$. So a nondegenerate curve of degree 6 in $\mathbb{P}^3$ has genus at most 4, and the canonical curve of a non-hyperelliptic curve of genus 4, of degree $2g-2=6$ in $\mathbb{P}^{g-1}=\mathbb{P}^3$ (theorem 7.3.8), attains the bound.

The proof chooses a hyperplane section D of X whose d points are in linearly general position, bounds $\ell(nD) - \ell((n-1)D)$ from below for each $n \geq 1$, and compares the sum with Riemann-Roch for large n. The case $r=3$ is [36, Theorem IV.6.4], and the general case is [1, Chapter III]. The Hilbert schemes that parametrize curves in $\mathbb{P}^r$ are constructed in *Moduli, Deformation Theory, and Stacks* [20].

7.7.1 Riemann-Roch in Families and Semicontinuity

The cohomology of line bundles on curves can be studied in *families*. Let $f\colon \mathcal{X} \to S$ be a flat projective morphism of Noetherian schemes whose geometric fibers are smooth curves of genus g, and let $\mathcal{L}$ be a line bundle on $\mathcal{X}$. Each point $s \in S$ gives a curve $X_s = f^{-1}(s)$ and a line bundle $\mathcal{L}_s = \mathcal{L}|_{X_s}$, and the question is how $h^0(X_s, \mathcal{L}_s) = \ell(\mathcal{L}_s)$ varies with s.

The Euler characteristic is locally constant. Both $\mathcal{L}$ and $\mathcal{O}_{\mathcal{X}}$ are flat over S, so by theorem 6.8.17(2) the functions $s \mapsto \chi(\mathcal{L}_s)$ and $s \mapsto \chi(\mathcal{O}_{X_s}) = 1-g$ are locally constant on S, and hence so is $\deg \mathcal{L}_s = \chi(\mathcal{L}_s) - \chi(\mathcal{O}_{X_s})$, by Riemann-Roch on the geometric fibers. The individual dimensions $h^i(X_s, \mathcal{L}_s)$ can jump, in a controlled direction:

1. By theorem 6.8.17(1), the function $s \mapsto h^i(X_s, \mathcal{L}_s)$ is upper semicontinuous: for each n, the set where $h^i(X_s, \mathcal{L}_s) \geq n$ is closed in S.
2. Since $\chi(\mathcal{L}_s) = h^0(X_s, \mathcal{L}_s) - h^1(X_s, \mathcal{L}_s)$ is locally constant, h^0 jumps up at a point exactly when h^1 does.

Grauert's theorem (theorem 6.8.19) complements this: if S is reduced and $h^i(X_s, \mathcal{L}_s)$ is constant on S, then $R^i f_* \mathcal{L}$ is locally free of that rank and its formation commutes with every base change from a locally Noetherian scheme, so its fiber at s is $H^i(X_s, \mathcal{L}_s)$.

Example 7.7.3: Semicontinuity for Curves
Let $\mathcal{X} \to S$ be a family as above with $g = 3$ and S irreducible, and let $\mathcal{L}$ have degree 3 on each fiber. Then $\chi(\mathcal{L}_s) = 3 + 1 - 3 = 1$ for all s. Suppose $\mathcal{L}_s$ is non-special for some s, that is, $h^1(X_s, \mathcal{L}_s) = \ell(K - D_s) = 0$ for a divisor D_s with $\mathcal{L}_s \cong \mathcal{L}(D_s)$. Then by theorem 6.8.17(1), the locus where $h^1 \geq 1$ is a proper closed subset of S, and off it $h^0 = 1$. At its points D_s is special, $h^1 \geq 1$ and $h^0 = 1 + h^1 \geq 2$. The hypothesis cannot be dropped: if $\mathcal{L}_s \cong \omega_{X_s}(-P_s)$ for a point P_s of each fiber, then $h^0 = g - 1 = 2$ on every fiber by theorem 7.3.8(1), so every fiber is special. Loci of this kind, where h^0 of a line bundle of fixed degree jumps, are the subject of *Brill-Noether theory*.

7.7.4 Forward Reference: Deformation Theory of Curves The tangent space to the deformation functor of a smooth curve X of genus g is $H^1(X, T_X)$, where $T_X = \Omega^\vee_{X/k}$ is the tangent sheaf. By Serre duality, $H^1(X, T_X) \cong H^0(X, \omega_X^{\otimes 2})^\vee$, and by Riemann-Roch

$$\dim H^0(X, \omega_X^{\otimes 2}) = \begin{cases} 0 & \text{if } g = 0, \\ 1 & \text{if } g = 1, \\ 3g - 3 & \text{if } g \geq 2 \end{cases}$$

For $g \geq 2$ this is the dimension $3g - 3$ of the moduli space $\mathcal{M}_g$. The obstructions to deformation lie in $H^2(X, T_X)$, which vanishes because $\dim X = 1$, so deformations of X are unobstructed. *Moduli, Deformation Theory, and Stacks* [20] makes this precise with Schlessinger's theorem and the cotangent complex.

7.7.5 Singular Curves in Moduli: Stable Curves and $\overline{\mathcal{M}}_g$ The moduli space $\mathcal{M}_g$ of smooth curves of genus $g \geq 2$ is not compact: its coarse moduli space is not proper. A family of smooth curves can degenerate, with smooth fibers acquiring singularities. Semicontinuity (theorem 6.8.17) controls how cohomology jumps, and the arithmetic genus stays constant in a flat family, because $\chi(\mathcal{O}_{X_s})$ is locally constant by theorem 6.8.17(2), but the fibers leave the class of smooth curves.
Compactifying $\mathcal{M}_g$ requires parametrizing more curves. The *Deligne-Mumford compactification* $\overline{\mathcal{M}}_g$ adds the singular *stable curves*, where a stable curve is a connected complete curve X of arithmetic genus g whose only singularities are *nodes* (ordinary double points) and whose automorphism group $\mathrm{Aut}(X)$ is finite.
Three facts single out nodes:

1. A node has $\delta = 1$ (box 7.2.9), so a rational curve with g nodes has $p_a = g$, the genus of the smooth curves it can be a limit of. A cusp also has $\delta = 1$, but admitting cusps would make limits non-unique. By the stable reduction theorem [20, Stable Reduction], a family of smooth curves of genus $g \geq 2$ over the generic point of a discrete valuation ring extends, after a finite extension of the ring, to a family of stable curves, and this extension is unique. A family with a cuspidal special fiber therefore also has a second limit, a stable one, after such an extension.

2. On a nodal curve X of arithmetic genus $g \geq 2$, the dualizing sheaf ω_X° (definition 6.5.1) is an ample line bundle if and only if $\mathrm{Aut}(X)$ is finite. So stable curves are exactly the nodal curves for which ω_X° is a polarization.

3. In a family of plane cubics degenerating an elliptic curve, a nodal fiber has smooth locus $\mathbb{G}_m$ (multiplicative reduction) and a cuspidal fiber has smooth locus $\mathbb{G}_a$ (additive reduction), as described in section 7.6. For the action of SL_3 on plane cubics, the nodal cubic is semistable and the cuspidal cubic is unstable in the sense of geometric invariant theory [56, Chapter 4, §2], matching the names *semistable* and *unstable* reduction.

Moduli, Deformation Theory, and Stacks constructs $\overline{\mathcal{M}}_g$ as a proper Deligne-Mumford stack and proves, following Knudsen and Mumford, that its coarse moduli space is a projective variety [20, The Coarse Space $\overline{M}_g$ is a Projective Variety]. Of the singular curves in box 7.2.9 and example 5.4.37, the nodal cubic with a smooth point marked is the one curve that $\overline{\mathcal{M}}_{1,1}$ adds to $\mathcal{M}_{1,1}$, and the cuspidal cubic is not stable.

7.7.2 Curves over Non-Algebraically-Closed Fields

The standing assumption of this chapter is $k = \overline{k}$. Over a general field k, three things change.

1. Rational points may not exist. A smooth projective geometrically integral curve of genus 0 over k is isomorphic to a conic in $\mathbb{P}^2_k$, and it is isomorphic to $\mathbb{P}^1_k$ if and only if it has a k-rational point [47, Proposition 7.4.1]. Riemann-Roch over k [47, Theorem 7.3.26] gives $\ell(P) = [\kappa(P) : k] + 1$ for a closed point P of such a curve, so the argument of proposition 7.3.6 produces an isomorphism with $\mathbb{P}^1_k$ exactly when some P has $\kappa(P) = k$. Over $\mathbb{Q}$, whether a conic has a rational point is decided by the Hasse-Minkowski theorem (the local-global principle).

2. A curve of genus 1 without a k-rational point is not an elliptic curve, as it has no origin for the

group law. Such a curve C is a *torsor* (principal homogeneous space) for its Jacobian E, and its class as an E-torsor is an element of the Galois cohomology group $H^1(k, E)$ [15]. For k a number field, the Shafarevich-Tate group $Ш(E/k)$ classifies the torsors that have a point over every completion of k.

3. Over imperfect fields, a regular scheme need not be smooth (example 2.7.16), which complicates the theory of differentials and the adjunction formula.

Over a finite field $\mathbb{F}_q$, the genus bounds the number of rational points of a curve: a smooth projective geometrically integral curve of genus g over $\mathbb{F}_q$ has between $q+1-2g\sqrt{q}$ and $q+1+2g\sqrt{q}$ rational points. This is Weil's theorem (theorem 8.7.12), proved in section 8.7 with the intersection theory of the product of the curve with itself over $\overline{\mathbb{F}}_q$.

Riemann-Roch, obtained from Serre duality, computed the dimensions $\ell(D)$ and gave the embeddings by linear systems (corollary 7.3.4 and theorem 7.3.8). Hurwitz's formula, obtained from the sheaf of differentials, related the genera of curves under finite maps (theorem 7.4.5), and semicontinuity controlled how h^0 and h^1 vary in families (section 7.7.1). The genus alone classifies curves only in genus 0 (proposition 7.3.6): elliptic curves vary with j (proposition 7.6.5), and curves of genus $g \geq 2$ have $3g-3$ moduli (box 7.7.4).

8

Application: Algebraic Surfaces

Two features separate surfaces from curves. First, a divisor on a surface is a formal sum of curves, and two curves without a common component meet in finitely many points. Counting those points with multiplicity extends to a symmetric bilinear pairing on the Picard group of the surface, the intersection pairing, which has no counterpart on a curve. It is the first case of the intersection product of *Intersection Theory* [19], which also proves the Hirzebruch-Riemann-Roch and Grothendieck-Riemann-Roch theorems. Second, birational equivalence no longer implies isomorphism. Two smooth projective curves with isomorphic function fields are isomorphic (theorem 7.1.4(3)), but blowing up a closed point of a smooth projective surface produces a smooth projective surface that is birational to the original and not isomorphic to it. The contraction of (-1)-curves to a minimal surface, which reverses such blow-ups, is the two-dimensional case of the minimal model program of *Birational Geometry* [8].

Section 8.1 studies the Picard and Néron-Severi groups of a surface, and section 8.2 constructs the intersection pairing from Euler characteristics and proves the adjunction formula. Section 8.3 proves the Riemann-Roch theorem for surfaces and the Hodge index theorem, which determines the signature of the pairing in every characteristic, and states Noether's formula. Over $\mathbb{C}$ the Néron-Severi group embeds in $H^2(X;\mathbb{Z})$, and *Hodge Theory* [17] proves the Lefschetz $(1,1)$-theorem, which describes the image, and the analytic form of the Hodge index theorem. Section 8.4 constructs blow-ups, computes the intersection pairing on the blow-up of a surface at a point, and gives Castelnuovo's criterion for contracting a smooth rational curve of self-intersection -1. Section 8.5 treats ruled and rational surfaces, and section 8.6 sorts minimal surfaces by Kodaira dimension in the Enriques-Kodaira classification. For a curve of genus g defined over $\mathbb{F}_q$, section 8.7 applies the Hodge index theorem on the product of the curve with itself to prove Weil's bound $|N-(q+1)| \leq 2g\sqrt{q}$ for the number N of its points over $\mathbb{F}_q$.

8.1 Surfaces and the Néron-Severi Group

The intersection pairing of section 8.2 is defined on $\mathrm{Pic}\,(X)$, and it depends only on the image of a class in the Néron-Severi group $\mathrm{NS}(X)$, the quotient of $\mathrm{Pic}\,(X)$ by the classes algebraically equivalent to zero. This section defines surfaces, introduces the surfaces $\mathbb{P}^2$, $\mathbb{P}^1 \times \mathbb{P}^1$, the smooth surfaces in $\mathbb{P}^3$ and the Hirzebruch surfaces that recur through the chapter, and states the theorem of the base, by which $\mathrm{NS}(X)$ is finitely generated. It also defines the ample, nef, big and effective divisors on a surface.

The field k and every curve considered on its own are as in box 7.0.1. Curves lying on a surface X can be singular, and the irreducible ones are the prime divisors of X. A result that needs a curve $C \subseteq X$ to be nonsingular says so.

Definition 8.1.1: Surface

A *surface over* k is a Noetherian, integral, separated scheme X of dimension 2 and of finite type over k. The surface X is

1. *projective* if there exists a closed immersion $X \hookrightarrow \mathbb{P}^n_k$ for some n,
2. *nonsingular* (or *smooth*) if all local rings $\mathcal{O}_{X,x}$ are regular, or equivalently (over $k = \overline{k}$) if the structure morphism is smooth.

Unless otherwise stated, in this chapter a *surface* is a *smooth projective surface over* $k = \overline{k}$.

For curves properness implies projectivity (corollary 7.3.5), and the same holds for smooth surfaces: every proper smooth surface over k is projective, a theorem of Zariski quoted in [36, §V.1]. This book does not prove it, so projectivity is assumed, which keeps an ample line bundle available. In dimension 3 the implication fails (section 8.4.1).

Example 8.1.2: The Basic Surfaces

1. The projective plane $\mathbb{P}^2$ has $\mathrm{Pic}\,(\mathbb{P}^2) \cong \mathbb{Z}$, generated by the class H of a line, and a curve of degree d has class dH. Taking determinants in the Euler sequence (theorem 5.5.19) gives $\omega_{\mathbb{P}^2} \cong \mathcal{O}(-3)$, so the canonical class $K_{\mathbb{P}^2} = -3H$ is anti-ample.
2. The product $\mathbb{P}^1 \times \mathbb{P}^1$ is the simplest *ruled surface*: the first projection exhibits it as a family of copies of $\mathbb{P}^1$ parametrized by $\mathbb{P}^1$. Its Picard group is $\mathrm{Pic}\,(\mathbb{P}^1 \times \mathbb{P}^1) \cong \mathbb{Z}^2$, generated by the classes of the two rulings $F = \{p\} \times \mathbb{P}^1$ and $G = \mathbb{P}^1 \times \{q\}$. A divisor of *bidegree* (a, b) is one of class $aF + bG$, and the canonical class is $K = -2F - 2G$. This surface is not isomorphic to $\mathbb{P}^2$, since the Picard groups differ, but it is *birational* to $\mathbb{P}^2$: blowing up two distinct points of $\mathbb{P}^2$ and then contracting the strict transform of the line through them, a smooth rational curve of self-intersection -1, produces $\mathbb{P}^1 \times \mathbb{P}^1$ (section 8.4).
3. A smooth surface $S \subseteq \mathbb{P}^3$ of degree d has canonical class $K_S = (d-4)H$ by the adjunction formula (proposition 6.5.7), which separates three cases:
   (a) $d \leq 3$: K_S is anti-ample, and S (a plane, a quadric or a cubic surface) is rational.
   (b) $d = 4$: $K_S = 0$, and S is a K3 surface.
   (c) $d \geq 5$: K_S is ample, and S is of general type.

Over $\mathbb{C}$, a *very general* such surface with $d \geq 4$ has $\operatorname{Pic}(S) = \mathbb{Z}H$, by the Noether-Lefschetz theorem[a]. Special surfaces can have larger Picard groups (example 8.1.7).

4. For each $n \geq 0$, the nth *Hirzebruch surface* is $\mathbb{F}_n = \mathbb{P}(\mathcal{O}_{\mathbb{P}^1} \oplus \mathcal{O}_{\mathbb{P}^1}(-n))$, the projectivization of a rank-2 bundle on $\mathbb{P}^1$. It is a ruled surface with a section C_0 satisfying $C_0^2 = -n$ and fibers F with $F^2 = 0$ and $C_0 \cdot F = 1$. In particular $\mathbb{F}_0 = \mathbb{P}^1 \times \mathbb{P}^1$ and $\mathbb{F}_1 \cong \operatorname{Bl}_p \mathbb{P}^2$, the blow-up of $\mathbb{P}^2$ at a point (section 8.4).

[a] That is, the surfaces of degree $d \geq 4$ with $\operatorname{Pic}(S) \neq \mathbb{Z}H$ lie in a countable union of proper closed subsets of the space of all smooth surfaces of degree d. Here $\operatorname{Pic}(S) = \operatorname{NS}(S)$ because $H^1(S, \mathcal{O}_S) = 0$, so by theorem 8.1.6 the statement is $H^{1,1}(S) \cap H^2(S; \mathbb{Z}) = \mathbb{Z}H$. Its proof is Hodge-theoretic: see *Hodge Theory* [17, The Noether-Lefschetz Locus for Surfaces in $\mathbb{C}\mathrm{P}^3$].

8.1.3 Birational Models of Surfaces Every birational map between smooth projective curves extends to an isomorphism. For surfaces this fails: blowing up a closed point of a smooth surface X gives a smooth surface containing an *exceptional curve* $E \cong \mathbb{P}^1$ with $E^2 = -1$, birational to X (the blow-up is an isomorphism away from the point) and not isomorphic to it (section 8.4). A smooth projective surface therefore has many smooth projective models in its birational class, and the question is whether some of them are distinguished.
Contracting curves reverses blowing up. Castelnuovo's criterion (theorem 8.4.8) contracts any smooth rational curve E with $E^2 = -1$, each contraction lowers the Picard number by one, and so after finitely many contractions one reaches a *minimal surface*, one containing no such curve. When the Kodaira dimension (section 8.6.1) is ≥ 0, the minimal model in a birational class is unique. When it is $-\infty$ it is not: a rational surface is birational to $\mathbb{P}^2$ and to every Hirzebruch surface $\mathbb{F}_n$ with $n \neq 1$, and all of these are minimal.
In dimension ≥ 3 contracting curves can produce singular varieties, and the minimal model program has to allow singularities and *flips*. That theory is the subject of *Birational Geometry* [8].

Divisors on a surface behave as on any smooth variety. Let X be a smooth projective surface over $k = \overline{k}$. Every local ring $\mathcal{O}_{X,x}$ is regular, hence a UFD (proposition 2.7.27), so X is locally factorial (definition 2.4.30). By proposition 5.4.33 and theorem 5.4.44, Weil and Cartier divisors coincide and

$$\operatorname{CaCl} X \;\cong\; \operatorname{Cl} X \;\cong\; \operatorname{Pic} X$$

As with curves, divisors and line bundles are identified, and additive and multiplicative notation are used interchangeably.

A divisor on X is thus a formal sum $D = \sum n_i C_i$ of irreducible curves $C_i \subseteq X$ with $n_i \in \mathbb{Z}$, where on a curve it was a formal sum of points. The invertible sheaf $\mathcal{L}(D)$, its spaces of global sections, and the complete linear system $|D| \cong \mathbb{P}(H^0(X, \mathcal{L}(D)))$ of effective divisors linearly equivalent to D are defined as for curves.

For a curve, $\operatorname{Pic}(X) \cong \operatorname{Pic}^0(X) \oplus \mathbb{Z}$, where $\operatorname{Pic}^0(X)$ is the group of points of the Jacobian and the factor $\mathbb{Z}$ records the degree. On a surface the quotient of $\operatorname{Pic}(X)$ by $\operatorname{Pic}^0(X)$ is the Néron-Severi group, and its rank can exceed 1 (example 8.1.7).

Definition 8.1.4: Néron-Severi Group and Picard Number

Let X be a smooth projective variety. The case of interest in this chapter is a surface, but nothing below needs the dimension. The *Néron-Severi group* of X is the quotient

$$\mathrm{NS}(X) \;=\; \mathrm{Pic}\,(X)/\mathrm{Pic}^{\,0}(X)$$

where $\mathrm{Pic}^{\,0}(X)$ is the group of line bundles that are *algebraically equivalent* to zero (equivalently, the group of k-points of the identity component of the Picard scheme $\mathrm{Pic}_{X/k}$). The *Picard number* (or *rank of the Néron-Severi group*) is $\rho(X) = \mathrm{rk}\mathrm{NS}(X)$.

The group $\mathrm{Pic}\,(X)$ is in general not finitely generated: for an abelian surface, $\mathrm{Pic}^{\,0}$ is the group of points of a two-dimensional abelian variety (example 8.1.7). The quotient $\mathrm{NS}(X)$ is finitely generated.

Theorem 8.1.5: Theorem of the Base

Let X be a smooth projective variety over an algebraically closed field. Then $\mathrm{NS}(X)$ is a finitely generated abelian group. Its rank $\rho(X)$ is the *Picard number*, and over $\mathbb{C}$ it satisfies $\rho(X) \le h^{1,1}(X)$.

***Proof Cited*:**
This is the Néron-Severi theorem, and neither available proof is scheme-theoretic. Over $\mathbb{C}$, the exponential sequence $0 \to \mathbb{Z} \to \mathcal{O}_X \to \mathcal{O}_X^{\times} \to 0$ on the complex manifold $X(\mathbb{C})$ has a connecting map $\mathrm{Pic}\,(X) \to H^2(X;\mathbb{Z})$ with kernel $\mathrm{Pic}^{\,0}(X)$, so $\mathrm{NS}(X)$ embeds in the finitely generated group $H^2(X;\mathbb{Z})$. That argument needs the singular cohomology of $X(\mathbb{C})$ and its finite generation, which this book does not develop. The image consists of classes of type $(1,1)$ in the Hodge decomposition of corollary 6.9.10, which gives $\rho \le h^{1,1}$. Over an arbitrary algebraically closed field the theorem is due to Néron, by an arithmetic argument over a finitely generated field. The note to [36, Exercise V.1.7] gives references.

The bound $\rho \le h^{1,1}$ can be strict: a very general quartic surface in $\mathbb{P}^3_{\mathbb{C}}$ has $\rho = 1$ and $h^{1,1} = 20$ (examples 8.1.7 and 8.3.6). Over $\mathbb{C}$ the Lefschetz theorem describes exactly which cohomology classes are classes of divisors.

Theorem 8.1.6: Lefschetz Theorem on $(1,1)$-Classes

Let X be a smooth projective variety over $\mathbb{C}$. A class in $H^2(X;\mathbb{Z})$ is the class of a divisor if and only if its image in $H^2(X;\mathbb{C})$ lies in $H^{1,1}(X)$. More precisely, the cycle class map $\mathrm{NS}(X) \to H^2(X;\mathbb{Z})$ is injective with image $H^{1,1}(X) \cap H^2(X;\mathbb{Z})$.

***Proof Cited*:**
The proof is analytic. The connecting map $\mathrm{Pic}\,(X) \to H^2(X;\mathbb{Z})$ of the exponential sequence $0 \to \mathbb{Z} \to \mathcal{O}_X \to \mathcal{O}_X^{\times} \to 0$ has image the kernel of $H^2(X;\mathbb{Z}) \to H^2(X;\mathcal{O}_X)$, and Hodge theory identifies that kernel with the integral classes of type $(1,1)$. Both steps use the sheaf cohomology of the complex manifold $X(\mathbb{C})$ and the analytic form of the Hodge decomposition of corollary 6.9.10. A proof is in *Hodge Theory* [17, The Lefschetz $(1,1)$-Theorem]. The statement is the codimension-one case of the Hodge conjecture (box 8.3.14).

Example 8.1.7: Picard Groups of Basic Surfaces

1. $\mathrm{Pic}\,(\mathbb{P}^2) \cong \mathbb{Z}$, generated by the hyperplane class H. Here $\mathrm{Pic}^{\,0} = 0$, $\mathrm{NS} \cong \mathbb{Z}$ and $\rho = 1$.
2. $\mathrm{Pic}\,(\mathbb{P}^1 \times \mathbb{P}^1) \cong \mathbb{Z}^2$, generated by the two rulings F and G. Again $\mathrm{Pic}^{\,0} = 0$, $\mathrm{NS} \cong \mathbb{Z}^2$ and $\rho = 2$.
3. For a smooth quartic surface $S \subseteq \mathbb{P}^3$ (a K3 surface), $\mathrm{Pic}^{\,0}(S) = 0$ since $H^1(S, \mathcal{O}_S) = 0$, so $\mathrm{Pic}\,(S) = \mathrm{NS}(S)$. Over $\mathbb{C}$, a very general quartic has $\rho = 1$, while special quartics can have ρ as large as 20, the maximum for complex K3 surfaces because $h^{1,1} = 20$ (example 8.3.6). The Fermat quartic $x^4 + y^4 + z^4 + w^4 = 0$ attains it.
4. For a complex abelian surface A (a 2-dimensional abelian variety), $\mathrm{Pic}^{\,0}(A) \cong \widehat{A}(\mathbb{C})$ is the group of points of the dual abelian variety, which has dimension 2, and $1 \leq \rho \leq h^{1,1} = 4$.

On a curve a divisor is ample exactly when it has positive degree. On a surface positivity has several inequivalent forms. The intersection number $D \cdot C$ used below is defined in section 8.2.

1. A divisor D is *ample* if $\mathcal{O}_X(D)$ is ample in the sense of definition 5.3.30. By theorem 5.3.41, proposition 5.3.31 and lemma 5.3.33, this holds exactly when some positive multiple $\mathcal{O}_X(nD)$ is very ample, that is, when some $|nD|$ gives a closed immersion into projective space. By the Nakai-Moishezon criterion (theorem 8.2.11), D is ample if and only if $D^2 > 0$ and $D \cdot C > 0$ for every irreducible curve $C \subseteq X$.
2. A divisor D is *nef* ("numerically effective") if $D \cdot C \geq 0$ for every irreducible curve $C \subseteq X$.
3. A divisor D is *big* if $\ell(nD) \geq c\,n^2$ for some $c > 0$ and infinitely many $n \geq 1$. Quadratic growth is the maximum possible on a surface.
4. A divisor D is *effective* if $D \geq 0$ (all coefficients non-negative).

These notions recur in the Enriques-Kodaira classification (section 8.6). On a curve, ample means positive degree and nef means non-negative degree, but effectivity is not a condition on the degree: by lemma 7.2.2, a divisor of degree 0 is linearly equivalent to an effective divisor only if it is principal.

8.1.8 The Cone of Curves and Duality Ampleness and nefness are dual to the *cone of curves* $\mathrm{NE}(X)$, the convex cone in $\mathrm{Num}(X) \otimes \mathbb{R}$ (definition 8.2.5) spanned by the classes of irreducible curves. A divisor is nef exactly when its class pairs non-negatively with every element of $\mathrm{NE}(X)$, that is, when it lies in the dual cone $\mathrm{NE}(X)^\vee$, the *nef cone*. By *Kleiman's criterion*, a divisor is ample exactly when its class lies in the interior of the nef cone.
Mori's *Cone Theorem* describes the closure $\overline{\mathrm{NE}}(X)$: it is the sum of its part where $K_X \geq 0$ and of countably many rays, each spanned by the class of a rational curve C with $K_X \cdot C < 0$, and these rays accumulate only towards the hyperplane where K_X vanishes. Each such extremal ray is contracted by a morphism, and these contractions are the steps of the minimal model program of *Birational Geometry* [8].

8.2 Intersection Theory on Surfaces

This section attaches to any two divisors D_1, D_2 on a smooth projective surface an integer $D_1 \cdot D_2$, defined through Euler characteristics, proves that it is bilinear and depends only on linear equivalence classes, and shows that for curves without a common component it is the number of intersection points counted with multiplicity.

Definition 8.2.1: Intersection Number (Cohomological)

Let X be a smooth projective surface and D_1, D_2 divisors on X. The *intersection number* of D_1 and D_2 is the integer

$$D_1 \cdot D_2 \;=\; \chi(\mathcal{O}_X) - \chi(\mathcal{O}_X(-D_1)) - \chi(\mathcal{O}_X(-D_2)) + \chi(\mathcal{O}_X(-D_1 - D_2))$$

and $D^2 = D \cdot D$ is the *self-intersection* of D.

For effective D_1 and D_2 without a common component, the right-hand side is the Euler characteristic of the finite scheme $D_1 \cap D_2$, whose length counts the intersection points with multiplicity (proposition 8.2.2(5)).

Proposition 8.2.2: Properties of the Intersection Pairing

Let X be a smooth projective surface over $k = \overline{k}$.

1. The intersection number $D_1 \cdot D_2$ is $\mathbb{Z}$-bilinear in D_1 and D_2.
2. It is symmetric: $D_1 \cdot D_2 = D_2 \cdot D_1$.
3. It depends only on linear equivalence classes: if $D_1 \sim D_1'$ and $D_2 \sim D_2'$, then $D_1 \cdot D_2 = D_1' \cdot D_2'$.
4. If C_1 and C_2 are effective divisors with no common component that meet transversally, meaning that at each point of $C_1 \cap C_2$ their local equations generate the maximal ideal, then $C_1 \cdot C_2 = \#(C_1 \cap C_2)$, the number of intersection points.
5. If C_1 and C_2 are effective divisors with no common component, then
$$C_1 \cdot C_2 \;=\; \sum_{P \in C_1 \cap C_2} i_P(C_1, C_2)$$
where $i_P(C_1, C_2) = \dim_k \mathcal{O}_{X,P}/(f_1, f_2) \geq 1$ is the *local intersection multiplicity* at P, and f_1, f_2 are local equations of C_1 and C_2 at P.
6. If C is an effective divisor and D is any divisor, then $D \cdot C = \chi(\mathcal{O}_C) - \chi(\mathcal{O}_C \otimes \mathcal{O}_X(-D))$. If C is a smooth irreducible curve, then $D \cdot C = \deg(\mathcal{O}_X(D)|_C)$.

***Proof*:**
Step 1: (2), (3) and (6). The expression defining $D_1 \cdot D_2$ is symmetric in D_1 and D_2, which is (2). The Euler characteristic $\chi(\mathcal{O}_X(-D))$ depends only on the isomorphism class of $\mathcal{O}_X(-D)$,

which depends only on the linear equivalence class of D (theorem 5.4.44), and this gives (3). Now let C be an effective divisor. Its ideal sheaf is $\mathcal{O}_X(-C)$ (proposition 5.4.47), so there is an exact sequence

$$0 \to \mathcal{O}_X(-C) \to \mathcal{O}_X \to \mathcal{O}_C \to 0 \tag{8.1}$$

Tensoring with the invertible, hence flat, sheaf $\mathcal{O}_X(-D)$ gives

$$0 \to \mathcal{O}_X(-C-D) \to \mathcal{O}_X(-D) \to \mathcal{O}_C(-D) \to 0$$

where $\mathcal{O}_C(-D) = \mathcal{O}_C \otimes \mathcal{O}_X(-D)$. Additivity of χ (proposition 6.7.2) gives

$$\begin{aligned}\chi(\mathcal{O}_X(-C)) &= \chi(\mathcal{O}_X) - \chi(\mathcal{O}_C),\\ \chi(\mathcal{O}_X(-C-D)) &= \chi(\mathcal{O}_X(-D)) - \chi(\mathcal{O}_C(-D))\end{aligned}$$

and substituting into the definition,

$$\begin{aligned}D \cdot C &= \chi(\mathcal{O}_X) - \chi(\mathcal{O}_X(-D)) - \chi(\mathcal{O}_X(-C)) + \chi(\mathcal{O}_X(-C-D))\\ &= \chi(\mathcal{O}_C) - \chi(\mathcal{O}_C(-D))\end{aligned}$$

which is the first formula of (6). If C is a smooth irreducible curve, lemma 6.7.11 gives $\chi(\mathcal{O}_C(-D)) = \deg(\mathcal{O}_X(-D)|_C) + \chi(\mathcal{O}_C)$, and $\deg(\mathcal{O}_X(-D)|_C) = -\deg(\mathcal{O}_X(D)|_C)$ because the degree is additive and $\mathcal{O}_X(-D) \cong \mathcal{O}_X(D)^\vee$ (proposition 5.4.43). Hence $D \cdot C = \deg(\mathcal{O}_X(D)|_C)$, which completes (6).

Step 2: (5) and (4). Let C_1, C_2 be effective with no common component, let $P \in X$, and let $f_1, f_2 \in \mathcal{O}_{X,P}$ be local equations. The ring $\mathcal{O}_{X,P}$ is a UFD (proposition 2.7.27), and a prime element dividing both f_1 and f_2 would define an irreducible curve through P contained in both C_1 and C_2, hence a common component. So f_1 and f_2 have no common prime factor, and multiplication by f_1 is injective on $\mathcal{O}_{X,P}/(f_2)$: if $f_1 g \in (f_2)$ then f_2 divides g. Tensoring (8.1) for $C = C_1$ with $\mathcal{O}_{C_2}$ therefore gives an exact sequence

$$0 \to \mathcal{O}_{C_2}(-C_1) \to \mathcal{O}_{C_2} \to \mathcal{O}_{C_1 \cap C_2} \to 0 \tag{8.2}$$

where $C_1 \cap C_2$ is the closed subscheme with ideal sheaf $\mathcal{I}_{C_1} + \mathcal{I}_{C_2}$, whose stalk at P is $\mathcal{O}_{X,P}/(f_1, f_2)$. Its support is finite. Indeed, each irreducible component Γ of C_2 is an irreducible curve not contained in C_1 (otherwise Γ would be a component of C_1), so $\Gamma \cap C_1$ is a proper closed subset of the one-dimensional irreducible space Γ, hence a finite set of closed points. A coherent sheaf supported on finitely many closed points is flasque, its sections over an open set U being the product of its stalks at the points of the support lying in U. So it has no higher cohomology (corollary 6.1.7), and its global sections are the direct sum of its stalks, which gives $\chi(\mathcal{O}_{C_1 \cap C_2}) = \sum_P \dim_k \mathcal{O}_{X,P}/(f_1, f_2)$. Taking Euler characteristics in (8.2) and using (6) with $C = C_2$ and $D = C_1$,

$$C_1 \cdot C_2 = \chi(\mathcal{O}_{C_2}) - \chi(\mathcal{O}_{C_2}(-C_1)) = \chi(\mathcal{O}_{C_1 \cap C_2}) = \sum_{P \in C_1 \cap C_2} i_P(C_1, C_2)$$

which is (5). Each term is at least 1 because P lies in the support. If C_1 and C_2 meet transversally, then $(f_1, f_2) = \mathfrak{m}_P$ and $i_P = \dim_k \mathcal{O}_{X,P}/\mathfrak{m}_P = 1$, since k is algebraically closed, and this is (4).

Step 3: additivity against a fixed effective divisor. Fix an effective divisor C and write $\varphi_C(D) = D \cdot C$, which by Step 1 equals $\chi(\mathcal{O}_C) - \chi(\mathcal{O}_C(-D))$ and depends only on the linear equivalence

class of D. Let A be an effective divisor with no component in common with C. Tensoring (8.2), for $C_1 = A$ and $C_2 = C$, with $\mathcal{O}_X(-D)$ gives an exact sequence

$$0 \to \mathcal{O}_C(-A-D) \to \mathcal{O}_C(-D) \to \mathcal{O}_{A\cap C} \otimes \mathcal{O}_X(-D) \to 0$$

The last sheaf is supported on the finite set $A \cap C$, near each point of which $\mathcal{O}_X(-D)$ is free of rank one. It therefore has the same stalks as $\mathcal{O}_{A\cap C}$, and by Step 2 the same Euler characteristic $\chi(\mathcal{O}_{A\cap C}) = A \cdot C = \varphi_C(A)$. Hence $\chi(\mathcal{O}_C(-D)) - \chi(\mathcal{O}_C(-D-A)) = \varphi_C(A)$, that is,

$$\varphi_C(D+A) = \varphi_C(D) + \varphi_C(A) \tag{8.3}$$

for every divisor D and every effective A without a component in common with C.

Step 4: moving a divisor off C. Every divisor E is linearly equivalent to $A - B$ with A and B effective and without a component in common with C. Fix a very ample sheaf $\mathcal{O}_X(1)$ and a divisor H_1 with $\mathcal{O}_X(H_1) \cong \mathcal{O}_X(1)$. By theorem 5.3.26 there is an $n \geq 1$ such that $\mathcal{O}_X(E)(n)$ and $\mathcal{O}_X(n)$ are both generated by global sections. Pick a closed point x_j on each irreducible component Γ_j of C. For a globally generated invertible sheaf $\mathcal{L}$, the sections vanishing at x_j form a proper subspace of $H^0(X, \mathcal{L})$, and a vector space over the infinite field k is not a union of finitely many proper subspaces. So some $s \in H^0(X, \mathcal{L})$ vanishes at no x_j, and s can be taken nonzero (a globally generated invertible sheaf on the nonempty X has a nonzero section, which covers the case $C = 0$). Its divisor of zeros is an effective divisor in $|\mathcal{L}|$ (proposition 5.4.47) containing no Γ_j. Applying this to $\mathcal{L} = \mathcal{O}_X(E)(n)$ and to $\mathcal{L} = \mathcal{O}_X(n)$ gives effective $A \sim E + nH_1$ and $B \sim nH_1$ without a component in common with C, and $E \sim A - B$.

Step 5: (1). For fixed effective C and arbitrary D, E, write $E \sim A - B$ as in Step 4. Applying (8.3) to the sums $(D-B)+A$, $(D-B)+B$, $(-B)+A$ and $(-B)+B$ gives

$$\begin{aligned} \varphi_C(D+E) &= \varphi_C(D-B) + \varphi_C(A), & \varphi_C(D) &= \varphi_C(D-B) + \varphi_C(B), \\ \varphi_C(E) &= \varphi_C(-B) + \varphi_C(A), & 0 = \varphi_C(0) &= \varphi_C(-B) + \varphi_C(B) \end{aligned}$$

so $\varphi_C(D+E) = \varphi_C(D) + \varphi_C(A) - \varphi_C(B) = \varphi_C(D) + \varphi_C(E)$. Thus $D \mapsto D \cdot C$ is additive whenever C is effective.

To remove the effectivity, write $\chi(D)$ for $\chi(\mathcal{O}_X(D))$ and set $s(D_1, D_2, D_3) = (D_1 + D_2) \cdot D_3 - D_1 \cdot D_3 - D_2 \cdot D_3$. Expanding the definition,

$$s(D_1, D_2, D_3) = -\chi(0) + \sum_i \chi(-D_i) - \sum_{i<j} \chi(-D_i - D_j) + \chi(-D_1 - D_2 - D_3)$$

which is symmetric in D_1, D_2, D_3. The previous paragraph shows $s(D_1, D_2, C) = 0$ for C effective, so by symmetry and (3), s vanishes whenever one of its arguments is linearly equivalent to an effective divisor. Given arbitrary D_1, D_2, D_3, write $D_1 \sim A - B$ with A, B effective (Step 4 with $C = 0$). The vanishing of $s(A, D_2 - B, D_3)$ and of $s(B, D_2 - B, D_3)$ gives

$$(D_1 + D_2) \cdot D_3 = A \cdot D_3 + (D_2 - B) \cdot D_3, \qquad D_2 \cdot D_3 = B \cdot D_3 + (D_2 - B) \cdot D_3$$

and the case $D_2 = 0$ of these two equations, together with $0 \cdot D_3 = 0$, gives $D_1 \cdot D_3 = A \cdot D_3 - B \cdot D_3$. Subtracting, $(D_1 + D_2) \cdot D_3 = D_1 \cdot D_3 + D_2 \cdot D_3$. With (2) this is (1), completing the proof.

Example 8.2.3: Intersection Numbers on Basic Surfaces

1. Two distinct lines $L_1, L_2 \subseteq \mathbb{P}^2$, both of class H, meet transversally in one point, so $H^2 = 1$ by proposition 8.2.2(4). By bilinearity, curves of degrees d_1 and d_2 have intersection number $d_1 d_2$, and for curves without a common component proposition 8.2.2(5) turns this into Bézout's theorem ([22, Bézout's Theorem]).

2. On $\mathbb{P}^1 \times \mathbb{P}^1$ let $F = \{p\} \times \mathbb{P}^1$ and $G = \mathbb{P}^1 \times \{q\}$ be the two rulings. Then

$$F \cdot G = 1, \qquad F \cdot F = 0, \qquad G \cdot G = 0$$

   The curves F and G meet transversally in the single point (p, q). For $p' \neq p$ the fiber $F' = \{p'\} \times \mathbb{P}^1$ is linearly equivalent to F and disjoint from it, so $F^2 = F \cdot F' = 0$, and similarly $G^2 = 0$. Linearly equivalent effective divisors on a surface can therefore be disjoint.

3. If $C \subseteq \mathbb{P}^1 \times \mathbb{P}^1$ has class $aF + bG$, then

$$C \cdot F = b, \qquad C \cdot G = a, \qquad C^2 = 2ab$$

   For instance, the diagonal $\Delta \subseteq \mathbb{P}^1 \times \mathbb{P}^1$ has bidegree $(1, 1)$, so $\Delta^2 = 2$.

An ample divisor meets every curve positively, which is the "only if" half of the Nakai-Moishezon criterion (theorem 8.2.11).

Lemma 8.2.4: Ample Divisors Meet Curves Positively

Let H be an ample divisor on a smooth projective surface X.

1. $H \cdot C > 0$ for every irreducible curve $C \subseteq X$.
2. $H \cdot E \geq 0$ for every effective divisor E, with strict inequality if $E \neq 0$.
3. $H^2 > 0$.

***Proof*:**
By theorem 5.3.41 some multiple mH with $m \geq 1$ is very ample. Fix the corresponding closed immersion $X \hookrightarrow \mathbb{P}^N_k$. For a linear form λ not vanishing identically on X, the hyperplane section $X \cap V_+(\lambda)$ is an effective divisor linearly equivalent to mH (proposition 5.4.47).

(1) Choose a closed point $x \in C$ and a linear form λ with $\lambda(x) \neq 0$, and let $H' = X \cap V_+(\lambda)$. Then $C \not\subseteq H'$, so the effective divisors C and H' have no common component. They meet. Otherwise C is a closed subscheme of $X \cap D_+(\lambda)$, which is closed in the affine space $D_+(\lambda) \cong \mathbb{A}^N_k$, so C is affine. Being also proper and integral over $k = \overline{k}$, it has $\Gamma(C, \mathcal{O}_C) = k$ (theorem 3.5.33), so $C \cong \operatorname{Spec} k$ is a point, contradicting $\dim C = 1$. By proposition 8.2.2(5), $m\,(H \cdot C) = H' \cdot C = \sum_{P \in C \cap H'} i_P(C, H') \geq 1$, so $H \cdot C > 0$.

(2) Write $E = \sum n_i C_i$ with $n_i \geq 0$ and C_i irreducible curves. By bilinearity and (1), $H \cdot E = \sum n_i\,(H \cdot C_i) \geq 0$, with strict inequality if some $n_i > 0$.

(3) A hyperplane section $H' = X \cap V_+(\lambda)$ as in (1) is a nonzero divisor, since otherwise $X \subseteq D_+(\lambda)$ would be affine and the argument of (1), applied to X, would make X a point. So H' is a

nonzero effective divisor linearly equivalent to mH, and $m^2H^2 = (mH)\cdot H' = m\,(H\cdot H') > 0$ by (2), as claimed.

Numerical equivalence identifies divisors that the intersection pairing cannot distinguish.

Definition 8.2.5: Numerical Equivalence

A divisor D on a smooth projective surface X is *numerically equivalent to zero*, written $D \equiv 0$, if $D\cdot E = 0$ for every divisor E. Divisors D and D' are *numerically equivalent*, written $D \equiv D'$, if $D - D' \equiv 0$. The group of divisors modulo numerical equivalence is denoted $\mathrm{Num}(X)$.

Every divisor is a $\mathbb{Z}$-combination of irreducible curves, so by bilinearity $D \equiv 0$ if and only if $D\cdot C = 0$ for every irreducible curve C. Linearly equivalent divisors are numerically equivalent by proposition 8.2.2(3), so $\mathrm{Num}(X)$ is a quotient of $\mathrm{Pic}\,(X)$, and the intersection pairing descends to a nondegenerate symmetric pairing $\mathrm{Num}(X)\times\mathrm{Num}(X)\to\mathbb{Z}$. The group $\mathrm{Num}(X)$ is torsion-free: if $nD\equiv 0$ with $n\neq 0$, then $D\cdot E = 0$ for every E. A nonzero effective divisor E is never numerically equivalent to zero, since $E\cdot H > 0$ for H ample (lemma 8.2.4).

For a curve C on X the self-intersection C^2 can be positive, zero or negative, and for a smooth curve it is the degree of the normal bundle.

Proposition 8.2.6: Self-Intersection via the Normal Bundle

Let C be a smooth irreducible curve on a smooth projective surface X. Then

$$C^2 \;=\; \deg \mathcal{N}_{C/X}$$

where $\mathcal{N}_{C/X} = (\mathcal{I}_C/\mathcal{I}_C^2)^\vee$ is the normal sheaf, an invertible sheaf on C because C has codimension 1 in X (definition 5.5.32).

***Proof*:**
The ideal sheaf of C is $\mathcal{I}_C \cong \mathcal{O}_X(-C)$ (proposition 5.4.47), so $\mathcal{I}_C/\mathcal{I}_C^2 \cong \mathcal{I}_C\otimes\mathcal{O}_C \cong \mathcal{O}_X(-C)|_C$, and dualizing gives $\mathcal{N}_{C/X}\cong\mathcal{O}_X(C)|_C$. By proposition 8.2.2(6) with $D = C$, $C^2 = \deg(\mathcal{O}_X(C)|_C) = \deg\mathcal{N}_{C/X}$, as claimed.

The sign of C^2 separates three kinds of behaviour.

1. If $C^2 > 0$ and C is irreducible, the multiples of C move in large linear systems: $\ell(nC)\geq\chi(\mathcal{O}_X(nC))$ for $n\gg 0$, and the right-hand side grows like $n^2C^2/2$ (corollary 8.3.8, which applies because $C\cdot H > 0$ for H ample by lemma 8.2.4(1)). Lines in $\mathbb{P}^2$ have $L^2 = 1$.

2. Fibers of a morphism from X onto $\mathbb{P}^1$ have self-intersection 0, since two distinct fibers are linearly equivalent and disjoint. The rulings of $\mathbb{P}^1\times\mathbb{P}^1$ are an example (example 8.2.3).

3. If $C^2 < 0$ and C is irreducible, then C is the only effective divisor numerically equivalent to C (corollary 8.3.13), so it does not move in a linear system. The exceptional curve E of a blow-up, with $E^2 = -1$, is the basic example (section 8.4).

Castelnuovo's criterion (theorem 8.4.8) contracts every smooth rational curve of self-intersection -1 to a smooth point, and corollary 8.3.13 shows that an irreducible curve orthogonal to a divisor of positive square has negative self-intersection.

Example 8.2.7: Self-Intersection Numbers

1. Lines in $\mathbb{P}^2$ have $L^2 = 1$, and a curve of degree d has $C^2 = d^2$.
2. If $\pi\colon \widetilde{X} \to X$ is the blow-up of a point p and $E = \pi^{-1}(p)$ is the exceptional divisor, then $E^2 = -1$ (section 8.4).
3. For $n > 0$ the section $C_0 \subseteq \mathbb{F}_n$ with $C_0^2 = -n$ is the only effective divisor in its numerical class (corollary 8.3.13). For $n = 0$, $C_0^2 = 0$ and C_0 moves in a pencil of disjoint curves, one of the two rulings of $\mathbb{F}_0 = \mathbb{P}^1 \times \mathbb{P}^1$.
4. On an abelian surface A every irreducible curve C has $C^2 \geq 0$. For a general $a \in A$ the translate $C + a$ is an irreducible curve different from C. The translates of C form a family parametrized by the connected variety A, so $C + a$ is algebraically, hence numerically, equivalent to C [36, Exercise V.1.7], and $C^2 = C \cdot (C + a) \geq 0$ by proposition 8.2.2(5).

The adjunction formula expresses the canonical class of a curve on a surface in terms of the canonical class of the surface and the curve's self-intersection, and so computes the genus of a smooth curve from its divisor class.

Theorem 8.2.8: The Adjunction Formula

Let C be a smooth irreducible curve on a smooth projective surface X. Then

$$2g(C) - 2 \;=\; C \cdot (C + K_X)$$

where $g(C)$ is the genus of C and K_X is a canonical divisor of X. More precisely, $\omega_C \cong (\omega_X \otimes \mathcal{L}(C))|_C$, that is, $K_C \sim (K_X + C)|_C$.

***Proof*:**
The codimension-one case of proposition 5.5.33, applied to the smooth curve C in the smooth surface X, gives $\omega_C \cong (\omega_X \otimes \mathcal{L}(C))|_C$, which in divisor language is $K_C \sim (K_X + C)|_C$. Taking degrees on C and using proposition 8.2.2(6),

$$\deg K_C = \deg(\mathcal{O}_X(K_X)|_C) + \deg(\mathcal{O}_X(C)|_C) = K_X \cdot C + C^2 = C \cdot (C + K_X)$$

Since $\deg K_C = 2g(C) - 2$ (proposition 7.3.1), this is $2g(C) - 2 = C \cdot (C + K_X)$, as claimed.

The isomorphism is the case of proposition 5.5.33 for a curve on a surface (and proposition 6.5.7 covers subvarieties of projective space). The intersection pairing turns it into a formula for the genus of C in terms of the class of C alone.

Example 8.2.9: The Adjunction Formula in Action

1. Let $C \subseteq \mathbb{P}^2$ be a smooth curve of degree d. Then $C \sim dH$, so $C \cdot C = d^2$ and $C \cdot K_{\mathbb{P}^2} = C \cdot (-3H) = -3d$, and adjunction gives
$$2g - 2 = d^2 - 3d = d(d-3), \qquad g = \frac{(d-1)(d-2)}{2}$$
This is the genus formula of proposition 7.2.6, now recovered from the intersection pairing.

2. Let C be a smooth curve of bidegree (a, b) on $\mathbb{P}^1 \times \mathbb{P}^1$, where $K = -2F - 2G$. Then $C^2 = 2ab$ and $C \cdot K = (aF + bG) \cdot (-2F - 2G) = -2b - 2a$, so
$$2g - 2 = 2ab - 2a - 2b, \qquad g = (a-1)(b-1)$$
A smooth $(2,2)$-curve has $g = 1$, and a smooth $(3,3)$-curve has $g = 4$.

3. Let S be a K3 surface ($K_S = 0$) and $C \subseteq S$ a smooth irreducible curve. Adjunction gives $2g(C) - 2 = C^2$, so $g(C) = C^2/2 + 1$. In particular C^2 is even, and $C^2 \geq -2$ with equality if and only if $g(C) = 0$, that is, $C \cong \mathbb{P}^1$. Smooth rational curves on a K3 surface, which have $C^2 = -2$, are called (-2)-*curves* or *nodal curves*.

4. If $E \cong \mathbb{P}^1$ is the exceptional curve of the blow-up $\mathrm{Bl}_p X \to X$ of a smooth surface at a point, then $g(E) = 0$, and adjunction gives $-2 = E^2 + K_{\mathrm{Bl}_p X} \cdot E$. Section 8.4 shows that $K_{\mathrm{Bl}_p X} = \pi^* K_X + E$ and $(\pi^* K_X) \cdot E = 0$, so $K_{\mathrm{Bl}_p X} \cdot E = E^2$, hence $-2 = 2E^2$ and $E^2 = -1$.

8.2.10 Forward Reference: Chow Rings and the General Intersection Product
The intersection pairing on a smooth projective surface is the simplest case of the *intersection product* on the Chow ring of a smooth variety,
$$A^p(X) \otimes A^q(X) \to A^{p+q}(X)$$
where $A^p(X)$ is the group of codimension-p algebraic cycles modulo rational equivalence ([19, Chow group]). For a surface, $A^0(X) = \mathbb{Z}$ is generated by the class of X, $A^1(X) = \mathrm{Pic}\,(X)$, and $A^2(X)$ is the group of zero-cycles modulo rational equivalence, which carries a degree map $\deg\colon A^2(X) \to \mathbb{Z}$ sending the class of a closed point to 1. The intersection pairing is the composite $A^1(X) \times A^1(X) \to A^2(X) \xrightarrow{\deg} \mathbb{Z}$.
Intersection Theory [19] constructs the Chow ring of a smooth variety of any dimension, proves the *moving lemma* (which replaces cycles by rationally equivalent ones meeting properly), defines Chern classes as elements of $A^*(X)$, and proves the Grothendieck-Riemann-Roch theorem.

Positivity of the pairing on curves characterizes ample divisors.

Theorem 8.2.11: Nakai-Moishezon Criterion

Let D be a divisor on a smooth projective surface X. Then D is ample if and only if
$$D^2 > 0 \qquad \text{and} \qquad D \cdot C > 0 \ \text{ for every irreducible curve } C \subseteq X$$

***Proof Key ideas*:**
Necessity is proved in full by lemma 8.2.4(1) and (3). Sufficiency needs three inputs this book does not prove: that an invertible sheaf $\mathcal{L}$ on a projective curve, possibly singular and non-reduced, is ample when $\chi(\mathcal{L}|_{C_i}) - \chi(\mathcal{O}_{C_i}) > 0$ on each integral component C_i; that a proper morphism with finite fibers is finite (a consequence of theorem 3.4.38); and that the pullback of an ample sheaf along a finite morphism is ample. The argument is [36, Theorem V.1.10], and it runs as follows. A very ample divisor is linearly equivalent to an effective divisor H, nonzero because $H^2 > 0$ (lemma 8.2.4(3)), and $D \cdot H > 0$ by the hypothesis applied to the components of H. Together with $D^2 > 0$, corollary 8.3.8, proved below from Riemann-Roch without this theorem, makes some multiple of D effective, and D may be replaced by it. The restriction $\mathcal{O}_X(D)|_D$ satisfies $\chi(\mathcal{O}_{C_i}(D)) - \chi(\mathcal{O}_{C_i}) = D \cdot C_i > 0$ on each component C_i of D (proposition 8.2.2(6) applied to $-D$), hence is ample on the curve D, so $H^1(D, \mathcal{O}_X(nD)|_D) = 0$ for $n \gg 0$. The sequence $0 \to \mathcal{O}_X((n-1)D) \to \mathcal{O}_X(nD) \to \mathcal{O}_X(nD)|_D \to 0$ then makes $h^1(\mathcal{O}_X(nD))$ non-increasing, hence eventually constant, so $H^0(X, \mathcal{O}_X(nD)) \to H^0(D, \mathcal{O}_X(nD)|_D)$ is eventually surjective. Sections then generate $\mathcal{O}_X(nD)$ along D, and the section vanishing exactly on nD generates it off D. The resulting morphism $\varphi\colon X \to \mathbb{P}^M$ contracts no curve C, since a hyperplane missing $\varphi(C)$ would give a member of $|nD|$ disjoint from C, against $nD \cdot C > 0$. So φ has finite fibers, it is finite, and $\mathcal{O}_X(nD) = \varphi^*\mathcal{O}(1)$ is ample. Since D was replaced by a multiple, the original D has an ample multiple, and it is ample by lemma 5.3.33, as claimed.

On a curve the analogous statement is that a divisor is ample exactly when it has positive degree. On a projective variety X of any dimension, a divisor D is ample exactly when $D^{\dim V} \cdot V > 0$ for every closed subvariety $V \subseteq X$ of positive dimension [36, Theorem A.5.1], the intersection numbers being those of *Intersection Theory* [19].

8.3 Riemann-Roch and the Hodge Index Theorem

The Riemann-Roch theorem for curves (theorem 7.2.10) computes $\chi(\mathcal{L}(D))$ from $\deg D$ and the genus g. On a surface the intersection pairing takes the place of the degree, and the Euler characteristic becomes a quadratic function of D. Its constant term $\chi(\mathcal{O}_X)$ is computed over $\mathbb{C}$ by Noether's formula (section 8.3.2). When $D^2 > 0$ and $D \cdot H > 0$ for an ample divisor H, the quadratic growth of $\chi(\mathcal{O}_X(nD))$ in n makes nD effective for large n (corollary 8.3.8), and the Hodge index theorem (section 8.3.3) is deduced from this. Section 8.3.4 compares linear, algebraic and numerical equivalence of divisors through the Picard scheme.

8.3.1 The Riemann-Roch Theorem for Surfaces

Theorem 8.3.1: Riemann-Roch for Surfaces

Let X be a smooth projective surface over $k = \overline{k}$, and let D be a divisor on X. Then

$$\chi(\mathcal{L}(D)) \;=\; \frac{1}{2}\, D \cdot (D - K_X) \;+\; \chi(\mathcal{O}_X)$$

Equivalently, expanding the intersection product,

$$\chi(\mathcal{L}(D)) \;=\; \frac{D^2 - D \cdot K_X}{2} + \chi(\mathcal{O}_X)$$

For a curve of genus g, Riemann-Roch says $\chi(\mathcal{L}(D)) = \deg D + 1 - g = \deg D + \chi(\mathcal{O}_X)$, a linear function of D. On a surface $\chi(\mathcal{L}(D))$ is quadratic in D: it involves the self-intersection D^2 in addition to the linear term $D \cdot K_X$.

***Proof*:**
Here $\mathcal{L}(D) = \mathcal{O}_X(D)$. For a locally free sheaf on the two-dimensional smooth projective X, Serre duality (corollary 6.5.10) gives $H^i(X, \mathcal{O}_X(D)) \cong H^{2-i}(X, \mathcal{O}_X(K_X - D))^*$, so

$$\chi(\mathcal{O}_X(D)) = \sum_{i=0}^{2} (-1)^i\, h^{2-i}(\mathcal{O}_X(K_X - D)) = \chi(\mathcal{O}_X(K_X - D))$$

for every divisor D. In particular $\chi(\mathcal{O}_X(K_X)) = \chi(\mathcal{O}_X)$. Apply the definition of the intersection number to $-D$ and $D - K_X$:

$$(-D)\cdot(D-K_X) = \chi(\mathcal{O}_X) - \chi(\mathcal{O}_X(D)) - \chi(\mathcal{O}_X(K_X - D)) + \chi(\mathcal{O}_X(K_X)) = 2\chi(\mathcal{O}_X) - 2\chi(\mathcal{O}_X(D))$$

By bilinearity (proposition 8.2.2) the left-hand side is $-D \cdot (D - K_X)$, so $\chi(\mathcal{O}_X(D)) = \frac{1}{2}\, D \cdot (D - K_X) + \chi(\mathcal{O}_X)$, as we sought to show.

The proof uses only Serre duality and the bilinearity of the intersection number. The comparison of Euler characteristics on X with Euler characteristics on curves, which the proof of Riemann-Roch for curves carries out one point at a time, takes place in the proof of proposition 8.2.2. In every dimension the corresponding statement is the Hirzebruch-Riemann-Roch theorem, a case of Grothendieck-Riemann-Roch (*Intersection Theory* [19]).

Example 8.3.2: Riemann-Roch for Surfaces in Practice

1. On $\mathbb{P}^2$, $K_{\mathbb{P}^2} = -3H$ and $\chi(\mathcal{O}_{\mathbb{P}^2}) = 1$, since $H^i(\mathbb{P}^2, \mathcal{O}) = 0$ for $i > 0$ (theorem 6.3.1). For $D = dH$,
$$\chi(\mathcal{O}(d)) = \frac{d^2 - d(-3)}{2} + 1 = \frac{d^2 + 3d + 2}{2} = \binom{d+2}{2}$$
For $d \geq 0$ the higher cohomology of $\mathcal{O}(d)$ vanishes (theorem 6.3.1), and this is $\dim_k k[x, y, z]_d = \binom{d+2}{2}$.

2. On $\mathbb{P}^1 \times \mathbb{P}^1$, $K = -2F - 2G$ and $\chi(\mathcal{O}) = 1$. For $D = aF + bG$,

$$\chi(\mathcal{O}(a,b)) = \frac{2ab - (aF+bG)\cdot(-2F-2G)}{2} + 1 = \frac{2ab+2a+2b}{2} + 1 = (a+1)(b+1)$$

For $a, b \geq 0$ the Künneth formula gives $H^0(\mathbb{P}^1 \times \mathbb{P}^1, \mathcal{O}(a,b)) = H^0(\mathbb{P}^1, \mathcal{O}(a)) \otimes H^0(\mathbb{P}^1, \mathcal{O}(b))$, of dimension $(a+1)(b+1)$, and vanishing higher cohomology.

3. A smooth quartic surface $S \subseteq \mathbb{P}^3$ (a K3 surface) has $K_S = 0$, so

$$\chi(\mathcal{L}(D)) = \frac{D^2}{2} + \chi(\mathcal{O}_S)$$

and $\chi(\mathcal{O}_S) = 2$ (example 8.3.5). For an ample divisor D with $D^2 = 2d$, $\chi(\mathcal{L}(D)) = d + 2$.

By analogy with curves, the *arithmetic genus* of the surface X is

$$p_a(X) \;=\; (-1)^{\dim X}(\chi(\mathcal{O}_X) - 1) \;=\; \chi(\mathcal{O}_X) - 1$$

Since $h^0(\mathcal{O}_X) = 1$ (theorem 3.5.33), this is $p_a(X) = h^2(\mathcal{O}_X) - h^1(\mathcal{O}_X)$. Thus $\chi(\mathcal{O}_X) = 1 + p_a$, and Riemann-Roch becomes

$$\chi(\mathcal{L}(D)) = \frac{D\cdot(D - K_X)}{2} + 1 + p_a$$

Setting $D = K_X$ gives

$$\chi(\omega_X) = \frac{K_X^2 - K_X \cdot K_X}{2} + \chi(\mathcal{O}_X) = \chi(\mathcal{O}_X) \tag{8.4}$$

which is also the case $D = 0$ of the identity $\chi(\mathcal{O}_X(D)) = \chi(\mathcal{O}_X(K_X - D))$ from the proof of theorem 8.3.1.

The adjunction formula extends to effective divisors that may be singular, reducible or non-reduced. For such a divisor C, define the *arithmetic genus* by $p_a(C) = 1 - \chi(\mathcal{O}_C)$. Then

$$2p_a(C) - 2 \;=\; C\cdot(C + K_X) \tag{8.5}$$

Indeed, (8.1) gives $\chi(\mathcal{O}_C) = \chi(\mathcal{O}_X) - \chi(\mathcal{O}_X(-C))$, and theorem 8.3.1 for $-C$ gives $\chi(\mathcal{O}_X(-C)) = \frac{1}{2}C\cdot(C+K_X) + \chi(\mathcal{O}_X)$, so $2p_a(C) - 2 = -2\chi(\mathcal{O}_C) = C\cdot(C+K_X)$. When C is a smooth irreducible curve, $p_a(C) = g(C)$ and this is theorem 8.2.8.

8.3.2 Noether's Formula

Riemann-Roch expresses $\chi(\mathcal{L}(D))$ through intersection numbers and the single invariant $\chi(\mathcal{O}_X)$.

Theorem 8.3.3: Noether's Formula

Let X be a smooth projective surface over $\mathbb{C}$. Then

$$\chi(\mathcal{O}_X) \;=\; \frac{1}{12}\big(K_X^2 + \chi_{\text{top}}(X)\big)$$

where $\chi_{\text{top}}(X) = \sum_{i=0}^{4}(-1)^i b_i(X)$ is the topological Euler characteristic, the alternating sum of the Betti numbers of $X(\mathbb{C})$.

***Proof Key ideas*:**
The proof needs two results this book does not prove, both in [40]: the Hirzebruch-Riemann-Roch theorem (proved algebraically in *Intersection Theory* [19, Hirzebruch-Riemann-Roch]), and the equality of the degree of the top Chern class of the tangent bundle of a compact complex manifold with its topological Euler characteristic. Over any algebraically closed field, the first of these gives $\chi(\mathcal{O}_X) = \frac{1}{12}(K_X^2 + \deg c_2(T_X))$ by the computation below, and that statement is proved in *Intersection Theory* [19, Noether's Formula]. Granting both, Hirzebruch-Riemann-Roch for $\mathcal{O}_X$ gives $\chi(\mathcal{O}_X) = \int_X \mathrm{td}(T_X)$, and for a surface

$$\mathrm{td}(T_X) = 1 + \frac{1}{2}c_1(T_X) + \frac{1}{12}\big(c_1(T_X)^2 + c_2(T_X)\big)$$

Here $c_1(T_X) = -c_1(\Omega_X) = -K_X$ in $A^1(X)$, so $c_1(T_X)^2 = K_X^2$, and $\deg c_2(T_X) = \chi_{\mathrm{top}}(X)$. Taking the degree of the component in $A^2(X)$ gives $\chi(\mathcal{O}_X) = \frac{1}{12}(K_X^2 + \chi_{\mathrm{top}}(X))$, as claimed.

Over $\mathbb{C}$ the Hodge numbers $h^{p,q} = \dim H^q(X, \Omega_X^p)$ of a smooth projective surface satisfy $h^{p,q} = h^{q,p}$, $h^{p,q} = h^{2-p,2-q}$ and $b_k = \sum_{p+q=k} h^{p,q}$ (corollary 6.9.10, which rests on theorem 0.6.28 and theorem 6.9.2). For a surface this gives

$$b_0 = 1, \quad b_1 = 2h^{0,1}, \quad b_2 = h^{2,0} + h^{1,1} + h^{0,2} = 2h^{2,0} + h^{1,1},$$
$$b_3 = 2h^{0,1}, \quad b_4 = 1$$

so $\chi_{\mathrm{top}} = 2 - 4h^{0,1} + 2h^{2,0} + h^{1,1}$. The numbers $h^{0,1}$ and $h^{2,0}$ have algebraic counterparts over any algebraically closed field.

Definition 8.3.4: Surface Invariants

For a smooth projective surface X over $k = \overline{k}$:

1. The *irregularity* is $q = q(X) = \dim H^1(X, \mathcal{O}_X)$.
2. The *geometric genus* is $p_g = p_g(X) = \dim H^0(X, \omega_X)$, which equals $\dim H^2(X, \mathcal{O}_X)$ by Serre duality (corollary 6.5.10).
3. The *nth plurigenus* is $P_n = P_n(X) = \dim H^0(X, \omega_X^{\otimes n})$ for $n \geq 1$, so $P_1 = p_g$.

Over $\mathbb{C}$, $q = h^{0,1}$ and $p_g = h^{2,0} = h^{0,2}$.

Since $h^0(\mathcal{O}_X) = 1$, $\chi(\mathcal{O}_X) = 1 - q + p_g$ and $p_a(X) = p_g - q$.

Example 8.3.5: Noether's Formula in Practice

1. For $\mathbb{P}^2$, $K = -3H$, $K^2 = 9$ and $\chi_{\mathrm{top}} = 3$ (Betti numbers $1, 0, 1, 0, 1$). Noether's formula gives $\chi(\mathcal{O}) = \frac{9+3}{12} = 1$.
2. For $\mathbb{P}^1 \times \mathbb{P}^1$, $K = -2F - 2G$, $K^2 = 8$ and $\chi_{\mathrm{top}} = 4$ ($b_0 = b_4 = 1$, $b_2 = 2$, $b_1 = b_3 = 0$). Noether's formula gives $\chi(\mathcal{O}) = \frac{8+4}{12} = 1$.
3. A smooth quartic surface $S \subseteq \mathbb{P}^3$ has $K_S = 0$, so $K_S^2 = 0$ and $p_g = h^0(\mathcal{O}_S) = 1$. The sequence $0 \to \mathcal{O}_{\mathbb{P}^3}(-4) \to \mathcal{O}_{\mathbb{P}^3} \to \mathcal{O}_S \to 0$ and theorem 6.3.1 give $h^1(\mathcal{O}_S) = 0$, so $q = 0$ and $\chi(\mathcal{O}_S) = 1 - 0 + 1 = 2$. Over $\mathbb{C}$, Noether's formula then gives $\chi_{\mathrm{top}}(S) = 12 \cdot 2 - 0 = 24$, and since $b_1 = b_3 = 2q = 0$ the Betti numbers of S are $1, 0, 22, 0, 1$.

4. An abelian surface A has $K_A = 0$ (a translation-invariant 2-form trivializes ω_A), so $K^2 = 0$. The Betti numbers of the real 4-torus $A(\mathbb{C}) \cong \mathbb{C}^2/\Lambda$ are $1, 4, 6, 4, 1$, so $\chi_{\text{top}} = 0$, and Noether's formula gives $\chi(\mathcal{O}_A) = 0$, in agreement with $q = 2$, $p_g = 1$ and $\chi = 1 - 2 + 1 = 0$.

Over $\mathbb{C}$ the Hodge numbers of a smooth projective surface form its *Hodge diamond* (definition 0.6.29):

$$\begin{array}{ccccc} & & h^{0,0} & & \\ & h^{1,0} & & h^{0,1} & \\ h^{2,0} & & h^{1,1} & & h^{0,2} \\ & h^{2,1} & & h^{1,2} & \\ & & h^{2,2} & & \end{array} \quad = \quad \begin{array}{ccccc} & & 1 & & \\ & q & & q & \\ p_g & & h^{1,1} & & p_g \\ & q & & q & \\ & & 1 & & \end{array}$$

The symmetries are $h^{p,q} = h^{q,p}$ (Hodge symmetry) and $h^{p,q} = h^{2-p,2-q}$ (Serre duality). The diamond is determined by $(q, p_g, h^{1,1})$, subject to $h^{1,1} \geq \rho$ (theorem 8.1.5) and $h^{1,1} \geq 1$, the class of an ample divisor being a nonzero $(1,1)$-class.

Example 8.3.6: Hodge Diamonds of Basic Surfaces

1. The plane $\mathbb{P}^2$ has diamond $\begin{smallmatrix} & & 1 & & \\ & 0 & & 0 & \\ 0 & & 1 & & 0 \\ & 0 & & 0 & \\ & & 1 & & \end{smallmatrix}$, so $q = 0$, $p_g = 0$ and $h^{1,1} = 1 = \rho$.

2. A K3 surface has diamond $\begin{smallmatrix} & & 1 & & \\ & 0 & & 0 & \\ 1 & & 20 & & 1 \\ & 0 & & 0 & \\ & & 1 & & \end{smallmatrix}$, so $q = 0$, $p_g = 1$ and $h^{1,1} = 20$, with $1 \leq \rho \leq 20$.

3. An abelian surface has diamond $\begin{smallmatrix} & & 1 & & \\ & 2 & & 2 & \\ 1 & & 4 & & 1 \\ & 2 & & 2 & \\ & & 1 & & \end{smallmatrix}$, so $q = 2$, $p_g = 1$ and $h^{1,1} = 4$, with $1 \leq \rho \leq 4$.

4. A smooth quintic surface $S \subseteq \mathbb{P}^3$ is of general type, with $K_S = H$ and $K_S^2 = 5$. The sequence $0 \to \mathcal{O}(-5) \to \mathcal{O} \to \mathcal{O}_S \to 0$ gives $q = 0$ and $p_g = h^2(\mathcal{O}_S) = h^3(\mathcal{O}_{\mathbb{P}^3}(-5)) = 4$, so $\chi(\mathcal{O}_S) = 5$. Noether's formula gives $\chi_{\text{top}} = 60 - 5 = 55$, and then $h^{1,1} = \chi_{\text{top}} - 2 - 2p_g = 45$.

8.3.3 The Hodge Index Theorem

The intersection pairing descends to a nondegenerate symmetric bilinear form on $\text{Num}(X)$ (definition 8.2.5), and an ample class H has $H^2 > 0$ (lemma 8.2.4). The Hodge index theorem says that the form is negative definite on the orthogonal complement of H, which determines its signature. The proof below, due to Grothendieck [36, Theorem V.1.9], uses Riemann-Roch and Serre duality and no Hodge theory, so it holds in every characteristic.

Lemma 8.3.7: Vanishing of H^2 in Large Degree

Let H be an ample divisor on a smooth projective surface X. If D is a divisor with $D \cdot H > K_X \cdot H$, then $H^2(X, \mathcal{O}_X(D)) = 0$.

***Proof*:**
By Serre duality (corollary 6.5.10), $h^2(\mathcal{O}_X(D)) = h^0(\mathcal{O}_X(K_X - D))$. If this were nonzero, a nonzero section of $\mathcal{O}_X(K_X - D)$ would give an effective divisor $E \sim K_X - D$ (proposition 5.4.47), and lemma 8.2.4(2) with proposition 8.2.2(3) would give $(K_X - D) \cdot H = E \cdot H \geq 0$, contradicting $D \cdot H > K_X \cdot H$. Hence $H^2(X, \mathcal{O}_X(D)) = 0$, as claimed.

On a curve the analogous statement is the vanishing of $H^1(X, \mathcal{L}(D))$ for $\deg D > 2g - 2$ (proposition 7.3.2), with $D \cdot H$ in the role of the degree.

Corollary 8.3.8: Divisors of Positive Square Have Effective Multiples

Let H be an ample divisor on a smooth projective surface X, and let D be a divisor with $D \cdot H > 0$ and $D^2 > 0$. Then $\ell(nD) \geq \chi(\mathcal{O}_X(nD))$ for all $n \gg 0$, the right-hand side tends to ∞ with n, and nD is linearly equivalent to an effective divisor for all $n \gg 0$.

***Proof*:**
For $n\,(D \cdot H) > K_X \cdot H$ we have $nD \cdot H > K_X \cdot H$, so $h^2(\mathcal{O}_X(nD)) = 0$ by lemma 8.3.7, and $\ell(nD) = h^0(\mathcal{O}_X(nD)) \geq h^0 - h^1 = \chi(\mathcal{O}_X(nD))$. By theorem 8.3.1,

$$\chi(\mathcal{O}_X(nD)) = \frac{n^2 D^2 - n\, D \cdot K_X}{2} + \chi(\mathcal{O}_X)$$

which tends to ∞ with n because $D^2 > 0$. So $\ell(nD) > 0$ for $n \gg 0$, and a nonzero section of $\mathcal{O}_X(nD)$ gives an effective divisor linearly equivalent to nD (proposition 5.4.47), completing the proof.

Theorem 8.3.9: Hodge Index Theorem

Let X be a smooth projective surface over an algebraically closed field k of any characteristic, and let H be an ample divisor on X. If D is a divisor with $D \cdot H = 0$, then $D^2 \leq 0$, and $D^2 = 0$ if and only if $D \equiv 0$.

***Proof*:**
If $D \equiv 0$ then $D^2 = 0$. It remains to show that $D \cdot H = 0$ excludes $D^2 > 0$, and that $D \cdot H = 0$ and $D^2 = 0$ together force $D \equiv 0$.

Step 1: $D \cdot H = 0$ implies $D^2 \leq 0$. Suppose $D \cdot H = 0$ and $D^2 > 0$. Since H is ample, definition 5.3.30 applied to $\mathscr{F} = \mathcal{O}_X(D)$ gives an m_0 such that $\mathcal{O}_X(D + mH)$ is generated by global sections for all $m \geq m_0$. Fix $n > m_0$ and put $H' = D + nH$, so that $\mathcal{O}_X(H') = \mathcal{M} \otimes \mathcal{O}_X(H)$ with $\mathcal{M} = \mathcal{O}_X(D + (n-1)H)$ generated by global sections. For a coherent sheaf $\mathscr{F}$,

$$\mathscr{F} \otimes \mathcal{O}_X(H')^{\otimes r} \;\cong\; \mathcal{M}^{\otimes r} \otimes \left(\mathscr{F} \otimes \mathcal{O}_X(H)^{\otimes r}\right)$$

is a tensor product of sheaves generated by global sections once r is large, and products of generating sections generate the stalks of a tensor product, so it is generated by global sections. Hence H' is ample. Now $D \cdot H' = D^2 + n\,(D \cdot H) = D^2 > 0$, and corollary 8.3.8, applied to D and the ample divisor H', gives an $m \geq 1$ and an effective divisor $E \sim mD$. If $E = 0$, then $mD \sim 0$, so $D \equiv 0$ and $D^2 = 0$, which is false. So E is a nonzero effective divisor, and lemma 8.2.4(2)

gives $m\,(D \cdot H) = E \cdot H > 0$, contradicting $D \cdot H = 0$.

Step 2: $D \cdot H = 0$ and $D^2 = 0$ imply $D \equiv 0$. Suppose not, so that some divisor E has $D \cdot E \neq 0$. Put $E' = (H^2)E - (E \cdot H)H$. Then $E' \cdot H = 0$, and $D \cdot E' = (H^2)(D \cdot E) \neq 0$, because $D \cdot H = 0$ and $H^2 > 0$ (lemma 8.2.4(3)). For $t \in \mathbb{Z}$ the divisor $D_t = tD + E'$ satisfies $D_t \cdot H = 0$ and

$$D_t^2 = t^2 D^2 + 2t\,(D \cdot E') + E'^2 = 2t\,(D \cdot E') + E'^2$$

which is positive for a suitable integer t, since $D \cdot E' \neq 0$. This contradicts Step 1 applied to D_t, completing the proof.

The theorem determines the signature of the intersection form once $\mathrm{Num}(X)$ is known to have finite rank.

Corollary 8.3.10: Signature of the Intersection Form

Let X be a smooth projective surface, and suppose that $\mathrm{Num}(X)$ has finite rank r. Then the intersection form on $\mathrm{Num}(X) \otimes_{\mathbb{Z}} \mathbb{R}$ is nondegenerate of signature $(1, r-1)$.

***Proof*:**
Fix an ample divisor H, for instance one with $\mathcal{O}_X(H)$ very ample (proposition 5.3.31). Then $H^2 > 0$ by lemma 8.2.4(3). Let $V = \mathrm{Num}(X) \otimes \mathbb{Q}$ and $H^\perp = \{w \in V : w \cdot H = 0\}$. Every $v \in V$ decomposes as

$$v \;=\; \frac{v \cdot H}{H^2}\,H + \Big(v - \frac{v \cdot H}{H^2}\,H\Big)$$

with the second term in $H^\perp$, so $V = \mathbb{Q}H \oplus H^\perp$, an orthogonal decomposition. If $0 \neq w \in H^\perp$, some multiple mw with $m \geq 1$ is the class of a divisor D with $D \cdot H = 0$, and $D \not\equiv 0$ because its class maps to $mw \neq 0$ in V, so theorem 8.3.9 gives $w^2 = D^2/m^2 < 0$. The form is therefore negative definite on $H^\perp$. In particular no nonzero vector of $H^\perp$ is isotropic, so Gram-Schmidt over $\mathbb{Q}$ produces an orthogonal basis $e_2, \ldots, e_r$ of $H^\perp$ with every $e_i^2 < 0$. Then $H, e_2, \ldots, e_r$ is an orthogonal basis of V, and of $\mathrm{Num}(X) \otimes \mathbb{R} = V \otimes_{\mathbb{Q}} \mathbb{R}$, with $H^2 > 0$ and $e_i^2 < 0$ for $i \geq 2$, so the form on $\mathrm{Num}(X) \otimes \mathbb{R}$ is nondegenerate of signature $(1, r-1)$, as claimed.

The hypothesis always holds. Algebraically equivalent divisors are numerically equivalent [36, Exercise V.1.7], so $\mathrm{Num}(X)$ is a quotient of $\mathrm{NS}(X)$, finitely generated by theorem 8.1.5, and torsion-free, hence free of rank $r \leq \rho(X)$. Over $\mathbb{C}$, numerical and homological equivalence of divisors coincide [36, Appendix A, §5], and $\mathrm{NS}(X)$ embeds in $H^2(X;\mathbb{Z})$ (theorem 8.1.6), so the kernel of $\mathrm{NS}(X) \to \mathrm{Num}(X)$ is the torsion subgroup of $\mathrm{NS}(X)$, $r = \rho(X)$, and the form on $\mathrm{NS}(X) \otimes \mathbb{R}$ has signature $(1, \rho - 1)$.

8.3.11 Remark: The Analytic Proof over $\mathbb{C}$ Over $\mathbb{C}$ the theorem also follows from Hodge theory. The cup product on $H^{1,1}(X) \cap H^2(X;\mathbb{R})$ has signature $(1, h^{1,1}-1)$, with the class of an ample divisor in the positive direction (*Hodge Theory* [17, Hodge Index Theorem]), and on classes of divisors the cup product computes the intersection number. Neither input is proved in this book, and the proof of theorem 8.3.9 uses neither, which is why it holds in every characteristic. In characteristic p, corollary 8.3.12(1), applied on $C \times C$ with the sum of the two fibers through a point as the divisor of positive square, gives the Castelnuovo-Severi inequality, and from it Weil's bound on the number of rational points of a curve C defined over a finite field $\mathbb{F}_q$ (the product taken over $\overline{\mathbb{F}}_q$; section 8.7, following [36, Exercises V.1.9 and V.1.10]).

Corollary 8.3.12: Algebraic Index Inequality

Let C be a divisor on a smooth projective surface X with $C^2 > 0$.

1. If F is a divisor with $F \cdot C = 0$, then $F^2 \leq 0$, with equality if and only if $F \equiv 0$.
2. For every divisor D,
$$(C \cdot D)^2 \;\geq\; C^2 \cdot D^2$$
with equality if and only if $(C^2)\,D \equiv (C \cdot D)\,C$, that is, D is numerically proportional to C.

***Proof*:**
(1) Fix an ample divisor H. By theorem 8.3.9, $C \cdot H \neq 0$, since otherwise $C^2 \leq 0$. Put $G = (F \cdot H)\,C - (C \cdot H)\,F$. Then $G \cdot H = 0$, so $G^2 \leq 0$ by theorem 8.3.9, and since $F \cdot C = 0$,

$$G^2 = (F \cdot H)^2\,C^2 + (C \cdot H)^2\,F^2 \leq 0$$

As $C^2 > 0$ and $(C \cdot H)^2 > 0$, this gives $F^2 \leq 0$. If $F^2 = 0$, then $(F \cdot H)^2 C^2 \leq 0$ forces $F \cdot H = 0$, and theorem 8.3.9 applied to F gives $F \equiv 0$. Conversely $F \equiv 0$ gives $F^2 = 0$.

(2) Put $F = (C^2)\,D - (C \cdot D)\,C$. Then $F \cdot C = 0$ and

$$F^2 = (C^2)^2 D^2 - 2\,C^2 (C \cdot D)^2 + (C \cdot D)^2 C^2 = C^2\big(C^2 D^2 - (C \cdot D)^2\big)$$

By (1), $F^2 \leq 0$, and $C^2 > 0$ gives $C^2 D^2 \leq (C \cdot D)^2$. Equality holds if and only if $F^2 = 0$, which by (1) means $F \equiv 0$, that is, $(C^2)\,D \equiv (C \cdot D)\,C$, completing the proof.

Taking $C = H$ ample in (2) gives $(D^2)(H^2) \leq (D \cdot H)^2$ for every divisor D.

Corollary 8.3.13: Negative Self-Intersection Criterion

Let C be an irreducible curve on a smooth projective surface X.

1. If H is a divisor with $H^2 > 0$ and $C \cdot H = 0$, then $C^2 < 0$.
2. If $C^2 < 0$, then C is the only effective divisor numerically equivalent to C.

***Proof*:**
(1) By corollary 8.3.12(1), with H in the role of C there and C in the role of F, we get $C^2 \leq 0$, with equality only if $C \equiv 0$. A nonzero effective divisor is not numerically equivalent to zero (lemma 8.2.4(2)), so $C^2 < 0$.

(2) Let $C' \neq C$ be an effective divisor with $C' \equiv C$, and fix an ample divisor A. Write $C' = aC + C''$ with $a \geq 0$ and C'' effective, not having C as a component. Then $C \cdot C'' \geq 0$ by proposition 8.2.2(5). If $a = 0$, then $C^2 = C \cdot C' = C \cdot C'' \geq 0$, contradicting $C^2 < 0$. If $a \geq 1$, then $C'' \equiv (1-a)C$, so $C'' \cdot A = (1-a)(C \cdot A) \leq 0$, where $C \cdot A > 0$ by lemma 8.2.4(1). Since C'' is effective, lemma 8.2.4(2) forces $C'' \cdot A = 0$, hence $C'' = 0$ and $a = 1$, so $C' = C$, a contradiction. Hence C is the only effective divisor numerically equivalent to C, completing the proof.

In particular $|C| = \{C\}$ when C is irreducible with $C^2 < 0$: such a curve does not move in its linear system.

8.3.14 Forward Reference: The Hodge Conjecture and Higher-Dimensional Hodge Theory Over $\mathbb{C}$, theorem 8.3.9 is also a consequence of the Hodge-Riemann bilinear relations (box 8.3.11), which hold on every compact Kähler manifold. On a compact Kähler manifold of dimension n with Kähler class ω, they determine the signature of the form $(\alpha, \beta) \mapsto \int_X \alpha \wedge \beta \wedge \omega^{n-2}$ on the real $(1,1)$-classes, using the decomposition into primitive classes that the Hard Lefschetz theorem provides.
The *Hodge conjecture* asserts that for a smooth projective variety X over $\mathbb{C}$, every class in $H^{2p}(X;\mathbb{Q}) \cap H^{p,p}(X)$ is a $\mathbb{Q}$-linear combination of classes of algebraic cycles. For $p = 1$ this is the Lefschetz $(1,1)$ theorem (theorem 8.1.6), so the conjecture holds for surfaces. In general it is open, and it is one of the Clay Millennium Problems. Hodge theory and its relation to algebraic cycles are developed in *Hodge Theory* [17].

8.3.4 The Picard Scheme and Algebraic Equivalence

For a curve X, the line bundles of degree 0 are the k-points of the Jacobian $\mathrm{Jac}(X)$, an abelian variety. For a surface the corresponding object is the *Picard scheme* $\mathrm{Pic}_{X/k}$, a group scheme whose k-points form $\mathrm{Pic}(X)$. Over $\mathbb{C}$ its identity component $\mathrm{Pic}^0_{X/k}$ is an abelian variety of dimension $q = h^{0,1}(X)$ (the irregularity). In characteristic p the scheme $\mathrm{Pic}^0_{X/k}$ can be non-reduced (Igusa), and its reduction is an abelian variety of dimension at most q. The connected components of $\mathrm{Pic}_{X/k}$ are the cosets of $\mathrm{Pic}^0_{X/k}$, and they are indexed by the Néron-Severi group $\mathrm{NS}(X)$ of definition 8.1.4.

Three equivalence relations on divisors are in use, each coarser than the one before:

1. *Linear equivalence* ($\sim$): $D_1 \sim D_2$ if $D_1 - D_2 = \mathrm{div}(f)$. The classes form $\mathrm{Pic}(X)$.
2. *Algebraic equivalence* ($\sim_{\mathrm{alg}}$): the equivalence relation generated by declaring $D_1 \sim_{\mathrm{alg}} D_2$ when there are a connected variety T, points $t_1, t_2 \in T$, and a line bundle $\mathcal{L}$ on $X \times T$ with $\mathcal{L}|_{X \times \{t_i\}} \cong \mathcal{O}_X(D_i)$ for $i = 1, 2$. The classes form the Néron-Severi group $\mathrm{NS}(X) = \mathrm{Pic}(X)/\sim_{\mathrm{alg}}$.
3. *Numerical equivalence* ($\equiv$, definition 8.2.5). The classes form $\mathrm{Num}(X)$.

By the theorem of the base (theorem 8.1.5) the group $\mathrm{NS}(X)$ is finitely generated, and Matsusaka proved that numerical and algebraic equivalence agree up to torsion: the kernel of $\mathrm{NS}(X) \to \mathrm{Num}(X)$

is the torsion subgroup of $\mathrm{NS}(X)$, so $\mathrm{Num}(X) \cong \mathrm{NS}(X)/\text{torsion}$. If $q = 0$, then $\mathrm{Pic}^0_{X/k}$ is trivial, because its tangent space at the identity is $H^1(X, \mathcal{O}_X) = 0$. Linear and algebraic equivalence then coincide, and numerical equivalence differs from them only on the torsion of $\mathrm{Pic}\,(X)$. On an Enriques surface (section 8.6.3), for example, K_X is a nonzero torsion class that is numerically trivial.

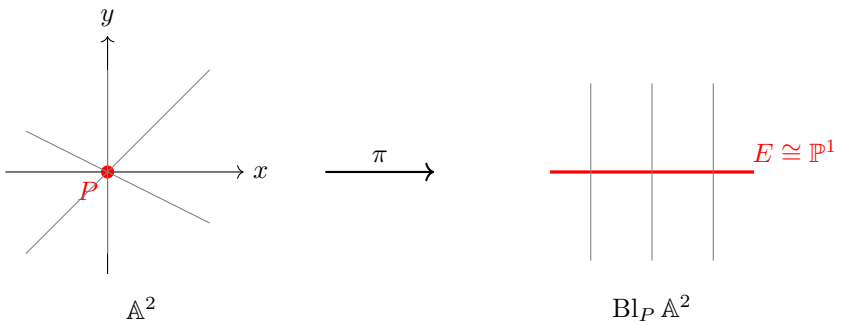
8.4 Blow-ups

Blowing up a closed point of a smooth projective surface raises its Picard number by one (theorem 8.4.6(2) and (3)), which is why the blow-up is not isomorphic to the surface (box 8.1.3). By a theorem of Zariski (theorem 8.4.11), every birational morphism between smooth projective surfaces is a composition of blow-ups at points, and every birational map between them factors into blow-ups at points and their inverses [36, Theorem V.5.5].

The model case is the plane. Consider $\mathbb{A}^2_k$ with coordinates (x, y) and the origin $P = (0, 0)$. The *blow-up* of $\mathbb{A}^2$ at P replaces P by a copy of $\mathbb{P}^1$ parametrizing the *directions* through P. Every line through the origin has a well-defined direction $[x : y] \in \mathbb{P}^1$ (the vertical line included), and the blow-up has one point over the origin for each of these directions. Concretely, the blow-up is the closed subvariety

$$\mathrm{Bl}_P \mathbb{A}^2 \;=\; \{((x, y), [s : t]) \in \mathbb{A}^2 \times \mathbb{P}^1 : xt = ys\} \;\subseteq\; \mathbb{A}^2 \times \mathbb{P}^1$$

The projection $\pi\colon \mathrm{Bl}_P \mathbb{A}^2 \to \mathbb{A}^2$ onto the first factor is an isomorphism away from the origin, with inverse $(x, y) \mapsto ((x, y), [x : y])$ on $\mathbb{A}^2 \backslash \{(0, 0)\}$. Over the origin, the fiber $\pi^{-1}(0, 0) = \{(0, 0)\} \times \mathbb{P}^1 \cong \mathbb{P}^1$ contains every direction. This fiber is the *exceptional divisor* E.



The strict transforms of distinct lines through the origin, the closures of their preimages away from P, are disjoint and meet E at the distinct points given by their directions. So the blow-up separates curves through P with distinct tangent directions. In particular a node, two branches crossing at a point with distinct tangent lines, is resolved by one blow-up (example 8.4.7(2)). Example 3.1.32 exhibits the two branches of the same node in the completed local ring.

The same construction applies to any closed subscheme of a Noetherian scheme, with the Rees algebra of its ideal sheaf in place of the incidence equation $xt = ys$.

Definition 8.4.1: Blow-up

Let X be a Noetherian scheme and $\mathcal{I} \subseteq \mathcal{O}_X$ a coherent ideal sheaf defining a closed subscheme $Z \subseteq X$. The *blow-up of X along Z* (or *along $\mathcal{I}$*) is

$$\mathrm{Bl}_Z X \;=\; \mathrm{Proj}\left(\bigoplus_{n\geq 0} \mathcal{I}^n\right)$$

where $\bigoplus_{n\geq 0}\mathcal{I}^n$ is the *Rees algebra* of $\mathcal{I}$, the graded $\mathcal{O}_X$-algebra whose degree-n piece is the nth power of the ideal sheaf (with $\mathcal{I}^0 = \mathcal{O}_X$). The natural map $\pi\colon \mathrm{Bl}_Z X \to X$ is called the *blow-up morphism*. The *exceptional divisor* is the closed subscheme $E = \pi^{-1}(Z)$ of $\mathrm{Bl}_Z X$ defined by the inverse image ideal sheaf $\pi^{-1}\mathcal{I}\cdot\mathcal{O}_{\mathrm{Bl}_Z X}$. Its underlying set is the preimage of Z.

The Proj construction here is the *relative Proj* over X (theorem 5.3.49, which applies as the Rees algebra is generated in degree one): locally on an affine open $U = \operatorname{Spec} A$ of X with $\mathcal{I}|_U = \widetilde{I}$ for an ideal $I \subseteq A$, the blow-up restricts to $\operatorname{Proj}\bigoplus_{n\geq 0} I^n$, the classical blow-up algebra of I.

Proposition 8.4.2: Basic Properties of Blow-ups

Let $\pi\colon \mathrm{Bl}_Z X \to X$ be the blow-up of a Noetherian scheme X along a closed subscheme Z defined by $\mathcal{I}$.

1. π is proper, and projective over every affine open of X. If X admits an ample invertible sheaf, then π is projective.
2. π is an isomorphism over $X \setminus Z$: the restriction $\pi^{-1}(X\setminus Z) \xrightarrow{\sim} X \setminus Z$ is an isomorphism. If Z has codimension ≥ 1, then π is birational.
3. The inverse image ideal sheaf $\pi^{-1}\mathcal{I}\cdot\mathcal{O}_{\mathrm{Bl}_Z X}$ is invertible, so the exceptional divisor $E = V(\pi^{-1}\mathcal{I}\cdot\mathcal{O}_{\mathrm{Bl}_Z X})$ is an effective Cartier divisor.
4. If $f\colon Y \to X$ is a morphism such that $f^{-1}\mathcal{I}\cdot\mathcal{O}_Y$ is an invertible ideal sheaf, then f factors uniquely through π.

***Proof*:**

1. The Rees algebra is generated in degree one (example 5.3.48(2)) and $\mathcal{I}$ is coherent, so theorem 5.3.49 makes π proper and projective over every affine open of X. When X admits an ample invertible sheaf, π is projective by [36, Proposition II.7.10(b)].
2. Over the complement of Z, $\mathcal{I} = \mathcal{O}_X$, so $\mathcal{I}^n = \mathcal{O}_X$ for all n, and Proj of the polynomial algebra $\mathcal{O}_X[t]$ (in a single variable of degree 1) is X itself. If Z has codimension ≥ 1, it contains no irreducible component of X, so $X \setminus Z$ is dense in X. Its preimage $\pi^{-1}(X\setminus Z)$ is the complement of E, and it is dense in $\mathrm{Bl}_Z X$: by item (3), whose proof does not use this one, E is cut out on each chart by a nonzerodivisor of a Noetherian ring, and a nonzerodivisor lies in no minimal prime. So π restricts to an isomorphism between dense opens, and it is birational.
3. Let $U = \operatorname{Spec} A$ be an affine open with $\mathcal{I}|_U = \widetilde{I}$. The ring A is Noetherian (ex-

ample 2.2.32(2)), so $I = (a_0, \dots, a_m)$ for some $a_i \in A$, and $\pi^{-1}(U) = \operatorname{Proj} S$ for $S = \bigoplus_{n\geq 0} I^n t^n \subseteq A[t]$, covered by the charts $D_+(a_i t) = \operatorname{Spec} S_{(a_i t)}$ (lemma 2.3.2). In $S_{(a_i t)}$ each a_j equals $\big((a_j t)/(a_i t)\big)\, a_i$, so $IS_{(a_i t)} = a_i S_{(a_i t)}$. The element a_i is a nonzerodivisor of $S_{(a_i t)}$: if $a_i \cdot s/(a_i t)^n = 0$ with $s \in I^n t^n$, then $(a_i t)^N a_i s = a_i^{N+1} t^N s = 0$ in $A[t]$ for some N, so $a_i^{N+1} s = 0$, hence $(a_i t)^{N+1} s = 0$ and $s/(a_i t)^n = 0$. Thus $\pi^{-1}\mathcal{I} \cdot \mathcal{O}_{\operatorname{Bl}_Z X}$ is locally generated by a single nonzerodivisor, which is to say that it is invertible.

4. This is the universal property of blowing up, [36, Proposition II.7.14] or [71, § 22]. Its proof reduces to the affine case and applies theorem 5.3.43 to the generators of $f^{-1}\mathcal{I} \cdot \mathcal{O}_Y$, completing the proof.

By proposition 8.4.2(3), the ideal $\mathcal{I}$, which need not be locally principal, becomes locally principal on the blow-up, generated on each chart by a nonzerodivisor, and E is the effective Cartier divisor it cuts out.

When the centre is nonsingular and sits inside a nonsingular variety, the blow-up is again nonsingular. This is the case used below, for the blow-up of a surface at a point (theorem 8.4.6) and for a nonsingular base locus of a pencil (the paragraph after lemma 8.4.4).

Proposition 8.4.3: Blow-up Along a Nonsingular Centre

Let X be a nonsingular variety over an algebraically closed field k and let $Z \subseteq X$ be a nonsingular closed subvariety of codimension $d \geq 1$. Then $\operatorname{Bl}_Z X$ is a nonsingular variety of the same dimension as X, and the blow-down $\pi\colon \operatorname{Bl}_Z X \to X$ is proper and birational. If X is quasi-projective, then π is projective.

***Proof*:**
Properness and birationality are proposition 8.4.2(1) and (2), and $\operatorname{Bl}_Z X$ is separated and of finite type over k because π is proper and X is a variety. If X is quasi-projective, π is projective by [36, Proposition II.7.16(c)].

Step 1: $\operatorname{Bl}_Z X$ is a variety of dimension $n = \dim X$. Let $\operatorname{Spec} A \subseteq X$ be an affine open meeting Z, with $\mathcal{I}|_{\operatorname{Spec} A} = \widetilde{I}$ and $I = (h_1, \dots, h_m)$, every h_i taken nonzero (zero generators may be discarded). The blow-up over $\operatorname{Spec} A$ is Proj of the Rees algebra $S = \bigoplus_{n\geq 0} I^n t^n \subseteq A[t]$, generated in degree one by the $h_i t$, so it is covered by the charts $D_+(h_i t) = \operatorname{Spec} S_{(h_i t)}$. Here A is a domain, X being integral, so S is a graded subring of a domain and every $S_{(h_i t)}$ is a domain. Hence $\operatorname{Bl}_Z X$ is reduced. Moreover

$$A \;\subseteq\; S_{(h_i t)} \;=\; A\big[h_1/h_i, \dots, h_m/h_i\big] \;\subseteq\; A_{h_i} \;\subseteq\; \operatorname{Frac} A$$

the middle equality because S is generated in degree one by the $h_j t$. So each chart is a finitely generated k-domain with fraction field $\operatorname{Frac} A$ and therefore has dimension $\operatorname{trdeg}_k \operatorname{Frac} A = n$ by theorem 2.6.11. Over an affine open of X missing Z the blow-down is an isomorphism (proposition 8.4.2(2)), so those opens have dimension n too. As $\operatorname{Bl}_Z X$ is covered by charts of these two kinds, proposition 2.6.2 gives $\dim \operatorname{Bl}_Z X = n$.

For irreducibility, let $U = \pi^{-1}(X \setminus Z)$, isomorphic to $X \setminus Z$ by proposition 8.4.2(2) and hence irreducible and nonempty, X being a variety. On the chart $D_+(h_i t)$ the ideal $IS_{(h_i t)}$ is generated by h_i, since $h_j = h_i \cdot (h_j/h_i)$ there, so U meets that chart in $D(h_i)$, a nonempty open of an irreducible scheme and hence dense in it. So U is dense in every chart, hence dense in $\operatorname{Bl}_Z X$,

which is therefore the closure of an irreducible set and so irreducible (lemma 2.1.24(1)). Thus $\mathrm{Bl}_Z X$ is an integral scheme, birational to X.

Step 2: local generators with independent differentials. Fix a closed point $z \in Z$. Since X and Z are nonsingular, theorem 5.5.27 makes $\mathcal{I}/\mathcal{I}^2$ locally free of rank d and the conormal sequence $0 \to \mathcal{I}/\mathcal{I}^2 \to \Omega_{X/k} \otimes \mathcal{O}_Z \to \Omega_{Z/k} \to 0$ exact with all three terms locally free. A short exact sequence with locally free quotient is locally split (lemma 5.5.20), so it stays exact after $\otimes\, \kappa(w)$ at every point $w \in Z$. Choose $a_1, \dots, a_d \in \mathcal{I}$ whose classes form a basis of $\mathcal{I}/\mathcal{I}^2$ near z. After shrinking, they generate $\mathcal{I}$ by Nakayama, and since the conormal sequence stays exact after $\otimes\, \kappa(w)$, their differentials $da_1, \dots, da_d$ are linearly independent in $\Omega_{X/k} \otimes \kappa(w)$ at every closed point $w \in Z$ of that neighbourhood. Being members of a basis of a free module of rank $d \geq 1$ over a nonzero ring, each a_i is nonzero. Shrink further so that the neighbourhood is an affine $U_z = \operatorname{Spec} A$ realized as a closed subscheme $\operatorname{Spec} k[x_1, \dots, x_N]/(f_1, \dots, f_r)$ of $\mathbb{A}^N_k$. At a closed point of $\operatorname{Spec} A$ the local ring has dimension n by theorem 2.6.12, so nonsingularity of X and proposition 2.7.7 make the Jacobian of the f_i of rank $N - n$ there.

Step 3: the charts. The algebra S is generated in degree one by $a_1 t, \dots, a_d t$, so the charts $D_+(a_i t)$ cover $\pi^{-1}(U_z)$. By symmetry it is enough to treat $i = 1$. Since $I = (a_1, \dots, a_d)$, every element of I^n is an A-linear combination of degree-n monomials in the a_j, so

$$S_{(a_1 t)} \;=\; A\big[a_2/a_1,\ \dots,\ a_d/a_1\big] \;\subseteq\; A_{a_1}$$

Let $C = A[y_2, \dots, y_d]/(g_2, \dots, g_d)$ with $g_j = a_j - a_1 y_j$, and let $\varphi\colon C \to S_{(a_1 t)}$ send $y_j \mapsto a_j/a_1$. This is well defined and surjective. Presenting A as in Step 2 realizes $\operatorname{Spec} C$ as the closed subscheme of $\mathbb{A}^M_k$, $M = N + d - 1$, cut out by $f_1, \dots, f_r, g_2, \dots, g_d$, where each a_j is read through a polynomial lift, well defined modulo $(f_1, \dots, f_r)$. Every computation below is made modulo the span of $df_1, \dots, df_r$, so the choice of lift does not matter.

Step 4: $\operatorname{Spec} C$ is regular, and each of its components has dimension at least n. Let $P \in \operatorname{Spec} C$ be a closed point, lying over $Q \in \operatorname{Spec} A$. Since k is algebraically closed and P is closed, $\kappa(P) = k$. Then $k \subseteq \kappa(Q) \subseteq \kappa(P)$ forces $\kappa(Q) = k$, so Q is closed too. In $\Omega_{\mathbb{A}^M} \otimes \kappa(P)$, spanned by $dx_1, \dots, dx_N, dy_2, \dots, dy_d$, we have $dg_j = da_j - y_j(P)\, da_1 - a_1(Q)\, dy_j$. If $a_1(Q) \neq 0$ the dy-components of $dg_2, \dots, dg_d$ are $-a_1(Q)$ times the identity matrix, so these $d - 1$ forms are independent modulo the span of $df_1, \dots, df_r$, which involves no dy. If $a_1(Q) = 0$, then $a_j(Q) = a_1(Q) y_j(P) = 0$ for every j, so $Q \in Z$. There $dg_j = da_j - y_j(P)\, da_1$, and since $da_1, \dots, da_d$ are independent in $\Omega_{X/k} \otimes \kappa(Q)$, that is modulo the span of the df_i, reading off the coefficient of da_j shows $dg_2, \dots, dg_d$ are again independent modulo that span. Either way the Jacobian of $(f_1, \dots, f_r, g_2, \dots, g_d)$ at P has rank $(N - n) + (d - 1) = M - n$.

That rank computes the cotangent space. Because $\kappa(P) = k$ is separable over k, the map $\mathfrak{n}/\mathfrak{n}^2 \to \Omega_{\mathbb{A}^M/k} \otimes \kappa(P)$ is an isomorphism (proposition 5.5.10), where $\mathfrak{n}$ is the maximal ideal of P in $\mathbb{A}^M_k$. The surjection $\mathfrak{n}/\mathfrak{n}^2 \to \mathfrak{m}_P/\mathfrak{m}_P^2$ has kernel the span of the images of the defining equations, of dimension the Jacobian rank, so $\dim_k \mathfrak{m}_P/\mathfrak{m}_P^2 = M - (M - n) = n$ and hence $\dim \mathcal{O}_{C,P} \leq n$ by [10, Dimension and Embedding Dimension].

For the reverse, work inside $A[y_2, \dots, y_d]$. That ring is a finitely generated k-domain whose fraction field has transcendence degree $n + (d - 1)$ over k, so it has dimension $n + d - 1$ by theorem 2.6.11, and $\operatorname{Spec} C$ is cut out in it by the $d - 1$ equations $g_2, \dots, g_d$. So every irreducible component of $\operatorname{Spec} C$ has codimension at most $d - 1$ by corollary 2.6.15, hence dimension at least n by theorem 2.6.12. If W is a component through the closed point P, then applying theorem 2.6.12 to $\{P\} \subseteq W$ gives $\dim \mathcal{O}_{W,P} = \dim W \geq n$, so $\dim \mathcal{O}_{C,P} \geq n$. Therefore $\dim \mathcal{O}_{C,P} = n = \dim_k \mathfrak{m}_P/\mathfrak{m}_P^2$, which is regularity at P (definition 2.7.3). This holds at every

closed point, so $\operatorname{Spec} C$ is regular by box 5.5.24.

Step 5: the presentation is an isomorphism. By Step 2 the ring A is a domain and $a_1 \neq 0$. On the locus $a_1 \neq 0$ the equations $g_j = a_j - a_1 y_j$ say exactly that $y_j = a_j/a_1$, so $C_{a_1} \cong A_{a_1}$ and $\operatorname{Spec} C$ meets that locus in a copy of $\operatorname{Spec} A_{a_1}$: irreducible, and of dimension n, being a nonempty open of the n-dimensional irreducible $\operatorname{Spec} A$. On the complementary locus $a_1 = 0$ every a_j vanishes as well, so $C/(a_1) = (A/I)[y_2, \dots, y_d]$. That ring is a finitely generated k-domain of transcendence degree $(n-d)+(d-1)$, using $\dim A/I = \dim Z = n-d$ from theorem 2.6.12, hence of dimension $n-1$ by theorem 2.6.11. By Step 4 every component of $\operatorname{Spec} C$ has dimension at least n, so none is contained in the locus $a_1 = 0$. Every component therefore meets the irreducible open $\{a_1 \neq 0\}$, and the argument of Step 1 makes $\operatorname{Spec} C$ irreducible. Being regular, $\operatorname{Spec} C$ is reduced (proposition 2.7.4), hence integral, so C is a domain, of dimension n because it contains the dense open $\operatorname{Spec} A_{a_1}$ of that dimension.

Now $\varphi\colon C \to S_{(a_1 t)}$ is a surjection onto a domain, itself of dimension n by Step 1. So $\ker \varphi$ is a prime $\mathfrak{p}$ with $\dim C/\mathfrak{p} = \dim C$, hence of codimension zero by theorem 2.6.12, hence zero. Thus $S_{(a_1 t)} \cong C$ is regular.

Step 6: globalization. Every closed point of Z has a neighbourhood U_z as in Step 2, and these cover Z. Over each, Steps 3 to 5 make the charts $D_+(a_i t)$ regular, and those charts cover $\pi^{-1}(U_z)$. Over the complement, $\pi^{-1}(X \setminus Z) \cong X \setminus Z$ is regular because X is. Every closed point of $\operatorname{Bl}_Z X$ lies over a closed point of X, so every closed point is covered by one of these cases. Regularity at closed points suffices (box 5.5.24), so $\operatorname{Bl}_Z X$ is regular. Being in addition an integral separated finite-type k-scheme, it is a nonsingular variety of dimension n, as claimed.

Two sections s_0, s_1 of an invertible sheaf $\mathcal{L}$ span a pencil when they are linearly independent, and theorem 5.3.43 turns them into the morphism $x \mapsto [s_0(x) : s_1(x)]$ to $\mathbb{P}^1$ away from the *scheme-theoretic base locus* $B = V(s_0, s_1)$ on which both vanish. Blowing up B extends this morphism to all of $\operatorname{Bl}_B X$, and the blow-up is the closure in $X \times \mathbb{P}^1$ of the graph of the morphism.

Lemma 8.4.4: The Blow-up as the Closure of a Graph

Let R be a ring, X an integral Noetherian R-scheme, $\mathcal{L}$ an invertible sheaf on X, and $s_0, s_1 \in \Gamma(X, \mathcal{L})$ two nonzero sections. Let $\mathcal{I} \subseteq \mathcal{O}_X$ be the image of

$$(s_0, s_1)\colon \mathcal{L}^\vee \oplus \mathcal{L}^\vee \longrightarrow \mathcal{O}_X, \qquad (\lambda, \mu) \longmapsto \lambda(s_0) + \mu(s_1)$$

a coherent ideal sheaf, let $B = V(s_0, s_1)$ be the closed subscheme of X it defines (proposition 5.1.34), and let $\pi\colon \widetilde{X} := \mathrm{Bl}_B X \to X$ be the blow-up along $\mathcal{I}$ (definition 8.4.1). Write $\mathbb{P}^1_R = \operatorname{Proj} R[T_0, T_1]$. The open $X \setminus B$ is nonempty and s_0, s_1 generate $\mathcal{L}$ over it, so theorem 5.3.43(2) gives an R-morphism

$$\varphi\colon X \setminus B \longrightarrow \mathbb{P}^1_R, \qquad x \longmapsto [s_0(x) : s_1(x)]$$

with $\varphi^*\mathcal{O}(1) \cong \mathcal{L}|_{X\setminus B}$ and $\varphi^* T_i = s_i$. Let $\Gamma_\varphi \subseteq (X \setminus B) \times_R \mathbb{P}^1_R \subseteq X \times_R \mathbb{P}^1_R$ be the image of (id, φ), the graph of φ, the first inclusion an open subscheme by proposition 3.2.5(1), applied to the projection to X. Then:

1. The $\mathcal{O}_{\widetilde{X}}$-submodule $\mathcal{M} \subseteq \pi^*\mathcal{L}$ generated by $\pi^* s_0$ and $\pi^* s_1$ is invertible and equals $(\mathcal{I}\mathcal{O}_{\widetilde{X}}) \cdot \pi^*\mathcal{L}$, so theorem 5.3.43(2) gives an R-morphism $g\colon \widetilde{X} \to \mathbb{P}^1_R$ with $g^*\mathcal{O}(1) \cong \mathcal{M}$ and $g^* T_i = \pi^* s_i$. Moreover π restricts to an isomorphism $\pi_W\colon W := \pi^{-1}(X\setminus B) \to X \setminus B$, and $g|_W = \varphi \circ \pi_W$.
2. The morphism $j := (\pi, g)\colon \widetilde{X} \to X \times_R \mathbb{P}^1_R$ is a closed immersion.
3. $\widetilde{X}$ is integral, and j identifies $\widetilde{X}$ with the closure $\overline{\Gamma_\varphi}$ of the graph in $X \times_R \mathbb{P}^1_R$, carrying its reduced induced structure (definition 3.1.26).

***Proof*:**

Step 1: the local model. Let $U = \operatorname{Spec} A \subseteq X$ be a nonempty affine open on which $\mathcal{L}$ is free with generator $e \in \mathcal{L}(U)$. Such opens cover X, since $\mathcal{L}$ is invertible and the affine opens form a basis. Write $s_i|_U = a_i e$ with $a_i \in A$. Under $\mathcal{L}^\vee|_U \cong \mathcal{O}_U$, $\lambda \mapsto \lambda(e)$, the map (s_0, s_1) becomes $\widetilde{(a_0, a_1)}\colon \widetilde{A^2} \to \widetilde{A}$, so $\mathcal{I}|_U = \widetilde{I}$ with $I = (a_0, a_1)$ (proposition 5.1.14(3)). Thus $\mathcal{I}$ is coherent, A being Noetherian (example 2.2.32(2)), and $B \cap U = V(I)$ (corollary 5.1.35), so $U \setminus B = D(a_0) \cup D(a_1)$ is the set of points at which some s_i generates $\mathcal{L}$. The ring A is a domain (definition 2.4.19) and $a_0 a_1 \neq 0$: if $a_i = 0$ then s_i vanishes on U, hence on every trivializing open V of $\mathcal{L}$, because $V \cap U$ is nonempty (X is irreducible) and $\mathcal{L}(V) \cong \mathcal{O}_X(V) \to \mathcal{O}_X(V \cap U)$ is injective (proposition 2.4.23), contradicting $s_i \neq 0$. In particular $X \setminus B \supseteq D(a_0) \neq \emptyset$.

By definition 8.4.1, example 5.3.48(2) and theorem 5.3.49(1), $\widetilde{U} := \pi^{-1}(U) \cong \operatorname{Proj} S$ over U, where $S = \bigoplus_{n\geq 0} I^n t^n \subseteq A[t]$ is the Rees algebra of I, a domain generated over A by $a_0 t$ and $a_1 t$. Hence $\widetilde{U}$ is covered by the charts $D_+(a_i t) = \operatorname{Spec} S_{(a_i t)}$ (lemma 2.3.2). The generators lie in degree one, so an element $s/(a_i t)^n$ of $S_{(a_i t)}$ with $s \in I^n t^n$ is a polynomial in a_{1-i}/a_i with coefficients in A, and conversely $a_{1-i}/a_i = (a_{1-i}t)/(a_i t)$ lies in $S_{(a_i t)}$. Hence

$$A \subseteq S_{(a_i t)} = A[a_{1-i}/a_i] \subseteq A_{a_i} \subseteq \operatorname{Frac}(A)$$

the inclusion $A \subseteq S_{(a_i t)}$ being $\pi^\#$ (example 3.1.11). Both charts are nonempty and they meet in $D_+(a_0 a_1 t^2) = \operatorname{Spec} S_{(a_0 a_1 t^2)}$, which is nonempty because $a_0 a_1 t^2 \neq 0$ in the domain S. In $S_{(a_i t)}$

we have $a_{1-i} = (a_{1-i}/a_i)\,a_i$, so $IS_{(a_it)} = a_i S_{(a_it)}$ is free of rank one on the nonzerodivisor a_i.

Step 2: the invertible sheaf $\mathcal{M}$ and the morphism g. On $\widetilde{U}$ the section π^*e generates $\pi^*\mathcal{L}$ and $\pi^*s_i = a_i\,\pi^*e$, so under $\pi^*\mathcal{L}|_{\widetilde{U}} \cong \mathcal{O}_{\widetilde{U}}$ the submodule $\mathcal{M}|_{\widetilde{U}}$ is the ideal sheaf generated by a_0, a_1, namely $I\mathcal{O}_{\widetilde{U}} = (\mathcal{I}\mathcal{O}_{\widetilde{X}})|_{\widetilde{U}}$, whose restriction to the chart $D_+(a_it)$ is $\widetilde{IS_{(a_it)}}$ (proposition 5.1.14(3),(7)), free of rank one on a_i by Step 1. Hence $\mathcal{M} = (\mathcal{I}\mathcal{O}_{\widetilde{X}}) \cdot \pi^*\mathcal{L}$ is invertible and generated by π^*s_0, π^*s_1, and theorem 5.3.43(2) yields g. On the chart $D_+(a_kt) = \operatorname{Spec} S_{(a_kt)}$, where $\mathcal{M}$ is free on a_k, the section $\pi^*s_i = (a_i/a_k)\,a_k\,\pi^*e$ generates $\mathcal{M}$ at a point if and only if a_i/a_k is a unit there, that is, on $D(a_i/a_k) = D_+(a_it) \cap D_+(a_kt)$ (a homogeneous prime $\mathfrak{p} \in D_+(a_kt)$ contains a_it if and only if $a_i/a_k = (a_it)/(a_kt)$ lies in $(\mathfrak{p}S_{a_kt})_0$). Taking the union over $k = 0, 1$, the non-vanishing locus of π^*s_i in $\widetilde{U}$ is $D_+(a_it)$. The equality $\widetilde{U} \cap g^{-1}(D_+(T_i)) = D_+(a_it)$ follows from theorem 5.3.43(2) alone. In one direction, T_i generates $\mathcal{O}(1)$ over $D_+(T_i)$ (proposition 5.3.5) and $g^*T_i = \pi^*s_i$, so π^*s_i generates $\mathcal{M}$ over $g^{-1}(D_+(T_i))$. In the other, a point of $\mathbb{P}^1_R$ outside $D_+(T_i)$ lies in $D_+(T_{1-i})$, where $\mathcal{O}(1)$ is free on T_{1-i} (proposition 5.3.5) and T_i corresponds to the function T_i/T_{1-i}, whose non-vanishing locus is $D_+(T_i) \cap D_+(T_{1-i})$. So if $g(z) \notin D_+(T_i)$ then T_i does not generate $\mathcal{O}(1)$ at $g(z)$, its germ lies in $\mathfrak{m}_{g(z)}\mathcal{O}(1)_{g(z)}$, the germ of $g^*T_i = \pi^*s_i$ lies in $\mathfrak{m}_z\mathcal{M}_z$, because $g^\#$ is a local homomorphism, and z is outside the non-vanishing locus of π^*s_i. Finally, over $D_+(T_i)$ the section T_{1-i} equals $(T_{1-i}/T_i)\,T_i$ in $\mathcal{O}(1)$, so pulling back, $\pi^*s_{1-i} = g^\#(T_{1-i}/T_i)\,\pi^*s_i$ in $\mathcal{M}$. On $D_+(a_it)$, where $\mathcal{M}$ is free on $a_i\,\pi^*e$ and $\pi^*s_{1-i} = (a_{1-i}/a_i)(a_i\,\pi^*e)$, comparing coefficients of the generator gives $g^\#(T_{1-i}/T_i) = a_{1-i}/a_i \in S_{(a_it)}$.

Step 3: π over $X \setminus B$. Let $V = \operatorname{Spec} A' \subseteq X \setminus B$ be affine open. Then $\mathcal{I}|_V = \mathcal{O}_V$, since $\mathcal{I}_x = \mathcal{O}_{X,x}$ for $x \notin B$, so the Rees algebra restricts to $\widetilde{A'[t]}$ with t in degree one. A homogeneous prime of $A'[t]$ not containing $A'[t]_+ = tA'[t]$ omits t, so $\pi^{-1}(V) \cong \operatorname{Proj} A'[t] = D_+(t) = \operatorname{Spec}(A'[t]_t)_0 = \operatorname{Spec} A'$ over V, the structure map being induced by the identity $A' \to (A'[t]_t)_0 = A'$. Thus π is an isomorphism over each such V, and the local inverses glue (lemma 3.1.9) to an inverse of π_W. Pullback preserves surjections, so π^*s_0, π^*s_1 generate $\pi^*\mathcal{L}$ over W and $\mathcal{M}|_W = \pi^*\mathcal{L}|_W$. Hence $g|_W$ and $\varphi \circ \pi_W$ are R-morphisms $W \to \mathbb{P}^1_R$ pulling $(\mathcal{O}(1), T_0, T_1)$ back to $(\pi^*\mathcal{L}|_W, \pi^*s_0, \pi^*s_1)$, and the uniqueness in theorem 5.3.43(2) gives $g|_W = \varphi \circ \pi_W$.

Step 4: j is a closed immersion. A morphism that restricts to a closed immersion over every member of an open cover of the target is a closed immersion (lemma 5.3.39). Now the opens $V_{U,i} := U \times_R D_+(T_i) \cong \operatorname{Spec} A[T_{1-i}/T_i]$, for U as in Step 1 and $i = 0, 1$, cover $X \times_R \mathbb{P}^1_R$ (proposition 3.2.5(1), applied to each projection in turn), and $j^{-1}(V_{U,i}) = \pi^{-1}(U) \cap g^{-1}(D_+(T_i)) = D_+(a_it)$ by Step 2. The restriction $D_+(a_it) \to V_{U,i}$ of j is the morphism of affine schemes induced by the ring map $A[T_{1-i}/T_i] \to S_{(a_it)}$ that is $\pi^\#$, the inclusion, on A and sends T_{1-i}/T_i to $g^\#(T_{1-i}/T_i) = a_{1-i}/a_i$ (Steps 1 and 2). Its image is $A[a_{1-i}/a_i] = S_{(a_it)}$, so it is surjective and $j^{-1}(V_{U,i}) \to V_{U,i}$ is a closed immersion (proposition 3.1.24(2)). Hence j is a closed immersion.

Step 5: $\widetilde{X}$ is integral and $j(\widetilde{X}) = \overline{\Gamma_\varphi}$. Every chart $D_+(a_it) = \operatorname{Spec} S_{(a_it)}$ is the spectrum of a domain, so the stalks of $\mathcal{O}_{\widetilde{X}}$ have no nonzero nilpotents and $\widetilde{X}$ is reduced (proposition 2.4.15). Each $\widetilde{U}$ is irreducible: a nonempty open $O \subseteq \widetilde{U}$ meets one of the two irreducible charts in a nonempty open, which is dense in that chart and so meets the nonempty open $D_+(a_0a_1t^2)$ of it (Step 1). Two nonempty opens of $\widetilde{U}$ therefore meet $D_+(a_0a_1t^2)$ in nonempty opens of the irreducible chart $D_+(a_0t)$, and these intersect. Now $W \cong X \setminus B$ is a nonempty open of the irreducible X, hence irreducible, and $W \cap \widetilde{U} = \pi^{-1}(U \setminus B)$ is nonempty, π being an isomorphism over the nonempty $U \setminus B$ (Steps 1 and 3), and it is a nonempty open of the irreducible $\widetilde{U}$, hence

dense. The $\widetilde{U}$ cover $\widetilde{X}$, so W is dense in $\widetilde{X}$ and $\widetilde{X} = \overline{W}$ is irreducible, being the closure of an irreducible subspace (lemma 2.1.24(1)). Thus $\widetilde{X}$ is integral (proposition 2.4.26).

By Step 3, $j|_W = (\pi_W, \varphi \circ \pi_W) = (\mathrm{id}, \varphi) \circ \pi_W$ with π_W an isomorphism, so $j(W) = \Gamma_\varphi$. As j is a homeomorphism onto the closed set $j(\widetilde{X})$ and W is dense in $\widetilde{X}$, we get $j(\widetilde{X}) = \overline{j(W)} = \overline{\Gamma_\varphi}$. The closed subscheme $j(\widetilde{X}) \cong \widetilde{X}$ is reduced, and a reduced closed subscheme is determined by its underlying closed set (the uniqueness clause of proposition 3.1.25). Hence j identifies $\widetilde{X}$ with $\overline{\Gamma_\varphi}$ carrying its reduced induced structure, completing the proof.

When X is a nonsingular projective variety over an algebraically closed field $k = R$ and the base locus B is a nonsingular closed subvariety of codimension two, proposition 8.4.3 shows that $\overline{\Gamma_\varphi} = \mathrm{Bl}_B X$ is again a nonsingular variety, birational to X through π. This is the situation of a Lefschetz pencil, whose blow-up along the base locus is studied in *Hodge Theory* [17, The Fibration Attached to a Pencil]. Example 8.4.5 is the case $X = \mathbb{A}^2_k$, $\mathcal{L} = \mathcal{O}_X$, and $(s_0, s_1) = (x, y)$.

Example 8.4.5: Blow-up of $\mathbb{A}^2$ at the Origin

Take $X = \mathbb{A}^2_k = \operatorname{Spec} k[x, y]$ and $Z = V(x, y)$ (the origin), with $\mathcal{I} = (x, y)$. The Rees algebra $\bigoplus_{n \geq 0} (x, y)^n$ is the subring $k[x, y][xu, yu]$ of $k[x, y][u]$, and sending $s \mapsto xu$, $t \mapsto yu$ defines a surjection of graded rings

$$k[x, y, s, t] \longrightarrow \bigoplus_{n \geq 0} (x, y)^n, \qquad \deg s = \deg t = 1, \quad \deg x = \deg y = 0$$

whose kernel is generated by $xt - ys$ [36, Example II.7.12.1]. Hence $\mathrm{Bl}_P \mathbb{A}^2 = \operatorname{Proj} k[x, y, s, t]/(xt - ys)$, which is the closed subvariety of $\mathbb{A}^2 \times \mathbb{P}^1$ described at the start of section 8.4.

The exceptional divisor $E \cong \mathbb{P}^1$ is the fiber $\pi^{-1}(0, 0) = \operatorname{Proj} k[s, t]$. On the blow-up, the ideal (x, y) becomes the ideal sheaf $\mathcal{O}(-E)$, which is invertible (generated by x on the chart $s \neq 0$ and by y on the chart $t \neq 0$).

On a smooth projective surface X, the blow-up $\pi \colon \widetilde{X} = \mathrm{Bl}_P X \to X$ at a closed point P changes the Picard group, the intersection pairing and the canonical class in a way computed entirely from X.

Theorem 8.4.6: Intersection Theory on the Blow-up

Let $\pi\colon \widetilde{X} \to X$ be the blow-up of a smooth projective surface X at a closed point P, with exceptional divisor $E = \pi^{-1}(P)$. Then:

1. $\widetilde{X}$ is a smooth projective surface, and $E \cong \mathbb{P}^1$.
2. $\operatorname{Pic}(\widetilde{X}) \cong \operatorname{Pic}(X) \oplus \mathbb{Z} \cdot E$, the first summand embedded by π^*. Every divisor on $\widetilde{X}$ is linearly equivalent to $\pi^* D + mE$ for a unique class $D \in \operatorname{Pic}(X)$ and a unique $m \in \mathbb{Z}$.
3. The intersection numbers on $\widetilde{X}$ are determined by
$$E^2 = -1, \qquad (\pi^* D_1) \cdot (\pi^* D_2) = D_1 \cdot D_2, \qquad (\pi^* D) \cdot E = 0$$
for any $D, D_1, D_2 \in \operatorname{Pic}(X)$.
4. The canonical class satisfies
$$K_{\widetilde{X}} \;=\; \pi^* K_X + E$$
5. Let $C \subseteq X$ be an irreducible curve through P, with local equation f at P, and let $m = \operatorname{mult}_P(C)$ be the *multiplicity* of C at P, the largest integer with $f \in \mathfrak{m}_P^m$. The *strict transform* $\widetilde{C} \subseteq \widetilde{X}$, the closure of $\pi^{-1}(C \setminus \{P\})$, satisfies
$$\widetilde{C} \;=\; \pi^* C - mE, \qquad \widetilde{C} \cdot E = m, \qquad \widetilde{C}^2 = C^2 - m^2$$

***Proof*:**

1. A closed point of X is a nonsingular closed subvariety $\operatorname{Spec} k$ of codimension 2, so $\widetilde{X}$ is a nonsingular surface by proposition 8.4.3, and it is projective because X is [36, Proposition II.7.16(c)]. The exceptional divisor is the fiber $\widetilde{X} \times_X \operatorname{Spec} \kappa(P)$, and relative Proj commutes with base change (proposition 5.3.50). The pullback of the Rees algebra of the ideal of P to $\operatorname{Spec} \kappa(P)$ is $\bigoplus_{n \geq 0} \mathfrak{m}_P^n / \mathfrak{m}_P^{n+1} = \operatorname{gr}_{\mathfrak{m}_P} \mathcal{O}_{X,P}$, so
$$E \;\cong\; \operatorname{Proj} \operatorname{gr}_{\mathfrak{m}_P} \mathcal{O}_{X,P}$$
compatibly with the twisting sheaves. The local ring $\mathcal{O}_{X,P}$ is regular of dimension 2, so $\operatorname{gr}_{\mathfrak{m}_P} \mathcal{O}_{X,P} \cong k[t_1, t_2]$, with t_1, t_2 the classes of any two generators u, v of $\mathfrak{m}_P$ [10, Characterizations of a Regular Local Ring]. Therefore $E \cong \mathbb{P}^1$ with $\mathcal{O}_{\widetilde{X}}(1)|_E \cong \mathcal{O}_{\mathbb{P}^1}(1)$. Moreover $\mathcal{O}_{\widetilde{X}}(1)$ is the inverse image ideal sheaf of P [36, Proposition II.7.13(a)], which is $\mathcal{O}_{\widetilde{X}}(-E)$, so $\mathcal{O}_{\widetilde{X}}(E)|_E \cong \mathcal{O}_{\mathbb{P}^1}(-1)$.
2. The surfaces X and $\widetilde{X}$ are nonsingular, so Pic $=$ Cl on both. By proposition 5.4.18(3), restriction gives an exact sequence $\mathbb{Z} \to \operatorname{Cl}(\widetilde{X}) \to \operatorname{Cl}(\widetilde{X} \setminus E) \to 0$, the first map sending 1 to $[E]$, and π identifies $\widetilde{X} \setminus E$ with $X \setminus \{P\}$. Restriction $\operatorname{Cl}(X) \to \operatorname{Cl}(X \setminus \{P\})$ is an isomorphism by proposition 5.4.18(2), as P has codimension 2, and it factors as π^* followed by restriction to $\widetilde{X} \setminus E$. So π^* splits the sequence. Its first map is injective because $(mE) \cdot E = -m$ by item (3), whose proof does not use this one. Hence $\operatorname{Pic}(\widetilde{X}) = \pi^* \operatorname{Pic}(X) \oplus \mathbb{Z} \cdot E$.

3. The curve E is nonsingular, so proposition 8.2.6 and item (1) give $E^2 = \deg \mathcal{O}_{\widetilde{X}}(E)|_E = -1$. For $D \in \operatorname{Pic}(X)$, the restriction $\pi^*\mathcal{O}_X(D)|_E$ is pulled back from $\mathcal{O}_X(D)|_{\{P\}}$ along $E \to \operatorname{Spec}\kappa(P)$, so it is trivial, and $(\pi^*D)\cdot E = \deg(\pi^*\mathcal{O}_X(D)|_E) = 0$ by the formula $D'\cdot C = \deg(\mathcal{L}(D')|_C)$ for a divisor D' and a smooth irreducible curve C (proposition 8.2.2(6), applied with $C = E \cong \mathbb{P}^1$). For the third rule, every divisor on X is linearly equivalent to a difference of nonsingular curves avoiding P, and for D_1 and D_2 these curves can be chosen so that each curve of the first difference meets each curve of the second transversally, by a Bertini argument [36, Proposition V.3.2]. Since π is an isomorphism over $X \setminus \{P\}$, the preimages of such curves meet in the same points, again transversally, and proposition 8.2.2(1), (3) and (4) give $(\pi^*D_1)\cdot(\pi^*D_2) = D_1\cdot D_2$.

4. The sheaves $\omega_{\widetilde{X}}$ and $\pi^*\omega_X$ agree on $\widetilde{X}\setminus E$, which π maps isomorphically onto $X\setminus\{P\}$, so by the exact sequence in item (2), $K_{\widetilde{X}} = \pi^*K_X + aE$ for some $a\in\mathbb{Z}$. The adjunction formula (theorem 8.2.8) for $E\cong\mathbb{P}^1$ and item (3) give

$$-2 \;=\; E\cdot(E+K_{\widetilde{X}}) \;=\; -1+(\pi^*K_X)\cdot E + aE^2 \;=\; -1-a$$

so $a = 1$.

5. Let $\mathcal{I}_P$ be the ideal sheaf of P. Since $\mathfrak{m}_P$ is generated by two elements, there is an affine open $U = \operatorname{Spec}A$ containing P on which $\mathcal{I}_P$ is generated by two elements $u, v \in A$ and $C\cap U = V(f)$. The section f lies in $\mathcal{I}_P^m(U) = (u,v)^m$, because its image in $\mathcal{O}_X/\mathcal{I}_P^m$, a sheaf supported at P alone, has zero stalk at P. So $f = \sum_{i+j=m} a_{ij}\,u^iv^j$ with $a_{ij}\in A$. The leading form $f_m = \sum \bar{a}_{ij}\,t_1^it_2^j$, the image of f in $\mathfrak{m}_P^m/\mathfrak{m}_P^{m+1} \cong k[t_1,t_2]_m$, is nonzero because $f\notin\mathfrak{m}_P^{m+1}$. On the chart $D_+(ut) = \operatorname{Spec}A[v/u]$ of $\pi^{-1}(U)$, the exceptional divisor is $V(u)$, with coordinate ring $A[v/u]/(u) = k[s]$ for s the image of v/u (the chart $D_+(t_1)$ of $E = \operatorname{Proj}k[t_1,t_2]$), and

$$\pi^*f \;=\; u^mg, \qquad g \;=\; \sum_{i+j=m} a_{ij}\,(v/u)^j, \qquad g|_E \;=\; f_m(1,s)$$

The polynomial $f_m(1,s)$ is nonzero, since $f_m(t_1,t_2) = t_1^m f_m(1,t_2/t_1)$. So g does not vanish on E, and E, whose local equation is u, occurs in π^*C with multiplicity exactly m. Every other prime divisor of $\widetilde{X}$ is the closure of a prime divisor of $\widetilde{X}\setminus E\cong X\setminus\{P\}$, on which π^*C restricts to $C\setminus\{P\}$ with multiplicity one. Hence $\pi^*C = \widetilde{C} + mE$. By item (3), $\widetilde{C}\cdot E = (\pi^*C)\cdot E - mE^2 = m$ and $\widetilde{C}^2 = (\pi^*C)^2 - 2m\,(\pi^*C)\cdot E + m^2E^2 = C^2 - m^2$, as claimed.

Example 8.4.7: Blow-ups in Practice

1. Let $P\in\mathbb{P}^2$ and let H be the class of a line in $\mathbb{P}^2$. Then $\operatorname{Pic}(\operatorname{Bl}_P\mathbb{P}^2) = \mathbb{Z}H\oplus\mathbb{Z}E$ (identifying π^*H with H by abuse). A line through P has strict transform $\widetilde{L} = H-E$ with $\widetilde{L}^2 = 1-1 = 0$ and $\widetilde{L}\cdot E = 1$. A smooth conic through P has $\widetilde{C} = 2H-E$ with $\widetilde{C}^2 = 4-1 = 3$.

   The surface $\operatorname{Bl}_P\mathbb{P}^2$ is isomorphic to the Hirzebruch surface $\mathbb{F}_1$ (proposition 8.5.4(5)): the pencil of lines through P becomes a ruling of $\mathbb{F}_1$, and E is the negative section with $E^2 = -1$.

   The canonical class is $K_{\operatorname{Bl}_P\mathbb{P}^2} = \pi^*(-3H)+E = -3H+E$, so $K^2 = 9-1 = 8$. Over $\mathbb{C}$, replacing the point P by $E\cong\mathbb{P}^1$ raises the topological Euler characteristic by 1, to $\chi_{\text{top}} = 4$,

and Noether's formula (theorem 8.3.3) gives $\chi(\mathcal{O}_{\mathrm{Bl}_P \mathbb{P}^2}) = \frac{8+4}{12} = 1 = \chi(\mathcal{O}_{\mathbb{P}^2})$.

2. Let $\operatorname{char} k \neq 2$ and let $C\colon y^2 = x^3 + x^2$ be the nodal cubic in $\mathbb{A}^2$. Its singular point is the origin, where $\operatorname{mult}_0(C) = 2$. In the chart $y = xt$ of the blow-up of the origin, the total transform is $x^2(t^2 - x - 1) = 0$, so the strict transform $\widetilde{C}\colon t^2 = x + 1$ is smooth and meets $E = \{x = 0\}$ in the two points $t = \pm 1$, the directions of the two tangent lines $y = \pm x$. The other chart adds no point of $\widetilde{C} \cap E$, and $\widetilde{C}$ is smooth there too. One blow-up has *resolved* the singularity.

3. The cuspidal cubic $y^2 = x^3$ also has $\operatorname{mult}_0 = 2$. In the chart $y = xt$ the total transform is $x^2(t^2 - x) = 0$, so the strict transform $t^2 = x$ is already smooth, and it misses E in the other chart. It meets $E = \{x = 0\}$ only at $t = 0$, with $\widetilde{C} \cdot E = 2$ there, so $\widetilde{C}$ is tangent to E. One blow-up resolves the cusp, and making the total transform a divisor with normal crossings takes three blow-ups in all [36, Example V.3.9.1].

Blowing up a point of a smooth projective surface creates a (-1)-*curve*, a curve $E \cong \mathbb{P}^1$ with $E^2 = -1$ (theorem 8.4.6). Castelnuovo's criterion is the converse: every (-1)-curve on a smooth projective surface is the exceptional curve of a blow-up.

Theorem 8.4.8: Castelnuovo's Contractibility Criterion

Let X be a smooth projective surface and $E \subseteq X$ a smooth rational curve (i.e. $E \cong \mathbb{P}^1$) with $E^2 = -1$. Then there exists a smooth projective surface Y and a point $Q \in Y$ such that $X \cong \mathrm{Bl}_Q Y$, with E as the exceptional divisor. So E can be *contracted*: there exists a morphism $\pi\colon X \to Y$ that maps E to the point Q and restricts to an isomorphism $X \setminus E \to Y \setminus \{Q\}$.

***Proof Key ideas*:**
The proof that the image point is nonsingular uses the theorem on formal functions, which this book does not develop. The full argument is [36, Theorem V.5.7]. Choose a very ample divisor H on X with $H^1(X, \mathcal{O}_X(H)) = 0$ and $k = H \cdot E \geq 2$, and put $L = H + kE$, so that $L \cdot E = k - k = 0$. Since $\mathcal{O}_X(H + iE)|_E \cong \mathcal{O}_{\mathbb{P}^1}(k - i)$, induction on i gives $H^1(X, \mathcal{O}_X(H + iE)) = 0$ for $0 \leq i \leq k$, and from this the global sections of $\mathcal{O}_X(L)$ generate it along E as well as away from E. The resulting morphism to $\mathbb{P}^N$ maps E to a point Q, because $L \cdot E = 0$, and is an isomorphism of $X \setminus E$ onto the complement of Q in its image, because $|L|$ contains $|H| + kE$. Let Y be the normalization of the image. The theorem on formal functions identifies $\widehat{\mathcal{O}}_{Y,Q}$ with $\lim H^0(\mathcal{O}_{E_n})$, where E_n runs over the infinitesimal neighbourhoods of E. Since $E^2 = -1$, the conormal sheaf of E is $\mathcal{O}_{\mathbb{P}^1}(1)$, which makes each $H^0(\mathcal{O}_{E_n})$ a truncated power series ring $k[[u, v]]/(u, v)^n$. So $\widehat{\mathcal{O}}_{Y,Q} \cong k[[u, v]]$ and Y is nonsingular at Q. The morphism $X \to Y$ contracts exactly one curve, and in the factorization of birational morphisms (theorem 8.4.11, whose proof does not use this theorem) each blow-up accounts for one contracted curve, so $X \to Y$ is the blow-up of Q, as claimed.

The value $E^2 = -1$ is what makes Q nonsingular. A curve $E \cong \mathbb{P}^1$ with $E^2 \leq -2$ can still be contracted to a point of a projective surface [36, Exercise V.5.2], but that point is singular [36, Remark V.5.7.2]. Over $\mathbb{C}$, Grauert's theorem contracts any connected curve whose components have a negative definite intersection matrix, but only to a point of a complex analytic space: a curve of genus one with self-intersection -1 constructed by Hironaka [36, Example V.5.7.3] cannot be contracted to a point of an algebraic surface.

Corollary 8.4.9: Detecting (-1)-Curves

Let E be an irreducible curve on a smooth projective surface X. Then E is a (-1)-curve (i.e. $E \cong \mathbb{P}^1$ with $E^2 = -1$) if and only if

$$E^2 = -1 \qquad \text{and} \qquad K_X \cdot E = -1$$

Equivalently, $E^2 = -1$ and the arithmetic genus $p_a(E)$ is 0.

***Proof*:**
The adjunction formula for the integral curve E, (8.5), gives $2p_a(E) - 2 = E^2 + K_X \cdot E$, so when $E^2 = -1$ the conditions $K_X \cdot E = -1$ and $p_a(E) = 0$ are equivalent. If $E \cong \mathbb{P}^1$, then $p_a(E) = 1 - \chi(\mathcal{O}_{\mathbb{P}^1}) = 0$. Conversely, suppose $p_a(E) = 0$. By box 7.2.9, $p_a(E) = g(\widetilde{E}) + \sum_Q \delta_Q$, where $\widetilde{E}$ is the normalization, $g(\widetilde{E}) \geq 0$, and $\delta_Q \geq 1$ at every singular point Q, since the local ring there is not integrally closed. So E is nonsingular of genus 0, and $E \cong \mathbb{P}^1$ by proposition 7.3.6, completing the proof.

8.4.10 Forward Reference: Extremal Contractions and the MMP Castelnuovo's contractibility criterion is the surface case of the *extremal contractions* of the Minimal Model Program (MMP), developed in *Birational Geometry* [8].

1. The *Cone Theorem* (Mori) describes the K_X-negative part of the Mori cone $\overline{\mathrm{NE}}(X)$ by extremal rays spanned by rational curves. On a smooth projective surface such a ray is spanned by a (-1)-curve, by a fiber of a ruled surface $X \to C$ over a curve, or, when $X \cong \mathbb{P}^2$, by a line.

2. Each such ray corresponds to a *contraction morphism*: for a (-1)-curve, this is exactly the blow-down of Castelnuovo's criterion.

3. The process of successively contracting (-1)-curves terminates (since ρ decreases by 1 at each step) and produces a surface with no (-1)-curves.

In dimension ≥ 3 the contraction of an extremal ray can have a singular target. When it contracts no divisor, the canonical divisor of the target is not $\mathbb{Q}$-Cartier, and the MMP continues with a *flip*, a birational map that is an isomorphism in codimension one. Flips exist by Mori in dimension 3 and by Birkar-Cascini-Hacon-McKernan in every dimension.

The proof of theorem 8.4.8 uses the following theorem of Zariski.

Theorem 8.4.11: Factorization of Birational Morphisms

Let $f\colon X \to Y$ be a birational morphism between smooth projective surfaces. Then f is, up to an isomorphism $X \cong X_n$, a composition of finitely many blow-ups at closed points

$$X_n \xrightarrow{\pi_n} X_{n-1} \xrightarrow{\pi_{n-1}} \cdots \xrightarrow{\pi_1} X_0 = Y$$

where each π_i is the blow-up of X_{i-1} at a closed point, and n is the number of irreducible curves of X contracted by f.

***Proof Key ideas*:**
The proof uses Zariski's Main Theorem for birational transformations and a computation in local coordinates. The full argument is [36, Proposition V.5.3, Corollary V.5.4]. If f is not an isomorphism, its rational inverse is undefined at some closed point $Q \in Y$, and then f factors as $\pi_1 \circ f_1$ through the blow-up $\pi_1\colon \mathrm{Bl}_Q Y \to Y$. Only finitely many irreducible curves of X map to points under f, and f_1 maps one fewer of them to points, since exactly one of them maps onto the exceptional curve of π_1. After finitely many steps no curve is contracted, and a birational morphism contracting no curve has an inverse defined everywhere, so it is an isomorphism, completing the factorization.

In dimension ≥ 3 a birational morphism between nonsingular projective varieties need not be a composition of blow-ups along nonsingular centres [36, Remark V.5.4.4], and the minimal model program needs flips, birational maps that are morphisms in neither direction (*Birational Geometry* [8]).

8.4.1 Hironaka's Example

Every complete variety in the examples so far has been projective. For smooth curves the two notions coincide (proposition 7.1.7), and every smooth complete surface is projective as well, by a theorem of Zariski [36, Remark II.4.10.2]. In dimension three they differ: Hironaka's example below is a smooth complete threefold with no ample line bundle. It is built by blowing up two curves in opposite orders over two open sets and gluing the results. The two orders give different fibres over the two points where the curves meet, and comparing these fibres produces a nonzero effective 1-cycle whose degree is zero for every line bundle, which no ample line bundle allows.

Proposition 8.4.12: Hironaka's Example

Let X be a smooth projective threefold over an algebraically closed field, and let $C_1, C_2 \subseteq X$ be smooth curves meeting transversally at exactly two points P and Q, and nowhere else. Construct $\pi\colon M \to X$ by gluing two blow-ups:

- over $X \setminus \{Q\}$, blow up C_1 first, then the strict transform of C_2,
- over $X \setminus \{P\}$, blow up C_2 first, then the strict transform of C_1.

Over the overlap $X \setminus \{P, Q\}$ the two curves are disjoint, so the two orders give canonically isomorphic results and the pieces glue. Then M is a smooth complete threefold, and M is *not* projective.

***Proof*:**
Each piece of M is obtained from a smooth variety by blowing up two smooth curves in succession, because the strict transform of either curve under the blow-up of the other is isomorphic to the original curve [36, Corollary II.7.15], the inverse-image ideal being invertible on the smooth curve, so that blowing it up is an isomorphism (proposition 8.4.2(4) applied to the identity). So M is smooth by proposition 8.4.3. The morphism π is proper over each of the open sets $X \setminus \{Q\}$ and $X \setminus \{P\}$ (proposition 8.4.2(1)), and properness is local on the target, so π is proper and M is complete.
For a line bundle $\mathcal{L}$ on M and an integral complete curve $\gamma \subseteq M$, write $\mathcal{L} \cdot \gamma = \deg(\mathcal{L}|_\gamma)$, and extend this linearly to 1-cycles.

(a) A morphism from an integral scheme onto a nonsingular curve is flat [36, Proposition III.9.7].

(b) If $g\colon W \to T$ is a flat proper morphism onto a nonsingular connected curve T whose fibres have dimension 1, then for every line bundle $\mathcal{L}$ on W the number $\mathcal{L} \cdot [g^{-1}(t)]$ is independent of $t \in T$, where $[g^{-1}(t)]$ is the 1-cycle of the scheme-theoretic fibre, each component counted with its multiplicity (conservation of number, [29, § 10.2]).

For $c \in C_1 \setminus \{P, Q\}$ the fibre $\pi^{-1}(c)$ is a line ℓ_c, and ℓ denotes any one of these. Similarly m denotes the line $\pi^{-1}(c')$ over a point $c' \in C_2 \setminus \{P, Q\}$.
Step 1: the fibre over P. Over $X \setminus \{Q\}$ the morphism π is a composite $\alpha \circ \beta$, where $\alpha\colon X_1 \to X \setminus \{Q\}$ is the blow-up along C_1 and β is the blow-up of X_1 along the strict transform C_2' of C_2. The exceptional divisor $E_1 = \alpha^{-1}(C_1)$ is the projectivized normal bundle of C_1, a $\mathbb{P}^1$-bundle over $C_1 \setminus \{Q\}$, and $E_2 = \beta^{-1}(C_2')$ is likewise a $\mathbb{P}^1$-bundle over C_2' [36, Theorem II.8.24]. The curve C_2' meets the line $F_P = \alpha^{-1}(P)$ in a single point x: identifying F_P with the projectivized normal space of C_1 at P, the point x is the image of the tangent line of C_2 at P, which is not tangent to C_1 because the curves meet transversally. Hence $\pi^{-1}(P) = \beta^{-1}(F_P)$ is the union of two irreducible curves: ℓ_0, the closure of $\beta^{-1}(F_P \setminus \{x\})$, and $\ell_1 = \beta^{-1}(x)$, a fibre of $E_2 \to C_2'$.

Let Z be the closure of $\pi^{-1}(C_1 \setminus \{P, Q\})$ in $\pi^{-1}(X \setminus \{Q\})$, and let $g = \pi|_Z\colon Z \to C_1 \setminus \{Q\}$. Then Z is an integral surface, being the closure of a $\mathbb{P}^1$-bundle over an irreducible curve, g is proper and surjective, and g is flat by (a). Its fibre over $c \neq P$ is the line ℓ_c. Its fibre over P is cut out on Z by one equation, so every component of it has dimension 1, and it lies in $\ell_0 \cup \ell_1$. Thus $[g^{-1}(P)] = a\ell_0 + b\ell_1$ for integers $a, b \geq 0$, and in fact both are positive.

- The closed set $\beta(Z)$ contains $\alpha^{-1}(C_1 \setminus \{P, Q\})$, which is dense in the irreducible surface E_1, so $\beta(Z)$ contains F_P. As β is an isomorphism over $X_1 \setminus C_2'$ and $F_P \setminus \{x\}$ does not meet C_2', the surface Z contains $\beta^{-1}(F_P \setminus \{x\})$, hence contains ℓ_0. So $a \geq 1$.
- The exceptional divisor E_2 is an effective Cartier divisor (proposition 8.4.2(3)), and it is disjoint from ℓ_c for $c \neq P$ since $\pi(E_2) \subseteq C_2$. By (b) applied to $\mathcal{O}(E_2)$, $a\,\mathcal{O}(E_2) \cdot \ell_0 + b\,\mathcal{O}(E_2) \cdot \ell_1 = \mathcal{O}(E_2) \cdot \ell_c = 0$. The curve ℓ_0 is not contained in E_2, and it meets E_2 because $\beta(\ell_0)$ is closed and contains $F_P \setminus \{x\}$, hence contains x. So E_2 restricts to a nonzero effective Cartier divisor on ℓ_0, and $\mathcal{O}(E_2) \cdot \ell_0 > 0$. With $a \geq 1$ this forces $b \geq 1$.

The $\mathbb{P}^1$-bundle $E_2 \to C_2'$ is flat by (a), and its fibres include ℓ_1 and m. Applying (b) to it and to g gives, for every line bundle $\mathcal{L}$ on M,

$$\mathcal{L} \cdot \ell \;=\; a\,\mathcal{L} \cdot \ell_0 + b\,\mathcal{L} \cdot m$$

Step 2: the fibre over Q. Over $X \setminus \{P\}$ the roles of C_1 and C_2 are exchanged, and the same argument gives an irreducible curve $m_0 \subseteq \pi^{-1}(Q)$ and integers $a', b' \geq 1$ with

$$\mathcal{L} \cdot m \;=\; a'\,\mathcal{L} \cdot m_0 + b'\,\mathcal{L} \cdot \ell$$

for every line bundle $\mathcal{L}$ on M.

Step 3: no ample line bundle. Substituting Step 2 into Step 1, the 1-cycle

$$\gamma \;=\; a\ell_0 + ba'\,m_0 + (bb' - 1)\ell$$

satisfies $\mathcal{L} \cdot \gamma = 0$ for every line bundle $\mathcal{L}$ on M. It is effective because $bb' \geq 1$, and nonzero because $a \geq 1$. An ample line bundle $\mathcal{H}$ on M would restrict to an ample line bundle, of positive degree, on every integral curve, giving $\mathcal{H} \cdot \gamma \geq a\,\mathcal{H} \cdot \ell_0 > 0$. Hence M carries no ample line bundle and is not projective, as claimed.

The variety M is smooth and complete, so its failure to be projective comes from neither singularities nor non-completeness. It comes from the two orders of blowing up, which make the fibres over P and over Q asymmetric. A local computation shows that all four multiplicities a, b, a', b' equal 1, so that $\gamma = \ell_0 + m_0$, and that ℓ is algebraically equivalent to $\ell_0 + m$ and m to $m_0 + \ell$ [36, Appendix B, Example 3.4.1]. The proof above uses only that the multiplicities are positive. Hironaka's theorem on resolution of singularities [39] is a different result and is not proved in this book.

8.5 Ruled Surfaces and Rational Surfaces

The *ruled surfaces*, fibered over a curve with every fiber isomorphic to $\mathbb{P}^1$, and the *rational surfaces*, birational to $\mathbb{P}^2$, have Kodaira dimension $\kappa = -\infty$ (definition 8.6.1). Up to birational equivalence they are all the surfaces with $\kappa = -\infty$: such a surface is rational or birational to $C \times \mathbb{P}^1$ for a curve C [36, Theorem V.6.1].

Definition 8.5.1: Ruled Surface

A *ruled surface* over a smooth projective curve C is a smooth projective surface S together with a surjective morphism $p\colon S \to C$ whose fibers are all isomorphic to $\mathbb{P}^1$.

Every ruled surface is isomorphic over C to the projective bundle $\mathbb{P}(\mathscr{E}) = \underline{\mathrm{Proj}}_C(\mathrm{Sym}^\bullet \mathscr{E})$ of a locally free sheaf $\mathscr{E}$ of rank 2 on C (definition 5.3.51), whose fiber over $c \in C$ is $\mathbb{P}(\mathscr{E}(c)) \cong \mathbb{P}^1$, and conversely every such $\mathbb{P}(\mathscr{E})$ is a ruled surface [36, Proposition V.2.2]. Hartshorne's definition also requires a section of p, which exists by Tsen's theorem (see the note following that definition). In the convention of definition 5.3.51 (Grothendieck's), a point of $\mathbb{P}(\mathscr{E})$ over c is a one-dimensional *quotient* of $\mathscr{E}(c)$, and the sections σ of p correspond to the surjections $\mathscr{E} \twoheadrightarrow \mathcal{L}$ onto invertible sheaves on C, with $\mathcal{L} = \sigma^* \mathcal{O}_{\mathbb{P}(\mathscr{E})}(1)$ [36, Proposition V.2.6].

8.5.2 Warning: The Other $\mathbb{P}(\mathscr{E})$ The quotient convention of definition 5.3.51 is Grothendieck's and is used throughout this book. The opposite convention, under which $\mathbb{P}(\mathscr{E})$ is the bundle of one-dimensional *subspaces* of $\mathscr{E}$, is equally standard and is the one Fulton uses. *Intersection Theory* [19] follows it, since its Segre and Chern classes are built that way. The two differ by a dual, $\mathbb{P}_{\text{quot}}(\mathscr{E}) = \mathbb{P}_{\text{sub}}(\mathscr{E}^\vee)$, so a formula carried between the two books must be dualized. The point matters wherever $\mathcal{O}_{\mathbb{P}(\mathscr{E})}(1)$ appears: the pushforward $f_*\mathcal{O}(d) \cong \operatorname{Sym}^d \mathscr{E}$ of example 6.6.17(2) holds in this convention and acquires a dual in the other.

Two ruled surfaces $\mathbb{P}(\mathscr{E})$ and $\mathbb{P}(\mathscr{E}')$ over C are isomorphic (as surfaces over C) if and only if $\mathscr{E}' \cong \mathscr{E} \otimes \mathcal{L}$ for some invertible sheaf $\mathcal{L}$ on C [36, Proposition V.2.2]. Twisting by an invertible sheaf does not change the projectivization (example 5.3.55(3)), which is the "if" direction.

Over $\mathbb{P}^1$ every ruled surface is a Hirzebruch surface. A locally free sheaf of rank 2 on $\mathbb{P}^1$ splits as a direct sum of invertible sheaves (by the Birkhoff-Grothendieck theorem, whose rank-2 case is [36, Corollary V.2.14]), so $\mathscr{E} \cong \mathcal{O}(a) \oplus \mathcal{O}(b)$. Ordering the summands so that $a \geq b$ and twisting by $\mathcal{O}(-a)$ gives $\mathscr{E} \otimes \mathcal{O}(-a) \cong \mathcal{O} \oplus \mathcal{O}(-n)$ with $n = a - b \geq 0$, so every ruled surface over $\mathbb{P}^1$ is isomorphic to one of the following.

Definition 8.5.3: Hirzebruch Surface

For $n \geq 0$, the nth *Hirzebruch surface* is

$$\mathbb{F}_n \;=\; \mathbb{P}(\mathcal{O}_{\mathbb{P}^1} \oplus \mathcal{O}_{\mathbb{P}^1}(-n))$$

Proposition 8.5.4: Invariants of $\mathbb{F}_n$

The Hirzebruch surface $\mathbb{F}_n$ has the following properties:

1. $\operatorname{Pic}(\mathbb{F}_n) \cong \mathbb{Z} \cdot C_0 \oplus \mathbb{Z} \cdot F$, where F is the class of a fiber of $p\colon \mathbb{F}_n \to \mathbb{P}^1$ and C_0 is the section corresponding to the surjection $\mathcal{O} \oplus \mathcal{O}(-n) \twoheadrightarrow \mathcal{O}(-n)$ onto the second summand.
2. Intersection numbers: $F^2 = 0$, $C_0 \cdot F = 1$, $C_0^2 = -n$.
3. Canonical class: $K_{\mathbb{F}_n} = -2C_0 - (n+2)F$.
4. $\mathbb{F}_0 = \mathbb{P}^1 \times \mathbb{P}^1$ (with C_0 and F fibers of the two projections).
5. $\mathbb{F}_1 \cong \operatorname{Bl}_P \mathbb{P}^2$, with $C_0 = E$ (the exceptional divisor) and F the strict transform of a line through P.
6. For $n \geq 1$, C_0 is the unique irreducible curve with negative self-intersection on $\mathbb{F}_n$, and $\mathbb{F}_0$ has none. In particular, $\mathbb{F}_n \ncong \mathbb{F}_m$ for $n \neq m$.

***Proof*:**

1. For a ruled surface $p\colon S \to C$ with a section C_0, $\operatorname{Pic}(S) = \mathbb{Z} \cdot C_0 \oplus p^*\operatorname{Pic}(C)$ [36, Proposition V.2.3]. Here $\operatorname{Pic}(\mathbb{P}^1) = \mathbb{Z}$ is generated by the class of a point, whose pullback is F.
2. Two distinct fibers are disjoint and linearly equivalent, so $F^2 = 0$, and a section meets

each fiber in one point, transversally, so $C_0 \cdot F = 1$ [36, Proposition V.2.3]. A section D corresponding to a surjection $\mathscr{E} \twoheadrightarrow \mathcal{L}$ with kernel $\mathcal{N}$ satisfies $\mathcal{O}_S(D)|_D \cong \mathcal{L} \otimes \mathcal{N}^\vee$ under $D \cong C$ [36, Proposition V.2.6], so proposition 8.2.6 gives $D^2 = \deg \mathcal{L} - \deg \mathcal{N} = 2\deg \mathcal{L} - \deg \mathscr{E}$. For C_0 on $\mathbb{F}_n$, $\mathcal{L} = \mathcal{O}(-n)$ and $\deg \mathscr{E} = -n$, so $C_0^2 = -2n + n = -n$.

3. By item (1), $K_{\mathbb{F}_n} = aC_0 + bF$ for some $a, b \in \mathbb{Z}$. The fiber F and the section C_0 are smooth rational curves, so the adjunction formula (theorem 8.2.8) and item (2) give

$$-2 = F \cdot (F + K_{\mathbb{F}_n}) = a, \qquad -2 = C_0 \cdot (C_0 + K_{\mathbb{F}_n}) = -n - an + b$$

The first gives $a = -2$, and then the second reads $-2 = n + b$, so $b = -(n+2)$.

4. The sheaf $\mathcal{O} \oplus \mathcal{O}$ is free, so $\mathbb{F}_0 = \mathbb{P}^1_{\mathbb{P}^1} = \mathbb{P}^1 \times \mathbb{P}^1$ with p the first projection (example 5.3.55(1)). The section corresponding to the quotient onto a summand is a fiber of the second projection.

5. The blow-up $\mathrm{Bl}_P \mathbb{P}^2$ is the closed subvariety of $\mathbb{P}^2 \times \mathbb{P}^1$ described in example 4.7.12. Its second projection to $\mathbb{P}^1$ has as fibers the strict transforms of the lines through P, each isomorphic to $\mathbb{P}^1$. So $\mathrm{Bl}_P \mathbb{P}^2$ is a ruled surface over $\mathbb{P}^1$, hence isomorphic to some $\mathbb{F}_m$. The curve E is irreducible with $E^2 = -1$ (theorem 8.4.6), so item (6), proved below without using this one, forces $m = 1$ and $C_0 = E$. The class F is that of a fiber.

6. Let $C \neq C_0$ be an irreducible curve, with class $aC_0 + bF$. Then $a = C \cdot F \geq 0$: if C is a fiber then $a = 0$, and otherwise C and a fiber are distinct irreducible curves, so proposition 8.2.2(5) applies. Likewise $b - an = C \cdot C_0 \geq 0$. Hence $C^2 = -a^2 n + 2ab = a^2 n + 2a(b - an) \geq 0$. So for $n \geq 1$ the curve C_0, with $C_0^2 = -n$, is the only irreducible curve of negative self-intersection, and $\mathbb{F}_0$ has none. The self-intersection of that curve recovers n from $\mathbb{F}_n$, which proves the last assertion and completes the proof.

Example 8.5.5: Ruled Surfaces over Higher-Genus Curves

If C has genus $g \geq 1$, rank-2 bundles on C need not split, and the ruled surfaces $\mathbb{P}(\mathscr{E}) \to C$ vary in positive-dimensional families.

8.5.1 Rational Surfaces

Definition 8.5.6: Rational Surface

A smooth projective surface X is *rational* if it is birational to $\mathbb{P}^2$.

Contracting (-1)-curves one at a time (theorem 8.4.8) stops, because the Picard number drops by one at each step (theorem 8.4.6(2) and (3)), and takes a rational surface X to a rational surface without (-1)-curves. Such a surface is $\mathbb{P}^2$ or some $\mathbb{F}_n$ with $n \neq 1$ [36, Example V.5.8.2, Remark V.5.8.4]. Reversing the contractions, every rational surface is obtained from $\mathbb{P}^2$ or from some $\mathbb{F}_n$ by a sequence of blow-ups at points.

Example 8.5.7: Rational Surfaces in Practice

1. $\mathbb{P}^2$ itself and all $\mathbb{F}_n$ (including $\mathbb{P}^1 \times \mathbb{P}^1 = \mathbb{F}_0$).

2. The blow-up of $\mathbb{P}^2$ at $r \leq 8$ points in *general position* (no three collinear, no six on a conic, no eight on a cubic that is singular at one of them) has $K^2 = 9 - r$ and $-K$ ample, so it is a *del Pezzo surface*, a surface with $-K$ ample.

3. A smooth cubic surface $S \subseteq \mathbb{P}^3$ is isomorphic to $\mathrm{Bl}_{P_1,\dots,P_6} \mathbb{P}^2$ (the blow-up of $\mathbb{P}^2$ at 6 points in general position). It has $K_S = -H$ (the hyperplane class), $K^2 = 3$, and $\rho = 7$. The 27 lines on S correspond to the 6 exceptional divisors E_i, the 15 strict transforms $\widetilde{L_{ij}}$ of the lines through pairs of blown-up points, and the 6 strict transforms $\widetilde{Q_i}$ of the conics through 5 of the 6 points. Each line ℓ satisfies $\ell^2 = -1$ and $K \cdot \ell = -1$, so every line is a (-1)-curve. The permutations of the 27 lines that preserve their incidences form a group of order 51840, isomorphic to the Weyl group $W(E_6)$ [36, § V.4].

The irregularity q and the second plurigenus P_2 (definition 8.3.4) vanish for $\mathbb{P}^2$ and are birational invariants, so they vanish on every rational surface. Castelnuovo proved the converse.

Theorem 8.5.8: Castelnuovo's Rationality Criterion

A smooth projective surface X over $\mathbb{C}$ is rational if and only if $q(X) = 0$ and $P_2(X) = 0$ (irregularity zero and second plurigenus zero).

For the "only if" direction, a blow-up at a point does not change q [36, Remark V.3.5.1], so [36, Theorem V.5.5] makes q a birational invariant, and the argument of [36, Theorem II.8.19] for the geometric genus applies unchanged to P_2. The "if" direction is not proved in this book. A proof over $\mathbb{C}$ is in Griffiths and Harris [31, Rational Surfaces II, the Castelnuovo-Enriques Theorem]. Over an algebraically closed field of any characteristic the theorem holds in the form $p_a = P_2 = 0$, the case of positive characteristic being Zariski's [36, Theorem V.6.2]. The two forms agree because $P_2 = 0$ forces $p_g = 0$, and then $p_a = p_g - q = -q$.

8.6 The Enriques-Kodaira Classification

Castelnuovo and Enriques classified smooth projective surfaces in the early twentieth century, Kodaira completed the classification over $\mathbb{C}$ in the 1960s, and Mumford and Bombieri extended it to characteristic p. The classification sorts surfaces by their *Kodaira dimension*, which measures the growth of the spaces of pluricanonical forms, and it works with one representative of each birational class: a minimal surface, reached by contracting (-1)-curves with theorem 8.4.8. The same two steps, a birational invariant and a reduction to minimal models, organize the classification of varieties in higher dimension (*Birational Geometry* [8]). The section also states Kodaira vanishing and its extension by Kawamata and Viehweg to nef and big divisors, a class that contains the canonical divisor of every minimal surface of general type (section 8.6.5).

8.6.1 Kodaira Dimension

Definition 8.6.1: Kodaira Dimension

Let X be a smooth projective variety of dimension d over k. For each $n \geq 1$, the *nth plurigenus* is $P_n(X) = \dim_k H^0(X, \omega_X^{\otimes n})$. The *Kodaira dimension* $\kappa(X)$ is defined by the growth rate of P_n:

$$\kappa(X) = \begin{cases} -\infty & \text{if } P_n = 0 \text{ for all } n \geq 1, \\ \min\left\{ j \geq 0 : \limsup_{n\to\infty} \dfrac{P_n}{n^j} < \infty \right\} & \text{otherwise.} \end{cases}$$

By a theorem of Iitaka, $\kappa(X)$ is also the dimension of the image of the *pluricanonical map* $\varphi_{|nK_X|} \colon X \dashrightarrow \mathbb{P}^{P_n - 1}$ for n sufficiently large and divisible (or $-\infty$ if all $P_n = 0$).

For a surface, $\kappa \in \{-\infty, 0, 1, 2\}$:

1. $\kappa = -\infty$: $P_n = 0$ for all n, that is, no positive multiple of K_X is linearly equivalent to an effective divisor.
2. $\kappa = 0$: $P_n \in \{0, 1\}$ for every n, and $P_n = 1$ for some n. On a minimal surface (definition 8.6.2) the canonical class is then numerically trivial (section 8.6.3).
3. $\kappa = 1$: P_n grows linearly in n, and for n sufficiently divisible the image of the pluricanonical map is a curve, so the map fibres the surface over a curve.
4. $\kappa = 2$: P_n grows quadratically in n, and the pluricanonical map is birational onto its image for $n \gg 0$. These are the *surfaces of general type*.

The plurigenera, and hence the Kodaira dimension, are birational invariants of smooth projective varieties. The proof of theorem 5.5.31 applies with $\omega_X^{\otimes n}$ in place of ω_X: Step 2 takes the nth tensor power of the map $f^*\omega_{X'} \to \omega_V$, which is still an isomorphism on a dense open set, and Step 3 extends sections of any invertible sheaf across a closed subset of codimension at least 2. So if X and Y are birational smooth projective varieties, then $P_n(X) = P_n(Y)$ for every $n \geq 1$, and $\kappa(X) = \kappa(Y)$.

8.6.2 Minimal Models

Definition 8.6.2: Minimal Surface

A smooth projective surface X is *minimal* if it contains no (-1)-curves (smooth rational curves E with $E^2 = -1$). Equivalently, X is not the blow-up of a smooth projective surface at a point: such a blow-up contains its exceptional curve, a (-1)-curve (theorem 8.4.6), and every (-1)-curve can be contracted (theorem 8.4.8).

Theorem 8.6.3: Existence and Uniqueness of Minimal Models

Let X be a smooth projective surface.

1. *Existence.* There is a birational morphism from X to a minimal surface, obtained by successively contracting (-1)-curves (theorem 8.4.8). The process terminates since the Picard number ρ drops by 1 at each step.

2. *Uniqueness for $\kappa \geq 0$.* If $\kappa(X) \geq 0$, any two minimal surfaces birational to X are isomorphic, so X has a unique minimal model.

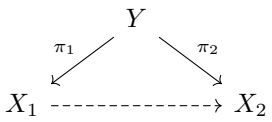
3. *Non-uniqueness for $\kappa = -\infty$.* If X is rational, its minimal models are $\mathbb{P}^2$ and the Hirzebruch surfaces $\mathbb{F}_n$ with $n \neq 1$. If X is birational to $C \times \mathbb{P}^1$ for a smooth projective curve C of genus $g(C) \geq 1$, its minimal models are the ruled surfaces $\mathbb{P}(\mathscr{E}) \to C$ of definition 8.5.1. In both cases there are infinitely many, pairwise non-isomorphic.

By Enriques' theorem (section 8.6.3), the two cases in (3) are all the surfaces with $\kappa = -\infty$. The surface $\mathbb{F}_1$ does not appear in (3) because its section C_0 is a (-1)-curve (proposition 8.5.4). The proof below omits two inputs that this book does not prove, the elimination of indeterminacy of a birational map between surfaces and the factorization of a birational morphism through the contraction of a curve it contracts, both in [36, § V.5], and it omits the proof that the lists in (3) are complete [36, Remark V.5.8.4].

***Proof Key ideas*:**
(1) is theorem 8.4.8 applied repeatedly. Each contraction lowers ρ by one (theorem 8.4.6), and $\rho \geq 1$ on a projective surface, so the process stops.

For (2), a minimal surface X_1 with $\kappa(X_1) \geq 0$ has K_{X_1} nef. Suppose C is an irreducible curve with $K_{X_1} \cdot C < 0$, and choose $n \geq 1$ and an effective $D \in |nK_{X_1}|$, which exists as $\kappa \geq 0$. Writing $D = aC + D'$ with D' effective and not containing C, the inequality $D \cdot C < 0$ forces $a \geq 1$ and $C^2 < 0$, because $D' \cdot C \geq 0$. Adjunction (8.5) gives $2p_a(C) - 2 = C^2 + K_{X_1} \cdot C \leq -2$, so $p_a(C) = 0$ and $C^2 = K_{X_1} \cdot C = -1$. An integral curve with $p_a = 0$ is smooth and rational [36, Exercise IV.1.8], so C is a (-1)-curve, contradicting minimality.

Now let X_1 and X_2 be minimal surfaces birational to X, and resolve the birational map between them by a smooth projective surface Y:



By theorem 8.4.11 each π_i is a composite of blow-ups at points, so iterating theorem 8.4.6(4) gives $K_Y = \pi_2^* K_{X_2} + Z_2$ with Z_2 effective and supported on the curves contracted by π_2. If π_1 is not an isomorphism, let $E \subseteq Y$ be the exceptional curve of its last blow-up, so that $K_Y \cdot E = -1$. If π_2 did not contract E, then $Z_2 \cdot E \geq 0$ and $\pi_2^* K_{X_2} \cdot E = K_{X_2} \cdot \pi_{2*} E \geq 0$ by nefness, so $K_Y \cdot E \geq 0$, a contradiction. Hence π_2 contracts E as well, both π_i factor through the contraction of E, and induction on the number of blow-ups in π_1 reduces to the case where π_1 is an isomorphism. Then $\pi_2 \circ \pi_1^{-1} \colon X_1 \to X_2$ is a birational morphism from a minimal surface, which is an isomorphism because otherwise X_1 would contain the (-1)-curve of its last blow-up.

For (3), the surfaces listed are minimal: $\mathbb{P}^2$ and $\mathbb{F}_0$ contain no curve of negative self-intersection,

the only such curve on $\mathbb{F}_n$ for $n \geq 2$ is C_0, with $C_0^2 = -n$ (proposition 8.5.4), and on a ruled surface over a curve of genus $g \geq 1$ every rational curve maps to a point of the base, so it is a fibre and has self-intersection 0. They are birational to one another, and there are infinitely many pairwise non-isomorphic ones: the $\mathbb{F}_n$ are distinguished by n (proposition 8.5.4), and over C the surfaces $\mathbb{P}(\mathcal{O}_C \oplus \mathcal{O}_C(-D))$ with $\deg D = e \geq 0$ by the invariant e [36, § V.2], which gives non-uniqueness, completing the sketch.

8.6.3 Overview of the Classification

Over $\mathbb{C}$, the Enriques-Kodaira classification sorts minimal surfaces into four classes by κ and divides the class $\kappa = 0$ into four types. The ground field is $\mathbb{C}$ from here to the end of section 8.6.4, and box 8.6.7 records what changes in characteristic p. The main theorems below are stated, with references for their proofs, in [36, § V.6].

A minimal surface with $\kappa = -\infty$ is either $\mathbb{P}^2$ or a ruled surface $\mathbb{P}(\mathscr{E}) \to C$ over a smooth projective curve C of genus $g \geq 0$ (definition 8.5.1), and Enriques' theorem identifies the class among all surfaces [36, Theorem V.6.1]:

$$\kappa(X) = -\infty \iff P_{12}(X) = 0 \iff X \text{ is birational to } C \times \mathbb{P}^1 \text{ for a smooth projective curve } C$$

The genus of C equals the irregularity $q(X)$, and X is rational exactly when $g(C) = 0$.

Minimal surfaces with $\kappa = 0$ satisfy $12K_X \sim 0$, so K_X is numerically trivial ($K_X \cdot C = 0$ for every curve C) and has order dividing 12 in $\operatorname{Pic}(X)$. They fall into exactly four families, distinguished by the irregularity q and the geometric genus p_g [36, Theorem V.6.3].

1. *Abelian surfaces* ($q = 2$, $p_g = 1$, $K_X \sim 0$) are the two-dimensional abelian varieties. For each of them $X(\mathbb{C})$ is a complex torus $\mathbb{C}^2/\Lambda$ for a lattice Λ of rank 4, the group law is addition on the torus, $h^{1,1} = 4$, and $\chi(\mathcal{O}_X) = 0$ (example 8.3.5).

2. *K3 surfaces* ($q = 0$, $p_g = 1$, $K_X \sim 0$) are defined by $K_X \sim 0$ and $q = 0$. They are simply connected, with $h^{1,1} = 20$ and $\chi(\mathcal{O}_X) = 2$, and a smooth quartic surface in $\mathbb{P}^3$ is one (example 8.3.5). Example 8.6.4 collects more of their properties.

3. *Enriques surfaces* ($q = 0$, $p_g = 0$, $K_X \not\sim 0$, $2K_X \sim 0$) are exactly the quotients of K3 surfaces by a fixed-point-free involution. They have $\chi(\mathcal{O}_X) = 1$ and $h^{1,1} = 10$.

4. *Bielliptic surfaces*, also called hyperelliptic surfaces ($q = 1$, $p_g = 0$), are the quotients $(E_1 \times E_2)/G$ in which E_1 and E_2 are elliptic curves and G is a finite group acting on E_1 by translations and on E_2 with $E_2/G \cong \mathbb{P}^1$. The projections give two fibrations: $X \to E_1/G$, onto an elliptic curve, with every fibre isomorphic to E_2, and $X \to E_2/G \cong \mathbb{P}^1$, with general fibre E_1. Here $\chi(\mathcal{O}_X) = 0$, $h^{1,1} = 2$, and K_X has order 2, 3, 4 or 6 in $\operatorname{Pic}(X)$.

Example 8.6.4: K3 Surfaces: A Closer Look

Huybrechts [42] treats all four items.

1. Smooth quartic surfaces in $\mathbb{P}^3$, smooth complete intersections of type $(2,3)$ in $\mathbb{P}^4$ and of type $(2,2,2)$ in $\mathbb{P}^5$, double covers of $\mathbb{P}^2$ branched along a smooth sextic curve, and Kummer surfaces (the minimal resolutions of $A/\{\pm 1\}$ for an abelian surface A) are K3 surfaces.

2. All complex K3 surfaces, projective or not, are deformation-equivalent as complex manifolds (Kodaira), and their local deformation spaces are 20-dimensional. The projective ones form countably many 19-dimensional families, indexed by the degree $2d$ of a primitive polarization, and a very general complex K3 surface is not projective.

3. With its intersection form, $H^2(S,\mathbb{Z})$ is the even unimodular lattice $U^{\oplus 3} \oplus E_8(-1)^{\oplus 2}$ of signature $(3,19)$, so $b_2(S) = 22$. By the global Torelli theorem, two complex K3 surfaces S and S' are isomorphic if and only if there is an isometry $H^2(S,\mathbb{Z}) \cong H^2(S',\mathbb{Z})$ that respects the Hodge decompositions (Piatetski-Shapiro and Shafarevich for projective K3 surfaces, Burns and Rapoport in general). The periods of K3 surfaces lie in the period domain $\mathrm{SO}(3,19)/(\mathrm{SO}(2)\times\mathrm{SO}(1,19))$.

4. Every projective complex K3 surface contains infinitely many rational curves. The last cases were proved by Chen, Gounelas and Liedtke in 2022, after work of Bogomolov, Hassett and Tschinkel and of Li and Liedtke. Since $K_S = 0$, adjunction (8.5) gives $C^2 = 2p_a(C) - 2 \geq -2$ for an integral curve $C \subseteq S$, with $C^2 = -2$ exactly for the smooth rational curves.

A minimal surface with $\kappa = 1$ is *properly elliptic*: it admits a fibration $f\colon X \to C$ onto a smooth projective curve whose general fibre is a smooth curve of genus 1 [36, Theorem V.6.4], and f is the Iitaka fibration, the map given by $|nK_X|$ for n sufficiently divisible. Kodaira's *canonical bundle formula* describes K_X through the fibration:

$$K_X \sim f^*(K_C + \mathcal{L}) + \sum_i (m_i - 1)F_i$$

where $\mathcal{L}$ is a line bundle on C of degree $\chi(\mathcal{O}_X)$, and the sum runs over the *multiple fibres* $f^*(c_i) = m_i F_i$ with $m_i \geq 2$. Since $m_i F_i = f^*(c_i)$, a suitable multiple of K_X is the pullback of a divisor on C whose degree is a positive multiple of

$$2g(C) - 2 + \chi(\mathcal{O}_X) + \sum_i \left(1 - \frac{1}{m_i}\right)$$

and $\kappa(X) = 1$ exactly when this number is positive. Elliptic surfaces connect to the arithmetic of elliptic curves over function fields, through the Mordell-Weil group of the generic fibre, and to Kodaira's classification of the singular fibres of elliptic fibrations.

A surface is of *general type* if $\kappa = 2$. Then K_X is big, P_n grows like n^2, and the pluricanonical map $\varphi_{|nK_X|}$ is birational onto its image for some n [36, Theorem V.6.5], and Bombieri proved that every $n \geq 5$ works. The numerical invariants of a minimal surface of general type satisfy three inequalities, proved in the chapter on surfaces of general type of Barth, Hulek, Peters and Van de Ven, *Compact Complex Surfaces*.

Theorem 8.6.5: Geography of Surfaces of General Type

Let X be a minimal surface of general type over $\mathbb{C}$.

1. *Bogomolov-Miyaoka-Yau inequality.* $K_X^2 \leq 9\chi(\mathcal{O}_X)$, with equality if and only if the universal cover of $X(\mathbb{C})$ is the unit ball in $\mathbb{C}^2$.
2. *Noether inequality.* $p_g(X) \leq \frac{1}{2}K_X^2 + 2$. Since $\chi(\mathcal{O}_X) = 1 - q + p_g \leq 1 + p_g$, this gives $K_X^2 \geq 2\chi(\mathcal{O}_X) - 6$.
3. $K_X^2 \geq 1$ and $\chi(\mathcal{O}_X) \geq 1$.

The region of the (K^2, χ)-plane cut out by these inequalities is the *geography* of minimal surfaces of general type, and which of its lattice points are realized by such surfaces is not completely known. Horikawa classified the surfaces with $K^2 = 2p_g - 4$: their canonical map has degree 2 onto a rational surface. The surfaces with $K^2 = 9\chi$ are the ball quotients of item (1).

Example 8.6.6: Surfaces of General Type

1. A smooth surface $S \subseteq \mathbb{P}^3$ of degree $d \geq 5$ has $K_S = (d-4)H$ (proposition 6.5.7) and $K_S^2 = d(d-4)^2$. The sequence $0 \to \mathcal{O}_{\mathbb{P}^3}(-d) \to \mathcal{O}_{\mathbb{P}^3} \to \mathcal{O}_S \to 0$ gives $q = 0$ and $p_g = \binom{d-1}{3}$, so $\chi(\mathcal{O}_S) = 1 + \binom{d-1}{3}$. For $d = 5$ this is $K^2 = 5$, $p_g = 4$ and $\chi = 5$, as in example 8.3.6, and both inequalities hold: $5 \leq 9 \cdot 5$ and $4 \leq \frac{5}{2} + 2$.
2. For smooth projective curves C_1, C_2 of genera $g_1, g_2 \geq 2$, the product $X = C_1 \times C_2$ has $K_X = p_1^* K_{C_1} + p_2^* K_{C_2}$, $K^2 = 8(g_1 - 1)(g_2 - 1)$ and $\chi(\mathcal{O}_X) = (g_1 - 1)(g_2 - 1)$, so $K^2/\chi = 8$ for every such product.
3. The *Godeaux surface* is the quotient of the Fermat quintic $\{x_0^5 + \cdots + x_3^5 = 0\} \subseteq \mathbb{P}^3$ by the group of order 5 generated by $(x_0 : x_1 : x_2 : x_3) \mapsto (x_0 : \zeta x_1 : \zeta^2 x_2 : \zeta^3 x_3)$, for ζ a primitive fifth root of unity. Every nontrivial element of the group has four distinct eigenvalues, so its only fixed points in $\mathbb{P}^3$ are the coordinate points, which do not lie on the quintic, and the action is free. The quotient map is therefore étale of degree 5, which divides K^2 and χ by 5: the Godeaux surface has $K^2 = 1$ and $\chi = 1$, the smallest values allowed by theorem 8.6.5(3).

8.6.4 The Classification Table

The table lists the irregularity, the geometric genus and K^2 of each class of minimal surfaces over $\mathbb{C}$.

κ	q	p_g	K^2	Type
$-\infty$	any	0	$8(1-q)$, or 9 for $\mathbb{P}^2$	Ruled / Rational
0	2	1	0	Abelian surface
0	0	1	0	K3 surface
0	0	0	0	Enriques surface ($2K \sim 0$)
0	1	0	0	Bielliptic surface
1	any	any	0	Properly elliptic surface
2	any	any	> 0	General type

1. On a minimal surface the canonical class becomes more positive as κ increases. For $\kappa = -\infty$ it is not nef: $K_X \cdot F = -2$ for a fibre F of a ruling, and $K_{\mathbb{P}^2} = -3H$. For $\kappa = 0$ it is numerically trivial. For $\kappa = 1$ it is nef with $K_X^2 = 0$, and by the canonical bundle formula a multiple of it is the pullback of a divisor of positive degree on the base curve. For $\kappa = 2$ it is nef and big.

2. The self-intersection K^2 of a minimal surface is 0 for $\kappa = 0$ and $\kappa = 1$, and positive for $\kappa = 2$. For $\kappa = -\infty$ it is $8(1-q)$ for a ruled surface over a curve of genus q, and 9 for $\mathbb{P}^2$.

3. The pair (q, p_g) distinguishes the four types with $\kappa = 0$. For $\kappa = 1$ and $\kappa = 2$ more invariants are needed: the elliptic fibration and its multiple fibres, and the position in the geography, respectively.

8.6.7 Forward Reference: The Classification in Higher Dimensions and Positive Characteristic The Enriques-Kodaira classification of surfaces is the prototype for the *classification of algebraic varieties* in all dimensions, which also sorts varieties by Kodaira dimension and uses the Minimal Model Program to reduce to minimal models. *Birational Geometry* [8] develops this program:

1. The *cone theorem* and the *contraction theorem* generalize the theory of (-1)-curves to extremal rays of the Mori cone in any dimension.

2. *Flips* are needed when an extremal contraction is small, that is, contracts no divisor. This never happens on surfaces and does happen in every dimension ≥ 3.

3. The *abundance conjecture*, a theorem in dimension at most 3 and open in general, predicts that the canonical class of a minimal variety is semi-ample.

Bombieri and Mumford extended the classification of surfaces to characteristic $p > 0$, and new phenomena appear there. With $\kappa = 0$ there are quasi-hyperelliptic surfaces (in characteristics 2 and 3), K3 surfaces of Picard number 22 (the supersingular K3 surfaces, impossible over $\mathbb{C}$, where $\rho \leq h^{1,1} = 20$), and in characteristic 2 Enriques surfaces with $K_X \sim 0$. For general type, the Bogomolov-Miyaoka-Yau inequality $K^2 \leq 9\chi$ fails in characteristic p, and Bombieri's proof that $|5K_X|$ gives a birational map uses vanishing theorems that fail in characteristic p (theorem 8.6.8 and the remark after it). Ekedahl studied pluricanonical maps in characteristic p by other methods.

8.6.5 Vanishing Theorems

Kodaira vanishing and its extension by Kawamata and Viehweg control the higher cohomology of positive line bundles. Neither is proved in this book. Kodaira proved the first with harmonic forms, Deligne and Illusie gave an algebraic proof by reduction to characteristic p [26], and Kawamata-Viehweg vanishing is deduced from Kodaira vanishing by cyclic covering constructions.

Theorem 8.6.8: Kodaira Vanishing

Let X be a smooth projective variety over $\mathbb{C}$ and $\mathcal{L}$ an ample line bundle on X. Then

$$H^i(X, \omega_X \otimes \mathcal{L}) = 0 \qquad \text{for all } i > 0$$

Equivalently (by Serre duality), $H^i(X, \mathcal{L}^{-1}) = 0$ for $i < \dim X$.

The theorem is [36, Remark III.7.15]. The hypothesis $k = \mathbb{C}$ (more generally, characteristic 0) is needed: Raynaud constructed, in every characteristic $p > 0$, a smooth projective surface with an ample line bundle $\mathcal{L}$ such that $H^1(X, \mathcal{L}^{-1}) \neq 0$. In characteristic p Kodaira vanishing does hold for a smooth projective X of dimension at most p that lifts to the ring $W_2(k)$ of Witt vectors of length 2, by the theorem of Deligne and Illusie.

For a surface, Kodaira vanishing gives $H^1(X, \omega_X \otimes \mathcal{L}) = H^2(X, \omega_X \otimes \mathcal{L}) = 0$ for $\mathcal{L}$ ample, so Riemann-Roch for surfaces (theorem 8.3.1) computes the dimension of the space of sections:

$$h^0(X, \omega_X \otimes \mathcal{L}) = \chi(\omega_X \otimes \mathcal{L}) = \frac{D \cdot (D + K_X)}{2} + \chi(\mathcal{O}_X)$$

where D is a divisor with $\mathcal{L} \cong \mathcal{O}_X(D)$.

Theorem 8.6.9: Kawamata-Viehweg Vanishing

Let X be a smooth projective variety over $\mathbb{C}$ and D a nef and big divisor on X. Then

$$H^i(X, \omega_X \otimes \mathcal{L}(D)) = 0 \qquad \text{for all } i > 0$$

Kawamata and Viehweg proved this in 1982. It extends Kodaira vanishing from ample to nef and big divisors, and the extension is needed already for surfaces: on a minimal surface of general type the canonical divisor is nef and big (by the first paragraph of the proof of part (2) of theorem 8.6.3), and it fails to be ample exactly when X contains a smooth rational curve C with $C^2 = -2$. Indeed, if $K_X \cdot C = 0$ for an irreducible curve C, then $C^2 < 0$ by corollary 8.3.12 (as $K_X^2 > 0$), and adjunction (8.5) gives $2p_a(C) - 2 = C^2 < 0$, so $p_a(C) = 0$ and $C^2 = -2$. If no such curve exists, K_X is ample by theorem 8.2.11.

8.6.10 Forward Reference: Vanishing Theorems in the MMP Kawamata-Viehweg vanishing and its extensions (Nadel vanishing, Kollár vanishing) are the main technical input of the Minimal Model Program, and *Birational Geometry* [8] uses them to prove the following theorems, stated here for a smooth projective variety X over $\mathbb{C}$ (they hold more generally when X has klt singularities).

1. The *basepoint-free theorem* (Kawamata, Shokurov): if D is a nef Cartier divisor and $aD - K_X$ is nef and big for some $a > 0$, then $|mD|$ is base-point-free for all $m \gg 0$.
2. The *rationality theorem* (Kawamata): if K_X is not nef and H is ample, then $r = \max\{t \in \mathbb{R} : H + tK_X \text{ is nef}\}$ is a rational number.
3. The *non-vanishing theorem* (Shokurov): if D is a nef Cartier divisor and $aD - K_X$ is nef and big for some $a > 0$, then $H^0(X, \mathcal{O}_X(mD)) \neq 0$ for all $m \gg 0$.

In these proofs, vanishing of the higher cohomology lets Riemann-Roch compute dimensions of spaces of sections, as in the surface computation after theorem 8.6.8.

8.7 Application: Counting Points on Curves over Finite Fields

A system of homogeneous polynomial equations with coefficients in a finite field $\mathbb{F}_q$ has finitely many solutions in $\mathbb{P}^m(\mathbb{F}_q)$. When the equations cut out a smooth projective curve C_0, the number N of solutions is controlled by the genus g of the curve: the projective line has exactly $q+1$ points over $\mathbb{F}_q$, and every curve of genus g has $q+1$ points up to an error of at most $2g\sqrt{q}$. Hasse proved this for $g = 1$ in the 1930s, and Weil proved it for every g in 1948. We prove Weil's theorem with the surface theory of this chapter. The points of C_0 over $\mathbb{F}_q$ are the fixed points of the q-power Frobenius morphism F of the curve C obtained from C_0 by extending scalars to $\overline{\mathbb{F}}_q$. Their number is the intersection number of the graph of F with the diagonal on the surface $C \times C$, and the Hodge index theorem (theorem 8.3.9), which holds in every characteristic, bounds that intersection number in terms of g and q. Over a number field the number of points can be infinite, and section 8.7.1 records how the Euler characteristic of the curve governs the finiteness of its integral points.

Only the curve C_0 and its points are defined over $\mathbb{F}_q$. The curve C and the surface $C \times C$ are defined over the algebraically closed field $\overline{\mathbb{F}}_q$, where the conventions of box 7.0.1 hold.

8.7.1 Conventions Throughout this section $q = p^r$ is a power of a prime p, and $k = \overline{\mathbb{F}}_q$ is an algebraic closure of $\mathbb{F}_q$. For $n \geq 1$, $\mathbb{F}_{q^n}$ is the subfield of k formed by the q^n roots of $T^{q^n} - T$. The letter C_0 denotes a projective $\mathbb{F}_q$-scheme whose base change $C = C_0 \times_{\mathbb{F}_q} k$ is a curve over k in the sense of box 7.0.1: integral, of dimension 1, complete and nonsingular. The *genus* of C_0 is $g = g(C)$.

A closed immersion $C_0 \hookrightarrow \mathbb{P}^m_{\mathbb{F}_q}$ is fixed. On a standard chart $D_+(x_j) = \operatorname{Spec} \mathbb{F}_q[y]$, where y is the tuple of the m coordinates x_i/x_j with $i \neq j$, the closed subscheme $C_0 \cap D_+(x_j)$ is $\operatorname{Spec}(\mathbb{F}_q[y]/I_j)$ for an ideal I_j (proposition 3.1.24(1)), and $C \cap D_+(x_j) = \operatorname{Spec}(k[y]/I_j k[y])$, because $(\mathbb{F}_q[y]/I_j) \otimes_{\mathbb{F}_q} k \cong k[y]/I_j k[y]$. By theorem 0.1.12(3) the closed points of $C \cap D_+(x_j)$ are the points $a \in k^m$ at which every element of I_j vanishes, so every closed point of C has homogeneous coordinates $[a_0 : \cdots : a_m]$. For a subfield $L \subseteq k$ containing $\mathbb{F}_q$, $C_0(L)$ denotes the set of closed points of C that have homogeneous coordinates in L, and

$$N_n = \#C_0(\mathbb{F}_{q^n}), \qquad N = N_1$$

The set $C_0(\mathbb{F}_q)$ is in bijection with the set of $\mathbb{F}_q$-rational points of C_0, the points z with $\kappa(z) = \mathbb{F}_q$. Such a point of $C_0 \cap D_+(x_j)$ is the kernel of a surjection $\mathbb{F}_q[y]/I_j \to \mathbb{F}_q$, which is determined by the images $a \in \mathbb{F}_q^m$ of the coordinates, and a is a zero of every element of I_j, that is, a closed point of C with coordinates in $\mathbb{F}_q$. Conversely, every such closed point a defines the surjection $\mathbb{F}_q[y]/I_j \to \mathbb{F}_q$, $y \mapsto a$. In particular N does not depend on the embedding.

Raising the homogeneous coordinates of a closed point of C to the q-th power fixes exactly the points of $C_0(\mathbb{F}_q)$, and this operation comes from the q-th power map on the functions of C_0.

Definition 8.7.2: The q-Power Frobenius Morphism

The *q-power absolute Frobenius* of C_0 is the morphism $\Phi\colon C_0 \to C_0$ that is the identity on the underlying topological space and is $h \mapsto h^q$ on $\mathcal{O}_{C_0}$. The *q-power Frobenius* of C is the morphism of k-schemes

$$F = \Phi \times_{\mathbb{F}_q} \mathrm{id}_k \colon C = C_0 \times_{\mathbb{F}_q} k \longrightarrow C_0 \times_{\mathbb{F}_q} k = C$$

The map $h \mapsto h^q$ is a ring endomorphism of each $\mathcal{O}_{C_0}(V)$, because q is a power of the characteristic p, and it commutes with restriction. On stalks it is a local homomorphism, since h^q lies in a prime ideal exactly when h does. It fixes every $c \in \mathbb{F}_q$, as $c^q = c$, so Φ is a morphism of $\mathbb{F}_q$-schemes and its fiber product with id_k over $\mathbb{F}_q$ is defined. On an affine open $U_0 = \operatorname{Spec} A_0$ of C_0, the morphism Φ restricts to Spec of the ring map $a \mapsto a^q$: this map is the identity on prime ideals, and on each $D(b)$ it is $a/b^n \mapsto a^q/b^{qn}$. Hence on the affine open $U = U_0 \times_{\mathbb{F}_q} k = \operatorname{Spec}(A_0 \otimes_{\mathbb{F}_q} k)$ of C, the morphism F restricts to Spec of the k-algebra endomorphism

$$F^{\#}\colon A_0 \otimes_{\mathbb{F}_q} k \longrightarrow A_0 \otimes_{\mathbb{F}_q} k, \qquad a \otimes \lambda \longmapsto a^q \otimes \lambda \tag{8.6}$$

The q-th power map $h \mapsto h^q$ on $\mathcal{O}_C$ sends a constant $\lambda \in k$ to λ^q, so it is not a morphism of k-schemes. The morphism F raises to the q-th power only the functions that come from C_0, and it is k-linear. Under the identification of C with its Frobenius twist, which holds because C comes from $\mathbb{F}_q$, it is the k-linear Frobenius morphism of [36, Remark IV.2.4.1 and Exercise V.1.10] for the power q.

Example 8.7.3: The Frobenius of the Projective Line

Let $C_0 = \mathbb{P}^1_{\mathbb{F}_q}$, so that $C = \mathbb{P}^1_k$. On the chart $D_+(x_0) = \operatorname{Spec} \mathbb{F}_q[t]$ with $t = x_1/x_0$, (8.6) is the k-algebra map $k[t] \to k[t]$, $t \mapsto t^q$. The closed point $t = c$ corresponds to the kernel of $t \mapsto c$, and composing with $t \mapsto t^q$ gives $t \mapsto c^q$, so F sends it to $t = c^q$. On the other chart the same computation applies, and in homogeneous coordinates $F[a_0 : a_1] = [a_0^q : a_1^q]$. The point $[1 : c]$ is fixed exactly when $c^q = c$, that is, when $c \in \mathbb{F}_q$, and $[0 : 1]$ is fixed, so F has $q + 1 = \#\mathbb{P}^1(\mathbb{F}_q)$ fixed points. Every $c \in k$ has a unique q-th root in k, so F is a bijection on closed points. On function fields F induces the map $k(t) \to k(t)$, $t \mapsto t^q$, whose image is $k(t^q) = k(t)^q$, the subfield of q-th powers.

The degree of a finite morphism of curves is the degree of an extension of function fields (definition 7.1.8). For $\mathbb{P}^1$ the extension is $k(t)$ over its subfield of q-th powers. Over a perfect field, the function field of every curve has degree q over its subfield of q-th powers.

Lemma 8.7.4: A Function Field Has Degree q over Its q-th Powers

Let k be a perfect field of characteristic $p > 0$, let q be a power of p, and let K be a finitely generated field extension of k of transcendence degree 1. Then $K^q = \{h^q : h \in K\}$ is a subfield of K containing k, and $[K : K^q] = q$.

***Proof*:**

The map $\mathrm{Fr}\colon K \to K$, $h \mapsto h^q$, is an injective ring homomorphism, so its image K^q is a subfield of K isomorphic to K. It contains $k = k^q$, since k is perfect.

Choose $t \in K$ transcendental over k. The extension $K/k(t)$ is finitely generated and algebraic, hence finite, of some degree n. The isomorphism $\mathrm{Fr}\colon K \to K^q$ carries $k(t)$ onto $k^q(t^q) = k(t^q)$, so $[K^q : k(t^q)] = n$. The element t is a root of $T^q - t^q \in k(t^q)[T]$, so $[k(t) : k(t^q)] \leq q$. The powers $1, t, \ldots, t^{q-1}$ are linearly independent over $k(t^q)$: after clearing denominators, a relation $\sum_{i=0}^{q-1} c_i t^i = 0$ has coefficients $c_i \in k[t^q]$, every monomial occurring in $c_i t^i$ has exponent congruent to i modulo q, and since t is transcendental the relation holds in the polynomial ring $k[t]$, so each $c_i t^i$ vanishes and every c_i is zero. Hence $[k(t) : k(t^q)] = q$. Computing $[K : k(t^q)]$ through the two towers $K \supseteq K^q \supseteq k(t^q)$ and $K \supseteq k(t) \supseteq k(t^q)$ gives

$$[K : K^q] \cdot n = [K : k(t^q)] = n \cdot q$$

so $[K : K^q] = q$, as claimed.

For transcendence degree d the same two towers, built on a transcendence basis $t_1, \ldots, t_d$, give $[K : K^q] = q^d$.

Proposition 8.7.5: Fixed Points, Degree and Differential of the Frobenius

Let $F\colon C \to C$ be the q-power Frobenius.

1. For every closed point $[a_0 : \cdots : a_m]$ of C,
$$F[a_0 : \cdots : a_m] = [a_0^q : \cdots : a_m^q]$$
The closed points fixed by F are exactly the points of $C_0(\mathbb{F}_q)$, so F has exactly N fixed closed points.
2. F is a finite surjective morphism of degree q, and the map $F^*\colon K(C) \to K(C)$ it induces on the function field has image $K(C)^q$.
3. For every closed point x of C and every $h \in \mathcal{O}_{C,F(x)}$, the differential $d(F^\# h) \in \Omega_{C/k,x}$ is zero. Equivalently, the map $F^*\Omega_{C/k} \to \Omega_{C/k}$ of the cotangent sequence (proposition 5.5.17) is zero.

***Proof*:**
Let $U_0 = \operatorname{Spec} A_0$ be a nonempty affine open of C_0 and $U = \operatorname{Spec} A$ with $A = A_0 \otimes_{\mathbb{F}_q} k$. By (8.6), F maps U into U and is given there by $F^\#(a \otimes \lambda) = a^q \otimes \lambda$.

(1) Take $U_0 = C_0 \cap D_+(x_j)$, so $A_0 = \mathbb{F}_q[y]/I_j$ and $A = k[y]/I_j k[y]$, with $F^\#(y_i) = y_i^q$ for each coordinate. A closed point a of U, with $a_j = 1$, corresponds to the kernel of the evaluation $\operatorname{ev}_a\colon A \to k$, $y_i \mapsto a_i$ (theorem 0.1.12(3)). Its image $F(a)$ corresponds to the prime ideal $(F^\#)^{-1}(\ker \operatorname{ev}_a) = \ker(\operatorname{ev}_a \circ F^\#)$, and $\operatorname{ev}_a \circ F^\#$ is the k-algebra map with $y_i \mapsto a_i^q$. So $F(a)$ is the point with coordinates a_i^q, which in homogeneous coordinates is $[a_0^q : \cdots : a_m^q]$, since $a_j = 1 = a_j^q$. The point a is fixed exactly when $a_i^q = a_i$ for every i. The q elements of $\mathbb{F}_q$ are roots of $T^q - T$, because $c^{q-1} = 1$ for $c \in \mathbb{F}_q^\times$, and the polynomial has at most q roots in k, so its roots are exactly the elements of $\mathbb{F}_q$. Hence a is fixed if and only if every a_i lies in $\mathbb{F}_q$, that is, $a \in C_0(\mathbb{F}_q)$. The charts $D_+(x_j)$ cover C, which proves (1).

(2) The Frobenius $\sigma\colon k \to k$, $\lambda \mapsto \lambda^q$, is a field automorphism fixing $\mathbb{F}_q$ pointwise: it is injective, and surjective because $T^q - \lambda$ has a root in the algebraically closed field k. So $\psi = \mathrm{id}_{A_0} \otimes \sigma^{-1}$ is a ring automorphism of A. For $h = \sum_i a_i \otimes \lambda_i$, the q-th power map is additive in characteristic p, so
$$\psi(h)^q = \sum_i a_i^q \otimes \lambda_i = F^\#(h)$$
The ring A is a domain, as C is integral, so $F^\#(h) = 0$ forces $\psi(h) = 0$ and $h = 0$. Thus $F^\#$ is injective, F maps the generic point of C to itself, and F is not constant. By proposition 7.1.6, F is finite and surjective. The map F^* on $K = K(C) = \operatorname{Frac} A$ extends $F^\#$, and $F^\#(A) = \psi(A)^q = A^q$, so $F^*K = \operatorname{Frac}(A^q) = K^q$. The field K is finitely generated of transcendence degree 1 over the perfect field k (theorem 2.6.11), so definition 7.1.8 and lemma 8.7.4 give $\deg F = [K : F^*K] = [K : K^q] = q$.

(3) Let x be a closed point and choose U_0 with $x \in U$. Then $F(x) \in U$, and $F^\#\colon \mathcal{O}_{C,F(x)} \to \mathcal{O}_{C,x}$ is the map induced by $F^\#\colon A \to A$ on localizations. An element of $\mathcal{O}_{C,F(x)}$ has the form $h = a/b$ with $a, b \in A$ and b outside the prime ideal of $F(x)$. Since $F^\#$ is local, $\psi(b)^q = F^\# b$ is a unit of $\mathcal{O}_{C,x}$, hence so is $\psi(b)$, and $F^\# h = \psi(a)^q/\psi(b)^q = u^q$ with $u = \psi(a)/\psi(b)$. Hence $d(F^\# h) = q\, u^{q-1}\, du = 0$, because $q = 0$ in k. The map $F^*\Omega_{C/k} \to \Omega_{C/k}$ sends $F^*(dh)$ to

$d(F^{\#}h)$ (proposition 5.5.5), and the elements $F^*(dh)$ generate $F^*\Omega_{C/k}$, so the map is zero, completing the proof.

As on $\mathbb{P}^1$, the morphism F is a bijection on closed points: coordinatewise q-th roots invert the formula of (1), and they preserve the zero set of each I_j, because an element h of I_j has coefficients c with $c^q = c$, so that $h(b)^q = h(b^q)$ for every $b \in k^m$. Nevertheless F has degree q, so over each closed point there is a single point, with ramification index q (proposition 7.1.9). The extension K/K^q is purely inseparable, and (3) is the vanishing of (7.3) for the inseparable morphism F (box 7.4.4).

By proposition 8.7.5(1), N is the number of closed points x with $F(x) = x$, that is, the number of points where the graph of F meets the diagonal of the surface $C \times C$.

Proposition 8.7.6: The Product of Two Curves

Let C and C' be curves over k, and let $X = C \times C' = C \times_k C'$, with projections p_1 and p_2.

1. X is a surface: a smooth projective surface over k in the sense of definition 8.1.1.
2. The map
$$\alpha\colon p_1^*\Omega_{C/k} \oplus p_2^*\Omega_{C'/k} \longrightarrow \Omega_{X/k}$$
whose two components are the first maps of the cotangent sequences (proposition 5.5.17) of p_1 and p_2, is an isomorphism.
3. $\omega_X \cong p_1^*\omega_C \otimes p_2^*\omega_{C'}$.
4. Let P be a closed point of X, with $x = p_1(P)$ and $y = p_2(P)$. For $h \in \mathcal{O}_{C,x}$ and $h' \in \mathcal{O}_{C',y}$ write $h \otimes 1 = p_1^{\#}h$ and $1 \otimes h' = p_2^{\#}h'$ in $\mathcal{O}_{X,P}$, and write $dh(x)$ for the image of dh in $\Omega_{C/k,x} \otimes \kappa(x)$, and likewise $dh'(y)$. There is an isomorphism of k-vector spaces
$$\mathfrak{m}_P/\mathfrak{m}_P^2 \;\cong\; \big(\Omega_{C/k,x} \otimes \kappa(x)\big) \oplus \big(\Omega_{C'/k,y} \otimes \kappa(y)\big)$$
sending the class of $h \otimes 1 + 1 \otimes h'$, for $h \in \mathfrak{m}_x$ and $h' \in \mathfrak{m}_y$, to $(dh(x), dh'(y))$. Each summand is one-dimensional, spanned by $dt(x)$, respectively $dt'(y)$, for uniformizers t at x and t' at y.

***Proof*:**

Step 1: (1). The curves C and C' are regular, hence smooth over the perfect field k (theorem 2.7.15), of relative dimension 1, since by proposition 2.7.12 the relative dimension of a smooth morphism to $\operatorname{Spec} k$ is the dimension of its source. By proposition 2.7.11(3), X is smooth over k of relative dimension 2. So X has pure dimension 2 (proposition 2.7.12) and is regular (theorem 2.7.15), and it is integral, separated and of finite type over k by proposition 3.7.10. Choose closed immersions $C \hookrightarrow \mathbb{P}^m_k$ and $C' \hookrightarrow \mathbb{P}^{m'}_k$ (proposition 7.1.7). The morphisms $C \times C' \to \mathbb{P}^m_k \times C'$ and $\mathbb{P}^m_k \times C' \to \mathbb{P}^m_k \times \mathbb{P}^{m'}_k$ are base changes of these closed immersions, hence closed immersions (proposition 3.2.5(2)), and the Segre embedding $\mathbb{P}^m_k \times \mathbb{P}^{m'}_k \to \mathbb{P}^M_k$ is a closed immersion (example 5.3.45(5)). The composite is a closed immersion (proposition 3.1.24(3)), so X is projective.

Step 2: (2). By proposition 5.5.25, $\Omega_{X/k}$ is locally free of rank 2, and the source of α is a direct sum of two invertible sheaves. Let $U = \operatorname{Spec} A \subseteq C$ and $U' = \operatorname{Spec} B \subseteq C'$ be affine opens.

Then $U \times U' = \operatorname{Spec}(A \otimes_k B)$, such opens cover X, and $\Omega_{X/k}|_{U\times U'}$ is the sheaf associated to $\Omega_{A\otimes_k B/k}$ (proposition 5.5.15). On $U \times U'$ the first component of α sends $da \otimes c$ to $c\,d(a \otimes 1)$ and the second sends $db \otimes c$ to $c\,d(1 \otimes b)$ (proposition 5.5.5). The module $\Omega_{A\otimes_k B/k}$ is generated by the differentials of the elements of $A \otimes_k B$ (definition 5.5.2, Construction 1), and by the Leibniz rule

$$d(a \otimes b) = (1 \otimes b)\, d(a \otimes 1) + (a \otimes 1)\, d(1 \otimes b)$$

so α is surjective. At a point P of X, α_P is then a surjection $\mathcal{O}_{X,P}^2 \to \mathcal{O}_{X,P}^2$ of free modules. It has a right inverse β, the target being free, and $\det(\alpha_P)\det(\beta) = 1$, so $\det(\alpha_P)$ is a unit and α_P is an isomorphism. Hence α is an isomorphism.

Step 3: (3). For a curve $\omega_C = \Omega_{C/k}$, and $\omega_X = \bigwedge^2 \Omega_{X/k}$ (definition 5.5.28). Put $\mathcal{L}_1 = p_1^*\omega_C$ and $\mathcal{L}_2 = p_2^*\omega_{C'}$. The bilinear map $(s_1, s_2) \mapsto s_1 \wedge s_2$ induces $\mathcal{L}_1 \otimes \mathcal{L}_2 \to \bigwedge^2(\mathcal{L}_1 \oplus \mathcal{L}_2)$. Where $\mathcal{L}_1$ and $\mathcal{L}_2$ are free on generators u_1 and u_2, the sheaf $\mathcal{L}_1 \oplus \mathcal{L}_2$ is free on u_1, u_2, its second exterior power is free on $u_1 \wedge u_2$, and the map sends the generator $u_1 \otimes u_2$ to it. So the map is an isomorphism, and with (2) it gives $\omega_X \cong \bigwedge^2(\mathcal{L}_1 \oplus \mathcal{L}_2) \cong \mathcal{L}_1 \otimes \mathcal{L}_2$.

Step 4: (4). The residue fields $\kappa(P)$, $\kappa(x)$, $\kappa(y)$ are all k. By proposition 5.5.10, the map $\mathfrak{m}_P/\mathfrak{m}_P^2 \to \Omega_{X/k,P} \otimes \kappa(P)$ sending the class of u to $du \otimes 1$ is an isomorphism. Composing with the inverse of $\alpha_P \otimes \kappa(P)$ and using $(p_1^*\Omega_{C/k})_P \otimes \kappa(P) = \Omega_{C/k,x} \otimes \kappa(x)$, and likewise for p_2, gives the isomorphism of (4). Under it $d(h \otimes 1) = \alpha(p_1^* dh, 0)$ goes to $(dh(x), 0)$ and $d(1 \otimes h')$ to $(0, dh'(y))$. The same proposition on C identifies $\Omega_{C/k,x} \otimes \kappa(x)$ with $\mathfrak{m}_x/\mathfrak{m}_x^2$, which is spanned by the class of t, so it is spanned by $dt(x)$, and likewise for C', completing the proof.

The proposition is the algebraic form of the splitting of the cotangent space of a product into the cotangent spaces of its factors, and (3) says that a canonical divisor of $C \times C'$ is $p_1^*K_C + p_2^*K_{C'}$.

From now on C is a curve of genus g and $X = C \times C$. For a closed point P of C, the fibers of p_1 and p_2 over P are

$$e = P \times_k C = \{P\} \times C, \qquad f = C \times_k P = C \times \{P\}$$

For a morphism $\varphi\colon C \to C$, the morphism $\gamma_\varphi = (\mathrm{id}_C, \varphi)\colon C \to X$ is a closed immersion (proposition 3.5.9, C being separated over k), and its image Γ_φ is the *graph* of φ. Its closed points are the $(x, \varphi(x))$. The graph $\Delta = \Gamma_{\mathrm{id}_C}$ is the diagonal. The fiber f is the graph of the constant morphism c_P with value P.

Proposition 8.7.7: Intersection Numbers of Fibers and Graphs

Let P be a closed point of C, let e and f be the fibers over P, and let $\varphi\colon C \to C$ be a finite morphism of degree d.

1. The fibers e and f are smooth irreducible curves on X, with $\mathcal{O}_X(e) \cong p_1^*\mathcal{L}(P)$ and $\mathcal{O}_X(f) \cong p_2^*\mathcal{L}(P)$, and
$$e^2 = f^2 = 0, \qquad e \cdot f = 1$$
2. The graph Γ_φ is a smooth irreducible curve isomorphic to C, and
$$\Gamma_\varphi \cdot e = 1, \qquad \Gamma_\varphi \cdot f = d$$
3. $\Gamma_\varphi^2 = d(2-2g)$. In particular $\Delta^2 = 2 - 2g$.

***Proof*:**
Step 1: degrees along graphs. Let $\psi\colon C \to C$ be a morphism. Its graph Γ_ψ is a closed subscheme isomorphic to C through γ_ψ, hence a smooth irreducible curve on X. For a divisor D on X, proposition 8.2.2(6) and this isomorphism give

$$D \cdot \Gamma_\psi = \deg\bigl(\gamma_\psi^* \mathcal{O}_X(D)\bigr) \tag{8.7}$$

For an invertible sheaf $\mathcal{L}$ on C: if ψ is finite of degree d_ψ, then $\deg \psi^*\mathcal{L} = d_\psi \deg \mathcal{L}$, because $\mathcal{L} \cong \mathcal{L}(D_0)$ for a divisor D_0 (corollary 5.4.45), $\psi^*\mathcal{L}(D_0) \cong \mathcal{L}(\psi^* D_0)$ (as shown before (7.4)), and $\deg \psi^* D_0 = d_\psi \deg D_0$ (proposition 7.1.9). If $\psi = c_P$ is constant, then ψ factors through $\operatorname{Spec} k$, on which every invertible sheaf is trivial, so $\deg \psi^*\mathcal{L} = 0$.

Step 2: (1). The point P is an effective Cartier divisor on C with ideal sheaf $\mathcal{I}_P \cong \mathcal{L}(-P)$ (proposition 5.4.47(1)), and it is flat over $\operatorname{Spec} k$, as every module over a field is. The fiber $e = P \times_k C$ is its base change along the second factor $C \to \operatorname{Spec} k$, and the projection from $X = C \times_k C$ to the factor containing P is p_1. By proposition 5.4.31, e is an effective Cartier divisor on X with $\mathcal{I}_e \cong p_1^*\mathcal{I}_P$, so $\mathcal{O}_X(-e) \cong p_1^*\mathcal{L}(-P)$, and dualizing gives $\mathcal{O}_X(e) \cong p_1^*\mathcal{L}(P)$. Exchanging the factors, $\mathcal{O}_X(f) \cong p_2^*\mathcal{L}(P)$. The closed immersion $f = C \times_k P \hookrightarrow X$, composed with the isomorphism $C \cong C \times_k P$, is $\gamma_{c_P} = (\mathrm{id}_C, c_P)\colon x \mapsto (x, P)$, and in the same way e is the image of the closed immersion $\varepsilon = (c_P, \mathrm{id}_C)\colon x \mapsto (P, x)$. Both fibers are smooth irreducible curves isomorphic to C, and (8.7), together with its analogue for ε, gives

$$\begin{aligned} f^2 &= \deg \gamma_{c_P}^* p_2^* \mathcal{L}(P) = \deg c_P^* \mathcal{L}(P) = 0, \\ e \cdot f &= \deg \gamma_{c_P}^* p_1^* \mathcal{L}(P) = \deg \mathcal{L}(P) = 1, \\ e^2 &= \deg \varepsilon^* p_1^* \mathcal{L}(P) = \deg c_P^* \mathcal{L}(P) = 0 \end{aligned}$$

using $p_1 \circ \gamma_{c_P} = \mathrm{id}_C = p_2 \circ \varepsilon$ and $p_2 \circ \gamma_{c_P} = c_P = p_1 \circ \varepsilon$.

Step 3: (2). Since $p_1 \circ \gamma_\varphi = \mathrm{id}_C$ and $p_2 \circ \gamma_\varphi = \varphi$, (8.7) and Step 1 give

$$\Gamma_\varphi \cdot e = \deg \mathcal{L}(P) = 1, \qquad \Gamma_\varphi \cdot f = \deg \varphi^* \mathcal{L}(P) = d$$

Step 4: (3). Let K_X be a canonical divisor of X. By proposition 8.7.6(3), $\mathcal{O}_X(K_X) \cong p_1^*\omega_C \otimes p_2^*\omega_C$, so by (8.7), Step 1 and proposition 7.3.1,

$$K_X \cdot \Gamma_\varphi = \deg \omega_C + \deg \varphi^* \omega_C = (2g-2) + d(2g-2)$$

The curve $\Gamma_\varphi \cong C$ has genus g, and the adjunction formula (theorem 8.2.8) gives

$$2g - 2 = \Gamma_\varphi^2 + K_X \cdot \Gamma_\varphi = \Gamma_\varphi^2 + (1+d)(2g-2)$$

so $\Gamma_\varphi^2 = -d(2g-2) = d(2-2g)$. The diagonal is the case $\varphi = \mathrm{id}_C$, $d = 1$, completing the proof.

As sets, Γ_φ meets the fiber e of the first projection in the single point $(P, \varphi(P))$, and the fiber f of the second projection in the points (x, P) with $x \in \varphi^{-1}(P)$, of which there are d counted with ramification. For $g \geq 2$ the diagonal has negative self-intersection, so by corollary 8.3.13(2) it is the only effective divisor in its numerical class.

The intersection number $\Gamma_F \cdot \Delta$ is a sum of local intersection multiplicities (proposition 8.2.2(5)), and computing those requires a local equation of each graph.

Lemma 8.7.8: Local Equation of a Graph

Let $\varphi\colon C \to C$ be a morphism, x a closed point of C, $y = \varphi(x)$, $P = (x, y)$, and t a uniformizer at y. Then the ideal of Γ_φ in $\mathcal{O}_{X,P}$ is generated by

$$1 \otimes t - \varphi^{\#} t \otimes 1$$

Under the isomorphism of proposition 8.7.6(4), the class of this element in $\mathfrak{m}_P/\mathfrak{m}_P^2$ is $\big(-d(\varphi^{\#}t)(x),\, dt(y)\big)$, whose second component is nonzero.

***Proof*:**

Write $g_P = 1\otimes t - \varphi^{\#}t \otimes 1$ and let $J \subseteq \mathcal{O}_{X,P}$ be the ideal of Γ_φ. Choose an affine open $V = \operatorname{Spec} B$ containing y with $t \in B$, and an affine open $U = \operatorname{Spec} A$ containing x with $\varphi(U) \subseteq V$. Then $\gamma_\varphi^{-1}(U \times V) = U$, and $\gamma_\varphi|_U \colon U \to U \times V = \operatorname{Spec}(A \otimes_k B)$ is Spec of the surjection $A \otimes_k B \to A$, $a \otimes b \mapsto a\,\varphi^{\#}(b)$. So $\Gamma_\varphi \cap (U \times V)$ is the closed subscheme defined by the kernel of this surjection, which contains $1 \otimes t - \varphi^{\#}t \otimes 1$, and $g_P \in J$.

Here $t \in \mathfrak{m}_y$ and $\varphi^{\#}t \in \mathfrak{m}_x$, as $(\varphi^{\#}t)(x) = t(\varphi(x)) = t(y) = 0$, so proposition 8.7.6(4) applies with $h = -\varphi^{\#}t$ and $h' = t$, and gives the class $\big(-d(\varphi^{\#}t)(x), dt(y)\big)$. Its second component is nonzero because $dt(y)$ spans $\Omega_{C/k,y} \otimes \kappa(y)$. In particular $g_P \notin \mathfrak{m}_P^2$ and $g_P \neq 0$.

The local ring $\mathcal{O}_{X,P}$ is regular of dimension 2 (proposition 8.7.6(1) and theorem 2.6.12), hence a unique factorization domain (proposition 2.7.27). The ideal J is prime, and $\mathcal{O}_{X,P}/J \cong \mathcal{O}_{C,x}$ through γ_φ, a discrete valuation ring, so $J \neq \mathfrak{m}_P$. The element g_P is a nonzero nonunit of the prime ideal J, so one of its prime factors π lies in J. The chain $0 \subsetneq (\pi) \subseteq J \subsetneq \mathfrak{m}_P$ of prime ideals in a local ring of dimension 2 forces $(\pi) = J$. Write $g_P = \pi u$. If $u \in \mathfrak{m}_P$ then $g_P \in \mathfrak{m}_P^2$, which is false, so u is a unit and $J = (g_P)$, as claimed.

Proposition 8.7.9: The Rational Points as an Intersection Number

Let $\Gamma_F \subseteq X = C \times C$ be the graph of the q-power Frobenius and Δ the diagonal. Then Γ_F and Δ meet transversally, and

$$\Gamma_F \cdot \Delta = N$$

***Proof*:**

By proposition 8.7.5(2), F is finite of degree $q \geq 2$, so proposition 8.7.7(2) gives $\Gamma_F \cdot f = q$ while $\Delta \cdot f = 1$. Hence $\Gamma_F \neq \Delta$, and the two irreducible curves have no common component. The closed points of Γ_F are the $(x, F(x))$ and those of Δ are the (x, x), so the closed points of $\Gamma_F \cap \Delta$ are the (x, x) with $F(x) = x$. By proposition 8.7.5(1) there are N of them, and by the proof of proposition 8.2.2(5) the intersection consists of closed points only.

Let $P = (x, x)$ with $F(x) = x$, and let t be a uniformizer at x. By lemma 8.7.8, applied to id_C and to F, local equations of Δ and Γ_F at P are

$$\delta = 1 \otimes t - t \otimes 1, \qquad \gamma = 1 \otimes t - F^{\#}t \otimes 1$$

and their classes in $\mathfrak{m}_P/\mathfrak{m}_P^2$ are $(-dt(x), dt(x))$ and $(-d(F^{\#}t)(x), dt(x))$. By proposition 8.7.5(3), $d(F^{\#}t) = 0$, so the second class is $(0, dt(x))$. As $dt(x) \neq 0$, the two classes are linearly independent, so they span the two-dimensional space $\mathfrak{m}_P/\mathfrak{m}_P^2$, and by Nakayama's lemma δ and

γ generate $\mathfrak{m}_P$. So Γ_F and Δ meet transversally at every point of $\Gamma_F \cap \Delta$, and proposition 8.2.2(4) gives $\Gamma_F \cdot \Delta = \#(\Gamma_F \cap \Delta) = N$, as claimed.

For a smooth map $\theta\colon M \to M$ of a compact oriented manifold, the Lefschetz number is the oriented intersection number of the diagonal with the graph of θ, and when $d\theta_p - \mathrm{id}$ is invertible at every fixed point p, the Lefschetz-Hopf theorem expresses it as a sum of local contributions ± 1, one for each fixed point (*Differential Topology* [59, Lefschetz Number, The Lefschetz-Hopf Theorem]). Proposition 8.7.9 is the algebraic counterpart for F: its differential vanishes, so $dF - \mathrm{id} = -\mathrm{id}$ is invertible at every fixed point, and each fixed point contributes local intersection multiplicity 1.

Example 8.7.10: The Graph of Frobenius on $\mathbb{P}^1 \times \mathbb{P}^1$

Let $C_0 = \mathbb{P}^1_{\mathbb{F}_q}$ as in example 8.7.3, so that $g = 0$ and $X = \mathbb{P}^1_k \times \mathbb{P}^1_k$. The fibers e and f are members of the two rulings, and $\operatorname{Pic} X = \mathbb{Z}e \oplus \mathbb{Z}f$ (example 8.1.7(2)). Write $\Gamma_F \sim \alpha e + \beta f$. Since $e^2 = f^2 = 0$ and $e \cdot f = 1$, the numbers $\Gamma_F \cdot e = 1$ and $\Gamma_F \cdot f = q$ of proposition 8.7.7(2) give $\beta = 1$ and $\alpha = q$, so $\Gamma_F \sim qe + f$. In the same way $\Delta \sim e + f$, and

$$\Gamma_F \cdot \Delta = (qe + f) \cdot (e + f) = q + 1, \qquad \Gamma_F^2 = 2q, \qquad \Delta^2 = 2$$

This agrees with proposition 8.7.9 and $\#\mathbb{P}^1(\mathbb{F}_q) = q + 1$, with proposition 8.7.7(3) for $g = 0$, and with the self-intersection 2 of a curve of bidegree $(1, 1)$ (example 8.2.3(3)).

The Hodge index theorem bounds the self-intersection of every divisor on X by its intersection numbers with the two fibers, in an inequality due to Castelnuovo and Severi.

Theorem 8.7.11: Castelnuovo-Severi Inequality

Let C be a curve, $X = C \times C$, P a closed point of C, and e and f the fibers over P. For a divisor D on X put $a = D \cdot e$ and $b = D \cdot f$. Then

$$D^2 \;\leq\; 2ab$$

with equality if and only if $D \equiv be + af$.

***Proof*:**

Put $D' = D - be - af$. By proposition 8.7.7(1), $e^2 = f^2 = 0$ and $e \cdot f = 1$, so

$$D' \cdot e = a - b\,e^2 - a\,(f \cdot e) = 0, \qquad D' \cdot f = b - b\,(e \cdot f) - a\,f^2 = 0$$

and therefore $D' \cdot (e + f) = 0$, while $(e + f)^2 = 2 > 0$. By corollary 8.3.12(1), applied to the divisor $e + f$ of positive square and to the divisor D' orthogonal to it, $D'^2 \leq 0$, with equality if and only if $D' \equiv 0$. Expanding,

$$D'^2 = D^2 - 2b\,(D \cdot e) - 2a\,(D \cdot f) + b^2 e^2 + 2ab\,(e \cdot f) + a^2 f^2 = D^2 - 2ab - 2ab + 2ab = D^2 - 2ab$$

so $D^2 \leq 2ab$, with equality if and only if $D \equiv be + af$, as claimed.

The proof uses only that $e + f$ has positive square. On $\mathbb{P}^1 \times \mathbb{P}^1$ every divisor is linearly equivalent to $be + af$, because $\operatorname{Pic}(\mathbb{P}^1 \times \mathbb{P}^1) = \mathbb{Z}e \oplus \mathbb{Z}f$, so equality holds for all D, as in example 8.2.3(3). For $g \geq 1$ the inequality is strict for the diagonal, since $\Delta \cdot e = \Delta \cdot f = 1$ while $\Delta^2 = 2 - 2g < 2$.

Applied to the divisors $r\Gamma_F + s\Delta$, the inequality becomes a family of quadratic inequalities whose only unknown is N.

Theorem 8.7.12: The Weil Bound for Curves over Finite Fields

Let C_0 be as in box 8.7.1, of genus g, and let $N = \#C_0(\mathbb{F}_q)$. Then

$$\big|N - (q+1)\big| \;\le\; 2g\sqrt{q}$$

***Proof*:**
Step 1: a quadratic form. Fix a closed point P of C, let e and f be the fibers over P, and write $\Gamma = \Gamma_F$. By proposition 8.7.7, applied to F (finite of degree q by proposition 8.7.5(2)) and to id_C, and by proposition 8.7.9,

$$\Gamma\cdot e = \Delta\cdot e = \Delta\cdot f = 1, \qquad \Gamma\cdot f = q, \qquad \Gamma^2 = q(2-2g), \qquad \Delta^2 = 2-2g, \qquad \Gamma\cdot\Delta = N$$

For integers r and s let $D = r\Gamma + s\Delta$. Then $a = D\cdot e = r+s$, $b = D\cdot f = qr+s$, and

$$D^2 = q(2-2g)\,r^2 + 2N\,rs + (2-2g)\,s^2, \qquad 2ab = 2q\,r^2 + 2(q+1)\,rs + 2\,s^2$$

By theorem 8.7.11, $D^2 \le 2ab$. Subtracting and dividing by 2,

$$gq\,r^2 + (q+1-N)\,rs + g\,s^2 \;\ge\; 0 \qquad \text{for all } r,s\in\mathbb{Z} \tag{8.8}$$

Step 2: the discriminant. Put $B = q+1-N$. If $g=0$, then (8.8) at $(r,s)=(1,-B)$ reads $-B^2\ge 0$, so $B=0$. If $g\ge 1$, then at $(r,s)=(B,-2gq)$ it reads

$$gq\,B^2 - 2gq\,B^2 + 4g^3q^2 = gq\,(4g^2q - B^2) \;\ge\; 0$$

and $gq>0$ gives $B^2\le 4g^2q$. In both cases $|N-(q+1)| = |B| \le 2g\sqrt{q}$, as we sought to show.

Applied over the extensions of $\mathbb{F}_q$, theorem 8.7.12 bounds every N_n.

Corollary 8.7.13: The Weil Bound over Every Finite Extension

Let C_0 be as in box 8.7.1, of genus g. For every $n\ge 1$,

$$\big|N_n - (q^n+1)\big| \;\le\; 2g\,q^{n/2}$$

***Proof*:**
Let $C_0' = C_0 \times_{\mathbb{F}_q} \mathbb{F}_{q^n}$, a closed subscheme of $\mathbb{P}^m_{\mathbb{F}_{q^n}}$ whose chart ideals are generated by the same I_j. Its base change $C_0' \times_{\mathbb{F}_{q^n}} k$ is $C_0 \times_{\mathbb{F}_q} k = C$, as $(A_0 \otimes_{\mathbb{F}_q} \mathbb{F}_{q^n}) \otimes_{\mathbb{F}_{q^n}} k \cong A_0 \otimes_{\mathbb{F}_q} k$ on each chart, and k is an algebraic closure of $\mathbb{F}_{q^n}$. So C_0' satisfies the conventions of box 8.7.1 over $\mathbb{F}_{q^n}$, with the same curve C and genus g, and its $\mathbb{F}_{q^n}$-points are the closed points of C with coordinates in $\mathbb{F}_{q^n}$, which number N_n. Theorem 8.7.12 for C_0', with q^n in place of q, gives $|N_n - (q^n+1)| \le 2g\sqrt{q^n}$, as claimed.

In particular $N_n/q^n \to 1$, so for large n the number of points of C_0 over $\mathbb{F}_{q^n}$ is close to that of $\mathbb{P}^1$. Over a single field the theorem forces rational points when q is large compared with g.

Corollary 8.7.14: Curves over Large Finite Fields Have Rational Points

Let C_0 be as in box 8.7.1, of genus g. If $q+1 > 2g\sqrt{q}$, which holds whenever $q \geq 4g^2$, then $C_0(\mathbb{F}_q)$ is nonempty. In particular $C_0(\mathbb{F}_q)$ is nonempty when $g \leq 1$.

***Proof*:**

By theorem 8.7.12, $N \geq q+1-2g\sqrt{q} > 0$. If $q \geq 4g^2$, then $2g\sqrt{q} \leq \sqrt{q}\cdot\sqrt{q} = q < q+1$. For $g=0$ the condition is $q+1>0$, and for $g=1$ it is $(\sqrt{q}-1)^2 > 0$, which holds since $q \geq 2$, completing the proof.

For $g=2$ the condition $q+1 > 4\sqrt{q}$ is $\sqrt{q} > 2+\sqrt{3}$, that is, $q > 7+4\sqrt{3} \approx 13.9$, and the least prime power in that range is 16, so every curve of genus 2 over $\mathbb{F}_q$ with $q \geq 16$ has a rational point. For $g=0$ the theorem gives $N = q+1$ exactly, the number of points of $\mathbb{P}^1_{\mathbb{F}_q}$. For $g=1$ the curve C_0 has a rational point O by corollary 8.7.14, so (C_0, O) is an elliptic curve over $\mathbb{F}_q$, and the theorem is Hasse's bound $|q+1-N| \leq 2\sqrt{q}$ of box 7.6.12, proved in *Elliptic Curves* [15, Hasse's Theorem].

For a plane curve the genus is determined by the degree (proposition 7.2.6), so the bound depends only on the degree and q once the curve is known to be smooth, and smoothness can be checked on the partial derivatives of the equation.

Lemma 8.7.15: A Plane Curve with Nowhere-Vanishing Gradient Is Smooth

Let $G \in k[x,y,z]$ be homogeneous of degree $d \geq 1$, and suppose that $\partial G/\partial x$, $\partial G/\partial y$ and $\partial G/\partial z$ have no common zero in $\mathbb{P}^2(k)$. Then $V_+(G) \subseteq \mathbb{P}^2_k$ is a smooth projective curve of genus $(d-1)(d-2)/2$.

***Proof*:**

Step 1: $V_+(G)$ is integral of dimension 1. A homogeneous $H \in k[x,y,z]$ of positive degree has a zero in $\mathbb{P}^2(k)$: either $H(x,y,0)=0$, and every $[a:b:0]$ is a zero, or $H(x,y,0)$ is a nonzero binary form, which is a product of linear forms over the algebraically closed field k and vanishes at some $[a:b:0]$. Suppose $G = G_1G_2$ with G_1 and G_2 homogeneous of positive degree. They have a common zero: a zero of a common irreducible factor if they have one, and otherwise one of the $\deg G_1 \cdot \deg G_2 > 0$ intersection points counted by Bézout's theorem (*Classical Algebraic Geometry* [22, Bézout's Theorem]). At a common zero, the product rule $\partial G = G_1\,\partial G_2 + G_2\,\partial G_1$ makes all three partial derivatives of G vanish, against the hypothesis. So G is irreducible, (G) is a prime ideal, and $V_+(G)$ is integral. It is nonempty, and by lemma 3.7.11, applied to $\mathbb{P}^2_k$, it has dimension 1. Being closed in $\mathbb{P}^2_k$, it is projective and complete.

Step 2: nonsingularity. Let a be a closed point of $V_+(G)$. Permuting the variables, we may assume a lies in $D_+(z)$, with affine coordinates (a_1,a_2). There $V_+(G)$ is $\operatorname{Spec} k[x,y]/(g)$ with $g(x,y) = G(x,y,1)$, and $\partial g/\partial x(a) = \partial G/\partial x(a_1,a_2,1)$ and $\partial g/\partial y(a) = \partial G/\partial y(a_1,a_2,1)$. These do not both vanish: Euler's identity $x\,\partial G/\partial x + y\,\partial G/\partial y + z\,\partial G/\partial z = dG$, which holds monomial by monomial, evaluated at $(a_1,a_2,1)$, where G vanishes, would otherwise force $\partial G/\partial z(a_1,a_2,1) = 0$ as well. So the Jacobian matrix of g has rank 1 at a. The local ring of $V_+(G)$ at a has dimension 1 (theorem 2.6.12), and $1 = 2-1$, so $V_+(G)$ is regular at a by

proposition 2.7.7. Regularity at every closed point gives regularity everywhere (box 5.5.24), so $V_+(G)$ is a smooth projective curve, and proposition 7.2.6 gives its genus, as claimed.

Corollary 8.7.16: Points on Smooth Plane Curves

Let $G \in \mathbb{F}_q[x,y,z]$ be homogeneous of degree d, suppose that $V_+(G) \subseteq \mathbb{P}^2_k$ is a smooth curve (by lemma 8.7.15, this holds when the partial derivatives of G have no common zero in $\mathbb{P}^2(k)$), and let $C_0 = V_+(G) \subseteq \mathbb{P}^2_{\mathbb{F}_q}$. Then

$$\big|N - (q+1)\big| \;\leq\; (d-1)(d-2)\sqrt{q}$$

***Proof*:**
The scheme C_0 is projective over $\mathbb{F}_q$, and on each chart its base change to k is cut out by the same equation, so $C = V_+(G) \subseteq \mathbb{P}^2_k$, a smooth curve of genus $(d-1)(d-2)/2$ by proposition 7.2.6. Theorem 8.7.12 gives the bound, as claimed.

The bound of theorem 8.7.12 cannot be improved in general: for every square q there is a plane curve that attains the upper bound.

Example 8.7.17: The Hermitian Curve Attains the Bound
Let Q be a power of p, let $q = Q^2$, and let $C_0 = V_+(G) \subseteq \mathbb{P}^2_{\mathbb{F}_q}$ with

$$G = x^{Q+1} + y^{Q+1} + z^{Q+1}$$

the *Hermitian curve*. Since $Q+1 = 1$ in k, the partial derivatives of G are x^Q, y^Q and z^Q, which have no common zero in $\mathbb{P}^2(k)$. By lemma 8.7.15, $C = V_+(G) \subseteq \mathbb{P}^2_k$ is a smooth curve of genus $g = Q(Q-1)/2$, and, as in the proof of corollary 8.7.16, C_0 satisfies the conventions of box 8.7.1.

The count uses the map $u \mapsto u^{Q+1}$ on $\mathbb{F}_q$. The group $\mathbb{F}_q^\times$ is cyclic of order $Q^2-1 = (Q-1)(Q+1)$, so this map sends it onto its unique subgroup of order $Q-1$, which is $\mathbb{F}_Q^\times$, with fibers of $Q+1$ elements. Hence for $c \in \mathbb{F}_Q$ the equation $u^{Q+1} = c$ has one solution $u \in \mathbb{F}_q$ if $c = 0$ and $Q+1$ solutions if $c \neq 0$.

1. Points $[a:b:1]$. Put $s = a^{Q+1}$ and $t = b^{Q+1}$, so that $s,t \in \mathbb{F}_Q$ and $s+t = -1$. The pairs $(s,t) = (0,-1)$ and $(-1,0)$ contribute $Q+1$ solutions (a,b) each, and each of the $Q-2$ values $s \in \mathbb{F}_Q \setminus \{0,-1\}$ contributes $(Q+1)^2$. The total is $2(Q+1) + (Q-2)(Q+1)^2 = (Q+1)(Q^2-Q) = Q^3 - Q$.
2. Points $[a:b:0]$. Here $b \neq 0$, and with $b = 1$ the condition is $a^{Q+1} = -1$, which has $Q+1$ solutions.

So $N = Q^3 - Q + Q + 1 = Q^3 + 1$, while

$$q + 1 + 2g\sqrt{q} = Q^2 + 1 + Q(Q-1)\cdot Q = Q^3 + 1$$

The Hermitian curve attains the upper bound of theorem 8.7.12. For $Q = 2$ it is the plane cubic $x^3 + y^3 + z^3 = 0$ over $\mathbb{F}_4$, of genus 1, with $9 = 4 + 1 + 2\cdot 2$ points, and for $Q = 3$ it is a plane quartic of genus 3 over $\mathbb{F}_9$ with $28 = 9 + 1 + 6\cdot 3$ points.

Taken together for all n, the bounds of corollary 8.7.13 are equivalent to the statement that the $2g$ eigenvalues of the q-power Frobenius on the first ℓ-adic cohomology group of C have absolute value $\sqrt{q}$, which is the Riemann hypothesis of theorem 8.7.18 for a curve. Weil formulated these conjectures in 1949 for every smooth projective variety over $\mathbb{F}_q$.

Theorem 8.7.18: The Weil Conjectures (Dwork, Grothendieck, Deligne)

Let X be a smooth projective variety of dimension n over $\mathbb{F}_q$, let $\overline{X} = X \times_{\mathbb{F}_q} \overline{\mathbb{F}}_q$, let ℓ be a prime not dividing q, and let F be the endomorphism of $H^i = H^i(\overline{X}, \mathbb{Q}_\ell)$ induced by the q-power Frobenius morphism of $\overline{X}$. Write $Z(X,t) = \exp\left(\sum_{m=1}^{\infty} \frac{\#X(\mathbb{F}_{q^m})}{m} t^m\right)$.

1. *Rationality.* $Z(X,t) = \prod_{i=0}^{2n} P_i(t)^{(-1)^{i+1}}$ with $P_i(t) = \det(1 - tF \,|\, H^i)$. In particular $Z(X,t) \in \mathbb{Q}(t)$.
2. *Functional equation.* $Z(X, q^{-n}t^{-1}) = \pm\, q^{nE/2}\, t^E\, Z(X,t)$, where $E = \sum_i (-1)^i \dim H^i$.
3. *Riemann hypothesis.* The eigenvalues of F on H^i are algebraic numbers all of whose complex conjugates have absolute value $q^{i/2}$.

Dwork proved rationality in 1960 by p-adic analysis, and Grothendieck proved rationality and the functional equation in 1964-1965 from the Lefschetz trace formula and Poincaré duality in étale cohomology. Deligne proved the Riemann hypothesis in 1974 [25]. *Étale Cohomology* [16] develops étale cohomology and gives Deligne's proof.

8.7.1 Arithmetic Surfaces and Integral Points

The theory of surfaces has an arithmetic counterpart over the rings of integers of number fields. An *arithmetic surface* is a 2-dimensional integral Noetherian scheme $\mathcal{X}$ equipped with a proper flat morphism

$$\pi \colon \mathcal{X} \to \operatorname{Spec} \mathcal{O}_K$$

where $\mathcal{O}_K$ is the ring of integers of a number field K. The generic fiber $\mathcal{X}_\eta = \mathcal{X} \times_{\mathcal{O}_K} K$ is a curve over K. The fiber $\mathcal{X}_\mathfrak{p} = \mathcal{X} \times_{\mathcal{O}_K} \mathcal{O}_K/\mathfrak{p}$ over a maximal ideal $\mathfrak{p}$ is a proper one-dimensional scheme over the finite field $\mathcal{O}_K/\mathfrak{p}$, which may be singular, reducible or non-reduced. An arithmetic surface is thus a family of curves over $\operatorname{Spec} \mathcal{O}_K$: the generic fiber is a curve over a number field, and the closed fibers are its reductions modulo the primes of $\mathcal{O}_K$.

Example 8.7.19: Arithmetic Surfaces in Practice

1. The projective line $\mathcal{X} = \mathbb{P}^1_{\mathbb{Z}}$, with $\pi \colon \mathbb{P}^1_{\mathbb{Z}} \to \operatorname{Spec} \mathbb{Z}$, has generic fiber $\mathbb{P}^1_{\mathbb{Q}}$ and fiber $\mathbb{P}^1_{\mathbb{F}_p}$ over (p).
2. Let $E \colon y^2 = x^3 - x$ over $\mathbb{Q}$, and let $\mathcal{E} \subseteq \mathbb{P}^2_{\mathbb{Z}}$ be the closed subscheme defined by $y^2 z = x^3 - xz^2$. The polynomial is irreducible and its coefficients have no common prime factor, so $\mathcal{E}$ is integral and $\mathbb{Z}[x,y,z]/(y^2z - x^3 + xz^2)$ has no $\mathbb{Z}$-torsion, which makes $\mathcal{E}$ flat over $\operatorname{Spec} \mathbb{Z}$. It is an arithmetic surface with generic fiber E. Its fiber over an odd prime p is the curve $y^2z = x^3 - xz^2$ over $\mathbb{F}_p$, an elliptic curve since the discriminant 64 of the Weierstrass equation is a unit modulo p. Its fiber over 2 is the curve $y^2 z = x(x+z)^2$ over $\mathbb{F}_2$, which is singular at the point $(1:0:1)$, reflecting the bad reduction of E at 2. The Néron model of E is a

different object: it is smooth over $\mathbb{Z}$, so none of its fibers is singular, and since E has bad reduction at 2 it is not proper over $\mathbb{Z}$.

Intersection theory on arithmetic surfaces is *Arakelov theory*, introduced by Arakelov in 1974. The scheme $\operatorname{Spec} \mathcal{O}_K$ has no points corresponding to the archimedean places of K, and Arakelov theory adds data at these places: an Arakelov divisor carries a real coefficient for each archimedean place, and line bundles carry Hermitian metrics on the Riemann surfaces $\mathcal{X}_\sigma(\mathbb{C})$ for the embeddings $\sigma\colon K \to \mathbb{C}$. The resulting intersection pairing

$$\widehat{\operatorname{Div}}(\mathcal{X}) \times \widehat{\operatorname{Div}}(\mathcal{X}) \to \mathbb{R}$$

is the arithmetic counterpart of the intersection pairing on a surface, and it takes values in $\mathbb{R}$ because of the archimedean contributions. Arakelov proved an arithmetic adjunction formula, Faltings an arithmetic Riemann-Roch theorem, and Faltings and Hriljac the arithmetic Hodge index theorem. Gillet and Soulé proved an arithmetic Riemann-Roch theorem in all dimensions. Vojta's 1991 proof of the Mordell conjecture (Faltings' theorem that a curve of genus at least 2 over a number field has finitely many rational points) uses Arakelov intersection theory and the arithmetic Riemann-Roch theorem of Gillet and Soulé. Faltings' original proof of 1983 used different methods.

Over $\mathbb{F}_q$ every curve has finitely many points, and the genus bounds their number. The Euler characteristic of the associated complex Riemann surface determines whether the set of integral points becomes infinite after the field and the allowed denominators are enlarged.

8.7.20 Integral Points and the Euler Characteristic Let K be a number field, S a finite set of places of K containing the archimedean ones, and $\mathcal{O}_{K,S}$ the ring of S-integers. Let $C^\circ \subseteq \mathbb{A}^m_K$ be a smooth geometrically integral affine curve cut out by polynomials with coefficients in $\mathcal{O}_{K,S}$, let C be the smooth projective curve containing C° as a dense open subscheme, of genus g, and let $s \geq 1$ be the number of points of $C \setminus C^\circ$ over $\overline{K}$. The *S-integral points* of C° are the points of $C^\circ(K)$ whose coordinates lie in $\mathcal{O}_{K,S}$. For a smooth projective geometrically integral curve C over K, put $C^\circ = C$ and $s = 0$, and let all of $C(K)$ play the role of the S-integral points. For every embedding $K \hookrightarrow \mathbb{C}$, the space $C^\circ(\mathbb{C})$ is a compact Riemann surface of genus g with s points removed, and its topological Euler characteristic is

$$\chi = 2 - 2g - s$$

If $\chi \geq 0$, the set of S-integral points can be finite, and it becomes infinite once K is replaced by a suitable finite extension and S by a suitable larger set. If $\chi < 0$, it is finite for every number field and every S.

1. $\chi > 0$, that is, $g = 0$ and $s \leq 1$. For $C = \mathbb{P}^1_K$ the set $C(K)$ is infinite, and the S-integral points of the affine line $\mathbb{A}^1_K = \mathbb{P}^1_K \setminus \{\infty\}$ form the infinite set $\mathcal{O}_{K,S}$.
2. $\chi = 0$, that is, $(g, s) = (0, 2)$ or $(1, 0)$. The S-integral points of $\mathbb{G}_m = \mathbb{P}^1_K \setminus \{0, \infty\}$, embedded in $\mathbb{A}^2_K$ as $xy = 1$, form the unit group $\mathcal{O}^\times_{K,S}$, a finitely generated abelian group of rank $\#S - 1$ by Dirichlet's unit theorem in its S-unit form (for S the set of archimedean places, *Algebraic Number Theory* [7, Dirichlet Unit Theorem]). It is finite for $K = \mathbb{Q}$ and S the archimedean place, where it is $\mathbb{Z}^\times = \{\pm 1\}$. For an elliptic curve E over K, the group $E(K)$ is finitely generated (*Elliptic Curves* [15, The Mordell-Weil Theorem]), and its rank can be 0, while it is positive over a suitable finite extension of K.
3. $\chi < 0$. If $s \geq 1$, the set of S-integral points of C° is finite. Siegel proved this for integral points in 1929 (reprinted in [67]), and Lang proved it for S-integral points in 1960 [46], after Mahler's case of genus 1 over $\mathbb{Q}$ (1934) [48]. For an elliptic curve minus its origin, given by a Weierstrass equation, this is *Elliptic Curves* [15, Siegel's Theorem (S-Integral Version)]. If $s = 0$, then $g \geq 2$, and $C(K)$ is finite. Mordell conjectured this in 1922 [54] and Faltings proved it in 1983 [27].

Exercise 8.7.1

1. Let C be a curve of genus g and $\Delta \subseteq C \times C$ the diagonal, with ideal sheaf $\mathcal{I}_\Delta$. Show that the conormal sheaf $\mathcal{I}_\Delta/\mathcal{I}^2_\Delta$ corresponds to $\Omega_{C/k}$ under the isomorphism $C \cong \Delta$ given by the diagonal morphism (definition 5.5.14), and use proposition 8.2.6 to give a second proof that $\Delta^2 = 2 - 2g$.
2. Let $\sigma \neq \mathrm{id}_C$ be an automorphism of a curve C of genus g.
    1. Apply theorem 8.7.11 to $D = \Gamma_\sigma + \Delta$ to show that $\Gamma_\sigma \cdot \Delta \leq 2g + 2$, and deduce that σ has at most $2g + 2$ fixed points.
    2. Let C be hyperelliptic, with $\mathrm{char} k \neq 2$, and let ι be its hyperelliptic involution, whose fixed points are the $2g + 2$ Weierstrass points (definition 7.5.3). Show that $\Gamma_\iota \cdot \Delta = 2g + 2$ and that Γ_ι meets Δ transversally.

3. Let $q \equiv 2 \pmod 3$ and $C_0 = V_+(x^3 + y^3 + z^3) \subseteq \mathbb{P}^2_{\mathbb{F}_q}$. Show that C_0 satisfies the conventions of box 8.7.1 with $g = 1$, and that $N = q + 1$. (*Hint:* $u \mapsto u^3$ is a bijection of $\mathbb{F}_q$.) Compare with theorem 8.7.12.

4. Let C_0 be as in box 8.7.1. Show that for every sufficiently large n there is a closed point of C whose homogeneous coordinates can be chosen in $\mathbb{F}_{q^n}$ but not in $\mathbb{F}_{q^d}$ for any proper divisor d of n. (*Hint:* compare N_n with $\sum_{d \mid n,\, d<n} N_d$ using corollary 8.7.13.)

5. Let $C_0 = V_+(G) \subseteq \mathbb{P}^2_{\mathbb{F}_q}$ be as in corollary 8.7.16, of degree d, and suppose that $C_0(\mathbb{F}_q)$ is all of $\mathbb{P}^2(\mathbb{F}_q)$. Show that $(d-1)(d-2) \geq q^{3/2}$, and deduce that no smooth plane cubic over $\mathbb{F}_q$ passes through every point of $\mathbb{P}^2(\mathbb{F}_q)$.

9

What Next?

Sheaves, schemes, quasi-coherent sheaves and their cohomology, built in chapters 1 to 3, 5 and 6, were applied in chapters 7 and 8 to curves and surfaces over an algebraically closed field. Those two chapters worked under four restrictions. The varieties studied had dimension one or two, apart from Hironaka's threefold (section 8.4.1). No moduli space was constructed: $\mathcal{M}_g$ and $\overline{\mathcal{M}}_g$ are described without proof (boxes 7.2.8 and 7.7.5), and families of curves entered through semicontinuity (section 7.7.1) and through the fibrations of ruled, elliptic and arithmetic surfaces. The ground field was algebraically closed, with sections 7.7.2 and 8.7 the main exceptions. The cohomology computed was the cohomology of coherent sheaves, and the singular cohomology of $X(\mathbb{C})$ entered the two chapters only through cited results such as theorem 8.1.6.

Intersection theory and Riemann-Roch in every dimension. On a smooth variety X of dimension n, the classes of subvarieties of every codimension modulo rational equivalence form the Chow ring $A^*(X)$, and the intersection pairing of a surface is its product $A^1(X) \times A^1(X) \to A^2(X)$ followed by the degree (box 8.2.10). Riemann-Roch generalizes to the Hirzebruch-Riemann-Roch theorem, which expresses $\chi(\mathscr{E})$ for a vector bundle $\mathscr{E}$ through the Chern classes of $\mathscr{E}$ and of the tangent bundle of X. For line bundles on a curve and on a surface it gives theorem 7.2.10 and theorem 8.3.1, and the proof of Noether's formula (theorem 8.3.3) applies it to $\mathcal{O}_X$. The Grothendieck-Riemann-Roch theorem extends it to a proper morphism. *Intersection Theory* [19] constructs the Chow ring and Chern classes and proves both theorems. Fulton's *Intersection Theory* is the standard reference, and Eisenbud and Harris's *3264 and All That* develops the subject through enumerative problems.

Birational geometry and the minimal model program. On a surface, contracting (-1)-curves (theorem 8.4.8) reaches a minimal model, which is unique when $\kappa \geq 0$ (theorem 8.6.3), and every surface in the process is smooth. From dimension three on, the contraction of a K_X-negative extremal ray can have a singular image, and when it contracts no divisor the canonical divisor of the image is not $\mathbb{Q}$-Cartier, so the program admits mild singularities and continues by a flip (box 8.4.10). Flips exist in every dimension (Birkar, Cascini, Hacon and McKernan, 2010), their termination is known in dimension three and open in general, and the abundance conjecture of box 8.6.7 is open from

dimension four on. *Birational Geometry* [8] develops the cone and contraction theorems, flips, and the vanishing theorems used to prove them (box 8.6.10). Kollár and Mori's *Birational Geometry of Algebraic Varieties* is the standard text, and Debarre's *Higher-Dimensional Algebraic Geometry* treats the rational curves that span the extremal rays of the cone theorem.

Toric varieties. The surfaces $\mathbb{P}^2$, $\mathbb{P}^1 \times \mathbb{P}^1$ and $\mathbb{F}_n$ of example 8.1.2 each contain the torus $\mathbb{G}_m^2$ as a dense open subset, and the action of $\mathbb{G}_m^2$ on itself by translation extends to the whole surface. Varieties with this property are *toric*. They exist in every dimension, and a normal toric variety of dimension n is determined by a fan of cones in $\mathbb{R}^n$. Its divisors, Picard group, intersection numbers and cohomology are computed from the fan, and so is projectivity: a complete toric variety is projective exactly when its fan is the normal fan of a polytope, and, as with Hironaka's threefold (section 8.4.1), completeness does not imply projectivity: some complete toric threefolds are not projective. *Toric Geometry* [21] develops the theory. Fulton's *Introduction to Toric Varieties* is a short introduction, and Cox, Little and Schenck's *Toric Varieties* is a comprehensive account.

Abelian varieties and Jacobians. Theorem 7.6.3 identified an elliptic curve E with $\mathrm{Pic}^0(E)$. For a curve X of genus $g \geq 2$, the group $\mathrm{Pic}^0(X)$ is the group of points of the Jacobian $\mathrm{Jac}(X)$, a projective group variety of dimension g into which X embeds by the Abel-Jacobi map, and by the Torelli theorem $\mathrm{Jac}(X)$ with its theta divisor determines X. The complete connected group varieties are the *abelian varieties*. They are commutative and projective, and those of dimension 1 are the elliptic curves. *Abelian Varieties* [5] develops them over an arbitrary field, with the theorem of the cube, the dual abelian variety, polarizations, Jacobians and the Torelli theorem. Mumford's *Abelian Varieties* is the standard text, and Birkenhake and Lange's *Complex Abelian Varieties* treats the theory over $\mathbb{C}$.

Moduli of curves and stacks. Curves of genus $g \geq 2$ depend on $3g-3$ parameters (box 7.7.4), and elliptic curves are classified by the j-line (box 7.6.6). Some curves of every genus have nontrivial automorphisms, so no scheme represents the moduli functor of curves of genus g, and for $g \geq 2$ the moduli problem is a Deligne-Mumford stack $\mathcal{M}_g$ (box 7.2.8). Its coarse moduli space is not proper, and the Deligne-Mumford compactification $\overline{\mathcal{M}}_g$ adds the stable curves of box 7.7.5. *Moduli, Deformation Theory, and Stacks* [20] constructs Hilbert schemes, Picard schemes and $\overline{\mathcal{M}}_g$, and develops the deformation theory behind the count $3g-3$. Harris and Morrison's *Moduli of Curves* is the standard introduction, and Arbarello, Cornalba and Griffiths's *Geometry of Algebraic Curves, Volume II* is the comprehensive reference.

The arithmetic of elliptic curves. Over a field that is not algebraically closed, a curve of genus 1 can have no rational point, and it is then a torsor under its Jacobian E, with a class in $H^1(k, E)$ (section 7.7.2). Over a number field K the group $E(K)$ is finitely generated (example 7.6.13), the Birch and Swinnerton-Dyer conjecture predicts its rank from the L-function of E, and the Galois action on the Tate module (box 7.6.12) is a basic example of an ℓ-adic Galois representation. *Elliptic Curves* [15] develops the Mordell-Weil and Siegel theorems, Selmer and Shafarevich-Tate groups, and complex multiplication. Silverman's *The Arithmetic of Elliptic Curves* is the standard text, and his *Advanced Topics in the Arithmetic of Elliptic Curves* continues it.

Étale cohomology and the Weil conjectures. Section 8.7 bounded the number of points of a curve over $\mathbb{F}_q$ by the Hodge index theorem on the product of the curve with itself, and the bounds over all the extensions $\mathbb{F}_{q^n}$ together are equivalent to the Riemann hypothesis of the Weil conjectures (theorem 8.7.18) for curves. For a variety X of higher dimension, an argument of the same kind on $X \times X$ requires Grothendieck's standard conjectures, which are open. The Weil conjectures were proved with ℓ-adic étale cohomology, which attaches to a variety over $\overline{\mathbb{F}}_q$ cohomology groups with coefficients in $\mathbb{Q}_\ell$ and a Lefschetz trace formula for the Frobenius morphism. *Étale Cohomology* [16]

develops the étale site, the étale fundamental group of box 7.4.10, ℓ-adic cohomology and Deligne's proof. Milne's *Étale Cohomology* is the standard reference, and Freitag and Kiehl's *Étale Cohomology and the Weil Conjecture* is organized around Deligne's theorem.

Hodge theory and the Hodge conjecture. Over $\mathbb{C}$ the Néron-Severi group of a surface is identified with the integral cohomology classes of type $(1,1)$ (theorem 8.1.6), and the Hodge decomposition (corollary 6.9.10) gives the Hodge diamond of a surface (example 8.3.6). On a compact Kähler manifold of any dimension the Hard Lefschetz theorem and the Hodge-Riemann bilinear relations hold, and on a surface the second gives the Hodge index theorem over $\mathbb{C}$ (box 8.3.14). The Hodge conjecture asks whether every rational cohomology class of type (p,p) on a smooth projective complex variety is a rational combination of classes of subvarieties of codimension p. Its case $p = 1$ is theorem 8.1.6, and it is open in general. *Hodge Theory* [17] develops the Hodge decomposition, the Hard Lefschetz theorem and the Hodge-Riemann relations, and proves the Lefschetz $(1,1)$-theorem. Voisin's *Hodge Theory and Complex Algebraic Geometry* is the standard modern text, and Griffiths and Harris's *Principles of Algebraic Geometry* the classical one.

Derived categories of coherent sheaves. Sheaf cohomology consists of the right derived functors of global sections (chapter 6), and for a smooth projective variety Serre duality, used in the proofs of theorem 7.2.10 and theorem 8.3.1, is the case of the structure morphism $X \to \operatorname{Spec} k$ of Grothendieck duality, a statement about derived categories. The bounded derived category $D^b(\mathbf{Coh}X)$ of coherent sheaves is an invariant of X, and it does not always determine X. By Mukai's theorem an abelian variety and its dual, which need not be isomorphic, have equivalent derived categories through the Fourier-Mukai transform. By the Bondal-Orlov theorem a smooth projective variety whose canonical divisor is ample or anti-ample is determined by its derived category. Derived categories of abelian categories are constructed in *Homological Algebra* [18], and *Derived Algebraic Geometry* [14] develops derived categories of coherent sheaves, Fourier-Mukai transforms and the Bondal-Orlov theorem. Huybrechts's *Fourier-Mukai Transforms in Algebraic Geometry* is the standard introduction.

Index

$\mathcal{O}_X$-module, 412
Étale Morphism, 183
    generic, 448
    smooth of relative dimension zero, 184
Čech Cohomology, 584
    degree-one comparison, 591
    Leray's acyclic covering theorem, 588
    of a space, 590

Acyclic Covering, 588
Affine Group Scheme
    closed subgroup schemes, 386
Affine Morphism
    stability, 263
Affine Scheme, 92
Affine-local Property, 121
Affinization, 136
Algebraic de Rham Theorem
    affine case, 675
Algebraic Index Inequality, 756
algebraizable, 572
Ample
    affine non-vanishing loci, 478
    invertible sheaf, 474
    very ample power, 480
Ample Divisor
    positivity on curves, 745

Base Change, 231
    Along the Algebraic Closure, 272
    of a relative Cartier divisor, 517
Base Locus, 534
    of a pair of sections, 762
Bertini's Theorem, 327
    Irreducibility II, 351
    Irreducibility of General Sections, 327
    Irreducibility of Hyperplane Preimages, 339
    Smoothness of General Sections, 333
Birational Map, 315
    isomorphism on dense opens, 543
Birational Transform, 531
    exceptional terms, 544
    functoriality, 532
    of a linear system, 542
Blow-up, 758
    along a nonsingular centre, 760
    as closure of a graph, 762
    intersection theory on, 765
Bott Vanishing, 607
Branch Divisor, 714

Canonical Divisor, 696
Cartier Divisor
    base change of, 517
    families of Weil divisors as, 529
    fibers of a relative divisor, 515
    fiberwise criterion for, 516
    relative effective, 514
Castelnuovo's Contractibility Criterion, 768
Castelnuovo's Rationality Criterion, 775
Castelnuovo-Severi Inequality, 791
Chevalley's Theorem, 442

Closed Immersion
local on the target, 479
Closed Subscheme
regularly embedded, 433
Compatible Germs, 15
Conormal Sheaf, 434
Constructible Set, 439
Locus of Irreducible Fibers, 346
stability under generization, 656
Curve, 682
(−1)-curve, 768
complete is projective, 690
function field of, 686
Joining Two Closed Points, 328, 329
projective completion of, 684

Degree
Morphism of Curves, 691
Projective Hypersurface, 468
Delta invariant, 700
Density
fiberwise, 444
Descent
of properties of morphisms, 376
Differentials
cohomology of $\Omega^p(m)$ on $\mathbb{P}^n$, 607
Directed Set, 360
Distinguished Open Set, 73
Divisor
birational transform, 531
Extension by Closure, 506
family is relative Cartier, 529
family of Weil divisors, 528
fibers of a family over a dense open, 537
Sheaf Associated to a Weil Divisor, 501
Domain of Definition, 399
Dominant Morphism, 220
Dual Sheaf
of an exact sequence, 562

Effective Cartier Divisor
relative to a base, 514
Effective Divisor
multiples of a divisor of positive square, 754
Elliptic Curve, 724
equidimensional, 168
Essentially Smooth, 378
Euler Sequence, 560
tangent form, 562
Evaluation on Locally Ringed Space, 54
Exactness of Global Section, 420
Exceptional Divisor, 758

Factorization of Birational Morphisms, 769
Family of Schemes, 653
Fibers of Scheme, 234
Filtered Colimit
of Sets and Rings, 361
Finite Morphism
closed maps, 270
is projective, 492
stability, 270
Finite Presentation
stability, 276
versus finite type, 276
Finitely Presented Algebra
compactness, 368
Finitely Presented Chart, 369
Finiteness of Higher Direct Images
proper case, 638
Flasque Sheaf, 579
Flatness
affine-local, 654
and non-zero-divisors, 514
criterion via twisted global sections, 668
dimension criterion, 449
lifts generizations, 656
local criterion and relative divisors, 516
openness of flat morphisms, 657
openness over Noetherian bases, 658
stability under composition and base change, 655
transitivity and base change, 654
Frobenius Morphism
q-power, 784
fixed points and degree, 785
fixed points as an intersection number, 790
Function Field
of a curve, 686
subfield of q-th powers, 785

General Member, 331
Generic Finiteness, 448
Generic Freeness, 440
fiberwise nonzerodivisors, 443
Generic Point, 72

Generization, 71
Genus
    Curve, 696
Geometric Genus, 752
Geometrically
    Connected, 273
    Integral, 234
Geometrically Integral, 234
Global Section Functor, 578
Graded Algebra
    quasi-coherent, 486
Graph of a Morphism
    intersection numbers, 788
    local equation, 789
Grassmannian
    Linear Subspaces Inside a Closed Set, 344
    Of a Linear System, 344
    tautological sequence, 492
Grothendieck's Comparison Theorem, 675
Group Scheme
    closed subgroup scheme, 386
    Geometrically Integral, 383
    Image of a Homomorphism, 452
    projective linear, 384
    Translates of an Open Subset, 405

Hermitian Curve, 794
Hirzebruch Surface, 773
Hodge Index Theorem, 754
Homomorphism of Group Schemes
    Closed Image, 452
    From a Rational Map, 407
Hyperelliptic Curve, 721
Hyperelliptic Involution, 722
Hyperplane, 221
    General, 331
Hypersurface
    Sections of a Variety, 325

Incidence Correspondence, 535
Injective Presheaf Morphism, 22
Injective Sheaf Morphism, 22
Integral Morphism
    closed maps, 269
Integral Points
    and the Euler characteristic, 796
Intersection Form
    signature, 755
Intersection Multiplicity, 742
Intersection Number, 742
    properties, 742
Inverse System
    of Schemes, 360
Irreducible
    images and closures, 79
    topological space, 78
Irregularity, 752
Isogeny, 730
Isomorphism
    of Ringed Spaces, 55
    of Schemes, 102

Jacobson Ring, 321
Jacobson Scheme, 321
Jacobson Topology, 321
Jet
    conditions after re-embedding, 612
    of a section of an invertible sheaf, 609
    on projective space, 609
    separation by large twists, 610

Kodaira Dimension, 776

Lefschetz Theorem on $(1,1)$-Classes, 740
Leray's Theorem, 588
Limit
    Base Change, 365
    morphisms from a limit, 370
    of Schemes, 363
Line Bundle, 428
Linear Space, 221
Linear System
    birational transform of, 542
    Connectedness of Members, 348
    exceptional terms of a transform, 544
    incidence correspondence of, 535
    irreducibly parametrized, 547
    linear span in, 547
    normalized, 540
    of Weil divisors, 534
    transform of an irreducibly parametrized system, 548
    without fixed components, 540
Local Complete Intersection Morphism, 436
Local Scheme, 114
Locally Ringed Space, 54
Locally Split Exact Sequence, 562

Miracle Flatness Theorem, 449
Morphism
of Schemes, 102
nonconstant, 686
of Presheaves, 20
of Sheaves, 20
morphism
Ringed Space, 53
Multiplicity
of a curve at a point, 765

Néron-Severi Group, 739
Néron-Severi Theorem, 740
Nakai-Moishezon Criterion, 748
Noether's Formula, 751
Nonzerodivisor
on nearby fibers, 443
Normal Bundle, 434
Normal Scheme
regular in codimension one, 498
Normalization
of a projective scheme, 495
of a Variety, 280
projectivity of, 495
Numerical Equivalence, 746

Picard Number, 739
Plane Curve
smoothness criterion, 793
Plurigenus, 752
Poincaré Lemma
holomorphic, 673
Polar Locus, 401
Presheaf
Axiomatic, 4
Functorial, 6
Presheaf Pushforward, 43
Principal Open Set, 73
Product of Curves, 787
Proj
of a finite algebra, 492
relative, 487
relative, base change, 489
Projective Bundle, 489
tautological quotient, 490
Projective Completion
of a smooth curve, 684
Projective Hypersurface, 221
Projective Linear Group Scheme, 384
Projective Morphism
charts of, 479
from a finite morphism, 492
Projective Scheme
complete curves, 690
from proper and ample, 481
Projective Space – as a Set, 138
Proper Transform, *see* Birational Transform
Pullback
Scheme-Theoretic Pullback, 231

Quasi-Coherent Sheaf
graded algebra, 486
Quasi-compact
and quasi-separated, 365
open of a limit, 367
Quasicompact, 76
Quasiprojective Scheme, 142
affine schemes of finite type, 467

Ramification Divisor, 714
Ramification Index, 712
Rational Homomorphism, 407
Rational Map, 315
Domain of Definition, 399
Extension to a Proper Target, 400
from a Homomorphism of Function Fields, 317
Multiplicative, 407
Rational Surface, 774
Reduced Induced Structure, 215
Rees Algebra, 758
Refinement of a Covering, 590
Regular Embedding, 433
conormal sequence of a chain, 435
Regular Function, 93
Relative Proj, 487
Residue Field, 115
Riemann Hypothesis
for curves over finite fields, 792
Riemann-Roch Theorem
for surfaces, 750
Ringed Space, 52
Ringed Space Morphism, 53
Ruled Surface, 772

Scheme, 102
S-Scheme, 108, 207

Factorial, 160
Finite, 108
Finite type, 108
Integral, 155
Jacobson, 321
Noetherian, 147
Normal, 159
Open Subscheme, 127
Quasiprojective, 142
Quasiseparated, 151
Reduced, 153
Separated, 283
Vanishing, 430
Scheme Morphism
Affine, 261
Finite, 265
Finite Type, 210, 274
Finite Type over a Noetherian base, 305
Formal Completion, 571
Integral, 268
Locally Finite Presentation, 275
Locally Finite Type, 210, 273
Quasicompact, 260
Quasifinite, 278
Quasiseparated, 260
Section, 423
Self-Intersection
negative, 756
Serre Vanishing
and jet separation, 610
on projective space over a Noetherian ring, 667
Sheaf
Axiomatic Definition, 8
Extension by Zero, 49
Functorial Definition, 9
Inverse Image Sheaf, 45
Lower Shriek, 49
Pullback, 45
Pushforward, 43
Quasi-coherent Sheaf, 415
Sections with Support, 51
Sheaf Associated to Graded Module, 456
Sheaf Associated to Module, 415
Structure Sheaf, 52
Unit Sheaf, 53
Upper Shriek, 51
Sheaf Cohomology, 578
Sheaf Hom, 35
Sheaf on a Base, 16
Singular Locus
codimension of, 498
Smooth Morphism
implies flat, 180
stability, 179
standard smooth chart, 375
Soft Sheaf, 592
Specialization, 71
via discrete valuation rings, 306
Spectra Map, 80
Spectrum, 59
of a Filtered Colimit, 362
Spreading Out
fiberwise density, 444
Stalk
Equivalence Relationship, 6
Functorial, 6
Stalk-local Property, 121
Strict Transform, 765
Support of Section, 15
Surface, 738
Surjective Presheaf Morphism, 24
Surjective Sheaf Morphism, 29

Tangent Sheaf
of projective space, 562
Theorem of the Base, 740

Universal Family
Of Curve Sections, 344
Universally Closed Morphism
detected by affine spaces, 308
Unramified Morphism, 183
fibre structure, 182

Valuation Criterion
Properness via discrete valuation rings, 310
Separatedness via discrete valuation rings, 307
Value of an Element [on a spectrum], 61
Vanishing
of H^2 on a surface, 753
Variety, 320
classical, iii
Complete, 329
Curves Joining Two Points, 328
Vector Bundle, 423

associated projective bundle, 489
Veronese Embedding, 223
Very Ample Sheaf
from an ample sheaf, 480

Weierstrass Point
of a hyperelliptic curve, 722
Weighted Projective Space, 223
Weil Bound, 792
Weil Divisor
family of, 528
linear system of, 534
Weil's Extension Theorem, 402

Z-valued Points, 243
Zariski Topology, 68
Zariski's Main Theorem, 278
Birational Form, 279
Zariski-local Property, 121

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